

1 Introduction to Differential Equations

Exercises 1.1

1. Second-order; linear.
2. Third-order; nonlinear because of $(dy/dx)^4$.
3. The differential equation is first-order. Writing it in the form $x(dy/dx) + y^2 = 1$, we see that it is nonlinear in y because of y^2 . However, writing it in the form $(y^2 - 1)(dx/dy) + x = 0$, we see that it is linear in x .
4. The differential equation is first-order. Writing it in the form $u(dv/du) + (1 + u)v = ue^u$ we see that it is linear in v . However, writing it in the form $(v + uv - ue^u)(du/dv) + u = 0$, we see that it is nonlinear in u .
5. Fourth-order; linear
6. Second-order; nonlinear because of $\cos(r + u)$
7. Second-order; nonlinear because of $\sqrt{1 + (dy/dx)^2}$
8. Second-order; nonlinear because of $1/R^2$
9. Third-order; linear
10. Second-order; nonlinear because of $\dot{x}^2$
11. From $y = e^{-x/2}$ we obtain $y' = -\frac{1}{2}e^{-x/2}$. Then $2y' + y = -e^{-x/2} + e^{-x/2} = 0$.
12. From $y = \frac{6}{5} - \frac{6}{5}e^{-20t}$ we obtain $dy/dt = 24e^{-20t}$, so that

$$\frac{dy}{dt} + 20y = 24e^{-20t} + 20\left(\frac{6}{5} - \frac{6}{5}e^{-20t}\right) = 24.$$

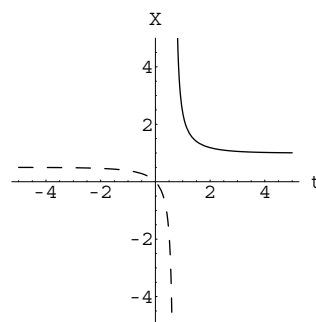
13. From $y = e^{3x} \cos 2x$ we obtain $y' = 3e^{3x} \cos 2x - 2e^{3x} \sin 2x$ and $y'' = 5e^{3x} \cos 2x - 12e^{3x} \sin 2x$, so that $y'' - 6y' + 13y = 0$.
14. From $y = -\cos x \ln(\sec x + \tan x)$ we obtain $y' = -1 + \sin x \ln(\sec x + \tan x)$ and $y'' = \tan x + \cos x \ln(\sec x + \tan x)$. Then $y'' + y = \tan x$.
15. Writing $\ln(2X - 1) - \ln(X - 1) = t$ and differentiating implicitly we obtain

$$\frac{2}{2X - 1} \frac{dX}{dt} - \frac{1}{X - 1} \frac{dX}{dt} = 1$$

$$\left(\frac{2}{2X - 1} - \frac{1}{X - 1}\right) \frac{dX}{dt} = 1$$

$$\frac{2X - 2 - 2X + 1}{(2X - 1)(X - 1)} \frac{dX}{dt} = 1$$

$$\frac{dX}{dt} = -(2X - 1)(X - 1) = (X - 1)(1 - 2X).$$



Exponentiating both sides of the implicit solution we obtain

$$\frac{2X - 1}{X - 1} = e^t \implies 2X - 1 = Xe^t - e^t \implies (e^t - 1) = (e^t - 2)X \implies X = \frac{e^t - 1}{e^t - 2}.$$

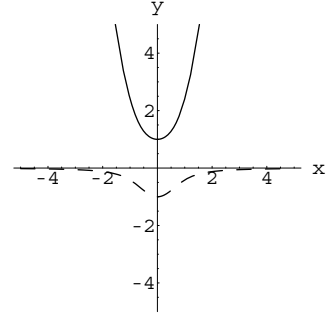
Solving $e^t - 2 = 0$ we get $t = \ln 2$. Thus, the solution is defined on $(-\infty, \ln 2)$ or on $(\ln 2, \infty)$. The graph of the solution defined on $(-\infty, \ln 2)$ is dashed, and the graph of the solution defined on $(\ln 2, \infty)$ is solid.

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16. Implicitly differentiating the solution we obtain

$$\begin{aligned} -2x^2 \frac{dy}{dx} - 4xy + 2y \frac{dy}{dx} &= 0 \implies -x^2 dy - 2xy dx + y dy = 0 \\ \implies 2xy dx + (x^2 - y) dy &= 0. \end{aligned}$$

Using the quadratic formula to solve $y^2 - 2x^2y - 1 = 0$ for y , we get $y = (2x^2 \pm \sqrt{4x^4 + 4})/2 = x^2 \pm \sqrt{x^4 + 1}$. Thus, two explicit solutions are $y_1 = x^2 + \sqrt{x^4 + 1}$ and $y_2 = x^2 - \sqrt{x^4 + 1}$. Both solutions are defined on $(-\infty, \infty)$. The graph of $y_1(x)$ is solid and the graph of y_2 is dashed.



17. Differentiating $P = c_1 e^t / (1 + c_1 e^t)$ we obtain

$$\begin{aligned} \frac{dP}{dt} &= \frac{(1 + c_1 e^t) c_1 e^t - c_1 e^t \cdot c_1 e^t}{(1 + c_1 e^t)^2} \\ &= \frac{c_1 e^t}{1 + c_1 e^t} \frac{[(1 + c_1 e^t) - c_1 e^t]}{1 + c_1 e^t} = P(1 - P). \end{aligned}$$

18. Differentiating $y = e^{-x^2} \int_0^x e^{t^2} dt + c_1 e^{-x^2}$ we obtain

$$y' = e^{-x^2} e^{x^2} - 2xe^{-x^2} \int_0^x e^{t^2} dt - 2c_1 x e^{-x^2} = 1 - 2xe^{-x^2} \int_0^x e^{t^2} dt - 2c_1 x e^{-x^2}.$$

Substituting into the differential equation, we have

$$y' + 2xy = 1 - 2xe^{-x^2} \int_0^x e^{t^2} dt - 2c_1 x e^{-x^2} + 2xe^{-x^2} \int_0^x e^{t^2} dt + 2c_1 x e^{-x^2} = 1.$$

19. From $y = c_1 e^{2x} + c_2 x e^{2x}$ we obtain $\frac{dy}{dx} = (2c_1 + c_2) e^{2x} + 2c_2 x e^{2x}$ and $\frac{d^2 y}{dx^2} = (4c_1 + 4c_2) e^{2x} + 4c_2 x e^{2x}$, so that

$$\frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + 4y = (4c_1 + 4c_2 - 8c_1 - 4c_2 + 4c_1) e^{2x} + (4c_2 - 8c_2 + 4c_2) x e^{2x} = 0.$$

20. From $y = c_1 x^{-1} + c_2 x + c_3 x \ln x + 4x^2$ we obtain

$$\frac{dy}{dx} = -c_1 x^{-2} + c_2 + c_3 + c_3 \ln x + 8x,$$

$$\frac{d^2 y}{dx^2} = 2c_1 x^{-3} + c_3 x^{-1} + 8,$$

and

$$\frac{d^3 y}{dx^3} = -6c_1 x^{-4} - c_3 x^{-2},$$

so that

$$\begin{aligned} x^3 \frac{d^3 y}{dx^3} + 2x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + y &= (-6c_1 + 4c_1 + c_1 + c_1) x^{-1} + (-c_3 + 2c_3 - c_2 - c_3 + c_2) x \\ &\quad + (-c_3 + c_3) x \ln x + (16 - 8 + 4) x^2 \\ &= 12x^2. \end{aligned}$$

21. From $y = \begin{cases} -x^2, & x < 0 \\ x^2, & x \geq 0 \end{cases}$ we obtain $y' = \begin{cases} -2x, & x < 0 \\ 2x, & x \geq 0 \end{cases}$ so that $xy' - 2y = 0$.

22. The function $y(x)$ is not continuous at $x = 0$ since $\lim_{x \rightarrow 0^-} y(x) = 5$ and $\lim_{x \rightarrow 0^+} y(x) = -5$. Thus, $y'(x)$ does not exist at $x = 0$.

23. From $x = e^{-2t} + 3e^{6t}$ and $y = -e^{-2t} + 5e^{6t}$ we obtain

$$\frac{dx}{dt} = -2e^{-2t} + 18e^{6t} \quad \text{and} \quad \frac{dy}{dt} = 2e^{-2t} + 30e^{6t}.$$

Then

$$\begin{aligned} x + 3y &= (e^{-2t} + 3e^{6t}) + 3(-e^{-2t} + 5e^{6t}) \\ &= -2e^{-2t} + 18e^{6t} = \frac{dx}{dt} \end{aligned}$$

and

$$\begin{aligned} 5x + 3y &= 5(e^{-2t} + 3e^{6t}) + 3(-e^{-2t} + 5e^{6t}) \\ &= 2e^{-2t} + 30e^{6t} = \frac{dy}{dt}. \end{aligned}$$

24. From $x = \cos 2t + \sin 2t + \frac{1}{5}e^t$ and $y = -\cos 2t - \sin 2t - \frac{1}{5}e^t$ we obtain

$$\frac{dx}{dt} = -2\sin 2t + 2\cos 2t + \frac{1}{5}e^t \quad \text{and} \quad \frac{dy}{dt} = 2\sin 2t - 2\cos 2t - \frac{1}{5}e^t$$

and

$$\frac{d^2x}{dt^2} = -4\cos 2t - 4\sin 2t + \frac{1}{5}e^t \quad \text{and} \quad \frac{d^2y}{dt^2} = 4\cos 2t + 4\sin 2t - \frac{1}{5}e^t.$$

Then

$$\begin{aligned} 4y + e^t &= 4(-\cos 2t - \sin 2t - \frac{1}{5}e^t) + e^t \\ &= -4\cos 2t - 4\sin 2t + \frac{1}{5}e^t = \frac{d^2x}{dt^2} \end{aligned}$$

and

$$\begin{aligned} 4x - e^t &= 4(\cos 2t + \sin 2t + \frac{1}{5}e^t) - e^t \\ &= 4\cos 2t + 4\sin 2t - \frac{1}{5}e^t = \frac{d^2y}{dt^2}. \end{aligned}$$

25. An interval on which $\tan 5t$ is continuous is $-\pi/2 < 5t < \pi/2$, so $5 \tan 5t$ will be a solution on $(-\pi/10, \pi/10)$.

26. For $(1 - \sin t)^{-1/2}$ to be continuous we must have $1 - \sin t > 0$ or $\sin t < 1$. Thus, $(1 - \sin t)^{-1/2}$ will be a solution on $(\pi/2, 5\pi/2)$.

27. $(y')^2 + 1 = 0$ has no real solution.

28. The only solution of $(y')^2 + y^2 = 0$ is $y = 0$, since if $y \neq 0$, $y^2 > 0$ and $(y')^2 + y^2 \geq y^2 > 0$.

29. The first derivative of $f(t) = e^t$ is e^t . The first derivative of $f(t) = e^{kt}$ is ke^{kt} . The differential equations are $y' = y$ and $y' = ky$, respectively.

30. Any function of the form $y = ce^t$ or $y = ce^{-t}$ is its own second derivative. The corresponding differential equation is $y'' - y = 0$. Functions of the form $y = c \sin t$ or $y = c \cos t$ have second derivatives that are the negatives of themselves. The differential equation is $y'' + y = 0$.

31. Since the n th derivative of $\phi(x)$ must exist if $\phi(x)$ is a solution of the n th order differential equation, all lower-order derivatives of $\phi(x)$ must exist and be continuous. [Recall that a differentiable function is continuous.]

32. Solving the system

$$\begin{aligned} c_1 y_1(0) + c_2 y_2(0) &= 2 \\ c_1 y_1'(0) + c_2 y_2'(0) &= 0 \end{aligned}$$

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for c_1 and c_2 we get

$$c_1 = \frac{2y_2'(0)}{y_1(0)y_2'(0) - y_1'(0)y_2(0)} \quad \text{and} \quad c_2 = -\frac{2y_1'(0)}{y_1(0)y_2'(0) - y_1'(0)y_2(0)}.$$

Thus, a particular solution is

$$y = \frac{2y_2'(0)}{y_1(0)y_2'(0) - y_1'(0)y_2(0)} y_1 - \frac{2y_1'(0)}{y_1(0)y_2'(0) - y_1'(0)y_2(0)} y_2,$$

where we assume that $y_1(0)y_2'(0) - y_1'(0)y_2(0) \neq 0$.

- 33.** For the first-order differential equation integrate $f(x)$. For the second-order differential equation integrate twice. In the latter case we get $y = \int(\int f(t)dt)dt + c_1t + c_2$.

- 34.** Solving for y' using the quadratic formula we obtain the two differential equations

$$y' = \frac{1}{t} \left(2 + 2\sqrt{1 + 3t^6} \right) \quad \text{and} \quad y' = \frac{1}{t} \left(2 - 2\sqrt{1 + 3t^6} \right),$$

so the differential equation cannot be put in the form $dy/dt = f(t, y)$.

- 35.** The differential equation $yy' - ty = 0$ has normal form $dy/dt = t$. These are not equivalent because $y = 0$ is a solution of the first differential equation but not a solution of the second.
- 36.** Differentiating we get $y' = c_1 + 3c_2t^2$ and $y'' = 6c_2t$. Then $c_2 = y''/6t$ and $c_1 = y' - ty''/2$, so

$$y = \left(y' - \frac{ty''}{2} \right) t + \left(\frac{y''}{6t} \right) t^3 = ty' - \frac{1}{3}t^2y''$$

and the differential equation is $t^2y'' - 3ty' + 3y = 0$.

- 37. (a)** From $y = e^{mt}$ we obtain $y' = me^{mt}$. Then $y' + 2y = 0$ implies

$$me^{mt} + 2e^{mt} = (m + 2)e^{mt} = 0.$$

Since $e^{mt} > 0$ for all t , $m = -2$. Thus $y = e^{-2t}$ is a solution.

- (b)** From $y = e^{mt}$ we obtain $y' = me^{mt}$ and $y'' = m^2e^{mt}$. Then $y'' - 5y' + 6y = 0$ implies

$$m^2e^{mt} - 5me^{mt} + 6e^{mt} = (m - 2)(m - 3)e^{mt} = 0.$$

Since $e^{mt} > 0$ for all t , $m = 2$ and $m = 3$. Thus $y = e^{2t}$ and $y = e^{3t}$ are solutions.

- (c)** From $y = t^m$ we obtain $y' = mt^{m-1}$ and $y'' = m(m-1)t^{m-2}$. Then $ty'' + 2y' = 0$ implies

$$\begin{aligned} tm(m-1)t^{m-2} + 2mt^{m-1} &= [m(m-1) + 2m]t^{m-1} = (m^2 + m)t^{m-1} \\ &= m(m+1)t^{m-1} = 0. \end{aligned}$$

Since $t^{m-1} > 0$ for $t > 0$, $m = 0$ and $m = -1$. Thus $y = 1$ and $y = t^{-1}$ are solutions.

- (d)** From $y = t^m$ we obtain $y' = mt^{m-1}$ and $y'' = m(m-1)t^{m-2}$. Then $t^2y'' - 7ty' + 15y = 0$ implies

$$\begin{aligned} t^2m(m-1)t^{m-2} - 7tmt^{m-1} + 15t^m &= [m(m-1) - 7m + 15]t^m \\ &= (m^2 - 8m + 15)t^m = (m-3)(m-5)t^m = 0. \end{aligned}$$

Since $t^m > 0$ for $t > 0$, $m = 3$ and $m = 5$. Thus $y = t^3$ and $y = t^5$ are solutions.

- 38.** When $g(t) = 0$, $y = 0$ is a solution of a linear equation.

- 39. (a)** Solving $(10 - 5y)/3x = 0$ we see that $y = 2$ is a constant solution.

- (b)** Solving $y^2 + 2y - 3 = (y + 3)(y - 1) = 0$ we see that $y = -3$ and $y = 1$ are constant solutions.

- (c)** Since $1/(y - 1) = 0$ has no solutions, the differential equation has no constant solutions.

- (d) Setting $y' = 0$ we have $y'' = 0$ and $6y = 10$. Thus $y = 5/3$ is a constant solution.
40. From $y' = (1 - y)/(x - 2)$ we see that a tangent line to the graph of $y(x)$ is possibly vertical at $x = 2$. Intervals of existence could be $(-\infty, 2)$ and $(2, \infty)$.
41. One solution is given by the upper portion of the graph with domain approximately $(0, 2.6)$. The other solution is given by the lower portion of the graph, also with domain approximately $(0, 2.6)$.
42. One solution, with domain approximately $(-\infty, 1.6)$ is the portion of the graph in the second quadrant together with the lower part of the graph in the first quadrant. A second solution, with domain approximately $(0, 1.6)$ is the upper part of the graph in the first quadrant. The third solution, with domain $(0, \infty)$, is the part of the graph in the fourth quadrant.
43. Differentiating $(x^3 + y^3)/xy = 3c$ we obtain

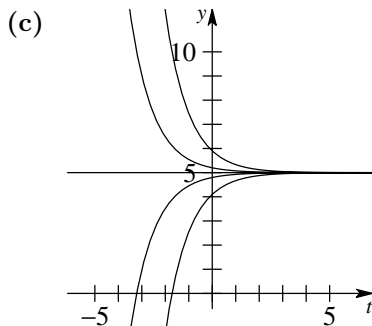
$$\begin{aligned}\frac{xy(3x^2 + 3y^2y') - (x^3 + y^3)(xy' + y)}{x^2y^2} &= 0 \\ 3x^3 + 3xy^3y' - x^4y' - x^3y - xy^3y' - y^4 &= 0 \\ (3xy^3 - x^4 - xy^3)y' &= -3x^3y + x^3y + y^4 \\ y' &= \frac{y^4 - 2x^3y}{2xy^3 - x^4} = \frac{y(y^3 - 2x^3)}{x(2y^3 - x^3)}.\end{aligned}$$

44. A tangent line will be vertical where y' is undefined, or in this case, where $x(2y^3 - x^3) = 0$. This gives $x = 0$ and $2y^3 = x^3$. Substituting $y^3 = x^3/2$ into $x^3 + y^3 = 3xy$ we get

$$\begin{aligned}x^3 + \frac{1}{2}x^3 &= 3x \left(\frac{1}{2^{1/3}}x \right) \\ \frac{3}{2}x^3 &= \frac{3}{2^{1/3}}x^2 \\ x^3 &= 2^{2/3}x^2 \\ x^2(x - 2^{2/3}) &= 0.\end{aligned}$$

Thus, there are vertical tangent lines at $x = 0$ and $x = 2^{2/3}$, or at $(0, 0)$ and $(2^{2/3}, 2^{1/3})$. Since $2^{2/3} \approx 1.59$, the estimates of the domains in Problem 42 were close.

45. Since $\phi'(x) > 0$ for all x in I , $\phi(x)$ is an increasing function on I . Hence, it can have no relative extrema on I .
46. (a) When $y = 5$, $y' = 0$, so $y = 5$ is a solution of $y' = 5 - y$.
- (b) When $y > 5$, $y' < 0$, and the solution must be decreasing. When $y < 5$, $y' > 0$, and the solution must be increasing. Thus, none of the curves in color can be solutions.



47. (a) $y = 0$ and $y = a/b$.

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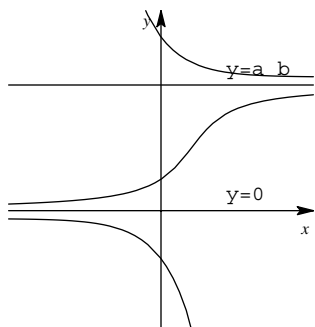
(b) Since $dy/dx = y(a - by) > 0$ for $0 < y < a/b$, $y = \phi(x)$ is increasing on this interval. Since $dy/dx < 0$ for $y < 0$ or $y > a/b$, $y = \phi(x)$ is decreasing on these intervals.

(c) Using implicit differentiation we compute

$$\frac{d^2y}{dx^2} = y(-by') + y'(a - by) = y'(a - 2by).$$

Solving $d^2y/dx^2 = 0$ we obtain $y = a/2b$. Since $d^2y/dx^2 > 0$ for $0 < y < a/2b$ and $d^2y/dx^2 < 0$ for $a/2b < y < a/b$, the graph of $y = \phi(x)$ has a point of inflection at $y = a/2b$.

(d)



48. (a) In *Mathematica* use

```
Clear[y]
y[x_]:= x Exp[5x] Cos[2x]
y[x]
y''''[x] - 20 y'''[x] + 158 y''[x] - 580 y'[x] + 841 y[x] // Simplify
```

(b) In *Mathematica* use

```
Clear[y]
y[x_]:= 20 Cos[5 Log[x]]/x - 3 Sin[5 Log[x]]/x
y[x]
x^3 y''''[x] + 2x^2 y'''[x] + 20 x y''[x] - 78 y[x] // Simplify
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Exercises 1.2

1. Solving $-\frac{1}{3} = \frac{1}{1 + c_1}$ we get $c_1 = -4$. The solution is $y = \frac{1}{1 - 4e^{-t}}$.
2. Solving $2 = \frac{1}{1 + c_1 e}$ we get $c_1 = -\frac{1}{2}e^{-1}$. The solution is $y = \frac{2}{2 - e^{-(t+1)}}$.
3. Using $x' = -c_1 \sin t + c_2 \cos t$ we obtain $c_1 = -1$ and $c_2 = 8$. The solution is $x = -\cos t + 8 \sin t$.
4. Using $x' = -c_1 \sin t + c_2 \cos t$ we obtain $c_2 = 0$ and $-c_1 = 1$. The solution is $x = -\cos t$.
5. Using $x' = -c_1 \sin t + c_2 \cos t$ we obtain

$$\begin{aligned} \frac{\sqrt{3}}{2}c_1 + \frac{1}{2}c_2 &= \frac{1}{2} \\ -\frac{1}{2}c_1 + \frac{\sqrt{3}}{2}c_2 &= 0. \end{aligned}$$

Solving we find $c_1 = \frac{\sqrt{3}}{4}$ and $c_2 = \frac{1}{4}$. The solution is $x = \frac{\sqrt{3}}{4} \cos t + \frac{1}{4} \sin t$.

6. Using $x' = -c_1 \sin t + c_2 \cos t$ we obtain

$$\begin{aligned}\frac{\sqrt{2}}{2} c_1 + \frac{\sqrt{2}}{2} c_2 &= \sqrt{2} \\ -\frac{\sqrt{2}}{2} c_1 + \frac{\sqrt{2}}{2} c_2 &= 2\sqrt{2}.\end{aligned}$$

Solving we find $c_1 = -1$ and $c_2 = 3$. The solution is $x = -\cos t + 3 \sin t$.

7. From the initial conditions we obtain the system

$$\begin{aligned}c_1 + c_2 &= 1 \\ c_1 - c_2 &= 2.\end{aligned}$$

Solving we get $c_1 = \frac{3}{2}$ and $c_2 = -\frac{1}{2}$. A solution of the initial-value problem is $y = \frac{3}{2}e^x - \frac{1}{2}e^{-x}$.

8. From the initial conditions we obtain the system

$$\begin{aligned}c_1 e + c_2 e^{-1} &= 0 \\ c_1 e - c_2 e^{-1} &= e.\end{aligned}$$

Solving we get $c_1 = \frac{1}{2}$ and $c_2 = -\frac{1}{2}e^2$. A solution of the initial-value problem is $y = \frac{1}{2}e^x - \frac{1}{2}e^{2-x}$.

9. From the initial conditions we obtain

$$\begin{aligned}c_1 e^{-1} + c_2 e &= 5 \\ c_1 e^{-1} - c_2 e &= -5.\end{aligned}$$

Solving we get $c_1 = 0$ and $c_2 = 5e^{-1}$. A solution of the initial-value problem is $y = 5e^{-x-1}$.

10. From the initial conditions we obtain

$$\begin{aligned}c_1 + c_2 &= 0 \\ c_1 - c_2 &= 0.\end{aligned}$$

Solving we get $c_1 = c_2 = 0$. A solution of the initial-value problem is $y = 0$.

11. Two solutions are $y = 0$ and $y = x^3$.
12. Two solutions are $y = 0$ and $y = x^2$. (Also, any constant multiple of x^2 is a solution.)
13. For $f(x, y) = y^{2/3}$ we have $\frac{\partial f}{\partial y} = \frac{2}{3}y^{-1/3}$. Thus the differential equation will have a unique solution in any rectangular region of the plane where $y \neq 0$.
14. For $f(x, y) = \sqrt{xy}$ we have $\frac{\partial f}{\partial y} = \frac{1}{2}\sqrt{\frac{x}{y}}$. Thus the differential equation will have a unique solution in any region where $x > 0$ and $y > 0$ or where $x < 0$ and $y < 0$.
15. For $f(x, y) = \frac{y}{x}$ we have $\frac{\partial f}{\partial y} = \frac{1}{x}$. Thus the differential equation will have a unique solution in any region where $x \neq 0$.
16. For $f(x, y) = x + y$ we have $\frac{\partial f}{\partial y} = 1$. Thus the differential equation will have a unique solution in the entire plane.
17. For $f(x, y) = \frac{x^2}{4 - y^2}$ we have $\frac{\partial f}{\partial y} = \frac{2x^2 y}{(4 - y^2)^2}$. Thus the differential equation will have a unique solution in any region where $y < -2$, $-2 < y < 2$, or $y > 2$.

Exercises 1.2

18. For $f(x, y) = \frac{x^2}{1+y^3}$ we have $\frac{\partial f}{\partial y} = \frac{-3x^2y^2}{(1+y^3)^2}$. Thus the differential equation will have a unique solution in any region where $y \neq -1$.
19. For $f(x, y) = \frac{y^2}{x^2+y^2}$ we have $\frac{\partial f}{\partial y} = \frac{2x^2y}{(x^2+y^2)^2}$. Thus the differential equation will have a unique solution in any region not containing $(0, 0)$.
20. For $f(x, y) = \frac{y+x}{y-x}$ we have $\frac{\partial f}{\partial y} = \frac{-2x}{(y-x)^2}$. Thus the differential equation will have a unique solution in any region where $y < x$ or where $y > x$.
21. The differential equation has a unique solution at $(1, 4)$.
22. The differential equation is not guaranteed to have a unique solution at $(5, 3)$.
23. The differential equation is not guaranteed to have a unique solution at $(2, -3)$.
24. The differential equation is not guaranteed to have a unique solution at $(-1, 1)$.
25. (a) A one-parameter family of solutions is $y = cx$. Since $y' = c$, $xy' = xc = y$ and $y(0) = c \cdot 0 = 0$.
- (b) Writing the equation in the form $y' = y/x$ we see that R cannot contain any point on the y -axis. Thus, any rectangular region disjoint from the y -axis and containing (x_0, y_0) will determine an interval around x_0 and a unique solution through (x_0, y_0) . Since $x_0 = 0$ in part (a) we are not guaranteed a unique solution through $(0, 0)$.
- (c) The piecewise-defined function which satisfies $y(0) = 0$ is not a solution since it is not differentiable at $x = 0$.
26. (a) Since $\frac{d}{dx} \tan(x+c) = \sec^2(x+c) = 1 + \tan^2(x+c)$, we see that $y = \tan(x+c)$ satisfies the differential equation.
- (b) Solving $y(0) = \tan c = 0$ we obtain $c = 0$ and $y = \tan x$. Since $\tan x$ is discontinuous at $x = \pm\pi/2$, the solution is not defined on $(-2, 2)$ because it contains $\pm\pi/2$.
- (c) The largest interval on which the solution can exist is $(-\pi/2, \pi/2)$.
27. (a) Since $\frac{d}{dt} \left(-\frac{1}{t+c} \right) = \frac{1}{(t+c)^2} = y^2$, we see that $y = -\frac{1}{t+c}$ is a solution of the differential equation.
- (b) Solving $y(0) = -1/c = 1$ we obtain $c = -1$ and $y = 1/(1-t)$. Solving $y(0) = -1/c = -1$ we obtain $c = 1$ and $y = -1/(1+t)$. Being sure to include $t = 0$, we see that the interval of existence of $y = 1/(1-t)$ is $(-\infty, 1)$, while the interval of existence of $y = -1/(1+t)$ is $(-1, \infty)$.
- (c) Solving $y(0) = -1/c = y_0$ we obtain $c = -1/y_0$ and
- $$y = -\frac{1}{-1/y_0 + t} = \frac{y_0}{1 - y_0 t}, \quad y_0 \neq 0.$$
- Since we must have $-1/y_0 + t \neq 0$, the largest interval of existence (which must contain 0) is either $(-\infty, 1/y_0)$ when $y_0 > 0$ or $(1/y_0, \infty)$ when $y_0 < 0$.
- (d) By inspection we see that $y = 0$ is a solution on $(-\infty, \infty)$.
28. (a) Differentiating $3x^2 - y^2 = c$ we get $6x - 2yy' = 0$ or $yy' = 3x$.

(b) Solving $3x^2 - y^2 = 3$ for y we get

$$y = \phi_1(x) = \sqrt{3(x^2 - 1)}, \quad 1 < x < \infty,$$

$$y = \phi_2(x) = -\sqrt{3(x^2 - 1)}, \quad 1 < x < \infty,$$

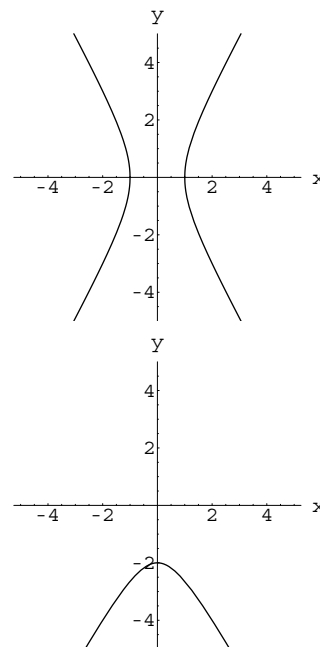
$$y = \phi_3(x) = \sqrt{3(x^2 - 1)}, \quad -\infty < x < -1,$$

$$y = \phi_4(x) = -\sqrt{3(x^2 - 1)}, \quad -\infty < x < -1.$$

Only $y = \phi_3(x)$ satisfies $y(-2) = 3$.

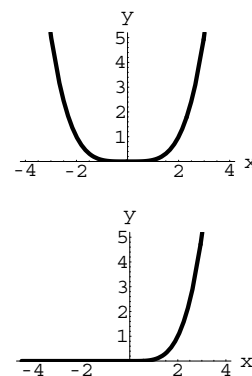
(c) Setting $x = 2$ and $y = -4$ in $3x^2 - y^2 = c$ we get $12 - 16 = -4 = c$, so the explicit solution is

$$y = -\sqrt{3x^2 + 4}, \quad -\infty < x < \infty.$$



(d) Setting $c = 0$ we have $y = \sqrt{3}x$ and $y = -\sqrt{3}x$, both defined on $(-\infty, \infty)$.

29. When $x = 0$ and $y = \frac{1}{2}$, $y' = -1$, so the only plausible solution curve is the one with negative slope at $(0, \frac{1}{2})$, or the black curve.
30. If the solution is tangent to the x -axis at $(x_0, 0)$, then $y' = 0$ when $x = x_0$ and $y = 0$. Substituting these values into $y' + 2y = 3x - 6$ we get $0 + 0 = 3x_0 - 6$ or $x_0 = 2$.
31. The theorem guarantees a unique (meaning single) solution through any point. Thus, there cannot be two distinct solutions through any point.
32. Differentiating $y = x^4/16$ we obtain $y' = x^3/4 = x(x^4/16)^{1/2}$, so the first function is a solution of the differential equation. Since $y = 0$ satisfies the differential equation and $\lim_{x \rightarrow 0^+} x^3/4 = 0$, the second function is also a solution of the differential equation. Both functions satisfy the condition $y(2) = 1$. Theorem 1.1 simply guarantees the existence of a unique solution on *some* interval containing $(2, 1)$. In this, such an interval could be $(0, 4)$, where the two functions are identical.



33. The antiderivative of $y' = 8e^{2x} + 6x$ is $y = 4e^{2x} + 3x^2 + c$. Setting $x = 0$ and $y = 9$ we get $9 = 4 + c$, so $c = 5$ and $y = 4e^{2x} + 3x^2 + 5$.
34. The antiderivative of $y'' = 12x - 2$ is $y' = 6x^2 - 2x + c_1$. From the equation of the tangent line we see that when $x = 1$, $y = 4$ and $y' = -1$ (the slope of the tangent line). Solving $-1 = 6(1)^2 - 2(1) + c_1$ we get $c_1 = -5$. The antiderivative of $y' = 6x^2 - 2x - 5$ is $y = 2x^3 - x^2 - 5x + c_2$. Setting $x = 1$ and $y = 4$ we get $4 = -4 + c_2$, so $c_2 = 8$ and $y = 2x^3 - x^2 - 5x + 8$.

Exercises 1.3

1. $\frac{dP}{dt} = kP + r$; $\frac{dP}{dt} = kP - r$
2. Let b be the rate of births and d the rate of deaths. Then $b = k_1P$ and $d = k_2P$. Since $dP/dt = b - d$, the differential equation is $dP/dt = k_1P - k_2P$.
3. Let b be the rate of births and d the rate of deaths. Then $b = k_1P$ and $d = k_2P^2$. Since $dP/dt = b - d$, the differential equation is $dP/dt = k_1P - k_2P^2$.
4. Let $P(t)$ be the number of owls present at time t . Then $dP/dt = k(P - 200 + 10t)$.
5. From the graph we estimate $T_0 = 180^\circ$ and $T_m = 75^\circ$. We observe that when $T = 85$, $dT/dt \approx -1$. From the differential equation we then have

$$k = \frac{dT/dt}{T - T_m} = \frac{-1}{85 - 75} = -0.1.$$

6. By inspecting the graph we take T_m to be $T_m(t) = 80 - 30 \cos \pi t / 12$. Then the temperature of the body at time t is determined by the differential equation

$$\frac{dT}{dt} = k \left[T - \left(80 - 30 \cos \frac{\pi}{12} t \right) \right], \quad t > 0.$$

7. The number of students with the flu is x and the number not infected is $1000 - x$, so $dx/dt = kx(1000 - x)$.
8. By analogy with differential equation modeling the spread of a disease we assume that the rate at which the technological innovation is adopted is proportional to the number of people who have adopted the innovation and also to the number of people, $y(t)$, who have not yet adopted it. If one person who has adopted the innovation is introduced into the population then $x + y = n + 1$ and

$$\frac{dx}{dt} = kx(n + 1 - x), \quad x(0) = 1.$$

9. The rate at which salt is leaving the tank is

$$(3 \text{ gal/min}) \cdot \left(\frac{A}{300} \text{ lb/gal} \right) = \frac{A}{100} \text{ lb/min}.$$

Thus $dA/dt = A/100$.

10. The rate at which salt is entering the tank is

$$R_1 = (3 \text{ gal/min}) \cdot (2 \text{ lb/gal}) = 6 \text{ lb/min}.$$

Since the solution is pumped out at a slower rate, it is accumulating at the rate of $(3 - 2)\text{gal/min} = 1 \text{ gal/min}$. After t minutes there are $300 + t$ gallons of brine in the tank. The rate at which salt is leaving is

$$R_2 = (2 \text{ gal/min}) \cdot \left(\frac{A}{300 + t} \text{ lb/gal} \right) = \frac{2A}{300 + t} \text{ lb/min}.$$

The differential equation is

$$\frac{dA}{dt} = 6 - \frac{2A}{300 + t}.$$

11. The volume of water in the tank at time t is $V = A_w h$. The differential equation is then

$$\frac{dh}{dt} = \frac{1}{A_w} \frac{dV}{dt} = \frac{1}{A_w} \left(-cA_0 \sqrt{2gh} \right) = -\frac{cA_0}{A_w} \sqrt{2gh}.$$

Using $A_0 = \pi \left(\frac{2}{12}\right)^2 = \frac{\pi}{36}$, $A_w = 10^2 = 100$, and $g = 32$, this becomes

$$\frac{dh}{dt} = -\frac{c\pi/36}{100} \sqrt{64h} = -\frac{c\pi}{450} \sqrt{h}.$$

12. The volume of water in the tank at time t is $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}A_w h$. Using the formula from Problem 11 for the volume of water leaving the tank we see that the differential equation is

$$\frac{dh}{dt} = \frac{3}{A_w} \frac{dV}{dt} = \frac{3}{A_w} (-cA_h \sqrt{2gh}) = -\frac{3cA_h}{A_w} \sqrt{2gh}.$$

Using $A_h = \pi(2/12)^2 = \pi/36$, $g = 32$, and $c = 0.6$, this becomes

$$\frac{dh}{dt} = -\frac{3(0.6)\pi/36}{A_w} \sqrt{64h} = -\frac{0.4\pi}{A_w} h^{1/2}.$$

To find A_w we let r be the radius of the top of the water. Then $r/h = 8/20$, so $r = 2h/5$ and $A_w = \pi(2h/5)^2 = 4\pi h^2/25$. Thus

$$\frac{dh}{dt} = -\frac{0.4\pi}{4\pi h^2/25} h^{1/2} = -2.5h^{-3/2}.$$

13. Since $i = \frac{dq}{dt}$ and $L \frac{d^2q}{dt^2} + R \frac{dq}{dt} = E(t)$ we obtain $L \frac{di}{dt} + Ri = E(t)$.

14. By Kirchoff's second law we obtain $R \frac{dq}{dt} + \frac{1}{C} q = E(t)$.

15. From Newton's second law we obtain $m \frac{dv}{dt} = -kv^2 + mg$.

16. We have from Archimedes' principle

$$\begin{aligned} \text{upward force of water on barrel} &= \text{weight of water displaced} \\ &= (62.4) \times (\text{volume of water displaced}) \\ &= (62.4)\pi(s/2)^2 y = 15.6\pi s^2 y. \end{aligned}$$

It then follows from Newton's second law that $\frac{w}{g} \frac{d^2y}{dt^2} = -15.6\pi s^2 y$ or $\frac{d^2y}{dt^2} + \frac{15.6\pi s^2 g}{w} y = 0$, where $g = 32$ and w is the weight of the barrel in pounds.

17. The net force acting on the mass is

$$F = ma = m \frac{d^2x}{dt^2} = -k(s+x) + mg = -kx + mg - ks.$$

Since the condition of equilibrium is $mg = ks$, the differential equation is

$$m \frac{d^2x}{dt^2} = -kx.$$

18. From Problem 17, without a damping force, the differential equation is $m d^2x/dt^2 = -kx$. With a damping force proportional to velocity the differential equation becomes

$$m \frac{d^2x}{dt^2} = -kx - \beta \frac{dx}{dt} \quad \text{or} \quad m \frac{d^2x}{dt^2} + \beta \frac{dx}{dt} + kx = 0.$$

19. Let $x(t)$ denote the height of the top of the chain at time t with the positive direction upward. The weight of the portion of chain off the ground is $W = (x \text{ ft}) \cdot (1 \text{ lb/ft}) = x$. The mass of the chain is $m = W/g = x/32$. The net force is $F = 5 - W = 5 - x$. By Newton's second law,

$$\frac{d}{dt} \left(\frac{x}{32} v \right) = 5 - x \quad \text{or} \quad x \frac{dv}{dt} + v \frac{dx}{dt} = 160 - 32x.$$

Exercises 1.3

Thus, the differential equation is

$$x \frac{d^2x}{dt^2} + \left(\frac{dx}{dt}\right)^2 + 32x = 160.$$

20. The force is the weight of the chain, $2L$, so by Newton's second law, $\frac{d}{dt}[mv] = 2L$. Since the mass of the portion of chain off the ground is $m = 2(L - x)/g$, we have

$$\frac{d}{dt}\left[\frac{2(L - x)}{g}v\right] = 2L \quad \text{or} \quad (L - x)\frac{dv}{dt} + v\left(-\frac{dx}{dt}\right) = Lg.$$

Thus, the differential equation is

$$(L - x)\frac{d^2x}{dt^2} - \left(\frac{dx}{dt}\right)^2 = Lg.$$

21. From $g = k/R^2$ we find $k = gR^2$. Using $a = d^2r/dt^2$ and the fact that the positive direction is upward we get

$$\frac{d^2r}{dt^2} = -a = -\frac{k}{r^2} = -\frac{gR^2}{r^2} \quad \text{or} \quad \frac{d^2r}{dt^2} + \frac{gR^2}{r^2} = 0.$$

22. The gravitational force on m is $F = -kM_r m/r^2$. Since $M_r = 4\pi\delta r^3/3$ and $M = 4\pi\delta R^3/3$ we have $M_r = r^3 M/R^3$ and

$$F = -k \frac{M_r m}{r^2} = -k \frac{r^3 M m / R^3}{r^2} = -k \frac{mM}{R^3} r.$$

Now from $F = ma = d^2r/dt^2$ we have

$$m \frac{d^2r}{dt^2} = -k \frac{mM}{R^3} r \quad \text{or} \quad \frac{d^2r}{dt^2} = -\frac{kM}{R^3} r.$$

23. The differential equation is $\frac{dA}{dt} = k(M - A)$.

24. The differential equation is $\frac{dA}{dt} = k_1(M - A) - k_2A$.

25. The differential equation is $x'(t) = r - kx(t)$ where $k > 0$.

26. By the Pythagorean Theorem the slope of the tangent line is $y' = \frac{-y}{\sqrt{s^2 - y^2}}$.

27. We see from the figure that $2\theta + \alpha = \pi$. Thus

$$\frac{y}{-x} = \tan \alpha = \tan(\pi - 2\theta) = -\tan 2\theta = -\frac{2 \tan \theta}{1 - \tan^2 \theta}.$$

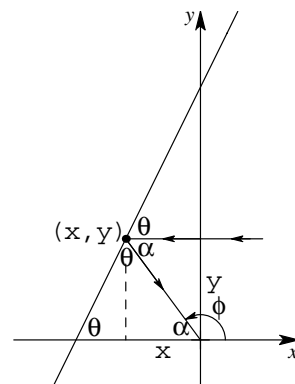
Since the slope of the tangent line is $y' = \tan \theta$ we have $y/x = 2y'[1 - (y')^2]$ or $y - y(y')^2 = 2xy'$, which is the quadratic equation $y(y')^2 + 2xy' - y = 0$ in y' .

Using the quadratic formula we get

$$y' = \frac{-2x \pm \sqrt{4x^2 + 4y^2}}{2y} = \frac{-x \pm \sqrt{x^2 + y^2}}{y}.$$

Since $dy/dx > 0$, the differential equation is

$$\frac{dy}{dx} = \frac{-x + \sqrt{x^2 + y^2}}{y} \quad \text{or} \quad y \frac{dy}{dx} - \sqrt{x^2 + y^2} + x = 0.$$



28. The differential equation is $dP/dt = kP$, so from Problem 29 in Exercises 1.1, $P = e^{kt}$, and a one-parameter family of solutions is $P = ce^{kt}$.

29. The differential equation in (3) is $dT/dt = k(T - T_m)$. When the body is cooling, $T > T_m$, so $T - T_m > 0$. Since T is decreasing, $dT/dt < 0$ and $k < 0$. When the body is warming, $T < T_m$, so $T - T_m < 0$. Since T is increasing, $dT/dt > 0$ and $k < 0$.
30. The differential equation in (8) is $dA/dt = 6 - A/100$. If $A(t)$ attains a maximum, then $dA/dt = 0$ at this time and $A = 600$. If $A(t)$ continues to increase without reaching a maximum then $A'(t) > 0$ for $t > 0$ and A cannot exceed 600. In this case, if $A'(t)$ approaches 0 as t increases to infinity, we see that $A(t)$ approaches 600 as t increases to infinity.
31. This differential equation could describe a population that undergoes periodic fluctuations.

32. (1): $\frac{dP}{dt} = kP$ is linear (2): $\frac{dA}{dt} = kA$ is linear
 (3): $\frac{dT}{dt} = k(T - T_m)$ is linear (5): $\frac{dx}{dt} = kx(n + 1 - x)$ is nonlinear
 (6): $\frac{dX}{dt} = k(\alpha - X)(\beta - X)$ is nonlinear (8): $\frac{dA}{dt} = 6 - \frac{A}{100}$ is linear
 (10): $\frac{dh}{dt} = -\frac{A_h}{A_w} \sqrt{2gh}$ is nonlinear (11): $L \frac{d^2q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C}q = E(t)$ is linear
 (12): $\frac{d^2s}{dt^2} = -g$ is linear (14): $m \frac{dv}{dt} = mg - kv$ is linear
 (15): $m \frac{d^2s}{dt^2} + k \frac{ds}{dt} = mg$ is linear (16): $\frac{d^2x}{dt^2} - \frac{64}{L}x = 0$ is linear

33. From Problem 21, $d^2r/dt^2 = -gR^2/r^2$. Since R is a constant, if $r = R + s$, then $d^2r/dt^2 = d^2s/dt^2$ and, using a Taylor series, we get

$$\frac{d^2s}{dt^2} = -g \frac{R^2}{(R+s)^2} = -gR^2(R+s)^{-2} \approx -gR^2[R^{-2} - 2sR^{-3} + \cdots] = -g + \frac{2gs}{R^3} + \cdots$$

Thus, for R much larger than s , the differential equation is approximated by $d^2s/dt^2 = -g$.

34. If ρ is the mass density of the raindrop, then $m = \rho V$ and

$$\frac{dm}{dt} = \rho \frac{dV}{dt} = \rho \frac{d}{dt} \left[\frac{4}{3} \pi r^3 \right] = \rho \left(4\pi r^2 \frac{dr}{dt} \right) = \rho S \frac{dr}{dt}.$$

If dr/dt is a constant, then $dm/dt = kS$ where $\rho dr/dt = k$ or $dr/dt = k/\rho$. Since the radius is decreasing, $k < 0$. Solving $dr/dt = k/\rho$ we get $r = (k/\rho)t + c_0$. Since $r(0) = r_0$, $c_0 = r_0$ and $r = kt/\rho + r_0$.

From Newton's second law, $\frac{d}{dt}[mv] = mg$, where v is the velocity of the raindrop. Then

$$m \frac{dv}{dt} + v \frac{dm}{dt} = mg \quad \text{or} \quad \rho \left(\frac{4}{3} \pi r^3 \right) \frac{dv}{dt} + v(k4\pi r^2) = \rho \left(\frac{4}{3} \pi r^3 \right) g.$$

Dividing by $4\rho\pi r^3/3$ we get

$$\frac{dv}{dt} + \frac{3k}{\rho r} v = g \quad \text{or} \quad \frac{dv}{dt} + \frac{3k/\rho}{kt/\rho + r_0} v = g, \quad k < 0.$$

35. We assume that the plow clears snow at a constant rate of k cubic miles per hour. Let t be the time in hours after noon, $x(t)$ the depth in miles of the snow at time t , and $y(t)$ the distance the plow has moved in t hours. Then dy/dt is the velocity of the plow and the assumption gives

$$wx \frac{dy}{dt} = k$$

Exercises 1.3

where w is the width of the plow. Each side of this equation simply represents the volume of snow plowed in one hour. Now let t_0 be the number of hours before noon when it started snowing and let s be the constant rate in miles per hour at which x increases. Then for $t > -t_0$, $x = s(t + t_0)$. The differential equation then becomes

$$\frac{dy}{dt} = \frac{k}{ws} \frac{1}{t + t_0}.$$

Integrating we obtain

$$y = \frac{k}{ws} [\ln(t + t_0) + c]$$

where c is a constant. Now when $t = 0$, $y = 0$ so $c = -\ln t_0$ and

$$y = \frac{k}{ws} \ln \left(1 + \frac{t}{t_0} \right).$$

Finally, from the fact that when $t = 1$, $y = 2$ and when $t = 2$, $y = 3$, we obtain

$$\left(1 + \frac{2}{t_0} \right)^2 = \left(1 + \frac{1}{t_0} \right)^3.$$

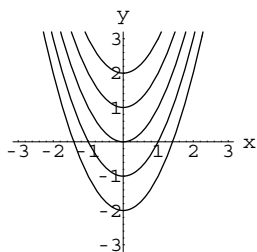
Expanding and simplifying gives $t_0^2 + t_0 - 1 = 0$. Since $t_0 > 0$, we find $t_0 \approx 0.618$ hours ≈ 37 minutes. Thus it started snowing at about 11:23 in the morning.

Chapter 1 Review Exercises

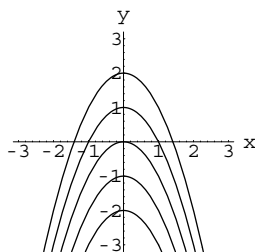
- $\frac{d}{dx} \frac{c_1}{x} = -\frac{c_1}{x^2} = -\frac{c_1/x}{x}$; $\frac{dy}{dx} = -\frac{y}{x}$
- $\frac{d}{dx}(5 + c_1 e^{-2x}) = -2c_1 e^{-2x} = -2(5 + c_1 e^{-2x} - 5)$; $\frac{dy}{dx} = -2(y - 5)$ or $\frac{dy}{dx} = -2y + 10$
- $\frac{d}{dx}(c_1 \cos kx + c_2 \sin kx) = -kc_1 \sin kx + kc_2 \cos kx$;
 $\frac{d^2}{dx^2}(c_1 \cos kx + c_2 \sin kx) = -k^2 c_1 \cos kx - k^2 c_2 \sin kx = -k^2(c_1 \cos kx + c_2 \sin kx)$;
 $\frac{d^2 y}{dx^2} = -k^2 y$ or $\frac{d^2 y}{dx^2} + k^2 y = 0$
- $\frac{d}{dx}(c_1 \cosh kx + c_2 \sinh kx) = kc_1 \sinh kx + kc_2 \cosh kx$;
 $\frac{d^2}{dx^2}(c_1 \cosh kx + c_2 \sinh kx) = k^2 c_1 \cosh kx + k^2 c_2 \sinh kx = k^2(c_1 \cosh kx + c_2 \sinh kx)$;
 $\frac{d^2 y}{dx^2} = k^2 y$ or $\frac{d^2 y}{dx^2} - k^2 y = 0$
- $y' = c_1 e^x + c_2 x e^x + c_2 e^x$; $y'' = c_1 e^x + c_2 x e^x + 2c_2 e^x$;
 $y'' + y = 2(c_1 e^x + c_2 x e^x) + 2c_2 e^x = 2(c_1 e^x + c_2 x e^x + c_2 e^x) = 2y'$; $y'' - 2y' + y = 0$
- $y' = -c_1 e^x \sin x + c_1 e^x \cos x + c_2 e^x \cos x + c_2 e^x \sin x$;
 $y'' = -c_1 e^x \cos x - c_1 e^x \sin x - c_1 e^x \sin x + c_1 e^x \cos x - c_2 e^x \sin x + c_2 e^x \cos x + c_2 e^x \cos x + c_2 e^x \sin x$
 $= -2c_1 e^x \sin x + 2c_2 e^x \cos x$;
 $y'' - 2y' = -2c_1 e^x \cos x - 2c_2 e^x \sin x = -2y$; $y'' - 2y' + 2y = 0$
- a,d
- c
- b
- a,c
- b
- a,b,d
- A few solutions are $y = 0$, $y = c$, and $y = e^x$.

14. Easy solutions to see are $y = 0$ and $y = 3$.
15. The slope of the tangent line at (x, y) is y' , so the differential equation is $y' = x^2 + y^2$.
16. The rate at which the slope changes is $dy'/dx = y''$, so the differential equation is $y'' = -y'$ or $y'' + y' = 0$.
17. (a) The domain is all real numbers.
 (b) Since $y' = 2/3x^{1/3}$, the solution $y = x^{2/3}$ is undefined at $x = 0$. This function is a solution of the differential equation on $(-\infty, 0)$ and also on $(0, \infty)$.
18. (a) Differentiating $y^2 - 2y = x^2 - x + c$ we obtain $2yy' - 2y' = 2x - 1$ or $(2y - 2)y' = 2x - 1$.
 (b) Setting $x = 0$ and $y = 1$ in the solution we have $1 - 2 = 0 - 0 + c$ or $c = -1$. Thus, a solution of the initial-value problem is $y^2 - 2y = x^2 - x - 1$.
 (c) Using the quadratic formula to solve $y^2 - 2y - (x^2 - x - 1) = 0$ we get $y = (2 \pm \sqrt{4 + 4(x^2 - x - 1)})/2 = 1 \pm \sqrt{x^2 - x} = 1 \pm \sqrt{x(x-1)}$. Since $x(x-1) \geq 0$ for $x \leq 0$ or $x \geq 1$, we see that neither $y = 1 + \sqrt{x(x-1)}$ nor $y = 1 - \sqrt{x(x-1)}$ is differentiable at $x = 0$. Thus, both functions are solutions of the differential equation, but neither is a solution of the initial-value problem.

19. (a)



$$y = x^2 + c_1$$



$$y = -x^2 + c_2$$

- (b) When $y = x^2 + c_1$, $y' = 2x$ and $(y')^2 = 4x^2$. When $y = -x^2 + c_2$, $y' = -2x$ and $(y')^2 = 4x^2$.
- (c) Pasting together x^2 , $x \geq 0$, and $-x^2$, $x \leq 0$, we get $y = \begin{cases} -x^2, & x \leq 0 \\ x^2, & x > 0 \end{cases}$.
20. The slope of the tangent line is $y' \big|_{(-1,4)} = 6\sqrt{4} + 5(-1)^3 = 7$.
21. Differentiating $y = \sin(\ln x)$ we obtain $y' = \cos(\ln x)/x$ and $y'' = -[\sin(\ln x) + \cos(\ln x)]/x^2$. Then

$$x^2 y'' + x y' + y = x^2 \left(-\frac{\sin(\ln x) + \cos(\ln x)}{x^2} \right) + x \frac{\cos(\ln x)}{x} + \sin(\ln x) = 0.$$

22. Differentiating $y = \cos(\ln x) \ln(\cos(\ln x)) + (\ln x) \sin(\ln x)$ we obtain

$$\begin{aligned} y' &= \cos(\ln x) \frac{1}{\cos(\ln x)} \left(-\frac{\sin(\ln x)}{x} \right) + \ln(\cos(\ln x)) \left(-\frac{\sin(\ln x)}{x} \right) + \ln x \frac{\cos(\ln x)}{x} + \frac{\sin(\ln x)}{x} \\ &= -\frac{\ln(\cos(\ln x)) \sin(\ln x)}{x} + \frac{(\ln x) \cos(\ln x)}{x} \end{aligned}$$

and

$$\begin{aligned} y'' &= -x \left[\ln(\cos(\ln x)) \frac{\cos(\ln x)}{x} + \sin(\ln x) \frac{1}{\cos(\ln x)} \left(-\frac{\sin(\ln x)}{x} \right) \right] \frac{1}{x^2} \\ &\quad + \ln(\cos(\ln x)) \sin(\ln x) \frac{1}{x^2} + x \left[(\ln x) \left(-\frac{\sin(\ln x)}{x} \right) + \frac{\cos(\ln x)}{x} \right] \frac{1}{x^2} - (\ln x) \cos(\ln x) \frac{1}{x^2} \\ &= \frac{1}{x^2} \left[-\ln(\cos(\ln x)) \cos(\ln x) + \frac{\sin^2(\ln x)}{\cos(\ln x)} + \ln(\cos(\ln x)) \sin(\ln x) \right. \\ &\quad \left. - (\ln x) \sin(\ln x) + \cos(\ln x) - (\ln x) \cos(\ln x) \right]. \end{aligned}$$

Chapter 1 Review Exercises

Then

$$\begin{aligned}
 x^2 y'' + xy' + y &= -\ln(\cos(\ln x)) \cos(\ln x) + \frac{\sin^2(\ln x)}{\cos(\ln x)} + \ln(\cos(\ln x)) \sin(\ln x) - (\ln x) \sin(\ln x) \\
 &\quad + \cos(\ln x) - (\ln x) \cos(\ln x) - \ln(\cos(\ln x)) \sin(\ln x) \\
 &\quad + (\ln x) \cos(\ln x) + \cos(\ln x) \ln(\cos(\ln x)) + (\ln x) \sin(\ln x) \\
 &= \frac{\sin^2(\ln x)}{\cos(\ln x)} + \cos(\ln x) = \frac{\sin^2(\ln x) + \cos^2(\ln x)}{\cos(\ln x)} = \frac{1}{\cos(\ln x)} = \sec(\ln x).
 \end{aligned}$$

(This problem is easily done using *Mathematica*.)

23. From the graph we see that estimates for y_0 and y_1 are $y_0 = -3$ and $y_1 = 0$.

24. The differential equation is

$$\frac{dh}{dt} = -\frac{cA_0}{A_w} \sqrt{2gh}.$$

Using $A_0 = \pi(1/24)^2 = \pi/576$, $A_w = \pi(2)^2 = 4\pi$, and $g = 32$, this becomes

$$\frac{dh}{dt} = -\frac{c\pi/576}{4\pi} \sqrt{64h} = \frac{c}{288} \sqrt{h}.$$

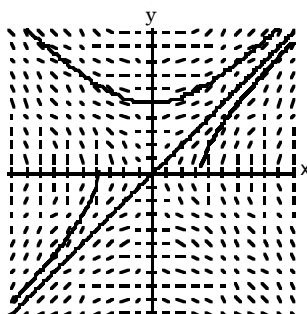
25. From Newton's second law we obtain

$$m \frac{dv}{dt} = \frac{1}{2} mg - \mu \frac{\sqrt{3}}{2} mg \quad \text{or} \quad \frac{dv}{dt} = 16 \left(1 - \sqrt{3} \mu \right).$$

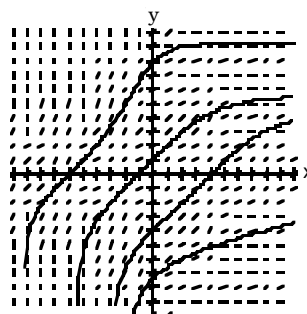
2 First-Order Differential Equations

Exercises 2.1

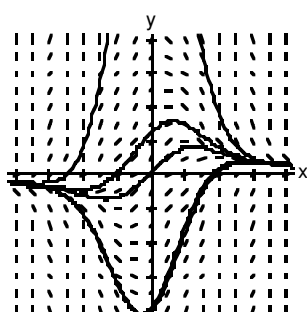
1.



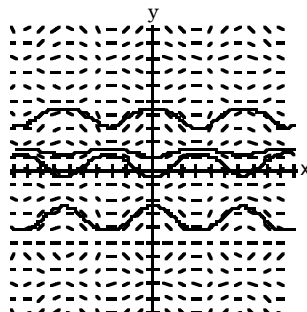
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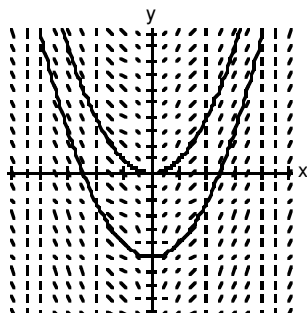
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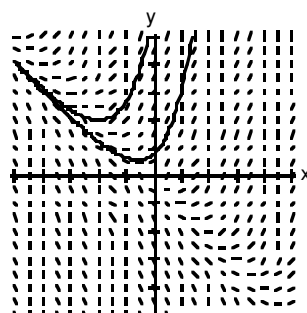
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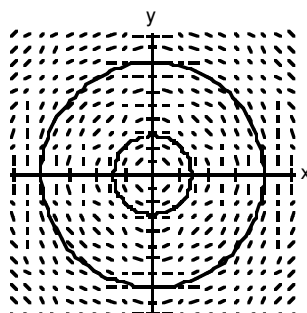
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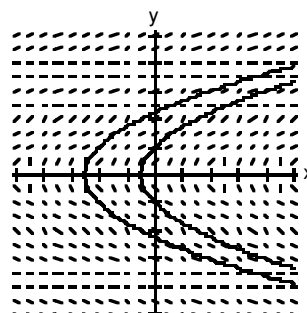
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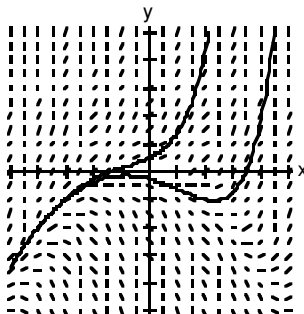


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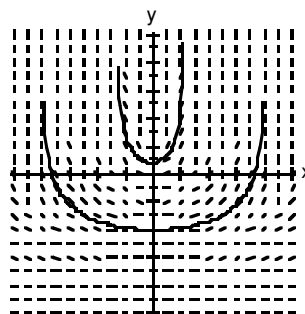


Exercises 2.1

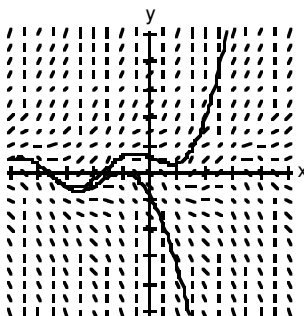
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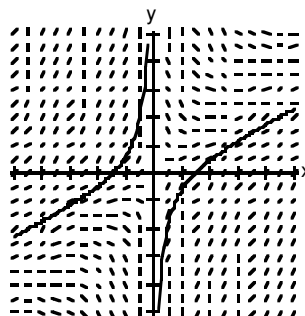
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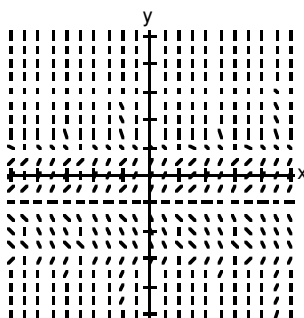
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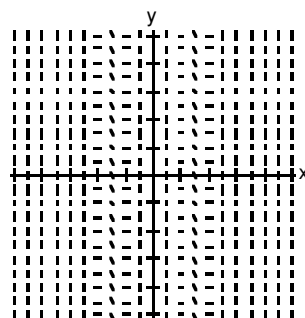
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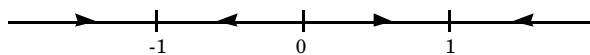
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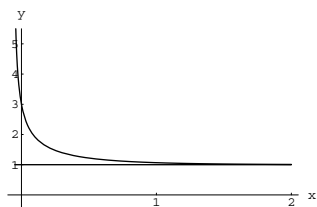
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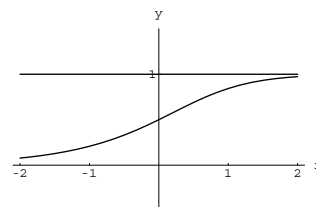
15. Writing the differential equation in the form $dy/dx = y(1-y)(1+y)$ we see that critical points are located at $y = -1$, $y = 0$, and $y = 1$. The phase portrait is shown below.



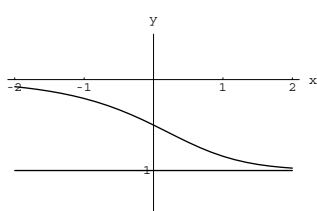
(a)



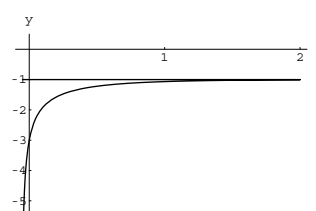
(b)



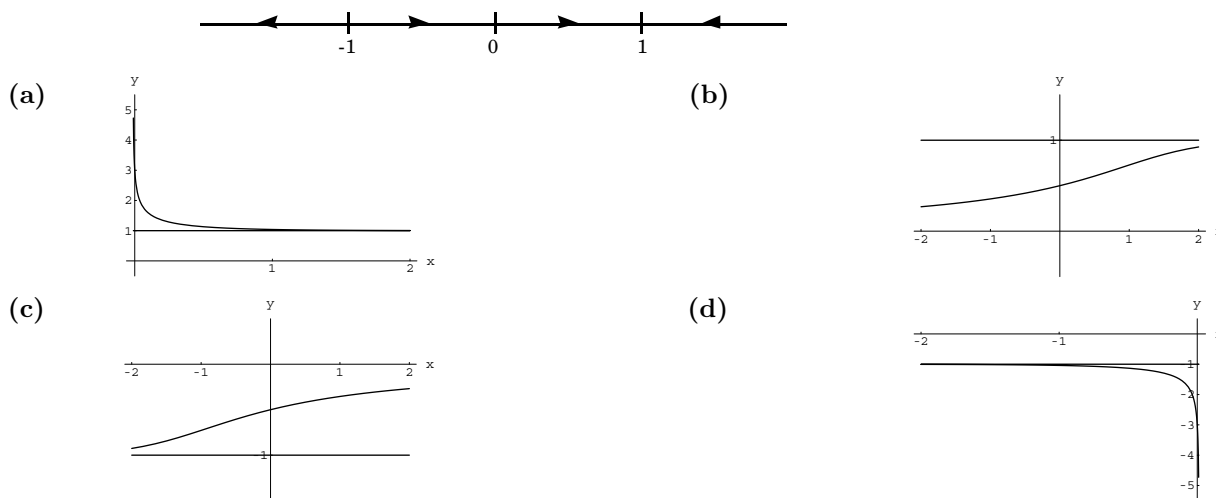
(c)



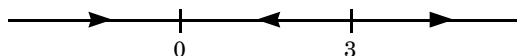
(d)



16. Writing the differential equation in the form $dy/dx = y^2(1-y)(1+y)$ we see that critical points are located at $y = -1$, $y = 0$, and $y = 1$. The phase portrait is shown below.

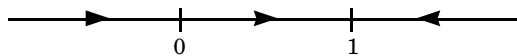


17. Solving $y^2 - 3y = y(y - 3) = 0$ we obtain the critical points 0 and 3.



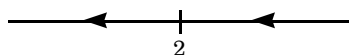
From the phase portrait we see that 0 is asymptotically stable and 3 is unstable.

18. Solving $y^2 - y^3 = y^2(1 - y) = 0$ we obtain the critical points 0 and 1.



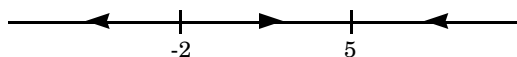
From the phase portrait we see that 1 is asymptotically stable and 0 is semi-stable.

19. Solving $(y - 2)^2 = 0$ we obtain the critical point 2.



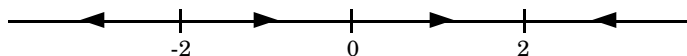
From the phase portrait we see that 2 is semi-stable.

20. Solving $10 + 3y - y^2 = (5 - y)(2 + y) = 0$ we obtain the critical points -2 and 5.



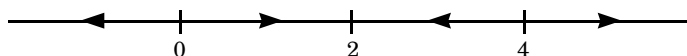
From the phase portrait we see that 5 is asymptotically stable and -2 is unstable.

21. Solving $y^2(4 - y^2) = y^2(2 - y)(2 + y) = 0$ we obtain the critical points -2, 0, and 2.



From the phase portrait we see that 2 is asymptotically stable, 0 is semi-stable, and -2 is unstable.

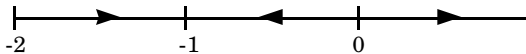
22. Solving $y(2 - y)(4 - y) = 0$ we obtain the critical points 0, 2, and 4.



From the phase portrait we see that 2 is asymptotically stable and 0 and 4 are unstable.

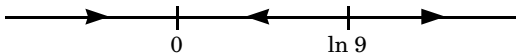
Exercises 2.1

23. Solving $y \ln(y + 2) = 0$ we obtain the critical points -1 and 0 .



From the phase portrait we see that -1 is asymptotically stable and 0 is unstable.

24. Solving $ye^y - 9y = y(e^y - 9) = 0$ we obtain the critical points 0 and $\ln 9$.

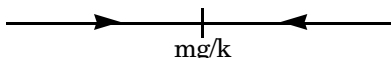


From the phase portrait we see that 0 is asymptotically stable and $\ln 9$ is unstable.

25. (a) Writing the differential equation in the form

$$\frac{dv}{dt} = \frac{k}{m} \left(\frac{mg}{k} - v \right)$$

we see that a critical point is mg/k .

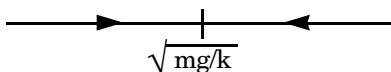


From the phase portrait we see that mg/k is an asymptotically stable critical point. Thus, $\lim_{t \rightarrow \infty} v = mg/k$.

- (b) Writing the differential equation in the form

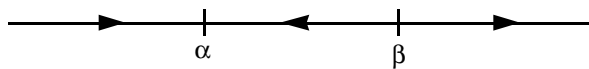
$$\frac{dv}{dt} = \frac{k}{m} \left(\frac{mg}{k} - v^2 \right) = \frac{k}{m} \left(\sqrt{\frac{mg}{k}} - v \right) \left(\sqrt{\frac{mg}{k}} + v \right)$$

we see that a critical point is $\sqrt{mg/k}$.



From the phase portrait we see that $\sqrt{mg/k}$ is an asymptotically stable critical point. Thus, $\lim_{t \rightarrow \infty} v = \sqrt{mg/k}$.

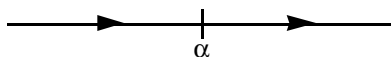
26. (a) From the phase portrait we see that critical points are α and β . Let $X(0) = X_0$.



If $X_0 < \alpha$, we see that $X \rightarrow \alpha$ as $t \rightarrow \infty$. If $\alpha < X_0 < \beta$, we see that $X \rightarrow \alpha$ as $t \rightarrow \infty$.

If $X_0 > \beta$, we see that $X(t)$ increases in an unbounded manner, but more specific behavior of $X(t)$ as $t \rightarrow \infty$ is not known.

- (b) When $\alpha = \beta$ the phase portrait is as shown.



If $X_0 < \alpha$, then $X(t) \rightarrow \alpha$ as $t \rightarrow \infty$. If $X_0 > \alpha$, then $X(t)$ increases in an unbounded manner. This could happen in a finite amount of time. That is, the phase portrait does not indicate that X becomes unbounded as $t \rightarrow \infty$.

- (c) When $k = 1$ and $\alpha = \beta$ the differential equation is $dX/dt = (\alpha - X)^2$. Separating variables and integrating we have

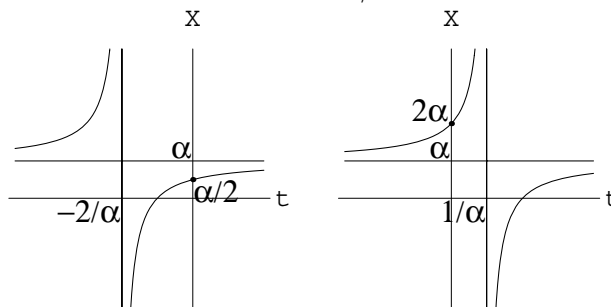
$$\begin{aligned}\frac{dX}{(\alpha - X)^2} &= dt \\ \frac{1}{\alpha - X} &= t + c \\ \alpha - X &= \frac{1}{t + c} \\ X &= \alpha - \frac{1}{t + c}.\end{aligned}$$

For $X(0) = \alpha/2$ we obtain

$$X(t) = \alpha - \frac{1}{t + 2/\alpha}.$$

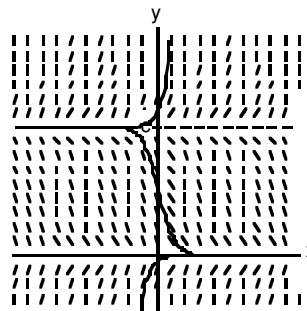
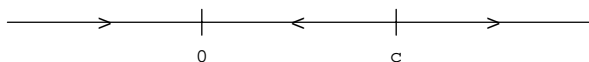
For $X(0) = 2\alpha$ we obtain

$$X(t) = \alpha - \frac{1}{t - 1/\alpha}.$$

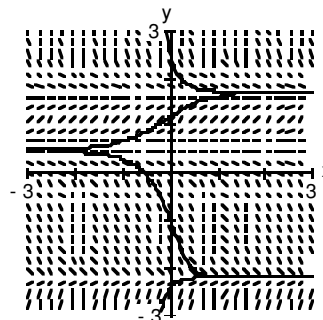
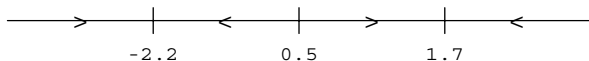


For $X_0 > \alpha$, $X(t)$ increases without bound up to $t = 1/\alpha$. For $t > 1/\alpha$, $X(t)$ increases but $X \rightarrow \alpha$ as $t \rightarrow \infty$.

27. Critical points are $y = 0$ and $y = c$.



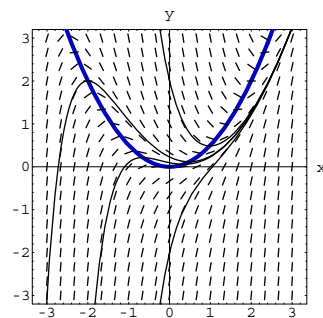
28. Critical points are $y = -2.2$, $y = 0.5$, and $y = 1.7$.



29. At each point on the circle of radius c the lineal element has slope c^2 .

Exercises 2.1

30. When $y < \frac{1}{2}x^2$, $y' = x^2 - 2y$ is positive and the portions of solution curves “inside” the nullcline parabola are increasing. When $y > \frac{1}{2}x^2$, $y' = x^2 - 2y$ is negative and the portions of the solution curves “outside” the nullcline parabola are decreasing.

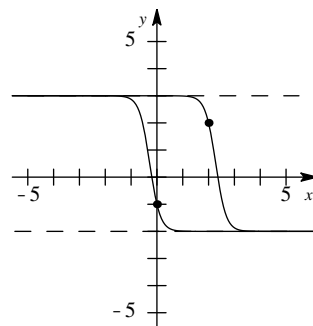


31. Recall that for $dy/dx = f(y)$ we are assuming that f and f' are continuous functions of y on some interval I . Now suppose that the graph of a nonconstant solution of the differential equation crosses the line $y = c$. If the point of intersection is taken as an initial condition we have two distinct solutions of the initial-value problem. This violates uniqueness, so the graph of any nonconstant solution must lie entirely on one side of any equilibrium solution. Since f is continuous it can only change signs around a point where it is 0. But this is a critical point. Thus, $f(y)$ is completely positive or completely negative in each region R_i . If $y(x)$ is oscillatory or has a relative extremum, then it must have a horizontal tangent line at some point (x_0, y_0) . In this case y_0 would be a critical point of the differential equation, but we saw above that the graph of a nonconstant solution cannot intersect the graph of the equilibrium solution $y = y_0$.
32. By Problem 31, a solution $y(x)$ of $dy/dx = f(y)$ cannot have relative extrema and hence must be monotone. Since $y'(x) = f(y) > 0$, $y(x)$ is monotone increasing, and since $y(x)$ is bounded above by c_2 , $\lim_{x \rightarrow \infty} y(x) = L$, where $L \leq c_2$. We want to show that $L = c_2$. Since L is a horizontal asymptote of $y(x)$, $\lim_{x \rightarrow \infty} y'(x) = 0$. Using the fact that $f(y)$ is continuous we have

$$f(L) = f(\lim_{x \rightarrow \infty} y(x)) = \lim_{x \rightarrow \infty} f(y(x)) = \lim_{x \rightarrow \infty} y'(x) = 0.$$

But then L is a critical point of f . Since $c_1 < L \leq c_2$, and f has no critical points between c_1 and c_2 , $L = c_2$.

33. (a) Assuming the existence of the second derivative, points of inflection of $y(x)$ occur where $y''(x) = 0$. From $dy/dx = g(y)$ we have $d^2y/dx^2 = g'(y) dy/dx$. Thus, the y -coordinate of a point of inflection can be located by solving $g'(y) = 0$. (Points where $dy/dx = 0$ correspond to constant solutions of the differential equation.)
- (b) Solving $y^2 - y - 6 = (y - 3)(y + 2) = 0$ we see that 3 and -2 are critical points. Now $d^2y/dx^2 = (2y - 1) dy/dx = (2y - 1)(y - 3)(y + 2)$, so the only possible point of inflection is at $y = \frac{1}{2}$, although the concavity of solutions can be different on either side of $y = -2$ and $y = 3$. Since $y''(x) < 0$ for $y < -2$ and $\frac{1}{2} < y < 3$, and $y''(x) > 0$ for $-2 < y < \frac{1}{2}$ and $y > 3$, we see that solution curves are concave down for $y < -2$ and $\frac{1}{2} < y < 3$ and concave up for $-2 < y < \frac{1}{2}$ and $y > 3$. Points of inflection of solutions of autonomous differential equations will have the same y -coordinates because between critical points they are horizontal translates of each other.



Exercises 2.2

In many of the following problems we will encounter an expression of the form $\ln |g(y)| = f(x) + c$. To solve for $g(y)$ we exponentiate both sides of the equation. This yields $|g(y)| = e^{f(x)+c} = e^c e^{f(x)}$ which implies $g(y) = \pm e^c e^{f(x)}$. Letting $c_1 = \pm e^c$ we obtain $g(y) = c_1 e^{f(x)}$.

1. From $dy = \sin 5x \, dx$ we obtain $y = -\frac{1}{5} \cos 5x + c$.
2. From $dy = (x+1)^2 \, dx$ we obtain $y = \frac{1}{3}(x+1)^3 + c$.
3. From $dy = -e^{-3x} \, dx$ we obtain $y = \frac{1}{3}e^{-3x} + c$.
4. From $\frac{1}{(y-1)^2} \, dy = dx$ we obtain $-\frac{1}{y-1} = x + c$ or $y = 1 - \frac{1}{x+c}$.
5. From $\frac{1}{y} \, dy = \frac{4}{x} \, dx$ we obtain $\ln |y| = 4 \ln |x| + c$ or $y = c_1 x^4$.
6. From $\frac{1}{y} \, dy = -2x \, dx$ we obtain $\ln |y| = -x^2 + c$ or $y = c_1 e^{-x^2}$.
7. From $e^{-2y} \, dy = e^{3x} \, dx$ we obtain $3e^{-2y} + 2e^{3x} = c$.
8. From $ye^y \, dy = (e^{-x} + e^{-3x}) \, dx$ we obtain $ye^y - e^y + e^{-x} + \frac{1}{3}e^{-3x} = c$.
9. From $\left(y + 2 + \frac{1}{y}\right) \, dy = x^2 \ln x \, dx$ we obtain $\frac{y^2}{2} + 2y + \ln |y| = \frac{x^3}{3} \ln |x| - \frac{1}{9}x^3 + c$.
10. From $\frac{1}{(2y+3)^2} \, dy = \frac{1}{(4x+5)^2} \, dx$ we obtain $\frac{2}{2y+3} = \frac{1}{4x+5} + c$.
11. From $\frac{1}{\csc y} \, dy = -\frac{1}{\sec^2 x} \, dx$ or $\sin y \, dy = -\cos^2 x \, dx = -\frac{1}{2}(1 + \cos 2x) \, dx$ we obtain $-\cos y = -\frac{1}{2}x - \frac{1}{4} \sin 2x + c$ or $4 \cos y = 2x + \sin 2x + c_1$.
12. From $2y \, dy = -\frac{\sin 3x}{\cos^3 3x} \, dx = -\tan 3x \sec^2 3x \, dx$ we obtain $y^2 = -\frac{1}{6} \sec^2 3x + c$.
13. From $\frac{e^y}{(e^y+1)^2} \, dy = \frac{-e^x}{(e^x+1)^3} \, dx$ we obtain $-(e^y+1)^{-1} = \frac{1}{2}(e^x+1)^{-2} + c$.
14. From $\frac{y}{(1+y^2)^{1/2}} \, dy = \frac{x}{(1+x^2)^{1/2}} \, dx$ we obtain $(1+y^2)^{1/2} = (1+x^2)^{1/2} + c$.
15. From $\frac{1}{S} \, dS = k \, dr$ we obtain $S = ce^{kr}$.
16. From $\frac{1}{Q-70} \, dQ = k \, dt$ we obtain $\ln |Q-70| = kt + c$ or $Q-70 = c_1 e^{kt}$.
17. From $\frac{1}{P-P^2} \, dP = \left(\frac{1}{P} + \frac{1}{1-P}\right) \, dP = dt$ we obtain $\ln |P| - \ln |1-P| = t + c$ so that $\ln \frac{P}{1-P} = t + c$ or $\frac{P}{1-P} = c_1 e^t$. Solving for P we have $P = \frac{c_1 e^t}{1 + c_1 e^t}$.
18. From $\frac{1}{N} \, dN = (te^{t+2} - 1) \, dt$ we obtain $\ln |N| = te^{t+2} - e^{t+2} - t + c$.
19. From $\frac{y-2}{y+3} \, dy = \frac{x-1}{x+4} \, dx$ or $\left(1 - \frac{5}{y+3}\right) \, dy = \left(1 - \frac{5}{x+4}\right) \, dx$ we obtain

$$y - 5 \ln |y+3| = x - 5 \ln |x+4| + c \quad \text{or} \quad \left(\frac{x+4}{y+3}\right)^5 = c_1 e^{x-y}.$$

Exercises 2.2

20. From $\frac{y+1}{y-1} dy = \frac{x+2}{x-3} dx$ or $\left(1 + \frac{2}{y-1}\right) dy = \left(1 + \frac{5}{x-3}\right) dx$ we obtain

$$y + 2 \ln |y-1| = x + 5 \ln |x-3| + c \quad \text{or} \quad \frac{(y-1)^2}{(x-3)^5} = c_1 e^{x-y}.$$

21. From $x dx = \frac{1}{\sqrt{1-y^2}} dy$ we obtain $\frac{1}{2}x^2 = \sin^{-1} y + c$ or $y = \sin\left(\frac{x^2}{2} + c_1\right)$.

22. From $\frac{1}{y^2} dy = \frac{1}{e^x + e^{-x}} dx = \frac{e^x}{(e^x)^2 + 1} dx$ we obtain $-\frac{1}{y} = \tan^{-1} e^x + c$ or $y = -\frac{1}{\tan^{-1} e^x + c}$.

23. From $\frac{1}{x^2+1} dx = 4 dt$ we obtain $\tan^{-1} x = 4t + c$. Using $x(\pi/4) = 1$ we find $c = -3\pi/4$. The solution of the initial-value problem is $\tan^{-1} x = 4t - \frac{3\pi}{4}$ or $x = \tan\left(4t - \frac{3\pi}{4}\right)$.

24. From $\frac{1}{y^2-1} dy = \frac{1}{x^2-1} dx$ or $\frac{1}{2}\left(\frac{1}{y-1} - \frac{1}{y+1}\right) dy = \frac{1}{2}\left(\frac{1}{x-1} - \frac{1}{x+1}\right) dx$ we obtain

$$\ln |y-1| - \ln |y+1| = \ln |x-1| - \ln |x+1| + \ln c \quad \text{or} \quad \frac{y-1}{y+1} = \frac{c(x-1)}{x+1}.$$

Using $y(2) = 2$ we find $c = 1$. The solution of the initial-value problem is $\frac{y-1}{y+1} = \frac{x-1}{x+1}$ or $y = x$.

25. From $\frac{1}{y} dy = \frac{1-x}{x^2} dx = \left(\frac{1}{x^2} - \frac{1}{x}\right) dx$ we obtain $\ln |y| = -\frac{1}{x} - \ln |x| + c$ or $xy = c_1 e^{-1/x}$. Using $y(-1) = -1$ we find $c_1 = e^{-1}$. The solution of the initial-value problem is $xy = e^{-1-1/x}$.

26. From $\frac{1}{1-2y} dy = dt$ we obtain $-\frac{1}{2} \ln |1-2y| = t + c$ or $1-2y = c_1 e^{-2t}$. Using $y(0) = 5/2$ we find $c_1 = -4$.

The solution of the initial-value problem is $1-2y = -4e^{-2t}$ or $y = 2e^{-2t} + \frac{1}{2}$.

27. Separating variables and integrating we obtain

$$\frac{dx}{\sqrt{1-x^2}} - \frac{dy}{\sqrt{1-y^2}} = 0 \quad \text{and} \quad \sin^{-1} x - \sin^{-1} y = c.$$

Setting $x = 0$ and $y = \sqrt{3}/2$ we obtain $c = -\pi/3$. Thus, an implicit solution of the initial-value problem is $\sin^{-1} x - \sin^{-1} y = \pi/3$. Solving for y and using a trigonometric identity we get

$$y = \sin\left(\sin^{-1} x + \frac{\pi}{3}\right) = x \cos \frac{\pi}{3} + \sqrt{1-x^2} \sin \frac{\pi}{3} = \frac{x}{2} + \frac{\sqrt{3}\sqrt{1-x^2}}{2}.$$

28. From $\frac{1}{1+(2y)^2} dy = \frac{-x}{1+(x^2)^2} dx$ we obtain

$$\frac{1}{2} \tan^{-1} 2y = -\frac{1}{2} \tan^{-1} x^2 + c \quad \text{or} \quad \tan^{-1} 2y + \tan^{-1} x^2 = c_1.$$

Using $y(1) = 0$ we find $c_1 = \pi/4$. The solution of the initial-value problem is

$$\tan^{-1} 2y + \tan^{-1} x^2 = \frac{\pi}{4}.$$

29. (a) The equilibrium solutions $y(x) = 2$ and $y(x) = -2$ satisfy the initial conditions $y(0) = 2$ and $y(0) = -2$, respectively. Setting $x = \frac{1}{4}$ and $y = 1$ in $y = 2(1 + ce^{4x})/(1 - ce^{4x})$ we obtain

$$1 = 2 \frac{1+ce}{1-ce}, \quad 1-ce = 2+2ce, \quad -1 = 3ce, \quad \text{and} \quad c = -\frac{1}{3e}.$$

The solution of the corresponding initial-value problem is

$$y = 2 \frac{1 - \frac{1}{3}e^{4x-1}}{1 + \frac{1}{3}e^{4x-1}} = 2 \frac{3 - e^{4x-1}}{3 + e^{4x-1}}.$$

(b) Separating variables and integrating yields

$$\begin{aligned} \frac{1}{4} \ln |y-2| - \frac{1}{4} \ln |y+2| + \ln c_1 &= x \\ \ln |y-2| - \ln |y+2| + \ln c &= 4x \\ \ln \left| \frac{c(y-2)}{y+2} \right| &= 4x \\ c \frac{y-2}{y+2} &= e^{4x}. \end{aligned}$$

Solving for y we get $y = 2(c + e^{4x})/(c - e^{4x})$. The initial condition $y(0) = -2$ can be solved for, yielding $c = 0$ and $y(x) = -2$. The initial condition $y(0) = 2$ does not correspond to a value of c , and it must simply be recognized that $y(x) = 2$ is a solution of the initial-value problem. Setting $x = \frac{1}{4}$ and $y = 1$ in $y = 2(c + e^{4x})/(c - e^{4x})$ leads to $c = -3e$. Thus, a solution of the initial-value problem is

$$y = 2 \frac{-3e + e^{4x}}{-3e - e^{4x}} = 2 \frac{3 - e^{4x-1}}{3 + e^{4x-1}}.$$

30. From $\left(\frac{1}{y-1} + \frac{-1}{y}\right) dy = \frac{1}{x} dx$ we obtain $\ln |y-1| - \ln |y| = \ln |x| + c$ or $y = \frac{1}{1 - c_1 x}$. Another solution is $y = 0$.

(a) If $y(0) = 1$ then $y = 1$.

(b) If $y(0) = 0$ then $y = 0$.

(c) If $y(1/2) = 1/2$ then $y = \frac{1}{1 + 2x}$.

(d) Setting $x = 2$ and $y = \frac{1}{4}$ we obtain

$$\frac{1}{4} = \frac{1}{1 - c_1(2)}, \quad 1 - 2c_1 = 4, \quad \text{and} \quad c_1 = -\frac{3}{2}.$$

$$\text{Thus, } y = \frac{1}{1 + \frac{3}{2}x} = \frac{2}{2 + 3x}.$$

31. Singular solutions of $dy/dx = x\sqrt{1-y^2}$ are $y = -1$ and $y = 1$. A singular solution of $(e^x + e^{-x})dy/dx = y^2$ is $y = 0$.

32. Differentiating $\ln(x^2 + 10) + \csc y = c$ we get

$$\frac{2x}{x^2 + 10} - \cot y \csc y \frac{dy}{dx} = 0,$$

$$\frac{2x}{x^2 + 10} - \frac{\cos y}{\sin y} \frac{1}{\sin y} \frac{dy}{dx} = 0,$$

or

$$2x \sin^2 y \, dx - (x^2 + 10) \cos y \, dy = 0.$$

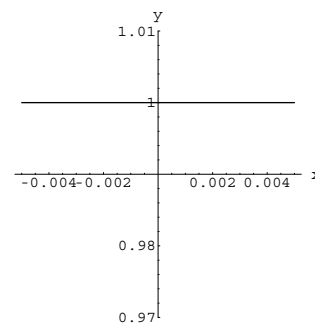
Writing the differential equation in the form

$$\frac{dy}{dx} = \frac{2x \sin^2 y}{(x^2 + 10) \cos y}$$

we see that singular solutions occur when $\sin^2 y = 0$, or $y = k\pi$, where k is an integer.

Exercises 2.2

33. The singular solution $y = 1$ satisfies the initial-value problem.

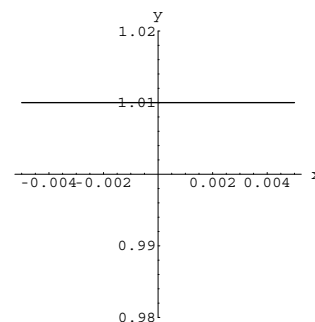


34. Separating variables we obtain $\frac{dy}{(y-1)^2} = dx$. Then

$$-\frac{1}{y-1} = x + c \quad \text{and} \quad y = \frac{x + c - 1}{x + c}.$$

Setting $x = 0$ and $y = 1.01$ we obtain $c = -100$. The solution is

$$y = \frac{x - 101}{x - 100}.$$

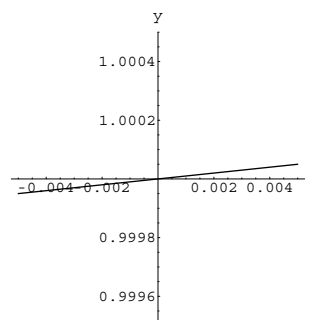


35. Separating variables we obtain $\frac{dy}{(y-1)^2 + 0.01} = dx$. Then

$$10 \tan^{-1} 10(y-1) = x + c \quad \text{and} \quad y = 1 + \frac{1}{10} \tan \frac{x + c}{10}.$$

Setting $x = 0$ and $y = 1$ we obtain $c = 0$. The solution is

$$y = 1 + \frac{1}{10} \tan \frac{x}{10}.$$

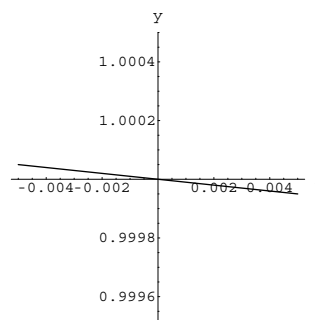


36. Separating variables we obtain $\frac{dy}{(y-1)^2 - 0.01} = dx$. Then

$$5 \ln \left| \frac{10y - 11}{10y - 9} \right| = x + c.$$

Setting $x = 0$ and $y = 1$ we obtain $c = 5 \ln 1 = 0$. The solution is

$$5 \ln \left| \frac{10y - 11}{10y - 9} \right| = x.$$



37. Separating variables, we have

$$\frac{dy}{y - y^3} = \frac{dy}{y(1 - y)(1 + y)} = \left(\frac{1}{y} + \frac{1/2}{1 - y} - \frac{1/2}{1 + y} \right) dy = dx.$$

Integrating, we get

$$\ln |y| - \frac{1}{2} \ln |1 - y| - \frac{1}{2} \ln |1 + y| = x + c.$$

When $y > 1$, this becomes

$$\ln y - \frac{1}{2} \ln(y-1) - \frac{1}{2} \ln(y+1) = \ln \frac{y}{\sqrt{y^2-1}} = x + c.$$

Letting $x = 0$ and $y = 2$ we find $c = \ln(2/\sqrt{3})$. Solving for y we get $y_1(x) = 2e^x/\sqrt{4e^{2x}-3}$, where $x > \ln(\sqrt{3}/2)$.

When $0 < y < 1$ we have

$$\ln y - \frac{1}{2} \ln(1-y) - \frac{1}{2} \ln(1+y) = \ln \frac{y}{\sqrt{1-y^2}} = x + c.$$

Letting $x = 0$ and $y = \frac{1}{2}$ we find $c = \ln(1/\sqrt{3})$. Solving for y we get $y_2(x) = e^x/\sqrt{e^{2x}+3}$, where $-\infty < x < \infty$.

When $-1 < y < 0$ we have

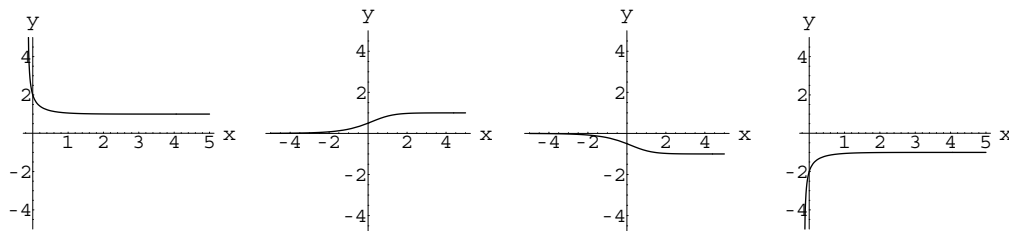
$$\ln(-y) - \frac{1}{2} \ln(1-y) - \frac{1}{2} \ln(1+y) = \ln \frac{-y}{\sqrt{1-y^2}} = x + c.$$

Letting $x = 0$ and $y = -\frac{1}{2}$ we find $c = \ln(1/\sqrt{3})$. Solving for y we get $y_3(x) = -e^x/\sqrt{e^{2x}+3}$, where $-\infty < x < \infty$.

When $y < -1$ we have

$$\ln(-y) - \frac{1}{2} \ln(1-y) - \frac{1}{2} \ln(-1-y) = \ln \frac{-y}{\sqrt{y^2-1}} = x + c.$$

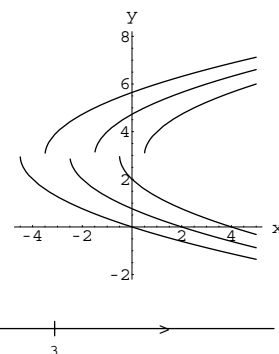
Letting $x = 0$ and $y = -2$ we find $c = \ln(2/\sqrt{3})$. Solving for y we get $y_4(x) = -2e^x/\sqrt{4e^{2x}-3}$, where $x > \ln(\sqrt{3}/2)$.



38. (a) The second derivative of y is

$$\frac{d^2y}{dx^2} = -\frac{dy/dx}{(y-1)^2} = -\frac{1/(y-3)}{(y-3)^2} = -\frac{1}{(y-3)^3}.$$

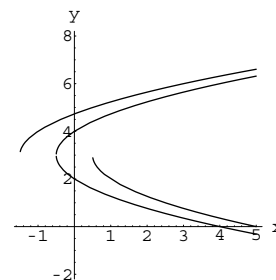
The solution curve is concave up when $d^2y/dx^2 > 0$ or $y > 3$, and concave down when $d^2y/dx^2 < 0$ or $y < 3$. From the phase portrait we see that the solution curve is decreasing when $y < 3$ and increasing when $y > 3$.



Exercises 2.2

- (b) Separating variables and integrating we obtain

$$\begin{aligned}(y-3) dy &= dx \\ \frac{1}{2}y^2 - 3y &= x + c \\ y^2 - 6y + 9 &= 2x + c_1 \\ (y-3)^2 &= 2x + c_1 \\ y &= 3 \pm \sqrt{2x + c_1}.\end{aligned}$$



The initial condition dictates whether to use the plus or minus sign.

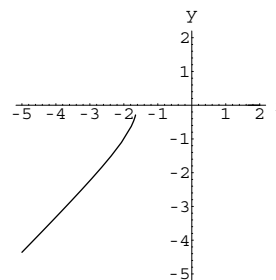
When $y_1(0) = 4$ we have $c_1 = 1$ and $y_1(x) = 3 + \sqrt{2x + 1}$.

When $y_2(0) = 2$ we have $c_1 = 1$ and $y_2(x) = 3 - \sqrt{2x + 1}$.

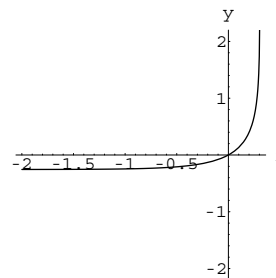
When $y_3(1) = 2$ we have $c_1 = -1$ and $y_3(x) = 3 - \sqrt{2x - 1}$.

When $y_4(-1) = 4$ we have $c_1 = 3$ and $y_4(x) = 3 + \sqrt{2x + 3}$.

39. (a) Separating variables we have $2y dy = (2x + 1)dx$. Integrating gives $y^2 = x^2 + x + c$. When $y(-2) = -1$ we find $c = -1$, so $y^2 = x^2 + x - 1$ and $y = -\sqrt{x^2 + x - 1}$. The negative square root is chosen because of the initial condition.
- (b) The interval of definition appears to be approximately $(-\infty, -1.65)$.



- (c) Solving $x^2 + x - 1 = 0$ we get $x = -\frac{1}{2} \pm \frac{1}{2}\sqrt{5}$, so the exact interval of definition is $(-\infty, -\frac{1}{2} - \frac{1}{2}\sqrt{5})$.
40. (a) From Problem 7 the general solution is $3e^{-2y} + 2e^{3x} = c$. When $y(0) = 0$ we find $c = 5$, so $3e^{-2y} + 2e^{3x} = 5$. Solving for y we get $y = -\frac{1}{2} \ln \frac{1}{3}(5 - 2e^{3x})$.
- (b) The interval of definition appears to be approximately $(-\infty, 0.3)$.



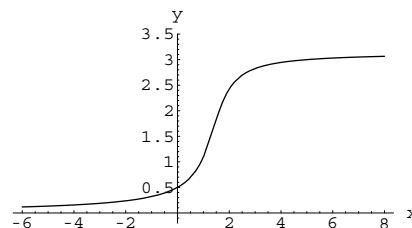
- (c) Solving $\frac{1}{3}(5 - 2e^{3x}) = 0$ we get $x = \frac{1}{3} \ln(\frac{5}{2})$, so the exact interval of definition is $(-\infty, \frac{1}{3} \ln(\frac{5}{2}))$.
41. (a) While $y_2(x) = -\sqrt{25 - x^2}$ is defined at $x = -5$ and $x = 5$, $y'_1(x)$ is not defined at these values, and so the interval of definition is the open interval $(-5, 5)$.
- (b) At any point on the x -axis the derivative of $y(x)$ is undefined, so no solution curve can cross the x -axis. Since $-x/y$ is not defined when $y = 0$, the initial-value problem has no solution.
42. (a) Separating variables and integrating we obtain $x^2 - y^2 = c$. For $c \neq 0$ the graph is a square hyperbola centered at the origin. All four initial conditions imply $c = 0$ and $y = \pm x$. Since the differential equation is

not defined for $y = 0$, solutions are $y = \pm x$, $x < 0$ and $y = \pm x$, $x > 0$. The solution for $y(a) = a$ is $y = x$, $x > 0$; for $y(a) = -a$ is $y = -x$; for $y(-a) = a$ is $y = -x$, $x < 0$; and for $y(-a) = -a$ is $y = x$, $x < 0$.

(b) Since x/y is not defined when $y = 0$, the initial-value problem has no solution.

(c) Setting $x = 1$ and $y = 2$ in $x^2 - y^2 = c$ we get $c = -3$, so $y^2 = x^2 + 3$ and $y(x) = \sqrt{x^2 + 3}$, where the positive square root is chosen because of the initial condition. The domain is all real numbers since $x^2 + 3 > 0$ for all x .

43. Separating variables we have $dy/(\sqrt{1+y^2} \sin^2 y) = dx$ which is not readily integrated (even by a CAS). We note that $dy/dx \geq 0$ for all values of x and y and that $dy/dx = 0$ when $y = 0$ and $y = \pi$, which are equilibrium solutions.



44. Separating variables we have $dy/(\sqrt{y} + y) = dx/(\sqrt{x} + x)$. To integrate $\int dt/(\sqrt{t} + t)$ we substitute $u^2 = t$ and get

$$\int \frac{2u}{u+u^2} du = \int \frac{2}{1+u} du = 2 \ln|1+u| + c = 2 \ln(1+\sqrt{x}) + c.$$

Integrating the separated differential equation we have

$$2 \ln(1+\sqrt{y}) = 2 \ln(1+\sqrt{x}) + c \quad \text{or} \quad \ln(1+\sqrt{y}) = \ln(1+\sqrt{x}) + \ln c_1.$$

Solving for y we get $y = [c_1(1+\sqrt{x}) - 1]^2$.

45. We are looking for a function $y(x)$ such that

$$y^2 + \left(\frac{dy}{dx}\right)^2 = 1.$$

Using the positive square root gives

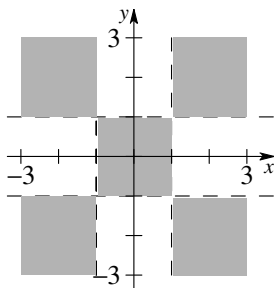
$$\frac{dy}{dx} = \sqrt{1-y^2} \implies \frac{dy}{\sqrt{1-y^2}} = dx \implies \sin^{-1} y = x + c.$$

Thus a solution is $y = \sin(x+c)$. If we use the negative square root we obtain

$$y = \sin(c-x) = -\sin(x-c) = -\sin(x+c_1).$$

Note also that $y = 1$ and $y = -1$ are solutions.

46. (a)



- (b) For $|x| > 1$ and $|y| > 1$ the differential equation is $dy/dx = \sqrt{y^2-1}/\sqrt{x^2-1}$. Separating variables and integrating, we obtain

$$\frac{dy}{\sqrt{y^2-1}} = \frac{dx}{\sqrt{x^2-1}} \quad \text{and} \quad \cosh^{-1} y = \cosh^{-1} x + c.$$

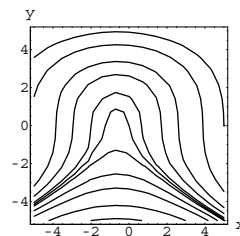
Exercises 2.2

Setting $x = 2$ and $y = 2$ we find $c = \cosh^{-1} 2 - \cosh^{-1} 2 = 0$ and $\cosh^{-1} y = \cosh^{-1} x$. An explicit solution is $y = x$.

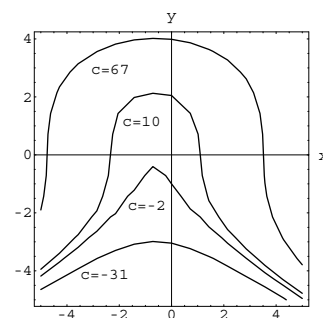
47. (a) Separating variables and integrating, we have

$$(3y^2 + 1)dy = -(8x + 5)dx \quad \text{and} \quad y^3 + y = -4x^2 - 5x + c.$$

Using a CAS we show various contours of $f(x, y) = y^3 + y + 4x^2 + 5x$. The plots shown on $[-5, 5] \times [-5, 5]$ correspond to c -values of 0, ± 5 , ± 20 , ± 40 , ± 80 , and ± 125 .



- (b) The value of c corresponding to $y(0) = -1$ is $f(0, -1) = -2$; to $y(0) = 2$ is $f(0, 2) = 10$; to $y(-1) = 4$ is $f(-1, 4) = 67$; and to $y(-1) = -3$ is -31 .



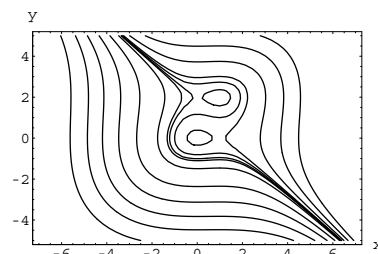
48. (a) Separating variables and integrating, we have

$$(-2y + y^2)dy = (x - x^2)dx$$

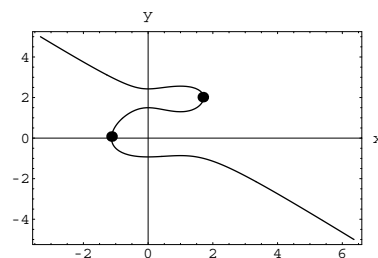
and

$$-y^2 + \frac{1}{3}y^3 = \frac{1}{2}x^2 - \frac{1}{3}x^3 + c.$$

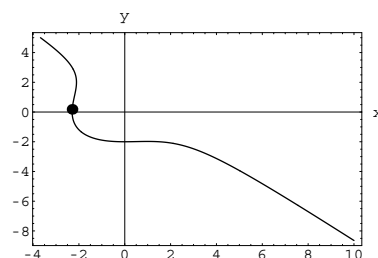
Using a CAS we show some contours of $f(x, y) = 2y^3 - 6y^2 + 2x^3 - 3x^2$. The plots shown on $[-7, 7] \times [-5, 5]$ correspond to c -values of -450 , -300 , -200 , -120 , -60 , -20 , -10 , -8.1 , -5 , -0.8 , 20 , 60 , and 120 .



- (b) The value of c corresponding to $y(0) = \frac{3}{2}$ is $f(0, \frac{3}{2}) = -\frac{27}{4}$. The portion of the graph between the dots corresponds to the solution curve satisfying the initial condition. To determine the interval of definition we find dy/dx for $2y^3 - 6y^2 + 2x^3 - 3x^2 = -\frac{27}{4}$. Using implicit differentiation we get $y' = (x - x^2)/(y^2 - 2y)$, which is infinite when $y = 0$ and $y = 2$. Letting $y = 0$ in $2y^3 - 6y^2 + 2x^3 - 3x^2 = -\frac{27}{4}$ and using a CAS to solve for x we get $x = -1.13232$. Similarly, letting $y = 2$, we find $x = 1.71299$. The largest interval of definition is approximately $(-1.13232, 1.71299)$.



- (c) The value of c corresponding to $y(0) = -2$ is $f(0, -2) = -40$. The portion of the graph to the right of the dot corresponds to the solution curve satisfying the initial condition. To determine the interval of definition we find dy/dx for $2y^3 - 6y^2 + 2x^3 - 3x^2 = -40$. Using implicit differentiation we get $y' = (x - x^2)/(y^2 - 2y)$, which is infinite when $y = 0$ and $y = 2$. Letting $y = 0$ in $2y^3 - 6y^2 + 2x^3 - 3x^2 = -40$ and using a CAS to solve for x we get $x = -2.29551$. The largest interval of definition is approximately $(-2.29551, \infty)$.



Exercises 2.3

1. For $y' - 5y = 0$ an integrating factor is $e^{-\int 5 dx} = e^{-5x}$ so that $\frac{d}{dx} [e^{-5x}y] = 0$ and $y = ce^{5x}$ for $-\infty < x < \infty$.
2. For $y' + 2y = 0$ an integrating factor is $e^{\int 2 dx} = e^{2x}$ so that $\frac{d}{dx} [e^{2x}y] = 0$ and $y = ce^{-2x}$ for $-\infty < x < \infty$.
The transient term is ce^{-2x} .
3. For $y' + y = e^{3x}$ an integrating factor is $e^{\int dx} = e^x$ so that $\frac{d}{dx} [e^x y] = e^{4x}$ and $y = \frac{1}{4}e^{3x} + ce^{-x}$ for $-\infty < x < \infty$.
The transient term is ce^{-x} .
4. For $y' + 4y = \frac{4}{3}$ an integrating factor is $e^{\int 4 dx} = e^{4x}$ so that $\frac{d}{dx} [e^{4x}y] = \frac{4}{3}e^{4x}$ and $y = \frac{1}{3} + ce^{-4x}$ for $-\infty < x < \infty$. The transient term is ce^{-4x} .
5. For $y' + 3x^2y = x^2$ an integrating factor is $e^{\int 3x^2 dx} = e^{x^3}$ so that $\frac{d}{dx} [e^{x^3}y] = x^2e^{x^3}$ and $y = \frac{1}{3} + ce^{-x^3}$ for $-\infty < x < \infty$. The transient term is ce^{-x^3} .
6. For $y' + 2xy = x^3$ an integrating factor is $e^{\int 2x dx} = e^{x^2}$ so that $\frac{d}{dx} [e^{x^2}y] = x^3e^{x^2}$ and $y = \frac{1}{2}x^2 - \frac{1}{2} + ce^{-x^2}$ for $-\infty < x < \infty$. The transient term is ce^{-x^2} .
7. For $y' + \frac{1}{x}y = \frac{1}{x^2}$ an integrating factor is $e^{\int (1/x) dx} = x$ so that $\frac{d}{dx} [xy] = \frac{1}{x}$ and $y = \frac{1}{x} \ln x + \frac{c}{x}$ for $0 < x < \infty$.
8. For $y' - 2y = x^2 + 5$ an integrating factor is $e^{-\int 2 dx} = e^{-2x}$ so that $\frac{d}{dx} [e^{-2x}y] = x^2e^{-2x} + 5e^{-2x}$ and $y = -\frac{1}{2}x^2 - \frac{1}{2}x - \frac{11}{4} + ce^{2x}$ for $-\infty < x < \infty$.
9. For $y' - \frac{1}{x}y = x \sin x$ an integrating factor is $e^{-\int (1/x) dx} = \frac{1}{x}$ so that $\frac{d}{dx} \left[\frac{1}{x} y \right] = \sin x$ and $y = cx - x \cos x$ for $0 < x < \infty$.
10. For $y' + \frac{2}{x}y = \frac{3}{x}$ an integrating factor is $e^{\int (2/x) dx} = x^2$ so that $\frac{d}{dx} [x^2y] = 3x$ and $y = \frac{3}{2} + cx^{-2}$ for $0 < x < \infty$.
11. For $y' + \frac{4}{x}y = x^2 - 1$ an integrating factor is $e^{\int (4/x) dx} = x^4$ so that $\frac{d}{dx} [x^4y] = x^6 - x^4$ and $y = \frac{1}{7}x^3 - \frac{1}{5}x + cx^{-4}$ for $0 < x < \infty$.
12. For $y' - \frac{x}{(1+x)}y = x$ an integrating factor is $e^{-\int [x/(1+x)] dx} = (x+1)e^{-x}$ so that $\frac{d}{dx} [(x+1)e^{-x}y] = x(x+1)e^{-x}$ and $y = -x - \frac{2x+3}{x+1} + \frac{ce^x}{x+1}$ for $-1 < x < \infty$.
13. For $y' + \left(1 + \frac{2}{x}\right)y = \frac{e^x}{x^2}$ an integrating factor is $e^{\int [1+(2/x)] dx} = x^2e^x$ so that $\frac{d}{dx} [x^2e^xy] = e^{2x}$ and $y = \frac{1}{2} \frac{e^x}{x^2} + \frac{ce^{-x}}{x^2}$ for $0 < x < \infty$. The transient term is $\frac{ce^{-x}}{x^2}$.
14. For $y' + \left(1 + \frac{1}{x}\right)y = \frac{1}{x}e^{-x} \sin 2x$ an integrating factor is $e^{\int [1+(1/x)] dx} = xe^x$ so that $\frac{d}{dx} [xe^xy] = \sin 2x$ and $y = -\frac{1}{2x}e^{-x} \cos 2x + \frac{ce^{-x}}{x}$ for $0 < x < \infty$. The entire solution is transient.
15. For $\frac{dx}{dy} - \frac{4}{y}x = 4y^5$ an integrating factor is $e^{-\int (4/y) dy} = y^{-4}$ so that $\frac{d}{dy} [y^{-4}x] = 4y$ and $x = 2y^6 + cy^4$ for $0 < y < \infty$.

Exercises 2.3

16. For $\frac{dx}{dy} + \frac{2}{y}x = e^y$ an integrating factor is $e^{\int (2/y)dy} = y^2$ so that $\frac{d}{dy}[y^2x] = y^2e^y$ and $x = e^y - \frac{2}{y}e^y + \frac{2}{y^2} + \frac{c}{y^2}$ for $0 < y < \infty$. The transient term is $\frac{2+c}{y^2}$.
17. For $y' + (\tan x)y = \sec x$ an integrating factor is $e^{\int \tan x dx} = \sec x$ so that $\frac{d}{dx}[(\sec x)y] = \sec^2 x$ and $y = \sin x + c \cos x$ for $-\pi/2 < x < \pi/2$.
18. For $y' + (\cot x)y = \sec^2 x \csc x$ an integrating factor is $e^{\int \cot x dx} = \sin x$ so that $\frac{d}{dx}[(\sin x)y] = \sec^2 x$ and $y = \sec x + c \csc x$ for $0 < x < \pi/2$.
19. For $y' + \frac{x+2}{x+1}y = \frac{2xe^{-x}}{x+1}$ an integrating factor is $e^{\int [(x+2)/(x+1)]dx} = (x+1)e^x$ so that $\frac{d}{dx}[(x+1)e^xy] = 2x$ and $y = \frac{x^2}{x+1}e^{-x} + \frac{c}{x+1}e^{-x}$ for $-1 < x < \infty$. The entire solution is transient.
20. For $y' + \frac{4}{x+2}y = \frac{5}{(x+2)^2}$ an integrating factor is $e^{\int [4/(x+2)]dx} = (x+2)^4$ so that $\frac{d}{dx}[(x+2)^4y] = 5(x+2)^2$ and $y = \frac{5}{3}(x+2)^{-1} + c(x+2)^{-4}$ for $-2 < x < \infty$. The entire solution is transient.
21. For $\frac{dr}{d\theta} + r \sec \theta = \cos \theta$ an integrating factor is $e^{\int \sec \theta d\theta} = \sec \theta + \tan \theta$ so that $\frac{d}{d\theta}[r(\sec \theta + \tan \theta)] = 1 + \sin \theta$ and $r(\sec \theta + \tan \theta) = \theta - \cos \theta + c$ for $-\pi/2 < \theta < \pi/2$.
22. For $\frac{dP}{dt} + (2t-1)P = 4t-2$ an integrating factor is $e^{\int (2t-1)dt} = e^{t^2-t}$ so that $\frac{d}{dt}[Pe^{t^2-t}] = (4t-2)e^{t^2-t}$ and $P = 2 + ce^{t-t^2}$ for $-\infty < t < \infty$. The transient term is ce^{t-t^2} .
23. For $y' + \left(3 + \frac{1}{x}\right)y = \frac{e^{-3x}}{x}$ an integrating factor is $e^{\int [3+(1/x)]dx} = xe^{3x}$ so that $\frac{d}{dx}[xe^{3x}y] = 1$ and $y = e^{-3x} + \frac{ce^{-3x}}{x}$ for $0 < x < \infty$. The transient term is ce^{-3x}/x .
24. For $y' + \frac{2}{x^2-1}y = \frac{x+1}{x-1}$ an integrating factor is $e^{\int [2/(x^2-1)]dx} = \frac{x-1}{x+1}$ so that $\frac{d}{dx}\left[\frac{x-1}{x+1}y\right] = 1$ and $(x-1)y = x(x+1) + c(x+1)$ for $-1 < x < 1$.
25. For $y' + \frac{1}{x}y = \frac{1}{x}e^x$ an integrating factor is $e^{\int (1/x)dx} = x$ so that $\frac{d}{dx}[xy] = e^x$ and $y = \frac{1}{x}e^x + \frac{c}{x}$ for $0 < x < \infty$. If $y(1) = 2$ then $c = 2 - e$ and $y = \frac{1}{x}e^x + \frac{2-e}{x}$.
26. For $\frac{dx}{dy} - \frac{1}{y}x = 2y$ an integrating factor is $e^{-\int (1/y)dy} = \frac{1}{y}$ so that $\frac{d}{dy}\left[\frac{1}{y}x\right] = 2$ and $x = 2y^2 + cy$ for $-\infty < y < \infty$. If $y(1) = 5$ then $c = -49/5$ and $x = 2y^2 - \frac{49}{5}y$.
27. For $\frac{di}{dt} + \frac{R}{L}i = \frac{E}{L}$ an integrating factor is $e^{\int (R/L)dt} = e^{Rt/L}$ so that $\frac{d}{dt}[ie^{Rt/L}] = \frac{E}{L}e^{Rt/L}$ and $i = \frac{E}{R} + ce^{-Rt/L}$ for $-\infty < t < \infty$. If $i(0) = i_0$ then $c = i_0 - E/R$ and $i = \frac{E}{R} + \left(i_0 - \frac{E}{R}\right)e^{-Rt/L}$.
28. For $\frac{dT}{dt} - kT = -T_mk$ an integrating factor is $e^{\int (-k)dt} = e^{-kt}$ so that $\frac{d}{dt}[Te^{-kt}] = -T_mk e^{-kt}$ and $T = T_m + ce^{kt}$ for $-\infty < t < \infty$. If $T(0) = T_0$ then $c = T_0 - T_m$ and $T = T_m + (T_0 - T_m)e^{kt}$.
29. For $y' + \frac{1}{x+1}y = \frac{\ln x}{x+1}$ an integrating factor is $e^{\int [1/(x+1)]dx} = x+1$ so that $\frac{d}{dx}[(x+1)y] = \ln x$ and $y = \frac{x}{x+1} \ln x - \frac{x}{x+1} + \frac{c}{x+1}$ for $0 < x < \infty$. If $y(1) = 10$ then $c = 21$ and $y = \frac{x}{x+1} \ln x - \frac{x}{x+1} + \frac{21}{x+1}$.

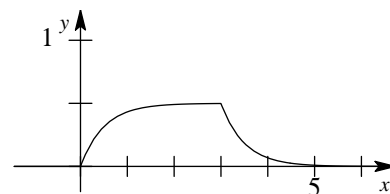
30. For $y' + (\tan x)y = \cos^2 x$ an integrating factor is $e^{\int \tan x dx} = \sec x$ so that $\frac{d}{dx}[(\sec x)y] = \cos x$ and $y = \sin x \cos x + c \cos x$ for $-\pi/2 < x < \pi/2$. If $y(0) = -1$ then $c = -1$ and $y = \sin x \cos x - \cos x$.

31. For $y' + 2y = f(x)$ an integrating factor is e^{2x} so that

$$ye^{2x} = \begin{cases} \frac{1}{2}e^{2x} + c_1, & 0 \leq x \leq 3; \\ c_2, & x > 3. \end{cases}$$

If $y(0) = 0$ then $c_1 = -1/2$ and for continuity we must have $c_2 = \frac{1}{2}e^6 - \frac{1}{2}$ so that

$$y = \begin{cases} \frac{1}{2}(1 - e^{-2x}), & 0 \leq x \leq 3; \\ \frac{1}{2}(e^6 - 1)e^{-2x}, & x > 3. \end{cases}$$

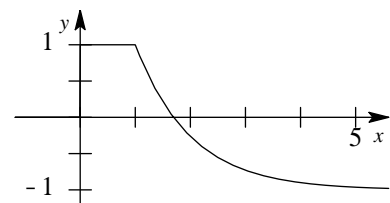


32. For $y' + y = f(x)$ an integrating factor is e^x so that

$$ye^x = \begin{cases} e^x + c_1, & 0 \leq x \leq 1; \\ -e^x + c_2, & x > 1. \end{cases}$$

If $y(0) = 1$ then $c_1 = 0$ and for continuity we must have $c_2 = 2e$ so that

$$y = \begin{cases} 1, & 0 \leq x \leq 1; \\ 2e^{1-x} - 1, & x > 1. \end{cases}$$

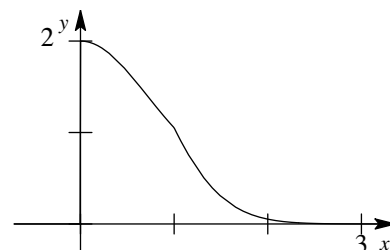


33. For $y' + 2xy = f(x)$ an integrating factor is e^{x^2} so that

$$ye^{x^2} = \begin{cases} \frac{1}{2}e^{x^2} + c_1, & 0 \leq x \leq 1; \\ c_2, & x > 1. \end{cases}$$

If $y(0) = 2$ then $c_1 = 3/2$ and for continuity we must have $c_2 = \frac{1}{2}e + \frac{3}{2}$ so that

$$y = \begin{cases} \frac{1}{2} + \frac{3}{2}e^{-x^2}, & 0 \leq x \leq 1; \\ (\frac{1}{2}e + \frac{3}{2})e^{-x^2}, & x > 1. \end{cases}$$



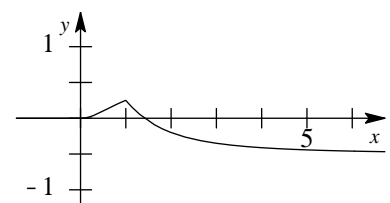
34. For $y' + \frac{2x}{1+x^2}y = \begin{cases} \frac{x}{1+x^2}, & 0 \leq x \leq 1; \\ \frac{-x}{1+x^2}, & x > 1 \end{cases}$

an integrating factor is $1+x^2$ so that

$$(1+x^2)y = \begin{cases} \frac{1}{2}x^2 + c_1, & 0 \leq x \leq 1; \\ -\frac{1}{2}x^2 + c_2, & x > 1. \end{cases}$$

If $y(0) = 0$ then $c_1 = 0$ and for continuity we must have $c_2 = 1$ so that

$$y = \begin{cases} \frac{1}{2} - \frac{1}{2(1+x^2)}, & 0 \leq x \leq 1; \\ \frac{3}{2(1+x^2)} - \frac{1}{2}, & x > 1. \end{cases}$$



Exercises 2.3

35. We need

$$\int P(x)dx = \begin{cases} 2x, & 0 \leq x \leq 1 \\ -2 \ln x, & x > 1 \end{cases}.$$

An integrating factor is

$$e^{\int P(x)dx} = \begin{cases} e^{2x}, & 0 \leq x \leq 1 \\ 1/x^2, & x > 1 \end{cases}$$

and

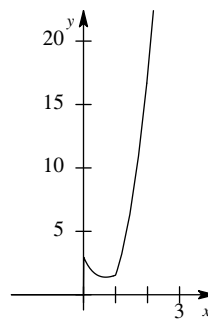
$$\frac{d}{dx} \left[\begin{cases} ye^{2x}, & 0 \leq x \leq 1 \\ y/x^2, & x > 1 \end{cases} \right] = \begin{cases} 4xe^{2x}, & 0 \leq x \leq 1 \\ 4/x, & x > 1 \end{cases}.$$

Integrating we get

$$\begin{cases} ye^{2x}, & 0 \leq x \leq 1 \\ y/x^2, & x > 1 \end{cases} = \begin{cases} 2xe^{2x} - e^{2x} + c_1, & 0 \leq x \leq 1 \\ 4 \ln x + c_2, & x > 1 \end{cases}.$$

Using $y(0) = 3$ we find $c_1 = 4$. For continuity we must have $c_2 = 2 - 1 + 4e^{-2} = 1 + 4e^{-2}$. Then

$$y = \begin{cases} 2x - 1 + 4e^{-2x}, & 0 \leq x \leq 1 \\ 4x^2 \ln x + (1 + 4e^{-2})x^2, & x > 1 \end{cases}.$$



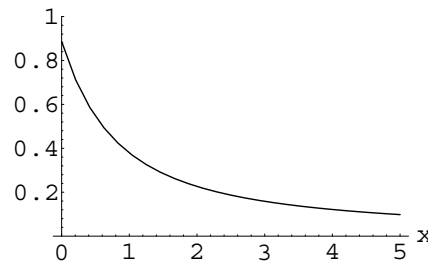
36. (a) An integrating factor for $y' - 2xy = -1$ is e^{-x^2} . Thus

$$\begin{aligned} \frac{d}{dx}[e^{-x^2}y] &= -e^{-x^2} \\ e^{-x^2}y &= -\int_0^x e^{-t^2} dt = -\frac{\sqrt{\pi}}{2} \operatorname{erf}(x) + c. \end{aligned}$$

From $y(0) = \sqrt{\pi}/2$, and noting that $\operatorname{erf}(0) = 0$, we get $c = \sqrt{\pi}/2$. Thus

$$y = e^{x^2} \left(-\frac{\sqrt{\pi}}{2} \operatorname{erf}(x) + \frac{\sqrt{\pi}}{2} \right) = \frac{\sqrt{\pi}}{2} e^{x^2} (1 - \operatorname{erf}(x)) = \frac{\sqrt{\pi}}{2} e^{x^2} \operatorname{erfc}(x).$$

(b) Using *Mathematica* we find $y(2) \approx 0.226339$.



37. An integrating factor for $y' - 2xy = 1$ is e^{-x^2} . Thus

$$\begin{aligned} \frac{d}{dx}[e^{-x^2}y] &= e^{-x^2} \\ e^{-x^2}y &= \int_0^x e^{-t^2} dt = \operatorname{erf}(x) + c \end{aligned}$$

and

$$y = e^{x^2} \operatorname{erf}(x) + ce^{x^2}.$$

From $y(1) = 1$ we get $1 = e \operatorname{erf}(1) + ce$, so that $c = e^{-1} - \operatorname{erf}(1)$. Thus

$$y = e^{x^2} \operatorname{erf}(x) + (e^{-1} - \operatorname{erf}(1))e^{x^2} = e^{x^2-1} + e^{x^2}(\operatorname{erf}(x) - \operatorname{erf}(1)).$$

38. (a) An integrating factor for

$$y' + \frac{2}{x}y = \frac{10 \sin x}{x^3}$$

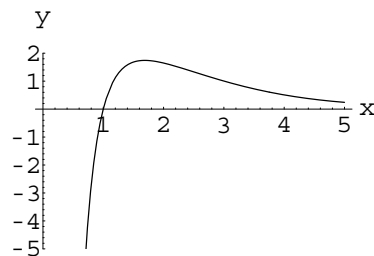
is x^2 . Thus

$$\begin{aligned}\frac{d}{dx}[x^2 y] &= 10 \frac{\sin x}{x} \\ x^2 y &= 10 \int_0^x \frac{\sin t}{t} dt + c \\ y &= 10x^{-2} \text{Si}(x) + cx^{-2}.\end{aligned}$$

From $y(1) = 0$ we get $c = -10\text{Si}(1)$. Thus

$$y = 10x^{-2}\text{Si}(x) - 10x^{-2}\text{Si}(1) = 10x^{-2}(\text{Si}(x) - \text{Si}(1)).$$

(b) Using *Mathematica* we find $y(2) \approx 1.64832$.



39. (a) Separating variables and integrating, we have

$$\frac{dy}{y} = \sin x^2 dx \quad \text{and} \quad \ln |y| = \int_0^x \sin t^2 dt + c.$$

Now, letting $t = \sqrt{\pi/2} u$ we have

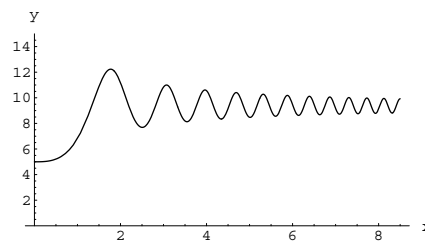
$$\int_0^x \sin t^2 dt = \sqrt{\frac{\pi}{2}} \int_0^{\sqrt{2/\pi} x} \sin\left(\frac{\pi}{2} u^2\right) du,$$

so

$$y = c_1 e^{\int_0^x \sin t^2 dt} = c_1 e^{\sqrt{\pi/2} \int_0^{\sqrt{2/\pi} x} \sin(\pi u^2/2) du} = c_1 e^{\sqrt{\pi/2} S(\sqrt{2/\pi} x)}.$$

Using $S(0) = 0$ and $y(0) = 5$ we see that $c_1 = 5$ and $y = 5e^{\sqrt{\pi/2} S(\sqrt{2/\pi} x)}$.

(b) Using *Mathematica* we find $y(2) \approx 11.181$. From the graph we see that as $x \rightarrow \infty$, $y(x)$ oscillates with decreasing amplitude approaching 9.35672.



40. For $y' + e^x y = 1$ an integrating factor is e^{e^x} . Thus

$$\frac{d}{dx} [e^{e^x} y] = e^{e^x} \quad \text{and} \quad e^{e^x} y = \int_0^x e^{e^t} dt + c.$$

From $y(0) = 1$ we get $c = e$, so $y = e^{-e^x} \int_0^x e^{e^t} dt + e^{1-e^x}$.

When $y' + e^x y = 0$ we can separate variables and integrate:

$$\frac{dy}{y} = -e^x dx \quad \text{and} \quad \ln |y| = -e^x + c.$$

Thus $y = c_1 e^{-e^x}$. From $y(0) = 1$ we get $c_1 = e$, so $y = e^{1-e^x}$.

When $y' + e^x y = e^x$ we can see by inspection that $y = 1$ is a solution.

Exercises 2.3

41. On the interval $(-3, 3)$ the integrating factor is

$$e^{\int x \, dx / (x^2 - 9)} = e^{-\int x \, dx / (9 - x^2)} = e^{\frac{1}{2} \ln(9 - x^2)} = \sqrt{9 - x^2}$$

and so

$$\frac{d}{dx} \left[\sqrt{9 - x^2} y \right] = 0 \quad \text{and} \quad y = \frac{c}{\sqrt{9 - x^2}}.$$

42. We want 4 to be a critical point, so use $y' = 4 - y$.
43. (a) All solutions of the form $y = x^5 e^x - x^4 e^x + cx^4$ satisfy the initial condition. In this case, since $4/x$ is discontinuous at $x = 0$, the hypotheses of Theorem 1.1 are not satisfied and the initial-value problem does not have a unique solution.
- (b) The differential equation has no solution satisfying $y(0) = y_0$, $y_0 \neq 0$.
- (c) In this case, since $x_0 \neq 0$, Theorem 1.1 applies and the initial-value problem has a unique solution given by $y = x^5 e^x - x^4 e^x + cx^4$ where $c = y_0/x_0^4 - x_0 e^{x_0} + e^{x_0}$.
44. We want the general solution to be $y = 3x - 5 + ce^{-x}$. (Rather than e^{-x} , any function that approaches 0 as $x \rightarrow \infty$ could be used.) Differentiating we get

$$y' = 3 - ce^{-x} = 3 - (y - 3x + 5) = -y + 3x - 2,$$

so the differential equation $y' + y = 3x - 2$ has solutions asymptotic to the line $y = 3x - 5$.

45. The left-hand derivative of the function at $x = 1$ is $1/e$ and the right-hand derivative at $x = 1$ is $1 - 1/e$. Thus, y is not differentiable at $x = 1$.
46. Differentiating $y_c = c/x^3$ we get

$$y'_c = -\frac{3c}{x^4} = -\frac{3}{x} \frac{c}{x^3} = -\frac{3}{x} y_c$$

so a differential equation with general solution $y_c = c/x^3$ is $xy' + 3y = 0$. Now

$$xy'_p + 3y_p = x(3x^2) + 3(x^3) = 6x^3$$

so a differential equation with general solution $y = c/x^3 + x^3$ is $xy' + 3y = 6x^3$.

47. Since $e^{\int P(x)dx+c} = e^c e^{\int P(x)dx} = c_1 e^{\int P(x)dx}$, we would have

$$c_1 e^{\int P(x)dx} y = c_2 + \int c_1 e^{\int P(x)dx} f(x) dx \quad \text{and} \quad e^{\int P(x)dx} y = c_3 + \int e^{\int P(x)dx} f(x) dx,$$

which is the same as (6) in the text.

48. We see by inspection that $y = 0$ is a solution.

Exercises 2.4

1. Let $M = 2x - 1$ and $N = 3y + 7$ so that $M_y = 0 = N_x$. From $f_x = 2x - 1$ we obtain $f = x^2 - x + h(y)$, $h'(y) = 3y + 7$, and $h(y) = \frac{3}{2}y^2 + 7y$. The solution is $x^2 - x + \frac{3}{2}y^2 + 7y = c$.
2. Let $M = 2x + y$ and $N = -x - 6y$. Then $M_y = 1$ and $N_x = -1$, so the equation is not exact.
3. Let $M = 5x + 4y$ and $N = 4x - 8y^3$ so that $M_y = 4 = N_x$. From $f_x = 5x + 4y$ we obtain $f = \frac{5}{2}x^2 + 4xy + h(y)$, $h'(y) = -8y^3$, and $h(y) = -2y^4$. The solution is $\frac{5}{2}x^2 + 4xy - 2y^4 = c$.
4. Let $M = \sin y - y \sin x$ and $N = \cos x + x \cos y - y$ so that $M_y = \cos y - \sin x = N_x$. From $f_x = \sin y - y \sin x$ we obtain $f = x \sin y + y \cos x + h(y)$, $h'(y) = -y$, and $h(y) = -\frac{1}{2}y^2$. The solution is $x \sin y + y \cos x - \frac{1}{2}y^2 = c$.
5. Let $M = 2y^2x - 3$ and $N = 2yx^2 + 4$ so that $M_y = 4xy = N_x$. From $f_x = 2y^2x - 3$ we obtain $f = x^2y^2 - 3x + h(y)$, $h'(y) = 4$, and $h(y) = 4y$. The solution is $x^2y^2 - 3x + 4y = c$.
6. Let $M = 4x^3 - 3y \sin 3x - y/x^2$ and $N = 2y - 1/x + \cos 3x$ so that $M_y = -3 \sin 3x - 1/x^2$ and $N_x = 1/x^2 - 3 \sin 3x$. The equation is not exact.
7. Let $M = x^2 - y^2$ and $N = x^2 - 2xy$ so that $M_y = -2y$ and $N_x = 2x - 2y$. The equation is not exact.
8. Let $M = 1 + \ln x + y/x$ and $N = -1 + \ln x$ so that $M_y = 1/x = N_x$. From $f_y = -1 + \ln x$ we obtain $f = -y + y \ln x + h(y)$, $h'(x) = 1 + \ln x$, and $h(y) = x \ln x$. The solution is $-y + y \ln x + x \ln x = c$.
9. Let $M = y^3 - y^2 \sin x - x$ and $N = 3xy^2 + 2y \cos x$ so that $M_y = 3y^2 - 2y \sin x = N_x$. From $f_x = y^3 - y^2 \sin x - x$ we obtain $f = xy^3 + y^2 \cos x - \frac{1}{2}x^2 + h(y)$, $h'(y) = 0$, and $h(y) = 0$. The solution is $xy^3 + y^2 \cos x - \frac{1}{2}x^2 = c$.
10. Let $M = x^3 + y^3$ and $N = 3xy^2$ so that $M_y = 3y^2 = N_x$. From $f_x = x^3 + y^3$ we obtain $f = \frac{1}{4}x^4 + xy^3 + h(y)$, $h'(y) = 0$, and $h(y) = 0$. The solution is $\frac{1}{4}x^4 + xy^3 = c$.
11. Let $M = y \ln y - e^{-xy}$ and $N = 1/y + x \ln y$ so that $M_y = 1 + \ln y + ye^{-xy}$ and $N_x = \ln y$. The equation is not exact.
12. Let $M = 3x^2y + e^y$ and $N = x^3 + xe^y - 2y$ so that $M_y = 3x^2 + e^y = N_x$. From $f_x = 3x^2y + e^y$ we obtain $f = x^3y + xe^y + h(y)$, $h'(y) = -2y$, and $h(y) = -y^2$. The solution is $x^3y + xe^y - y^2 = c$.
13. Let $M = y - 6x^2 - 2xe^x$ and $N = x$ so that $M_y = 1 = N_x$. From $f_x = y - 6x^2 - 2xe^x$ we obtain $f = xy - 2x^3 - 2xe^x + h(y)$, $h'(y) = 0$, and $h(y) = 0$. The solution is $xy - 2x^3 - 2xe^x + 2e^x = c$.
14. Let $M = 1 - 3/x + y$ and $N = 1 - 3/y + x$ so that $M_y = 1 = N_x$. From $f_x = 1 - 3/x + y$ we obtain $f = x - 3 \ln |x| + xy + h(y)$, $h'(y) = 1 - \frac{3}{y}$, and $h(y) = y - 3 \ln |y|$. The solution is $x + y + xy - 3 \ln |xy| = c$.
15. Let $M = x^2y^3 - 1/(1 + 9x^2)$ and $N = x^3y^2$ so that $M_y = 3x^2y^2 = N_x$. From $f_x = x^2y^3 - 1/(1 + 9x^2)$ we obtain $f = \frac{1}{3}x^3y^3 - \frac{1}{3} \arctan(3x) + h(y)$, $h'(y) = 0$, and $h(y) = 0$. The solution is $x^3y^3 - \arctan(3x) = c$.
16. Let $M = -2y$ and $N = 5y - 2x$ so that $M_y = -2 = N_x$. From $f_x = -2y$ we obtain $f = -2xy + h(y)$, $h'(y) = 5y$, and $h(y) = \frac{5}{2}y^2$. The solution is $-2xy + \frac{5}{2}y^2 = c$.
17. Let $M = \tan x - \sin x \sin y$ and $N = \cos x \cos y$ so that $M_y = -\sin x \cos y = N_x$. From $f_x = \tan x - \sin x \sin y$ we obtain $f = \ln |\sec x| + \cos x \sin y + h(y)$, $h'(y) = 0$, and $h(y) = 0$. The solution is $\ln |\sec x| + \cos x \sin y = c$.
18. Let $M = 2y \sin x \cos x - y + 2y^2e^{xy^2}$ and $N = -x + \sin^2 x + 4xye^{xy^2}$ so that

$$M_y = 2 \sin x \cos x - 1 + 4xy^3e^{xy^2} + 4ye^{xy^2} = N_x.$$

From $f_x = 2y \sin x \cos x - y + 2y^2e^{xy^2}$ we obtain $f = y \sin^2 x - xy + 2e^{xy^2} + h(y)$, $h'(y) = 0$, and $h(y) = 0$. The solution is $y \sin^2 x - xy + 2e^{xy^2} = c$.

Exercises 2.4

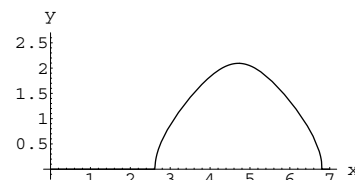
19. Let $M = 4t^3y - 15t^2 - y$ and $N = t^4 + 3y^2 - t$ so that $M_y = 4t^3 - 1 = N_t$. From $f_t = 4t^3y - 15t^2 - y$ we obtain $f = t^4y - 5t^3 - ty + h(y)$, $h'(y) = 3y^2$, and $h(y) = y^3$. The solution is $t^4y - 5t^3 - ty + y^3 = c$.
20. Let $M = 1/t + 1/t^2 - y/(t^2 + y^2)$ and $N = ye^y + t/(t^2 + y^2)$ so that $M_y = (y^2 - t^2)/(t^2 + y^2)^2 = N_t$. From $f_t = 1/t + 1/t^2 - y/(t^2 + y^2)$ we obtain $f = \ln|t| - \frac{1}{t} - \arctan\left(\frac{t}{y}\right) + h(y)$, $h'(y) = ye^y$, and $h(y) = ye^y - e^y$. The solution is
- $$\ln|t| - \frac{1}{t} - \arctan\left(\frac{t}{y}\right) + ye^y - e^y = c.$$
21. Let $M = x^2 + 2xy + y^2$ and $N = 2xy + x^2 - 1$ so that $M_y = 2(x + y) = N_x$. From $f_x = x^2 + 2xy + y^2$ we obtain $f = \frac{1}{3}x^3 + x^2y + xy^2 + h(y)$, $h'(y) = -1$, and $h(y) = -y$. The general solution is $\frac{1}{3}x^3 + x^2y + xy^2 - y = c$. If $y(1) = 1$ then $c = 4/3$ and the solution of the initial-value problem is $\frac{1}{3}x^3 + x^2y + xy^2 - y = \frac{4}{3}$.
22. Let $M = e^x + y$ and $N = 2 + x + ye^y$ so that $M_y = 1 = N_x$. From $f_x = e^x + y$ we obtain $f = e^x + xy + h(y)$, $h'(y) = 2 + ye^y$, and $h(y) = 2y + ye^y - e^y$. The general solution is $e^x + xy + 2y + ye^y - e^y = c$. If $y(0) = 1$ then $c = 3$ and the solution of the initial-value problem is $e^x + xy + 2y + ye^y - e^y = 3$.
23. Let $M = 4y + 2t - 5$ and $N = 6y + 4t - 1$ so that $M_y = 4 = N_t$. From $f_t = 4y + 2t - 5$ we obtain $f = 4ty + t^2 - 5t + h(y)$, $h'(y) = 6y - 1$, and $h(y) = 3y^2 - y$. The general solution is $4ty + t^2 - 5t + 3y^2 - y = c$. If $y(-1) = 2$ then $c = 8$ and the solution of the initial-value problem is $4ty + t^2 - 5t + 3y^2 - y = 8$.
24. Let $M = t/2y^4$ and $N = (3y^2 - t^2)/y^5$ so that $M_y = -2t/y^5 = N_t$. From $f_t = t/2y^4$ we obtain $f = \frac{t^2}{4y^4} + h(y)$, $h'(y) = \frac{3}{y^3}$, and $h(y) = -\frac{3}{2y^2}$. The general solution is $\frac{t^2}{4y^4} - \frac{3}{2y^2} = c$. If $y(1) = 1$ then $c = -5/4$ and the solution of the initial-value problem is $\frac{t^2}{4y^4} - \frac{3}{2y^2} = -\frac{5}{4}$.
25. Let $M = y^2 \cos x - 3x^2y - 2x$ and $N = 2y \sin x - x^3 + \ln y$ so that $M_y = 2y \cos x - 3x^2 = N_x$. From $f_x = y^2 \cos x - 3x^2y - 2x$ we obtain $f = y^2 \sin x - x^3y - x^2 + h(y)$, $h'(y) = \ln y$, and $h(y) = y \ln y - y$. The general solution is $y^2 \sin x - x^3y - x^2 + y \ln y - y = c$. If $y(0) = e$ then $c = 0$ and the solution of the initial-value problem is $y^2 \sin x - x^3y - x^2 + y \ln y - y = 0$.
26. Let $M = y^2 + y \sin x$ and $N = 2xy - \cos x - 1/(1 + y^2)$ so that $M_y = 2y + \sin x = N_x$. From $f_x = y^2 + y \sin x$ we obtain $f = xy^2 - y \cos x + h(y)$, $h'(y) = \frac{-1}{1 + y^2}$, and $h(y) = -\tan^{-1} y$. The general solution is $xy^2 - y \cos x - \tan^{-1} y = c$. If $y(0) = 1$ then $c = -1 - \pi/4$ and the solution of the initial-value problem is $xy^2 - y \cos x - \tan^{-1} y = -1 - \frac{\pi}{4}$.
27. Equating $M_y = 3y^2 + 4kxy^3$ and $N_x = 3y^2 + 40xy^3$ we obtain $k = 10$.
28. Equating $M_y = 18xy^2 - \sin y$ and $N_x = 4kxy^2 - \sin y$ we obtain $k = 9/2$.
29. Let $M = -x^2y^2 \sin x + 2xy^2 \cos x$ and $N = 2x^2y \cos x$ so that $M_y = -2x^2y \sin x + 4xy \cos x = N_x$. From $f_y = 2x^2y \cos x$ we obtain $f = x^2y^2 \cos x + h(y)$, $h'(y) = 0$, and $h(y) = 0$. The solution of the differential equation is $x^2y^2 \cos x = c$.
30. Let $M = (x^2 + 2xy - y^2)/(x^2 + 2xy + y^2)$ and $N = (y^2 + 2xy - x^2)/(y^2 + 2xy + x^2)$ so that $M_y = -4xy/(x + y)^3 = N_x$. From $f_x = (x^2 + 2xy + y^2 - 2y^2)/(x + y)^2$ we obtain $f = x + \frac{2y^2}{x + y} + h(y)$, $h'(y) = -1$, and $h(y) = -y$. The solution of the differential equation is $x^2 + y^2 = c(x + y)$.

31. We note that $(M_y - N_x)/N = 1/x$, so an integrating factor is $e^{\int dx/x} = x$. Let $M = 2xy^2 + 3x^2$ and $N = 2x^2y$ so that $M_y = 4xy = N_x$. From $f_x = 2xy^2 + 3x^2$ we obtain $f = x^2y^2 + x^3 + h(y)$, $h'(y) = 0$, and $h(y) = 0$. The solution of the differential equation is $x^2y^2 + x^3 = c$.
32. We note that $(M_y - N_x)/N = 1$, so an integrating factor is $e^{\int dx} = e^x$. Let $M = xye^x + y^2e^x + ye^x$ and $N = xe^x + 2ye^x$ so that $M_y = xe^x + 2ye^x + e^x = N_x$. From $f_y = xye^x + y^2e^x$ we obtain $f = xye^x + y^2e^x + h(x)$, $h'(y) = 0$, and $h(y) = 0$. The solution of the differential equation is $xye^x + y^2e^x = c$.
33. We note that $(N_x - M_y)/M = 2/y$, so an integrating factor is $e^{\int 2dy/y} = y^2$. Let $M = 6xy^3$ and $N = 4y^3 + 9x^2y^2$ so that $M_y = 18xy^2 = N_x$. From $f_x = 6xy^3$ we obtain $f = 3x^2y^3 + h(y)$, $h'(y) = 4y^3$, and $h(y) = y^4$. The solution of the differential equation is $3x^2y^3 + y^4 = c$.
34. We note that $(M_y - N_x)/N = -\cot x$, so an integrating factor is $e^{-\int \cot x dx} = \csc x$. Let $M = \cos x \csc x = \cot x$ and $N = (1 + 2/y) \sin x \csc x = 1 + 2/y$, so that $M_y = 0 = N_x$. From $f_x = \cot x$ we obtain $f = \ln(\sin x) + h(y)$, $h'(y) = 1 + 2/y$, and $h(y) = y + \ln y^2$. The solution of the differential equation is $\ln(\sin x) + y + \ln y^2 = c$.
35. (a) Write the separable equation as $-g(x) dx + dy/h(y)$. Identifying $M = -g(x)$ and $N = 1/h(y)$, we see that $M_y = 0 = N_x$, so the differential equation is exact.
- (b) Separating variables and integrating we have

$$-(\sin y) dy = (\cos x) dx \quad \text{and} \quad \cos y = \sin x + c.$$

Using $y(7\pi/6) = \pi/2$ we get $c = 1/2$, so the solution of the initial-value problem is $\cos y = \sin x + \frac{1}{2}$.

- (c) Solving for y we have $y = \cos^{-1}(\sin x + \frac{1}{2})$. Since the domain of $\cos^{-1} t$ is $[-1, 1]$ we see that $-1 \leq \sin x + \frac{1}{2} \leq 1$ or $-\frac{3}{2} \leq \sin x \leq \frac{1}{2}$, which is equivalent to $\sin x \leq \frac{1}{2}$. We also want the interval to contain $7\pi/6$, so the interval of definition is $(5\pi/6, 13\pi/6)$.



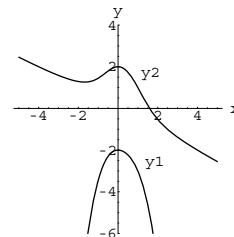
36. (a) Implicitly differentiating $x^3 + 2x^2y + y^2 = c$ and solving for dy/dx we obtain

$$3x^2 + 2x^2 \frac{dy}{dx} + 4xy + 2y \frac{dy}{dx} = 0 \quad \text{and} \quad \frac{dy}{dx} = -\frac{3x^2 + 4xy}{2x^2 + 2y}.$$

Separating variables we get $(4xy + 3x^2)dx + (2y + 2x^2)dy = 0$.

- (b) Setting $x = 0$ and $y = -2$ in $x^3 + 2x^2y + y^2 = c$ we find $c = 4$, and setting $x = y = 1$ we also find $c = 4$. Thus, both initial conditions determine the same implicit solution.
- (c) Solving $x^3 + 2x^2y + y^2 = 4$ for y we get

$$y_1(x) = -x^2 - \sqrt{4 - x^3 + x^4} \quad \text{and} \quad y_2(x) = -x^2 + \sqrt{4 - x^3 + x^4}.$$



37. To see that the equations are not equivalent consider $dx = (x/y)dy = 0$. An integrating factor is $\mu(x, y) = y$ resulting in $y dx + x dy = 0$. A solution of the latter equation is $y = 0$, but this is not a solution of the original equation.
38. The explicit solution is $y = \sqrt{(3 + \cos^2 x)/(1 - x^2)}$. Since $3 + \cos^2 x > 0$ for all x we must have $1 - x^2 > 0$ or $-1 < x < 1$. Thus, the interval of definition is $(-1, 1)$.

Exercises 2.4

39. (a) Since $f_y = N(x, y) = xe^{xy} + 2xy + 1/x$ we obtain $f = e^{xy} + xy^2 + \frac{y}{x} + h(x)$ so that $f_x = ye^{xy} + y^2 - \frac{y}{x^2} + h'(x)$.
Let $M(x, y) = ye^{xy} + y^2 - \frac{y}{x^2}$.

(b) Since $f_x = M(x, y) = y^{1/2}x^{-1/2} + x(x^2 + y)^{-1}$ we obtain $f = 2y^{1/2}x^{1/2} + \frac{1}{2}\ln|x^2 + y| + g(y)$ so that
 $f_y = y^{-1/2}x^{1/2} + \frac{1}{2}(x^2 + y)^{-1} + g'(y)$. Let $N(x, y) = y^{-1/2}x^{1/2} + \frac{1}{2}(x^2 + y)^{-1}$.

40. First note that

$$d(\sqrt{x^2 + y^2}) = \frac{x}{\sqrt{x^2 + y^2}} dx + \frac{y}{\sqrt{x^2 + y^2}} dy.$$

Then $x dx + y dy = \sqrt{x^2 + y^2} dx$ becomes

$$\frac{x}{\sqrt{x^2 + y^2}} dx + \frac{y}{\sqrt{x^2 + y^2}} dy = d(\sqrt{x^2 + y^2}) = dx.$$

The left side is the total differential of $\sqrt{x^2 + y^2}$ and the right side is the total differential of $x + c$. Thus $\sqrt{x^2 + y^2} = x + c$ is a solution of the differential equation.

Exercises 2.5

1. Letting $y = ux$ we have

$$\begin{aligned}(x - ux) dx + x(u dx + x du) &= 0 \\ dx + x du &= 0 \\ \frac{dx}{x} + du &= 0 \\ \ln|x| + u &= c \\ x \ln|x| + y &= cx.\end{aligned}$$

2. Letting $y = ux$ we have

$$\begin{aligned}(x + ux) dx + x(u dx + x du) &= 0 \\ (1 + 2u) dx + x du &= 0 \\ \frac{dx}{x} + \frac{du}{1 + 2u} &= 0 \\ \ln|x| + \frac{1}{2} \ln|1 + 2u| &= c \\ x^2 \left(1 + 2\frac{y}{x}\right) &= c_1 \\ x^2 + 2xy &= c_1.\end{aligned}$$

3. Letting $x = vy$ we have

$$\begin{aligned}vy(v dy + y dv) + (y - 2vy) dy &= 0 \\ vy dv + (v^2 - 2v + 1) dy &= 0 \\ \frac{v dv}{(v - 1)^2} + \frac{dy}{y} &= 0 \\ \ln|v - 1| - \frac{1}{v - 1} + \ln|y| &= c \\ \ln\left|\frac{x}{y} - 1\right| - \frac{1}{x/y - 1} + \ln y &= c \\ (x - y) \ln|x - y| - y &= c(x - y).\end{aligned}$$

4. Letting $x = vy$ we have

$$\begin{aligned}
 y(v \, dy + y \, dv) - 2(vy + y) \, dy &= 0 \\
 y \, dv - (v + 2) \, dy &= 0 \\
 \frac{dv}{v + 2} - \frac{dy}{y} &= 0 \\
 \ln |v + 2| - \ln |y| &= c \\
 \ln \left| \frac{x}{y} + 2 \right| - \ln |y| &= c \\
 x + 2y &= c_1 y^2.
 \end{aligned}$$

5. Letting $y = ux$ we have

$$\begin{aligned}
 (u^2 x^2 + ux^2) \, dx - x^2(u \, dx + x \, du) &= 0 \\
 u^2 \, dx - x \, du &= 0 \\
 \frac{dx}{x} - \frac{du}{u^2} &= 0 \\
 \ln |x| + \frac{1}{u} &= c \\
 \ln |x| + \frac{x}{y} &= c \\
 y \ln |x| + x &= cy.
 \end{aligned}$$

6. Letting $y = ux$ we have

$$\begin{aligned}
 (u^2 x^2 + ux^2) \, dx + x^2(u \, dx + x \, du) &= 0 \\
 (u^2 + 2u) \, dx + x \, du &= 0 \\
 \frac{dx}{x} + \frac{du}{u(u + 2)} &= 0 \\
 \ln |x| + \frac{1}{2} \ln |u| - \frac{1}{2} \ln |u + 2| &= c \\
 \frac{x^2 u}{u + 2} &= c_1 \\
 x^2 \frac{y}{x} &= c_1 \left(\frac{y}{x} + 2 \right) \\
 x^2 y &= c_1 (y + 2x).
 \end{aligned}$$

7. Letting $y = ux$ we have

$$\begin{aligned}
 (ux - x) \, dx - (ux + x)(u \, dx + x \, du) &= 0 \\
 (u^2 + 1) \, dx + x(u + 1) \, du &= 0 \\
 \frac{dx}{x} + \frac{u + 1}{u^2 + 1} \, du &= 0 \\
 \ln |x| + \frac{1}{2} \ln (u^2 + 1) + \tan^{-1} u &= c \\
 \ln x^2 \left(\frac{y^2}{x^2} + 1 \right) + 2 \tan^{-1} \frac{y}{x} &= c_1 \\
 \ln (x^2 + y^2) + 2 \tan^{-1} \frac{y}{x} &= c_1.
 \end{aligned}$$

Exercises 2.5

8. Letting $y = ux$ we have

$$\begin{aligned}
 (x + 3ux) dx - (3x + ux)(u dx + x du) &= 0 \\
 (u^2 - 1) dx + x(u + 3) du &= 0 \\
 \frac{dx}{x} + \frac{u + 3}{(u - 1)(u + 1)} du &= 0 \\
 \ln |x| + 2 \ln |u - 1| - \ln |u + 1| &= c \\
 \frac{x(u - 1)^2}{u + 1} &= c_1 \\
 x \left(\frac{y}{x} - 1 \right)^2 &= c_1 \left(\frac{y}{x} + 1 \right) \\
 (y - x)^2 &= c_1(y + x).
 \end{aligned}$$

9. Letting $y = ux$ we have

$$\begin{aligned}
 -ux dx + (x + \sqrt{u}x)(u dx + x du) &= 0 \\
 (x + x\sqrt{u}) du + u^{3/2} dx &= 0 \\
 \left(u^{-3/2} + \frac{1}{u} \right) du + \frac{dx}{x} &= 0 \\
 -2u^{-1/2} + \ln |u| + \ln |x| &= c \\
 \ln |y/x| + \ln |x| &= 2\sqrt{x/y} + c \\
 y(\ln |y| - c)^2 &= 4x.
 \end{aligned}$$

10. Letting $y = ux$ we have

$$\begin{aligned}
 (ux + \sqrt{x^2 + u^2x^2}) dx - x(u dx + x du) &= 0 \\
 x\sqrt{1 + u^2} dx - x^2 du &= 0 \\
 \frac{dx}{x} - \frac{du}{\sqrt{1 + u^2}} &= 0 \\
 \ln |x| - \ln |u + \sqrt{1 + u^2}| &= c \\
 u + \sqrt{1 + u^2} &= c_1 x \\
 y + \sqrt{y^2 + x^2} &= c_1 x^2.
 \end{aligned}$$

11. Letting $y = ux$ we have

$$\begin{aligned}
 (x^3 - u^3x^3) dx + u^2x^3(u dx + x du) &= 0 \\
 dx + u^2x du &= 0 \\
 \frac{dx}{x} + u^2 du &= 0 \\
 \ln |x| + \frac{1}{3}u^3 &= c \\
 3x^3 \ln |x| + y^3 &= c_1 x^3.
 \end{aligned}$$

Using $y(1) = 2$ we find $c_1 = 8$. The solution of the initial-value problem is $3x^3 \ln |x| + y^3 = 8x^3$.

12. Letting $y = ux$ we have

$$\begin{aligned}(x^2 + 2u^2x^2) dx - ux^2(u dx + x du) &= 0 \\(1 + u^2) dx - ux du &= 0 \\ \frac{dx}{x} - \frac{u du}{1 + u^2} &= 0 \\ \ln|x| - \frac{1}{2} \ln(1 + u^2) &= c \\ \frac{x^2}{1 + u^2} &= c_1 \\ x^4 &= c_1(y^2 + x^2).\end{aligned}$$

Using $y(-1) = 1$ we find $c_1 = 1/2$. The solution of the initial-value problem is $2x^4 = y^2 + x^2$.

13. Letting $y = ux$ we have

$$\begin{aligned}(x + uxe^u) dx - xe^u(u dx + x du) &= 0 \\ dx - xe^u du &= 0 \\ \frac{dx}{x} - e^u du &= 0 \\ \ln|x| - e^u &= c \\ \ln|x| - e^{y/x} &= c.\end{aligned}$$

Using $y(1) = 0$ we find $c = -1$. The solution of the initial-value problem is $\ln|x| = e^{y/x} - 1$.

14. Letting $x = vy$ we have

$$\begin{aligned}y(v dy + y dv) + vy(\ln vy - \ln y - 1) dy &= 0 \\ y dv + v \ln v dy &= 0 \\ \frac{dv}{v \ln v} + \frac{dy}{y} &= 0 \\ \ln|\ln|v|| + \ln|y| &= c \\ y \ln \left| \frac{x}{y} \right| &= c_1.\end{aligned}$$

Using $y(1) = e$ we find $c_1 = -e$. The solution of the initial-value problem is $y \ln \left| \frac{x}{y} \right| = -e$.

15. From $y' + \frac{1}{x}y = \frac{1}{x}y^{-2}$ and $w = y^3$ we obtain $\frac{dw}{dx} + \frac{3}{x}w = \frac{3}{x}$. An integrating factor is x^3 so that $x^3w = x^3 + c$ or $y^3 = 1 + cx^{-3}$.
16. From $y' - y = e^xy^2$ and $w = y^{-1}$ we obtain $\frac{dw}{dx} + w = -e^x$. An integrating factor is e^x so that $e^xw = -\frac{1}{2}e^{2x} + c$ or $y^{-1} = -\frac{1}{2}e^x + ce^{-x}$.
17. From $y' + y = xy^4$ and $w = y^{-3}$ we obtain $\frac{dw}{dx} - 3w = -3x$. An integrating factor is e^{-3x} so that $e^{-3x} = xe^{-3x} + \frac{1}{3}e^{-3x} + c$ or $y^{-3} = x + \frac{1}{3} + ce^{3x}$.
18. From $y' - \left(1 + \frac{1}{x}\right)y = y^2$ and $w = y^{-1}$ we obtain $\frac{dw}{dx} + \left(1 + \frac{1}{x}\right)w = -1$. An integrating factor is xe^x so that $xe^xw = -xe^x + e^x + c$ or $y^{-1} = -1 + \frac{1}{x} + \frac{c}{x}e^{-x}$.

Exercises 2.5

19. From $y' - \frac{1}{t}y = -\frac{1}{t^2}y^2$ and $w = y^{-1}$ we obtain $\frac{dw}{dt} + \frac{1}{t}w = \frac{1}{t^2}$. An integrating factor is t so that $tw = \ln t + c$ or $y^{-1} = \frac{1}{t} \ln t + \frac{c}{t}$. Writing this in the form $\frac{t}{y} = \ln t + c$, we see that the solution can also be expressed in the form $e^{t/y} = c_1 t$.
20. From $y' + \frac{2}{3(1+t^2)}y = \frac{2t}{3(1+t^2)}y^4$ and $w = y^{-3}$ we obtain $\frac{dw}{dt} - \frac{2t}{1+t^2}w = \frac{-2t}{1+t^2}$. An integrating factor is $\frac{1}{1+t^2}$ so that $\frac{w}{1+t^2} = \frac{1}{1+t^2} + c$ or $y^{-3} = 1 + c(1+t^2)$.
21. From $y' - \frac{2}{x}y = \frac{3}{x^2}y^4$ and $w = y^{-3}$ we obtain $\frac{dw}{dx} + \frac{6}{x}w = -\frac{9}{x^2}$. An integrating factor is x^6 so that $x^6 w = -\frac{9}{5}x^5 + c$ or $y^{-3} = -\frac{9}{5}x^{-1} + cx^{-6}$. If $y(1) = \frac{1}{2}$ then $c = \frac{49}{5}$ and $y^{-3} = -\frac{9}{5}x^{-1} + \frac{49}{5}x^{-6}$.
22. From $y' + y = y^{-1/2}$ and $w = y^{3/2}$ we obtain $\frac{dw}{dx} + \frac{3}{2}w = \frac{3}{2}$. An integrating factor is $e^{3x/2}$ so that $e^{3x/2}w = e^{3x/2} + c$ or $y^{3/2} = 1 + ce^{-3x/2}$. If $y(0) = 4$ then $c = 7$ and $y^{3/2} = 1 + 7e^{-3x/2}$.
23. Let $u = x + y + 1$ so that $du/dx = 1 + dy/dx$. Then $\frac{du}{dx} - 1 = u^2$ or $\frac{1}{1+u^2} du = dx$. Thus $\tan^{-1} u = x + c$ or $u = \tan(x + c)$, and $x + y + 1 = \tan(x + c)$ or $y = \tan(x + c) - x - 1$.
24. Let $u = x + y$ so that $du/dx = 1 + dy/dx$. Then $\frac{du}{dx} - 1 = \frac{1-u}{u}$ or $u du = dx$. Thus $\frac{1}{2}u^2 = x + c$ or $u^2 = 2x + c_1$, and $(x + y)^2 = 2x + c_1$.
25. Let $u = x + y$ so that $du/dx = 1 + dy/dx$. Then $\frac{du}{dx} - 1 = \tan^2 u$ or $\cos^2 u du = dx$. Thus $\frac{1}{2}u + \frac{1}{4}\sin 2u = x + c$ or $2u + \sin 2u = 4x + c_1$, and $2(x + y) + \sin 2(x + y) = 4x + c_1$ or $2y + \sin 2(x + y) = 2x + c_1$.
26. Let $u = x + y$ so that $du/dx = 1 + dy/dx$. Then $\frac{du}{dx} - 1 = \sin u$ or $\frac{1}{1 + \sin u} du = dx$. Multiplying by $(1 - \sin u)/(1 - \sin u)$ we have $\frac{1 - \sin u}{\cos^2 u} du = dx$ or $(\sec^2 u - \tan u \sec u) du = dx$. Thus $\tan u - \sec u = x + c$ or $\tan(x + y) - \sec(x + y) = x + c$.
27. Let $u = y - 2x + 3$ so that $du/dx = dy/dx - 2$. Then $\frac{du}{dx} + 2 = 2 + \sqrt{u}$ or $\frac{1}{\sqrt{u}} du = dx$. Thus $2\sqrt{u} = x + c$ and $2\sqrt{y - 2x + 3} = x + c$.
28. Let $u = y - x + 5$ so that $du/dx = dy/dx - 1$. Then $\frac{du}{dx} + 1 = 1 + e^u$ or $e^{-u} du = dx$. Thus $-e^{-u} = x + c$ and $-e^{y-x+5} = x + c$.
29. Let $u = x + y$ so that $du/dx = 1 + dy/dx$. Then $\frac{du}{dx} - 1 = \cos u$ and $\frac{1}{1 + \cos u} du = dx$. Now
- $$\frac{1}{1 + \cos u} = \frac{1 - \cos u}{1 - \cos^2 u} = \frac{1 - \cos u}{\sin^2 u} = \csc^2 u - \csc u \cot u$$
- so we have $\int (\csc^2 u - \csc u \cot u) du = \int dx$ and $-\cot u + \csc u = x + c$. Thus $-\cot(x + y) + \csc(x + y) = x + c$. Setting $x = 0$ and $y = \pi/4$ we obtain $c = \sqrt{2} - 1$. The solution is

$$\csc(x + y) - \cot(x + y) = x + \sqrt{2} - 1.$$

30. Let $u = 3x + 2y$ so that $du/dx = 3 + 2 dy/dx$. Then $\frac{du}{dx} = 3 + \frac{2u}{u+2} = \frac{5u+6}{u+2}$ and $\frac{u+2}{5u+6} du = dx$. Now
- $$\frac{u+2}{5u+6} = \frac{1}{5} + \frac{4}{25u+30}$$

so we have

$$\int \left(\frac{1}{5} + \frac{4}{25u + 30} \right) du = dx$$

and $\frac{1}{5}u + \frac{4}{25} \ln |25u + 30| = x + c$. Thus

$$\frac{1}{5}(3x + 2y) + \frac{4}{25} \ln |75x + 50y + 30| = x + c.$$

Setting $x = -1$ and $y = -1$ we obtain $c = \frac{4}{5} \ln 95$. The solution is

$$\frac{1}{5}(3x + 2y) + \frac{4}{25} \ln |75x + 50y + 30| = x + \frac{4}{5} \ln 95$$

or

$$5y - 5x + 2 \ln |75x + 50y + 30| = 10 \ln 95.$$

- 31.** We write the differential equation $M(x, y)dx + N(x, y)dy = 0$ as $dy/dx = f(x, y)$ where

$$f(x, y) = -\frac{M(x, y)}{N(x, y)}.$$

The function $f(x, y)$ must necessarily be homogeneous of degree 0 when M and N are homogeneous of degree α . Since M is homogeneous of degree α , $M(tx, ty) = t^\alpha M(x, y)$, and letting $t = 1/x$ we have

$$M(1, y/x) = \frac{1}{x^\alpha} M(x, y) \quad \text{or} \quad M(x, y) = x^\alpha M(1, y/x).$$

Thus

$$\frac{dy}{dx} = f(x, y) = -\frac{x^\alpha M(1, y/x)}{x^\alpha N(1, y/x)} = -\frac{M(1, y/x)}{N(1, y/x)} = F\left(\frac{y}{x}\right).$$

To show that the differential equation also has the form

$$\frac{dy}{dx} = G\left(\frac{x}{y}\right)$$

we use the fact that $M(x, y) = y^\alpha M(x/y, 1)$. The forms $F(y/x)$ and $G(x/y)$ suggest, respectively, the substitutions $u = y/x$ or $y = ux$ and $v = x/y$ or $x = vy$.

- 32.** As $x \rightarrow -\infty$, $e^{6x} \rightarrow 0$ and $y \rightarrow 2x + 3$. Now write $(1 + ce^{6x})/(1 - ce^{6x})$ as $(e^{-6x} + c)/(e^{-6x} - c)$. Then, as $x \rightarrow \infty$, $e^{-6x} \rightarrow 0$ and $y \rightarrow 2x - 3$.

- 33. (a)** The substitutions $y = y_1 + u$ and

$$\frac{dy}{dx} = \frac{dy_1}{dx} + \frac{du}{dx}$$

lead to

$$\begin{aligned} \frac{dy_1}{dx} + \frac{du}{dx} &= P + Q(y_1 + u) + R(y_1 + u)^2 \\ &= P + Qy_1 + Ry_1^2 + Qu + 2y_1Ru + Ru^2 \end{aligned}$$

or

$$\frac{du}{dx} - (Q + 2y_1R)u = Ru^2.$$

This is a Bernoulli equation with $n = 2$ which can be reduced to the linear equation

$$\frac{dw}{dx} + (Q + 2y_1R)w = -R$$

by the substitution $w = u^{-1}$.

Exercises 2.5

(b) Identify $P(x) = -4/x^2$, $Q(x) = -1/x$, and $R(x) = 1$. Then $\frac{dw}{dx} + \left(-\frac{1}{x} + \frac{4}{x}\right)w = -1$. An integrating factor is x^3 so that $x^3w = -\frac{1}{4}x^4 + c$ or $u = \left[-\frac{1}{4}x + cx^{-3}\right]^{-1}$. Thus, $y = \frac{2}{x} + u$.

34. Write the differential equation in the form $x(y'/y) = \ln x + \ln y$ and let $u = \ln y$. Then $du/dx = y'/y$ and the differential equation becomes $x(du/dx) = \ln x + u$ or $du/dx - u/x = (\ln x)/x$, which is first-order, linear. An integrating factor is $e^{-\int dx/x} = 1/x$, so that (using integration by parts)

$$\frac{d}{dx} \left[\frac{1}{x} u \right] = \frac{\ln x}{x^2} \quad \text{and} \quad \frac{u}{x} = -\frac{1}{x} - \frac{\ln x}{x} + c.$$

The solution is

$$\ln y = -1 - \ln x + cx \quad \text{or} \quad y = \frac{e^{-cx-1}}{x}.$$

Exercises 2.6

1. We identify $f(x, y) = 2x - 3y + 1$. Then, for $h = 0.1$,

$$y_{n+1} = y_n + 0.1(2x_n - 3y_n + 1) = 0.2x_n + 0.7y_n + 0.1,$$

and

$$y(1.1) \approx y_1 = 0.2(1) + 0.7(5) + 0.1 = 3.8$$

$$y(1.2) \approx y_2 = 0.2(1.1) + 0.7(3.8) + 0.1 = 2.98.$$

For $h = 0.05$

$$y_{n+1} = y_n + 0.05(2x_n - 3y_n + 1) = 0.1x_n + 0.85y_n + 0.1,$$

and

$$y(1.05) \approx y_1 = 0.1(1) + 0.85(5) + 0.1 = 4.4$$

$$y(1.1) \approx y_2 = 0.1(1.05) + 0.85(4.4) + 0.1 = 3.895$$

$$y(1.15) \approx y_3 = 0.1(1.1) + 0.85(3.895) + 0.1 = 3.47075$$

$$y(1.2) \approx y_4 = 0.1(1.15) + 0.85(3.47075) + 0.1 = 3.11514$$

2. We identify $f(x, y) = x + y^2$. Then, for $h = 0.1$,

$$y_{n+1} = y_n + 0.1(x_n + y_n^2) = 0.1x_n + y_n + 0.1y_n^2,$$

and

$$y(0.1) \approx y_1 = 0.1(0) + 0 + 0.1(0)^2 = 0$$

$$y(0.2) \approx y_2 = 0.1(0.1) + 0 + 0.1(0)^2 = 0.01.$$

For $h = 0.05$

$$y_{n+1} = y_n + 0.05(x_n + y_n^2) = 0.05x_n + y_n + 0.05y_n^2,$$

and

$$y(0.05) \approx y_1 = 0.05(0) + 0 + 0.05(0)^2 = 0$$

$$y(0.1) \approx y_2 = 0.05(0.05) + 0 + 0.05(0)^2 = 0.0025$$

$$y(0.15) \approx y_3 = 0.05(0.1) + 0.0025 + 0.05(0.0025)^2 = 0.0075$$

$$y(0.2) \approx y_4 = 0.05(0.15) + 0.0075 + 0.05(0.0075)^2 = 0.0150.$$

3. Separating variables and integrating, we have

$$\frac{dy}{y} = dx \quad \text{and} \quad \ln|y| = x + c.$$

Thus $y = c_1 e^x$ and, using $y(0) = 1$, we find $c = 1$, so $y = e^x$ is the solution of the initial-value problem.

h=0.1

x_n	y_n	True Value	Abs. Error	% Rel. Error
0.00	1.0000	1.0000	0.0000	0.00
0.10	1.1000	1.1052	0.0052	0.47
0.20	1.2100	1.2214	0.0114	0.93
0.30	1.3310	1.3499	0.0189	1.40
0.40	1.4641	1.4918	0.0277	1.86
0.50	1.6105	1.6487	0.0382	2.32
0.60	1.7716	1.8221	0.0506	2.77
0.70	1.9487	2.0138	0.0650	3.23
0.80	2.1436	2.2255	0.0820	3.68
0.90	2.3579	2.4596	0.1017	4.13
1.00	2.5937	2.7183	0.1245	4.58

h=0.05

x_n	y_n	True Value	Abs. Error	% Rel. Error
0.00	1.0000	1.0000	0.0000	0.00
0.05	1.0500	1.0513	0.0013	0.12
0.10	1.1025	1.1052	0.0027	0.24
0.15	1.1576	1.1618	0.0042	0.36
0.20	1.2155	1.2214	0.0059	0.48
0.25	1.2763	1.2840	0.0077	0.60
0.30	1.3401	1.3499	0.0098	0.72
0.35	1.4071	1.4191	0.0120	0.84
0.40	1.4775	1.4918	0.0144	0.96
0.45	1.5513	1.5683	0.0170	1.08
0.50	1.6289	1.6487	0.0198	1.20
0.55	1.7103	1.7333	0.0229	1.32
0.60	1.7959	1.8221	0.0263	1.44
0.65	1.8856	1.9155	0.0299	1.56
0.70	1.9799	2.0138	0.0338	1.68
0.75	2.0789	2.1170	0.0381	1.80
0.80	2.1829	2.2255	0.0427	1.92
0.85	2.2920	2.3396	0.0476	2.04
0.90	2.4066	2.4596	0.0530	2.15
0.95	2.5270	2.5857	0.0588	2.27
1.00	2.6533	2.7183	0.0650	2.39

4. Separating variables and integrating, we have

$$\frac{dy}{y} = 2x dx \quad \text{and} \quad \ln|y| = x^2 + c.$$

Thus $y = c_1 e^{x^2}$ and, using $y(1) = 1$, we find $c = e^{-1}$, so $y = e^{x^2-1}$ is the solution of the initial-value problem.

h=0.1

x_n	y_n	True Value	Abs. Error	% Rel. Error
1.00	1.0000	1.0000	0.0000	0.00
1.10	1.2000	1.2337	0.0337	2.73
1.20	1.4640	1.5527	0.0887	5.71
1.30	1.8154	1.9937	0.1784	8.95
1.40	2.2874	2.6117	0.3243	12.42
1.50	2.9278	3.4903	0.5625	16.12

h=0.05

x_n	y_n	True Value	Abs. Error	% Rel. Error
1.00	1.0000	1.0000	0.0000	0.00
1.05	1.1000	1.1079	0.0079	0.72
1.10	1.2155	1.2337	0.0182	1.47
1.15	1.3492	1.3806	0.0314	2.27
1.20	1.5044	1.5527	0.0483	3.11
1.25	1.6849	1.7551	0.0702	4.00
1.30	1.8955	1.9937	0.0982	4.93
1.35	2.1419	2.2762	0.1343	5.90
1.40	2.4311	2.6117	0.1806	6.92
1.45	2.7714	3.0117	0.2403	7.98
1.50	3.1733	3.4903	0.3171	9.08

5.

h=0.1

x_n	y_n
0.00	0.0000
0.10	0.1000
0.20	0.1905
0.30	0.2731
0.40	0.3492
0.50	0.4198

h=0.05

x_n	y_n
0.00	0.0000
0.05	0.0500
0.10	0.0976
0.15	0.1429
0.20	0.1863
0.25	0.2278
0.30	0.2676
0.35	0.3058
0.40	0.3427
0.45	0.3782
0.50	0.4124

6.

h=0.1

x_n	y_n
0.00	1.0000
0.10	1.1000
0.20	1.2220
0.30	1.3753
0.40	1.5735
0.50	1.8371

h=0.05

x_n	y_n
0.00	1.0000
0.05	1.0500
0.10	1.1053
0.15	1.1668
0.20	1.2360
0.25	1.3144
0.30	1.4039
0.35	1.5070
0.40	1.6267
0.45	1.7670
0.50	1.9332

Exercises 2.6

7. $h=0.1$

x_n	y_n
0.00	0.5000
0.10	0.5250
0.20	0.5431
0.30	0.5548
0.40	0.5613
0.50	0.5639

$h=0.05$

x_n	y_n
0.00	0.5000
0.05	0.5125
0.10	0.5232
0.15	0.5322
0.20	0.5395
0.25	0.5452
0.30	0.5496
0.35	0.5527
0.40	0.5547
0.45	0.5559
0.50	0.5565

8. $h=0.1$

x_n	y_n
0.00	1.0000
0.10	1.1000
0.20	1.2159
0.30	1.3505
0.40	1.5072
0.50	1.6902

$h=0.05$

x_n	y_n
0.00	1.0000
0.05	1.0500
0.10	1.1039
0.15	1.1619
0.20	1.2245
0.25	1.2921
0.30	1.3651
0.35	1.4440
0.40	1.5293
0.45	1.6217
0.50	1.7219

9. $h=0.1$

x_n	y_n
1.00	1.0000
1.10	1.0000
1.20	1.0191
1.30	1.0588
1.40	1.1231
1.50	1.2194

$h=0.05$

x_n	y_n
1.00	1.0000
1.05	1.0000
1.10	1.0049
1.15	1.0147
1.20	1.0298
1.25	1.0506
1.30	1.0775
1.35	1.1115
1.40	1.1538
1.45	1.2057
1.50	1.2696

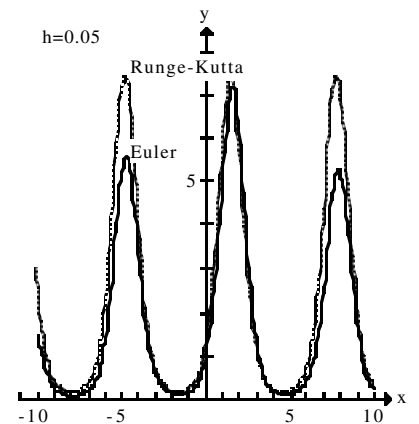
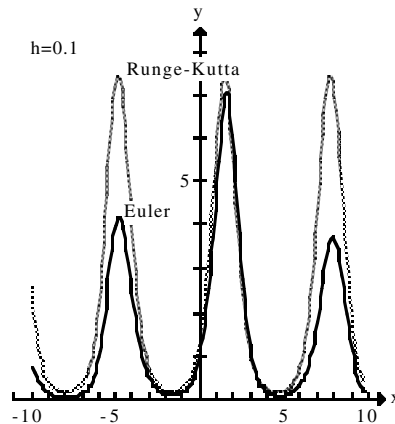
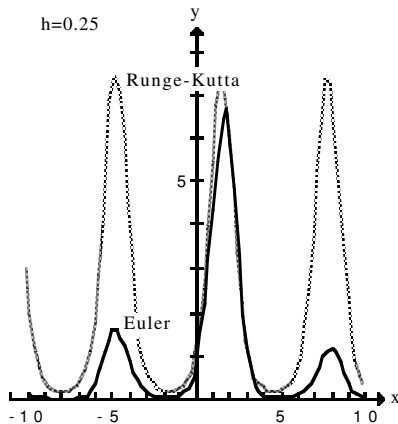
10. $h=0.1$

x_n	y_n
0.00	0.5000
0.10	0.5250
0.20	0.5499
0.30	0.5747
0.40	0.5991
0.50	0.6231

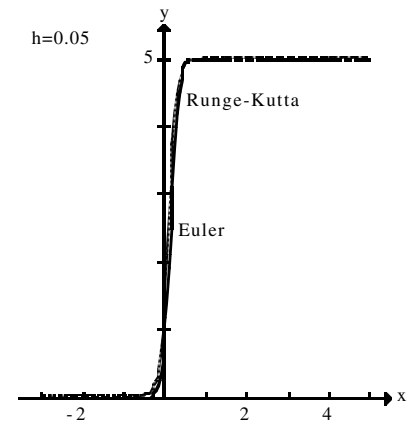
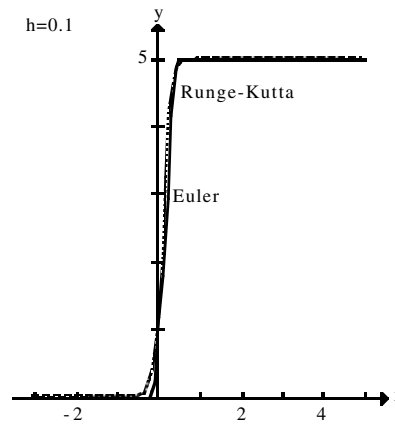
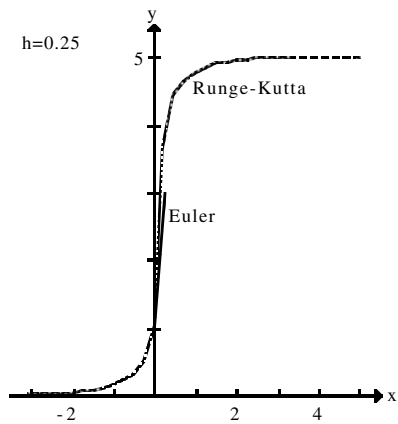
$h=0.05$

x_n	y_n
0.00	0.5000
0.05	0.5125
0.10	0.5250
0.15	0.5375
0.20	0.5499
0.25	0.5623
0.30	0.5746
0.35	0.5868
0.40	0.5989
0.45	0.6109
0.50	0.6228

11.



12.



13. Using separation of variables we find that the solution of the differential equation is $y = 1/(1 - x^2)$, which is undefined at $x = 1$, where the graph has a vertical asymptote.

h=0.1 Euler		h=0.05 Euler		h=0.1 R-K		h=0.05 R-K	
x_n	y_n	x_n	y_n	x_n	y_n	x_n	y_n
0.00	1.0000	0.00	1.0000	0.00	1.0000	0.00	1.0000
0.10	1.0000	0.05	1.0000	0.10	1.0101	0.05	1.0025
0.20	1.0200	0.10	1.0050	0.20	1.0417	0.10	1.0101
0.30	1.0616	0.15	1.0151	0.30	1.0989	0.15	1.0230
0.40	1.1292	0.20	1.0306	0.40	1.1905	0.20	1.0417
0.50	1.2313	0.25	1.0518	0.50	1.3333	0.25	1.0667
0.60	1.3829	0.30	1.0795	0.60	1.5625	0.30	1.0989
0.70	1.6123	0.35	1.1144	0.70	1.9607	0.35	1.1396
0.80	1.9763	0.40	1.1579	0.80	2.7771	0.40	1.1905
0.90	2.6012	0.45	1.2115	0.90	5.2388	0.45	1.2539
1.00	3.8191	0.50	1.2776	1.00	42.9931	0.50	1.3333
		0.55	1.3592			0.55	1.4337
		0.60	1.4608			0.60	1.5625
		0.65	1.5888			0.65	1.7316
		0.70	1.7529			0.70	1.9608
		0.75	1.9679			0.75	2.2857
		0.80	2.2584			0.80	2.7777
		0.85	2.6664			0.85	3.6034
		0.90	3.2708			0.90	5.2609
		0.95	4.2336			0.95	10.1973
		1.00	5.9363			1.00	84.0132

Exercises 2.7

1. Let $P = P(t)$ be the population at time t , and P_0 the initial population. From $dP/dt = kP$ we obtain $P = P_0 e^{kt}$. Using $P(5) = 2P_0$ we find $k = \frac{1}{5} \ln 2$ and $P = P_0 e^{(\ln 2)t/5}$. Setting $P(t) = 3P_0$ we have

$$3 = e^{(\ln 2)t/5} \implies \ln 3 = \frac{(\ln 2)t}{5} \implies t = \frac{5 \ln 3}{\ln 2} \approx 7.9 \text{ years.}$$

Setting $P(t) = 4P_0$ we have

$$4 = e^{(\ln 2)t/5} \implies \ln 4 = \frac{(\ln 2)t}{5} \implies t = 10 \text{ years.}$$

2. Setting $P = 10,000$ and $t = 3$ in Problem 1 we obtain

$$10,000 = P_0 e^{(\ln 2)3/5} \implies P_0 = 10,000 e^{-0.6 \ln 2} \approx 6597.5.$$

Then $P(10) = P_0 e^{2 \ln 2} = 4P_0 \approx 26,390$.

3. Let $P = P(t)$ be the population at time t . From $dP/dt = kt$ and $P(0) = P_0 = 500$ we obtain $P = 500e^{kt}$. Using $P(10) = 575$ we find $k = \frac{1}{10} \ln 1.15$. Then $P(30) = 500e^{3 \ln 1.15} \approx 760$ years.
4. Let $N = N(t)$ be the number of bacteria at time t and N_0 the initial number. From $dN/dt = kN$ we obtain $N = N_0 e^{kt}$. Using $N(3) = 400$ and $N(10) = 2000$ we find $400 = N_0 e^{3k}$ or $e^k = (400/N_0)^{1/3}$. From $N(10) = 2000$ we then have

$$2000 = N_0 e^{10k} = N_0 \left(\frac{400}{N_0} \right)^{10/3} \implies \frac{2000}{400^{10/3}} = N_0^{-7/3} \implies N_0 = \left(\frac{2000}{400^{10/3}} \right)^{-3/7} \approx 201.$$

5. Let $I = I(t)$ be the intensity, t the thickness, and $I(0) = I_0$. If $dI/dt = kI$ and $I(3) = 0.25I_0$ then $I = I_0 e^{kt}$, $k = \frac{1}{3} \ln 0.25$, and $I(15) = 0.00098I_0$.
6. From $dS/dt = rS$ we obtain $S = S_0 e^{rt}$ where $S(0) = S_0$.

Exercises 2.7

(a) If $S_0 = \$5000$ and $r = 5.75\%$ then $S(5) = \$6665.45$.

(b) If $S(t) = \$10,000$ then $t = 12$ years.

(c) $S \approx \$6651.82$

7. Let $N = N(t)$ be the amount of lead at time t . From $dN/dt = kN$ and $N(0) = 1$ we obtain $N = e^{kt}$. Using $N(3.3) = 1/2$ we find $k = \frac{1}{3.3} \ln 1/2$. When 90% of the lead has decayed, 0.1 grams will remain. Setting $N(t) = 0.1$ we have

$$e^{t(1/3.3) \ln(1/2)} = 0.1 \implies \frac{t}{3.3} \ln \frac{1}{2} = \ln 0.1 \implies t = \frac{3.3 \ln 0.1}{\ln 1/2} \approx 10.96 \text{ hours.}$$

8. Let $N = N(t)$ be the amount at time t . From $dN/dt = kt$ and $N(0) = 100$ we obtain $N = 100e^{kt}$. Using $N(6) = 97$ we find $k = \frac{1}{6} \ln 0.97$. Then $N(24) = 100e^{(1/6)(\ln 0.97)24} = 100(0.97)^4 \approx 88.5$ mg.

9. Setting $N(t) = 50$ in Problem 8 we obtain

$$50 = 100e^{kt} \implies kt = \ln \frac{1}{2} \implies t = \frac{\ln 1/2}{(1/6) \ln 0.97} \approx 136.5 \text{ hours.}$$

10. (a) The solution of $dA/dt = kA$ is $A(t) = A_0 e^{kt}$. Letting $A = \frac{1}{2}A_0$ and solving for t we obtain the half-life $T = -(\ln 2)/k$.

(b) Since $k = -(\ln 2)/T$ we have

$$A(t) = A_0 e^{-(\ln 2)t/T} = A_0 2^{-t/T}.$$

11. Assume that $A = A_0 e^{kt}$ and $k = -0.00012378$. If $A(t) = 0.145A_0$ then $t \approx 15,600$ years.
12. From Example 3, the amount of carbon present at time t is $A(t) = A_0 e^{-0.00012378t}$. Letting $t = 660$ and solving for A_0 we have $A(660) = A_0 e^{-0.00012378(660)} = 0.921553A_0$. Thus, approximately 92% of the original amount of C-14 remained in the cloth as of 1988.
13. Assume that $dT/dt = k(T - 10)$ so that $T = 10 + ce^{kt}$. If $T(0) = 70^\circ$ and $T(1/2) = 50^\circ$ then $c = 60$ and $k = 2 \ln(2/3)$ so that $T(1) = 36.67^\circ$. If $T(t) = 15^\circ$ then $t = 3.06$ minutes.
14. Assume that $dT/dt = k(T - 5)$ so that $T = 5 + ce^{kt}$. If $T(1) = 55^\circ$ and $T(5) = 30^\circ$ then $k = -\frac{1}{4} \ln 2$ and $c = 59.4611$ so that $T(0) = 64.4611^\circ$.
15. Assume that $dT/dt = k(T - 100)$ so that $T = 100 + ce^{kt}$. If $T(0) = 20^\circ$ and $T(1) = 22^\circ$ then $c = -80$ and $k = \ln(39/40)$ so that $T(t) = 90^\circ$ implies $t = 82.1$ seconds. If $T(t) = 98^\circ$ then $t = 145.7$ seconds.
16. Using separation of variables to solve $dT/dt = k(T - T_m)$ we get $T(t) = T_m + ce^{kt}$. Using $T(0) = 70$ we find $c = 70 - T_m$, so $T(t) = T_m + (70 - T_m)e^{kt}$. Using the given observations, we obtain

$$T\left(\frac{1}{2}\right) = T_m + (70 - T_m)e^{k/2} = 110$$

$$T(1) = T_m + (70 - T_m)e^k = 145.$$

Then $e^{k/2} = (110 - T_m)/(70 - T_m)$ and

$$e^k = (e^{k/2})^2 = \left(\frac{110 - T_m}{70 - T_m}\right)^2 = \frac{145 - T_m}{70 - T_m}$$

$$\frac{(110 - T_m)^2}{70 - T_m} = 145 - T_m$$

$$12100 - 220T_m + T_m^2 = 10150 - 250T_m + T_m^2$$

$$T_m = 390.$$

The temperature in the oven is 390° .

17. We are given $T_m = 65$. By Newton's law of cooling we then have

$$\frac{dT}{dt} = k(T - 65), \quad T(0) = 82.$$

Using linearity or separation of variables we obtain $T = 65 + ce^{kt}$. From $T(0) = 82$ we find $c = 17$, so that $T = 65 + 17e^{kt}$. To find k we use the fact that the body temperature at $t = 2$ hours was 75°F . Then $75 = 65 + 17e^{2k}$ and $k = -0.2653$ and

$$T(t) = 65 + 17e^{-0.2653t}.$$

At the time of death, t_0 , $T(t_0) = 98.6^\circ\text{F}$, so $98.6 = 65 + 17e^{-0.2653t_0}$, which gives $t = -2.568$. Thus, the murder took place about 2.568 hours prior to the discovery of the body.

18. We will assume that the temperature of both the room and the cream is 72°F , and that the temperature of the coffee when it is first put on the table is 175°F . If we let $T_1(t)$ represent the temperature of the coffee in Mr. Jones' cup at time t , then

$$\frac{dT_1}{dt} = k(T_1 - 72),$$

which implies $T_1 = 72 + c_1e^{kt}$. At time $t = 0$ Mr. Jones adds cream to his coffee which immediately reduces its temperature by an amount α , so that $T_1(0) = 175 - \alpha$. Thus $175 - \alpha = T_1(0) = 72 + c_1$, which implies $c_1 = 103 - \alpha$, so that $T_1(t) = 72 + (103 - \alpha)e^{kt}$. At $t = 5$, $T_1(5) = 72 + (103 - \alpha)e^{5k}$. Now we let $T_2(t)$ represent the temperature of the coffee in Mrs. Jones' cup. From $T_2 = 72 + c_2e^{kt}$ and $T_2(0) = 175$ we obtain $c_2 = 103$, so that $T_2(t) = 72 + 103e^{kt}$. At $t = 5$, $T_2(5) = 72 + 103e^{5k}$. When cream is added to Mrs. Jones' coffee the temperature is reduced by an amount α . Using the fact that $k < 0$ we have

$$\begin{aligned} T_2(5) - \alpha &= 72 + 103e^{5k} - \alpha < 72 + 103e^{5k} - \alpha e^{5k} \\ &= 72 + (103 - \alpha)e^{5k} = T_1(5). \end{aligned}$$

Thus, the temperature of the coffee in Mr. Jones' cup is hotter.

19. From $dA/dt = 4 - A/50$ we obtain $A = 200 + ce^{-t/50}$. If $A(0) = 30$ then $c = -170$ and $A = 200 - 170e^{-t/50}$.
20. From $dA/dt = 0 - A/50$ we obtain $A = ce^{-t/50}$. If $A(0) = 30$ then $c = 30$ and $A = 30e^{-t/50}$.
21. From $dA/dt = 10 - A/100$ we obtain $A = 1000 + ce^{-t/100}$. If $A(0) = 0$ then $c = -1000$ and $A = 1000 - 1000e^{-t/100}$. At $t = 5$, $A(5) \approx 48.77$ pounds.
22. From $\frac{dA}{dt} = 10 - \frac{10A}{500 - (10 - 5)t} = 10 - \frac{2A}{100 - t}$ we obtain $A = 1000 - 10t + c(100 - t)^2$. If $A(0) = 0$ then $c = -\frac{1}{10}$. The tank is empty in 100 minutes.
23. From $\frac{dA}{dt} = 3 - \frac{4A}{100 + (6 - 4)t} = 3 - \frac{2A}{50 + t}$ we obtain $A = 50 + t + c(50 + t)^{-2}$. If $A(0) = 10$ then $c = -100,000$ and $A(30) = 64.38$ pounds.
24. (a) Initially the tank contains 300 gallons of solution. Since brine is pumped in at a rate of 3 gal/min and the solution is pumped out at a rate of 2 gal/min, the net change is an increase of 1 gal/min. Thus, in 100 minutes the tank will contain its capacity of 400 gallons.
- (b) The differential equation describing the amount of salt in the tank is $A'(t) = 6 - 2A/(300 + t)$ with solution

$$A(t) = 600 + 2t - (4.95 \times 10^7)(300 + t)^{-2}, \quad 0 \leq t \leq 100,$$

Exercises 2.7

as noted in the discussion following Example 5 in the text. Thus, the amount of salt in the tank when it overflows is

$$A(100) = 800 - (4.95 \times 10^7)(400)^{-2} = 490.625 \text{ lbs.}$$

(c) When the tank is overflowing the amount of salt in the tank is governed by the differential equation

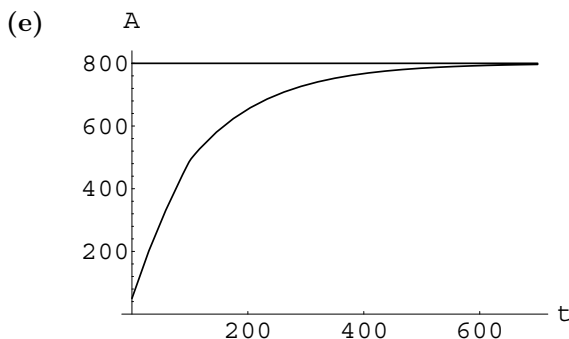
$$\begin{aligned} \frac{dA}{dt} &= (3 \text{ gal/min})(2 \text{ lb/gal}) - \left(\frac{A}{400} \text{ lb/gal}\right)(3 \text{ gal/min}) \\ &= 6 - \frac{3A}{400}, \quad A(100) = 490.625. \end{aligned}$$

Solving the equation we obtain $A(t) = 800 + ce^{-3t/400}$. The initial condition yields $c = -654.947$, so that

$$A(t) = 800 - 654.947e^{-3t/400}.$$

When $t = 150$, $A(150) = 587.37$ lbs.

(d) As $t \rightarrow \infty$, the amount of salt is 800 lbs, which is to be expected since $(400 \text{ gal})(2 \text{ lbs/gal}) = 800 \text{ lbs}$.



25. Assume $L di/dt + Ri = E(t)$, $L = 0.1$, $R = 50$, and $E(t) = 50$ so that $i = \frac{3}{5} + ce^{-500t}$. If $i(0) = 0$ then $c = -3/5$ and $\lim_{t \rightarrow \infty} i(t) = 3/5$.

26. Assume $L di/dt + Ri = E(t)$, $E(t) = E_0 \sin \omega t$, and $i(0) = i_0$ so that

$$i = \frac{E_0 R}{L^2 \omega^2 + R^2} \sin \omega t - \frac{E_0 L \omega}{L^2 \omega^2 + R^2} \cos \omega t + ce^{-Rt/L}.$$

Since $i(0) = i_0$ we obtain $c = i_0 + \frac{E_0 L \omega}{L^2 \omega^2 + R^2}$.

27. Assume $R dq/dt + (1/c)q = E(t)$, $R = 200$, $C = 10^{-4}$, and $E(t) = 100$ so that $q = 1/100 + ce^{-50t}$. If $q(0) = 0$ then $c = -1/100$ and $i = \frac{1}{2}e^{-50t}$.

28. Assume $R dq/dt + (1/c)q = E(t)$, $R = 1000$, $C = 5 \times 10^{-6}$, and $E(t) = 200$. Then $q = \frac{1}{1000} + ce^{-200t}$ and $i = -200ce^{-200t}$. If $i(0) = 0.4$ then $c = -\frac{1}{500}$, $q(0.005) = 0.003$ coulombs, and $i(0.005) = 0.1472$ amps. As $t \rightarrow \infty$ we have $q \rightarrow \frac{1}{1000}$.

29. For $0 \leq t \leq 20$ the differential equation is $20 di/dt + 2i = 120$. An integrating factor is $e^{t/10}$, so $\frac{d}{dt} [e^{t/10} i] = 6e^{t/10}$ and $i = 60 + c_1 e^{-t/10}$. If $i(0) = 0$ then $c_1 = -60$ and $i = 60 - 60e^{-t/10}$.

For $t > 20$ the differential equation is $20 di/dt + 2i = 0$ and $i = c_2 e^{-t/10}$.

At $t = 20$ we want $c_2 e^{-2} = 60 - 60e^{-2}$ so that $c_2 = 60(e^2 - 1)$. Thus

$$i(t) = \begin{cases} 60 - 60e^{-t/10}, & 0 \leq t \leq 20; \\ 60(e^2 - 1)e^{-t/10}, & t > 20. \end{cases}$$

30. Separating variables we obtain

$$\frac{dq}{E_0 - q/C} = \frac{dt}{k_1 + k_2 t} \implies -C \ln \left| E_0 - \frac{q}{C} \right| = \frac{1}{k_2} \ln |k_1 + k_2 t| + c_1 \implies \frac{(E_0 - q/C)^{-C}}{(k_1 + k_2 t)^{1/k_2}} = c_2.$$

Setting $q(0) = q_0$ we find $c_2 = \frac{(E_0 - q_0/C)^{-C}}{k_1^{1/k_2}}$, so

$$\begin{aligned} \frac{(E_0 - q/C)^{-C}}{(k_1 + k_2 t)^{1/k_2}} &= \frac{(E_0 - q_0/C)^{-C}}{k_1^{1/k_2}} \implies \left(E_0 - \frac{q}{C} \right)^{-C} = \left(E_0 - \frac{q_0}{C} \right)^{-C} \left(\frac{k_1}{k_1 + k_2 t} \right)^{-1/k_2} \\ &\implies E_0 - \frac{q}{C} = \left(E_0 - \frac{q_0}{C} \right) \left(\frac{k_1}{k_1 + k_2 t} \right)^{1/Ck_2} \\ &\implies q = E_0 C + (q_0 - E_0 C) \left(\frac{k_1}{k_1 + k_2 t} \right)^{1/Ck_2}. \end{aligned}$$

31. (a) From $m dv/dt = mg - kv$ we obtain $v = gm/k + ce^{-kt/m}$. If $v(0) = v_0$ then $c = v_0 - gm/k$ and the solution of the initial-value problem is

$$v = \frac{gm}{k} + \left(v_0 - \frac{gm}{k} \right) e^{-kt/m}.$$

(b) As $t \rightarrow \infty$ the limiting velocity is gm/k .

(c) From $ds/dt = v$ and $s(0) = 0$ we obtain

$$s = \frac{gm}{k} t - \frac{m}{k} \left(v_0 - \frac{gm}{k} \right) e^{-kt/m} + \frac{m}{k} \left(v_0 - \frac{gm}{k} \right).$$

32. (a) Integrating $d^2s/dt^2 = -g$ we get $v(t) = ds/dt = -gt + c$. From $v(0) = 300$ we find $c = 300$, so the velocity is $v(t) = -32t + 300$. Integrating again and using $s(0) = 0$ we get $s(t) = -16t^2 + 300t$. The maximum height is attained when $v = 0$, that is, at $t_a = 9.375$. The maximum height will be $s(9.375) = 1406.25$ ft.

(b) When air resistance is proportional to velocity the model for the velocity is $m dv/dt = -mg - kv$ (using the fact that the positive direction is upward.) Solving the differential equation using separation of variables we obtain $v(t) = -mg/k + ce^{-kt/m}$. From $v(0) = 300$ we get

$$v(t) = -\frac{mg}{k} + \left(300 + \frac{mg}{k} \right) e^{-kt/m}.$$

Integrating and using $s(0) = 0$ we find

$$s(t) = -\frac{mg}{k} t + \frac{m}{k} \left(300 + \frac{mg}{k} \right) (1 - e^{-kt/m}).$$

Setting $k = 0.0025$, $m = 16/32 = 0.5$, and $g = 32$ we have

$$s(t) = 1,340,000 - 6,400t - 1,340,000e^{-0.005t}$$

and

$$v(t) = -6,400 + 6,700e^{-0.005t}.$$

The maximum height is attained when $v = 0$, that is, at $t_a = 9.162$. The maximum height will be $s(9.162) = 1363.79$ ft, which is less than the maximum height in part (a).

33. We saw in part (a) of Problem 31 that the ascent time is $t_a = 9.375$. To find when the cannonball hits the ground we solve $s(t) = -16t^2 + 300t = 0$, getting a total time in flight of $t = 18.75$. Thus, the time of descent is $t_d = 18.75 - 9.375 = 9.375$. The impact velocity is $v_i = v(18.75) = -300$, which has the same magnitude as the initial velocity.

Exercises 2.7

We saw in part (b) of Problem 31 that the ascent time in the case of air resistance is $t_a = 9.162$. Solving $s(t) = 1,340,000 - 6,400t - 1,340,000e^{-0.005t} = 0$ we see that the total time of flight is 18.466. Thus, the descent time is $t_d = 18.466 - 9.162 = 9.304$. The impact velocity is $v_i = v(18.466) = -290.91$, compared to an initial velocity of $v_0 = 300$.

- 34.** Assuming that air resistance is proportional to velocity and the positive direction is downward, the model for the velocity is $m dv/dt = mg - kv$. Using separation of variables to solve this differential equation we obtain $v(t) = mg/k + ce^{-kt/m}$. From $v(0) = 0$ we get $v(t) = (mg/k)(1 - e^{-kt/m})$. Letting $k = 0.5$, $m = 160/32 = 5$, and $g = 32$ we have $v(t) = 320(1 - e^{-0.1t})$. Integrating, we find $s(t) = 320t + 3200e^{-0.1t}$. At $t = 15$, when the parachute opens, $v(15) = 248.598$ and $s(15) = 5514.02$. At this point the value of k changes to $k = 10$ and the new initial velocity is $v_0 = 248.598$. Her velocity with the parachute open (with time measured from the instant of opening) is $v_p(t) = 16 + 232.598e^{-2t}$. Integrating, we find $s_p(t) = 16t - 116.299e^{-2t}$. Twenty seconds after leaving the plane is five seconds after the parachute opens. Her velocity at this time is $v_p(5) = 16.0106$ ft/sec and she has fallen $s(15) + s_p(5) = 5514.02 + 79.9947 = 5594.01$ ft. Her terminal velocity is $\lim_{t \rightarrow \infty} v_p(t) = 16$, so she has very nearly reached her terminal velocity five seconds after the parachute opens. When the parachute opens, the distance to the ground is $15,000 - 5514.02 = 9485.98$ ft. Solving $s_p(t) = 9485.98$ we get $t = 592.874$ s = 9.88 min. Thus, it will take her approximately 9.88 minutes to reach the ground after her parachute has opened and a total of $(592.874 + 15)/60 = 10.13$ minutes after she exits the plane.

- 35. (a)** The differential equation is first-order, linear. Letting $b = k/\rho$, the integrating factor is $e^{\int 3b dt/(bt+r_0)} = (r_0 + bt)^3$. Then

$$\frac{d}{dt}[(r_0 + bt)^3 v] = g(r_0 + bt)^3 \quad \text{and} \quad (r_0 + bt)^3 v = \frac{g}{4b}(r_0 + bt)^4 + c.$$

The solution of the differential equation is $v(t) = (g/4b)(r_0 + bt) + c(r_0 + bt)^{-3}$. Using $v(0) = 0$ we find $c = -gr_0^4/4b$, so that

$$v(t) = \frac{g}{4b}(r_0 + bt) - \frac{gr_0^4}{4b(r_0 + bt)^3} = \frac{g\rho}{4k}\left(r_0 + \frac{k}{\rho}t\right) - \frac{g\rho r_0^4}{4k(r_0 + kt/\rho)^3}.$$

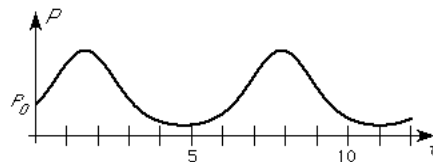
- (b)** Integrating $dr/dt = k/\rho$ we get $r = kt/\rho + c$. Using $r(0) = r_0$ we have $c = r_0$, so $r(t) = kt/\rho + r_0$.

- (c)** If $r = 0.007$ ft when $t = 10$ s, then solving $r(10) = 0.007$ for k/ρ , we obtain $k/\rho = -0.0003$ and $r(t) = 0.01 - 0.0003t$. Solving $r(t) = 0$ we get $t = 33.3$, so the raindrop will have evaporated completely at 33.3 seconds.

- 36.** Separating variables we obtain

$$\frac{dP}{P} = k \cos t \, dt \implies \ln |P| = k \sin t + c \implies P = c_1 e^{k \sin t}.$$

If $P(0) = P_0$ then $c_1 = P_0$ and $P = P_0 e^{k \sin t}$.



- 37. (a)** From $dP/dt = (k_1 - k_2)P$ we obtain $P = P_0 e^{(k_1 - k_2)t}$ where $P_0 = P(0)$.

- (b)** If $k_1 > k_2$ then $P \rightarrow \infty$ as $t \rightarrow \infty$. If $k_1 = k_2$ then $P = P_0$ for every t . If $k_1 < k_2$ then $P \rightarrow 0$ as $t \rightarrow \infty$.

- 38.** The first equation can be solved by separation of variables. We obtain $x = c_1 e^{-\lambda_1 t}$. From $x(0) = x_0$ we obtain $c_1 = x_0$ and so $x = x_0 e^{-\lambda_1 t}$. The second equation then becomes

$$\frac{dy}{dt} = x_0 \lambda_1 e^{-\lambda_1 t} - \lambda_2 y \quad \text{or} \quad \frac{dy}{dt} + \lambda_2 y = x_0 \lambda_1 e^{-\lambda_1 t}$$

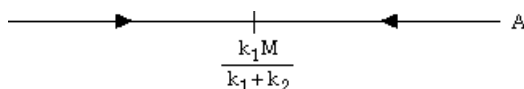
which is linear. An integrating factor is $e^{\lambda_2 t}$. Thus

$$\begin{aligned}\frac{d}{dt} [e^{\lambda_2 t} y] &= x_0 \lambda_1 e^{-\lambda_1 t} e^{\lambda_2 t} = x_0 \lambda_1 e^{(\lambda_2 - \lambda_1)t} \\ e^{\lambda_2 t} y &= \frac{x_0 \lambda_1}{\lambda_2 - \lambda_1} e^{(\lambda_2 - \lambda_1)t} + c_2 \\ y &= \frac{x_0 \lambda_1}{\lambda_2 - \lambda_1} e^{-\lambda_1 t} + c_2 e^{-\lambda_2 t}.\end{aligned}$$

From $y(0) = y_0$ we obtain $c_2 = (y_0 \lambda_2 - y_0 \lambda_1 - x_0 \lambda_1) / (\lambda_2 - \lambda_1)$. The solution is

$$y = \frac{x_0 \lambda_1}{\lambda_2 - \lambda_1} e^{-\lambda_1 t} + \frac{y_0 \lambda_2 - y_0 \lambda_1 - x_0 \lambda_1}{\lambda_2 - \lambda_1} e^{-\lambda_2 t}.$$

39. (a) Solving $k_1(M - A) - k_2 A = 0$ for A we find the equilibrium solution $A = k_1 M / (k_1 + k_2)$. From the phase portrait we see that $\lim_{t \rightarrow \infty} A(t) = k_1 M / (k_1 + k_2)$.

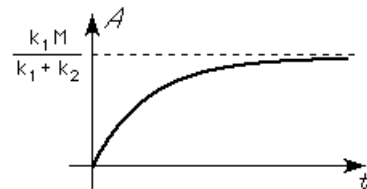


Since $k_2 > 0$, the material will never be completely memorized and the larger k_2 is, the less the amount of material will be memorized over time.

- (b) Write the differential equation in the form $dA/dt + (k_1 + k_2)A = k_1 M$.

Then an integrating factor is $e^{(k_1 + k_2)t}$, and

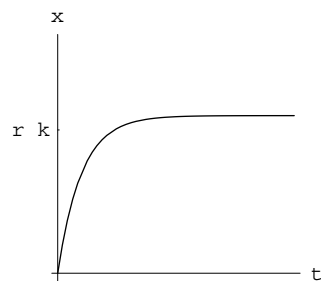
$$\begin{aligned}\frac{d}{dt} [e^{(k_1 + k_2)t} A] &= k_1 M e^{(k_1 + k_2)t} \\ \implies e^{(k_1 + k_2)t} A &= \frac{k_1 M}{k_1 + k_2} e^{(k_1 + k_2)t} + c \\ \implies A &= \frac{k_1 M}{k_1 + k_2} + c e^{-(k_1 + k_2)t}.\end{aligned}$$



Using $A(0) = 0$ we find $c = -\frac{k_1 M}{k_1 + k_2}$ and $A = \frac{k_1 M}{k_1 + k_2} (1 - e^{-(k_1 + k_2)t})$. As $t \rightarrow \infty$, $A \rightarrow \frac{k_1 M}{k_1 + k_2}$.

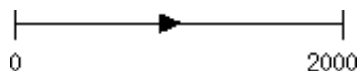
40. (a) Solving $r - kx = 0$ for x we find the equilibrium solution $x = r/k$. When $x < r/k$, $dx/dt > 0$ and when $x > r/k$, $dx/dt < 0$. From the phase portrait we see that $\lim_{t \rightarrow \infty} x(t) = r/k$.

- (b) From $dx/dt = r - kx$ and $x(0) = 0$ we obtain $x = r/k - (r/k)e^{-kt}$ so that $x \rightarrow r/k$ as $t \rightarrow \infty$. If $x(T) = r/2k$ then $T = (\ln 2)/k$.



Exercises 2.8

1. (a) Solving $N(1 - 0.0005N) = 0$ for N we find the equilibrium solutions $N = 0$ and $N = 2000$. When $0 < N < 2000$, $dN/dt > 0$. From the phase portrait we see that $\lim_{t \rightarrow \infty} N(t) = 2000$.



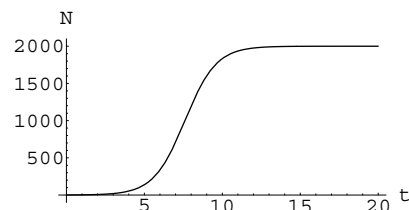
A graph of the solution is shown in part (b).

- (b) Separating variables and integrating we have

$$\frac{dN}{N(1 - 0.0005N)} = \left(\frac{1}{N} - \frac{1}{N - 2000} \right) dN = dt$$

and

$$\ln N - \ln(N - 2000) = t + c.$$



Solving for N we get $N(t) = 2000e^{c+t}/(1 + e^{c+t}) = 2000e^c e^t/(1 + e^c e^t)$. Using $N(0) = 1$ and solving for e^c we find $e^c = 1/1999$ and so $N(t) = 2000e^t/(1999 + e^t)$. Then $N(10) = 1833.59$, so 1834 companies are expected to adopt the new technology when $t = 10$.

2. From $\frac{dN}{dt} = N(a - bN)$ and $N(0) = 500$ we obtain $N = \frac{500a}{500b + (a - 500b)e^{-at}}$. Since $\lim_{t \rightarrow \infty} N = \frac{a}{b} = 50,000$

and $N(1) = 1000$ we have $a = 0.7033$, $b = 0.00014$, and $N = \frac{50,000}{1 + 99e^{-0.7033t}}$.

3. From $\frac{dP}{dt} = P(10^{-1} - 10^{-7}P)$ and $P(0) = 5000$ we obtain $P = \frac{500}{0.0005 + 0.0995e^{-0.1t}}$ so that $P \rightarrow 1,000,000$ as $t \rightarrow \infty$. If $P(t) = 500,000$ then $t = 52.9$ months.

4. (a) We have $dP/dt = P(a - bP)$ with $P(0) = 3.929$ million. Using separation of variables we obtain

$$\begin{aligned} P(t) &= \frac{3.929a}{3.929b + (a - 3.929b)e^{-at}} = \frac{a/b}{1 + (a/3.929b - 1)e^{-at}} \\ &= \frac{c}{1 + (c/3.929 - 1)e^{-at}}. \end{aligned}$$

At $t = 60(1850)$ the population is 23.192 million, so

$$23.192 = \frac{c}{1 + (c/3.929 - 1)e^{-60a}}$$

or $c = 23.192 + 23.192(c/3.929 - 1)e^{-60a}$. At $t = 120(1910)$

$$91.972 = \frac{c}{1 + (c/3.929 - 1)e^{-120a}}$$

or $c = 91.972 + 91.972(c/3.929 - 1)(e^{-60a})^2$. Combining the two equations for c we get

$$\left(\frac{(c - 23.192)/23.192}{c/3.929 - 1} \right)^2 \left(\frac{c}{3.929} - 1 \right) = \frac{c - 91.972}{91.972}$$

or

$$91.972(3.929)(c - 23.192)^2 = (23.192)^2(c - 91.972)(c - 3.929).$$

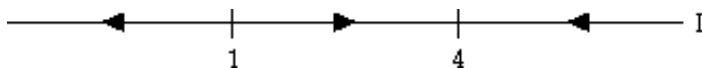
The solution of this quadratic equation is $c = 197.274$. This in turn gives $a = 0.0313$. Therefore

$$P(t) = \frac{197.274}{1 + 49.21e^{-0.0313t}}.$$

(b)

Year	Census Population	Predicted Population	Error	% Error
1790	3.929	3.929	0.000	0.00
1800	5.308	5.334	-0.026	-0.49
1810	7.240	7.222	0.018	0.24
1820	9.638	9.746	-0.108	-1.12
1830	12.866	13.090	-0.224	-1.74
1840	17.069	17.475	-0.406	-2.38
1850	23.192	23.143	0.049	0.21
1860	31.433	30.341	1.092	3.47
1870	38.558	39.272	-0.714	-1.85
1880	50.156	50.044	0.112	0.22
1890	62.948	62.600	0.348	0.55
1900	75.996	76.666	-0.670	-0.88
1910	91.972	91.739	0.233	0.25
1920	105.711	107.143	-1.432	-1.35
1930	122.775	122.140	0.635	0.52
1940	131.669	136.068	-4.399	-3.34
1950	150.697	148.445	2.252	1.49

- (c) The model predicts a population of 159.0 million for 1960 and 167.8 million for 1970. The census populations for these years were 179.3 and 203.3, respectively. The percentage errors are 12.8 and 21.2, respectively.
5. (a) The differential equation is $dP/dt = P(5 - P) - 4$. Solving $P(5 - P) - 4 = 0$ for P we obtain equilibrium solutions $P = 1$ and $P = 4$. The phase portrait is shown below and solution curves are shown in part (b).



We see that for $P_0 > 4$ and $1 < P_0 < 4$ the population approaches 4 as t increases. For $0 < P_0 < 1$ the population decreases to 0 in finite time.

- (b) The differential equation is

$$\frac{dP}{dt} = P(5 - P) - 4 = -(P^2 - 5P + 4) = -(P - 4)(P - 1).$$

Separating variables and integrating, we obtain

$$\begin{aligned} \frac{dP}{(P - 4)(P - 1)} &= -dt \\ \left(\frac{1/3}{P - 4} - \frac{1/3}{P - 1} \right) dP &= -dt \\ \frac{1}{3} \ln \left| \frac{P - 4}{P - 1} \right| &= -t + c \\ \frac{P - 4}{P - 1} &= c_1 e^{-3t}. \end{aligned}$$

Setting $t = 0$ and $P = P_0$ we find $c_1 = (P_0 - 4)/(P_0 - 1)$. Solving for P we obtain

$$P(t) = \frac{4(P_0 - 1) - (P_0 - 4)e^{-3t}}{(P_0 - 1) - (P_0 - 4)e^{-3t}}.$$

- (c) To find when the population becomes extinct in the case $0 < P_0 < 1$ we set $P = 0$ in

$$\frac{P - 4}{P - 1} = \frac{P_0 - 4}{P_0 - 1} e^{-3t}$$

Exercises 2.8

from part (a) and solve for t . This gives the time of extinction

$$t = -\frac{1}{3} \ln \frac{4(P_0 - 1)}{P_0 - 4}.$$

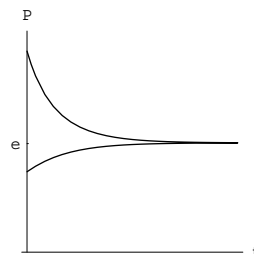
6. (a) Solving $P(5 - P) - \frac{25}{4} = 0$ for P we obtain the equilibrium solution $P = \frac{5}{2}$. For $P \neq \frac{5}{2}$, $dP/dt < 0$. Thus, if $P_0 < \frac{5}{2}$, the population becomes extinct (otherwise there would be another equilibrium solution.) Using separation of variables to solve the initial-value problem we get $P(t) = [4P_0 + (10P_0 - 25)t]/[4 + (4P_0 - 10)t]$. To find when the population becomes extinct for $P_0 < \frac{5}{2}$ we solve $P(t) = 0$ for t . We see that the time of extinction is $t = 4P_0/5(5 - 2P_0)$.
- (b) Solving $P(5 - P) - 7 = 0$ for P we obtain complex roots, so there are no equilibrium solutions. Since $dP/dt < 0$ for all values of P , the population becomes extinct for any initial condition. Using separation of variables to solve the initial-value problem we get

$$P(t) = \frac{5}{2} + \frac{\sqrt{3}}{2} \tan \left[\tan^{-1} \left(\frac{2P_0 - 5}{\sqrt{3}} \right) - \frac{\sqrt{3}}{2} t \right].$$

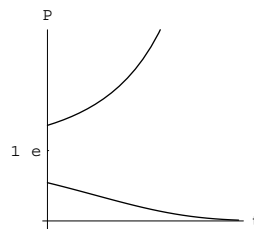
Solving $P(t) = 0$ for t we see that the time of extinction is

$$t = \frac{2}{3} \left(\sqrt{3} \tan^{-1}(5/\sqrt{3}) + \sqrt{3} \tan^{-1}[(2P_0 - 5)/\sqrt{3}] \right).$$

7. (a) The differential equation is $dP/dt = P(1 - \ln P)$, which has equilibrium solution $P = e$. When $P_0 > e$, $dP/dt < 0$, and when $P_0 < e$, $dP/dt > 0$.



- (b) The differential equation is $dP/dt = P(1 + \ln P)$, which has equilibrium solution $P = 1/e$. When $P_0 > 1/e$, $dP/dt > 0$, and when $P_0 < 1/e$, $dP/dt < 0$.

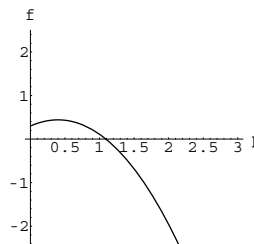


- (c) From $\frac{dP}{dt} = P(a - b \ln P)$ we obtain $\frac{-1}{b} \ln |a - b \ln P| = t + c_1$ so that $P = e^{a/b} e^{-ce^{-bt}}$. If $P(0) = P_0$ then $c = \frac{a}{b} - \ln P_0$.

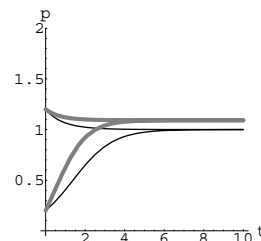
8. (a) Using a CAS to solve $P(1 - P) + 0.3e^{-P} = 0$ for P we see that $P = 1.09216$ is an equilibrium solution.
- (b) Since $f(P) > 0$ for $0 < P < 1.09216$, the solution $P(t)$ of

$$dP/dt = P(1 - P) + 0.3e^{-P}, \quad P(0) = P_0,$$

is increasing for $P_0 < 1.09216$. Since $f(P) < 0$ for $P > 1.09216$, the solution $P(t)$ is decreasing for $P_0 > 1.09216$. Thus $P = 1.09216$ is an attractor.

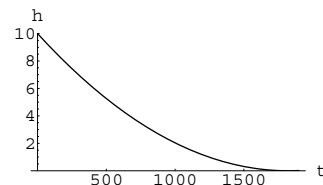


- (c) The curves for the second initial-value problem are thicker. The equilibrium solution for the logic model is $P = 1$. Comparing 1.09216 and 1, we see that the percentage increase is 9.216%.



9. Let $X = X(t)$ be the amount of C at time t and $\frac{dX}{dt} = k(120 - 2X)(150 - X)$. If $X(0) = 0$ and $X(5) = 10$ then $X = \frac{150 - 150e^{180kt}}{1 - 2.5e^{180kt}}$ where $k = .0001259$, and $X(20) = 29.3$ grams. Now $X \rightarrow 60$ as $t \rightarrow \infty$, so that the amount of $A \rightarrow 0$ and the amount of $B \rightarrow 30$ as $t \rightarrow \infty$.
10. From $\frac{dX}{dt} = k(150 - X)^2$, $X(0) = 0$, and $X(5) = 10$ we obtain $X = 150 - \frac{150}{150kt + 1}$ where $k = .000095238$. Then $X(20) = 33.3$ grams and $X \rightarrow 150$ as $t \rightarrow \infty$ so that the amount of $A \rightarrow 0$ and the amount of $B \rightarrow 0$ as $t \rightarrow \infty$. If $X(t) = 75$ then $t = 70$ minutes.
11. (a) The initial-value problem is $dh/dt = -8A_h\sqrt{h}/A_w$, $h(0) = H$. Separating variables and integrating we have

$$\frac{dh}{\sqrt{h}} = -\frac{8A_h}{A_w} dt \quad \text{and} \quad 2\sqrt{h} = -\frac{8A_h}{A_w}t + c.$$



Using $h(0) = H$ we find $c = 2\sqrt{H}$, so the solution of the initial-value problem is $\sqrt{h(t)} = (A_w\sqrt{H} - 4A_ht)/A_w$, where $A_w\sqrt{H} - 4A_ht \geq 0$. Thus, $h(t) = (A_w\sqrt{H} - 4A_ht)^2/A_w^2$ for $0 \leq t \leq A_wH/4A_h$.

- (b) Identifying $H = 10$, $A_w = 4\pi$, and $A_h = \pi/576$ we have $h(t) = t^2/331,776 - (\sqrt{5/2}/144)t + 10$. Solving $h(t) = 0$ we see that the tank empties in $576\sqrt{10}$ seconds or 30.36 minutes.
12. To obtain the solution of this differential equation we use $h(t)$ from part (a) in Problem 11 with A_h replaced by cA_h . Then $h(t) = (A_w\sqrt{H} - 4cA_ht)^2/A_w^2$. Solving $h(t) = 0$ with $c = 0.6$ and the values from Problem 11 we see that the tank empties in 3035.79 seconds or 50.6 minutes.
13. (a) Separating variables and integrating we have

$$6h^{3/2}dh = -5t \quad \text{and} \quad \frac{12}{5}h^{5/2} = -5t + c.$$

Using $h(0) = 20$ we find $c = 1920\sqrt{5}$, so the solution of the initial-value problem is $h(t) = (800\sqrt{5} - \frac{25}{12}t)^{2/5}$. Solving $h(t) = 0$ we see that the tank empties in $384\sqrt{5}$ seconds or 14.31 minutes.

- (b) When the height of the water is h , the radius of the top of the water is $r = h \tan 30^\circ = h/\sqrt{3}$ and $A_w = \pi h^2/3$. The differential equation is

$$\frac{dh}{dt} = -c \frac{A_h}{A_w} \sqrt{2gh} = -0.6 \frac{\pi(2/12)^2}{\pi h^2/3} \sqrt{64h} = -\frac{2}{5h^{3/2}}.$$

Separating variables and integrating we have

$$5h^{3/2}dh = -2dt \quad \text{and} \quad 2h^{5/2} = -2t + c.$$

Using $h(0) = 9$ we find $c = 486$, so the solution of the initial-value problem is $h(t) = (243 - t)^{2/5}$. Solving $h(t) = 0$ we see that the tank empties in 24.3 seconds or 4.05 minutes.

Exercises 2.8

14. When the height of the water is h , the radius of the top of the water is $\frac{2}{5}(20 - h)$ and $A_w = 4\pi(20 - h)^2/25$. The differential equation is

$$\frac{dh}{dt} = -c \frac{A_h}{A_w} \sqrt{2gh} = -0.6 \frac{\pi(2/12)^2}{4\pi(20 - h)^2/25} \sqrt{64h} = -\frac{5}{6} \frac{\sqrt{h}}{(20 - h)^2}.$$

Separating variables and integrating we have

$$\frac{(20 - h)^2}{\sqrt{h}} dh = -\frac{5}{6} dt \quad \text{and} \quad 800\sqrt{h} - \frac{80}{3}h^{3/2} + \frac{2}{5}h^{5/2} = -\frac{5}{6}t + c.$$

Using $h(0) = 20$ we find $c = 2560\sqrt{5}/3$, so an implicit solution of the initial-value problem is

$$800\sqrt{h} - \frac{80}{3}h^{3/2} + \frac{2}{5}h^{5/2} = -\frac{5}{6}t + \frac{2560\sqrt{5}}{3}.$$

To find the time it takes the tank to empty we set $h = 0$ and solve for t . The tank empties in $1024\sqrt{5}$ seconds or 38.16 minutes. Thus, the tank empties more slowly when the base of the cone is on the bottom.

15. (a) Separating variables we obtain

$$\begin{aligned} \frac{m dv}{mg - kv^2} &= dt \\ \frac{1}{g} \frac{dv}{1 - (kv/mg)^2} &= dt \\ \frac{\sqrt{mg}}{\sqrt{k}g} \frac{\sqrt{k/mg} dv}{1 - (\sqrt{k}v/\sqrt{mg})^2} &= dt \\ \sqrt{\frac{m}{kg}} \tanh^{-1} \frac{\sqrt{k}v}{\sqrt{mg}} &= t + c \\ \tanh^{-1} \frac{\sqrt{k}v}{\sqrt{mg}} &= \sqrt{\frac{kg}{m}} t + c_1. \end{aligned}$$

Thus the velocity at time t is

$$v(t) = \sqrt{\frac{mg}{k}} \tanh \left(\sqrt{\frac{kg}{m}} t + c_1 \right).$$

Setting $t = 0$ and $v = v_0$ we find $c_1 = \tanh^{-1}(\sqrt{k}v_0/\sqrt{mg})$.

- (b) Since $\tanh t \rightarrow 1$ as $t \rightarrow \infty$, we have $v \rightarrow \sqrt{mg/k}$ as $t \rightarrow \infty$.
(c) Integrating the expression for $v(t)$ in part (a) we obtain

$$s(t) = \sqrt{\frac{mg}{k}} \int \tanh \left(\sqrt{\frac{kg}{m}} t + c_1 \right) dt = \frac{m}{k} \ln \left[\cosh \left(\sqrt{\frac{kg}{m}} t + c_1 \right) \right] + c_2.$$

Setting $t = 0$ and $s = s_0$ we find $c_2 = s_0 - \frac{m}{k} \ln(\cosh c_1)$.

16. We solve

$$m \frac{dv}{dt} = -mg - kv^2, \quad v(0) = 300$$

using separation of variables. This gives

$$v(t) = \sqrt{\frac{mg}{k}} \tan \left(\tan^{-1} 300 \sqrt{\frac{k}{mg}} - \sqrt{\frac{kg}{m}} t \right).$$

Integrating and using $s(0) = 0$ we find

$$s(t) = \frac{m}{k} \ln \left| \cos \sqrt{\frac{kg}{m}} t - \tan^{-1} 300 \sqrt{\frac{k}{mg}} \right| + \frac{m}{2k} \ln \left(1 + \frac{90000k}{mg} \right).$$

Solving $v(t) = 0$ we see that $t_a = 6.60159$. The maximum height is $s(t_a) = 823.84$ ft.

17. To find t_d we solve

$$m \frac{dv}{dt} = mg - kv^2, \quad v(0) = 0$$

using separation of variables. This gives

$$v(t) = \sqrt{\frac{mg}{k}} \tanh \sqrt{\frac{kg}{m}} t.$$

Integrating and using $s(0) = 0$ gives

$$s(t) = \frac{m}{k} \ln \left(\cosh \sqrt{\frac{kg}{m}} t \right).$$

To find the time of descent we solve $s(t) = 823.84$ and find $t_d = 7.77882$. The impact velocity is $v(t_d) = 182.998$, which is positive because the positive direction is downward.

18. (a) Solving $v_t = \sqrt{mg/k}$ for k we obtain $k = mg/v_t^2$. The differential equation then becomes

$$m \frac{dv}{dt} = mg - \frac{mg}{v_t^2} v^2 \quad \text{or} \quad \frac{dv}{dt} = g \left(1 - \frac{1}{v_t^2} v^2 \right).$$

Separating variables and integrating gives

$$v_t \tanh^{-1} \frac{v}{v_t} = gt + c_1.$$

The initial condition $v(0) = 0$ implies $c_1 = 0$, so

$$v(t) = v_t \tanh \frac{gt}{v_t}.$$

We find the distance by integrating:

$$s(t) = \int v_t \tanh \frac{gt}{v_t} dt = \frac{v_t^2}{g} \ln \left(\cosh \frac{gt}{v_t} \right) + c_2.$$

The initial condition $s(0) = 0$ implies $c_2 = 0$, so

$$s(t) = \frac{v_t^2}{g} \ln \left(\cosh \frac{gt}{v_t} \right).$$

In 25 seconds she has fallen $20,000 - 14,800 = 5,200$ feet. Solving $5200 = (v_t^2/32) \ln \left(\cosh \frac{32(25)}{v_t} \right)$ for v_t gives $v_t \approx 271.711$ ft/s. Then

$$s(t) = \frac{v_t^2}{g} \ln \left(\cosh \frac{gt}{v_t} \right) = 2307.08 \ln(\cosh 0.117772t).$$

- (b) At $t = 15$, $s(15) = 2,542.94$ ft and $v(15) = s'(15) = 256.287$ ft/sec.

19. While the object is in the air its velocity is modeled by the linear differential equation $m dv/dt = mg - kv$. Using $m = 160$, $k = \frac{1}{4}$, and $g = 32$, the differential equation becomes $dv/dt + (1/640)v = 32$. The integrating factor is $e^{\int dt/640} = e^{t/640}$ and the solution of the differential equation is $e^{t/640}v = \int 32e^{t/640}dt = 20,480e^{t/640} + c$. Using $v(0) = 0$ we see that $c = -20,480$ and $v(t) = 20,480 - 20,480e^{-t/640}$. Integrating we get $s(t) = 20,480t + 13,107,200e^{-t/640} + c$. Since $s(0) = 0$, $c = -13,107,200$ and $s(t) = -13,107,200 + 20,480t + 13,107,200e^{-t/640}$.

Exercises 2.8

To find when the object hits the liquid we solve $s(t) = 500 - 75 = 425$, obtaining $t_a = 5.16018$. The velocity at the time of impact with the liquid is $v_a = v(t_a) = 164.482$. When the object is in the liquid its velocity is modeled by $m dv/dt = mg - kv^2$. Using $m = 160$, $g = 32$, and $k = 0.1$ this becomes $dv/dt = (51,200 - v^2)/1600$. Separating variables and integrating we have

$$\frac{dv}{51,200 - v^2} = \frac{dt}{1600} \quad \text{and} \quad \frac{\sqrt{2}}{640} \ln \left| \frac{v - 160\sqrt{2}}{v + 160\sqrt{2}} \right| = \frac{1}{1600}t + c.$$

Solving $v(0) = v_a = 164.482$ we obtain $c = -0.00407537$. Then, for $v < 160\sqrt{2} = 226.274$,

$$\left| \frac{v - 160\sqrt{2}}{v + 160\sqrt{2}} \right| = e^{\sqrt{2}t/5 - 1.8443} \quad \text{or} \quad -\frac{v - 160\sqrt{2}}{v + 160\sqrt{2}} = e^{\sqrt{2}t/5 - 1.8443}.$$

Solving for v we get

$$v(t) = \frac{13964.6 - 2208.29e^{\sqrt{2}t/5}}{61.7153 + 9.75937e^{\sqrt{2}t/5}}.$$

Integrating we find

$$s(t) = 226.275t - 1600 \ln(6.3237 + e^{\sqrt{2}t/5}) + c.$$

Solving $s(0) = 0$ we see that $c = 3185.78$, so

$$s(t) = 3185.78 + 226.275t - 1600 \ln(6.3237 + e^{\sqrt{2}t/5}).$$

To find when the object hits the bottom of the tank we solve $s(t) = 75$, obtaining $t_b = 0.466273$. The time from when the object is dropped from the helicopter to when it hits the bottom of the tank is $t_a + t_b = 5.62708$ seconds.

- 20. (a)** Let ρ be the weight density of the water and V the volume of the object. Archimedes' principle states that the upward buoyant force has magnitude equal to the weight of the water displaced. Taking the positive direction to be down, the differential equation is

$$m \frac{dv}{dt} = mg - kv^2 - \rho V.$$

- (b)** Using separation of variables we have

$$\begin{aligned} \frac{m dv}{(mg - \rho V) - kv^2} &= dt \\ \frac{m}{\sqrt{k}} \frac{\sqrt{k} dv}{(\sqrt{mg - \rho V})^2 - (\sqrt{k} v)^2} &= dt \\ \frac{m}{\sqrt{k}} \frac{1}{\sqrt{mg - \rho V}} \tanh^{-1} \frac{\sqrt{k} v}{\sqrt{mg - \rho V}} &= t + c. \end{aligned}$$

Thus

$$v(t) = \sqrt{\frac{mg - \rho V}{k}} \tanh \left(\frac{\sqrt{kmg - k\rho V}}{m} t + c_1 \right).$$

- (c)** Since $\tanh t \rightarrow 1$ as $t \rightarrow \infty$, the terminal velocity is $\sqrt{(mg - \rho V)/k}$.

- 21. (a)** Writing the equation in the form $(x - \sqrt{x^2 + y^2})dx + y dy$ we identify $M = x - \sqrt{x^2 + y^2}$ and $N = y$. Since

M and N are both homogeneous of degree 1 we use the substitution $y = ux$. It follows that

$$\begin{aligned} (x - \sqrt{x^2 + u^2 x^2}) dx + ux(u dx + x du) &= 0 \\ x \left[(1 - \sqrt{1 + u^2}) + u^2 \right] dx + x^2 u du &= 0 \\ -\frac{u du}{1 + u^2 - \sqrt{1 + u^2}} &= \frac{dx}{x} \\ \frac{u du}{\sqrt{1 + u^2} (1 - \sqrt{1 + u^2})} &= \frac{dx}{x}. \end{aligned}$$

Letting $w = 1 - \sqrt{1 + u^2}$ we have $dw = -u du / \sqrt{1 + u^2}$ so that

$$\begin{aligned} -\ln(1 - \sqrt{1 + u^2}) &= \ln x + c \\ \frac{1}{1 - \sqrt{1 + u^2}} &= c_1 x \\ 1 - \sqrt{1 + u^2} &= -\frac{c_2}{x} \quad (-c_2 = 1/c_1) \\ 1 + \frac{c_2}{x} &= \sqrt{1 + \frac{y^2}{x^2}} \\ 1 + \frac{2c_2}{x} + \frac{c_2^2}{x^2} &= 1 + \frac{y^2}{x^2}. \end{aligned}$$

Solving for y^2 we have

$$y^2 = 2c_2 x + c_2^2 = 4 \left(\frac{c_2}{2} \right) \left(x + \frac{c_2}{2} \right)$$

which is a family of parabolas symmetric with respect to the x -axis with vertex at $(-c_2/2, 0)$ and focus at the origin.

(b) Let $u = x^2 + y^2$ so that

$$\frac{du}{dx} = 2x + 2y \frac{dy}{dx}.$$

Then

$$y \frac{dy}{dx} = \frac{1}{2} \frac{du}{dx} - x$$

and the differential equation can be written in the form

$$\frac{1}{2} \frac{du}{dx} - x = -x + \sqrt{u} \quad \text{or} \quad \frac{1}{2} \frac{du}{dx} = \sqrt{u}.$$

Separating variables and integrating we have

$$\begin{aligned} \frac{du}{2\sqrt{u}} &= dx \\ \sqrt{u} &= x + c \\ u &= x^2 + 2cx + c^2 \\ x^2 + y^2 &= x^2 + 2cx + c^2 \\ y^2 &= 2cx + c^2. \end{aligned}$$

22. (a) From $2W^2 - W^3 = W^2(2 - W) = 0$ we see that $W = 0$ and $W = 2$ are constant solutions.

(b) Separating variables and using a CAS to integrate we get

$$\frac{dW}{W\sqrt{4-2W}} = dx \quad \text{and} \quad -\tanh^{-1}\left(\frac{1}{2}\sqrt{4-2W}\right) = x + c.$$

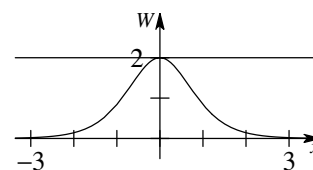
Exercises 2.8

Using the facts that the hyperbolic tangent is an odd function and $1 - \tanh^2 x = \operatorname{sech}^2 x$ we have

$$\begin{aligned}\frac{1}{2}\sqrt{4-2W} &= \tanh(-x-c) = -\tanh(x+c) \\ \frac{1}{4}(4-2W) &= \tanh^2(x+c) \\ 1 - \frac{1}{2}W &= \tanh^2(x+c) \\ \frac{1}{2}W &= 1 - \tanh^2(x+c) = \operatorname{sech}^2(x+c).\end{aligned}$$

Thus, $W(x) = 2\operatorname{sech}^2(x+c)$.

(c) Letting $x = 0$ and $W = 2$ we find that $\operatorname{sech}^2(c) = 1$ and $c = 0$.



23. (a) Solving $r^2 + (10 - h)^2 = 10^2$ for r^2 we see that $r^2 = 20h - h^2$. Combining the rate of input of water, π , with the rate of output due to evaporation, $k\pi r^2 = k\pi(20h - h^2)$, we have $dV/dt = \pi - k\pi(20h - h^2)$. Using $V = 10\pi h^2 - \frac{1}{3}\pi h^3$, we see also that $dV/dt = (20\pi h - \pi h^2)dh/dt$. Thus,

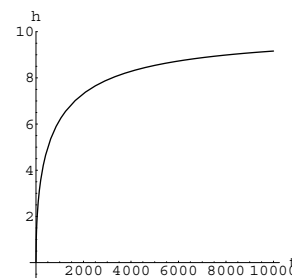
$$(20\pi h - \pi h^2)\frac{dh}{dt} = \pi - k\pi(20h - h^2) \quad \text{and} \quad \frac{dh}{dt} = \frac{1 - 20kh + kh^2}{20h - h^2}.$$

- (b) Letting $k = 1/100$, separating variables and integrating (with the help of a CAS), we get

$$\frac{100h(h-20)}{(h-10)^2} dh = dt \quad \text{and} \quad \frac{100(h^2 - 10h + 100)}{10 - h} = t + c.$$

Using $h(0) = 0$ we find $c = 1000$, and solving for h we get

$h(t) = 0.005(\sqrt{t^2 + 4000t} - t)$, where the positive square root is chosen because $h \geq 0$.

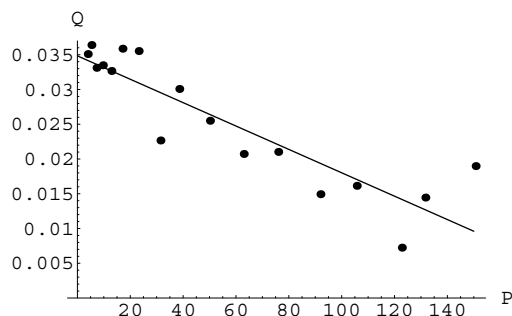


- (c) The volume of the tank is $V = \frac{2}{3}\pi(10)^3$ feet, so at a rate of π cubic feet per minute, the tank will fill in $\frac{2}{3}(10)^3 \approx 666.67$ minutes ≈ 11.11 hours.
- (d) At 666.67 minutes, the depth of the water is $h(666.67) = 5.486$ feet. From the graph in (b) we suspect that $\lim_{t \rightarrow \infty} h(t) = 10$, in which case the tank will never completely fill. To prove this we compute the limit of $h(t)$:

$$\begin{aligned}\lim_{t \rightarrow \infty} h(t) &= 0.005 \lim_{t \rightarrow \infty} \left(\sqrt{t^2 + 4000t} - t \right) = 0.005 \lim_{t \rightarrow \infty} \frac{t^2 + 4000t - t^2}{\sqrt{t^2 + 4000t} + t} \\ &= 0.005 \lim_{t \rightarrow \infty} \frac{4000t}{t\sqrt{1 + 4000/t} + t} = 0.005 \lim_{t \rightarrow \infty} \frac{4000}{1 + 1} = 0.005(2000) = 10.\end{aligned}$$

24. (a)

t	P(t)	Q(t)
0	3.929	0.035
10	5.308	0.036
20	7.240	0.033
30	9.638	0.033
40	12.866	0.033
50	17.069	0.036
60	23.192	0.036
70	31.433	0.023
80	38.558	0.030
90	50.156	0.026
100	62.948	0.021
110	75.996	0.021
120	91.972	0.015
130	105.711	0.016
140	122.775	0.007
150	131.669	0.014
160	150.697	0.019
170	179.300	

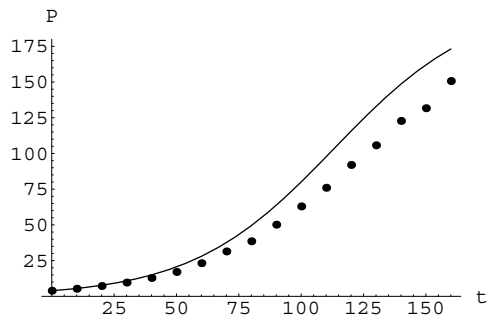
(b) The regression line is $Q = 0.0348391 - 0.000168222P$.(c) The solution of the logistic equation is given in equation (5) in the text. Identifying $a = 0.0348391$ and $b = 0.000168222$ we have

$$P(t) = \frac{aP_0}{bP_0 + (a - bP_0)e^{-at}}.$$

(d) With $P_0 = 3.929$ the solution becomes

$$P(t) = \frac{0.136883}{0.000660944 + 0.0341781e^{-0.0348391t}}.$$

(e)

(f) We identify $t = 180$ with 1970, $t = 190$ with 1980, and $t = 200$ with 1990. The model predicts $P(180) = 188.661$, $P(190) = 193.735$, and $P(200) = 197.485$. The actual population figures for these years are 203.303, 226.542, and 248.765 millions. As $t \rightarrow \infty$, $P(t) \rightarrow a/b = 207.102$.

Exercises 2.9

1. The equation $dx/dt = -\lambda_1 x$ can be solved by separation of variables. Integrating both sides of $dx/x = -\lambda_1 dt$ we obtain $\ln|x| = -\lambda_1 t + c$ from which we get $x = c_1 e^{-\lambda_1 t}$. Using $x(0) = x_0$ we find $c_1 x_0$ so that $x = x_0 e^{-\lambda_1 t}$. Substituting this result into the second differential equation we have

$$\frac{dy}{dt} + \lambda_2 y = \lambda_1 x_0 e^{-\lambda_1 t}$$

which is linear. An integrating factor is $e^{\lambda_2 t}$ so that

$$\frac{d}{dt} [e^{\lambda_2 t} y] = \lambda_1 x_0 e^{(\lambda_2 - \lambda_1)t} + c_2$$

$$y = \frac{\lambda_1 x_0}{\lambda_2 - \lambda_1} e^{(\lambda_2 - \lambda_1)t} e^{-\lambda_2 t} + c_2 e^{-\lambda_2 t} = \frac{\lambda_1 x_0}{\lambda_2 - \lambda_1} e^{-\lambda_1 t} + c_2 e^{-\lambda_2 t}.$$

Using $y(0) = 0$ we find $c_2 = -\lambda_1 x_0 / (\lambda_2 - \lambda_1)$. Thus

$$y = \frac{\lambda_1 x_0}{\lambda_2 - \lambda_1} (e^{-\lambda_1 t} - e^{-\lambda_2 t}).$$

Substituting this result into the third differential equation we have

$$\frac{dz}{dt} = \frac{\lambda_1 \lambda_2 x_0}{\lambda_2 - \lambda_1} (e^{-\lambda_1 t} - e^{-\lambda_2 t}).$$

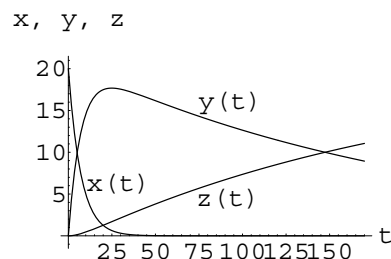
Integrating we find

$$z = -\frac{\lambda_2 x_0}{\lambda_2 - \lambda_1} e^{-\lambda_1 t} + \frac{\lambda_1 x_0}{\lambda_2 - \lambda_1} e^{-\lambda_2 t} + c_3.$$

Using $z(0) = 0$ we find $c_3 = x_0$. Thus

$$z = x \left(1 - \frac{\lambda_2}{\lambda_2 - \lambda_1} e^{-\lambda_1 t} + \frac{\lambda_1}{\lambda_2 - \lambda_1} e^{-\lambda_2 t} \right).$$

2. We see from the graph that the half-life of A is approximately 4.7 days. To determine the half-life of B we use $t = 50$ as a base, since at this time the amount of substance A is so small that it contributes very little to substance B . Now we see from the graph that $y(50) \approx 16.2$ and $y(191) \approx 8.1$. Thus, the half-life of B is approximately 141 days.



3. The amounts of x and y are the same at about $t = 5$ days. The amounts of x and z are the same at about $t = 20$ days. The amounts of y and z are the same at about $t = 147$ days. The time when y and z are the same makes sense because most of A and half of B are gone, so half of C should have been formed.
4. Suppose that the series is described schematically by $W \Rightarrow -\lambda_1 X \Rightarrow -\lambda_2 Y \Rightarrow -\lambda_3 Z$ where $-\lambda_1$, $-\lambda_2$, and $-\lambda_3$ are the decay constants for W , X and Y , respectively, and Z is a stable element. Let $w(t)$, $x(t)$, $y(t)$, and $z(t)$ denote the amounts of substances W , X , Y , and Z , respectively. A model for the radioactive series is

$$\frac{dw}{dt} = -\lambda_1 w$$

$$\frac{dx}{dt} = \lambda_1 w - \lambda_2 x$$

$$\frac{dy}{dt} = \lambda_2 x - \lambda_3 y$$

$$\frac{dz}{dt} = \lambda_3 y.$$

5. The system is

$$\begin{aligned}x_1' &= 2 \cdot 3 + \frac{1}{50}x_2 - \frac{1}{50}x_1 \cdot 4 = -\frac{2}{25}x_1 + \frac{1}{50}x_2 + 6 \\x_2' &= \frac{1}{50}x_1 \cdot 4 - \frac{1}{50}x_2 - \frac{1}{50}x_2 \cdot 3 = \frac{2}{25}x_1 - \frac{2}{25}x_2.\end{aligned}$$

6. Let x_1 , x_2 , and x_3 be the amounts of salt in tanks A, B, and C, respectively, so that

$$\begin{aligned}x_1' &= \frac{1}{100}x_2 \cdot 2 - \frac{1}{100}x_1 \cdot 6 = \frac{1}{50}x_2 - \frac{3}{50}x_1 \\x_2' &= \frac{1}{100}x_1 \cdot 6 + \frac{1}{100}x_3 - \frac{1}{100}x_2 \cdot 2 - \frac{1}{100}x_2 \cdot 5 = \frac{3}{50}x_1 - \frac{7}{100}x_2 + \frac{1}{100}x_3 \\x_3' &= \frac{1}{100}x_2 \cdot 5 - \frac{1}{100}x_3 - \frac{1}{100}x_3 \cdot 4 = \frac{1}{20}x_2 - \frac{1}{20}x_3.\end{aligned}$$

7. (a) A model is

$$\begin{aligned}\frac{dx_1}{dt} &= 3 \cdot \frac{x_2}{100-t} - 2 \cdot \frac{x_1}{100+t}, & x_1(0) &= 100 \\ \frac{dx_2}{dt} &= 2 \cdot \frac{x_1}{100+t} - 3 \cdot \frac{x_2}{100-t}, & x_2(0) &= 50.\end{aligned}$$

(b) Since the system is closed, no salt enters or leaves the system and $x_1(t) + x_2(t) = 100 + 50 = 150$ for all time. Thus $x_1 = 150 - x_2$ and the second equation in part (a) becomes

$$\frac{dx_2}{dt} = \frac{2(150 - x_2)}{100 + t} - \frac{3x_2}{100 - t} = \frac{300}{100 + t} - \frac{2x_2}{100 + t} - \frac{3x_2}{100 - t}$$

or

$$\frac{dx_2}{dt} + \left(\frac{2}{100 + t} + \frac{3}{100 - t} \right) x_2 = \frac{300}{100 + t},$$

which is linear in x_2 . An integrating factor is

$$e^{2 \ln(100+t) - 3 \ln(100-t)} = (100 + t)^2 (100 - t)^{-3}$$

so

$$\frac{d}{dt}[(100 + t)^2 (100 - t)^{-3} x_2] = 300(100 + t)(100 - t)^{-3}.$$

Using integration by parts, we obtain

$$(100 + t)^2 (100 - t)^{-3} x_2 = 300 \left[\frac{1}{2} (100 + t)(100 - t)^{-2} - \frac{1}{2} (100 - t)^{-1} + c \right].$$

Thus

$$\begin{aligned}x_2 &= \frac{300}{(100 + t)^2} \left[c(100 - t)^3 - \frac{1}{2}(100 - t)^2 + \frac{1}{2}(100 + t)(100 - t) \right] \\ &= \frac{300}{(100 + t)^2} [c(100 - t)^3 + t(100 - t)].\end{aligned}$$

Using $x_2(0) = 50$ we find $c = 5/3000$. At $t = 30$, $x_2 = (300/130^2)(70^3 c + 30 \cdot 70) \approx 47.4$ lbs.

8. A model is

$$\begin{aligned}\frac{dx_1}{dt} &= (4 \text{ gal/min})(0 \text{ lb/gal}) - (4 \text{ gal/min}) \left(\frac{1}{200} x_1 \text{ lb/gal} \right) \\ \frac{dx_2}{dt} &= (4 \text{ gal/min}) \left(\frac{1}{200} x_1 \text{ lb/gal} \right) - (4 \text{ gal/min}) \left(\frac{1}{150} x_2 \text{ lb/gal} \right) \\ \frac{dx_3}{dt} &= (4 \text{ gal/min}) \left(\frac{1}{150} x_2 \text{ lb/gal} \right) - (4 \text{ gal/min}) \left(\frac{1}{100} x_3 \text{ lb/gal} \right)\end{aligned}$$

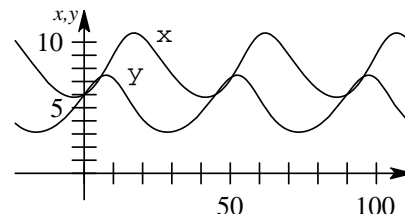
Exercises 2.9

or

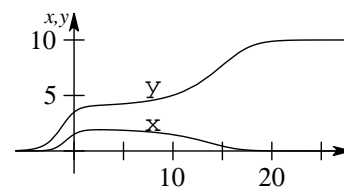
$$\begin{aligned}\frac{dx_1}{dt} &= -\frac{1}{50}x_1 \\ \frac{dx_2}{dt} &= \frac{1}{50}x_1 - \frac{2}{75}x_2 \\ \frac{dx_3}{dt} &= \frac{2}{75}x_2 - \frac{1}{25}x_3.\end{aligned}$$

Over a long period of time we would expect x_1 , x_2 , and x_3 to approach 0 because the entering pure water should flush the salt out of all three tanks.

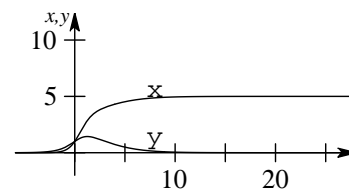
9. From the graph we see that the populations are first equal at about $t = 5.6$. The approximate periods of x and y are both 45.



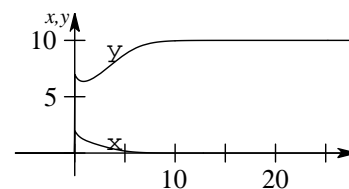
10. (a) The population $y(t)$ approaches 10,000, while the population $x(t)$ approaches extinction.



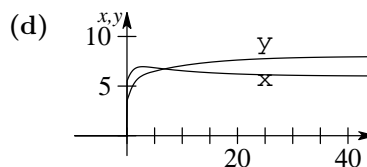
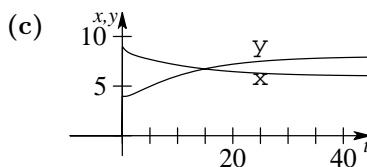
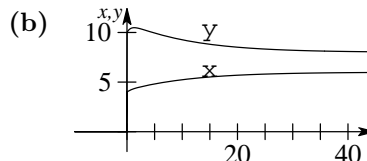
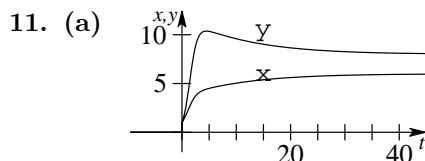
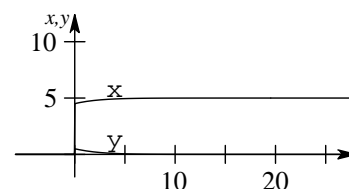
- (b) The population $x(t)$ approaches 5,000, while the population $y(t)$ approaches extinction.



- (c) The population $y(t)$ approaches 10,000, while the population $x(t)$ approaches extinction.



- (d) The population $x(t)$ approaches 5,000, while the population $y(t)$ approaches extinction.



In each case the population $x(t)$ approaches 6,000, while the population $y(t)$ approaches 8,000.

12. By Kirchoff's first law we have $i_1 = i_2 + i_3$. By Kirchoff's second law, on each loop we have $E(t) = Li'_1 + R_1i_2$ and $E(t) = Li'_1 + R_2i_3 + \frac{1}{C}q$ so that $q = CR_1i_2 - CR_2i_3$. Then $i_3 = q' = CR_1i'_2 - CR_2i'_3$ so that the system is

$$\begin{aligned} Li'_2 + Li'_3 + R_1i_2 &= E(t) \\ -R_1i'_2 + R_2i'_3 + \frac{1}{C}i_3 &= 0. \end{aligned}$$

13. By Kirchoff's first law we have $i_1 = i_2 + i_3$. Applying Kirchoff's second law to each loop we obtain

$$E(t) = i_1R_1 + L_1\frac{di_2}{dt} + i_2R_2$$

and

$$E(t) = i_1R_1 + L_2\frac{di_3}{dt} + i_3R_3.$$

Combining the three equations, we obtain the system

$$\begin{aligned} L_1\frac{di_2}{dt} + (R_1 + R_2)i_2 + R_1i_3 &= E \\ L_2\frac{di_3}{dt} + R_1i_2 + (R_1 + R_3)i_3 &= E. \end{aligned}$$

14. By Kirchoff's first law we have $i_1 = i_2 + i_3$. By Kirchoff's second law, on each loop we have $E(t) = Li'_1 + Ri_2$ and $E(t) = Li'_1 + \frac{1}{C}q$ so that $q = CRi_2$. Then $i_3 = q' = CRi'_2$ so that system is

$$\begin{aligned} Li' + Ri_2 &= E(t) \\ CRi'_2 + i_2 - i_1 &= 0. \end{aligned}$$

15. We first note that $s(t) + i(t) + r(t) = n$. Now the rate of change of the number of susceptible persons, $s(t)$, is proportional to the number of contacts between the number of people infected and the number who are susceptible; that is, $ds/dt = -k_1si$. We use $-k_1$ because $s(t)$ is decreasing. Next, the rate of change of the number of persons who have recovered is proportional to the number infected; that is, $dr/dt = k_2i$ where k_2 is positive since r is increasing. Finally, to obtain di/dt we use

$$\frac{d}{dt}(s + i + r) = \frac{d}{dt}n = 0.$$

This gives

$$\frac{di}{dt} = -\frac{dr}{dt} - \frac{ds}{dt} = -k_2i + k_1si.$$

The system of equations is then

$$\begin{aligned} \frac{ds}{dt} &= -k_1si \\ \frac{di}{dt} &= -k_2i + k_1si \\ \frac{dr}{dt} &= k_2i. \end{aligned}$$

A reasonable set of initial conditions is $i(0) = i_0$, the number of infected people at time 0, $s(0) = n - i_0$, and $r(0) = 0$.

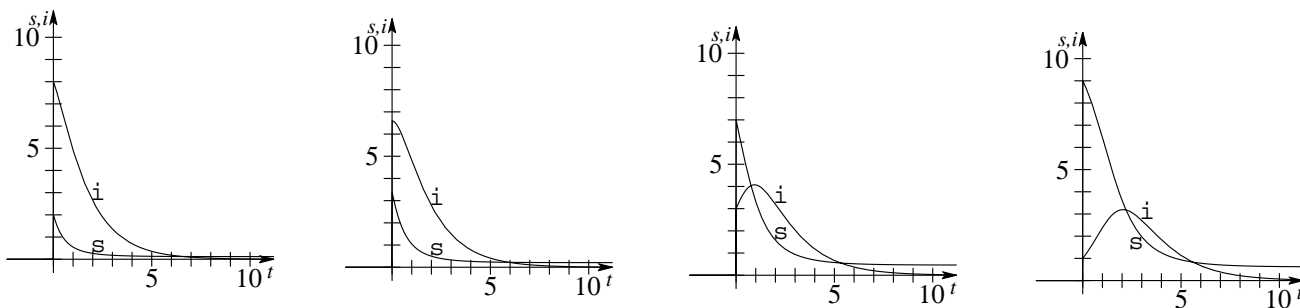
16. (a) If we know $s(t)$ and $i(t)$ then we can determine $r(t)$ from $s + i + r = n$.

Exercises 2.9

(b) In this case the system is

$$\begin{aligned}\frac{ds}{dt} &= -0.2si \\ \frac{di}{dt} &= -0.7i + 0.2si.\end{aligned}$$

We also note that when $i(0) = i_0$, $s(0) = 10 - i_0$ since $r(0) = 0$ and $i(t) + s(t) + r(t) = 10$ for all values of t . Now $k_2/k_1 = 0.7/0.2 = 3.5$, so we consider initial conditions $s(0) = 2, i(0) = 8$; $s(0) = 3.4, i(0) = 6.6$; $s(0) = 7, i(0) = 3$; and $s(0) = 9, i(0) = 1$.



We see that an initial susceptible population greater than k_2/k_1 results in an epidemic in the sense that the number of infected persons increases to a maximum before decreasing to 0. On the other hand, when $s(0) < k_2/k_1$, the number of infected persons decreases from the start and there is no epidemic.

Chapter 2 Review Exercises

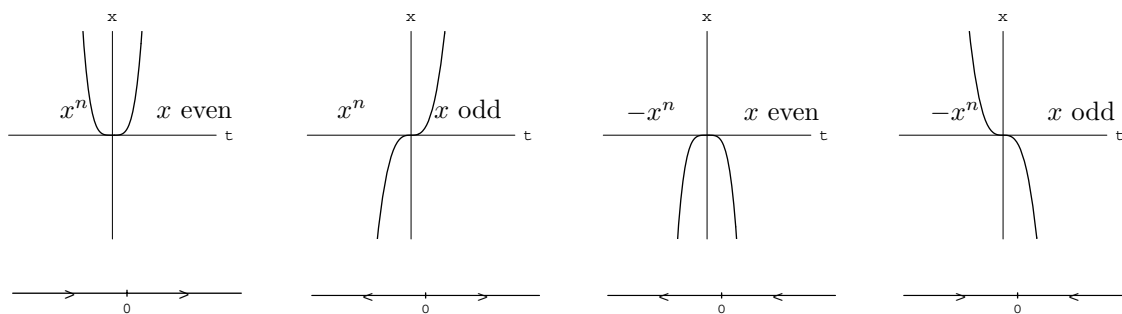
- Writing the differential equation in the form $y' = k(y + A/k)$ we see that the critical point $-A/k$ is a repeller for $k > 0$ and an attractor for $k < 0$.
- Separating variables and integrating we have

$$\begin{aligned}\frac{dy}{y} &= \frac{4}{x} dx \\ \ln y &= 4 \ln x + c = \ln x^4 + c \\ y &= c_1 x^4.\end{aligned}$$

We see that when $x = 0, y = 0$, so the initial-value problem has an infinite number of solutions for $k = 0$ and no solutions for $k \neq 0$.

- $\frac{dy}{dx} = (y - 1)^2(y - 3)^2$
- $\frac{dy}{dx} = y(y - 2)^2(y - 4)$

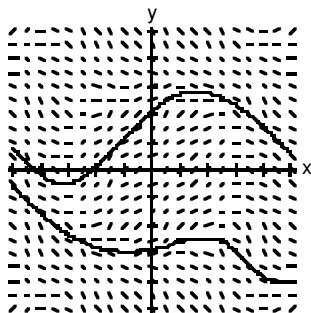
5.



For $dx/dt = x^n$, when n is even, 0 is semi-stable; when n is odd, 0 is unstable. For $dx/dt = -x^n$, when n is even, 0 is semi-stable; when n is odd, 0 is stable.

6. The zero of f occurs at approximately 1.3. Since $P'(t) = f(P) > 0$ for $P < 1.3$ and $P'(t) = f(P) < 0$ for $P > 1.3$, $\lim_{t \rightarrow \infty} P(t) = 1.3$.

7.



8. (a) linear in y , homogeneous, exact
 (b) linear in x
 (c) separable, exact, linear in x and y
 (d) Bernoulli in x
 (e) separable
 (f) separable, linear in x , Bernoulli
 (g) linear in x
 (h) homogeneous
 (i) Bernoulli
 (j) homogeneous, exact, Bernoulli
 (k) linear in x and y , exact, separable, homogeneous
 (l) exact, linear in y
 (m) homogeneous
 (n) separable

9. Separating variables we obtain

$$\cos^2 x \, dx = \frac{y}{y^2 + 1} \, dy \implies \frac{1}{2}x + \frac{1}{4}\sin 2x = \frac{1}{2}\ln(y^2 + 1) + c \implies 2x + \sin 2x = 2\ln(y^2 + 1) + c.$$

Chapter 2 Review Exercises

10. Write the differential equation in the form $y \ln \frac{x}{y} dx = \left(x \ln \frac{x}{y} - y \right) dy$. This is a homogeneous equation, so let $x = uy$. Then $dx = u dy + y du$ and the differential equation becomes

$$y \ln u (u dy + y du) = (uy \ln u - y) dy \quad \text{or} \quad y \ln u du = -dy.$$

Separating variables we obtain

$$\begin{aligned} \ln u du = -\frac{dy}{y} &\implies u \ln |u| - u = -\ln |y| + c \implies \frac{x}{y} \ln \left| \frac{x}{y} \right| - \frac{x}{y} = -\ln |y| + c \\ &\implies x(\ln x - \ln y) - x = -y \ln |y| + cy. \end{aligned}$$

11. The differential equation $\frac{dy}{dx} + \frac{2}{6x+1}y = -\frac{3x^2}{6x+1}y^{-2}$ is Bernoulli. Using $w = y^3$ we obtain $\frac{dw}{dx} + \frac{6}{6x+1}w = -\frac{9x^2}{6x+1}$. An integrating factor is $6x+1$, so

$$\frac{d}{dx}[(6x+1)w] = -9x^2 \implies w = -\frac{3x^3}{6x+1} + \frac{c}{6x+1} \implies (6x+1)y^3 = -3x^3 + c.$$

(Note: The differential equation is also exact.)

12. Write the differential equation in the form $(3y^2 + 2x)dx + (4y^2 + 6xy)dy = 0$. Letting $M = 3y^2 + 2x$ and $N = 4y^2 + 6xy$ we see that $M_y = 6y = N_x$ so the differential equation is exact. From $f_x = 3y^2 + 2x$ we obtain $f = 3xy^2 + x^2 + h(y)$. Then $f_y = 6xy + h'(y) = 4y^2 + 6xy$ and $h'(y) = 4y^2$ so $h(y) = \frac{4}{3}y^3$. The general solution is

$$3xy^2 + x^2 + \frac{4}{3}y^3 = c.$$

13. Write the equation in the form

$$\frac{dQ}{dt} + \frac{1}{t}Q = t^3 \ln t.$$

An integrating factor is $e^{\int \frac{1}{t} dt} = t$, so

$$\begin{aligned} \frac{d}{dt}[tQ] &= t^4 \ln t \implies tQ = -\frac{1}{25}t^5 + \frac{1}{5}t^5 \ln t + c \\ &\implies Q = -\frac{1}{25}t^4 + \frac{1}{5}t^4 \ln t + \frac{c}{t}. \end{aligned}$$

14. Letting $u = 2x + y + 1$ we have

$$\frac{du}{dx} = 2 + \frac{dy}{dx},$$

and so the given differential equation is transformed into

$$u \left(\frac{du}{dx} - 2 \right) = 1 \quad \text{or} \quad \frac{du}{dx} = \frac{2u+1}{u}.$$

Separating variables and integrating we get

$$\begin{aligned} \frac{u}{2u+1} du &= dx \\ \left(\frac{1}{2} - \frac{1}{2} \frac{1}{2u+1} \right) du &= dx \\ \frac{1}{2}u - \frac{1}{4} \ln |2u+1| &= x + c \\ 2u - \ln |2u+1| &= 2x + c_1. \end{aligned}$$

Resubstituting for u gives the solution

$$4x + 2y + 2 - \ln|4x + 2y + 3| = 2x + c_1$$

or

$$2x + 2y + 2 - \ln|4x + 2y + 3| = c_1.$$

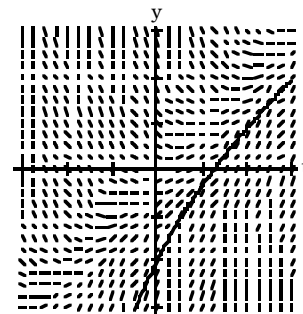
15. Write the equation in the form $\frac{dy}{dx} + \frac{8x}{x^2 + 4}y = \frac{2x}{x^2 + 4}$. An integrating factor is $(x^2 + 4)^4$, so

$$\frac{d}{dx} \left[(x^2 + 4)^4 y \right] = 2x (x^2 + 4)^3 \implies (x^2 + 4)^4 y = \frac{1}{4} (x^2 + 4)^4 + c \implies y = \frac{1}{4} + c (x^2 + 4)^{-4}.$$

16. Letting $M = 2r^2 \cos \theta \sin \theta + r \cos \theta$ and $N = 4r + \sin \theta - 2r \cos^2 \theta$ we see that $M_r = 4r \cos \theta \sin \theta + \cos \theta = N_\theta$ so the differential equation is exact. From $f_\theta = 2r^2 \cos \theta \sin \theta + r \cos \theta$ we obtain $f = -r^2 \cos^2 \theta + r \sin \theta + h(r)$. Then $f_r = -2r \cos^2 \theta + \sin \theta + h'(r) = 4r + \sin \theta - 2r \cos^2 \theta$ and $h'(r) = 4r$ so $h(r) = 2r^2$. The general solution is

$$-r^2 \cos^2 \theta + r \sin \theta + 2r^2 = c.$$

17. The differential equation has the form $\frac{d}{dx}[(\sin x)y] = 0$. Integrating we have $(\sin x)y = c$ or $y = c/\sin x$. The initial condition implies $c = -2 \sin(7\pi/6) = 1$. Thus, $y = 1/\sin x$, where $\pi < x < 2\pi$ is chosen to include $x = 7\pi/6$.
18. For $y > t$ the slopes are negative and for $y < t$ the slopes are positive, so the differential equation is (a).



19. (a) For $y < 0$, $\sqrt{y}$ is not a real number.
- (b) Separating variables and integrating we have

$$\frac{dy}{\sqrt{y}} = dx \quad \text{and} \quad 2\sqrt{y} = x + c.$$

Letting $y(x_0) = y_0$ we get $c = 2\sqrt{y_0} - x_0$, so that

$$2\sqrt{y} = x + 2\sqrt{y_0} - x_0 \quad \text{and} \quad y = \frac{1}{4}(x + 2\sqrt{y_0} - x_0)^2.$$

Since $\sqrt{y} > 0$ for $y \neq 0$, we see that $dy/dx = \frac{1}{2}(x + 2\sqrt{y_0} - x_0)$ must be positive. Thus, the interval on which the solution is defined is $(x_0 - 2\sqrt{y_0}, \infty)$.

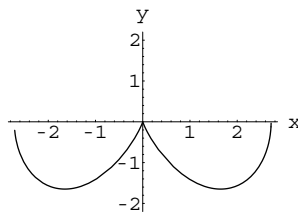
Chapter 2 Review Exercises

20. (a) The differential equation is homogeneous and we let $y = ux$. Then

$$\begin{aligned}(x^2 - y^2) dx + xy dy &= 0 \\(x^2 - u^2 x^2) dx + ux^2(u dx + x du) &= 0 \\dx + ux du &= 0 \\u du &= -\frac{dx}{x} \\\frac{1}{2}u^2 &= -\ln|x| + c \\\frac{y^2}{x^2} &= -2\ln|x| + c_1.\end{aligned}$$

The initial condition gives $c_1 = 2$, so an implicit solution is $y^2 = x^2(2 - 2\ln|x|)$.

- (b) Solving for y in part (a) and being sure that the initial condition is still satisfied, we have $y = -\sqrt{2}|x|(1 - \ln|x|)^{1/2}$, where $-e \leq x \leq e$ so that $1 - \ln|x| \geq 0$. A graph of this function shown below indicates that the derivative is not defined at $x = 0$ and $x = e$. Thus, the solution of the initial-value problem is $y = -\sqrt{2}x(1 - \ln x)^{1/2}$, for $0 < x < e$.



21. The graph of $y_1(x)$ is the portion of the closed black curve lying in the fourth quadrant. Its interval of definition is approximately $(0.7, 4.3)$. The graph of $y_2(x)$ is the portion of the left-hand black curve lying in the third quadrant. Its interval of definition is $(-\infty, 0)$.
22. The first step of Euler's method gives $y(1.1) \approx 9 + 0.1(1 + 3) = 9.4$. Applying Euler's method one more time gives $y(1.2) \approx 9.4 + 0.1(1 + 1.1\sqrt{9.4}) \approx 9.8373$.
23. From $\frac{dP}{dt} = 0.018P$ and $P(0) = 4$ billion we obtain $P = 4e^{0.018t}$ so that $P(45) = 8.99$ billion.
24. Let $A = A(t)$ be the volume of CO_2 at time t . From $\frac{dA}{dt} = 1.2 - \frac{A}{4}$ and $A(0) = 16 \text{ ft}^3$ we obtain $A = 4.8 + 11.2e^{-t/4}$. Since $A(10) = 5.7 \text{ ft}^3$, the concentration is 0.017%. As $t \rightarrow \infty$ we have $A \rightarrow 4.8 \text{ ft}^3$ or 0.06%.
25. From $dE/dt = -E/RC$ and $E(t_1) = E_0$ we obtain $E = E_0 e^{(t_1 - t)/RC}$.
26. From $V dC/dt = kA(C_s - C)$ and $C(0) = C_0$ we obtain $C = C_s + (C_0 - C_s)e^{-kAt/V}$.
27. (a) The differential equation is

$$\frac{dT}{dt} = k[T - T_2 - B(T_1 - T)] = k[(1 + B)T - (BT_1 + T_2)].$$

Separating variables we obtain $\frac{dT}{(1 + B)T - (BT_1 + T_2)} = k dt$. Then

$$\frac{1}{1 + B} \ln|(1 + B)T - (BT_1 + T_2)| = kt + c \quad \text{and} \quad T(t) = \frac{BT_1 + T_2}{1 + B} + c_3 e^{k(1+B)t}.$$

Since $T(0) = T_1$ we must have $c_3 = \frac{T_1 - T_2}{1 + B}$ and so

$$T(t) = \frac{BT_1 + T_2}{1 + B} + \frac{T_1 - T_2}{1 + B} e^{k(1+B)t}.$$

(b) Since $k < 0$, $\lim_{t \rightarrow \infty} e^{k(1+B)t} = 0$ and $\lim_{t \rightarrow \infty} T(t) = \frac{BT_1 + T_2}{1 + B}$.

(c) Since $T_s = T_2 + B(T_1 - T)$, $\lim_{t \rightarrow \infty} T_s = T_2 + BT_1 - B \left(\frac{BT_1 + T_2}{1 + B} \right) = \frac{BT_1 + T_2}{1 + B}$.

28. We first solve $\left(1 - \frac{t}{10}\right) \frac{di}{dt} + 0.2i = 4$. Separating variables we obtain $\frac{di}{40 - 2i} = \frac{dt}{10 - t}$.

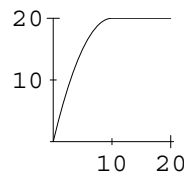
Then

$$-\frac{1}{2} \ln |40 - 2i| = -\ln |10 - t| + c \quad \text{or} \quad \sqrt{40 - 2i} = c_1(10 - t).$$

Since $i(0) = 0$ we must have $c_1 = 2/\sqrt{10}$. Solving for i we get $i(t) = 4t - \frac{1}{5}t^2$,

$0 \leq t < 10$. For $t \geq 10$ the equation for the current becomes $0.2i = 4$ or $i = 20$. Thus

$$i(t) = \begin{cases} 4t - \frac{1}{5}t^2, & 0 \leq t < 10 \\ 20, & t \geq 10 \end{cases}.$$



29. From $y[1 + (y')^2] = k$ we obtain $dx = \frac{\sqrt{y}}{\sqrt{k-y}} dy$. If $y = k \sin^2 \theta$ then

$$dy = 2k \sin \theta \cos \theta d\theta, \quad dx = 2k \left(\frac{1}{2} - \frac{1}{2} \cos 2\theta \right) d\theta, \quad \text{and} \quad x = k\theta - \frac{k}{2} \sin 2\theta + c.$$

If $x = 0$ when $\theta = 0$ then $c = 0$.

30. (a) From $y = -x - 1 + c_1 e^x$ we obtain $y' = y + x$ so that the differential equation of the orthogonal family is $\frac{dy}{dx} = -\frac{1}{y+x}$ or $\frac{dx}{dy} + x = -y$. An integrating factor is e^y , so

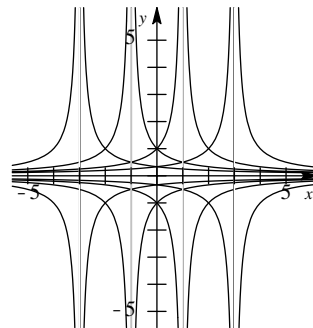
$$\frac{d}{dy}[e^y x] = -ye^y \implies e^y x = -ye^y + e^y + c \implies x = -y + 1 + ce^{-y}.$$

(b) Differentiating the family of curves, we have

$$y' = -\frac{1}{(x + c_1)^2} = -\frac{1}{y^2}.$$

The differential equation for the family of orthogonal trajectories is then $y' = y^2$. Separating variables and integrating we get

$$\begin{aligned} \frac{dy}{y^2} &= dx \\ -\frac{1}{y} &= x + c_1 \\ y &= -\frac{1}{x + c_1}. \end{aligned}$$



31. From $\frac{dx}{dt} = k_1 x(\alpha - x)$ we obtain $\left(\frac{1/\alpha}{x} + \frac{1/\alpha}{\alpha - x} \right) dx = k_1 dt$ so that $x = \frac{\alpha c_1 e^{\alpha k_1 t}}{1 + c_1 e^{\alpha k_1 t}}$. From $\frac{dy}{dt} = k_2 xy$ we obtain

$$\ln |y| = \frac{k_2}{k_1} \ln |1 + c_1 e^{\alpha k_1 t}| + c \quad \text{or} \quad y = c_2 (1 + c_1 e^{\alpha k_1 t})^{k_2/k_1}.$$

Chapter 2 Review Exercises

32. In tank A the salt input is

$$\left(7 \frac{\text{gal}}{\text{min}}\right) \left(2 \frac{\text{lb}}{\text{gal}}\right) + \left(1 \frac{\text{gal}}{\text{min}}\right) \left(\frac{x_2}{100} \frac{\text{lb}}{\text{gal}}\right) = \left(14 + \frac{1}{100}x_2\right) \frac{\text{lb}}{\text{min}}.$$

The salt output is

$$\left(3 \frac{\text{gal}}{\text{min}}\right) \left(\frac{x_1}{100} \frac{\text{lb}}{\text{gal}}\right) + \left(5 \frac{\text{gal}}{\text{min}}\right) \left(\frac{x_1}{100} \frac{\text{lb}}{\text{gal}}\right) = \frac{2}{25}x_1 \frac{\text{lb}}{\text{min}}.$$

In tank B the salt input is

$$\left(5 \frac{\text{gal}}{\text{min}}\right) \left(\frac{x_1}{100} \frac{\text{lb}}{\text{gal}}\right) = \frac{1}{20}x_1 \frac{\text{lb}}{\text{min}}.$$

The salt output is

$$\left(1 \frac{\text{gal}}{\text{min}}\right) \left(\frac{x_2}{100} \frac{\text{lb}}{\text{gal}}\right) + \left(4 \frac{\text{gal}}{\text{min}}\right) \left(\frac{x_2}{100} \frac{\text{lb}}{\text{gal}}\right) = \frac{1}{20}x_2 \frac{\text{lb}}{\text{min}}.$$

The system of differential equations is then

$$\begin{aligned}\frac{dx_1}{dt} &= 14 + \frac{1}{100}x_2 - \frac{2}{25}x_1 \\ \frac{dx_2}{dt} &= \frac{1}{20}x_1 - \frac{1}{20}x_2.\end{aligned}$$

3 Higher-Order Differential Equations

Exercises 3.1

1. From $y = c_1 e^x + c_2 e^{-x}$ we find $y' = c_1 e^x - c_2 e^{-x}$. Then $y(0) = c_1 + c_2 = 0$, $y'(0) = c_1 - c_2 = 1$ so that $c_1 = 1/2$ and $c_2 = -1/2$. The solution is $y = \frac{1}{2}e^x - \frac{1}{2}e^{-x}$.
2. From $y = c_1 e^{4x} + c_2 e^{-x}$ we find $y' = 4c_1 e^{4x} - c_2 e^{-x}$. Then $y(0) = c_1 + c_2 = 1$, $y'(0) = 4c_1 - c_2 = 2$ so that $c_1 = 3/5$ and $c_2 = 2/5$. The solution is $y = \frac{3}{5}e^{4x} + \frac{2}{5}e^{-x}$.
3. From $y = c_1 x + c_2 x \ln x$ we find $y' = c_1 + c_2(1 + \ln x)$. Then $y(1) = c_1 = 3$, $y'(1) = c_1 + c_2 = -1$ so that $c_1 = 3$ and $c_2 = -4$. The solution is $y = 3x - 4x \ln x$.
4. From $y = c_1 + c_2 \cos x + c_3 \sin x$ we find $y' = -c_2 \sin x + c_3 \cos x$ and $y'' = -c_2 \cos x - c_3 \sin x$. Then $y(\pi) = c_1 - c_2 = 0$, $y'(\pi) = -c_3 = 2$, $y''(\pi) = c_2 = -1$ so that $c_1 = -1$, $c_2 = -1$, and $c_3 = -2$. The solution is $y = -1 - \cos x - 2 \sin x$.
5. From $y = c_1 + c_2 x^2$ we find $y' = 2c_2 x$. Then $y(0) = c_1 = 0$, $y'(0) = 2c_2 \cdot 0 = 0$ and $y'(0) = 1$ is not possible. Since $a_2(x) = x$ is 0 at $x = 0$, Theorem 3.1 is not violated.
6. In this case we have $y(0) = c_1 = 0$, $y'(0) = 2c_2 \cdot 0 = 0$ so $c_1 = 0$ and c_2 is arbitrary. Two solutions are $y = x^2$ and $y = 2x^2$.
7. From $x(0) = x_0 = c_1$ we see that $x(t) = x_0 \cos \omega t + c_2 \sin \omega t$ and $x'(t) = -x_0 \sin \omega t + c_2 \omega \cos \omega t$. Then $x'(0) = x_1 = c_2 \omega$ implies $c_2 = x_1/\omega$. Thus

$$x(t) = x_0 \cos \omega t + \frac{x_1}{\omega} \sin \omega t.$$

8. Solving the system

$$x(t_0) = c_1 \cos \omega t_0 + c_2 \sin \omega t_0 = x$$

$$x'(t_0) = -c_1 \omega \sin \omega t_0 + c_2 \omega \cos \omega t_0 = x_1$$

for c_1 and c_2 gives

$$c_1 = \frac{\omega x_0 \cos \omega t_0 - x_1 \sin \omega t_0}{\omega} \quad \text{and} \quad c_2 = \frac{x_1 \cos \omega t_0 + \omega x_0 \sin \omega t_0}{\omega}.$$

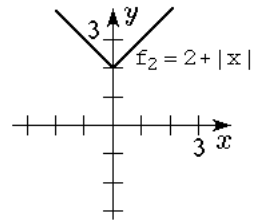
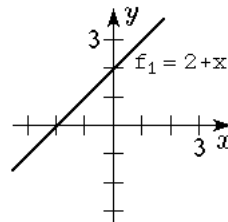
Thus

$$\begin{aligned} x(t) &= \frac{\omega x_0 \cos \omega t_0 - x_1 \sin \omega t_0}{\omega} \cos \omega t + \frac{x_1 \cos \omega t_0 + \omega x_0 \sin \omega t_0}{\omega} \sin \omega t \\ &= x_0 (\cos \omega t \cos \omega t_0 + \sin \omega t \sin \omega t_0) + \frac{x_1}{\omega} (\sin \omega t \cos \omega t_0 - \cos \omega t \sin \omega t_0) \\ &= x_0 \cos \omega(t - t_0) + \frac{x_1}{\omega} \sin \omega(t - t_0). \end{aligned}$$

9. Since $a_2(x) = x - 2$ and $x_0 = 0$ the problem has a unique solution for $-\infty < x < 2$.
10. Since $a_0(x) = \tan x$ and $x_0 = 0$ the problem has a unique solution for $-\pi/2 < x < \pi/2$.
11. We have $y(0) = c_1 + c_2 = 0$, $y'(1) = c_1 e + c_2 e^{-1} = 1$ so that $c_1 = e/(e^2 - 1)$ and $c_2 = -e/(e^2 - 1)$. The solution is $y = e(e^x - e^{-x})/(e^2 - 1)$.
12. In this case we have $y(0) = c_1 = 1$, $y'(1) = 2c_2 = 6$ so that $c_1 = 1$ and $c_2 = 3$. The solution is $y = 1 + 3x^2$.

Exercises 3.1

13. From $y = c_1 e^x \cos x + c_2 e^x \sin x$ we find $y' = c_1 e^x (-\sin x + \cos x) + c_2 e^x (\cos x + \sin x)$.
- (a) We have $y(0) = c_1 = 1$, $y'(0) = c_1 + c_2 = 0$ so that $c_1 = 1$ and $c_2 = -1$. The solution is $y = e^x \cos x - e^x \sin x$.
- (b) We have $y(0) = c_1 = 1$, $y(\pi) = -c_1 e^\pi = -1$, which is not possible.
- (c) We have $y(0) = c_1 = 1$, $y(\pi/2) = c_2 e^{\pi/2} = 1$ so that $c_1 = 1$ and $c_2 = e^{-\pi/2}$. The solution is $y = e^x \cos x + e^{-\pi/2} e^x \sin x$.
- (d) We have $y(0) = c_1 = 0$, $y(\pi) = -c_1 e^\pi = 0$ so that $c_1 = 0$ and c_2 is arbitrary. Solutions are $y = c_2 e^x \sin x$, for any real numbers c_2 .
14. (a) We have $y(-1) = c_1 + c_2 + 3 = 0$, $y(1) = c_1 + c_2 + 3 = 4$, which is not possible.
- (b) We have $y(0) = c_1 \cdot 0 + c_2 \cdot 0 + 3 = 1$, which is not possible.
- (c) We have $y(0) = c_1 \cdot 0 + c_2 \cdot 0 + 3 = 3$, $y(1) = c_1 + c_2 + 3 = 0$ so that c_1 is arbitrary and $c_2 = -3 - c_1$. Solutions are $y = c_1 x^2 - (c_1 + 3)x^4 + 3$.
- (d) We have $y(1) = c_1 + c_2 + 3 = 3$, $y(2) = 4c_1 + 16c_2 + 3 = 15$ so that $c_1 = -1$ and $c_2 = 1$. The solution is $y = -x^2 + x^4 + 3$.
15. Since $(-4)x + (3)x^2 + (1)(4x - 3x^2) = 0$ the functions are linearly dependent.
16. Since $(1)0 + (0)x + (0)e^x = 0$ the functions are linearly dependent. A similar argument shows that any set of functions containing $f(x) = 0$ will be linearly dependent.
17. Since $(-1/5)5 + (1)\cos^2 x + (1)\sin^2 x = 0$ the functions are linearly dependent.
18. Since $(1)\cos 2x + (1)1 + (-2)\cos^2 x = 0$ the functions are linearly dependent.
19. Since $(-4)x + (3)(x-1) + (1)(x+3) = 0$ the functions are linearly dependent.
20. From the graphs of $f_1(x) = 2 + x$ and $f_2(x) = 2 + |x|$ we see that the functions are linearly independent since they cannot be multiples of each other.



21. The functions are linearly independent since $W(1+x, x, x^2) = \begin{vmatrix} 1+x & x & x^2 \\ 1 & 1 & 2x \\ 0 & 0 & 2 \end{vmatrix} = 2 \neq 0$.
22. Since $(-1/2)e^x + (1/2)e^{-x} + (1)\sinh x = 0$ the functions are linearly dependent.
23. The functions satisfy the differential equation and are linearly independent since

$$W(e^{-3x}, e^{4x}) = 7e^x \neq 0$$

for $-\infty < x < \infty$. The general solution is

$$y = c_1 e^{-3x} + c_2 e^{4x}.$$

24. The functions satisfy the differential equation and are linearly independent since

$$W(\cosh 2x, \sinh 2x) = 2$$

for $-\infty < x < \infty$. The general solution is

$$y = c_1 \cosh 2x + c_2 \sinh 2x.$$

25. The functions satisfy the differential equation and are linearly independent since

$$W(e^x \cos 2x, e^x \sin 2x) = 2e^{2x} \neq 0$$

for $-\infty < x < \infty$. The general solution is $y = c_1 e^x \cos 2x + c_2 e^x \sin 2x$.

26. The functions satisfy the differential equation and are linearly independent since

$$W(e^{x/2}, xe^{x/2}) = e^x \neq 0$$

for $-\infty < x < \infty$. The general solution is

$$y = c_1 e^{x/2} + c_2 x e^{x/2}.$$

27. The functions satisfy the differential equation and are linearly independent since

$$W(x^3, x^4) = x^6 \neq 0$$

for $0 < x < \infty$. The general solution is

$$y = c_1 x^3 + c_2 x^4.$$

28. The functions satisfy the differential equation and are linearly independent since

$$W(\cos(\ln x), \sin(\ln x)) = 1/x \neq 0$$

for $0 < x < \infty$. The general solution is

$$y = c_1 \cos(\ln x) + c_2 \sin(\ln x).$$

29. The functions satisfy the differential equation and are linearly independent since

$$W(x, x^{-2}, x^{-2} \ln x) = 9x^{-6} \neq 0$$

for $0 < x < \infty$. The general solution is

$$y = c_1 x + c_2 x^{-2} + c_3 x^{-2} \ln x.$$

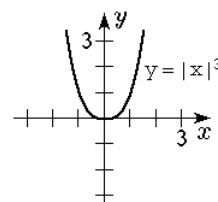
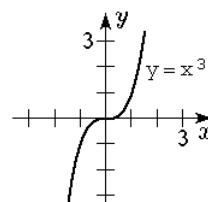
30. The functions satisfy the differential equation and are linearly independent since

$$W(1, x, \cos x, \sin x) = 1$$

for $-\infty < x < \infty$. The general solution is

$$y = c_1 + c_2 x + c_3 \cos x + c_4 \sin x.$$

31. (a) From the graphs of $y_1 = x^3$ and $y_2 = |x|^3$ we see that the functions are linearly independent since they cannot be multiples of each other. It is easily shown that $y_1 = x^3$ solves $x^2 y'' - 4xy' + 6y = 0$. To show that $y_2 = |x|^3$ is a solution let $y_2 = x^3$ for $x \geq 0$ and let $y_2 = -x^3$ for $x < 0$.



- (b) If $x \geq 0$ then $y_2 = x^3$ and $W(y_1, y_2) = \begin{vmatrix} x^3 & x^3 \\ 3x^2 & 3x^2 \end{vmatrix} = 0$. If $x < 0$ then $y_2 = -x^3$ and

$$W(y_1, y_2) = \begin{vmatrix} x^3 & -x^3 \\ 3x^2 & -3x^2 \end{vmatrix} = 0.$$

This does not violate Theorem 3.3 since $a_2(x) = x^2$ is zero at $x = 0$.

Exercises 3.1

- (c) The functions $Y_1 = x^3$ and $Y_2 = x^2$ are solutions of $x^2y'' - 4xy' + 6y = 0$. They are linearly independent since $W(x^3, x^2) = x^4 \neq 0$ for $-\infty < x < \infty$.
- (d) The function $y = x^3$ satisfies $y(0) = 0$ and $y'(0) = 0$.
- (e) Neither is the general solution since we form a general solution on an interval for which $a_2(x) \neq 0$ for every x in the interval.
32. By the superposition principle, if $y_1 = e^x$ and $y_2 = e^{-x}$ are both solutions of a homogeneous linear differential equation, then so are

$$\frac{1}{2}(y_1 + y_2) = \frac{e^x + e^{-x}}{2} = \cosh x \quad \text{and} \quad \frac{1}{2}(y_1 - y_2) = \frac{e^x - e^{-x}}{2} = \sinh x.$$

33. Since $e^{x-3} = e^{-3}e^x = (e^{-5}e^2)e^x = e^{-5}e^{x+2}$, we see that e^{x-3} is a constant multiple of e^{x+2} and the functions are linearly dependent.
34. Since $0y_1 + 0y_2 + \cdots + 0y_k + 1y_{k+1} = 0$, the set of solutions is linearly dependent.
35. The functions $y_1 = e^{2x}$ and $y_2 = e^{5x}$ form a fundamental set of solutions of the homogeneous equation, and $y_p = 6e^x$ is a particular solution of the nonhomogeneous equation.
36. The functions $y_1 = \cos x$ and $y_2 = \sin x$ form a fundamental set of solutions of the homogeneous equation, and $y_p = x \sin x + (\cos x) \ln(\cos x)$ is a particular solution of the nonhomogeneous equation.
37. The functions $y_1 = e^{2x}$ and $y_2 = xe^{2x}$ form a fundamental set of solutions of the homogeneous equation, and $y_p = x^2e^{2x} + x - 2$ is a particular solution of the nonhomogeneous equation.
38. The functions $y_1 = x^{-1/2}$ and $y_2 = x^{-1}$ form a fundamental set of solutions of the homogeneous equation, and $y_p = \frac{1}{15}x^2 - \frac{1}{6}x$ is a particular solution of the nonhomogeneous equation.
39. (a) We have $y'_{p_1} = 6e^{2x}$ and $y''_{p_1} = 12e^{2x}$, so

$$y''_{p_1} - 6y'_{p_1} + 5y_{p_1} = 12e^{2x} - 36e^{2x} + 15e^{2x} = -9e^{2x}.$$

Also, $y'_{p_2} = 2x + 3$ and $y''_{p_2} = 2$, so

$$y''_{p_2} - 6y'_{p_2} + 5y_{p_2} = 2 - 6(2x + 3) + 5(x^2 + 3x) = 5x^2 + 3x - 16.$$

- (b) By the superposition principle for nonhomogeneous equations a particular solution of $y'' - 6y' + 5y = 5x^2 + 3x - 16 - 9e^{2x}$ is $y_p = x^2 + 3x + 3e^{2x}$. A particular solution of the second equation is

$$y_p = -2y_{p_2} - \frac{1}{9}y_{p_1} = -2x^2 - 6x - \frac{1}{3}e^{2x}.$$

40. (a) $y_{p_1} = 5$
- (b) $y_{p_2} = -2x$
- (c) $y_p = y_{p_1} + y_{p_2} = 5 - 2x$
- (d) $y_p = \frac{1}{2}y_{p_1} - 2y_{p_2} = \frac{5}{2} + 4x$

Exercises 3.2

In Problems 1-8 we use reduction of order to find a second solution. In Problems 9-16 we use formula (5) from the text.

1. Define $y = u(x)e^{2x}$ so

$$y' = 2ue^{2x} + u'e^{2x}, \quad y'' = e^{2x}u'' + 4e^{2x}u' + 4e^{2x}u, \quad \text{and} \quad y'' - 4y' + 4y = 4e^{2x}u'' = 0.$$

Therefore $u'' = 0$ and $u = c_1x + c_2$. Taking $c_1 = 1$ and $c_2 = 0$ we see that a second solution is $y_2 = xe^{2x}$.

2. Define $y = u(x)xe^{-x}$ so

$$y' = (1-x)e^{-x}u + xe^{-x}u', \quad y'' = xe^{-x}u'' + 2(1-x)e^{-x}u' - (2-x)e^{-x}u,$$

and

$$y'' + 2y' + y = e^{-x}(xu'' + 2u') = 0 \quad \text{or} \quad u'' + \frac{2}{x}u' = 0.$$

If $w = u'$ we obtain the first-order equation $w' + \frac{2}{x}w = 0$ which has the integrating factor $e^{\int 2/x \, dx} = x^2$. Now

$$\frac{d}{dx}[x^2w] = 0 \quad \text{gives} \quad x^2w = c.$$

Therefore $w = u' = c/x^2$ and $u = c_1/x$. A second solution is $y_2 = \frac{1}{x}xe^{-x} = e^{-x}$.

3. Define $y = u(x)\cos 4x$ so

$$y' = -4u\sin 4x + u'\cos 4x, \quad y'' = u''\cos 4x - 8u'\sin 4x - 16u\cos 4x$$

and

$$y'' + 16y = (\cos 4x)u'' - 8(\sin 4x)u' = 0 \quad \text{or} \quad u'' - 8(\tan 4x)u' = 0.$$

If $w = u'$ we obtain the first-order equation $w' - 8(\tan 4x)w = 0$ which has the integrating factor $e^{-8\int \tan 4x \, dx} = \cos^2 4x$. Now

$$\frac{d}{dx}[(\cos^2 4x)w] = 0 \quad \text{gives} \quad (\cos^2 4x)w = c.$$

Therefore $w = u' = c \sec^2 4x$ and $u = c_1 \tan 4x$. A second solution is $y_2 = \tan 4x \cos 4x = \sin 4x$.

4. Define $y = u(x)\sin 3x$ so

$$y' = 3u\cos 3x + u'\sin 3x, \quad y'' = u''\sin 3x + 6u'\cos 3x - 9u\sin 3x,$$

and

$$y'' + 9y = (\sin 3x)u'' + 6(\cos 3x)u' = 0 \quad \text{or} \quad u'' + 6(\cot 3x)u' = 0.$$

If $w = u'$ we obtain the first-order equation $w' + 6(\cot 3x)w = 0$ which has the integrating factor $e^{\int 6 \cot 3x \, dx} = \sin^2 3x$. Now

$$\frac{d}{dx}[(\sin^2 3x)w] = 0 \quad \text{gives} \quad (\sin^2 3x)w = c.$$

Therefore $w = u' = c \csc^2 3x$ and $u = c_1 \cot 3x$. A second solution is $y_2 = \cot 3x \sin 3x = \cos 3x$.

5. Define $y = u(x)\cosh x$ so

$$y' = u\sinh x + u'\cosh x, \quad y'' = u''\cosh x + 2u'\sinh x + u\cosh x$$

and

$$y'' - y = (\cosh x)u'' + 2(\sinh x)u' = 0 \quad \text{or} \quad u'' + 2(\tanh x)u' = 0.$$

Exercises 3.2

If $w = u'$ we obtain the first-order equation $w' + 2(\tanh x)w = 0$ which has the integrating factor $e^{2 \int \tanh x \, dx} = \cosh^2 x$. Now

$$\frac{d}{dx}[(\cosh^2 x)w] = 0 \quad \text{gives} \quad (\cosh^2 x)w = c.$$

Therefore $w = u' = c \operatorname{sech}^2 x$ and $u = c_1 \tanh x$. A second solution is $y_2 = \tanh x \cosh x = \sinh x$.

6. Define $y = u(x)e^{5x}$ so

$$y' = 5e^{5x}u + e^{5x}u', \quad y'' = e^{5x}u'' + 10e^{5x}u' + 25e^{5x}u$$

and

$$y'' - 25y = e^{5x}(u'' + 10u') = 0 \quad \text{or} \quad u'' + 10u' = 0.$$

If $w = u'$ we obtain the first-order equation $w' + 10w = 0$ which has the integrating factor $e^{10 \int dx} = e^{10x}$. Now

$$\frac{d}{dx}[e^{10x}w] = 0 \quad \text{gives} \quad e^{10x}w = c.$$

Therefore $w = u' = ce^{-10x}$ and $u = c_1 e^{-10x}$. A second solution is $y_2 = e^{-10x}e^{5x} = e^{-5x}$.

7. Define $y = u(x)e^{2x/3}$ so

$$y' = \frac{2}{3}e^{2x/3}u + e^{2x/3}u', \quad y'' = e^{2x/3}u'' + \frac{4}{3}e^{2x/3}u' + \frac{4}{9}e^{2x/3}u$$

and

$$9y'' - 12y' + 4y = 9e^{2x/3}u'' = 0.$$

Therefore $u'' = 0$ and $u = c_1 x + c_2$. Taking $c_1 = 1$ and $c_2 = 0$ we see that a second solution is $y_2 = xe^{2x/3}$.

8. Define $y = u(x)e^{x/3}$ so

$$y' = \frac{1}{3}e^{x/3}u + e^{x/3}u', \quad y'' = e^{x/3}u'' + \frac{2}{3}e^{x/3}u' + \frac{1}{9}e^{x/3}u$$

and

$$6y'' + y' - y = e^{x/3}(6u'' + 5u') = 0 \quad \text{or} \quad u'' + \frac{5}{6}u' = 0.$$

If $w = u'$ we obtain the first-order equation $w' + \frac{5}{6}w = 0$ which has the integrating factor $e^{(5/6) \int dx} = e^{5x/6}$. Now

$$\frac{d}{dx}[e^{5x/6}w] = 0 \quad \text{gives} \quad e^{5x/6}w = c.$$

Therefore $w = u' = ce^{-5x/6}$ and $u = c_1 e^{-5x/6}$. A second solution is $y_2 = e^{-5x/6}e^{x/3} = e^{-x/2}$.

9. Identifying $P(x) = -7/x$ we have

$$y_2 = x^4 \int \frac{e^{-\int (7/x) \, dx}}{x^8} \, dx = x^4 \int \frac{1}{x} \, dx = x^4 \ln |x|.$$

A second solution is $y_2 = x^4 \ln |x|$.

10. Identifying $P(x) = 2/x$ we have

$$y_2 = x^2 \int \frac{e^{-\int (2/x) \, dx}}{x^4} \, dx = x^2 \int x^{-6} \, dx = -\frac{1}{5}x^{-3}.$$

A second solution is $y_2 = x^{-3}$.

11. Identifying $P(x) = 1/x$ we have

$$y_2 = \ln x \int \frac{e^{-\int dx/x}}{(\ln x)^2} \, dx = \ln x \int \frac{dx}{x(\ln x)^2} = \ln x \left(-\frac{1}{\ln x} \right) = -1.$$

A second solution is $y_2 = 1$.

12. Identifying $P(x) = 0$ we have

$$y_2 = x^{1/2} \ln x \int \frac{e^{-\int 0 dx}}{x(\ln x)^2} dx = x^{1/2} \ln x \left(-\frac{1}{\ln x} \right) = -x^{1/2}.$$

A second solution is $y_2 = x^{1/2}$.

13. Identifying $P(x) = -1/x$ we have

$$\begin{aligned} y_2 &= x \sin(\ln x) \int \frac{e^{-\int -dx/x}}{x^2 \sin^2(\ln x)} dx = x \sin(\ln x) \int \frac{x}{x^2 \sin^2(\ln x)} dx \\ &= [x \sin(\ln x)] [-\cot(\ln x)] = -x \cos(\ln x). \end{aligned}$$

A second solution is $y_2 = x \cos(\ln x)$.

14. Identifying $P(x) = -3/x$ we have

$$\begin{aligned} y_2 &= x^2 \cos(\ln x) \int \frac{e^{-\int -3 dx/x}}{x^4 \cos^2(\ln x)} dx = x^2 \cos(\ln x) \int \frac{x^3}{x^4 \cos^2(\ln x)} dx \\ &= x^2 \cos(\ln x) \tan(\ln x) = x^2 \sin(\ln x). \end{aligned}$$

A second solution is $y_2 = x^2 \sin(\ln x)$.

15. Identifying $P(x) = 2(1+x)/(1-2x-x^2)$ we have

$$\begin{aligned} y_2 &= (x+1) \int \frac{e^{-\int 2(1+x)dx/(1-2x-x^2)}}{(x+1)^2} dx = (x+1) \int \frac{e^{\ln(1-2x-x^2)}}{(x+1)^2} dx \\ &= (x+1) \int \frac{1-2x-x^2}{(x+1)^2} dx = (x+1) \int \left[\frac{2}{(x+1)^2} - 1 \right] dx \\ &= (x+1) \left[-\frac{2}{x+1} - x \right] = -2 - x^2 - x. \end{aligned}$$

A second solution is $y_2 = x^2 + x + 2$.

16. Identifying $P(x) = -2x/(1-x^2)$ we have

$$y_2 = \int e^{-\int -2x dx/(1-x^2)} dx = \int e^{-\ln(1-x^2)} dx = \int \frac{1}{1-x^2} dx = \frac{1}{2} \ln \left| \frac{1+x}{1-x} \right|.$$

A second solution is $y_2 = \ln |(1+x)/(1-x)|$.

17. Define $y = u(x)e^{-2x}$ so

$$y' = -2ue^{-2x} + u'e^{-2x}, \quad y'' = u''e^{-2x} - 4u'e^{-2x} + 4ue^{-2x}$$

and

$$y'' - 4y = e^{-2x}u'' - 4e^{-2x}u' = 0 \quad \text{or} \quad u'' - 4u' = 0.$$

If $w = u'$ we obtain the first order equation $w' - 4w = 0$ which has the integrating factor $e^{-4} \int dx = e^{-4x}$. Now

$$\frac{d}{dx}[e^{-4x}w] = 0 \quad \text{gives} \quad e^{-4x}w = c.$$

Therefore $w = u' = ce^{4x}$ and $u = c_1e^{4x}$. A second solution is $y_2 = e^{-2x}e^{4x} = e^{2x}$. We see by observation that a particular solution is $y_p = -1/2$. The general solution is

$$y = c_1e^{-2x} + c_2e^{2x} - \frac{1}{2}.$$

Exercises 3.2

18. Define $y = u(x) \cdot 1$ so

$$y' = u', \quad y'' = u'' \quad \text{and} \quad y'' + y' = u'' + u' = 0.$$

If $w = u'$ we obtain the first order equation $w' + w = 0$ which has the integrating factor $e^{\int dx} = e^x$. Now

$$\frac{d}{dx}[e^x w] = 0 \quad \text{gives} \quad e^x w = c.$$

Therefore $w = u' = ce^{-x}$ and $u = c_1 e^{-x}$. A second solution is $y_2 = 1 \cdot e^{-x} = e^{-x}$. We see by observation that a particular solution is $y_p = x$. The general solution is

$$y = c_1 + c_2 e^{-x} + x.$$

19. Define $y = u(x)e^x$ so

$$y' = ue^x + u'e^x, \quad y'' = u''e^x + 2u'e^x + ue^x$$

and

$$y'' - 3y' + 2y = e^x u'' - e^x u' = 0 \quad \text{or} \quad u'' - u' = 0.$$

If $w = u'$ we obtain the first order equation $w' - w = 0$ which has the integrating factor $e^{-\int dx} = e^{-x}$. Now

$$\frac{d}{dx}[e^{-x} w] = 0 \quad \text{gives} \quad e^{-x} w = c.$$

Therefore $w = u' = ce^x$ and $u = ce^x$. A second solution is $y_2 = e^x e^x = e^{2x}$. To find a particular solution we try $y_p = Ae^{3x}$. Then $y' = 3Ae^{3x}$, $y'' = 9Ae^{3x}$, and $9Ae^{3x} - 3(3Ae^{3x}) + 2Ae^{3x} = 5e^{3x}$. Thus $A = 5/2$ and $y_p = \frac{5}{2}e^{3x}$. The general solution is

$$y = c_1 e^x + c_2 e^{2x} + \frac{5}{2}e^{3x}.$$

20. Define $y = u(x)e^x$ so

$$y' = ue^x + u'e^x, \quad y'' = u''e^x + 2u'e^x + ue^x$$

and

$$y'' - 4y' + 3y = e^x u'' - 2e^x u' = 0 \quad \text{or} \quad u'' - 2u' = 0.$$

If $w = u'$ we obtain the first order equation $w' - 2w = 0$ which has the integrating factor $e^{-2\int dx} = e^{-2x}$. Now

$$\frac{d}{dx}[e^{-2x} w] = 0 \quad \text{gives} \quad e^{-2x} w = c.$$

Therefore $w = u' = ce^{2x}$ and $u = c_1 e^{2x}$. A second solution is $y_2 = e^x e^{2x} = e^{3x}$. To find a particular solution we try $y_p = ax + b$. Then $y'_p = a$, $y''_p = 0$, and $0 - 4a + 3(ax + b) = 3ax - 4a + 3b = x$. Then $3a = 1$ and $-4a + 3b = 0$ so $a = 1/3$ and $b = 4/9$. A particular solution is $y_p = \frac{1}{3}x + \frac{4}{9}$ and the general solution is

$$y = c_1 e^x + c_2 e^{3x} + \frac{1}{3}x + \frac{4}{9}.$$

21. (a) For m_1 constant, let $y_1 = e^{m_1 x}$. Then $y'_1 = m_1 e^{m_1 x}$ and $y''_1 = m_1^2 e^{m_1 x}$. Substituting into the differential equation we obtain

$$\begin{aligned} ay''_1 + by'_1 + cy_1 &= am_1^2 e^{m_1 x} + bm_1 e^{m_1 x} + ce^{m_1 x} \\ &= e^{m_1 x}(am_1^2 + bm_1 + c) = 0. \end{aligned}$$

Thus, $y_1 = e^{m_1 x}$ will be a solution of the differential equation whenever $am_1^2 + bm_1 + c = 0$. Since a quadratic equation always has at least one real or complex root, the differential equation must have a solution of the form $y_1 = e^{m_1 x}$.

- (b) Write the differential equation in the form

$$y'' + \frac{b}{a}y' + \frac{c}{a}y = 0,$$

and let $y_1 = e^{m_1 x}$ be a solution. Then a second solution is given by

$$\begin{aligned} y_2 &= e^{m_1 x} \int \frac{e^{-bx/a}}{e^{2m_1 x}} dx \\ &= e^{m_1 x} \int e^{-(b/a+2m_1)x} dx \\ &= -\frac{1}{b/a+2m_1} e^{m_1 x} e^{-(b/a+2m_1)x} \quad (m_1 \neq -b/2a) \\ &= -\frac{1}{b/a+2m_1} e^{-(b/a+m_1)x}. \end{aligned}$$

Thus, when $m_1 \neq -b/2a$, a second solution is given by $y_2 = e^{m_2 x}$ where $m_2 = -b/a - m_1$. When $m_1 = -b/2a$ a second solution is given by

$$y_2 = e^{m_1 x} \int dx = x e^{m_1 x}.$$

(c) The functions

$$\begin{aligned} \sin x &= \frac{1}{2i}(e^{ix} - e^{-ix}) & \cos x &= \frac{1}{2}(e^{ix} + e^{-ix}) \\ \sinh x &= \frac{1}{2}(e^x - e^{-x}) & \cosh x &= \frac{1}{2}(e^x + e^{-x}) \end{aligned}$$

are all expressible in terms of exponential functions.

- 22.** We have $y'_1 = 1$ and $y''_1 = 0$, so $xy''_1 - xy'_1 + y_1 = 0 - x + x = 0$ and $y_1(x) = x$ is a solution of the differential equation. Letting $y = u(x)y_1(x) = xu(x)$ we get

$$y' = xu'(x) + u(x) \quad \text{and} \quad y'' = xu''(x) + 2u'(x).$$

Then $xy'' - xy' + y = x^2u'' + 2xu' - x^2u' - xu + xu = x^2u'' - (x^2 - 2x)u' = 0$. If we make the substitution $w = u'$, the second-order linear differential equation becomes $x^2w' - (x^2 - x)w = 0$, which is separable:

$$\begin{aligned} \frac{dw}{dx} &= \left(1 - \frac{1}{x}\right)w \\ \frac{dw}{w} &= \left(1 - \frac{1}{x}\right)dx \\ \ln w &= x - \ln x + c \\ w &= c_1 \frac{e^x}{x}. \end{aligned}$$

Then $u' = c_1 e^x/x$ and $u = c_1 \int e^x dx/x$. To integrate e^x/x we use the series representation for e^x . Thus, a second solution is

$$\begin{aligned} y_2 &= xu(x) = c_1 x \int \frac{e^x}{x} dx \\ &= c_1 x \int \frac{1}{x} \left(1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \cdots\right) dx \\ &= c_1 x \int \left(\frac{1}{x} + 1 + \frac{1}{2!}x + \frac{1}{3!}x^2 + \cdots\right) dx \\ &= c_1 x \left(\ln x + x + \frac{1}{2(2!)}x^2 + \frac{1}{3(3!)}x^3 + \cdots\right) \\ &= c_1 \left(x \ln x + x^2 + \frac{1}{2(2!)}x^3 + \frac{1}{3(3!)}x^4 + \cdots\right). \end{aligned}$$

An interval of definition is probably $(0, \infty)$ because of the $\ln x$ term.

Exercises 3.3

1. From $4m^2 + m = 0$ we obtain $m = 0$ and $m = -1/4$ so that $y = c_1 + c_2e^{-x/4}$.
2. From $m^2 - 36 = 0$ we obtain $m = 6$ and $m = -6$ so that $y = c_1e^{6x} + c_2e^{-6x}$.
3. From $m^2 - m - 6 = 0$ we obtain $m = 3$ and $m = -2$ so that $y = c_1e^{3x} + c_2e^{-2x}$.
4. From $m^2 - 3m + 2 = 0$ we obtain $m = 1$ and $m = 2$ so that $y = c_1e^x + c_2e^{2x}$.
5. From $m^2 + 8m + 16 = 0$ we obtain $m = -4$ and $m = -4$ so that $y = c_1e^{-4x} + c_2xe^{-4x}$.
6. From $m^2 - 10m + 25 = 0$ we obtain $m = 5$ and $m = 5$ so that $y = c_1e^{5x} + c_2xe^{5x}$.
7. From $12m^2 - 5m - 2 = 0$ we obtain $m = -1/4$ and $m = 2/3$ so that $y = c_1e^{-x/4} + c_2e^{2x/3}$.
8. From $m^2 + 4m - 1 = 0$ we obtain $m = -2 \pm \sqrt{5}$ so that $y = c_1e^{(-2+\sqrt{5})x} + c_2e^{(-2-\sqrt{5})x}$.
9. From $m^2 + 9 = 0$ we obtain $m = 3i$ and $m = -3i$ so that $y = c_1 \cos 3x + c_2 \sin 3x$.
10. From $3m^2 + 1 = 0$ we obtain $m = i/\sqrt{3}$ and $m = -i/\sqrt{3}$ so that $y = c_1 \cos x/\sqrt{3} + c_2 \sin x/\sqrt{3}$.
11. From $m^2 - 4m + 5 = 0$ we obtain $m = 2 \pm i$ so that $y = e^{2x}(c_1 \cos x + c_2 \sin x)$.
12. From $2m^2 + 2m + 1 = 0$ we obtain $m = -1/2 \pm i/2$ so that

$$y = e^{-x/2}(c_1 \cos x/2 + c_2 \sin x/2).$$

13. From $3m^2 + 2m + 1 = 0$ we obtain $m = -1/3 \pm \sqrt{2}i/3$ so that

$$y = e^{-x/3} \left(c_1 \cos \sqrt{2}x/3 + c_2 \sin \sqrt{2}x/3 \right).$$

14. From $2m^2 - 3m + 4 = 0$ we obtain $m = 3/4 \pm \sqrt{23}i/4$ so that

$$y = e^{3x/4} \left(c_1 \cos \sqrt{23}x/4 + c_2 \sin \sqrt{23}x/4 \right).$$

15. From $m^3 - 4m^2 - 5m = 0$ we obtain $m = 0$, $m = 5$, and $m = -1$ so that

$$y = c_1 + c_2e^{5x} + c_3e^{-x}.$$

16. From $m^3 - 1 = 0$ we obtain $m = 1$ and $m = -1/2 \pm \sqrt{3}i/2$ so that

$$y = c_1e^x + e^{-x/2} \left(c_2 \cos \sqrt{3}x/2 + c_3 \sin \sqrt{3}x/2 \right).$$

17. From $m^3 - 5m^2 + 3m + 9 = 0$ we obtain $m = -1$, $m = 3$, and $m = 3$ so that

$$y = c_1e^{-x} + c_2e^{3x} + c_3xe^{3x}.$$

18. From $m^3 + 3m^2 - 4m - 12 = 0$ we obtain $m = -2$, $m = 2$, and $m = -3$ so that

$$y = c_1e^{-2x} + c_2e^{2x} + c_3e^{-3x}.$$

19. From $m^3 + m^2 - 2 = 0$ we obtain $m = 1$ and $m = -1 \pm i$ so that

$$u = c_1e^t + e^{-t}(c_2 \cos t + c_3 \sin t).$$

20. From $m^3 - m^2 - 4 = 0$ we obtain $m = 2$ and $m = -1/2 \pm \sqrt{7}i/2$ so that

$$x = c_1e^{2t} + e^{-t/2} \left(c_2 \cos \sqrt{7}t/2 + c_3 \sin \sqrt{7}t/2 \right).$$

21. From $m^3 + 3m^2 + 3m + 1 = 0$ we obtain $m = -1$, $m = -1$, and $m = -1$ so that

$$y = c_1 e^{-x} + c_2 x e^{-x} + c_3 x^2 e^{-x}.$$

22. From $m^3 - 6m^2 + 12m - 8 = 0$ we obtain $m = 2$, $m = 2$, and $m = 2$ so that

$$y = c_1 e^{2x} + c_2 x e^{2x} + c_3 x^2 e^{2x}.$$

23. From $m^4 + m^3 + m^2 = 0$ we obtain $m = 0$, $m = 0$, and $m = -1/2 \pm \sqrt{3}i/2$ so that

$$y = c_1 + c_2 x + e^{-x/2} \left(c_3 \cos \sqrt{3}x/2 + c_4 \sin \sqrt{3}x/2 \right).$$

24. From $m^4 - 2m^2 + 1 = 0$ we obtain $m = 1$, $m = 1$, $m = -1$, and $m = -1$ so that

$$y = c_1 e^x + c_2 x e^x + c_3 e^{-x} + c_4 x e^{-x}.$$

25. From $16m^4 + 24m^2 + 9 = 0$ we obtain $m = \pm\sqrt{3}i/2$ and $m = \pm\sqrt{3}i/2$ so that

$$y = c_1 \cos \sqrt{3}x/2 + c_2 \sin \sqrt{3}x/2 + c_3 x \cos \sqrt{3}x/2 + c_4 x \sin \sqrt{3}x/2.$$

26. From $m^4 - 7m^2 - 18 = 0$ we obtain $m = 3$, $m = -3$, and $m = \pm\sqrt{2}i$ so that

$$y = c_1 e^{3x} + c_2 e^{-3x} + c_3 \cos \sqrt{2}x + c_4 \sin \sqrt{2}x.$$

27. From $m^5 + 5m^4 - 2m^3 - 10m^2 + m + 5 = 0$ we obtain $m = -1$, $m = -1$, $m = 1$, and $m = 1$, and $m = -5$ so that

$$u = c_1 e^{-r} + c_2 r e^{-r} + c_3 e^r + c_4 r e^r + c_5 e^{-5r}.$$

28. From $2m^5 - 7m^4 + 12m^3 + 8m^2 = 0$ we obtain $m = 0$, $m = 0$, $m = -1/2$, and $m = 2 \pm 2i$ so that

$$x = c_1 + c_2 s + c_3 e^{-s/2} + e^{2s} (c_4 \cos 2s + c_5 \sin 2s).$$

29. From $m^2 + 16 = 0$ we obtain $m = \pm 4i$ so that $y = c_1 \cos 4x + c_2 \sin 4x$. If $y(0) = 2$ and $y'(0) = -2$ then $c_1 = 2$, $c_2 = -1/2$, and $y = 2 \cos 4x - \frac{1}{2} \sin 4x$.

30. From $m^2 + 1 = 0$ we obtain $m = \pm i$ so that $y = c_1 \cos \theta + c_2 \sin \theta$. If $y(\pi/3) = 0$ and $y'(\pi/3) = 2$ then $\frac{1}{2}c_1 + \frac{\sqrt{3}}{2}c_2 = 0$, $-\frac{\sqrt{3}}{2}c_1 + \frac{1}{2}c_2 = 2$, so $c_1 = -\sqrt{3}$, $c_2 = 1$, and $y = -\sqrt{3} \cos \theta + \sin \theta$.

31. From $m^2 - 4m - 5 = 0$ we obtain $m = -1$ and $m = 5$, so that $y = c_1 e^{-x} + c_2 e^{5x}$. If $y(1) = 0$ and $y'(1) = 2$, then $c_1 e^{-1} + c_2 e^5 = 0$, $-c_1 e^{-1} + 5c_2 e^5 = 2$, so $c_1 = -e/3$, $c_2 = e^{-5}/3$, and $y = -\frac{1}{3}e^{1-x} + \frac{1}{3}e^{5x-5}$.

32. From $4m^2 - 4m - 3 = 0$ we obtain $m = -1/2$ and $m = 3/2$ so that $y = c_1 e^{-x/2} + c_2 e^{3x/2}$. If $y(0) = 1$ and $y'(0) = 5$ then $c_1 + c_2 = 1$, $-\frac{1}{2}c_1 + \frac{3}{2}c_2 = 5$, so $c_1 = -7/4$, $c_2 = 11/4$, and $y = -\frac{7}{4}e^{-x/2} + \frac{11}{4}e^{3x/2}$.

33. From $m^2 + m + 2 = 0$ we obtain $m = -1/2 \pm \sqrt{7}i/2$ so that $y = e^{-x/2} (c_1 \cos \sqrt{7}x/2 + c_2 \sin \sqrt{7}x/2)$. If $y(0) = 0$ and $y'(0) = 0$ then $c_1 = 0$ and $c_2 = 0$ so that $y = 0$.

34. From $m^2 - 2m + 1 = 0$ we obtain $m = 1$ and $m = 1$ so that $y = c_1 e^x + c_2 x e^x$. If $y(0) = 5$ and $y'(0) = 10$ then $c_1 = 5$, $c_1 + c_2 = 10$ so $c_1 = 5$, $c_2 = 5$, and $y = 5e^x + 5xe^x$.

35. From $m^3 + 12m^2 + 36m = 0$ we obtain $m = 0$, $m = -6$, and $m = -6$ so that $y = c_1 + c_2 e^{-6x} + c_3 x e^{-6x}$. If $y(0) = 0$, $y'(0) = 1$, and $y''(0) = -7$ then

$$c_1 + c_2 = 0, \quad -6c_2 + c_3 = 1, \quad 36c_2 - 12c_3 = -7,$$

$$\text{so } c_1 = 5/36, \quad c_2 = -5/36, \quad c_3 = 1/6, \quad \text{and } y = \frac{5}{36} - \frac{5}{36}e^{-6x} + \frac{1}{6}xe^{-6x}.$$

Exercises 3.3

36. From $m^3 + 2m^2 - 5m - 6 = 0$ we obtain $m = -1$, $m = 2$, and $m = -3$ so that

$$y = c_1 e^{-x} + c_2 e^{2x} + c_3 e^{-3x}.$$

If $y(0) = 0$, $y'(0) = 0$, and $y''(0) = 1$ then

$$c_1 + c_2 + c_3 = 0, \quad -c_1 + 2c_2 - 3c_3 = 0, \quad c_1 + 4c_2 + 9c_3 = 1,$$

so $c_1 = -1/6$, $c_2 = 1/15$, $c_3 = 1/10$, and

$$y = -\frac{1}{6}e^{-x} + \frac{1}{15}e^{2x} + \frac{1}{10}e^{-3x}.$$

37. From $m^4 - 3m^3 + 3m^2 - m = 0$ we obtain $m = 0$, $m = 1$, $m = 1$, and $m = 1$ so that $y = c_1 + c_2 e^x + c_3 x e^x + c_4 x^2 e^x$. If $y(0) = 0$, $y'(0) = 0$, $y''(0) = 1$, and $y'''(0) = 1$ then

$$c_1 + c_2 = 0, \quad c_2 + c_3 = 0, \quad c_2 + 2c_3 + 2c_4 = 1, \quad c_2 + 3c_3 + 6c_4 = 1,$$

so $c_1 = 2$, $c_2 = -2$, $c_3 = 2$, $c_4 = -1/2$, and

$$y = 2 - 2e^x + 2xe^x - \frac{1}{2}x^2 e^x.$$

38. From $m^4 - 1 = 0$ we obtain $m = 1$, $m = -1$, and $m = \pm i$ so that $y = c_1 e^x + c_2 e^{-x} + c_3 \cos x + c_4 \sin x$. If $y(0) = 0$, $y'(0) = 0$, $y''(0) = 0$, and $y'''(0) = 1$ then

$$c_1 + c_2 + c_3 = 0, \quad c_1 - c_2 + c_4 = 0, \quad c_1 + c_2 - c_3 = 0, \quad c_1 - c_2 - c_4 = 1,$$

so $c_1 = 1/4$, $c_2 = -1/4$, $c_3 = 0$, $c_4 = -1/2$, and

$$y = \frac{1}{4}e^x - \frac{1}{4}e^{-x} - \frac{1}{2}\sin x.$$

39. From $m^2 - 10m + 25 = 0$ we obtain $m = 5$ and $m = 5$ so that $y = c_1 e^{5x} + c_2 x e^{5x}$. If $y(0) = 1$ and $y(1) = 0$ then $c_1 = 1$, $c_1 e^5 + c_2 e^5 = 0$, so $c_1 = 1$, $c_2 = -1$, and $y = e^{5x} - x e^{5x}$.
40. From $m^2 + 4 = 0$ we obtain $m = \pm 2i$ so that $y = c_1 \cos 2x + c_2 \sin 2x$. If $y(0) = 0$ and $y(\pi) = 0$ then $c_1 = 0$ and $y = c_2 \sin 2x$.
41. From $m^2 + 1 = 0$ we obtain $m = \pm i$ so that $y = c_1 \cos x + c_2 \sin x$. If $y'(0) = 0$ and $y'(\pi/2) = 2$ then $c_1 = -2$, $c_2 = 0$, and $y = -2 \cos x$.
42. From $m^2 - 2m + 2 = 0$ we obtain $m = 1 \pm i$ so that $y = e^x(c_1 \cos x + c_2 \sin x)$. If $y(0) = 1$ and $y(\pi) = 1$ then $c_1 = 1$ and $y(\pi) = e^\pi \cos \pi = -e^\pi$. Since $-e^\pi \neq 1$, the boundary-value problem has no solution.
43. The auxiliary equation is $m^2 - 3 = 0$ which has roots $-\sqrt{3}$ and $\sqrt{3}$. By (10) the general solution is $y = c_1 e^{\sqrt{3}x} + c_2 e^{-\sqrt{3}x}$. By (11) the general solution is $y = c_1 \cosh \sqrt{3}x + c_2 \sinh \sqrt{3}x$. For $y = c_1 e^{\sqrt{3}x} + c_2 e^{-\sqrt{3}x}$ the initial conditions imply $c_1 + c_2 = 1$, $\sqrt{3}c_1 - \sqrt{3}c_2 = 5$. Solving for $c_1 + c_2$ we find $c_1 = \frac{1}{6}(3 + 5\sqrt{3})$ and $c_2 = \frac{1}{6}(3 - 5\sqrt{3})$ so $y = \frac{1}{6}(3 + 5\sqrt{3})e^{\sqrt{3}x} + \frac{1}{6}(3 - 5\sqrt{3})e^{-\sqrt{3}x}$. For $y = c_1 \cosh \sqrt{3}x + c_2 \sinh \sqrt{3}x$ the initial conditions imply $c_1 = 1$, $\sqrt{3}c_2 = 5$. Solving for c_1 and c_2 we find $c_1 = 1$ and $c_2 = \frac{5}{3}\sqrt{3}$ so $y = \cosh \sqrt{3}x + \frac{5}{3}\sqrt{3} \sinh \sqrt{3}x$.
44. The auxiliary equation is $m^2 - 1 = 0$ which has roots -1 and 1 . By (10) the general solution is $y = c_1 e^x + c_2 e^{-x}$. By (11) the general solution is $y = c_1 \cosh x + c_2 \sinh x$. For $y = c_1 e^x + c_2 e^{-x}$ the boundary conditions imply $c_1 + c_2 = 1$, $c_1 e - c_2 e^{-1} = 0$. Solving for c_1 and c_2 we find $c_1 = 1/(1 + e^2)$ and $c_2 = e^2/(1 + e^2)$ so $y = e^x/(1 + e^2) + e^2 e^{-x}/(1 + e^2)$. For $y = c_1 \cosh x + c_2 \sinh x$ the boundary conditions imply $c_1 = 1$, $c_2 = -\tanh 1$, so $y = \cosh x - (\tanh 1) \sinh x$.
45. Using a CAS to solve the auxiliary equation $m^3 - 6m^2 + 2m + 1 = 0$ we find $m_1 = -0.270534$,

$m_2 = 0.658675$, and $m_3 = 5.61186$. The general solution is

$$y = c_1 e^{-0.270534x} + c_2 e^{0.658675x} + c_3 e^{5.61186x}.$$

46. Using a CAS to solve the auxiliary equation $6.11m^3 + 8.59m^2 + 7.93m + 0.778 = 0$ we find $m_1 = -0.110241$, $m_2 = -0.647826 + 0.857532i$, and $m_3 = -0.647826 - 0.857532i$. The general solution is

$$y = c_1 e^{-0.110241x} + e^{-0.647826x} (c_2 \cos 0.857532x + c_3 \sin 0.857532x).$$

47. Using a CAS to solve the auxiliary equation $3.15m^4 - 5.34m^2 + 6.33m - 2.03 = 0$ we find $m_1 = -1.74806$, $m_2 = 0.501219$, $m_3 = 0.62342 + 0.588965i$, and $m_4 = 0.62342 - 0.588965i$. The general solution is

$$y = c_1 e^{-1.74806x} + c_2 e^{0.501219x} + e^{0.62342x} (c_3 \cos 0.588965x + c_4 \sin 0.588965x).$$

48. Using a CAS to solve the auxiliary equation $m^4 + 2m^2 - m + 2 = 0$ we find $m_1 = 1/2 + \sqrt{3}i/2$, $m_2 = 1/2 - \sqrt{3}i/2$, $m_3 = -1/2 + \sqrt{7}i/2$, and $m_4 = -1/2 - \sqrt{7}i/2$. The general solution is

$$y = e^{x/2} \left(c_1 \cos \frac{\sqrt{3}}{2}x + c_2 \sin \frac{\sqrt{3}}{2}x \right) + e^{-x/2} \left(c_3 \cos \frac{\sqrt{7}}{2}x + c_4 \sin \frac{\sqrt{7}}{2}x \right).$$

49. The auxiliary equation should have two positive roots, so that the solution has the form $y = c_1 e^{k_1 x} + c_2 e^{k_2 x}$. Thus, the differential equation is (f).
50. The auxiliary equation should have one positive and one negative root, so that the solution has the form $y = c_1 e^{k_1 x} + c_2 e^{-k_2 x}$. Thus, the differential equation is (a).
51. The auxiliary equation should have a pair of complex roots $a \pm bi$ where $a < 0$, so that the solution has the form $e^{ax}(c_1 \cos bx + c_2 \sin bx)$. Thus, the differential equation is (e).
52. The auxiliary equation should have a repeated negative root, so that the solution has the form $y = c_1 e^{-x} + c_2 x e^{-x}$. Thus, the differential equation is (c).
53. The differential equation should have the form $y'' + k^2 y = 0$ where $k = 1$ so that the period of the solution is 2π . Thus, the differential equation is (d).
54. The differential equation should have the form $y'' + k^2 y = 0$ where $k = 2$ so that the period of the solution is π . Thus, the differential equation is (b).
55. (a) The auxiliary equation is $m^2 - 64/L = 0$ which has roots $\pm 8/\sqrt{L}$. Thus, the general solution of the differential equation is $x = c_1 \cosh(8t/\sqrt{L}) + c_2 \sinh(8t/\sqrt{L})$.
- (b) Setting $x(0) = x_0$ and $x'(0) = 0$ we have $c_1 = x_0$, $8c_2/\sqrt{L} = 0$. Solving for c_1 and c_2 we get $c_1 = x_0$ and $c_2 = 0$, so $x(t) = x_0 \cosh(8t/\sqrt{L})$.
- (c) When $L = 20$ and $x_0 = 1$, $x(t) = \cosh(4t\sqrt{5})$. The chain will last touch the peg when $x(t) = 10$. Solving $x(t) = 10$ for t we get $t_1 = \frac{1}{4}\sqrt{5} \cosh^{-1} 10 \approx 1.67326$. The velocity of the chain at this instant is $x'(t_1) = 12\sqrt{11/5} \approx 17.7989$ ft/s.
56. Both $-C[1]$ and c_1 represent arbitrary constants, and each may take on any real value.
57. Since $(m-4)(m+5)^2 = m^3 + 6m^2 - 15m - 100$ the differential equation is $y''' + 6y'' - 15y' - 100y = 0$. The differential equation is not unique since any constant multiple of the left-hand side of the differential equation would lead to the auxiliary roots.
58. A third root must be $m_3 = 3 - i$ and the auxiliary equation is

$$\left(m + \frac{1}{2}\right)[m - (3 + i)][m - (3 - i)] = \left(m + \frac{1}{2}\right)(m^2 - 6x + 10) = m^3 - \frac{11}{2}m^2 + 7m + 5.$$

Exercises 3.3

The differential equation is

$$y''' - \frac{11}{2}y'' + 7y' + 5y = 0.$$

59. From the solution $y_1 = e^{-4x} \cos x$ we conclude that $m_1 = -4 + i$ and $m_2 = -4 - i$ are roots of the auxiliary equation. Hence another solution must be $y_2 = e^{-4x} \sin x$. Now dividing the polynomial $m^3 + 6m^2 + m - 34$ by $[m - (-4 + i)][m - (-4 - i)] = m^2 + 8m + 17$ gives $m - 2$. Therefore $m_3 = 2$ is the third root of the auxiliary equation, and the general solution of the differential equation is

$$y = c_1 e^{-4x} \cos x + c_2 e^{-4x} \sin x + c_3 e^{2x}.$$

60. Factoring the difference of two squares we obtain

$$m^4 + 1 = (m^2 + 1)^2 - 2m^2 = (m^2 + 1 - \sqrt{2}m)(m^2 + 1 + \sqrt{2}m) = 0.$$

Using the quadratic formula on each factor we get $m = \pm\sqrt{2}/2 \pm \sqrt{2}i/2$. The solution of the differential equation is

$$y(x) = e^{\sqrt{2}x/2} \left(c_1 \cos \frac{\sqrt{2}}{2}x + c_2 \sin \frac{\sqrt{2}}{2}x \right) + e^{-\sqrt{2}x/2} \left(c_3 \cos \frac{\sqrt{2}}{2}x + c_4 \sin \frac{\sqrt{2}}{2}x \right).$$

61. The auxiliary equation is $m^2 + \lambda = 0$ and we consider three cases.

Case I When $\lambda = 0$ the general solution of the differential equation is $y = c_1 + c_2x$. The boundary conditions imply $0 = y(0) = c_1$ and $0 = y(\pi/2) = c_2\pi/2$, so that $c_1 = c_2 = 0$ and the problem possesses only the trivial solution.

Case II When $\lambda < 0$ the general solution of the differential equation is $y = c_1 e^{\sqrt{-\lambda}x} + c_2 e^{-\sqrt{-\lambda}x}$, or alternatively, $y = c_1 \cosh \sqrt{-\lambda}x + c_2 \sinh \sqrt{-\lambda}x$. Again, $y(0) = 0$ implies $c_1 = 0$ so $y = c_2 \sinh \sqrt{-\lambda}x$. The second boundary condition implies $0 = y(\pi/2) = c_2 \sinh \sqrt{-\lambda}\pi/2$ or $c_2 = 0$. In this case also, the problem possesses only the trivial solution.

Case III When $\lambda > 0$ the general solution of the differential equation is $y = c_1 \cos \sqrt{\lambda}x + c_2 \sin \sqrt{\lambda}x$. In this case also, $y(0) = 0$ yields $c_1 = 0$, so that $y = c_2 \sin \sqrt{\lambda}x$. The second boundary condition implies $0 = c_2 \sin \sqrt{\lambda}\pi/2$. When $\sqrt{\lambda}\pi/2$ is an integer multiple of π , that is, when $\sqrt{\lambda} = 2k$ for k a nonzero integer, the problem will have nontrivial solutions. Thus, for $\lambda = 4k^2$ the boundary-value problem will have nontrivial solutions $y = c_2 \sin 2kx$, where k is a nonzero integer. On the other hand, when $\sqrt{\lambda}$ is not an even integer, the boundary-value problem will have only the trivial solution.

62. Applying integration by parts twice we have

$$\begin{aligned} \int e^{ax} f(x) dx &= \frac{1}{a} e^{ax} f(x) - \frac{1}{a} \int e^{ax} f'(x) dx \\ &= \frac{1}{a} e^{ax} f(x) - \frac{1}{a} \left[\frac{1}{a} e^{ax} f'(x) - \frac{1}{a} \int e^{ax} f''(x) dx \right] \\ &= \frac{1}{a} e^{ax} f(x) - \frac{1}{a^2} e^{ax} f'(x) + \frac{1}{a^2} \int e^{ax} f''(x) dx. \end{aligned}$$

Collecting the integrals we get

$$\int e^{ax} \left(f(x) - \frac{1}{a^2} f''(x) \right) dx = \frac{1}{a} e^{ax} f(x) - \frac{1}{a^2} e^{ax} f'(x).$$

In order for the technique to work we need to have

$$\int e^{ax} \left(f(x) - \frac{1}{a^2} f''(x) \right) dx = k \int e^{ax} f(x) dx$$

or

$$f(x) - \frac{1}{a^2} f''(x) = kf(x),$$

where $k \neq 0$. This is the second-order differential equation

$$f''(x) + a^2(k-1)f(x) = 0.$$

If $k < 1$, $k \neq 0$, the solution of the differential equation is a pair of exponential functions, in which case the original integrand is an exponential function and does not require integration by parts for its evaluation. Similarly, if $k = 1$, $f''(x) = 0$ and $f(x)$ has the form $f(x) = ax + b$. In this case a single application of integration by parts will suffice. Finally, if $k > 1$, the solution of the differential equation is

$$f(x) = c_1 \cos a\sqrt{k-1}x + c_2 \sin a\sqrt{k-1}x,$$

and we see that the technique will work for linear combinations of $\cos \alpha x$ and $\sin \alpha x$.

Exercises 3.4

1. From $m^2 + 3m + 2 = 0$ we find $m_1 = -1$ and $m_2 = -2$. Then $y_c = c_1 e^{-x} + c_2 e^{-2x}$ and we assume $y_p = A$. Substituting into the differential equation we obtain $2A = 6$. Then $A = 3$, $y_p = 3$ and

$$y = c_1 e^{-x} + c_2 e^{-2x} + 3.$$

2. From $4m^2 + 9 = 0$ we find $m_1 = -\frac{3}{2}i$ and $m_2 = \frac{3}{2}i$. Then $y_c = c_1 \cos \frac{3}{2}x + c_2 \sin \frac{3}{2}x$ and we assume $y_p = A$. Substituting into the differential equation we obtain $9A = 15$. Then $A = \frac{5}{3}$, $y_p = \frac{5}{3}$ and

$$y = c_1 \cos \frac{3}{2}x + c_2 \sin \frac{3}{2}x + \frac{5}{3}.$$

3. From $m^2 - 10m + 25 = 0$ we find $m_1 = m_2 = 5$. Then $y_c = c_1 e^{5x} + c_2 x e^{5x}$ and we assume $y_p = Ax + B$. Substituting into the differential equation we obtain $25A = 30$ and $-10A + 25B = 3$. Then $A = \frac{6}{5}$, $B = \frac{6}{5}$, $y_p = \frac{6}{5}x + \frac{6}{5}$, and

$$y = c_1 e^{5x} + c_2 x e^{5x} + \frac{6}{5}x + \frac{6}{5}.$$

4. From $m^2 + m - 6 = 0$ we find $m_1 = -3$ and $m_2 = 2$. Then $y_c = c_1 e^{-3x} + c_2 e^{2x}$ and we assume $y_p = Ax + B$. Substituting into the differential equation we obtain $-6A = 2$ and $A - 6B = 0$. Then $A = -\frac{1}{3}$, $B = -\frac{1}{18}$, $y_p = -\frac{1}{3}x - \frac{1}{18}$, and

$$y = c_1 e^{-3x} + c_2 e^{2x} - \frac{1}{3}x - \frac{1}{18}.$$

5. From $\frac{1}{4}m^2 + m + 1 = 0$ we find $m_1 = m_2 = 0$. Then $y_c = c_1 e^{-2x} + c_2 x e^{-2x}$ and we assume $y_p = Ax^2 + Bx + C$. Substituting into the differential equation we obtain $A = 1$, $2A + B = -2$, and $\frac{1}{2}A + B + C = 0$. Then $A = 1$, $B = -4$, $C = \frac{7}{2}$, $y_p = x^2 - 4x + \frac{7}{2}$, and

$$y = c_1 e^{-2x} + c_2 x e^{-2x} + x^2 - 4x + \frac{7}{2}.$$

Exercises 3.4

6. From $m^2 - 8m + 20 = 0$ we find $m_1 = 2 + 4i$ and $m_2 = 2 - 4i$. Then $y_c = e^{2x}(c_1 \cos 4x + c_2 \sin 4x)$ and we assume $y_p = Ax^2 + Bx + C + (Dx + E)e^x$. Substituting into the differential equation we obtain

$$2A - 8B + 20C = 0$$

$$-6D + 13E = 0$$

$$-16A + 20B = 0$$

$$13D = -26$$

$$20A = 100.$$

Then $A = 5$, $B = 4$, $C = \frac{11}{10}$, $D = -2$, $E = -\frac{12}{13}$, $y_p = 5x^2 + 4x + \frac{11}{10} + (-2x - \frac{12}{13})e^x$ and

$$y = e^{2x}(c_1 \cos 4x + c_2 \sin 4x) + 5x^2 + 4x + \frac{11}{10} + \left(-2x - \frac{12}{13}\right)e^x.$$

7. From $m^2 + 3 = 0$ we find $m_1 = \sqrt{3}i$ and $m_2 = -\sqrt{3}i$. Then $y_c = c_1 \cos \sqrt{3}x + c_2 \sin \sqrt{3}x$ and we assume $y_p = (Ax^2 + Bx + C)e^{3x}$. Substituting into the differential equation we obtain $2A + 6B + 12C = 0$, $12A + 12B = 0$, and $12A = -48$. Then $A = -4$, $B = 4$, $C = -\frac{4}{3}$, $y_p = (-4x^2 + 4x - \frac{4}{3})e^{3x}$ and

$$y = c_1 \cos \sqrt{3}x + c_2 \sin \sqrt{3}x + \left(-4x^2 + 4x - \frac{4}{3}\right)e^{3x}.$$

8. From $4m^2 - 4m - 3 = 0$ we find $m_1 = \frac{3}{2}$ and $m_2 = -\frac{1}{2}$. Then $y_c = c_1 e^{3x/2} + c_2 e^{-x/2}$ and we assume $y_p = A \cos 2x + B \sin 2x$. Substituting into the differential equation we obtain $-19 - 8B = 1$ and $8A - 19B = 0$. Then $A = -\frac{19}{425}$, $B = -\frac{8}{425}$, $y_p = -\frac{19}{425} \cos 2x - \frac{8}{425} \sin 2x$, and

$$y = c_1 e^{3x/2} + c_2 e^{-x/2} - \frac{19}{425} \cos 2x - \frac{8}{425} \sin 2x.$$

9. From $m^2 - m = 0$ we find $m_1 = 1$ and $m_2 = 0$. Then $y_c = c_1 e^x + c_2$ and we assume $y_p = Ax$. Substituting into the differential equation we obtain $-A = -3$. Then $A = 3$, $y_p = 3x$ and $y = c_1 e^x + c_2 + 3x$.
10. From $m^2 + 2m = 0$ we find $m_1 = -2$ and $m_2 = 0$. Then $y_c = c_1 e^{-2x} + c_2$ and we assume $y_p = Ax^2 + Bx + Cx e^{-2x}$. Substituting into the differential equation we obtain $2A + 2B = 5$, $4A = 2$, and $-2C = -1$. Then $A = \frac{1}{2}$, $B = 2$, $C = \frac{1}{2}$, $y_p = \frac{1}{2}x^2 + 2x + \frac{1}{2}x e^{-2x}$, and

$$y = c_1 e^{-2x} + c_2 + \frac{1}{2}x^2 + 2x + \frac{1}{2}x e^{-2x}.$$

11. From $m^2 - m + \frac{1}{4} = 0$ we find $m_1 = m_2 = \frac{1}{2}$. Then $y_c = c_1 e^{x/2} + c_2 x e^{x/2}$ and we assume $y_p = A + Bx^2 e^{x/2}$. Substituting into the differential equation we obtain $\frac{1}{4}A = 3$ and $2B = 1$. Then $A = 12$, $B = \frac{1}{2}$, $y_p = 12 + \frac{1}{2}x^2 e^{x/2}$, and

$$y = c_1 e^{x/2} + c_2 x e^{x/2} + 12 + \frac{1}{2}x^2 e^{x/2}.$$

12. From $m^2 - 16 = 0$ we find $m_1 = 4$ and $m_2 = -4$. Then $y_c = c_1 e^{4x} + c_2 e^{-4x}$ and we assume $y_p = A x e^{4x}$. Substituting into the differential equation we obtain $8A = 2$. Then $A = \frac{1}{4}$, $y_p = \frac{1}{4}x e^{4x}$ and

$$y = c_1 e^{4x} + c_2 e^{-4x} + \frac{1}{4}x e^{4x}.$$

13. From $m^2 + 4 = 0$ we find $m_1 = 2i$ and $m_2 = -2i$. Then $y_c = c_1 \cos 2x + c_2 \sin 2x$ and we assume $y_p = Ax \cos 2x + Bx \sin 2x$. Substituting into the differential equation we obtain $4B = 0$ and $-4A = 3$. Then $A = -\frac{3}{4}$, $B = 0$, $y_p = -\frac{3}{4}x \cos 2x$, and

$$y = c_1 \cos 2x + c_2 \sin 2x - \frac{3}{4}x \cos 2x.$$

14. From $m^2 + 4 = 0$ we find $m_1 = 2i$ and $m_2 = -2i$. Then $y_c = c_1 \cos 2x + c_2 \sin 2x$ and we assume $y_p = (Ax^3 + Bx^2 + Cx) \cos 2x + (Dx^3 + Ex^2 + Fx) \sin 2x$. Substituting into the differential equation we obtain

$$\begin{aligned} 2B + 4F &= 0 \\ 6A + 8E &= 0 \\ 12D &= 0 \\ -4C + 2E &= -3 \\ -8B + 6D &= 0 \\ -12A &= 1. \end{aligned}$$

Then $A = -\frac{1}{12}$, $B = 0$, $C = \frac{25}{32}$, $D = 0$, $E = \frac{1}{16}$, $F = 0$, $y_p = \left(-\frac{1}{12}x^3 + \frac{25}{32}x\right) \cos 2x + \frac{1}{16}x^2 \sin 2x$, and

$$y = c_1 \cos 2x + c_2 \sin 2x + \left(-\frac{1}{12}x^3 + \frac{25}{32}x\right) \cos 2x + \frac{1}{16}x^2 \sin 2x.$$

15. From $m^2 + 1 = 0$ we find $m_1 = i$ and $m_2 = -i$. Then $y_c = c_1 \cos x + c_2 \sin x$ and we assume $y_p = (Ax^2 + Bx) \cos x + (Cx^2 + Dx) \sin x$. Substituting into the differential equation we obtain $4C = 0$, $2A + 2D = 0$, $-4A = 2$, and $-2B + 2C = 0$. Then $A = -\frac{1}{2}$, $B = 0$, $C = 0$, $D = \frac{1}{2}$, $y_p = -\frac{1}{2}x^2 \cos x + \frac{1}{2}x \sin x$, and

$$y = c_1 \cos x + c_2 \sin x - \frac{1}{2}x^2 \cos x + \frac{1}{2}x \sin x.$$

16. From $m^2 - 5m = 0$ we find $m_1 = 5$ and $m_2 = 0$. Then $y_c = c_1 e^{5x} + c_2$ and we assume $y_p = Ax^4 + Bx^3 + Cx^2 + Dx$. Substituting into the differential equation we obtain $-20A = 2$, $12A - 15B = -4$, $6B - 10C = -1$, and $2C - 5D = 6$. Then $A = -\frac{1}{10}$, $B = \frac{14}{75}$, $C = \frac{53}{250}$, $D = -\frac{697}{625}$, $y_p = -\frac{1}{10}x^4 + \frac{14}{75}x^3 + \frac{53}{250}x^2 - \frac{697}{625}x$, and

$$y = c_1 e^{5x} + c_2 - \frac{1}{10}x^4 + \frac{14}{75}x^3 + \frac{53}{250}x^2 - \frac{697}{625}x.$$

17. From $m^2 - 2m + 5 = 0$ we find $m_1 = 1 + 2i$ and $m_2 = 1 - 2i$. Then $y_c = e^x(c_1 \cos 2x + c_2 \sin 2x)$ and we assume $y_p = Axe^x \cos 2x + Bxe^x \sin 2x$. Substituting into the differential equation we obtain $4B = 1$ and $-4A = 0$. Then $A = 0$, $B = \frac{1}{4}$, $y_p = \frac{1}{4}xe^x \sin 2x$, and

$$y = e^x(c_1 \cos 2x + c_2 \sin 2x) + \frac{1}{4}xe^x \sin 2x.$$

18. From $m^2 - 2m + 2 = 0$ we find $m_1 = 1 + i$ and $m_2 = 1 - i$. Then $y_c = e^x(c_1 \cos x + c_2 \sin x)$ and we assume $y_p = Ae^{2x} \cos x + Be^{2x} \sin x$. Substituting into the differential equation we obtain $A + 2B = 1$ and $-2A + B = -3$. Then $A = \frac{7}{5}$, $B = -\frac{1}{5}$, $y_p = \frac{7}{5}e^{2x} \cos x - \frac{1}{5}e^{2x} \sin x$ and

$$y = e^x(c_1 \cos x + c_2 \sin x) + \frac{7}{5}e^{2x} \cos x - \frac{1}{5}e^{2x} \sin x.$$

19. From $m^2 + 2m + 1 = 0$ we find $m_1 = m_2 = -1$. Then $y_c = c_1 e^{-x} + c_2 x e^{-x}$ and we assume $y_p = A \cos x + B \sin x + C \cos 2x + D \sin 2x$. Substituting into the differential equation we obtain $2B = 0$, $-2A = 1$, $-3C + 4D = 3$, and $-4C - 3D = 0$. Then $A = -\frac{1}{2}$, $B = 0$, $C = -\frac{9}{25}$, $D = \frac{12}{25}$, $y_p = -\frac{1}{2} \cos x - \frac{9}{25} \cos 2x + \frac{12}{25} \sin 2x$, and

$$y = c_1 e^{-x} + c_2 x e^{-x} - \frac{1}{2} \cos x - \frac{9}{25} \cos 2x + \frac{12}{25} \sin 2x.$$

20. From $m^2 + 2m - 24 = 0$ we find $m_1 = -6$ and $m_2 = 4$. Then $y_c = c_1 e^{-6x} + c_2 e^{4x}$ and we assume $y_p = A + (Bx^2 + Cx)e^{4x}$. Substituting into the differential equation we obtain $-24A = 16$, $2B + 10C = -2$, and $20B = -1$. Then $A = -\frac{2}{3}$, $B = -\frac{1}{20}$, $C = -\frac{19}{100}$, $y_p = -\frac{2}{3} - \left(\frac{1}{20}x^2 + \frac{19}{100}x\right)e^{4x}$, and

$$y = c_1 e^{-6x} + c_2 e^{4x} - \frac{2}{3} - \left(\frac{1}{20}x^2 + \frac{19}{100}x\right)e^{4x}.$$

Exercises 3.4

21. From $m^3 - 6m^2 = 0$ we find $m_1 = m_2 = 0$ and $m_3 = 6$. Then $y_c = c_1 + c_2x + c_3e^{6x}$ and we assume $y_p = Ax^2 + B\cos x + C\sin x$. Substituting into the differential equation we obtain $-12A = 3$, $6B - C = -1$, and $B + 6C = 0$. Then $A = -\frac{1}{4}$, $B = -\frac{6}{37}$, $C = \frac{1}{37}$, $y_p = -\frac{1}{4}x^2 - \frac{6}{37}\cos x + \frac{1}{37}\sin x$, and

$$y = c_1 + c_2x + c_3e^{6x} - \frac{1}{4}x^2 - \frac{6}{37}\cos x + \frac{1}{37}\sin x.$$

22. From $m^3 - 2m^2 - 4m + 8 = 0$ we find $m_1 = m_2 = 2$ and $m_3 = -2$. Then $y_c = c_1e^{2x} + c_2xe^{2x} + c_3e^{-2x}$ and we assume $y_p = (Ax^3 + Bx^2)e^{2x}$. Substituting into the differential equation we obtain $24A = 6$ and $6A + 8B = 0$. Then $A = \frac{1}{4}$, $B = -\frac{3}{16}$, $y_p = (\frac{1}{4}x^3 - \frac{3}{16}x^2)e^{2x}$, and

$$y = c_1e^{2x} + c_2xe^{2x} + c_3e^{-2x} + \left(\frac{1}{4}x^3 - \frac{3}{16}x^2\right)e^{2x}.$$

23. From $m^3 - 3m^2 + 3m - 1 = 0$ we find $m_1 = m_2 = m_3 = 1$. Then $y_c = c_1e^x + c_2xe^x + c_3x^2e^x$ and we assume $y_p = Ax + B + Cx^3e^x$. Substituting into the differential equation we obtain $-A = 1$, $3A - B = 0$, and $6C = -4$. Then $A = -1$, $B = -3$, $C = -\frac{2}{3}$, $y_p = -x - 3 - \frac{2}{3}x^3e^x$, and

$$y = c_1e^x + c_2xe^x + c_3x^2e^x - x - 3 - \frac{2}{3}x^3e^x.$$

24. From $m^3 - m^2 - 4m + 4 = 0$ we find $m_1 = 1$, $m_2 = 2$, and $m_3 = -2$. Then $y_c = c_1e^x + c_2e^{2x} + c_3e^{-2x}$ and we assume $y_p = A + Bxe^x + Cxe^{2x}$. Substituting into the differential equation we obtain $4A = 5$, $-3B = -1$, and $4C = 1$. Then $A = \frac{5}{4}$, $B = \frac{1}{3}$, $C = \frac{1}{4}$, $y_p = \frac{5}{4} + \frac{1}{3}xe^x + \frac{1}{4}xe^{2x}$, and

$$y = c_1e^x + c_2e^{2x} + c_3e^{-2x} + \frac{5}{4} + \frac{1}{3}xe^x + \frac{1}{4}xe^{2x}.$$

25. From $m^4 + 2m^2 + 1 = 0$ we find $m_1 = m_3 = i$ and $m_2 = m_4 = -i$. Then $y_c = c_1\cos x + c_2\sin x + c_3x\cos x + c_4x\sin x$ and we assume $y_p = Ax^2 + Bx + C$. Substituting into the differential equation we obtain $A = 1$, $B = -2$, and $4A + C = 1$. Then $A = 1$, $B = -2$, $C = -3$, $y_p = x^2 - 2x - 3$, and

$$y = c_1\cos x + c_2\sin x + c_3x\cos x + c_4x\sin x + x^2 - 2x - 3.$$

26. From $m^4 - m^2 = 0$ we find $m_1 = m_2 = 0$, $m_3 = 1$, and $m_4 = -1$. Then $y_c = c_1 + c_2x + c_3e^x + c_4e^{-x}$ and we assume $y_p = Ax^3 + Bx^2 + (Cx^2 + Dx)e^{-x}$. Substituting into the differential equation we obtain $-6A = 4$, $-2B = 0$, $10C - 2D = 0$, and $-4C = 2$. Then $A = -\frac{2}{3}$, $B = 0$, $C = -\frac{1}{2}$, $D = -\frac{5}{2}$, $y_p = -\frac{2}{3}x^3 - (\frac{1}{2}x^2 + \frac{5}{2}x)e^{-x}$, and

$$y = c_1 + c_2x + c_3e^x + c_4e^{-x} - \frac{2}{3}x^3 - \left(\frac{1}{2}x^2 + \frac{5}{2}x\right)e^{-x}.$$

27. We have $y_c = c_1\cos 2x + c_2\sin 2x$ and we assume $y_p = A$. Substituting into the differential equation we find $A = -\frac{1}{2}$. Thus $y = c_1\cos 2x + c_2\sin 2x - \frac{1}{2}$. From the initial conditions we obtain $c_1 = 0$ and $c_2 = \sqrt{2}$, so $y = \sqrt{2}\sin 2x - \frac{1}{2}$.

28. We have $y_c = c_1e^{-2x} + c_2e^{x/2}$ and we assume $y_p = Ax^2 + Bx + C$. Substituting into the differential equation we find $A = -7$, $B = -19$, and $C = -37$. Thus $y = c_1e^{-2x} + c_2e^{x/2} - 7x^2 - 19x - 37$. From the initial conditions we obtain $c_1 = -\frac{1}{5}$ and $c_2 = \frac{186}{5}$, so

$$y = -\frac{1}{5}e^{-2x} + \frac{186}{5}e^{x/2} - 7x^2 - 19x - 37.$$

29. We have $y_c = c_1e^{-x/5} + c_2$ and we assume $y_p = Ax^2 + Bx$. Substituting into the differential equation we find $A = -3$ and $B = 30$. Thus $y = c_1e^{-x/5} + c_2 - 3x^2 + 30x$. From the initial conditions we obtain $c_1 = 200$ and $c_2 = -200$, so

$$y = 200e^{-x/5} - 200 - 3x^2 + 30x.$$

30. We have $y_c = c_1 e^{-2x} + c_2 x e^{-2x}$ and we assume $y_p = (Ax^3 + Bx^2)e^{-2x}$. Substituting into the differential equation we find $A = \frac{1}{6}$ and $B = \frac{3}{2}$. Thus $y = c_1 e^{-2x} + c_2 x e^{-2x} + (\frac{1}{6}x^3 + \frac{3}{2}x^2)e^{-2x}$. From the initial conditions we obtain $c_1 = 2$ and $c_2 = 9$, so

$$y = 2e^{-2x} + 9xe^{-2x} + \left(\frac{1}{6}x^3 + \frac{3}{2}x^2\right)e^{-2x}.$$

31. We have $y_c = e^{-2x}(c_1 \cos x + c_2 \sin x)$ and we assume $y_p = Ae^{-4x}$. Substituting into the differential equation we find $A = 5$. Thus $y = e^{-2x}(c_1 \cos x + c_2 \sin x) + 7e^{-4x}$. From the initial conditions we obtain $c_1 = -10$ and $c_2 = 9$, so

$$y = e^{-2x}(-10 \cos x + 9 \sin x + 7e^{-4x}).$$

32. We have $y_c = c_1 \cosh x + c_2 \sinh x$ and we assume $y_p = Ax \cosh x + Bx \sinh x$. Substituting into the differential equation we find $A = 0$ and $B = \frac{1}{2}$. Thus

$$y = c_1 \cosh x + c_2 \sinh x + \frac{1}{2}x \sinh x.$$

From the initial conditions we obtain $c_1 = 2$ and $c_2 = 12$, so

$$y = 2 \cosh x + 12 \sinh x + \frac{1}{2}x \sinh x.$$

33. We have $x_c = c_1 \cos \omega t + c_2 \sin \omega t$ and we assume $x_p = At \cos \omega t + Bt \sin \omega t$. Substituting into the differential equation we find $A = -F_0/2\omega$ and $B = 0$. Thus $x = c_1 \cos \omega t + c_2 \sin \omega t - (F_0/2\omega)t \cos \omega t$. From the initial conditions we obtain $c_1 = 0$ and $c_2 = F_0/2\omega^2$, so

$$x = (F_0/2\omega^2) \sin \omega t - (F_0/2\omega)t \cos \omega t.$$

34. We have $x_c = c_1 \cos \omega t + c_2 \sin \omega t$ and we assume $x_p = A \cos \gamma t + B \sin \gamma t$, where $\gamma \neq \omega$. Substituting into the differential equation we find $A = F_0/(\omega^2 - \gamma^2)$ and $B = 0$. Thus

$$x = c_1 \cos \omega t + c_2 \sin \omega t + \frac{F_0}{(\omega^2 - \gamma^2)} \cos \gamma t.$$

From the initial conditions we obtain $c_1 = F_0/(\omega^2 - \gamma^2)$ and $c_2 = 0$, so

$$x = \frac{F_0}{(\omega^2 - \gamma^2)} \cos \omega t + \frac{F_0}{(\omega^2 - \gamma^2)} \cos \gamma t.$$

35. We have $y_c = c_1 + c_2 e^x + c_3 x e^x$ and we assume $y_p = Ax + Bx^2 e^x + Ce^{5x}$. Substituting into the differential equation we find $A = 2$, $B = -12$, and $C = \frac{1}{2}$. Thus

$$y = c_1 + c_2 e^x + c_3 x e^x + 2x - 12x^2 e^x + \frac{1}{2}e^{5x}.$$

From the initial conditions we obtain $c_1 = 11$, $c_2 = -11$, and $c_3 = 9$, so

$$y = 11 - 11e^x + 9xe^x + 2x - 12x^2 e^x + \frac{1}{2}e^{5x}.$$

36. We have $y_c = c_1 e^{-2x} + e^x(c_2 \cos \sqrt{3}x + c_3 \sin \sqrt{3}x)$ and we assume $y_p = Ax + B + Cx e^{-2x}$. Substituting into the differential equation we find $A = \frac{1}{4}$, $B = -\frac{5}{8}$, and $C = \frac{2}{3}$. Thus

$$y = c_1 e^{-2x} + e^x(c_2 \cos \sqrt{3}x + c_3 \sin \sqrt{3}x) + \frac{1}{4}x - \frac{5}{8} + \frac{2}{3}x e^{-2x}.$$

From the initial conditions we obtain $c_1 = -\frac{23}{12}$, $c_2 = -\frac{59}{24}$, and $c_3 = \frac{17}{72}\sqrt{3}$, so

$$y = -\frac{23}{12}e^{-2x} + e^x\left(-\frac{59}{24}\cos \sqrt{3}x + \frac{17}{72}\sqrt{3}\sin \sqrt{3}x\right) + \frac{1}{4}x - \frac{5}{8} + \frac{2}{3}x e^{-2x}.$$

Exercises 3.4

37. We have $y_c = c_1 \cos x + c_2 \sin x$ and we assume $y_p = A^2 + Bx + C$. Substituting into the differential equation we find $A = 1$, $B = 0$, and $C = -1$. Thus $y = c_1 \cos x + c_2 \sin x + x^2 - 1$. From $y(0) = 5$ and $y(1) = 0$ we obtain

$$\begin{aligned}c_1 - 1 &= 5 \\(\cos 1)c_1 + \sin(1)c_2 &= 0.\end{aligned}$$

Solving this system we find $c_1 = 6$ and $c_2 = -6 \cot 1$. The solution of the boundary-value problem is

$$y = 6 \cos x - 6(\cot 1) \sin x + x^2 - 1.$$

38. We have $y_c = e^x(c_1 \cos x + c_2 \sin x)$ and we assume $y_p = Ax + B$. Substituting into the differential equation we find $A = 1$ and $B = 0$. Thus $y = e^x(c_1 \cos x + c_2 \sin x) + x$. From $y(0) = 0$ and $y(\pi) = \pi$ we obtain

$$\begin{aligned}c_1 &= 0 \\ \pi - e^\pi c_1 &= \pi.\end{aligned}$$

Solving this system we find $c_1 = 0$ and c_2 is any real number. The solution of the boundary-value problem is

$$y = c_2 e^x \sin x + x.$$

39. We have $y_c = c_1 \cos 2x + c_2 \sin 2x$ and we assume $y_p = A \cos x + B \sin x$ on $[0, \pi/2]$. Substituting into the differential equation we find $A = 0$ and $B = \frac{1}{3}$. Thus $y = c_1 \cos 2x + c_2 \sin 2x + \frac{1}{3} \sin x$ on $[0, \pi/2]$. On $(\pi/2, \infty)$ we have $y = c_3 \cos 2x + c_4 \sin 2x$. From $y(0) = 1$ and $y'(0) = 2$ we obtain

$$\begin{aligned}c_1 &= 1 \\ \frac{1}{3} + 2c_2 &= 2.\end{aligned}$$

Solving this system we find $c_1 = 1$ and $c_2 = \frac{5}{6}$. Thus $y = \cos 2x + \frac{5}{6} \sin 2x + \frac{1}{3} \sin x$ on $[0, \pi/2]$. Now continuity of y at $x = \pi/2$ implies

$$\cos \pi + \frac{5}{6} \sin \pi + \frac{1}{3} \sin \frac{\pi}{2} = c_3 \cos \pi + c_4 \sin \pi$$

or $-1 + \frac{1}{3} = -c_3$. Hence $c_3 = \frac{2}{3}$. Continuity of y' at $x = \pi/2$ implies

$$-2 \sin \pi + \frac{5}{3} \cos \pi + \frac{1}{3} \cos \frac{\pi}{2} = -2c_3 \sin \pi + 2c_4 \cos \pi$$

or $-\frac{5}{3} = -2c_4$. Then $c_4 = \frac{5}{6}$ and the solution of the boundary-value problem is

$$y(x) = \begin{cases} \cos 2x + \frac{5}{6} \sin 2x + \frac{1}{3} \sin x, & 0 \leq x \leq \pi/2 \\ \frac{2}{3} \cos 2x + \frac{5}{6} \sin 2x, & x > \pi/2. \end{cases}$$

40. The complementary function is $y_c = e^{2x}(c_1 \cos 2x + c_2 \sin 2x)$. We assume a particular solution of the form $y_p = (Ax^3 + Bx^2 + Cx)e^{2x} \cos 2x + (Dx^3 + Ex^2 + F)e^{2x} \sin 2x$. Substituting into the differential equation and using a CAS to simplify yields

$$\begin{aligned}[12Dx^2 + (6A + 8E)x + (2B + 4F)]e^{2x} \cos 2x + [-12Ax^2 + (-8B + 6D)x + (-4C + 2E)]e^{2x} \sin 2x \\ = (2x^2 - 3x)e^{2x} \cos 2x + (10x^2 - x - 1)e^{2x} \sin 2x.\end{aligned}$$

This gives the system of equations

$$\begin{aligned}12D &= 2, & 6A + 8E &= -3, & 2B + 4F &= 0, \\ -12A &= 10, & -8B + 6D &= -1, & -4C + 2E &= -1,\end{aligned}$$

from which we find $A = -\frac{5}{6}$, $B = \frac{1}{4}$, $C = \frac{3}{8}$, $D = \frac{1}{6}$, $E = \frac{1}{4}$, and $F = -\frac{1}{8}$. Thus, a particular solution of the differential equation is

$$y_p = \left(-\frac{5}{6}x^3 + \frac{1}{4}x^2 + \frac{3}{8}x\right)e^{2x} \cos 2x + \left(\frac{1}{6}x^3 + \frac{1}{4}x^2 - \frac{1}{8}x\right)e^{2x} \sin 2x.$$

41. $f(t) = e^t \sin t$. We see that $y_p \rightarrow \infty$ as $t \rightarrow \infty$ and $y_p \rightarrow 0$ as $t \rightarrow -\infty$.
42. $f(t) = e^{-t}$. We see that $y_p \rightarrow \infty$ as $t \rightarrow \infty$ and $y_p \rightarrow \infty$ as $t \rightarrow -\infty$.
43. $f(t) = \sin 2t$. We see that y_p is sinusoidal.
44. $f(t) = 1$. We see that y_p is constant and simply translates y_c vertically.

Exercises 3.5

The particular solution, $y_p = u_1 y_1 + u_2 y_2$, in the following problems can take on a variety of forms, especially where trigonometric functions are involved. The validity of a particular form can best be checked by substituting it back into the differential equation.

1. The auxiliary equation is $m^2 + 1 = 0$, so $y_c = c_1 \cos x + c_2 \sin x$ and

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = 1.$$

Identifying $f(x) = \sec x$ we obtain

$$u_1' = -\frac{\sin x \sec x}{1} = -\tan x$$

$$u_2' = \frac{\cos x \sec x}{1} = 1.$$

Then $u_1 = \ln |\cos x|$, $u_2 = x$, and

$$y = c_1 \cos x + c_2 \sin x + \cos x \ln |\cos x| + x \sin x.$$

2. The auxiliary equation is $m^2 + 1 = 0$, so $y_c = c_1 \cos x + c_2 \sin x$ and

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = 1.$$

Identifying $f(x) = \tan x$ we obtain

$$u_1' = -\sin x \tan x = \frac{\cos^2 x - 1}{\cos x} = \cos x - \sec x$$

$$u_2' = \sin x.$$

Then $u_1 = \sin x - \ln |\sec x + \tan x|$, $u_2 = -\cos x$, and

$$y = c_1 \cos x + c_2 \sin x + \cos x (\sin x - \ln |\sec x + \tan x|) - \cos x \sin x.$$

3. The auxiliary equation is $m^2 + 1 = 0$, so $y_c = c_1 \cos x + c_2 \sin x$ and

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = 1.$$

Identifying $f(x) = \sin x$ we obtain

$$u_1' = -\sin^2 x$$

$$u_2' = \cos x \sin x.$$

Exercises 3.5

Then

$$u_1 = \frac{1}{4} \sin 2x - \frac{1}{2}x = \frac{1}{2} \sin x \cos x - \frac{1}{2}x$$

$$u_2 = -\frac{1}{2} \cos^2 x.$$

and

$$y = c_1 \cos x + c_2 \sin x + \frac{1}{2} \sin x \cos^2 x - \frac{1}{2}x \cos x - \frac{1}{2} \cos^2 x \sin x$$

$$= c_1 \cos x + c_2 \sin x - \frac{1}{2}x \cos x.$$

4. The auxiliary equation is $m^2 + 1 = 0$, so $y_c = c_1 \cos x + c_2 \sin x$ and

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = 1.$$

Identifying $f(x) = \sec x \tan x$ we obtain

$$u_1' = -\sin x (\sec x \tan x) = -\tan^2 x = 1 - \sec^2 x$$

$$u_2' = \cos x (\sec x \tan x) = \tan x.$$

Then $u_1 = x - \tan x$, $u_2 = -\ln |\cos x|$, and

$$y = c_1 \cos x + c_2 \sin x + x \cos x - \sin x - \sin x \ln |\cos x|$$

$$= c_1 \cos x + c_3 \sin x + x \cos x - \sin x \ln |\cos x|.$$

5. The auxiliary equation is $m^2 + 1 = 0$, so $y_c = c_1 \cos x + c_2 \sin x$ and

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = 1.$$

Identifying $f(x) = \cos^2 x$ we obtain

$$u_1' = -\sin x \cos^2 x$$

$$u_2' = \cos^3 x = \cos x (1 - \sin^2 x).$$

Then $u_1 = \frac{1}{3} \cos^3 x$, $u_2 = \sin x - \frac{1}{3} \sin^3 x$, and

$$y = c_1 \cos x + c_2 \sin x + \frac{1}{3} \cos^4 x + \sin^2 x - \frac{1}{3} \sin^4 x$$

$$= c_1 \cos x + c_2 \sin x + \frac{1}{3} (\cos^2 x + \sin^2 x) (\cos^2 x - \sin^2 x) + \sin^2 x$$

$$= c_1 \cos x + c_2 \sin x + \frac{1}{3} \cos^2 x + \frac{2}{3} \sin^2 x$$

$$= c_1 \cos x + c_2 \sin x + \frac{1}{3} + \frac{1}{3} \sin^2 x.$$

6. The auxiliary equation is $m^2 + 1 = 0$, so $y_c = c_1 \cos x + c_2 \sin x$ and

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = 1.$$

Identifying $f(x) = \sec^2 x$ we obtain

$$u_1' = -\frac{\sin x}{\cos^2 x}$$

$$u_2' = \sec x.$$

Then

$$u_1 = -\frac{1}{\cos x} = -\sec x$$

$$u_2 = \ln |\sec x + \tan x|$$

and

$$\begin{aligned} y &= c_1 \cos x + c_2 \sin x - \cos x \sec x + \sin x \ln |\sec x + \tan x| \\ &= c_1 \cos x + c_2 \sin x - 1 + \sin x \ln |\sec x + \tan x|. \end{aligned}$$

7. The auxiliary equation is $m^2 - 1 = 0$, so $y_c = c_1 e^x + c_2 e^{-x}$ and

$$W = \begin{vmatrix} e^x & e^{-x} \\ e^x & -e^{-x} \end{vmatrix} = -2.$$

Identifying $f(x) = \cosh x = \frac{1}{2}(e^{-x} + e^x)$ we obtain

$$u_1' = \frac{1}{4}e^{2x} + \frac{1}{4}$$

$$u_2' = -\frac{1}{4} - \frac{1}{4}e^{2x}.$$

Then

$$u_1 = -\frac{1}{8}e^{-2x} + \frac{1}{4}x$$

$$u_2 = -\frac{1}{8}e^{2x} - \frac{1}{4}x$$

and

$$\begin{aligned} y &= c_1 e^x + c_2 e^{-x} - \frac{1}{8}e^{-x} + \frac{1}{4}xe^x - \frac{1}{8}e^x - \frac{1}{4}xe^{-x} \\ &= c_3 e^x + c_4 e^{-x} + \frac{1}{4}x(e^x - e^{-x}) \\ &= c_3 e^x + c_4 e^{-x} + \frac{1}{2}x \sinh x. \end{aligned}$$

8. The auxiliary equation is $m^2 - 1 = 0$, so $y_c = c_1 e^x + c_2 e^{-x}$ and

$$W = \begin{vmatrix} e^x & e^{-x} \\ e^x & -e^{-x} \end{vmatrix} = -2.$$

Identifying $f(x) = \sinh 2x$ we obtain

$$u_1' = -\frac{1}{4}e^{-3x} + \frac{1}{4}e^x$$

$$u_2' = \frac{1}{4}e^{-x} - \frac{1}{4}e^{3x}.$$

Then

$$u_1 = \frac{1}{12}e^{-3x} + \frac{1}{4}e^x$$

$$u_2 = -\frac{1}{4}e^{-x} - \frac{1}{12}e^{3x}.$$

and

$$\begin{aligned} y &= c_1 e^x + c_2 e^{-x} + \frac{1}{12}e^{-2x} + \frac{1}{4}e^{2x} - \frac{1}{4}e^{-2x} - \frac{1}{12}e^{2x} \\ &= c_1 e^x + c_2 e^{-x} + \frac{1}{6}(e^{2x} - e^{-2x}) \\ &= c_1 e^x + c_2 e^{-x} + \frac{1}{3} \sinh 2x. \end{aligned}$$

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9. The auxiliary equation is $m^2 - 4 = 0$, so $y_c = c_1 e^{2x} + c_2 e^{-2x}$ and

$$W = \begin{vmatrix} e^{2x} & e^{-2x} \\ 2e^{2x} & -2e^{-2x} \end{vmatrix} = -4.$$

Identifying $f(x) = e^{2x}/x$ we obtain $u'_1 = 1/4x$ and $u'_2 = -e^{4x}/4x$. Then

$$u_1 = \frac{1}{4} \ln |x|,$$

$$u_2 = -\frac{1}{4} \int_{x_0}^x \frac{e^{4t}}{t} dt$$

and

$$y = c_1 e^{2x} + c_2 e^{-2x} + \frac{1}{4} \left(e^{2x} \ln |x| - e^{-2x} \int_{x_0}^x \frac{e^{4t}}{t} dt \right), \quad x_0 > 0.$$

10. The auxiliary equation is $m^2 - 9 = 0$, so $y_c = c_1 e^{3x} + c_2 e^{-3x}$ and

$$W = \begin{vmatrix} e^{3x} & e^{-3x} \\ 3e^{3x} & -3e^{-3x} \end{vmatrix} = -6.$$

Identifying $f(x) = 9x/e^{3x}$ we obtain $u'_1 = \frac{3}{2}x e^{-6x}$ and $u'_2 = -\frac{3}{2}x$. Then

$$u_1 = -\frac{1}{24} e^{-6x} - \frac{1}{4} x e^{-6x},$$

$$u_2 = -\frac{3}{4} x^2$$

and

$$y = c_1 e^{3x} + c_2 e^{-3x} - \frac{1}{24} e^{-3x} - \frac{1}{4} x e^{-3x} - \frac{3}{4} x^2 e^{-3x}$$

$$= c_1 e^{3x} + c_3 e^{-3x} - \frac{1}{4} x e^{-3x} (1 - 3x).$$

11. The auxiliary equation is $m^2 + 3m + 2 = (m+1)(m+2) = 0$, so $y_c = c_1 e^{-x} + c_2 e^{-2x}$ and

$$W = \begin{vmatrix} e^{-x} & e^{-2x} \\ -e^{-x} & -2e^{-2x} \end{vmatrix} = -e^{-3x}.$$

Identifying $f(x) = 1/(1+e^x)$ we obtain

$$u'_1 = \frac{e^x}{1+e^x}$$

$$u'_2 = -\frac{e^{2x}}{1+e^x} = \frac{e^x}{1+e^x} - e^x.$$

Then $u_1 = \ln(1+e^x)$, $u_2 = \ln(1+e^x) - e^x$, and

$$y = c_1 e^{-x} + c_2 e^{-2x} + e^{-x} \ln(1+e^x) + e^{-2x} \ln(1+e^x) - e^{-x}$$

$$= c_3 e^{-x} + c_2 e^{-2x} + (1+e^{-x}) e^{-x} \ln(1+e^x).$$

12. The auxiliary equation is $m^2 - 2m + 1 = (m-1)^2 = 0$, so $y_c = c_1 e^x + c_2 x e^x$ and

$$W = \begin{vmatrix} e^x & x e^x \\ e^x & x e^x + e^x \end{vmatrix} = e^{2x}.$$

Identifying $f(x) = e^x/(1+x^2)$ we obtain

$$u'_1 = -\frac{x e^x e^x}{e^{2x} (1+x^2)} = -\frac{x}{1+x^2}$$

$$u'_2 = \frac{e^x e^x}{e^{2x} (1+x^2)} = \frac{1}{1+x^2}.$$

Then $u_1 = -\frac{1}{2} \ln(1+x^2)$, $u_2 = \tan^{-1} x$, and

$$y = c_1 e^x + c_2 x e^x - \frac{1}{2} e^x \ln(1+x^2) + x e^x \tan^{-1} x.$$

13. The auxiliary equation is $m^2 + 3m + 2 = (m+1)(m+2) = 0$, so $y_c = c_1 e^{-x} + c_2 e^{-2x}$ and

$$W = \begin{vmatrix} e^{-x} & e^{-2x} \\ -e^{-x} & -2e^{-2x} \end{vmatrix} = -e^{-3x}.$$

Identifying $f(x) = \sin e^x$ we obtain

$$u'_1 = \frac{e^{-2x} \sin e^x}{e^{-3x}} = e^x \sin e^x$$

$$u'_2 = \frac{e^{-x} \sin e^x}{-e^{-3x}} = -e^{2x} \sin e^x.$$

Then $u_1 = -\cos e^x$, $u_2 = e^x \cos x - \sin e^x$, and

$$y = c_1 e^{-x} + c_2 e^{-2x} - e^{-x} \cos e^x + e^{-x} \cos e^x - e^{-2x} \sin e^x$$

$$= c_1 e^{-x} + c_2 e^{-2x} - e^{-2x} \sin e^x.$$

14. The auxiliary equation is $m^2 - 2m + 1 = (m-1)^2 = 0$, so $y_c = c_1 e^t + c_2 t e^t$ and

$$W = \begin{vmatrix} e^t & t e^t \\ e^t & t e^t + e^t \end{vmatrix} = e^{2t}.$$

Identifying $f(t) = e^t \tan^{-1} t$ we obtain

$$u'_1 = -\frac{t e^t e^t \tan^{-1} t}{e^{2t}} = -t \tan^{-1} t$$

$$u'_2 = \frac{e^t e^t \tan^{-1} t}{e^{2t}} = \tan^{-1} t.$$

Then

$$u_1 = -\frac{1+t^2}{2} \tan^{-1} t + \frac{t}{2}$$

$$u_2 = t \tan^{-1} t - \frac{1}{2} \ln(1+t^2)$$

and

$$y = c_1 e^t + c_2 t e^t + \left(-\frac{1+t^2}{2} \tan^{-1} t + \frac{t}{2} \right) e^t + \left(t \tan^{-1} t - \frac{1}{2} \ln(1+t^2) \right) t e^t$$

$$= c_1 e^t + c_3 t e^t + \frac{1}{2} e^t [(t^2 - 1) \tan^{-1} t - \ln(1+t^2)].$$

15. The auxiliary equation is $m^2 + 2m + 1 = (m+1)^2 = 0$, so $y_c = c_1 e^{-t} + c_2 t e^{-t}$ and

$$W = \begin{vmatrix} e^{-t} & t e^{-t} \\ -e^{-t} & -t e^{-t} + e^{-t} \end{vmatrix} = e^{-2t}.$$

Identifying $f(t) = e^{-t} \ln t$ we obtain

$$u'_1 = -\frac{t e^{-t} e^{-t} \ln t}{e^{-2t}} = -t \ln t$$

$$u'_2 = \frac{e^{-t} e^{-t} \ln t}{e^{-2t}} = \ln t.$$

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Then

$$u_1 = -\frac{1}{2}t^2 \ln t + \frac{1}{4}t^2$$

$$u_2 = t \ln t - t$$

and

$$\begin{aligned} y &= c_1 e^{-t} + c_2 t e^{-t} - \frac{1}{2}t^2 e^{-t} \ln t + \frac{1}{4}t^2 e^{-t} + t^2 e^{-t} \ln t - t^2 e^{-t} \\ &= c_1 e^{-t} + c_2 t e^{-t} + \frac{1}{2}t^2 e^{-t} \ln t - \frac{3}{4}t^2 e^{-t}. \end{aligned}$$

16. The auxiliary equation is $2m^2 + 2m + 1 = 0$, so $y_c = e^{-x/2}(c_1 \cos x/2 + c_2 \sin x/2)$ and

$$W = \begin{vmatrix} e^{-x/2} \cos \frac{x}{2} & e^{-x/2} \sin \frac{x}{2} \\ -\frac{1}{2}e^{-x/2} \cos \frac{x}{2} - \frac{1}{2}e^{-x/2} \sin \frac{x}{2} & \frac{1}{2}e^{-x/2} \cos \frac{x}{2} - \frac{1}{2}e^{-x/2} \sin \frac{x}{2} \end{vmatrix} = \frac{1}{2}e^{-x}.$$

Identifying $f(x) = 2\sqrt{x}$ we obtain

$$u_1' = -\frac{e^{-x/2} \sin(x/2) 2\sqrt{x}}{e^{-x/2}} = -4e^{x/2} \sqrt{x} \sin \frac{x}{2}$$

$$u_2' = -\frac{e^{-x/2} \cos(x/2) 2\sqrt{x}}{e^{-x/2}} = 4e^{x/2} \sqrt{x} \cos \frac{x}{2}.$$

Then

$$u_1 = -4 \int_{x_0}^x e^{t/2} \sqrt{t} \sin \frac{t}{2} dt$$

$$u_2 = 4 \int_{x_0}^x e^{t/2} \sqrt{t} \cos \frac{t}{2} dt$$

and

$$y = e^{-x/2} \left(c_1 \cos \frac{x}{2} + c_2 \sin \frac{x}{2} \right) - 4e^{-x/2} \cos \frac{x}{2} \int_{x_0}^x e^{t/2} \sqrt{t} \sin \frac{t}{2} dt + 4e^{-x/2} \sin \frac{x}{2} \int_{x_0}^x e^{t/2} \sqrt{t} \cos \frac{t}{2} dt.$$

17. The auxiliary equation is $3m^2 - 6m + 6 = 0$, so $y_c = e^x(c_1 \cos x + c_2 \sin x)$ and

$$W = \begin{vmatrix} e^x \cos x & e^x \sin x \\ e^x \cos x - e^x \sin x & e^x \cos x + e^x \sin x \end{vmatrix} = e^{2x}.$$

Identifying $f(x) = \frac{1}{3}e^x \sec x$ we obtain

$$u_1' = -\frac{(e^x \sin x)(e^x \sec x)/3}{e^{2x}} = -\frac{1}{3} \tan x$$

$$u_2' = \frac{(e^x \cos x)(e^x \sec x)/3}{e^{2x}} = \frac{1}{3}.$$

Then $u_1 = \frac{1}{3} \ln(\cos x)$, $u_2 = \frac{1}{3}x$, and

$$y = c_1 e^x \cos x + c_2 e^x \cos x + \frac{1}{3} \ln(\cos x) e^x \cos x + \frac{1}{3} x e^x \sin x.$$

18. The auxiliary equation is $4m^2 - 4m + 1 = (2m - 1)^2 = 0$, so $y_c = c_1 e^{x/2} + c_2 x e^{x/2}$ and

$$W = \begin{vmatrix} e^{x/2} & x e^{x/2} \\ \frac{1}{2} e^{x/2} & \frac{1}{2} x e^{x/2} + e^{x/2} \end{vmatrix} = e^x.$$

Identifying $f(x) = \frac{1}{4}e^{x/2}\sqrt{1-x^2}$ we obtain

$$u'_1 = -\frac{xe^{x/2}e^{x/2}\sqrt{1-x^2}}{4e^x} = -\frac{1}{4}x\sqrt{1-x^2}$$

$$u'_2 = \frac{e^{x/2}e^{x/2}\sqrt{1-x^2}}{4e^x} = \frac{1}{4}\sqrt{1-x^2}.$$

Then

$$u_1 = \frac{1}{12}(1-x^2)^{3/2}$$

$$u_2 = \frac{x}{8}\sqrt{1-x^2} + \frac{1}{8}\sin^{-1}x$$

and

$$y = c_1e^{x/2} + c_2xe^{x/2} + \frac{1}{12}e^{x/2}(1-x^2)^{3/2} + \frac{1}{8}x^2e^{x/2}\sqrt{1-x^2} + \frac{1}{8}xe^{x/2}\sin^{-1}x.$$

19. The auxiliary equation is $4m^2 - 1 = (2m-1)(2m+1) = 0$, so $y_c = c_1e^{x/2} + c_2e^{-x/2}$ and

$$W = \begin{vmatrix} e^{x/2} & e^{-x/2} \\ \frac{1}{2}e^{x/2} & -\frac{1}{2}e^{-x/2} \end{vmatrix} = -1.$$

Identifying $f(x) = xe^{x/2}/4$ we obtain $u'_1 = x/4$ and $u'_2 = -xe^x/4$. Then $u_1 = x^2/8$ and $u_2 = -xe^x/4 + e^x/4$.

Thus

$$\begin{aligned} y &= c_1e^{x/2} + c_2e^{-x/2} + \frac{1}{8}x^2e^{x/2} - \frac{1}{4}xe^{x/2} + \frac{1}{4}e^{x/2} \\ &= c_3e^{x/2} + c_2e^{-x/2} + \frac{1}{8}x^2e^{x/2} - \frac{1}{4}xe^{x/2} \end{aligned}$$

and

$$y' = \frac{1}{2}c_3e^{x/2} - \frac{1}{2}c_2e^{-x/2} + \frac{1}{16}x^2e^{x/2} + \frac{1}{8}xe^{x/2} - \frac{1}{4}e^{x/2}.$$

The initial conditions imply

$$c_3 + c_2 = 1$$

$$\frac{1}{2}c_3 - \frac{1}{2}c_2 - \frac{1}{4} = 0.$$

Thus $c_3 = 3/4$ and $c_2 = 1/4$, and

$$y = \frac{3}{4}e^{x/2} + \frac{1}{4}e^{-x/2} + \frac{1}{8}x^2e^{x/2} - \frac{1}{4}xe^{x/2}.$$

20. The auxiliary equation is $2m^2 + m - 1 = (2m-1)(m+1) = 0$, so $y_c = c_1e^{x/2} + c_2e^{-x}$ and

$$W = \begin{vmatrix} e^{x/2} & e^{-x} \\ \frac{1}{2}e^{x/2} & -e^{-x} \end{vmatrix} = -\frac{3}{2}e^{-x/2}.$$

Identifying $f(x) = (x+1)/2$ we obtain

$$u'_1 = \frac{1}{3}e^{-x/2}(x+1)$$

$$u'_2 = -\frac{1}{3}e^x(x+1).$$

Then

$$u_1 = -e^{-x/2}\left(\frac{2}{3}x - 2\right)$$

$$u_2 = -\frac{1}{3}xe^x.$$

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Thus

$$y = c_1 e^{x/2} + c_2 e^{-x} - x - 2$$

and

$$y' = \frac{1}{2} c_1 e^{x/2} - c_2 e^{-x} - 1.$$

The initial conditions imply

$$c_1 - c_2 - 2 = 1$$

$$\frac{1}{2} c_1 - c_2 - 1 = 0.$$

Thus $c_1 = 8/3$ and $c_2 = 1/3$, and

$$y = \frac{8}{3} e^{x/2} + \frac{1}{3} e^{-x} - x - 2.$$

- 21.** The auxiliary equation is $m^2 + 2m - 8 = (m - 2)(m + 4) = 0$, so $y_c = c_1 e^{2x} + c_2 e^{-4x}$ and

$$W = \begin{vmatrix} e^{2x} & e^{-4x} \\ 2e^{2x} & -4e^{-4x} \end{vmatrix} = -6e^{-2x}.$$

Identifying $f(x) = 2e^{-2x} - e^{-x}$ we obtain

$$u_1' = \frac{1}{3} e^{-4x} - \frac{1}{6} e^{-3x}$$

$$u_2' = -\frac{1}{6} e^{3x} - \frac{1}{3} e^{2x}.$$

Then

$$u_1 = -\frac{1}{12} e^{-4x} + \frac{1}{18} e^{-3x}$$

$$u_2 = \frac{1}{18} e^{3x} - \frac{1}{6} e^{2x}.$$

Thus

$$y = c_1 e^{2x} + c_2 e^{-4x} - \frac{1}{12} e^{-2x} + \frac{1}{18} e^{-x} + \frac{1}{18} e^{-x} - \frac{1}{6} e^{-2x}$$

$$= c_1 e^{2x} + c_2 e^{-4x} - \frac{1}{4} e^{-2x} + \frac{1}{9} e^{-x}$$

and

$$y' = 2c_1 e^{2x} - 4c_2 e^{-4x} + \frac{1}{2} e^{-2x} - \frac{1}{9} e^{-x}.$$

The initial conditions imply

$$c_1 + c_2 - \frac{5}{36} = 1$$

$$2c_1 - 4c_2 + \frac{7}{18} = 0.$$

Thus $c_1 = 25/36$ and $c_2 = 4/9$, and

$$y = \frac{25}{36} e^{2x} + \frac{4}{9} e^{-4x} - \frac{1}{4} e^{-2x} + \frac{1}{9} e^{-x}.$$

- 22.** The auxiliary equation is $m^2 - 4m + 4 = (m - 2)^2 = 0$, so $y_c = c_1 e^{2x} + c_2 x e^{2x}$ and

$$W = \begin{vmatrix} e^{2x} & x e^{2x} \\ 2e^{2x} & 2x e^{2x} + e^{2x} \end{vmatrix} = e^{4x}.$$

Identifying $f(x) = (12x^2 - 6x)e^{2x}$ we obtain

$$u'_1 = 6x^2 - 12x^3$$

$$u'_2 = 12x^2 - 6x.$$

Then

$$u_1 = 2x^3 - 3x^4$$

$$u_2 = 4x^3 - 3x^2.$$

Thus

$$\begin{aligned} y &= c_1 e^{2x} + c_2 x e^{2x} + (2x^3 - 3x^4) e^{2x} + (4x^3 - 3x^2) x e^{2x} \\ &= c_1 e^{2x} + c_2 x e^{2x} + e^{2x} (x^4 - x^3) \end{aligned}$$

and

$$y' = 2c_1 e^{2x} + c_2 (2x e^{2x} + e^{2x}) + e^{2x} (4x^3 - 3x^2) + 2e^{2x} (x^4 - x^3).$$

The initial conditions imply

$$c_1 = 1$$

$$2c_1 + c_2 = 0.$$

Thus $c_1 = 1$ and $c_2 = -2$, and

$$y = e^{2x} - 2x e^{2x} + e^{2x} (x^4 - x^3) = e^{2x} (x^4 - x^3 - 2x + 1).$$

23. Write the equation in the form

$$y'' + \frac{1}{x}y' + \left(1 - \frac{1}{4x^2}\right)y = x^{-1/2}$$

and identify $f(x) = x^{-1/2}$. From $y_1 = x^{-1/2} \cos x$ and $y_2 = x^{-1/2} \sin x$ we compute

$$W(y_1, y_2) = \begin{vmatrix} x^{-1/2} \cos x & x^{-1/2} \sin x \\ -x^{-1/2} \sin x - \frac{1}{2}x^{-3/2} \cos x & x^{-1/2} \cos x - \frac{1}{2}x^{-3/2} \sin x \end{vmatrix} = \frac{1}{x}.$$

Now

$$u'_1 = \sin x \quad \text{so} \quad u_1 = \cos x,$$

and

$$u'_2 = \cos x \quad \text{so} \quad u_2 = \sin x.$$

Thus

$$\begin{aligned} y &= c_1 x^{-1/2} \cos x + c_2 x^{-1/2} \sin x + x^{-1/2} \cos^2 x + x^{-1/2} \sin^2 x \\ &= c_1 x^{-1/2} \cos x + c_2 x^{-1/2} \sin x + x^{-1/2}. \end{aligned}$$

24. Write the equation in the form

$$y'' + \frac{1}{x}y' + \frac{1}{x^2}y = \frac{\sec(\ln x)}{x^2}$$

and identify $f(x) = \sec(\ln x)/x^2$. From $y_1 = \cos(\ln x)$ and $y_2 = \sin(\ln x)$ we compute

$$W = \begin{vmatrix} \cos(\ln x) & \sin(\ln x) \\ -\frac{\sin(\ln x)}{x} & \frac{\cos(\ln x)}{x} \end{vmatrix} = \frac{1}{x}.$$

Now

$$u'_1 = -\frac{\tan(\ln x)}{x} \quad \text{so} \quad u_1 = \ln |\cos(\ln x)|,$$

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and

$$u_2' = \frac{1}{x} \quad \text{so} \quad u_2 = \ln x.$$

Thus, a particular solution is

$$y_p = \cos(\ln x) \ln |\cos(\ln x)| + (\ln x) \sin(\ln x).$$

- 25.** The auxiliary equation is $m^3 + m = m(m^2 + 1) = 0$, so $y_c = c_1 + c_2 \cos x + c_3 \sin x$ and

$$W = \begin{vmatrix} 1 & \cos x & \sin x \\ 0 & -\sin x & \cos x \\ 0 & -\cos x & -\sin x \end{vmatrix} = 1.$$

Identifying $f(x) = \tan x$ we obtain

$$\begin{aligned} u_1' = W_1 &= \begin{vmatrix} 0 & \cos x & \sin x \\ 0 & -\sin x & \cos x \\ \tan x & -\cos x & -\sin x \end{vmatrix} = \tan x \\ u_2' = W_2 &= \begin{vmatrix} 1 & 0 & \sin x \\ 0 & 0 & \cos x \\ 0 & \tan x & -\sin x \end{vmatrix} = -\sin x \\ u_3' = W_3 &= \begin{vmatrix} 1 & \cos x & 0 \\ 0 & -\sin x & 0 \\ 0 & -\cos x & \tan x \end{vmatrix} = -\sin x \tan x = \frac{\cos^2 x - 1}{\cos x} = \cos x - \sec x. \end{aligned}$$

Then

$$\begin{aligned} u_1 &= -\ln |\cos x| \\ u_2 &= \cos x \\ u_3 &= \sin x - \ln |\sec x + \tan x| \end{aligned}$$

and

$$\begin{aligned} y &= c_1 + c_2 \cos x + c_3 \sin x - \ln |\cos x| + \cos^2 x \\ &\quad + \sin^2 x - \sin x \ln |\sec x + \tan x| \\ &= c_4 + c_2 \cos x + c_3 \sin x - \ln |\cos x| - \sin x \ln |\sec x + \tan x| \end{aligned}$$

for $-\infty < x < \infty$.

- 26.** The auxiliary equation is $m^3 + 4m = m(m^2 + 4) = 0$, so $y_c = c_1 + c_2 \cos 2x + c_3 \sin 2x$ and

$$W = \begin{vmatrix} 1 & \cos 2x & \sin 2x \\ 0 & -2 \sin 2x & 2 \cos 2x \\ 0 & -4 \cos 2x & -4 \sin 2x \end{vmatrix} = 8.$$

Identifying $f(x) = \sec 2x$ we obtain

$$u'_1 = \frac{1}{8}W_1 = \frac{1}{8} \begin{vmatrix} 0 & \cos 2x & \sin 2x \\ 0 & -2 \sin 2x & 2 \cos 2x \\ \sec 2x & -4 \cos 2x & -4 \sin 2x \end{vmatrix} = \frac{1}{4} \sec 2x$$

$$u'_2 = \frac{1}{8}W_2 = \frac{1}{8} \begin{vmatrix} 1 & 0 & \sin 2x \\ 0 & 0 & 2 \cos 2x \\ 0 & \sec 2x & -4 \sin 2x \end{vmatrix} = -\frac{1}{4}$$

$$u'_3 = \frac{1}{8}W_3 = \frac{1}{8} \begin{vmatrix} 1 & \cos 2x & 0 \\ 0 & -2 \sin 2x & 0 \\ 0 & -4 \cos 2x & \sec 2x \end{vmatrix} = -\frac{1}{4} \tan 2x.$$

Then

$$u_1 = \frac{1}{8} \ln |\sec 2x + \tan 2x|$$

$$u_2 = -\frac{1}{4}x$$

$$u_3 = \frac{1}{8} \ln |\cos 2x|$$

and

$$y = c_1 + c_2 \cos 2x + c_3 \sin 2x + \frac{1}{8} \ln |\sec 2x + \tan 2x| - \frac{1}{4}x \cos 2x + \frac{1}{8} \sin 2x \ln |\cos 2x|$$

for $-\pi/4 < x < \pi/4$.

- 27.** The auxiliary equation is $3m^2 - 6m + 30 = 0$, which has roots $1 + 3i$, so $y_c = e^x(c_1 \cos 3x + c_2 \sin 3x)$. We consider first the differential equation $3y'' - 6y' + 30y = 15 \sin x$, which can be solved using undetermined coefficients. Letting $y_{p_1} = A \cos x + B \sin x$ and substituting into the differential equation we get

$$(27A - 6B) \cos x + (6A + 27B) \sin x = 15 \sin x.$$

Then

$$27A - 6B = 0 \quad \text{and} \quad 6A + 27B = 15,$$

so $A = \frac{2}{17}$ and $B = \frac{9}{17}$. Thus, $y_{p_1} = \frac{2}{17} \cos x + \frac{9}{17} \sin x$. Next, we consider the differential equation $3y'' - 6y' + 30y$, for which a particular solution y_{p_2} can be found using variation of parameters. The Wronskian is

$$W = \begin{vmatrix} e^x \cos 3x & e^x \sin 3x \\ e^x \cos 3x - 3e^x \sin 3x & 3e^x \cos 3x + e^x \sin 3x \end{vmatrix} = 3e^{2x}.$$

Identifying $f(x) = \frac{1}{3}e^x \tan x$ we obtain

$$u'_1 = -\frac{1}{9} \sin 3x \tan 3x \quad \text{and} \quad u'_2 = \frac{1}{9} \sin 3x.$$

Then

$$u_1 = \frac{1}{27} \sin 3x + \frac{1}{27} \left[\ln \left(\cos \frac{3x}{2} - \sin \frac{3x}{2} \right) - \ln \left(\cos \frac{3x}{2} + \sin \frac{3x}{2} \right) \right]$$

$$u_2 = -\frac{1}{27} \cos 3x.$$

Thus

$$y_{p_2} = \frac{1}{27} e^x \cos 3x \left[\ln \left(\cos \frac{3x}{2} - \sin \frac{3x}{2} \right) - \ln \left(\cos \frac{3x}{2} + \sin \frac{3x}{2} \right) \right]$$

Exercises 3.5

and the general solution of the original differential equation is

$$y = e^x(c_1 \cos 3x + c_2 \sin 3x) + y_{p_1}(x) + y_{p_2}(x).$$

- 28.** The auxiliary equation is $m^2 - 2m + 1 = (m - 1)^2 = 0$, which has repeated root 1, so $y_c = c_1 e^x + c_2 x e^x$. We consider first the differential equation $y'' - 2y' + y = 4x^2 - 3$, which can be solved using undetermined coefficients. Letting $y_{p_1} = Ax^2 + Bx + C$ and substituting into the differential equation we get

$$Ax^2 + (-4A + B)x + (2A - 2B + C) = 4x^2 - 3.$$

Then

$$A = 4, \quad -4A + B = 0, \quad \text{and} \quad 2A - 2B + C = -3,$$

so $A = 4$, $B = 16$, and $C = 21$. Thus, $y_{p_1} = 4x^2 + 16x + 21$. Next we consider the differential equation $y'' - 2y' + y = x^{-1}e^x$, for which a particular solution y_{p_2} can be found using variation of parameters. The Wronskian is

$$W = \begin{vmatrix} e^x & x e^x \\ e^x & x e^x + e^x \end{vmatrix} = e^{2x}.$$

Identifying $f(x) = e^x/x$ we obtain $u'_1 = -1$ and $u'_2 = 1/x$. Then $u_1 = -x$ and $u_2 = \ln x$, so that

$$y_{p_2} = -x e^x + x e^x \ln x,$$

and the general solution of the original differential equation is

$$y = y_c + y_{p_1} + y_{p_2} = c_1 e^x + c_2 x e^x + 4x^2 + 16x + 21 - x e^x + x e^x \ln x.$$

- 29.** The interval of definition for Problem 1 is $(-\pi/2, \pi/2)$, for Problem 7 is $(-\infty, \infty)$, for Problem 9 is $(0, \infty)$, and for Problem 18 is $(-1, 1)$. In Problem 24 the general solution is

$$y = c_1 \cos(\ln x) + c_2 \sin(\ln x) + \cos(\ln x) \ln |\cos(\ln x)| + (\ln x) \sin(\ln x)$$

for $-\pi/2 < \ln x < \pi/2$ or $e^{-\pi/2} < x < e^{\pi/2}$. The bounds on $\ln x$ are due to the presence of $\sec(\ln x)$ in the differential equation.

- 30.** We are given that $y_1 = x^2$ is a solution of $x^4 y'' + x^3 y' - 4x^2 y = 0$. To find a second solution we use reduction of order. Let $y = x^2 u(x)$. Then the product rule gives

$$y' = x^2 u' + 2xu \quad \text{and} \quad y'' = x^2 u'' + 4xu' + 2u,$$

so

$$x^4 y'' + x^3 y' - 4x^2 y = x^5(xu'' + 5u') = 0.$$

Letting $w = u'$, this becomes $xw' + 5w = 0$. Separating variables and integrating we have

$$\frac{dw}{w} = -\frac{5}{x} dx \quad \text{and} \quad \ln |w| = -5 \ln x + c.$$

Thus, $w = x^{-5}$ and $u = -\frac{1}{4}x^{-4}$. A second solution is then $y_2 = x^2 x^{-4} = 1/x^2$, and the general solution of the homogeneous differential equation is $y_c = c_1 x^2 + c_2/x^2$. To find a particular solution, y_p , we use variation of parameters. The Wronskian is

$$W = \begin{vmatrix} x^2 & 1/x^2 \\ 2x & -2/x^3 \end{vmatrix} = -\frac{4}{x}.$$

Identifying $f(x) = 1/x^4$ we obtain $u'_1 = \frac{1}{4}x^{-5}$ and $u'_2 = -\frac{1}{4}x^{-1}$. Then $u_1 = -\frac{1}{16}x^{-4}$ and $u_2 = -\frac{1}{4} \ln x$, so

$$y_p = -\frac{1}{16}x^{-4}x^2 - \frac{1}{4}(\ln x)x^{-2} = -\frac{1}{16}x^{-2} - \frac{1}{4}x^{-2} \ln x.$$

The general solution is

$$y = c_1 x^2 + \frac{c_2}{x^2} - \frac{1}{16x^2} - \frac{1}{4x^2} \ln x.$$

Exercises 3.6

1. The auxiliary equation is $m^2 - m - 2 = (m + 1)(m - 2) = 0$ so that $y = c_1 x^{-1} + c_2 x^2$.
2. The auxiliary equation is $4m^2 - 4m + 1 = (2m - 1)^2 = 0$ so that $y = c_1 x^{1/2} + c_2 x^{1/2} \ln x$.
3. The auxiliary equation is $m^2 = 0$ so that $y = c_1 + c_2 \ln x$.
4. The auxiliary equation is $m^2 - 4m = m(m - 4) = 0$ so that $y = c_1 + c_2 x^4$.
5. The auxiliary equation is $m^2 + 4 = 0$ so that $y = c_1 \cos(2 \ln x) + c_2 \sin(2 \ln x)$.
6. The auxiliary equation is $m^2 + 4m + 3 = (m + 1)(m + 3) = 0$ so that $y = c_1 x^{-1} + c_2 x^{-3}$.
7. The auxiliary equation is $m^2 - 4m - 2 = 0$ so that $y = c_1 x^{2-\sqrt{6}} + c_2 x^{2+\sqrt{6}}$.
8. The auxiliary equation is $m^2 + 2m - 4 = 0$ so that $y = c_1 x^{-1+\sqrt{5}} + c_2 x^{-1-\sqrt{5}}$.
9. The auxiliary equation is $25m^2 + 1 = 0$ so that $y = c_1 \cos\left(\frac{1}{5} \ln x\right) + c_2 \left(\frac{1}{5} \ln x\right)$.
10. The auxiliary equation is $4m^2 - 1 = (2m - 1)(2m + 1) = 0$ so that $y = c_1 x^{1/2} + c_2 x^{-1/2}$.
11. The auxiliary equation is $m^2 + 4m + 4 = (m + 2)^2 = 0$ so that $y = c_1 x^{-2} + c_2 x^{-2} \ln x$.
12. The auxiliary equation is $m^2 + 7m + 6 = (m + 1)(m + 6) = 0$ so that $y = c_1 x^{-1} + c_2 x^{-6}$.
13. The auxiliary equation is $3m^2 + 3m + 1 = 0$ so that $y = x^{-1/2} \left[c_1 \cos\left(\frac{\sqrt{3}}{6} \ln x\right) + c_2 \sin\left(\frac{\sqrt{3}}{6} \ln x\right) \right]$.
14. The auxiliary equation is $m^2 - 8m + 41 = 0$ so that $y = x^4 [c_1 \cos(5 \ln x) + c_2 \sin(5 \ln x)]$.
15. Assuming that $y = x^m$ and substituting into the differential equation we obtain

$$m(m - 1)(m - 2) - 6 = m^3 - 3m^2 + 2m - 6 = (m - 3)(m^2 + 2) = 0.$$

Thus

$$y = c_1 x^3 + c_2 \cos\left(\sqrt{2} \ln x\right) + c_3 \sin\left(\sqrt{2} \ln x\right).$$

16. Assuming that $y = x^m$ and substituting into the differential equation we obtain

$$m(m - 1)(m - 2) + m - 1 = m^3 - 3m^2 + 3m - 1 = (m - 1)^3 = 0.$$

Thus

$$y = c_1 x + c_2 x \ln x + c_3 x (\ln x)^2.$$

17. Assuming that $y = x^m$ and substituting into the differential equation we obtain

$$m(m - 1)(m - 2)(m - 3) + 6m(m - 1)(m - 2) = m^4 - 7m^2 + 6m = m(m - 1)(m - 2)(m + 3) = 0.$$

Thus

$$y = c_1 + c_2 x + c_3 x^2 + c_4 x^{-3}.$$

18. Assuming that $y = x^m$ and substituting into the differential equation we obtain

$$m(m - 1)(m - 2)(m - 3) + 6m(m - 1)(m - 2) + 9m(m - 1) + 3m + 1 = m^4 + 2m^2 + 1 = (m^2 + 1)^2 = 0.$$

Thus

$$y = c_1 \cos(\ln x) + c_2 \sin(\ln x) + c_3 \ln x \cos(\ln x) + c_4 \ln x \sin(\ln x).$$

Exercises 3.6

19. The auxiliary equation is $m^2 - 5m = m(m - 5) = 0$ so that $y_c = c_1 + c_2x^5$ and

$$W(1, x^5) = \begin{vmatrix} 1 & x^5 \\ 0 & 5x^4 \end{vmatrix} = 5x^4.$$

Identifying $f(x) = x^3$ we obtain $u'_1 = -\frac{1}{5}x^4$ and $u'_2 = 1/5x$. Then $u_1 = -\frac{1}{25}x^5$, $u_2 = \frac{1}{5}\ln x$, and

$$y = c_1 + c_2x^5 - \frac{1}{25}x^5 + \frac{1}{5}x^5 \ln x = c_1 + c_3x^5 + \frac{1}{5}x^5 \ln x.$$

20. The auxiliary equation is $2m^2 + 3m + 1 = (2m + 1)(m + 1) = 0$ so that $y_c = c_1x^{-1} + c_2x^{-1/2}$ and

$$W(x^{-1}, x^{-1/2}) = \begin{vmatrix} x^{-1} & x^{-1/2} \\ -x^{-2} & -\frac{1}{2}x^{-3/2} \end{vmatrix} = \frac{1}{2}x^{-5/2}.$$

Identifying $f(x) = \frac{1}{2} - \frac{1}{2x}$ we obtain $u'_1 = x - x^2$ and $u'_2 = x^{3/2} - x^{1/2}$. Then $u_1 = \frac{1}{2}x^2 - \frac{1}{3}x^3$, $u_2 = \frac{2}{5}x^{5/2} - \frac{2}{3}x^{3/2}$, and

$$y = c_1x^{-1} + c_2x^{-1/2} + \frac{1}{2}x - \frac{1}{3}x^2 + \frac{2}{5}x^2 - \frac{2}{3}x = c_1x^{-1} + c_2x^{-1/2} - \frac{1}{6}x + \frac{1}{15}x^2.$$

21. The auxiliary equation is $m^2 - 2m + 1 = (m - 1)^2 = 0$ so that $y_c = c_1x + c_2x \ln x$ and

$$W(x, x \ln x) = \begin{vmatrix} x & x \ln x \\ 1 & 1 + \ln x \end{vmatrix} = x.$$

Identifying $f(x) = 2/x$ we obtain $u'_1 = -2 \ln x/x$ and $u'_2 = 2/x$. Then $u_1 = -(\ln x)^2$, $u_2 = 2 \ln x$, and

$$\begin{aligned} y &= c_1x + c_2x \ln x - x(\ln x)^2 + 2x(\ln x)^2 \\ &= c_1x + c_2x \ln x + x(\ln x)^2. \end{aligned}$$

22. The auxiliary equation is $m^2 - 3m + 2 = (m - 1)(m - 2) = 0$ so that $y_c = c_1x + c_2x^2$ and

$$W(x, x^2) = \begin{vmatrix} x & x^2 \\ 1 & 2x \end{vmatrix} = x^2.$$

Identifying $f(x) = x^2e^x$ we obtain $u'_1 = -x^2e^x$ and $u'_2 = xe^x$. Then $u_1 = -x^2e^x + 2xe^x - 2e^x$, $u_2 = xe^x - e^x$, and

$$\begin{aligned} y &= c_1x + c_2x^2 - x^3e^x + 2x^2e^x - 2xe^x + x^3e^x - x^2e^x \\ &= c_1x + c_2x^2 + x^2e^x - 2xe^x. \end{aligned}$$

23. The auxiliary equation is $m^2 + 2m = m(m + 2) = 0$, so that

$$y = c_1 + c_2x^{-2} \quad \text{and} \quad y' = -2c_2x^{-3}.$$

The initial conditions imply

$$\begin{aligned} c_1 + c_2 &= 0 \\ -2c_2 &= 4. \end{aligned}$$

Thus, $c_1 = 2$, $c_2 = -2$, and $y = 2 - 2x^{-2}$.

24. The auxiliary equation is $m^2 - 6m + 8 = (m - 2)(m - 4) = 0$, so that

$$y = c_1x^2 + c_2x^4 \quad \text{and} \quad y' = 2c_1x + 4c_2x^3.$$

The initial conditions imply

$$\begin{aligned} 4c_1 + 16c_2 &= 32 \\ 4c_1 + 32c_2 &= 0. \end{aligned}$$

Thus, $c_1 = 16$, $c_2 = -2$, and $y = 16x^2 - 2x^4$.

25. The auxiliary equation is $m^2 + 1 = 0$, so that

$$y = c_1 \cos(\ln x) + c_2 \sin(\ln x) \quad \text{and} \quad y' = -c_1 \frac{1}{x} \sin(\ln x) + c_2 \frac{1}{x} \cos(\ln x).$$

The initial conditions imply $c_1 = 1$ and $c_2 = 2$. Thus $y = \cos(\ln x) + 2 \sin(\ln x)$.

26. The auxiliary equation is $m^2 - 4m + 4 = (m - 2)^2 = 0$, so that

$$y = c_1 x^2 + c_2 x^2 \ln x \quad \text{and} \quad y' = 2c_1 x + c_2(x + 2x \ln x).$$

The initial conditions imply $c_1 = 5$ and $c_2 + 10 = 3$. Thus $y = 5x^2 - 7x^2 \ln x$.

27. The auxiliary equation is $m^2 = 0$ so that $y_c = c_1 + c_2 \ln x$ and

$$W(1, \ln x) = \begin{vmatrix} 1 & \ln x \\ 0 & 1/x \end{vmatrix} = \frac{1}{x}.$$

Identifying $f(x) = 1$ we obtain $u'_1 = -x \ln x$ and $u'_2 = x$. Then $u_1 = \frac{1}{4}x^2 - \frac{1}{2}x^2 \ln x$, $u_2 = \frac{1}{2}x^2$, and

$$y = c_1 + c_2 \ln x + \frac{1}{4}x^2 - \frac{1}{2}x^2 \ln x + \frac{1}{2}x^2 \ln x = c_1 + c_2 \ln x + \frac{1}{4}x^2.$$

The initial conditions imply $c_1 + \frac{1}{4} = 1$ and $c_2 + \frac{1}{2} = -\frac{1}{2}$. Thus, $c_1 = \frac{3}{4}$, $c_2 = -1$, and $y = \frac{3}{4} - \ln x + \frac{1}{4}x^2$.

28. The auxiliary equation is $m^2 - 6m + 8 = (m - 2)(m - 4) = 0$, so that $y_c = c_1 x^2 + c_2 x^4$ and

$$W = \begin{vmatrix} x^2 & x^4 \\ 2x & 4x^3 \end{vmatrix} = 2x^5.$$

Identifying $f(x) = 8x^4$ we obtain $u'_1 = -4x^3$ and $u'_2 = 4x$. Then $u_1 = -x^4$, $u_2 = 2x^2$, and $y = c_1 x^2 + c_2 x^4 + x^6$. The initial conditions imply

$$\begin{aligned} \frac{1}{4}c_1 + \frac{1}{16}c_2 &= -\frac{1}{64} \\ c_1 + \frac{1}{2}c_2 &= -\frac{3}{16}. \end{aligned}$$

Thus $c_1 = \frac{1}{16}$, $c_2 = -\frac{1}{2}$, and $y = \frac{1}{16}x^2 - \frac{1}{2}x^4 + x^6$.

29. Substituting into the differential equation we obtain

$$\frac{d^2 y}{dt^2} + 9 \frac{dy}{dt} + 8y = e^{2t}.$$

The auxiliary equation is $m^2 + 9 + 8 = (m + 1)(m + 8) = 0$ so that $y_c = c_1 e^{-t} + c_2 e^{-8t}$. Using undetermined coefficients we try $y_p = Ae^{2t}$. This leads to $30Ae^{2t} = e^{2t}$, so that $A = 1/30$ and

$$y = c_1 e^{-t} + c_2 e^{-8t} + \frac{1}{30}e^{2t} = c_1 x^{-1} + c_2 x^{-8} + \frac{1}{30}x^2.$$

30. Substituting into the differential equation we obtain

$$\frac{d^2 y}{dt^2} - 5 \frac{dy}{dt} + 6y = 2t.$$

The auxiliary equation is $m^2 - 5m + 6 = (m - 2)(m - 3) = 0$ so that $y_c = c_1 e^{2t} + c_2 e^{3t}$. Using undetermined coefficients we try $y_p = At + B$. This leads to $(-5A + 6B) + 6At = 2t$, so that $A = 1/3$, $B = 5/18$, and

$$y = c_1 e^{2t} + c_2 e^{3t} + \frac{1}{3}t + \frac{5}{18} = c_1 x^2 + c_2 x^3 + \frac{1}{3} \ln x + \frac{5}{18}.$$

31. Substituting into the differential equation we obtain

$$\frac{d^2 y}{dt^2} - 4 \frac{dy}{dt} + 13y = 4 + 3e^t.$$

Exercises 3.6

The auxiliary equation is $m^2 - 4m + 13 = 0$ so that $y_c = e^{2t}(c_1 \cos 3t + c_2 \sin 3t)$. Using undetermined coefficients we try $y_p = A + Be^t$. This leads to $13A + 10Be^t = 4 + 3e^t$, so that $A = 4/13$, $B = 3/10$, and

$$\begin{aligned} y &= e^{2t}(c_1 \cos 3t + c_2 \sin 3t) + \frac{4}{13} + \frac{3}{10}e^t \\ &= x^2 [c_1 \cos(3 \ln x) + c_2 \sin(3 \ln x)] + \frac{4}{13} + \frac{3}{10}x. \end{aligned}$$

32. From

$$\frac{d^2 y}{dx^2} = \frac{1}{x^2} \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right)$$

it follows that

$$\begin{aligned} \frac{d^3 y}{dx^3} &= \frac{1}{x^2} \frac{d}{dx} \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right) - \frac{2}{x^3} \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right) \\ &= \frac{1}{x^2} \frac{d}{dx} \left(\frac{d^2 y}{dt^2} \right) - \frac{1}{x^2} \frac{d}{dx} \left(\frac{dy}{dt} \right) - \frac{2}{x^3} \frac{d^2 y}{dt^2} + \frac{2}{x^3} \frac{dy}{dt} \\ &= \frac{1}{x^2} \frac{d^3 y}{dt^3} \left(\frac{1}{x} \right) - \frac{1}{x^2} \frac{d^2 y}{dt^2} \left(\frac{1}{x} \right) - \frac{2}{x^3} \frac{d^2 y}{dt^2} + \frac{2}{x^3} \frac{dy}{dt} \\ &= \frac{1}{x^3} \left(\frac{d^3 y}{dt^3} - 3 \frac{d^2 y}{dt^2} + 2 \frac{dy}{dt} \right). \end{aligned}$$

Substituting into the differential equation we obtain

$$\frac{d^3 y}{dt^3} - 3 \frac{d^2 y}{dt^2} + 2 \frac{dy}{dt} - 3 \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right) + 6 \frac{dy}{dt} - 6y = 3 + 3t$$

or

$$\frac{d^3 y}{dt^3} - 6 \frac{d^2 y}{dt^2} + 11 \frac{dy}{dt} - 6y = 3 + 3t.$$

The auxiliary equation is $m^3 - 6m^2 + 11m - 6 = (m-1)(m-2)(m-3) = 0$ so that $y_c = c_1 e^t + c_2 e^{2t} + c_3 e^{3t}$. Using undetermined coefficients we try $y_p = A + Bt$. This leads to $(11B - 6A) - 6Bt = 3 + 3t$, so that $A = -17/12$, $B = -1/2$, and

$$y = c_1 e^t + c_2 e^{2t} + c_3 e^{3t} - \frac{17}{12} - \frac{1}{2}t = c_1 x + c_2 x^2 + c_3 x^3 - \frac{17}{12} - \frac{1}{2} \ln x.$$

33. The auxiliary equation is $2m(m-1)(m-2) - 10.98m(m-1) + 8.5m + 1.3 = 0$, so that $m_1 = -0.053299$, $m_2 = 1.81164$, $m_3 = 6.73166$, and

$$y = c_1 x^{-0.053299} + c_2 x^{1.81164} + c_3 x^{6.73166}.$$

34. The auxiliary equation is $m(m-1)(m-2) + 4m(m-1) + 5m - 9 = 0$, so that $m_1 = 1.40819$ and the two complex roots are $-1.20409 \pm 2.22291i$. The general solution of the differential equation is

$$y = c_1 x^{1.40819} + x^{-1.20409} [c_2 \cos(2.22291 \ln x) + c_3 \sin(2.22291 \ln x)].$$

35. The auxiliary equation is $m(m-1)(m-2)(m-3) + 6m(m-1)(m-2) + 3m(m-1) - 3m + 4 = 0$, so that $m_1 = m_2 = \sqrt{2}$ and $m_3 = m_4 = -\sqrt{2}$. The general solution of the differential equation is

$$y = c_1 x^{\sqrt{2}} + c_2 x^{\sqrt{2}} \ln x + c_3 x^{-\sqrt{2}} + c_4 x^{-\sqrt{2}} \ln x.$$

36. The auxiliary equation is $m(m-1)(m-2)(m-3) - 6m(m-1)(m-2) + 33m(m-1) - 105m + 169 = 0$, so that $m_1 = m_2 = 3 + 2i$ and $m_3 = m_4 = 3 - 2i$. The general solution of the differential equation is

$$y = x^3 [c_1 \cos(2 \ln x) + c_2 \sin(2 \ln x)] + x^3 \ln x [c_3 \cos(2 \ln x) + c_4 \sin(2 \ln x)].$$

In the next two problems we use the substitution $t = -x$ since the initial conditions are on the interval $(-\infty, 0)$. In this case

$$\frac{dy}{dt} = \frac{dy}{dx} \frac{dx}{dt} = -\frac{dy}{dx}$$

and

$$\frac{d^2y}{dt^2} = \frac{d}{dt} \left(\frac{dy}{dt} \right) = \frac{d}{dt} \left(-\frac{dy}{dx} \right) = -\frac{d}{dt} (y') = -\frac{dy'}{dx} \frac{dx}{dt} = -\frac{d^2y}{dx^2} \frac{dx}{dt} = \frac{d^2y}{dx^2}.$$

37. The differential equation and initial conditions become

$$4t^2 \frac{d^2y}{dt^2} + y = 0; \quad y(t) \Big|_{t=1} = 2, \quad y'(t) \Big|_{t=1} = -4.$$

The auxiliary equation is $4m^2 - 4m + 1 = (2m - 1)^2 = 0$, so that

$$y = c_1 t^{1/2} + c_2 t^{1/2} \ln t \quad \text{and} \quad y' = \frac{1}{2} c_1 t^{-1/2} + c_2 \left(t^{-1/2} + \frac{1}{2} t^{-1/2} \ln t \right).$$

The initial conditions imply $c_1 = 2$ and $1 + c_2 = -4$. Thus

$$y = 2t^{1/2} - 5t^{1/2} \ln t = 2(-x)^{1/2} - 5(-x)^{1/2} \ln(-x), \quad x < 0.$$

38. The differential equation and initial conditions become

$$t^2 \frac{d^2y}{dt^2} - 4t \frac{dy}{dt} + 6y = 0; \quad y(t) \Big|_{t=2} = 8, \quad y'(t) \Big|_{t=2} = 0.$$

The auxiliary equation is $m^2 - 5m + 6 = (m - 2)(m - 3) = 0$, so that

$$y = c_1 t^2 + c_2 t^3 \quad \text{and} \quad y' = 2c_1 t + 3c_2 t^2.$$

The initial conditions imply

$$4c_1 + 8c_2 = 8$$

$$4c_1 + 12c_2 = 0$$

from which we find $c_1 = 6$ and $c_2 = -2$. Thus

$$y = 6t^2 - 2t^3 = 6x^2 + 2x^3, \quad x < 0.$$

39. Letting $u = x + 2$ we obtain

$$\frac{dy}{dx} = \frac{dy}{du}$$

and, using the chain rule,

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{du} \right) = \frac{d^2y}{du^2} \frac{du}{dx} = \frac{d^2y}{du^2} (1) = \frac{d^2y}{du^2}.$$

Substituting into the differential equation we obtain

$$u^2 \frac{d^2y}{du^2} + u \frac{dy}{du} + y = 0.$$

The auxiliary equation is $m^2 + 1 = 0$ so that

$$y = c_1 \cos(\ln u) + c_2 \sin(\ln u) = c_1 \cos[\ln(x + 2)] + c_2 \sin[\ln(x + 2)].$$

40. If $1 - i$ is a root of the auxiliary equation then so is $1 + i$, and the auxiliary equation is

$$(m - 2)[m - (1 + i)][m - (1 - i)] = m^3 - 4m^2 + 6m - 4 = 0.$$

Exercises 3.6

We need $m^3 - 4m^2 + 6m - 4$ to have the form $m(m-1)(m-2) + bm(m-1) + cm + d$. Expanding this last expression and equating coefficients we get $b = -1$, $c = 3$, and $d = -4$. Thus, the differential equation is

$$x^3 y''' - x^2 y'' + 3xy' - 4y = 0.$$

41. For $x^2 y'' = 0$ the auxiliary equation is $m(m-1) = 0$ and the general solution is $y = c_1 + c_2 x$. The initial conditions imply $c_1 = y_0$ and $c_2 = y_1$, so $y = y_0 + y_1 x$. The initial conditions are satisfied for all real values of y_0 and y_1 .

For $x^2 y'' - 2xy' + 2y = 0$ the auxiliary equation is $m^2 - 3m + 2 = (m-1)(m-2) = 0$ and the general solution is $y = c_1 x + c_2 x^2$. The initial condition $y(0) = y_0$ implies $0 = y_0$ and the condition $y'(0) = y_1$ implies $c_1 = y_1$. Thus, the initial conditions are satisfied for $y_0 = 0$ and for all real values of y_1 .

For $x^2 y'' - 4xy' + 6y = 0$ the auxiliary equation is $m^2 - 5m + 6 = (m-2)(m-3) = 0$ and the general solution is $y = c_1 x^2 + c_2 x^3$. The initial conditions imply $y(0) = 0 = y_0$ and $y'(0) = 0$. Thus, the initial conditions are satisfied only for $y_0 = y_1 = 0$.

42. The function $y(x) = -\sqrt{x} \cos(\ln x)$ is defined for $x > 0$ and has x -intercepts where $\ln x = \pi/2 + k\pi$ for k an integer or where $x = e^{\pi/2 + k\pi}$. Solving $\pi/2 + k\pi = 0.5$ we get $k \approx -0.34$, so $e^{\pi/2 + k\pi} < 0.5$ for all negative integers and the graph has infinitely many x -intercepts in $(0, 0.5)$.

Exercises 3.7

1. We have $y'_1 = y''_1 = e^x$, so

$$(y''_1)^2 = (e^x)^2 = e^{2x} = y_1^2.$$

Also, $y'_2 = -\sin x$ and $y''_2 = -\cos x$, so

$$(y''_2)^2 = (-\cos x)^2 = \cos^2 x = y_2^2.$$

However, if $y = c_1 y_1 + c_2 y_2$, we have $(y'')^2 = (c_1 e^x - c_2 \cos x)^2$ and $y^2 = (c_1 e^x + c_2 \cos x)^2$. Thus $(y'')^2 \neq y^2$.

2. We have $y'_1 = y''_1 = 0$, so

$$y_1 y''_1 = 1 \cdot 0 = 0 = \frac{1}{2}(0)^2 = \frac{1}{2}(y'_1)^2.$$

Also, $y'_2 = 2x$ and $y''_2 = 2$, so

$$y_2 y''_2 = x^2(2) = 2x^2 = \frac{1}{2}(2x)^2 = \frac{1}{2}(y'_2)^2.$$

However, if $y = c_1 y_1 + c_2 y_2$, we have $yy'' = (c_1 \cdot 1 + c_2 x^2)(c_1 \cdot 0 + 2c_2) = 2c_2(c_1 + c_2 x^2)$ and $\frac{1}{2}(y')^2 = \frac{1}{2}[c_1 \cdot 0 + c_2(2x)]^2 = 2c_2^2 x^2$. Thus $yy'' \neq \frac{1}{2}(y')^2$.

3. Let $u = y'$ so that $u' = y''$. The equation becomes $u' = -u - 1$ which is separable. Thus

$$\frac{du}{u^2 + 1} = -dx \implies \tan^{-1} u = -x + c_1 \implies y' = \tan(c_1 - x) \implies y = \ln |\cos(c_1 - x)| + c_2.$$

4. Let $u = y'$ so that $u' = y''$. The equation becomes $u' = 1 + u^2$. Separating variables we obtain

$$\frac{du}{1 + u^2} = dx \implies \tan^{-1} u = x + c_1 \implies u = \tan(x + c_1) \implies y = -\ln |\cos(x + c_1)| + c_2.$$

5. Let $u = y'$ so that $u' = y''$. The equation becomes $x^2 u' + u^2 = 0$. Separating variables we obtain

$$\begin{aligned}\frac{du}{u^2} &= -\frac{dx}{x^2} \implies -\frac{1}{u} = \frac{1}{x} + c_1 = \frac{c_1 x + 1}{x} \implies u = -\frac{1}{c_1} \left(\frac{x}{x + 1/c_1} \right) = \frac{1}{c_1} \left(\frac{1}{c_1 x + 1} - 1 \right) \\ \implies y &= \frac{1}{c_1^2} \ln |c_1 x + 1| - \frac{1}{c_1} x + c_2.\end{aligned}$$

6. Let $u = y'$ so that $y'' = u \frac{du}{dy}$. The equation becomes $(y + 1)u \frac{du}{dy} = u^2$. Separating variables we obtain

$$\begin{aligned}\frac{du}{u} &= \frac{dy}{y + 1} \implies \ln |u| = \ln |y + 1| + \ln c_1 \implies u = c_1(y + 1) \\ \implies \frac{dy}{dx} &= c_1(y + 1) \implies \frac{dy}{y + 1} = c_1 dx \\ \implies \ln |y + 1| &= c_1 x + c_2 \implies y + 1 = c_3 e^{c_1 x}.\end{aligned}$$

7. Let $u = y'$ so that $y'' = u \frac{du}{dy}$. The equation becomes $u \frac{du}{dy} + 2yu^3 = 0$. Separating variables we obtain

$$\begin{aligned}\frac{du}{u^2} + 2y dy &= 0 \implies -\frac{1}{u} + y^2 = c \implies u = \frac{1}{y^2 + c_1} \implies y' = \frac{1}{y^2 + c_1} \\ \implies (y^2 + c_1) dy &= dx \implies \frac{1}{3} y^3 + c_1 y = x + c_2.\end{aligned}$$

8. Let $u = y'$ so that $y'' = u \frac{du}{dy}$. The equation becomes $y^2 u \frac{du}{dy} = u$. Separating variables we obtain

$$\begin{aligned}du &= \frac{dy}{y^2} \implies u = -\frac{1}{y} + c_1 \implies y' = \frac{c_1 y - 1}{y} \implies \frac{y}{c_1 y - 1} dy = dx \\ \implies \frac{1}{c_1} \left(1 + \frac{1}{c_1 y - 1} \right) dy &= dx \text{ (for } c_1 \neq 0) \implies \frac{1}{c_1} y + \frac{1}{c_1^2} \ln |y - 1| = x + c_2.\end{aligned}$$

If $c_1 = 0$, then $y dy = -dx$ and another solution is $\frac{1}{2} y^2 = -x + c_2$.

9. Let $u = y'$ so that $y'' = u \frac{du}{dy}$. The equation becomes $u \frac{du}{dy} + yu = 0$. Separating variables we obtain

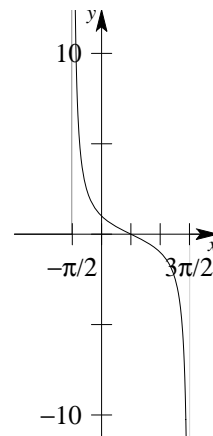
$$du = -y dy \implies u = -\frac{1}{2} y^2 + c_1 \implies y' = -\frac{1}{2} y^2 + c_1.$$

When $x = 0$, $y = 1$ and $y' = -1$ so $-1 = -\frac{1}{2} + c_1$ and $c_1 = -\frac{1}{2}$. Then

$$\begin{aligned}\frac{dy}{dx} &= -\frac{1}{2} y^2 - \frac{1}{2} \implies \frac{dy}{y^2 + 1} = -\frac{1}{2} dx \implies \tan^{-1} y = -\frac{1}{2} x + c_2 \\ \implies y &= \tan \left(-\frac{1}{2} x + c_2 \right).\end{aligned}$$

When $x = 0$, $y = 1$ so $1 = \tan c_2$ and $c_2 = \pi/4$. The solution of the initial-value problem is

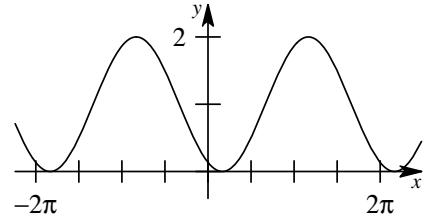
$$y = \tan \left(\frac{\pi}{4} - \frac{1}{2} x \right), \quad -\frac{\pi}{2} < x < \frac{3\pi}{2}.$$



Exercises 3.7

10. Let $u = y'$ so that $u' = y''$. The equation becomes $(u')^2 + u^2 = 1$ which results in $u' = \pm\sqrt{1-u^2}$. To solve $u' = \sqrt{1-u^2}$ we separate variables:

$$\begin{aligned}\frac{du}{\sqrt{1-u^2}} &= dx \implies \sin^{-1} u = x + c_1 \implies u = \sin(x + c_1) \\ &\implies y' = \sin(x + c_1).\end{aligned}$$



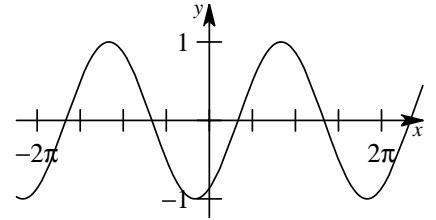
When $x = \frac{\pi}{2}$, $y' = \frac{\sqrt{3}}{2}$, so $\frac{\sqrt{3}}{2} = \sin\left(\frac{\pi}{2} + c_1\right)$ and $c_1 = -\frac{\pi}{6}$. Thus

$$y' = \sin\left(x - \frac{\pi}{6}\right) \implies y = -\cos\left(x - \frac{\pi}{6}\right) + c_2.$$

When $x = \frac{\pi}{2}$, $y = \frac{1}{2}$, so $\frac{1}{2} = -\cos\left(\frac{\pi}{2} - \frac{\pi}{6}\right) + c_2 = -\frac{1}{2} + c_2$ and $c_2 = 1$. The solution of the initial-value problem is $y = 1 - \cos\left(x - \frac{\pi}{6}\right)$.

To solve $u' = -\sqrt{1-u^2}$ we separate variables:

$$\begin{aligned}\frac{du}{\sqrt{1-u^2}} &= -dx \implies \cos^{-1} u = x + c_1 \\ &\implies u = \cos(x + c_1) \implies y' = \cos(x + c_1).\end{aligned}$$



When $x = \frac{\pi}{2}$, $y' = \frac{\sqrt{3}}{2}$, so $\frac{\sqrt{3}}{2} = \cos\left(\frac{\pi}{2} + c_1\right)$ and $c_1 = -\frac{\pi}{3}$. Thus

$$y' = \cos\left(x - \frac{\pi}{3}\right) \implies y = \sin\left(x - \frac{\pi}{3}\right) + c_2.$$

When $x = \frac{\pi}{2}$, $y = \frac{1}{2}$, so $\frac{1}{2} = \sin\left(\frac{\pi}{2} - \frac{\pi}{3}\right) + c_2 = \frac{1}{2} + c_2$ and $c_2 = 0$. The solution of the initial-value problem is $y = \sin\left(x - \frac{\pi}{3}\right)$.

11. Let $u = y'$ so that $u' = y''$. The equation becomes $u' - \frac{1}{x}u = \frac{1}{x}u^3$, which is Bernoulli. Using $w = u^{-2}$ we obtain $\frac{dw}{dx} + \frac{2}{x}w = -\frac{2}{x}$. An integrating factor is x^2 , so

$$\begin{aligned}\frac{d}{dx}[x^2w] &= -2x \implies x^2w = -x^2 + c_1 \implies w = -1 + \frac{c_1}{x^2} \\ &\implies u^{-2} = -1 + \frac{c_1}{x^2} \implies u = \frac{x}{\sqrt{c_1 - x^2}} \\ &\implies \frac{dy}{dx} = \frac{x}{\sqrt{c_1 - x^2}} \implies y = -\sqrt{c_1 - x^2} + c_2 \\ &\implies c_1 - x^2 = (c_2 - y)^2 \implies x^2 + (c_2 - y)^2 = c_1.\end{aligned}$$

12. Let $u = y'$ so that $u' = y''$. The equation becomes $u' - \frac{1}{x}u = u^2$, which is Bernoulli. Using the substitution $w = u^{-1}$ we obtain $\frac{dw}{dx} + \frac{1}{x}w = -1$. An integrating factor is x , so

$$\frac{d}{dx}[xw] = -x \implies w = -\frac{1}{2}x + \frac{1}{x}c \implies \frac{1}{u} = \frac{c_1 - x^2}{2x} \implies u = \frac{2x}{c_1 - x^2} \implies y = -\ln|c_1 - x^2| + c_2.$$

In Problems 13-16 the thinner curve is obtained using a numerical solver, while the thicker curve is the graph of the Taylor polynomial.

13. We look for a solution of the form

$$y(x) = y(0) + y'(0)x + \frac{1}{2}y''(0)x^2 + \frac{1}{3!}y'''(0)x^3 + \frac{1}{4!}y^{(4)}(0)x^4 + \frac{1}{5!}y^{(5)}(0)x^5.$$

From $y''(x) = x + y^2$ we compute

$$y'''(x) = 1 + 2yy'$$

$$y^{(4)}(x) = 2yy'' + 2(y')^2$$

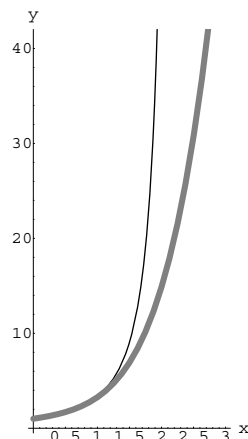
$$y^{(5)}(x) = 2yy''' + 6y'y''.$$

Using $y(0) = 1$ and $y'(0) = 1$ we find

$$y''(0) = 1, \quad y'''(0) = 3, \quad y^{(4)}(0) = 4, \quad y^{(5)}(0) = 12.$$

An approximate solution is

$$y(x) = 1 + x + \frac{1}{2}x^2 + \frac{1}{2}x^3 + \frac{1}{6}x^4 + \frac{1}{10}x^5.$$



14. We look for a solution of the form

$$y(x) = y(0) + y'(0)x + \frac{1}{2}y''(0)x^2 + \frac{1}{3!}y'''(0)x^3 + \frac{1}{4!}y^{(4)}(0)x^4 + \frac{1}{5!}y^{(5)}(0)x^5.$$

From $y''(x) = 1 - y^2$ we compute

$$y'''(x) = -2yy'$$

$$y^{(4)}(x) = -2yy'' - 2(y')^2$$

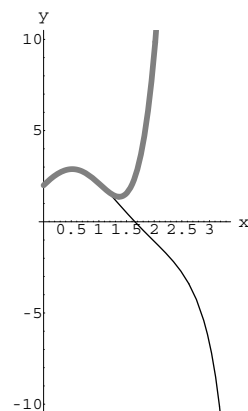
$$y^{(5)}(x) = -2yy''' - 6y'y''.$$

Using $y(0) = 2$ and $y'(0) = 3$ we find

$$y''(0) = -3, \quad y'''(0) = -12, \quad y^{(4)}(0) = -6, \quad y^{(5)}(0) = 102.$$

An approximate solution is

$$y(x) = 2 + 3x - \frac{3}{2}x^2 - 2x^3 - \frac{1}{4}x^4 + \frac{17}{20}x^5.$$



15. We look for a solution of the form

$$y(x) = y(0) + y'(0)x + \frac{1}{2}y''(0)x^2 + \frac{1}{3!}y'''(0)x^3 + \frac{1}{4!}y^{(4)}(0)x^4 + \frac{1}{5!}y^{(5)}(0)x^5.$$

From $y''(x) = x^2 + y^2 - 2y'$ we compute

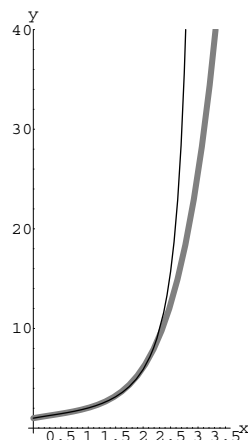
$$y'''(x) = 2x + 2yy' - 2y''$$

$$y^{(4)}(x) = 2 + 2(y')^2 + 2yy'' - 2y'''$$

$$y^{(5)}(x) = 6y'y'' + 2yy''' - 2y^{(4)}.$$

Using $y(0) = 1$ and $y'(0) = 1$ we find

$$y''(0) = -1, \quad y'''(0) = 4, \quad y^{(4)}(0) = -6, \quad y^{(5)}(0) = 14.$$



Exercises 3.7

An approximate solution is

$$y(x) = 1 + x - \frac{1}{2}x^2 + \frac{2}{3}x^3 - \frac{1}{4}x^4 + \frac{7}{60}x^5.$$

16. We look for a solution of the form

$$y(x) = y(0) + y'(0)x + \frac{1}{2}y''(0)x^2 + \frac{1}{3!}y'''(0)x^3 + \frac{1}{4!}y^{(4)}(0)x^4 + \frac{1}{5!}y^{(5)}(0)x^5 + \frac{1}{6!}y^{(6)}(0)x^6.$$

From $y''(x) = e^y$ we compute

$$y'''(x) = e^y y'$$

$$y^{(4)}(x) = e^y (y')^2 + e^y y''$$

$$y^{(5)}(x) = e^y (y')^3 + 3e^y y' y'' + e^y y'''$$

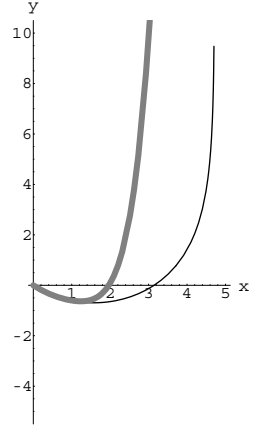
$$y^{(6)}(x) = e^y (y')^4 + 6e^y (y')^2 y'' + 3e^y (y'')^2 + 4e^y y' y''' + e^y y^{(4)}.$$

Using $y(0) = 0$ and $y'(0) = -1$ we find

$$y''(0) = 1, \quad y'''(0) = -1, \quad y^{(4)}(0) = 2, \quad y^{(5)}(0) = -5, \quad y^{(6)}(0) = 16.$$

An approximate solution is

$$y(x) = -x + \frac{1}{2}x^2 - \frac{1}{6}x^3 + \frac{1}{12}x^4 + \frac{1}{24}x^5 + \frac{1}{45}x^6.$$



17. We need to solve $[1 + (y')^2]^{3/2} = y''$. Let $u = y'$ so that $u' = y''$. The equation becomes $(1 + u^2)^{3/2} = u'$ or $(1 + u^2)^{3/2} = \frac{du}{dx}$. Separating variables and using the substitution $u = \tan \theta$ we have

$$\begin{aligned} \frac{du}{(1 + u^2)^{3/2}} = dx &\implies \int \frac{\sec^2 \theta}{(1 + \tan^2 \theta)^{3/2}} d\theta = x \implies \int \frac{\sec^2 \theta}{\sec^3 \theta} d\theta = x \\ &\implies \int \cos \theta d\theta = x \implies \sin \theta = x \implies \frac{u}{\sqrt{1 + u^2}} = x \\ &\implies \frac{y'}{\sqrt{1 + (y')^2}} = x \implies (y')^2 = x^2 [1 + (y')^2] = \frac{x^2}{1 - x^2} \\ &\implies y' = \frac{x}{\sqrt{1 - x^2}} \quad (\text{for } x > 0) \implies y = -\sqrt{1 - x^2}. \end{aligned}$$

18. Let $u = \frac{dx}{dt}$ so that $\frac{d^2x}{dt^2} = u \frac{du}{dx}$. The equation becomes $u \frac{du}{dx} = \frac{-k^2}{x^2}$. Separating variables we obtain

$$u du = -\frac{k^2}{x^2} dx \implies \frac{1}{2}u^2 = \frac{k^2}{x} + c \implies \frac{1}{2}v^2 = \frac{k^2}{x} + c.$$

When $t = 0$, $x = x_0$ and $v = 0$ so $0 = \frac{k^2}{x_0} + c$ and $c = -\frac{k^2}{x_0}$. Then

$$\frac{1}{2}v^2 = k^2 \left(\frac{1}{x} - \frac{1}{x_0} \right) \quad \text{and} \quad \frac{dx}{dt} = -k\sqrt{2} \sqrt{\frac{x_0 - x}{xx_0}}.$$

Separating variables we have

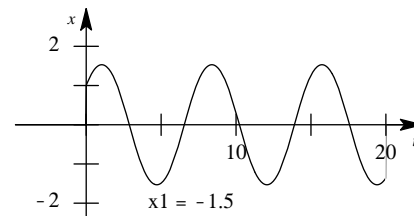
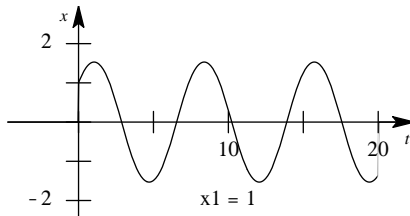
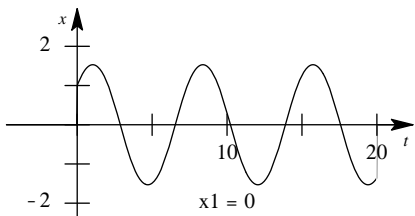
$$-\sqrt{\frac{xx_0}{x_0 - x}} dx = k\sqrt{2} dt \implies t = -\frac{1}{k} \sqrt{\frac{x_0}{2}} \int \sqrt{\frac{x}{x_0 - x}} dx.$$

Using *Mathematica* to integrate we obtain

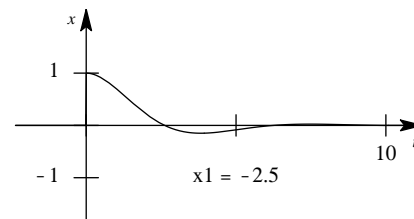
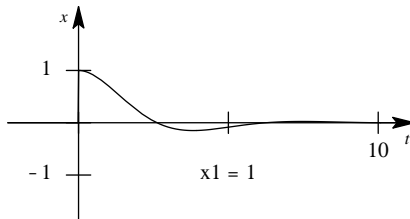
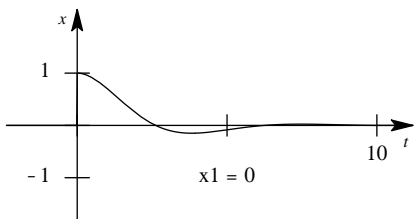
$$t = -\frac{1}{k}\sqrt{\frac{x_0}{2}} \left[-\sqrt{x(x_0-x)} - \frac{x_0}{2} \tan^{-1} \frac{(x_0-2x)}{2x} \sqrt{\frac{x}{x_0-x}} \right]$$

$$= \frac{1}{k}\sqrt{\frac{x_0}{2}} \left[\sqrt{x(x_0-x)} + \frac{x_0}{2} \tan^{-1} \frac{x_0-2x}{2\sqrt{x(x_0-x)}} \right].$$

19.



For $\frac{d^2x}{dt^2} + \sin x = 0$ the motion appears to be periodic with amplitude 1 when $x_1 = 0$. The amplitude and period are larger for larger magnitudes of x_1 .



For $\frac{d^2x}{dt^2} + \frac{dx}{dt} + \sin x = 0$ the motion appears to be periodic with decreasing amplitude. The dx/dt term could be said to have a damping effect.

20. From $(y'')^2 - y^2 = 0$ we have $y'' = \pm y$, which can be treated as two linear equations. Since linear combinations of solutions of linear homogeneous differential equations are also solutions, we see that $y = c_1 e^x + c_2 e^{-x}$ and $y = c_3 \cos x + c_4 \sin x$ must satisfy the differential equation.
21. Letting $u = y''$, separating variables, and integrating we have

$$\frac{du}{dx} = \sqrt{1+u^2}, \quad \frac{du}{\sqrt{1+u^2}} = dx, \quad \text{and} \quad \sinh^{-1} u = x + c_1.$$

Then

$$u = y'' = \sinh(x + c_1), \quad y' = \cosh(x + c_1) + c_2, \quad \text{and} \quad y = \sinh(x + c_1) + c_2 x + c_3.$$

Exercises 3.8

1. From $\frac{1}{8}x' + 16x = 0$ we obtain

$$x = c_1 \cos 8\sqrt{2}t + c_2 \sin 8\sqrt{2}t$$

so that the period of motion is $2\pi/8\sqrt{2} = \sqrt{2}\pi/8$ seconds.

2. From $20x'' + kx = 0$ we obtain

$$x = c_1 \cos \frac{1}{2}\sqrt{\frac{k}{5}}t + c_2 \sin \frac{1}{2}\sqrt{\frac{k}{5}}t$$

Exercises 3.8

so that the frequency $2/\pi = \frac{1}{4}\sqrt{k/5}\pi$ and $k = 320$ N/m. If $80x'' + 320x = 0$ then $x = c_1 \cos 2t + c_2 \sin 2t$ so that the frequency is $2/2\pi = 1/\pi$ vibrations/second.

3. From $\frac{3}{4}x'' + 72x = 0$, $x(0) = -1/4$, and $x'(0) = 0$ we obtain $x = -\frac{1}{4} \cos 4\sqrt{6}t$.

4. From $\frac{3}{4}x'' + 72x = 0$, $x(0) = 0$, and $x'(0) = 2$ we obtain $x = \frac{\sqrt{6}}{12} \sin 4\sqrt{6}t$.

5. From $\frac{5}{8}x'' + 40x = 0$, $x(0) = 1/2$, and $x'(0) = 0$ we obtain $x = \frac{1}{2} \cos 8t$.

(a) $x(\pi/12) = -1/4$, $x(\pi/8) = -1/2$, $x(\pi/6) = -1/4$, $x(\pi/4) = 1/2$, $x(9\pi/32) = \sqrt{2}/4$.

(b) $x' = -4 \sin 8t$ so that $x'(3\pi/16) = 4$ ft/s directed downward.

(c) If $x = \frac{1}{2} \cos 8t = 0$ then $t = (2n+1)\pi/16$ for $n = 0, 1, 2, \dots$.

6. From $50x'' + 200x = 0$, $x(0) = 0$, and $x'(0) = -10$ we obtain $x = -5 \sin 2t$ and $x' = -10 \cos 2t$.

7. From $20x'' + 20x = 0$, $x(0) = 0$, and $x'(0) = -10$ we obtain $x = -10 \sin t$ and $x' = -10 \cos t$.

(a) The 20 kg mass has the larger amplitude.

(b) 20 kg: $x'(\pi/4) = -5\sqrt{2}$ m/s, $x'(\pi/2) = 0$ m/s; 50 kg: $x'(\pi/4) = 0$ m/s, $x'(\pi/2) = 10$ m/s

(c) If $-5 \sin 2t = -10 \sin t$ then $2 \sin t(\cos t - 1) = 0$ so that $t = n\pi$ for $n = 0, 1, 2, \dots$, placing both masses at the equilibrium position. The 50 kg mass is moving upward; the 20 kg mass is moving upward when n is even and downward when n is odd.

8. From $x'' + 16x = 0$, $x(0) = -1$, and $x'(0) = -2$ we obtain

$$x = -\cos 4t - \frac{1}{2} \sin 4t = \frac{\sqrt{5}}{2} \cos(4t - 3.6).$$

The period is $\pi/2$ seconds and the amplitude is $\sqrt{5}/2$ feet. In 4π seconds it will make 8 complete vibrations.

9. From $\frac{1}{4}x'' + x = 0$, $x(0) = 1/2$, and $x'(0) = 3/2$ we obtain

$$x = \frac{1}{2} \cos 2t + \frac{3}{4} \sin 2t = \frac{\sqrt{13}}{4} \sin(2t + 0.588).$$

10. From $1.6x'' + 40x = 0$, $x(0) = -1/3$, and $x'(0) = 5/4$ we obtain

$$x = -\frac{1}{3} \cos 5t + \frac{1}{4} \sin 5t = \frac{5}{12} \sin(5t + 0.927).$$

If $x = 5/24$ then $t = \frac{1}{5} \left(\frac{\pi}{6} + 0.927 + 2n\pi \right)$ and $t = \frac{1}{5} \left(\frac{5\pi}{6} + 0.927 + 2n\pi \right)$ for $n = 0, 1, 2, \dots$.

11. From $2x'' + 200x = 0$, $x(0) = -2/3$, and $x'(0) = 5$ we obtain

(a) $x = -\frac{2}{3} \cos 10t + \frac{1}{2} \sin 10t = \frac{5}{6} \sin(10t - 0.927)$.

(b) The amplitude is $5/6$ ft and the period is $2\pi/10 = \pi/5$

(c) $3\pi = \pi k/5$ and $k = 15$ cycles.

(d) If $x = 0$ and the weight is moving downward for the second time, then $10t - 0.927 = 2\pi$ or $t = 0.721$ s.

- (e) If $x' = \frac{25}{3} \cos(10t - 0.927) = 0$ then $10t - 0.927 = \pi/2 + n\pi$ or $t = (2n + 1)\pi/20 + 0.0927$ for $n = 0, 1, 2, \dots$.
- (f) $x(3) = -0.597$ ft
- (g) $x'(3) = -5.814$ ft/s
- (h) $x''(3) = 59.702$ ft/s²
- (i) If $x = 0$ then $t = \frac{1}{10}(0.927 + n\pi)$ for $n = 0, 1, 2, \dots$ and $x'(t) = \pm \frac{25}{3}$ ft/s.
- (j) If $x = 5/12$ then $t = \frac{1}{10}(\pi/6 + 0.927 + 2n\pi)$ and $t = \frac{1}{10}(5\pi/6 + 0.927 + 2n\pi)$ for $n = 0, 1, 2, \dots$.
- (k) If $x = 5/12$ and $x' < 0$ then $t = \frac{1}{10}(5\pi/6 + 0.927 + 2n\pi)$ for $n = 0, 1, 2, \dots$.

12. From $x' + 9x = 0$, $x(0) = -1$, and $x'(0) = -\sqrt{3}$ we obtain

$$x = -\cos 3t - \frac{\sqrt{3}}{3} \sin 3t = \frac{2}{\sqrt{3}} \sin \left(3t + \frac{4\pi}{3} \right)$$

and $x' = 2\sqrt{3} \cos(3t + 4\pi/3)$. If $x' = 3$ then $t = -7\pi/18 + 2n\pi/3$ and $t = -\pi/2 + 2n\pi/3$ for $n = 1, 2, 3, \dots$.

13. From $k_1 = 40$ and $k_2 = 120$ we compute the effective spring constant $k = 4(40)(120)/160 = 120$. Now, $m = 20/32$ so $k/m = 120(32)/20 = 192$ and $x'' + 192x = 0$. Using $x(0) = 0$ and $x'(0) = 2$ we obtain $x(t) = \frac{\sqrt{3}}{12} \sin 8\sqrt{3}t$.

14. Let m denote the mass in slugs of the first weight. Let k_1 and k_2 be the spring constants and $k = 4k_1k_2/(k_1 + k_2)$ the effective spring constant of the system. Now, the numerical value of the first weight is $W = mg = 32m$, so

$$32m = k_1 \left(\frac{1}{3} \right) \quad \text{and} \quad 32m = k_2 \left(\frac{1}{2} \right).$$

From these equations we find $2k_1 = 3k_2$. The given period of the combined system is $2\pi/\omega = \pi/15$, so $\omega = 30$. Since the mass of an 8-pound weight is $1/4$ slug, we have from $\omega^2 = k/m$

$$30^2 = \frac{k}{1/4} = 4k \quad \text{or} \quad k = 225.$$

We now have the system of equations

$$\frac{4k_1k_2}{k_1 + k_2} = 225$$

$$2k_1 = 3k_2.$$

Solving the second equation for k_1 and substituting in the first equation, we obtain

$$\frac{4(3k_2/2)k_2}{3k_2/2 + k_2} = \frac{12k_2^2}{5k_2} = \frac{12k_2}{5} = 225.$$

Thus, $k_2 = 375/4$ and $k_1 = 1125/8$. Finally, the value of the first weight is

$$W = 32m = \frac{k_1}{3} = \frac{1125/8}{3} = \frac{375}{8} \approx 46.88 \text{ lb.}$$

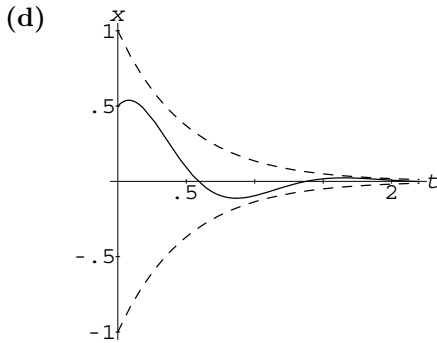
15. For large values of t the differential equation is approximated by $x'' = 0$. The solution of this equation is the linear function $x = c_1t + c_2$. Thus, for large time, the restoring force will have decayed to the point where the spring is incapable of returning the mass, and the spring will simply keep on stretching.
16. As t becomes larger the spring constant increases; that is, the spring is stiffening. It would seem that the oscillations would become periodic and the spring would oscillate more rapidly. It is likely that the amplitudes of the oscillations would decrease as t increases.

17. (a) above

- (b) heading upward

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18. (a) below (b) from rest
19. (a) below (b) heading upward
20. (a) above (b) heading downward
21. From $\frac{1}{8}x'' + x' + 2x = 0$, $x(0) = -1$, and $x'(0) = 8$ we obtain $x = 4te^{-4t} - e^{-4t}$ and $x' = 8e^{-4t} - 16te^{-4t}$. If $x = 0$ then $t = 1/4$ second. If $x' = 0$ then $t = 1/2$ second and the extreme displacement is $x = e^{-2}$ feet.
22. From $\frac{1}{4}x'' + \sqrt{2}x' + 2x = 0$, $x(0) = 0$, and $x'(0) = 5$ we obtain $x = 5te^{-2\sqrt{2}t}$ and $x' = 5e^{-2\sqrt{2}t}(1 - 2\sqrt{2}t)$. If $x' = 0$ then $t = \sqrt{2}/4$ second and the extreme displacement is $x = 5\sqrt{2}e^{-1}/4$ feet.
23. (a) From $x'' + 10x' + 16x = 0$, $x(0) = 1$, and $x'(0) = 0$ we obtain $x = \frac{4}{3}e^{-2t} - \frac{1}{3}e^{-8t}$.
 (b) From $x'' + x' + 16x = 0$, $x(0) = 1$, and $x'(0) = -12$ then $x = -\frac{2}{3}e^{-2t} + \frac{5}{3}e^{-8t}$.
24. (a) $x = \frac{1}{3}e^{-8t}(4e^{6t} - 1)$ is never zero; the extreme displacement is $x(0) = 1$ meter.
 (b) $x = \frac{1}{3}e^{-8t}(5 - 2e^{6t}) = 0$ when $t = \frac{1}{6}\ln \frac{5}{2} \approx 0.153$ second; if $x' = \frac{4}{3}e^{-8t}(e^{6t} - 10) = 0$ then $t = \frac{1}{6}\ln 10 \approx 0.384$ second and the extreme displacement is $x = -0.232$ meter.
25. (a) From $0.1x'' + 0.4x' + 2x = 0$, $x(0) = -1$, and $x'(0) = 0$ we obtain $x = e^{-2t}[-\cos 4t - \frac{1}{2}\sin 4t]$.
 (b) $x = e^{-2t}\frac{\sqrt{5}}{2}\left[-\frac{2}{\sqrt{5}}\cos 4t - \frac{1}{\sqrt{5}}\sin 4t\right] = \frac{\sqrt{5}}{2}e^{-2t}\sin(4t + 4.25)$.
 (c) If $x = 0$ then $4t + 4.25 = 2\pi, 3\pi, 4\pi, \dots$ so that the first time heading upward is $t = 1.294$ seconds.
26. (a) From $\frac{1}{4}x'' + x' + 5x = 0$, $x(0) = 1/2$, and $x'(0) = 1$ we obtain $x = e^{-2t}(\frac{1}{2}\cos 4t + \frac{1}{2}\sin 4t)$.
 (b) $x = e^{-2t}\frac{1}{\sqrt{2}}\left(\frac{\sqrt{2}}{2}\cos 4t + \frac{\sqrt{2}}{2}\sin 4t\right) = \frac{1}{\sqrt{2}}e^{-2t}\sin\left(4t + \frac{\pi}{4}\right)$.
 (c) If $x = 0$ then $4t + \pi/4 = \pi, 2\pi, 3\pi, \dots$ so that the times heading downward are $t = (7 + 8n)\pi/16$ for $n = 0, 1, 2, \dots$.



27. From $\frac{5}{16}x'' + \beta x' + 5x = 0$ we find that the roots of the auxiliary equation are $m = -\frac{8}{5}\beta \pm \frac{4}{5}\sqrt{4\beta^2 - 25}$.
 (a) If $4\beta^2 - 25 > 0$ then $\beta > 5/2$.
 (b) If $4\beta^2 - 25 = 0$ then $\beta = 5/2$.
 (c) If $4\beta^2 - 25 < 0$ then $0 < \beta < 5/2$.
28. From $0.75x'' + \beta x' + 6x = 0$ and $\beta > 3\sqrt{2}$ we find that the roots of the auxiliary equation are $m = -\frac{2}{3}\beta \pm \frac{2}{3}\sqrt{\beta^2 - 18}$ and

$$x = e^{-2\beta t/3} \left[c_1 \cosh \frac{2}{3}\sqrt{\beta^2 - 18}t + c_2 \sinh \frac{2}{3}\sqrt{\beta^2 - 18}t \right].$$

If $x(0) = 0$ and $x'(0) = -2$ then $c_1 = 0$ and $c_2 = -3/\sqrt{\beta^2 - 18}$.

29. If $\frac{1}{2}x'' + \frac{1}{2}x' + 6x = 10 \cos 3t$, $x(0) = -2$, and $x'(0) = 0$ then

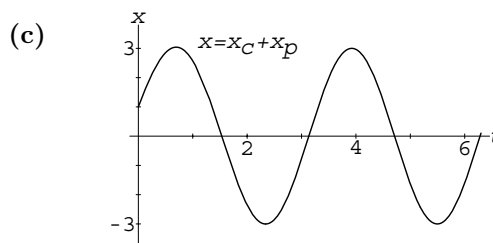
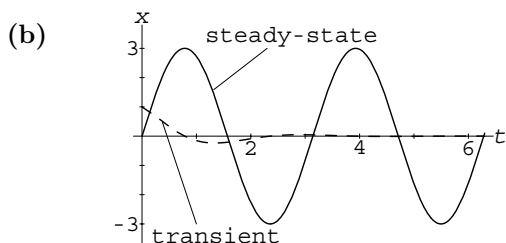
$$x_c = e^{-t/2} \left(c_1 \cos \frac{\sqrt{47}}{2} t + c_2 \sin \frac{\sqrt{47}}{2} t \right)$$

and $x_p = \frac{10}{3}(\cos 3t + \sin 3t)$ so that the equation of motion is

$$x = e^{-t/2} \left(-\frac{4}{3} \cos \frac{\sqrt{47}}{2} t - \frac{64}{3\sqrt{47}} \sin \frac{\sqrt{47}}{2} t \right) + \frac{10}{3}(\cos 3t + \sin 3t).$$

30. (a) If $x'' + 2x' + 5x = 12 \cos 2t + 3 \sin 2t$, $x(0) = -1$, and $x'(0) = 5$ then $x_c = e^{-t}(c_1 \cos 2t + c_2 \sin 2t)$ and $x_p = 3 \sin 2t$ so that the equation of motion is

$$x = e^{-t} \cos 2t + 3 \sin 2t.$$



31. From $x'' + 8x' + 16x = 8 \sin 4t$, $x(0) = 0$, and $x'(0) = 0$ we obtain $x_c = c_1 e^{-4t} + c_2 t e^{-4t}$ and $x_p = -\frac{1}{4} \cos 4t$ so that the equation of motion is

$$x = \frac{1}{4} e^{-4t} + t e^{-4t} - \frac{1}{4} \cos 4t.$$

32. From $x'' + 8x' + 16x = e^{-t} \sin 4t$, $x(0) = 0$, and $x'(0) = 0$ we obtain $x_c = c_1 e^{-4t} + c_2 t e^{-4t}$ and $x_p = -\frac{24}{625} e^{-t} \cos 4t - \frac{7}{625} e^{-t} \sin 4t$ so that

$$x = \frac{1}{625} e^{-4t} (24 + 100t) - \frac{1}{625} e^{-t} (24 \cos 4t + 7 \sin 4t).$$

As $t \rightarrow \infty$ the displacement $x \rightarrow 0$.

33. From $2x'' + 32x = 68e^{-2t} \cos 4t$, $x(0) = 0$, and $x'(0) = 0$ we obtain $x_c = c_1 \cos 4t + c_2 \sin 4t$ and $x_p = \frac{1}{2} e^{-2t} \cos 4t - 2e^{-2t} \sin 4t$ so that

$$x = -\frac{1}{2} \cos 4t + \frac{9}{4} \sin 4t + \frac{1}{2} e^{-2t} \cos 4t - 2e^{-2t} \sin 4t.$$

34. Since $x = \frac{\sqrt{85}}{4} \sin(4t - 0.219) - \frac{\sqrt{17}}{2} e^{-2t} \sin(4t - 2.897)$, the amplitude approaches $\sqrt{85}/4$ as $t \rightarrow \infty$.

35. (a) By Hooke's law the external force is $F(t) = kh(t)$ so that $mx'' + \beta x' + kx = kh(t)$.

- (b) From $\frac{1}{2}x'' + 2x' + 4x = 20 \cos t$, $x(0) = 0$, and $x'(0) = 0$ we obtain $x_c = e^{-2t}(c_1 \cos 2t + c_2 \sin 2t)$ and $x_p = \frac{56}{13} \cos t + \frac{32}{13} \sin t$ so that

$$x = e^{-2t} \left(-\frac{56}{13} \cos 2t - \frac{72}{13} \sin 2t \right) + \frac{56}{13} \cos t + \frac{32}{13} \sin t.$$

36. (a) From $100x'' + 1600x = 1600 \sin 8t$, $x(0) = 0$, and $x'(0) = 0$ we obtain $x_c = c_1 \cos 4t + c_2 \sin 4t$ and $x_p = -\frac{1}{3} \sin 8t$ so that

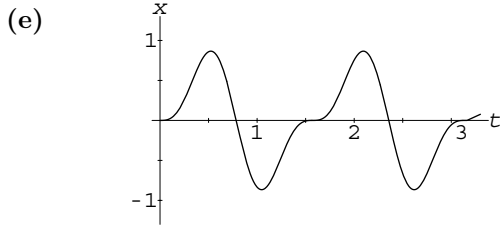
$$x = \frac{2}{3} \sin 4t - \frac{1}{3} \sin 8t.$$

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(b) If $x = \frac{1}{3} \sin 4t(2 - 2 \cos 4t) = 0$ then $t = n\pi/4$ for $n = 0, 1, 2, \dots$.

(c) If $x' = \frac{8}{3} \cos 4t - \frac{8}{3} \cos 8t = \frac{8}{3}(1 - \cos 4t)(1 + 2 \cos 4t) = 0$ then $t = \pi/3 + n\pi/2$ and $t = \pi/6 + n\pi/2$ for $n = 0, 1, 2, \dots$ at the extreme values. *Note:* There are many other values of t for which $x' = 0$.

(d) $x(\pi/6 + n\pi/2) = \sqrt{3}/2$ cm. and $x(\pi/3 + n\pi/2) = -\sqrt{3}/2$ cm.



37. From $x'' + 4x = -5 \sin 2t + 3 \cos 2t$, $x(0) = -1$, and $x'(0) = 1$ we obtain $x_c = c_1 \cos 2t + c_2 \sin 2t$, $x_p = \frac{3}{4}t \sin 2t + \frac{5}{4}t \cos 2t$, and

$$x = -\cos 2t - \frac{1}{8} \sin 2t + \frac{3}{4}t \sin 2t + \frac{5}{4}t \cos 2t.$$

38. From $x'' + 9x = 5 \sin 3t$, $x(0) = 2$, and $x'(0) = 0$ we obtain $x_c = c_1 \cos 3t + c_2 \sin 3t$, $x_p = -\frac{5}{6}t \cos 3t$, and

$$x = 2 \cos 3t + \frac{5}{18} \sin 3t - \frac{5}{6}t \cos 3t.$$

39. (a) From $x'' + \omega^2 x = F_0 \cos \gamma t$, $x(0) = 0$, and $x'(0) = 0$ we obtain $x_c = c_1 \cos \omega t + c_2 \sin \omega t$ and $x_p = (F_0 \cos \gamma t) / (\omega^2 - \gamma^2)$ so that

$$x = -\frac{F_0}{\omega^2 - \gamma^2} \cos \omega t + \frac{F_0}{\omega^2 - \gamma^2} \cos \gamma t.$$

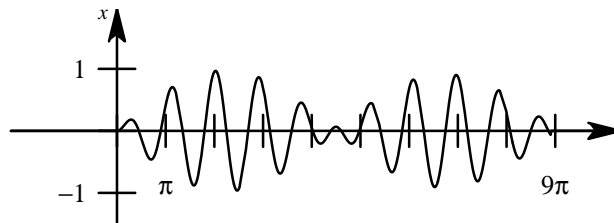
$$(b) \lim_{\gamma \rightarrow \omega} \frac{F_0}{\omega^2 - \gamma^2} (\cos \gamma t - \cos \omega t) = \lim_{\gamma \rightarrow \omega} \frac{-F_0 t \sin \gamma t}{-2\gamma} = \frac{F_0}{2\omega} t \sin \omega t.$$

40. From $x'' + \omega^2 x = F_0 \cos \omega t$, $x(0) = 0$, and $x'(0) = 0$ we obtain $x_c = c_1 \cos \omega t + c_2 \sin \omega t$ and $x_p = (F_0 t / 2\omega) \sin \omega t$ so that $x = (F_0 t / 2\omega) \sin \omega t$ and $\lim_{\gamma \rightarrow \omega} \frac{F_0}{2\omega} t \sin \omega t = \frac{F_0}{2\omega} t \sin \omega t$.

41. (a) From $\cos(u - v) = \cos u \cos v + \sin u \sin v$ and $\cos(u + v) = \cos u \cos v - \sin u \sin v$ we obtain $\sin u \sin v = \frac{1}{2}[\cos(u - v) - \cos(u + v)]$. Letting $u = \frac{1}{2}(\gamma - \omega)t$ and $v = \frac{1}{2}(\gamma + \omega)t$, the result follows.

(b) If $\epsilon = \frac{1}{2}(\gamma - \omega)$ then $\gamma \approx \omega$ so that $x = (F_0 / 2\epsilon\gamma) \sin \epsilon t \sin \gamma t$.

42. See the article "Distinguished Oscillations of a Forced Harmonic Oscillator" by T.G. Procter in *The College Mathematics Journal*, March, 1995. In this article the author illustrates that for $F_0 = 1$, $\lambda = 0.01$, $\gamma = 22/9$, and $\omega = 2$ the system exhibits beats oscillations on the interval $[0, 9\pi]$, but that this phenomenon is transient as $t \rightarrow \infty$.



43. (a) The general solution of the homogeneous equation is

$$\begin{aligned}x_c(t) &= c_1 e^{-\lambda t} \cos(\sqrt{\omega^2 - \lambda^2} t) + c_2 e^{-\lambda t} \sin(\sqrt{\omega^2 - \lambda^2} t) \\&= A e^{-\lambda t} \sin[\sqrt{\omega^2 - \lambda^2} t + \phi],\end{aligned}$$

where $A = \sqrt{c_1^2 + c_2^2}$, $\sin \phi = c_1/A$, and $\cos \phi = c_2/A$. Now

$$x_p(t) = \frac{F_0(\omega^2 - \gamma^2)}{(\omega^2 - \gamma^2)^2 + 4\lambda^2\gamma^2} \sin \gamma t + \frac{F_0(-2\lambda\gamma)}{(\omega^2 - \gamma^2)^2 + 4\lambda^2\gamma^2} \cos \gamma t = A \sin(\gamma t + \theta),$$

where

$$\sin \theta = \frac{\frac{F_0(-2\lambda\gamma)}{(\omega^2 - \gamma^2)^2 + 4\lambda^2\gamma^2}}{\frac{F_0}{\sqrt{(\omega^2 - \gamma^2)^2 + 4\lambda^2\gamma^2}}} = \frac{-2\lambda\gamma}{\sqrt{(\omega^2 - \gamma^2)^2 + 4\lambda^2\gamma^2}}$$

and

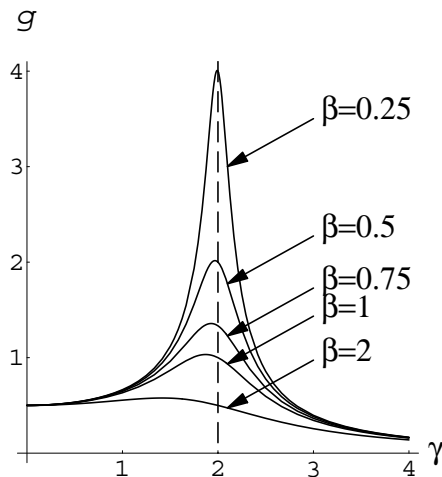
$$\cos \theta = \frac{\frac{F_0(\omega^2 - \gamma^2)}{(\omega^2 - \gamma^2)^2 + 4\lambda^2\gamma^2}}{\frac{F_0}{\sqrt{(\omega^2 - \gamma^2)^2 + 4\lambda^2\gamma^2}}} = \frac{\omega^2 - \gamma^2}{\sqrt{(\omega^2 - \gamma^2)^2 + 4\lambda^2\gamma^2}}.$$

- (b) If $g'(\gamma) = 0$ then $\gamma(\gamma^2 + 2\lambda^2 - \omega^2) = 0$ so that $\gamma = 0$ or $\gamma = \sqrt{\omega^2 - 2\lambda^2}$. The first derivative test shows that g has a maximum value at $\gamma = \sqrt{\omega^2 - 2\lambda^2}$. The maximum value of g is

$$g(\sqrt{\omega^2 - 2\lambda^2}) = F_0/2\lambda\sqrt{\omega^2 - \lambda^2}.$$

- (c) We identify $\omega^2 = k/m = 4$, $\lambda = \beta/2$, and $\gamma_1 = \sqrt{\omega^2 - 2\lambda^2} = \sqrt{4 - \beta^2/2}$. As $\beta \rightarrow 0$, $\gamma_1 \rightarrow 2$ and the resonance curve grows without bound at $\gamma_1 = 2$. That is, the system approaches pure resonance.

β	γ_1	g
2.00	1.41	0.58
1.00	1.87	1.03
0.75	1.93	1.36
0.50	1.97	2.02
0.25	1.99	4.01



44. (a) For $n = 2$, $\sin^2 \gamma t = \frac{1}{2}(1 - \cos 2\gamma t)$. The system is in pure resonance when $2\gamma_1/2\pi = \omega/2\pi$, or when $\gamma_1 = \omega/2$.
- (b) Note that

$$\sin^3 \gamma t = \sin \gamma t \sin^2 \gamma t = \frac{1}{2}[\sin \gamma t - \sin \gamma t \cos 2\gamma t].$$

Now

$$\sin(A + B) + \sin(A - B) = 2 \sin A \cos B$$

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so

$$\sin \gamma t \cos 2\gamma t = \frac{1}{2}[\sin 3\gamma t - \sin \gamma t]$$

and

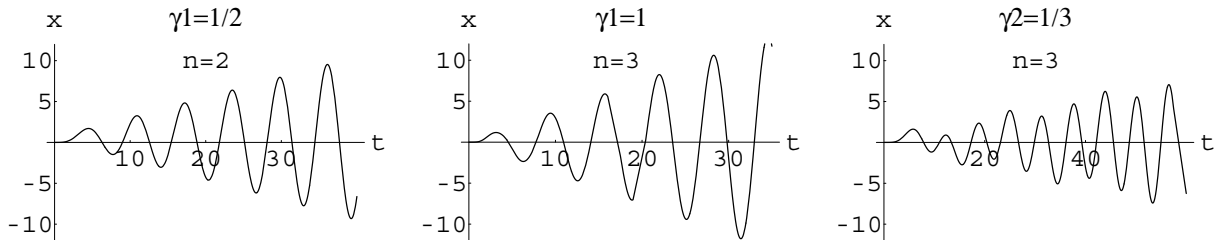
$$\sin^3 \gamma t = \frac{3}{4} \sin \gamma t - \frac{1}{4} \sin 3\gamma t.$$

Thus

$$x'' + \omega^2 x = \frac{3}{4} \sin \gamma t - \frac{1}{4} \sin 3\gamma t.$$

The frequency of free vibration is $\omega/2\pi$. Thus, when $\gamma_1/2\pi = \omega/2\pi$ or $\gamma_1 = \omega$, and when $3\gamma_2/2\pi = \omega/2\pi$ or $3\gamma_2 = \omega$ or $\gamma_3 = \omega/3$, the system will be in pure resonance.

(c)



45. Solving $\frac{1}{20}q'' + 2q' + 100q = 0$ we obtain $q(t) = e^{-20t}(c_1 \cos 40t + c_2 \sin 40t)$. The initial conditions $q(0) = 5$ and $q'(0) = 0$ imply $c_1 = 5$ and $c_2 = 5/2$. Thus

$$q(t) = e^{-20t} \left(5 \cos 40t + \frac{5}{2} \sin 40t \right) \approx \sqrt{25 + 25/4} e^{-20t} \sin(40t + 1.1071)$$

and $q(0.01) \approx 4.5676$ coulombs. The charge is zero for the first time when $40t + 0.4636 = \pi$ or $t \approx 0.0509$ second.

46. Solving $\frac{1}{4}q'' + 20q' + 300q = 0$ we obtain $q(t) = c_1 e^{-20t} + c_2 e^{-60t}$. The initial conditions $q(0) = 4$ and $q'(0) = 0$ imply $c_1 = 6$ and $c_2 = -2$. Thus

$$q(t) = 6e^{-20t} - 2e^{-60t}.$$

Setting $q = 0$ we find $e^{40t} = 1/3$ which implies $t < 0$. Therefore the charge is never 0.

47. Solving $\frac{5}{3}q'' + 10q' + 30q = 300$ we obtain $q(t) = e^{-3t}(c_1 \cos 3t + c_2 \sin 3t) + 10$. The initial conditions $q(0) = q'(0) = 0$ imply $c_1 = c_2 = -10$. Thus

$$q(t) = 10 - 10e^{-3t}(\cos 3t + \sin 3t) \quad \text{and} \quad i(t) = 60e^{3t} \sin 3t.$$

Solving $i(t) = 0$ we see that the maximum charge occurs when $t = \pi/3$ and $q(\pi/3) \approx 10.432$ coulombs.

48. Solving $q'' + 100q' + 2500q = 30$ we obtain $q(t) = c_1 e^{-50t} + c_2 t e^{-50t} + 0.012$. The initial conditions $q(0) = 0$ and $q'(0) = 2$ imply $c_1 = -0.012$ and $c_2 = 1.4$. Thus

$$q(t) = -0.012e^{-50t} + 1.4te^{-50t} + 0.012 \quad \text{and} \quad i(t) = 2e^{-50t} - 70te^{-50t}.$$

Solving $i(t) = 0$ we see that the maximum charge occurs when $t = 1/35$ and $q(1/35) \approx 0.01871$.

49. Solving $q'' + 2q' + 4q = 0$ we obtain $y_c = e^{-t}(\cos \sqrt{3}t + \sin \sqrt{3}t)$. The steady-state charge has the form $y_p = A \cos t + B \sin t$. Substituting into the differential equation we find

$$(3A + 2B) \cos t + (3B - 2A) \sin t = 50 \cos t.$$

Thus, $A = 150/13$ and $B = 100/13$. The steady-state charge is

$$q_p(t) = \frac{150}{13} \cos t + \frac{100}{13} \sin t$$

and the steady-state current is

$$i_p(t) = -\frac{150}{13} \sin t + \frac{100}{13} \cos t.$$

50. From

$$i_p(t) = \frac{E_0}{Z} \left(\frac{R}{Z} \sin \gamma t - \frac{X}{Z} \cos \gamma t \right)$$

and $Z = \sqrt{X^2 + R^2}$ we see that the amplitude of $i_p(t)$ is

$$A = \sqrt{\frac{E_0^2 R^2}{Z^4} + \frac{E_0^2 X^2}{Z^4}} = \frac{E_0}{Z^2} \sqrt{R^2 + X^2} = \frac{E_0}{Z}.$$

51. The differential equation is $\frac{1}{2}q'' + 20q' + 1000q = 100 \sin t$. To use Example 11 in the text we identify $E_0 = 100$ and $\gamma = 60$. Then

$$X = L\gamma - \frac{1}{C\gamma} = \frac{1}{2}(60) - \frac{1}{0.001(60)} \approx 13.3333,$$

$$Z = \sqrt{X^2 + R^2} = \sqrt{X^2 + 400} \approx 24.0370,$$

and

$$\frac{E_0}{Z} = \frac{100}{Z} \approx 4.1603.$$

From Problem 50, then

$$i_p(t) \approx 4.1603(60t + \phi)$$

where $\sin \phi = -X/Z$ and $\cos \phi = R/Z$. Thus $\tan \phi = -X/R \approx -0.6667$ and ϕ is a fourth quadrant angle. Now $\phi \approx -0.5880$ and

$$i_p(t) \approx 4.1603(60t - 0.5880).$$

52. Solving $\frac{1}{2}q'' + 20q' + 1000q = 0$ we obtain $q_c(t) = (c_1 \cos 40t + c_2 \sin 40t)$. The steady-state charge has the form $q_p(t) = A \sin 60t + B \cos 60t + C \sin 40t + D \cos 40t$. Substituting into the differential equation we find

$$\begin{aligned} & (-1600A - 2400B) \sin 60t + (2400A - 1600B) \cos 60t \\ & + (400C - 1600D) \sin 40t + (1600C + 400D) \cos 40t \\ & = 200 \sin 60t + 400 \cos 40t. \end{aligned}$$

Equating coefficients we obtain $A = -1/26$, $B = -3/52$, $C = 4/17$, and $D = 1/17$. The steady-state charge is

$$q_p(t) = -\frac{1}{26} \sin 60t - \frac{3}{52} \cos 60t + \frac{4}{17} \sin 40t + \frac{1}{17} \cos 40t$$

and the steady-state current is

$$i_p(t) = -\frac{30}{13} \cos 60t + \frac{45}{13} \sin 60t + \frac{160}{17} \cos 40t - \frac{40}{17} \sin 40t.$$

53. Solving $\frac{1}{2}q'' + 10q' + 100q = 150$ we obtain $q(t) = e^{-10t}(c_1 \cos 10t + c_2 \sin 10t) + 3/2$. The initial conditions $q(0) = 1$ and $q'(0) = 0$ imply $c_1 = c_2 = -1/2$. Thus

$$q(t) = -\frac{1}{2}e^{-10t}(\cos 10t + \sin 10t) + \frac{3}{2}.$$

As $t \rightarrow \infty$, $q(t) \rightarrow 3/2$.

54. By Problem 50 the amplitude of the steady-state current is E_0/Z , where $Z = \sqrt{X^2 + R^2}$ and $X = L\gamma - 1/C\gamma$. Since E_0 is constant the amplitude will be a maximum when Z is a minimum. Since R is constant, Z will be a

Exercises 3.8

minimum when $X = 0$. Solving $L\gamma - 1/C\gamma = 0$ for γ we obtain $\gamma = 1/\sqrt{LC}$. The maximum amplitude will be E_0/R .

55. By Problem 50 the amplitude of the steady-state current is E_0/Z , where $Z = \sqrt{X^2 + R^2}$ and $X = L\gamma - 1/C\gamma$. Since E_0 is constant the amplitude will be a maximum when Z is a minimum. Since R is constant, Z will be a minimum when $X = 0$. Solving $L\gamma - 1/C\gamma = 0$ for C we obtain $C = 1/L\gamma^2$.

56. Solving $0.1q'' + 10q = 100 \sin \gamma t$ we obtain $q(t) = c_1 \cos 10t + c_2 \sin 10t + q_p(t)$ where $q_p(t) = A \sin \gamma t + B \cos \gamma t$. Substituting $q_p(t)$ into the differential equation we find

$$(100 - \gamma^2)A \sin \gamma t + (100 - \gamma^2)B \cos \gamma t = 100 \sin \gamma t.$$

Equating coefficients we obtain $A = 100/(100 - \gamma^2)$ and $B = 0$. Thus, $q_p(t) = \frac{100}{100 - \gamma^2} \sin \gamma t$. The initial conditions $q(0) = q'(0) = 0$ imply $c_1 = 0$ and $c_2 = -10\gamma/(100 - \gamma^2)$. The charge is

$$q(t) = \frac{10}{100 - \gamma^2} (10 \sin \gamma t - \gamma \sin 10t)$$

and the current is

$$i(t) = \frac{100\gamma}{100 - \gamma^2} (\cos \gamma t - \cos 10t).$$

57. In an L - C series circuit there is no resistor, so the differential equation is

$$L \frac{d^2 q}{dt^2} + \frac{1}{C} q = E(t).$$

Then $q(t) = c_1 \cos(t/\sqrt{LC}) + c_2 \sin(t/\sqrt{LC}) + q_p(t)$ where $q_p(t) = A \sin \gamma t + B \cos \gamma t$. Substituting $q_p(t)$ into the differential equation we find

$$\left(\frac{1}{C} - L\gamma^2\right) A \sin \gamma t + \left(\frac{1}{C} - L\gamma^2\right) B \cos \gamma t = E_0 \cos \gamma t.$$

Equating coefficients we obtain $A = 0$ and $B = E_0 C / (1 - LC\gamma^2)$. Thus, the charge is

$$q(t) = c_1 \cos \frac{1}{\sqrt{LC}} t + c_2 \sin \frac{1}{\sqrt{LC}} t + \frac{E_0 C}{1 - LC\gamma^2} \cos \gamma t.$$

The initial conditions $q(0) = q_0$ and $q'(0) = i_0$ imply $c_1 = q_0 - E_0 C / (1 - LC\gamma^2)$ and $c_2 = i_0 \sqrt{LC}$. The current is

$$\begin{aligned} i(t) &= -\frac{c_1}{\sqrt{LC}} \sin \frac{1}{\sqrt{LC}} t + \frac{c_2}{\sqrt{LC}} \cos \frac{1}{\sqrt{LC}} t - \frac{E_0 C \gamma}{1 - LC\gamma^2} \sin \gamma t \\ &= i_0 \cos \frac{1}{\sqrt{LC}} t - \frac{1}{\sqrt{LC}} \left(q_0 - \frac{E_0 C}{1 - LC\gamma^2} \right) \sin \frac{1}{\sqrt{LC}} t - \frac{E_0 C \gamma}{1 - LC\gamma^2} \sin \gamma t. \end{aligned}$$

58. When the circuit is in resonance the form of $q_p(t)$ is $q_p(t) = At \cos kt + Bt \sin kt$ where $k = 1/\sqrt{LC}$. Substituting $q_p(t)$ into the differential equation we find

$$q_p'' + k^2 q_p = -2kA \sin kt + 2kB \cos kt = \frac{E_0}{L} \cos kt.$$

Equating coefficients we obtain $A = 0$ and $B = E_0 / 2kL$. The charge is

$$q(t) = c_1 \cos kt + c_2 \sin kt + \frac{E_0}{2kL} t \sin kt.$$

The initial conditions $q(0) = q_0$ and $q'(0) = i_0$ imply $c_1 = q_0$ and $c_2 = i_0/k$. The current is

$$\begin{aligned} i(t) &= -c_1 k \sin kt + c_2 k \cos kt + \frac{E_0}{2kL} (kt \cos kt + \sin kt) \\ &= \left(\frac{E_0}{2kL} - q_0 k \right) \sin kt + i_0 \cos kt + \frac{E_0}{2L} t \cos kt. \end{aligned}$$

Exercises 3.9

1. (a) The general solution is

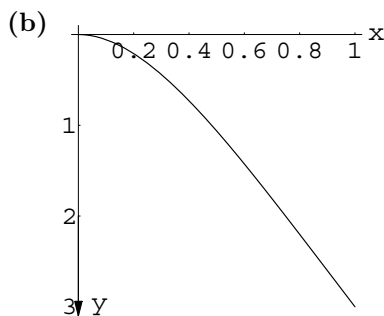
$$y(x) = c_1 + c_2 x + c_3 x^2 + c_4 x^3 + \frac{w_0}{24EI} x^4.$$

The boundary conditions are $y(0) = 0$, $y'(0) = 0$, $y''(L) = 0$, $y'''(L) = 0$. The first two conditions give $c_1 = 0$ and $c_2 = 0$. The conditions at $x = L$ give the system

$$\begin{aligned} 2c_3 + 6c_4 L + \frac{w_0}{2EI} L^2 &= 0 \\ 6c_4 + \frac{w_0}{EI} L &= 0. \end{aligned}$$

Solving, we obtain $c_3 = w_0 L^2 / 4EI$ and $c_4 = -w_0 L / 6EI$. The deflection is

$$y(x) = \frac{w_0}{24EI} (6L^2 x^2 - 4Lx^3 + x^4).$$



2. (a) The general solution is

$$y(x) = c_1 + c_2 x + c_3 x^2 + c_4 x^3 + \frac{w_0}{24EI} x^4.$$

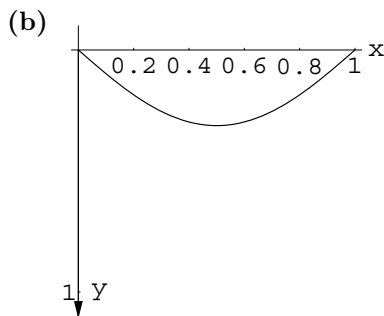
The boundary conditions are $y(0) = 0$, $y''(0) = 0$, $y(L) = 0$, $y''(L) = 0$. The first two conditions give $c_1 = 0$ and $c_3 = 0$. The conditions at $x = L$ give the system

$$\begin{aligned} c_2 L + c_4 L^3 + \frac{w_0}{24EI} L^4 &= 0 \\ 6c_4 L + \frac{w_0}{2EI} L^2 &= 0. \end{aligned}$$

Solving, we obtain $c_2 = w_0 L^3 / 24EI$ and $c_4 = -w_0 L / 12EI$. The deflection is

$$y(x) = \frac{w_0}{24EI} (L^3 x - 2Lx^3 + x^4).$$

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3. (a) The general solution is

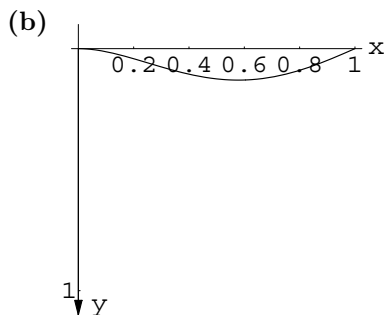
$$y(x) = c_1 + c_2x + c_3x^2 + c_4x^3 + \frac{w_0}{24EI}x^4.$$

The boundary conditions are $y(0) = 0$, $y'(0) = 0$, $y(L) = 0$, $y''(L) = 0$. The first two conditions give $c_1 = 0$ and $c_2 = 0$. The conditions at $x = L$ give the system

$$\begin{aligned} c_3L^2 + c_4L^3 + \frac{w_0}{24EI}L^4 &= 0 \\ 2c_3 + 6c_4L + \frac{w_0}{2EI}L^2 &= 0. \end{aligned}$$

Solving, we obtain $c_3 = w_0L^2/16EI$ and $c_4 = -5w_0L/48EI$. The deflection is

$$y(x) = \frac{w_0}{48EI}(3L^2x^2 - 5Lx^3 + 2x^4).$$



4. (a) The general solution is

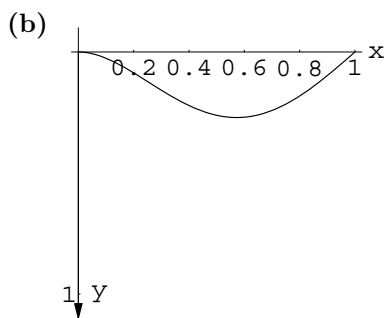
$$y(x) = c_1 + c_2x + c_3x^2 + c_4x^3 + \frac{w_0L^4}{EI\pi^4} \sin \frac{\pi}{L}x.$$

The boundary conditions are $y(0) = 0$, $y'(0) = 0$, $y(L) = 0$, $y''(L) = 0$. The first two conditions give $c_1 = 0$ and $c_2 = -w_0L^3/EI\pi^3$. The conditions at $x = L$ give the system

$$\begin{aligned} c_3L^2 + c_4L^3 + \frac{w_0}{EI\pi^3}L^4 &= 0 \\ 2c_3 + 6c_4L &= 0. \end{aligned}$$

Solving, we obtain $c_3 = 3w_0L^2/2EI\pi^3$ and $c_4 = -w_0L/2EI\pi^3$. The deflection is

$$y(x) = \frac{w_0L}{2EI\pi^3} \left(-2L^2x + 3Lx^2 - x^3 + \frac{2L^3}{\pi} \sin \frac{\pi}{L}x \right).$$



5. (a) $y_{\max} = y(L) = \frac{w_0 L^4}{8EI}$
 (b) Replacing both L and x by $L/2$ in $y(x)$ we obtain $w_0 L^4 / 128EI$, which is $1/16$ of the maximum deflection when the length of the beam is L .
6. (a) $y_{\max} = y(L/2) = \frac{5w_0 L^4}{384EI}$
 (b) The maximum deflection of the beam in Example 1 is $y(L/2) = (w_0/24EI)L^4/16 = w_0 L^4 / 384EI$, which is $1/5$ of the maximum displacement of the beam in Problem 2.
7. The general solution of the differential equation is

$$y = c_1 \cosh \sqrt{\frac{P}{EI}} x + c_2 \sinh \sqrt{\frac{P}{EI}} x + \frac{w_0}{2P} x^2 + \frac{w_0 EI}{P^2}.$$

Setting $y(0) = 0$ we obtain $c_1 = -w_0 EI / P^2$, so that

$$y = -\frac{w_0 EI}{P^2} \cosh \sqrt{\frac{P}{EI}} x + c_2 \sinh \sqrt{\frac{P}{EI}} x + \frac{w_0}{2P} x^2 + \frac{w_0 EI}{P^2}.$$

Setting $y'(L) = 0$ we find

$$c_2 = \left(\sqrt{\frac{P}{EI}} \frac{w_0 EI}{P^2} \sinh \sqrt{\frac{P}{EI}} L - \frac{w_0 L}{P} \right) / \left(\sqrt{\frac{P}{EI}} \cosh \sqrt{\frac{P}{EI}} L \right).$$

8. The general solution of the differential equation is

$$y = c_1 \cos \sqrt{\frac{P}{EI}} x + c_2 \sin \sqrt{\frac{P}{EI}} x + \frac{w_0}{2P} x^2 + \frac{w_0 EI}{P^2}.$$

Setting $y(0) = 0$ we obtain $c_1 = -w_0 EI / P^2$, so that

$$y = -\frac{w_0 EI}{P^2} \cos \sqrt{\frac{P}{EI}} x + c_2 \sin \sqrt{\frac{P}{EI}} x + \frac{w_0}{2P} x^2 + \frac{w_0 EI}{P^2}.$$

Setting $y'(L) = 0$ we find

$$c_2 = \left(-\sqrt{\frac{P}{EI}} \frac{w_0 EI}{P^2} \sin \sqrt{\frac{P}{EI}} L - \frac{w_0 L}{P} \right) / \left(\sqrt{\frac{P}{EI}} \cos \sqrt{\frac{P}{EI}} L \right).$$

9. For $\lambda \leq 0$ the only solution of the boundary-value problem is $y = 0$. For $\lambda > 0$ we have

$$y = c_1 \cos \sqrt{\lambda} x + c_2 \sin \sqrt{\lambda} x.$$

Now $y(0) = 0$ implies $c_1 = 0$, so

$$y(\pi) = c_2 \sin \sqrt{\lambda} \pi = 0$$

gives

$$\sqrt{\lambda} \pi = n\pi \quad \text{or} \quad \lambda = n^2, \quad n = 1, 2, 3, \dots$$

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The eigenvalues n^2 correspond to the eigenfunctions $\sin nx$ for $n = 1, 2, 3, \dots$.

10. For $\lambda \leq 0$ the only solution of the boundary-value problem is $y = 0$. For $\lambda > 0$ we have

$$y = c_1 \cos \sqrt{\lambda} x + c_2 \sin \sqrt{\lambda} x.$$

Now $y(0) = 0$ implies $c_1 = 0$, so

$$y\left(\frac{\pi}{4}\right) = c_2 \sin \sqrt{\lambda} \frac{\pi}{4} = 0$$

gives

$$\sqrt{\lambda} \frac{\pi}{4} = n\pi \quad \text{or} \quad \lambda = 16n^2, \quad n = 1, 2, 3, \dots$$

The eigenvalues $16n^2$ correspond to the eigenfunctions $\sin 4nx$ for $n = 1, 2, 3, \dots$.

11. For $\lambda \leq 0$ the only solution of the boundary-value problem is $y = 0$. For $\lambda > 0$ we have

$$y = c_1 \cos \sqrt{\lambda} x + c_2 \sin \sqrt{\lambda} x.$$

Now

$$y'(x) = -c_1 \sqrt{\lambda} \sin \sqrt{\lambda} x + c_2 \sqrt{\lambda} \cos \sqrt{\lambda} x$$

and $y'(0) = 0$ implies $c_2 = 0$, so

$$y(L) = c_1 \cos \sqrt{\lambda} L = 0$$

gives

$$\sqrt{\lambda} L = \frac{(2n-1)\pi}{2} \quad \text{or} \quad \lambda = \frac{(2n-1)^2 \pi^2}{4L^2}, \quad n = 1, 2, 3, \dots$$

The eigenvalues $(2n-1)^2 \pi^2 / 4L^2$ correspond to the eigenfunctions $\cos \frac{(2n-1)\pi}{2L} x$ for $n = 1, 2, 3, \dots$.

12. For $\lambda \leq 0$ the only solution of the boundary-value problem is $y = 0$. For $\lambda > 0$ we have

$$y = c_1 \cos \sqrt{\lambda} x + c_2 \sin \sqrt{\lambda} x.$$

Now $y(0) = 0$ implies $c_1 = 0$, so

$$y'\left(\frac{\pi}{2}\right) = c_2 \sqrt{\lambda} \cos \sqrt{\lambda} \frac{\pi}{2} = 0$$

gives

$$\sqrt{\lambda} \frac{\pi}{2} = \frac{(2n-1)\pi}{2} \quad \text{or} \quad \lambda = (2n-1)^2, \quad n = 1, 2, 3, \dots$$

The eigenvalues $(2n-1)^2$ correspond to the eigenfunctions $\sin(2n-1)x$.

13. For $\lambda < 0$ the only solution of the boundary-value problem is $y = 0$. For $\lambda = 0$ we have $y = c_1 x + c_2$. Now $y' = c_1$ and $y'(0) = 0$ implies $c_1 = 0$. Then $y = c_2$ and $y'(\pi) = 0$. Thus, $\lambda = 0$ is an eigenvalue with corresponding eigenfunction $y = 1$.

For $\lambda > 0$ we have

$$y = c_1 \cos \sqrt{\lambda} x + c_2 \sin \sqrt{\lambda} x.$$

Now

$$y'(x) = -c_1 \sqrt{\lambda} \sin \sqrt{\lambda} x + c_2 \sqrt{\lambda} \cos \sqrt{\lambda} x$$

and $y'(0) = 0$ implies $c_2 = 0$, so

$$y'(\pi) = -c_1 \sqrt{\lambda} \sin \sqrt{\lambda} \pi = 0$$

gives

$$\sqrt{\lambda} \pi = n\pi \quad \text{or} \quad \lambda = n^2, \quad n = 1, 2, 3, \dots$$

The eigenvalues n^2 correspond to the eigenfunctions $\cos nx$ for $n = 0, 1, 2, \dots$.

14. For $\lambda \leq 0$ the only solution of the boundary-value problem is $y = 0$. For $\lambda > 0$ we have

$$y = c_1 \cos \sqrt{\lambda} x + c_2 \sin \sqrt{\lambda} x.$$

Now $y(-\pi) = y(\pi) = 0$ implies

$$\begin{aligned} c_1 \cos \sqrt{\lambda} \pi - c_2 \sin \sqrt{\lambda} \pi &= 0 \\ c_1 \cos \sqrt{\lambda} \pi + c_2 \sin \sqrt{\lambda} \pi &= 0. \end{aligned} \tag{1}$$

This homogeneous system will have a nontrivial solution when

$$\begin{vmatrix} \cos \sqrt{\lambda} \pi & -\sin \sqrt{\lambda} \pi \\ \cos \sqrt{\lambda} \pi & \sin \sqrt{\lambda} \pi \end{vmatrix} = 2 \sin \sqrt{\lambda} \pi \cos \sqrt{\lambda} \pi = \sin 2\sqrt{\lambda} \pi = 0.$$

Then

$$2\sqrt{\lambda} \pi = n\pi \quad \text{or} \quad \lambda = \frac{n^2}{4}; \quad n = 1, 2, 3, \dots$$

When $n = 2k - 1$ is odd, the eigenvalues are $(2k - 1)^2/4$. Since $\cos(2k - 1)\pi/2 = 0$ and $\sin(2k - 1)\pi/2 \neq 0$, we see from either equation in (1) that $c_2 = 0$. Thus, the eigenfunctions corresponding to the eigenvalues $(2k - 1)^2/4$ are $y = \cos(2k - 1)x/2$ for $k = 1, 2, 3, \dots$. Similarly, when $n = 2k$ is even, the eigenvalues are k^2 with corresponding eigenfunctions $y = \sin kx$ for $k = 1, 2, 3, \dots$.

15. The auxiliary equation has solutions

$$m = \frac{1}{2} \left(-2 \pm \sqrt{4 - 4(\lambda + 1)} \right) = -1 \pm \sqrt{-\lambda}.$$

For $\lambda < 0$ we have

$$y = e^{-x} \left(c_1 \cosh \sqrt{-\lambda} x + c_2 \sinh \sqrt{-\lambda} x \right).$$

The boundary conditions imply

$$\begin{aligned} y(0) &= c_1 = 0 \\ y(5) &= c_2 e^{-5} \sinh 5\sqrt{-\lambda} = 0 \end{aligned}$$

so $c_1 = c_2 = 0$ and the only solution of the boundary-value problem is $y = 0$.

For $\lambda = 0$ we have

$$y = c_1 e^{-x} + c_2 x e^{-x}$$

and the only solution of the boundary-value problem is $y = 0$.

For $\lambda > 0$ we have

$$y = e^{-x} \left(c_1 \cos \sqrt{\lambda} x + c_2 \sin \sqrt{\lambda} x \right).$$

Now $y(0) = 0$ implies $c_1 = 0$, so

$$y(5) = c_2 e^{-5} \sin 5\sqrt{\lambda} = 0$$

gives

$$5\sqrt{\lambda} = n\pi \quad \text{or} \quad \lambda = \frac{n^2 \pi^2}{25}, \quad n = 1, 2, 3, \dots$$

The eigenvalues $n^2 \pi^2 / 25$ correspond to the eigenfunctions $e^{-x} \sin \frac{n\pi}{5} x$ for $n = 1, 2, 3, \dots$.

16. For $\lambda < -1$ the only solution of the boundary-value problem is $y = 0$. For $\lambda = -1$ we have $y = c_1 x + c_2$. Now $y' = c_1$ and $y'(0) = 0$ implies $c_1 = 0$. Then $y = c_2$ and $y'(1) = 0$. Thus, $\lambda = -1$ is an eigenvalue with corresponding eigenfunction $y = 1$.

For $\lambda > -1$ we have

$$y = c_1 \cos \sqrt{\lambda + 1} x + c_2 \sin \sqrt{\lambda + 1} x.$$

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Now

$$y' = -c_1\sqrt{\lambda+1}\sin\sqrt{\lambda+1}x + c_2\sqrt{\lambda+1}\cos\sqrt{\lambda+1}x$$

and $y'(0) = 0$ implies $c_2 = 0$, so

$$y'(1) = -c_1\sqrt{\lambda+1}\sin\sqrt{\lambda+1} = 0$$

gives

$$\sqrt{\lambda+1} = n\pi \quad \text{or} \quad \lambda = n^2\pi^2 - 1, \quad n = 1, 2, 3, \dots$$

The eigenvalues $n^2\pi^2 - 1$ correspond to the eigenfunctions $\cos n\pi x$ for $n = 0, 1, 2, \dots$.

17. For $\lambda = 0$ the only solution of the boundary-value problem is $y = 0$. For $\lambda \neq 0$ we have

$$y = c_1 \cos \lambda x + c_2 \sin \lambda x.$$

Now $y(0) = 0$ implies $c_1 = 0$, so

$$y(L) = c_2 \sin \lambda L = 0$$

gives

$$\lambda L = n\pi \quad \text{or} \quad \lambda = \frac{n\pi}{L}, \quad n = 1, 2, 3, \dots$$

The eigenvalues $n\pi/L$ correspond to the eigenfunctions $\sin \frac{n\pi}{L}x$ for $n = 1, 2, 3, \dots$.

18. For $\lambda = 0$ the only solution of the boundary-value problem is $y = 0$. For $\lambda \neq 0$ we have

$$y = c_1 \cos \lambda x + c_2 \sin \lambda x.$$

Now $y(0) = 0$ implies $c_1 = 0$, so

$$y'(3\pi) = c_2 \lambda \cos 3\pi \lambda = 0$$

gives

$$3\pi\lambda = \frac{(2n-1)\pi}{2} \quad \text{or} \quad \lambda = \frac{2n-1}{6}, \quad n = 1, 2, 3, \dots$$

The eigenvalues $(2n-1)/6$ correspond to the eigenfunctions $\sin \frac{2n-1}{6}x$ for $n = 1, 2, 3, \dots$.

19. For $\lambda > 0$ a general solution of the given differential equation is

$$y = c_1 \cos(\sqrt{\lambda} \ln x) + c_2 \sin(\sqrt{\lambda} \ln x).$$

Since $\ln 1 = 0$, the boundary condition $y(1) = 0$ implies $c_1 = 0$. Therefore

$$y = c_2 \sin(\sqrt{\lambda} \ln x).$$

Using $\ln e^\pi = \pi$ we find that $y(e^\pi) = 0$ implies

$$c_2 \sin \sqrt{\lambda} \pi = 0$$

or $\sqrt{\lambda} \pi = n\pi$, $n = 1, 2, 3, \dots$. The eigenvalues and eigenfunctions are, in turn,

$$\lambda = n^2, \quad n = 1, 2, 3, \dots \quad \text{and} \quad y = \sin(n \ln x).$$

For $\lambda \leq 0$ the only solution of the boundary-value problem is $y = 0$.

To obtain the self-adjoint form we note that the integrating factor is $(1/x^2)e^{\int dx/x} = 1/x$. That is, the self-adjoint form is

$$\frac{d}{dx}[xy'] + \frac{\lambda}{x}y = 0.$$

Identifying the weight function $p(x) = 1/x$ we can then write the orthogonality relation

$$\int_1^{e^\pi} \frac{1}{x} \sin(n \ln x) \sin(m \ln x) dx = 0, \quad m \neq n.$$

- 20.** For $\lambda = 0$ the general solution is $y = c_1 + c_2 \ln x$. Now $y' = c_2/x$, so $y'(e^{-1}) = c_2 e = 0$ implies $c_2 = 0$. Then $y = c_1$ and $y(1) = 0$ gives $c_1 = 0$. Thus $y(x) = 0$.

For $\lambda < 0$, $y = c_1 x^{-\sqrt{-\lambda}} + c_2 x^{\sqrt{-\lambda}}$. The initial conditions give $c_2 = c_1 e^{2\sqrt{-\lambda}}$ and $c_1 = 0$, so that $c_2 = 0$ and $y(x) = 0$.

For $\lambda > 0$, $y = c_1 \cos(\sqrt{\lambda} \ln x) + c_2 \sin(\sqrt{\lambda} \ln x)$. From $y(1) = 0$ we obtain $c_1 = 0$ and $y = c_2 \sin(\sqrt{\lambda} \ln x)$. Now $y' = c_2(\sqrt{\lambda}/x) \cos(\sqrt{\lambda} \ln x)$, so $y'(e^{-1}) = c_2 e \sqrt{\lambda} \cos \sqrt{\lambda} = 0$ implies $\cos \sqrt{\lambda} = 0$ or $\lambda = (2n-1)^2 \pi^2 / 4$ for $n = 1, 2, 3, \dots$. The corresponding eigenfunctions are

$$y = \sin \left(\frac{2n-1}{2} \pi \ln x \right).$$

- 21.** For $\lambda = 0$ the general solution is $y = c_1 + c_2 \ln x$. Now $y' = c_2/x$, so $y'(1) = c_2 = 0$ and $y = c_1$. Since $y'(e^2) = 0$ for any c_1 we see that $y(x) = 1$ is an eigenfunction corresponding to the eigenvalue $\lambda = 0$.

For $\lambda < 0$, $y = c_1 x^{-\sqrt{-\lambda}} + c_2 x^{\sqrt{-\lambda}}$. The initial conditions imply $c_1 = c_2 = 0$, so $y(x) = 0$.

For $\lambda > 0$, $y = c_1 \cos(\sqrt{\lambda} \ln x) + c_2 \sin(\sqrt{\lambda} \ln x)$. Now

$$y' = -c_1 \frac{\sqrt{\lambda}}{x} \sin(\sqrt{\lambda} \ln x) + c_2 \frac{\sqrt{\lambda}}{x} \cos(\sqrt{\lambda} \ln x),$$

and $y'(1) = c_2 \sqrt{\lambda} = 0$ implies $c_2 = 0$. Finally, $y'(e^2) = -(c_1 \sqrt{\lambda}/e^2) \sin(2\sqrt{\lambda}) = 0$ implies $\lambda = n^2 \pi^2 / 4$ for $n = 1, 2, 3, \dots$. The corresponding eigenfunctions are

$$y = \cos \left(\frac{n\pi}{2} \ln x \right).$$

- 22.** For $\lambda > 1/4$ a general solution of the given differential equation is

$$y = c_1 x^{-1/2} \cos \left(\frac{\sqrt{4\lambda-1}}{2} \ln x \right) + c_2 x^{-1/2} \sin \left(\frac{\sqrt{4\lambda-1}}{2} \ln x \right).$$

Since $\ln 1 = 0$, the boundary condition $y(1) = 0$ implies $c_1 = 0$. Therefore

$$y = c_2 x^{-1/2} \sin \left(\frac{\sqrt{4\lambda-1}}{2} \ln x \right).$$

Using $\ln e^2 = 2$ we find that $y(e^2) = 0$ implies

$$c_2 e^{-1} \sin \left(\sqrt{4\lambda-1} \right) = 0$$

or $\sqrt{4\lambda-1} = n\pi$, $n = 1, 2, 3, \dots$. The eigenvalues and eigenfunctions are, in turn,

$$\lambda = (n^2 \pi^2 + 1) / 4, \quad n = 1, 2, 3, \dots \quad \text{and} \quad y = x^{-1/2} \sin \left(\frac{n\pi}{2} \ln x \right).$$

For $\lambda < 0$ the only solution of the boundary-value problem is $y = 0$.

For $\lambda = 1/4$ a general solution of the differential equation is

$$y = c_1 x^{-1/2} + c_2 x^{-1/2} \ln x.$$

From $y(1) = 0$ we obtain $c_1 = 0$, so $y = c_2 x^{-1/2} \ln x$. From $y(e^2) = 0$ we obtain $2c_2 e^{-1} = 0$ or $c_2 = 0$. Thus, there are no eigenvalues and eigenfunctions in this case.

Exercises 3.9

To obtain the self-adjoint form we note that the integrating factor is $(1/x^2)e^{\int (2/x) dx} = (1/x^2) \cdot x^2 = 1$. That is, the self-adjoint form is

$$\frac{d}{dx} [x^2 y'] + \lambda y = 0.$$

Identifying the weight function $p(x) = 1$ we can then write the orthogonality relation

$$\int_1^{e^2} 1 \cdot x^{-1/2} \sin\left(\frac{m\pi}{2} \ln x\right) x^{-1/2} \sin\left(\frac{n\pi}{2} \ln x\right) dx = 0, \quad m \neq n,$$

or

$$\int_1^{e^2} x^{-1} \sin\left(\frac{m\pi}{2} \ln x\right) \sin\left(\frac{n\pi}{2} \ln x\right) dx = 0, \quad m \neq n.$$

- 23.** For $\lambda > 0$ the general solution is $y = c_1 \cos \sqrt{\lambda} x + c_2 \sin \sqrt{\lambda} x$. Setting $y(0) = 0$ we find $c_1 = 0$, so that $y = c_2 \sin \sqrt{\lambda} x$. The boundary condition $y(1) + y'(1) = 0$ implies

$$c_2 \sin \sqrt{\lambda} + c_2 \sqrt{\lambda} \cos \sqrt{\lambda} = 0.$$

Taking $c_2 \neq 0$, this equation is equivalent to $\tan \sqrt{\lambda} = -\sqrt{\lambda}$. Thus, the eigenvalues are $\lambda_n = x_n^2$, $n = 1, 2, 3, \dots$, where the x_n are the consecutive positive roots of $\tan \sqrt{\lambda} = -\sqrt{\lambda}$.

- 24. (a)** Since $\lambda_n = x_n^2$, there are no new eigenvalues when $x_n < 0$. For $\lambda = 0$, the differential equation $y'' = 0$ has general solution $y = c_1 x + c_2$. The boundary conditions imply $c_1 = c_2 = 0$, so $y = 0$.

(b) $\lambda_1 = 4.1159$, $\lambda_2 = 24.1393$, $\lambda_3 = 63.6591$, $\lambda_4 = 122.8892$.

- 25.** If restraints are put on the column at $x = L/4$, $x = L/2$, and $x = 3L/4$, then the critical load will be P_4 .



- 26. (a)** The general solution of the differential equation is

$$y = c_1 \cos \sqrt{\frac{P}{EI}} x + c_2 \sin \sqrt{\frac{P}{EI}} x + \delta.$$

Since the column is embedded at $x = 0$, the initial conditions are $y(0) = y'(0) = 0$. If $\delta = 0$ this implies that $c_1 = c_2 = 0$ and $y(x) = 0$. That is, there is no deflection.

- (b)** If $\delta \neq 0$, the initial conditions give, in turn, $c_1 = -\delta$ and $c_2 = 0$. Then

$$y = \delta \left(1 - \cos \sqrt{\frac{P}{EI}} x \right).$$

In order to satisfy the condition $y(L) = \delta$ we must have

$$\delta = \delta \left(1 - \cos \sqrt{\frac{P}{EI}} L \right) \quad \text{or} \quad \cos \sqrt{\frac{P}{EI}} L = 0.$$

This gives $\sqrt{P/EI} L = n\pi/2$ for $n = 1, 2, 3, \dots$. The smallest value of P_n , the Euler load, is then

$$\sqrt{\frac{P_1}{EI}} L = \frac{\pi}{2} \quad \text{or} \quad P_1 = \frac{1}{4} \left(\frac{\pi^2 EI}{L^2} \right).$$

27. The general solution is

$$y = c_1 \cos \sqrt{\frac{\rho\omega^2}{T}} x + c_2 \sin \sqrt{\frac{\rho\omega^2}{T}} x.$$

From $y(0) = 0$ we obtain $c_1 = 0$. Setting $y(L) = 0$ we find $\sqrt{\rho\omega^2/T} L = n\pi$, $n = 1, 2, 3, \dots$. Thus, critical speeds are $\omega_n = n\pi\sqrt{T/L}\sqrt{\rho}$, $n = 1, 2, 3, \dots$. The corresponding deflection curves are

$$y(x) = c_2 \sin \frac{n\pi}{L} x, \quad n = 1, 2, 3, \dots,$$

where $c_2 \neq 0$.

28. (a) When $T(x) = x^2$ the given differential equation is the Cauchy-Euler equation

$$x^2 y'' + 2xy' + \rho\omega^2 y = 0.$$

The solutions of the auxiliary equation

$$m(m-1) + 2m + \rho\omega^2 = m^2 + m + \rho\omega^2 = 0$$

are

$$m_1 = -\frac{1}{2} - \frac{1}{2}\sqrt{4\rho\omega^2 - 1}i, \quad m_2 = -\frac{1}{2} + \frac{1}{2}\sqrt{4\rho\omega^2 - 1}i$$

when $\rho\omega^2 > 0.25$. Thus

$$y = c_1 x^{-1/2} \cos(\lambda \ln x) + c_2 x^{-1/2} \sin(\lambda \ln x)$$

where $\lambda = \sqrt{4\rho\omega^2 - 1}/2$. Applying $y(1) = 0$ gives $c_1 = 0$ and consequently

$$y = c_2 x^{-1/2} \sin(\lambda \ln x).$$

The condition $y(e) = 0$ requires $c_2 e^{-1/2} \sin \lambda = 0$. We obtain a nontrivial solution when $\lambda_n = n\pi$, $n = 1, 2, 3, \dots$. But

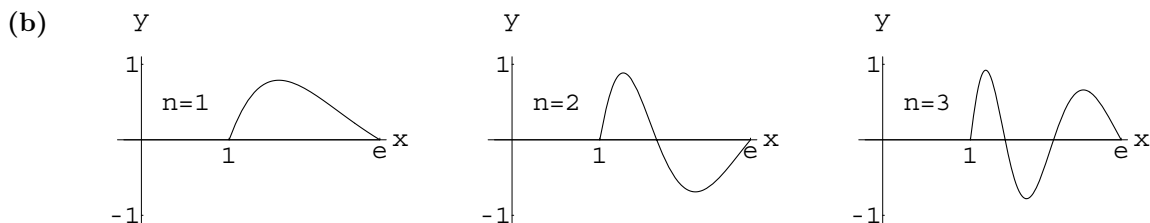
$$\lambda_n = \sqrt{4\rho\omega_n^2 - 1}/2 = n\pi.$$

Solving for ω_n gives

$$\omega_n = \frac{1}{2} \sqrt{(4n^2\pi^2 + 1)/\rho}.$$

The corresponding solutions are

$$y_n(x) = c_2 x^{-1/2} \sin(n\pi \ln x).$$



29. The auxiliary equation is $m^2 + m = m(m+1) = 0$ so that $u(r) = c_1 r^{-1} + c_2$. The boundary conditions $u(a) = u_0$ and $u(b) = u_1$ yield the system $c_1 a^{-1} + c_2 = u_0$, $c_1 b^{-1} + c_2 = u_1$. Solving gives

$$c_1 = \left(\frac{u_0 - u_1}{b - a} \right) ab \quad \text{and} \quad c_2 = \frac{u_1 b - u_0 a}{b - a}.$$

Exercises 3.9

Thus

$$u(r) = \left(\frac{u_0 - u_1}{b - a} \right) \frac{ab}{r} + \frac{u_1 b - u_0 a}{b - a}.$$

30. The auxiliary equation is $m^2 = 0$ so that $u(r) = c_1 + c_2 \ln r$. The boundary conditions $u(a) = u_0$ and $u(b) = u_1$ yield the system $c_1 + c_2 \ln a = u_0$, $c_1 + c_2 \ln b = u_1$. Solving gives

$$c_1 = \frac{u_1 \ln a - u_0 \ln b}{\ln(a/b)} \quad \text{and} \quad c_2 = \frac{u_0 - u_1}{\ln(a/b)}.$$

Thus

$$u(r) = \frac{u_1 \ln a - u_0 \ln b}{\ln(a/b)} + \frac{u_0 - u_1}{\ln(a/b)} \ln r = \frac{u_0 \ln(r/b) - u_1 \ln(r/a)}{\ln(a/b)}.$$

31. (a) The general solution of the differential equation is $y = c_1 \cos 4x + c_2 \sin 4x$. From $y_0 = y(0) = c_1$ we see that $y = y_0 \cos 4x + c_2 \sin 4x$. From $y_1 = y(\pi/2) = y_0$ we see that any solution must satisfy $y_0 = y_1$. We also see that when $y_0 = y_1$, $y = y_0 \cos 4x + c_2 \sin 4x$ is a solution of the boundary-value problem for any choice of c_2 . Thus, the boundary-value problem does not have a unique solution for any choice of y_0 and y_1 .
- (b) Whenever $y_0 = y_1$ there are infinitely many solutions.
- (c) When $y_0 \neq y_1$ there will be no solutions.
- (d) The boundary-value problem will have the trivial solution when $y_0 = y_1 = 0$. This solution will not be unique.
32. (a) The general solution of the differential equation is $y = c_1 \cos 4x + c_2 \sin 4x$. From $1 = y(0) = c_1$ we see that $y = \cos 4x + c_2 \sin 4x$. From $1 = y(L) = \cos 4L + c_2 \sin 4L$ we see that $c_2 = (1 - \cos 4L)/\sin 4L$. Thus,

$$y = \cos 4x + \left(\frac{1 - \cos 4L}{\sin 4L} \right) \sin 4x$$

will be a unique solution when $\sin 4L \neq 0$; that is, when $L \neq k\pi/4$ where $k = 1, 2, 3, \dots$.

- (b) There will be infinitely many solutions when $\sin 4L = 0$ and $1 - \cos 4L = 0$; that is, when $L = k\pi/2$ where $k = 1, 2, 3, \dots$.
- (c) There will be no solution when $\sin 4L \neq 0$ and $1 - \cos 4L \neq 0$; that is, when $L = k\pi/4$ where $k = 1, 3, 5, \dots$.
- (d) There can be no trivial solution since it would fail to satisfy the boundary conditions.
33. (a) A solution curve has the same y -coordinate at both ends of the interval $[-\pi, \pi]$ and the tangent lines at the endpoints of the interval are parallel.
- (b) For $\lambda = 0$ the solution of $y'' = 0$ is $y = c_1 x + c_2$. From the first boundary condition we have

$$y(-\pi) = -c_1\pi + c_2 = y(\pi) = c_1\pi + c_2$$

or $2c_1\pi = 0$. Thus, $c_1 = 0$ and $y = c_2$. This constant solution is seen to satisfy the boundary-value problem.

For $\lambda < 0$ we have $y = c_1 \cosh \lambda x + c_2 \sinh \lambda x$. In this case the first boundary condition gives

$$\begin{aligned} y(-\pi) &= c_1 \cosh(-\lambda\pi) + c_2 \sinh(-\lambda\pi) \\ &= c_1 \cosh \lambda\pi - c_2 \sinh \lambda\pi \\ &= y(\pi) = c_1 \cosh \lambda\pi + c_2 \sinh \lambda\pi \end{aligned}$$

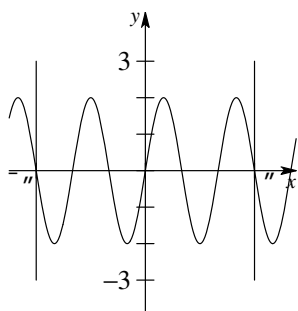
or $2c_2 \sinh \lambda\pi = 0$. Thus $c_2 = 0$ and $y = c_1 \cosh \lambda x$. The second boundary condition implies in a similar fashion that $c_1 = 0$. Thus, for $\lambda < 0$, the only solution of the boundary-value problem is $y = 0$.

For $\lambda > 0$ we have $y = c_1 \cos \lambda x + c_2 \sin \lambda x$. The first boundary condition implies

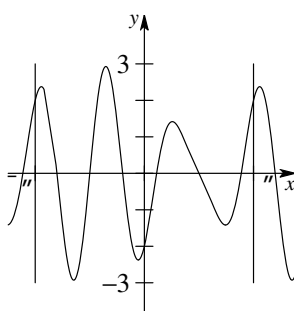
$$\begin{aligned} y(-\pi) &= c_1 \cos(-\lambda\pi) + c_2 \sin(-\lambda\pi) \\ &= c_1 \cos \lambda\pi - c_2 \sin \lambda\pi \\ &= y(\pi) = c_1 \cos \lambda\pi + c_2 \sin \lambda\pi \end{aligned}$$

or $2c_2 \sin \lambda\pi = 0$. Similarly, the second boundary condition implies $2c_1 \lambda \sin \lambda\pi = 0$. If $c_1 = c_2 = 0$ the solution is $y = 0$. However, if $c_1 \neq 0$ or $c_2 \neq 0$, then $\sin \lambda\pi = 0$, which implies that λ must be an integer, n . Therefore, for c_1 and c_2 not both 0, $y = c_1 \cos nx + c_2 \sin nx$ is a nontrivial solution of the boundary-value problem. Since $\cos(-nx) = \cos nx$ and $\sin(-nx) = -\sin nx$, we may assume without loss of generality that the eigenvalues are $\lambda_n = n$, for n a positive integer. The corresponding eigenfunctions are $y_n = \cos nx$ and $y_n = \sin nx$.

(c)



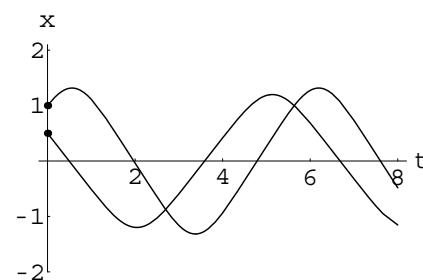
$$y = 2 \sin 3x$$



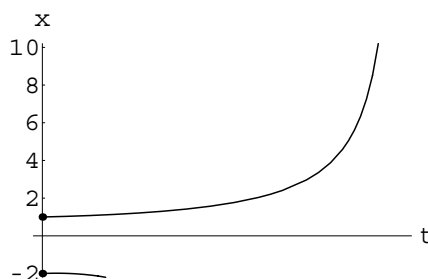
$$y = \sin 4x - 2 \cos 3x$$

Exercises 3.10

1. The period corresponding to $x(0) = 1$, $x'(0) = 1$ is approximately 5.6. The period corresponding to $x(0) = 1/2$, $x'(0) = -1$ is approximately 6.2.

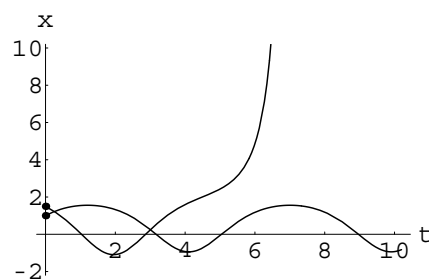


2. The solutions are not periodic.

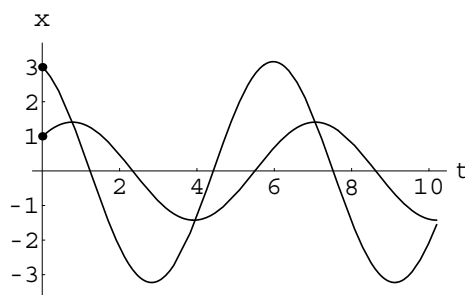


Exercises 3.10

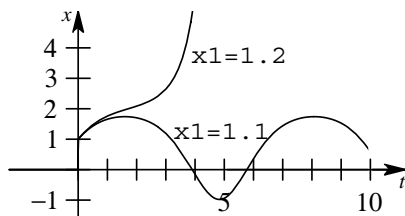
3. The period corresponding to $x(0) = 1$, $x'(0) = 1$ is approximately 5.8.
The second initial-value problem does not have a periodic solution.



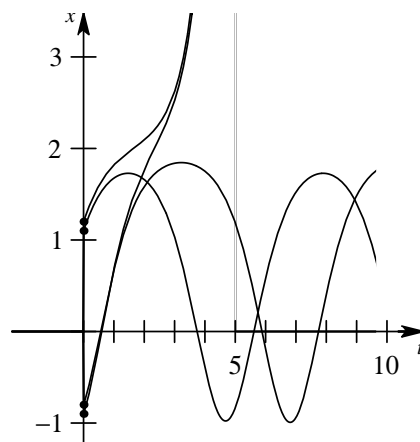
4. Both solutions have periods of approximately 6.3.



5. From the graph we see that $|x_1| \approx 1.2$.



6. From the graphs we see that the interval is approximately $(-0.8, 1.1)$.

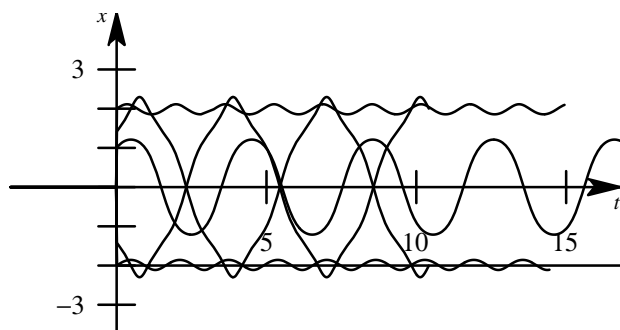


7. Since

$$xe^{0.01x} = x[1 + 0.01x + \frac{1}{2!}(0.01x)^2 + \cdots] \approx x$$

for small values of x , a linearization is $\frac{d^2x}{dt^2} + x = 0$.

8.



For $x(0) = 1$ and $x'(0) = 1$ the oscillations are symmetric about the line $x = 0$ with amplitude slightly greater than 1.

For $x(0) = -2$ and $x'(0) = 0.5$ the oscillations are symmetric about the line $x = -2$ with small amplitude.

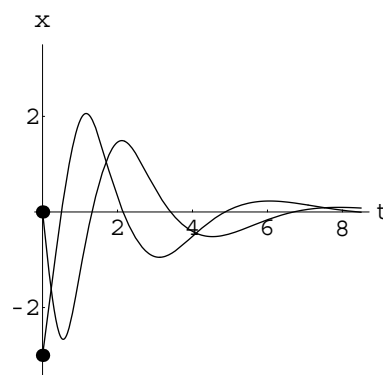
For $x(0) = \sqrt{2}$ and $x'(0) = 1$ the oscillations are symmetric about the line $x = 0$ with amplitude a little greater than 2.

For $x(0) = 2$ and $x'(0) = 0.5$ the oscillations are symmetric about the line $x = 2$ with small amplitude.

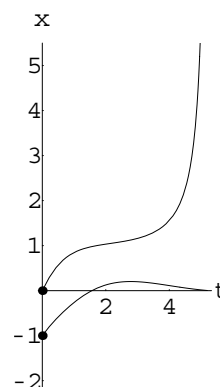
For $x(0) = -2$ and $x'(0) = 0$ there is no oscillation; the solution is constant.

For $x(0) = -\sqrt{2}$ and $x'(0) = -1$ the oscillations are symmetric about the line $x = 0$ with amplitude a little greater than 2.

9. This is a damped hard spring, so all solutions should be oscillatory with $x \rightarrow 0$ as $t \rightarrow \infty$.

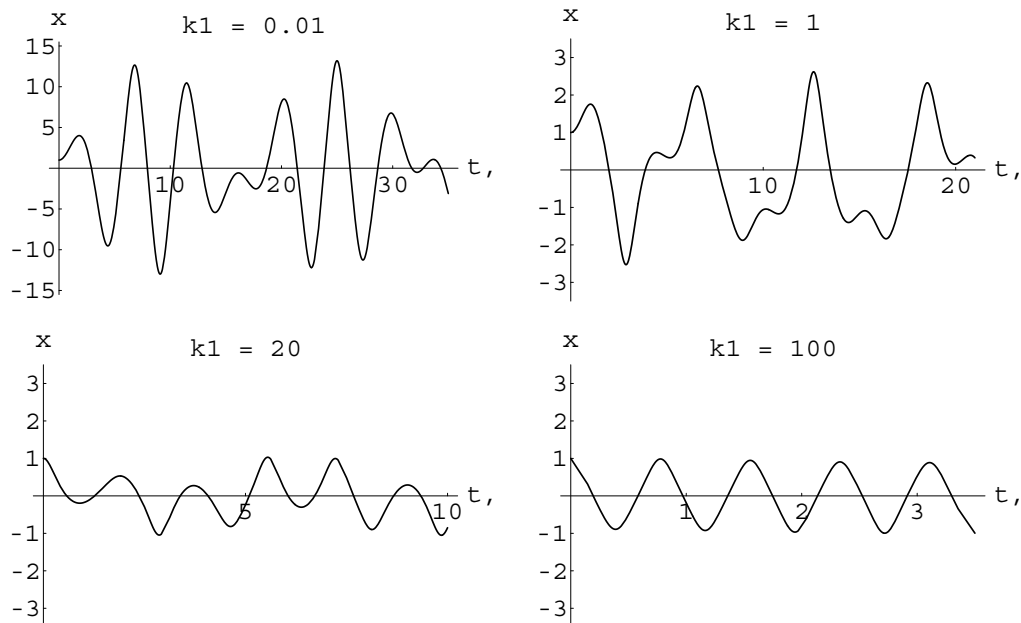


10. This is a damped soft spring, so so we expect no oscillatory solutions.



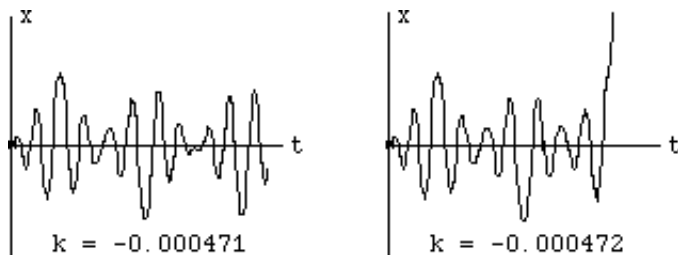
Exercises 3.10

11.



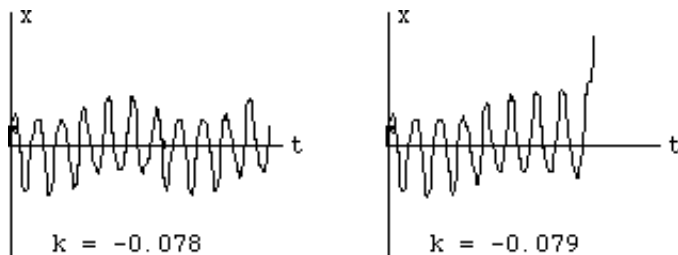
When k_1 is very small the effect of the nonlinearity is greatly diminished, and the system is close to pure resonance.

12. (a)



The system appears to be oscillatory for $-0.000471 \leq k_1 < 0$ and nonoscillatory for $k_1 \leq -0.000472$.

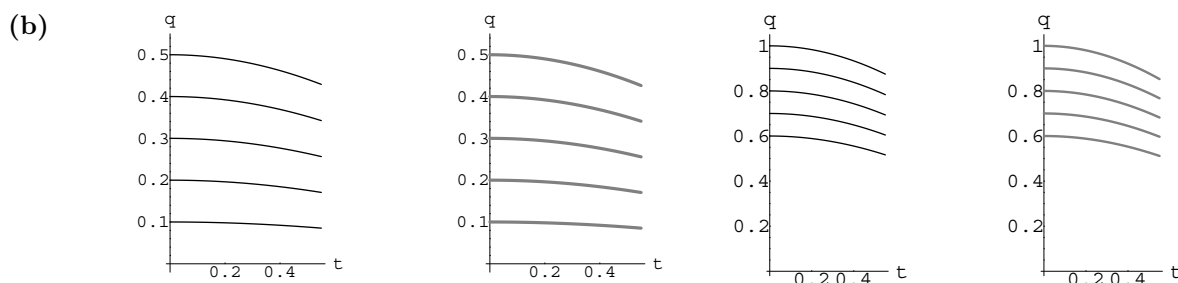
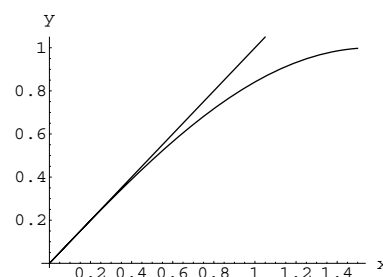
(b)



The system appears to be oscillatory for $-0.077 \leq k_1 < 0$ and nonoscillatory for $k_1 \leq -0.078$.

13. Since $(dx/dt)^2$ is always positive, it is necessary to use $|dx/dt|(dx/dt)$ in order to account for the fact that the motion is oscillatory and the velocity (or its square) should be negative when the spring is contracting.

14. (a) The approximation is accurate to two decimal places for $\theta_1 = 0.3$, and accurate to one decimal place for $\theta_1 = 0.6$.

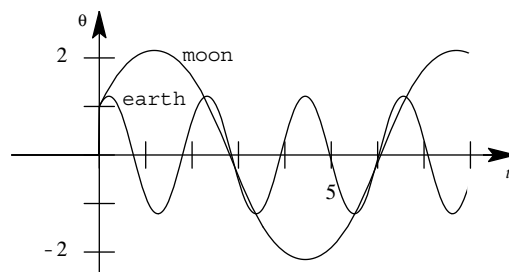


The thinner curves are solutions of the nonlinear differential equation, while the thicker curves are solutions of the linear differential equation.

15. (a) Write the differential equation as

$$\frac{d^2\theta}{dt^2} + \omega^2 \sin \theta = 0,$$

where $\omega^2 = g/\ell$. To test for differences between the earth and the moon we take $\ell = 3$, $\theta(0) = 1$, and $\theta'(0) = 2$. Using $g = 32$ on the earth and $g = 5.5$ on the moon we obtain the graphs shown in the figure. Comparing the apparent periods of the graphs, we see that the pendulum oscillates faster on the earth than on the moon.



- (b) The amplitude is greater on the moon than on the earth.
 (c) The linear model is

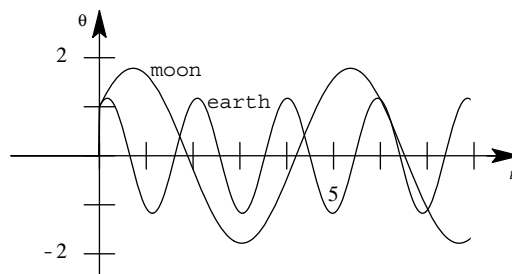
$$\frac{d^2\theta}{dt^2} + \omega^2 \theta = 0,$$

where $\omega^2 = g/\ell$. When $g = 32$, $\ell = 3$, $\theta(0) = 1$, and $\theta'(0) = 2$, the general solution is

$$\theta(t) = \cos 3.266t + 0.612 \sin 3.266t.$$

When $g = 5.5$ the general solution is

$$\theta(t) = \cos 1.354t + 1.477 \sin 1.354t.$$



As in the nonlinear case, the pendulum oscillates faster on the earth than on the moon and still has greater amplitude on the moon.

Exercises 3.10

16. (a) The general solution of

$$\frac{d^2\theta}{dt^2} + \theta = 0$$

is $\theta(t) = c_1 \cos t + c_2 \sin t$. From $\theta(0) = \pi/12$ and $\theta'(0) = -1/3$ we find $\theta(t) = (\pi/12) \cos t - (1/3) \sin t$. Setting $\theta(t) = 0$ we have $\tan t = \pi/4$ which implies $t_1 = \tan^{-1}(\pi/4) \approx 0.66577$.

- (b) We set $\theta(t) = \theta(0) + \theta'(0)t + \frac{1}{2}\theta''(0)t^2 + \frac{1}{6}\theta'''(0)t^3 + \dots$ and use $\theta''(t) = -\sin \theta(t)$ together with $\theta(0) = \pi/12$ and $\theta'(0) = -1/3$. Then $\theta''(0) = -\sin(\pi/12) = -\sqrt{2}(\sqrt{3}-1)/4$ and $\theta'''(0) = -\cos \theta(0) \cdot \theta'(0) = -\cos(\pi/12)(-1/3) = \sqrt{2}(\sqrt{3}+1)/12$. Thus

$$\theta(t) = \frac{\pi}{12} - \frac{1}{3}t - \frac{\sqrt{2}(\sqrt{3}-1)}{8}t^2 + \frac{\sqrt{2}(\sqrt{3}+1)}{72}t^3 + \dots$$

- (c) Setting $\pi/12 - t/3 = 0$ we obtain $t_1 = \pi/4 \approx 0.785398$.

- (d) Setting

$$\frac{\pi}{12} - \frac{1}{3}t - \frac{\sqrt{2}(\sqrt{3}-1)}{8}t^2 = 0$$

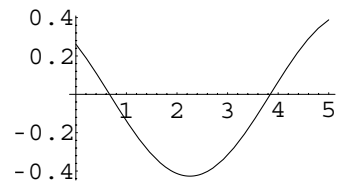
and using the positive root we obtain $t_1 \approx 0.63088$.

- (e) Setting

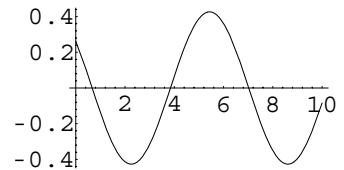
$$\frac{\pi}{12} - \frac{1}{3}t - \frac{\sqrt{2}(\sqrt{3}-1)}{8}t^2 + \frac{\sqrt{2}(\sqrt{3}+1)}{72}t^3 = 0$$

we find $t_1 \approx 0.661973$ to be the first positive root.

- (f) From the output we see that $y(t)$ is an interpolating function on the interval $0 \leq t \leq 5$, whose graph is shown. The positive root of $y(t) = 0$ near $t = 1$ is $t_1 = 0.666404$.



- (g) To find the next two positive roots we change the interval used in **NDSolve** and **Plot** from $\{t, 0, 5\}$ to $\{t, 0, 10\}$. We see from the graph that the second and third positive roots are near 4 and 7, respectively. Replacing $\{t, 1\}$ in **FindRoot** with $\{t, 4\}$ and then $\{t, 7\}$ we obtain $t_2 = 3.84411$ and $t_3 = 7.0218$.



17. From the table below we see that the pendulum first passes the vertical position between 1.7 and 1.8 seconds. To refine our estimate of t_1 we estimate the solution of the differential equation on $[1.7, 1.8]$ using a step size of $h = 0.01$. From the resulting table we see that t_1 is between 1.76 and 1.77 seconds. Repeating the process with $h = 0.001$ we conclude that $t_1 \approx 1.767$. Then the period of the pendulum is approximately $4t_1 = 7.068$. The error when using $t_1 = 2\pi$ is $7.068 - 6.283 = 0.785$ and the percentage relative error is $(0.785/7.068)100 = 11.1$.

h=0.1		h=0.01	
t_n	θ_n	t_n	θ_n
0.00	0.78540	1.70	0.07706
0.10	0.78523	1.71	0.06572
0.20	0.78407	1.72	0.05428
0.30	0.78092	1.73	0.04275
0.40	0.77482	1.74	0.03111
0.50	0.76482	1.75	0.01938
0.60	0.75004	1.76	0.00755
0.70	0.72962	1.77	-0.00438
0.80	0.70275	1.78	-0.01641
0.90	0.66872	1.79	-0.02854
1.00	0.62687	1.80	-0.04076
1.10	0.57660		
1.20	0.51744	h=0.001	
1.30	0.44895	1.763	0.00398
1.40	0.37085	1.764	0.00279
1.50	0.28289	1.765	0.00160
1.60	0.18497	1.766	0.00040
1.70	0.07706	1.767	-0.00079
1.80	-0.04076	1.768	-0.00199
1.90	-0.16831	1.769	-0.00318
2.00	-0.30531	1.770	-0.00438

18. (a) Setting $dy/dt = v$, the differential equation in (13) becomes $dv/dt = -gR^2/y^2$. But, by the chain rule, $dv/dt = (dv/dy)(dy/dt) = v dv/dy$, so $v dv/dy = -gR^2/y^2$. Separating variables and integrating we obtain

$$v dv = -gR^2 \frac{dy}{y^2} \quad \text{and} \quad \frac{1}{2}v^2 = \frac{gR^2}{y} + c.$$

Setting $v = v_0$ and $y = R$ we find $c = -gR + \frac{1}{2}v_0^2$ and

$$v^2 = 2g \frac{R^2}{y} - 2gR + v_0^2.$$

- (b) As $y \rightarrow \infty$ we assume that $v \rightarrow 0^+$. Then $v_0^2 = 2gR$ and $v_0 = \sqrt{2gR}$.

- (c) Using $g = 32$ ft/s and $R = 4000(5280)$ ft we find

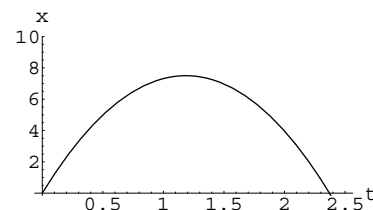
$$v_0 = \sqrt{2(32)(4000)(5280)} \approx 36765.2 \text{ ft/s} \approx 25067 \text{ mi/hr.}$$

- (d) $v_0 = \sqrt{2(0.165)(32)(1080)} \approx 7760 \text{ ft/s} \approx 5291 \text{ mi/hr}$

19. (a) Intuitively, one might expect that only half of a 10-pound chain could be lifted by a 5-pound force.

- (b) Since $x = 0$ when $t = 0$, and $v = dx/dt = \sqrt{160 - 64x/3}$, we have $v(0) = \sqrt{160} \approx 12.65$ ft/s.

- (c) Since x should always be positive, we solve $x(t) = 0$, getting $t = 0$ and $t = \frac{3}{2}\sqrt{5/2} \approx 2.3717$. Since the graph of $x(t)$ is a parabola, the maximum value occurs at $t_m = \frac{3}{4}\sqrt{5/2}$. (This can also be obtained by solving $x'(t) = 0$.) At this time the height of the chain is $x(t_m) \approx 7.5$ ft. This is higher than predicted because of the momentum generated by the force. When the chain is 5 feet high it still has a positive velocity of about 7.3 ft/s, which keeps it going higher for a while.



20. (a) Setting $dx/dt = v$, the differential equation becomes $(L - x)dv/dt - v^2 = Lg$. But, by the chain rule, $dv/dt = (dv/dx)(dx/dt) = v dv/dx$, so $(L - x)v dv/dx - v^2 = Lg$. Separating variables and integrating we

Exercises 3.10

obtain

$$\frac{v}{v^2 + Lg} dv = \frac{1}{L - x} dx \quad \text{and} \quad \frac{1}{2} \ln(v^2 + Lg) = -\ln(L - x) + \ln c,$$

so $\sqrt{v^2 + Lg} = c/(L - x)$. When $x = 0$, $v = 0$, and $c = L\sqrt{Lg}$. Solving for v and simplifying we get

$$\frac{dx}{dt} = v(x) = \frac{\sqrt{Lg(2Lx - x^2)}}{L - x}.$$

Again, separating variables and integrating we obtain

$$\frac{L - x}{\sqrt{Lg(2Lx - x^2)}} dx = dt \quad \text{and} \quad \frac{\sqrt{2Lx - x^2}}{\sqrt{Lg}} = t + c_1.$$

Since $x(0) = 0$, we have $c_1 = 0$ and $\sqrt{2Lx - x^2}/\sqrt{Lg} = t$. Solving for x we get

$$x(t) = L - \sqrt{L^2 - Lgt^2} \quad \text{and} \quad v(t) = \frac{dx}{dt} = \frac{\sqrt{Lgt}}{\sqrt{L - gt^2}}.$$

(b) The chain will be completely on the ground when $x(t) = L$ or $t = \sqrt{L/g}$.

(c) The predicted velocity of the upper end of the chain when it hits the ground is infinity.

21. (a) The weight of x feet of the chain is $2x$, so the corresponding mass is $m = 2x/32 = x/16$. The only force acting on the chain is the weight of the portion of the chain hanging over the edge of the platform. Thus, by Newton's second law,

$$\frac{d}{dt}(mv) = \frac{d}{dt}\left(\frac{x}{16}v\right) = \frac{1}{16}\left(x\frac{dv}{dt} + v\frac{dx}{dt}\right) = \frac{1}{16}\left(x\frac{dv}{dt} + v^2\right) = 2x$$

and $x dv/dt + v^2 = 32x$. Now, by the chain rule, $dv/dt = (dv/dx)(dx/dt) = v dv/dx$, so $xv dv/dx + v^2 = 32x$.

- (b) We separate variables and write the differential equation as $(v^2 - 32x) dx + xv dv = 0$. This is not an exact form, but $\mu(x) = x$ is an integrating factor. Multiplying by x we get $(xv^2 - 32x^2) dx + x^2v dv = 0$. This form is the total differential of $u = \frac{1}{2}x^2v^2 - \frac{32}{3}x^3$, so an implicit solution is $\frac{1}{2}x^2v^2 - \frac{32}{3}x^3 = c$. Letting $x = 3$ and $v = 0$ we find $c = -288$. Solving for v we get

$$\frac{dx}{dt} = v = \frac{8\sqrt{x^3 - 27}}{\sqrt{3}x}, \quad 3 \leq x \leq 8.$$

- (c) Separating variables and integrating we obtain

$$\frac{x}{\sqrt{x^3 - 27}} dx = \frac{8}{\sqrt{3}} dt \quad \text{and} \quad \int_3^x \frac{s}{\sqrt{s^3 - 27}} ds = \frac{8}{\sqrt{3}} t + c.$$

Since $x = 3$ when $t = 0$, we see that $c = 0$ and

$$t = \frac{\sqrt{3}}{8} \int_3^x \frac{s}{\sqrt{s^3 - 27}} ds.$$

We want to find t when $x = 7$. Using a CAS we find $t(7) = 0.576$ seconds.

22. (a) There are two forces acting on the chain as it falls from the platform. One is the force due to gravity on the portion of the chain hanging over the edge of the platform. This is $F_1 = 2x$. The second is due to the motion of the portion of the chain stretched out on the platform. By Newton's second law this is

$$\begin{aligned} F_2 &= \frac{d}{dt}[mv] = \frac{d}{dt}\left[\frac{(8-x)2}{32}v\right] = \frac{d}{dt}\left[\frac{8-x}{16}v\right] \\ &= \frac{8-x}{16} \frac{dv}{dt} - \frac{1}{16}v \frac{dx}{dt} = \frac{1}{16}\left[(8-x)\frac{dv}{dt} - v^2\right]. \end{aligned}$$

From $\frac{d}{dt}[mv] = F_1 - F_2$ we have

$$\begin{aligned}\frac{d}{dt}\left[\frac{2x}{32}v\right] &= 2x - \frac{1}{16}\left[(8-x)\frac{dv}{dt} - v^2\right] \\ \frac{x}{16}\frac{dv}{dt} + \frac{1}{16}v\frac{dx}{dt} &= 2x - \frac{1}{16}\left[(8-x)\frac{dv}{dt} - v^2\right] \\ x\frac{dv}{dt} + v^2 &= 32x - (8-x)\frac{dv}{dt} + v^2 \\ x\frac{dv}{dt} &= 32x - 8\frac{dv}{dt} + x\frac{dv}{dt} \\ 8\frac{dv}{dt} &= 32x.\end{aligned}$$

By the chain rule, $dv/dt = (dv/dx)(dx/dt) = v dv/dx$, so

$$8\frac{dv}{dt} = 8v\frac{dv}{dx} = 32x \quad \text{and} \quad v\frac{dv}{dx} = 4x.$$

- (b) Integrating $v dv = 4x dx$ we get $\frac{1}{2}v^2 = 2x^2 + c$. Since $v = 0$ when $x = 3$, we have $c = -18$. Then $v^2 = 4x^2 - 36$ and $v = \sqrt{4x^2 - 36}$. Using $v = dx/dt$, separating variables, and integrating we obtain

$$\frac{dx}{\sqrt{x^2 - 9}} = 2 dt \quad \text{and} \quad \cosh^{-1} \frac{x}{3} = 2t + c_1.$$

Solving for x we get $x(t) = 3 \cosh(2t + c_1)$. Since $x = 3$ when $t = 0$, we have $\cosh c_1 = 1$ and $c_1 = 0$. Thus, $x(t) = 3 \cosh 2t$. Differentiating, we find $v(t) = dx/dt = 6 \sinh 2t$.

- (c) To find when the back end of the chain will leave the platform we solve $x(t) = 3 \cosh 2t = 8$. This gives $t_1 = \frac{1}{2} \cosh^{-1} \frac{8}{3} \approx 0.8184$ seconds. The velocity at this instant is $v(t_1) = 6 \sinh(\cosh^{-1} \frac{8}{3}) = 2\sqrt{55} \approx 14.83$ ft/s.
- (d) Replacing 8 with L and 32 with g in part (a) we have $L dv/dt = gx$. Then

$$L\frac{dv}{dt} = Lv\frac{dv}{dx} = gx \quad \text{and} \quad v\frac{dv}{dx} = \frac{g}{L}x.$$

Integrating we get $\frac{1}{2}v^2 = \frac{g}{2L}x^2 + c$. Setting $x = x_0$ and $v = 0$, we find $c = -\frac{g}{2L}x_0^2$. Solving for v we find

$$v(x) = \sqrt{\frac{g}{L}x^2 - \frac{g}{L}x_0^2}.$$

Then the velocity at which the end of the chain leaves the edge of the platform is

$$v(L) = \sqrt{\frac{g}{L}(L^2 - x_0^2)}.$$

23. (a) Let (x, y) be the coordinates of S_2 on the curve C . The slope at (x, y) is then

$$dy/dx = (v_1 t - y)/(0 - x) = (y - v_1 t)/x \quad \text{or} \quad xy' - y = -v_1 t.$$

- (b) Differentiating with respect to x gives

$$\begin{aligned}xy'' + y' - y' &= -v_1 \frac{dt}{dx} \\ xy'' &= -v_1 \frac{dt}{ds} \frac{ds}{dx} \\ xy'' &= -v_1 \frac{1}{v_2} (-\sqrt{1 + (y')^2}) \\ xy'' &= r\sqrt{1 + (y')^2}.\end{aligned}$$

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Letting $u = y'$ and separating variables, we obtain

$$\begin{aligned}x \frac{du}{dx} &= r\sqrt{1+u^2} \\ \frac{du}{\sqrt{1+u^2}} &= \frac{r}{x} dx \\ \sinh^{-1} u &= r \ln x + \ln c = \ln(cx^r) \\ u &= \sinh(\ln cx^r) \\ \frac{dy}{dx} &= \frac{1}{2} \left(cx^r - \frac{1}{cx^r} \right).\end{aligned}$$

At $t = 0$, $dy/dx = 0$ and $x = a$, so $0 = ca^r - 1/ca^r$. Thus $c = 1/a^r$ and

$$\frac{dy}{dx} = \frac{1}{2} \left[\left(\frac{x}{a} \right)^r - \left(\frac{a}{x} \right)^r \right] = \frac{1}{2} \left[\left(\frac{x}{a} \right)^r - \left(\frac{x}{a} \right)^{-r} \right].$$

If $r > 1$ or $r < 1$, integrating gives

$$y = \frac{a}{2} \left[\frac{1}{1+r} \left(\frac{x}{a} \right)^{1+r} - \frac{1}{1-r} \left(\frac{x}{a} \right)^{1-r} \right] + c_1.$$

When $t = 0$, $y = 0$ and $x = a$, so $0 = (a/2)[1/(1+r) - 1/(1-r)] + c_1$. Thus $c_1 = ar/(1-r^2)$ and

$$y = \frac{a}{2} \left[\frac{1}{1+r} \left(\frac{x}{a} \right)^{1+r} - \frac{1}{1-r} \left(\frac{x}{a} \right)^{1-r} \right] + \frac{ar}{1-r^2}.$$

- (c) If $r > 1$, $v_1 > v_2$ and $y \rightarrow \infty$ as $x \rightarrow 0^+$. In other words, S_2 always lags behind S_1 . If $r < 1$, $v_1 < v_2$ and $y = ar/(1-r^2)$ when $x = 0$. In other words, when the submarine's speed is greater than the ship's, their paths will intersect at the point $(0, ar/(1-r^2))$.

If $r = 1$, integration gives

$$y = \frac{1}{2} \left[\frac{x^2}{2a} - \frac{1}{a} \ln x \right] + c_2.$$

When $t = 0$, $y = 0$ and $x = a$, so $0 = (1/2)[a/2 - (1/a) \ln a] + c_2$. Thus $c_2 = -(1/2)[a/2 - (1/a) \ln a]$ and

$$y = \frac{1}{2} \left[\frac{x^2}{2a} - \frac{1}{a} \ln x \right] - \frac{1}{2} \left[\frac{a}{2} - \frac{1}{a} \ln a \right] = \frac{1}{2} \left[\frac{1}{2a}(x^2 - a^2) + \frac{1}{a} \ln \frac{a}{x} \right].$$

Since $y \rightarrow \infty$ as $x \rightarrow 0^+$, S_2 will never catch up with S_1 .

24. (a) Let (r, θ) denote the polar coordinates of the destroyer S_1 . When S_1 travels the 6 miles from $(9, 0)$ to $(3, 0)$ it stands to reason, since S_2 travels half as fast as S_1 , that the polar coordinates of S_2 are $(3, \theta_2)$, where θ_2 is unknown. In other words, the distances of the ships from $(0, 0)$ are the same and $r(t) = 15t$ then gives the radial distance of both ships. This is necessary if S_1 is to intercept S_2 .
- (b) The differential of arc length in polar coordinates is $(ds)^2 = (r d\theta)^2 + (dr)^2$, so that

$$\left(\frac{ds}{dt} \right)^2 = r^2 \left(\frac{d\theta}{dt} \right)^2 + \left(\frac{dr}{dt} \right)^2.$$

Using $ds/dt = 30$ and $dr/dt = 15$ then gives

$$900 = 225t^2 \left(\frac{d\theta}{dt} \right)^2 + 225$$

$$675 = 225t^2 \left(\frac{d\theta}{dt} \right)^2$$

$$\frac{d\theta}{dt} = \frac{\sqrt{3}}{t}$$

$$\theta(t) = \sqrt{3} \ln t + c = \sqrt{3} \ln \frac{r}{15} + c.$$

When $r = 3$, $\theta = 0$, so $c = -\sqrt{3} \ln(1/5)$ and

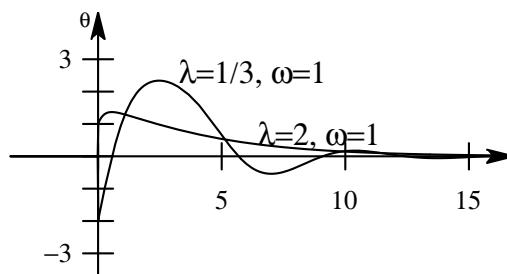
$$\theta(t) = \sqrt{3} \left(\ln \frac{r}{15} - \ln \frac{1}{5} \right) = \sqrt{3} \ln \frac{r}{3}.$$

Thus $r = 3e^{\theta/\sqrt{3}}$, whose graph is a logarithmic spiral.

- (c) The time for S_1 to go from $(9, 0)$ to $(3, 0) = \frac{1}{5}$ hour. Now S_1 must intercept the path of S_2 for some angle β , where $0 < \beta < 2\pi$. At the time of interception t_2 we have $15t_2 = 3e^{\beta/\sqrt{3}}$ or $t = e^{\beta/\sqrt{3}}/5$. The total time is then

$$t = \frac{1}{5} + \frac{1}{5}e^{\beta/\sqrt{3}} < \frac{1}{5}(1 + e^{2\pi/\sqrt{3}}).$$

25. For $\lambda^2 - \omega^2 > 0$ we choose $\lambda = 2$ and $\omega = 1$ with $x(0) = 1$ and $x'(0) = 2$. For $\lambda^2 - \omega^2 < 0$ we choose $\lambda = 1/3$ and $\omega = 1$ with $x(0) = -2$ and $x'(0) = 4$. In both cases the motion corresponds to the overdamped and underdamped cases for spring/mass systems.



Exercises 3.11

1. From $Dx = 2x - y$ and $Dy = x$ we obtain $y = 2x - Dx$, $Dy = 2Dx - D^2x$, and $(D^2 - 2D + 1)x = 0$. Then

$$x = c_1 e^t + c_2 t e^t \quad \text{and} \quad y = (c_1 - c_2)e^t + c_2 t e^t.$$

2. From $Dx = 4x + 7y$ and $Dy = x - 2y$ we obtain $y = \frac{1}{7}Dx - \frac{4}{7}x$, $Dy = \frac{1}{7}D^2x - \frac{4}{7}Dx$, and $(D^2 - 2D - 15)x = 0$. Then

$$x = c_1 e^{5t} + c_2 e^{-3t} \quad \text{and} \quad y = \frac{1}{7}c_1 e^{5t} - c_2 e^{-3t}.$$

3. From $Dx = -y + t$ and $Dy = x - t$ we obtain $y = t - Dx$, $Dy = 1 - D^2x$, and $(D^2 + 1)x = 1 + t$. Then

$$x = c_1 \cos t + c_2 \sin t + 1 + t$$

and

$$y = c_1 \sin t - c_2 \cos t + t - 1.$$

4. From $Dx - 4y = 1$ and $x + Dy = 2$ we obtain $y = \frac{1}{4}Dx - \frac{1}{4}$, $Dy = \frac{1}{4}D^2x$, and $(D^2 + 1)x = 2$. Then

$$x = c_1 \cos t + c_2 \sin t + 2$$

and

$$y = \frac{1}{4}c_2 \cos t - \frac{1}{4}c_1 \sin t - \frac{1}{4}c_1 \sin t - \frac{1}{4}.$$

5. From $(D^2 + 5)x - 2y = 0$ and $-2x + (D^2 + 2)y = 0$ we obtain $y = \frac{1}{2}(D^2 + 5)x$, $D^2y = \frac{1}{2}(D^4 + 5D^2)x$, and $(D^2 + 1)(D^2 + 6)x = 0$. Then

$$x = c_1 \cos t + c_2 \sin t + c_3 \cos \sqrt{6}t + c_4 \sin \sqrt{6}t$$

and

$$y = 2c_1 \cos t + 2c_2 \sin t - \frac{1}{2}c_3 \cos \sqrt{6}t - \frac{1}{2}c_4 \sin \sqrt{6}t.$$

Exercises 3.11

6. From $(D+1)x + (D-1)y = 2$ and $3x + (D+2)y = -1$ we obtain $x = -\frac{1}{3} - \frac{1}{3}(D+2)y$, $Dx = -\frac{1}{3}(D^2+2D)y$, and $(D^2+5)y = -7$. Then

$$y = c_1 \cos \sqrt{5}t + c_2 \sin \sqrt{5}t - \frac{7}{5}$$

and

$$x = \left(-\frac{2}{3}c_1 - \frac{\sqrt{5}}{3}c_2\right) \cos \sqrt{5}t + \left(\frac{\sqrt{5}}{3}c_1 - \frac{2}{3}c_2\right) \sin \sqrt{5}t + \frac{3}{5}.$$

7. From $D^2x = 4y + e^t$ and $D^2y = 4x - e^t$ we obtain $y = \frac{1}{4}D^2x - \frac{1}{4}e^t$, $D^2y = \frac{1}{4}D^4x - \frac{1}{4}e^t$, and $(D^2+4)(D-2)(D+2)x = -3e^t$. Then

$$x = c_1 \cos 2t + c_2 \sin 2t + c_3 e^{2t} + c_4 e^{-2t} + \frac{1}{5}e^t$$

and

$$y = -c_1 \cos 2t - c_2 \sin 2t + c_3 e^{2t} + c_4 e^{-2t} - \frac{1}{5}e^t.$$

8. From $(D^2+5)x + Dy = 0$ and $(D+1)x + (D-4)y = 0$ we obtain $(D-5)(D^2+4)x = 0$ and $(D-5)(D^2+4)y = 0$. Then

$$x = c_1 e^{5t} + c_2 \cos 2t + c_3 \sin 2t$$

and

$$y = c_4 e^{5t} + c_5 \cos 2t + c_6 \sin 2t.$$

Substituting into $(D+1)x + (D-4)y = 0$ gives

$$(6c_1 + c_4)e^{5t} + (c_2 + 2c_3 - 4c_5 + 2c_6) \cos 2t + (-2c_2 + c_3 - 2c_5 - 4c_6) \sin 2t = 0$$

so that $c_4 = -6c_1$, $c_5 = \frac{1}{2}c_3$, $c_6 = -\frac{1}{2}c_2$, and

$$y = -6c_1 e^{5t} + \frac{1}{2}c_3 \cos 2t - \frac{1}{2}c_2 \sin 2t.$$

9. From $Dx + D^2y = e^{3t}$ and $(D+1)x + (D-1)y = 4e^{3t}$ we obtain $D(D^2+1)x = 34e^{3t}$ and $D(D^2+1)y = -8e^{3t}$. Then

$$y = c_1 + c_2 \sin t + c_3 \cos t - \frac{4}{15}e^{3t}$$

and

$$x = c_4 + c_5 \sin t + c_6 \cos t + \frac{17}{15}e^{3t}.$$

Substituting into $(D+1)x + (D-1)y = 4e^{3t}$ gives

$$(c_4 - c_1) + (c_5 - c_6 - c_3 - c_2) \sin t + (c_6 + c_5 + c_2 - c_3) \cos t = 0$$

so that $c_4 = c_1$, $c_5 = c_3$, $c_6 = -c_2$, and

$$x = c_1 - c_2 \cos t + c_3 \sin t + \frac{17}{15}e^{3t}.$$

10. From $D^2x - Dy = t$ and $(D+3)x + (D+3)y = 2$ we obtain $D(D+1)(D+3)x = 1+3t$ and $D(D+1)(D+3)y = -1-3t$. Then

$$x = c_1 + c_2 e^{-t} + c_3 e^{-3t} - t + \frac{1}{2}t^2$$

and

$$y = c_4 + c_5 e^{-t} + c_6 e^{-3t} + t - \frac{1}{2}t^2.$$

Substituting into $(D+3)x + (D+3)y = 2$ and $D^2x - Dy = t$ gives

$$3(c_1 + c_4) + 2(c_2 + c_5)e^{-t} = 2$$

and

$$(c_2 + c_5)e^{-t} + 3(3c_3 + c_6)e^{-3t} = 0$$

so that $c_4 = -c_1$, $c_5 = -c_2$, $c_6 = -3c_3$, and

$$y = -c_1 - c_2e^{-t} - 3c_3e^{-3t} + t - \frac{1}{2}t^2.$$

11. From $(D^2 - 1)x - y = 0$ and $(D - 1)x + Dy = 0$ we obtain $y = (D^2 - 1)x$, $Dy = (D^3 - D)x$, and $(D - 1)(D^2 + D + 1)x = 0$. Then

$$x = c_1e^t + e^{-t/2} \left[c_2 \cos \frac{\sqrt{3}}{2}t + c_3 \sin \frac{\sqrt{3}}{2}t \right]$$

and

$$y = \left(-\frac{3}{2}c_2 - \frac{\sqrt{3}}{2}c_3 \right) e^{-t/2} \cos \frac{\sqrt{3}}{2}t + \left(\frac{\sqrt{3}}{2}c_2 - \frac{3}{2}c_3 \right) e^{-t/2} \sin \frac{\sqrt{3}}{2}t.$$

12. From $(2D^2 - D - 1)x - (2D + 1)y = 1$ and $(D - 1)x + Dy = -1$ we obtain $(2D + 1)(D - 1)(D + 1)x = -1$ and $(2D + 1)(D + 1)y = -2$. Then

$$x = c_1e^{-t/2} + c_2e^{-t} + c_3e^t + 1$$

and

$$y = c_4e^{-t/2} + c_5e^{-t} - 2.$$

Substituting into $(D - 1)x + Dy = -1$ gives

$$\left(-\frac{3}{2}c_1 - \frac{1}{2}c_4 \right) e^{-t/2} + (-2c_2 - c_5)e^{-t} = 0$$

so that $c_4 = -3c_1$, $c_5 = -2c_2$, and

$$y = -3c_1e^{-t/2} - 2c_2e^{-t} - 2.$$

13. From $(2D - 5)x + Dy = e^t$ and $(D - 1)x + Dy = 5e^t$ we obtain $Dy = (5 - 2D)x + e^t$ and $(4 - D)x = 4e^t$. Then

$$x = c_1e^{4t} + \frac{4}{3}e^t$$

and $Dy = -3c_1e^{4t} + 5e^t$ so that

$$y = -\frac{3}{4}c_1e^{4t} + c_2 + 5e^t.$$

14. From $Dx + Dy = e^t$ and $(-D^2 + D + 1)x + y = 0$ we obtain $y = (D^2 - D - 1)x$, $Dy = (D^3 - D^2 - D)x$, and $D^2(D - 1)x = e^t$. Then

$$x = c_1 + c_2t + c_3e^t + te^t$$

and

$$y = -c_1 - c_2 - c_2t - c_3e^t - te^t + e^t.$$

15. Multiplying the first equation by $D + 1$ and the second equation by $D^2 + 1$ and subtracting we obtain $(D^4 - D^2)x = 1$. Then

$$x = c_1 + c_2t + c_3e^t + c_4e^{-t} - \frac{1}{2}t^2.$$

Exercises 3.11

Multiplying the first equation by $D + 1$ and subtracting we obtain $D^2(D + 1)y = 1$. Then

$$y = c_5 + c_6 t + c_7 e^{-t} - \frac{1}{2} t^2.$$

Substituting into $(D - 1)x + (D^2 + 1)y = 1$ gives

$$(-c_1 + c_2 + c_5 - 1) + (-2c_4 + 2c_7)e^{-t} + (-1 - c_2 + c_6)t = 1$$

so that $c_5 = c_1 - c_2 + 2$, $c_6 = c_2 + 1$, and $c_7 = c_4$. The solution of the system is

$$\begin{aligned} x &= c_1 + c_2 t + c_3 e^t + c_4 e^{-t} - \frac{1}{2} t^2 \\ y &= (c_1 - c_2 + 2) + (c_2 + 1)t + c_4 e^{-t} - \frac{1}{2} t^2. \end{aligned}$$

16. From $D^2 x - 2(D^2 + D)y = \sin t$ and $x + Dy = 0$ we obtain $x = -Dy$, $D^2 x = -D^3 y$, and $D(D^2 + 2D + 2)y = -\sin t$. Then

$$y = c_1 + c_2 e^{-t} \cos t + c_3 e^{-t} \sin t + \frac{1}{5} \cos t + \frac{2}{5} \sin t$$

and

$$x = (c_2 + c_3)e^{-t} \sin t + (c_2 - c_3)e^{-t} \cos t + \frac{1}{5} \sin t - \frac{2}{5} \cos t.$$

17. From $Dx = y$, $Dy = z$, and $Dz = x$ we obtain $x = D^2 y = D^3 x$ so that $(D - 1)(D^2 + D + 1)x = 0$,

$$\begin{aligned} x &= c_1 e^t + e^{-t/2} \left[c_2 \sin \frac{\sqrt{3}}{2} t + c_3 \cos \frac{\sqrt{3}}{2} t \right], \\ y &= c_1 e^t + \left(-\frac{1}{2} c_2 - \frac{\sqrt{3}}{2} c_3 \right) e^{-t/2} \sin \frac{\sqrt{3}}{2} t + \left(\frac{\sqrt{3}}{2} c_2 - \frac{1}{2} c_3 \right) e^{-t/2} \cos \frac{\sqrt{3}}{2} t, \end{aligned}$$

and

$$z = c_1 e^t + \left(-\frac{1}{2} c_2 + \frac{\sqrt{3}}{2} c_3 \right) e^{-t/2} \sin \frac{\sqrt{3}}{2} t + \left(-\frac{\sqrt{3}}{2} c_2 - \frac{1}{2} c_3 \right) e^{-t/2} \cos \frac{\sqrt{3}}{2} t.$$

18. From $Dx + z = e^t$, $(D - 1)x + Dy + Dz = 0$, and $x + 2y + Dz = e^t$ we obtain $z = -Dx + e^t$, $Dz = -D^2 x + e^t$, and the system $(-D^2 + D - 1)x + Dy = -e^t$ and $(-D^2 + 1)x + 2y = 0$. Then $y = \frac{1}{2}(D^2 - 1)x$, $Dy = \frac{1}{2}D(D^2 - 1)x$, and $(D - 2)(D^2 + 1)x = -2e^t$ so that

$$\begin{aligned} x &= c_1 e^{2t} + c_2 \cos t + c_3 \sin t + e^t, \\ y &= \frac{3}{2} c_1 e^{2t} - c_2 \cos t - c_3 \sin t, \end{aligned}$$

and

$$z = -2c_1 e^{2t} - c_3 \cos t + c_2 \sin t.$$

19. Write the system in the form

$$\begin{aligned} Dx - 6y &= 0 \\ x - Dy + z &= 0 \\ x + y - Dz &= 0. \end{aligned}$$

Multiplying the second equation by D and adding to the third equation we obtain $(D + 1)x - (D^2 - 1)y = 0$. Eliminating y between this equation and $Dx - 6y = 0$ we find

$$(D^3 - D - 6D - 6)x = (D + 1)(D + 2)(D - 3)x = 0.$$

Thus

$$x = c_1 e^{-t} + c_2 e^{-2t} + c_3 e^{3t},$$

and, successively substituting into the first and second equations, we get

$$\begin{aligned} y &= -\frac{1}{6}c_1 e^{-t} - \frac{1}{3}c_2 e^{-2t} + \frac{1}{2}c_3 e^{3t} \\ z &= -\frac{5}{6}c_1 e^{-t} - \frac{1}{3}c_2 e^{-2t} + \frac{1}{2}c_3 e^{3t}. \end{aligned}$$

20. Write the system in the form

$$\begin{aligned} (D+1)x - z &= 0 \\ (D+1)y - z &= 0 \\ x - y + Dz &= 0. \end{aligned}$$

Multiplying the third equation by $D+1$ and adding to the second equation we obtain $(D+1)x + (D^2 + D - 1)z = 0$. Eliminating z between this equation and $(D+1)x - z = 0$ we find $D(D+1)^2 x = 0$. Thus

$$x = c_1 + c_2 e^{-t} + c_3 t e^{-t},$$

and, successively substituting into the first and third equations, we get

$$\begin{aligned} y &= c_1 + (c_2 - c_3)e^{-t} + c_3 t e^{-t} \\ z &= c_1 + c_3 e^{-t}. \end{aligned}$$

21. From $(D+5)x + y = 0$ and $4x - (D+1)y = 0$ we obtain $y = -(D+5)x$ so that $Dy = -(D^2 + 5D)x$. Then $4x + (D^2 + 5D)x + (D+5)x = 0$ and $(D+3)^2 x = 0$. Thus

$$x = c_1 e^{-3t} + c_2 t e^{-3t}$$

and

$$y = -(2c_1 + c_2)e^{-3t} - 2c_2 t e^{-3t}.$$

Using $x(1) = 0$ and $y(1) = 1$ we obtain

$$\begin{aligned} c_1 e^{-3} + c_2 e^{-3} &= 0 \\ -(2c_1 + c_2)e^{-3} - 2c_2 e^{-3} &= 1 \end{aligned}$$

or

$$\begin{aligned} c_1 + c_2 &= 0 \\ 2c_1 + 3c_2 &= -e^3. \end{aligned}$$

Thus $c_1 = e^3$ and $c_2 = -e^3$. The solution of the initial value problem is

$$\begin{aligned} x &= e^{-3t+3} - t e^{-3t+3} \\ y &= -e^{-3t+3} + 2t e^{-3t+3}. \end{aligned}$$

22. From $Dx - y = -1$ and $3x + (D-2)y = 0$ we obtain $x = -\frac{1}{3}(D-2)y$ so that $Dx = -\frac{1}{3}(D^2 - 2D)y$. Then $-\frac{1}{3}(D^2 - 2D)y = y - 1$ and $(D^2 - 2D + 3)y = 3$. Thus

$$y = e^t \left(c_1 \cos \sqrt{2}t + c_2 \sin \sqrt{2}t \right) + 1$$

and

$$x = \frac{1}{3}e^t \left[\left(c_1 - \sqrt{2}c_2 \right) \cos \sqrt{2}t + \left(\sqrt{2}c_1 + c_2 \right) \sin \sqrt{2}t \right] + \frac{2}{3}.$$

Exercises 3.11

Using $x(0) = y(0) = 0$ we obtain

$$\begin{aligned} c_1 + 1 &= 0 \\ \frac{1}{3} (c_1 - \sqrt{2} c_2) + \frac{2}{3} &= 0. \end{aligned}$$

Thus $c_1 = -1$ and $c_2 = \sqrt{2}/2$. The solution of the initial value problem is

$$\begin{aligned} x &= e^t \left(-\frac{2}{3} \cos \sqrt{2} t - \frac{\sqrt{2}}{6} \sin \sqrt{2} t \right) + \frac{2}{3} \\ y &= e^t \left(-\cos \sqrt{2} t + \frac{\sqrt{2}}{2} \sin \sqrt{2} t \right) + 1. \end{aligned}$$

23. (a) Write the system as

$$\begin{aligned} (D^2 + 3)x_1 - x_2 &= 0 \\ -2x_1 + (D^2 + 2)x_2 &= 0. \end{aligned}$$

Then

$$(D^2 + 2)(D^2 + 3)x_1 - 2x_1 = (D^2 + 1)(D^2 + 4)x_1 = 0,$$

and

$$x_1(t) = c_1 \cos t + c_2 \sin t + c_3 \cos 2t + c_4 \sin 2t.$$

Since $x_2 = (D^2 + 3)x_1$, we have

$$x_2(t) = 2c_1 \cos t + 2c_2 \sin t - c_3 \cos 2t - c_4 \sin 2t.$$

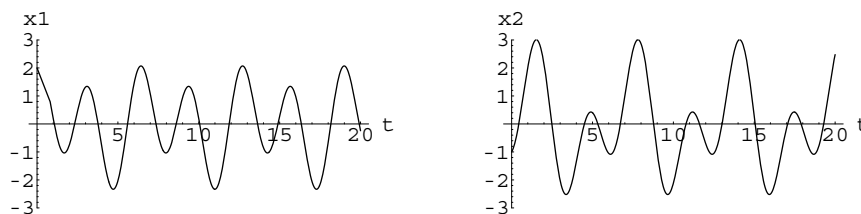
The initial conditions imply

$$\begin{aligned} x_1(0) &= c_1 + c_3 = 2 \\ x_1'(0) &= c_1 + 2c_4 = 1 \\ x_2(0) &= 2c_1 - c_3 = -1 \\ x_2'(0) &= 2c_2 - 2c_4 = 1, \end{aligned}$$

so $c_1 = \frac{1}{3}$, $c_2 = \frac{2}{3}$, $c_3 = \frac{5}{3}$, and $c_4 = \frac{1}{6}$. Thus

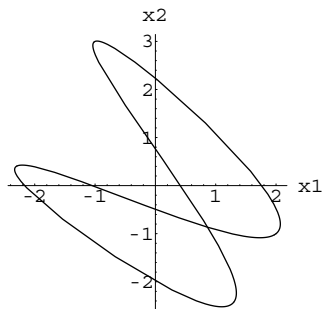
$$\begin{aligned} x_1(t) &= \frac{1}{3} \cos t + \frac{2}{3} \sin t + \frac{5}{3} \cos 2t + \frac{1}{6} \sin 2t \\ x_2(t) &= \frac{2}{3} \cos t + \frac{4}{3} \sin t - \frac{5}{3} \cos 2t - \frac{1}{6} \sin 2t. \end{aligned}$$

(b)



In this problem the motion appears to be periodic with period 2π . In Example 4 the motion does not appear to be periodic.

(c)



24. Write the system as

$$\begin{aligned} 0.04i_2' + 30i_2 + 10i_3 &= 50 & \text{or} & & i_2' + 750i_2 + 250i_3 &= 1250 \\ 0.02i_3' + 10i_2 + 10i_3 &= 50 & & & i_3' + 500i_2 + 500i_3 &= 2500. \end{aligned}$$

In operator notation this becomes

$$(D + 750)i_2 + 250i_3 = 1250$$

$$500i_2 + (D + 500)i_3 = 2500.$$

Eliminating i_3 we get

$$\left[(D + 500)(D + 750) - 250(500) \right] i_2 = (D + 250)(D + 1000)i_2 = 0.$$

Thus

$$i_2(t) = c_1 e^{-250t} + c_2 e^{-1000t}.$$

Since $i_3 = \frac{1}{250}[1250 - (D + 750)i_2]$, we have

$$i_3(t) = -2c_1 e^{-250t} + c_2 e^{-1000t} + 5.$$

The initial conditions imply

$$i_2(0) = c_1 + c_2 = 0$$

$$i_3(0) = -2c_1 + c_2 + 5 = 0,$$

so $c_1 = \frac{5}{3}$ and $c_2 = -\frac{5}{3}$. Thus

$$\begin{aligned} i_2(t) &= \frac{5}{3} e^{-250t} - \frac{5}{3} e^{-1000t} \\ i_3(t) &= -\frac{10}{3} e^{-250t} - \frac{5}{3} e^{-1000t} + 5. \end{aligned}$$

25. Equating Newton's law with the net forces in the x - and y -directions gives $m \frac{d^2 x}{dt^2} = 0$ and $m \frac{d^2 y}{dt^2} = -mg$, respectively. From $mD^2 x = 0$ we obtain $x(t) = c_1 t + c_2$, and from $mD^2 y = -mg$ or $D^2 y = -g$ we obtain $y(t) = -\frac{1}{2}gt^2 + c_3 t + c_4$.

26. From Newton's second law in the x -direction we have

$$m \frac{d^2 x}{dt^2} = -k \cos \theta = -k \frac{1}{v} \frac{dx}{dt} = -|c| \frac{dx}{dt}.$$

In the y -direction we have

$$m \frac{d^2 y}{dt^2} = -mg - k \sin \theta = -mg - k \frac{1}{v} \frac{dy}{dt} = -mg - |c| \frac{dy}{dt}.$$

Exercises 3.11

From $mD^2x + |c|Dx = 0$ we have $D(mD + |c|)x = 0$ so that $(mD + |c|)x = c_1$. This is a first-order linear equation. An integrating factor is $e^{\int |c|dt/m} e^{|c|t/m}$ so that

$$\frac{d}{dt}[e^{|c|t/m}x] = c_1 e^{|c|t/m}$$

and $e^{|c|t}x = (c_1m/|c|)e^{|c|t/m} + c_2$. The general solution of this equation is $x(t) = c_3 + c_2e^{|c|t/m}$. From $(mD^2 + |c|D)y = -mg$ we have $D(mD + |c|)y = -mg$ so that $(mD + |c|)y = -mgt + c_1$. This is a first-order linear equation with integrating factor $e^{|c|t/m}$. Thus

$$\frac{d}{dt}[e^{|c|t/m}y] = (-mgt + c_1)e^{|c|t/m}$$

$$e^{|c|t/m}y = -\frac{m^2g}{|c|}te^{|c|t/m} + \frac{m^3g}{c^2}e^{|c|t/m} + \frac{c_1m}{|c|}e^{|c|t/m} + c_2$$

and

$$y(t) = -\frac{m^2g}{|c|}t + \frac{m^3g}{c^2} + c_3 + c_2e^{-|c|t/m}.$$

Chapter 3 Review Exercises

- $y = 0$
- Since $y_c = c_1e^x + c_2e^{-x}$, a particular solution for $y'' - y = 1 + e^x$ is $y_p = A + Bxe^x$.
- True
- True
- 8 ft., since $k = 4$.
- $2\pi/5$, since $\frac{1}{4}x'' + 6.25x = 0$.
- $5/4$ m, since $x = -\cos 4t + \frac{3}{4}\sin 4t$.
- From $x(0) = (\sqrt{2}/2)\sin\phi = -1/2$ we see that $\sin\phi = -1/\sqrt{2}$, so ϕ is an angle in the third or fourth quadrant. Since $x'(t) = \sqrt{2}\cos(2t + \phi)$, $x'(0) = \sqrt{2}\cos\phi = 1$ and $\cos\phi > 0$. Thus ϕ is in the fourth quadrant and $\phi = -\pi/4$.
- They are linearly independent over $(-\infty, \infty)$ and linearly dependent over $(0, \infty)$.
- (a) Since $f_2(x) = 2\ln x = 2f_1(x)$, the functions are linearly dependent.
 (b) Since x^{n+1} is not a constant multiple of x^n , the functions are linearly independent.
 (c) Since $x + 1$ is not a constant multiple of x , the functions are linearly independent.
 (d) Since $f_1(x) = \cos x \cos(\pi/2) - \sin x \sin(\pi/2) = -\sin x = -f_2(x)$, the functions are linearly dependent.
 (e) Since $f_1(x) = 0 \cdot f_2(x)$, the functions are linearly dependent.
 (f) Since $2x$ is not a constant multiple of 2 , the functions are linearly independent.
 (g) Since $3(x^2) + 2(1 - x^2) - (2 + x^2) = 0$, the functions are linearly dependent.
 (h) since $xe^{x+1} + 0(4x - 5)e^x - exe^x = 0$, the functions are linearly dependent.
- (a) The auxiliary equation is $(m - 3)(m + 5)(m - 1) = m^3 + m^2 - 17m + 15 = 0$, so the differential equation is $y''' + y'' - 17y' + 15y = 0$.
 (b) The form of the auxiliary equation is

$$m(m - 1)(m - 2) + bm(m - 1) + cm + d = m^3 + (b - 3)m^2 + (c - b + 2)m + d = 0.$$

Since $(m-3)(m+5)(m-1) = m^3 + m^2 - 17m + 15 = 0$, we have $b-3=1$, $c-b+2=-17$, and $d=15$. Thus, $b=4$ and $c=-15$, so the differential equation is $y''' + 4y'' - 15y' + 15y = 0$.

12. (a) The auxiliary equation is $am(m-1) + bm + c = am^2 + (b-a)m + c = 0$. If the roots are 3 and -1 , then we want $(m-3)(m+1) = m^2 - 2m - 3 = 0$. Thus, let $a=1$, $b=-1$, and $c=-3$, so that the differential equation is $x^2y'' - xy' - 3y = 0$.

- (b) In this case we want the auxiliary equation to be $m^2 + 1 = 0$, so let $a=1$, $b=1$, and $c=1$. Then the differential equation is $x^2y'' + xy' + y = 0$.

13. From $m^2 - 2m - 2 = 0$ we obtain $m = 1 \pm \sqrt{3}$ so that

$$y = c_1 e^{(1+\sqrt{3})x} + c_2 e^{(1-\sqrt{3})x}.$$

14. From $2m^2 + 2m + 3 = 0$ we obtain $m = -1/2 \pm \sqrt{5}/2$ so that

$$y = e^{-x/2} \left(c_1 \cos \frac{\sqrt{5}}{2}x + c_2 \sin \frac{\sqrt{5}}{2}x \right).$$

15. From $m^3 + 10m^2 + 25m = 0$ we obtain $m=0$, $m=-5$, and $m=-5$ so that

$$y = c_1 + c_2 e^{-5x} + c_3 x e^{-5x}.$$

16. From $2m^3 + 9m^2 + 12m + 5 = 0$ we obtain $m=-1$, $m=-1$, and $m=-5/2$ so that

$$y = c_1 e^{-5x/2} + c_2 e^{-x} + c_3 x e^{-x}.$$

17. From $3m^3 + 10m^2 + 15m + 4 = 0$ we obtain $m=-1/3$ and $m=-3/2 \pm \sqrt{7}/2$ so that

$$y = c_1 e^{-x/3} + e^{-3x/2} \left(c_2 \cos \frac{\sqrt{7}}{2}x + c_3 \sin \frac{\sqrt{7}}{2}x \right).$$

18. From $2m^4 + 3m^3 + 2m^2 + 6m - 4 = 0$ we obtain $m=1/2$, $m=-2$, and $m=\pm\sqrt{2}i$ so that

$$y = c_1 e^{x/2} + c_2 e^{-2x} + c_3 \cos \sqrt{2}x + c_4 \sin \sqrt{2}x.$$

19. Applying D^4 to the differential equation we obtain $D^4(D^2 - 3D + 5) = 0$. Then

$$y = \underbrace{e^{3x/2} \left(c_1 \cos \frac{\sqrt{11}}{2}x + c_2 \sin \frac{\sqrt{11}}{2}x \right)}_{y_c} + c_3 + c_4 x + c_5 x^2 + c_6 x^3$$

and $y_p = A + Bx + Cx^2 + Dx^3$. Substituting y_p into the differential equation yields

$$(5A - 3B + 2C) + (5B - 6C + 6D)x + (5C - 9D)x^2 + 5Dx^3 = -2x + 4x^3.$$

Equating coefficients gives $A = -222/625$, $B = 46/125$, $C = 36/25$, and $D = 4/5$. The general solution is

$$y = e^{3x/2} \left(c_1 \cos \frac{\sqrt{11}}{2}x + c_2 \sin \frac{\sqrt{11}}{2}x \right) - \frac{222}{625} + \frac{46}{125}x + \frac{36}{25}x^2 + \frac{4}{5}x^3.$$

20. Applying $(D-1)^3$ to the differential equation we obtain $(D-1)^3(D-2D+1) = (D-1)^5 = 0$. Then

$$y = \underbrace{c_1 e^x + c_2 x e^x}_{y_c} + c_3 x^2 e^x + c_4 x^3 e^x + c_5 x^4 e^x$$

Chapter 3 Review Exercises

and $y_p = Ax^2e^x + Bx^3e^x + Cx^4e^x$. Substituting y_p into the differential equation yields

$$12Cx^2e^x + 6Bxe^x + 2Ae^x = x^2e^x.$$

Equating coefficients gives $A = 0$, $B = 0$, and $C = 1/12$. The general solution is

$$y = c_1e^x + c_2xe^x + \frac{1}{12}x^4e^x.$$

21. Applying $D(D^2 + 1)$ to the differential equation we obtain

$$D(D^2 + 1)(D^3 - 5D^2 + 6D) = D^2(D^2 + 1)(D - 2)(D - 3) = 0.$$

Then

$$y = \underbrace{c_1 + c_2e^{2x} + c_3e^{3x}}_{y_c} + c_4x + c_5 \cos x + c_6 \sin x$$

and $y_p = Ax + B \cos x + C \sin x$. Substituting y_p into the differential equation yields

$$6A + (5B + 5C) \cos x + (-5B + 5C) \sin x = 8 + 2 \sin x.$$

Equating coefficients gives $A = 4/3$, $B = -1/5$, and $C = 1/5$. The general solution is

$$y = c_1 + c_2e^{2x} + c_3e^{3x} + \frac{4}{3}x - \frac{1}{5} \cos x + \frac{1}{5} \sin x.$$

22. Applying D to the differential equation we obtain $D(D^3 - D^2) = D^3(D - 1) = 0$. Then

$$y = \underbrace{c_1 + c_2x + c_3e^x}_{y_c} + c_4x^2$$

and $y_p = Ax^2$. Substituting y_p into the differential equation yields $-2A = 6$. Equating coefficients gives $A = -3$. The general solution is

$$y = c_1 + c_2x + c_3e^x - 3x^2.$$

23. The auxiliary equation is $m^2 - 2m + 2 = [m - (1 + i)][m - (1 - i)] = 0$, so $y_c = c_1e^x \sin x + c_2e^x \cos x$ and

$$W = \begin{vmatrix} e^x \sin x & e^x \cos x \\ e^x \cos x + e^x \sin x & -e^x \sin x + e^x \cos x \end{vmatrix} = -e^{2x}.$$

Identifying $f(x) = e^x \tan x$ we obtain

$$u'_1 = -\frac{(e^x \cos x)(e^x \tan x)}{-e^{2x}} = \sin x$$

$$u'_2 = \frac{(e^x \sin x)(e^x \tan x)}{-e^{2x}} = -\frac{\sin^2 x}{\cos x} = \cos x - \sec x.$$

Then $u_1 = -\cos x$, $u_2 = \sin x - \ln |\sec x + \tan x|$, and

$$y = c_1e^x \sin x + c_2e^x \cos x - e^x \sin x \cos x + e^x \sin x \cos x - e^x \cos x \ln |\sec x + \tan x|$$

$$= c_1e^x \sin x + c_2e^x \cos x - e^x \cos x \ln |\sec x + \tan x|.$$

24. The auxiliary equation is $m^2 - 1 = 0$, so $y_c = c_1e^x + c_2e^{-x}$ and

$$W = \begin{vmatrix} e^x & e^{-x} \\ e^x & -e^{-x} \end{vmatrix} = -2.$$

Identifying $f(x) = 2e^x/(e^x + e^{-x})$ we obtain

$$u'_1 = \frac{1}{e^x + e^{-x}} = \frac{e^x}{1 + e^{2x}}$$

$$u'_2 = -\frac{e^{2x}}{e^x + e^{-x}} = -\frac{e^{3x}}{1 + e^{2x}} = -e^x + \frac{e^x}{1 + e^{2x}}.$$

Then $u_1 = \tan^{-1} e^x$, $u_2 = -e^x + \tan^{-1} e^x$, and

$$y = c_1 e^x + c_2 e^{-x} + e^x \tan^{-1} e^x - 1 + e^{-x} \tan^{-1} e^x.$$

25. The auxiliary equation is $6m^2 - m - 1 = 0$ so that

$$y = c_1 x^{1/2} + c_2 x^{-1/3}.$$

26. The auxiliary equation is $2m^3 + 13m^2 + 24m + 9 = (m + 3)^2(m + 1/2) = 0$ so that

$$y = c_1 x^{-3} + c_2 x^{-3} \ln x + \frac{1}{4} x^3.$$

27. The auxiliary equation is $m^2 - 5m + 6 = (m - 2)(m - 3) = 0$ and a particular solution is $y_p = x^4 - x^2 \ln x$ so that

$$y = c_1 x^2 + c_2 x^3 + x^4 - x^2 \ln x.$$

28. The auxiliary equation is $m^2 - 2m + 1 = (m - 1)^2 = 0$ and a particular solution is $y_p = \frac{1}{4} x^3$ so that

$$y = c_1 x + c_2 x \ln x + \frac{1}{4} x^3.$$

29. (a) The auxiliary equation is $m^2 + \omega^2 = 0$, so $y_c = c_1 \cos \omega t + c_2 \sin \omega t$. When $\omega \neq \alpha$, $y_p = A \cos \alpha t + B \sin \alpha t$ and

$$y = c_1 \cos \omega t + c_2 \sin \omega t + A \cos \alpha t + B \sin \alpha t.$$

When $\omega = \alpha$, $y_p = At \cos \alpha t + Bt \sin \alpha t$ and

$$y = c_1 \cos \omega t + c_2 \sin \omega t + At \cos \alpha t + Bt \sin \alpha t.$$

- (b) The auxiliary equation is $m^2 - \omega^2 = 0$, so $y_c = c_1 e^{\omega t} + c_2 e^{-\omega t}$. When $\omega \neq \alpha$, $y_p = Ae^{\alpha t}$ and

$$y = c_1 e^{\omega t} + c_2 e^{-\omega t} + Ae^{\alpha t}.$$

When $\omega = \alpha$, $y_p = Ate^{\alpha t}$ and

$$y = c_1 e^{\omega t} + c_2 e^{-\omega t} + Ate^{\alpha t}.$$

30. (a) If $y = \sin x$ is a solution then so is $y = \cos x$ and $m^2 + 1$ is a factor of the auxiliary equation $m^4 + 2m^3 + 11m^2 + 2m + 10 = 0$. Dividing by $m^2 + 1$ we get $m^2 + 2m + 10$, which has roots $-1 \pm 3i$. The general solution of the differential equation is

$$y = c_1 \cos x + c_2 \sin x + e^{-x}(c_3 \cos 3x + c_4 \sin 3x).$$

- (b) The auxiliary equation is $m(m + 1) = m^2 + m = 0$, so the associated homogeneous differential equation is $y'' + y' = 0$. Letting $y = c_1 + c_2 e^{-x} + \frac{1}{2} x^2 - x$ and computing $y'' + y'$ we get x . Thus, the differential equation is $y'' + y' = x$.

31. (a) The auxiliary equation is $m^4 - 2m^2 + 1 = (m^2 - 1)^2 = 0$, so the general solution of the differential equation is

$$y = c_1 \sinh x + c_2 \cosh x + c_3 x \sinh x + c_4 x \cosh x.$$

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(b) Since both $\sinh x$ and $x \sinh x$ are solutions of the associated homogeneous differential equation, a particular solution of $y^{(4)} - 2y'' + y = \sinh x$ has the form $y_p = Ax^2 \sinh x + Bx^2 \cosh x$.

32. Since $y'_1 = 1$ and $y''_1 = 0$, $x^2 y''_1 - (x^2 + 2x)y'_1 + (x+2)y_1 = -x^2 - 2x + x^2 + 2x = 0$, and $y_1 = x$ is a solution of the associated homogeneous equation. Using the method of reduction of order, we let $y = ux$. Then $y' = xu' + u$ and $y'' = xu'' + 2u'$, so

$$\begin{aligned} x^2 y'' - (x^2 + 2x)y' + (x+2)y &= x^3 u'' + 2x^2 u' - x^3 u' - 2x^2 u' - x^2 u - 2xu + x^2 u + 2xu \\ &= x^3 u'' - x^3 u' = x^3(u'' - u'). \end{aligned}$$

To find a second solution of the homogeneous equation we note that $u = e^x$ is a solution of $u'' - u' = 0$. Thus, $y_c = c_1 x + c_2 x e^x$. To find a particular solution we set $x^3(u'' - u') = x^3$ so that $u'' - u' = 1$. This differential equation has a particular solution of the form Ax . Substituting, we find $A = -1$, so a particular solution of the original differential equation is $y_p = -x^2$ and the general solution is $y = c_1 x + c_2 x e^x - x^2$.

33. The auxiliary equation is $m^2 - 2m + 2 = 0$ so that $m = 1 \pm i$ and $y = e^x(c_1 \cos x + c_2 \sin x)$. Setting $y(\pi/2) = 0$ and $y(\pi) = -1$ we obtain $c_1 = e^{-\pi}$ and $c_2 = 0$. Thus, $y = e^{x-\pi} \cos x$.
34. The auxiliary equation is $m^2 + 2m + 1 = (m+1)^2 = 0$, so that $y = c_1 e^{-x} + c_2 x e^{-x}$. Setting $y(-1) = 0$ and $y'(0) = 0$ we get $c_1 e - c_2 e = 0$ and $-c_1 + c_2 = 0$. Thus $c_1 = c_2$ and $y = c e^{-x} + c x e^{-x}$ is a solution of the boundary-value problem for any real number c .
35. The auxiliary equation is $m^2 - 1 = (m-1)(m+1) = 0$ so that $m = \pm 1$ and $y = c_1 e^x + c_2 e^{-x}$. Assuming $y_p = Ax + B + C \sin x$ and substituting into the differential equation we find $A = -1$, $B = 0$, and $C = -\frac{1}{2}$. Thus $y_p = -x - \frac{1}{2} \sin x$ and

$$y = c_1 e^x + c_2 e^{-x} - x - \frac{1}{2} \sin x.$$

Setting $y(0) = 2$ and $y'(0) = 3$ we obtain

$$\begin{aligned} c_1 + c_2 &= 2 \\ c_1 - c_2 - \frac{3}{2} &= 3. \end{aligned}$$

Solving this system we find $c_1 = \frac{13}{4}$ and $c_2 = -\frac{5}{4}$. The solution of the initial-value problem is

$$y = \frac{13}{4} e^x - \frac{5}{4} e^{-x} - x - \frac{1}{2} \sin x.$$

36. The auxiliary equation is $m^2 + 1 = 0$, so $y_c = c_1 \cos x + c_2 \sin x$ and

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = 1.$$

Identifying $f(x) = \sec^3 x$ we obtain

$$u'_1 = -\sin x \sec^3 x = -\frac{\sin x}{\cos^3 x}$$

$$u'_2 = \cos x \sec^3 x = \sec^2 x.$$

Then

$$u_1 = -\frac{1}{2} \frac{1}{\cos^2 x} = -\frac{1}{2} \sec^2 x$$

$$u_2 = \tan x.$$

Thus

$$y = c_1 \cos x + c_2 \sin x - \frac{1}{2} \cos x \sec^2 x + \sin x \tan x$$

$$\begin{aligned}
&= c_1 \cos x + c_2 \sin x - \frac{1}{2} \sec x + \frac{1 - \cos^2 x}{\cos x} \\
&= c_3 \cos x + c_2 \sin x + \frac{1}{2} \sec x.
\end{aligned}$$

and

$$y'_p = -c_3 \sin x + c_2 \cos x + \frac{1}{2} \sec x \tan x.$$

The initial conditions imply

$$\begin{aligned}
c_3 + \frac{1}{2} &= 1 \\
c_2 &= \frac{1}{2}.
\end{aligned}$$

Thus $c_3 = c_2 = 1/2$ and

$$y = \frac{1}{2} \cos x + \frac{1}{2} \sin x + \frac{1}{2} \sec x.$$

- 37.** Let $u = y'$ so that $u' = y''$. The equation becomes $u \frac{du}{dx} = 4x$. Separating variables we obtain

$$u \, du = 4x \, dx \implies \frac{1}{2} u^2 = 2x^2 + c_1 \implies u^2 = 4x^2 + c_2.$$

When $x = 1$, $y' = u = 2$, so $4 = 4 + c_2$ and $c_2 = 0$. Then

$$\begin{aligned}
u^2 = 4x^2 &\implies \frac{dy}{dx} = 2x \quad \text{or} \quad \frac{dy}{dx} = -2x \\
&\implies y = x^2 + c_3 \quad \text{or} \quad y = -x^2 + c_4.
\end{aligned}$$

When $x = 1$, $y = 5$, so $5 = 1 + c_3$ and $5 = -1 + c_4$. Thus $c_3 = 4$ and $c_4 = 6$. We have $y = x^2 + 4$ and $y = -x^2 + 6$. Note however that when $y = -x^2 + 6$, $y' = -2x$ and $y'(1) = -2 \neq 2$. Thus, the solution of the initial-value problem is $y = x^2 + 4$.

- 38.** Let $u = y'$ so that $y'' = u \frac{du}{dy}$. The equation becomes $2u \frac{du}{dy} = 3y^2$. Separating variables we obtain

$$2u \, dy = 3y^2 \, dy \implies u^2 = y^3 + c_1.$$

When $x = 0$, $y = 1$ and $y' = u = 1$ so $1 = 1 + c_1$ and $c_1 = 0$. Then

$$\begin{aligned}
u^2 = y^3 &\implies \left(\frac{dy}{dx} \right)^2 = y^3 \implies \frac{dy}{dx} = y^{3/2} \implies y^{-3/2} dy = dx \\
&\implies -2y^{-1/2} = x + c_2 \implies y = \frac{4}{(x + c_2)^2}.
\end{aligned}$$

When $x = 0$, $y = 1$, so $1 = \frac{4}{c_2^2} \implies c_2 = \pm 2$. Thus, $y = \frac{4}{(x+2)^2}$ and $y = \frac{4}{(x-2)^2}$. Note however that when $y = \frac{4}{(x+2)^2}$, $y' = -\frac{8}{(x+2)^3}$ and $y'(0) = -1 \neq 1$. Thus, the solution of the initial-value problem is $y = \frac{4}{(x-2)^2}$.

- 39. (a)** The auxiliary equation is $12m^4 + 64m^3 + 59m^2 - 23m - 12 = 0$ and has roots -4 , $-3/2$, $-1/3$, and $1/2$.

The general solution is

$$y = c_1 e^{-4x} + c_2 e^{-3x/2} + c_3 e^{-x/3} + c_4 e^{x/2}.$$

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(b) The system of equations is

$$\begin{aligned}c_1 + c_2 + c_3 + c_4 &= -1 \\-4c_1 - \frac{3}{2}c_2 - \frac{1}{3}c_3 + \frac{1}{2}c_4 &= 2 \\16c_1 + \frac{9}{4}c_2 + \frac{1}{9}c_3 + \frac{1}{4}c_4 &= 5 \\-64c_1 - \frac{27}{8}c_2 - \frac{1}{27}c_3 + \frac{1}{8}c_4 &= 0.\end{aligned}$$

Using a CAS we find $c_1 = -\frac{73}{495}$, $c_2 = \frac{109}{35}$, $c_3 = -\frac{3726}{385}$, and $c_4 = \frac{257}{45}$. The solution of the initial-value problem is

$$y = -\frac{73}{495}e^{-4x} + \frac{109}{35}e^{-3x/2} - \frac{3726}{385}e^{-x/3} + \frac{257}{45}e^{x/2}.$$

40. Consider $xy'' + y' = 0$ and look for a solution of the form $y = x^m$. Substituting into the differential equation we have

$$xy'' + y' = m(m-1)x^{m-1} + mx^{m-1} = m^2x.$$

Thus, the general solution of $xy'' + y' = 0$ is $y_c = c_1 + c_2 \ln x$. To find a particular solution of $xy'' + y' = -\sqrt{x}$ we use variation of parameters. The Wronskian is

$$W = \begin{vmatrix} 1 & \ln x \\ 0 & 1/x \end{vmatrix} = \frac{1}{x}.$$

Identifying $f(x) = -x^{-1/2}$ we obtain

$$u'_1 = \frac{x^{-1/2} \ln x}{1/x} = \sqrt{x} \ln x \quad \text{and} \quad u'_2 = \frac{-x^{-1/2}}{1/x} = -\sqrt{x},$$

so that

$$u_1 = x^{3/2} \left(\frac{2}{3} \ln x - \frac{4}{9} \right) \quad \text{and} \quad u_2 = -\frac{2}{3}x^{3/2}.$$

Then $y_p = x^{3/2} \left(\frac{2}{3} \ln x - \frac{4}{9} \right) - \frac{2}{3}x^{3/2} \ln x = -\frac{4}{9}x^{3/2}$ and the general solution of the differential equation is $y = c_1 + c_2 \ln x - \frac{4}{9}x^{3/2}$. The initial conditions are $y(1) = 0$ and $y'(1) = 0$. These imply that $c_1 = \frac{4}{9}$ and $c_2 = \frac{2}{3}$. The solution of the initial-value problem is $y = \frac{4}{9} + \frac{2}{3} \ln x - \frac{4}{9}x^{3/2}$.

41. From $(D-2)x + (D-2)y = 1$ and $Dx + (2D-1)y = 3$ we obtain $(D-1)(D-2)y = -6$ and $Dx = 3 - (2D-1)y$. Then

$$y = c_1 e^{2t} + c_2 e^t - 3 \quad \text{and} \quad x = -c_2 e^t - \frac{3}{2}c_1 e^{2t} + c_3.$$

Substituting into $(D-2)x + (D-2)y = 1$ gives $c_3 = 5/2$ so that

$$x = -c_2 e^t - \frac{3}{2}c_1 e^{2t} + \frac{5}{2}.$$

42. From $(D-2)x - y = t - 2$ and $-3x + (D-4)y = -4t$ we obtain $(D-1)(D-5)x = 9 - 8t$. Then

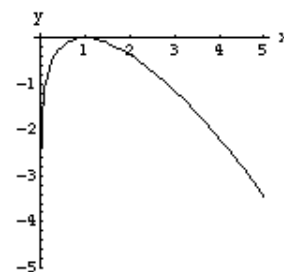
$$x = c_1 e^t + c_2 e^{5t} - \frac{8}{5}t - \frac{3}{25}$$

and

$$y = (D-2)x - t + 2 = -c_1 e^t + 3c_2 e^{5t} + \frac{16}{25} + \frac{11}{25}t.$$

43. From $(D-2)x - y = -e^t$ and $-3x + (D-4)y = -7e^t$ we obtain $(D-1)(D-5)x = -4e^t$ so that

$$x = c_1 e^t + c_2 e^{5t} + t e^t.$$



Then

$$y = (D - 2)x + e^t = -c_1 e^t + 3c_2 e^{5t} - t e^t + 2e^t.$$

44. From $(D + 2)x + (D + 1)y = \sin 2t$ and $5x + (D + 3)y = \cos 2t$ we obtain $(D^2 + 5)y = 2 \cos 2t - 7 \sin 2t$. Then

$$y = c_1 \cos t + c_2 \sin t - \frac{2}{3} \cos 2t + \frac{7}{3} \sin 2t$$

and

$$\begin{aligned} x &= -\frac{1}{5}(D + 3)y + \frac{1}{5} \cos 2t \\ &= \left(\frac{1}{5}c_1 - \frac{3}{5}c_2\right) \sin t + \left(-\frac{1}{5}c_2 - \frac{3}{5}c_1\right) \cos t - \frac{5}{3} \sin 2t - \frac{1}{3} \cos 2t. \end{aligned}$$

45. The period of a spring mass system is given by $T = 2\pi/\omega$ where $\omega^2 = k/m = kg/W$, where k is the spring constant, W is the weight of the mass attached to the spring, and g is the acceleration due to gravity. Thus, the period of oscillation is $T = (2\pi/\sqrt{kg})\sqrt{W}$. If the weight of the original mass is W , then $(2\pi/\sqrt{kg})\sqrt{W} = 3$ and $(2\pi/\sqrt{kg})\sqrt{W-8} = 2$. Dividing, we get $\sqrt{W}/\sqrt{W-8} = 3/2$ or $W = \frac{9}{4}(W-8)$. Solving for W we find that the weight of the original mass was 14.4 pounds.

46. (a) Solving $\frac{3}{8}x'' + 6x = 0$ subject to $x(0) = 1$ and $x'(0) = -4$ we obtain

$$x = \cos 4t - \sin 4t = \sqrt{2} \sin(4t + 3\pi/4).$$

(b) The amplitude is $\sqrt{2}$, period is $\pi/2$, and frequency is $2/\pi$.

(c) If $x = 1$ then $t = n\pi/2$ and $t = -\pi/8 + n\pi/2$ for $n = 1, 2, 3, \dots$.

(d) If $x = 0$ then $t = \pi/16 + n\pi/4$ for $n = 0, 1, 2, \dots$. The motion is upward for n even and downward for n odd.

(e) $x'(3\pi/16) = 0$

(f) If $x' = 0$ then $4t + 3\pi/4 = \pi/2 + n\pi$ or $t = 3\pi/16 + n\pi$.

47. From $mx'' + 4x' + 2x = 0$ we see that non-oscillatory motion results if $16 - 8m \geq 0$ or $0 < m \leq 2$.

48. From $x'' + \beta x' + 64x = 0$ we see that oscillatory motion results if $\beta^2 - 256 < 0$ or $0 \leq |\beta| < 16$.

49. From $q'' + 10^4 q = 100 \sin 50t$, $q(0) = 0$, and $q'(t) = 0$ we obtain $q_c = c_1 \cos 100t + c_2 \sin 100t$, $q_p = \frac{1}{75} \sin 50t$, and

(a) $q = -\frac{1}{150} \sin 100t + \frac{1}{75} \sin 50t$,

(b) $i = -\frac{2}{3} \cos 100t + \frac{2}{3} \cos 50t$, and

(c) $q = 0$ when $\sin 50t(1 - \cos 50t) = 0$ or $t = n\pi/50$ for $n = 0, 1, 2, \dots$.

50. Differentiate $L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q = E(t)$ and use $q'(t) = i(t)$ to obtain the desired result.

51. For $\lambda > 0$ the general solution is $y = c_1 \cos \sqrt{\lambda} x + c_2 \sin \sqrt{\lambda} x$. Now $y(0) = c_1$ and $y(2\pi) = c_1 \cos 2\pi\sqrt{\lambda} + c_2 \sin 2\pi\sqrt{\lambda}$, so the condition $y(0) = y(2\pi)$ implies

$$c_1 = c_1 \cos 2\pi\sqrt{\lambda} + c_2 \sin 2\pi\sqrt{\lambda}$$

which is true when $\sqrt{\lambda} = n$ or $\lambda = n^2$ for $n = 1, 2, 3, \dots$. Since $y' = -\sqrt{\lambda} c_1 \sin \sqrt{\lambda} x + \sqrt{\lambda} c_2 \cos \sqrt{\lambda} x = -nc_1 \sin nx + nc_2 \cos nx$, we see that $y'(0) = nc_2 = y'(2\pi)$ for $n = 1, 2, 3, \dots$. Thus, the eigenvalues are n^2 for $n = 1, 2, 3, \dots$, with corresponding eigenfunctions $\cos nx$ and $\sin nx$. When $\lambda = 0$, the general solution is $y = c_1 x + c_2$ and the corresponding eigenfunction is $y = 1$. For $\lambda < 0$ the general solution is $y = c_1 \cosh \sqrt{-\lambda} x + c_2 \sinh \sqrt{-\lambda} x$. In this case $y(0) = c_1$ and $y(2\pi) = c_1 \cosh 2\pi\sqrt{-\lambda} + c_2 \sinh 2\pi\sqrt{-\lambda}$, so $y(0) = y(2\pi)$ can only be valid for $\lambda = 0$. Thus, there are no eigenvalues corresponding to $\lambda < 0$.

Chapter 3 Review Exercises

52. (a) The differential equation is $d^2r/dt^2 - \omega^2 r = -g \sin \omega t$. The auxiliary equation is $m^2 - \omega^2 = 0$, so $r_c = c_1 e^{\omega t} + c_2 e^{-\omega t}$. A particular solution has the form $r_p = A \sin \omega t + B \cos \omega t$. Substituting into the differential equation we find $-2A\omega^2 \sin \omega t - 2B\omega^2 \cos \omega t = -g \sin \omega t$. Thus, $B = 0$, $A = g/2\omega^2$, and $r_p = (g/2\omega^2) \sin \omega t$. The general solution of the differential equation is $r(t) = c_1 e^{\omega t} + c_2 e^{-\omega t} + (g/2\omega^2) \sin \omega t$. The initial conditions imply $c_1 + c_2 = r_0$ and $g/2\omega - \omega c_1 + \omega c_2 = v_0$. Solving for c_1 and c_2 we get

$$c_1 = (2\omega^2 r_0 + 2\omega v_0 - g)/4\omega^2 \quad \text{and} \quad c_2 = (2\omega^2 r_0 - 2\omega v_0 + g)/4\omega^2,$$

so that

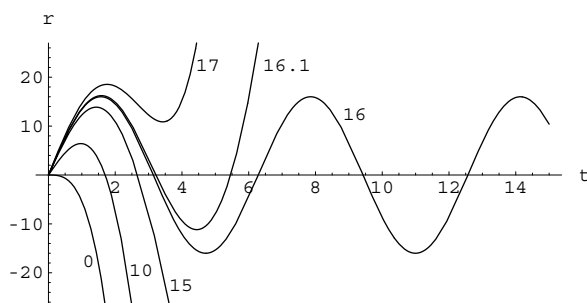
$$r(t) = \frac{2\omega^2 r_0 + 2\omega v_0 - g}{4\omega^2} e^{\omega t} + \frac{2\omega^2 r_0 - 2\omega v_0 + g}{4\omega^2} e^{-\omega t} + \frac{g}{2\omega^2} \sin \omega t.$$

- (b) The bead will exhibit simple harmonic motion when the exponential terms are missing. Solving $c_1 = 0$, $c_2 = 0$ for r_0 and v_0 we find $r_0 = 0$ and $v_0 = g/2\omega$.

To find the minimum length of rod that will accommodate simple harmonic motion we determine the amplitude of $r(t)$ and double it. Thus $L = g/\omega^2$.

- (c) As t increases, $e^{\omega t}$ approaches infinity and $e^{-\omega t}$ approaches 0. Since $\sin \omega t$ is bounded, the distance, $r(t)$, of the bead from the pivot point increases without bound and the distance of the bead from P will eventually exceed $L/2$.

(d)



- (e) For each v_0 we want to find the smallest value of t for which $r(t) = \pm 20$. Whether we look for $r(t) = -20$ or $r(t) = 20$ is determined by looking at the graphs in part (d). The total times that the bead stays on the rod is shown in the table below.

v_0	0	10	15	16.1	17
r	-20	-20	-20	20	20
t	1.55007	2.35494	3.43088	6.11627	4.22339

When $v_0 = 16$ the bead never leaves the rod.

4 The Laplace Transform

Exercises 4.1

1. $\mathcal{L}\{f(t)\} = \int_0^1 -e^{-st} dt + \int_1^\infty e^{-st} dt = \frac{1}{s} e^{-st} \Big|_0^1 - \frac{1}{s} e^{-st} \Big|_1^\infty$
 $= \frac{1}{s} e^{-s} - \frac{1}{s} - \left(0 - \frac{1}{s} e^{-s}\right) = \frac{2}{s} e^{-s} - \frac{1}{s}, \quad s > 0$
2. $\mathcal{L}\{f(t)\} = \int_0^2 4e^{-st} dt = -\frac{4}{s} e^{-st} \Big|_0^2 = -\frac{4}{s} (e^{-2s} - 1), \quad s > 0$
3. $\mathcal{L}\{f(t)\} = \int_0^1 t e^{-st} dt + \int_1^\infty e^{-st} dt = \left(-\frac{1}{s} t e^{-st} - \frac{1}{s^2} e^{-st}\right) \Big|_0^1 - \frac{1}{s} e^{-st} \Big|_1^\infty$
 $= \left(-\frac{1}{s} e^{-s} - \frac{1}{s^2} e^{-s}\right) - \left(0 - \frac{1}{s^2}\right) - \frac{1}{s} (0 - e^{-s}) = \frac{1}{s^2} (1 - e^{-s}), \quad s > 0$
4. $\mathcal{L}\{f(t)\} = \int_0^1 (2t+1) e^{-st} dt = \left(-\frac{2}{s} t e^{-st} - \frac{2}{s^2} e^{-st} - \frac{1}{s} e^{-st}\right) \Big|_0^1$
 $= \left(-\frac{2}{s} e^{-s} - \frac{2}{s^2} e^{-s} - \frac{1}{s} e^{-s}\right) - \left(0 - \frac{2}{s^2} - \frac{1}{s}\right) = \frac{1}{s} (1 - 3e^{-s}) + \frac{2}{s^2} (1 - e^{-s}), \quad s > 0$
5. $\mathcal{L}\{f(t)\} = \int_0^\pi (\sin t) e^{-st} dt = \left(-\frac{s}{s^2+1} e^{-st} \sin t - \frac{1}{s^2+1} e^{-st} \cos t\right) \Big|_0^\pi$
 $= \left(0 + \frac{1}{s^2+1} e^{-\pi s}\right) - \left(0 - \frac{1}{s^2+1}\right) = \frac{1}{s^2+1} (e^{-\pi s} + 1), \quad s > 0$
6. $\mathcal{L}\{f(t)\} = \int_{\pi/2}^\infty (\cos t) e^{-st} dt = \left(-\frac{s}{s^2+1} e^{-st} \cos t + \frac{1}{s^2+1} e^{-st} \sin t\right) \Big|_{\pi/2}^\infty$
 $= 0 - \left(0 + \frac{1}{s^2+1} e^{-\pi s/2}\right) = -\frac{1}{s^2+1} e^{-\pi s/2}, \quad s > 0$
7. $f(t) = \begin{cases} 0, & 0 < t < 1 \\ t, & t > 1 \end{cases}$
 $\mathcal{L}\{f(t)\} = \int_1^\infty t e^{-st} dt = \left(-\frac{1}{s} t e^{-st} - \frac{1}{s^2} e^{-st}\right) \Big|_1^\infty = \frac{1}{s} e^{-s} + \frac{1}{s^2} e^{-s}, \quad s > 0$
8. $f(t) = \begin{cases} 0, & 0 < t < 1 \\ 2t-2, & t > 1 \end{cases}$
 $\mathcal{L}\{f(t)\} = 2 \int_1^\infty (t-1) e^{-st} dt = 2 \left(-\frac{1}{s} (t-1) e^{-st} - \frac{1}{s^2} e^{-st}\right) \Big|_1^\infty = \frac{2}{s^2} e^{-s}, \quad s > 0$
9. $f(t) = \begin{cases} 1-t, & 0 < t < 1 \\ 0, & t > 0 \end{cases}$
 $\mathcal{L}\{f(t)\} = \int_0^1 (1-t) e^{-st} dt = \left(-\frac{1}{s} (1-t) e^{-st} + \frac{1}{s^2} e^{-st}\right) \Big|_0^1 = \frac{1}{s^2} e^{-s} + \frac{1}{s} - \frac{1}{s^2}, \quad s > 0$
10. $f(t) = \begin{cases} 0, & 0 < t < a \\ c, & a < t < b \\ 0, & t > b \end{cases} \quad \mathcal{L}\{f(t)\} = \int_a^b c e^{-st} dt = -\frac{c}{s} e^{-st} \Big|_a^b = \frac{c}{s} (e^{-sa} - e^{-sb}), \quad s > 0$

Exercises 4.1

$$11. \quad \mathcal{L}\{f(t)\} = \int_0^\infty e^{t+7} e^{-st} dt = e^7 \int_0^\infty e^{(1-s)t} dt = \frac{e^7}{1-s} e^{(1-s)t} \Big|_0^\infty = 0 - \frac{e^7}{1-s} = \frac{e^7}{s-1}, \quad s > 1$$

$$12. \quad \mathcal{L}\{f(t)\} = \int_0^\infty e^{-2t-5} e^{-st} dt = e^{-5} \int_0^\infty e^{-(s+2)t} dt = -\frac{e^{-5}}{s+2} e^{-(s+2)t} \Big|_0^\infty = \frac{e^{-5}}{s+2}, \quad s > -2$$

$$13. \quad \mathcal{L}\{f(t)\} = \int_0^\infty t e^{4t} e^{-st} dt = \int_0^\infty t e^{(4-s)t} dt = \left(\frac{1}{4-s} t e^{(4-s)t} - \frac{1}{(4-s)^2} e^{(4-s)t} \right) \Big|_0^\infty = \frac{1}{(4-s)^2}, \quad s > 4$$

$$14. \quad \mathcal{L}\{f(t)\} = \int_0^\infty t^2 e^{-2t} e^{-st} dt = \int_0^\infty t^2 e^{-(s+2)t} dt \\ = \left(-\frac{1}{s+2} t^2 e^{-(s+2)t} - \frac{2}{(s+2)^2} t e^{-(s+2)t} - \frac{2}{(s+2)^3} e^{-(s+2)t} \right) \Big|_0^\infty = \frac{2}{(s+2)^3}, \quad s > -2$$

$$15. \quad \mathcal{L}\{f(t)\} = \int_0^\infty e^{-t} (\sin t) e^{-st} dt = \int_0^\infty (\sin t) e^{-(s+1)t} dt \\ = \left(\frac{-(s+1)}{(s+1)^2 + 1} e^{-(s+1)t} \sin t - \frac{1}{(s+1)^2 + 1} e^{-(s+1)t} \cos t \right) \Big|_0^\infty \\ = \frac{1}{(s+1)^2 + 1} = \frac{1}{s^2 + 2s + 2}, \quad s > -1$$

$$16. \quad \mathcal{L}\{f(t)\} = \int_0^\infty e^t (\cos t) e^{-st} dt = \int_0^\infty (\cos t) e^{(1-s)t} dt \\ = \left(\frac{1-s}{(1-s)^2 + 1} e^{(1-s)t} \cos t + \frac{1}{(1-s)^2 + 1} e^{(1-s)t} \sin t \right) \Big|_0^\infty \\ = -\frac{1-s}{(1-s)^2 + 1} = \frac{s-1}{s^2 - 2s + 2}, \quad s > 1$$

$$17. \quad \mathcal{L}\{f(t)\} = \int_0^\infty t (\cos t) e^{-st} dt \\ = \left[\left(-\frac{st}{s^2 + 1} - \frac{s^2 - 1}{(s^2 + 1)^2} \right) (\cos t) e^{-st} + \left(\frac{t}{s^2 + 1} + \frac{2s}{(s^2 + 1)^2} \right) (\sin t) e^{-st} \right]_0^\infty \\ = \frac{s^2 - 1}{(s^2 + 1)^2}, \quad s > 0$$

$$18. \quad \mathcal{L}\{f(t)\} = \int_0^\infty t (\sin t) e^{-st} dt \\ = \left[\left(-\frac{t}{s^2 + 1} - \frac{2s}{(s^2 + 1)^2} \right) (\cos t) e^{-st} - \left(\frac{st}{s^2 + 1} + \frac{s^2 - 1}{(s^2 + 1)^2} \right) (\sin t) e^{-st} \right]_0^\infty \\ = \frac{2s}{(s^2 + 1)^2}, \quad s > 0$$

$$19. \quad \mathcal{L}\{2t^4\} = 2 \frac{4!}{s^5}$$

$$20. \quad \mathcal{L}\{t^5\} = \frac{5!}{s^6}$$

$$21. \quad \mathcal{L}\{4t - 10\} = \frac{4}{s^2} - \frac{10}{s}$$

$$22. \quad \mathcal{L}\{7t + 3\} = \frac{7}{s^2} + \frac{3}{s}$$

$$23. \quad \mathcal{L}\{t^2 + 6t - 3\} = \frac{2}{s^3} + \frac{6}{s^2} - \frac{3}{s}$$

$$24. \quad \mathcal{L}\{-4t^2 + 16t + 9\} = -4 \frac{2}{s^3} + \frac{16}{s^2} + \frac{9}{s}$$

$$25. \quad \mathcal{L}\{t^3 + 3t^2 + 3t + 1\} = \frac{3!}{s^4} + 3 \frac{2}{s^3} + \frac{3}{s^2} + \frac{1}{s}$$

$$26. \quad \mathcal{L}\{8t^3 - 12t^2 + 6t - 1\} = 8 \frac{3!}{s^4} - 12 \frac{2}{s^3} + \frac{6}{s^2} - \frac{1}{s}$$

$$27. \quad \mathcal{L}\{1 + e^{4t}\} = \frac{1}{s} + \frac{1}{s-4}$$

$$28. \quad \mathcal{L}\{t^2 - e^{-9t} + 5\} = \frac{2}{s^3} - \frac{1}{s+9} + \frac{5}{s}$$

$$29. \quad \mathcal{L}\{1 + 2e^{2t} + e^{4t}\} = \frac{1}{s} + \frac{2}{s-2} + \frac{1}{s-4}$$

$$30. \quad \mathcal{L}\{e^{2t} - 2 + e^{-2t}\} = \frac{1}{s-2} - \frac{2}{s} + \frac{1}{s+2}$$

$$31. \quad \mathcal{L}\{4t^2 - 5\sin 3t\} = 4\frac{2}{s^3} - 5\frac{3}{s^2+9}$$

$$32. \quad \mathcal{L}\{\cos 5t + \sin 2t\} = \frac{s}{s^2+25} + \frac{2}{s^2+4}$$

$$33. \quad \mathcal{L}\{\sinh kt\} = \frac{k}{s^2 - k^2}$$

$$34. \quad \mathcal{L}\{\cosh kt\} = \frac{s}{s^2 - k^2}$$

$$35. \quad \mathcal{L}\{e^t \sinh t\} = \mathcal{L}\left\{e^t \frac{e^t - e^{-t}}{2}\right\} = \mathcal{L}\left\{\frac{1}{2}e^{2t} - \frac{1}{2}\right\} = \frac{1}{2(s-2)} - \frac{1}{2s}$$

$$36. \quad \mathcal{L}\{e^{-t} \cosh t\} = \mathcal{L}\left\{e^{-t} \frac{e^t + e^{-t}}{2}\right\} = \mathcal{L}\left\{\frac{1}{2} + \frac{1}{2}e^{-2t}\right\} = \frac{1}{2s} + \frac{1}{2(s+2)}$$

$$37. \quad \mathcal{L}\{\sin 2t \cos 2t\} = \mathcal{L}\left\{\frac{1}{2} \sin 4t\right\} = \frac{2}{s^2 + 16}$$

$$38. \quad \mathcal{L}\{\cos^2 t\} = \mathcal{L}\left\{\frac{1}{2} + \frac{1}{2} \cos 2t\right\} = \frac{1}{2s} + \frac{1}{2} \frac{s}{s^2 + 4}$$

39. (a) Using integration by parts for $\alpha > 0$,

$$\Gamma(\alpha + 1) = \int_0^\infty t^\alpha e^{-t} dt = -t^\alpha e^{-t} \Big|_0^\infty + \alpha \int_0^\infty t^{\alpha-1} e^{-t} dt = \alpha \Gamma(\alpha).$$

(b) Let $u = st$ so that $du = s dt$. Then

$$\mathcal{L}\{t^\alpha\} = \int_0^\infty e^{-st} t^\alpha dt = \int_0^\infty e^{-u} \left(\frac{u}{s}\right)^\alpha \frac{1}{s} du = \frac{1}{s^{\alpha+1}} \Gamma(\alpha + 1), \quad \alpha > -1.$$

$$40. \quad (a) \quad \mathcal{L}\{t^{-1/2}\} = \frac{\Gamma(1/2)}{s^{1/2}} = \sqrt{\frac{\pi}{s}}$$

$$(b) \quad \mathcal{L}\{t^{1/2}\} = \frac{\Gamma(3/2)}{s^{3/2}} = \frac{\sqrt{\pi}}{2s^{3/2}}$$

$$(c) \quad \mathcal{L}\{t^{3/2}\} = \frac{\Gamma(5/2)}{s^{5/2}} = \frac{3\sqrt{\pi}}{4s^{5/2}}$$

41. Identifying $f(t) = t^n$ we have $f'(t) = nt^{n-1}$, $n = 1, 2, 3, \dots$. Then, since $f(0) = 0$,

$$n \mathcal{L}\{t^{n-1}\} = \mathcal{L}\{nt^{n-1}\} = s \mathcal{L}\{t^n\} \quad \text{and} \quad \mathcal{L}\{t^n\} = \frac{n}{s} \mathcal{L}\{t^{n-1}\}.$$

$$\text{For } n = 1, \quad \mathcal{L}\{t\} = \frac{1}{s} \mathcal{L}\{1\} = \frac{1}{s^2}.$$

$$\text{For } n = 2, \quad \mathcal{L}\{t^2\} = \frac{2}{s} \mathcal{L}\{t\} = \frac{2}{s^3}.$$

$$\text{For } n = 3, \quad \mathcal{L}\{t^3\} = \frac{3}{s} \mathcal{L}\{t^2\} = \frac{6}{s^4}.$$

42. Let $F(t) = t^{1/3}$. Then $F(t)$ is of exponential order, but $f(t) = F'(t) = \frac{1}{3}t^{-2/3}$ is unbounded near $t = 0$ and hence is not of exponential order.

Exercises 4.1

Let $f(t) = 2te^{t^2} \cos e^{t^2} = \frac{d}{dt} \sin e^{t^2}$. This function is not of exponential order, but we can show that its Laplace transform exists. Using integration by parts we have

$$\begin{aligned}\mathcal{L}\{2te^{t^2} \cos e^{t^2}\} &= \int_0^\infty e^{-st} \left(\frac{d}{dt} \sin e^{t^2} \right) dt = \lim_{a \rightarrow \infty} \left[e^{-st} \sin e^{t^2} \Big|_0^a + s \int_0^a e^{-st} \sin e^{t^2} dt \right] \\ &= s \int_0^\infty e^{-st} \sin e^{t^2} dt = s \mathcal{L}\{\sin e^{t^2}\}.\end{aligned}$$

Since $\sin e^{t^2}$ is continuous and of exponential order, $\mathcal{L}\{\sin e^{t^2}\}$ exists, and therefore $\mathcal{L}\{2te^{t^2} \cos e^{t^2}\}$ exists.

43. The relation will be valid when s is greater than the maximum of c_1 and c_2 .

44. Since e^t is an increasing function and $t^2 > \ln M + ct$ for $M > 0$ we have $e^{t^2} > e^{\ln M + ct} = Me^{ct}$ for t sufficiently large and for any c . Thus, e^{t^2} is not of exponential order.

45. By part (c) of Theorem 4.1

$$\mathcal{L}\{e^{(a+ib)t}\} = \frac{1}{s - (a+ib)} = \frac{1}{(s-a) - ib} \frac{(s-a) + ib}{(s-a) + ib} = \frac{s-a+ib}{(s-a)^2 + b^2}.$$

By Euler's formula, $e^{i\theta} = \cos \theta + i \sin \theta$, so

$$\begin{aligned}\mathcal{L}\{e^{(a+ib)t}\} &= \mathcal{L}\{e^{at} e^{ibt}\} \mathcal{L}\{e^{at}(\cos bt + i \sin bt)\} \\ &= \mathcal{L}\{e^{at} \cos bt\} + i \mathcal{L}\{e^{at} \sin bt\} \\ &= \frac{s-a}{(s-a)^2 + b^2} + i \frac{b}{(s-a)^2 + b^2}.\end{aligned}$$

Equating real and imaginary parts we get

$$\mathcal{L}\{e^{at} \cos bt\} = \frac{s-a}{(s-a)^2 + b^2} \quad \text{and} \quad \mathcal{L}\{e^{at} \sin bt\} = \frac{b}{(s-a)^2 + b^2}.$$

46. We want $f(\alpha x + \beta y) = \alpha f(x) + \beta f(y)$ or

$$m(\alpha x + \beta y) + b = \alpha(mx + b) + \beta(my + b) = m(\alpha x + \beta y) + (\alpha + \beta)b$$

for all real numbers α and β . Taking $\alpha = \beta = 1$ we see that $b = 2b$, so $b = 0$. Thus, $f(x) = mx + b$ will be a linear transformation when $b = 0$.

Exercises 4.2

$$1. \mathcal{L}^{-1}\left\{\frac{1}{s^3}\right\} = \frac{1}{2} \mathcal{L}^{-1}\left\{\frac{2}{s^3}\right\} = \frac{1}{2}t^2$$

$$2. \mathcal{L}^{-1}\left\{\frac{1}{s^4}\right\} = \frac{1}{6} \mathcal{L}^{-1}\left\{\frac{3!}{s^4}\right\} = \frac{1}{6}t^3$$

$$3. \mathcal{L}^{-1}\left\{\frac{1}{s^2} - \frac{48}{s^5}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s^2} - \frac{48}{24} \cdot \frac{4!}{s^5}\right\} = t - 2t^4$$

$$4. \mathcal{L}^{-1}\left\{\left(\frac{2}{s} - \frac{1}{s^3}\right)^2\right\} = \mathcal{L}^{-1}\left\{4 \cdot \frac{1}{s^2} - \frac{4}{6} \cdot \frac{3!}{s^4} + \frac{1}{120} \cdot \frac{5!}{s^6}\right\} = 4t - \frac{2}{3}t^3 + \frac{1}{120}t^5$$

$$5. \mathcal{L}^{-1}\left\{\frac{(s+1)^3}{s^4}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s} + 3 \cdot \frac{1}{s^2} + \frac{3}{2} \cdot \frac{2}{s^3} + \frac{1}{6} \cdot \frac{3!}{s^4}\right\} = 1 + 3t + \frac{3}{2}t^2 + \frac{1}{6}t^3$$

$$6. \mathcal{L}^{-1}\left\{\frac{(s+2)^2}{s^3}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s} + 4 \cdot \frac{1}{s^2} + 2 \cdot \frac{2}{s^3}\right\} = 1 + 4t + 2t^2$$

7. $\mathcal{L}^{-1}\left\{\frac{1}{s^2} - \frac{1}{s} + \frac{1}{s-2}\right\} = t - 1 + e^{2t}$
8. $\mathcal{L}^{-1}\left\{\frac{4}{s} + \frac{6}{s^5} - \frac{1}{s+8}\right\} = \mathcal{L}^{-1}\left\{4 \cdot \frac{1}{s} + \frac{1}{4} \cdot \frac{4!}{s^5} - \frac{1}{s+8}\right\} = 4 + \frac{1}{4}t^4 - e^{-8t}$
9. $\mathcal{L}^{-1}\left\{\frac{1}{4s+1}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{4} \cdot \frac{1}{s+1/4}\right\} = \frac{1}{4}e^{-t/4}$
10. $\mathcal{L}^{-1}\left\{\frac{1}{5s-2}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{5} \cdot \frac{1}{s-2/5}\right\} = \frac{1}{5}e^{2t/5}$
11. $\mathcal{L}^{-1}\left\{\frac{5}{s^2+49}\right\} = \mathcal{L}^{-1}\left\{\frac{5}{7} \cdot \frac{7}{s^2+49}\right\} = \frac{5}{7}\sin 7t$
12. $\mathcal{L}^{-1}\left\{\frac{10s}{s^2+16}\right\} = 10\cos 4t$
13. $\mathcal{L}^{-1}\left\{\frac{4s}{4s^2+1}\right\} = \mathcal{L}^{-1}\left\{\frac{s}{s^2+1/4}\right\} = \cos \frac{1}{2}t$
14. $\mathcal{L}^{-1}\left\{\frac{1}{4s^2+1}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{2} \cdot \frac{1/2}{s^2+1/4}\right\} = \frac{1}{2}\sin \frac{1}{2}t$
15. $\mathcal{L}^{-1}\left\{\frac{2s-6}{s^2+9}\right\} = \mathcal{L}^{-1}\left\{2 \cdot \frac{s}{s^2+9} - 2 \cdot \frac{3}{s^2+9}\right\} = 2\cos 3t - 2\sin 3t$
16. $\mathcal{L}^{-1}\left\{\frac{s+1}{s^2+2}\right\} = \mathcal{L}^{-1}\left\{\frac{s}{s^2+2} + \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{2}}{s^2+2}\right\} = \cos \sqrt{2}t + \frac{\sqrt{2}}{2}\sin \sqrt{2}t$
17. $\mathcal{L}^{-1}\left\{\frac{1}{s^2+3s}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{3} \cdot \frac{1}{s} - \frac{1}{3} \cdot \frac{1}{s+3}\right\} = \frac{1}{3} - \frac{1}{3}e^{-3t}$
18. $\mathcal{L}^{-1}\left\{\frac{s+1}{s^2-4s}\right\} = \mathcal{L}^{-1}\left\{-\frac{1}{4} \cdot \frac{1}{s} + \frac{5}{4} \cdot \frac{1}{s-4}\right\} = -\frac{1}{4} + \frac{5}{4}e^{4t}$
19. $\mathcal{L}^{-1}\left\{\frac{s}{s^2+2s-3}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{4} \cdot \frac{1}{s-1} + \frac{3}{4} \cdot \frac{1}{s+3}\right\} = \frac{1}{4}e^t + \frac{3}{4}e^{-3t}$
20. $\mathcal{L}^{-1}\left\{\frac{1}{s^2+s-20}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{9} \cdot \frac{1}{s-4} - \frac{1}{9} \cdot \frac{1}{s+5}\right\} = \frac{1}{9}e^{4t} - \frac{1}{9}e^{-5t}$
21. $\mathcal{L}^{-1}\left\{\frac{0.9s}{(s-0.1)(s+0.2)}\right\} = \mathcal{L}^{-1}\left\{(0.3) \cdot \frac{1}{s-0.1} + (0.6) \cdot \frac{1}{s+0.2}\right\} = 0.3e^{0.1t} + 0.6e^{-0.2t}$
22. $\mathcal{L}^{-1}\left\{\frac{s-3}{(s-\sqrt{3})(s+\sqrt{3})}\right\} = \mathcal{L}^{-1}\left\{\frac{s}{s^2-3} - \sqrt{3} \cdot \frac{\sqrt{3}}{s^2-3}\right\} = \cosh \sqrt{3}t - \sqrt{3}\sinh \sqrt{3}t$
23. $\mathcal{L}^{-1}\left\{\frac{s}{(s-2)(s-3)(s-6)}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{2} \cdot \frac{1}{s-2} - \frac{1}{s-3} + \frac{1}{2} \cdot \frac{1}{s-6}\right\} = \frac{1}{2}e^{2t} - e^{3t} + \frac{1}{2}e^{6t}$
24. $\mathcal{L}^{-1}\left\{\frac{s^2+1}{s(s-1)(s+1)(s-2)}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{2} \cdot \frac{1}{s} - \frac{1}{s-1} - \frac{1}{3} \cdot \frac{1}{s+1} + \frac{5}{6} \cdot \frac{1}{s-2}\right\}$

$$= \frac{1}{2} - e^t - \frac{1}{3}e^{-t} + \frac{5}{6}e^{2t}$$
25. $\mathcal{L}^{-1}\left\{\frac{1}{s^3+5s}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s(s^2+5)}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{5} \cdot \frac{1}{s} - \frac{1}{5} \cdot \frac{s}{s^2+5}\right\} = \frac{1}{5} - \frac{1}{5}\cos \sqrt{5}t$

Exercises 4.2

$$26. \quad \mathcal{L}^{-1}\left\{\frac{s}{(s^2+4)(s+2)}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{4} \cdot \frac{s}{s^2+4} + \frac{1}{4} \cdot \frac{2}{s^2+4} - \frac{1}{4} \cdot \frac{1}{s+2}\right\} = \frac{1}{4} \cos 2t + \frac{1}{4} \sin 2t - \frac{1}{4} e^{-2t}$$

$$27. \quad \mathcal{L}^{-1}\left\{\frac{2s-4}{(s^2+s)(s^2+1)}\right\} = \mathcal{L}^{-1}\left\{\frac{2s-4}{s(s^2+1)^2}\right\} = \mathcal{L}^{-1}\left\{-\frac{4}{s} + \frac{3}{s+1} + \frac{s}{s^2+1} + \frac{3}{s^2+1}\right\}$$

$$= -4 + 3e^{-t} + \cos t + 3 \sin t$$

$$28. \quad \mathcal{L}^{-1}\left\{\frac{1}{s^4-9}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{6\sqrt{3}} \cdot \frac{\sqrt{3}}{s^2-3} - \frac{1}{6\sqrt{3}} \cdot \frac{\sqrt{3}}{s^2-3}\right\} = \frac{1}{6\sqrt{3}} \sinh \sqrt{3}t - \frac{1}{6\sqrt{3}} \sin \sqrt{3}t$$

$$29. \quad \mathcal{L}^{-1}\left\{\frac{s}{(s^2+4)(s+2)}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{4} \cdot \frac{s}{s^2+4} + \frac{1}{4} \cdot \frac{2}{s^2+4} - \frac{1}{4} \cdot \frac{1}{s+2}\right\} = \frac{1}{4} \cos 2t + \frac{1}{4} \sin 2t - \frac{1}{4} e^{-2t}$$

$$30. \quad \mathcal{L}^{-1}\left\{\frac{6s+3}{(s^2+1)(s^2+4)}\right\} = \mathcal{L}^{-1}\left\{2 \cdot \frac{s}{s^2+1} + \frac{1}{s^2+1} - 2 \cdot \frac{s}{s^2+4} - \frac{1}{2} \cdot \frac{2}{s^2+4}\right\}$$

$$= 2 \cos t + \sin t - 2 \cos 2t - \frac{1}{2} \sin 2t$$

31. The Laplace transform of the differential equation is

$$s\mathcal{L}\{y\} - y(0) - \mathcal{L}\{y\} = \frac{1}{s}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = -\frac{1}{s} + \frac{1}{s-1}.$$

Thus

$$y = -1 + e^t.$$

32. The Laplace transform of the differential equation is

$$2s\mathcal{L}\{y\} - 2y(0) = \mathcal{L}\{y\} = 0.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{6}{2s+1} = \frac{3}{s+1/2}.$$

Thus

$$y = 3e^{-t/2}.$$

33. The Laplace transform of the differential equation is

$$s\mathcal{L}\{y\} - y(0) + 6\mathcal{L}\{y\} = \frac{1}{s-4}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{1}{(s-4)(s+6)} + \frac{2}{s+6} = \frac{1}{10} \cdot \frac{1}{s-4} + \frac{19}{10} \cdot \frac{1}{s+6}.$$

Thus

$$y = \frac{1}{10}e^{4t} + \frac{19}{10}e^{-6t}.$$

34. The Laplace transform of the differential equation is

$$s\mathcal{L}\{y\} - \mathcal{L}\{y\} = \frac{2s}{s^2+25}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{2}{(s-1)(s^2+25)} = \frac{1}{13} \cdot \frac{1}{s-1} - \frac{1}{13} \frac{s}{s^2+25} + \frac{5}{13} \cdot \frac{5}{s^2+25}.$$

Thus

$$y = \frac{1}{13}e^t - \frac{1}{13}\cos 5t + \frac{5}{13}\sin 5t.$$

35. The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{y\} - sy(0) - y'(0) + 5[s \mathcal{L}\{y\} - y(0)] + 4 \mathcal{L}\{y\} = 0.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{s+5}{s^2+5s+4} = \frac{4}{3} \frac{1}{s+1} - \frac{1}{3} \frac{1}{s+4}.$$

Thus

$$y = \frac{4}{3}e^{-t} - \frac{1}{3}e^{-4t}.$$

36. The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{y\} - sy(0) - y'(0) - 4[s \mathcal{L}\{y\} - y(0)] = \frac{6}{s-3} - \frac{3}{s+1}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\begin{aligned} \mathcal{L}\{y\} &= \frac{6}{(s-3)(s^2-4s)} - \frac{3}{(s+1)(s^2-4s)} + \frac{s-5}{s^2-4s} \\ &= \frac{5}{2} \cdot \frac{1}{s} - \frac{2}{s-3} - \frac{3}{5} \cdot \frac{1}{s+1} + \frac{11}{10} \cdot \frac{1}{s-4}. \end{aligned}$$

Thus

$$y = \frac{5}{2} - 2e^{3t} - \frac{3}{5}e^{-t} + \frac{11}{10}e^{4t}.$$

37. The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{y\} - sy(0) + \mathcal{L}\{y\} = \frac{2}{s^2+2}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{2}{(s^2+1)(s^2+2)} + \frac{10s}{s^2+1} = \frac{10s}{s^2+1} + \frac{2}{s^2+1} - \frac{2}{s^2+2}.$$

Thus

$$y = 10 \cos t + 2 \sin t - \sqrt{2} \sin \sqrt{2}t.$$

38. The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{y\} + 9 \mathcal{L}\{y\} = \frac{1}{s-1}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{1}{(s-1)(s^2+9)} = \frac{1}{10} \cdot \frac{1}{s-1} - \frac{1}{10} \cdot \frac{1}{s^2+9} - \frac{1}{10} \cdot \frac{s}{s^2+9}.$$

Thus

$$y = \frac{1}{10}e^t - \frac{1}{30}\sin 3t - \frac{1}{10}\cos 3t.$$

39. The Laplace transform of the differential equation is

$$2[s^3 \mathcal{L}\{y\} - s^2(0) - sy'(0) - y''(0)] + 3[s^2 \mathcal{L}\{y\} - sy(0) - y'(0)] - 3[s \mathcal{L}\{y\} - y(0)] - 2 \mathcal{L}\{y\} = \frac{1}{s+1}.$$

Exercises 4.2

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{2s+3}{(s+1)(s-1)(2s+1)(s+2)} = \frac{1}{2} \frac{1}{s+1} + \frac{5}{18} \frac{1}{s-1} - \frac{8}{9} \frac{1}{s+1/2} + \frac{1}{9} \frac{1}{s+2}.$$

Thus

$$y = \frac{1}{2}e^{-t} + \frac{5}{18}e^t - \frac{8}{9}e^{-t/2} + \frac{1}{9}e^{-2t}.$$

40. The Laplace transform of the differential equation is

$$s^3 \mathcal{L}\{y\} - s^2(0) - sy'(0) - y''(0) + 2[s^2 \mathcal{L}\{y\} - sy(0) - y'(0)] - [s \mathcal{L}\{y\} - y(0)] - 2 \mathcal{L}\{y\} = \frac{3}{s^2 + 9}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\begin{aligned} \mathcal{L}\{y\} &= \frac{s^2 + 12}{(s-1)(s+1)(s+2)(s^2+9)} \\ &= \frac{13}{60} \frac{1}{s-1} - \frac{13}{20} \frac{1}{s+1} + \frac{16}{39} \frac{1}{s+2} + \frac{3}{130} \frac{s}{s^2+9} - \frac{1}{65} \frac{3}{s^2+9}. \end{aligned}$$

Thus

$$y = \frac{13}{60}e^t - \frac{13}{20}e^{-t} + \frac{16}{39}e^{-2t} + \frac{3}{130}\cos 3t - \frac{1}{65}\sin 3t.$$

41. For $y' + 6y = e^{4t}$ the transfer function is $W(s) = 1/(s+6)$. The zero-input response is

$$y_0(t) = \mathcal{L}^{-1}\left\{\frac{2}{s+6}\right\} = 2e^{-6t},$$

and the zero-state response is

$$y_1(t) = \mathcal{L}^{-1}\left\{\frac{1}{(s-4)(s+6)}\right\} = \mathcal{L}^{-1}\left\{-\frac{1}{10} \cdot \frac{1}{s+6} + \frac{1}{10} \cdot \frac{1}{s-4}\right\} = -\frac{1}{10}e^{-6t} + \frac{1}{10}e^{4t}.$$

42. For $y'' - 4y' = 6e^{3t} - 3e^{-t}$ the transfer function is $W(s) = 1/(s^2 - 4s)$. The zero-input response is

$$y_0(t) = \mathcal{L}^{-1}\left\{\frac{s-5}{s^2-4s}\right\} = \mathcal{L}^{-1}\left\{\frac{5}{4} \cdot \frac{1}{s} - \frac{1}{4} \cdot \frac{1}{s-4}\right\} = \frac{5}{4} - \frac{1}{4}e^{4t},$$

and the zero-state response is

$$\begin{aligned} y_1(t) &= \mathcal{L}^{-1}\left\{\frac{6}{(s-3)(s^2-4s)} - \frac{3}{(s+1)(s^2-4s)}\right\} \\ &= \mathcal{L}^{-1}\left\{\frac{27}{20} \cdot \frac{1}{s-4} - \frac{2}{s-3} + \frac{5}{4} \cdot \frac{1}{s} - \frac{3}{5} \cdot \frac{1}{s+1}\right\} \\ &= \frac{27}{20}e^{4t} - 2e^{3t} + \frac{5}{4} - \frac{3}{5}e^{-t}. \end{aligned}$$

43. The Laplace transform of the differential equation is

$$s \mathcal{L}\{y\} + \mathcal{L}\{y\} = \frac{s+3}{(s+3)^2 + 4} = \frac{s+3}{s^2 + 6s + 13}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\begin{aligned} \mathcal{L}\{y\} &= \frac{s+3}{(s+1)(s^2+6s+13)} = \frac{1}{4} \cdot \frac{1}{s+1} - \frac{1}{4} \cdot \frac{s+1}{s^2+6s+13} \\ &= \frac{1}{4} \cdot \frac{1}{s+1} - \frac{1}{4} \left(\frac{s+3}{(s+3)^2+4} - \frac{2}{(s+3)^2+4} \right). \end{aligned}$$

Thus

$$y = \frac{1}{4}e^{-t} - e^{-3t} \cos 2t + \frac{1}{4}e^{-3t} \sin 2t.$$

44. Let $f(t) = 1$ and $g(t) = \begin{cases} 1, & t \geq 0, \\ 0, & t = 1 \end{cases}$. Then $\mathcal{L}\{f(t)\} = \mathcal{L}\{g(t)\} = 1$, but $f(t) \neq g(t)$.

Exercises 4.3

1. $\mathcal{L}\{te^{10t}\} = \frac{1}{(s-10)^2}$
2. $\mathcal{L}\{te^{-6t}\} = \frac{1}{(s+6)^2}$
3. $\mathcal{L}\{t^3e^{-2t}\} = \frac{3!}{(s+2)^4}$
4. $\mathcal{L}\{t^{10}e^{-7t}\} = \frac{10!}{(s+7)^{11}}$
5. $\mathcal{L}\{t(e^t + e^{2t})^2\} = \mathcal{L}\{te^{2t} + 2te^{3t} + te^{4t}\} = \frac{1}{(s-2)^2} + \frac{2}{(s-3)^2} + \frac{1}{(s-4)^2}$
6. $\mathcal{L}\{e^{2t}(t-1)^2\} = \mathcal{L}\{t^2e^{2t} - 2te^{2t} + e^{2t}\} = \frac{2}{(s-2)^3} - \frac{2}{(s-2)^2} + \frac{1}{s-2}$
7. $\mathcal{L}\{e^t \sin 3t\} = \frac{3}{(s-1)^2 + 9}$
8. $\mathcal{L}\{e^{-2t} \cos 4t\} = \frac{s+2}{(s+2)^2 + 16}$
9. $\mathcal{L}\{(1 - e^t + 3e^{-4t}) \cos 5t\} = \mathcal{L}\{\cos 5t - e^t \cos 5t + 3e^{-4t} \cos 5t\} = \frac{s}{s^2 + 25} - \frac{s-1}{(s-1)^2 + 25} + \frac{3(s+4)}{(s+4)^2 + 25}$
10. $\mathcal{L}\left\{e^{3t}\left(9 - 4t + 10 \sin \frac{t}{2}\right)\right\} = \mathcal{L}\left\{9e^{3t} - 4te^{3t} + 10e^{3t} \sin \frac{t}{2}\right\} = \frac{9}{s-3} - \frac{4}{(s-3)^2} + \frac{5}{(s-3)^2 + 1/4}$
11. $\mathcal{L}^{-1}\left\{\frac{1}{(s+2)^3}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{2} \frac{2}{(s+2)^3}\right\} = \frac{1}{2}t^2e^{-2t}$
12. $\mathcal{L}^{-1}\left\{\frac{1}{(s-1)^4}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{6} \frac{3!}{(s-1)^4}\right\} = \frac{1}{6}t^3e^t$
13. $\mathcal{L}^{-1}\left\{\frac{1}{s^2 - 6s + 10}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{(s-3)^2 + 1^2}\right\} = e^{3t} \sin t$
14. $\mathcal{L}^{-1}\left\{\frac{1}{s^2 + 2s + 5}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{2} \frac{2}{(s+1)^2 + 2^2}\right\} = \frac{1}{2}e^{-t} \sin 2t$
15. $\mathcal{L}^{-1}\left\{\frac{s}{s^2 + 4s + 5}\right\} = \mathcal{L}^{-1}\left\{\frac{(s+2)}{(s+2)^2 + 1^2} - 2 \frac{1}{(s+2)^2 + 1^2}\right\} = e^{-2t} \cos t - 2e^{-2t} \sin t$
16. $\mathcal{L}^{-1}\left\{\frac{2s+5}{s^2 + 6s + 34}\right\} = \mathcal{L}^{-1}\left\{2 \frac{(s+3)}{(s+3)^2 + 5^2} - \frac{1}{5} \frac{5}{(s+3)^2 + 5^2}\right\} = 2e^{-3t} \cos 5t - \frac{1}{5}e^{-3t} \sin 5t$
17. $\mathcal{L}^{-1}\left\{\frac{s}{(s+1)^2}\right\} = \mathcal{L}^{-1}\left\{\frac{s+1-1}{(s+1)^2}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s+1} - \frac{1}{(s+1)^2}\right\} = e^{-t} - te^{-t}$

Exercises 4.3

18. $\mathcal{L}^{-1}\left\{\frac{5s}{(s-2)^2}\right\} = \mathcal{L}^{-1}\left\{\frac{5(s-2)+10}{(s-2)^2}\right\} = \mathcal{L}^{-1}\left\{\frac{5}{s-2} + \frac{10}{(s-2)^2}\right\} = 5e^{2t} + 10te^{2t}$
19. $\mathcal{L}^{-1}\left\{\frac{2s-1}{s^2(s+1)^3}\right\} = \mathcal{L}^{-1}\left\{\frac{5}{s} - \frac{1}{s^2} - \frac{5}{s+1} - \frac{4}{(s+1)^2} - \frac{3}{2} \frac{2}{(s+1)^3}\right\} = 5 - t - 5e^{-t} - 4te^{-t} - \frac{3}{2}t^2e^{-t}$
20. $\mathcal{L}^{-1}\left\{\frac{(s+1)^2}{(s+2)^4}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{(s+2)^2} - \frac{2}{(s+2)^3} + \frac{1}{6} \frac{3!}{(s+2)^4}\right\} = te^{-2t} - t^2e^{-2t} + \frac{1}{6}t^3e^{-2t}$
21. The Laplace transform of the differential equation is

$$s\mathcal{L}\{y\} - y(0) + 4\mathcal{L}\{y\} = \frac{1}{s+4}.$$

Solving for $\mathcal{L}\{y\}$ we obtain $\mathcal{L}\{y\} = \frac{1}{(s+4)^2} + \frac{2}{s+4}$. Thus

$$y = te^{-4t} + 2e^{-4t}.$$

22. The Laplace transform of the differential equation is

$$s\mathcal{L}\{y\} - \mathcal{L}\{y\} = \frac{1}{s} + \frac{1}{(s-1)^2}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{1}{s(s-1)} + \frac{1}{(s-1)^3} = -\frac{1}{s} + \frac{1}{s-1} + \frac{1}{(s-1)^3}.$$

Thus

$$y = -1 + e^t + \frac{1}{2}t^2e^t.$$

23. The Laplace transform of the differential equation is

$$s^2\mathcal{L}\{y\} - sy(0) - y'(0) + 2[s\mathcal{L}\{y\} - y(0)] + \mathcal{L}\{y\} = 0.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{s+3}{(s+1)^2} = \frac{1}{s+1} + \frac{2}{(s+1)^2}.$$

Thus

$$y = e^{-t} + 2te^{-t}.$$

24. The Laplace transform of the differential equation is

$$s^2\mathcal{L}\{y\} - sy(0) - y'(0) - 4[s\mathcal{L}\{y\} - y(0)] + 4\mathcal{L}\{y\} = \frac{6}{(s-2)^4}.$$

Solving for $\mathcal{L}\{y\}$ we obtain $\mathcal{L}\{y\} = \frac{1}{20} \frac{5!}{(s-2)^6}$. Thus, $y = \frac{1}{20}t^5e^{2t}$.

25. The Laplace transform of the differential equation is

$$s^2\mathcal{L}\{y\} - sy(0) - y'(0) - 6[s\mathcal{L}\{y\} - y(0)] + 9\mathcal{L}\{y\} = \frac{1}{s^2}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{1+s^2}{s^2(s-3)^2} = \frac{2}{27} \frac{1}{s} + \frac{1}{9} \frac{1}{s^2} - \frac{2}{27} \frac{1}{s-3} + \frac{10}{9} \frac{1}{(s-3)^2}.$$

Thus

$$y = \frac{2}{27} + \frac{1}{9}t - \frac{2}{27}e^{3t} + \frac{10}{9}te^{3t}.$$

26. The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{y\} - sy(0) - y'(0) - 4[s \mathcal{L}\{y\} - y(0)] + 4 \mathcal{L}\{y\} = \frac{6}{s^4}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{s^5 - 4s^4 + 6}{s^4(s-2)^2} = \frac{3}{4} \frac{1}{s} + \frac{9}{8} \frac{1}{s^2} + \frac{3}{4} \frac{2}{s^3} + \frac{1}{4} \frac{3!}{s^4} + \frac{1}{4} \frac{1}{s-2} - \frac{13}{8} \frac{1}{(s-2)^2}.$$

Thus

$$y = \frac{3}{4} + \frac{9}{8}t + \frac{3}{4}t^2 + \frac{1}{4}t^3 + \frac{1}{4}e^{2t} - \frac{13}{8}te^{2t}.$$

27. The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{y\} - sy(0) - y'(0) - 6[s \mathcal{L}\{y\} - y(0)] + 13 \mathcal{L}\{y\} = 0.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = -\frac{3}{s^2 - 6s + 13} = -\frac{3}{2} \frac{2}{(s-3)^2 + 2^2}.$$

Thus

$$y = -\frac{3}{2}e^{3t} \sin 2t.$$

28. The Laplace transform of the differential equation is

$$2[s^2 \mathcal{L}\{y\} - sy(0)] + 20[s \mathcal{L}\{y\} - y(0)] + 51 \mathcal{L}\{y\} = 0.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{4s + 40}{2s^2 + 20s + 51} = \frac{2s + 20}{(s+5)^2 + 1/2} = \frac{2(s+5)}{(s+5)^2 + 1/2} + \frac{10}{(s+5)^2 + 1/2}.$$

Thus

$$y = 2e^{-5t} \cos(t/\sqrt{2}) + 10\sqrt{2}e^{-5t} \sin(t/\sqrt{2}).$$

29. The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{y\} - sy(0) - y'(0) - [s \mathcal{L}\{y\} - y(0)] = \frac{s-1}{(s-1)^2 + 1}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{1}{s(s^2 - 2s + 2)} = \frac{1}{2} \frac{1}{s} - \frac{1}{2} \frac{s-1}{(s-1)^2 + 1} + \frac{1}{2} \frac{1}{(s-1)^2 + 1}.$$

Thus

$$y = \frac{1}{2} - \frac{1}{2}e^t \cos t + \frac{1}{2}e^t \sin t.$$

30. The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{y\} - sy(0) - y'(0) - 2[s \mathcal{L}\{y\} - y(0)] + 5 \mathcal{L}\{y\} = \frac{1}{s} + \frac{1}{s^2}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\begin{aligned} \mathcal{L}\{y\} &= \frac{4s^2 + s + 1}{s^2(s^2 - 2s + 5)} = \frac{7}{25} \frac{1}{s} + \frac{1}{5} \frac{1}{s^2} + \frac{-7s/25 + 109/25}{s^2 - 2s + 5} \\ &= \frac{7}{25} \frac{1}{s} + \frac{1}{5} \frac{1}{s^2} - \frac{7}{25} \frac{s-1}{(s-1)^2 + 2^2} + \frac{51}{25} \frac{2}{(s-1)^2 + 2^2}. \end{aligned}$$

Exercises 4.3

Thus

$$y = \frac{7}{25} + \frac{1}{5}t - \frac{7}{25}e^t \cos 2t + \frac{51}{25}e^t \sin 2t.$$

31. Taking the Laplace transform of both sides of the differential equation and letting $c = y(0)$ we obtain

$$\begin{aligned}\mathcal{L}\{y''\} + \mathcal{L}\{2y'\} + \mathcal{L}\{y\} &= 0 \\ s^2 \mathcal{L}\{y\} - sy(0) - y'(0) + 2s \mathcal{L}\{y\} - 2y(0) + \mathcal{L}\{y\} &= 0 \\ s^2 \mathcal{L}\{y\} - cs - 2 + 2s \mathcal{L}\{y\} - 2c + \mathcal{L}\{y\} &= 0 \\ (s^2 + 2s + 1) \mathcal{L}\{y\} &= cs + 2c + 2 \\ \mathcal{L}\{y\} &= \frac{cs}{(s+1)^2} + \frac{2c+2}{(s+1)^2} \\ &= c \frac{s+1-1}{(s+1)^2} + \frac{2c+2}{(s+1)^2} \\ &= \frac{c}{s+1} + \frac{c+2}{(s+1)^2}.\end{aligned}$$

Therefore,

$$y(t) = c \mathcal{L}^{-1} \left\{ \frac{1}{s+1} \right\} + (c+2) \mathcal{L}^{-1} \left\{ \frac{1}{(s+1)^2} \right\} = ce^{-t} + (c+2)te^{-t}.$$

To find c we let $y(1) = 2$. Then $2 = ce^{-1} + (c+2)e^{-1} = 2(c+1)e^{-1}$ and $c = e - 1$. Thus

$$y(t) = (e-1)e^{-t} + (e+1)te^{-t}.$$

32. Taking the Laplace transform of both sides of the differential equation and letting $c = y'(0)$ we obtain

$$\begin{aligned}\mathcal{L}\{y''\} + \mathcal{L}\{8y'\} + \mathcal{L}\{20y\} &= 0 \\ s^2 \mathcal{L}\{y\} - y'(0) + 8s \mathcal{L}\{y\} + 20 \mathcal{L}\{y\} &= 0 \\ s^2 \mathcal{L}\{y\} - c + 8s \mathcal{L}\{y\} + 20 \mathcal{L}\{y\} &= 0 \\ (s^2 + 8s + 20) \mathcal{L}\{y\} &= c \\ \mathcal{L}\{y\} &= \frac{c}{s^2 + 8s + 20} = \frac{c}{(s+4)^2 + 4}.\end{aligned}$$

Therefore,

$$y(t) = \mathcal{L}^{-1} \left\{ \frac{c}{(s+4)^2 + 4} \right\} = ce^{-4t} \sin 2t.$$

To find c we let $y'(\pi) = 0$. Then $0 = y'(\pi) = ce^{-4\pi}$ and $c = 0$. Thus, $y(t) = 0$. (Since the differential equation is homogeneous and both boundary conditions are 0, we can see immediately that $y(t) = 0$ is a solution. We have shown that it is the only solution.)

33. Recall from Chapter 3 that $mx'' = -kx - \beta x'$. Now $m = W/g = 4/32 = \frac{1}{8}$ slug, and $4 = 2k$ so that $k = 2$ lb/ft. Thus, the differential equation is $x'' + 7x' + 16x = 0$. The initial conditions are $x(0) = -3/2$ and $x'(0) = 0$. The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{x\} + \frac{3}{2}s + 7s \mathcal{L}\{x\} + \frac{21}{2} + 16 \mathcal{L}\{x\} = 0.$$

Solving for $\mathcal{L}\{x\}$ we obtain

$$\mathcal{L}\{x\} = \frac{-3s/2 - 21/2}{s^2 + 7s + 16} = -\frac{3}{2} \frac{s + 7/2}{(s + 7/2)^2 + (\sqrt{15}/2)^2} - \frac{7\sqrt{15}}{10} \frac{\sqrt{15}/2}{(s + 7/2)^2 + (\sqrt{15}/2)^2}.$$

Thus

$$x = -\frac{3}{2}e^{-7t/2} \cos \frac{\sqrt{15}}{2}t - \frac{7\sqrt{15}}{10}e^{-7t/2} \sin \frac{\sqrt{15}}{2}t.$$

34. The differential equation is

$$\frac{d^2q}{dt^2} + 20\frac{dq}{dt} + 200q = 150, \quad q(0) = q'(0) = 0.$$

The Laplace transform of this equation is

$$s^2\mathcal{L}\{q\} + 20s\mathcal{L}\{q\} + 200\mathcal{L}\{q\} = \frac{150}{s}.$$

Solving for $\mathcal{L}\{q\}$ we obtain

$$\mathcal{L}\{q\} = \frac{150}{s(s^2 + 20s + 200)} = \frac{3}{4} \frac{1}{s} - \frac{3}{4} \frac{s + 10}{(s + 10)^2 + 10^2} - \frac{3}{4} \frac{10}{(s + 10)^2 + 10^2}.$$

Thus

$$q(t) = \frac{3}{4} - \frac{3}{4}e^{-10t} \cos 10t - \frac{3}{4}e^{-10t} \sin 10t$$

and

$$i(t) = q'(t) = 15e^{-10t} \sin 10t.$$

35. The differential equation is

$$\frac{d^2q}{dt^2} + 2\lambda\frac{dq}{dt} + \omega^2q = \frac{E_0}{L}, \quad q(0) = q'(0) = 0.$$

The Laplace transform of this equation is

$$s^2\mathcal{L}\{q\} + 2\lambda s\mathcal{L}\{q\} + \omega^2\mathcal{L}\{q\} = \frac{E_0}{L} \frac{1}{s}$$

or

$$(s^2 + 2\lambda s + \omega^2)\mathcal{L}\{q\} = \frac{E_0}{L} \frac{1}{s}.$$

Solving for $\mathcal{L}\{q\}$ and using partial fractions we obtain

$$\mathcal{L}\{q\} = \frac{E_0}{L} \left(\frac{1/\omega^2}{s} - \frac{(1/\omega^2)s + 2\lambda/\omega^2}{s^2 + 2\lambda s + \omega^2} \right) = \frac{E_0}{L\omega^2} \left(\frac{1}{s} - \frac{s + 2\lambda}{s^2 + 2\lambda s + \omega^2} \right).$$

For $\lambda > \omega$ we write $s^2 + 2\lambda s + \omega^2 = (s + \lambda)^2 - (\lambda^2 - \omega^2)$, so (recalling that $\omega^2 = 1/LC$),

$$\mathcal{L}\{q\} = E_0 C \left(\frac{1}{s} - \frac{s + \lambda}{(s + \lambda)^2 - (\lambda^2 - \omega^2)} - \frac{\lambda}{(s + \lambda)^2 - (\lambda^2 - \omega^2)} \right).$$

Thus for $\lambda > \omega$,

$$q(t) = E_0 C \left(1 - e^{-\lambda t} \cosh \sqrt{\lambda^2 - \omega^2} t - \frac{\lambda}{\sqrt{\lambda^2 - \omega^2}} \sinh \sqrt{\lambda^2 - \omega^2} t \right).$$

For $\lambda < \omega$ we write $s^2 + 2\lambda s + \omega^2 = (s + \lambda)^2 + (\omega^2 - \lambda^2)$, so

$$\mathcal{L}\{q\} = E_0 \left(\frac{1}{s} - \frac{s + \lambda}{(s + \lambda)^2 + (\omega^2 - \lambda^2)} - \frac{\lambda}{(s + \lambda)^2 + (\omega^2 - \lambda^2)} \right).$$

Thus for $\lambda < \omega$,

$$q(t) = E_0 C \left(1 - e^{-\lambda t} \cos \sqrt{\omega^2 - \lambda^2} t - \frac{\lambda}{\sqrt{\omega^2 - \lambda^2}} \sin \sqrt{\omega^2 - \lambda^2} t \right).$$

For $\lambda = \omega$, $s^2 + 2\lambda s + \omega^2 = (s + \lambda)^2$ and

$$\mathcal{L}\{q\} = \frac{E_0}{L} \frac{1}{s(s + \lambda)^2} = \frac{E_0}{L} \left(\frac{1/\lambda^2}{s} - \frac{1/\lambda^2}{s + \lambda} - \frac{1/\lambda}{(s + \lambda)^2} \right) = \frac{E_0}{L\lambda^2} \left(\frac{1}{s} - \frac{1}{s + \lambda} - \frac{\lambda}{(s + \lambda)^2} \right).$$

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Thus for $\lambda = \omega$,

$$q(t) = E_0 C (1 - e^{-\lambda t} - \lambda t e^{-\lambda t}).$$

36. The differential equation is

$$R \frac{dq}{dt} + \frac{1}{C} q = E_0 e^{-kt}, \quad q(0) = 0.$$

The Laplace transform of this equation is

$$R \mathcal{L}\{q\} + \frac{1}{C} \mathcal{L}\{q\} = E_0 \frac{1}{s+k}.$$

Solving for $\mathcal{L}\{q\}$ we obtain

$$\mathcal{L}\{q\} = \frac{E_0 C}{(s+k)(RCs+1)} = \frac{E_0/R}{(s+k)(s+1/RC)}.$$

When $1/RC \neq k$ we have by partial fractions

$$\mathcal{L}\{q\} = \frac{E_0}{R} \left(\frac{1/(1/RC - k)}{s+k} - \frac{1/(1/RC - k)}{s+1/RC} \right) = \frac{E_0}{R} \frac{1}{1/RC - k} \left(\frac{1}{s+k} - \frac{1}{s+1/RC} \right).$$

Thus

$$q(t) = \frac{E_0 C}{1 - kRC} (e^{-kt} - e^{-t/RC}).$$

When $1/RC = k$ we have

$$\mathcal{L}\{q\} = \frac{E_0}{R} \frac{1}{(s+k)^2}.$$

Thus

$$q(t) = \frac{E_0}{R} t e^{-kt} = \frac{E_0}{R} t e^{-t/RC}.$$

$$37. \mathcal{L}\{(t-1)\mathcal{U}(t-1)\} = \frac{e^{-s}}{s^2}$$

$$38. \mathcal{L}\{e^{2-t}\mathcal{U}(t-2)\} = \mathcal{L}\{e^{-(t-2)}\mathcal{U}(t-2)\} = \frac{e^{-2s}}{s+1}$$

$$39. \mathcal{L}\{t\mathcal{U}(t-2)\} = \mathcal{L}\{(t-2)\mathcal{U}(t-2) + 2\mathcal{U}(t-2)\} = \frac{e^{-2s}}{s^2} + \frac{2e^{-2s}}{s}$$

$$40. \mathcal{L}\{(3t+1)\mathcal{U}(t-1)\} = 3\mathcal{L}\{(t-1)\mathcal{U}(t-1)\} + 4\mathcal{L}\{\mathcal{U}(t-1)\} = \frac{e^{-s}}{s^2} + \frac{4e^{-s}}{s}$$

$$41. \mathcal{L}\{\cos 2t\mathcal{U}(t-\pi)\} = \mathcal{L}\{\cos 2(t-\pi)\mathcal{U}(t-\pi)\} = \frac{se^{-\pi s}}{s^2+4}$$

$$42. \mathcal{L}\left\{\sin t\mathcal{U}\left(t-\frac{\pi}{2}\right)\right\} = \mathcal{L}\left\{\cos\left(t-\frac{\pi}{2}\right)\mathcal{U}\left(t-\frac{\pi}{2}\right)\right\} = \frac{se^{-\pi s}}{s^2+1}$$

$$43. \mathcal{L}^{-1}\left\{\frac{e^{-2s}}{s^3}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{2} \cdot \frac{2}{s^3}e^{-2s}\right\} = \frac{1}{2}(t-2)^2\mathcal{U}(t-2)$$

$$44. \mathcal{L}^{-1}\left\{\frac{(1+e^{-2s})^2}{s+2}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s+2} + \frac{2e^{-2s}}{s+2} + \frac{e^{-4s}}{s+2}\right\} = e^{-2t} + 2e^{-2(t-2)}\mathcal{U}(t-2) + e^{-2(t-4)}\mathcal{U}(t-4)$$

$$45. \mathcal{L}^{-1}\left\{\frac{e^{-\pi s}}{s^2+1}\right\} = \sin(t-\pi)\mathcal{U}(t-\pi)$$

$$46. \mathcal{L}^{-1}\left\{\frac{se^{-\pi s/2}}{s^2+4}\right\} = \cos 2\left(t-\frac{\pi}{2}\right)\mathcal{U}\left(t-\frac{\pi}{2}\right)$$

$$47. \mathcal{L}^{-1}\left\{\frac{e^{-s}}{s(s+1)}\right\} = \mathcal{L}^{-1}\left\{\frac{e^{-s}}{s} - \frac{e^{-s}}{s+1}\right\} = \mathcal{U}(t-1) - e^{-(t-1)}\mathcal{U}(t-1)$$

$$48. \mathcal{L}^{-1}\left\{\frac{e^{-2s}}{s^2(s-1)}\right\} = \mathcal{L}^{-1}\left\{-\frac{e^{-2s}}{s} - \frac{e^{-2s}}{s^2} + \frac{e^{-2s}}{s-1}\right\} = -\mathcal{U}(t-2) - (t-2)\mathcal{U}(t-2) + e^{t-2}\mathcal{U}(t-2)$$

$$49. \text{ (c)} \quad 50. \text{ (e)} \quad 51. \text{ (f)} \quad 52. \text{ (b)} \quad 53. \text{ (a)} \quad 54. \text{ (d)}$$

$$55. \mathcal{L}\{2 - 4\mathcal{U}(t-3)\} = \frac{2}{s} - \frac{4}{s}e^{-3s}$$

$$56. \mathcal{L}\{1 - \mathcal{U}(t-4) + \mathcal{U}(t-5)\} = \frac{1}{s} - \frac{e^{-4s}}{s} + \frac{e^{-5s}}{s}$$

$$57. \mathcal{L}\{t^2\mathcal{U}(t-1)\} = \mathcal{L}\{[(t-1)^2 + 2(t-1) - 1]\mathcal{U}(t-1)\} = \mathcal{L}\{[(t-1)^2 + 2(t-1) - 1]\mathcal{U}(t-1)\} \\ = \left(\frac{2}{s^3} + \frac{2}{s^2} + \frac{1}{s}\right)e^{-s}$$

$$58. \mathcal{L}\left\{\sin t \mathcal{U}\left(t - \frac{3\pi}{2}\right)\right\} = \mathcal{L}\left\{-\cos\left(t - \frac{3\pi}{2}\right)\mathcal{U}\left(t - \frac{3\pi}{2}\right)\right\} = -\frac{se^{-3\pi s/2}}{s^2 + 1}$$

$$59. \mathcal{L}\{t - t\mathcal{U}(t-2)\} = \mathcal{L}\{t - (t-2)\mathcal{U}(t-2) - 2\mathcal{U}(t-2)\} = \frac{1}{s^2} - \frac{e^{-2s}}{s^2} - \frac{2e^{-2s}}{s}$$

$$60. \mathcal{L}\{\sin t - \sin t \mathcal{U}(t-2\pi)\} = \mathcal{L}\{\sin t - \sin(t-2\pi)\mathcal{U}(t-2\pi)\} = \frac{1}{s^2 + 1} - \frac{e^{-2\pi s}}{s^2 + 1}$$

$$61. \mathcal{L}\{f(t)\} = \mathcal{L}\{\mathcal{U}(t-a) - \mathcal{U}(t-b)\} = \frac{e^{-as}}{s} - \frac{e^{-bs}}{s}$$

$$62. \mathcal{L}\{f(t)\} = \mathcal{L}\{\mathcal{U}(t-1) + \mathcal{U}(t-2) + \mathcal{U}(t-3) + \cdots\} = \frac{e^{-s}}{s} + \frac{e^{-2s}}{s} + \frac{e^{-3s}}{s} + \cdots = \frac{1}{s} \frac{e^{-s}}{1 - e^{-s}}$$

63. The Laplace transform of the differential equation is

$$s\mathcal{L}\{y\} - y(0) + \mathcal{L}\{y\} = \frac{5}{s}e^{-s}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{5e^{-s}}{s(s+1)} = 5e^{-s} \left[\frac{1}{s} - \frac{1}{s+1} \right].$$

Thus

$$y = 5\mathcal{U}(t-1) - 5e^{-(t-1)}\mathcal{U}(t-1).$$

64. The Laplace transform of the differential equation is

$$s\mathcal{L}\{y\} - y(0) + \mathcal{L}\{y\} = \frac{1}{s} - \frac{2}{s}e^{-s}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{1}{s(s+1)} - \frac{2e^{-s}}{s(s+1)} = \frac{1}{s} - \frac{1}{s+1} - 2e^{-s} \left[\frac{1}{s} - \frac{1}{s+1} \right].$$

Thus

$$y = 1 - e^{-t} - 2[1 - e^{-(t-1)}]\mathcal{U}(t-1).$$

65. The Laplace transform of the differential equation is

$$s\mathcal{L}\{y\} - y(0) + 2\mathcal{L}\{y\} = \frac{1}{s^2} - e^{-s}\frac{s+1}{s^2}.$$

Exercises 4.3

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{1}{s^2(s+2)} - e^{-s} \frac{s+1}{s^2(s+1)} = -\frac{1}{4} \frac{1}{s} + \frac{1}{2} \frac{1}{s^2} + \frac{1}{4} \frac{1}{s+2} - e^{-s} \left[\frac{1}{4} \frac{1}{s} + \frac{1}{2} \frac{1}{s^2} - \frac{1}{4} \frac{1}{s+2} \right].$$

Thus

$$y = -\frac{1}{4} + \frac{1}{2}t + \frac{1}{4}e^{-2t} - \left[\frac{1}{4} + \frac{1}{2}(t-1) - \frac{1}{4}e^{-2(t-1)} \right] \mathcal{U}(t-1).$$

66. The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{y\} - sy(0) - y'(0) + 4 \mathcal{L}\{y\} = \frac{1}{s} - \frac{e^{-s}}{s}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{1-s}{s(s^2+4)} - e^{-s} \frac{1}{s(s^2+4)} = \frac{1}{4} \frac{1}{s} - \frac{1}{4} \frac{s}{s^2+4} - \frac{1}{2} \frac{2}{s^2+4} - e^{-s} \left[\frac{1}{4} \frac{1}{s} - \frac{1}{4} \frac{s}{s^2+4} \right].$$

Thus

$$y = \frac{1}{4} - \frac{1}{4} \cos 2t - \frac{1}{2} \sin 2t - \left[\frac{1}{4} - \frac{1}{4} \cos 2(t-1) \right] \mathcal{U}(t-1).$$

67. The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{y\} - sy(0) - y'(0) + 4 \mathcal{L}\{y\} = e^{-2\pi s} \frac{1}{s^2+1}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{s}{s^2+4} + e^{-2\pi s} \left[\frac{1}{3} \frac{1}{s^2+1} - \frac{1}{6} \frac{2}{s^2+4} \right].$$

Thus

$$y = \cos 2t + \left[\frac{1}{3} \sin(t-2\pi) - \frac{1}{6} \sin 2(t-2\pi) \right] \mathcal{U}(t-2\pi).$$

68. The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{y\} - sy(0) - y'(0) - 5[s \mathcal{L}\{y\} - y(0)] + 6 \mathcal{L}\{y\} = \frac{e^{-s}}{s}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\begin{aligned} \mathcal{L}\{y\} &= e^{-s} \frac{1}{s(s-2)(s-3)} + \frac{1}{(s-2)(s-3)} \\ &= e^{-s} \left[\frac{1}{6} \frac{1}{s} - \frac{1}{2} \frac{1}{s-2} + \frac{1}{3} \frac{1}{s-3} \right] - \frac{1}{s-2} + \frac{1}{s-3}. \end{aligned}$$

Thus

$$y = \left[\frac{1}{6} - \frac{1}{2}e^{2(t-1)} + \frac{1}{3}e^{3(t-1)} \right] \mathcal{U}(t-1) + e^{3t} - e^{2t}.$$

69. The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{y\} - sy(0) - y'(0) + \mathcal{L}\{y\} = \frac{e^{-\pi s}}{s} - \frac{e^{-2\pi s}}{s}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = e^{-\pi s} \left[\frac{1}{s} - \frac{s}{s^2+1} \right] - e^{-2\pi s} \left[\frac{1}{s} - \frac{s}{s^2+1} \right] + \frac{1}{s^2+1}.$$

Thus

$$y = [1 - \cos(t-\pi)] \mathcal{U}(t-\pi) - [1 - \cos(t-2\pi)] \mathcal{U}(t-2\pi) + \sin t.$$

70. The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{y\} - sy(0) - y'(0) + 4[s \mathcal{L}\{y\} - y(0)] + 3 \mathcal{L}\{y\} = \frac{1}{s} - \frac{e^{-2s}}{s} - \frac{e^{-4s}}{s} + \frac{e^{-6s}}{s}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\begin{aligned} \mathcal{L}\{y\} = & \frac{1}{3} \frac{1}{s} - \frac{1}{2} \frac{1}{s+1} + \frac{1}{6} \frac{1}{s+3} - e^{-2s} \left[\frac{1}{3} \frac{1}{s} - \frac{1}{2} \frac{1}{s+1} + \frac{1}{6} \frac{1}{s+3} \right] \\ & - e^{-4s} \left[\frac{1}{3} \frac{1}{s} - \frac{1}{2} \frac{1}{s+1} + \frac{1}{6} \frac{1}{s+3} \right] + e^{-6s} \left[\frac{1}{3} \frac{1}{s} - \frac{1}{2} \frac{1}{s+1} + \frac{1}{6} \frac{1}{s+3} \right]. \end{aligned}$$

Thus

$$\begin{aligned} y = & \frac{1}{3} - \frac{1}{2}e^{-t} + \frac{1}{6}e^{-3t} - \left[\frac{1}{3} - \frac{1}{2}e^{-(t-2)} + \frac{1}{6}e^{-3(t-2)} \right] \mathcal{U}(t-2) \\ & - \left[\frac{1}{3} - \frac{1}{2}e^{-(t-4)} + \frac{1}{6}e^{-3(t-4)} \right] \mathcal{U}(t-4) + \left[\frac{1}{3} - \frac{1}{2}e^{-(t-6)} + \frac{1}{6}e^{-3(t-6)} \right] \mathcal{U}(t-6). \end{aligned}$$

71. Recall from Chapter 3 that $mx'' = -kx + f(t)$. Now $m = W/g = 32/32 = 1$ slug, and $32 = 2k$ so that $k = 16$ lb/ft. Thus, the differential equation is $x'' + 16x = f(t)$. The initial conditions are $x(0) = 0$, $x'(0) = 0$. Also, since

$$f(t) = \begin{cases} 20t, & 0 \leq t < 5 \\ 0, & t \geq 5 \end{cases}$$

and $20t = 20(t-5) + 100$ we can write

$$f(t) = 20t - 20t \mathcal{U}(t-5) = 20t - 20(t-5) \mathcal{U}(t-5) - 100 \mathcal{U}(t-5).$$

The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{x\} + 16 \mathcal{L}\{x\} = \frac{20}{s^2} - \frac{20}{s^2} e^{-5s} - \frac{100}{s} e^{-5s}.$$

Solving for $\mathcal{L}\{x\}$ we obtain

$$\begin{aligned} \mathcal{L}\{x\} = & \frac{20}{s^2(s^2+16)} - \frac{20}{s^2(s^2+16)} e^{-5s} - \frac{100}{s(s^2+16)} e^{-5s} \\ = & \left(\frac{5}{4} \cdot \frac{1}{s^2} - \frac{5}{16} \cdot \frac{4}{s^2+16} \right) (1 - e^{-5s}) - \left(\frac{25}{4} \cdot \frac{1}{s} - \frac{25}{4} \cdot \frac{s}{s^2+16} \right) e^{-5s}. \end{aligned}$$

Thus

$$\begin{aligned} x(t) = & \frac{5}{4}t - \frac{5}{16} \sin 4t - \left[\frac{5}{4}(t-5) - \frac{5}{16} \sin 4(t-5) \right] \mathcal{U}(t-5) - \left[\frac{25}{4} - \frac{25}{4} \cos 4(t-5) \right] \mathcal{U}(t-5) \\ = & \frac{5}{4}t - \frac{5}{16} \sin 4t - \frac{5}{4}t \mathcal{U}(t-5) + \frac{5}{16} \sin 4(t-5) \mathcal{U}(t-5) + \frac{25}{4} \cos 4(t-5) \mathcal{U}(t-5). \end{aligned}$$

72. Recall from Chapter 3 that $mx'' = -kx + f(t)$. Now $m = W/g = 32/32 = 1$ slug, and $32 = 2k$ so that $k = 16$ lb/ft. Thus, the differential equation is $x'' + 16x = f(t)$. The initial conditions are $x(0) = 0$, $x'(0) = 0$. Also, since

$$f(t) = \begin{cases} \sin t, & 0 \leq t < 2\pi \\ 0, & t \geq 2\pi \end{cases}$$

and $\sin t = \sin(t-2\pi)$ we can write

$$f(t) = \sin t - \sin(t-2\pi) \mathcal{U}(t-2\pi).$$

Exercises 4.3

The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{x\} + 16 \mathcal{L}\{x\} = \frac{1}{s^2 + 1} - \frac{1}{s^2 + 1} e^{-2\pi s}.$$

Solving for $\mathcal{L}\{x\}$ we obtain

$$\begin{aligned} \mathcal{L}\{x\} &= \frac{1}{(s^2 + 16)(s^2 + 1)} - \frac{1}{(s^2 + 16)(s^2 + 1)} e^{-2\pi s} \\ &= \frac{-1/15}{s^2 + 16} + \frac{1/15}{s^2 + 1} - \left[\frac{-1/15}{s^2 + 16} + \frac{1/15}{s^2 + 1} \right] e^{-2\pi s}. \end{aligned}$$

Thus

$$\begin{aligned} x(t) &= -\frac{1}{60} \sin 4t + \frac{1}{15} \sin t + \frac{1}{60} \sin 4(t - 2\pi) \mathcal{U}(t - 2\pi) - \frac{1}{15} \sin(t - 2\pi) \mathcal{U}(t - 2\pi) \\ &= \begin{cases} -\frac{1}{60} \sin 4t + \frac{1}{15} \sin t, & 0 \leq t < 2\pi \\ 0, & t \geq 2\pi. \end{cases} \end{aligned}$$

73. The differential equation is

$$2.5 \frac{dq}{dt} + 12.5q = 5 \mathcal{U}(t - 3).$$

The Laplace transform of this equation is

$$s \mathcal{L}\{q\} + 5 \mathcal{L}\{q\} = \frac{2}{s} e^{-3s}.$$

Solving for $\mathcal{L}\{q\}$ we obtain

$$\mathcal{L}\{q\} = \frac{2}{s(s + 5)} e^{-3s} = \left(\frac{2}{5} \cdot \frac{1}{s} - \frac{2}{5} \cdot \frac{1}{s + 5} \right) e^{-3s}.$$

Thus

$$q(t) = \frac{2}{5} \mathcal{U}(t - 3) - \frac{2}{5} e^{-5(t-3)} \mathcal{U}(t - 3).$$

74. The differential equation is

$$10 \frac{dq}{dt} + 10q = 30e^t - 30e^t \mathcal{U}(t - 1.5).$$

The Laplace transform of this equation is

$$s \mathcal{L}\{q\} - q_0 + \mathcal{L}\{q\} = \frac{3}{s - 1} - \frac{3e^{1.5}}{s - 1.5} e^{-1.5s}.$$

Solving for $\mathcal{L}\{q\}$ we obtain

$$\mathcal{L}\{q\} = \left(q_0 - \frac{3}{2} \right) \cdot \frac{1}{s + 1} + \frac{3}{2} \cdot \frac{1}{s - 1} 3e^{1.5} \left(\frac{-2/5}{s + 1} + \frac{2/5}{s - 1.5} \right) e^{-1.5s}.$$

Thus

$$q(t) = \left(q_0 - \frac{3}{2} \right) e^{-t} + \frac{3}{2} e^t + \frac{6}{5} e^{1.5} \left(e^{-(t-1.5)} - e^{1.5(t-1.5)} \right) \mathcal{U}(t - 1.5).$$

75. (a) The differential equation is

$$\frac{di}{dt} + 10i = \sin t + \cos \left(t - \frac{3\pi}{2} \right) \mathcal{U} \left(t - \frac{3\pi}{2} \right), \quad i(0) = 0.$$

The Laplace transform of this equation is

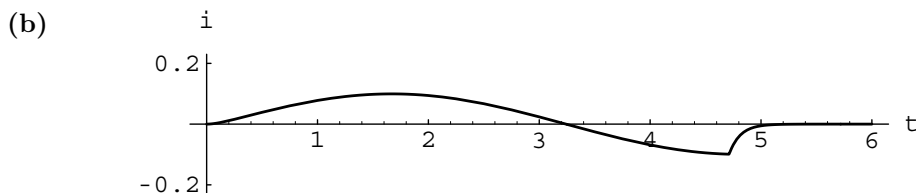
$$s \mathcal{L}\{i\} + 10 \mathcal{L}\{i\} = \frac{1}{s^2 + 1} + \frac{se^{-3\pi s/2}}{s^2 + 1}.$$

Solving for $\mathcal{L}\{i\}$ we obtain

$$\begin{aligned}\mathcal{L}\{i\} &= \frac{1}{(s^2+1)(s+10)} + \frac{s}{(s^2+1)(s+10)} e^{-3\pi s/2} \\ &= \frac{1}{101} \left(\frac{1}{s+10} - \frac{s}{s^2+1} + \frac{10}{s^2+1} \right) + \frac{1}{101} \left(\frac{-10}{s+10} + \frac{10s}{s^2+1} + \frac{1}{s^2+1} \right) e^{-3\pi s/2}.\end{aligned}$$

Thus

$$\begin{aligned}i(t) &= \frac{1}{101} (e^{-10t} - \cos t + 10 \sin t) \\ &\quad + \frac{1}{101} \left[-10e^{-10(t-3\pi/2)} + 10 \cos \left(t - \frac{3\pi}{2} \right) + \sin \left(t - \frac{3\pi}{2} \right) \right] \mathcal{U} \left(t - \frac{3\pi}{2} \right).\end{aligned}$$



The maximum value of $i(t)$ is approximately 0.1 at $t = 1.7$, the minimum is approximately -0.1 at 4.7 .

76. (a) The differential equation is

$$50 \frac{dq}{dt} + \frac{1}{0.01} q = E_0 [\mathcal{U}(t-1) - \mathcal{U}(t-3)], \quad q(0) = 0$$

or

$$50 \frac{dq}{dt} + 100q = E_0 [\mathcal{U}(t-1) - \mathcal{U}(t-3)], \quad q(0) = 0.$$

The Laplace transform of this equation is

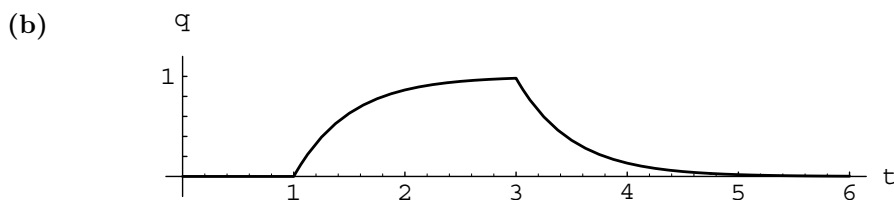
$$50s \mathcal{L}\{q\} + 100 \mathcal{L}\{q\} = E_0 \left(\frac{1}{s} e^{-s} - \frac{1}{s} e^{-3s} \right).$$

Solving for $\mathcal{L}\{q\}$ we obtain

$$\mathcal{L}\{q\} = \frac{E_0}{50} \left[\frac{e^{-s}}{s(s+2)} - \frac{e^{-3s}}{s(s+2)} \right] = \frac{E_0}{50} \left[\frac{1}{2} \left(\frac{1}{s} - \frac{1}{s+2} \right) e^{-s} - \frac{1}{2} \left(\frac{1}{s} - \frac{1}{s+2} \right) e^{-3s} \right].$$

Thus

$$q(t) = \frac{E_0}{100} \left[\left(1 - e^{-2(t-1)} \right) \mathcal{U}(t-1) - \left(1 - e^{-2(t-3)} \right) \mathcal{U}(t-3) \right].$$



The maximum value of $q(t)$ is approximately 1 at $t = 3$.

77. The differential equation is

$$EI \frac{d^4 y}{dx^4} = w_0 [1 - \mathcal{U}(x - L/2)].$$

Taking the Laplace transform of both sides and using $y(0) = y'(0) = 0$ we obtain

$$s^4 \mathcal{L}\{y\} - sy''(0) - y'''(0) = \frac{w_0}{EI} \frac{1}{s} \left(1 - e^{-Ls/2} \right).$$

Exercises 4.3

Letting $y''(0) = c_1$ and $y'''(0) = c_2$ we have

$$\mathcal{L}\{y\} = \frac{c_1}{s^3} + \frac{c_2}{s^4} + \frac{w_0}{EI} \frac{1}{s^5} (1 - e^{-Ls/2})$$

so that

$$y(x) = \frac{1}{2}c_1x^2 + \frac{1}{6}c_2x^3 + \frac{1}{24} \frac{w_0}{EI} \left[x^4 - \left(x - \frac{L}{2}\right)^4 \mathcal{U}\left(x - \frac{L}{2}\right) \right].$$

To find c_1 and c_2 we compute

$$y''(x) = c_1 + c_2x + \frac{1}{2} \frac{w_0}{EI} \left[x^2 - \left(x - \frac{L}{2}\right)^2 \mathcal{U}\left(x - \frac{L}{2}\right) \right]$$

and

$$y'''(x) = c_2 + \frac{w_0}{EI} \left[x - \left(x - \frac{L}{2}\right) \mathcal{U}\left(x - \frac{L}{2}\right) \right].$$

Then $y''(L) = y'''(L) = 0$ yields the system

$$\begin{aligned} c_1 + c_2L + \frac{1}{2} \frac{w_0}{EI} \left[L^2 - \left(\frac{L}{2}\right)^2 \right] &= c_1 + c_2L + \frac{3}{8} \frac{w_0L^2}{EI} = 0 \\ c_2 + \frac{w_0}{EI} \left(\frac{L}{2}\right) &= c_2 + \frac{1}{2} \frac{w_0L}{EI} = 0. \end{aligned}$$

Solving for c_1 and c_2 we obtain $c_1 = \frac{1}{8}w_0L^2/EI$ and $c_2 = -\frac{1}{2}w_0L/EI$. Thus

$$y(x) = \frac{w_0}{EI} \left(\frac{1}{16}L^2x^2 - \frac{1}{12}Lx^3 + \frac{1}{24}x^4 - \frac{1}{24} \left(x - \frac{L}{2}\right)^4 \mathcal{U}\left(x - \frac{L}{2}\right) \right).$$

78. The differential equation is

$$EI \frac{d^4y}{dx^4} = w_0[\mathcal{U}(x - L/3) - \mathcal{U}(x - 2L/3)].$$

Taking the Laplace transform of both sides and using $y(0) = y'(0) = 0$ we obtain

$$s^4 \mathcal{L}\{y\} - sy''(0) - y'''(0) = \frac{w_0}{EI} \frac{1}{s} (e^{-Ls/3} - e^{-2Ls/3}).$$

Letting $y''(0) = c_1$ and $y'''(0) = c_2$ we have

$$\mathcal{L}\{y\} = \frac{c_1}{s^3} + \frac{c_2}{s^4} + \frac{w_0}{EI} \frac{1}{s^5} (e^{-Ls/3} - e^{-2Ls/3})$$

so that

$$y(x) = \frac{1}{2}c_1x^2 + \frac{1}{6}c_2x^3 + \frac{1}{24} \frac{w_0}{EI} \left[\left(x - \frac{L}{3}\right)^4 \mathcal{U}\left(x - \frac{L}{3}\right) - \left(x - \frac{2L}{3}\right)^4 \mathcal{U}\left(x - \frac{2L}{3}\right) \right].$$

To find c_1 and c_2 we compute

$$y''(x) = c_1 + c_2x + \frac{1}{2} \frac{w_0}{EI} \left[\left(x - \frac{L}{3}\right)^2 \mathcal{U}\left(x - \frac{L}{3}\right) - \left(x - \frac{2L}{3}\right)^2 \mathcal{U}\left(x - \frac{2L}{3}\right) \right]$$

and

$$y'''(x) = c_2 + \frac{w_0}{EI} \left[\left(x - \frac{L}{3}\right) \mathcal{U}\left(x - \frac{L}{3}\right) - \left(x - \frac{2L}{3}\right) \mathcal{U}\left(x - \frac{2L}{3}\right) \right].$$

Then $y''(L) = y'''(L) = 0$ yields the system

$$\begin{aligned} c_1 + c_2L + \frac{1}{2} \frac{w_0}{EI} \left[\left(\frac{2L}{3}\right)^2 - \left(\frac{L}{3}\right)^2 \right] &= c_1 + c_2L + \frac{1}{6} \frac{w_0L^2}{EI} = 0 \\ c_2 + \frac{w_0}{EI} \left[\frac{2L}{3} - \frac{L}{3} \right] &= c_2 + \frac{1}{3} \frac{w_0L}{EI} = 0. \end{aligned}$$

Solving for c_1 and c_2 we obtain $c_1 = \frac{1}{6}w_0L^2/EI$ and $c_2 = -\frac{1}{3}w_0L/EI$. Thus

$$y(x) = \frac{w_0}{EI} \left(\frac{1}{12}L^2x^2 - \frac{1}{18}Lx^3 + \frac{1}{24} \left[\left(x - \frac{L}{3}\right)^4 \mathcal{U}\left(x - \frac{L}{3}\right) - \left(x - \frac{2L}{3}\right)^4 \mathcal{U}\left(x - \frac{2L}{3}\right) \right] \right).$$

79. The differential equation is

$$EI \frac{d^4y}{dx^4} = \frac{2w_0}{L} \left[\frac{L}{2} - x + \left(x - \frac{L}{2}\right) \mathcal{U}\left(x - \frac{L}{2}\right) \right].$$

Taking the Laplace transform of both sides and using $y(0) = y'(0) = 0$ we obtain

$$s^4 \mathcal{L}\{y\} - sy''(0) - y'''(0) = \frac{2w_0}{EIL} \left[\frac{L}{2s} - \frac{1}{s^2} + \frac{1}{s^2} e^{-Ls/2} \right].$$

Letting $y''(0) = c_1$ and $y'''(0) = c_2$ we have

$$\mathcal{L}\{y\} = \frac{c_1}{s^3} + \frac{c_2}{s^4} + \frac{2w_0}{EIL} \left[\frac{L}{2s^5} - \frac{1}{s^6} + \frac{1}{s^6} e^{-Ls/2} \right]$$

so that

$$\begin{aligned} y(x) &= \frac{1}{2}c_1x^2 + \frac{1}{6}c_2x^3 + \frac{2w_0}{EIL} \left[\frac{L}{48}x^4 - \frac{1}{120}x^5 + \frac{1}{120} \left(x - \frac{L}{2}\right) \mathcal{U}\left(x - \frac{L}{2}\right) \right] \\ &= \frac{1}{2}c_1x^2 + \frac{1}{6}c_2x^3 + \frac{w_0}{60EIL} \left[\frac{5L}{2}x^4 - x^5 + \left(x - \frac{L}{2}\right)^5 \mathcal{U}\left(x - \frac{L}{2}\right) \right]. \end{aligned}$$

To find c_1 and c_2 we compute

$$y''(x) = c_1 + c_2x + \frac{w_0}{60EIL} \left[30Lx^2 - 20x^3 + 20 \left(x - \frac{L}{2}\right)^3 \mathcal{U}\left(x - \frac{L}{2}\right) \right]$$

and

$$y'''(x) = c_2 + \frac{w_0}{60EIL} \left[60Lx - 60x^2 + 60 \left(x - \frac{L}{2}\right)^2 \mathcal{U}\left(x - \frac{L}{2}\right) \right].$$

Then $y''(L) = y'''(L) = 0$ yields the system

$$\begin{aligned} c_1 + c_2L + \frac{w_0}{60EIL} \left[30L^3 - 20L^3 + \frac{5}{2}L^3 \right] &= c_1 + c_2L + \frac{5w_0L^2}{24EI} = 0 \\ c_2 + \frac{w_0}{60EIL} [60L^2 - 60L^2 + 15L^2] &= c_2 + \frac{w_0L}{4EI} = 0. \end{aligned}$$

Solving for c_1 and c_2 we obtain $c_1 = w_0L^2/24EI$ and $c_2 = -w_0L/4EI$. Thus

$$y(x) = \frac{w_0L^2}{48EI}x^2 - \frac{w_0L}{24EI} + \frac{w_0}{60EIL} \left[\frac{5L}{2}x^4 - x^5 + \left(x - \frac{L}{2}\right)^5 \mathcal{U}\left(x - \frac{L}{2}\right) \right].$$

80. The differential equation is

$$EI \frac{d^4y}{dx^4} = w_0[1 - \mathcal{U}(x - L/2)].$$

Taking the Laplace transform of both sides and using $y(0) = y'(0) = 0$ we obtain

$$s^4 \mathcal{L}\{y\} - sy''(0) - y'''(0) = \frac{w_0}{EI} \frac{1}{s} (1 - e^{-Ls/2}).$$

Letting $y''(0) = c_1$ and $y'''(0) = c_2$ we have

$$\mathcal{L}\{y\} = \frac{c_1}{s^3} + \frac{c_2}{s^4} + \frac{w_0}{EI} \frac{1}{s^5} (1 - e^{-Ls/2})$$

Exercises 4.3

so that

$$y(x) = \frac{1}{2}c_1x^2 + \frac{1}{6}c_2x^3 + \frac{1}{24}\frac{w_0}{EI}\left[x^4 - \left(x - \frac{L}{2}\right)^4 \mathcal{U}\left(x - \frac{L}{2}\right)\right].$$

To find c_1 and c_2 we compute

$$y''(x) = c_1 + c_2x + \frac{1}{2}\frac{w_0}{EI}\left[x^2 - \left(x - \frac{L}{2}\right)^2 \mathcal{U}\left(x - \frac{L}{2}\right)\right].$$

Then $y(L) = y''(L) = 0$ yields the system

$$\begin{aligned}\frac{1}{2}c_1L^2 + \frac{1}{6}c_2L^3 + \frac{1}{24}\frac{w_0}{EI}\left[L^4 - \left(\frac{L}{2}\right)^4\right] &= \frac{1}{2}c_1L^2 + \frac{1}{6}c_2L^3 + \frac{5w_0}{128EI}L^4 = 0 \\ c_1 + c_2L + \frac{1}{2}\frac{w_0}{EI}\left[L^2 - \left(\frac{L}{2}\right)^2\right] &= c_1 + c_2L + \frac{3w_0}{8EI}L^2 = 0.\end{aligned}$$

Solving for c_1 and c_2 we obtain $c_1 = \frac{9}{128}w_0L^2/EI$ and $c_2 = -\frac{57}{128}w_0L/EI$. Thus

$$y(x) = \frac{w_0}{EI}\left(\frac{9}{256}L^2x^2 - \frac{19}{256}Lx^3 + \frac{1}{24}x^4 - \frac{1}{24}\left(x - \frac{L}{2}\right)^4 \mathcal{U}\left(x - \frac{L}{2}\right)\right).$$

81. From Theorem 4.6 we have $\mathcal{L}\{te^{kti}\} = 1/(s - ki)^2$. Then, using Euler's formula,

$$\begin{aligned}\mathcal{L}\{te^{kti}\} &= \mathcal{L}\{t \cos kt + it \sin kt\} = \mathcal{L}\{t \cos kt\} + i \mathcal{L}\{t \sin kt\} \\ &= \frac{1}{(s - ki)^2} = \frac{(s + ki)^2}{(s^2 + k^2)^2} = \frac{s^2 - k^2}{(s^2 + k^2)^2} + i \frac{2ks}{(s^2 + k^2)^2}.\end{aligned}$$

Equating real and imaginary parts we have

$$\mathcal{L}\{t \cos kt\} = \frac{s^2 - k^2}{(s^2 + k^2)^2} \quad \text{and} \quad \mathcal{L}\{t \sin kt\} = \frac{2ks}{(s^2 + k^2)^2}.$$

82. The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{x\} + \omega^2 \mathcal{L}\{x\} = \frac{s}{s^2 + \omega^2}.$$

Solving for $\mathcal{L}\{x\}$ we obtain $\mathcal{L}\{x\} = s/(s^2 + \omega^2)^2$. Thus $x = (1/2\omega)t \sin \omega t$.

Exercises 4.4

1. $\mathcal{L}\{t \cos 2t\} = -\frac{d}{ds}\left(\frac{s}{s^2 + 4}\right) = \frac{s^2 - 4}{(s^2 + 4)^2}$
2. $\mathcal{L}\{t \sinh 3t\} = -\frac{d}{ds}\left(\frac{3}{s^2 - 9}\right) = \frac{6s}{(s^2 - 9)^2}$
3. $\mathcal{L}\{t^2 \sinh t\} = \frac{d^2}{ds^2}\left(\frac{1}{s^2 - 1}\right) = \frac{6s^2 + 2}{(s^2 - 1)^3}$
4. $\mathcal{L}\{t^2 \cos t\} = \frac{d^2}{ds^2}\left(\frac{s}{s^2 + 1}\right) = \frac{d}{ds}\left(\frac{1 - s^2}{(s^2 + 1)^2}\right) = \frac{2s(s^2 - 3)}{(s^2 + 1)^3}$
5. $\mathcal{L}\{te^{2t} \sin 6t\} = -\frac{d}{ds}\left(\frac{6}{(s - 2)^2 + 36}\right) = \frac{12(s - 2)}{[(s - 2)^2 + 36]^2}$

6. $\mathcal{L}\{te^{-3t} \cos 3t\} = -\frac{d}{ds} \left(\frac{s+3}{(s+3)^2+9} \right) = \frac{(s+3)^2-9}{[(s+3)^2+9]^2}$
7. $\mathcal{L}\{1 * t^3\} = \frac{1}{s} \frac{3!}{s^4} = \frac{6}{s^5}$
8. $\mathcal{L}\{t^2 * te^t\} = \frac{2}{s^3(s-1)^2}$
9. $\mathcal{L}\{e^{-t} * e^t \cos t\} = \frac{s-1}{(s+1)[(s-1)^2+1]}$
10. $\mathcal{L}\{e^{2t} * \sin t\} = \frac{1}{(s-2)(s^2+1)}$
11. $\mathcal{L}\left\{\int_0^t e^\tau d\tau\right\} = \frac{1}{s} \mathcal{L}\{e^t\} = \frac{1}{s(s-1)}$
12. $\mathcal{L}\left\{\int_0^t \cos \tau d\tau\right\} = \frac{1}{s} \mathcal{L}\{\cos t\} = \frac{s}{s(s^2+1)} = \frac{1}{s^2+1}$
13. $\mathcal{L}\left\{\int_0^t e^{-\tau} \cos \tau d\tau\right\} = \frac{1}{s} \mathcal{L}\{e^{-t} \cos t\} = \frac{1}{s} \frac{s+1}{(s+1)^2+1} = \frac{s+1}{s(s^2+2s+2)}$
14. $\mathcal{L}\left\{\int_0^t \tau \sin \tau d\tau\right\} = \frac{1}{s} \mathcal{L}\{t \sin t\} = \frac{1}{s} \left(-\frac{d}{ds} \frac{1}{s^2+1} \right) = -\frac{1}{s} \frac{-2s}{(s^2+1)^2} = \frac{2}{(s^2+1)^2}$
15. $\mathcal{L}\left\{\int_0^t \tau e^{t-\tau} d\tau\right\} = \mathcal{L}\{t\} \mathcal{L}\{e^t\} = \frac{1}{s^2(s-1)}$
16. $\mathcal{L}\left\{\int_0^t \sin \tau \cos(t-\tau) d\tau\right\} = \mathcal{L}\{\sin t\} \mathcal{L}\{\cos t\} = \frac{s}{(s^2+1)^2}$
17. $\mathcal{L}\left\{t \int_0^t \sin \tau d\tau\right\} = -\frac{d}{ds} \mathcal{L}\left\{\int_0^t \sin \tau d\tau\right\} = -\frac{d}{ds} \left(\frac{1}{s} \frac{1}{s^2+1} \right) = \frac{3s^2+1}{s^2(s^2+1)^2}$
18. $\mathcal{L}\left\{t \int_0^t \tau e^{-\tau} d\tau\right\} = -\frac{d}{ds} \mathcal{L}\left\{t \int_0^t \tau e^{-\tau} d\tau\right\} = -\frac{d}{ds} \left(\frac{1}{s} \frac{1}{(s+1)^2} \right) = \frac{3s+1}{s^2(s+1)^3}$
19. (a) $\mathcal{L}^{-1}\left\{\frac{1}{s(s-1)}\right\} = \mathcal{L}^{-1}\left\{\frac{1/(s-1)}{s}\right\} = \int_0^t e^\tau d\tau = e^t - 1$
 (b) $\mathcal{L}^{-1}\left\{\frac{1}{s^2(s-1)}\right\} = \mathcal{L}^{-1}\left\{\frac{1/s(s-1)}{s}\right\} = \int_0^t (e^\tau - 1) d\tau = e^t - t - 1$
 (c) $\mathcal{L}^{-1}\left\{\frac{1}{s^3(s-1)}\right\} = \mathcal{L}^{-1}\left\{\frac{1/s^2(s-1)}{s}\right\} = \int_0^t (e^\tau - \tau - 1) d\tau = e^t - \frac{1}{2}t^2 - t - 1$
20. (a) The result in (4) is $\mathcal{L}^{-1}\{F(s)G(s)\} = f * g$, so identify

$$F(s) = \frac{2k^3}{(s^2+k^2)^2} \quad \text{and} \quad G(s) = \frac{4s}{s^2+k^2}.$$

Then

$$f(t) = \sin kt - kt \cos kt \quad \text{and} \quad g(t) = 4 \cos kt$$

so

$$\begin{aligned} \mathcal{L}^{-1}\left\{\frac{8k^3s}{(s^2+k^2)^2}\right\} &= \mathcal{L}^{-1}\{F(s)G(s)\} = f * g = 4 \int_0^t f(\tau)g(t-\tau)d\tau \\ &= 4 \int_0^t (\sin k\tau - k\tau \cos k\tau) \cos k(t-\tau)d\tau. \end{aligned}$$

Exercises 4.4

Using a CAS to evaluate the integral we get

$$\mathcal{L}^{-1}\left\{\frac{8k^3s}{(s^2+k^2)^3}\right\} = t \sin kt - kt^2 \cos kt.$$

(b) Observe from part (a) that

$$\mathcal{L}\{t(\sin kt - kt \cos kt)\} = \frac{8k^3s}{(s^2+k^2)^3},$$

and from Theorem 4.8 that $\mathcal{L}\{tf(t)\} = -F'(s)$. We saw in (5) that $\mathcal{L}\{\sin kt - kt \cos kt\} = 2k^3/(s^2+k^2)^2$, so

$$\mathcal{L}\{t(\sin kt - kt \cos kt)\} = -\frac{d}{ds} \frac{2k^3}{(s^2+k^2)^2} = \frac{8k^3s}{(s^2+k^2)^3}.$$

$$21. f(t) = -\frac{1}{t} \mathcal{L}^{-1}\left\{\frac{d}{ds} [\ln(s-3) - \ln(s+1)]\right\} = -\frac{1}{t} \mathcal{L}^{-1}\left\{\frac{1}{s-3} - \frac{1}{s+1}\right\} = -\frac{1}{t} (e^{3t} - e^{-t})$$

$$22. f(t) = -\frac{1}{t} \mathcal{L}^{-1}\left\{\frac{d}{ds} [\ln(s^2+1) - \ln(s^2+4)]\right\} = -\frac{1}{t} \mathcal{L}^{-1}\left\{\frac{2s}{s^2+1} - \frac{2s}{s^2+2^2}\right\} = -\frac{1}{t} (2 \cos t - 2 \cos 2t)$$

$$23. \mathcal{L}\{f(t)\} = \frac{1}{1-e^{-2as}} \left[\int_0^a e^{-st} dt - \int_a^{2a} e^{-st} dt \right] = \frac{(1-e^{-as})^2}{s(1-e^{-2as})} = \frac{1-e^{-as}}{s(1+e^{-as})}$$

$$24. \mathcal{L}\{f(t)\} = \frac{1}{1-e^{-2as}} \int_0^a e^{-st} dt = \frac{1}{s(1+e^{-as})}$$

$$25. \mathcal{L}\{f(t)\} = \frac{1}{1-e^{-bs}} \int_0^b \frac{a}{b} t e^{-st} dt = \frac{a}{s} \left(\frac{1}{bs} - \frac{1}{e^{bs}-1} \right)$$

$$26. \mathcal{L}\{f(t)\} = \frac{1}{1-e^{-2s}} \left[\int_0^1 t e^{-st} dt + \int_1^2 (2-t) e^{-st} dt \right] = \frac{1-e^{-s}}{s^2(1-e^{-2s})}$$

$$27. \mathcal{L}\{f(t)\} = \frac{1}{1-e^{-\pi s}} \int_0^\pi e^{-st} \sin t dt = \frac{1}{s^2+1} \cdot \frac{e^{\pi s/2} + e^{-\pi s/2}}{e^{\pi s/2} - e^{-\pi s/2}} = \frac{1}{s^2+1} \coth \frac{\pi s}{2}$$

$$28. \mathcal{L}\{f(t)\} = \frac{1}{1-e^{-2\pi s}} \int_0^\pi e^{-st} \sin t dt = \frac{1}{s^2+1} \cdot \frac{1}{1-e^{-\pi s}}$$

29. The Laplace transform of the differential equation is

$$s \mathcal{L}\{y\} + \mathcal{L}\{y\} = \frac{2s}{(s^2+1)^2}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{2s}{(s+1)(s^2+1)^2} = -\frac{1}{2} \frac{1}{s+1} - \frac{1}{2} \frac{1}{s^2+1} + \frac{1}{2} \frac{s}{s^2+1} + \frac{1}{(s^2+1)^2} + \frac{s}{(s^2+1)^2}.$$

Thus

$$\begin{aligned} y(t) &= -\frac{1}{2} e^{-t} - \frac{1}{2} \sin t + \frac{1}{2} \cos t + \frac{1}{2} (\sin t - t \cos t) + \frac{1}{2} t \sin t \\ &= -\frac{1}{2} e^{-t} + \frac{1}{2} \cos t - \frac{1}{2} t \cos t + \frac{1}{2} t \sin t. \end{aligned}$$

30. The Laplace transform of the differential equation is

$$s \mathcal{L}\{y\} - \mathcal{L}\{y\} = \frac{2(s-1)}{((s^2-1)^2+1)^2}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{2}{((s-1)^2+1)^2}.$$

Thus

$$y = e^t \sin t - te^t \cos t.$$

31. The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{y\} - sy(0) - y'(0) + 9 \mathcal{L}\{y\} = \frac{s}{s^2 + 9}.$$

Letting $y(0) = 2$ and $y'(0) = 5$ and solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{2s^3 + 5s^2 + 19s - 45}{(s^2 + 9)^2} = \frac{2s}{s^2 + 9} + \frac{5}{s^2 + 9} + \frac{s}{(s^2 + 9)^2}.$$

Thus

$$y = 2 \cos 3t + \frac{5}{3} \sin 3t + \frac{1}{6} t \sin 3t.$$

32. The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{y\} - sy(0) - y'(0) + \mathcal{L}\{y\} = \frac{1}{s^2 + 1}.$$

Solving for $\mathcal{L}\{y\}$ we obtain

$$\mathcal{L}\{y\} = \frac{s^3 - s^2 + s}{(s^2 + 1)^2} = \frac{s}{s^2 + 1} - \frac{1}{s^2 + 1} + \frac{1}{(s^2 + 1)^2}.$$

Thus

$$y = \cos t - \frac{1}{2} \sin t - \frac{1}{2} t \cos t.$$

33. The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{y\} - sy(0) - y'(0) + 16 \mathcal{L}\{y\} = \mathcal{L}\{\cos 4t - \cos 4t \mathcal{U}(t - \pi)\}$$

or

$$\begin{aligned} (s^2 + 16) \mathcal{L}\{y\} &= 1 + \frac{s}{s^2 + 16} - e^{-\pi s} \mathcal{L}\{\cos 4(t + \pi)\} \\ &= 1 + \frac{s}{s^2 + 16} - e^{-\pi s} \mathcal{L}\{\cos 4t\} \\ &= 1 + \frac{s}{s^2 + 16} - \frac{s}{s^2 + 16} e^{-\pi s}. \end{aligned}$$

Thus

$$\mathcal{L}\{y\} = \frac{1}{s^2 + 16} + \frac{s}{(s^2 + 16)^2} - \frac{s}{(s^2 + 16)^2} e^{-\pi s}$$

and

$$y = \frac{1}{4} \sin 4t + \frac{1}{8} t \sin 4t - \frac{1}{8} (t - \pi) \sin 4(t - \pi) \mathcal{U}(t - \pi).$$

34. The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{y\} - sy(0) - y'(0) + \mathcal{L}\{y\} = \mathcal{L}\left\{1 - \mathcal{U}\left(t - \frac{\pi}{2}\right) + \sin t \mathcal{U}\left(t - \frac{\pi}{2}\right)\right\}$$

or

$$\begin{aligned} (s^2 + 1) \mathcal{L}\{y\} &= s + \frac{1}{s} - \frac{1}{s} e^{-\pi s/2} + e^{-\pi s/2} \mathcal{L}\left\{\sin\left(t + \frac{\pi}{2}\right)\right\} \\ &= s + \frac{1}{s} - \frac{1}{s} e^{-\pi s/2} + e^{-\pi s/2} \mathcal{L}\{\cos t\} \\ &= s + \frac{1}{s} - \frac{1}{s} e^{-\pi s/2} + \frac{s}{s^2 + 1} e^{-\pi s/2}. \end{aligned}$$

Exercises 4.4

Thus

$$\begin{aligned}\mathcal{L}\{y\} &= \frac{s}{s^2+1} + \frac{1}{s(s^2+1)} - \frac{1}{s(s^2+1)}e^{-\pi s/2} + \frac{s}{(s^2+1)^2}e^{-\pi s/2} \\ &= \frac{s}{s^2+1} + \frac{1}{s} - \frac{s}{s^2+1} - \left(\frac{1}{s} - \frac{s}{s^2+1}\right)e^{-\pi s/2} + \frac{s}{(s^2+1)^2}e^{-\pi s/2} \\ &= \frac{1}{s} - \left(\frac{1}{s} - \frac{s}{s^2+1}\right)e^{-\pi s/2} + \frac{s}{(s^2+1)^2}e^{-\pi s/2}\end{aligned}$$

and

$$\begin{aligned}y &= 1 - \left[1 - \cos\left(t - \frac{\pi}{2}\right)\right]\mathcal{U}\left(t - \frac{\pi}{2}\right) + \frac{1}{2}\left(t - \frac{\pi}{2}\right)\sin\left(t - \frac{\pi}{2}\right)\mathcal{U}\left(t - \frac{\pi}{2}\right) \\ &= 1 - (1 - \sin t)\mathcal{U}\left(t - \frac{\pi}{2}\right) + \frac{1}{2}\left(t - \frac{\pi}{2}\right)\cos t\mathcal{U}\left(t - \frac{\pi}{2}\right).\end{aligned}$$

35. The Laplace transform of the differential equation is

$$s^2 \mathcal{L}\{y\} + \mathcal{L}\{y\} = \frac{1}{(s^2+1)} + \frac{2s}{(s^2+1)^2}.$$

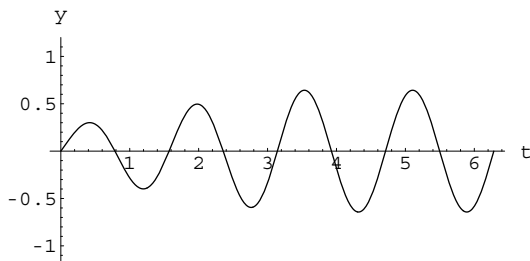
Thus

$$\mathcal{L}\{y\} = \frac{1}{(s^2+1)^2} + \frac{2s}{(s^2+1)^3}$$

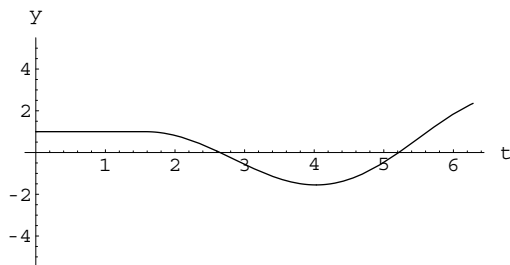
and, using Problem 20,

$$y = \frac{1}{2}(\sin t - t \cos t) + \frac{1}{4}(t \sin t - t^2 \cos t).$$

36. (a)



(b)



37. The Laplace transform of the given equation is

$$\mathcal{L}\{f\} + \mathcal{L}\{t\}\mathcal{L}\{f\} = \mathcal{L}\{t\}.$$

Solving for $\mathcal{L}\{f\}$ we obtain $\mathcal{L}\{f\} = \frac{1}{s^2+1}$. Thus, $f(t) = \sin t$.

38. The Laplace transform of the given equation is

$$\mathcal{L}\{f\} = \mathcal{L}\{2t\} - 4\mathcal{L}\{\sin t\}\mathcal{L}\{f\}.$$

Solving for $\mathcal{L}\{f\}$ we obtain

$$\mathcal{L}\{f\} = \frac{2s^2 + 2}{s^2(s^2 + 5)} = \frac{2}{5} \frac{1}{s^2} + \frac{8}{5\sqrt{5}} \frac{\sqrt{5}}{s^2 + 5}.$$

Thus

$$f(t) = \frac{2}{5}t + \frac{8}{5\sqrt{5}} \sin \sqrt{5}t.$$

39. The Laplace transform of the given equation is

$$\mathcal{L}\{f\} = \mathcal{L}\{te^t\} + \mathcal{L}\{t\} \mathcal{L}\{f\}.$$

Solving for $\mathcal{L}\{f\}$ we obtain

$$\mathcal{L}\{f\} = \frac{s^2}{(s-1)^3(s+1)} = \frac{1}{8} \frac{1}{s-1} + \frac{3}{4} \frac{1}{(s-1)^2} + \frac{1}{4} \frac{2}{(s-1)^3} - \frac{1}{8} \frac{1}{s+1}.$$

Thus

$$f(t) = \frac{1}{8}e^t + \frac{3}{4}te^t + \frac{1}{4}t^2e^t - \frac{1}{8}e^{-t}$$

40. The Laplace transform of the given equation is

$$\mathcal{L}\{f\} + 2\mathcal{L}\{\cos t\} \mathcal{L}\{f\} = 4\mathcal{L}\{e^{-t}\} + \mathcal{L}\{\sin t\}.$$

Solving for $\mathcal{L}\{f\}$ we obtain

$$\mathcal{L}\{f\} = \frac{4s^2 + s + 5}{(s+1)^3} = \frac{4}{s+1} - \frac{7}{(s+1)^2} + 4\frac{2}{(s+1)^3}.$$

Thus

$$f(t) = 4e^{-t} - 7te^{-t} + 4t^2e^{-t}.$$

41. The Laplace transform of the given equation is

$$\mathcal{L}\{f\} + \mathcal{L}\{1\} \mathcal{L}\{f\} = \mathcal{L}\{1\}.$$

Solving for $\mathcal{L}\{f\}$ we obtain $\mathcal{L}\{f\} = \frac{1}{s+1}$. Thus, $f(t) = e^{-t}$.

42. The Laplace transform of the given equation is

$$\mathcal{L}\{f\} = \mathcal{L}\{\cos t\} + \mathcal{L}\{e^{-t}\} \mathcal{L}\{f\}.$$

Solving for $\mathcal{L}\{f\}$ we obtain

$$\mathcal{L}\{f\} = \frac{s}{s^2 + 1} + \frac{1}{s^2 + 1}.$$

Thus

$$f(t) = \cos t + \sin t.$$

43. The Laplace transform of the given equation is

$$\mathcal{L}\{f\} = \mathcal{L}\{1\} + \mathcal{L}\{t\} - \frac{8}{3} \mathcal{L}\{t^3\} \mathcal{L}\{f\}.$$

Solving for $\mathcal{L}\{f\}$ we obtain

$$\mathcal{L}\{f\} = \frac{s^2(s+1)}{s^4 + 16} = \frac{s^3}{s^4 + 16} + \frac{s^2}{s^4 + 16}.$$

Thus

$$f(t) = \cos \sqrt{2}t \cosh \sqrt{2}t + \frac{1}{2\sqrt{2}}(\sin \sqrt{2}t \cosh \sqrt{2}t + \cos \sqrt{2}t \sinh \sqrt{2}t).$$

Exercises 4.4

44. The Laplace transform of the given equation is

$$\mathcal{L}\{t\} - 2\mathcal{L}\{f\} = \mathcal{L}\{e^t - e^{-t}\}\mathcal{L}\{f\}.$$

Solving for $\mathcal{L}\{f\}$ we obtain

$$\mathcal{L}\{f\} = \frac{s^2 - 1}{2s^4} = \frac{1}{2} \frac{1}{s^2} - \frac{1}{12} \frac{3!}{s^4}.$$

Thus

$$f(t) = \frac{1}{2}t - \frac{1}{12}t^3.$$

45. The Laplace transform of the given equation is

$$s\mathcal{L}\{y\} - y(0) = \mathcal{L}\{1\} - \mathcal{L}\{\sin t\} - \mathcal{L}\{1\}\mathcal{L}\{y\}.$$

Solving for $\mathcal{L}\{f\}$ we obtain

$$\mathcal{L}\{y\} = \frac{s^3 - s^2 + s}{s(s^2 + 1)^2} = \frac{1}{s^2 + 1} + \frac{1}{2} \frac{2s}{(s^2 + 1)^2}.$$

Thus

$$y = \sin t - \frac{1}{2}t \sin t.$$

46. The Laplace transform of the given equation is

$$s\mathcal{L}\{y\} - y(0) + 6\mathcal{L}\{y\} + 9\mathcal{L}\{1\}\mathcal{L}\{y\} = \mathcal{L}\{1\}.$$

Solving for $\mathcal{L}\{f\}$ we obtain $\mathcal{L}\{y\} = \frac{1}{(s+3)^2}$. Thus, $y = te^{-3t}$.

47. The differential equation is

$$0.1 \frac{di}{dt} + 3i + \frac{1}{0.05} \int_0^t i(\tau) d\tau = 100[\mathcal{U}(t-1) - \mathcal{U}(t-2)]$$

or

$$\frac{di}{dt} + 30i + 200 \int_0^t i(\tau) d\tau = 1000[\mathcal{U}(t-1) - \mathcal{U}(t-2)],$$

where $i(0) = 0$. The Laplace transform of the differential equation is

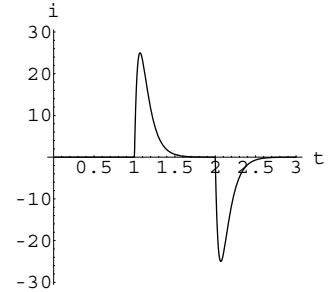
$$s\mathcal{L}\{i\} - y(0) + 30\mathcal{L}\{i\} + \frac{200}{s}\mathcal{L}\{i\} = \frac{1000}{s}(e^{-s} - e^{-2s}).$$

Solving for $\mathcal{L}\{i\}$ we obtain

$$\mathcal{L}\{i\} = \frac{1000e^{-s} - 1000e^{-2s}}{s^2 + 30s + 200} = \left(\frac{100}{s+10} - \frac{100}{s+20} \right) (e^{-s} - e^{-2s}).$$

Thus

$$i(t) = 100(e^{-10(t-1)} - e^{-20(t-1)})\mathcal{U}(t-1) - 100(e^{-10(t-2)} - e^{-20(t-2)})\mathcal{U}(t-2).$$



48. The differential equation is

$$0.005 \frac{di}{dt} + i + \frac{1}{0.02} \int_0^t i(\tau) d\tau = 100[t - (t-1)\mathcal{U}(t-1)]$$

or

$$\frac{di}{dt} + 200i + 10,000 \int_0^t i(\tau) d\tau = 20,000[t - (t-1)\mathcal{U}(t-1)],$$

where $i(0) = 0$. The Laplace transform of the differential equation is

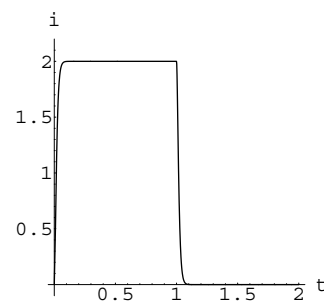
$$s \mathcal{L}\{i\} + 200 \mathcal{L}\{i\} + \frac{10,000}{s} \mathcal{L}\{i\} = 20,000 \left(\frac{1}{s^2} - \frac{1}{s^2} e^{-s} \right).$$

Solving for $\mathcal{L}\{i\}$ we obtain

$$\mathcal{L}\{i\} = \frac{20,000}{s(s+100)^2} (1 - e^{-s}) = \left[\frac{2}{s} - \frac{2}{s+100} - \frac{200}{(s+100)^2} \right] (1 - e^{-s}).$$

Thus

$$i(t) = 2 - 2e^{-100t} - 200te^{-100t} - 2\mathcal{U}(t-1) + 2e^{-100(t-1)}\mathcal{U}(t-1) + 200(t-1)e^{-100(t-1)}\mathcal{U}(t-1).$$



49. The differential equation is

$$\frac{di}{dt} + i = E(t),$$

where $i(0) = 0$. The Laplace transform of this equation is

$$s \mathcal{L}\{i\} + \mathcal{L}\{i\} = \mathcal{L}\{E(t)\}.$$

From Problem 23 we have

$$\mathcal{L}\{E(t)\} = \frac{1 - e^{-s}}{s(1 + e^{-s})}.$$

Thus

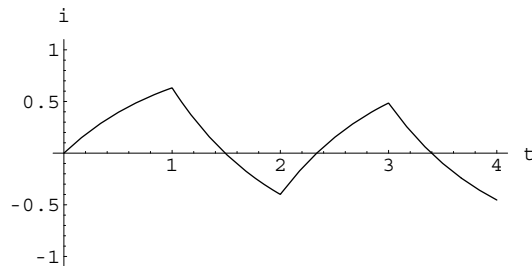
$$(s+1) \mathcal{L}\{i\} = \frac{1 - e^{-s}}{s(1 + e^{-s})}$$

and

$$\begin{aligned} \mathcal{L}\{i\} &= \frac{1 - e^{-s}}{s(s+1)(1 + e^{-s})} = \frac{1 - e^{-s}}{s(s+1)} \frac{1}{1 + e^{-s}} \\ &= \left(\frac{1}{s} - \frac{1}{s+1} \right) (1 - e^{-s})(1 - e^{-s} + e^{-2s} - e^{-3s} + e^{-4s} - \dots) \\ &= \left(\frac{1}{s} - \frac{1}{s+1} \right) (1 - 2e^{-s} + 2e^{-2s} - 2e^{-3s} + 2e^{-4s} - \dots). \end{aligned}$$

Therefore

$$\begin{aligned} i(t) &= [1 - 2\mathcal{U}(t-1) + 2\mathcal{U}(t-2) - 2\mathcal{U}(t-3) + 2\mathcal{U}(t-4) - \dots] \\ &\quad - [e^{-t} + 2e^{-(t-1)}\mathcal{U}(t-1) - 2e^{-(t-2)}\mathcal{U}(t-2) \\ &\quad + 2e^{-(t-3)}\mathcal{U}(t-3) - 2e^{-(t-4)}\mathcal{U}(t-4) + \dots] \\ &= 1 - e^{-t} + 2 \sum_{n=1}^{\infty} (-1)^n (1 - e^{-(t-n)}) \mathcal{U}(t-n). \end{aligned}$$



Exercises 4.4

50. The differential equation is

$$\frac{di}{dt} + i = E(t),$$

where $i(0) = 0$. The Laplace transform of this equation is

$$s \mathcal{L}\{i\} + \mathcal{L}\{i\} = \mathcal{L}\{E(t)\}.$$

From Problem 25 we have

$$\mathcal{L}\{E(t)\} = \frac{1}{s} \left(\frac{1}{s} - \frac{1}{e^s - 1} \right) = \frac{1}{s^2} - \frac{1}{s} \frac{1}{e^s - 1}.$$

Thus

$$(s+1) \mathcal{L}\{i\} = \frac{1}{s^2} - \frac{1}{s} \frac{1}{e^s - 1}$$

and

$$\begin{aligned} \mathcal{L}\{i\} &= \frac{1}{s^2(s+1)} - \frac{1}{s(s+1)} \frac{1}{e^s - 1} \\ &= \left(\frac{1}{s^2} - \frac{1}{s} + \frac{1}{s+1} \right) - \left(\frac{1}{s} - \frac{1}{s+1} \right) \frac{1}{e^s - 1} \\ &= \left(\frac{1}{s^2} - \frac{1}{s} + \frac{1}{s+1} \right) - \left(\frac{1}{s} - \frac{1}{s+1} \right) (e^{-s} + e^{-2s} + e^{-3s} + e^{-4s} + \dots). \end{aligned}$$

Therefore

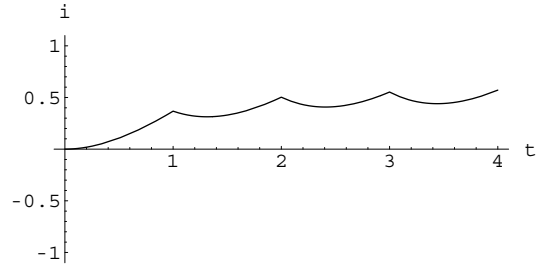
$$\begin{aligned} i(t) &= (t-1+e^{-t}) - (1-e^{-(t-1)}) \mathcal{U}(t-1) - (1-e^{-(t-2)}) \mathcal{U}(t-2) \\ &\quad - (1-e^{-(t-3)}) \mathcal{U}(t-3) - (1-e^{-(t-4)}) \mathcal{U}(t-4) - \dots \\ &= (t-1+e^{-t}) - \sum_{n=1}^{\infty} (1-e^{-(t-n)}) \mathcal{U}(t-n). \end{aligned}$$

51. The differential equation is $x'' + 2x' + 10x = 20f(t)$, where $f(t)$ is the meander function with $a = \pi$. Using the initial conditions $x(0) = x'(0) = 0$ and taking the Laplace transform we obtain

$$\begin{aligned} (s^2 + 2s + 10) \mathcal{L}\{x(t)\} &= \frac{20}{s} (1 - e^{-\pi s}) \frac{1}{1 + e^{-\pi s}} \\ &= \frac{20}{s} (1 - e^{-\pi s}) (1 - e^{-\pi s} + e^{-2\pi s} - e^{-3\pi s} + \dots) \\ &= \frac{20}{s} (1 - 2e^{-\pi s} + 2e^{-2\pi s} - 2e^{-3\pi s} + \dots) \\ &= \frac{20}{s} + \frac{40}{s} \sum_{n=1}^{\infty} (-1)^n e^{-n\pi s}. \end{aligned}$$

Then

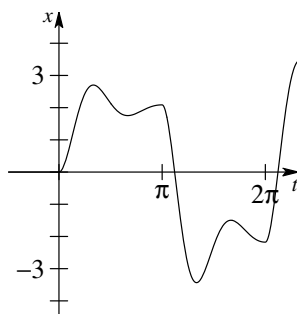
$$\begin{aligned} \mathcal{L}\{x(t)\} &= \frac{20}{s(s^2 + 2s + 10)} + \frac{40}{s(s^2 + 2s + 10)} \sum_{n=1}^{\infty} (-1)^n e^{-n\pi s} \\ &= \frac{2}{s} - \frac{2s+4}{s^2 + 2s + 10} + \sum_{n=1}^{\infty} (-1)^n \left[\frac{4}{s} - \frac{4s+8}{s^2 + 2s + 10} \right] e^{-n\pi s} \\ &= \frac{2}{s} - \frac{2(s+1)+2}{(s+1)^2 + 9} + 4 \sum_{n=1}^{\infty} (-1)^n \left[\frac{1}{s} - \frac{(s+1)+1}{(s+1)^2 + 9} \right] e^{-n\pi s} \end{aligned}$$



and

$$x(t) = 2 \left(1 - e^{-t} \cos 3t - \frac{1}{3} e^{-t} \sin 3t \right) + 4 \sum_{n=1}^{\infty} (-1)^n \left[1 - e^{-(t-n\pi)} \cos 3(t-n\pi) - \frac{1}{3} e^{-(t-n\pi)} \sin 3(t-n\pi) \right] \mathcal{U}(t-n\pi).$$

The graph of $x(t)$ on the interval $[0, 2\pi)$ is shown below.



52. The differential equation is $x'' + 2x' + x = 5f(t)$, where $f(t)$ is the square wave function with $a = \pi$. Using the initial conditions $x(0) = x'(0) = 0$ and taking the Laplace transform, we obtain

$$\begin{aligned} (s^2 + 2s + 1) \mathcal{L}\{x(t)\} &= \frac{5}{s} \frac{1}{1 + e^{-\pi s}} = \frac{5}{s} (1 - e^{-\pi s} + e^{-2\pi s} - e^{-3\pi s} + e^{-4\pi s} - \dots) \\ &= \frac{5}{s} \sum_{n=0}^{\infty} (-1)^n e^{-n\pi s}. \end{aligned}$$

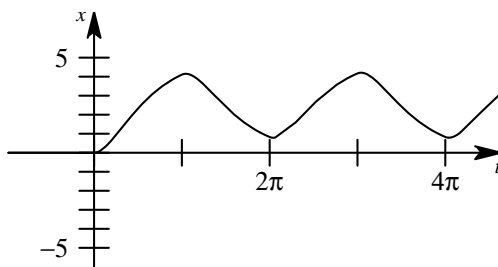
Then

$$\mathcal{L}\{x(t)\} = \frac{5}{s(s+1)^2} \sum_{n=0}^{\infty} (-1)^n e^{-n\pi s} = 5 \sum_{n=0}^{\infty} (-1)^n \left(\frac{1}{s} - \frac{1}{s+1} - \frac{1}{(s+1)^2} \right) e^{-n\pi s}$$

and

$$x(t) = 5 \sum_{n=0}^{\infty} (-1)^n (1 - e^{-(t-n\pi)} - (t-n\pi)e^{-(t-n\pi)}) \mathcal{U}(t-n\pi).$$

The graph of $x(t)$ on the interval $[0, 4\pi)$ is shown below.



53. Let $u = t - \tau$ so that $du = d\tau$ and

$$f * g = \int_0^t f(\tau)g(t-\tau) d\tau = - \int_t^0 f(t-u)g(u) du = g * f.$$

54.
$$\begin{aligned} f * (g + h) &= \int_0^t f(\tau)[g(t-\tau) + h(t-\tau)] d\tau = \int_0^t f(\tau)g(t-\tau) d\tau + \int_0^t f(\tau)h(t-\tau) d\tau \\ &= \int_0^t f(\tau)[g(t-\tau) + h(t-\tau)] d\tau = f * g + f * h \end{aligned}$$

Exercises 4.4

55. (a) The output for the first three lines of the program are

$$\begin{aligned} 9y[t] + 6y'[t] + y''[t] &== t \sin[t] \\ 1 - 2s + 9Y + s^2Y + 6(-2 + sY) &== \frac{2s}{(1 + s^2)^2} \\ Y &\rightarrow -\left(\frac{-11 - 4s - 22s^2 - 4s^3 - 11s^4 - 2s^5}{(1 + s^2)^2(9 + 6s + s^2)}\right) \end{aligned}$$

The fourth line is the same as the third line with $Y \rightarrow$ removed. The final line of output shows a solution involving complex coefficients of e^{it} and e^{-it} . To get the solution in more standard form write the last line as two lines:

```
euler = {E^(I t) -> Cos[t] + I Sin[t], E^(-I t) -> Cos[t] - I Sin[t]}
InverseLaplaceTransform[Y,s,t]/.euler//Expand
```

We see that the solution is

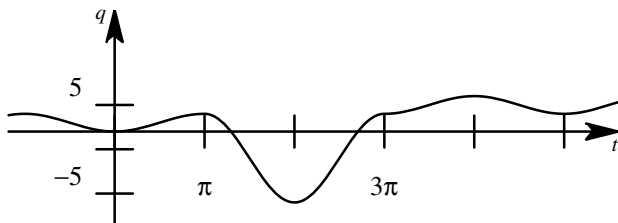
$$y(t) = \left(\frac{487}{250} + \frac{247}{50}t\right)e^{-3t} + \frac{1}{250}(13 \cos t - 15t \cos t - 9 \sin t + 20t \sin t).$$

- (b) The solution is

$$y(t) = \frac{1}{6}e^t - \frac{1}{6}e^{-t/2} \cos \sqrt{15}t - \frac{\sqrt{3/5}}{6}e^{-t/2} \sin \sqrt{15}t.$$

- (c) The solution is

$$q(t) = 1 - \cos t + (6 - 6 \cos t) \mathcal{U}(t - 3\pi) - (4 + 4 \cos t) \mathcal{U}(t - \pi).$$



Exercises 4.5

1. The Laplace transform of the differential equation yields

$$\mathcal{L}\{y\} = \frac{1}{s-3}e^{-2s}$$

so that

$$y = e^{3(t-2)} \mathcal{U}(t-2).$$

2. The Laplace transform of the differential equation yields

$$\mathcal{L}\{y\} = \frac{2}{s+1} + \frac{e^{-s}}{s+1}$$

so that

$$y = 2e^{-t} + e^{-(t-1)} \mathcal{U}(t-1).$$

3. The Laplace transform of the differential equation yields

$$\mathcal{L}\{y\} = \frac{1}{s^2+1}(1 + e^{-2\pi s})$$

so that

$$y = \sin t + \sin t \mathcal{U}(t - 2\pi).$$

4. The Laplace transform of the differential equation yields

$$\mathcal{L}\{y\} = \frac{1}{4} \frac{4}{s^2 + 16} e^{-2\pi s}$$

so that

$$y = \frac{1}{4} \sin 4(t - 2\pi) \mathcal{U}(t - 2\pi).$$

5. The Laplace transform of the differential equation yields

$$\mathcal{L}\{y\} = \frac{1}{s^2 + 1} (e^{-\pi s/2} + e^{-3\pi s/2})$$

so that

$$\begin{aligned} y &= \sin\left(t - \frac{\pi}{2}\right) \mathcal{U}\left(t - \frac{\pi}{2}\right) + \sin\left(t - \frac{3\pi}{2}\right) \mathcal{U}\left(t - \frac{3\pi}{2}\right) \\ &= -\cos t \mathcal{U}\left(t - \frac{\pi}{2}\right) + \cos t \mathcal{U}\left(t - \frac{3\pi}{2}\right). \end{aligned}$$

6. The Laplace transform of the differential equation yields

$$\mathcal{L}\{y\} = \frac{s}{s^2 + 1} + \frac{1}{s^2 + 1} (e^{-2\pi s} + e^{-4\pi s})$$

so that

$$y = \cos t + \sin t [\mathcal{U}(t - 2\pi) + \mathcal{U}(t - 4\pi)].$$

7. The Laplace transform of the differential equation yields

$$\mathcal{L}\{y\} = \frac{1}{s^2 + 2s} (1 + e^{-s}) = \left[\frac{1}{2} \frac{1}{s} - \frac{1}{2} \frac{1}{s + 2} \right] (1 + e^{-s})$$

so that

$$y = \frac{1}{2} - \frac{1}{2} e^{-2t} + \left[\frac{1}{2} - \frac{1}{2} e^{-2(t-1)} \right] \mathcal{U}(t - 1).$$

8. The Laplace transform of the differential equation yields

$$\mathcal{L}\{y\} = \frac{s+1}{s^2(s-2)} + \frac{1}{s(s-2)} e^{-2s} = \frac{3}{4} \frac{1}{s-2} - \frac{3}{4} \frac{1}{s} - \frac{1}{2} \frac{1}{s^2} + \left[\frac{1}{2} \frac{1}{s-2} - \frac{1}{2} \frac{1}{s} \right] e^{-2s}$$

so that

$$y = \frac{3}{4} e^{2t} - \frac{3}{4} - \frac{1}{2} t + \left[\frac{1}{2} e^{2(t-2)} - \frac{1}{2} \right] \mathcal{U}(t - 2).$$

9. The Laplace transform of the differential equation yields

$$\mathcal{L}\{y\} = \frac{1}{(s+2)^2 + 1} e^{-2\pi s}$$

so that

$$y = e^{-2(t-2\pi)} \sin t \mathcal{U}(t - 2\pi).$$

10. The Laplace transform of the differential equation yields

$$\mathcal{L}\{y\} = \frac{1}{(s+1)^2} e^{-s}$$

so that

$$y = (t-1) e^{-(t-1)} \mathcal{U}(t-1).$$

Exercises 4.5

11. The Laplace transform of the differential equation yields

$$\begin{aligned}\mathcal{L}\{y\} &= \frac{4+s}{s^2+4s+13} + \frac{e^{-\pi s} + e^{-3\pi s}}{s^2+4s+13} \\ &= \frac{2}{3} \frac{3}{(s+2)^2+3^2} + \frac{s+2}{(s+2)^2+3^2} + \frac{1}{3} \frac{3}{(s+2)^2+3^2} (e^{-\pi s} + e^{-3\pi s})\end{aligned}$$

so that

$$\begin{aligned}y &= \frac{2}{3} e^{-2t} \sin 3t + e^{-2t} \cos 3t + \frac{1}{3} e^{-2(t-\pi)} \sin 3(t-\pi) \mathcal{U}(t-\pi) \\ &\quad + \frac{1}{3} e^{-2(t-3\pi)} \sin 3(t-3\pi) \mathcal{U}(t-3\pi).\end{aligned}$$

12. The Laplace transform of the differential equation yields

$$\begin{aligned}\mathcal{L}\{y\} &= \frac{1}{(s-1)^2(s-6)} + \frac{e^{-2s} + e^{-4s}}{(s-1)(s-6)} \\ &= -\frac{1}{25} \frac{1}{s-1} - \frac{1}{5} \frac{1}{(s-1)^2} + \frac{1}{25} \frac{1}{s-6} + \left[-\frac{1}{5} \frac{1}{s-1} + \frac{1}{5} \frac{1}{s-6} \right] (e^{-2s} + e^{-4s})\end{aligned}$$

so that

$$\begin{aligned}y &= -\frac{1}{25} e^t - \frac{1}{5} t e^t + \frac{1}{25} e^{6t} + \left[-\frac{1}{5} e^{t-2} + \frac{1}{5} e^{6(t-2)} \right] \mathcal{U}(t-2) \\ &\quad + \left[-\frac{1}{5} e^{t-4} + \frac{1}{5} e^{6(t-4)} \right] \mathcal{U}(t-4).\end{aligned}$$

13. The Laplace transform of the differential equation yields

$$\mathcal{L}\{y\} = \frac{1}{2} \frac{2}{s^3} y''(0) + \frac{1}{6} \frac{3!}{s^4} y'''(0) + \frac{1}{6} \frac{P_0}{EI} \frac{3!}{s^4} e^{-Ls/2}$$

so that

$$y = \frac{1}{2} y''(0) x^2 + \frac{1}{6} y'''(0) x^3 + \frac{1}{6} \frac{P_0}{EI} \left(X - \frac{L}{2} \right)^3 \mathcal{U}\left(x - \frac{L}{2} \right).$$

Using $y''(L) = 0$ and $y'''(L) = 0$ we obtain

$$\begin{aligned}y &= \frac{1}{4} \frac{P_0 L}{EI} x^2 - \frac{1}{6} \frac{P_0}{EI} x^3 + \frac{1}{6} \frac{P_0}{EI} \left(x - \frac{L}{2} \right)^3 \mathcal{U}\left(x - \frac{L}{2} \right) \\ &= \begin{cases} \frac{P_0}{EI} \left(\frac{L}{4} x^2 - \frac{1}{6} x^3 \right), & 0 \leq x < \frac{L}{2} \\ \frac{P_0 L^2}{4EI} \left(\frac{1}{2} x - \frac{L}{12} \right), & \frac{L}{2} \leq x \leq L. \end{cases}\end{aligned}$$

14. From Problem 13 we know that

$$y = \frac{1}{2} y''(0) x^2 + \frac{1}{6} y'''(0) x^3 + \frac{1}{6} \frac{P_0}{EI} \left(X - \frac{L}{2} \right)^3 \mathcal{U}\left(x - \frac{L}{2} \right).$$

Using $y(L) = 0$ and $y'(L) = 0$ we obtain

$$\begin{aligned}y &= \frac{1}{16} \frac{P_0 L}{EI} x^2 - \frac{1}{12} \frac{P_0}{EI} x^3 + \frac{1}{6} \frac{P_0}{EI} \left(x - \frac{L}{2} \right)^3 \mathcal{U}\left(x - \frac{L}{2} \right) \\ &= \begin{cases} \frac{P_0}{EI} \left(\frac{L}{16} x^2 - \frac{1}{12} x^3 \right), & 0 \leq x < \frac{L}{2} \\ \frac{P_0}{EI} \left(\frac{L}{16} x^2 - \frac{1}{12} x^3 \right) + \frac{1}{6} \frac{P_0}{EI} \left(x - \frac{L}{2} \right)^3, & \frac{L}{2} \leq x \leq L. \end{cases}\end{aligned}$$

15. The Laplace transform of the differential equation yields

$$\mathcal{L}\{y\} = \frac{1}{s^2 + \omega^2}$$

so that $y(t) = \sin \omega t$. Note that $y'(0) = 1$, even though the initial condition was $y'(0) = 0$.

Exercises 4.6

1. Taking the Laplace transform of the system gives

$$\begin{aligned} s \mathcal{L}\{x\} &= -\mathcal{L}\{x\} + \mathcal{L}\{y\} \\ s \mathcal{L}\{y\} - 1 &= 2 \mathcal{L}\{x\} \end{aligned}$$

so that

$$\mathcal{L}\{x\} = \frac{1}{(s-1)(s+2)} = \frac{1}{3} \frac{1}{s-1} - \frac{1}{3} \frac{1}{s+2}$$

and

$$\mathcal{L}\{y\} = \frac{1}{s} + \frac{2}{s(s-1)(s+2)} = \frac{2}{3} \frac{1}{s-1} + \frac{1}{3} \frac{1}{s+2}.$$

Then

$$x = \frac{1}{3}e^t - \frac{1}{3}e^{-2t} \quad \text{and} \quad y = \frac{2}{3}e^t + \frac{1}{3}e^{-2t}.$$

2. Taking the Laplace transform of the system gives

$$\begin{aligned} s \mathcal{L}\{x\} - 1 &= 2 \mathcal{L}\{y\} + \frac{1}{s-1} \\ s \mathcal{L}\{y\} - 1 &= 8 \mathcal{L}\{x\} - \frac{1}{s^2} \end{aligned}$$

so that

$$\mathcal{L}\{y\} = \frac{s^3 + 7s^2 - s + 1}{s(s-1)(s^2-16)} = \frac{1}{16} \frac{1}{s} - \frac{8}{15} \frac{1}{s-1} + \frac{173}{96} \frac{1}{s-4} - \frac{53}{160} \frac{1}{s+4}$$

and

$$y = \frac{1}{16} - \frac{8}{15}e^t + \frac{173}{96}e^{4t} - \frac{53}{160}e^{-4t}.$$

Then

$$x = \frac{1}{8}y' + \frac{1}{8}t = \frac{1}{8}t - \frac{1}{15}e^t + \frac{173}{192}e^{4t} + \frac{53}{320}e^{-4t}.$$

3. Taking the Laplace transform of the system gives

$$\begin{aligned} s \mathcal{L}\{x\} + 1 &= \mathcal{L}\{x\} - 2 \mathcal{L}\{y\} \\ s \mathcal{L}\{y\} - 2 &= 5 \mathcal{L}\{x\} - \mathcal{L}\{y\} \end{aligned}$$

so that

$$\mathcal{L}\{x\} = \frac{-s-5}{s^2+9} = -\frac{s}{s^2+9} - \frac{5}{3} \frac{3}{s^2+9}$$

and

$$x = -\cos 3t - \frac{5}{3} \sin 3t.$$

Then

$$y = \frac{1}{2}x - \frac{1}{2}x' = 2 \cos 3t - \frac{7}{3} \sin 3t.$$

4. Taking the Laplace transform of the system gives

$$\begin{aligned} (s+3) \mathcal{L}\{x\} + s \mathcal{L}\{y\} &= \frac{1}{s} \\ (s-1) \mathcal{L}\{x\} + (s-1) \mathcal{L}\{y\} &= \frac{1}{s-1} \end{aligned}$$

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so that
$$\mathcal{L}\{y\} = \frac{5s-1}{3s(s-1)^2} = -\frac{1}{3} \frac{1}{s} + \frac{1}{3} \frac{1}{s-1} + \frac{4}{3} \frac{1}{(s-1)^2}$$

and
$$\mathcal{L}\{x\} = \frac{1-2s}{3s(s-1)^2} = \frac{1}{3} \frac{1}{s} - \frac{1}{3} \frac{1}{s-1} - \frac{1}{3} \frac{1}{(s-1)^2}.$$

Then
$$x = \frac{1}{3} - \frac{1}{3}e^t - \frac{1}{3}te^t \quad \text{and} \quad y = -\frac{1}{3} + \frac{1}{3}e^t + \frac{4}{3}te^t.$$

5. Taking the Laplace transform of the system gives

$$\begin{aligned}(2s-2)\mathcal{L}\{x\} + s\mathcal{L}\{y\} &= \frac{1}{s} \\ (s-3)\mathcal{L}\{x\} + (s-3)\mathcal{L}\{y\} &= \frac{2}{s}\end{aligned}$$

so that
$$\mathcal{L}\{x\} = \frac{-s-3}{s(s-2)(s-3)} = -\frac{1}{2} \frac{1}{s} + \frac{5}{2} \frac{1}{s-2} - \frac{2}{s-3}$$

and
$$\mathcal{L}\{y\} = \frac{3s-1}{s(s-2)(s-3)} = -\frac{1}{6} \frac{1}{s} - \frac{5}{2} \frac{1}{s-2} + \frac{8}{3} \frac{1}{s-3}.$$

Then
$$x = -\frac{1}{2} + \frac{5}{2}e^{2t} - 2e^{3t} \quad \text{and} \quad y = -\frac{1}{6} - \frac{5}{2}e^{2t} + \frac{8}{3}e^{3t}.$$

6. Taking the Laplace transform of the system gives

$$\begin{aligned}(s+1)\mathcal{L}\{x\} - (s-1)\mathcal{L}\{y\} &= -1 \\ s\mathcal{L}\{x\} + (s+2)\mathcal{L}\{y\} &= 1\end{aligned}$$

so that
$$\mathcal{L}\{y\} = \frac{s+1/2}{s^2+s+1} = \frac{s+1/2}{(s+1/2)^2 + (\sqrt{3}/2)^2}$$

and
$$\mathcal{L}\{x\} = \frac{-3/2}{s^2+s+1} = \frac{-3/2}{(s+1/2)^2 + (\sqrt{3}/2)^2}.$$

Then
$$y = e^{-t/2} \cos \frac{\sqrt{3}}{2}t \quad \text{and} \quad x = e^{-t/2} \sin \frac{\sqrt{3}}{2}t.$$

7. Taking the Laplace transform of the system gives

$$\begin{aligned}(s^2+1)\mathcal{L}\{x\} - \mathcal{L}\{y\} &= -2 \\ -\mathcal{L}\{x\} + (s^2+1)\mathcal{L}\{y\} &= 1\end{aligned}$$

so that
$$\mathcal{L}\{x\} = \frac{-2s^2-1}{s^4+2s^2} = -\frac{1}{2} \frac{1}{s^2} - \frac{3}{2} \frac{1}{s^2+2}$$

and
$$x = -\frac{1}{2}t - \frac{3}{2\sqrt{2}} \sin \sqrt{2}t.$$

Then
$$y = x'' + x = -\frac{1}{2}t + \frac{3}{2\sqrt{2}} \sin \sqrt{2}t.$$

8. Taking the Laplace transform of the system gives

$$\begin{aligned}(s+1)\mathcal{L}\{x\} + \mathcal{L}\{y\} &= 1 \\ 4\mathcal{L}\{x\} - (s+1)\mathcal{L}\{y\} &= 1\end{aligned}$$

so that
$$\mathcal{L}\{x\} = \frac{s+2}{s^2+2s+5} = \frac{s+1}{(s+1)^2+2^2} + \frac{1}{2} \frac{2}{(s+1)^2+2^2}$$

and
$$\mathcal{L}\{y\} = \frac{-s+3}{s^2+2s+5} = -\frac{s+1}{(s+1)^2+2^2} + 2 \frac{2}{(s+1)^2+2^2}.$$

Then
$$x = e^{-t} \cos 2t + \frac{1}{2} e^{-t} \sin 2t \quad \text{and} \quad y = -e^{-t} \cos 2t + 2e^{-t} \sin 2t.$$

9. Adding the equations and then subtracting them gives

$$\frac{d^2x}{dt^2} = \frac{1}{2}t^2 + 2t$$

$$\frac{d^2y}{dt^2} = \frac{1}{2}t^2 - 2t.$$

Taking the Laplace transform of the system gives

and
$$\mathcal{L}\{x\} = 8 \frac{1}{s} + \frac{1}{24} \frac{4!}{s^5} + \frac{1}{3} \frac{3!}{s^4}$$

$$\mathcal{L}\{y\} = \frac{1}{24} \frac{4!}{s^5} - \frac{1}{3} \frac{3!}{s^4}$$

so that
$$x = 8 + \frac{1}{24}t^4 + \frac{1}{3}t^3 \quad \text{and} \quad y = \frac{1}{24}t^4 - \frac{1}{3}t^3.$$

10. Taking the Laplace transform of the system gives

$$(s-4) \mathcal{L}\{x\} + s^3 \mathcal{L}\{y\} = \frac{6}{s^2+1}$$

$$(s+2) \mathcal{L}\{x\} - 2s^3 \mathcal{L}\{y\} = 0$$

so that
$$\mathcal{L}\{x\} = \frac{4}{(s-2)(s^2+1)} = \frac{4}{5} \frac{1}{s-2} - \frac{4}{5} \frac{s}{s^2+1} - \frac{8}{5} \frac{1}{s^2+1}$$

and
$$\mathcal{L}\{y\} = \frac{2s+4}{s^3(s-2)(s^2+1)} = \frac{1}{s} - \frac{2}{s^2} - 2 \frac{2}{s^3} + \frac{1}{5} \frac{1}{s-2} - \frac{6}{5} \frac{s}{s^2+1} + \frac{8}{5} \frac{1}{s^2+1}.$$

Then
$$x = \frac{4}{5}e^{2t} - \frac{4}{5} \cos t - \frac{8}{5} \sin t$$

and
$$y = 1 - 2t - 2t^2 + \frac{1}{5}e^{2t} - \frac{6}{5} \cos t + \frac{8}{5} \sin t.$$

11. Taking the Laplace transform of the system gives

$$s^2 \mathcal{L}\{x\} + 3(s+1) \mathcal{L}\{y\} = 2$$

$$s^2 \mathcal{L}\{x\} + 3 \mathcal{L}\{y\} = \frac{1}{(s+1)^2}$$

so that
$$\mathcal{L}\{x\} = -\frac{2s+1}{s^3(s+1)} = \frac{1}{s} + \frac{1}{s^2} + \frac{1}{2} \frac{2}{s^3} - \frac{1}{s+1}.$$

Then
$$x = 1 + t + \frac{1}{2}t^2 - e^{-t}$$

and
$$y = \frac{1}{3}te^{-t} - \frac{1}{3}x'' = \frac{1}{3}te^{-t} + \frac{1}{3}e^{-t} - \frac{1}{3}.$$

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12. Taking the Laplace transform of the system gives

$$\begin{aligned}(s-4)\mathcal{L}\{x\} + 2\mathcal{L}\{y\} &= \frac{2e^{-s}}{s} \\ -3\mathcal{L}\{x\} + (s+1)\mathcal{L}\{y\} &= \frac{1}{2} + \frac{e^{-s}}{s}\end{aligned}$$

so that

$$\begin{aligned}\mathcal{L}\{x\} &= \frac{-1/2}{(s-1)(s-2)} + e^{-s} \frac{1}{(s-1)(s-2)} \\ &= \left[\frac{1}{2} \frac{1}{s-1} - \frac{1}{2} \frac{1}{s-2} \right] + e^{-s} \left[-\frac{1}{s-1} + \frac{1}{s-2} \right]\end{aligned}$$

and

$$\begin{aligned}\mathcal{L}\{y\} &= \frac{e^{-s}}{s} + \frac{s/4-1}{(s-1)(s-2)} + e^{-s} \frac{-s/2+2}{(s-1)(s-2)} \\ &= \frac{3}{4} \frac{1}{s-1} - \frac{1}{2} \frac{1}{s-2} + e^{-s} \left[\frac{1}{s} - \frac{3}{2} \frac{1}{s-1} + \frac{1}{s-2} \right].\end{aligned}$$

Then

$$x = \frac{1}{2}e^t - \frac{1}{2}e^{2t} + \left[-e^{t-1} + e^{2(t-1)} \right] \mathcal{U}(t-1)$$

and

$$y = \frac{3}{4}e^t - \frac{1}{2}e^{2t} + \left[1 - \frac{3}{2}e^{t-1} + e^{2(t-1)} \right] \mathcal{U}(t-1).$$

13. The system is

$$\begin{aligned}x_1'' &= -3x_1 + 2(x_2 - x_1) \\ x_2'' &= -2(x_2 - x_1) \\ x_1(0) &= 0 \\ x_1'(0) &= 1 \\ x_2(0) &= 1 \\ x_2'(0) &= 0.\end{aligned}$$

Taking the Laplace transform of the system gives

$$\begin{aligned}(s^2+5)\mathcal{L}\{x_1\} - 2\mathcal{L}\{x_2\} &= 1 \\ -2\mathcal{L}\{x_1\} + (s^2+2)\mathcal{L}\{x_2\} &= s\end{aligned}$$

so that

$$\mathcal{L}\{x_1\} = \frac{s^2+2s+2}{s^4+7s^2+6} = \frac{2}{5} \frac{s}{s^2+1} + \frac{1}{5} \frac{1}{s^2+1} - \frac{2}{5} \frac{s}{s^2+6} + \frac{4}{5\sqrt{6}} \frac{\sqrt{6}}{s^2+6}$$

and

$$\mathcal{L}\{x_2\} = \frac{s^3+5s+2}{(s^2+1)(s^2+6)} = \frac{4}{5} \frac{s}{s^2+1} + \frac{2}{5} \frac{1}{s^2+1} + \frac{1}{5} \frac{s}{s^2+6} - \frac{2}{5\sqrt{6}} \frac{\sqrt{6}}{s^2+6}.$$

Then

$$x_1 = \frac{2}{5} \cos t + \frac{1}{5} \sin t - \frac{2}{5} \cos \sqrt{6}t + \frac{4}{5\sqrt{6}} \sin \sqrt{6}t$$

and

$$x_2 = \frac{4}{5} \cos t + \frac{2}{5} \sin t + \frac{1}{5} \cos \sqrt{6}t - \frac{2}{5\sqrt{6}} \sin \sqrt{6}t.$$

14. In this system x_1 and x_2 represent displacements of masses m_1 and m_2 from their equilibrium positions. Since the net forces acting on m_1 and m_2 are

$$-k_1x_1 + k_2(x_2 - x_1) \quad \text{and} \quad -k_2(x_2 - x_1) - k_3x_2,$$

respectively, Newton's second law of motion gives

$$m_1 x_1'' = -k_1 x_1 + k_2(x_2 - x_1)$$

$$m_2 x_2'' = -k_2(x_2 - x_1) - k_3 x_2.$$

Using $k_1 = k_2 = k_3 = 1$, $m_1 = m_2 = 1$, $x_1(0) = 0$, $x_1'(0) = -1$, $x_2(0) = 0$, and $x_2'(0) = 1$, and taking the Laplace transform of the system, we obtain

$$(2 + s^2) \mathcal{L}\{x_1\} - \mathcal{L}\{x_2\} = -1$$

$$\mathcal{L}\{x_1\} - (2 + s^2) \mathcal{L}\{x_2\} = -1$$

so that
$$\mathcal{L}\{x_1\} = -\frac{1}{s^2 + 3} \quad \text{and} \quad \mathcal{L}\{x_2\} = \frac{1}{s^2 + 3}.$$

Then
$$x_1 = -\frac{1}{\sqrt{3}} \sin \sqrt{3} t \quad \text{and} \quad x_2 = \frac{1}{\sqrt{3}} \sin \sqrt{3} t.$$

15. (a) By Kirchoff's first law we have $i_1 = i_2 + i_3$. By Kirchoff's second law, on each loop we have $E(t) = Ri_1 + L_1 i_2'$ and $E(t) = Ri_1 + L_2 i_3'$ or $L_1 i_2' + Ri_2 + Ri_3 = E(t)$ and $L_2 i_3' + Ri_2 + Ri_3 = E(t)$.

- (b) Taking the Laplace transform of the system

$$0.01 i_2' + 5i_2 + 5i_3 = 100$$

$$0.0125 i_3' + 5i_2 + 5i_3 = 100$$

gives
$$(s + 500) \mathcal{L}\{i_2\} + 500 \mathcal{L}\{i_3\} = \frac{10,000}{s}$$

$$400 \mathcal{L}\{i_2\} + (s + 400) \mathcal{L}\{i_3\} = \frac{8,000}{s}$$

so that
$$\mathcal{L}\{i_3\} = \frac{8,000}{s^2 + 900s} = \frac{80}{9} \frac{1}{s} - \frac{80}{9} \frac{1}{s + 900}.$$

Then

$$i_3 = \frac{80}{9} - \frac{80}{9} e^{-900t} \quad \text{and} \quad i_2 = 20 - 0.0025 i_3' - i_3 = \frac{100}{9} - \frac{100}{9} e^{-900t}.$$

- (c) $i_1 = i_2 + i_3 = 20 - 20e^{-900t}$

16. (a) Taking the Laplace transform of the system

$$i_2' + i_3' + 10i_2 = 120 - 120 \mathcal{U}(t - 2)$$

$$-10i_2' + 5i_3' + 5i_3 = 0$$

gives
$$(s + 10) \mathcal{L}\{i_2\} + s \mathcal{L}\{i_3\} = \frac{120}{s} (1 - e^{-2s})$$

$$-10s \mathcal{L}\{i_2\} + 5(s + 1) \mathcal{L}\{i_3\} = 0$$

so that

$$\mathcal{L}\{i_2\} = \frac{120(s + 1)}{(3s^2 + 11s + 10)s} (1 - e^{-2s}) = \left[\frac{48}{s + 5/3} - \frac{60}{s + 2} + \frac{12}{s} \right] (1 - e^{-2s})$$

and

$$\mathcal{L}\{i_3\} = \frac{240}{3s^2 + 11s + 10} (1 - e^{-2s}) = \left[\frac{240}{s + 5/3} - \frac{240}{s + 2} \right] (1 - e^{-2s}).$$

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Then $i_2 = 12 + 48e^{-5t/3} - 60e^{-2t} - \left[12 + 48e^{-5(t-2)/3} - 60e^{-2(t-2)}\right] \mathcal{U}(t-2)$

and $i_3 = 240e^{-5t/3} - 240e^{-2t} - \left[240e^{-5(t-2)/3} - 240e^{-2(t-2)}\right] \mathcal{U}(t-2).$

(b) $i_1 = i_2 + i_3 = 12 + 288e^{-5t/3} - 300e^{-2t} - \left[12 + 288e^{-5(t-2)/3} - 300e^{-2(t-2)}\right] \mathcal{U}(t-2)$

17. Taking the Laplace transform of the system

$$i_2' + 11i_2 + 6i_3 = 50 \sin t$$

$$i_3' + 6i_2 + 6i_3 = 50 \sin t$$

gives

$$(s+11) \mathcal{L}\{i_2\} + 6 \mathcal{L}\{i_3\} = \frac{50}{s^2+1}$$

$$6 \mathcal{L}\{i_2\} + (s+6) \mathcal{L}\{i_3\} = \frac{50}{s^2+1}$$

so that

$$\mathcal{L}\{i_2\} = \frac{50s}{(s+2)(s+15)(s^2+1)} = -\frac{20}{13} \frac{1}{s+2} + \frac{375}{1469} \frac{1}{s+15} + \frac{145}{113} \frac{s}{s^2+1} + \frac{85}{113} \frac{1}{s^2+1}.$$

Then

$$i_2 = -\frac{20}{13} e^{-2t} + \frac{375}{1469} e^{-15t} + \frac{145}{113} \cos t + \frac{85}{113} \sin t$$

and

$$i_3 = \frac{25}{3} \sin t - \frac{1}{6} i_2' - \frac{11}{6} i_2 = \frac{30}{13} e^{-2t} + \frac{250}{1469} e^{-15t} - \frac{280}{113} \cos t + \frac{810}{113} \sin t.$$

18. Taking the Laplace transform of the system

$$0.5i_1' + 50i_2 = 60$$

$$0.005i_2' + i_2 - i_1 = 0$$

gives

$$s \mathcal{L}\{i_1\} + 100 \mathcal{L}\{i_2\} = \frac{120}{s}$$

$$-200 \mathcal{L}\{i_1\} + (s+200) \mathcal{L}\{i_2\} = 0$$

so that

$$\mathcal{L}\{i_2\} = \frac{24,000}{s(s^2+200s+20,000)} = \frac{6}{5} \frac{1}{s} - \frac{6}{5} \frac{s+100}{(s+100)^2+100^2} - \frac{6}{5} \frac{100}{(s+100)^2+100^2}.$$

Then

$$i_2 = \frac{6}{5} - \frac{6}{5} e^{-100t} \cos 100t - \frac{6}{5} e^{-100t} \sin 100t$$

and

$$i_1 = 0.005i_2' + i_2 = \frac{6}{5} - \frac{6}{5} e^{-100t} \cos 100t.$$

19. Taking the Laplace transform of the system

$$2i_1' + 50i_2 = 60$$

$$0.005i_2' + i_2 - i_1 = 0$$

gives

$$2s \mathcal{L}\{i_1\} + 50 \mathcal{L}\{i_2\} = \frac{60}{s}$$

$$-200 \mathcal{L}\{i_1\} + (s+200) \mathcal{L}\{i_2\} = 0$$

so that

$$\mathcal{L}\{i_2\} = \frac{6,000}{s(s^2+200s+5,000)}$$

$$= \frac{6}{5} \frac{1}{s} - \frac{6}{5} \frac{s+100}{(s+100)^2 - (50\sqrt{2})^2} - \frac{6\sqrt{2}}{5} \frac{50\sqrt{2}}{(s+100)^2 - (50\sqrt{2})^2}.$$

Then
$$i_2 = \frac{6}{5} - \frac{6}{5}e^{-100t} \cosh 50\sqrt{2}t - \frac{6\sqrt{2}}{5}e^{-100t} \sinh 50\sqrt{2}t$$

and
$$i_1 = 0.005i_2' + i_2 = \frac{6}{5} - \frac{6}{5}e^{-100t} \cosh 50\sqrt{2}t - \frac{9\sqrt{2}}{10}e^{-100t} \sinh 50\sqrt{2}t.$$

- 20. (a)** Using Kirchoff's first law we write $i_1 = i_2 + i_3$. Since $i_2 = dq/dt$ we have $i_1 - i_3 = dq/dt$. Using Kirchoff's second law and summing the voltage drops across the shorter loop gives

$$E(t) = iR_1 + \frac{1}{C}q, \quad (1)$$

so that

$$i_1 = \frac{1}{R_1}E(t) - \frac{1}{R_1C}q.$$

Then

$$\frac{dq}{dt} = i_1 - i_3 = \frac{1}{R_1}E(t) - \frac{1}{R_1C}q - i_3$$

and

$$R_1 \frac{dq}{dt} + \frac{1}{C}q + R_1 i_3 = E(t).$$

Summing the voltage drops across the longer loop gives

$$E(t) = i_1 R_1 + L \frac{di_3}{dt} + R_2 i_3.$$

Combining this with (1) we obtain

$$i_1 R_1 + L \frac{di_3}{dt} + R_2 i_3 = i_1 R_1 + \frac{1}{C}q$$

or

$$L \frac{di_3}{dt} + R_2 i_3 - \frac{1}{C}q = 0.$$

- (b)** Using $L = R_1 = R_2 = C = 1$, $E(t) = 50e^{-t} \mathcal{U}(t-1) = 50e^{-1}e^{-(t-1)} \mathcal{U}(t-1)$, $q(0) = i_3(0) = 0$, and taking the Laplace transform of the system we obtain

$$(s+1) \mathcal{L}\{q\} + \mathcal{L}\{i_3\} = \frac{50e^{-1}}{s+1}e^{-s}$$

$$(s+1) \mathcal{L}\{i_3\} - \mathcal{L}\{q\} = 0,$$

so that

$$\mathcal{L}\{q\} = \frac{50e^{-1}e^{-s}}{(s+1)^2 + 1}$$

and

$$q(t) = 50e^{-1}e^{-(t-1)} \sin(t-1) \mathcal{U}(t-1) = 50e^{-t} \sin(t-1) \mathcal{U}(t-1).$$

- 21. (a)** Taking the Laplace transform of the system

$$4\theta_1'' + \theta_2'' + 8\theta_1 = 0$$

$$\theta_1'' + \theta_2'' + 2\theta_2 = 0$$

gives

$$4(s^2 + 2) \mathcal{L}\{\theta_1\} + s^2 \mathcal{L}\{\theta_2\} = 3s$$

$$s^2 \mathcal{L}\{\theta_1\} + (s^2 + 2) \mathcal{L}\{\theta_2\} = 0$$

so that

$$(3s^2 + 4)(s^2 + 4) \mathcal{L}\{\theta_2\} = -3s^3$$

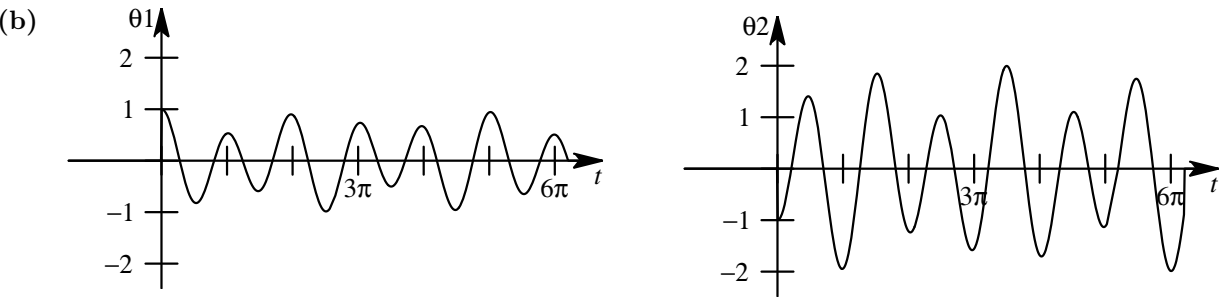
or

$$\mathcal{L}\{\theta_2\} = \frac{1}{2} \frac{s}{s^2 + 4/3} - \frac{3}{2} \frac{s}{s^2 + 4}.$$

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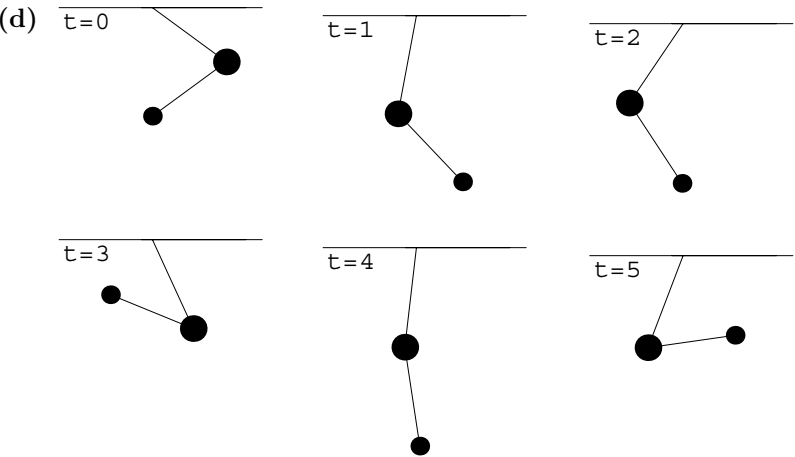
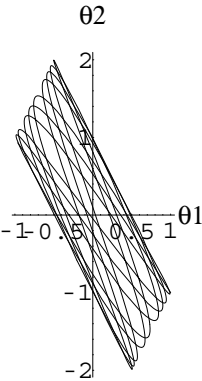
Then
$$\theta_2 = \frac{1}{2} \cos \frac{2}{\sqrt{3}}t - \frac{3}{2} \cos 2t \quad \text{and} \quad \theta_1'' = -\theta_2'' - 2\theta_2$$

so that
$$\theta_1 = \frac{1}{4} \cos \frac{2}{\sqrt{3}}t + \frac{3}{4} \cos 2t.$$

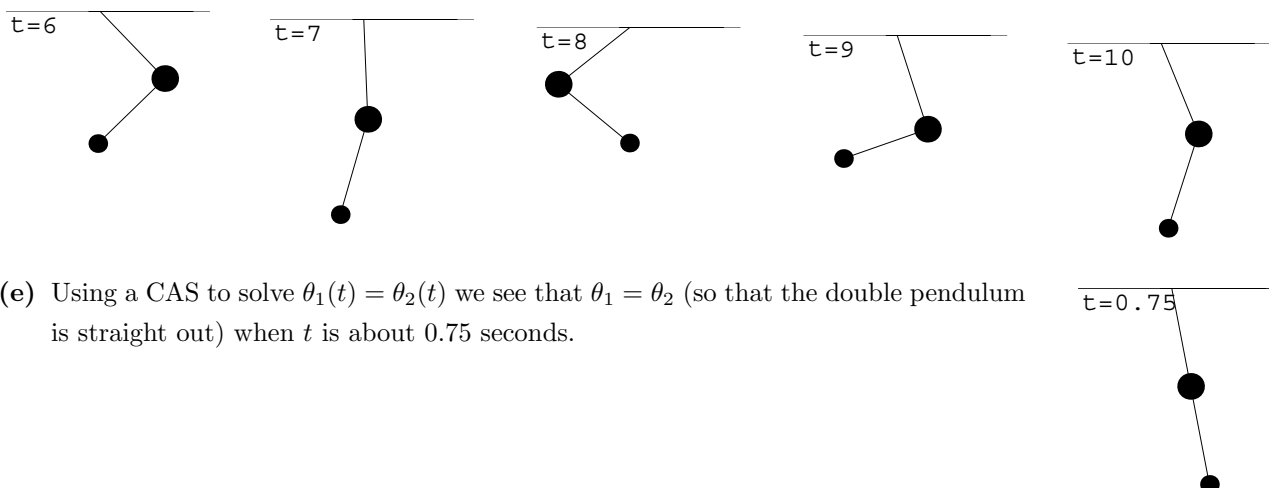


Mass m_2 has extreme displacements of greater magnitude. Mass m_1 first passes through its equilibrium position at about $t = 0.87$, and mass m_2 first passes through its equilibrium position at about $t = 0.66$. The motion of the pendulums is not periodic since $\cos 2t/\sqrt{3}$ has period $\sqrt{3}\pi$, $\cos 2t$ has period π , and the ratio of these periods is $\sqrt{3}$, which is not a rational number.

(c) The Lissajous curve is plotted for $0 \leq t \leq 30$.



t	θ_1	θ_2
1	-0.2111	0.8263
2	-0.6585	0.6438
3	0.4830	-1.9145
4	-0.1325	0.1715
5	-0.4111	1.6951
6	0.8327	-0.8662
7	0.0458	-0.3186
8	-0.9639	0.9452
9	0.3534	-1.2741
10	0.4370	-0.3502



- (e) Using a CAS to solve $\theta_1(t) = \theta_2(t)$ we see that $\theta_1 = \theta_2$ (so that the double pendulum is straight out) when t is about 0.75 seconds.

- (f) To make a movie of the pendulum it is necessary to locate the mass in the plane as a function of time. Suppose that the upper arm is attached to the origin and that the equilibrium position lies along the negative y -axis. Then mass m_1 is at $(x_1(t), y_1(t))$ and mass m_2 is at $(x_2(t), y_2(t))$, where

$$x_1(t) = 16 \sin \theta_1(t) \quad \text{and} \quad y_1(t) = -16 \cos \theta_1(t)$$

and

$$x_2(t) = x_1(t) + 16 \sin \theta_2(t) \quad \text{and} \quad y_2(t) = y_1(t) - 16 \cos \theta_2(t).$$

A reasonable movie can be constructed by letting t range from 0 to 10 in increments of 0.1 seconds.

Chapter 4 Review Exercises

1. $\mathcal{L}\{f(t)\} = \int_0^1 t e^{-st} dt + \int_1^\infty (2-t)e^{-st} dt = \frac{1}{s^2} - \frac{2}{s^2} e^{-s}$
2. $\mathcal{L}\{f(t)\} = \int_2^4 e^{-st} dt = \frac{1}{s} (e^{-2s} - e^{-4s})$
3. False; consider $f(t) = t^{-1/2}$.
4. False, since $f(t) = (e^t)^{10} = e^{10t}$.
5. True, since $\lim_{s \rightarrow \infty} F(s) = 1 \neq 0$. (See Theorem 4.5 in the text.)
6. False; consider $f(t) = 1$ and $g(t) = 1$.
7. $\mathcal{L}\{e^{-7t}\} = \frac{1}{s+7}$
8. $\mathcal{L}\{te^{-7t}\} = \frac{1}{(s+7)^2}$
9. $\mathcal{L}\{\sin 2t\} = \frac{2}{s^2+4}$
10. $\mathcal{L}\{e^{-3t} \sin 2t\} = \frac{2}{(s+3)^2+4}$
11. $\mathcal{L}\{t \sin 2t\} = -\frac{d}{ds} \left[\frac{2}{s^2+4} \right] = \frac{4s}{(s^2+4)^2}$

Chapter 4 Review Exercises

12. $\mathcal{L}\{\sin 2t \mathcal{U}(t - \pi)\} = \mathcal{L}\{\sin 2(t - \pi) \mathcal{U}(t - \pi)\} = \frac{2}{s^2 + 4} e^{-\pi s}$
13. $\mathcal{L}^{-1}\left\{\frac{20}{s^6}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{6} \frac{5!}{s^6}\right\} = \frac{1}{6} t^5$
14. $\mathcal{L}^{-1}\left\{\frac{1}{3s - 1}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{3} \frac{1}{s - 1/3}\right\} = \frac{1}{3} e^{t/3}$
15. $\mathcal{L}^{-1}\left\{\frac{1}{(s - 5)^3}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{2} \frac{2}{(s - 5)^3}\right\} = \frac{1}{2} t^2 e^{5t}$
16. $\mathcal{L}^{-1}\left\{\frac{1}{s^2 - 5}\right\} = \mathcal{L}^{-1}\left\{-\frac{1}{\sqrt{5}} \frac{1}{s + \sqrt{5}} + \frac{1}{\sqrt{5}} \frac{1}{s - \sqrt{5}}\right\} = -\frac{1}{\sqrt{5}} e^{-\sqrt{5}t} + \frac{1}{\sqrt{5}} e^{\sqrt{5}t}$
17. $\mathcal{L}^{-1}\left\{\frac{s}{s^2 - 10s + 29}\right\} = \mathcal{L}^{-1}\left\{\frac{s - 5}{(s - 5)^2 + 2^2} + \frac{5}{2} \frac{2}{(s - 5)^2 + 2^2}\right\} = e^{5t} \cos 2t + \frac{5}{2} e^{5t} \sin 2t$
18. $\mathcal{L}^{-1}\left\{\frac{1}{s^2} e^{-5s}\right\} = (t - 5) \mathcal{U}(t - 5)$
19. $\mathcal{L}^{-1}\left\{\frac{s + \pi}{s^2 + \pi^2} e^{-s}\right\} = \mathcal{L}^{-1}\left\{\frac{s}{s^2 + \pi^2} e^{-s} + \frac{\pi}{s^2 + \pi^2} e^{-s}\right\}$
 $= \cos \pi(t - 1) \mathcal{U}(t - 1) + \sin \pi(t - 1) \mathcal{U}(t - 1)$
20. $\mathcal{L}^{-1}\left\{\frac{1}{L^2 s^2 + n^2 \pi^2}\right\} = \frac{1}{L^2} \frac{L}{n\pi} \mathcal{L}^{-1}\left\{\frac{n\pi/L}{s^2 + (n^2 \pi^2)/L^2}\right\} = \frac{1}{Ln\pi} \sin \frac{n\pi}{L} t$
21. $\mathcal{L}\{e^{-5t}\}$ exists for $s > -5$.
22. $\mathcal{L}\{te^{8t} f(t)\} = -\frac{d}{ds} F(s - 8)$.
23. $\mathcal{L}\{e^{at} f(t - k) \mathcal{U}(t - k)\} = e^{-ks} \mathcal{L}\{e^{a(t+k)} f(t)\} = e^{-ks} e^{ak} \mathcal{L}\{e^{at} f(t)\} = e^{-k(s-a)} F(s - a)$
24. $\mathcal{L}\left\{\int_0^t e^{a\tau} f(\tau) d\tau\right\} = \frac{1}{s} \mathcal{L}\{e^{at} f(t)\} = \frac{F(s - a)}{s}$, whereas
 $\mathcal{L}\left\{e^{at} \int_0^t f(\tau) d\tau\right\} = \mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} \Big|_{s \rightarrow s-a} = \frac{F(s)}{s} \Big|_{s \rightarrow s-a} = \frac{F(s - a)}{s - a}$.
25. $f(t) \mathcal{U}(t - t_0)$
26. $f(t) - f(t) \mathcal{U}(t - t_0)$
27. $f(t - t_0) \mathcal{U}(t - t_0)$
28. $f(t) - f(t) \mathcal{U}(t - t_0) + f(t) \mathcal{U}(t - t_1)$
29. $f(t) = t - [(t - 1) + 1] \mathcal{U}(t - 1) + \mathcal{U}(t - 1) - \mathcal{U}(t - 4) = t - (t - 1) \mathcal{U}(t - 1) - \mathcal{U}(t - 4)$
 $\mathcal{L}\{f(t)\} = \frac{1}{s^2} - \frac{1}{s^2} e^{-s} - \frac{1}{s} e^{-4s}$
 $\mathcal{L}\{e^t f(t)\} = \frac{1}{(s - 1)^2} - \frac{1}{(s - 1)^2} e^{-(s-1)} - \frac{1}{s - 1} e^{-4(s-1)}$
30. $f(t) = \sin t \mathcal{U}(t - \pi) - \sin t \mathcal{U}(t - 3\pi) = -\sin(t - \pi) \mathcal{U}(t - \pi) + \sin(t - 3\pi) \mathcal{U}(t - 3\pi)$
 $\mathcal{L}\{f(t)\} = -\frac{1}{s^2 + 1} e^{-\pi s} + \frac{1}{s^2 + 1} e^{-3\pi s}$
 $\mathcal{L}\{e^t f(t)\} = -\frac{1}{(s - 1)^2 + 1} e^{-\pi(s-1)} + \frac{1}{(s - 1)^2 + 1} e^{-3\pi(s-1)}$

31. $f(t) = 2 - 2\mathcal{U}(t-2) + [(t-2) + 2]\mathcal{U}(t-2) = 2 + (t-2)\mathcal{U}(t-2)$

$$\mathcal{L}\{f(t)\} = \frac{2}{s} + \frac{1}{s^2}e^{-2s}$$

$$\mathcal{L}\{e^t f(t)\} = \frac{2}{s-1} + \frac{1}{(s-1)^2}e^{-2(s-1)}$$

32. $f(t) = t - t\mathcal{U}(t-1) + (2-t)\mathcal{U}(t-1) - (2-t)\mathcal{U}(t-2) = t - 2(t-1)\mathcal{U}(t-1) + (t-2)\mathcal{U}(t-2)$

$$\mathcal{L}\{f(t)\} = \frac{1}{s^2} - \frac{2}{s^2}e^{-s} + \frac{1}{s^2}e^{-2s}$$

$$\mathcal{L}\{e^t f(t)\} = \frac{1}{(s-1)^2} - \frac{2}{(s-1)^2}e^{-(s-1)} + \frac{1}{(s-1)^2}e^{-2(s-1)}$$

33. Taking the Laplace transform of the differential equation we obtain

$$\mathcal{L}\{y\} = \frac{5}{(s-1)^2} + \frac{1}{2} \frac{2}{(s-1)^3}$$

so that

$$y = 5te^t + \frac{1}{2}t^2e^t.$$

34. Taking the Laplace transform of the differential equation we obtain

$$\begin{aligned}\mathcal{L}\{y\} &= \frac{1}{(s-1)^2(s^2-8s+20)} \\ &= \frac{6}{169} \frac{1}{s-1} + \frac{1}{13} \frac{1}{(s-1)^2} - \frac{6}{169} \frac{s-4}{(s-4)^2+2^2} + \frac{5}{338} \frac{2}{(s-4)^2+2^2}\end{aligned}$$

so that

$$y = \frac{6}{169}e^t + \frac{1}{13}te^t - \frac{6}{169}e^{4t}\cos 2t + \frac{5}{338}e^{4t}\sin 2t.$$

35. Taking the Laplace transform of the given differential equation we obtain

$$\begin{aligned}\mathcal{L}\{y\} &= \frac{s^3+6s^2+1}{s^2(s+1)(s+5)} - \frac{1}{s^2(s+1)(s+5)}e^{-2s} - \frac{2}{s(s+1)(s+5)}e^{-2s} \\ &= -\frac{6}{25} \cdot \frac{1}{s} + \frac{1}{5} \cdot \frac{1}{s^2} + \frac{3}{2} \cdot \frac{1}{s+1} - \frac{13}{50} \cdot \frac{1}{s+5} \\ &\quad - \left(-\frac{6}{25} \cdot \frac{1}{s} + \frac{1}{5} \cdot \frac{1}{s^2} + \frac{1}{4} \cdot \frac{1}{s+1} - \frac{1}{100} \cdot \frac{1}{s+5} \right) e^{-2s} \\ &\quad - \left(\frac{2}{5} \cdot \frac{1}{s} - \frac{1}{2} \cdot \frac{1}{s+1} + \frac{1}{10} \cdot \frac{1}{s+5} \right) e^{-2s}\end{aligned}$$

so that

$$\begin{aligned}y &= -\frac{6}{25} + \frac{1}{5}t^2 + \frac{3}{2}e^{-t} - \frac{13}{50}e^{-5t} - \frac{4}{25}\mathcal{U}(t-2) - \frac{1}{5}(t-2)^2\mathcal{U}(t-2) \\ &\quad + \frac{1}{4}e^{-(t-2)}\mathcal{U}(t-2) - \frac{9}{100}e^{-5(t-2)}\mathcal{U}(t-2).\end{aligned}$$

36. Taking the Laplace transform of the differential equation we obtain

$$\begin{aligned}\mathcal{L}\{y\} &= \frac{s^3+2}{s^3(s-5)} - \frac{2+2s+s^2}{s^3(s-5)}e^{-s} \\ &= -\frac{2}{125} \frac{1}{s} - \frac{2}{25} \frac{1}{s^2} - \frac{1}{5} \frac{2}{s^3} + \frac{127}{125} \frac{1}{s-5} - \left[-\frac{37}{125} \frac{1}{s} - \frac{12}{25} \frac{1}{s^2} - \frac{1}{5} \frac{2}{s^3} + \frac{37}{125} \frac{1}{s-5} \right] e^{-s}\end{aligned}$$

so that

$$y = -\frac{2}{125} - \frac{2}{25}t - \frac{1}{5}t^2 + \frac{127}{125}e^{5t} - \left[-\frac{37}{125} - \frac{12}{25}(t-1) - \frac{1}{5}(t-1)^2 + \frac{37}{125}e^{5(t-1)} \right] \mathcal{U}(t-1).$$

Chapter 4 Review Exercises

37. Taking the Laplace transform of the integral equation we obtain

$$\mathcal{L}\{y\} = \frac{1}{s} + \frac{1}{s^2} + \frac{1}{2} \frac{2}{s^3}$$

so that

$$y(t) = 1 + t + \frac{1}{2}t^2.$$

38. Taking the Laplace transform of the integral equation we obtain

$$(\mathcal{L}\{f\})^2 = 6 \cdot \frac{6}{s^4} \quad \text{or} \quad \mathcal{L}\{f\} = \pm 6 \cdot \frac{1}{s^2}$$

so that $f(t) = \pm 6t$.

39. Taking the Laplace transform of the system gives

$$s\mathcal{L}\{x\} + \mathcal{L}\{y\} = \frac{1}{s^2} + 1$$

$$4\mathcal{L}\{x\} + s\mathcal{L}\{y\} = 2$$

so that

$$\mathcal{L}\{x\} = \frac{s^2 - 2s + 1}{s(s-2)(s+2)} = -\frac{1}{4} \frac{1}{s} + \frac{1}{8} \frac{1}{s-2} + \frac{9}{8} \frac{1}{s+2}.$$

Then

$$x = -\frac{1}{4} + \frac{1}{8}e^{2t} + \frac{9}{8}e^{-2t} \quad \text{and} \quad y = -x' + t = \frac{9}{4}e^{-2t} - \frac{1}{4}e^{2t} + t.$$

40. Taking the Laplace transform of the system gives

$$s^2\mathcal{L}\{x\} + s^2\mathcal{L}\{y\} = \frac{1}{s-2}$$

$$2s\mathcal{L}\{x\} + s^2\mathcal{L}\{y\} = -\frac{1}{s-2}$$

so that

$$\mathcal{L}\{x\} = \frac{2}{s(s-2)^2} = \frac{1}{2} \frac{1}{s} - \frac{1}{2} \frac{1}{s-2} + \frac{1}{(s-2)^2}$$

and

$$\mathcal{L}\{y\} = \frac{-s-2}{s^2(s-2)^2} = -\frac{3}{4} \frac{1}{s} - \frac{1}{2} \frac{1}{s^2} + \frac{3}{4} \frac{1}{s-2} - \frac{1}{(s-2)^2}.$$

Then

$$x = \frac{1}{2} - \frac{1}{2}e^{2t} + te^{2t} \quad \text{and} \quad y = -\frac{3}{4} - \frac{1}{2}t + \frac{3}{4}e^{2t} - te^{2t}.$$

41. The integral equation is

$$10i + 2 \int_0^t i(\tau) d\tau = 2t^2 + 2t.$$

Taking the Laplace transform we obtain

$$\mathcal{L}\{i\} = \left(\frac{4}{s^3} + \frac{2}{s^2} \right) \frac{s}{10s+2} = \frac{s+2}{s^2(5s+2)} = -\frac{9}{s} + \frac{2}{s^2} + \frac{45}{5s+1} = -\frac{9}{s} + \frac{2}{s^2} + \frac{9}{s+1/5}.$$

Thus

$$i(t) = -9 + 2t + 9e^{-t/5}.$$

42. The differential equation is

$$\frac{1}{2} \frac{d^2q}{dt^2} + 10 \frac{dq}{dt} + 100q = 10 - 10\mathcal{U}(t-5).$$

Taking the Laplace transform we obtain

$$\begin{aligned}\mathcal{L}\{q\} &= \frac{20}{2(s^2 + 20s + 200)} (1 - e^{-5s}) \\ &= \left[\frac{1}{10} \frac{1}{s} - \frac{1}{10} \frac{s + 10}{(s + 10)^2 + 10^2} - \frac{1}{10} \frac{10}{(s + 10)^2 + 10^2} \right] (1 - e^{-5s})\end{aligned}$$

so that

$$\begin{aligned}q(t) &= \frac{1}{10} - \frac{1}{10} e^{-10t} \cos 10t - \frac{1}{10} e^{-10t} \sin 10t \\ &\quad - \left[\frac{1}{10} - \frac{1}{10} e^{-10(t-5)} \cos 10(t-5) - \frac{1}{10} e^{-10(t-5)} \sin 10(t-5) \right] \mathcal{U}(t-5).\end{aligned}$$

43. Taking the Laplace transform of the given differential equation we obtain

$$\mathcal{L}\{y\} = \frac{2w_0}{EIL} \left(\frac{L}{48} \cdot \frac{4!}{s^5} - \frac{1}{120} \cdot \frac{5!}{s^6} + \frac{1}{120} \cdot \frac{5!}{s^6} e^{-sL/2} \right) + \frac{c_1}{2} \cdot \frac{2!}{s^3} + \frac{c_2}{6} \cdot \frac{3!}{s^4}$$

so that

$$y = \frac{2w_0}{EIL} \left[\frac{L}{48} x^4 - \frac{1}{120} x^5 + \frac{1}{120} \left(x - \frac{L}{2} \right)^5 \mathcal{U} \left(x - \frac{L}{2} \right) + \frac{c_1}{2} x^2 + \frac{c_2}{6} x^3 \right]$$

where $y''(0) = c_1$ and $y'''(0) = c_2$. Using $y''(L) = 0$ and $y'''(L) = 0$ we find

$$c_1 = w_0 L^2 / 24EI, \quad c_2 = -w_0 L / 4EI.$$

Hence

$$y = \frac{w_0}{12EIL} \left[-\frac{1}{5} x^5 + \frac{L}{2} x^4 - \frac{L^2}{2} x^3 + \frac{L^3}{4} x^2 + \frac{1}{5} \left(x - \frac{L}{2} \right)^5 \mathcal{U} \left(x - \frac{L}{2} \right) \right].$$

44. Taking the Laplace transform of the given differential equation we obtain

$$\mathcal{L}\{y\} = \frac{c_1}{2} \cdot \frac{2s}{s^4 + 4} + \frac{c_2}{4} \cdot \frac{4}{s^4 + 4} + \frac{w_0}{4EI} \cdot \frac{4}{s^4 + 4} e^{-s\pi/2}$$

so that

$$\begin{aligned}y &= \frac{c_1}{2} \sin x \sinh x + \frac{c_2}{4} (\sin x \cosh x - \cos x \sinh x) \\ &\quad + \frac{w_0}{4EI} \left[\sin \left(x - \frac{\pi}{2} \right) \cosh \left(x - \frac{\pi}{2} \right) - \cos \left(x - \frac{\pi}{2} \right) \sinh \left(x - \frac{\pi}{2} \right) \right] \mathcal{U} \left(x - \frac{\pi}{2} \right)\end{aligned}$$

where $y''(0) = c_1$ and $y'''(0) = c_2$. Using $y(\pi) = 0$ and $y'(\pi) = 0$ we find

$$c_1 = \frac{w_0}{EI} \frac{\sinh \frac{\pi}{2}}{\sinh \pi}, \quad c_2 = -\frac{w_0}{EI} \frac{\cosh \frac{\pi}{2}}{\sinh \pi}.$$

Hence

$$\begin{aligned}y &= \frac{w_0}{2EI} \frac{\sinh \frac{\pi}{2}}{\sinh \pi} \sin x \sinh x - \frac{w_0}{4EI} \frac{\cosh \frac{\pi}{2}}{\sinh \pi} (\sin x \cosh x - \cos x \sinh x) \\ &\quad + \frac{w_0}{4EI} \left[\sin \left(x - \frac{\pi}{2} \right) \cosh \left(x - \frac{\pi}{2} \right) - \cos \left(x - \frac{\pi}{2} \right) \sinh \left(x - \frac{\pi}{2} \right) \right] \mathcal{U} \left(x - \frac{\pi}{2} \right).\end{aligned}$$

5 Series Solutions of Linear Equations

Exercises 5.1

$$1. \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{2^{n+1}x^{n+1}/(n+1)}{2^n x^n/n} \right| = \lim_{n \rightarrow \infty} \frac{2n}{n+1} |x| = 2|x|$$

The series is absolutely convergent for $2|x| < 1$ or $|x| < 1/2$. At $x = -1/2$, the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ converges by the alternating series test. At $x = 1/2$, the series $\sum_{n=1}^{\infty} \frac{1}{n}$ is the harmonic series which diverges. Thus, the given series converges on $[-1/2, 1/2)$.

$$2. \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{100^{n+1}(x+7)^{n+1}/(n+1)!}{100^n(x+7)^n/n!} \right| = \lim_{n \rightarrow \infty} \frac{100}{n+1} |x+7| = 0$$

The series is absolutely convergent on $(-\infty, \infty)$.

$$3. \lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| = \lim_{k \rightarrow \infty} \left| \frac{(x-5)^{k+1}/10^{k+1}}{(x-5)^k/10^k} \right| = \lim_{k \rightarrow \infty} \frac{1}{10} |x-5| = \frac{1}{10} |x-5|$$

The series is absolutely convergent for $\frac{1}{10} |x-5| < 1$, $|x-5| < 10$, or on $(-5, 15)$. At $x = -5$, the series $\sum_{k=1}^{\infty} \frac{(-1)^k(-10)^k}{10^k} = \sum_{k=1}^{\infty} 1$ diverges by the k -th term test. At $x = 15$, the series $\sum_{k=1}^{\infty} \frac{(-1)^k 10^k}{10^k} = \sum_{k=1}^{\infty} (-1)^k$ diverges by the k -th term test. Thus, the series converges on $(-5, 15)$.

$$4. \lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| = \lim_{k \rightarrow \infty} \left| \frac{(k+1)!(x-1)^{k+1}}{k!(x-1)^k} \right| = \lim_{k \rightarrow \infty} (k+1) |x-1| = \infty, \quad x \neq 1$$

The radius of convergence is 0 and the series converges only for $x = 1$.

$$5. \sin x \cos x = \left(x - \frac{x^3}{6} + \frac{x^5}{120} - \frac{x^7}{5040} + \cdots \right) \left(1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} + \cdots \right) = x - \frac{2x^3}{3} + \frac{2x^5}{15} - \frac{4x^7}{315} + \cdots$$

$$6. e^{-x} \cos x = \left(1 - x + \frac{x^2}{2} - \frac{x^3}{6} + \frac{x^4}{24} - \cdots \right) \left(1 - \frac{x^2}{2} + \frac{x^4}{24} - \cdots \right) = 1 - x + \frac{x^3}{3} - \frac{x^4}{6} + \cdots$$

$$7. \frac{1}{\cos x} = \frac{1}{1 - \frac{x^2}{2} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots} = 1 + \frac{x^2}{2} + \frac{5x^4}{4!} + \frac{61x^6}{6!} + \cdots$$

Since $\cos(\pi/2) = \cos(-\pi/2) = 0$, the series converges on $(-\pi/2, \pi/2)$.

$$8. \frac{1-x}{2+x} = \frac{1}{2} - \frac{3}{4}x + \frac{3}{8}x^2 - \frac{3}{16}x^3 + \cdots$$

Since the function is undefined at $x = -2$, the series converges on $(-2, 2)$.

$$\begin{aligned}
9. \quad \sum_{n=1}^{\infty} 2nc_n x^{n-1} + \sum_{n=0}^{\infty} 6c_n x^{n+1} &= 2 \cdot 1 \cdot c_1 x^0 + \underbrace{\sum_{n=2}^{\infty} 2nc_n x^{n-1}}_{k=n-1} + \underbrace{\sum_{n=0}^{\infty} 6c_n x^{n+1}}_{k=n+1} \\
&= 2c_1 + \sum_{k=1}^{\infty} 2(k+1)c_{k+1}x^k + \sum_{k=1}^{\infty} 6c_{k-1}x^k \\
&= 2c_1 + \sum_{k=1}^{\infty} [2(k+1)c_{k+1} + 6c_{k-1}]x^k
\end{aligned}$$

$$\begin{aligned}
10. \quad \sum_{n=2}^{\infty} n(n-1)c_n x^n + 2 \sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} + 3 \sum_{n=1}^{\infty} nc_n x^n \\
&= 2 \cdot 2 \cdot 1c_2 x^0 + 2 \cdot 3 \cdot 2c_3 x' + 3 \cdot 1 \cdot c_1 x' + \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^n}_{k=n} + 2 \underbrace{\sum_{n=4}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} + 3 \underbrace{\sum_{n=2}^{\infty} nc_n x^n}_{k=n} \\
&= 4c_2 + (12c_3 + (12c_3 + 3c_1)x + \sum_{n=2}^{\infty} k(k-1)c_k x^k + 2 \sum_{n=2}^{\infty} (k+2)(k+1)c_{k+2}x^k + 3 \sum_{n=2}^{\infty} kc_k x^k \\
&= 4c_2 + (3c_1 + 12c_3)x + \sum_{n=2}^{\infty} ([k(k-1) + 3k]c_k + 2(k+2)(k+1)c_{k+2})x^k \\
&= 4c_2 + (3c_1 + 12c_3)x + \sum_{n=2}^{\infty} [k(k+2)c_k + 2(k+1)(k+2)c_{k+2}]x^k
\end{aligned}$$

$$\begin{aligned}
11. \quad y' &= \sum_{n=1}^{\infty} (-1)^{n+1} x^{n-1}, \quad y'' = \sum_{n=2}^{\infty} (-1)^{n+1} (n-1) x^{n-2} \\
(x+1)y'' + y' &= (x+1) \sum_{n=2}^{\infty} (-1)^{n+1} (n-1) x^{n-2} + \sum_{n=1}^{\infty} (-1)^{n+1} x^{n-1} \\
&= \sum_{n=2}^{\infty} (-1)^{n+1} (n-1) x^{n-1} + \sum_{n=2}^{\infty} (-1)^{n+1} (n-1) x^{n-2} + \sum_{n=1}^{\infty} (-1)^{n+1} x^{n-1} \\
&= -x^0 + x^0 + \underbrace{\sum_{n=2}^{\infty} (-1)^{n+1} (n-1) x^{n-1}}_{k=n-1} + \underbrace{\sum_{n=3}^{\infty} (-1)^{n+1} (n-1) x^{n-2}}_{k=n-2} + \underbrace{\sum_{n=2}^{\infty} (-1)^{n+1} x^{n-1}}_{k=n-1} \\
&= \sum_{k=1}^{\infty} (-1)^{k+2} k x^k + \sum_{k=1}^{\infty} (-1)^{k+3} (k+1) x^k + \sum_{k=1}^{\infty} (-1)^{k+2} x^k \\
&= \sum_{k=1}^{\infty} [(-1)^{k+2} k - (-1)^{k+2} k - (-1)^{k+2} + (-1)^{k+2}] x^k = 0
\end{aligned}$$

$$\begin{aligned}
12. \quad y' &= \sum_{n=1}^{\infty} \frac{(-1)^n 2n}{2^{2n} (n!)^2} x^{2n-1}, \quad y'' = \sum_{n=1}^{\infty} \frac{(-1)^n 2n(2n-1)}{2^{2n} (n!)^2} x^{2n-2} \\
xy'' + y' + xy &= \underbrace{\sum_{n=1}^{\infty} \frac{(-1)^n 2n(2n-1)}{2^{2n} (n!)^2} x^{2n-1}}_{k=n} + \underbrace{\sum_{n=1}^{\infty} \frac{(-1)^n 2n}{2^{2n} (n!)^2} x^{2n-1}}_{k=n} + \underbrace{\sum_{n=0}^{\infty} \frac{(-1)^n}{2^{2n} (n!)^2} x^{2n+1}}_{k=n+1}
\end{aligned}$$

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$$\begin{aligned}
&= \sum_{k=1}^{\infty} \left[\frac{(-1)^k 2k(2k-1)}{2^{2k} (k!)^2} + \frac{(-1)^k 2k}{2^{2k} (k!)^2} + \frac{(-1)^{k-1}}{2^{2k-2} [(k-1)!]^2} \right] x^{2k-1} \\
&= \sum_{k=1}^{\infty} \left[\frac{(-1)^k (2k)^2}{2^{2k} (k!)^2} - \frac{(-1)^k}{2^{2k-2} [(k-1)!]^2} \right] x^{2k-1} \\
&= \sum_{k=1}^{\infty} (-1)^k \left[\frac{(2k)^2 - 2^2 k^2}{2^{2k} (k!)^2} \right] x^{2k-1} = 0
\end{aligned}$$

13. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned}
y'' - xy &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} - \underbrace{\sum_{n=0}^{\infty} c_n x^{n+1}}_{k=n+1} = \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k - \sum_{k=1}^{\infty} c_{k-1} x^k \\
&= 2c_2 + \sum_{k=1}^{\infty} [(k+2)(k+1)c_{k+2} - c_{k-1}] x^k = 0.
\end{aligned}$$

Thus

$$c_2 = 0$$

$$(k+2)(k+1)c_{k+2} - c_{k-1} = 0$$

and

$$c_{k+2} = \frac{1}{(k+2)(k+1)} c_{k-1}, \quad k = 1, 2, 3, \dots$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$c_3 = \frac{1}{6}$$

$$c_4 = c_5 = 0$$

$$c_6 = \frac{1}{180}$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$c_3 = 0$$

$$c_4 = \frac{1}{12}$$

$$c_5 = c_6 = 0$$

$$c_7 = \frac{1}{504}$$

and so on. Thus, two solutions are

$$y_1 = 1 + \frac{1}{6}x^3 + \frac{1}{180}x^6 + \dots \quad \text{and} \quad y_2 = x + \frac{1}{12}x^4 + \frac{1}{504}x^7 + \dots$$

14. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned}
y'' + x^2 y &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} + \underbrace{\sum_{n=0}^{\infty} c_n x^{n+2}}_{k=n+2} = \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k + \sum_{k=2}^{\infty} c_{k-2} x^k \\
&= 2c_2 + 6c_3 x + \sum_{k=2}^{\infty} [(k+2)(k+1)c_{k+2} + c_{k-2}] x^k = 0.
\end{aligned}$$

Thus

$$c_2 = c_3 = 0$$

$$(k+2)(k+1)c_{k+2} + c_{k-2} = 0$$

and

$$c_{k+2} = -\frac{1}{(k+2)(k+1)} c_{k-2}, \quad k = 2, 3, 4, \dots$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$c_4 = -\frac{1}{12}$$

$$c_5 = c_6 = c_7 = 0$$

$$c_8 = \frac{1}{672}$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$c_4 = 0$$

$$c_5 = -\frac{1}{20}$$

$$c_6 = c_7 = c_8 = 0$$

$$c_9 = \frac{1}{1440}$$

and so on. Thus, two solutions are

$$y_1 = 1 - \frac{1}{12}x^4 + \frac{1}{672}x^8 - \dots \quad \text{and} \quad y_2 = x - \frac{1}{20}x^5 + \frac{1}{1440}x^9 - \dots$$

15. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned} y'' - 2xy' + y &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} - 2 \underbrace{\sum_{n=1}^{\infty} n c_n x^n}_{k=n} + \underbrace{\sum_{n=0}^{\infty} c_n x^n}_{k=n} \\ &= \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k - 2 \sum_{k=1}^{\infty} k c_k x^k + \sum_{k=0}^{\infty} c_k x^k \\ &= 2c_2 + c_0 + \sum_{k=1}^{\infty} [(k+2)(k+1)c_{k+2} - (2k-1)c_k] x^k = 0. \end{aligned}$$

Thus

$$2c_2 + c_0 = 0$$

$$(k+2)(k+1)c_{k+2} - (2k-1)c_k = 0$$

and

$$c_2 = -\frac{1}{2}c_0$$

$$c_{k+2} = \frac{2k-1}{(k+2)(k+1)} c_k, \quad k = 1, 2, 3, \dots$$

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Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$\begin{aligned}c_2 &= -\frac{1}{2} \\c_3 &= c_5 = c_7 = \cdots = 0 \\c_4 &= -\frac{1}{8} \\c_6 &= -\frac{7}{336}\end{aligned}$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$\begin{aligned}c_2 &= c_4 = c_6 = \cdots = 0 \\c_3 &= \frac{1}{6} \\c_5 &= \frac{1}{24} \\c_7 &= \frac{1}{112}\end{aligned}$$

and so on. Thus, two solutions are

$$y_1 = 1 - \frac{1}{2}x^2 - \frac{1}{8}x^4 - \frac{7}{336}x^6 - \cdots \quad \text{and} \quad y_2 = x + \frac{1}{6}x^3 + \frac{1}{24}x^5 + \frac{1}{112}x^7 + \cdots.$$

16. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned}y'' - xy' + 2y &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} - \underbrace{\sum_{n=1}^{\infty} n c_n x^n}_{k=n} + 2 \underbrace{\sum_{n=0}^{\infty} c_n x^n}_{k=n} \\&= \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k - \sum_{k=1}^{\infty} k c_k x^k + 2 \sum_{k=0}^{\infty} c_k x^k \\&= 2c_2 + 2c_0 + \sum_{k=1}^{\infty} [(k+2)(k+1)c_{k+2} - (k-2)c_k] x^k = 0.\end{aligned}$$

Thus

$$\begin{aligned}2c_2 + 2c_0 &= 0 \\(k+2)(k+1)c_{k+2} - (k-2)c_k &= 0\end{aligned}$$

and

$$\begin{aligned}c_2 &= -c_0 \\c_{k+2} &= \frac{k-2}{(k+2)(k+1)} c_k, \quad k = 1, 2, 3, \dots\end{aligned}$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$\begin{aligned}c_2 &= -1 \\c_3 &= c_5 = c_7 = \cdots = 0 \\c_4 &= 0 \\c_6 &= c_8 = c_{10} = \cdots = 0.\end{aligned}$$

For $c_0 = 0$ and $c_1 = 1$ we obtain

$$\begin{aligned} c_2 &= c_4 = c_6 = \cdots = 0 \\ c_3 &= -\frac{1}{6} \\ c_5 &= -\frac{1}{120} \end{aligned}$$

and so on. Thus, two solutions are

$$y_1 = 1 - x^2 \quad \text{and} \quad y_2 = x - \frac{1}{6}x^3 - \frac{1}{120}x^5 - \cdots$$

17. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned} y'' + x^2 y' + xy &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} + \underbrace{\sum_{n=1}^{\infty} n c_n x^{n+1}}_{k=n+1} + \underbrace{\sum_{n=0}^{\infty} c_n x^{n+1}}_{k=n+1} \\ &= \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k + \sum_{k=2}^{\infty} (k-1)c_{k-1} x^k + \sum_{k=1}^{\infty} c_{k-1} x^k \\ &= 2c_2 + (6c_3 + c_0)x + \sum_{k=2}^{\infty} [(k+2)(k+1)c_{k+2} + k c_{k-1}] x^k = 0. \end{aligned}$$

Thus

$$\begin{aligned} c_2 &= 0, \quad 6c_3 + c_0 = 0 \\ (k+2)(k+1)c_{k+2} + k c_{k-1} &= 0 \end{aligned}$$

and

$$\begin{aligned} c_3 &= -\frac{1}{6}c_0 \\ c_{k+2} &= -\frac{k}{(k+2)(k+1)} c_{k-1}, \quad k = 2, 3, 4, \dots \end{aligned}$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$\begin{aligned} c_3 &= -\frac{1}{6} \\ c_4 &= c_5 = 0 \\ c_6 &= \frac{1}{45} \end{aligned}$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$\begin{aligned} c_3 &= 0 \\ c_4 &= -\frac{1}{6} \\ c_5 &= c_6 = 0 \\ c_7 &= \frac{5}{252} \end{aligned}$$

and so on. Thus, two solutions are

$$y_1 = 1 - \frac{1}{6}x^3 + \frac{1}{45}x^6 - \cdots \quad \text{and} \quad y_2 = x - \frac{1}{6}x^4 + \frac{5}{252}x^7 - \cdots$$

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18. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned}
 y'' + 2xy' + 2y &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} + 2 \underbrace{\sum_{n=1}^{\infty} n c_n x^n}_{k=n} + 2 \underbrace{\sum_{n=0}^{\infty} c_n x^n}_{k=n} \\
 &= \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k + 2 \sum_{k=1}^{\infty} k c_k x^k + 2 \sum_{k=0}^{\infty} c_k x^k \\
 &= 2c_2 + 2c_0 + \sum_{k=1}^{\infty} [(k+2)(k+1)c_{k+2} + 2(k+1)c_k] x^k = 0.
 \end{aligned}$$

Thus

$$2c_2 + 2c_0 = 0$$

$$(k+2)(k+1)c_{k+2} + 2(k+1)c_k = 0$$

and

$$c_2 = -c_0$$

$$c_{k+2} = -\frac{2}{k+2} c_k, \quad k = 1, 2, 3, \dots$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$c_2 = -1$$

$$c_3 = c_5 = c_7 = \dots = 0$$

$$c_4 = \frac{1}{2}$$

$$c_6 = -\frac{1}{6}$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$c_2 = c_4 = c_6 = \dots = 0$$

$$c_3 = -\frac{2}{3}$$

$$c_5 = \frac{4}{15}$$

$$c_7 = -\frac{8}{105}$$

and so on. Thus, two solutions are

$$y_1 = 1 - x^2 + \frac{1}{2}x^4 - \frac{1}{6}x^6 + \dots \quad \text{and} \quad y_2 = x - \frac{2}{3}x^3 + \frac{4}{15}x^5 - \frac{8}{105}x^7 + \dots$$

19. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned}
 (x-1)y'' + y' &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-1}}_{k=n-1} - \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} + \underbrace{\sum_{n=1}^{\infty} n c_n x^{n-1}}_{k=n-1} \\
 &= \sum_{k=1}^{\infty} (k+1)k c_{k+1} x^k - \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k + \sum_{k=0}^{\infty} (k+1)c_{k+1} x^k \\
 &= -2c_2 + c_1 + \sum_{k=1}^{\infty} [(k+1)k c_{k+1} - (k+2)(k+1)c_{k+2} + (k+1)c_{k+1}] x^k = 0.
 \end{aligned}$$

Thus

$$-2c_2 + c_1 = 0$$

$$(k+1)^2 c_{k+1} - (k+2)(k+1)c_{k+2} = 0$$

and

$$c_2 = \frac{1}{2}c_1$$

$$c_{k+2} = \frac{k+1}{k+2} c_{k+1}, \quad k = 1, 2, 3, \dots$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find $c_2 = c_3 = c_4 = \dots = 0$. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$c_2 = \frac{1}{2}, \quad c_3 = \frac{1}{3}, \quad c_4 = \frac{1}{4},$$

and so on. Thus, two solutions are

$$y_1 = 1 \quad \text{and} \quad y_2 = x + \frac{1}{2}x^2 + \frac{1}{3}x^3 + \frac{1}{4}x^4 + \dots$$

20. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned} (x+2)y'' + xy' - y &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-1}}_{k=n-1} + \underbrace{\sum_{n=2}^{\infty} 2n(n-1)c_n x^{n-2}}_{k=n-2} + \underbrace{\sum_{n=1}^{\infty} n c_n x^n}_{k=n} - \underbrace{\sum_{n=0}^{\infty} c_n x^n}_{k=n} \\ &= \sum_{k=1}^{\infty} (k+1)k c_{k+1} x^k + \sum_{k=0}^{\infty} 2(k+2)(k+1)c_{k+2} x^k + \sum_{k=1}^{\infty} k c_k x^k - \sum_{k=0}^{\infty} c_k x^k \\ &= 4c_2 - c_0 + \sum_{k=1}^{\infty} [(k+1)k c_{k+1} + 2(k+2)(k+1)c_{k+2} + (k-1)c_k] x^k = 0. \end{aligned}$$

Thus

$$4c_2 - c_0 = 0$$

$$(k+1)k c_{k+1} + 2(k+2)(k+1)c_{k+2} + (k-1)c_k = 0, \quad k = 1, 2, 3, \dots$$

and

$$c_2 = \frac{1}{4}c_0$$

$$c_{k+2} = -\frac{(k+1)k c_{k+1} + (k-1)c_k}{2(k+2)(k+1)}, \quad k = 1, 2, 3, \dots$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$c_1 = 0, \quad c_2 = \frac{1}{4}, \quad c_3 = -\frac{1}{24}, \quad c_4 = 0, \quad c_5 = \frac{1}{480}$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$c_2 = 0$$

$$c_3 = 0$$

$$c_4 = c_5 = c_6 = \dots = 0.$$

Thus, two solutions are

$$y_1 = c_0 \left[1 + \frac{1}{4}x^2 - \frac{1}{24}x^3 + \frac{1}{480}x^5 + \dots \right] \quad \text{and} \quad y_2 = c_1 x.$$

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21. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned}
 y'' - (x+1)y' - y &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} - \underbrace{\sum_{n=1}^{\infty} n c_n x^n}_{k=n} - \underbrace{\sum_{n=1}^{\infty} n c_n x^{n-1}}_{k=n-1} - \underbrace{\sum_{n=0}^{\infty} c_n x^n}_{k=n} \\
 &= \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2}x^k - \sum_{k=1}^{\infty} k c_k x^k - \sum_{k=0}^{\infty} (k+1)c_{k+1}x^k - \sum_{k=0}^{\infty} c_k x^k \\
 &= 2c_2 - c_1 - c_0 + \sum_{k=1}^{\infty} [(k+2)(k+1)c_{k+2} - (k+1)c_{k+1} - (k+1)c_k]x^k = 0.
 \end{aligned}$$

Thus

$$2c_2 - c_1 - c_0 = 0$$

$$(k+2)(k+1)c_{k+2} - (k+1)(c_{k+1} + c_k) = 0$$

and

$$\begin{aligned}
 c_2 &= \frac{c_1 + c_0}{2} \\
 c_{k+2} &= \frac{c_{k+1} + c_k}{k+2} c_k, \quad k = 2, 3, 4, \dots
 \end{aligned}$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$c_2 = \frac{1}{2}, \quad c_3 = \frac{1}{6}, \quad c_4 = \frac{1}{6}$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$c_2 = \frac{1}{2}, \quad c_3 = \frac{1}{2}, \quad c_4 = \frac{1}{4}$$

and so on. Thus, two solutions are

$$y_1 = 1 + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{6}x^4 + \dots \quad \text{and} \quad y_2 = x + \frac{1}{2}x^2 + \frac{1}{2}x^3 + \frac{1}{4}x^4 + \dots$$

22. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned}
 (x^2 + 1)y'' - 6y &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^n}_{k=n} + \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} - 6 \underbrace{\sum_{n=0}^{\infty} c_n x^n}_{k=n} \\
 &= \sum_{k=2}^{\infty} k(k-1)c_k x^k + \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2}x^k - 6 \sum_{k=0}^{\infty} c_k x^k \\
 &= 2c_2 - 6c_0 + (6c_3 - 6c_1)x + \sum_{k=2}^{\infty} [(k^2 - k - 6)c_k + (k+2)(k+1)c_{k+2}]x^k = 0.
 \end{aligned}$$

Thus

$$2c_2 - 6c_0 = 0$$

$$6c_3 - 6c_1 = 0$$

$$(k-3)(k+2)c_k + (k+2)(k+1)c_{k+2} = 0$$

and

$$\begin{aligned}c_2 &= 3c_0 \\c_3 &= c_1 \\c_{k+2} &= -\frac{k-3}{k+1}c_k, \quad k = 2, 3, 4, \dots\end{aligned}$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$\begin{aligned}c_2 &= 3 \\c_3 &= c_5 = c_7 = \dots = 0 \\c_4 &= 1 \\c_6 &= -\frac{1}{5}\end{aligned}$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$\begin{aligned}c_2 &= c_4 = c_6 = \dots = 0 \\c_3 &= 1 \\c_5 &= c_7 = c_9 = \dots = 0.\end{aligned}$$

Thus, two solutions are

$$y_1 = 1 + 3x^2 + x^4 - \frac{1}{5}x^6 + \dots \quad \text{and} \quad y_2 = x + x^3.$$

23. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned}(x^2 + 2)y'' + 3xy' - y &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^n}_{k=n} + 2 \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} + 3 \underbrace{\sum_{n=1}^{\infty} n c_n x^n}_{k=n} - \underbrace{\sum_{n=0}^{\infty} c_n x^n}_{k=n} \\&= \sum_{k=2}^{\infty} k(k-1)c_k x^k + 2 \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k + 3 \sum_{k=1}^{\infty} k c_k x^k - \sum_{k=0}^{\infty} c_k x^k \\&= (4c_2 - c_0) + (12c_3 + 2c_1)x + \sum_{k=2}^{\infty} [2(k+2)(k+1)c_{k+2} + (k^2 + 2k - 1)c_k] x^k = 0.\end{aligned}$$

Thus

$$\begin{aligned}4c_2 - c_0 &= 0 \\12c_3 + 2c_1 &= 0 \\2(k+2)(k+1)c_{k+2} + (k^2 + 2k - 1)c_k &= 0\end{aligned}$$

and

$$\begin{aligned}c_2 &= \frac{1}{4}c_0 \\c_3 &= -\frac{1}{6}c_1 \\c_{k+2} &= -\frac{k^2 + 2k - 1}{2(k+2)(k+1)}c_k, \quad k = 2, 3, 4, \dots\end{aligned}$$

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Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$\begin{aligned} c_2 &= \frac{1}{4} \\ c_3 &= c_5 = c_7 = \cdots = 0 \\ c_4 &= -\frac{7}{96} \end{aligned}$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$\begin{aligned} c_2 &= c_4 = c_6 = \cdots = 0 \\ c_3 &= -\frac{1}{6} \\ c_5 &= \frac{7}{120} \end{aligned}$$

and so on. Thus, two solutions are

$$y_1 = 1 + \frac{1}{4}x^2 - \frac{7}{96}x^4 + \cdots \quad \text{and} \quad y_2 = x - \frac{1}{6}x^3 + \frac{7}{120}x^5 - \cdots.$$

24. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned} (x^2 - 1)y'' + xy' - y &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^n}_{k=n} - \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} + \underbrace{\sum_{n=1}^{\infty} n c_n x^n}_{k=n} - \underbrace{\sum_{n=0}^{\infty} c_n x^n}_{k=n} \\ &= \sum_{k=2}^{\infty} k(k-1)c_k x^k - \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k + \sum_{k=1}^{\infty} k c_k x^k - \sum_{k=0}^{\infty} c_k x^k \\ &= (-2c_2 - c_0) - 6c_3 x + \sum_{k=2}^{\infty} [-(k+2)(k+1)c_{k+2} + (k^2 - 1)c_k] x^k = 0. \end{aligned}$$

Thus

$$\begin{aligned} -2c_2 - c_0 &= 0 \\ -6c_3 &= 0 \\ -(k+2)(k+1)c_{k+2} + (k-1)(k+1)c_k &= 0 \end{aligned}$$

and

$$\begin{aligned} c_2 &= -\frac{1}{2}c_0 \\ c_3 &= 0 \\ c_{k+2} &= \frac{k-1}{k+2} c_k, \quad k = 2, 3, 4, \dots \end{aligned}$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$\begin{aligned} c_2 &= -\frac{1}{2} \\ c_3 &= c_5 = c_7 = \cdots = 0 \\ c_4 &= -\frac{1}{8} \end{aligned}$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$\begin{aligned} c_2 &= c_4 = c_6 = \cdots = 0 \\ c_3 &= c_5 = c_7 = \cdots = 0. \end{aligned}$$

Thus, two solutions are

$$y_1 = 1 - \frac{1}{2}x^2 - \frac{1}{8}x^4 - \dots \quad \text{and} \quad y_2 = x.$$

25. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned} (x-1)y'' - xy' + y &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-1}}_{k=n-1} - \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} - \underbrace{\sum_{n=1}^{\infty} n c_n x^n}_{k=n} + \underbrace{\sum_{n=0}^{\infty} c_n x^n}_{k=n} \\ &= \sum_{k=1}^{\infty} (k+1)k c_{k+1} x^k - \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k - \sum_{k=1}^{\infty} k c_k x^k + \sum_{k=0}^{\infty} c_k x^k \\ &= -2c_2 + c_0 + \sum_{k=1}^{\infty} [-(k+2)(k+1)c_{k+2} + (k+1)k c_{k+1} - (k-1)c_k] x^k = 0. \end{aligned}$$

Thus

$$-2c_2 + c_0 = 0$$

$$-(k+2)(k+1)c_{k+2} + (k-1)k c_{k+1} - (k-1)c_k = 0$$

and

$$\begin{aligned} c_2 &= \frac{1}{2}c_0 \\ c_{k+2} &= \frac{k c_{k+1}}{k+2} - \frac{(k-1)c_k}{(k+2)(k+1)}, \quad k = 1, 2, 3, \dots \end{aligned}$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$c_2 = \frac{1}{2}, \quad c_3 = \frac{1}{6}, \quad c_4 = 0$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain $c_2 = c_3 = c_4 = \dots = 0$. Thus,

$$y = C_1 \left(1 + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \dots \right) + C_2 x$$

and

$$y' = C_1 \left(x + \frac{1}{2}x^2 + \dots \right) + C_2.$$

The initial conditions imply $C_1 = -2$ and $C_2 = 6$, so

$$y = -2 \left(1 + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \dots \right) + 6x = 8x - 2e^x.$$

26. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned} (x+1)y'' - (2-x)y' + y &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-1}}_{k=n-1} + \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} - 2 \underbrace{\sum_{n=1}^{\infty} n c_n x^{n-1}}_{k=n-1} + \underbrace{\sum_{n=1}^{\infty} n c_n x^n}_{k=n} + \underbrace{\sum_{n=0}^{\infty} c_n x^n}_{k=n} \\ &= \sum_{k=1}^{\infty} (k+1)k c_{k+1} x^k + \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k - 2 \sum_{k=0}^{\infty} (k+1)c_{k+1} x^k + \sum_{k=1}^{\infty} k c_k x^k + \sum_{k=0}^{\infty} c_k x^k \\ &= 2c_2 - 2c_1 + c_0 + \sum_{k=1}^{\infty} [(k+2)(k+1)c_{k+2} - (k+1)c_{k+1} + (k+1)c_k] x^k = 0. \end{aligned}$$

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Thus

$$2c_2 - 2c_1 + c_0 = 0$$

$$(k+2)(k+1)c_{k+2} - (k+1)c_{k+1} + (k+1)c_k = 0$$

and

$$c_2 = c_1 - \frac{1}{2}c_0$$

$$c_{k+2} = \frac{1}{k+2}c_{k+1} - \frac{1}{k+2}c_k, \quad k = 1, 2, 3, \dots$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$c_2 = -\frac{1}{2}, \quad c_3 = -\frac{1}{6}, \quad c_4 = \frac{1}{12}$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$c_2 = 1, \quad c_3 = 0, \quad c_4 = -\frac{1}{4}$$

and so on. Thus,

$$y = C_1 \left(1 - \frac{1}{2}x^2 - \frac{1}{6}x^3 + \frac{1}{12}x^4 + \dots \right) + C_2 \left(x + x^2 - \frac{1}{4}x^4 + \dots \right)$$

and

$$y' = C_1 \left(-x - \frac{1}{2}x^2 + \frac{1}{3}x^3 + \dots \right) + C_2 (1 + 2x - x^3 + \dots).$$

The initial conditions imply $C_1 = 2$ and $C_2 = -1$, so

$$y = 2 \left(1 - \frac{1}{2}x^2 - \frac{1}{6}x^3 + \frac{1}{12}x^4 + \dots \right) - \left(x + x^2 - \frac{1}{4}x^4 + \dots \right)$$

$$= 2 - x - 2x^2 - \frac{1}{3}x^3 + \frac{5}{12}x^4 + \dots$$

27. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$y'' - 2xy' + 8y = \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} - 2 \underbrace{\sum_{n=1}^{\infty} n c_n x^n}_{k=n} + 8 \underbrace{\sum_{n=0}^{\infty} c_n x^n}_{k=n}$$

$$= \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k - 2 \sum_{k=1}^{\infty} k c_k x^k + 8 \sum_{k=0}^{\infty} c_k x^k$$

$$= 2c_2 + 8c_0 + \sum_{k=1}^{\infty} [(k+2)(k+1)c_{k+2} + (8-2k)c_k] x^k = 0.$$

Thus

$$2c_2 + 8c_0 = 0$$

$$(k+2)(k+1)c_{k+2} + (8-2k)c_k = 0$$

and

$$c_2 = -4c_0$$

$$c_{k+2} = \frac{2k-8}{(k+2)(k+1)} c_k, \quad k = 1, 2, 3, \dots$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$\begin{aligned}c_2 &= -4 \\c_3 &= c_5 = c_7 = \cdots = 0 \\c_4 &= \frac{4}{3} \\c_6 &= c_8 = c_{10} = \cdots = 0.\end{aligned}$$

For $c_0 = 0$ and $c_1 = 1$ we obtain

$$\begin{aligned}c_2 &= c_4 = c_6 = \cdots = 0 \\c_3 &= -1 \\c_5 &= \frac{1}{10}\end{aligned}$$

and so on. Thus,

$$y = C_1 \left(1 - 4x^2 + \frac{4}{3}x^4 \right) + C_2 \left(x - x^3 + \frac{1}{10}x^5 + \cdots \right)$$

and

$$y' = C_1 \left(-8x + \frac{16}{3}x^3 \right) + C_2 \left(1 - 3x^2 + \frac{1}{2}x^4 + \cdots \right).$$

The initial conditions imply $C_1 = 3$ and $C_2 = 0$, so

$$y = 3 \left(1 - 4x^2 + \frac{4}{3}x^4 \right) = 3 - 12x^2 + 4x^4.$$

28. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned}(x^2 + 1)y'' + 2xy' &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^n}_{k=n} + \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} + \underbrace{\sum_{n=1}^{\infty} 2nc_n x^n}_{k=n} \\&= \sum_{k=2}^{\infty} k(k-1)c_k x^k + \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k + \sum_{k=1}^{\infty} 2kc_k x^k \\&= 2c_2 + (6c_3 + 2c_1)x + \sum_{k=2}^{\infty} [k(k+1)c_k + (k+2)(k+1)c_{k+2}] x^k = 0.\end{aligned}$$

Thus

$$\begin{aligned}2c_2 &= 0 \\6c_3 + 2c_1 &= 0 \\k(k+1)c_k + (k+2)(k+1)c_{k+2} &= 0\end{aligned}$$

and

$$\begin{aligned}c_2 &= 0 \\c_3 &= -\frac{1}{3}c_1 \\c_{k+2} &= -\frac{k}{k+2}c_k, \quad k = 2, 3, 4, \dots\end{aligned}$$

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Choosing $c_0 = 1$ and $c_1 = 0$ we find $c_3 = c_4 = c_5 = \cdots = 0$. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$\begin{aligned}c_3 &= -\frac{1}{3} \\c_4 &= c_6 = c_8 = \cdots = 0 \\c_5 &= -\frac{1}{5} \\c_7 &= \frac{1}{7}\end{aligned}$$

and so on. Thus

$$y = c_0 + c_1 \left(x - \frac{1}{3}x^3 + \frac{1}{5}x^5 - \frac{1}{7}x^7 + \cdots \right)$$

and

$$y' = c_1 (1 - x^2 + x^4 - x^6 + \cdots).$$

The initial conditions imply $c_0 = 0$ and $c_1 = 1$, so

$$y = x - \frac{1}{3}x^3 + \frac{1}{5}x^5 - \frac{1}{7}x^7 + \cdots.$$

29. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned}y'' + (\sin x)y &= \sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} + \left(x - \frac{1}{6}x^3 + \frac{1}{120}x^5 - \cdots \right) (c_0 + c_1 x + c_2 x^2 + \cdots) \\&= [2c_2 + 6c_3 x + 12c_4 x^2 + 20c_5 x^3 + \cdots] + \left[c_0 x + c_1 x^2 + \left(c_2 - \frac{1}{6}c_0 \right) x^3 + \cdots \right] \\&= 2c_2 + (6c_3 + c_0)x + (12c_4 + c_1)x^2 + \left(20c_5 + c_2 - \frac{1}{6}c_0 \right) x^3 + \cdots = 0.\end{aligned}$$

Thus

$$\begin{aligned}2c_2 &= 0 \\6c_3 + c_0 &= 0 \\12c_4 + c_1 &= 0 \\20c_5 + c_2 - \frac{1}{6}c_0 &= 0\end{aligned}$$

and

$$\begin{aligned}c_2 &= 0 \\c_3 &= -\frac{1}{6}c_0 \\c_4 &= -\frac{1}{12}c_1 \\c_5 &= -\frac{1}{20}c_2 + \frac{1}{120}c_0.\end{aligned}$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$c_2 = 0, \quad c_3 = -\frac{1}{6}, \quad c_4 = 0, \quad c_5 = \frac{1}{120}$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$c_2 = 0, \quad c_3 = 0, \quad c_4 = -\frac{1}{12}, \quad c_5 = 0$$

and so on. Thus, two solutions are

$$y_1 = 1 - \frac{1}{6}x^3 + \frac{1}{120}x^5 + \cdots \quad \text{and} \quad y_2 = x - \frac{1}{12}x^4 + \cdots.$$

30. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned} y'' + e^x y' - y &= \sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} \\ &\quad + \left(1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \cdots\right) (c_1 + 2c_2x + 3c_3x^2 + 4c_4x^3 + \cdots) - \sum_{n=0}^{\infty} c_n x^n \\ &= [2c_2 + 6c_3x + 12c_4x^2 + 20c_5x^3 + \cdots] \\ &\quad + \left[c_1 + (2c_2 + c_1)x + \left(3c_3 + 2c_2 + \frac{1}{2}c_1\right)x^2 + \cdots\right] - [c_0 + c_1x + c_2x^2 + \cdots] \\ &= (2c_2 + c_1 - c_0) + (6c_3 + 2c_2)x + \left(12c_4 + 3c_3 + c_2 + \frac{1}{2}c_1\right)x^2 + \cdots = 0. \end{aligned}$$

Thus

$$\begin{aligned} 2c_2 + c_1 - c_0 &= 0 \\ 6c_3 + 2c_2 &= 0 \\ 12c_4 + 3c_3 + c_2 + \frac{1}{2}c_1 &= 0 \end{aligned}$$

and

$$\begin{aligned} c_2 &= \frac{1}{2}c_0 - \frac{1}{2}c_1 \\ c_3 &= -\frac{1}{3}c_2 \\ c_4 &= -\frac{1}{4}c_3 + \frac{1}{12}c_2 - \frac{1}{24}c_1. \end{aligned}$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$c_2 = \frac{1}{2}, \quad c_3 = -\frac{1}{6}, \quad c_4 = 0$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$c_2 = -\frac{1}{2}, \quad c_3 = \frac{1}{6}, \quad c_4 = -\frac{1}{24}$$

and so on. Thus, two solutions are

$$y_1 = 1 + \frac{1}{2}x^2 - \frac{1}{6}x^3 + \cdots \quad \text{and} \quad y_2 = x - \frac{1}{2}x^2 + \frac{1}{6}x^3 - \frac{1}{24}x^4 + \cdots.$$

31. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the first differential equation leads to

$$\begin{aligned} y'' - xy &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} - \underbrace{\sum_{n=0}^{\infty} c_n x^{n+1}}_{k=n+1} = \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2}x^k - \sum_{k=1}^{\infty} c_{k-1}x^k \\ &= 2c_2 + \sum_{k=1}^{\infty} [(k+2)(k+1)c_{k+2} - c_{k-1}]x^k = 1. \end{aligned}$$

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Thus

$$2c_2 = 1$$

$$(k+2)(k+1)c_{k+2} - c_{k-1} = 0$$

and

$$c_2 = \frac{1}{2}$$

$$c_{k+2} = \frac{c_{k-1}}{(k+2)(k+1)}, \quad k = 1, 2, 3, \dots$$

Let c_0 and c_1 be arbitrary and iterate to find

$$c_2 = \frac{1}{2}$$

$$c_3 = \frac{1}{6}c_0$$

$$c_4 = \frac{1}{12}c_1$$

$$c_5 = \frac{1}{20}c_2 = \frac{1}{40}$$

and so on. The solution is

$$\begin{aligned} y &= c_0 + c_1x + \frac{1}{2}x^2 + \frac{1}{6}c_0x^3 + \frac{1}{12}c_1x^4 + \frac{1}{40}c_5 + \dots \\ &= c_0 \left(1 + \frac{1}{6}x^3 + \dots \right) + c_1 \left(x + \frac{1}{12}x^4 + \dots \right) + \frac{1}{2}x^2 + \frac{1}{40}x^5 + \dots \end{aligned}$$

Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the second differential equation leads to

$$\begin{aligned} y'' - 4xy' - 4y &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} - \underbrace{\sum_{n=1}^{\infty} 4nc_n x^n}_{k=n} - \underbrace{\sum_{n=0}^{\infty} 4c_n x^n}_{k=n} \\ &= \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2}x^k - \sum_{k=1}^{\infty} 4kc_k x^k - \sum_{k=0}^{\infty} 4c_k x^k \\ &= 2c_2 - 4c_0 + \sum_{k=1}^{\infty} [(k+2)(k+1)c_{k+2} - 4(k+1)c_k]x^k \\ &= e^x = 1 + \sum_{k=1}^{\infty} \frac{1}{k!}x^k. \end{aligned}$$

Thus

$$2c_2 - 4c_0 = 1$$

$$(k+2)(k+1)c_{k+2} - 4(k+1)c_k = \frac{1}{k!}$$

and

$$c_2 = \frac{1}{2} + 2c_0$$

$$c_{k+2} = \frac{1}{(k+2)!} + \frac{4}{k+2}c_k, \quad k = 1, 2, 3, \dots$$

Let c_0 and c_1 be arbitrary and iterate to find

$$\begin{aligned}c_2 &= \frac{1}{2} + 2c_0 \\c_3 &= \frac{1}{3!} + \frac{4}{3}c_1 = \frac{1}{3!} + \frac{4}{3}c_1 \\c_4 &= \frac{1}{4!} + \frac{4}{4}c_2 = \frac{1}{4!} + \frac{1}{2} + 2c_0 = \frac{13}{4!} + 2c_0 \\c_5 &= \frac{1}{5!} + \frac{4}{5}c_3 = \frac{1}{5!} + \frac{4}{5 \cdot 3!} + \frac{16}{15}c_1 = \frac{17}{5!} + \frac{16}{15}c_1 \\c_6 &= \frac{1}{6!} + \frac{4}{6}c_4 = \frac{1}{6!} + \frac{4 \cdot 13}{6 \cdot 4!} + \frac{8}{6}c_0 = \frac{261}{6!} + \frac{4}{3}c_0 \\c_7 &= \frac{1}{7!} + \frac{4}{7}c_5 = \frac{1}{7!} + \frac{4 \cdot 17}{7 \cdot 5!} + \frac{64}{105}c_1 = \frac{409}{7!} + \frac{64}{105}c_1\end{aligned}$$

and so on. The solution is

$$\begin{aligned}y &= c_0 + c_1x + \left(\frac{1}{2} + 2c_0\right)x^2 + \left(\frac{1}{3!} + \frac{4}{3}c_1\right)x^3 - \left(\frac{13}{4!} + 2c_0\right)x^4 + \left(\frac{17}{5!} + \frac{16}{15}c_1\right)x^5 \\&\quad + \left(\frac{261}{6!} + \frac{4}{3}c_0\right)x^6 + \left(\frac{409}{7!} + \frac{64}{105}c_1\right)x^7 + \cdots \\&= c_0 \left[1 + 2x^2 + 2x^4 + \frac{4}{3}x^6 + \cdots\right] + c_1 \left[x + \frac{4}{3}x^3 + \frac{16}{15}x^5 + \frac{64}{105}x^7 + \cdots\right] \\&\quad + \frac{1}{2}x^2 + \frac{1}{3!}x^3 + \frac{13}{4!}x^4 + \frac{17}{5!}x^5 + \frac{261}{6!}x^6 + \frac{409}{7!}x^7 + \cdots.\end{aligned}$$

32. We identify $P(x) = 0$ and $Q(x) = \sin x/x$. The Taylor series representation for $\sin x/x$ is $1 - x^2/3! + x^4/5! - \cdots$, for $|x| < \infty$. Thus, $Q(x)$ is analytic at $x = 0$ and $x = 0$ is an ordinary point of the differential equation.

33. (a) Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned}y'' + xy' + y &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} + \underbrace{\sum_{n=1}^{\infty} n c_n x^n}_{k=n} + \underbrace{\sum_{n=0}^{\infty} c_n x^n}_{k=n} \\&= \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k + \sum_{k=1}^{\infty} k c_k x^k + \sum_{k=0}^{\infty} c_k x^k \\&= (2c_2 + c_0) + \sum_{k=1}^{\infty} [(k+2)(k+1)c_{k+2} + (k+1)c_k] x^k = 0.\end{aligned}$$

Thus

$$\begin{aligned}2c_2 + c_0 &= 0 \\(k+2)(k+1)c_{k+2} + (k+1)c_k &= 0\end{aligned}$$

and

$$\begin{aligned}c_2 &= -\frac{1}{2}c_0 \\c_{k+2} &= -\frac{1}{k+2}c_k.\end{aligned}$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

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$$c_2 = -\frac{1}{2}$$

$$c_3 = c_5 = c_7 = \cdots = 0$$

$$c_4 = -\frac{1}{4} \left(-\frac{1}{2} \right) = \frac{1}{2^2 \cdot 2}$$

$$c_6 = -\frac{1}{6} \left(\frac{1}{2^2 \cdot 2} \right) = -\frac{1}{2^3 \cdot 3!}$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$c_2 = c_4 = c_6 = \cdots = 0$$

$$c_3 = -\frac{1}{3} = -\frac{2}{3!}$$

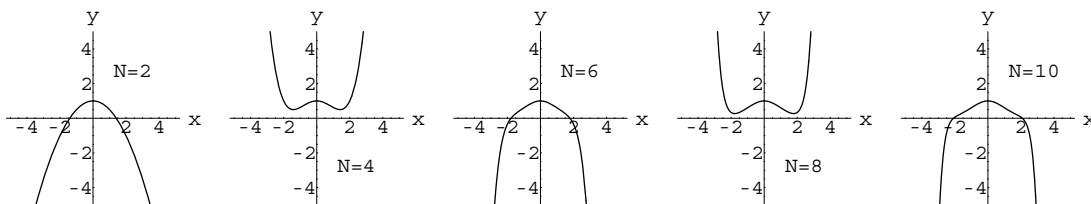
$$c_5 = -\frac{1}{5} \left(-\frac{1}{3} \right) = \frac{1}{5 \cdot 3} = \frac{4 \cdot 2}{5!}$$

$$c_7 = -\frac{1}{7} \left(\frac{4 \cdot 2}{5!} \right) = -\frac{6 \cdot 4 \cdot 2}{7!}$$

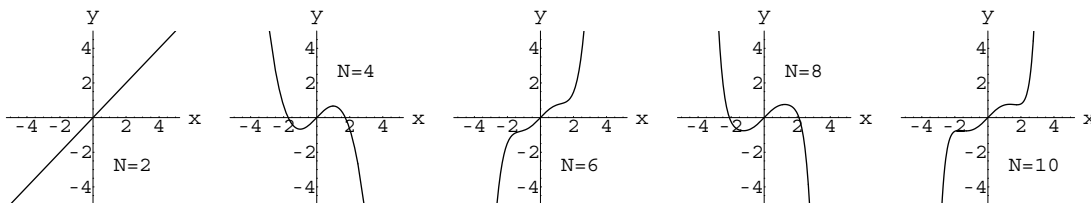
and so on. Thus, two solutions are

$$y_1 = \sum_{k=0}^{\infty} \frac{(-1)^k}{2^k \cdot k!} x^{2k} \quad \text{and} \quad y_2 = \sum_{k=0}^{\infty} \frac{(-1)^k 2^k k!}{(2k+1)!} x^{2k+1}.$$

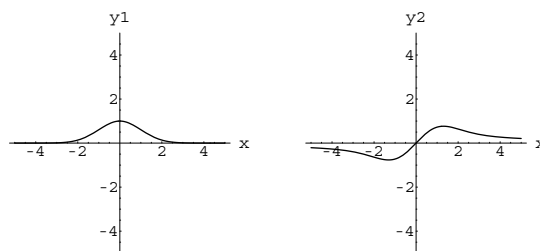
(b) For y_1 , $S_3 = S_2$ and $S_5 = S_4$, so we plot S_2, S_4, S_6, S_8 , and S_{10} .



For y_2 , $S_3 = S_4$ and $S_5 = S_6$, so we plot S_2, S_4, S_6, S_8 , and S_{10} .



(c) The graphs of y_1 and y_2 obtained from a numerical solver are shown. We see that the partial sum representations indicate the even and odd natures of the solution, but don't really give a very accurate representation of the true solution. Increasing N to about 20 gives a much more accurate representation on $[-4, 4]$.



34. (a) We have

$$\begin{aligned}
 y'' + (\cos x)y &= 2c_2 + 6c_3x + 12c_4x^2 + 20c_5x^3 + 30c_6x^4 + 42c_7x^5 \\
 &+ \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots\right)(c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5 + \cdots) \\
 &= (2c_2 + c_0) + (6c_3 + c_1)x + \left(12c_4 + c_2 - \frac{1}{2}c_0\right)x^2 + \left(20c_5 + c_3 - \frac{1}{2}c_1\right)x^3 \\
 &+ \left(30c_6 + c_4 + \frac{1}{24}c_0 - \frac{1}{2}c_2\right)x^4 + \left(42c_7 + c_5 + \frac{1}{24}c_1 - \frac{1}{2}c_3\right)x^5 + \cdots.
 \end{aligned}$$

Then

$$30c_6 + c_4 + \frac{1}{24}c_0 - \frac{1}{2}c_2 = 0 \quad \text{and} \quad 42c_7 + c_5 + \frac{1}{24}c_1 - \frac{1}{2}c_3 = 0,$$

which gives $c_6 = -c_0/80$ and $c_7 = -19c_1/5040$. Thus

$$y_1(x) = 1 - \frac{1}{2}x^2 + \frac{1}{12}x^4 - \frac{1}{80}x^6 + \cdots$$

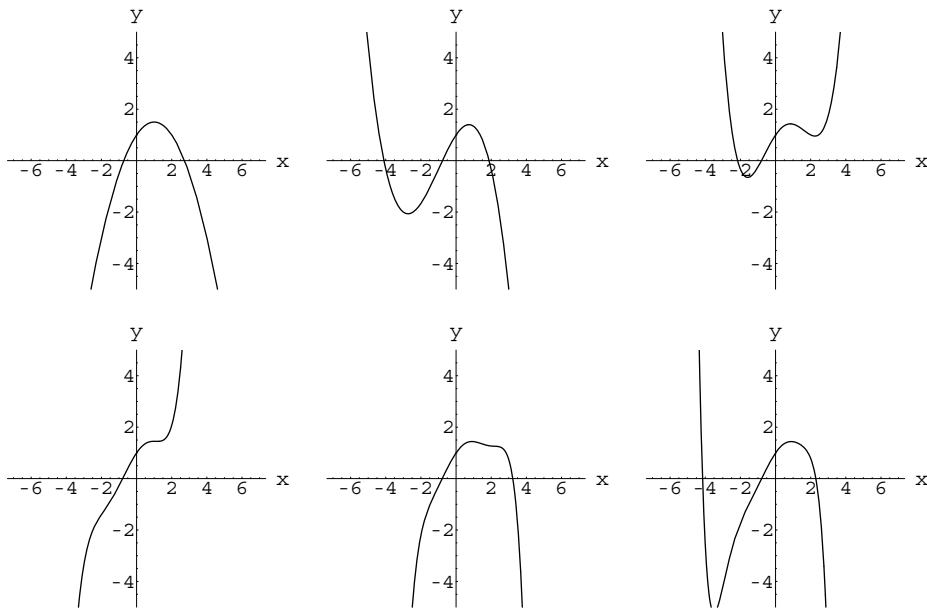
and

$$y_2(x) = x - \frac{1}{6}x^3 + \frac{1}{30}x^5 - \frac{19}{5040}x^7 + \cdots.$$

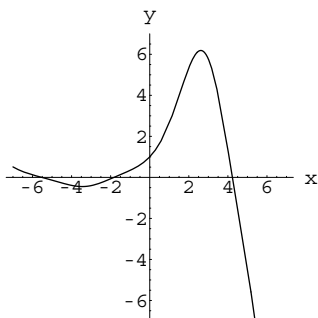
(b) From part (a) the general solution of the differential equation is $y = c_1y_1 + c_2y_2$. Then $y(0) = c_1 + c_2 \cdot 0 = c_1$ and $y'(0) = c_1 \cdot 0 + c_2 = c_2$, so the solution of the initial-value problem is

$$y = y_1 + y_2 = 1 + x - \frac{1}{2}x^2 - \frac{1}{6}x^3 + \frac{1}{12}x^4 + \frac{1}{30}x^5 - \frac{1}{80}x^6 - \frac{19}{5040}x^7 + \cdots.$$

(c)



(d)



Exercises 5.1

35. From $e^x = \sum_{k=0}^{\infty} x^k/k!$ we see that $e^{-x^2/2} = \sum_{k=0}^{\infty} (-x^2/2)^k/k! = \sum_{k=0}^{\infty} (-1)^k x^{2k}/2^k k!$. From (5) of Section 3.2 we have

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{-\int x dx}}{y_1^2} dx = e^{-x^2/2} \int \frac{e^{-x^2/2}}{(e^{-x^2/2})^2} dx = e^{-x^2/2} \int \frac{e^{-x^2/2}}{e^{-x^2}} dx = e^{-x^2/2} \int e^{x^2/2} dx \\ &= \sum_{k=0}^{\infty} \frac{(-1)^k}{2^k k!} x^{2k} \int \sum_{k=0}^{\infty} \frac{1}{2^k k!} x^{2k} dx = \left(\sum_{k=0}^{\infty} \frac{(-1)^k}{2^k k!} x^{2k} \right) \left(\sum_{k=0}^{\infty} \int \frac{1}{2^k k!} x^{2k} dx \right) \\ &= \left(\sum_{k=0}^{\infty} \frac{(-1)^k}{2^k k!} x^{2k} \right) \left(\sum_{k=0}^{\infty} \frac{1}{(2k+1)2^k k!} x^{2k+1} \right) \\ &= \left(1 - \frac{1}{2}x^2 + \frac{1}{2^2 \cdot 2}x^4 - \frac{1}{2^3 \cdot 3!}x^6 + \cdots \right) \left(x + \frac{1}{3 \cdot 2}x^3 + \frac{1}{5 \cdot 2^2 \cdot 2}x^5 + \frac{1}{7 \cdot 2^3 \cdot 3!}x^7 + \cdots \right) \\ &= x - \frac{2}{3!}x^3 + \frac{4 \cdot 2}{5!}x^5 - \frac{6 \cdot 4 \cdot 2}{7!}x^7 + \cdots = \sum_{k=0}^{\infty} \frac{(-1)^k 2^k k!}{(2k+1)!} x^{2k+1}. \end{aligned}$$

36. If $x > 0$ and $y > 0$, then $y'' = -xy < 0$ and the graph of a solution curve is concave down. Thus, whatever portion of a solution curve lies in the first quadrant is concave down. When $x > 0$ and $y < 0$, $y'' = -xy > 0$, so whatever portion of a solution curve lies in the fourth quadrant is concave up.

Exercises 5.2

- Irregular singular point: $x = 0$
- Regular singular points: $x = 0, -3$
- Irregular singular point: $x = 3$; regular singular point: $x = -3$
- Irregular singular point: $x = 1$; regular singular point: $x = 0$
- Regular singular points: $x = 0, \pm 2i$
- Irregular singular point: $x = 5$; regular singular point: $x = 0$
- Regular singular points: $x = -3, 2$
- Regular singular points: $x = 0, \pm i$
- Irregular singular point: $x = 0$; regular singular points: $x = 2, \pm 5$
- Irregular singular point: $x = -1$; regular singular points: $x = 0, 3$
- Writing the differential equation in the form

$$y'' + \frac{5}{x-1} y' + \frac{x}{x+1} y = 0$$

we see that $x_0 = 1$ and $x_0 = -1$ are regular singular points. For $x_0 = 1$ the differential equation can be put in the form

$$(x-1)^2 y'' + 5(x-1)y' + \frac{x(x-1)^2}{x+1} y = 0.$$

In this case $p(x) = 5$ and $q(x) = x(x-1)^2/(x+1)$. For $x_0 = -1$ the differential equation can be put in the form

$$(x+1)^2 y'' + 5(x+1)\frac{x+1}{x-1} y' + x(x+1)y = 0.$$

In this case $p(x) = (x+1)/(x-1)$ and $q(x) = x(x+1)$.

12. Writing the differential equation in the form

$$y'' + \frac{x+3}{x}y' + 7xy = 0$$

we see that $x_0 = 0$ is a regular singular point. Multiplying by x^2 , the differential equation can be put in the form

$$x^2y'' + x(x+3)y' + 7x^3y = 0.$$

We identify $p(x) = x+3$ and $q(x) = 7x^3$.

13. We identify $P(x) = 5/3x + 1$ and $Q(x) = -1/3x^2$, so that $p(x) = xP(x) = \frac{5}{3}x + x$ and $q(x) = x^2Q(x) = -\frac{1}{3}$. Then $a_0 = \frac{5}{3}$, $b_0 = -\frac{1}{3}$, and the indicial equation is

$$r(r-1) + \frac{5}{3}r - \frac{1}{3} = r^2 + \frac{2}{3}r - \frac{1}{3} = \frac{1}{3}(3r^2 + 2r - 1) = \frac{1}{3}(3r-1)(r+1) = 0.$$

The indicial roots are $\frac{1}{3}$ and -1 . Since these do not differ by an integer we expect to find two series solutions using the method of Frobenius.

14. We identify $P(x) = 1/x$ and $Q(x) = 10/x$, so that $p(x) = xP(x) = 1$ and $q(x) = x^2Q(x) = 10x$. Then $a_0 = 1$, $b_0 = 0$, and the indicial equation is

$$r(r-1) + r = r^2 = 0.$$

The indicial roots are 0 and 0. Since these are equal, we expect the method of Frobenius to yield a single series solution.

15. Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation and collecting terms, we obtain

$$2xy'' - y' + 2y = (2r^2 - 3r)c_0x^{r-1} + \sum_{k=1}^{\infty} [2(k+r-1)(k+r)c_k - (k+r)c_k + 2c_{k-1}]x^{k+r-1} = 0,$$

which implies

$$2r^2 - 3r = r(2r-3) = 0$$

and

$$(k+r)(2k+2r-3)c_k + 2c_{k-1} = 0.$$

The indicial roots are $r = 0$ and $r = 3/2$. For $r = 0$ the recurrence relation is

$$c_k = -\frac{2c_{k-1}}{k(2k-3)}, \quad k = 1, 2, 3, \dots,$$

and

$$c_1 = 2c_0, \quad c_2 = -2c_0, \quad c_3 = \frac{4}{9}c_0.$$

For $r = 3/2$ the recurrence relation is

$$c_k = -\frac{2c_{k-1}}{(2k+3)k}, \quad k = 1, 2, 3, \dots,$$

and

$$c_1 = -\frac{2}{5}c_0, \quad c_2 = \frac{2}{35}c_0, \quad c_3 = -\frac{4}{945}c_0.$$

The general solution on $(0, \infty)$ is

$$y = C_1 \left(1 + 2x - 2x^2 + \frac{4}{9}x^3 + \dots \right) + C_2 x^{3/2} \left(1 - \frac{2}{5}x + \frac{2}{35}x^2 - \frac{4}{945}x^3 + \dots \right).$$

Exercises 5.2

16. Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation and collecting terms, we obtain

$$\begin{aligned} 2xy'' + 5y' + xy &= (2r^2 + 3r) c_0 x^{r-1} + (2r^2 + 7r + 5) c_1 x^r \\ &+ \sum_{k=2}^{\infty} [2(k+r)(k+r-1)c_k + 5(k+r)c_k + c_{k-2}] x^{k+r-1} \\ &= 0, \end{aligned}$$

which implies

$$\begin{aligned} 2r^2 + 3r &= r(2r + 3) = 0, \\ (2r^2 + 7r + 5) c_1 &= 0, \end{aligned}$$

and

$$(k+r)(2k+2r+3)c_k + c_{k-2} = 0.$$

The indicial roots are $r = -3/2$ and $r = 0$, so $c_1 = 0$. For $r = -3/2$ the recurrence relation is

$$c_k = -\frac{c_{k-2}}{(2k-3)k}, \quad k = 2, 3, 4, \dots,$$

and

$$c_2 = -\frac{1}{2}c_0, \quad c_3 = 0, \quad c_4 = \frac{1}{40}c_0.$$

For $r = 0$ the recurrence relation is

$$c_k = -\frac{c_{k-2}}{k(2k+3)}, \quad k = 2, 3, 4, \dots,$$

and

$$c_2 = -\frac{1}{14}c_0, \quad c_3 = 0, \quad c_4 = \frac{1}{616}c_0.$$

The general solution on $(0, \infty)$ is

$$y = C_1 x^{-3/2} \left(1 - \frac{1}{2}x^2 + \frac{1}{40}x^4 + \dots \right) + C_2 \left(1 - \frac{1}{14}x^2 + \frac{1}{616}x^4 + \dots \right).$$

17. Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation and collecting terms, we obtain

$$\begin{aligned} 4xy'' + \frac{1}{2}y' + y &= \left(4r^2 - \frac{7}{2}r \right) c_0 x^{r-1} + \sum_{k=1}^{\infty} \left[4(k+r)(k+r-1)c_k + \frac{1}{2}(k+r)c_k + c_{k-1} \right] x^{k+r-1} \\ &= 0, \end{aligned}$$

which implies

$$4r^2 - \frac{7}{2}r = r \left(4r - \frac{7}{2} \right) = 0$$

and

$$\frac{1}{2}(k+r)(8k+8r-7)c_k + c_{k-1} = 0.$$

The indicial roots are $r = 0$ and $r = 7/8$. For $r = 0$ the recurrence relation is

$$c_k = -\frac{2c_{k-1}}{k(8k-7)}, \quad k = 1, 2, 3, \dots,$$

and

$$c_1 = -2c_0, \quad c_2 = \frac{2}{9}c_0, \quad c_3 = -\frac{4}{459}c_0.$$

For $r = 7/8$ the recurrence relation is

$$c_k = -\frac{2c_{k-1}}{(8k+7)k}, \quad k = 1, 2, 3, \dots,$$

and

$$c_1 = -\frac{2}{15}c_0, \quad c_2 = \frac{2}{345}c_0, \quad c_3 = -\frac{4}{32,085}c_0.$$

The general solution on $(0, \infty)$ is

$$y = C_1 \left(1 - 2x + \frac{2}{9}x^2 - \frac{4}{459}x^3 + \dots \right) + C_2 x^{7/8} \left(1 - \frac{2}{15}x + \frac{2}{345}x^2 - \frac{4}{32,085}x^3 + \dots \right).$$

18. Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation and collecting terms, we obtain

$$\begin{aligned} 2x^2 y'' - xy' + (x^2 + 1)y &= (2r^2 - 3r + 1)c_0 x^r + (2r^2 + r)c_1 x^{r+1} \\ &+ \sum_{k=2}^{\infty} [2(k+r)(k+r-1)c_k - (k+r)c_k + c_k + c_{k-2}]x^{k+r} \\ &= 0, \end{aligned}$$

which implies

$$\begin{aligned} 2r^2 - 3r + 1 &= (2r-1)(r-1) = 0, \\ (2r^2 + r)c_1 &= 0, \end{aligned}$$

and

$$[(k+r)(2k+2r-3)+1]c_k + c_{k-2} = 0.$$

The indicial roots are $r = 1/2$ and $r = 1$, so $c_1 = 0$. For $r = 1/2$ the recurrence relation is

$$c_k = -\frac{c_{k-2}}{k(2k-1)}, \quad k = 2, 3, 4, \dots,$$

and

$$c_2 = -\frac{1}{6}c_0, \quad c_3 = 0, \quad c_4 = \frac{1}{168}c_0.$$

For $r = 1$ the recurrence relation is

$$c_k = -\frac{c_{k-2}}{k(2k+1)}, \quad k = 2, 3, 4, \dots,$$

and

$$c_2 = -\frac{1}{10}c_0, \quad c_3 = 0, \quad c_4 = \frac{1}{360}c_0.$$

The general solution on $(0, \infty)$ is

$$y = C_1 x^{1/2} \left(1 - \frac{1}{6}x^2 + \frac{1}{168}x^4 + \dots \right) + C_2 x \left(1 - \frac{1}{10}x^2 + \frac{1}{360}x^4 + \dots \right).$$

19. Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation and collecting terms, we obtain

$$\begin{aligned} 3xy'' + (2-x)y' - y &= (3r^2 - r)c_0 x^{r-1} \\ &+ \sum_{k=1}^{\infty} [3(k+r-1)(k+r)c_k + 2(k+r)c_k - (k+r)c_{k-1}]x^{k+r-1} \\ &= 0, \end{aligned}$$

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which implies

$$3r^2 - r = r(3r - 1) = 0$$

and

$$(k + r)(3k + 3r - 1)c_k - (k + r)c_{k-1} = 0.$$

The indicial roots are $r = 0$ and $r = 1/3$. For $r = 0$ the recurrence relation is

$$c_k = \frac{c_{k-1}}{(3k - 1)}, \quad k = 1, 2, 3, \dots,$$

and

$$c_1 = \frac{1}{2}c_0, \quad c_2 = \frac{1}{10}c_0, \quad c_3 = \frac{1}{80}c_0.$$

For $r = 1/3$ the recurrence relation is

$$c_k = \frac{c_{k-1}}{3k}, \quad k = 1, 2, 3, \dots,$$

and

$$c_1 = \frac{1}{3}c_0, \quad c_2 = \frac{1}{18}c_0, \quad c_3 = \frac{1}{162}c_0.$$

The general solution on $(0, \infty)$ is

$$y = C_1 \left(1 + \frac{1}{2}x + \frac{1}{10}x^2 + \frac{1}{80}x^3 + \dots \right) + C_2 x^{1/3} \left(1 + \frac{1}{3}x + \frac{1}{18}x^2 + \frac{1}{162}x^3 + \dots \right).$$

- 20.** Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation and collecting terms, we obtain

$$\begin{aligned} x^2 y'' - \left(x - \frac{2}{9} \right) y &= \left(r^2 - r + \frac{2}{9} \right) c_0 x^r + \sum_{k=1}^{\infty} \left[(k+r)(k+r-1)c_k + \frac{2}{9}c_k - c_{k-1} \right] x^{k+r} \\ &= 0, \end{aligned}$$

which implies

$$r^2 - r + \frac{2}{9} = \left(r - \frac{2}{3} \right) \left(r - \frac{1}{3} \right) = 0$$

and

$$\left[(k+r)(k+r-1) + \frac{2}{9} \right] c_k - c_{k-1} = 0.$$

The indicial roots are $r = 2/3$ and $r = 1/3$. For $r = 2/3$ the recurrence relation is

$$c_k = \frac{3c_{k-1}}{3k^2 + k}, \quad k = 1, 2, 3, \dots,$$

and

$$c_1 = \frac{3}{4}c_0, \quad c_2 = \frac{9}{56}c_0, \quad c_3 = \frac{9}{560}c_0.$$

For $r = 1/3$ the recurrence relation is

$$c_k = \frac{3c_{k-1}}{3k^2 - k}, \quad k = 1, 2, 3, \dots,$$

and

$$c_1 = \frac{3}{2}c_0, \quad c_2 = \frac{9}{20}c_0, \quad c_3 = \frac{9}{160}c_0.$$

The general solution on $(0, \infty)$ is

$$y = C_1 x^{2/3} \left(1 + \frac{3}{4}x + \frac{9}{56}x^2 + \frac{9}{560}x^3 + \dots \right) + C_2 x^{1/3} \left(1 + \frac{3}{2}x + \frac{9}{20}x^2 + \frac{9}{160}x^3 + \dots \right).$$

21. Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation and collecting terms, we obtain

$$\begin{aligned} 2xy'' - (3+2x)y' + y &= (2r^2 - 5r)c_0 x^{r-1} + \sum_{k=1}^{\infty} [2(k+r)(k+r-1)c_k \\ &\quad - 3(k+r)c_k - 2(k+r-1)c_{k-1} + c_{k-1}]x^{k+r-1} \\ &= 0, \end{aligned}$$

which implies

$$2r^2 - 5r = r(2r - 5) = 0$$

and

$$(k+r)(2k+2r-5)c_k - (2k+2r-3)c_{k-1} = 0.$$

The indicial roots are $r = 0$ and $r = 5/2$. For $r = 0$ the recurrence relation is

$$c_k = \frac{(2k-3)c_{k-1}}{k(2k-5)}, \quad k = 1, 2, 3, \dots,$$

and

$$c_1 = \frac{1}{3}c_0, \quad c_2 = -\frac{1}{6}c_0, \quad c_3 = -\frac{1}{6}c_0.$$

For $r = 5/2$ the recurrence relation is

$$c_k = \frac{2(k+1)c_{k-1}}{k(2k+5)}, \quad k = 1, 2, 3, \dots,$$

and

$$c_1 = \frac{4}{7}c_0, \quad c_2 = \frac{4}{21}c_0, \quad c_3 = \frac{32}{693}c_0.$$

The general solution on $(0, \infty)$ is

$$y = C_1 \left(1 + \frac{1}{3}x - \frac{1}{6}x^2 - \frac{1}{6}x^3 + \dots \right) + C_2 x^{5/2} \left(1 + \frac{4}{7}x + \frac{4}{21}x^2 + \frac{32}{693}x^3 + \dots \right).$$

22. Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation and collecting terms, we obtain

$$\begin{aligned} x^2 y'' + xy' + \left(x^2 - \frac{4}{9}\right)y &= \left(r^2 - \frac{4}{9}\right)c_0 x^r + \left(r^2 + 2r + \frac{5}{9}\right)c_1 x^{r+1} \\ &\quad + \sum_{k=2}^{\infty} \left[(k+r)(k+r-1)c_k + (k+r)c_k - \frac{4}{9}c_k + c_{k-2} \right] x^{k+r} \\ &= 0, \end{aligned}$$

which implies

$$r^2 - \frac{4}{9} = \left(r + \frac{2}{3}\right)\left(r - \frac{2}{3}\right) = 0,$$

$$\left(r^2 + 2r + \frac{5}{9}\right)c_1 = 0,$$

and

$$\left[(k+r)^2 - \frac{4}{9} \right] c_k + c_{k-2} = 0.$$

The indicial roots are $r = -2/3$ and $r = 2/3$, so $c_1 = 0$. For $r = -2/3$ the recurrence relation is

$$c_k = -\frac{9c_{k-2}}{3k(3k-4)}, \quad k = 2, 3, 4, \dots,$$

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and

$$c_2 = -\frac{3}{4}c_0, \quad c_3 = 0, \quad c_4 = \frac{9}{128}c_0.$$

For $r = 2/3$ the recurrence relation is

$$c_k = -\frac{9c_{k-2}}{3k(3k+4)}, \quad k = 2, 3, 4, \dots,$$

and

$$c_2 = -\frac{3}{20}c_0, \quad c_3 = 0, \quad c_4 = \frac{9}{1,280}c_0.$$

The general solution on $(0, \infty)$ is

$$y = C_1 x^{-2/3} \left(1 - \frac{3}{4}x^2 + \frac{9}{128}x^4 + \dots \right) + C_2 x^{2/3} \left(1 - \frac{3}{20}x^2 + \frac{9}{1,280}x^4 + \dots \right).$$

- 23.** Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation and collecting terms, we obtain

$$\begin{aligned} 9x^2 y'' + 9x^2 y' + 2y &= (9r^2 - 9r + 2) c_0 x^r \\ &+ \sum_{k=1}^{\infty} [9(k+r)(k+r-1)c_k + 2c_k + 9(k+r-1)c_{k-1}] x^{k+r} \\ &= 0, \end{aligned}$$

which implies

$$9r^2 - 9r + 2 = (3r-1)(3r-2) = 0$$

and

$$[9(k+r)(k+r-1) + 2]c_k + 9(k+r-1)c_{k-1} = 0.$$

The indicial roots are $r = 1/3$ and $r = 2/3$. For $r = 1/3$ the recurrence relation is

$$c_k = -\frac{(3k-2)c_{k-1}}{k(3k-1)}, \quad k = 1, 2, 3, \dots,$$

and

$$c_1 = -\frac{1}{2}c_0, \quad c_2 = \frac{1}{5}c_0, \quad c_3 = -\frac{7}{120}c_0.$$

For $r = 2/3$ the recurrence relation is

$$c_k = -\frac{(3k-1)c_{k-1}}{k(3k+1)}, \quad k = 1, 2, 3, \dots,$$

and

$$c_1 = -\frac{1}{2}c_0, \quad c_2 = \frac{5}{28}c_0, \quad c_3 = -\frac{1}{21}c_0.$$

The general solution on $(0, \infty)$ is

$$y = C_1 x^{1/3} \left(1 - \frac{1}{2}x + \frac{1}{5}x^2 - \frac{7}{120}x^3 + \dots \right) + C_2 x^{2/3} \left(1 - \frac{1}{2}x + \frac{5}{28}x^2 - \frac{1}{21}x^3 + \dots \right).$$

- 24.** Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation and collecting terms, we obtain

$$\begin{aligned} 2x^2 y'' + 3xy' + (2x-1)y &= (2r^2 + r - 1) c_0 x^r \\ &+ \sum_{k=1}^{\infty} [2(k+r)(k+r-1)c_k + 3(k+r)c_k - c_k + 2c_{k-1}] x^{k+r} \\ &= 0, \end{aligned}$$

which implies

$$2r^2 + r - 1 = (2r - 1)(r + 1) = 0$$

and

$$[(k + r)(2k + 2r + 1) - 1]c_k + 2c_{k-1} = 0.$$

The indicial roots are $r = -1$ and $r = 1/2$. For $r = -1$ the recurrence relation is

$$c_k = -\frac{2c_{k-1}}{k(2k-3)}, \quad k = 1, 2, 3, \dots,$$

and

$$c_1 = 2c_0, \quad c_2 = -2c_0, \quad c_3 = \frac{4}{9}c_0.$$

For $r = 1/2$ the recurrence relation is

$$c_k = -\frac{2c_{k-1}}{k(2k+3)}, \quad k = 1, 2, 3, \dots,$$

and

$$c_1 = -\frac{2}{5}c_0, \quad c_2 = \frac{2}{35}c_0, \quad c_3 = -\frac{4}{945}c_0.$$

The general solution on $(0, \infty)$ is

$$y = C_1 x^{-1} \left(1 + 2x - 2x^2 + \frac{4}{9}x^3 + \dots \right) + C_2 x^{1/2} \left(1 - \frac{2}{5}x + \frac{2}{35}x^2 - \frac{4}{945}x^3 + \dots \right).$$

- 25.** Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation and collecting terms, we obtain

$$\begin{aligned} xy'' + 2y' - xy &= (r^2 + r)c_0 x^{r-1} + (r^2 + 3r + 2)c_1 x^r \\ &\quad + \sum_{k=2}^{\infty} [(k+r)(k+r-1)c_k + 2(k+r)c_k - c_{k-2}]x^{k+r-1} \\ &= 0, \end{aligned}$$

which implies

$$\begin{aligned} r^2 + r &= r(r+1) = 0, \\ (r^2 + 3r + 2)c_1 &= 0, \end{aligned}$$

and

$$(k+r)(k+r+1)c_k - c_{k-2} = 0.$$

The indicial roots are $r_1 = 0$ and $r_2 = -1$, so $c_1 = 0$. For $r_1 = 0$ the recurrence relation is

$$c_k = \frac{c_{k-2}}{k(k+1)}, \quad k = 2, 3, 4, \dots,$$

and

$$\begin{aligned} c_2 &= \frac{1}{3!}c_0 \\ c_3 &= c_5 = c_7 = \dots = 0 \\ c_4 &= \frac{1}{5!}c_0 \\ c_{2n} &= \frac{1}{(2n+1)!}c_0. \end{aligned}$$

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For $r_2 = -1$ the recurrence relation is

$$c_k = \frac{c_{k-2}}{k(k-1)}, \quad k = 2, 3, 4, \dots,$$

and

$$c_2 = \frac{1}{2!}c_0$$

$$c_3 = c_5 = c_7 = \dots = 0$$

$$c_4 = \frac{1}{4!}c_0$$

$$c_{2n} = \frac{1}{(2n)!}c_0.$$

The general solution on $(0, \infty)$ is

$$\begin{aligned} y &= C_1 \sum_{n=0}^{\infty} \frac{1}{(2n+1)!} x^{2n} + C_2 x^{-1} \sum_{n=0}^{\infty} \frac{1}{(2n)!} x^{2n} \\ &= \frac{1}{x} \left[C_1 \sum_{n=0}^{\infty} \frac{1}{(2n+1)!} x^{2n+1} + C_2 \sum_{n=0}^{\infty} \frac{1}{(2n)!} x^{2n} \right] \\ &= \frac{1}{x} [C_1 \sinh x + C_2 \cosh x]. \end{aligned}$$

26. Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation and collecting terms, we obtain

$$\begin{aligned} x^2 y'' + x y' + \left(x^2 - \frac{1}{4}\right) y &= \left(r^2 - \frac{1}{4}\right) c_0 x^r + \left(r^2 + 2r + \frac{3}{4}\right) c_1 x^{r+1} \\ &\quad + \sum_{k=2}^{\infty} \left[(k+r)(k+r-1)c_k + (k+r)c_k - \frac{1}{4}c_k + c_{k-2} \right] x^{k+r} \\ &= 0, \end{aligned}$$

which implies

$$r^2 - \frac{1}{4} = \left(r - \frac{1}{2}\right) \left(r + \frac{1}{2}\right) = 0,$$

$$\left(r^2 + 2r + \frac{3}{4}\right) c_1 = 0,$$

and

$$\left[(k+r)^2 - \frac{1}{4} \right] c_k + c_{k-2} = 0.$$

The indicial roots are $r_1 = 1/2$ and $r_2 = -1/2$, so $c_1 = 0$. For $r_1 = 1/2$ the recurrence relation is

$$c_k = -\frac{c_{k-2}}{k(k+1)}, \quad k = 2, 3, 4, \dots,$$

and

$$c_2 = -\frac{1}{3!}c_0$$

$$c_3 = c_5 = c_7 = \dots = 0$$

$$c_4 = \frac{1}{5!}c_0$$

$$c_{2n} = \frac{(-1)^n}{(2n+1)!}c_0.$$

For $r_2 = -1/2$ the recurrence relation is

$$c_k = -\frac{c_{k-2}}{k(k-1)}, \quad k = 2, 3, 4, \dots,$$

and

$$\begin{aligned} c_2 &= -\frac{1}{2!}c_0 \\ c_3 &= c_5 = c_7 = \dots = 0 \\ c_4 &= \frac{1}{4!}c_0 \\ c_{2n} &= \frac{(-1)^n}{(2n)!}c_0. \end{aligned}$$

The general solution on $(0, \infty)$ is

$$\begin{aligned} y &= C_1 x^{1/2} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n} + C_2 x^{-1/2} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} \\ &= C_1 x^{-1/2} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1} + C_2 x^{-1/2} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} \\ &= x^{-1/2} [C_1 \sin x + C_2 \cos x]. \end{aligned}$$

27. Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation and collecting terms, we obtain

$$xy'' - xy' + y = (r^2 - r) c_0 x^{r-1} + \sum_{k=0}^{\infty} [(k+r+1)(k+r)c_{k+1} - (k+r)c_k + c_k] x^{k+r} = 0$$

which implies

$$r^2 - r = r(r-1) = 0$$

and

$$(k+r+1)(k+r)c_{k+1} - (k+r-1)c_k = 0.$$

The indicial roots are $r_1 = 1$ and $r_2 = 0$. For $r_1 = 1$ the recurrence relation is

$$c_{k+1} = \frac{kc_k}{(k+2)(k+1)}, \quad k = 0, 1, 2, \dots,$$

and one solution is $y_1 = c_0 x$. A second solution is

$$\begin{aligned} y_2 &= x \int \frac{e^{-\int -dx}}{x^2} dx = x \int \frac{e^x}{x^2} dx = x \int \frac{1}{x^2} \left(1 + x + \frac{1}{2}x^2 + \frac{1}{3!}x^3 + \dots \right) dx \\ &= x \int \left(\frac{1}{x^2} + \frac{1}{x} + \frac{1}{2} + \frac{1}{3!}x + \frac{1}{4!}x^2 + \dots \right) dx = x \left[-\frac{1}{x} + \ln x + \frac{1}{2}x + \frac{1}{12}x^2 + \frac{1}{72}x^3 + \dots \right] \\ &= x \ln x - 1 + \frac{1}{2}x^2 + \frac{1}{12}x^3 + \frac{1}{72}x^4 + \dots \end{aligned}$$

The general solution on $(0, \infty)$ is

$$y = C_1 x + C_2 y_2(x).$$

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28. Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation and collecting terms, we obtain

$$\begin{aligned} y'' + \frac{3}{x}y' - 2y &= (r^2 + 2r)c_0 x^{r-2} + (r^2 + 4r + 3)c_1 x^{r-1} \\ &+ \sum_{k=2}^{\infty} [(k+r)(k+r-1)c_k + 3(k+r)c_k - 2c_{k-2}]x^{k+r-2} \\ &= 0, \end{aligned}$$

which implies

$$\begin{aligned} r^2 + 2r &= r(r+2) = 0 \\ (r^2 + 4r + 3)c_1 &= 0 \\ (k+r)(k+r+2)c_k - 2c_{k-2} &= 0. \end{aligned}$$

The indicial roots are $r_1 = 0$ and $r_2 = -2$, so $c_1 = 0$. For $r_1 = 0$ the recurrence relation is

$$c_k = \frac{2c_{k-2}}{k(k+2)}, \quad k = 2, 3, 4, \dots,$$

and

$$\begin{aligned} c_2 &= \frac{1}{4}c_0 \\ c_3 &= c_5 = c_7 = \dots = 0 \\ c_4 &= \frac{1}{48}c_0 \\ c_6 &= \frac{1}{1,152}c_0. \end{aligned}$$

The result is

$$y_1 = c_0 \left(1 + \frac{1}{4}x^2 + \frac{1}{48}x^4 + \frac{1}{1,152}c_6 + \dots \right).$$

A second solution is

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{-\int (3/x)dx}}{y_1^2} dx = y_1 \int \frac{dx}{x^3 \left(1 + \frac{1}{4}x^2 + \frac{1}{48}x^4 + \dots \right)^2} \\ &= y_1 \int \frac{dx}{x^3 \left(1 + \frac{1}{2}x^2 + \frac{5}{48}x^4 + \frac{7}{576}x^6 + \dots \right)} = y_1 \int \frac{1}{x^3} \left(1 - \frac{1}{2}x^2 + \frac{7}{48}x^4 + \frac{19}{576}x^6 + \dots \right) \\ &= y_1 \int \left(\frac{1}{x^3} - \frac{1}{2x} + \frac{7}{48}x - \frac{19}{576}x^3 + \dots \right) = y_1 \left[-\frac{1}{2x^2} - \frac{1}{2} \ln x + \frac{7}{96}x^2 - \frac{19}{2,304}x^4 + \dots \right] \\ &= -\frac{1}{2}y_1 \ln x + y_1 \left[-\frac{1}{2x^2} + \frac{7}{96}x^2 - \frac{19}{2,304}x^4 + \dots \right]. \end{aligned}$$

The general solution on $(0, \infty)$ is

$$y = C_1 y_1(x) + C_2 y_2(x).$$

29. Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation and collecting terms, we obtain

$$xy'' + (1-x)y' - y = r^2 c_0 x^{r-1} + \sum_{k=0}^{\infty} [(k+r)(k+r-1)c_k + (k+r)c_k - (k+r)c_{k-1}]x^{k+r-1} = 0,$$

which implies $r^2 = 0$ and

$$(k+r)^2 c_k - (k+r)c_{k-1} = 0.$$

The indicial roots are $r_1 = r_2 = 0$ and the recurrence relation is

$$c_k = \frac{c_{k-1}}{k}, \quad k = 1, 2, 3, \dots$$

One solution is

$$y_1 = c_0 \left(1 + x + \frac{1}{2}x^2 + \frac{1}{3!}x^3 + \dots \right) = c_0 e^x.$$

A second solution is

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{-\int (1/x-1)dx}}{e^{2x}} dx = e^x \int \frac{e^x/x}{e^{2x}} dx = e^x \int \frac{1}{x} e^{-x} dx \\ &= e^x \int \frac{1}{x} \left(1 - x + \frac{1}{2}x^2 - \frac{1}{3!}x^3 + \dots \right) dx = e^x \int \left(\frac{1}{x} - 1 + \frac{1}{2}x - \frac{1}{3!}x^2 + \dots \right) dx \\ &= e^x \left[\ln x - x + \frac{1}{2 \cdot 2}x^2 - \frac{1}{3 \cdot 3!}x^3 + \dots \right] = e^x \ln x - e^x \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n \cdot n!} x^n. \end{aligned}$$

The general solution on $(0, \infty)$ is

$$y = C_1 e^x + C_2 e^x \left(\ln x - \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n \cdot n!} x^n \right).$$

- 30.** Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation and collecting terms, we obtain

$$xy'' + y' + y = r^2 c_0 x^{r-1} + \sum_{k=1}^{\infty} [(k+r)(k+r-1)c_k + (k+r)c_k + c_{k-1}] x^{k+r-1} = 0$$

which implies $r^2 = 0$ and

$$(k+r)^2 c_k + c_{k-1} = 0.$$

The indicial roots are $r_1 = r_2 = 0$ and the recurrence relation is

$$c_k = -\frac{c_{k-1}}{k^2}, \quad k = 1, 2, 3, \dots$$

One solution is

$$y_1 = c_0 \left(1 - x + \frac{1}{2^2}x^2 - \frac{1}{(3!)^2}x^3 + \frac{1}{(4!)^2}x^4 - \dots \right) = c_0 \sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2} x^n.$$

A second solution is

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{-\int (1/x)dx}}{y_1^2} dx = y_1 \int \frac{dx}{x \left(1 - x + \frac{1}{4}x^2 - \frac{1}{36}x^3 + \dots \right)^2} \\ &= y_1 \int \frac{dx}{x \left(1 - 2x + \frac{3}{2}x^2 - \frac{5}{9}x^3 + \frac{35}{288}x^4 - \dots \right)} \\ &= y_1 \int \frac{1}{x} \left(1 + 2x + \frac{5}{2}x^2 + \frac{23}{9}x^3 + \frac{677}{288}x^4 + \dots \right) dx \\ &= y_1 \int \left(\frac{1}{x} + 2 + \frac{5}{2}x + \frac{23}{9}x^2 + \frac{677}{288}x^3 + \dots \right) dx \\ &= y_1 \left[\ln x + 2x + \frac{5}{4}x^2 + \frac{23}{27}x^3 + \frac{677}{1,152}x^4 + \dots \right] \\ &= y_1 \ln x + y_1 \left(2x + \frac{5}{4}x^2 + \frac{23}{27}x^3 + \frac{677}{1,152}x^4 + \dots \right). \end{aligned}$$

Exercises 5.2

The general solution on $(0, \infty)$ is

$$y = C_1 y_1(x) + C_2 y_2(x).$$

31. Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation and collecting terms, we obtain

$$\begin{aligned} xy'' + (x-6)y' - 3y &= (r^2 - 7r)c_0 x^{r-1} + \sum_{k=1}^{\infty} [(k+r)(k+r-1)c_k + (k+r-1)c_{k-1} \\ &\quad - 6(k+r)c_k - 3c_{k-1}] x^{k+r-1} = 0, \end{aligned}$$

which implies

$$r^2 - 7r = r(r-7) = 0$$

and

$$(k+r)(k+r-7)c_k + (k+r-4)c_{k-1} = 0.$$

The indicial roots are $r_1 = 7$ and $r_2 = 0$. For $r_1 = 7$ the recurrence relation is

$$(k+7)kc_k + (k+3)c_{k-1} = 0, \quad k = 1, 2, 3, \dots,$$

or

$$c_k = -\frac{k+3}{k+7} c_{k-1}, \quad k = 1, 2, 3, \dots$$

Taking $c_0 \neq 0$ we obtain

$$c_1 = -\frac{1}{2}c_0$$

$$c_2 = \frac{5}{18}c_0$$

$$c_3 = -\frac{1}{6}c_0,$$

and so on. Thus, the indicial root $r_1 = 7$ yields a single solution. Now, for $r_2 = 0$ the recurrence relation is

$$k(k-7)c_k + (k-4)c_{k-1} = 0, \quad k = 1, 2, 3, \dots$$

Then

$$-6c_1 - 3c_0 = 0$$

$$-10c_2 - 2c_1 = 0$$

$$-12c_3 - c_2 = 0$$

$$-12c_4 + 0c_3 = 0 \implies c_4 = 0$$

$$-10c_5 + c_4 = 0 \implies c_5 = 0$$

$$-6c_6 + 2c_5 = 0 \implies c_6 = 0$$

$$0c_7 + 3c_6 = 0 \implies c_7 \text{ is arbitrary}$$

and

$$c_k = -\frac{k-4}{k(k-7)} c_{k-1}, \quad k = 8, 9, 10, \dots$$

Taking $c_0 \neq 0$ and $c_7 = 0$ we obtain

$$c_1 = -\frac{1}{2}c_0$$

$$c_2 = \frac{1}{10}c_0$$

$$c_3 = -\frac{1}{120}c_0$$

$$c_4 = c_5 = c_6 = \dots = 0.$$

Taking $c_0 = 0$ and $c_7 \neq 0$ we obtain

$$\begin{aligned} c_1 &= c_2 = c_3 = c_4 = c_5 = c_6 = 0 \\ c_8 &= -\frac{1}{2}c_7 \\ c_9 &= \frac{5}{36}c_7 \\ c_{10} &= -\frac{1}{36}c_7. \end{aligned}$$

In this case we obtain the two solutions

$$y_1 = 1 - \frac{1}{2}x + \frac{1}{10}x^2 - \frac{1}{120}x^3 \quad \text{and} \quad y_2 = x^7 - \frac{1}{2}x^8 + \frac{5}{36}x^9 - \frac{1}{36}x^{10} + \cdots.$$

32. Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation and collecting terms, we obtain

$$\begin{aligned} x(x-1)y'' + 3y' - 2y &= (4r - r^2)c_0 x^{r-1} + \sum_{k=1}^{\infty} [(k+r-1)(k+r-12)c_{k-1} - (k+r)(k+r-1)c_k \\ &\quad + 3(k+r)c_k - 2c_{k-1}]x^{k+r-1} \\ &= 0, \end{aligned}$$

which implies

$$4r - r^2 = r(4 - r) = 0$$

and

$$-(k+r)(k+r-4)c_k + [(k+r-1)(k+r-2) - 2]c_{k-1} = 0.$$

The indicial roots are $r_1 = 4$ and $r_2 = 0$. For $r_1 = 4$ the recurrence relation is

$$-(k+4)kc_k + [(k+3)(k+2) - 2]c_{k-1} = 0$$

or

$$c_k = (k+1)c_{k-1}, \quad k = 1, 2, 3, \dots$$

Taking $c_0 \neq 0$ we obtain

$$\begin{aligned} c_1 &= 2c_0 \\ c_2 &= 3!c_0 \\ c_3 &= 4!c_0, \end{aligned}$$

and so on. Thus, the indicial root $r_1 = 4$ yields a single solution. For $r_2 = 0$ the recurrence relation is

$$-k(k-4)c_k + k(k-3)c_{k-1} = 0, \quad k = 1, 2, 3, \dots,$$

or

$$-(k-4)c_k + (k-3)c_{k-1} = 0, \quad k = 1, 2, 3, \dots$$

Then

$$\begin{aligned} 3c_1 - 2c_0 &= 0 \\ 2c_2 - c_1 &= 0 \\ c_3 + 0c_2 &= 0 \Rightarrow c_3 = 0 \\ 0c_4 + c_3 &= 0 \Rightarrow c_4 \text{ is arbitrary} \end{aligned}$$

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and

$$c_k = \frac{(k-3)c_{k-1}}{c-4}, \quad k = 5, 6, 7, \dots$$

Taking $c_0 \neq 0$ and $c_4 = 0$ we obtain

$$c_1 = \frac{2}{3}c_0$$

$$c_2 = \frac{1}{3}c_0$$

$$c_3 = c_4 = c_5 = \dots = 0.$$

Taking $c_0 = 0$ and $c_4 \neq 0$ we obtain

$$c_1 = c_2 = c_3 = 0$$

$$c_5 = 2c_4$$

$$c_6 = 3c_4$$

$$c_7 = 4c_4.$$

In this case we obtain the two solutions

$$y_1 = 1 + \frac{2}{3}x + \frac{1}{3}x^2 \quad \text{and} \quad y_2 = x^4 + 2x^5 + 3x^6 + 4x^7 + \dots$$

33. (a) From $t = 1/x$ we have $dt/dx = -1/x^2 = -t^2$. Then

$$\frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx} = -t^2 \frac{dy}{dt}$$

and

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} \left(-t^2 \frac{dy}{dt} \right) = -t^2 \frac{d^2y}{dt^2} \frac{dt}{dx} - \frac{dy}{dt} \left(2t \frac{dt}{dx} \right) = t^4 \frac{d^2y}{dt^2} + 2t^3 \frac{dy}{dt}.$$

Now

$$x^4 \frac{d^2y}{dx^2} + \lambda y = \frac{1}{t^4} \left(t^4 \frac{d^2y}{dt^2} + 2t^3 \frac{dy}{dt} \right) + \lambda y = \frac{d^2y}{dt^2} + \frac{2}{t} \frac{dy}{dt} + \lambda y = 0$$

becomes

$$t \frac{d^2y}{dt^2} + 2 \frac{dy}{dt} + \lambda ty = 0.$$

(b) Substituting $y = \sum_{n=0}^{\infty} c_n t^{n+r}$ into the differential equation and collecting terms, we obtain

$$\begin{aligned} t \frac{d^2y}{dt^2} + 2 \frac{dy}{dt} + \lambda ty &= (r^2 + r)c_0 t^{r-1} + (r^2 + 3r + 2)c_1 t^r \\ &\quad + \sum_{k=2}^{\infty} [(k+r)(k+r-1)c_k + 2(k+r)c_k + \lambda c_{k-2}] t^{k+r-1} \\ &= 0, \end{aligned}$$

which implies

$$r^2 + r = r(r+1) = 0,$$

$$(r^2 + 3r + 2)c_1 = 0,$$

and

$$(k+r)(k+r+1)c_k + \lambda c_{k-2} = 0.$$

The indicial roots are $r_1 = 0$ and $r_2 = -1$, so $c_1 = 0$. For $r_1 = 0$ the recurrence relation is

$$c_k = -\frac{\lambda c_{k-2}}{k(k+1)}, \quad k = 2, 3, 4, \dots,$$

and

$$\begin{aligned}c_2 &= -\frac{\lambda}{3!}c_0 \\c_3 &= c_5 = c_7 = \cdots = 0 \\c_4 &= \frac{\lambda^2}{5!}c_0 \\c_{2n} &= (-1)^n \frac{\lambda^n}{(2n+1)!}c_0.\end{aligned}$$

For $r_2 = -1$ the recurrence relation is

$$c_k = -\frac{\lambda c_{k-2}}{k(k-1)}, \quad k = 2, 3, 4, \dots,$$

and

$$\begin{aligned}c_2 &= -\frac{\lambda}{2!}c_0 \\c_3 &= c_5 = c_7 = \cdots = 0 \\c_4 &= \frac{\lambda^2}{4!}c_0 \\c_{2n} &= (-1)^n \frac{\lambda^n}{(2n)!}c_0.\end{aligned}$$

The general solution on $(0, \infty)$ is

$$\begin{aligned}y(t) &= C_1 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} (\sqrt{\lambda}t)^{2n} + C_2 t^{-1} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} (\sqrt{\lambda}t)^{2n} \\&= \frac{1}{t} \left[C_1 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} (\sqrt{\lambda}t)^{2n+1} + C_2 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} (\sqrt{\lambda}t)^{2n} \right] \\&= \frac{1}{t} [C_1 \sin \sqrt{\lambda}t + C_2 \cos \sqrt{\lambda}t].\end{aligned}$$

(c) Using $t = 1/x$, the solution of the original equation is

$$y(x)C_1x \sin \frac{\sqrt{\lambda}}{x} + C_2x \cos \frac{\sqrt{\lambda}}{x}.$$

34. (a) From the boundary conditions $y(a) = 0$, $y(b) = 0$ we find

$$\begin{aligned}C_1 \sin \frac{\sqrt{\lambda}}{a} + C_2 \cos \frac{\sqrt{\lambda}}{a} &= 0 \\C_1 \sin \frac{\sqrt{\lambda}}{b} + C_2 \cos \frac{\sqrt{\lambda}}{b} &= 0.\end{aligned}$$

Since this is a homogeneous system of linear equations, it will have nontrivial solutions if

$$\begin{aligned}\begin{vmatrix} \sin \frac{\sqrt{\lambda}}{a} & \cos \frac{\sqrt{\lambda}}{a} \\ \sin \frac{\sqrt{\lambda}}{b} & \cos \frac{\sqrt{\lambda}}{b} \end{vmatrix} &= \sin \frac{\sqrt{\lambda}}{a} \cos \frac{\sqrt{\lambda}}{b} - \cos \frac{\sqrt{\lambda}}{a} \sin \frac{\sqrt{\lambda}}{b} \\&= \sin \left(\frac{\sqrt{\lambda}}{a} - \frac{\sqrt{\lambda}}{b} \right) = \sin \sqrt{\lambda} \left(\frac{b-a}{ab} \right) = 0.\end{aligned}$$

This will be the case if

$$\sqrt{\lambda} \left(\frac{b-a}{ab} \right) = n\pi \quad \text{or} \quad \sqrt{\lambda} = \frac{n\lambda ab}{b-a} = \frac{n\lambda ab}{L}, \quad n = 1, 2, \dots,$$

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or, if

$$\lambda_n = \frac{n^2 \pi^2 a^2 b^2}{L^2} = \frac{P_n b^4}{EI}.$$

The critical loads are then $P_n = n^2 \pi^2 (a/b)^2 EI_0 / L^2$. Using $C_2 = -C_1 \sin(\sqrt{\lambda}/a) / \cos(\sqrt{\lambda}/a)$ we have

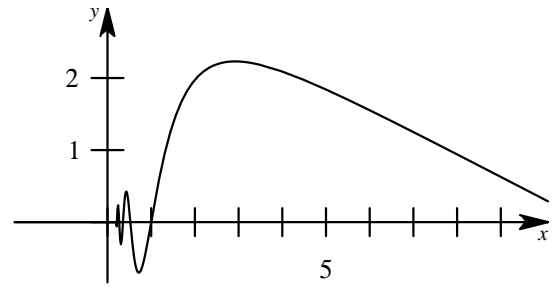
$$\begin{aligned} y &= C_1 x \left[\sin \frac{\sqrt{\lambda}}{x} - \frac{\sin(\sqrt{\lambda}/a)}{\cos(\sqrt{\lambda}/a)} \cos \frac{\sqrt{\lambda}}{x} \right] \\ &= C_3 x \left[\sin \frac{\sqrt{\lambda}}{x} \cos \frac{\sqrt{\lambda}}{a} - \cos \frac{\sqrt{\lambda}}{x} \sin \frac{\sqrt{\lambda}}{a} \right] \\ &= C_3 x \sin \sqrt{\lambda} \left(\frac{1}{x} - \frac{1}{a} \right), \end{aligned}$$

and

$$y_n(x) = C_3 x \sin \frac{n\pi ab}{L} \left(\frac{1}{x} - \frac{1}{a} \right) = C_4 x \sin \frac{n\pi ab}{L} \left(1 - \frac{a}{x} \right).$$

(b) When $n = 1$, $b = 11$, and $a = 1$, we have, for $C_4 = 1$,

$$y_1(x) = x \sin 1.1\pi \left(1 - \frac{1}{x} \right).$$



Exercises 5.3

1. Since $\nu^2 = 1/9$ the general solution is $y = c_1 J_{1/3}(x) + c_2 J_{-1/3}(x)$.
2. Since $\nu^2 = 1$ the general solution is $y = c_1 J_1(x) + c_2 Y_1(x)$.
3. Since $\nu^2 = 25/4$ the general solution is $y = c_1 J_{5/2}(x) + c_2 J_{-5/2}(x)$.
4. Since $\nu^2 = 1/16$ the general solution is $y = c_1 J_{1/4}(x) + c_2 J_{-1/4}(x)$.
5. Since $\nu^2 = 0$ the general solution is $y = c_1 J_0(x) + c_2 Y_0(x)$.
6. Since $\nu^2 = 4$ the general solution is $y = c_1 J_2(x) + c_2 Y_2(x)$.
7. Since $\nu^2 = 2$ the general solution is $y = c_1 J_2(3x) + c_2 Y_2(3x)$.
8. Since $\nu^2 = 1/4$ the general solution is $y = c_1 J_{1/2}(6x) + c_2 J_{-1/2}(6x)$.
9. If $y = x^{-1/2}v(x)$ then

$$\begin{aligned} y' &= x^{-1/2}v'(x) - \frac{1}{2}x^{-3/2}v(x), \\ y'' &= x^{-1/2}v''(x) - x^{-3/2}v'(x) + \frac{3}{4}x^{-5/2}v(x), \end{aligned}$$

and

$$x^2 y'' + 2xy' + \lambda^2 x^2 y = x^{3/2}v'' + x^{1/2}v' + \left(\lambda^2 x^{3/2} - \frac{1}{4}x^{-1/2} \right) v.$$

Multiplying by $x^{1/2}$ we obtain

$$x^2 v'' + xv' + \left(\lambda^2 x^2 - \frac{1}{4} \right) v = 0,$$

whose solution is $v = c_1 J_{1/2}(\lambda x) + c_2 J_{-1/2}(\lambda x)$. Then $y = c_1 x^{-1/2} J_{1/2}(\lambda x) + c_2 x^{-1/2} J_{-1/2}(\lambda x)$.

10. From $y = x^n J_n(x)$ we find

$$y' = x^n J'_n + n x^{n-1} J_n \quad \text{and} \quad y'' = x^n J''_n + 2n x^{n-1} J'_n + n(n-1) x^{n-2} J_n.$$

Substituting into the differential equation, we have

$$\begin{aligned} x^{n+1} J''_n + 2n x^n J'_n + n(n-1) x^{n-1} J_n + (1-2n)(x^n J'_n + n x^{n-1} J_n) + x^{n+1} J_n \\ = x^{n+1} J''_n + (2n+1-2n)x^n J'_n + (n^2-n+n-2n^2)x^{n-1} J_n + x^{n+1} J_n \\ = x^{n+1} [x^2 J''_n + x J'_n - n^2 J_n + x^2 J_n] \\ = x^{n+1} [x^2 J''_n + x J'_n + (x^2 - n^2) J_n] \\ = x^{n-1} \cdot 0 \quad (\text{since } J_n \text{ is a solution of Bessel's equation}) \\ = 0. \end{aligned}$$

Therefore, $x^n J_n$ is a solution of the original equation.

11. From $y = x^{-n} J_n$ we find

$$y' = x^{-n} J'_n - n x^{-n-1} J_n \quad \text{and} \quad y'' = x^{-n} J''_n - 2n x^{-n-1} J'_n + n(n+1) x^{-n-2} J_n.$$

Substituting into the differential equation, we have

$$\begin{aligned} x y'' + (1+2n) y' + x y = x^{-n-1} [x^2 J''_n + x J'_n + (x^2 - n^2) J_n] \\ = x^{-n-1} \cdot 0 \quad (\text{since } J_n \text{ is a solution of Bessel's equation}) \\ = 0. \end{aligned}$$

Therefore, $x^{-n} J_n$ is a solution of the original equation.

12. From $y = \sqrt{x} J_\nu(\lambda x)$ we find

$$y' = \lambda \sqrt{x} J'_\nu(\lambda x) + \frac{1}{2} x^{-1/2} J_\nu(\lambda x)$$

and

$$y'' = \lambda^2 \sqrt{x} J''_\nu(\lambda x) + \lambda x^{-1/2} J'_\nu(\lambda x) - \frac{1}{4} x^{-3/2} J_\nu(\lambda x).$$

Substituting into the differential equation, we have

$$\begin{aligned} x^2 y'' + \left(\lambda^2 x^2 - \nu^2 + \frac{1}{4} \right) y = \sqrt{x} [\lambda^2 x^2 J''_\nu(\lambda x) + \lambda x J'_\nu(\lambda x) + (\lambda^2 x^2 - \nu^2) J_\nu(\lambda x)] \\ = \sqrt{x} \cdot 0 \quad (\text{since } J_\nu \text{ is a solution of Bessel's equation}) \\ = 0. \end{aligned}$$

Therefore, $\sqrt{x} J_\nu(\lambda x)$ is a solution of the original equation.

13. From Problem 10 with $n = 1/2$ we find $y = x^{1/2} J_{1/2}(x)$. From Problem 11 with $n = -1/2$ we find $y = x^{1/2} J_{-1/2}(x)$.
14. From Problem 10 with $n = 1$ we find $y = x J_1(x)$. From Problem 11 with $n = -1$ we find $y = x J_{-1}(x) = -x J_1(x)$.
15. From Problem 10 with $n = -1$ we find $y = x^{-1} J_{-1}(x)$. From Problem 11 with $n = 1$ we find $y = x^{-1} J_1(x) = -x^{-1} J_{-1}(x)$.
16. From Problem 12 with $\lambda = 2$ and $\nu = 0$ we find $y = \sqrt{x} J_0(2x)$.
17. From Problem 12 with $\lambda = 1$ and $\nu = \pm 3/2$ we find $y = \sqrt{x} J_{3/2}(x)$ and $y = \sqrt{x} J_{-3/2}(x)$.

Exercises 5.3

18. From Problem 10 with $n = 3$ we find $y = x^3 J_3(x)$. From Problem 11 with $n = -3$ we find $y = x^3 J_{-3}(x) = -x^3 J_3(x)$.

19. (a) The recurrence relation follows from

$$\begin{aligned} -\nu J_\nu(x) + x J_{\nu-1}(x) &= -\sum_{n=0}^{\infty} \frac{(-1)^n \nu}{n! \Gamma(1+\nu+n)} \left(\frac{x}{2}\right)^{2n+\nu} + x \sum_{n=0}^{\infty} \frac{(-1)^n}{n! \Gamma(\nu+n)} \left(\frac{x}{2}\right)^{2n+\nu-1} \\ &= -\sum_{n=0}^{\infty} \frac{(-1)^n \nu}{n! \Gamma(1+\nu+n)} \left(\frac{x}{2}\right)^{2n+\nu} + \sum_{n=0}^{\infty} \frac{(-1)^n (\nu+n)}{n! \Gamma(1+\nu+n)} \cdot 2 \left(\frac{x}{2}\right) \left(\frac{x}{2}\right)^{2n+\nu-1} \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n (2n+\nu)}{n! \Gamma(1+\nu+n)} \left(\frac{x}{2}\right)^{2n+\nu} = x J'_\nu(x). \end{aligned}$$

(b) The formula in part (a) is a linear first-order differential equation in $J_\nu(x)$. An integrating factor for this equation is x^ν , so

$$\frac{d}{dx} [x^\nu J_\nu(x)] = x^\nu J_{\nu-1}(x).$$

20. Subtracting the formula in part (a) of Problem 19 from the formula in Example 4 we obtain

$$0 = 2\nu J_\nu(x) - x J_{\nu+1}(x) - x J_{\nu-1}(x) \quad \text{or} \quad 2\nu J_\nu(x) = x J_{\nu+1}(x) + x J_{\nu-1}(x).$$

21. Letting $\nu = 1$ in (15) we have

$$x J_0(x) = \frac{d}{dx} [x J_1(x)] \quad \text{so} \quad \int_0^x r J_0(r) dr = r J_1(r) \Big|_{r=0}^{r=x} = x J_1(x).$$

22. From (14) we obtain $J'_0(x) = -J_1(x)$, and from (15) we obtain $J'_0(x) = J_{-1}(x)$. Thus $J'_0(x) = J_{-1}(x) = -J_1(x)$.

23. Since

$$\Gamma\left(1 - \frac{1}{2} + n\right) = \frac{(2n-1)!}{(n-1)! 2^{2n-1}}$$

we obtain

$$J_{-1/2}(x) = \sum_{n=0}^{\infty} \frac{(-1)^n 2^{1/2} x^{-1/2}}{2n(2n-1)! \sqrt{\pi}} x^{2n} = \sqrt{\frac{2}{\pi x}} \cos x.$$

24. (a) By Problem 20, with $\nu = 1/2$, we obtain $J_{1/2}(x) = x J_{3/2}(x) + x J_{-1/2}(x)$ so that

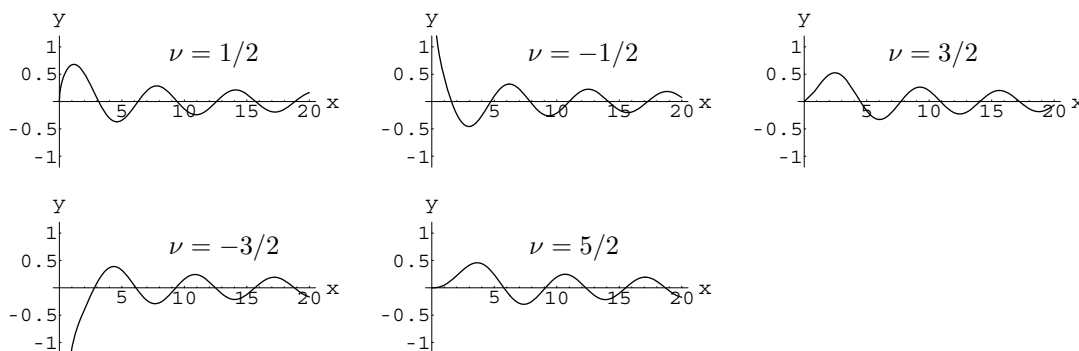
$$J_{3/2}(x) = \sqrt{\frac{2}{\pi x}} \left(\frac{\sin x}{x} - \cos x \right);$$

with $\nu = -1/2$ we obtain $-J_{-1/2}(x) = x J_{1/2}(x) + x J_{-3/2}(x)$ so that

$$J_{-3/2}(x) = -\sqrt{\frac{2}{\pi x}} \left(\frac{\cos x}{x} + \sin x \right);$$

and with $\nu = 3/2$ we obtain $3J_{3/2}(x) = x J_{5/2}(x) + x J_{1/2}(x)$ so that

$$J_{5/2}(x) = \sqrt{\frac{2}{\pi x}} \left(\frac{3 \sin x}{x^2} - \frac{3 \cos x}{x} - \sin x \right).$$



25. Rolle's theorem states that for a differentiable function $f(x)$, for which $f(a) = f(b) = 0$, there exists a number c between a and b such that $f'(c) = 0$. From Problem 20 with $\nu = 0$ we have $J'_0(x) = J_1(x)$. Thus, if a and b are successive zeros of $J_0(x)$, then there exists a c between a and b for which $J_1(x) = J'_0(x) = 0$.
26. Since $\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$, we have

$$|J_n(x)| \leq \frac{1}{\pi} \int_0^\pi |\cos(x \sin t - nt)| dt \leq \frac{1}{\pi} \int_0^\pi 1 dt = 1.$$

27. Letting

$$s = \frac{2}{\alpha} \sqrt{\frac{k}{m}} e^{-\alpha t/2},$$

we have

$$\frac{dx}{dt} = \frac{dx}{ds} \frac{ds}{dt} = \frac{dx}{ds} \left[\frac{2}{\alpha} \sqrt{\frac{k}{m}} \left(-\frac{\alpha}{2} \right) e^{-\alpha t/2} \right] = \frac{dx}{ds} \left(-\sqrt{\frac{k}{m}} e^{-\alpha t/2} \right)$$

and

$$\begin{aligned} \frac{d^2x}{dt^2} &= \frac{d}{dt} \left(\frac{dx}{dt} \right) = \frac{dx}{ds} \left(\frac{\alpha}{2} \sqrt{\frac{k}{m}} e^{-\alpha t/2} \right) + \frac{d}{dt} \left(\frac{dx}{ds} \right) \left(-\sqrt{\frac{k}{m}} e^{-\alpha t/2} \right) \\ &= \frac{dx}{ds} \left(\frac{\alpha}{2} \sqrt{\frac{k}{m}} e^{-\alpha t/2} \right) + \frac{d^2x}{ds^2} \frac{ds}{dt} \left(-\sqrt{\frac{k}{m}} e^{-\alpha t/2} \right) \\ &= \frac{dx}{ds} \left(\frac{\alpha}{2} \sqrt{\frac{k}{m}} e^{-\alpha t/2} \right) + \frac{d^2x}{ds^2} \left(\frac{k}{m} e^{-\alpha t} \right). \end{aligned}$$

Then

$$m \frac{d^2x}{dt^2} + k e^{-\alpha t} x = k e^{-\alpha t} \frac{d^2x}{ds^2} + \frac{m\alpha}{2} \sqrt{\frac{k}{m}} e^{-\alpha t/2} \frac{dx}{ds} + k e^{-\alpha t} x = 0.$$

Multiplying by $2^2/\alpha^2 m$ we have

$$\frac{2^2}{\alpha^2} \frac{k}{m} e^{-\alpha t} \frac{d^2x}{ds^2} + \frac{2}{\alpha} \sqrt{\frac{k}{m}} e^{-\alpha t/2} \frac{dx}{ds} + \frac{2}{\alpha^2} \frac{k}{m} e^{-\alpha t} x = 0$$

or, since $s = (2/\alpha) \sqrt{k/m} e^{-\alpha t/2}$,

$$s^2 \frac{d^2x}{ds^2} + s \frac{dx}{ds} + s^2 x = 0.$$

28. (a) We identify $m = 4$, $k = 1$, and $\alpha = 0.1$. Then

$$x(t) = c_1 J_0(10e^{-0.05t}) + c_2 Y_0(10e^{-0.05t})$$

and

$$x'(t) = -0.5c_1 J'_0(10e^{-0.05t}) - 0.5c_2 Y'_0(10e^{-0.05t}).$$

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Now $x(0) = 1$ and $x'(0) = -1/2$ imply

$$c_1 J_0(10) + c_2 Y_0(10) = 1$$

$$c_1 J'_0(10) + c_2 Y'_0(10) = 1.$$

Using Cramer's rule we obtain

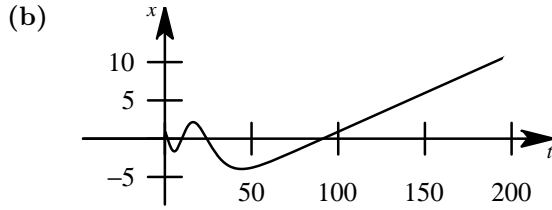
$$c_1 = \frac{Y'_0(10) - Y_0(10)}{J_0(10)Y'_0(10) - J'_0(10)Y_0(10)}$$

and

$$c_2 = \frac{J_0(10) - J'_0(10)}{J_0(10)Y'_0(10) - J'_0(10)Y_0(10)}.$$

Using $Y'_0 = -Y_1$ and $J'_0 = -J_1$ and Table 6.1 we find $c_1 = -4.7860$ and $c_2 = -3.1803$. Thus

$$x(t) = -4.7860 J_0(10e^{-0.05t}) - 3.1803 Y_0(10e^{-0.05t}).$$



29. Differentiating $y = x^{1/2}w\left(\frac{2}{3}\alpha x^{3/2}\right)$ with respect to $\frac{2}{3}\alpha x^{3/2}$ we obtain

$$y' = x^{1/2}w'\left(\frac{2}{3}\alpha x^{3/2}\right)\alpha x^{1/2} + \frac{1}{2}x^{-1/2}w\left(\frac{2}{3}\alpha x^{3/2}\right)$$

and

$$\begin{aligned} y'' &= \alpha x w''\left(\frac{2}{3}\alpha x^{3/2}\right)\alpha x^{1/2} + \alpha w'\left(\frac{2}{3}\alpha x^{3/2}\right) \\ &\quad + \frac{1}{2}\alpha w'\left(\frac{2}{3}\alpha x^{3/2}\right) - \frac{1}{4}x^{-3/2}w\left(\frac{2}{3}\alpha x^{3/2}\right). \end{aligned}$$

Then, after combining terms and simplifying, we have

$$y'' + \alpha^2 xy = \alpha \left[\alpha x^{3/2} w'' + \frac{3}{2} w' + \left(\alpha x^{3/2} - \frac{1}{4\alpha x^{3/2}} \right) w \right] = 0.$$

Letting $t = \frac{2}{3}\alpha x^{3/2}$ or $\alpha x^{3/2} = \frac{3}{2}t$ this differential equation becomes

$$\frac{3}{2} \frac{\alpha}{t} \left[t^2 w''(t) + t w'(t) + \left(t^2 - \frac{1}{9} \right) w(t) \right] = 0, \quad t > 0.$$

30. The general solution of Bessel's equation is

$$w(t) = c_1 J_{1/3}(t) + c_2 J_{-1/3}(t), \quad t > 0.$$

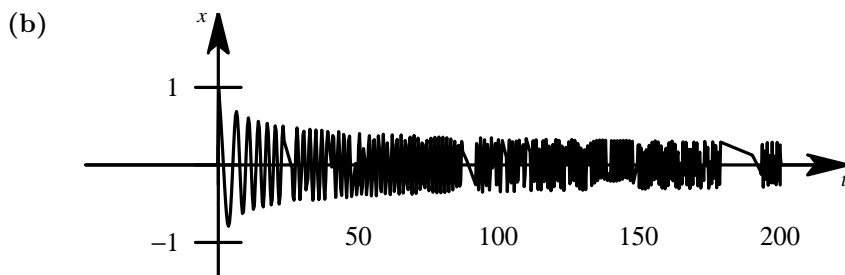
Thus, the general solution of Airy's equation for $x > 0$ is

$$y = x^{1/2}w\left(\frac{2}{3}\alpha x^{3/2}\right) = c_1 x^{1/2} J_{1/3}\left(\frac{2}{3}\alpha x^{3/2}\right) + c_2 x^{1/2} J_{-1/3}\left(\frac{2}{3}\alpha x^{3/2}\right).$$

31. (a) Identifying $\alpha = \frac{1}{2}$, the general solution of $x'' + \frac{1}{4}tx = 0$ is

$$x(t) = c_1 x^{1/2} J_{1/3}\left(\frac{1}{3}x^{3/2}\right) + c_2 x^{1/2} J_{-1/3}\left(\frac{1}{3}x^{3/2}\right).$$

Solving the system $x(0.1) = 1$, $x'(0.1) = -\frac{1}{2}$ we find $c_1 = -0.809264$ and $c_2 = 0.782397$.



32. (a) Letting $t = L - x$, the boundary-value problem becomes

$$\frac{d^2\theta}{dt^2} + \alpha^2 t\theta = 0, \quad \theta'(0) = 0, \quad \theta(L) = 0,$$

where $\alpha^2 = \delta g/EI$. This is Airy's differential equation, so by Problem 30 its solution is

$$y = c_1 t^{1/2} J_{1/3} \left(\frac{2}{3} \alpha t^{3/2} \right) + c_2 t^{1/2} J_{-1/3} \left(\frac{2}{3} \alpha t^{3/2} \right) = c_1 \theta_1(t) + c_2 \theta_2(t).$$

- (b) Looking at the series forms of θ_1 and θ_2 we see that $\theta_1'(0) \neq 0$, while $\theta_2'(0) = 0$. Thus, the boundary condition $\theta'(0) = 0$ implies $c_1 = 0$, and so

$$\theta(t) = c_2 \sqrt{t} J_{-1/3} \left(\frac{2}{3} \alpha t^{3/2} \right).$$

From $\theta(L) = 0$ we have

$$c_2 \sqrt{L} J_{-1/3} \left(\frac{2}{3} \alpha L^{3/2} \right) = 0,$$

so either $c_2 = 0$, in which case $\theta(t) = 0$, or $J_{-1/3}(\frac{2}{3}\alpha L^{3/2}) = 0$. The column will just start to bend when L is the length corresponding to the smallest positive zero of $J_{-1/3}$. Using *Mathematica*, the first positive root of $J_{-1/3}(x)$ is $x_1 \approx 1.86635$. Thus $\frac{2}{3}\alpha L^{3/2} = 1.86635$ implies

$$\begin{aligned} L &= \left(\frac{3(1.86635)}{2\alpha} \right)^{2/3} = \left[\frac{9EI}{4\delta g} (1.86635)^2 \right]^{1/3} \\ &= \left[\frac{9(2.6 \times 10^7) \pi (0.05)^4 / 4}{4(0.28) \pi (0.05)^2} (1.86635)^2 \right]^{1/3} \approx 76.9 \text{ in.} \end{aligned}$$

33. Setting $y = \sqrt{x} J_1(2\sqrt{x})$ and differentiating we obtain

$$y' = \sqrt{x} J_1'(2\sqrt{x}) \frac{2}{2\sqrt{x}} + \frac{1}{2\sqrt{x}} J_1(2\sqrt{x}) = J_1'(2\sqrt{x}) + \frac{1}{2\sqrt{x}} J_1(2\sqrt{x})$$

and

$$\begin{aligned} y'' &= J_1''(2\sqrt{x}) \frac{2}{2\sqrt{x}} + \frac{1}{2\sqrt{x}} J_1'(2\sqrt{x}) \frac{2}{2\sqrt{x}} - \frac{1}{4x^{3/2}} J_1(2\sqrt{x}) \\ &= \frac{1}{\sqrt{x}} J_1''(2\sqrt{x}) + \frac{1}{2x} J_1'(2\sqrt{x}) - \frac{1}{4x^{3/2}} J_1(2\sqrt{x}). \end{aligned}$$

Substituting into the differential equation and letting $t = 2\sqrt{x}$ we have

$$\begin{aligned} xy'' + y &= \sqrt{x} J_1''(2\sqrt{x}) + \frac{1}{2} J_1'(2\sqrt{x}) - \frac{1}{4\sqrt{x}} J_1(2\sqrt{x}) + \sqrt{x} J_1(2\sqrt{x}) \\ &= \frac{1}{\sqrt{x}} \left[x J_1''(2\sqrt{x}) + \frac{\sqrt{x}}{2} J_1'(2\sqrt{x}) + \left(x - \frac{1}{4} \right) J_1(2\sqrt{x}) \right] \\ &= \frac{2}{t} \left[\frac{t^2}{4} J_1''(t) + \frac{t}{4} J_1'(t) + \left(\frac{t^2}{4} - \frac{1}{4} \right) J_1(t) \right] \\ &= \frac{1}{2t} [t^2 J_1''(t) + t J_1'(t) + (t^2 - 1) J_1(t)]. \end{aligned}$$

Exercises 5.3

Since $J_1(t)$ is a solution of $t^2 y'' + ty' + (t^2 - 1)y = 0$, we see that the last expression above is 0 and $y = \sqrt{x} J_1(2\sqrt{x})$ is a solution of $xy'' + y = 0$.

34. (a) Writing the differential equation in the form $xy'' + (PL/M)y = 0$, we identify $\lambda = PL/M$. From Problem 33 the solution of this differential equation is

$$y = c_1 \sqrt{x} J_1 \left(2\sqrt{PLx/M} \right) + c_2 \sqrt{x} Y_1 \left(2\sqrt{PLx/M} \right).$$

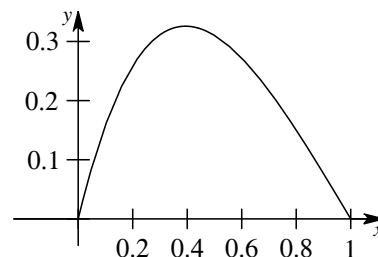
Now $J_1(0) = 0$, so $y(0) = 0$ implies $c_2 = 0$ and

$$y = c_1 \sqrt{x} J_1 \left(2\sqrt{PLx/M} \right).$$

- (b) From $y(L) = 0$ we have $y = J_1(2L\sqrt{PM}) = 0$. The first positive zero of J_1 is 3.8317 so, solving $2L\sqrt{P_1/M} = 3.8317$, we find $P_1 = 3.6705M/L^2$. Therefore,

$$y_1(x) = c_1 \sqrt{x} J_1 \left(2\sqrt{\frac{3.6705x}{L}} \right) = c_1 \sqrt{x} J_1 \left(\frac{3.8317}{L} \sqrt{x} \right).$$

- (c) For $c_1 = 1$ and $L = 1$ the graph of $y_1 = \sqrt{x} J_1(3.8317\sqrt{x})$ is shown.



35. (a) Using the expressions for the two linearly independent power series solutions, $y_1(x)$ and $y_2(x)$, given in the text we obtain

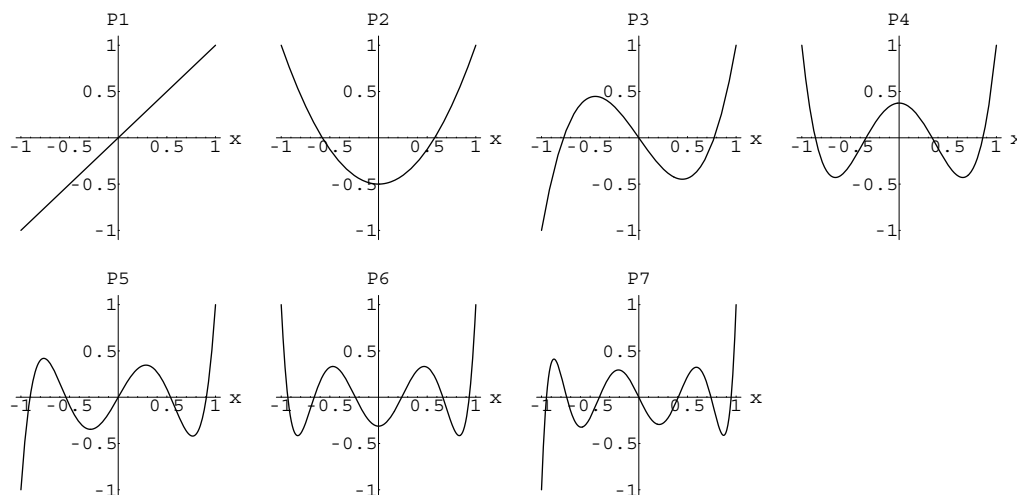
$$P_6(x) = \frac{1}{16} (231x^6 - 315x^4 + 105x^2 - 5)$$

and

$$P_7(x) = \frac{1}{16} (429x^7 - 693x^5 + 315x^3 - 35x).$$

- (b) $P_6(x)$ satisfies $(1 - x^2)y'' - 2xy' + 42y = 0$ and $P_7(x)$ satisfies $(1 - x^2)y'' - 2xy' + 56y = 0$.

36. (a)



(b) Zeros of Legendre polynomials

$$P_1(x) : 0$$

$$P_2(x) : \pm 0.57735$$

$$P_3(x) : 0, \pm 0.77460$$

$$P_4(x) : \pm 0.33998, \pm 0.86115$$

$$P_5(x) : 0, \pm 0.53847, \pm 0.90618$$

$$P_6(x) : \pm 0.23862, \pm 0.66121, \pm 0.93247$$

$$P_7(x) : 0, \pm 0.40585, \pm 0.74153, \pm 0.94911$$

$$P_{10}(x) : \pm 0.14887, \pm 0.43340, \pm 0.67941, \pm 0.86506, \pm 0.097391$$

The zeros of any Legendre polynomial are in the interval $(-1, 1)$ and are symmetric with respect to 0.

37. The recurrence relation can be written

$$P_{k+1}(x) = \frac{2k+1}{k+1} x P_k(x) - \frac{k}{k+1} P_{k-1}(x), \quad k = 2, 3, 4, \dots$$

$$k = 1: P_2(x) = \frac{3}{2}x^2 - \frac{1}{2}$$

$$k = 2: P_3(x) = \frac{5}{3}x \left(\frac{3}{2}x^2 - \frac{1}{2} \right) - \frac{2}{3}x = \frac{5}{2}x^3 - \frac{3}{2}x$$

$$k = 3: P_4(x) = \frac{7}{4}x \left(\frac{5}{2}x^3 - \frac{3}{2}x \right) - \frac{3}{4} \left(\frac{3}{2}x^2 - \frac{1}{2} \right) = \frac{35}{8}x^4 - \frac{30}{8}x^2 + \frac{3}{8}$$

$$k = 4: P_5(x) = \frac{9}{5}x \left(\frac{35}{8}x^4 - \frac{30}{8}x^2 + \frac{3}{8} \right) - \frac{4}{5} \left(\frac{5}{2}x^3 - \frac{3}{2}x \right) = \frac{63}{8}x^5 - \frac{35}{4}x^3 + \frac{15}{8}x$$

$$k = 5: P_6(x) = \frac{11}{6}x \left(\frac{63}{8}x^5 - \frac{35}{4}x^3 + \frac{15}{8}x \right) - \frac{5}{6} \left(\frac{35}{8}x^4 - \frac{30}{8}x^2 + \frac{3}{8} \right) = \frac{231}{16}x^6 - \frac{315}{16}x^4 + \frac{105}{16}x^2 - \frac{5}{16}$$

$$\begin{aligned} k = 6: P_7(x) &= \frac{13}{7}x \left(\frac{231}{16}x^6 - \frac{315}{16}x^4 + \frac{105}{16}x^2 - \frac{5}{16} \right) - \frac{5}{6} \left(\frac{63}{8}x^5 - \frac{35}{4}x^3 + \frac{15}{8}x \right) \\ &= \frac{429}{16}x^7 - \frac{693}{16}x^5 + \frac{315}{16}x^3 - \frac{35}{16}x \end{aligned}$$

38. If $x = \cos \theta$ then

$$\frac{dy}{d\theta} = -\sin \theta \frac{dy}{dx},$$

$$\frac{d^2y}{d\theta^2} = \sin^2 \theta \frac{d^2y}{dx^2} - \cos \theta \frac{dy}{dx},$$

and

$$\sin \theta \frac{d^2y}{d\theta^2} + \cos \theta \frac{dy}{d\theta} + n(n+1)(\sin \theta)y = \sin \theta \left[(1 - \cos^2 \theta) \frac{d^2y}{dx^2} - 2 \cos \theta \frac{dy}{dx} + n(n+1)y \right] = 0.$$

That is,

$$(1 - x^2) \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} + n(n+1)y = 0.$$

Chapter 5 Review Exercises

1. The interval of convergence is centered at 4. Since the series converges at -2 , it converges at least on the interval $[-2, 10)$. Since it diverges at 13, it converges at most on the interval $[-5, 13)$. Thus at 10 it might converge; at 7 it does converge; at -7 it does not converge; and at 11 it might converge.
2. We have

$$f(x) = \frac{\sin x}{\cos x} = \frac{x - \frac{x^3}{6} + \frac{x^5}{120} - \cdots}{1 - \frac{x^2}{2} + \frac{x^4}{24} - \cdots} = x + \frac{x^3}{3} + \frac{2x^5}{15} + \cdots.$$

3. Solving $x^2 - 2x + 10 = 0$ we obtain $x = 1 \pm \sqrt{11}$, which are singular points. Thus, the minimum radius of convergence is $|1 - \sqrt{11}| = \sqrt{11} - 1$.
4. Setting $1 - \sin x = 0$ we see the singular points closest to 0 are $-3\pi/2$ and $\pi/2$. Thus, the minimum radius of convergence is $\pi/2$.
5. The differential equation $(x^3 - x^2)y'' + y' + y = 0$ has a regular singular point at $x = 1$ and an irregular singular point at $x = 1$.
6. The differential equation $(x - 1)(x + 3)y'' + y = 0$ has regular singular points at $x = 1$ and $x = -3$.
7. Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation we obtain

$$2xy'' + y' + y = (2r^2 - r)c_0 x^{r-1} + \sum_{k=1}^{\infty} [2(k+r)(k+r-1)c_k + (k+r)c_k + c_{k-1}]x^{k+r-1} = 0$$

which implies

$$2r^2 - r = r(2r - 1) = 0$$

and

$$(k+r)(2k+2r-1)c_k + c_{k-1} = 0.$$

The indicial roots are $r = 0$ and $r = 1/2$. For $r = 0$ the recurrence relation is

$$c_k = -\frac{c_{k-1}}{k(2k-1)}, \quad k = 1, 2, 3, \dots,$$

so

$$c_1 = -c_0, \quad c_2 = \frac{1}{6}c_0, \quad c_3 = -\frac{1}{90}c_0.$$

For $r = 1/2$ the recurrence relation is

$$c_k = -\frac{c_{k-1}}{k(2k+1)}, \quad k = 1, 2, 3, \dots,$$

so

$$c_1 = -\frac{1}{3}c_0, \quad c_2 = \frac{1}{30}c_0, \quad c_3 = -\frac{1}{630}c_0.$$

Two linearly independent solutions are

$$y_1 = C_1 \left(1 - x + \frac{1}{6}x^2 - \frac{1}{90}x^3 + \cdots \right)$$

and

$$y_2 = C_2 x^{1/2} \left(1 - \frac{1}{3}x + \frac{1}{30}x^2 - \frac{1}{630}x^3 + \cdots \right).$$

8. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned}
 y'' - xy' - y &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} - \underbrace{\sum_{n=1}^{\infty} n c_n x^n}_{k=n} - \underbrace{\sum_{n=0}^{\infty} c_n x^n}_{k=n} \\
 &= \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k - \sum_{k=1}^{\infty} k c_k x^k - \sum_{k=0}^{\infty} c_k x^k \\
 &= 2c_2 - c_0 + \sum_{k=1}^{\infty} [(k+2)(k+1)c_{k+2} - (k+1)c_k] x^k = 0.
 \end{aligned}$$

Thus

$$2c_2 - c_0 = 0$$

$$(k+2)(k+1)c_{k+2} - (k+1)c_k = 0$$

and

$$\begin{aligned}
 c_2 &= \frac{1}{2}c_0 \\
 c_{k+2} &= \frac{1}{k+2}c_k, \quad k = 1, 2, 3, \dots
 \end{aligned}$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$\begin{aligned}
 c_2 &= \frac{1}{2} \\
 c_3 &= c_5 = c_7 = \dots = 0 \\
 c_4 &= \frac{1}{8} \\
 c_6 &= \frac{1}{48}
 \end{aligned}$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$\begin{aligned}
 c_2 &= c_4 = c_6 = \dots = 0 \\
 c_3 &= \frac{1}{3} \\
 c_5 &= \frac{1}{15} \\
 c_7 &= \frac{1}{105}
 \end{aligned}$$

and so on. Thus, two solutions are

$$y_1 = 1 + \frac{1}{2}x^2 + \frac{1}{8}x^4 + \frac{1}{48}x^6 + \dots$$

and

$$y_2 = x + \frac{1}{3}x^3 + \frac{1}{15}x^5 + \frac{1}{105}x^7 + \dots$$

9. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we obtain

$$(x-1)y'' + 3y = (-2c_2 + 3c_0) + \sum_{k=3}^{\infty} (k-1)(k-2)c_{k-1} - k(k-1)c_k + 3c_{k-2}]x^{k-2} = 0$$

which implies $c_2 = 3c_0/2$ and

$$c_k = \frac{(k-1)(k-2)c_{k-1} + 3c_{k-2}}{k(k-1)}, \quad k = 3, 4, 5, \dots$$

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Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$c_2 = \frac{3}{2}, \quad c_3 = \frac{1}{2}, \quad c_4 = \frac{5}{8}$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$c_2 = 0, \quad c_3 = \frac{1}{2}, \quad c_4 = \frac{1}{4}$$

and so on. Thus, two solutions are

$$y_1 = C_1 \left(1 + \frac{3}{2}x^2 + \frac{1}{2}x^3 + \frac{5}{8}x^4 + \cdots \right)$$

and

$$y_2 = C_2 \left(x + \frac{1}{2}x^3 + \frac{1}{4}x^4 + \cdots \right).$$

10. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we obtain

$$y'' - x^2 y' + xy = 2c_2 + (6c_3 + c_0)x + \sum_{k=1}^{\infty} [(k+3)(k+2)c_{k+3} - (k-1)c_k]x^{k+1} = 0$$

which implies $c_2 = 0$, $c_3 = -c_0/6$, and

$$c_{k+3} = \frac{k-1}{(k+3)(k+2)}c_k, \quad k = 1, 2, 3, \dots$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$c_3 = -\frac{1}{6}$$

$$c_4 = c_7 = c_{10} = \cdots = 0$$

$$c_5 = c_8 = c_{11} = \cdots = 0$$

$$c_6 = -\frac{1}{90}$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$c_3 = c_6 = c_9 = \cdots = 0$$

$$c_4 = c_7 = c_{10} = \cdots = 0$$

$$c_5 = c_8 = c_{11} = \cdots = 0$$

and so on. Thus, two solutions are

$$y_1 = c_0 \left(1 - \frac{1}{6}x^3 - \frac{1}{90}x^6 - \cdots \right) \quad \text{and} \quad y_2 = c_1 x.$$

11. Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation and collecting terms, we obtain

$$\begin{aligned} xy'' + (x-6)y' - 3y &= (r^2 - 7r)c_0 x^{r-1} + \sum_{k=1}^{\infty} [(k+r)(k+r-1)c_k + (k+r-1)c_{k-1} \\ &\quad - 6(k+r)c_k - 3c_{k-1}]x^{k+r-1} = 0, \end{aligned}$$

which implies

$$r^2 - 7r = r(r-7) = 0$$

and

$$(k+r)(k+r-7)c_k + (k+r-4)c_{k-1} = 0.$$

The indicial roots are $r_1 = 7$ and $r_2 = 0$. For $r_2 = 0$ the recurrence relation is

$$k(k-7)c_k + (k-4)c_{k-1} = 0, \quad k = 1, 2, 3, \dots$$

Then

$$\begin{aligned} -6c_1 - 3c_0 &= 0 \\ -10c_2 - 2c_1 &= 0 \\ -12c_3 - c_2 &= 0 \\ -12c_4 + 0c_3 &= 0 \implies c_4 = 0 \\ -10c_5 + c_4 &= 0 \implies c_5 = 0 \\ -6c_6 + 2c_5 &= 0 \implies c_6 = 0 \\ 0c_7 + 3c_6 &= 0 \implies c_7 \text{ is arbitrary} \end{aligned}$$

and

$$c_k = -\frac{k-4}{k(k-7)}c_{k-1} \quad k = 8, 9, 10, \dots$$

Taking $c_0 \neq 0$ and $c_7 = 0$ we obtain

$$\begin{aligned} c_1 &= -\frac{1}{2}c_0 \\ c_2 &= \frac{1}{10}c_0 \\ c_3 &= -\frac{1}{120}c_0 \\ c_4 &= c_5 = c_6 = \dots = 0. \end{aligned}$$

Taking $c_0 = 0$ and $c_7 \neq 0$ we obtain

$$\begin{aligned} c_1 &= c_2 = c_3 = c_4 = c_5 = c_6 = 0 \\ c_8 &= -\frac{1}{2}c_7 \\ c_9 &= \frac{5}{36}c_7 \\ c_{10} &= -\frac{1}{36}c_7. \end{aligned}$$

In this case we obtain the two solutions

$$y_1 = 1 - \frac{1}{2}x + \frac{1}{10}x^2 - \frac{1}{120}x^3 \quad \text{and} \quad y_2 = x^7 - \frac{1}{2}x^8 + \frac{5}{36}x^9 - \frac{1}{36}x^{10} + \dots$$

12. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned} (\cos x)y'' + y &= \left(1 - \frac{1}{2}x^2 + \frac{1}{24}x^4 - \frac{1}{720}x^6 + \dots\right)(2c_2 + 6c_3x + 12c_4x^2 + 20c_5x^3 + 30c_6x^4 + \dots) \\ &\quad + \sum_{n=0}^{\infty} c_n x^n \\ &= \left[2c_2 + 6c_3x + (12c_4 - c_2)x^2 + (20c_5 - 3c_3)x^3 + \left(30c_6 - 6c_4 + \frac{1}{12}c_2\right)x^4 + \dots\right] \\ &\quad + [c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + \dots] \\ &= (c_0 + 2c_2) + (c_1 + 6c_3)x + 12c_4x^2 + (20c_5 - 2c_3)x^3 + \left(30c_6 - 5c_4 + \frac{1}{12}c_2\right)x^4 + \dots \\ &= 0. \end{aligned}$$

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Thus

$$\begin{aligned}c_0 + 2c_2 &= 0 \\c_1 + 6c_3 &= 0 \\12c_4 &= 0 \\20c_5 - 2c_3 &= 0 \\30c_6 - 5c_4 + \frac{1}{12}c_2 &= 0\end{aligned}$$

and

$$\begin{aligned}c_2 &= -\frac{1}{2}c_0 \\c_3 &= -\frac{1}{6}c_1 \\c_4 &= 0 \\c_5 &= \frac{1}{10}c_3 \\c_6 &= \frac{1}{6}c_4 - \frac{1}{360}c_2.\end{aligned}$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$c_2 = -\frac{1}{2}, \quad c_3 = 0, \quad c_4 = 0, \quad c_5 = 0, \quad c_6 = \frac{1}{720}$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we find

$$c_2 = 0, \quad c_3 = -\frac{1}{6}, \quad c_4 = 0, \quad c_5 = -\frac{1}{60}, \quad c_6 = 0$$

and so on. Thus, two solutions are

$$y_1 = 1 - \frac{1}{2}x^2 + \frac{1}{720}x^6 + \cdots \quad \text{and} \quad y_2 = x - \frac{1}{6}x^3 - \frac{1}{60}x^5 + \cdots.$$

- 13.** Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned}y'' + xy' + 2y &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} + \underbrace{\sum_{n=1}^{\infty} n c_n x^n}_{k=n} + 2 \underbrace{\sum_{n=0}^{\infty} c_n x^n}_{k=n} \\&= \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k + \sum_{k=1}^{\infty} k c_k x^k + 2 \sum_{k=0}^{\infty} c_k x^k \\&= 2c_2 + 2c_0 + \sum_{k=1}^{\infty} [(k+2)(k+1)c_{k+2} + (k+2)c_k] x^k = 0.\end{aligned}$$

Thus

$$\begin{aligned}2c_2 + 2c_0 &= 0 \\(k+2)(k+1)c_{k+2} + (k+2)c_k &= 0\end{aligned}$$

and

$$\begin{aligned}c_2 &= -c_0 \\c_{k+2} &= -\frac{1}{k+1} c_k, \quad k = 1, 2, 3, \dots\end{aligned}$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$\begin{aligned}c_2 &= -1 \\c_3 &= c_5 = c_7 = \cdots = 0 \\c_4 &= \frac{1}{3} \\c_6 &= -\frac{1}{15}\end{aligned}$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$\begin{aligned}c_2 &= c_4 = c_6 = \cdots = 0 \\c_3 &= -\frac{1}{2} \\c_5 &= \frac{1}{8} \\c_7 &= -\frac{1}{48}\end{aligned}$$

and so on. Thus, the general solution is

$$y = c_0 \left(1 - x^2 + \frac{1}{3}x^4 - \frac{1}{15}x^6 + \cdots \right) + c_1 \left(x - \frac{1}{2}x^3 + \frac{1}{8}x^5 - \frac{1}{48}x^7 + \cdots \right)$$

and

$$y' = c_0 \left(-2x + \frac{4}{3}x^3 - \frac{2}{5}x^5 + \cdots \right) + c_1 \left(1 - \frac{3}{2}x^2 + \frac{5}{8}x^4 - \frac{7}{48}x^6 + \cdots \right).$$

Setting $y(0) = 3$ and $y'(0) = -2$ we find $c_0 = 3$ and $c_1 = -2$. Therefore, the solution of the initial-value problem is

$$y = 3 - 2x - 3x^2 + x^3 + x^4 - \frac{1}{4}x^5 - \frac{1}{5}x^6 + \frac{1}{24}x^7 + \cdots.$$

14. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned}(x+2)y'' + 3y &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-1}}_{k=n-1} + 2 \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} + 3 \underbrace{\sum_{n=0}^{\infty} c_n x^n}_{k=n} \\&= \sum_{k=1}^{\infty} (k+1)k c_{k+1} x^k + 2 \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k + 3 \sum_{k=0}^{\infty} c_k x^k \\&= 4c_2 + 3c_0 + \sum_{k=1}^{\infty} [(k+1)k c_{k+1} + 2(k+2)(k+1)c_{k+2} + 3c_k] x^k = 0.\end{aligned}$$

Thus

$$\begin{aligned}4c_2 + 3c_0 &= 0 \\(k+1)k c_{k+1} + 2(k+2)(k+1)c_{k+2} + 3c_k &= 0\end{aligned}$$

and

$$\begin{aligned}c_2 &= -\frac{3}{4}c_0 \\c_{k+2} &= -\frac{k}{2(k+2)} c_{k+1} - \frac{3}{2(k+2)(k+1)} c_k, \quad k = 1, 2, 3, \dots\end{aligned}$$

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Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$\begin{aligned}c_2 &= -\frac{3}{4} \\c_3 &= \frac{1}{8} \\c_4 &= \frac{1}{16} \\c_5 &= -\frac{9}{320}\end{aligned}$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$\begin{aligned}c_2 &= 0 \\c_3 &= -\frac{1}{4} \\c_4 &= \frac{1}{16} \\c_5 &= 0\end{aligned}$$

and so on. Thus, the general solution is

$$y = c_0 \left(1 - \frac{3}{4}x^2 + \frac{1}{8}x^3 + \frac{1}{16}x^4 - \frac{9}{320}x^5 + \cdots \right) + c_1 \left(x - \frac{1}{4}x^3 + \frac{1}{16}x^4 + \cdots \right)$$

and

$$y' = c_0 \left(-\frac{3}{2}x + \frac{3}{8}x^2 + \frac{1}{4}x^3 - \frac{9}{64}x^4 + \cdots \right) + c_1 \left(1 - \frac{3}{4}x^2 + \frac{1}{4}x^3 + \cdots \right).$$

Setting $y(0) = 0$ and $y'(0) = 1$ we find $c_0 = 0$ and $c_1 = 1$. Therefore, the solution of the initial-value problem is

$$y = x - \frac{1}{4}x^3 + \frac{1}{16}x^4 + \cdots.$$

15. Writing the differential equation in the form

$$y'' + \left(\frac{1 - \cos x}{x} \right) y' + xy = 0,$$

and noting that

$$\frac{1 - \cos x}{x} = \frac{x}{2} - \frac{x^3}{24} + \frac{x^5}{720} - \cdots$$

is analytic at $x = 0$, we conclude that $x = 0$ is an ordinary point of the differential equation.

16. Writing the differential equation in the form

$$y'' + \left(\frac{x}{e^x - 1 - x} \right) y = 0$$

and noting that

$$\frac{x}{e^x - 1 - x} = \frac{2}{x} - \frac{2}{3} + \frac{x}{18} + \frac{x^2}{270} - \cdots$$

we see that $x = 0$ is a singular point of the differential equation. Since

$$x^2 \left(\frac{x}{e^x - 1 - x} \right) = 2x - \frac{2x^2}{3} + \frac{x^3}{18} + \frac{x^4}{270} - \cdots,$$

we conclude that $x = 0$ is a regular singular point.

17. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned}
 y'' + x^2 y' + 2xy &= \sum_{n=2}^{\infty} \underbrace{n(n-1)c_n x^{n-2}}_{k=n-2} + \sum_{n=1}^{\infty} \underbrace{nc_n x^{n+1}}_{k=n+1} + 2 \sum_{n=0}^{\infty} \underbrace{c_n x^{n+1}}_{k=n+1} \\
 &= \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k + \sum_{k=2}^{\infty} (k-1)c_{k-1} x^k + 2 \sum_{k=1}^{\infty} c_{k-1} x^k \\
 &= 2c_2 + (6c_3 + 2c_0)x + \sum_{k=2}^{\infty} [(k+2)(k+1)c_{k+2} + (k+1)c_{k-1}]x^k = 5 - 2x + 10x^3.
 \end{aligned}$$

Thus

$$\begin{aligned}
 2c_2 &= 5 \\
 6c_3 + 2c_0 &= -2 \\
 12c_4 + 3c_1 &= 0 \\
 20c_5 + 4c_2 &= 10 \\
 (k+2)(k+1)c_{k+2} + (k+1)c_{k-1} &= 0, \quad k = 4, 5, 6, \dots,
 \end{aligned}$$

and

$$\begin{aligned}
 c_2 &= \frac{5}{2} \\
 c_3 &= -\frac{1}{3}c_0 - \frac{1}{3} \\
 c_4 &= -\frac{1}{4}c_1 \\
 c_5 &= \frac{1}{2} - \frac{1}{5}c_2 = \frac{1}{2} - \frac{1}{5}\left(\frac{5}{2}\right) = 0 \\
 c_{k+2} &= -\frac{1}{k+2}c_{k-1}.
 \end{aligned}$$

Using the recurrence relation, we find

$$\begin{aligned}
 c_6 &= -\frac{1}{6}c_3 = \frac{1}{3 \cdot 6}(c_0 + 1) = \frac{1}{3^2 \cdot 2!}c_0 + \frac{1}{3^2 \cdot 2!} \\
 c_7 &= -\frac{1}{7}c_4 = \frac{1}{4 \cdot 7}c_1 \\
 c_8 &= c_{11} = c_{14} = \dots = 0 \\
 c_9 &= -\frac{1}{9}c_6 = -\frac{1}{3^3 \cdot 3!}c_0 - \frac{1}{3^3 \cdot 3!} \\
 c_{10} &= -\frac{1}{10}c_7 = -\frac{1}{4 \cdot 7 \cdot 10}c_1 \\
 c_{12} &= -\frac{1}{12}c_9 = \frac{1}{3^4 \cdot 4!}c_0 + \frac{1}{3^4 \cdot 4!} \\
 c_{13} &= -\frac{1}{13}c_{10} = \frac{1}{4 \cdot 7 \cdot 10 \cdot 13}c_1
 \end{aligned}$$

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and so on. Thus

$$\begin{aligned} y = c_0 & \left[1 - \frac{1}{3}x^3 + \frac{1}{3^2 \cdot 2!}x^6 - \frac{1}{3^3 \cdot 3!}x^9 + \frac{1}{3^4 \cdot 4!}x^{12} - \dots \right] \\ & + c_1 \left[x - \frac{1}{4}x^4 + \frac{1}{4 \cdot 7}x^7 - \frac{1}{4 \cdot 7 \cdot 10}x^{10} + \frac{1}{4 \cdot 7 \cdot 10 \cdot 13}x^{13} - \dots \right] \\ & + \left[\frac{5}{2}x^2 - \frac{1}{3}x^3 + \frac{1}{3^2 \cdot 2!}x^6 - \frac{1}{3^3 \cdot 3!}x^9 + \frac{1}{3^4 \cdot 4!}x^{12} - \dots \right]. \end{aligned}$$

18. (a) From $y = -\frac{1}{u} \frac{du}{dx}$ we obtain

$$\frac{dy}{dx} = -\frac{1}{u} \frac{d^2u}{dx^2} + \frac{1}{u^2} \left(\frac{du}{dx} \right)^2.$$

Then $dy/dx = x^2 + y^2$ becomes

$$-\frac{1}{u} \frac{d^2u}{dx^2} + \frac{1}{u^2} \left(\frac{du}{dx} \right)^2 = x^2 + \frac{1}{u^2} \left(\frac{du}{dx} \right)^2,$$

so $\frac{d^2u}{dx^2} + x^2u = 0$.

- (b) If $u = x^{1/2}w(\frac{1}{2}x^2)$ then

$$u' = x^{3/2}w' \left(\frac{1}{2}x^2 \right) + \frac{1}{2}x^{-1/2}w \left(\frac{1}{2}x^2 \right)$$

and

$$u'' = x^{5/2}w'' \left(\frac{1}{2}x^2 \right) + 2x^{1/2}w' \left(\frac{1}{2}x^2 \right) - \frac{1}{4}x^{-3/2}w \left(\frac{1}{2}x^2 \right),$$

so

$$u'' + x^2u = x^{1/2} \left[x^2w'' \left(\frac{1}{2}x^2 \right) + 2w' \left(\frac{1}{2}x^2 \right) + \left(x^2 - \frac{1}{4}x^{-2} \right) w \left(\frac{1}{2}x^2 \right) \right] = 0.$$

Letting $t = \frac{1}{2}x^2$ we have

$$\sqrt{2t} \left[2tw''(t) + 2w'(t) + \left(2t - \frac{1}{4 \cdot 2t} \right) w(t) \right] = 0$$

or

$$t^2w''(t) + tw'(t) + \left(t^2 - \frac{1}{16} \right) w(t) = 0.$$

This is Bessel's equation with $\nu = 1/4$, so

$$w(t) = c_1J_{1/4}(t) + c_2J_{-1/4}(t).$$

- (c) We have

$$\begin{aligned} y &= -\frac{1}{u} \frac{du}{dx} = -\frac{1}{x^{1/2}w(t)} \frac{d}{dx} x^{1/2}w(t) \\ &= -\frac{1}{x^{1/2}w} \left[x^{1/2} \frac{dw}{dt} \frac{dt}{dx} + \frac{1}{2}x^{-1/2}w \right] \\ &= -\frac{1}{x^{1/2}w} \left[x^{3/2} \frac{dw}{dt} + \frac{1}{2x^{1/2}} w \right] \\ &= -\frac{1}{2xw} \left[2x^2 \frac{dw}{dt} + w \right] = -\frac{1}{2xw} \left[4t \frac{dw}{dt} + w \right]. \end{aligned}$$

Now

$$\begin{aligned}
 4t \frac{dw}{dt} + w &= 4t \frac{d}{dt} [c_1 J_{1/4}(t) + c_2 J_{-1/4}(t)] + c_1 J_{1/4}(t) + c_2 J_{-1/4}(t) \\
 &= 4t \left[c_1 \left(J_{-3/4}(t) - \frac{1}{4t} J_{1/4}(t) \right) + c_2 \left(-\frac{1}{4t} J_{-1/4}(t) - J_{3/4}(t) \right) \right] \\
 &\quad + c_1 J_{1/4}(t) + c_2 J_{-1/4}(t) \\
 &= 4c_1 t J_{-3/4}(t) - 4c_2 t J_{3/4}(t) \\
 &= 2c_1 x^2 J_{-3/4} \left(\frac{1}{2} x^2 \right) - 2c_2 x^2 J_{3/4} \left(\frac{1}{2} x^2 \right),
 \end{aligned}$$

so

$$\begin{aligned}
 y &= -\frac{2c_1 x^2 J_{-3/4}(\frac{1}{2}x^2) - 2c_2 x^2 J_{3/4}(\frac{1}{2}x^2)}{2x[c_1 J_{1/4}(\frac{1}{2}x^2) + c_2 J_{-1/4}(\frac{1}{2}x^2)]} \\
 &= x \frac{-c_1 J_{-3/4}(\frac{1}{2}x^2) + c_2 J_{3/4}(\frac{1}{2}x^2)}{c_1 J_{1/4}(\frac{1}{2}x^2) + c_2 J_{-1/4}(\frac{1}{2}x^2)}.
 \end{aligned}$$

Letting $c = c_1/c_2$ we have

$$y = x \frac{J_{3/4}(\frac{1}{2}x^2) - c J_{-3/4}(\frac{1}{2}x^2)}{c J_{1/4}(\frac{1}{2}x^2) + J_{-1/4}(\frac{1}{2}x^2)}.$$

19. Let

$$y_2 = \frac{1}{2}x[\ln(1+x) - \ln(1-x)] - 1$$

so that

$$y_2' = \frac{1}{2}x \left[\frac{1}{1+x} + \frac{1}{1-x} \right] + \frac{1}{2}[\ln(1+x) - \ln(1-x)]$$

and

$$\begin{aligned}
 y_2'' &= \frac{1}{2}x \left[-\frac{1}{(1+x)^2} + \frac{1}{(1-x)^2} \right] + \frac{1}{2} \left[\frac{1}{1+x} + \frac{1}{1-x} \right] + \frac{1}{2} \left[\frac{1}{1+x} + \frac{1}{1-x} \right] \\
 &= \frac{1}{2}x \left[-\frac{1}{(1+x)^2} + \frac{1}{(1-x)^2} \right] + \frac{1}{1+x} + \frac{1}{1-x}.
 \end{aligned}$$

Then

$$(1-x)(1+x)y_2'' - 2xy_2' + 2y_2 = 0.$$

20. $n = 0$: $P_0(x) = 1$

$$n = 1: P_1(x) = \frac{1}{2} \frac{d}{dx} (x^2 - 1) = x$$

$$n = 2: P_2(x) = \frac{1}{8} \frac{d^2}{dx^2} (x^2 - 1)^2 = \frac{1}{8} \frac{d^2}{dx^2} (x^4 - 2x^2 + 1) = \frac{1}{8} (12x^2 - 4) = \frac{3}{2}x^2 - \frac{1}{2}$$

$$n = 3: P_3(x) = \frac{1}{48} \frac{d^3}{dx^3} (x^2 - 1)^3 = \frac{1}{48} \frac{d^3}{dx^3} (x^6 - 3x^4 + 3x^2 - 3) = \frac{1}{48} (120x^3 - 72x) = \frac{5}{2}x^3 - \frac{3}{2}x$$

6 Numerical Solutions of Ordinary Differential Equations

Exercises 6.1

All tables in this chapter were constructed in a spreadsheet program which does not support subscripts. Consequently, x_n and y_n will be indicated as $x(n)$ and $y(n)$, respectively.

1.

$h = 0.1$	
x	y
1.00	5.0000
1.10	3.9900
1.20	3.2546
1.30	2.7236
1.40	2.3451
1.50	2.0801

$h = 0.05$	
$x(n)$	$y(n)$
1.00	5.0000
1.05	4.4475
1.10	3.9763
1.15	3.5751
1.20	3.2342
1.25	2.9452
1.30	2.7009
1.35	2.4952
1.40	2.3226
1.45	2.1786
1.50	2.0592

2.

$h = 0.1$	
$x(n)$	$y(n)$
0.00	2.0000
0.10	1.6600
0.20	1.4172
0.30	1.2541
0.40	1.1564
0.50	1.1122

$h = 0.05$	
$x(n)$	$y(n)$
0.00	2.0000
0.05	1.8150
0.10	1.6571
0.15	1.5237
0.20	1.4124
0.25	1.3212
0.30	1.2482
0.35	1.1916
0.40	1.1499
0.45	1.1217
0.50	1.1056

3.

$h = 0.1$	
$x(n)$	$y(n)$
0.00	0.0000
0.10	0.1005
0.20	0.2030
0.30	0.3098
0.40	0.4234
0.50	0.5470

$h = 0.05$	
$x(n)$	$y(n)$
0.00	0.0000
0.05	0.0501
0.10	0.1004
0.15	0.1512
0.20	0.2028
0.25	0.2554
0.30	0.3095
0.35	0.3652
0.40	0.4230
0.45	0.4832
0.50	0.5465

4.

$h = 0.1$		$h = 0.05$	
$x(n)$	$y(n)$	$x(n)$	$y(n)$
0.00	1.0000	0.00	1.0000
0.10	1.1110	0.05	1.0526
0.20	1.2515	0.10	1.1113
0.30	1.4361	0.15	1.1775
0.40	1.6880	0.20	1.2526
0.50	2.0488	0.25	1.3388
		0.30	1.4387
		0.35	1.5556
		0.40	1.6939
		0.45	1.8598
		0.50	2.0619

5.

$h = 0.1$		$h = 0.05$	
$x(n)$	$y(n)$	$x(n)$	$y(n)$
0.00	0.0000	0.00	0.0000
0.10	0.0952	0.05	0.0488
0.20	0.1822	0.10	0.0953
0.30	0.2622	0.15	0.1397
0.40	0.3363	0.20	0.1823
0.50	0.4053	0.25	0.2231
		0.30	0.2623
		0.35	0.3001
		0.40	0.3364
		0.45	0.3715
		0.50	0.4054

6.

$h = 0.1$		$h = 0.05$	
$x(n)$	$y(n)$	$x(n)$	$y(n)$
0.00	0.0000	0.00	0.0000
0.10	0.0050	0.05	0.0013
0.20	0.0200	0.10	0.0050
0.30	0.0451	0.15	0.0113
0.40	0.0805	0.20	0.0200
0.50	0.1266	0.25	0.0313
		0.30	0.0451
		0.35	0.0615
		0.40	0.0805
		0.45	0.1022
		0.50	0.1266

7.

$h = 0.1$		$h = 0.05$	
$x(n)$	$y(n)$	$x(n)$	$y(n)$
0.00	0.5000	0.00	0.5000
0.10	0.5215	0.05	0.5116
0.20	0.5362	0.10	0.5214
0.30	0.5449	0.15	0.5294
0.40	0.5490	0.20	0.5359
0.50	0.5503	0.25	0.5408
		0.30	0.5444
		0.35	0.5469
		0.40	0.5484
		0.45	0.5492
		0.50	0.5495

Exercises 6.1

8.

$h = 0.1$		$h = 0.05$	
$x(n)$	$y(n)$	$x(n)$	$y(n)$
0.00	1.0000	0.00	1.0000
0.10	1.1079	0.05	1.0519
0.20	1.2337	0.10	1.1079
0.30	1.3806	0.15	1.1684
0.40	1.5529	0.20	1.2337
0.50	1.7557	0.25	1.3043
		0.30	1.3807
		0.35	1.4634
		0.40	1.5530
		0.45	1.6503
		0.50	1.7560

9.

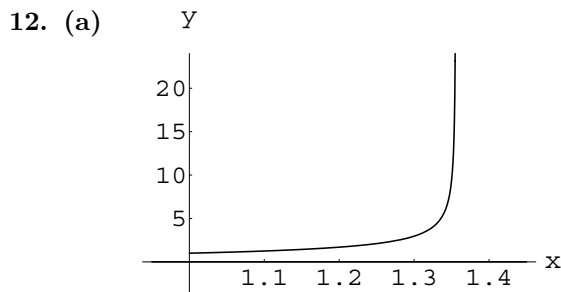
$h = 0.1$		$h = 0.05$	
$x(n)$	$y(n)$	$x(n)$	$y(n)$
1.00	1.0000	1.00	1.0000
1.10	1.0095	1.05	1.0024
1.20	1.0404	1.10	1.0100
1.30	1.0967	1.15	1.0228
1.40	1.1866	1.20	1.0414
1.50	1.3260	1.25	1.0663
		1.30	1.0984
		1.35	1.1389
		1.40	1.1895
		1.45	1.2526
		1.50	1.3315

10.

$h = 0.1$		$h = 0.05$	
$x(n)$	$y(n)$	$x(n)$	$y(n)$
0.00	0.5000	0.00	0.5000
0.10	0.5250	0.05	0.5125
0.20	0.5498	0.10	0.5250
0.30	0.5744	0.15	0.5374
0.40	0.5986	0.20	0.5498
0.50	0.6224	0.25	0.5622
		0.30	0.5744
		0.35	0.5866
		0.40	0.5987
		0.45	0.6106
		0.50	0.6224

11.

$h = 0.1$			$h = 0.05$		
$x(n)$	$y(n)$	exact	$x(n)$	$y(n)$	exact
0.00	2.0000	2.0000	0.00	2.0000	2.0000
0.10	2.1220	2.1230	0.05	2.0553	2.0554
0.20	2.3049	2.3085	0.10	2.1228	2.1230
0.30	2.5858	2.5958	0.15	2.2056	2.2061
0.40	3.0378	3.0650	0.20	2.3075	2.3085
0.50	3.8254	3.9082	0.25	2.4342	2.4358
			0.30	2.5931	2.5958
			0.35	2.7953	2.7997
			0.40	3.0574	3.0650
			0.45	3.4057	3.4189
			0.50	3.8840	3.9082



(b)

h=0.1	EULER	IMPROVED EULER
$x(n)$	$y(n)$	$y(n)$
1.00	1.0000	1.0000
1.10	1.2000	1.2469
1.20	1.4938	1.6668
1.30	1.9711	2.6427
1.40	2.9060	8.7988

13. (a) Using the Euler method we obtain $y(0.1) \approx y_1 = 1.2$.

(b) Using $y'' = 4e^{2x}$ we see that the local truncation error is

$$y''(c) \frac{h^2}{2} = 4e^{2c} \frac{(0.1)^2}{2} = 0.02e^{2c}.$$

Since e^{2x} is an increasing function, $e^{2c} \leq e^{2(0.1)} = e^{0.2}$ for $0 \leq c \leq 0.1$. Thus an upper bound for the local truncation error is $0.02e^{0.2} = 0.0244$.

(c) Since $y(0.1) = e^{0.2} = 1.2214$, the actual error is $y(0.1) - y_1 = 0.0214$, which is less than 0.0244.

(d) Using the Euler method with $h = 0.05$ we obtain $y(0.1) \approx y_2 = 1.21$.

(e) The error in (d) is $1.2214 - 1.21 = 0.0114$. With global truncation error $O(h)$, when the step size is halved we expect the error for $h = 0.05$ to be one-half the error when $h = 0.1$. Comparing 0.0114 with 0.214 we see that this is the case.

14. (a) Using the improved Euler method we obtain $y(0.1) \approx y_1 = 1.22$.

(b) Using $y''' = 8e^{2x}$ we see that the local truncation error is

$$y'''(c) \frac{h^3}{6} = 8e^{2c} \frac{(0.1)^3}{6} = 0.001333e^{2c}.$$

Since e^{2x} is an increasing function, $e^{2c} \leq e^{2(0.1)} = e^{0.2}$ for $0 \leq c \leq 0.1$. Thus an upper bound for the local truncation error is $0.001333e^{0.2} = 0.001628$.

(c) Since $y(0.1) = e^{0.2} = 1.221403$, the actual error is $y(0.1) - y_1 = 0.001403$ which is less than 0.001628.

(d) Using the improved Euler method with $h = 0.05$ we obtain $y(0.1) \approx y_2 = 1.221025$.

(e) The error in (d) is $1.221403 - 1.221025 = 0.000378$. With global truncation error $O(h^2)$, when the step size is halved we expect the error for $h = 0.05$ to be one-fourth the error for $h = 0.1$. Comparing 0.000378 with 0.001403 we see that this is the case.

15. (a) Using the Euler method we obtain $y(0.1) \approx y_1 = 0.8$.

(b) Using $y'' = 5e^{-2x}$ we see that the local truncation error is

$$5e^{-2c} \frac{(0.1)^2}{2} = 0.025e^{-2c}$$

Since e^{-2x} is a decreasing function, $e^{-2c} \leq e^0 = 1$ for $0 \leq c \leq 0.1$. Thus an upper bound for the local truncation error is $0.025(1) = 0.025$.

(c) Since $y(0.1) = 0.8234$, the actual error is $y(0.1) - y_1 = 0.0234$, which is less than 0.025.

(d) Using the Euler method with $h = 0.05$ we obtain $y(0.1) \approx y_2 = 0.8125$.

(e) The error in (d) is $0.8234 - 0.8125 = 0.0109$. With global truncation error $O(h)$, when the step size is halved we expect the error for $h = 0.05$ to be one-half the error when $h = 0.1$. Comparing 0.0109 with 0.0234 we see that this is the case.

Exercises 6.1

16. (a) Using the improved Euler method we obtain $y(0.1) \approx y_1 = 0.825$.

- (b) Using $y''' = -10e^{-2x}$ we see that the local truncation error is

$$10e^{-2c} \frac{(0.1)^3}{6} = 0.001667e^{-2c}.$$

Since e^{-2x} is a decreasing function, $e^{-2c} \leq e^0 = 1$ for $0 \leq c \leq 0.1$. Thus an upper bound for the local truncation error is $0.001667(1) = 0.001667$.

- (c) Since $y(0.1) = 0.823413$, the actual error is $y(0.1) - y_1 = 0.001587$, which is less than 0.001667.

- (d) Using the improved Euler method with $h = 0.05$ we obtain $y(0.1) \approx y_2 = 0.823781$.

- (e) The error in (d) is $|0.823413 - 0.823781| = 0.000368$. With global truncation error $O(h^2)$, when the step size is halved we expect the error for $h = 0.05$ to be one-fourth the error when $h = 0.1$. Comparing 0.000368 with 0.001587 we see that this is the case.

17. (a) Using $y'' = 38e^{-3(x-1)}$ we see that the local truncation error is

$$y''(c) \frac{h^2}{2} = 38e^{-3(c-1)} \frac{h^2}{2} = 19h^2e^{-3(c-1)}.$$

- (b) Since $e^{-3(x-1)}$ is a decreasing function for $1 \leq x \leq 1.5$, $e^{-3(c-1)} \leq e^{-3(1-1)} = 1$ for $1 \leq c \leq 1.5$ and

$$y''(c) \frac{h^2}{2} \leq 19(0.1)^2(1) = 0.19.$$

- (c) Using the Euler method with $h = 0.1$ we obtain $y(1.5) \approx 1.8207$. With $h = 0.05$ we obtain $y(1.5) \approx 1.9424$.

- (d) Since $y(1.5) = 2.0532$, the error for $h = 0.1$ is $E_{0.1} = 0.2325$, while the error for $h = 0.05$ is $E_{0.05} = 0.1109$. With global truncation error $O(h)$ we expect $E_{0.1}/E_{0.05} \approx 2$. We actually have $E_{0.1}/E_{0.05} = 2.10$.

18. (a) Using $y''' = -114e^{-3(x-1)}$ we see that the local truncation error is

$$\left| y'''(c) \frac{h^3}{6} \right| = 114e^{-3(c-1)} \frac{h^3}{6} = 19h^3e^{-3(c-1)}.$$

- (b) Since $e^{-3(x-1)}$ is a decreasing function for $1 \leq x \leq 1.5$, $e^{-3(c-1)} \leq e^{-3(1-1)} = 1$ for $1 \leq c \leq 1.5$ and

$$\left| y'''(c) \frac{h^3}{6} \right| \leq 19(0.1)^3(1) = 0.019.$$

- (c) Using the improved Euler method with $h = 0.1$ we obtain $y(1.5) \approx 2.080108$. With $h = 0.05$ we obtain $y(1.5) \approx 2.059166$.

- (d) Since $y(1.5) = 2.053216$, the error for $h = 0.1$ is $E_{0.1} = 0.026892$, while the error for $h = 0.05$ is $E_{0.05} = 0.005950$. With global truncation error $O(h^2)$ we expect $E_{0.1}/E_{0.05} \approx 4$. We actually have $E_{0.1}/E_{0.05} = 4.52$.

19. (a) Using $y'' = -\frac{1}{(x+1)^2}$ we see that the local truncation error is

$$\left| y''(c) \frac{h^2}{2} \right| = \frac{1}{(c+1)^2} \frac{h^2}{2}.$$

- (b) Since $\frac{1}{(x+1)^2}$ is a decreasing function for $0 \leq x \leq 0.5$, $\frac{1}{(c+1)^2} \leq \frac{1}{(0+1)^2} = 1$ for $0 \leq c \leq 0.5$ and

$$\left| y''(c) \frac{h^2}{2} \right| \leq (1) \frac{(0.1)^2}{2} = 0.005.$$

- (c) Using the Euler method with $h = 0.1$ we obtain $y(0.5) \approx 0.4198$. With $h = 0.05$ we obtain $y(0.5) \approx 0.4124$.
 (d) Since $y(0.5) = 0.4055$, the error for $h = 0.1$ is $E_{0.1} = 0.0143$, while the error for $h = 0.05$ is $E_{0.05} = 0.0069$.
 With global truncation error $O(h)$ we expect $E_{0.1}/E_{0.05} \approx 2$. We actually have $E_{0.1}/E_{0.05} = 2.06$.

20. (a) Using $y''' = \frac{2}{(x+1)^3}$ we see that the local truncation error is

$$y'''(c) \frac{h^3}{6} = \frac{1}{(c+1)^3} \frac{h^3}{3}.$$

- (b) Since $\frac{1}{(x+1)^3}$ is a decreasing function for $0 \leq x \leq 0.5$, $\frac{1}{(c+1)^3} \leq \frac{1}{(0+1)^3} = 1$ for $0 \leq c \leq 0.5$ and

$$y'''(c) \frac{h^3}{6} \leq (1) \frac{(0.1)^3}{3} = 0.000333.$$

- (c) Using the improved Euler method with $h = 0.1$ we obtain $y(0.5) \approx 0.405281$. With $h = 0.05$ we obtain $y(0.5) \approx 0.405419$.
 (d) Since $y(0.5) = 0.405465$, the error for $h = 0.1$ is $E_{0.1} = 0.000184$, while the error for $h = 0.05$ is $E_{0.05} = 0.000046$. With global truncation error $O(h^2)$ we expect $E_{0.1}/E_{0.05} \approx 4$. We actually have $E_{0.1}/E_{0.05} = 3.98$.

21. Because y_{n+1}^* depends on y_n and is used to determine y_{n+1} , all of the y_n^* cannot be computed at one time independently of the corresponding y_n values. For example, the computation of y_4^* involves the value of y_3 .

Exercises 6.2

1.

h = 0.1		
x(n)	y(n)	exact
0.00	2.0000	2.0000
0.10	2.1230	2.1230
0.20	2.3085	2.3085
0.30	2.5958	2.5958
0.40	3.0649	3.0650
0.50	3.9078	3.9082

2. Setting $\alpha = 1/4$ we find $b = 2$, $a = -1$ and $\beta = 1/4$. The resulting second-order Runge-Kutta method is

$$y_{n+1} = y_n - k_1 + 2k_2$$

$$k_1 = hf(x_n, y_n)$$

$$k_2 = hf(x_n + h/4, y_n + k_1/4).$$

h = 0.1	R-K	IMP EULER
x(n)	y(n)	y(n)
0.00	2.0000	2.0000
0.10	2.1205	2.1220
0.20	2.3006	2.3049
0.30	2.5759	2.5858
0.40	3.0152	3.0378
0.50	3.7693	3.8254

3.

x(n)	y(n)
1.00	5.0000
1.10	3.9724
1.20	3.2284
1.30	2.6945
1.40	2.3163
1.50	2.0533

4.

x(n)	y(n)
0.00	2.0000
0.10	1.6562
0.20	1.4110
0.30	1.2465
0.40	1.1480
0.50	1.1037

Exercises 6.2

5.

$x(n)$	$y(n)$
0.00	0.0000
0.10	0.1003
0.20	0.2027
0.30	0.3093
0.40	0.4228
0.50	0.5463

7.

$x(n)$	$y(n)$
0.00	0.0000
0.10	0.0953
0.20	0.1823
0.30	0.2624
0.40	0.3365
0.50	0.4055

9.

$x(n)$	$y(n)$
0.00	0.5000
0.10	0.5213
0.20	0.5358
0.30	0.5443
0.40	0.5482
0.50	0.5493

11.

$x(n)$	$y(n)$
1.00	1.0000
1.10	1.0101
1.20	1.0417
1.30	1.0989
1.40	1.1905
1.50	1.3333

6.

$x(n)$	$y(n)$
0.00	1.0000
0.10	1.1115
0.20	1.2530
0.30	1.4397
0.40	1.6961
0.50	2.0670

8.

$x(n)$	$y(n)$
0.00	0.0000
0.10	0.0050
0.20	0.0200
0.30	0.0451
0.40	0.0805
0.50	0.1266

10.

$x(n)$	$y(n)$
0.00	1.0000
0.10	1.1079
0.20	1.2337
0.30	1.3807
0.40	1.5531
0.50	1.7561

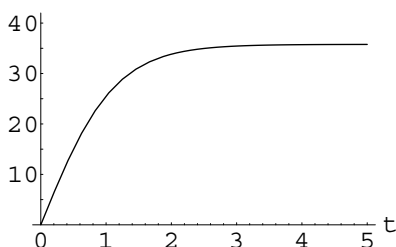
12.

$x(n)$	$y(n)$
0.00	0.5000
0.10	0.5250
0.20	0.5498
0.30	0.5744
0.40	0.5987
0.50	0.6225

13. (a) Write the equation in the form

$$\frac{dv}{dt} = 32 - 0.025v^2 = f(t, v).$$

- (b) $v(t)$



$t(n)$	$v(n)$
0.0	0.0000
1.0	25.2570
2.0	32.9390
3.0	34.9772
4.0	35.5503
5.0	35.7128

- (c) Separating variables and using partial fractions we have

$$\frac{1}{2\sqrt{32}} \left(\frac{1}{\sqrt{32} - \sqrt{0.025}v} + \frac{1}{\sqrt{32} + \sqrt{0.025}v} \right) dv = dt$$

and

$$\frac{1}{2\sqrt{32}\sqrt{0.025}} \left(\ln|\sqrt{32} + \sqrt{0.025}v| - \ln|\sqrt{32} - \sqrt{0.025}v| \right) = t + c.$$

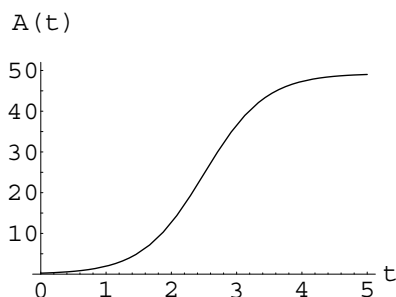
Since $v(0) = 0$ we find $c = 0$. Solving for v we obtain

$$v(t) = \frac{16\sqrt{5}(e^{\sqrt{3.2}t} - 1)}{e^{\sqrt{3.2}t} + 1}$$

and $v(5) \approx 35.7678$.

14. (a) See the table in part (c) of this problem.

(b) From the graph we estimate $A(1) \approx 1.68$, $A(2) \approx 13.2$, $A(3) \approx 36.8$, $A(4) \approx 46.9$, and $A(5) \approx 48.9$.



(c) Let $\alpha = 2.128$ and $\beta = 0.0432$. Separating variables we obtain

$$\frac{dA}{A(\alpha - \beta A)} = dt$$

$$\frac{1}{\alpha} \left(\frac{1}{A} + \frac{\beta}{\alpha - \beta A} \right) dA = dt$$

$$\frac{1}{\alpha} [\ln A - \ln(\alpha - \beta A)] = t + c$$

$$\ln \frac{A}{\alpha - \beta A} = \alpha(t + c)$$

$$\frac{A}{\alpha - \beta A} = e^{\alpha(t+c)}$$

$$A = \alpha e^{\alpha(t+c)} - \beta A e^{\alpha(t+c)}$$

$$\left[1 + \beta e^{\alpha(t+c)} \right] A = \alpha e^{\alpha(t+c)}.$$

Thus

$$A(t) = \frac{\alpha e^{\alpha(t+c)}}{1 + \beta e^{\alpha(t+c)}} = \frac{\alpha}{\beta + e^{-\alpha(t+c)}} = \frac{\alpha}{\beta + e^{-\alpha c} e^{-\alpha t}}.$$

From $A(0) = 0.24$ we obtain

$$0.24 = \frac{\alpha}{\beta + e^{-\alpha c}}$$

so that $e^{-\alpha c} = \alpha/0.24 - \beta \approx 8.8235$ and

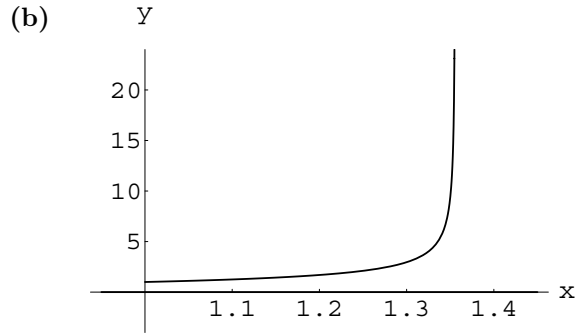
$$A(t) \approx \frac{2.128}{0.0432 + 8.8235 e^{-2.128t}}.$$

t (days)	1	2	3	4	5
A (observed)	2.78	13.53	36.30	47.50	49.40
A (approximated)	1.93	12.50	36.46	47.23	49.00
A (exact)	1.95	12.64	36.63	47.32	49.02

Exercises 6.2

15. (a)

	$h = 0.05$	$h = 0.1$
$x(n)$	$y(n)$	$y(n)$
1.00	1.0000	1.0000
1.05	1.1112	
1.10	1.2511	1.2511
1.15	1.4348	
1.20	1.6934	1.6934
1.25	2.1047	
1.30	2.9560	2.9425
1.35	7.8981	
1.40	1.06E+15	903.0282



16. (a) Using the fourth-order Runge-Kutta method we obtain $y(0.1) \approx y_1 = 1.2214$.

(b) Using $y^{(5)}(x) = 32e^{2x}$ we see that the local truncation error is

$$y^{(5)}(c) \frac{h^5}{120} = 32e^{2c} \frac{(0.1)^5}{120} = 0.000002667e^{2c}.$$

Since e^{2x} is an increasing function, $e^{2c} \leq e^{2(0.1)} = e^{0.2}$ for $0 \leq c \leq 0.1$. Thus an upper bound for the local truncation error is $0.000002667e^{0.2} = 0.000003257$.

(c) Since $y(0.1) = e^{0.2} = 1.221402758$, the actual error is $y(0.1) - y_1 = 0.000002758$ which is less than 0.000003257 .

(d) Using the fourth-order Runge-Kutta formula with $h = 0.05$ we obtain $y(0.1) \approx y_2 = 1.221402571$.

(e) The error in (d) is $1.221402758 - 1.221402571 = 0.000000187$. With global truncation error $O(h^4)$, when the step size is halved we expect the error for $h = 0.05$ to be one-sixteenth the error for $h = 0.1$. Comparing 0.000000187 with 0.000002758 we see that this is the case.

17. (a) Using the fourth-order Runge-Kutta method we obtain $y(0.1) \approx y_1 = 0.823416667$.

(b) Using $y^{(5)}(x) = -40e^{-2x}$ we see that the local truncation error is

$$40e^{-2c} \frac{(0.1)^5}{120} = 0.000003333.$$

Since e^{-2x} is a decreasing function, $e^{-2c} \leq e^0 = 1$ for $0 \leq c \leq 0.1$. Thus an upper bound for the local truncation error is $0.000003333(1) = 0.000003333$.

(c) Since $y(0.1) = 0.823413441$, the actual error is $|y(0.1) - y_1| = 0.000003225$, which is less than 0.000003333 .

(d) Using the fourth-order Runge-Kutta method with $h = 0.05$ we obtain $y(0.1) \approx y_2 = 0.823413627$.

(e) The error in (d) is $|0.823413441 - 0.823413627| = 0.000000185$. With global truncation error $O(h^4)$, when the step size is halved we expect the error for $h = 0.05$ to be one-sixteenth the error when $h = 0.1$. Comparing 0.000000185 with 0.000003225 we see that this is the case.

18. (a) Using $y^{(5)} = -1026e^{-3(x-1)}$ we see that the local truncation error is

$$\left| y^{(5)}(c) \frac{h^5}{120} \right| = 8.55h^5 e^{-3(c-1)}.$$

(b) Since $e^{-3(x-1)}$ is a decreasing function for $1 \leq x \leq 1.5$, $e^{-3(c-1)} \leq e^{-3(1-1)} = 1$ for $1 \leq c \leq 1.5$ and

$$y^{(5)}(c) \frac{h^5}{120} \leq 8.55(0.1)^5(1) = 0.0000855.$$

(c) Using the fourth-order Runge-Kutta method with $h = 0.1$ we obtain $y(1.5) \approx 2.053338827$. With $h = 0.05$ we obtain $y(1.5) \approx 2.053222989$.

19. (a) Using $y^{(5)} = \frac{24}{(x+1)^5}$ we see that the local truncation error is

$$y^{(5)}(c) \frac{h^5}{120} = \frac{1}{(c+1)^5} \frac{h^5}{5}.$$

- (b) Since $\frac{1}{(x+1)^5}$ is a decreasing function for $0 \leq x \leq 0.5$, $\frac{1}{(c+1)^5} \leq \frac{1}{(0+1)^5} = 1$ for $0 \leq c \leq 0.5$ and

$$y^{(5)}(c) \frac{h^5}{5} \leq (1) \frac{(0.1)^5}{5} = 0.000002.$$

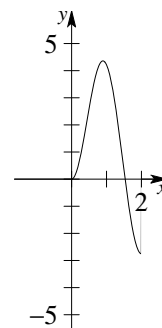
- (c) Using the fourth-order Runge-Kutta method with $h = 0.1$ we obtain $y(0.5) \approx 0.405465168$. With $h = 0.05$ we obtain $y(0.5) \approx 0.405465111$.

20. (a) For $y' + y = 10 \sin 3x$ an integrating factor is e^x so that

$$\begin{aligned} \frac{d}{dx}[e^x y] &= 10e^x \sin 3x \implies e^x y = e^x \sin 3x - 3e^x \cos 3x + c \\ \implies y &= \sin 3x - 3 \cos 3x + ce^{-x}. \end{aligned}$$

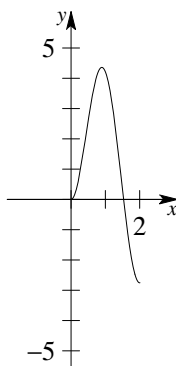
When $x = 0$, $y = 0$, so $0 = -3 + c$ and $c = 3$. The solution is $y = \sin 3x - 3 \cos 3x + 3e^{-x}$.

Using Newton's method we find that $x = 1.53235$ is the only positive root in $[0, 2]$.



- (b) Using the fourth-order Runge-Kutta method with $h = 0.1$ we obtain the table of values shown. These values are used to obtain an interpolating function in *Mathematica*. The graph of the interpolating function is shown. Using *Mathematica*'s root finding capability we see that the only positive root in $[0, 2]$ is $x = 1.53236$.

$x(n)$	$y(n)$	$x(n)$	$y(n)$
0.0	0.0000	1.0	4.2147
0.1	0.1440	1.1	3.8033
0.2	0.5448	1.2	3.1513
0.3	1.1409	1.3	2.3076
0.4	1.8559	1.4	1.3390
0.5	2.6049	1.5	0.3243
0.6	3.3019	1.6	-0.6530
0.7	3.8675	1.7	-1.5117
0.8	4.2356	1.8	-2.1809
0.9	4.3593	1.9	-2.6061
1.0	4.2147	2.0	-2.7539



Exercises 6.3

1. For $y' - y = x - 1$ an integrating factor is $e^{-\int dx} = e^{-x}$, so that

$$\frac{d}{dx}[e^{-x}y] = (x-1)e^{-x}$$

and

$$y = e^x(-xe^{-x} + c) = -x + ce^x.$$

From $y(0) = 1$ we find $c = 1$ and $y = -x + e^x$. Comparing exact values with approximations obtained in Example 1, we find $y(0.2) \approx 1.02140276$ compared to $y_1 = 1.02140000$, $y(0.4) \approx 1.09182470$ compared to

Exercises 6.3

$y_2 = 1.09181796$, $y(0.6) \approx 1.22211880$ compared to $y_3 = 1.22210646$, and $y(0.8) \approx 1.42554093$ compared to $y_4 = 1.42552788$.

```
2. 100 REM ADAMS-BASHFORTH/ADAMS-MOULTON
    110 REM METHOD TO SOLVE Y'=FNF(X,Y)
    120 REM DEFINE FNF(X,Y) HERE
    130 REM GET INPUTS
    140 PRINT
    150 INPUT "STEP SIZE=", H
    160 INPUT "NUMBER OF STEPS (AT LEAST 4)=", N
    170 IF N<4 GOTO 160
    180 INPUT "X0 =", X
    190 INPUT "Y0 =", Y
    200 PRINT
    210 REM SET UP TABLE
    220 PRINT "X", "Y"
    230 PRINT
    240 REM COMPUTE 3 ITERATES USING RUNGE-KUTTA
    250 DIM Z(4)
    260 Z(1)=Y
    270 FOR I=1 TO 3
    280 K1=H*FNF(X,Y)
    290 K2=H*FNF(X+H/2,Y+K1/2)
    300 K3=H*FNF(X+H/2,Y+K2/2)
    310 K4=H*FNF(X+H,Y+K3)
    320 Y=Y+(K1+2*K2+2*K3+K4)/6
    330 Z(I+1)=Y
    340 X=X+H
    350 PRINT X,Y
    360 NEXT I
    370 REM COMPUTE REMAINING X AND Y VALUES
    380 FOR I=4 TO N
    390 YP=Y+H*(55*FNF(X,Z(4))-59*FNF(X-H,Z(3))+37*FNF(X-2*H,Z(2))
        -9*FNF(X-3*H,Z(1)))/24
    400 Y=Y+H*(9*FNF(X+H,YP)+19*FNF(X,Z(4))-5*FNF(X-H,Z(3))+FNF(X-2*H,Z(2)))/24
    410 X=X+H
    420 PRINT X,Y
    430 Z(1)=Z(2)
    440 Z(2)=Z(3)
    450 Z(3)=Z(4)
    460 Z(4)=Y
    470 NEXT I
    480 END
```

3.

$x(n)$	$y(n)$	
0.00	1.0000	initial condition
0.20	0.7328	Runge-Kutta
0.40	0.6461	Runge-Kutta
0.60	0.6585	Runge-Kutta
	0.7332	<i>predictor</i>
0.80	0.7232	corrector

4.

$x(n)$	$y(n)$	
0.00	2.0000	initial condition
0.20	1.4112	Runge-Kutta
0.40	1.1483	Runge-Kutta
0.60	1.1039	Runge-Kutta
	1.2109	<i>predictor</i>
0.80	1.2049	corrector

5.

$x(n)$	$y(n)$	
0.00	0.0000	initial condition
0.20	0.2027	Runge-Kutta
0.40	0.4228	Runge-Kutta
0.60	0.6841	Runge-Kutta
	1.0234	<i>predictor</i>
0.80	1.0297	corrector
	1.5376	<i>predictor</i>
1.00	1.5569	corrector

$x(n)$	$y(n)$	
0.00	0.0000	initial condition
0.10	0.1003	Runge-Kutta
0.20	0.2027	Runge-Kutta
0.30	0.3093	Runge-Kutta
	0.4227	<i>predictor</i>
0.40	0.4228	corrector
	0.5462	<i>predictor</i>
0.50	0.5463	corrector
	0.6840	<i>predictor</i>
0.60	0.6842	corrector
	0.8420	<i>predictor</i>
0.70	0.8423	corrector
	1.0292	<i>predictor</i>
0.80	1.0297	corrector
	1.2592	<i>predictor</i>
0.90	1.2603	corrector
	1.5555	<i>predictor</i>
1.00	1.5576	corrector

Exercises 6.3

6.

$x(n)$	$y(n)$	
0.00	1.0000	initial condition
0.20	1.4414	Runge-Kutta
0.40	1.9719	Runge-Kutta
0.60	2.6028	Runge-Kutta
	3.3483	<i>predictor</i>
0.80	3.3486	corrector
	4.2276	<i>predictor</i>
1.00	4.2280	corrector

$x(n)$	$y(n)$	
0.00	1.0000	initial condition
0.10	1.2102	Runge-Kutta
0.20	1.4414	Runge-Kutta
0.30	1.6949	Runge-Kutta
	1.9719	<i>predictor</i>
0.40	1.9719	corrector
	2.2740	<i>predictor</i>
0.50	2.2740	corrector
	2.6028	<i>predictor</i>
0.60	2.6028	corrector
	2.9603	<i>predictor</i>
0.70	2.9603	corrector
	3.3486	<i>predictor</i>
0.80	3.3486	corrector
	3.7703	<i>predictor</i>
0.90	3.7703	corrector
	4.2280	<i>predictor</i>
1.00	4.2280	corrector

7.

$x(n)$	$y(n)$	
0.00	0.0000	initial condition
0.20	0.0026	Runge-Kutta
0.40	0.0201	Runge-Kutta
0.60	0.0630	Runge-Kutta
	0.1362	<i>predictor</i>
0.80	0.1360	corrector
	0.2379	<i>predictor</i>
1.00	0.2385	corrector

$x(n)$	$y(n)$	
0.00	0.0000	initial condition
0.10	0.0003	Runge-Kutta
0.20	0.0026	Runge-Kutta
0.30	0.0087	Runge-Kutta
	0.0201	<i>predictor</i>
0.40	0.0200	corrector
	0.0379	<i>predictor</i>
0.50	0.0379	corrector
	0.0630	<i>predictor</i>
0.60	0.0629	corrector
	0.0956	<i>predictor</i>
0.70	0.0956	corrector
	0.1359	<i>predictor</i>
0.80	0.1360	corrector
	0.1837	<i>predictor</i>
0.90	0.1837	corrector
	0.2384	<i>predictor</i>
1.00	0.2384	corrector

8.	$x(n)$	$y(n)$		$x(n)$	$y(n)$	
	0.00	1.0000	initial condition	0.00	1.0000	initial condition
	0.20	1.2337	Runge-Kutta	0.10	1.1079	Runge-Kutta
	0.40	1.5531	Runge-Kutta	0.20	1.2337	Runge-Kutta
	0.60	1.9961	Runge-Kutta	0.30	1.3807	Runge-Kutta
		2.6180	<i>predictor</i>		1.5530	<i>predictor</i>
	0.80	2.6214	corrector	0.40	1.5531	corrector
		3.5151	<i>predictor</i>		1.7560	<i>predictor</i>
	1.00	3.5208	corrector	0.50	1.7561	corrector
					1.9960	<i>predictor</i>
				0.60	1.9961	corrector
					2.2811	<i>predictor</i>
				0.70	2.2812	corrector
					2.6211	<i>predictor</i>
				0.80	2.6213	corrector
					3.0289	<i>predictor</i>
				0.90	3.0291	corrector
					3.5203	<i>predictor</i>
				1.00	3.5207	corrector

Exercises 6.4

1. The substitution $y' = u$ leads to the iteration formulas

$$y_{n+1} = y_n + hu_n, \quad u_{n+1} = u_n + h(4u_n - 4y_n).$$

The initial conditions are $y_0 = -2$ and $u_0 = 1$. Then

$$y_1 = y_0 + 0.1u_0 = -2 + 0.1(1) = -1.9$$

$$u_1 = u_0 + 0.1(4u_0 - 4y_0) = 1 + 0.1(4 + 8) = 2.2$$

$$y_2 = y_1 + 0.1u_1 = -1.9 + 0.1(2.2) = -1.68.$$

The general solution of the differential equation is $y = c_1 e^{2x} + c_2 x e^{2x}$. From the initial conditions we find $c_1 = -2$ and $c_2 = 5$. Thus $y = -2e^{2x} + 5xe^{2x}$ and $y(0.2) \approx 1.4918$.

2. The substitution $y' = u$ leads to the iteration formulas

$$y_{n+1} = y_n + hu_n, \quad u_{n+1} = u_n + h \left(\frac{2}{x} u_n - \frac{2}{x^2} y_n \right).$$

The initial conditions are $y_0 = 4$ and $u_0 = 9$. Then

$$y_1 = y_0 + 0.1u_0 = 4 + 0.1(9) = 4.9$$

$$u_1 = u_0 + 0.1 \left(\frac{2}{1} u_0 - \frac{2}{1} y_0 \right) = 9 + 0.1[2(9) - 2(4)] = 10$$

$$y_2 = y_1 + 0.1u_1 = 4.9 + 0.1(10) = 5.9.$$

The general solution of the Cauchy-Euler differential equation is $y = c_1 x + c_2 x^2$. From the initial conditions we find $c_1 = -1$ and $c_2 = 5$. Thus $y = -x + 5x^2$ and $y(1.2) = 6$.

3. The substitution $y' = u$ leads to the system

$$y' = u, \quad u' = 4u - 4y.$$

Using formula (4) in the text with x corresponding to t , y corresponding to x , and u corresponding to y , we obtain

Exercises 6.4

Runge-Kutta method with h=0.2

m1	m2	m3	m4	k1	k2	k3	k4	x	y	u
0.2000	0.4400	0.5280	0.9072	2.4000	3.2800	3.5360	4.8064	0.00	-2.0000	1.0000
								0.20	-1.4928	4.4731

Runge-Kutta method with h=0.1

m1	m2	m3	m4	k1	k2	k3	k4	x	y	u
								0.00	-2.0000	1.0000
0.1000	0.1600	0.1710	0.2452	1.2000	1.4200	1.4520	1.7124	0.10	-1.8321	2.4427
0.2443	0.3298	0.3444	0.4487	1.7099	2.0031	2.0446	2.3900	0.20	-1.4919	4.4753

4. The substitution $y' = u$ leads to the system

$$y' = u, \quad u' = \frac{2}{x}u - \frac{2}{x^2}y.$$

Using formula (4) in the text with x corresponding to t , y corresponding to x , and u corresponding to y , we obtain

Runge-Kutta method with h=0.2

m1	m2	m3	m4	k1	k2	k3	k4	x	y	u
								1.00	4.0000	9.0000
1.8000	2.0000	2.0017	2.1973	2.0000	2.0165	1.9865	1.9950	1.20	6.0001	11.0002

Runge-Kutta method with h=0.1

m1	m2	m3	m4	k1	k2	k3	k4	x	y	u
								1.00	4.0000	9.0000
0.9000	0.9500	0.9501	0.9998	1.0000	1.0023	0.9979	0.9996	1.10	4.9500	10.0000
1.0000	1.0500	1.0501	1.0998	1.0000	1.0019	0.9983	0.9997	1.20	6.0000	11.0000

5. The substitution $y' = u$ leads to the system

$$y' = u, \quad u' = 2u - 2y + e^t \cos t.$$

Using formula (4) in the text with y corresponding to x and u corresponding to y , we obtain

Runge-Kutta method with h=0.2

m1	m2	m3	m4	k1	k2	k3	k4	t	y	u
								0.00	1.0000	2.0000
0.4000	0.4600	0.4660	0.5320	0.6000	0.6599	0.6599	0.7170	0.20	1.4640	2.6594

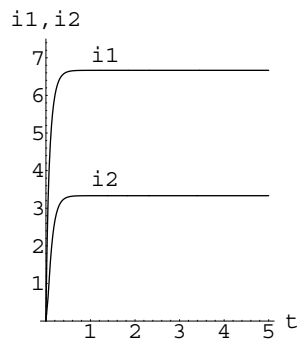
Runge-Kutta method with h=0.1

m1	m2	m3	m4	k1	k2	k3	k4	t	y	u
								0.00	1.0000	2.0000
0.2000	0.2150	0.2157	0.2315	0.3000	0.3150	0.3150	0.3298	0.10	1.2155	2.3150
0.2315	0.2480	0.2487	0.2659	0.3299	0.3446	0.3444	0.3587	0.20	1.4640	2.6594

- 6.

Runge-Kutta method with h=0.1

m1	m2	m3	m4	k1	k2	k3	k4	t	i1	i2
								0.00	0.0000	0.0000
10.0000	0.0000	12.5000	-20.0000	0.0000	5.0000	-5.0000	22.5000	0.10	2.5000	3.7500
8.7500	-2.5000	13.4375	-28.7500	-5.0000	4.3750	-10.6250	29.6875	0.20	2.8125	5.7813
10.1563	-4.3750	17.0703	-40.0000	-8.7500	5.0781	-16.0156	40.3516	0.30	2.0703	7.4023
13.2617	-6.3672	22.9443	-55.1758	-12.7344	6.6309	-22.5488	55.3076	0.40	0.6104	9.1919
17.9712	-8.8867	31.3507	-75.9326	-17.7734	8.9856	-31.2024	75.9821	0.50	-1.5619	11.4877



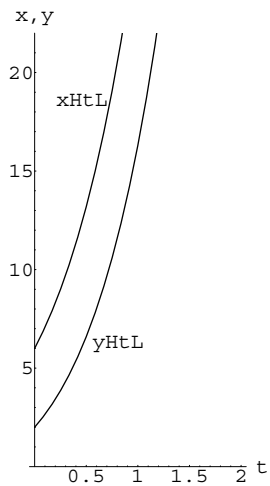
As $t \rightarrow \infty$ we see that $i_1(t) \rightarrow 6.75$ and $i_2(t) \rightarrow 3.4$.

7. Runge-Kutta method with $h=0.2$

$m1$	$m2$	$m3$	$m4$	$k1$	$k2$	$k3$	$k4$	t	x	y
								0.00	6.0000	2.0000
2.0000	2.2800	2.3160	2.6408	1.2000	1.4000	1.4280	1.6632	0.20	8.3055	3.4199

Runge-Kutta method with $h=0.1$

$m1$	$m2$	$m3$	$m4$	$k1$	$k2$	$k3$	$k4$	t	x	y
								0.00	6.0000	2.0000
1.0000	1.0700	1.0745	1.1496	0.6000	0.6500	0.6535	0.7075	0.10	7.0731	2.6524
1.1494	1.2289	1.2340	1.3193	0.7073	0.7648	0.7688	0.8307	0.20	8.3055	3.4199



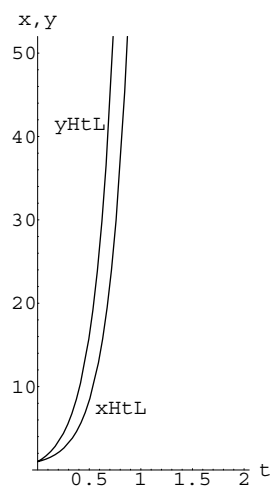
8. Runge-Kutta method with $h=0.2$

$m1$	$m2$	$m3$	$m4$	$k1$	$k2$	$k3$	$k4$	t	x	y
								0.00	1.0000	1.0000
0.6000	0.9400	1.1060	1.7788	1.4000	2.0600	2.3940	3.7212	0.20	2.0785	3.3382

Runge-Kutta method with $h=0.1$

$m1$	$m2$	$m3$	$m4$	$k1$	$k2$	$k3$	$k4$	t	x	y
								0.00	1.0000	1.0000
0.3000	0.3850	0.4058	0.5219	0.7000	0.8650	0.9068	1.1343	0.10	1.4006	1.8963
0.5193	0.6582	0.6925	0.8828	1.1291	1.4024	1.4711	1.8474	0.20	2.0845	3.3502

Exercises 6.4

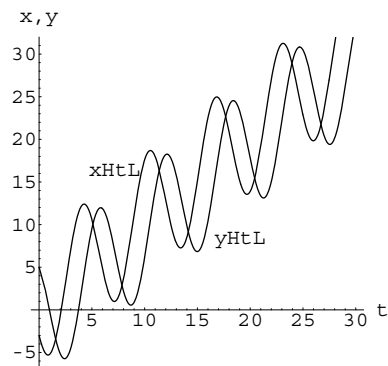


9. Runge-Kutta method with $h=0.2$

$m1$	$m2$	$m3$	$m4$	$k1$	$k2$	$k3$	$k4$	t	x	y
-1.0000	-0.9200	-0.9080	-0.8176	-0.6000	-0.7200	-0.7120	-0.8216	0.00	-3.0000	5.0000
								0.20	-3.9123	4.2857

Runge-Kutta method with $h=0.1$

$m1$	$m2$	$m3$	$m4$	$k1$	$k2$	$k3$	$k4$	t	x	y
-0.5000	-0.4800	-0.4785	-0.4571	-0.3000	-0.3300	-0.3290	-0.3579	0.00	-3.0000	5.0000
								0.10	-3.4790	4.6707
-0.4571	-0.4342	-0.4328	-0.4086	-0.3579	-0.3858	-0.3846	-0.4112	0.20	-3.9123	4.2857

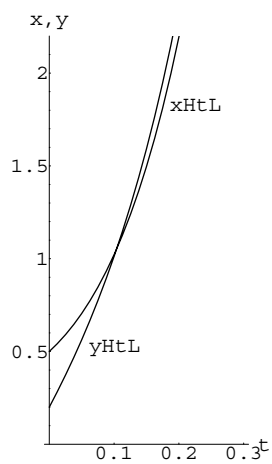


10. Runge-Kutta method with $h=0.2$

$m1$	$m2$	$m3$	$m4$	$k1$	$k2$	$k3$	$k4$	t	x	y
0.6400	1.2760	1.7028	3.3558	1.3200	1.7720	2.1620	3.5794	0.00	0.5000	0.2000
								0.20	2.1589	2.3279

Runge-Kutta method with $h=0.1$

$m1$	$m2$	$m3$	$m4$	$k1$	$k2$	$k3$	$k4$	t	x	y
0.3200	0.4790	0.5324	0.7816	0.6600	0.7730	0.8218	1.0195	0.00	0.5000	0.2000
								0.10	1.0207	1.0115
0.7736	1.0862	1.1929	1.6862	1.0117	1.2682	1.3692	1.7996	0.20	2.1904	2.3592



11. Solving for x' and y' we obtain the system

$$x' = -2x + y + 5t$$

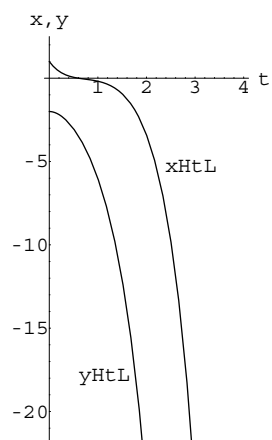
$$y' = 2x + y - 2t.$$

Runge-Kutta method with $h=0.2$

$m1$	$m2$	$m3$	$m4$	$k1$	$k2$	$k3$	$k4$	t	x	y
								0.00	1.0000	-2.0000
-0.8000	-0.5400	-0.6120	-0.3888	0.0000	-0.2000	-0.1680	-0.3584	0.20	0.4179	-2.1824

Runge-Kutta method with $h=0.1$

$m1$	$m2$	$m3$	$m4$	$k1$	$k2$	$k3$	$k4$	t	x	y
								0.00	1.0000	-2.0000
-0.4000	-0.3350	-0.3440	-0.2858	0.0000	-0.0500	-0.0460	-0.0934	0.10	0.6594	-2.0476
-0.2866	-0.2376	-0.2447	-0.2010	-0.0929	-0.1362	-0.1335	-0.1752	0.20	0.4173	-2.1821



Exercises 6.4

12. Solving for x' and y' we obtain the system

$$x' = \frac{1}{2}y - 3t^2 + 2t - 5$$

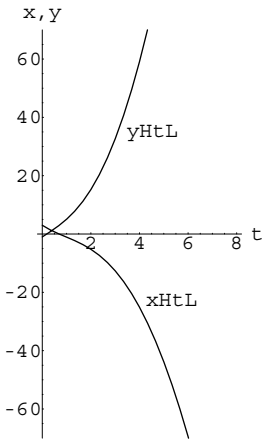
$$y' = -\frac{1}{2}y + 3t^2 + 2t + 5.$$

Runge-Kutta method with $h=0.2$

m1	m2	m3	m4	k1	k2	k3	k4	t	x	y
-1.1000	-1.0110	-1.0115	-0.9349	1.1000	1.0910	1.0915	1.0949	0.00	3.0000	-1.0000
								0.20	1.9867	0.0933

Runge-Kutta method with $h=0.1$

m1	m2	m3	m4	k1	k2	k3	k4	t	x	y
-0.5500	-0.5270	-0.5271	-0.5056	0.5500	0.5470	0.5471	0.5456	0.00	3.0000	-1.0000
-0.5506	-0.5270	-0.5271	-0.5056	0.5500	0.5470	0.5471	0.5456	0.10	2.4727	-0.4527
-0.5056	-0.4857	-0.4857	-0.4673	0.5456	0.5457	0.5457	0.5473	0.20	1.9867	0.0933



Exercises 6.5

1. We identify $P(x) = 0$, $Q(x) = 9$, $f(x) = 0$, and $h = (2 - 0)/4 = 0.5$. Then the finite difference equation is

$$y_{i+1} + 0.25y_i + y_{i-1} = 0.$$

The solution of the corresponding linear system gives

x	0.0	0.5	1.0	1.5	2.0
y	4.0000	-5.6774	-2.5807	6.3226	1.0000

2. We identify $P(x) = 0$, $Q(x) = -1$, $f(x) = x^2$, and $h = (1 - 0)/4 = 0.25$. Then the finite difference equation is

$$y_{i+1} - 2.0625y_i + y_{i-1} = 0.0625x_i^2.$$

The solution of the corresponding linear system gives

x	0.00	0.25	0.50	0.75	1.00
y	0.0000	-0.0172	-0.0316	-0.0324	0.0000

3. We identify $P(x) = 2$, $Q(x) = 1$, $f(x) = 5x$, and $h = (1 - 0)/5 = 0.2$. Then the finite difference equation is

$$1.2y_{i+1} - 1.96y_i + 0.8y_{i-1} = 0.04(5x_i).$$

The solution of the corresponding linear system gives

x	0.0	0.2	0.4	0.6	0.8	1.0
y	0.0000	-0.2259	-0.3356	-0.3308	-0.2167	0.0000

4. We identify $P(x) = -10$, $Q(x) = 25$, $f(x) = 1$, and $h = (1 - 0)/5 = 0.2$. Then the finite difference equation is

$$-y_i + 2y_{i-1} = 0.04.$$

The solution of the corresponding linear system gives

x	0.0	0.2	0.4	0.6	0.8	1.0
y	1.0000	1.9600	3.8800	7.7200	15.4000	0.0000

5. We identify $P(x) = -4$, $Q(x) = 4$, $f(x) = (1 + x)e^{2x}$, and $h = (1 - 0)/6 = 0.1667$. Then the finite difference equation is

$$0.6667y_{i+1} - 1.8889y_i + 1.3333y_{i-1} = 0.2778(1 + x_i)e^{2x_i}.$$

The solution of the corresponding linear system gives

x	0.0000	0.1667	0.3333	0.5000	0.6667	0.8333	1.0000
y	3.0000	3.3751	3.6306	3.6448	3.2355	2.1411	0.0000

6. We identify $P(x) = 5$, $Q(x) = 0$, $f(x) = 4\sqrt{x}$, and $h = (2 - 1)/6 = 0.1667$. Then the finite difference equation is

$$1.4167y_{i+1} - 2y_i + 0.5833y_{i-1} = 0.2778(4\sqrt{x_i}).$$

The solution of the corresponding linear system gives

x	1.0000	1.1667	1.3333	1.5000	1.6667	1.8333	2.0000
y	1.0000	-0.5918	-1.1626	-1.3070	-1.2704	-1.1541	-1.0000

7. We identify $P(x) = 3/x$, $Q(x) = 3/x^2$, $f(x) = 0$, and $h = (2 - 1)/8 = 0.125$. Then the finite difference equation is

$$\left(1 + \frac{0.1875}{x_i}\right)y_{i+1} + \left(-2 + \frac{0.0469}{x_i^2}\right)y_i + \left(1 - \frac{0.1875}{x_i}\right)y_{i-1} = 0.$$

The solution of the corresponding linear system gives

x	1.000	1.125	1.250	1.375	1.500	1.625	1.750	1.875	2.000
y	5.0000	3.8842	2.9640	2.2064	1.5826	1.0681	0.6430	0.2913	0.0000

8. We identify $P(x) = -1/x$, $Q(x) = x^{-2}$, $f(x) = \ln x/x^2$, and $h = (2 - 1)/8 = 0.125$. Then the finite difference equation is

$$\left(1 - \frac{0.0625}{x_i}\right)y_{i+1} + \left(-2 + \frac{0.0156}{x_i^2}\right)y_i + \left(1 + \frac{0.0625}{x_i}\right)y_{i-1} = 0.0156 \ln x_i.$$

The solution of the corresponding linear system gives

x	1.000	1.125	1.250	1.375	1.500	1.625	1.750	1.875	2.000
y	0.0000	-0.1988	-0.4168	-0.6510	-0.8992	-1.1594	-1.4304	-1.7109	-2.0000

9. We identify $P(x) = 1 - x$, $Q(x) = x$, $f(x) = x$, and $h = (1 - 0)/10 = 0.1$. Then the finite difference equation is

$$[1 + 0.05(1 - x_i)]y_{i+1} + [-2 + 0.01x_i]y_i + [1 - 0.05(1 - x_i)]y_{i-1} = 0.01x_i.$$

The solution of the corresponding linear system gives

Exercises 6.5

x	0.0	0.1	0.2	0.3	0.4	0.5	0.6
y	0.0000	0.2660	0.5097	0.7357	0.9471	1.1465	1.3353

0.7	0.8	0.9	1.0
1.5149	1.6855	1.8474	2.0000

10. We identify $P(x) = x$, $Q(x) = 1$, $f(x) = x$, and $h = (1 - 0)/10 = 0.1$. Then the finite difference equation is

$$(1 + 0.05x_i)y_{i+1} - 1.99y_i + (1 - 0.05x_i)y_{i-1} = 0.01x_i.$$

The solution of the corresponding linear system gives

x	0.0	0.1	0.2	0.3	0.4	0.5	0.6
y	1.0000	0.8929	0.7789	0.6615	0.5440	0.4296	0.3216

0.7	0.8	0.9	1.0
0.2225	0.1347	0.0601	0.0000

11. We identify $P(x) = 0$, $Q(x) = -4$, $f(x) = 0$, and $h = (1 - 0)/8 = 0.125$. Then the finite difference equation is

$$y_{i+1} - 2.0625y_i + y_{i-1} = 0.$$

The solution of the corresponding linear system gives

x	0.000	0.125	0.250	0.375	0.500	0.625	0.750	0.875	1.000
y	0.0000	0.3492	0.7202	1.1363	1.6233	2.2118	2.9386	3.8490	5.0000

12. We identify $P(r) = 2/r$, $Q(r) = 0$, $f(r) = 0$, and $h = (4 - 1)/6 = 0.5$. Then the finite difference equation is

$$\left(1 + \frac{0.5}{r_i}\right)u_{i+1} - 2u_i + \left(1 - \frac{0.5}{r_i}\right)u_{i-1} = 0.$$

The solution of the corresponding linear system gives

r	1.0	1.5	2.0	2.5	3.0	3.5	4.0
u	50.0000	72.2222	83.3333	90.0000	94.4444	97.6190	100.0000

13. (a) The difference equation

$$\left(1 + \frac{h}{2}P_i\right)y_{i+1} + (-2 + h^2Q_i)y_i + \left(1 - \frac{h}{2}P_i\right)y_{i-1} = h^2f_i$$

is the same as the one derived on page 383 in the text. The equations are the same because the derivation was based only on the differential equation, not the boundary conditions. If we allow i to range from 0 to $n - 1$ we obtain n equations in the $n + 1$ unknowns y_{-1} , y_0 , y_1 , $\dots$, y_{n-1} . Since y_n is one of the given boundary conditions, it is not an unknown.

- (b) Identifying $y_0 = y(0)$, $y_{-1} = y(0 - h)$, and $y_1 = y(0 + h)$ we have from (5) in the text

$$\frac{1}{2h}[y_1 - y_{-1}] = y'(0) = 1 \quad \text{or} \quad y_1 - y_{-1} = 2h.$$

The difference equation corresponding to $i = 0$,

$$\left(1 + \frac{h}{2}P_0\right)y_1 + (-2 + h^2Q_0)y_0 + \left(1 - \frac{h}{2}P_0\right)y_{-1} = h^2f_0$$

becomes, with $y_{-1} = y_1 - 2h$,

$$\left(1 + \frac{h}{2}P_0\right)y_1 + (-2 + h^2Q_0)y_0 + \left(1 - \frac{h}{2}P_0\right)(y_1 - 2h) = h^2f_0$$

or

$$2y_1 + (-2 + h^2Q_0)y_0 = h^2f_0 + 2h - P_0.$$

Alternatively, we may simply add the equation $y_1 - y_{-1} = 2h$ to the list of n difference equations obtaining $n + 1$ equations in the $n + 1$ unknowns $y_{-1}, y_0, y_1, \dots, y_{n-1}$.

(c) Using $n = 5$ we obtain

x	0.0	0.2	0.4	0.6	0.8	1.0
y	-2.2755	-2.0755	-1.8589	-1.6126	-1.3275	-1.0000

14. Using $h = 0.1$ and, after shooting a few times, $y'(0) = 0.43535$ we obtain the following table with the fourth-order Runge-Kutta method.

x	0.0	0.1	0.2	0.3	0.4	0.5	0.6
y	1.00000	1.04561	1.09492	1.14714	1.20131	1.25633	1.31096

0.7	0.8	0.9	1.0
1.36392	1.41388	1.45962	1.50003

Chapter 6 Review Exercises

- | h=0.1 | | | | h=0.05 | | | |
|-------|--------|----------------|-------------|--------|--------|----------------|-------------|
| x(n) | EULER | IMPROVED EULER | RUNGE KUTTA | x(n) | EULER | IMPROVED EULER | RUNGE KUTTA |
| 1.00 | 2.0000 | 2.0000 | 2.0000 | 1.00 | 2.0000 | 2.0000 | 2.0000 |
| 1.10 | 2.1386 | 2.1549 | 2.1556 | 1.05 | 2.0693 | 2.0735 | 2.0736 |
| 1.20 | 2.3097 | 2.3439 | 2.3454 | 1.10 | 2.1469 | 2.1554 | 2.1556 |
| 1.30 | 2.5136 | 2.5672 | 2.5695 | 1.15 | 2.2328 | 2.2459 | 2.2462 |
| 1.40 | 2.7504 | 2.8246 | 2.8278 | 1.20 | 2.3272 | 2.3450 | 2.3454 |
| 1.50 | 3.0201 | 3.1157 | 3.1197 | 1.25 | 2.4299 | 2.4527 | 2.4532 |
| | | | | 1.30 | 2.5409 | 2.5689 | 2.5695 |
| | | | | 1.35 | 2.6604 | 2.6937 | 2.6944 |
| | | | | 1.40 | 2.7883 | 2.8269 | 2.8278 |
| | | | | 1.45 | 2.9245 | 2.9686 | 2.9696 |
| | | | | 1.50 | 3.0690 | 3.1187 | 3.1197 |
- | h=0.1 | | | | h=0.05 | | | |
|-------|--------|----------------|-------------|--------|--------|----------------|-------------|
| x(n) | EULER | IMPROVED EULER | RUNGE KUTTA | x(n) | EULER | IMPROVED EULER | RUNGE KUTTA |
| 0.00 | 0.0000 | 0.0000 | 0.0000 | 0.00 | 0.0000 | 0.0000 | 0.0000 |
| 0.10 | 0.1000 | 0.1005 | 0.1003 | 0.05 | 0.0500 | 0.0501 | 0.0500 |
| 0.20 | 0.2010 | 0.2030 | 0.2026 | 0.10 | 0.1001 | 0.1004 | 0.1003 |
| 0.30 | 0.3049 | 0.3092 | 0.3087 | 0.15 | 0.1506 | 0.1512 | 0.1511 |
| 0.40 | 0.4135 | 0.4207 | 0.4201 | 0.20 | 0.2017 | 0.2027 | 0.2026 |
| 0.50 | 0.5279 | 0.5382 | 0.5376 | 0.25 | 0.2537 | 0.2552 | 0.2551 |
| | | | | 0.30 | 0.3067 | 0.3088 | 0.3087 |
| | | | | 0.35 | 0.3610 | 0.3638 | 0.3637 |
| | | | | 0.40 | 0.4167 | 0.4202 | 0.4201 |
| | | | | 0.45 | 0.4739 | 0.4782 | 0.4781 |
| | | | | 0.50 | 0.5327 | 0.5378 | 0.5376 |

Chapter 6 Review Exercises

3.	h=0.1		IMPROVED	RUNGE	h=0.05		IMPROVED	RUNGE
	x(n)	EULER	EULER	KUTTA	x(n)	EULER	EULER	KUTTA
	0.50	0.5000	0.5000	0.5000	0.50	0.5000	0.5000	0.5000
	0.60	0.6000	0.6048	0.6049	0.55	0.5500	0.5512	0.5512
	0.70	0.7095	0.7191	0.7194	0.60	0.6024	0.6049	0.6049
	0.80	0.8283	0.8427	0.8431	0.65	0.6573	0.6609	0.6610
	0.90	0.9559	0.9752	0.9757	0.70	0.7144	0.7193	0.7194
	1.00	1.0921	1.1163	1.1169	0.75	0.7739	0.7800	0.7801
					0.80	0.8356	0.8430	0.8431
					0.85	0.8996	0.9082	0.9083
					0.90	0.9657	0.9755	0.9757
					0.95	1.0340	1.0451	1.0452
					1.00	1.1044	1.1168	1.1169

4.	h=0.1		IMPROVED	RUNGE	h=0.05		IMPROVED	RUNGE
	x(n)	EULER	EULER	KUTTA	x(n)	EULER	EULER	KUTTA
	1.00	1.0000	1.0000	1.0000	1.00	1.0000	1.0000	1.0000
	1.10	1.2000	1.2380	1.2415	1.05	1.1000	1.1091	1.1095
	1.20	1.4760	1.5910	1.6036	1.10	1.2183	1.2405	1.2415
	1.30	1.8710	2.1524	2.1909	1.15	1.3595	1.4010	1.4029
	1.40	2.4643	3.1458	3.2745	1.20	1.5300	1.6001	1.6036
	1.50	3.4165	5.2510	5.8338	1.25	1.7389	1.8523	1.8586
					1.30	1.9988	2.1799	2.1911
					1.35	2.3284	2.6197	2.6401
					1.40	2.7567	3.2360	3.2755
					1.45	3.3296	4.1528	4.2363
					1.50	4.1253	5.6404	5.8446

5. Using

$$y_{n+1} = y_n + hu_n, \quad y_0 = 3$$

$$u_{n+1} = u_n + h(2x_n + 1)y_n, \quad u_0 = 1$$

we obtain (when $h = 0.2$) $y_1 = y(0.2) = y_0 + hu_0 = 3 + (0.2)1 = 3.2$. When $h = 0.1$ we have

$$y_1 = y_0 + 0.1u_0 = 3 + (0.1)1 = 3.1$$

$$u_1 = u_0 + 0.1(2x_0 + 1)y_0 = 1 + 0.1(1)3 = 1.3$$

$$y_2 = y_1 + 0.1u_1 = 3.1 + 0.1(1.3) = 3.23.$$

6.	x(n)	y(n)	
	0.00	2.0000	initial condition
	0.10	2.4734	Runge-Kutta
	0.20	3.1781	Runge-Kutta
	0.30	4.3925	Runge-Kutta
		6.7689	<i>predictor</i>
	0.40	7.0783	<i>corrector</i>

7. Using $x_0 = 1$, $y_0 = 2$, and $h = 0.1$ we have

$$x_1 = x_0 + h(x_0 + y_0) = 1 + 0.1(1 + 2) = 1.3$$

$$y_1 = y_0 + h(x_0 - y_0) = 2 + 0.1(1 - 2) = 1.9$$

and

$$x_2 = x_1 + h(x_1 + y_1) = 1.3 + 0.1(1.3 + 1.9) = 1.62$$

$$y_2 = y_1 + h(x_1 - y_1) = 1.9 + 0.1(1.3 - 1.9) = 1.84.$$

Thus, $x(0.2) \approx 1.62$ and $y(0.2) \approx 1.84$.

8. We identify $P(x) = 0$, $Q(x) = 6.55(1 + x)$, $f(x) = 1$, and $h = (1 - 0)/10 = 0.1$. Then the finite difference equation is

$$y_{i+1} + [-2 + 0.0655(1 + x_i)]y_i + y_{i-1} = 0.001$$

or

$$y_{i+1} + (0.0655x_i - 1.9345)y_i + y_{i-1} = 0.001.$$

The solution of the corresponding linear system gives

x	0.0	0.1	0.2	0.3	0.4	0.5	0.6
y	0.0000	4.1987	8.1049	11.3840	13.7038	14.7770	14.4083

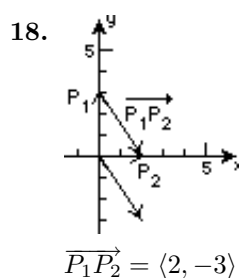
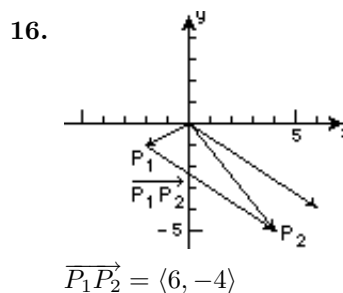
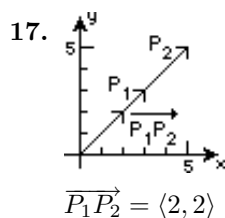
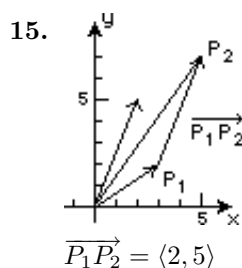
0.7	0.8	0.9	1.0
12.5396	9.2847	4.9450	0.0000

7 Vectors

Exercises 7.1

1. (a) $6\mathbf{i} + 12\mathbf{j}$ (b) $\mathbf{i} + 8\mathbf{j}$ (c) $3\mathbf{i}$ (d) $\sqrt{65}$ (e) 3
2. (a) $\langle 3, 3 \rangle$ (b) $\langle 3, 4 \rangle$ (c) $\langle -1, -2 \rangle$ (d) 5 (e) $\sqrt{5}$
3. (a) $\langle 12, 0 \rangle$ (b) $\langle 4, -5 \rangle$ (c) $\langle 4, 5 \rangle$ (d) $\sqrt{41}$ (e) $\sqrt{41}$
4. (a) $\frac{1}{2}\mathbf{i} - \frac{1}{2}\mathbf{j}$ (b) $\frac{2}{3}\mathbf{i} + \frac{2}{3}\mathbf{j}$ (c) $-\frac{1}{3}\mathbf{i} - \mathbf{j}$ (d) $2\sqrt{2}/3$ (e) $\sqrt{10}/3$
5. (a) $-9\mathbf{i} + 6\mathbf{j}$ (b) $-3\mathbf{i} + 9\mathbf{j}$ (c) $-3\mathbf{i} - 5\mathbf{j}$ (d) $3\sqrt{10}$ (e) $\sqrt{34}$
6. (a) $\langle 3, 9 \rangle$ (b) $\langle -4, -12 \rangle$ (c) $\langle 6, 18 \rangle$ (d) $4\sqrt{10}$ (e) $6\sqrt{10}$
7. (a) $-6\mathbf{i} + 27\mathbf{j}$ (b) $\mathbf{0}$ (c) $-4\mathbf{i} + 18\mathbf{j}$ (d) 0 (e) $2\sqrt{85}$
8. (a) $\langle 21, 30 \rangle$ (b) $\langle 8, 12 \rangle$ (c) $\langle 6, 8 \rangle$ (d) $4\sqrt{13}$ (e) 10

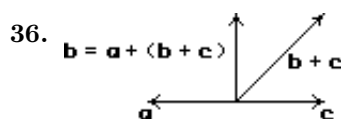
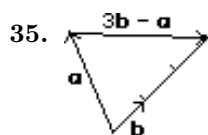
9. (a) $\langle 4, -12 \rangle - \langle -2, 2 \rangle = \langle 6, -14 \rangle$ (b) $\langle -3, 9 \rangle - \langle -5, 5 \rangle = \langle 2, 4 \rangle$
10. (a) $(4\mathbf{i} + 4\mathbf{j}) - (6\mathbf{i} - 4\mathbf{j}) = -2\mathbf{i} + 8\mathbf{j}$ (b) $(-3\mathbf{i} - 3\mathbf{j}) - (15\mathbf{i} - 10\mathbf{j}) = -18\mathbf{i} + 7\mathbf{j}$
11. (a) $(4\mathbf{i} - 4\mathbf{j}) - (-6\mathbf{i} + 8\mathbf{j}) = 10\mathbf{i} - 12\mathbf{j}$ (b) $(-3\mathbf{i} + 3\mathbf{j}) - (-15\mathbf{i} + 20\mathbf{j}) = 12\mathbf{i} - 17\mathbf{j}$
12. (a) $\langle 8, 0 \rangle - \langle 0, -6 \rangle = \langle 8, 6 \rangle$ (b) $\langle -6, 0 \rangle - \langle 0, -15 \rangle = \langle -6, 15 \rangle$
13. (a) $\langle 16, 40 \rangle - \langle -4, -12 \rangle = \langle 20, 52 \rangle$ (b) $\langle -12, -30 \rangle - \langle -10, -30 \rangle = \langle -2, 0 \rangle$
14. (a) $\langle 8, 12 \rangle - \langle 10, 6 \rangle = \langle -2, 6 \rangle$ (b) $\langle -6, -9 \rangle - \langle 25, 15 \rangle = \langle -31, -24 \rangle$



19. Since $\overrightarrow{P_1P_2} = \overrightarrow{OP_2} - \overrightarrow{OP_1}$, $\overrightarrow{OP_2} = \overrightarrow{P_1P_2} + \overrightarrow{OP_1} = (4\mathbf{i} + 8\mathbf{j}) + (-3\mathbf{i} + 10\mathbf{j}) = \mathbf{i} + 18\mathbf{j}$, and the terminal point is $(1, 18)$.
20. Since $\overrightarrow{P_1P_2} = \overrightarrow{OP_2} - \overrightarrow{OP_1}$, $\overrightarrow{OP_1} = \overrightarrow{OP_2} - \overrightarrow{P_1P_2} = \langle 4, 7 \rangle - \langle -5, -1 \rangle = \langle 9, 8 \rangle$, and the initial point is $(9, 8)$.
21. $a(= -\mathbf{a})$, $b(= -\frac{1}{4}\mathbf{a})$, $c(= \frac{5}{2}\mathbf{a})$, $e(= 2\mathbf{a})$, and $f(= -\frac{1}{2}\mathbf{a})$ are parallel to $\mathbf{a}$.
22. We want $-3\mathbf{b} = \mathbf{a}$, so $c = -3(9) = -27$.
23. $\langle 6, 15 \rangle$
24. $\langle 5, 2 \rangle$
25. $\|\mathbf{a}\| = \sqrt{4+4} = 2\sqrt{2}$; (a) $\mathbf{u} = \frac{1}{2\sqrt{2}}\langle 2, 2 \rangle = \langle \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \rangle$; (b) $-\mathbf{u} = \langle -\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \rangle$
26. $\|\mathbf{a}\| = \sqrt{9+16} = 5$; (a) $\mathbf{u} = \frac{1}{5}\langle -3, 4 \rangle = \langle -\frac{3}{5}, \frac{4}{5} \rangle$; (b) $-\mathbf{u} = \langle \frac{3}{5}, -\frac{4}{5} \rangle$
27. $\|\mathbf{a}\| = 5$; (a) $\mathbf{u} = \frac{1}{5}\langle 0, -5 \rangle = \langle 0, -1 \rangle$; (b) $-\mathbf{u} = \langle 0, 1 \rangle$
28. $\|\mathbf{a}\| = \sqrt{1+3} = 2$; (a) $\mathbf{u} = \frac{1}{2}\langle 1, -\sqrt{3} \rangle = \langle \frac{1}{2}, -\frac{\sqrt{3}}{2} \rangle$; (b) $-\mathbf{u} = \langle -\frac{1}{2}, \frac{\sqrt{3}}{2} \rangle$
29. $\|\mathbf{a} + \mathbf{b}\| = \|\langle 5, 12 \rangle\| = \sqrt{25+144} = 13$; $\mathbf{u} = \frac{1}{13}\langle 5, 12 \rangle = \langle \frac{5}{13}, \frac{12}{13} \rangle$
30. $\|\mathbf{a} + \mathbf{b}\| = \|\langle -5, 4 \rangle\| = \sqrt{25+16} = \sqrt{41}$; $\mathbf{u} = \frac{1}{\sqrt{41}}\langle -5, 4 \rangle = \langle -\frac{5}{\sqrt{41}}, \frac{4}{\sqrt{41}} \rangle$
31. $\|\mathbf{a}\| = \sqrt{9+49} = \sqrt{58}$; $\mathbf{b} = 2(\frac{1}{\sqrt{58}})(3\mathbf{i} + 7\mathbf{j}) = \frac{6}{\sqrt{58}}\mathbf{i} + \frac{14}{\sqrt{58}}\mathbf{j}$
32. $\|\mathbf{a}\| = \sqrt{\frac{1}{4} + \frac{1}{4}} = \frac{1}{\sqrt{2}}$; $\mathbf{b} = 3(\frac{1}{1/\sqrt{2}})(\frac{1}{2}\mathbf{i} - \frac{1}{2}\mathbf{j}) = \frac{3\sqrt{2}}{2}\mathbf{i} - \frac{3\sqrt{2}}{2}\mathbf{j}$

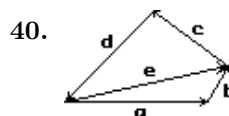
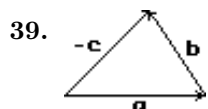
33. $-\frac{3}{4}\mathbf{a} = \langle -3, -15/2 \rangle$

34. $5(\mathbf{a} + \mathbf{b}) = 5\langle 0, 1 \rangle = \langle 0, 5 \rangle$



37. $\mathbf{x} = -(\mathbf{a} + \mathbf{b}) = -\mathbf{a} - \mathbf{b}$

38. $\mathbf{x} = 2(\mathbf{a} - \mathbf{b}) = 2\mathbf{a} - 2\mathbf{b}$



$\mathbf{b} = (-\mathbf{c}) - \mathbf{a}$; $(\mathbf{b} + \mathbf{c}) + \mathbf{a} = \mathbf{0}$; $\mathbf{a} + \mathbf{b} + \mathbf{c} = \mathbf{0}$

From Problem 39, $\mathbf{e} + \mathbf{c} + \mathbf{d} = \mathbf{0}$. But $\mathbf{b} = \mathbf{e} - \mathbf{a}$ and $\mathbf{e} = \mathbf{a} + \mathbf{b}$, so $(\mathbf{a} + \mathbf{b}) + \mathbf{c} + \mathbf{d} = \mathbf{0}$.

41. From $2\mathbf{i} + 3\mathbf{j} = k_1\mathbf{b} + k_2\mathbf{c} = k_1(\mathbf{i} + \mathbf{j}) + k_2(\mathbf{i} - \mathbf{j}) = (k_1 + k_2)\mathbf{i} + (k_1 - k_2)\mathbf{j}$ we obtain the system of equations $k_1 + k_2 = 2$, $k_1 - k_2 = 3$. Solving, we find $k_1 = \frac{5}{2}$ and $k_2 = -\frac{1}{2}$. Then $\mathbf{a} = \frac{5}{2}\mathbf{b} - \frac{1}{2}\mathbf{c}$.
42. From $2\mathbf{i} + 3\mathbf{j} = k_1\mathbf{b} + k_2\mathbf{c} = k_1(-2\mathbf{i} + 4\mathbf{j}) + k_2(5\mathbf{i} + 7\mathbf{j}) = (-2k_1 + 5k_2)\mathbf{i} + (4k_1 + 7k_2)\mathbf{j}$ we obtain the system of equations $-2k_1 + 5k_2 = 2$, $4k_1 + 7k_2 = 3$. Solving, we find $k_1 = \frac{1}{34}$ and $k_2 = \frac{7}{17}$.
43. From $y' = \frac{1}{2}x$ we see that the slope of the tangent line at $(2, 2)$ is 1. A vector with slope 1 is $\mathbf{i} + \mathbf{j}$. A unit vector is $(\mathbf{i} + \mathbf{j})/\|\mathbf{i} + \mathbf{j}\| = (\mathbf{i} + \mathbf{j})/\sqrt{2} = \frac{1}{\sqrt{2}}\mathbf{i} + \frac{1}{\sqrt{2}}\mathbf{j}$. Another unit vector tangent to the curve is $-\frac{1}{\sqrt{2}}\mathbf{i} - \frac{1}{\sqrt{2}}\mathbf{j}$.
44. From $y' = -2x + 3$ we see that the slope of the tangent line at $(0, 0)$ is 3. A vector with slope 3 is $\mathbf{i} + 3\mathbf{j}$. A unit vector is $(\mathbf{i} + 3\mathbf{j})/\|\mathbf{i} + 3\mathbf{j}\| = (\mathbf{i} + 3\mathbf{j})/\sqrt{10} = \frac{1}{\sqrt{10}}\mathbf{i} + \frac{3}{\sqrt{10}}\mathbf{j}$. Another unit vector is $-\frac{1}{\sqrt{10}}\mathbf{i} - \frac{3}{\sqrt{10}}\mathbf{j}$.
45. (a) Since $\mathbf{F}_f = -\mathbf{F}_g$, $\|\mathbf{F}_g\| = \|\mathbf{F}_f\| = \mu\|\mathbf{F}_n\|$ and $\tan \theta = \|\mathbf{F}_g\|/\|\mathbf{F}_n\| = \mu\|\mathbf{F}_n\|/\|\mathbf{F}_n\| = \mu$.

Exercises 7.1

(b) $\theta = \tan^{-1} 0.6 \approx 31^\circ$.

46. Since $\mathbf{w} + \mathbf{F}_1 + \mathbf{F}_2 = \mathbf{0}$,

$$-200\mathbf{j} + \|\mathbf{F}_1\| \cos 20^\circ \mathbf{i} + \|\mathbf{F}_1\| \sin 20^\circ \mathbf{j} - \|\mathbf{F}_2\| \cos 15^\circ \mathbf{i} + \|\mathbf{F}_2\| \sin 15^\circ \mathbf{j} = \mathbf{0}$$

or $(\|\mathbf{F}_1\| \cos 20^\circ - \|\mathbf{F}_2\| \cos 15^\circ)\mathbf{i} + (\|\mathbf{F}_1\| \sin 20^\circ + \|\mathbf{F}_2\| \sin 15^\circ - 200)\mathbf{j} = \mathbf{0}$.

Thus, $\|\mathbf{F}_1\| \cos 20^\circ - \|\mathbf{F}_2\| \cos 15^\circ = 0$; $\|\mathbf{F}_1\| \sin 20^\circ + \|\mathbf{F}_2\| \sin 15^\circ - 200 = 0$. Solving this system for $\|\mathbf{F}_1\|$ and $\|\mathbf{F}_2\|$, we obtain

$$\|\mathbf{F}_1\| = \frac{200 \cos 15^\circ}{\sin 15^\circ \cos 20^\circ + \cos 15^\circ \sin 20^\circ} = \frac{200 \cos 15^\circ}{\sin(15^\circ + 20^\circ)} = \frac{200 \cos 15^\circ}{\sin 35^\circ} \approx 336.8 \text{ lb}$$

and

$$\|\mathbf{F}_2\| = \frac{200 \cos 20^\circ}{\sin 15^\circ \cos 20^\circ + \cos 15^\circ \sin 20^\circ} = \frac{200 \cos 20^\circ}{\sin 35^\circ} \approx 327.7 \text{ lb}.$$

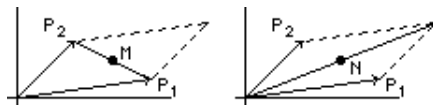
47. Since $y/2a(L^2 + y^2)^{3/2}$ is an odd function on $[-a, a]$, $F_y = 0$. Now, using the fact that $L/(L^2 + y^2)^{3/2}$ is an even function, we have

$$\begin{aligned} \int_{-a}^a \frac{L dy}{2a(L^2 + y^2)^{3/2}} &= \frac{L}{a} \int_0^a \frac{dy}{(L^2 + y^2)^{3/2}} \quad \boxed{y = L \tan \theta, \quad dy = L \sec^2 \theta d\theta} \\ &= \frac{L}{a} \int_0^{\tan^{-1} a/L} \frac{L \sec^2 \theta d\theta}{L^3(1 + \tan^2 \theta)^{3/2}} = \frac{1}{La} \int_0^{\tan^{-1} a/L} \frac{\sec^2 \theta d\theta}{\sec^3 \theta} \\ &= \frac{1}{La} \int_0^{\tan^{-1} a/L} \cos \theta d\theta = \frac{1}{La} \sin \theta \Big|_0^{\tan^{-1} a/L} \\ &= \frac{1}{La} \frac{a}{\sqrt{L^2 + a^2}} = \frac{1}{L\sqrt{L^2 + a^2}}. \end{aligned}$$

Then $F_x = qQ/4\pi\epsilon_0 L\sqrt{L^2 + a^2}$ and $\mathbf{F} = (qQ/4\pi\epsilon_0 L\sqrt{L^2 + a^2})\mathbf{i}$.

48. Place one corner of the parallelogram at the origin and let two adjacent sides be $\overrightarrow{OP_1}$ and $\overrightarrow{OP_2}$. Let M be the midpoint of the diagonal connecting P_1 and P_2 and N be the midpoint of the other diagonal.

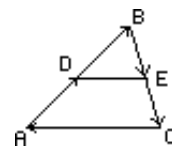
By Problem 37, $\overrightarrow{OM} = \frac{1}{2}(\overrightarrow{OP_1} + \overrightarrow{OP_2})$. Since $\overrightarrow{OP_1} + \overrightarrow{OP_2}$ is the main diagonal of the parallelogram and N is its midpoint, $\overrightarrow{ON} = \frac{1}{2}(\overrightarrow{OP_1} + \overrightarrow{OP_2})$. Thus, $\overrightarrow{OM} = \overrightarrow{ON}$ and the diagonals bisect each other.



49. By Problem 39, $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CA} = \mathbf{0}$ and $\overrightarrow{AD} + \overrightarrow{DE} + \overrightarrow{EC} + \overrightarrow{CA} = \mathbf{0}$. From the first equation, $\overrightarrow{AB} + \overrightarrow{BC} = -\overrightarrow{CA}$. Since D and E are midpoints, $\overrightarrow{AD} = \frac{1}{2}\overrightarrow{AB}$ and $\overrightarrow{EC} = \frac{1}{2}\overrightarrow{BC}$. Then, $\frac{1}{2}\overrightarrow{AB} + \overrightarrow{DE} + \frac{1}{2}\overrightarrow{BC} + \overrightarrow{CA} = \mathbf{0}$ and

$$\overrightarrow{DE} = -\overrightarrow{CA} - \frac{1}{2}(\overrightarrow{AB} + \overrightarrow{BC}) = -\overrightarrow{CA} - \frac{1}{2}(-\overrightarrow{CA}) = -\frac{1}{2}\overrightarrow{CA}.$$

Thus, the line segment joining the midpoints D and E is parallel to the side AC and half its length.



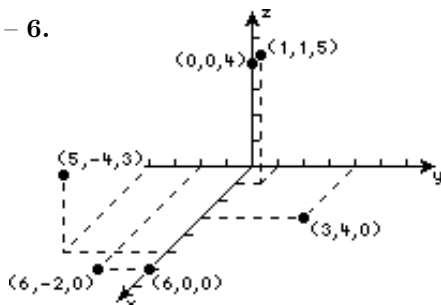
50. We have $\overrightarrow{OA} = 150 \cos 20^\circ \mathbf{i} + 150 \sin 20^\circ \mathbf{j}$, $\overrightarrow{AB} = 200 \cos 113^\circ \mathbf{i} + 200 \sin 113^\circ \mathbf{j}$, $\overrightarrow{BC} = 240 \cos 190^\circ \mathbf{i} + 240 \sin 190^\circ \mathbf{j}$. Then

$$\begin{aligned} \mathbf{r} &= (150 \cos 20^\circ + 200 \cos 113^\circ + 240 \cos 190^\circ)\mathbf{i} + (150 \sin 20^\circ + 200 \sin 113^\circ + 240 \sin 190^\circ)\mathbf{j} \\ &\approx -173.55\mathbf{i} + 193.73\mathbf{j} \end{aligned}$$

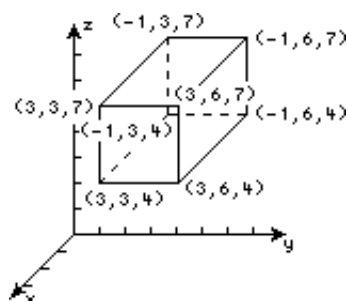
and $\|\mathbf{r}\| \approx 260.09$ miles.

Exercises 7.2

1. – 6.

7. A plane perpendicular to the z -axis, 5 units above the xy -plane8. A plane perpendicular to the x -axis, 1 unit in front of the yz -plane9. A line perpendicular to the xy -plane at $(2, 3, 0)$ 10. A single point located at $(4, -1, 7)$ 11. $(2, 0, 0)$, $(2, 5, 0)$, $(2, 0, 8)$, $(2, 5, 8)$, $(0, 5, 0)$, $(0, 5, 8)$, $(0, 0, 8)$, $(0, 0, 0)$

12.

13. (a) xy -plane: $(-2, 5, 0)$, xz -plane: $(-2, 0, 4)$, yz -plane: $(0, 5, 4)$; (b) $(-2, 5, -2)$ (c) Since the shortest distance between a point and a plane is a perpendicular line, the point in the plane $x = 3$ is $(3, 5, 4)$.14. We find planes that are parallel to coordinate planes: (a) $z = -5$; (b) $x = 1$ and $y = -1$; (c) $z = 2$ 15. The union of the planes $x = 0$, $y = 0$, and $z = 0$ 16. The origin $(0, 0, 0)$ 17. The point $(-1, 2, -3)$ 18. The union of the planes $x = 2$ and $z = 8$ 19. The union of the planes $z = 5$ and $z = -5$ 20. The line through the points $(1, 1, 1)$, $(-1, -1, -1)$, and the origin21. $d = \sqrt{(3-6)^2 + (-1-4)^2 + (2-8)^2} = \sqrt{70}$ 22. $d = \sqrt{(-1-0)^2 + (-3-4)^2 + (5-3)^2} = 3\sqrt{6}$ 23. (a) 7; (b) $d = \sqrt{(-3)^2 + (-4)^2} = 5$ 24. (a) 2; (b) $d = \sqrt{(-6)^2 + 2^2 + (-3)^2} = 7$ 25. $d(P_1, P_2) = \sqrt{3^2 + 6^2 + (-6)^2} = 9$; $d(P_1, P_3) = \sqrt{2^2 + 1^2 + 2^2} = 3$ $d(P_2, P_3) = \sqrt{(2-3)^2 + (1-6)^2 + (2-(-6))^2} = \sqrt{90}$; The triangle is a right triangle.

Exercises 7.2

$$26. d(P_1, P_2) = \sqrt{1^2 + 2^2 + 4^2} = \sqrt{21}; \quad d(P_1, P_3) = \sqrt{3^2 + 2^2 + (2\sqrt{2})^2} = \sqrt{21}$$

$$d(P_2, P_3) = \sqrt{(3-1)^2 + (2-2)^2 + (2\sqrt{2}-4)^2} = \sqrt{28-16\sqrt{2}}$$

The triangle is an isosceles triangle.

$$27. d(P_1, P_2) = \sqrt{(4-1)^2 + (1-2)^2 + (3-3)^2} = \sqrt{10}$$

$$d(P_1, P_3) = \sqrt{(4-1)^2 + (6-2)^2 + (4-3)^2} = \sqrt{26}$$

$$d(P_2, P_3) = \sqrt{(4-4)^2 + (6-1)^2 + (4-3)^2} = \sqrt{26}; \quad \text{The triangle is an isosceles triangle.}$$

$$28. d(P_1, P_2) = \sqrt{(1-1)^2 + (1-1)^2 + (1-(-1))^2} = 2$$

$$d(P_1, P_3) = \sqrt{(0-1)^2 + (-1-1)^2 + (1-(-1))^2} = 3$$

$$d(P_2, P_3) = \sqrt{(0-1)^2 + (-1-1)^2 + (1-1)^2} = \sqrt{5}; \quad \text{The triangle is a right triangle.}$$

$$29. d(P_1, P_2) = \sqrt{(-2-1)^2 + (-2-2)^2 + (-3-0)^2} = \sqrt{34}$$

$$d(P_1, P_3) = \sqrt{(7-1)^2 + (10-2)^2 + (6-0)^2} = 2\sqrt{34}$$

$$d(P_2, P_3) = \sqrt{(7-(-2))^2 + (10-(-2))^2 + (6-(-3))^2} = 3\sqrt{34}$$

Since $d(P_1, P_2) + d(P_1, P_3) = d(P_2, P_3)$, the points P_1 , P_2 , and P_3 are collinear.

$$30. d(P_1, P_2) = \sqrt{(1-2)^2 + (4-3)^2 + (4-2)^2} = \sqrt{6}$$

$$d(P_1, P_3) = \sqrt{(5-2)^2 + (0-3)^2 + (-4-2)^2} = 3\sqrt{6}$$

$$d(P_2, P_3) = \sqrt{(5-1)^2 + (0-4)^2 + (-4-4)^2} = 4\sqrt{6}$$

Since $d(P_1, P_2) + d(P_1, P_3) = d(P_2, P_3)$, the points P_1 , P_2 , and P_3 are collinear.

$$31. \sqrt{(2-x)^2 + (1-2)^2 + (1-3)^2} = \sqrt{21} \implies x^2 - 4x + 9 = 21 \implies x^2 - 4x + 4 = 16 \\ \implies (x-2)^2 = 16 \implies x = 2 \pm 4 \text{ or } x = 6, -2$$

$$32. \sqrt{(0-x)^2 + (3-x)^2 + (5-1)^2} = 5 \implies 2x^2 - 6x + 25 = 25 \implies x^2 - 3x = 0 \implies x = 0, 3$$

$$33. \left(\frac{1+7}{2}, \frac{3+(-2)}{2}, \frac{1/2+5/2}{2} \right) = (4, 1/2, 3/2)$$

$$34. \left(\frac{0+4}{2}, \frac{5+1}{2}, \frac{-8+(-6)}{2} \right) = (2, 3, -7)$$

$$35. (x_1 + 2)/2 = -1, \quad x_1 = -4; \quad (y_1 + 3)/2 = -4, \quad y_1 = -11; \quad (z_1 + 6)/2 = 8, \quad z_1 = 10$$

The coordinates of P_1 are $(-4, -11, 10)$.

$$36. (-3 + (-5))/2 = x_3 = -4; \quad (4 + 8)/2 = y_3 = 6; \quad (1 + 3)/2 = z_3 = 2.$$

The coordinates of P_3 are $(-4, 6, 2)$.

$$(a) \left(\frac{-3+(-4)}{2}, \frac{4+6}{2}, \frac{1+2}{2} \right) = (-7/2, 5, 3/2)$$

$$(b) \left(\frac{-4+(-5)}{2}, \frac{6+8}{2}, \frac{2+3}{2} \right) = (-9/2, 7, 5/2)$$

$$37. \overrightarrow{P_1P_2} = \langle -3, -6, 1 \rangle$$

$$38. \overrightarrow{P_1P_2} = \langle 8, -5/2, 8 \rangle$$

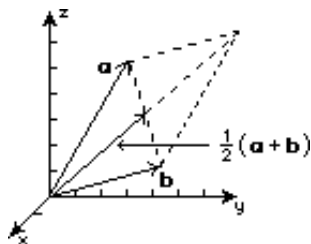
$$39. \overrightarrow{P_1P_2} = \langle 2, 1, 1 \rangle$$

$$40. \overrightarrow{P_1P_2} = \langle -3, -3, 7 \rangle$$

$$41. \mathbf{a} + (\mathbf{b} + \mathbf{c}) = \langle 2, 4, 12 \rangle$$

$$42. 2\mathbf{a} - (\mathbf{b} - \mathbf{c}) = \langle 2, -6, 4 \rangle - \langle -3, -5, -8 \rangle = \langle 5, -1, 12 \rangle$$

43. $\mathbf{b} + 2(\mathbf{a} - 3\mathbf{c}) = \langle -1, 1, 1 \rangle + 2\langle -5, -21, -25 \rangle = \langle -11, -41, -49 \rangle$
44. $4(\mathbf{a} + 2\mathbf{c}) - 6\mathbf{b} = 4\langle 5, 9, 20 \rangle - \langle -6, 6, 6 \rangle = \langle 26, 30, 74 \rangle$
45. $\|\mathbf{a} + \mathbf{c}\| = \|\langle 3, 3, 11 \rangle\| = \sqrt{9 + 9 + 121} = \sqrt{139}$
46. $\|\mathbf{c}\| \|\mathbf{2b}\| = (\sqrt{4 + 36 + 81})(2)(\sqrt{1 + 1 + 1}) = 22\sqrt{3}$
47. $\left\| \frac{\mathbf{a}}{\|\mathbf{a}\|} \right\| + 5 \left\| \frac{\mathbf{b}}{\|\mathbf{b}\|} \right\| = \frac{1}{\|\mathbf{a}\|} \|\mathbf{a}\| + 5 \frac{1}{\|\mathbf{b}\|} \|\mathbf{b}\| = 1 + 5 = 6$
48. $\|\mathbf{b}\|\mathbf{a} + \|\mathbf{a}\|\mathbf{b} = \sqrt{1 + 1 + 1} \langle 1, -3, 2 \rangle + \sqrt{1 + 9 + 4} \langle -1, 1, 1 \rangle = \langle \sqrt{3}, -3\sqrt{3}, 2\sqrt{3} \rangle + \langle -\sqrt{14}, \sqrt{14}, \sqrt{14} \rangle$
 $= \langle \sqrt{3} - \sqrt{14}, -3\sqrt{3} + \sqrt{14}, 2\sqrt{3} + \sqrt{14} \rangle$
49. $\|\mathbf{a}\| = \sqrt{100 + 25 + 100} = 15; \quad \mathbf{u} = -\frac{1}{15} \langle 10, -5, 10 \rangle = \langle -2/3, 1/3, -2/3 \rangle$
50. $\|\mathbf{a}\| = \sqrt{1 + 9 + 4} = \sqrt{14}; \quad \mathbf{u} = \frac{1}{\sqrt{14}} (\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}) = \frac{1}{\sqrt{14}} \mathbf{i} - \frac{3}{\sqrt{14}} \mathbf{j} + \frac{2}{\sqrt{14}} \mathbf{k}$
51. $\mathbf{b} = 4\mathbf{a} = 4\mathbf{i} - 4\mathbf{j} + 4\mathbf{k}$
52. $\|\mathbf{a}\| = \sqrt{36 + 9 + 4} = 7; \quad \mathbf{b} = -\frac{1}{2} \left(\frac{1}{7} \right) \langle -6, 3, -2 \rangle = \langle \frac{3}{7}, -\frac{3}{14}, \frac{1}{7} \rangle$
- 53.



Exercises 7.3

1. $\mathbf{a} \cdot \mathbf{b} = 10(5) \cos(\pi/4) = 25\sqrt{2}$
2. $\mathbf{a} \cdot \mathbf{b} = 6(12) \cos(\pi/6) = 36\sqrt{3}$
3. $\mathbf{a} \cdot \mathbf{b} = 2(-1) + (-3)2 + 4(5) = 12$
4. $\mathbf{b} \cdot \mathbf{c} = (-1)3 + 2(6) + 5(-1) = 4$
5. $\mathbf{a} \cdot \mathbf{c} = 2(3) + (-3)6 + 4(-1) = -16$
6. $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = 2(2) + (-3)8 + 4(4) = -4$
7. $\mathbf{a} \cdot (4\mathbf{b}) = 2(-4) + (-3)8 + 4(20) = 48$
8. $\mathbf{b} \cdot (\mathbf{a} - \mathbf{c}) = (-1)(-1) + 2(-9) + 5(5) = 8$
9. $\mathbf{a} \cdot \mathbf{a} = 2^2 + (-3)^2 + 4^2 = 29$
10. $(2\mathbf{b}) \cdot (3\mathbf{c}) = (-2)9 + 4(18) + 10(-3) = 24$
11. $\mathbf{a} \cdot (\mathbf{a} + \mathbf{b} + \mathbf{c}) = 2(4) + (-3)5 + 4(8) = 25$
12. $(2\mathbf{a}) \cdot (\mathbf{a} - 2\mathbf{b}) = 4(4) + (-6)(-7) + 8(-6) = 10$
13. $\left(\frac{\mathbf{a} \cdot \mathbf{b}}{\mathbf{b} \cdot \mathbf{b}} \right) \mathbf{b} = \left[\frac{2(-1) + (-3)2 + 4(5)}{(-1)^2 + 2^2 + 5^2} \right] \langle -1, 2, 5 \rangle = \frac{12}{30} \langle -1, 2, 5 \rangle = \langle -2/5, 4/5, 2 \rangle$
14. $(\mathbf{c} \cdot \mathbf{b})\mathbf{a} = [3(-1) + 6(2) + (-1)5] \langle 2, -3, 4 \rangle = 4 \langle 2, -3, 4 \rangle = \langle 8, -12, 16 \rangle$
15. $\mathbf{a}$ and $\mathbf{f}$, $\mathbf{b}$ and $\mathbf{e}$, $\mathbf{c}$ and $\mathbf{d}$
16. (a) $\mathbf{a} \cdot \mathbf{b} = 2 \cdot 3 + (-c)2 + 3(4) = 0 \implies c = 9$

Exercises 7.3

(b) $\mathbf{a} \cdot \mathbf{b} = c(-3) + \frac{1}{2}(4) + c^2 = c^2 - 3c + 2 = (c-2)(c-1) = 0 \implies c = 1, 2$

17. Solving the system of equations $3x_1 + y_1 - 1 = 0$, $-3x_1 + 2y_1 + 2 = 0$ gives $x_1 = 4/9$ and $y_1 = -1/3$. Thus, $\mathbf{v} = \langle 4/9, -1/3, 1 \rangle$.

18. If $\mathbf{a}$ and $\mathbf{b}$ represent adjacent sides of the rhombus, then $\|\mathbf{a}\| = \|\mathbf{b}\|$, the diagonals of the rhombus are $\mathbf{a} + \mathbf{b}$ and $\mathbf{a} - \mathbf{b}$, and

$$(\mathbf{a} + \mathbf{b}) \cdot (\mathbf{a} - \mathbf{b}) = \mathbf{a} \cdot \mathbf{a} - \mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{a} - \mathbf{b} \cdot \mathbf{b} = \mathbf{a} \cdot \mathbf{a} - \mathbf{b} \cdot \mathbf{b} = \|\mathbf{a}\|^2 - \|\mathbf{b}\|^2 = 0.$$

Thus, the diagonals are perpendicular.

19. Since

$$\mathbf{c} \cdot \mathbf{a} = \left(\mathbf{b} - \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|^2} \mathbf{a} \right) \cdot \mathbf{a} = \mathbf{b} \cdot \mathbf{a} - \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|^2} (\mathbf{a} \cdot \mathbf{a}) = \mathbf{b} \cdot \mathbf{a} - \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|^2} \|\mathbf{a}\|^2 = \mathbf{b} \cdot \mathbf{a} - \mathbf{a} \cdot \mathbf{b} = 0,$$

the vectors $\mathbf{c}$ and $\mathbf{a}$ are orthogonal.

20. $\mathbf{a} \cdot \mathbf{b} = 1(1) + c(1) = c + 1$; $\|\mathbf{a}\| = \sqrt{1+c^2}$, $\|\mathbf{b}\| = \sqrt{2}$

$$\cos 45^\circ = \frac{1}{\sqrt{2}} = \frac{c+1}{\sqrt{1+c^2}\sqrt{2}} \implies \sqrt{1+c^2} = c+1 \implies 1+c^2 = c^2+2c+1 \implies c=0$$

21. $\mathbf{a} \cdot \mathbf{b} = 3(2) + (-1)2 = 4$; $\|\mathbf{a}\| = \sqrt{10}$, $\|\mathbf{b}\| = 2\sqrt{2}$

$$\cos \theta = \frac{4}{(\sqrt{10})(2\sqrt{2})} = \frac{1}{\sqrt{5}} \implies \theta = \cos^{-1} \frac{1}{\sqrt{5}} \approx 1.11 \text{ rad} \approx 63.43^\circ$$

22. $\mathbf{a} \cdot \mathbf{b} = 2(-3) + 1(-4) = -10$; $\|\mathbf{a}\| = \sqrt{5}$, $\|\mathbf{b}\| = 5$

$$\cos \theta = \frac{-10}{(\sqrt{5})5} = -\frac{2}{\sqrt{5}} \implies \theta = \cos^{-1}(-2/\sqrt{5}) \approx 2.68 \text{ rad} \approx 153.43^\circ$$

23. $\mathbf{a} \cdot \mathbf{b} = 2(-1) + 4(-1) + 0(4) = -6$; $\|\mathbf{a}\| = 2\sqrt{5}$, $\|\mathbf{b}\| = 3\sqrt{2}$

$$\cos \theta = \frac{-6}{(2\sqrt{5})(3\sqrt{2})} = -\frac{1}{\sqrt{10}} \implies \theta = \cos^{-1}(-1/\sqrt{10}) \approx 1.89 \text{ rad} \approx 108.43^\circ$$

24. $\mathbf{a} \cdot \mathbf{b} = \frac{1}{2}(2) + \frac{1}{2}(-4) + \frac{3}{2}(6) = 8$; $\|\mathbf{a}\| = \sqrt{11}/2$, $\|\mathbf{b}\| = 2\sqrt{14}$

$$\cos \theta = \frac{8}{(\sqrt{11}/2)(2\sqrt{14})} = \frac{8}{\sqrt{154}} \implies \theta = \cos^{-1}(8/\sqrt{154}) \approx 0.87 \text{ rad} \approx 49.86^\circ$$

25. $\|\mathbf{a}\| = \sqrt{14}$; $\cos \alpha = 1/\sqrt{14}$, $\alpha \approx 74.50^\circ$; $\cos \beta = 2/\sqrt{14}$, $\beta \approx 57.69^\circ$; $\cos \gamma = 3/\sqrt{14}$, $\gamma \approx 36.70^\circ$

26. $\|\mathbf{a}\| = 9$; $\cos \alpha = 2/3$, $\alpha \approx 48.19^\circ$; $\cos \beta = 2/3$, $\beta \approx 48.19^\circ$; $\cos \gamma = -1/3$, $\gamma \approx 109.47^\circ$

27. $\|\mathbf{a}\| = 2$; $\cos \alpha = 1/2$, $\alpha = 60^\circ$; $\cos \beta = 0$, $\beta = 90^\circ$; $\cos \gamma = -\sqrt{3}/2$, $\gamma = 150^\circ$

28. $\|\mathbf{a}\| = \sqrt{78}$; $\cos \alpha = 5/\sqrt{78}$, $\alpha \approx 55.52^\circ$; $\cos \beta = 7/\sqrt{78}$, $\beta \approx 37.57^\circ$; $\cos \gamma = 2/\sqrt{78}$, $\gamma \approx 76.91^\circ$

29. Let θ be the angle between $\overrightarrow{AD}$ and $\overrightarrow{AB}$ and a be the length of an edge of the cube. Then $\overrightarrow{AD} = a\mathbf{i} + a\mathbf{j} + a\mathbf{k}$, $\overrightarrow{AB} = a\mathbf{i}$ and

$$\cos \theta = \frac{\overrightarrow{AD} \cdot \overrightarrow{AB}}{\|\overrightarrow{AD}\| \|\overrightarrow{AB}\|} = \frac{a^2}{\sqrt{3a^2} \sqrt{a^2}} = \frac{1}{\sqrt{3}}$$

so $\theta \approx 0.955317$ radian or 54.7356° . Letting ϕ be the angle between $\overrightarrow{AD}$ and $\overrightarrow{AC}$ and noting that $\overrightarrow{AC} = a\mathbf{i} + a\mathbf{j}$ we have

$$\cos \phi = \frac{\overrightarrow{AD} \cdot \overrightarrow{AC}}{\|\overrightarrow{AD}\| \|\overrightarrow{AC}\|} = \frac{a^2 + a^2}{\sqrt{3a^2} \sqrt{2a^2}} = \sqrt{\frac{2}{3}}$$

so $\phi \approx 0.61548$ radian or 35.2644° .

30. If $\mathbf{a}$ and $\mathbf{b}$ are orthogonal, then $\mathbf{a} \cdot \mathbf{b} = 0$ and

$$\begin{aligned}\cos \alpha_1 \cos \alpha_2 + \cos \beta_1 \cos \beta_2 + \cos \gamma_1 \cos \gamma_2 &= \frac{a_1}{\|\mathbf{a}\|} \frac{b_1}{\|\mathbf{b}\|} + \frac{a_2}{\|\mathbf{a}\|} \frac{b_2}{\|\mathbf{b}\|} + \frac{a_3}{\|\mathbf{a}\|} \frac{b_3}{\|\mathbf{b}\|} \\ &= \frac{1}{\|\mathbf{a}\| \|\mathbf{b}\|} (a_1 b_1 + a_2 b_2 + a_3 b_3) = \frac{1}{\|\mathbf{a}\| \|\mathbf{b}\|} (\mathbf{a} \cdot \mathbf{b}) = 0.\end{aligned}$$

31. $\mathbf{a} = \langle 5, 7, 4 \rangle$; $\|\mathbf{a}\| = 3\sqrt{10}$; $\cos \alpha = 5/3\sqrt{10}$, $\alpha \approx 58.19^\circ$; $\cos \beta = 7/3\sqrt{10}$, $\beta \approx 42.45^\circ$; $\cos \gamma = 4/3\sqrt{10}$, $\gamma \approx 65.06^\circ$

32. We want $\cos \alpha = \cos \beta = \cos \gamma$ or $a_1 = a_2 = a_3$. Letting $a_1 = a_2 = a_3 = 1$ we obtain the vector $\mathbf{i} + \mathbf{j} + \mathbf{k}$. A unit vector in the same direction is $\frac{1}{\sqrt{3}}\mathbf{i} + \frac{1}{\sqrt{3}}\mathbf{j} + \frac{1}{\sqrt{3}}\mathbf{k}$.

33. $\text{comp}_{\mathbf{b}}\mathbf{a} = \mathbf{a} \cdot \mathbf{b} / \|\mathbf{b}\| = \langle 1, -1, 3 \rangle \cdot \langle 2, 6, 3 \rangle / 7 = 5/7$

34. $\text{comp}_{\mathbf{a}}\mathbf{b} = \mathbf{b} \cdot \mathbf{a} / \|\mathbf{a}\| = \langle 2, 6, 3 \rangle \cdot \langle 1, -1, 3 \rangle / \sqrt{11} = 5/\sqrt{11}$

35. $\mathbf{b} - \mathbf{a} = \langle 1, 7, 0 \rangle$; $\text{comp}_{\mathbf{a}}(\mathbf{b} - \mathbf{a}) = (\mathbf{b} - \mathbf{a}) \cdot \mathbf{a} / \|\mathbf{a}\| = \langle 1, 7, 0 \rangle \cdot \langle 1, -1, 3 \rangle / \sqrt{11} = -6/\sqrt{11}$

36. $\mathbf{a} + \mathbf{b} = \langle 3, 5, 6 \rangle$; $2\mathbf{b} = \langle 4, 12, 6 \rangle$; $\text{comp}_{2\mathbf{b}}(\mathbf{a} + \mathbf{b}) \cdot 2\mathbf{b} / \|2\mathbf{b}\| = \langle 3, 5, 6 \rangle \cdot \langle 4, 12, 6 \rangle / 14 = 54/7$

37. $\overrightarrow{OP} = 3\mathbf{i} + 10\mathbf{j}$; $\|\overrightarrow{OP}\| = \sqrt{109}$; $\text{comp}_{\overrightarrow{OP}}\mathbf{a} = \mathbf{a} \cdot \overrightarrow{OP} / \|\overrightarrow{OP}\| = (4\mathbf{i} + 6\mathbf{j}) \cdot (3\mathbf{i} + 10\mathbf{j}) / \sqrt{109} = 72/\sqrt{109}$

38. $\overrightarrow{OP} = \langle 1, -1, 1 \rangle$; $\|\overrightarrow{OP}\| = \sqrt{3}$; $\text{comp}_{\overrightarrow{OP}}\mathbf{a} = \mathbf{a} \cdot \overrightarrow{OP} / \|\overrightarrow{OP}\| = \langle 2, 1, -1 \rangle \cdot \langle 1, -1, 1 \rangle / \sqrt{3} = 0$

39. (a) $\text{comp}_{\mathbf{b}}\mathbf{a} = \mathbf{a} \cdot \mathbf{b} / \|\mathbf{b}\| = (-5\mathbf{i} + 5\mathbf{j}) \cdot (-3\mathbf{i} + 4\mathbf{j}) / 5 = 7$

$$\text{proj}_{\mathbf{b}}\mathbf{a} = (\text{comp}_{\mathbf{b}}\mathbf{a})\mathbf{b} / \|\mathbf{b}\| = 7(-3\mathbf{i} + 4\mathbf{j}) / 5 = -\frac{21}{5}\mathbf{i} + \frac{28}{5}\mathbf{j}$$

(b) $\text{proj}_{\mathbf{b}^\perp}\mathbf{a} = \mathbf{a} - \text{proj}_{\mathbf{b}}\mathbf{a} = (-5\mathbf{i} + 5\mathbf{j}) - (-\frac{21}{5}\mathbf{i} + \frac{28}{5}\mathbf{j}) = -\frac{4}{5}\mathbf{i} - \frac{3}{5}\mathbf{j}$

40. (a) $\text{comp}_{\mathbf{b}}\mathbf{a} = \mathbf{a} \cdot \mathbf{b} / \|\mathbf{b}\| = (4\mathbf{i} + 2\mathbf{j}) \cdot (-3\mathbf{i} + \mathbf{j}) / \sqrt{10} = -\sqrt{10}$

$$\text{proj}_{\mathbf{b}}\mathbf{a} = (\text{comp}_{\mathbf{b}}\mathbf{a})\mathbf{b} / \|\mathbf{b}\| = -\sqrt{10}(-3\mathbf{i} + \mathbf{j}) / \sqrt{10} = 3\mathbf{i} - \mathbf{j}$$

(b) $\text{proj}_{\mathbf{b}^\perp}\mathbf{a} = \mathbf{a} - \text{proj}_{\mathbf{b}}\mathbf{a} = (4\mathbf{i} + 2\mathbf{j}) - (3\mathbf{i} - \mathbf{j}) = \mathbf{i} + 3\mathbf{j}$

41. (a) $\text{comp}_{\mathbf{b}}\mathbf{a} = \mathbf{a} \cdot \mathbf{b} / \|\mathbf{b}\| = (-\mathbf{i} - 2\mathbf{j} + 7\mathbf{k}) \cdot (6\mathbf{i} - 3\mathbf{j} - 2\mathbf{k}) / 7 = -2$

$$\text{proj}_{\mathbf{b}}\mathbf{a} = (\text{comp}_{\mathbf{b}}\mathbf{a})\mathbf{b} / \|\mathbf{b}\| = -2(6\mathbf{i} - 3\mathbf{j} - 2\mathbf{k}) / 7 = -\frac{12}{7}\mathbf{i} + \frac{6}{7}\mathbf{j} + \frac{4}{7}\mathbf{k}$$

(b) $\text{proj}_{\mathbf{b}^\perp}\mathbf{a} = \mathbf{a} - \text{proj}_{\mathbf{b}}\mathbf{a} = (-\mathbf{i} - 2\mathbf{j} + 7\mathbf{k}) - (-\frac{12}{7}\mathbf{i} + \frac{6}{7}\mathbf{j} + \frac{4}{7}\mathbf{k}) = \frac{5}{7}\mathbf{i} - \frac{20}{7}\mathbf{j} + \frac{45}{7}\mathbf{k}$

42. (a) $\text{comp}_{\mathbf{b}}\mathbf{a} = \mathbf{a} \cdot \mathbf{b} / \|\mathbf{b}\| = \langle 1, 1, 1 \rangle \cdot \langle -2, 2, -1 \rangle / 3 = -1/3$

$$\text{proj}_{\mathbf{b}}\mathbf{a} = (\text{comp}_{\mathbf{b}}\mathbf{a})\mathbf{b} / \|\mathbf{b}\| = -\frac{1}{3}\langle -2, 2, -1 \rangle / 3 = \langle 2/9, -2/9, 1/9 \rangle$$

(b) $\text{proj}_{\mathbf{b}^\perp}\mathbf{a} = \mathbf{a} - \text{proj}_{\mathbf{b}}\mathbf{a} = \langle 1, 1, 1 \rangle - \langle 2/9, -2/9, 1/9 \rangle = \langle 7/9, 11/9, 8/9 \rangle$

43. $\mathbf{a} + \mathbf{b} = 3\mathbf{i} + 4\mathbf{j}$; $\|\mathbf{a} + \mathbf{b}\| = 5$; $\text{comp}_{(\mathbf{a}+\mathbf{b})}\mathbf{a} = \mathbf{a} \cdot (\mathbf{a} + \mathbf{b}) / \|\mathbf{a} + \mathbf{b}\| = (4\mathbf{i} + 3\mathbf{j}) \cdot (3\mathbf{i} + 4\mathbf{j}) / 5 = 24/5$

$$\text{proj}_{(\mathbf{a}+\mathbf{b})}\mathbf{a} = (\text{comp}_{(\mathbf{a}+\mathbf{b})}\mathbf{a})(\mathbf{a} + \mathbf{b}) / \|\mathbf{a} + \mathbf{b}\| = \frac{24}{5}(3\mathbf{i} + 4\mathbf{j}) / 5 = \frac{72}{25}\mathbf{i} + \frac{96}{25}\mathbf{j}$$

44. $\mathbf{a} - \mathbf{b} = 5\mathbf{i} + 2\mathbf{j}$; $\|\mathbf{a} - \mathbf{b}\| = \sqrt{29}$; $\text{comp}_{(\mathbf{a}-\mathbf{b})}\mathbf{b} = \mathbf{b} \cdot (\mathbf{a} - \mathbf{b}) / \|\mathbf{a} - \mathbf{b}\| = (-\mathbf{i} + \mathbf{j}) \cdot (5\mathbf{i} + 2\mathbf{j}) / \sqrt{29} = -3/\sqrt{29}$

$$\text{proj}_{(\mathbf{a}-\mathbf{b})}\mathbf{b} = (\text{comp}_{(\mathbf{a}-\mathbf{b})}\mathbf{b})(\mathbf{a} - \mathbf{b}) / \|\mathbf{a} - \mathbf{b}\| = -\frac{3}{\sqrt{29}}(5\mathbf{i} + 2\mathbf{j}) / \sqrt{29} = -\frac{15}{29}\mathbf{i} - \frac{6}{29}\mathbf{j}$$

$$\text{proj}_{(\mathbf{a}-\mathbf{b})^\perp}\mathbf{b} = \mathbf{b} - \text{proj}_{(\mathbf{a}-\mathbf{b})}\mathbf{b} = (-\mathbf{i} + \mathbf{j}) - (-\frac{15}{29}\mathbf{i} - \frac{6}{29}\mathbf{j}) = -\frac{14}{29}\mathbf{i} + \frac{35}{29}\mathbf{j}$$

45. We identify $\|\mathbf{F}\| = 20$, $\theta = 60^\circ$ and $\|\mathbf{d}\| = 100$. Then $W = \|\mathbf{F}\| \|\mathbf{d}\| \cos \theta = 20(100)(\frac{1}{2}) = 1000$ ft-lb.

46. We identify $\mathbf{d} = -\mathbf{i} + 3\mathbf{j} + 8\mathbf{k}$. Then $W = \mathbf{F} \cdot \mathbf{d} = \langle 4, 3, 5 \rangle \cdot \langle -1, 3, 8 \rangle = 45$ N-m.

47. (a) Since $\mathbf{w}$ and $\mathbf{d}$ are orthogonal, $W = \mathbf{w} \cdot \mathbf{d} = 0$.

(b) We identify $\theta = 0^\circ$. Then $W = \|\mathbf{F}\| \|\mathbf{d}\| \cos \theta = 30(\sqrt{4^2 + 3^2}) = 150$ N-m.

48. Using $\mathbf{d} = 6\mathbf{i} + 2\mathbf{j}$ and $\mathbf{F} = 3(\frac{3}{5}\mathbf{i} + \frac{4}{5}\mathbf{j})$, $W = \mathbf{F} \cdot \mathbf{d} = \langle \frac{9}{5}, \frac{12}{5} \rangle \cdot \langle 6, 2 \rangle = \frac{78}{5}$ ft-lb.

Exercises 7.3

49. Let $\mathbf{a}$ and $\mathbf{b}$ be vectors from the center of the carbon atom to the centers of two distinct hydrogen atoms. The distance between two hydrogen atoms is then

$$\begin{aligned}\|\mathbf{b} - \mathbf{a}\| &= \sqrt{(\mathbf{b} - \mathbf{a}) \cdot (\mathbf{b} - \mathbf{a})} = \sqrt{\mathbf{b} \cdot \mathbf{b} - 2\mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{a}} \\ &= \sqrt{\|\mathbf{b}\|^2 + \|\mathbf{a}\|^2 - 2\|\mathbf{a}\|\|\mathbf{b}\|\cos\theta} = \sqrt{(1.1)^2 + (1.1)^2 - 2(1.1)(1.1)\cos 109.5^\circ} \\ &= \sqrt{1.21 + 1.21 - 2.42(-0.333807)} \approx 1.80 \text{ angstroms.}\end{aligned}$$

50. Using the fact that $|\cos\theta| \leq 1$, we have $|\mathbf{a} \cdot \mathbf{b}| = \|\mathbf{a}\|\|\mathbf{b}\|\cos\theta = \|\mathbf{a}\|\|\mathbf{b}\|\cos\theta \leq \|\mathbf{a}\|\|\mathbf{b}\|$.

$$\begin{aligned}51. \quad \|\mathbf{a} + \mathbf{b}\|^2 &= (\mathbf{a} + \mathbf{b}) \cdot (\mathbf{a} + \mathbf{b}) = \mathbf{a} \cdot \mathbf{a} + 2\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{b} = \|\mathbf{a}\|^2 + 2\mathbf{a} \cdot \mathbf{b} + \|\mathbf{b}\|^2 \\ &\leq \|\mathbf{a}\|^2 + 2|\mathbf{a} \cdot \mathbf{b}| + \|\mathbf{b}\|^2 \quad \boxed{\text{since } x \leq |x|} \\ &\leq \|\mathbf{a}\|^2 + 2\|\mathbf{a}\|\|\mathbf{b}\| + \|\mathbf{b}\|^2 = (\|\mathbf{a}\| + \|\mathbf{b}\|)^2 \quad \boxed{\text{by Problem 50}}\end{aligned}$$

Thus, since $\|\mathbf{a} + \mathbf{b}\|$ and $\|\mathbf{a}\| + \|\mathbf{b}\|$ are positive, $\|\mathbf{a} + \mathbf{b}\| \leq \|\mathbf{a}\| + \|\mathbf{b}\|$.

52. Let $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ be distinct points on the line $ax + by = -c$. Then

$$\begin{aligned}\mathbf{n} \cdot \overrightarrow{P_1P_2} &= \langle a, b \rangle \cdot \langle x_2 - x_1, y_2 - y_1 \rangle = ax_2 - ax_1 + by_2 - by_1 \\ &= (ax_2 + by_2) - (ax_1 + by_1) = -c - (-c) = 0,\end{aligned}$$

and the vectors are perpendicular. Thus, $\mathbf{n}$ is perpendicular to the line.

53. Let θ be the angle between $\mathbf{n}$ and $\overrightarrow{P_2P_1}$. Then

$$\begin{aligned}d = \|\overrightarrow{P_1P_2}\| \cos\theta &= \frac{|\mathbf{n} \cdot \overrightarrow{P_2P_1}|}{\|\mathbf{n}\|} = \frac{|\langle a, b \rangle \cdot \langle x_1 - x_2, y_1 - y_2 \rangle|}{\sqrt{a^2 + b^2}} = \frac{|ax_1 - ax_2 + by_1 - by_2|}{\sqrt{a^2 + b^2}} \\ &= \frac{|ax_1 + by_1 - (ax_2 + by_2)|}{\sqrt{a^2 + b^2}} = \frac{|ax_1 + by_1 - (-c)|}{\sqrt{a^2 + b^2}} = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}.\end{aligned}$$

Exercises 7.4

$$1. \quad \mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -1 & 0 \\ 0 & 3 & 5 \end{vmatrix} = \begin{vmatrix} -1 & 0 \\ 3 & 5 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 1 & 0 \\ 0 & 5 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 1 & -1 \\ 0 & 3 \end{vmatrix} \mathbf{k} = -5\mathbf{i} - 5\mathbf{j} + 3\mathbf{k}$$

$$2. \quad \mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & 0 \\ 4 & 0 & -1 \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & -1 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 2 & 0 \\ 4 & -1 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 2 & 1 \\ 4 & 0 \end{vmatrix} \mathbf{k} = -\mathbf{i} + 2\mathbf{j} - 4\mathbf{k}$$

$$3. \quad \mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -3 & 1 \\ 2 & 0 & 4 \end{vmatrix} = \begin{vmatrix} -3 & 1 \\ 0 & 4 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 1 & 1 \\ 2 & 4 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 1 & -3 \\ 2 & 0 \end{vmatrix} \mathbf{k} = \langle -12, -2, 6 \rangle$$

$$4. \quad \mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 1 \\ -5 & 2 & 3 \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 2 & 3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 1 & 1 \\ -5 & 3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 1 & 1 \\ -5 & 2 \end{vmatrix} \mathbf{k} = \langle 1, -8, 7 \rangle$$

$$5. \quad \mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -1 & 2 \\ -1 & 3 & -1 \end{vmatrix} = \begin{vmatrix} -1 & 2 \\ 3 & -1 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 2 & 2 \\ -1 & -1 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 2 & -1 \\ -1 & 3 \end{vmatrix} \mathbf{k} = -5\mathbf{i} + 5\mathbf{k}$$

$$6. \mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & 1 & -5 \\ 2 & 3 & -1 \end{vmatrix} = \begin{vmatrix} 1 & -5 \\ 3 & -1 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 4 & -5 \\ 2 & -1 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 4 & 1 \\ 2 & 3 \end{vmatrix} \mathbf{k} = 14\mathbf{i} - 6\mathbf{j} + 10\mathbf{k}$$

$$7. \mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1/2 & 0 & 1/2 \\ 4 & 6 & 0 \end{vmatrix} = \begin{vmatrix} 0 & 1/2 \\ 6 & 0 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 1/2 & 1/2 \\ 4 & 0 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 1/2 & 0 \\ 4 & 6 \end{vmatrix} \mathbf{k} = \langle -3, 2, 3 \rangle$$

$$8. \mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 5 & 0 \\ 2 & -3 & 4 \end{vmatrix} = \begin{vmatrix} 5 & 0 \\ -3 & 4 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 0 & 0 \\ 2 & 4 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 0 & 5 \\ 2 & -3 \end{vmatrix} \mathbf{k} = \langle 20, 0, -10 \rangle$$

$$9. \mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 2 & -4 \\ -3 & -3 & 6 \end{vmatrix} = \begin{vmatrix} 2 & -4 \\ -3 & 6 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 2 & -4 \\ -3 & 6 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 2 & 2 \\ -3 & -3 \end{vmatrix} \mathbf{k} = \langle 0, 0, 0 \rangle$$

$$10. \mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 8 & 1 & -6 \\ 1 & -2 & 10 \end{vmatrix} = \begin{vmatrix} 1 & -6 \\ -2 & 10 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 8 & -6 \\ 1 & 10 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 8 & 1 \\ 1 & -2 \end{vmatrix} \mathbf{k} = \langle -2, -86, -17 \rangle$$

$$11. \overrightarrow{P_1P_2} = \langle -2, 2, -4 \rangle; \overrightarrow{P_1P_3} = \langle -3, 1, 1 \rangle$$

$$\overrightarrow{P_1P_2} \times \overrightarrow{P_1P_3} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -2 & 2 & -4 \\ -3 & 1 & 1 \end{vmatrix} = \begin{vmatrix} 2 & -4 \\ 1 & 1 \end{vmatrix} \mathbf{i} - \begin{vmatrix} -2 & -4 \\ -3 & 1 \end{vmatrix} \mathbf{j} + \begin{vmatrix} -2 & 2 \\ -3 & 1 \end{vmatrix} \mathbf{k} = 6\mathbf{i} + 14\mathbf{j} + 4\mathbf{k}$$

$$12. \overrightarrow{P_1P_2} = \langle 0, 1, 1 \rangle; \overrightarrow{P_1P_3} = \langle 1, 2, 2 \rangle; \overrightarrow{P_1P_2} \times \overrightarrow{P_1P_3} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 1 & 1 \\ 1 & 2 & 2 \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 2 & 2 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 0 & 1 \\ 1 & 2 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 0 & 1 \\ 1 & 2 \end{vmatrix} \mathbf{k} = \mathbf{j} - \mathbf{k}$$

$$13. \mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 7 & -4 \\ 1 & 1 & -1 \end{vmatrix} = \begin{vmatrix} 7 & -4 \\ 1 & -1 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 2 & -4 \\ 1 & -1 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 2 & 7 \\ 1 & 1 \end{vmatrix} \mathbf{k} = -3\mathbf{i} - 2\mathbf{j} - 5\mathbf{k}$$

is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$.

$$14. \mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & -2 & 4 \\ 4 & -1 & 0 \end{vmatrix} = \begin{vmatrix} -2 & 4 \\ -1 & 0 \end{vmatrix} \mathbf{i} - \begin{vmatrix} -1 & 4 \\ 4 & 0 \end{vmatrix} \mathbf{j} + \begin{vmatrix} -1 & -2 \\ 4 & -1 \end{vmatrix} \mathbf{k} = \langle 4, 16, 9 \rangle$$

is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$.

$$15. \mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 5 & -2 & 1 \\ 2 & 0 & -7 \end{vmatrix} = \begin{vmatrix} -2 & 1 \\ 0 & -7 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 5 & 1 \\ 2 & -7 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 5 & -2 \\ 2 & 0 \end{vmatrix} \mathbf{k} = \langle 14, 37, 4 \rangle$$

$$\mathbf{a} \cdot (\mathbf{a} \times \mathbf{b}) = \langle 5, -2, -1 \rangle \cdot \langle 14, 37, 4 \rangle = 70 - 74 + 4 = 0; \mathbf{b} \cdot (\mathbf{a} \times \mathbf{b}) = \langle 2, 0, -7 \rangle \cdot \langle 14, 37, 4 \rangle = 28 + 0 - 28 = 0$$

$$16. \mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1/2 & -1/4 & 0 \\ 2 & -2 & 6 \end{vmatrix} = \begin{vmatrix} -1/4 & 0 \\ -2 & 6 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 1/2 & 0 \\ 2 & 6 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 1/2 & -1/4 \\ 2 & -2 \end{vmatrix} \mathbf{k} = -\frac{3}{2}\mathbf{i} - 3\mathbf{j} - \frac{1}{2}\mathbf{k}$$

$$\mathbf{a} \cdot (\mathbf{a} \times \mathbf{b}) = \left(\frac{1}{2}\mathbf{i} - \frac{1}{4}\mathbf{j}\right) \cdot \left(-\frac{3}{2}\mathbf{i} - 3\mathbf{j} - \frac{1}{2}\mathbf{k}\right) = -\frac{3}{4} + \frac{3}{4} + 0 = 0$$

$$\mathbf{b} \cdot (\mathbf{a} \times \mathbf{b}) = (2\mathbf{i} - 2\mathbf{j} + 6\mathbf{k}) \cdot \left(-\frac{3}{2}\mathbf{i} - 3\mathbf{j} - \frac{1}{2}\mathbf{k}\right) = -3 + 6 - 3 = 0$$

Exercises 7.4

$$17. (a) \mathbf{b} \times \mathbf{c} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & 1 \\ 3 & 1 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 2 & 1 \\ 3 & 1 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 2 & 1 \\ 3 & 1 \end{vmatrix} \mathbf{k} = \mathbf{j} - \mathbf{k}$$

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -1 & 2 \\ 0 & 1 & -1 \end{vmatrix} = \begin{vmatrix} -1 & 2 \\ 1 & -1 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 1 & 2 \\ 0 & -1 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 1 & -1 \\ 0 & 1 \end{vmatrix} \mathbf{k} = -\mathbf{i} + \mathbf{j} + \mathbf{k}$$

$$(b) \mathbf{a} \cdot \mathbf{c} = (\mathbf{i} - \mathbf{j} + 2\mathbf{k}) \cdot (3\mathbf{i} + \mathbf{j} + \mathbf{k}) = 4; (\mathbf{a} \cdot \mathbf{c})\mathbf{b} = 4(2\mathbf{i} + \mathbf{j} + \mathbf{k}) = 8\mathbf{i} + 4\mathbf{j} + 4\mathbf{k}$$

$$\mathbf{a} \cdot \mathbf{b} = (\mathbf{i} - \mathbf{j} + 2\mathbf{k}) \cdot (2\mathbf{i} + \mathbf{j} + \mathbf{k}) = 3; (\mathbf{a} \cdot \mathbf{b})\mathbf{c} = 3(3\mathbf{i} + \mathbf{j} + \mathbf{k}) = 9\mathbf{i} + 3\mathbf{j} + 3\mathbf{k}$$

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c} = (8\mathbf{i} + 4\mathbf{j} + 4\mathbf{k}) - (9\mathbf{i} + 3\mathbf{j} + 3\mathbf{k}) = -\mathbf{i} + \mathbf{j} + \mathbf{k}$$

$$18. (a) \mathbf{b} \times \mathbf{c} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & -1 \\ -1 & 5 & 8 \end{vmatrix} = \begin{vmatrix} 2 & -1 \\ 5 & 8 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 1 & -1 \\ -1 & 8 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 1 & 2 \\ -1 & 5 \end{vmatrix} \mathbf{k} = 21\mathbf{i} - 7\mathbf{j} + 7\mathbf{k}$$

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 0 & -4 \\ 21 & -7 & 7 \end{vmatrix} = \begin{vmatrix} 0 & -4 \\ -7 & 7 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 3 & -4 \\ 21 & 7 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 3 & 0 \\ 21 & -7 \end{vmatrix} \mathbf{k} = -28\mathbf{i} - 105\mathbf{j} - 21\mathbf{k}$$

$$(b) \mathbf{a} \cdot \mathbf{c} = (3\mathbf{i} - 4\mathbf{k}) \cdot (-\mathbf{i} + 5\mathbf{j} + 8\mathbf{k}) = -35; (\mathbf{a} \cdot \mathbf{c})\mathbf{b} = -35(\mathbf{i} + 2\mathbf{j} - \mathbf{k}) = -35\mathbf{i} - 70\mathbf{j} + 35\mathbf{k}$$

$$\mathbf{a} \cdot \mathbf{b} = (3\mathbf{i} - 4\mathbf{k}) \cdot (\mathbf{i} + 2\mathbf{j} - \mathbf{k}) = 7; (\mathbf{a} \cdot \mathbf{b})\mathbf{c} = 7(-\mathbf{i} + 5\mathbf{j} + 8\mathbf{k}) = -7\mathbf{i} + 35\mathbf{j} + 56\mathbf{k}$$

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c} = (-35\mathbf{i} - 70\mathbf{j} + 35\mathbf{k}) - (-7\mathbf{i} + 35\mathbf{j} + 56\mathbf{k}) = -28\mathbf{i} - 105\mathbf{j} - 21\mathbf{k}$$

$$19. (2\mathbf{i}) \times \mathbf{j} = 2(\mathbf{i} \times \mathbf{j}) = 2\mathbf{k}$$

$$20. \mathbf{i} \times (-3\mathbf{k}) = -3(\mathbf{i} \times \mathbf{k}) = -3(-\mathbf{j}) = 3\mathbf{j}$$

$$21. \mathbf{k} \times (2\mathbf{i} - \mathbf{j}) = \mathbf{k} \times (2\mathbf{i}) + \mathbf{k} \times (-\mathbf{j}) = 2(\mathbf{k} \times \mathbf{i}) - (\mathbf{k} \times \mathbf{j}) = 2\mathbf{j} - (-\mathbf{i}) = \mathbf{i} + 2\mathbf{j}$$

$$22. \mathbf{i} \times (\mathbf{j} \times \mathbf{k}) = \mathbf{i} \times \mathbf{i} = \mathbf{0}$$

$$23. [(2\mathbf{k}) \times (3\mathbf{j})] \times (4\mathbf{j}) = [2 \cdot 3(\mathbf{k} \times \mathbf{j}) \times (4\mathbf{j})] = 6(-\mathbf{i}) \times 4\mathbf{j} = (-6)(4)(\mathbf{i} \times \mathbf{j}) = -24\mathbf{k}$$

$$24. (2\mathbf{i} - \mathbf{j} + 5\mathbf{k}) \times \mathbf{i} = (2\mathbf{i} \times \mathbf{i}) + (-\mathbf{j} \times \mathbf{i}) + (5\mathbf{k} \times \mathbf{i}) = 2(\mathbf{i} \times \mathbf{i}) + (\mathbf{i} \times \mathbf{j}) + 5(\mathbf{k} \times \mathbf{i}) = 5\mathbf{j} + \mathbf{k}$$

$$25. (\mathbf{i} + \mathbf{j}) \times (\mathbf{i} + 5\mathbf{k}) = [(\mathbf{i} + \mathbf{j}) \times \mathbf{i}] + [(\mathbf{i} + \mathbf{j}) \times 5\mathbf{k}] = (\mathbf{i} \times \mathbf{i}) + (\mathbf{j} \times \mathbf{i}) + (\mathbf{i} \times 5\mathbf{k}) + (\mathbf{j} \times 5\mathbf{k}) \\ = -\mathbf{k} + 5(-\mathbf{j}) + 5\mathbf{i} = 5\mathbf{i} - 5\mathbf{j} - \mathbf{k}$$

$$26. \mathbf{i} \times \mathbf{k} - 2(\mathbf{j} \times \mathbf{i}) = -\mathbf{j} - 2(-\mathbf{k}) = -\mathbf{j} + 2\mathbf{k}$$

$$27. \mathbf{k} \cdot (\mathbf{j} \times \mathbf{k}) = \mathbf{k} \cdot \mathbf{i} = 0$$

$$28. \mathbf{i} \cdot [\mathbf{j} \times (-\mathbf{k})] = \mathbf{i} \cdot [-(\mathbf{j} \times \mathbf{k})] = \mathbf{i} \cdot (-\mathbf{i}) = -(\mathbf{i} \cdot \mathbf{i}) = -1$$

$$29. \|4\mathbf{j} - 5(\mathbf{i} \times \mathbf{j})\| = \|4\mathbf{j} - 5\mathbf{k}\| = \sqrt{41}$$

$$30. (\mathbf{i} \times \mathbf{j}) \cdot (3\mathbf{j} \times \mathbf{i}) = \mathbf{k} \cdot (-3\mathbf{k}) = -3(\mathbf{k} \cdot \mathbf{k}) = -3$$

$$31. \mathbf{i} \times (\mathbf{i} \times \mathbf{j}) = \mathbf{i} \times \mathbf{k} = -\mathbf{j}$$

$$32. (\mathbf{i} \times \mathbf{j}) \times \mathbf{i} = \mathbf{k} \times \mathbf{i} = \mathbf{j}$$

$$33. (\mathbf{i} \times \mathbf{i}) \times \mathbf{j} = \mathbf{0} \times \mathbf{j} = \mathbf{0}$$

$$34. (\mathbf{i} \cdot \mathbf{i})(\mathbf{i} \times \mathbf{j}) = 1(\mathbf{k}) = \mathbf{k}$$

$$35. 2\mathbf{j} \cdot [\mathbf{i} \times (\mathbf{j} - 3\mathbf{k})] = 2\mathbf{j} \cdot [(\mathbf{i} \times \mathbf{j}) + (\mathbf{i} \times (-3\mathbf{k}))] = 2\mathbf{j} \cdot [\mathbf{k} + 3(\mathbf{k} \times \mathbf{i})] = 2\mathbf{j} \cdot (\mathbf{k} + 3\mathbf{j}) = 2\mathbf{j} \cdot \mathbf{k} + 2\mathbf{j} \cdot 3\mathbf{j} \\ = 2(\mathbf{j} \cdot \mathbf{k}) + 6(\mathbf{j} \cdot \mathbf{j}) = 2(0) + 6(1) = 6$$

$$36. (\mathbf{i} \times \mathbf{k}) \times (\mathbf{j} \times \mathbf{i}) = (-\mathbf{j}) \times (-\mathbf{k}) = (-1)(-1)(\mathbf{j} \times \mathbf{k}) = \mathbf{j} \times \mathbf{k} = \mathbf{i}$$

$$37. \mathbf{a} \times (3\mathbf{b}) = 3(\mathbf{a} \times \mathbf{b}) = 3(4\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}) = 12\mathbf{i} - 9\mathbf{j} + 18\mathbf{k}$$

$$38. \mathbf{b} \times \mathbf{a} = -\mathbf{a} \times \mathbf{b} = -(\mathbf{a} \times \mathbf{b}) = -4\mathbf{i} + 3\mathbf{j} - 6\mathbf{k}$$

$$39. (-\mathbf{a}) \times \mathbf{b} = -(\mathbf{a} \times \mathbf{b}) = -4\mathbf{i} + 3\mathbf{j} - 6\mathbf{k}$$

$$40. |\mathbf{a} \times \mathbf{b}| = \sqrt{4^2 + (-3)^2 + 6^2} = \sqrt{61}$$

$$41. (\mathbf{a} \times \mathbf{b}) \times \mathbf{c} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & -3 & 6 \\ 2 & 4 & -1 \end{vmatrix} = \begin{vmatrix} -3 & 6 \\ 4 & -1 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 4 & 6 \\ 2 & -1 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 4 & -3 \\ 2 & -4 \end{vmatrix} \mathbf{k} = -21\mathbf{i} + 16\mathbf{j} + 22\mathbf{k}$$

$$42. (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = 4(2) + (-3)4 + 6(-1) = -10$$

$$43. \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = 4(2) + (-3)4 + 6(-1) = -10$$

$$44. (4\mathbf{a}) \cdot (\mathbf{b} \times \mathbf{c}) = (4\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = 4(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = 16(2) + (-12)4 + 24(-1) = -40$$

$$45. (\mathbf{a}) \text{ Let } A = (1, 3, 0), B = (2, 0, 0), C = (0, 0, 4), \text{ and } D = (1, -3, 4). \text{ Then } \overrightarrow{AB} = \mathbf{i} - 3\mathbf{j}, \overrightarrow{AC} = -\mathbf{i} - 3\mathbf{j} + 4\mathbf{k}, \\ \overrightarrow{CD} = \mathbf{i} - 3\mathbf{j}, \text{ and } \overrightarrow{BD} = -\mathbf{i} - 3\mathbf{j} + 4\mathbf{k}. \text{ Since } \overrightarrow{AB} = \overrightarrow{CD} \text{ and } \overrightarrow{AC} = \overrightarrow{BD}, \text{ the quadrilateral is a parallelogram.}$$

(b) Computing

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -3 & 0 \\ -1 & -3 & 4 \end{vmatrix} = -12\mathbf{i} - 4\mathbf{j} - 6\mathbf{k}$$

$$\text{we find that the area is } \|\overrightarrow{AB} \times \overrightarrow{AC}\| = \sqrt{144 + 16 + 36} = 14.$$

$$46. (\mathbf{a}) \text{ Let } A = (3, 4, 1), B = (-1, 4, 2), C = (2, 0, 2) \text{ and } D = (-2, 0, 3). \text{ Then } \overrightarrow{AB} = -4\mathbf{i} + \mathbf{k}, \overrightarrow{AC} = -\mathbf{i} - 4\mathbf{j} + \mathbf{k}, \\ \overrightarrow{CD} = -4\mathbf{i} + \mathbf{k}, \text{ and } \overrightarrow{BD} = -\mathbf{i} - 4\mathbf{j} + \mathbf{k}. \text{ Since } \overrightarrow{AB} = \overrightarrow{CD} \text{ and } \overrightarrow{AC} = \overrightarrow{BD}, \text{ the quadrilateral is a parallelogram.}$$

(b) Computing

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -4 & 0 & 1 \\ -1 & -4 & 1 \end{vmatrix} = 4\mathbf{i} + 3\mathbf{j} + 16\mathbf{k}$$

$$\text{we find that the area is } \|\overrightarrow{AB} \times \overrightarrow{AC}\| = \sqrt{16 + 9 + 256} = \sqrt{281} \approx 16.76.$$

$$47. \overrightarrow{P_1P_2} = \mathbf{j}; \overrightarrow{P_2P_3} = -\mathbf{j} + \mathbf{k}$$

$$\overrightarrow{P_1P_2} \times \overrightarrow{P_2P_3} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ -1 & 1 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 0 & 0 \\ 0 & 1 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 0 & 1 \\ 0 & -1 \end{vmatrix} \mathbf{k} = \mathbf{i}; \quad A = \frac{1}{2}\|\mathbf{i}\| = \frac{1}{2} \text{ sq. unit}$$

$$48. \overrightarrow{P_1P_2} = \mathbf{j} + 2\mathbf{k}; \overrightarrow{P_2P_3} = 2\mathbf{i} + \mathbf{j} - 2\mathbf{k}$$

$$\overrightarrow{P_1P_2} \times \overrightarrow{P_2P_3} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 1 & 2 \\ 2 & 1 & -2 \end{vmatrix} = \begin{vmatrix} 1 & 2 \\ 1 & -2 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 0 & 2 \\ 2 & -2 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 0 & 1 \\ 2 & 1 \end{vmatrix} \mathbf{k} = -4\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$$

$$A = \frac{1}{2}\|-4\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}\| = 3 \text{ sq. units}$$

$$49. \overrightarrow{P_1P_2} = -3\mathbf{j} - \mathbf{k}; \overrightarrow{P_2P_3} = -2\mathbf{i} - \mathbf{k}$$

$$\overrightarrow{P_1P_2} \times \overrightarrow{P_2P_3} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & -3 & -1 \\ -2 & 0 & -1 \end{vmatrix} = \begin{vmatrix} -3 & -1 \\ 0 & -1 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 0 & -1 \\ -2 & -1 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 0 & -3 \\ -2 & 0 \end{vmatrix} \mathbf{k} = 3\mathbf{i} + 2\mathbf{j} - 6\mathbf{k}$$

$$A = \frac{1}{2}\|3\mathbf{i} + 2\mathbf{j} - 6\mathbf{k}\| = \frac{7}{2} \text{ sq. units}$$

$$50. \overrightarrow{P_1P_2} = -\mathbf{i} + 3\mathbf{k}; \overrightarrow{P_2P_3} = 2\mathbf{i} + 4\mathbf{j} - \mathbf{k}$$

$$\overrightarrow{P_1P_2} \times \overrightarrow{P_2P_3} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & 0 & 3 \\ 2 & 4 & -1 \end{vmatrix} = \begin{vmatrix} 0 & 3 \\ 4 & -1 \end{vmatrix} \mathbf{i} - \begin{vmatrix} -1 & 3 \\ 2 & -1 \end{vmatrix} \mathbf{j} + \begin{vmatrix} -1 & 0 \\ 2 & 4 \end{vmatrix} \mathbf{k} = -12\mathbf{i} + 5\mathbf{j} - 4\mathbf{k}$$

$$A = \frac{1}{2}\|-12\mathbf{i} + 5\mathbf{j} - 4\mathbf{k}\| = \frac{\sqrt{185}}{2} \text{ sq. units}$$

Exercises 7.4

$$51. \mathbf{b} \times \mathbf{c} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & 4 & 0 \\ 2 & 2 & 2 \end{vmatrix} = \begin{vmatrix} 4 & 0 \\ 2 & 2 \end{vmatrix} \mathbf{i} - \begin{vmatrix} -1 & 0 \\ 2 & 2 \end{vmatrix} \mathbf{j} + \begin{vmatrix} -1 & 4 \\ 2 & 2 \end{vmatrix} \mathbf{k} = 8\mathbf{i} + 2\mathbf{j} - 10\mathbf{k}$$

$$\mathbf{v} = |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})| = |(\mathbf{i} + \mathbf{j}) \cdot (8\mathbf{i} + 2\mathbf{j} - 10\mathbf{k})| = |8 + 2 + 0| = 10 \text{ cu. units}$$

$$52. \mathbf{b} \times \mathbf{c} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 4 & 1 \\ 1 & 1 & 5 \end{vmatrix} = \begin{vmatrix} 4 & 1 \\ 1 & 5 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 1 & 1 \\ 1 & 5 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 1 & 4 \\ 1 & 1 \end{vmatrix} \mathbf{k} = 19\mathbf{i} - 4\mathbf{j} - 3\mathbf{k}$$

$$\mathbf{v} = |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})| = |(3\mathbf{i} + \mathbf{j} + \mathbf{k}) \cdot (19\mathbf{i} - 4\mathbf{j} - 3\mathbf{k})| = |57 - 4 - 3| = 50 \text{ cu. units}$$

$$53. \mathbf{b} \times \mathbf{c} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -2 & 6 & -6 \\ 5/2 & 3 & 1/2 \end{vmatrix} = \begin{vmatrix} 6 & -6 \\ 3 & 2/2 \end{vmatrix} \mathbf{i} - \begin{vmatrix} -2 & -6 \\ 5/2 & 1/2 \end{vmatrix} \mathbf{j} + \begin{vmatrix} -2 & 6 \\ 5/2 & 3 \end{vmatrix} \mathbf{k} = 21\mathbf{i} - 14\mathbf{j} - 21\mathbf{k}$$

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = (4\mathbf{i} + 6\mathbf{j}) \cdot (21\mathbf{i} - 14\mathbf{j} - 21\mathbf{k}) = 84 - 84 + 0 = 0. \text{ The vectors are coplanar.}$$

54. The four points will be coplanar if the three vectors $\overrightarrow{P_1P_2} = \langle 3, -1, -1 \rangle$, $\overrightarrow{P_2P_3} = \langle -3, -5, 13 \rangle$, and $\overrightarrow{P_3P_4} = \langle -8, 7, -6 \rangle$ are coplanar.

$$\overrightarrow{P_2P_3} \times \overrightarrow{P_3P_4} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -3 & -5 & 13 \\ -8 & 7 & -6 \end{vmatrix} = \begin{vmatrix} -5 & 13 \\ 7 & -6 \end{vmatrix} \mathbf{i} - \begin{vmatrix} -3 & 13 \\ -8 & -6 \end{vmatrix} \mathbf{j} + \begin{vmatrix} -3 & -5 \\ -8 & 7 \end{vmatrix} \mathbf{k} = \langle -61, -122, -61 \rangle$$

$$\overrightarrow{P_1P_2} \cdot (\overrightarrow{P_2P_3} \times \overrightarrow{P_3P_4}) = \langle 3, -1, -1 \rangle \cdot \langle -61, -122, -61 \rangle = -183 + 122 + 61 = 0$$

The four points are coplanar.

55. (a) Since $\theta = 90^\circ$, $\|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\| \|\mathbf{b}\| |\sin 90^\circ| = 6.4(5) = 32$.

(b) The direction of $\mathbf{a} \times \mathbf{b}$ is into the fourth quadrant of the xy -plane or to the left of the plane determined by $\mathbf{a}$ and $\mathbf{b}$ as shown in Figure 6.54 in the text. It makes an angle of 30° with the positive x -axis.

(c) We identify $\mathbf{n} = (\sqrt{3}\mathbf{i} - \mathbf{j})/2$. Then $\mathbf{a} \times \mathbf{b} = 32\mathbf{n} = 16\sqrt{3}\mathbf{i} - 16\mathbf{j}$.

56. Using Definition 7.4, $\mathbf{a} \times \mathbf{b} = \sqrt{27}(8) \sin 120^\circ \mathbf{n} = 24\sqrt{3}(\sqrt{3}/2)\mathbf{n} = 36\mathbf{n}$. By the right-hand rule, $\mathbf{n} = \mathbf{j}$ or $\mathbf{n} = -\mathbf{j}$. Thus, $\mathbf{a} \times \mathbf{b} = 36\mathbf{j}$ or $-36\mathbf{j}$.

57. (a) We note first that $\mathbf{a} \times \mathbf{b} = \mathbf{k}$, $\mathbf{b} \times \mathbf{c} = \frac{1}{2}(\mathbf{i} - \mathbf{k})$, $\mathbf{c} \times \mathbf{a} = \frac{1}{2}(\mathbf{j} - \mathbf{k})$, $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \frac{1}{2}$, $\mathbf{b} \cdot (\mathbf{c} \times \mathbf{a}) = \frac{1}{2}$, and $\mathbf{c} \cdot (\mathbf{a} \times \mathbf{b}) = \frac{1}{2}$. Then

$$\mathbf{A} = \frac{\frac{1}{2}(\mathbf{i} - \mathbf{k})}{\frac{1}{2}} = \mathbf{i} - \mathbf{k}, \quad \mathbf{B} = \frac{\frac{1}{2}(\mathbf{j} - \mathbf{k})}{\frac{1}{2}} = \mathbf{j} - \mathbf{k}, \quad \text{and} \quad \mathbf{C} = \frac{\mathbf{k}}{\frac{1}{2}} = 2\mathbf{k}.$$

(b) We need to compute $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$. Using formula (10) in the text we have

$$\begin{aligned} \mathbf{B} \times \mathbf{C} &= \frac{(\mathbf{c} \times \mathbf{a}) \times (\mathbf{a} \times \mathbf{b})}{[\mathbf{b} \cdot (\mathbf{c} \times \mathbf{a})][\mathbf{c} \cdot (\mathbf{a} \times \mathbf{b})]} = \frac{[(\mathbf{c} \times \mathbf{a}) \cdot \mathbf{b}]\mathbf{a} - [(\mathbf{c} \times \mathbf{a}) \cdot \mathbf{a}]\mathbf{b}}{[\mathbf{b} \cdot (\mathbf{c} \times \mathbf{a})][\mathbf{c} \cdot (\mathbf{a} \times \mathbf{b})]} \\ &= \frac{\mathbf{a}}{\mathbf{c} \cdot (\mathbf{a} \times \mathbf{b})} \quad \boxed{\text{since } (\mathbf{c} \times \mathbf{a}) \cdot \mathbf{a} = 0.} \end{aligned}$$

Then

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \frac{\mathbf{b} \times \mathbf{c}}{\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})} \cdot \frac{\mathbf{a}}{\mathbf{c} \cdot (\mathbf{a} \times \mathbf{b})} = \frac{1}{\mathbf{c} \cdot (\mathbf{a} \times \mathbf{b})}$$

and the volume of the unit cell of the reciprocal lattice is the reciprocal of the volume of the unit cell of the original lattice.

$$\begin{aligned}
58. \quad \mathbf{a} \times (\mathbf{b} + \mathbf{c}) &= \begin{vmatrix} a_2 & a_3 \\ b_2 + c_2 & b_3 + c_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} a_1 & a_3 \\ b_1 + c_1 & b_3 + c_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} a_1 & a_2 \\ b_1 + c_1 & b_2 + c_2 \end{vmatrix} \mathbf{k} \\
&= (a_2b_3 - a_3b_2)\mathbf{i} + (a_2c_3 - a_3c_2)\mathbf{i} - [(a_1b_3 - a_3b_1)\mathbf{j} + (a_1c_3 - a_3c_1)\mathbf{j}] + (a_1b_2 - a_2b_1)\mathbf{k} + (a_1c_2 - a_2c_1)\mathbf{k} \\
&= (a_2b_3 - a_3b_2)\mathbf{i} - (a_1b_3 - a_3b_1)\mathbf{j} + (a_1b_2 - a_2b_1)\mathbf{k} + (a_2c_3 - a_3c_2)\mathbf{i} - (a_1c_3 - a_3c_1)\mathbf{j} + (a_1c_2 - a_2c_1)\mathbf{k} \\
&= \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}
\end{aligned}$$

$$\begin{aligned}
59. \quad \mathbf{b} \times \mathbf{c} &= (b_2c_3 - b_3c_2)\mathbf{i} - (b_1c_3 - b_3c_1)\mathbf{j} + (b_1c_2 - b_2c_1)\mathbf{k} \\
\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) &= [a_2(b_1c_2 - b_2c_1) + a_3(b_1c_3 - b_3c_1)]\mathbf{i} - [a_1(b_1c_2 - b_2c_1) - a_3(b_2c_3 - b_3c_2)]\mathbf{j} \\
&\quad + [-a_1(b_1c_3 - b_3c_1) - a_2(b_2c_3 - b_3c_2)]\mathbf{k} \\
&= (a_2b_1c_2 - a_2b_2c_1 + a_3b_1c_3 - a_3b_3c_1)\mathbf{i} - (a_1b_1c_2 - a_1b_2c_1 - a_3b_2c_3 + a_3b_3c_2)\mathbf{j} \\
&\quad - (a_1b_1c_3 - a_1b_3c_1 + a_2b_2c_3 - a_2b_3c_2)\mathbf{k} \\
(\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c} &= (a_1c_1 + a_2c_2 + a_3c_3)(b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}) - (a_1b_1 + a_2b_2 + a_3b_3)(c_1\mathbf{i} + c_2\mathbf{j} + c_3\mathbf{k}) \\
&= (a_2b_1c_2 - a_2b_2c_1 + a_3b_1c_3 - a_3b_3c_1)\mathbf{i} - (a_1b_1c_2 - a_1b_2c_1 - a_3b_2c_3 + a_3b_3c_2)\mathbf{j} \\
&\quad - (a_1b_1c_3 - a_1b_3c_1 + a_2b_2c_3 - a_2b_3c_2)\mathbf{k}
\end{aligned}$$

60. The statement is false since $\mathbf{i} \times (\mathbf{i} \times \mathbf{j}) = \mathbf{i} \times \mathbf{k} = -\mathbf{j}$ and $(\mathbf{i} \times \mathbf{i}) \times \mathbf{j} = \mathbf{0} \times \mathbf{j} = \mathbf{0}$.

61. Using equation 9 in the text,

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \quad \text{and} \quad (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = \mathbf{c} \cdot (\mathbf{a} \times \mathbf{b}) = \begin{vmatrix} c_1 & c_2 & c_3 \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}.$$

The second determinant can be obtained from the first by an interchange of the second and third rows followed by an interchange of the new first and second rows. Using Theorem 8.11 in the text, we see that $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}$.

$$\begin{aligned}
62. \quad \mathbf{a} \times (\mathbf{b} \times \mathbf{c}) + \mathbf{b} \times (\mathbf{c} \times \mathbf{a}) + \mathbf{c} \times (\mathbf{a} \times \mathbf{b}) \\
&= (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c} + (\mathbf{b} \cdot \mathbf{a})\mathbf{c} - (\mathbf{b} \cdot \mathbf{c})\mathbf{a} + (\mathbf{c} \cdot \mathbf{b})\mathbf{a} - (\mathbf{c} \cdot \mathbf{a})\mathbf{b} \\
&= [(\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{c} \cdot \mathbf{a})\mathbf{b}] + [(\mathbf{b} \cdot \mathbf{a})\mathbf{c} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}] + [(\mathbf{c} \cdot \mathbf{b})\mathbf{a} - (\mathbf{b} \cdot \mathbf{c})\mathbf{a}] = \mathbf{0}
\end{aligned}$$

63. Since

$$\begin{aligned}
\|\mathbf{a} \times \mathbf{b}\|^2 &= (a_2b_3 - a_3b_2)^2 + (a_1b_3 - a_3b_1)^2 + (a_1b_2 - a_2b_1)^2 \\
&= a_2^2b_3^2 - 2a_2b_3a_3b_2 + a_3^2b_2^2 + a_1^2b_3^2 - 2a_1b_3a_3b_1 + a_3^2b_1^2 + a_1^2b_2^2 - 2a_1b_2a_2b_1 + a_2^2b_1^2
\end{aligned}$$

and

$$\begin{aligned}
\|\mathbf{a}\|^2\|\mathbf{b}\|^2 - (\mathbf{a} \cdot \mathbf{b})^2 &= (a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2) - (a_1b_1 + a_2b_2 + a_3b_3)^2 \\
&= a_1^2a_2^2 + a_1^2b_2^2 + a_1^2b_3^2 + a_2^2b_1^2 + a_2^2b_2^2 + a_2^2b_3^2 + a_3^2b_1^2 + a_3^2b_2^2 + a_3^2b_3^2 \\
&\quad - a_1^2b_1^2 - a_2^2b_2^2 - a_3^2b_3^2 - 2a_1b_1a_2b_2 - 2a_1b_1a_3b_3 - 2a_2b_2a_3b_3 \\
&= a_1^2b_2^2 + a_1^2b_3^2 + a_2^2b_1^2 + a_2^2b_3^2 + a_3^2b_1^2 + a_3^2b_2^2 - 2a_1a_2b_1b_2 - 2a_1a_3b_1b_3 - 2a_2a_3b_2b_3
\end{aligned}$$

we see that $\|\mathbf{a} \times \mathbf{b}\|^2 = \|\mathbf{a}\|^2\|\mathbf{b}\|^2 - (\mathbf{a} \cdot \mathbf{b})^2$.

64. No. For example $\mathbf{i} \times (\mathbf{i} + \mathbf{j}) = \mathbf{i} \times \mathbf{j}$ by the distributive law (iii) in the text, and the fact that $\mathbf{i} \times \mathbf{i} = \mathbf{0}$. But $\mathbf{i} + \mathbf{j}$ does not equal $\mathbf{j}$.

65. By the distributive law (iii) in the text:

$$(\mathbf{a} + \mathbf{b}) \times (\mathbf{a} - \mathbf{b}) = (\mathbf{a} + \mathbf{b}) \times \mathbf{a} - (\mathbf{a} + \mathbf{b}) \times \mathbf{b} = \mathbf{a} \times \mathbf{a} + \mathbf{b} \times \mathbf{a} - \mathbf{a} \times \mathbf{b} - \mathbf{b} \times \mathbf{b} = 2\mathbf{b} \times \mathbf{a}$$

since $\mathbf{a} \times \mathbf{a} = \mathbf{0}$, $\mathbf{b} \times \mathbf{b} = \mathbf{0}$, and $-\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{a}$.

Exercises 7.5

The equation of a line through P_1 and P_2 in 3-space with $\mathbf{r}_1 = \overrightarrow{OP_1}$ and $\mathbf{r}_2 = \overrightarrow{OP_2}$ can be expressed as $\mathbf{r} = \mathbf{r}_1 + t(k\mathbf{a})$ or $\mathbf{r} = \mathbf{r}_2 + t(k\mathbf{a})$ where $\mathbf{a} = \mathbf{r}_2 - \mathbf{r}_1$ and k is any non-zero scalar. Thus, the form of the equation of a line is not unique. (See the alternate solution to Problem 1.)

1. $\mathbf{a} = \langle 1 - 3, 2 - 5, 1 - (-2) \rangle = \langle -2, -3, 3 \rangle$; $\langle x, y, z \rangle = \langle 1, 2, 1 \rangle + t\langle -2, -3, 3 \rangle$
 Alternate Solution: $\mathbf{a} = \langle 3 - 1, 5 - 2, -2 - 1 \rangle = \langle 2, 3, -3 \rangle$; $\langle x, y, z \rangle = \langle 3, 5, -2 \rangle + t\langle 2, 3, -3 \rangle$
2. $\mathbf{a} = \langle 0 - (-2), 4 - 6, 5 - 3 \rangle = \langle 2, -2, 2 \rangle$; $\langle x, y, z \rangle = \langle 0, 4, 5 \rangle + t\langle 2, -2, 2 \rangle$
3. $\mathbf{a} = \langle 1/2 - (-3/2), -1/2 - 5/2, 1 - (-1/2) \rangle = \langle 2, -3, 3/2 \rangle$; $\langle x, y, z \rangle = \langle 1/2, -1/2, 1 \rangle + t\langle 2, -3, 3/2 \rangle$
4. $\mathbf{a} = \langle 10 - 5, 2 - (-3), -10 - 5 \rangle = \langle 5, 5, -15 \rangle$; $\langle x, y, z \rangle = \langle 10, 2, -10 \rangle + t\langle 5, 5, -15 \rangle$
5. $\mathbf{a} = \langle 1 - (-4), 1 - 1, -1 - (-1) \rangle = \langle 5, 0, 0 \rangle$; $\langle x, y, z \rangle = \langle 1, 1, -1 \rangle + t\langle 5, 0, 0 \rangle$
6. $\mathbf{a} = \langle 3 - 5/2, 2 - 1, 1 - (-2) \rangle = \langle 1/2, 1, 3 \rangle$; $\langle x, y, z \rangle = \langle 3, 2, 1 \rangle + t\langle 1/2, 1, 3 \rangle$
7. $\mathbf{a} = \langle 2 - 6, 3 - (-1), 5 - 8 \rangle = \langle -4, 4, -3 \rangle$; $x = 2 - 4t$, $y = 3 + 4t$, $z = 5 - 3t$
8. $\mathbf{a} = \langle 2 - 0, 0 - 4, 0 - 9 \rangle = \langle 2, -4, -9 \rangle$; $x = 2 + 2t$, $y = -4t$, $z = -9t$
9. $\mathbf{a} = \langle 1 - 3, 0 - (-2), 0 - (-7) \rangle = \langle -2, 2, 7 \rangle$; $x = 1 - 2t$, $y = 2t$, $z = 7t$
10. $\mathbf{a} = \langle 0 - (-2), 0 - 4, 5 - 0 \rangle = \langle 2, -4, 5 \rangle$; $x = 2t$, $y = -4t$, $z = 5 + 5t$
11. $\mathbf{a} = \langle 4 - (-6), 1/2 - (-1/4), 1/3 - 1/6 \rangle = \langle 10, 3/4, 1/6 \rangle$; $x = 4 + 10t$, $y = \frac{1}{2} + \frac{3}{4}t$, $z = \frac{1}{3} + \frac{1}{6}t$
12. $\mathbf{a} = \langle -3 - 4, 7 - (-8), 9 - (-1) \rangle = \langle -7, 15, 10 \rangle$; $x = -3 - 7t$, $y = 7 + 15t$, $z = 9 + 10t$
13. $a_1 = 10 - 1 = 9$, $a_2 = 14 - 4 = 10$, $a_3 = -2 - (-9) = 7$; $\frac{x - 10}{9} = \frac{y - 14}{10} = \frac{z + 2}{7}$
14. $a_1 = 1 - 2/3 = 1/3$, $a_2 = 3 - 0 = 3$, $a_3 = 1/4 - (-1/4) = 1/2$; $\frac{x - 1}{1/3} = \frac{y - 3}{3} = \frac{z - 1/4}{1/2}$
15. $a_1 = -7 - 4 = -11$, $a_2 = 2 - 2 = 0$, $a_3 = 5 - 1 = 4$; $\frac{x + 7}{-11} = \frac{z - 5}{4}$, $y = 2$
16. $a_1 = 1 - (-5) = 6$, $a_2 = 1 - (-2) = 3$, $a_3 = 2 - (-4) = 6$; $\frac{x - 1}{6} = \frac{y - 1}{3} = \frac{z - 2}{6}$
17. $a_1 = 5 - 5 = 0$, $a_2 = 10 - 1 = 9$, $a_3 = -2 - (-14) = 12$; $x = 5$, $\frac{y - 10}{9} = \frac{z + 2}{12}$
18. $a_1 = 5/6 - 1/3 = 1/2$; $a_2 = -1/4 - 3/8 = -5/8$; $a_3 = 1/5 - 1/10 = 1/10$
 $\frac{x - 5/6}{1/2} = \frac{y + 1/4}{-5/8} = \frac{z - 1/5}{1/10}$
19. parametric: $x = 4 + 3t$, $y = 6 + t/2$, $z = -7 - 3t/2$; symmetric: $\frac{x - 4}{3} = \frac{y - 6}{1/2} = \frac{z + 7}{-3/2}$
20. parametric: $x = 1 - 7t$, $y = 8 - 8t$, $z = -2$; symmetric: $\frac{x - 1}{-7} = \frac{y - 8}{-8}$, $z = -2$
21. parametric: $x = 5t$, $y = 9t$, $z = 4t$; symmetric: $\frac{x}{5} = \frac{y}{9} = \frac{z}{4}$
22. parametric: $x = 12t$, $y = -3 - 5t$, $z = 10 - 6t$; symmetric: $\frac{x}{12} = \frac{y + 3}{-5} = \frac{z - 10}{-6}$
23. Writing the given line in the form $x/2 = (y - 1)/(-3) = (z - 5)/6$, we see that a direction vector is $\langle 2, -3, 6 \rangle$.
 Parametric equations for the line are $x = 6 + 2t$, $y = 4 - 3t$, $z = -2 + 6t$.

24. A direction vector is $\langle 5, 1/3, -2 \rangle$. Symmetric equations for the line are $(x-4)/5 = (y+11)/(1/3) = (z+7)/(-2)$.
25. A direction vector parallel to both the xz - and xy -planes is $\mathbf{i} = \langle 1, 0, 0 \rangle$. Parametric equations for the line are $x = 2 + t$, $y = -2$, $z = 15$.
26. (a) Since the unit vector $\mathbf{j} = \langle 0, 1, 0 \rangle$ lies along the y -axis, we have $x = 1$, $y = 2 + t$, $z = 8$.
 (b) since the unit vector $\mathbf{k} = \langle 0, 0, 1 \rangle$ is perpendicular to the xy -plane, we have $x = 1$, $y = 2$, $z = 8 + t$.
27. Both lines go through the points $(0, 0, 0)$ and $(6, 6, 6)$. Since two points determine a line, the lines are the same.
28. $\mathbf{a}$ and $\mathbf{f}$ are parallel since $\langle 9, -12, 6 \rangle = -3\langle -3, 4, -2 \rangle$. $\mathbf{c}$ and $\mathbf{d}$ are orthogonal since $\langle 2, -3, 4 \rangle \cdot \langle 1, 4, 5/2 \rangle = 0$.
29. In the xy -plane, $z = 9 + 3t = 0$ and $t = -3$. Then $x = 4 - 2(-3) = 10$ and $y = 1 + 2(-3) = -5$. The point is $(10, -5, 0)$. In the xz -plane, $y = 1 + 2t = 0$ and $t = -1/2$. Then $x = 4 - 2(-1/2) = 5$ and $z = 9 + 3(-1/2) = 15/2$. The point is $(5, 0, 15/2)$. In the yz -plane, $x = 4 - 2t = 0$ and $t = 2$. Then $y = 1 + 2(2) = 5$ and $z = 9 + 3(2) = 15$. The point is $(0, 5, 15)$.
30. The parametric equations for the line are $x = 1 + 2t$, $y = -2 + 3t$, $z = 4 + 2t$. In the xy -plane, $z = 4 + 2t = 0$ and $t = -2$. Then $x = 1 + 2(-2) = -3$ and $y = -2 + 3(-2) = -8$. The point is $(-3, -8, 0)$. In the xz -plane, $y = -2 + 3t = 0$ and $t = 2/3$. Then $x = 1 + 2(2/3) = 7/3$ and $z = 4 + 2(2/3) = 16/3$. The point is $(7/3, 0, 16/3)$. In the yz -plane, $x = 1 + 2t = 0$ and $t = -1/2$. Then $y = -2 + 3(-1/2) = -7/2$ and $z = 4 + 2(-1/2) = 3$. The point is $(0, -7/2, 3)$.
31. Solving the system $4 + t = 6 + 2s$, $5 + t = 11 + 4s$, $-1 + 2t = -3 + s$, or $t - 2s = 2$, $t - 4s = 6$, $2t - s = -2$ yields $s = -2$ and $t = -2$ in all three equations. Thus, the lines intersect at the point $x = 4 + (-2) = 2$, $y = 5 + (-2) = 3$, $z = -1 + 2(-2) = -5$, or $(2, 3, -5)$.
32. Solving the system $1 + t = 2 - s$, $2 - t = 1 + s$, $3t = 6s$, or $t + s = 1$, $t + s = 1$, $t - 2s = 0$ yields $s = 1/3$ and $t = 2/3$ in all three equations. Thus, the lines intersect at the point $x = 1 + 2/3 = 5/3$, $y = 2 - 2/3 = 4/3$, $z = 3(2/3) = 2$, or $(5/3, 4/3, 2)$.
33. The system of equations $2 - t = 4 + s$, $3 + t = 1 + s$, $1 + t = 1 - s$, or $t + s = -2$, $t - s = -2$, $t + s = 0$ has no solution since $-2 \neq 0$. Thus, the lines do not intersect.
34. Solving the system $3 - t = 2 + 2s$, $2 + t = -2 + 3s$, $8 + 2t = -2 + 8s$, or $t + 2s = 1$, $t - 3s = -4$, $2t - 8s = -10$ yields $s = 1$ and $t = -1$ in all three equations. Thus, the lines intersect at the point $x = 3 - (-1) = 4$, $y = 2 + (-1) = 1$, $z = 8 + 2(-1) = 6$, or $(4, 1, 6)$.
35. $\mathbf{a} = \langle -1, 2, -2 \rangle$, $\mathbf{b} = \langle 2, 3, -6 \rangle$, $\mathbf{a} \cdot \mathbf{b} = 16$, $\|\mathbf{a}\| = 3$, $\|\mathbf{b}\| = 7$; $\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} = \frac{16}{3 \cdot 7}$;
 $\theta = \cos^{-1} \frac{16}{21} \approx 40.37^\circ$
36. $\mathbf{a} = \langle 2, 7, -1 \rangle$, $\mathbf{b} = \langle -2, 1, 4 \rangle$, $\mathbf{a} \cdot \mathbf{b} = -1$, $\|\mathbf{a}\| = 3\sqrt{6}$, $\|\mathbf{b}\| = \sqrt{21}$;
 $\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} = \frac{-1}{(3\sqrt{6})(\sqrt{21})} = -\frac{1}{9\sqrt{14}}$; $\theta = \cos^{-1}(-\frac{1}{9\sqrt{14}}) \approx 91.70^\circ$
37. A direction vector perpendicular to the given lines will be $\langle 1, 1, 1 \rangle \times \langle -2, 1, -5 \rangle = \langle -6, 3, 3 \rangle$. Equations of the line are $x = 4 - 6t$, $y = 1 + 3t$, $z = 6 + 3t$.
38. The direction vectors of the given lines are $\langle 3, 2, 4 \rangle$ and $\langle 6, 4, 8 \rangle = 2\langle 3, 2, 4 \rangle$. These are parallel, so we need a third vector parallel to the plane containing the lines which is not parallel to them. The point $(1, -1, 0)$ is on the first line and $(-4, 6, 10)$ is on the second line. A third vector is then $\langle 1, -1, 0 \rangle - \langle -4, 6, 10 \rangle = \langle 5, -7, -10 \rangle$. Now a direction vector perpendicular to the plane is $\langle 3, 2, 4 \rangle \times \langle 5, -7, -10 \rangle = \langle 8, 50, -31 \rangle$. Equations of the line through $(1, -1, 0)$ and perpendicular to the plane are $x = 1 + 8t$, $y = -1 + 50t$, $z = -31t$.

Exercises 7.5

39. $2(x-5) - 3(y-1) + 4(z-3) = 0$; $2x - 3y + 4z = 19$
40. $4(x-1) - 2(y-2) + 0(z-5) = 0$; $4x - 2y = 0$
41. $-5(x-6) + 0(y-10) + 3(z+7) = 0$; $-5x + 3z = -51$
42. $6x - y + 3z = 0$
43. $6(x-1/2) + 8(y-3/4) - 4(z+1/2) = 0$; $6x + 8y - 4z = 11$
44. $-(x+1) + (y-1) - (z-0) = 0$; $-x + y - z = 2$
45. From the points $(3, 5, 2)$ and $(2, 3, 1)$ we obtain the vector $\mathbf{u} = \mathbf{i} + 2\mathbf{j} + \mathbf{k}$. From the points $(2, 3, 1)$ and $(-1, -1, 4)$ we obtain the vector $\mathbf{v} = 3\mathbf{i} + 4\mathbf{j} - 3\mathbf{k}$. From the points $(-1, -1, 4)$ and (x, y, z) we obtain the vector $\mathbf{w} = (x+1)\mathbf{i} + (y+1)\mathbf{j} + (z-4)\mathbf{k}$. Then, a normal vector is

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & 1 \\ 3 & 4 & -3 \end{vmatrix} = -10\mathbf{i} + 6\mathbf{j} - 2\mathbf{k}.$$

A vector equation of the plane is $-10(x+1) + 6(y+1) - 2(z-4) = 0$ or $5x - 3y + z = 2$.

46. From the points $(0, 1, 0)$ and $(0, 1, 1)$ we obtain the vector $\mathbf{u} = \mathbf{k}$. From the points $(0, 1, 1)$ and $(1, 3, -1)$ we obtain the vector $\mathbf{v} = \mathbf{i} + 2\mathbf{j} - 2\mathbf{k}$. From the points $(1, 3, -1)$ and (x, y, z) we obtain the vector $\mathbf{w} = (x-1)\mathbf{i} + (y-3)\mathbf{j} + (z+1)\mathbf{k}$. Then, a normal vector is

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 0 & 1 \\ 1 & 2 & -2 \end{vmatrix} = -2\mathbf{i} + \mathbf{j}.$$

A vector equation of the plane is $-2(x-1) + (y-3) + 0(z+1) = 0$ or $-2x + y = 1$.

47. From the points $(0, 0, 0)$ and $(1, 1, 1)$ we obtain the vector $\mathbf{u} = \mathbf{i} + \mathbf{j} + \mathbf{k}$. From the points $(1, 1, 1)$ and $(3, 2, -1)$ we obtain the vector $\mathbf{v} = 2\mathbf{i} + \mathbf{j} - 2\mathbf{k}$. From the points $(3, 2, -1)$ and (x, y, z) we obtain the vector $\mathbf{w} = (x-3)\mathbf{i} + (y-2)\mathbf{j} + (z+1)\mathbf{k}$. Then, a normal vector is

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 1 \\ 2 & 1 & -2 \end{vmatrix} = -3\mathbf{i} + 4\mathbf{j} - \mathbf{k}.$$

A vector equation of the plane is $-3(x-3) + 4(y-2) - (z+1) = 0$ or $-3x + 4y - z = 0$.

48. The three points are not colinear and all satisfy $x = 0$, which is the equation of the plane.
49. From the points $(1, 2, -1)$ and $(4, 3, 1)$ we obtain the vector $\mathbf{u} = 3\mathbf{i} + \mathbf{j} + 2\mathbf{k}$. From the points $(4, 3, 1)$ and $(7, 4, 3)$ we obtain the vector $\mathbf{v} = 3\mathbf{i} + \mathbf{j} + 2\mathbf{k}$. From the points $(7, 4, 3)$ and (x, y, z) we obtain the vector $\mathbf{w} = (x-7)\mathbf{i} + (y-4)\mathbf{j} + (z-3)\mathbf{k}$. Since $\mathbf{u} \times \mathbf{v} = \mathbf{0}$, the points are colinear.
50. From the points $(2, 1, 2)$ and $(4, 1, 0)$ we obtain the vector $\mathbf{u} = 2\mathbf{i} - 2\mathbf{k}$. From the points $(4, 1, 0)$ and $(5, 0, -5)$ we obtain the vector $\mathbf{v} = \mathbf{i} - \mathbf{j} - 5\mathbf{k}$. From the points $(5, 0, -5)$ and (x, y, z) we obtain the vector $\mathbf{w} = (x-5)\mathbf{i} + y\mathbf{j} + (z+5)\mathbf{k}$. Then, a normal vector is

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 0 & -2 \\ 1 & -1 & -5 \end{vmatrix} = -2\mathbf{i} + 8\mathbf{j} - 2\mathbf{k}.$$

A vector equation of the plane is $-2(x-5) + 8y - 2(z+5) = 0$ or $x - 4y + z = 0$.

51. A normal vector to $x + y - 4z = 1$ is $\langle 1, 1, -4 \rangle$. The equation of the parallel plane is $(x - 2) + (y - 3) - 4(z + 5) = 0$ or $x + y - 4z = 25$.
52. A normal vector to $5x - y + z = 6$ is $\langle 5, -1, 1 \rangle$. The equation of the parallel plane is $5(x - 0) - (y - 0) + (z - 0) = 0$ or $5x - y + z = 0$.
53. A normal vector to the xy -plane is $\langle 0, 0, 1 \rangle$. The equation of the parallel plane is $z - 12 = 0$ or $z = 12$.
54. A normal vector is $\langle 0, 1, 0 \rangle$. The equation of the plane is $y + 5 = 0$ or $y = -5$.
55. Direction vectors of the lines are $\langle 3, -1, 1 \rangle$ and $\langle 4, 2, 1 \rangle$. A normal vector to the plane is $\langle 3, -1, 1 \rangle \times \langle 4, 2, 1 \rangle = \langle -3, 1, 10 \rangle$. A point on the first line, and thus in the plane, is $\langle 1, 1, 2 \rangle$. The equation of the plane is $-3(x - 1) + (y - 1) + 10(z - 2) = 0$ or $-3x + y + 10z = 18$.
56. Direction vectors of the lines are $\langle 2, -1, 6 \rangle$ and $\langle 1, 1, -3 \rangle$. A normal vector to the plane is $\langle 2, -1, 6 \rangle \times \langle 1, 1, -3 \rangle = \langle -3, 12, 3 \rangle$. A point on the first line, and thus in the plane, is $\langle 1, -1, 5 \rangle$. The equation of the plane is $-3(x - 1) + 12(y + 1) + 3(z - 5) = 0$ or $-x + 4y + z = 0$.
57. A direction vector for the two lines is $\langle 1, 2, 1 \rangle$. Points on the lines are $\langle 1, 1, 3 \rangle$ and $\langle 3, 0, -2 \rangle$. Thus, another vector parallel to the plane is $\langle 1 - 3, 1 - 0, 3 + 2 \rangle = \langle -2, 1, 5 \rangle$. A normal vector to the plane is $\langle 1, 2, 1 \rangle \times \langle -2, 1, 5 \rangle = \langle 9, -7, 5 \rangle$. Using the point $\langle 3, 0, -2 \rangle$ in the plane, the equation of the plane is $9(x - 3) - 7(y - 0) + 5(z + 2) = 0$ or $9x - 7y + 5z = 17$.
58. A direction vector for the line is $\langle 3, 2, -2 \rangle$. Letting $t = 0$, we see that the origin is on the line and hence in the plane. Thus, another vector parallel to the plane is $\langle 4 - 0, 0 - 0, -6 - 0 \rangle = \langle 4, 0, -6 \rangle$. A normal vector to the plane is $\langle 3, 2, -2 \rangle \times \langle 4, 0, -6 \rangle = \langle -12, 10, -8 \rangle$. The equation of the plane is $-12(x - 0) + 10(y - 0) - 8(z - 0) = 0$ or $6x - 5y + 4z = 0$.
59. A direction vector for the line, and hence a normal vector to the plane, is $\langle -3, 1, -1/2 \rangle$. The equation of the plane is $-3(x - 2) + (y - 4) - \frac{1}{2}(z - 8) = 0$ or $-3x + y - \frac{1}{2}z = -6$.
60. A normal vector to the plane is $\langle 2 - 1, 6 - 0, -3 + 2 \rangle = \langle 1, 6, -1 \rangle$. The equation of the plane is $(x - 1) + 6(y - 1) - (z - 1) = 0$ or $x + 6y - z = 6$.
61. Normal vectors to the planes are (a) $\langle 2, -1, 3 \rangle$, (b) $\langle 1, 2, 2 \rangle$, (c) $\langle 1, 1, -3/2 \rangle$, (d) $\langle -5, 2, 4 \rangle$, (e) $\langle -8, -8, 12 \rangle$, (f) $\langle -2, 1, -3 \rangle$. Parallel planes are (c) and (e), and (a) and (f). Perpendicular planes are (a) and (d), (b) and (c), (b) and (e), and (d) and (f).
62. A normal vector to the plane is $\langle -7, 2, 3 \rangle$. This is a direction vector for the line and the equations of the line are $x = -4 - 7t$, $y = 1 + 2t$, $z = 7 + 3t$.
63. A direction vector of the line is $\langle -6, 9, 3 \rangle$, and the normal vectors of the planes are (a) $\langle 4, 1, 2 \rangle$, (b) $\langle 2, -3, 1 \rangle$, (c) $\langle 10, -15, -5 \rangle$, (d) $\langle -4, 6, 2 \rangle$. Vectors (c) and (d) are multiples of the direction vector and hence the corresponding planes are perpendicular to the line.
64. A direction vector of the line is $\langle -2, 4, 1 \rangle$, and normal vectors to the planes are (a) $\langle 1, -1, 3 \rangle$, (b) $\langle 6, -3, 0 \rangle$, (c) $\langle 1, -2, 5 \rangle$, (d) $\langle -2, 1, -2 \rangle$. Since the dot product of each normal vector with the direction vector is non-zero, none of the planes are parallel to the line.
65. Letting $z = t$ in both equations and solving $5x - 4y = 8 + 9t$, $x + 4y = 4 - 3t$, we obtain $x = 2 + t$, $y = \frac{1}{2} - t$, $z = t$.
66. Letting $y = t$ in both equations and solving $x - z = 2 - 2t$, $3x + 2z = 1 + t$, we obtain $x = 1 - \frac{3}{5}t$, $y = t$, $z = -1 + \frac{7}{5}t$ or, letting $t = 5s$, $x = 1 - 3s$, $y = 5s$, $z = -1 + 7s$.
67. Letting $z = t$ in both equations and solving $4x - 2y = 1 + t$, $x + y = 1 - 2t$, we obtain $x = \frac{1}{2} - \frac{1}{2}t$, $y = \frac{1}{2} - \frac{3}{2}t$, $z = t$.

Exercises 7.5

68. Letting $z = t$ and using $y = 0$ in the first equation, we obtain $x = -\frac{1}{2}t$, $y = 0$, $z = t$.
69. Substituting the parametric equations into the equation of the plane, we obtain $2(1 + 2t) - 3(2 - t) + 2(-3t) = -7$ or $t = -3$. Letting $t = -3$ in the equation of the line, we obtain the point of intersection $(-5, 5, 9)$.
70. Substituting the parametric equations into the equation of the plane, we obtain $(3 - 2t) + (1 + 6t) + 4(2 - \frac{1}{2}t) = 12$ or $2t = 0$. Letting $t = 0$ in the equation of the line, we obtain the point of intersection $(3, 1, 2)$.
71. Substituting the parametric equations into the equation of the plane, we obtain $1 + 2 - (1 + t) = 8$ or $t = -6$. Letting $t = -6$ in the equation of the line, we obtain the point of intersection $(1, 2, -5)$.
72. Substituting the parametric equations into the equation of the plane, we obtain $4 + t - 3(2 + t) + 2(1 + 5t) = 0$ or $t = 0$. Letting $t = 0$ in the equation of the line, we obtain the point of intersection $(4, 2, 1)$.

In Problems 73 and 74, the cross product of the normal vectors to the two planes will be a vector parallel to both planes, and hence a direction vector for a line parallel to the two planes.

73. Normal vectors are $\langle 1, 1, -4 \rangle$ and $\langle 2, -1, 1 \rangle$. A direction vector is

$$\langle 1, 1, -4 \rangle \times \langle 2, -1, 1 \rangle = \langle -3, -9, -3 \rangle = -3\langle 1, 3, 1 \rangle.$$

Equations of the line are $x = 5 + t$, $y = 6 + 3t$, $z = -12 + t$.

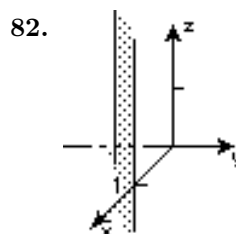
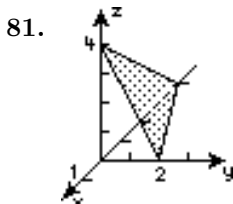
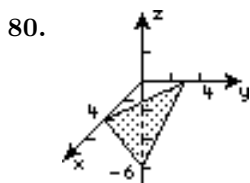
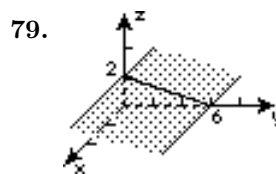
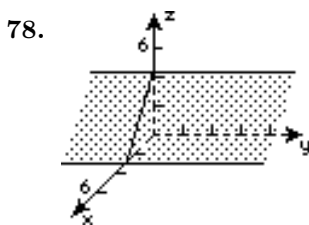
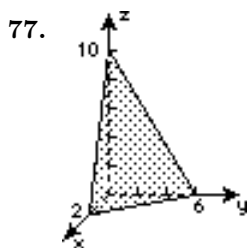
74. Normal vectors are $\langle 2, 0, 1 \rangle$ and $\langle -1, 3, 1 \rangle$. A direction vector is

$$\langle 2, 0, 1 \rangle \times \langle -1, 3, 1 \rangle = \langle -3, -3, 6 \rangle = -3\langle 1, 1, -2 \rangle.$$

Equations of the line are $x = -3 + t$, $y = 5 + t$, $z = -1 - 2t$.

In Problems 75 and 76, the cross product of the direction vector of the line with the normal vector of the given plane will be a normal vector to the desired plane.

75. A direction vector of the line is $\langle 3, -1, 5 \rangle$ and a normal vector to the given plane is $\langle 1, 1, 1 \rangle$. A normal vector to the desired plane is $\langle 3, -1, 5 \rangle \times \langle 1, 1, 1 \rangle = \langle -6, 2, 4 \rangle$. A point on the line, and hence in the plane, is $(4, 0, 1)$. The equation of the plane is $-6(x - 4) + 2(y - 0) + 4(z - 1) = 0$ or $3x - y - 2z = 10$.
76. A direction vector of the line is $\langle 3, 5, 2 \rangle$ and a normal vector to the given plane is $\langle 2, -4, -1 \rangle$. A normal vector to the desired plane is $\langle -3, 5, 2 \rangle \times \langle 2, -4, -1 \rangle = \langle 3, 1, 2 \rangle$. A point on the line, and hence in the plane, is $(2, -2, 8)$. The equation of the plane is $3(x - 2) + (y + 2) + 2(z - 8) = 0$ or $3x + y + 2z = 20$.



Exercises 7.6

1. Not a vector space. Axiom (vi) is not satisfied.
2. Not a vector space. Axiom (i) is not satisfied.
3. Not a vector space. Axiom (x) is not satisfied.
4. A vector space
5. A vector space
6. A vector space
7. Not a vector space. Axiom (ii) is not satisfied.
8. A vector space
9. A vector space
10. Not a vector space. Axiom (i) is not satisfied.
11. A subspace
12. Not a subspace. Axiom (i) is not satisfied.
13. Not a subspace. Axiom (ii) is not satisfied.
14. A subspace
15. A subspace
16. A subspace
17. A subspace
18. A subspace
19. Not a subspace. Neither axioms (i) nor(ii) are satisfied.
20. A subspace
21. Let (x_1, y_1, z_1) and (x_2, y_2, z_2) be in S . Then

$$(x_1, y_1, z_1) + (x_2, y_2, z_2) = (at_1, bt_1, ct_1) + (at_2, bt_2, ct_2) = (a(t_1 + t_2), b(t_1 + t_2), c(t_1 + t_2))$$

is in S . Also, for (x, y, z) in S then $k(x, y, z) = (kx, ky, kz) = (a(kt), b(kt), c(kt))$ is also in S .

22. Let (x_1, y_1, z_1) and (x_2, y_2, z_2) be in S . Then $ax_1 + by_1 + cz_1 = 0$ and $ax_2 + by_2 + cz_2 = 0$. Adding gives $a(x_1 + x_2) + b(y_1 + y_2) + c(z_1 + z_2) = 0$ and so $(x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2, y_1 + y_2, z_1 + z_2)$ is in S . Also, for (x, y, z) then $ax + by + cz = 0$ implies $k(ax + by + cz) = k \cdot 0 = 0$ and $a(kx) + b(ky) + c(kz) = 0$. this means $k(x, y, z) = (kx, ky, kz)$ is in S .
23. (a) $c_1 \mathbf{u}_1 + c_2 \mathbf{u}_2 + c_3 \mathbf{u}_3 = \mathbf{0}$ if and only if $c_1 + c_2 + c_3 = 0$, $c_2 + c_3 = 0$, $c_3 = 0$. The only solution of this system is $c_1 = 0$, $c_2 = 0$, $c_3 = 0$.
 (b) Solving the system $c_1 + c_2 + c_3 = 3$, $c_2 + c_3 = -4$, $c_3 = 8$ gives $c_1 = 7$, $c_2 = -12$, $c_3 = 8$. Thus $\mathbf{a} = 7\mathbf{u}_1 - 12\mathbf{u}_2 + 8\mathbf{u}_3$.
24. (a) The assumption $c_1 p_1 + c_2 p_2 = 0$ is equivalent to $(c_1 + c_2)x + (c_1 - c_2) = 0$. Thus $c_1 + c_2 = 0$, $c_1 - c_2 = 0$. The only solution of this system is $c_1 = 0$, $c_2 = 0$.
 (b) Solving the system $c_1 + c_2 = 5$, $c_1 - c_2 = 2$ gives $c_1 = \frac{7}{2}$, $c_2 = \frac{3}{2}$. Thus $p(x) = \frac{7}{2}p_1(x) + \frac{3}{2}p_2(x)$
25. Linearly dependent since $\langle -6, 12 \rangle = -\frac{3}{2}\langle 4, -8 \rangle$
26. Linearly dependent since $2\langle 1, 1 \rangle + 3\langle 0, 1 \rangle + (-1)\langle 2, 5 \rangle = \langle 0, 0 \rangle$
27. Linearly independent
28. Linearly dependent since for all x $(1) \cdot 1 + (-2)(x+1) + (1)(x+1)^2 + (-1)x^2 = 0$.
29. f is discontinuous at $x = -1$ and at $x = -3$.
30. $(x, \sin x) = \int_0^{2\pi} x \sin x \, dx = (-x \cos x + \sin x) \Big|_0^{2\pi} = -2\pi$

Exercises 7.6

31. $\|x\|^2 = \int_0^{2\pi} x^2 dx = \frac{1}{3}x^3 \Big|_0^{2\pi} = \frac{8}{3}\pi^3$ and so $\|x\| = 2\sqrt{\frac{2\pi^3}{3}}$. Now

$$\|\sin x\|^2 = \int_0^{2\pi} \sin^2 x dx = \frac{1}{2} \int_0^{2\pi} (1 - \cos 2x) dx = \frac{1}{2} \left(x - \frac{1}{2} \sin 2x \right) \Big|_0^{2\pi} = \pi$$

and so $\|\sin x\| = \sqrt{\pi}$.

32. A basis could be $1, x, e^x \cos 3x, e^x \sin 3x$.

33. We need to show that $\text{Span}\{x_1, x_2, \dots, x_n\}$ is closed under vector addition and scalar multiplication. Suppose $\mathbf{u}$ and $\mathbf{v}$ are in $\text{Span}\{x_1, x_2, \dots, x_n\}$. Then $\mathbf{u} = a_1x_1 + a_2x_2 + \dots + a_nx_n$ and $\mathbf{v} = b_1x_1 + b_2x_2 + \dots + b_nx_n$, so that

$$\mathbf{u} + \mathbf{v} = (a_1 + b_1)x_1 + (a_2 + b_2)x_2 + \dots + (a_n + b_n)x_n,$$

which is in $\text{Span}\{x_1, x_2, \dots, x_n\}$. Also, for any real number k ,

$$k\mathbf{u} = k(a_1x_1 + a_2x_2 + \dots + a_nx_n) = ka_1x_1 + ka_2x_2 + \dots + ka_nx_n,$$

which is in $\text{Span}\{x_1, x_2, \dots, x_n\}$. Thus, $\text{Span}\{x_1, x_2, \dots, x_n\}$ is a subspace of $\mathbf{V}$.

Chapter 7 Review Exercises

1. True
2. False; the points must be non-collinear.
3. False; since a normal to the plane is $\langle 2, 3, -4 \rangle$ which is not a multiple of the direction vector $\langle 5, -2, 1 \rangle$ of the line.
4. True
5. True
6. True
7. True
8. True
9. True
10. True; since $\mathbf{a} \times \mathbf{b}$ and $\mathbf{c} \times \mathbf{d}$ are both normal to the plane and hence parallel (unless $\mathbf{a} \times \mathbf{b} = \mathbf{0}$ or $\mathbf{c} \times \mathbf{d} = \mathbf{0}$.)
11. $9\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$
12. orthogonal
13. $-5(\mathbf{k} \times \mathbf{j}) = -5(-\mathbf{i}) = 5\mathbf{i}$
14. $\mathbf{i} \cdot (\mathbf{i} \times \mathbf{j}) = \mathbf{i} \times \mathbf{k} = \mathbf{0}$
15. $\sqrt{(-12)^2 + 4^2 + 6^2} = 14$
16. $(-1 - 20)\mathbf{i} - (-2 - 0)\mathbf{j} + (8 - 0)\mathbf{k} = -21\mathbf{i} + 2\mathbf{j} + 8\mathbf{k}$
17. $-6\mathbf{i} + \mathbf{j} - 7\mathbf{k}$
18. The coordinates of $(1, -2, -10)$ satisfy the given equation.
19. Writing the line in parametric form, we have $x = 1 + t, y = -2 + 3t, z = -1 + 2t$. Substituting into the equation of the plane yields $(1 + t) + 2(-2 + 3t) - (-1 + 2t) = 13$ or $t = 3$. Thus, the point of intersection is $x = 1 + 3 = 4, y = -2 + 3(3) = 7, z = -1 + 2(3) = 5$, or $(4, 7, 5)$.
20. $|\mathbf{a}| = \sqrt{4^2 + 3^2 + (-5)^2} = 5\sqrt{2}; \mathbf{u} = -\frac{1}{5\sqrt{2}}(4\mathbf{i} + 3\mathbf{j} - 5\mathbf{k}) = -\frac{4}{5\sqrt{2}}\mathbf{i} - \frac{3}{5\sqrt{2}}\mathbf{j} + \frac{1}{\sqrt{2}}\mathbf{k}$
21. $x_2 - 2 = 3, x_2 = 5; y_2 - 1 = 5, y_2 = 6; z_2 - 7 = -4, z_2 = 3; P_2 = (5, 6, 3)$
22. $(5, 1/2, 5/2)$
23. $(7.2)(10) \cos 135^\circ = -36\sqrt{2}$

24. $2\mathbf{b} = \langle -2, 4, 2 \rangle$; $4\mathbf{c} = \langle 0, -8, 8 \rangle$; $\mathbf{a} \cdot (2\mathbf{b} + 4\mathbf{c}) = \langle 3, 1, 0 \rangle \cdot \langle -2, -4, 10 \rangle = -10$

25. 12, -8, 6

26. $\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|\|\mathbf{b}\|} = \frac{1}{\sqrt{2}\sqrt{2}} = \frac{1}{2}$; $\theta = 60^\circ$

27. $A = \frac{1}{2}|5\mathbf{i} - 4\mathbf{j} - 7\mathbf{k}| = \frac{3\sqrt{10}}{2}$

28. From $3(x - 3) + 0(y - 6) + (1)(z - (-2)) = 0$ we obtain $3x + z = 7$.

29. $|-5 - (-3)| = 2$

30. parallel: $-2c = 5$, $c = -5/2$; orthogonal: $1(-2) + 3(-6) + c(5) = 0$, $c = 4$

31. $\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 0 \\ 1 & -2 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ -2 & 1 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 1 & 1 \\ 1 & -2 \end{vmatrix} \mathbf{k} = \mathbf{i} - \mathbf{j} - 3\mathbf{k}$

A unit vector perpendicular to both $\mathbf{a}$ and $\mathbf{b}$ is

$$\frac{\mathbf{a} \times \mathbf{b}}{\|\mathbf{a} \times \mathbf{b}\|} = \frac{1}{\sqrt{1+1+9}}(\mathbf{i} - \mathbf{j} - 3\mathbf{k}) = \frac{1}{\sqrt{11}}\mathbf{i} - \frac{1}{\sqrt{11}}\mathbf{j} - \frac{3}{\sqrt{11}}\mathbf{k}.$$

32. $\|\mathbf{a}\| = \sqrt{1/4 + 1/4 + 1/6} = \frac{3}{4}$; $\cos \alpha = \frac{1/2}{3/4} = \frac{2}{3}$, $\alpha \approx 48.19^\circ$; $\cos \beta = \frac{1/2}{3/4} = \frac{2}{3}$, $\beta \approx 48.19^\circ$;
 $\cos \gamma = \frac{-1/4}{3/4} = -\frac{1}{3}$, $\gamma \approx 109.47^\circ$

33. $\text{comp}_{\mathbf{b}} \mathbf{a} = \mathbf{a} \cdot \mathbf{b} / \|\mathbf{b}\| = \langle 1, 2, -2 \rangle \cdot \langle 4, 3, 0 \rangle / 5 = 2$

34. $\text{comp}_{\mathbf{a}} \mathbf{b} = \mathbf{b} \cdot \mathbf{a} / \|\mathbf{a}\| = \langle 4, 3, 0 \rangle \cdot \langle 1, 2, -2 \rangle / 3 = 10/3$
 $\text{proj}_{\mathbf{a}} \mathbf{b} = (\text{comp}_{\mathbf{a}} \mathbf{b})\mathbf{a} / \|\mathbf{a}\| = (10/3)\langle 1, 2, -2 \rangle / 3 = \langle 10/9, 20/9, -20/9 \rangle$

35. Using the result of Problem 34,

$$\text{proj}_{\mathbf{a}^\perp} \mathbf{b} = \mathbf{b} - \text{proj}_{\mathbf{a}} \mathbf{b} = \langle 4, 3, 0 \rangle - \langle 10/9, 20/9, -20/9 \rangle = \langle 26/9, 7/9, 20/9 \rangle.$$

36. $\text{comp}_{\mathbf{b}} (\mathbf{a} - \mathbf{b}) = (\mathbf{a} - \mathbf{b}) \cdot \mathbf{b} / \|\mathbf{b}\| = \langle -3, -1, -2 \rangle \cdot \langle 4, 3, 0 \rangle / 5 = -3$
 $\text{proj}_{\mathbf{b}} (\mathbf{a} - \mathbf{b}) = (\text{comp}_{\mathbf{b}} (\mathbf{a} - \mathbf{b}))\mathbf{b} / \|\mathbf{b}\| = -3\langle 4, 3, 0 \rangle / 5 = \langle -12/5, -9/5, 0 \rangle$
 $\text{proj}_{\mathbf{b}^\perp} (\mathbf{a} - \mathbf{b}) = (\mathbf{a} - \mathbf{b}) - \text{proj}_{\mathbf{b}} (\mathbf{a} - \mathbf{b}) = \langle -3, -1, -2 \rangle - \langle -12/5, -9/5, 0 \rangle = \langle -3/5, 4/5, -10/5 \rangle$

37. Let $\mathbf{a} = \langle a, b, c \rangle$ and $\mathbf{r} = \langle x, y, z \rangle$. Then

(a) $(\mathbf{r} - \mathbf{a}) \cdot \mathbf{r} = \langle x - a, y - b, z - c \rangle \cdot \langle x, y, z \rangle = x^2 - ax + y^2 - by + z^2 - cz = 0$ implies

$$(x - \frac{a}{2})^2 + (y - \frac{b}{2})^2 + (z - \frac{c}{2})^2 = \frac{a^2 + b^2 + c^2}{4}. \quad \text{The surface is a sphere.}$$

(b) $(\mathbf{r} - \mathbf{a}) \cdot \mathbf{a} = \langle x - a, y - b, z - c \rangle \cdot \langle a, b, c \rangle = a(x - a) + b(y - b) + c(z - c) = 0$
 The surface is a plane.

38. $\langle 4, 2, -2 \rangle - \langle 2, 4, -3 \rangle = \langle 2, -2, 1 \rangle$; $\langle 2, 4, -3 \rangle - \langle 6, 7, -5 \rangle = \langle -4, -3, 2 \rangle$; $\langle 2, -2, 1 \rangle \cdot \langle -4, -3, 2 \rangle = 0$
 The points are the vertices of a right triangle.

39. A direction vector of the given line is $\langle 4, -2, 6 \rangle$. A parallel line containing $(7, 3, -5)$ is $(x - 7)/4 = (y - 3)/(-2) = (z + 5)/6$.

40. A normal to the plane is $\langle 8, 3, -4 \rangle$. The line with this direction vector and through $(5, -9, 3)$ is $x = 5 + 8t$, $y = -9 + 3t$, $z = 3 - 4t$.

Chapter 7 Review Exercises

41. The direction vectors are $\langle -2, 3, 1 \rangle$ and $\langle 2, 1, 1 \rangle$. Since $\langle -2, 3, 1 \rangle \cdot \langle 2, 1, 1 \rangle = 0$, the lines are orthogonal. Solving $1 - 2t = x = 1 + 2s$, $3t = y = -4 + s$, we obtain $t = -1$ and $s = 1$. The point $(3, -3, 0)$ obtained by letting $t = -1$ and $s = 1$ is common to the two lines, so they do intersect.
42. Vectors in the plane are $\langle 2, 3, 1 \rangle$ and $\langle 1, 0, 2 \rangle$. A normal vector is $\langle 2, 3, 1 \rangle \times \langle 1, 0, 2 \rangle = \langle 6, -3, -3 \rangle = 3\langle 2, -1, -1 \rangle$. An equation of the plane is $2x - y - z = 0$.
43. The lines are parallel with direction vector $\langle 1, 4, -2 \rangle$. Since $(0, 0, 0)$ is on the first line and $(1, 1, 3)$ is on the second line, the vector $\langle 1, 1, 3 \rangle$ is in the plane. A normal vector to the plane is thus $\langle 1, 4, -2 \rangle \times \langle 1, 1, 3 \rangle = \langle 14, -5, -3 \rangle$. An equation of the plane is $14x - 5y - 3z = 0$.
44. Letting $z = t$ in the equations of the plane and solving $-x + y = 4 + 8t$, $3x - y = -2t$, we obtain $x = 2 + 3t$, $y = 6 + 11t$, $z = t$. Thus, a normal to the plane is $\langle 3, 11, 1 \rangle$ and an equation of the plane is

$$3(x - 1) + 11(y - 7) + (z + 1) = 0 \quad \text{or} \quad 3x + 11y + z = 79.$$

45. $\mathbf{F} = 10 \frac{\mathbf{a}}{\|\mathbf{a}\|} = \frac{10}{\sqrt{2}}(\mathbf{i} + \mathbf{j}) = 5\sqrt{2}\mathbf{i} + 5\sqrt{2}\mathbf{j}$; $\mathbf{d} = \langle 7, 4, 0 \rangle - \langle 4, 1, 0 \rangle = 3\mathbf{i} + 3\mathbf{j}$

$$W = \mathbf{F} \cdot \mathbf{d} = 15\sqrt{2} + 15\sqrt{2} = 30\sqrt{2} \text{ N-m}$$

46. $\mathbf{F} = 5\sqrt{2}\mathbf{i} + 5\sqrt{2}\mathbf{j} + 50\mathbf{i} = (5\sqrt{2} + 50)\mathbf{i} + 5\sqrt{2}\mathbf{j}$; $\mathbf{d} = 3\mathbf{i} + 3\mathbf{j}$

$$W = 15\sqrt{2} + 150 + 15\sqrt{2} = 30\sqrt{2} + 150 \text{ N-m} \approx 192.4 \text{ N-m}$$

47. Since $\mathbf{F}_2 = 200(\mathbf{i} + \mathbf{j})/\sqrt{2} = 100\sqrt{2}\mathbf{i} + 100\sqrt{2}\mathbf{j}$, $\mathbf{F}_3 = \mathbf{F}_2 - \mathbf{F}_1 = (100\sqrt{2} - 200)\mathbf{i} + 100\sqrt{2}\mathbf{j}$ and

$$\|\mathbf{F}_3\| = \sqrt{(100\sqrt{2} - 200)^2 + (100\sqrt{2})^2} = 200\sqrt{2 - \sqrt{2}} \approx 153 \text{ lb.}$$

48. Let $\|\mathbf{F}_1\| = F_1$ and $\|\mathbf{F}_2\| = F_2$. Then $\mathbf{F}_1 = F_1[(\cos 45^\circ)\mathbf{i} + (\sin 45^\circ)\mathbf{j}]$ and $\mathbf{F}_2 = F_2[(\cos 120^\circ)\mathbf{i} + (\sin 120^\circ)\mathbf{j}]$, or $\mathbf{F}_1 = F_1(\frac{1}{\sqrt{2}}\mathbf{i} + \frac{1}{\sqrt{2}}\mathbf{j})$ and $\mathbf{F}_2 = F_2(-\frac{1}{2}\mathbf{i} + \frac{\sqrt{3}}{2}\mathbf{j})$. Since $\mathbf{w} + \mathbf{F}_1 + \mathbf{F}_2 = \mathbf{0}$,

$$F_1(\frac{1}{\sqrt{2}}\mathbf{i} + \frac{1}{\sqrt{2}}\mathbf{j}) + F_2(-\frac{1}{2}\mathbf{i} + \frac{\sqrt{3}}{2}\mathbf{j}) = 50\mathbf{j}, \quad (\frac{1}{\sqrt{2}}F_1 - \frac{1}{2}F_2)\mathbf{i} + (\frac{1}{\sqrt{2}}F_1 + \frac{\sqrt{3}}{2}F_2)\mathbf{j} = 50\mathbf{j}$$

and

$$\frac{1}{\sqrt{2}}F_1 - \frac{1}{2}F_2 = 0, \quad \frac{1}{\sqrt{2}}F_1 + \frac{\sqrt{3}}{2}F_2 = 50.$$

Solving, we obtain $F_1 = 25(\sqrt{6} - \sqrt{2}) \approx 25.9 \text{ lb}$ and $F_2 = 50(\sqrt{3} - 1) \approx 36.6 \text{ lb}$.

49. Not a vector space. Axiom (viii) is not satisfied.

50. The vectors are linearly independent. The only solution of the system

$$c_1 = 0, \quad c_1 + 2c_2 + c_3 = 0, \quad 2c_1 + 3c_2 - c_3 = 0$$

is $c_1 = 0$, $c_2 = 0$, $c_3 = 0$.

51. Let p_1 and p_2 be in P_n such that $\frac{d^2 p_1}{dx^2} = 0$ and $\frac{d^2 p_2}{dx^2} = 0$. Since

$$0 = \frac{d^2 p_1}{dx^2} + \frac{d^2 p_2}{dx^2} = \frac{d^2}{dx^2}(p_1 + p_2) \quad \text{and} \quad 0 = k \frac{d^2 p_1}{dx^2} = \frac{d^2}{dx^2}(kp_1)$$

we conclude that the set of polynomials with the given property is a subspace of P_n . A basis for the subspace is $1, x$.

52. The intersection $W_1 \cap W_2$ is a subspace of V . If x and y are in $W_1 \cap W_2$ then $\mathbf{x}$ and $\mathbf{y}$ are in each subspace and so $\mathbf{x} + \mathbf{y}$ is in each subspace. That is, $\mathbf{x} + \mathbf{y}$ is in $W_1 \cap W_2$. Similarly, if $\mathbf{x}$ is in $W_1 \cap W_2$ then $\mathbf{x}$ is in each subspace and so $k\mathbf{x}$ is in each subspace. That is, $k\mathbf{x}$ is in $W_1 \cap W_2$ for any scalar k .

The union $W_1 \cup W_2$ is generally not a subspace. For example, $W_1 = \{\langle x, y \rangle \mid y = x\}$ and $W_2 = \{\langle x, y \rangle \mid y = 2x\}$ are subspaces of R^2 . Now $\langle 1, 1 \rangle$ is in W_1 and $\langle 1, 2 \rangle$ is in W_2 but $\langle 1, 1 \rangle + \langle 1, 2 \rangle = \langle 2, 3 \rangle$ is not in $W_1 \cup W_2$.

8 Matrices

Exercises 8.1

1. 2×4
2. 3×2
3. 3×3
4. 1×3
5. 3×4
6. 8×1
7. Not equal
8. Not equal
9. Not equal
10. Not equal
11. Solving $x = y - 2$, $y = 3x - 2$ we obtain $x = 2$, $y = 4$.
12. Solving $x^2 = 9$, $y = 4x$ we obtain $x = 3$, $y = 12$ and $x = -3$, $y = -12$.
13. $c_{23} = 2(0) - 3(-3) = 9$; $c_{12} = 2(3) - 3(-2) = 12$
14. $c_{23} = 2(1) - 3(0) = 2$; $c_{12} = 2(-1) - 3(0) = -2$
15. (a) $\mathbf{A} + \mathbf{B} = \begin{bmatrix} 4-2 & 5+6 \\ -6+8 & 9-10 \end{bmatrix} = \begin{bmatrix} 2 & 11 \\ 2 & -1 \end{bmatrix}$
 (b) $\mathbf{B} - \mathbf{A} = \begin{bmatrix} -2-4 & 6-5 \\ 8+6 & -10-9 \end{bmatrix} = \begin{bmatrix} -6 & 1 \\ 14 & -19 \end{bmatrix}$
 (c) $2\mathbf{A} + 3\mathbf{B} = \begin{bmatrix} 8 & 10 \\ -12 & 18 \end{bmatrix} + \begin{bmatrix} -6 & 18 \\ 24 & -30 \end{bmatrix} = \begin{bmatrix} 2 & 28 \\ 12 & -12 \end{bmatrix}$
16. (a) $\mathbf{A} - \mathbf{B} = \begin{bmatrix} -2-3 & 0+1 \\ 4-0 & 1-2 \\ 7+4 & 3+2 \end{bmatrix} = \begin{bmatrix} -5 & 1 \\ 4 & -1 \\ 11 & 5 \end{bmatrix}$
 (b) $\mathbf{B} - \mathbf{A} = \begin{bmatrix} 3+2 & -1-0 \\ 0-4 & 2-1 \\ -4-7 & -2-3 \end{bmatrix} = \begin{bmatrix} 5 & -1 \\ -4 & 1 \\ -11 & -5 \end{bmatrix}$
 (c) $2(\mathbf{A} + \mathbf{B}) = 2 \begin{bmatrix} 1 & -1 \\ 4 & 3 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ 8 & 6 \\ 6 & 2 \end{bmatrix}$
17. (a) $\mathbf{AB} = \begin{bmatrix} -2-9 & 12-6 \\ 5+12 & -30+8 \end{bmatrix} = \begin{bmatrix} -11 & 6 \\ 17 & -22 \end{bmatrix}$
 (b) $\mathbf{BA} = \begin{bmatrix} -2-30 & 3+24 \\ 6-10 & -9+8 \end{bmatrix} = \begin{bmatrix} -32 & 27 \\ -4 & -1 \end{bmatrix}$
 (c) $\mathbf{A}^2 = \begin{bmatrix} 4+15 & -6-12 \\ -10-20 & 15+16 \end{bmatrix} = \begin{bmatrix} 19 & -18 \\ -30 & 31 \end{bmatrix}$
 (d) $\mathbf{B}^2 = \begin{bmatrix} 1+18 & -6+12 \\ -3+6 & 18+4 \end{bmatrix} = \begin{bmatrix} 19 & 6 \\ 3 & 22 \end{bmatrix}$
18. (a) $\mathbf{AB} = \begin{bmatrix} -4+4 & 6-12 & -3+8 \\ -20+10 & 30-30 & -15+20 \\ -32+12 & 48-36 & -24+24 \end{bmatrix} = \begin{bmatrix} 0 & -6 & 5 \\ -10 & 0 & 5 \\ -20 & 12 & 0 \end{bmatrix}$

Exercises 8.1

$$(b) \mathbf{BA} = \begin{bmatrix} -4 + 30 - 24 & -16 + 60 - 36 \\ 1 - 15 + 16 & 4 - 30 + 24 \end{bmatrix} = \begin{bmatrix} 2 & 8 \\ 2 & -2 \end{bmatrix}$$

$$19. (a) \mathbf{BC} = \begin{bmatrix} 9 & 24 \\ 3 & 8 \end{bmatrix}$$

$$(b) \mathbf{A(BC)} = \begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} 9 & 24 \\ 3 & 8 \end{bmatrix} = \begin{bmatrix} 3 & 8 \\ -6 & -16 \end{bmatrix}$$

$$(c) \mathbf{C(BA)} = \begin{bmatrix} 0 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$(d) \mathbf{A(B + C)} = \begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} 6 & 5 \\ 5 & 5 \end{bmatrix} = \begin{bmatrix} -4 & -5 \\ 8 & 10 \end{bmatrix}$$

$$20. (a) \mathbf{AB} = \begin{bmatrix} 5 & -6 & 7 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \\ -1 \end{bmatrix} = (-16)$$

$$(b) \mathbf{BA} = \begin{bmatrix} 3 \\ 4 \\ -1 \end{bmatrix} \begin{bmatrix} 5 & -6 & 7 \end{bmatrix} = \begin{bmatrix} 15 & -18 & 21 \\ 20 & -24 & 28 \\ -5 & 6 & -7 \end{bmatrix}$$

$$(c) (\mathbf{BA})\mathbf{C} = \begin{bmatrix} 15 & -18 & 21 \\ 20 & -24 & 28 \\ -5 & 6 & -7 \end{bmatrix} \begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & -1 \\ 3 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 78 & 54 & 99 \\ 104 & 72 & 132 \\ -26 & -18 & -33 \end{bmatrix}$$

(d) Since $\mathbf{AB}$ is 1×1 and $\mathbf{C}$ is 3×3 the product $(\mathbf{AB})\mathbf{C}$ is not defined.

$$21. (a) \mathbf{A}^T \mathbf{A} = \begin{bmatrix} 4 & 8 & -10 \end{bmatrix} \begin{bmatrix} 4 \\ 8 \\ -10 \end{bmatrix} = (180)$$

$$(b) \mathbf{B}^T \mathbf{B} = \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix} \begin{bmatrix} 2 & 4 & 5 \end{bmatrix} = \begin{bmatrix} 4 & 8 & 10 \\ 8 & 16 & 20 \\ 10 & 20 & 25 \end{bmatrix}$$

$$(c) \mathbf{A} + \mathbf{B}^T = \begin{bmatrix} 4 \\ 8 \\ -10 \end{bmatrix} + \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 6 \\ 12 \\ -5 \end{bmatrix}$$

$$22. (a) \mathbf{A} + \mathbf{B}^T = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} + \begin{bmatrix} -2 & 5 \\ 3 & 7 \end{bmatrix} = \begin{bmatrix} -1 & 7 \\ 5 & 11 \end{bmatrix}$$

$$(b) 2\mathbf{A}^T - \mathbf{B}^T = \begin{bmatrix} 2 & 4 \\ 4 & 8 \end{bmatrix} - \begin{bmatrix} -2 & 5 \\ 3 & 7 \end{bmatrix} = \begin{bmatrix} 4 & -1 \\ 1 & 1 \end{bmatrix}$$

$$(c) \mathbf{A}^T(\mathbf{A} - \mathbf{B}) = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 3 & -1 \\ -3 & -3 \end{bmatrix} = \begin{bmatrix} -3 & -7 \\ -6 & -14 \end{bmatrix}$$

$$23. (a) (\mathbf{AB})^T = \begin{bmatrix} 7 & 10 \\ 38 & 75 \end{bmatrix}^T = \begin{bmatrix} 7 & 38 \\ 10 & 75 \end{bmatrix}$$

$$(b) \mathbf{B}^T \mathbf{A}^T = \begin{bmatrix} 5 & -2 \\ 10 & -5 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 4 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 38 \\ 10 & 75 \end{bmatrix}$$

$$24. (a) \mathbf{A}^T + \mathbf{B} = \begin{bmatrix} 5 & -4 \\ 9 & 6 \end{bmatrix} + \begin{bmatrix} -3 & 11 \\ -7 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 7 \\ 2 & 8 \end{bmatrix}$$

$$(b) 2\mathbf{A} + \mathbf{B}^T = \begin{bmatrix} 10 & 18 \\ -8 & 12 \end{bmatrix} + \begin{bmatrix} -3 & -7 \\ 11 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 11 \\ 3 & 14 \end{bmatrix}$$

$$25. \begin{bmatrix} -4 \\ 8 \end{bmatrix} - \begin{bmatrix} 4 \\ 16 \end{bmatrix} + \begin{bmatrix} -6 \\ 9 \end{bmatrix} = \begin{bmatrix} -14 \\ 1 \end{bmatrix}$$

$$26. \begin{bmatrix} 6 \\ 3 \\ -3 \end{bmatrix} + \begin{bmatrix} -5 \\ -5 \\ 15 \end{bmatrix} + \begin{bmatrix} -6 \\ -8 \\ 10 \end{bmatrix} = \begin{bmatrix} -5 \\ -10 \\ 22 \end{bmatrix}$$

$$27. \begin{bmatrix} -19 \\ 18 \end{bmatrix} - \begin{bmatrix} 19 \\ 20 \end{bmatrix} = \begin{bmatrix} -38 \\ -2 \end{bmatrix}$$

$$28. \begin{bmatrix} -7 \\ 17 \\ -6 \end{bmatrix} + \begin{bmatrix} -1 \\ 1 \\ 4 \end{bmatrix} - \begin{bmatrix} 2 \\ 8 \\ -6 \end{bmatrix} = \begin{bmatrix} -10 \\ 10 \\ 4 \end{bmatrix}$$

$$29. 4 \times 5$$

$$30. 3 \times 2$$

$$31. \mathbf{A}^T = \begin{bmatrix} 2 & -3 \\ 4 & 2 \end{bmatrix}; \quad (\mathbf{A}^T)^T = \begin{bmatrix} 2 & 4 \\ -3 & 2 \end{bmatrix} = \mathbf{A}$$

$$32. (\mathbf{A} + \mathbf{B})^T = \begin{bmatrix} 6 & -6 \\ 14 & 10 \end{bmatrix} = \mathbf{A}^T + \mathbf{B}^T$$

$$33. (\mathbf{AB})^T = \begin{bmatrix} 16 & 40 \\ -8 & -20 \end{bmatrix}^T = \begin{bmatrix} 16 & -8 \\ 40 & -20 \end{bmatrix}; \quad \mathbf{B}^T \mathbf{A}^T = \begin{bmatrix} 4 & 2 \\ 10 & 5 \end{bmatrix} \begin{bmatrix} 2 & -3 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} 16 & -8 \\ 40 & -20 \end{bmatrix}$$

$$34. (6\mathbf{A})^T = \begin{bmatrix} 12 & -18 \\ 24 & 12 \end{bmatrix} = 6\mathbf{A}^T$$

$$35. \mathbf{B} = \mathbf{AA}^T = \begin{bmatrix} 2 & 1 \\ 6 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 2 & 6 & 2 \\ 1 & 3 & 5 \end{bmatrix} = \begin{bmatrix} 5 & 15 & 9 \\ 15 & 39 & 27 \\ 9 & 27 & 29 \end{bmatrix} = \mathbf{B}^T$$

36. Using Problem 33 we have $(\mathbf{AA}^T)^T = (\mathbf{A}^T)^T \mathbf{A}^T = \mathbf{AA}^T$, so that $\mathbf{AA}^T$ is symmetric.

$$37. \text{ Let } \mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \text{ and } \mathbf{B} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}. \text{ Then } \mathbf{AB} = \mathbf{0}.$$

$$38. \text{ We see that } \mathbf{A} \neq \mathbf{B}, \text{ but } \mathbf{AC} = \begin{bmatrix} 2 & 3 & 4 \\ 4 & 6 & 8 \\ 6 & 9 & 12 \end{bmatrix} = \mathbf{BC}.$$

39. Since $(\mathbf{A} + \mathbf{B})^2 = (\mathbf{A} + \mathbf{B})(\mathbf{A} + \mathbf{B}) \neq \mathbf{A}^2 + \mathbf{AB} + \mathbf{BA} + \mathbf{B}^2$, and $\mathbf{AB} \neq \mathbf{BA}$ in general, $(\mathbf{A} + \mathbf{B})^2 \neq \mathbf{A}^2 + 2\mathbf{AB} + \mathbf{B}^2$.

40. Since $(\mathbf{A} + \mathbf{B})(\mathbf{A} - \mathbf{B}) = \mathbf{A}^2 - \mathbf{AB} + \mathbf{BA} - \mathbf{B}^2$, and $\mathbf{AB} \neq \mathbf{BA}$ in general, $(\mathbf{A} + \mathbf{B})(\mathbf{A} - \mathbf{B}) \neq \mathbf{A}^2 - \mathbf{B}^2$.

$$41. a_{11}x_1 + a_{12}x_2 = b_1; \quad a_{21}x_1 + a_{22}x_2 = b_2$$

$$42. \begin{bmatrix} 2 & 6 & 1 \\ 1 & 2 & -1 \\ 5 & 7 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 7 \\ -1 \\ 9 \end{bmatrix}$$

$$43. \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} a & b/2 \\ b/2 & c \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ax + by/2 & bx/2 + cy \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ax^2 + bxy/2 + bxy/2 + cy^2 \end{bmatrix} = \begin{bmatrix} ax^2 + bxy + cy^2 \end{bmatrix}$$

$$44. \begin{bmatrix} 0 & -\partial/\partial z & \partial/\partial y \\ \partial/\partial z & 0 & -\partial/\partial x \\ -\partial/\partial y & \partial/\partial x & 0 \end{bmatrix} \begin{bmatrix} P \\ Q \\ R \end{bmatrix} = \begin{bmatrix} -\partial Q/\partial z + \partial R/\partial y \\ \partial P/\partial z - \partial R/\partial x \\ -\partial P/\partial y + \partial Q/\partial x \end{bmatrix} = \text{curl } \mathbf{F}$$

Exercises 8.1

$$45. \text{ (a) } M_Y \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \cos \gamma & \sin \gamma & 0 \\ -\sin \gamma & \cos \gamma & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \cos \gamma + y \sin \gamma \\ -x \sin \gamma + y \cos \gamma \\ z \end{bmatrix} = \begin{bmatrix} x_Y \\ y_Y \\ z_Y \end{bmatrix}$$

$$\text{(b) } M_R = \begin{bmatrix} \cos \beta & 0 & -\sin \beta \\ 0 & 1 & 0 \\ \sin \beta & 0 & \cos \beta \end{bmatrix}; \quad M_P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha & \sin \alpha \\ 0 & -\sin \alpha & \cos \alpha \end{bmatrix}$$

$$\text{(c) } M_P \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos 30^\circ & \sin 30^\circ \\ 0 & -\sin 30^\circ & \cos 30^\circ \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{2} & \frac{1}{2} \\ 0 & -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ \frac{1}{2}(\sqrt{3} + 1) \\ \frac{1}{2}(\sqrt{3} - 1) \end{bmatrix}$$

$$\begin{aligned} M_R M_P \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} &= \begin{bmatrix} \cos 45^\circ & 0 & -\sin 45^\circ \\ 0 & 1 & 0 \\ \sin 45^\circ & 0 & \cos 45^\circ \end{bmatrix} \begin{bmatrix} 1 \\ \frac{1}{2}(\sqrt{3} + 1) \\ \frac{1}{2}(\sqrt{3} - 1) \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{2}}{2} & 0 & -\frac{\sqrt{2}}{2} \\ 0 & 1 & 0 \\ \frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} \end{bmatrix} \begin{bmatrix} 1 \\ \frac{1}{2}(\sqrt{3} + 1) \\ \frac{1}{2}(\sqrt{3} - 1) \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{4}(3\sqrt{2} - \sqrt{6}) \\ \frac{1}{2}(\sqrt{3} + 1) \\ \frac{1}{4}(\sqrt{2} + \sqrt{6}) \end{bmatrix} \end{aligned}$$

$$\begin{aligned} M_Y M_R M_P \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} &= \begin{bmatrix} \cos 60^\circ & \sin 60^\circ & 0 \\ -\sin 60^\circ & \cos 60^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{4}(3\sqrt{2} - \sqrt{6}) \\ \frac{1}{2}(\sqrt{3} + 1) \\ \frac{1}{4}(\sqrt{2} + \sqrt{6}) \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{4}(3\sqrt{2} - \sqrt{6}) \\ \frac{1}{2}(\sqrt{3} + 1) \\ \frac{1}{4}(\sqrt{2} + \sqrt{6}) \end{bmatrix} = \begin{bmatrix} \frac{1}{8}(3\sqrt{2} - \sqrt{6} + 6 + 2\sqrt{3}) \\ \frac{1}{8}(-3\sqrt{6} + 3\sqrt{2} + 2\sqrt{3} + 2) \\ \frac{1}{4}(\sqrt{2} + \sqrt{6}) \end{bmatrix} \end{aligned}$$

$$46. \text{ (a) } \mathbf{LU} = \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} 2 & -2 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ 1 & 2 \end{bmatrix} = \mathbf{A}$$

$$\text{(b) } \mathbf{LU} = \begin{bmatrix} 1 & 0 \\ \frac{2}{3} & 1 \end{bmatrix} \begin{bmatrix} 6 & 2 \\ 0 & -\frac{1}{3} \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 4 & 1 \end{bmatrix} = \mathbf{A}$$

$$\text{(c) } \mathbf{LU} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 10 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & -21 \end{bmatrix} = \begin{bmatrix} 1 & -2 & 1 \\ 0 & 1 & 2 \\ 2 & 6 & 1 \end{bmatrix} = \mathbf{A}$$

$$\text{(d) } \mathbf{LU} = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & -2 & -1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 3 & 1 & 2 \\ 1 & -1 & 1 \end{bmatrix} = \mathbf{A}$$

$$47. \text{ (a) } \mathbf{AB} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} = \begin{bmatrix} A_{11}B_1 + A_{12}B_2 \\ A_{21}B_1 + A_{22}B_2 \end{bmatrix} = \begin{bmatrix} 17 & 43 \\ 3 & 75 \\ -14 & 51 \end{bmatrix}$$

since

$$A_{11}B_1 + A_{12}B_2 = \begin{bmatrix} 13 & 25 \\ -9 & 49 \end{bmatrix} + \begin{bmatrix} 4 & 18 \\ 12 & 26 \end{bmatrix} = \begin{bmatrix} 17 & 43 \\ 3 & 75 \end{bmatrix}$$

and

$$A_{21}B_1 + A_{22}B_2 = \begin{bmatrix} -24 & 34 \\ 10 & 17 \end{bmatrix} + \begin{bmatrix} 10 & 17 \end{bmatrix} = \begin{bmatrix} -14 & 51 \end{bmatrix}.$$

- (b) It is easier to enter smaller strings of numbers and the chance of error is decreased. Also, if the large matrix has submatrices consisting of all zeros or diagonal matrices, these are easily entered without listing all of the entries.

Exercises 8.2

$$1. \left[\begin{array}{cc|c} 1 & -1 & 11 \\ 4 & 3 & -5 \end{array} \right] \xrightarrow{-4R_1+R_2} \left[\begin{array}{cc|c} 1 & -1 & 11 \\ 0 & 7 & -49 \end{array} \right] \xrightarrow{\frac{1}{7}R_2} \left[\begin{array}{cc|c} 1 & -1 & 11 \\ 0 & 1 & -7 \end{array} \right] \xrightarrow{R_3+R_1} \left[\begin{array}{cc|c} 1 & 0 & 4 \\ 0 & 1 & -7 \end{array} \right]$$

The solution is $x_1 = 4$, $x_2 = -7$.

$$2. \left[\begin{array}{cc|c} 3 & -2 & 4 \\ 1 & -1 & -2 \end{array} \right] \xrightarrow{R_{12}} \left[\begin{array}{cc|c} 1 & -1 & -2 \\ 3 & -2 & 4 \end{array} \right] \xrightarrow{-3R_1+R_2} \left[\begin{array}{cc|c} 1 & -1 & -2 \\ 0 & 1 & 10 \end{array} \right] \xrightarrow{R_2+R_1} \left[\begin{array}{cc|c} 1 & 0 & 8 \\ 0 & 1 & 10 \end{array} \right]$$

The solution is $x_1 = 8$, $x_2 = 10$.

$$3. \left[\begin{array}{cc|c} 9 & 3 & -5 \\ 2 & -1 & -1 \end{array} \right] \xrightarrow{\frac{1}{9}R_1} \left[\begin{array}{cc|c} 1 & \frac{1}{3} & -\frac{5}{9} \\ 2 & -1 & -1 \end{array} \right] \xrightarrow{-2R_1+R_2} \left[\begin{array}{cc|c} 1 & \frac{1}{3} & -\frac{5}{9} \\ 0 & \frac{1}{3} & \frac{1}{9} \end{array} \right] \xrightarrow{3R_2} \left[\begin{array}{cc|c} 1 & \frac{1}{3} & -\frac{5}{9} \\ 0 & 1 & \frac{1}{3} \end{array} \right]$$

$$\xrightarrow{-\frac{1}{3}R_2+R_1} \left[\begin{array}{cc|c} 1 & 0 & -\frac{2}{3} \\ 0 & 1 & \frac{1}{3} \end{array} \right]$$

The solution is $x_1 = -\frac{2}{3}$, $x_2 = \frac{1}{3}$.

$$4. \left[\begin{array}{cc|c} 10 & 15 & 1 \\ 3 & 2 & -1 \end{array} \right] \xrightarrow{\frac{1}{10}R_1} \left[\begin{array}{cc|c} 1 & \frac{3}{2} & \frac{1}{10} \\ 3 & 2 & -1 \end{array} \right] \xrightarrow{-3R_1+R_2} \left[\begin{array}{cc|c} 1 & \frac{3}{2} & \frac{1}{10} \\ 0 & -\frac{5}{2} & -\frac{13}{10} \end{array} \right] \xrightarrow{-\frac{2}{5}R_2} \left[\begin{array}{cc|c} 1 & \frac{3}{2} & \frac{1}{10} \\ 0 & 1 & \frac{13}{25} \end{array} \right]$$

$$\xrightarrow{-\frac{3}{2}R_2+R_1} \left[\begin{array}{cc|c} 1 & 0 & -\frac{17}{25} \\ 0 & 1 & \frac{13}{25} \end{array} \right]$$

The solution is $x_1 = -\frac{17}{25}$, $x_2 = \frac{13}{25}$.

$$5. \left[\begin{array}{ccc|c} 1 & -1 & -1 & -3 \\ 2 & 3 & 5 & 7 \\ 1 & -2 & 3 & -11 \end{array} \right] \xrightarrow{\begin{array}{l} -2R_1+R_2 \\ -R_1+R_3 \end{array}} \left[\begin{array}{ccc|c} 1 & -1 & -1 & -3 \\ 0 & 5 & 7 & 13 \\ 0 & -1 & 4 & -8 \end{array} \right] \xrightarrow{\frac{1}{5}R_2} \left[\begin{array}{ccc|c} 1 & -1 & -1 & -3 \\ 0 & 1 & \frac{7}{5} & \frac{13}{5} \\ 0 & -1 & 4 & -8 \end{array} \right]$$

$$\xrightarrow{\begin{array}{l} R_2+R_1 \\ R_2+R_3 \end{array}} \left[\begin{array}{ccc|c} 1 & 0 & \frac{2}{5} & -\frac{2}{5} \\ 0 & 1 & \frac{7}{5} & \frac{13}{5} \\ 0 & 0 & \frac{27}{5} & -\frac{27}{5} \end{array} \right] \xrightarrow{\frac{5}{27}R_3} \left[\begin{array}{ccc|c} 1 & 0 & \frac{2}{5} & -\frac{2}{5} \\ 0 & 1 & \frac{7}{5} & \frac{13}{5} \\ 0 & 0 & 1 & -1 \end{array} \right] \xrightarrow{\begin{array}{l} -\frac{2}{5}R_3+R_1 \\ -\frac{7}{5}R_3+R_2 \end{array}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & -1 \end{array} \right]$$

The solution is $x_1 = 0$, $x_2 = 4$, $x_3 = -1$.

$$6. \left[\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 2 & 1 & 2 & 9 \\ 1 & -1 & 1 & 3 \end{array} \right] \xrightarrow{\begin{array}{l} -2R_1+R_2 \\ -R_1+R_3 \end{array}} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & -3 & 4 & 9 \\ 0 & -3 & 2 & 3 \end{array} \right] \xrightarrow{-\frac{1}{3}R_2} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & 1 & -\frac{4}{3} & -3 \\ 0 & -3 & 2 & 3 \end{array} \right]$$

$$\xrightarrow{\begin{array}{l} -2R_2+R_1 \\ 3R_2+R_3 \end{array}} \left[\begin{array}{ccc|c} 1 & 0 & \frac{5}{3} & 6 \\ 0 & 1 & -\frac{4}{3} & -3 \\ 0 & 0 & -2 & -6 \end{array} \right] \xrightarrow{-\frac{1}{2}R_3} \left[\begin{array}{ccc|c} 1 & 0 & \frac{5}{3} & 6 \\ 0 & 1 & -\frac{4}{3} & -3 \\ 0 & 0 & 1 & 3 \end{array} \right] \xrightarrow{\begin{array}{l} -\frac{5}{3}R_3+R_1 \\ \frac{4}{3}R_3+R_2 \end{array}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

The solution is $x_1 = 1$, $x_2 = 1$, $x_3 = 3$.

$$7. \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 1 & 1 & 3 & 0 \end{array} \right] \xrightarrow{-R_1+R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 0 & 2 & 0 \end{array} \right]$$

Exercises 8.2

Since $x_3 = 0$, setting $x_2 = t$ we obtain $x_1 = -t$, $x_2 = t$, $x_3 = 0$.

$$8. \begin{bmatrix} 1 & 2 & -4 & | & 9 \\ 5 & -1 & 2 & | & 1 \end{bmatrix} \xrightarrow{-5R_1+R_2} \begin{bmatrix} 1 & 2 & -4 & | & 9 \\ 0 & -11 & 22 & | & -44 \end{bmatrix} \xrightarrow{-\frac{1}{11}R_2} \begin{bmatrix} 1 & 2 & -4 & | & 9 \\ 0 & 1 & -2 & | & 4 \end{bmatrix} \xrightarrow{-2R_2+R_1} \begin{bmatrix} 1 & 0 & 0 & | & 1 \\ 0 & 1 & -2 & | & 4 \end{bmatrix}$$

If $x_3 = t$, the solution is $x_1 = 1$, $x_2 = 4 + 2t$, $x_3 = t$

$$9. \begin{bmatrix} 1 & -1 & -1 & | & 8 \\ 1 & -1 & 1 & | & 3 \\ -1 & 1 & 1 & | & 4 \end{bmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{bmatrix} 1 & -1 & -1 & | & 8 \\ 0 & 0 & 2 & | & -5 \\ 0 & 0 & 0 & | & 12 \end{bmatrix}$$

Since the bottom row implies $0 = 12$, the system is inconsistent.

$$10. \begin{bmatrix} 3 & 1 & | & 4 \\ 4 & 3 & | & -3 \\ 2 & -1 & | & 11 \end{bmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{bmatrix} 1 & \frac{1}{3} & | & \frac{4}{3} \\ 0 & 1 & | & -5 \\ 0 & 0 & | & 0 \end{bmatrix}$$

The solution is $x_1 = 3$, $x_2 = -5$.

$$11. \begin{bmatrix} 2 & 2 & 0 & | & 0 \\ -2 & 1 & 1 & | & 0 \\ 3 & 0 & 1 & | & 0 \end{bmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{bmatrix} 1 & 1 & 0 & | & 0 \\ 0 & 1 & \frac{1}{3} & | & 0 \\ 0 & 0 & 1 & | & 0 \end{bmatrix}$$

The solution is $x_1 = x_2 = x_3 = 0$.

$$12. \begin{bmatrix} 1 & -1 & -2 & | & 0 \\ 2 & 4 & 5 & | & 0 \\ 6 & 0 & -3 & | & 0 \end{bmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{bmatrix} 1 & -1 & -2 & | & 0 \\ 0 & 1 & \frac{3}{2} & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

The solution is $x_1 = \frac{1}{2}t$, $x_2 = -\frac{3}{2}t$, $x_3 = t$.

$$13. \begin{bmatrix} 1 & 2 & 2 & | & 2 \\ 1 & 1 & 1 & | & 0 \\ 1 & -3 & -1 & | & 0 \end{bmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{bmatrix} 1 & 2 & 2 & | & 2 \\ 0 & 1 & 1 & | & 2 \\ 0 & 0 & 1 & | & 4 \end{bmatrix}$$

The solution is $x_1 = -2$, $x_2 = 2$, $x_3 = 4$.

$$14. \begin{bmatrix} 1 & -2 & 1 & | & 2 \\ 3 & -1 & 2 & | & 5 \\ 2 & 1 & 1 & | & 1 \end{bmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{bmatrix} 1 & -2 & 1 & | & 2 \\ 0 & 1 & -\frac{1}{5} & | & -\frac{1}{5} \\ 0 & 0 & 0 & | & -2 \end{bmatrix}$$

Since the bottom row implies $0 = -2$, the system is inconsistent.

$$15. \begin{bmatrix} 1 & 1 & 1 & | & 3 \\ 1 & -1 & -1 & | & -1 \\ 3 & 1 & 1 & | & 5 \end{bmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{bmatrix} 1 & 1 & 1 & | & 3 \\ 0 & 1 & 1 & | & 2 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

If $x_3 = t$ the solution is $x_1 = 1$, $x_2 = 2 - t$, $x_3 = t$.

$$16. \begin{bmatrix} 1 & -1 & -2 & | & -1 \\ -3 & -2 & 1 & | & -7 \\ 2 & 3 & 1 & | & 8 \end{bmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{bmatrix} 1 & -1 & -2 & | & -1 \\ 0 & 1 & 1 & | & 2 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

If $x_3 = t$ the solution is $x_1 = 1 + t$, $x_2 = 2 - t$, $x_3 = t$.

$$17. \left[\begin{array}{cccc|c} 1 & 0 & 1 & -1 & 1 \\ 0 & 2 & 1 & 1 & 3 \\ 1 & -1 & 0 & 1 & -1 \\ 1 & 1 & 1 & 1 & 2 \end{array} \right] \xrightarrow[\text{operations}]{\text{row}} \left[\begin{array}{cccc|c} 1 & 0 & 1 & -1 & 1 \\ 0 & 1 & \frac{1}{2} & \frac{1}{2} & \frac{3}{2} \\ 0 & 0 & 1 & -5 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{array} \right]$$

The solution is $x_1 = 0$, $x_2 = 1$, $x_3 = 1$, $x_4 = 0$.

$$18. \left[\begin{array}{cccc|c} 2 & 1 & 1 & 0 & 3 \\ 3 & 1 & 1 & 1 & 4 \\ 1 & 2 & 2 & 3 & 3 \\ 4 & 5 & -2 & 1 & 16 \end{array} \right] \xrightarrow[\text{operations}]{\text{row}} \left[\begin{array}{cccc|c} 1 & \frac{1}{2} & \frac{1}{2} & 0 & \frac{3}{2} \\ 0 & 1 & 1 & -2 & 1 \\ 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 1 & 0 \end{array} \right]$$

The solution is $x_1 = 1$, $x_2 = 2$, $x_3 = -1$, $x_4 = 0$.

$$19. \left[\begin{array}{cccc|c} 1 & 3 & 5 & -1 & 1 \\ 0 & 1 & 1 & -1 & 4 \\ 1 & 2 & 5 & -4 & -2 \\ 1 & 4 & 6 & -2 & 6 \end{array} \right] \xrightarrow[\text{operations}]{\text{row}} \left[\begin{array}{cccc|c} 1 & 3 & 5 & -1 & 1 \\ 0 & 1 & 1 & -1 & 4 \\ 0 & 0 & 1 & -4 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{array} \right]$$

Since the bottom row implies $0 = 1$, the system is inconsistent.

$$20. \left[\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 0 \\ 4 & 9 & 1 & 12 & 0 \\ 3 & 9 & 6 & 21 & 0 \\ 1 & 3 & 1 & 9 & 0 \end{array} \right] \xrightarrow[\text{operations}]{\text{row}} \left[\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 0 \\ 0 & 1 & 1 & 8 & 0 \\ 0 & 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

If $x_4 = t$ the solution is $x_1 = 19t$, $x_2 = -10t$, $x_3 = 2t$, $x_4 = t$.

$$21. \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4.280 \\ 0.2 & -0.1 & -0.5 & -1.978 \\ 4.1 & 0.3 & 0.12 & 1.686 \end{array} \right] \xrightarrow[\text{operations}]{\text{row}} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4.28 \\ 0 & 1 & 2.333 & 9.447 \\ 0 & 0 & 1 & 4.1 \end{array} \right]$$

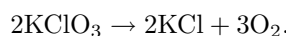
The solution is $x_1 = 0.3$, $x_2 = -0.12$, $x_3 = 4.1$.

$$22. \left[\begin{array}{ccc|c} 2.5 & 1.4 & 4.5 & 2.6170 \\ 1.35 & 0.95 & 1.2 & 0.7545 \\ 2.7 & 3.05 & -1.44 & -1.4292 \end{array} \right] \xrightarrow[\text{operations}]{\text{row}} \left[\begin{array}{ccc|c} 1 & 0.56 & 1.8 & 1.0468 \\ 0 & 1 & -6.3402 & -3.3953 \\ 0 & 0 & 1 & 0.28 \end{array} \right]$$

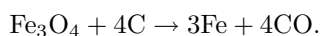
The solution is $x_1 = 1.45$, $x_2 = -1.62$, $x_3 = 0.28$.

23. From $x_1\text{Na} + x_2\text{H}_2\text{O} \rightarrow x_3\text{NaOH} + x_4\text{H}_2$ we obtain the system $x_1 = x_3$, $2x_2 = x_3 + 2x_4$, $x_2 = x_3$. We see that $x_1 = x_2 = x_3$, so the second equation becomes $2x_1 = x_1 + 2x_4$ or $x_1 = 2x_4$. A solution of the system is $x_1 = x_2 = x_3 = 2t$, $x_4 = t$. Letting $t = 1$ we obtain the balanced equation $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$.

24. From $x_1\text{KClO}_3 \rightarrow x_2\text{KCl} + x_3\text{O}_2$ we obtain the system $x_1 = x_2$, $x_1 = x_2$, $3x_1 = 2x_3$. Letting $x_3 = t$ we see that a solution of the system is $x_1 = x_2 = \frac{2}{3}t$, $x_3 = t$. Taking $t = 3$ we obtain the balanced equation

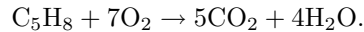


25. From $x_1\text{Fe}_3\text{O}_4 + x_2\text{C} \rightarrow x_3\text{Fe} + x_4\text{CO}$ we obtain the system $3x_1 = x_3$, $4x_1 = x_4$, $x_2 = x_4$. Letting $x_1 = t$ we see that $x_3 = 3t$ and $x_2 = x_4 = 4t$. Taking $t = 1$ we obtain the balanced equation



Exercises 8.2

26. From $x_1\text{C}_5\text{H}_8 + x_2\text{O}_2 \rightarrow x_3\text{CO}_2 + x_4\text{H}_2\text{O}$ we obtain the system $5x_1 = x_3$, $8x_1 = 2x_4$, $2x_2 = 2x_3 + x_4$. Letting $x_1 = t$ we see that $x_3 = 5t$, $x_4 = 4t$, and $x_2 = 7t$. Taking $t = 1$ we obtain the balanced equation



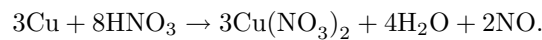
27. From $x_1\text{Cu} + x_2\text{HNO}_3 \rightarrow x_3\text{Cu}(\text{NO}_3)_2 + x_4\text{H}_2\text{O} + x_5\text{NO}$ we obtain the system

$$x_1 = 3, \quad x_2 = 2x_4, \quad x_2 = 2x_3 + x_5, \quad 3x_2 = 6x_3 + x_4 + x_5.$$

Letting $x_4 = t$ we see that $x_2 = 2t$ and

$$\begin{array}{rcl} 2t = 2x_3 + x_5 & & 2x_3 + x_5 = 2t \\ 6t = 6x_3 + t + x_5 & \text{or} & 6x_3 + x_5 = 5t. \end{array}$$

Then $x_3 = \frac{3}{4}t$ and $x_5 = \frac{1}{2}t$. Finally, $x_1 = x_3 = \frac{3}{4}t$. Taking $t = 4$ we obtain the balanced equation



28. From $x_1\text{Ca}_3(\text{PO}_4)_2 + x_2\text{H}_3\text{PO}_4 \rightarrow x_3\text{Ca}(\text{H}_2\text{PO}_4)_2$ we obtain the system

$$3x_1 = x_3, \quad 2x_1 + x_2 = 2x_3, \quad 8x_1 + 4x_2 = 8x_3, \quad 3x_2 = 4x_3.$$

Letting $x_1 = t$ we see from the first equation that $x_3 = 3t$ and from the fourth equation that $x_2 = 4t$. These choices also satisfy the second and third equations. Taking $t = 1$ we obtain the balanced equation



29. The system of equations is

$$\begin{array}{rcl} -i_1 + i_2 - i_3 = 0 & & -i_1 + i_2 - i_3 = 0 \\ 10 - 3i_1 + 5i_3 = 0 & \text{or} & 3i_1 - 5i_3 = 10 \\ 27 - 6i_2 - 5i_3 = 0 & & 6i_2 + 5i_3 = 27 \end{array}$$

Gaussian elimination gives

$$\left[\begin{array}{ccc|c} -1 & 1 & -1 & 0 \\ 3 & 0 & -5 & 10 \\ 0 & 6 & 5 & 27 \end{array} \right] \xrightarrow[\text{operations}]{\text{row}} \left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 1 & -8/3 & 10/3 \\ 0 & 0 & 1 & 1/3 \end{array} \right].$$

The solution is $i_1 = \frac{35}{9}$, $i_2 = \frac{38}{9}$, $i_3 = \frac{1}{3}$.

30. The system of equations is

$$\begin{array}{rcl} i_1 - i_2 - i_3 = 0 & & i_1 - i_2 - i_3 = 0 \\ 52 - i_1 - 5i_2 = 0 & \text{or} & i_1 + 5i_2 = 52 \\ -10i_3 + 5i_2 = 0 & & 5i_2 - 10i_3 = 0 \end{array}$$

Gaussian elimination gives

$$\left[\begin{array}{ccc|c} 1 & -1 & -1 & 0 \\ 1 & 5 & 0 & 52 \\ 0 & 5 & -10 & 0 \end{array} \right] \xrightarrow[\text{operations}]{\text{row}} \left[\begin{array}{ccc|c} 1 & -1 & -1 & 0 \\ 0 & 1 & 1/6 & 26/3 \\ 0 & 0 & 1 & 4 \end{array} \right].$$

The solution is $i_1 = 12$, $i_2 = 8$, $i_3 = 4$.

31. Interchange row 1 and row in $\mathbf{I}_3$.

32. Multiply row 3 by c in $\mathbf{I}_3$.

33. Add c times row 2 to row 3 in $\mathbf{I}_3$.

34. Add row 4 to row 1 in $\mathbf{I}_4$.

$$35. \mathbf{EA} = \begin{bmatrix} a_{21} & a_{22} & a_{23} \\ a_{11} & a_{12} & a_{13} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$36. \mathbf{EA} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ ca_{31} & ca_{32} & ca_{33} \end{bmatrix}$$

$$37. \mathbf{EA} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ ca_{21} + a_{31} & ca_{22} + a_{32} & ca_{23} + a_{33} \end{bmatrix}$$

$$38. \mathbf{E}_1 \mathbf{E}_2 \mathbf{A} = \mathbf{E}_1 \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ ca_{21} + a_{31} & ca_{22} + a_{32} & ca_{23} + a_{33} \end{bmatrix} = \begin{bmatrix} a_{21} & a_{22} & a_{23} \\ a_{11} & a_{12} & a_{13} \\ ca_{21} + a_{31} & ca_{22} + a_{32} & ca_{23} + a_{33} \end{bmatrix}$$

39. The system is equivalent to

$$\begin{bmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} 2 & -2 \\ 0 & 3 \end{bmatrix} \mathbf{X} = \begin{bmatrix} 2 \\ 6 \end{bmatrix}.$$

Letting

$$\mathbf{Y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ 0 & 3 \end{bmatrix} \mathbf{X}$$

we have

$$\begin{bmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 6 \end{bmatrix}.$$

This implies $y_1 = 2$ and $\frac{1}{2}y_1 + y_2 = 1 + y_2 = 6$ or $y_2 = 5$. Then

$$\begin{bmatrix} 2 & -2 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix},$$

which implies $3x_2 = 5$ or $x_2 = \frac{5}{3}$ and $2x_1 - 2x_2 = 2x_1 - \frac{10}{3} = 2$ or $x_1 = \frac{8}{3}$. The solution is $\mathbf{X} = (\frac{8}{3}, \frac{5}{3})$.

40. The system is equivalent to

$$\begin{bmatrix} 1 & 0 \\ \frac{2}{3} & 1 \end{bmatrix} \begin{bmatrix} 6 & 2 \\ 0 & -\frac{1}{3} \end{bmatrix} \mathbf{X} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Letting

$$\mathbf{Y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 0 & -\frac{1}{3} \end{bmatrix} \mathbf{X}$$

we have

$$\begin{bmatrix} 1 & 0 \\ \frac{2}{3} & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

This implies $y_1 = 1$ and $\frac{2}{3}y_1 + y_2 = \frac{2}{3} + y_2 = -1$ or $y_2 = -\frac{5}{3}$. Then

$$\begin{bmatrix} 6 & 2 \\ 0 & -\frac{1}{3} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -\frac{5}{3} \end{bmatrix},$$

which implies $-\frac{1}{3}x_2 = -\frac{5}{3}$ or $x_2 = 5$ and $6x_1 + 2x_2 = 6x_1 + 10 = 1$ or $x_1 = -\frac{3}{2}$. The solution is $\mathbf{X} = (-\frac{3}{2}, 5)$.

41. The system is equivalent to

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 10 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & -21 \end{bmatrix} \mathbf{X} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}.$$

Exercises 8.2

Letting

$$\mathbf{Y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1 & -2 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & -21 \end{bmatrix} \mathbf{X}$$

we have

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 10 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}.$$

This implies $y_1 = 2$, $y_2 = -1$, and $2y_1 + 10y_2 + y_3 = 4 - 10 + y_3 = 1$ or $y_3 = 7$. Then

$$\begin{bmatrix} 1 & -2 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & -21 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 7 \end{bmatrix},$$

which implies $-21x_3 = 7$ or $x_3 = -\frac{1}{3}$, $x_2 + 2x_3 = x_2 - \frac{2}{3} = -1$ or $x_2 = -\frac{1}{3}$, and $x_1 - 2x_2 + x_3 = x_1 + \frac{2}{3} - \frac{1}{3} = 2$ or $x_1 = \frac{5}{3}$. The solution is $\mathbf{X} = (\frac{5}{3}, -\frac{1}{3}, -\frac{1}{3})$.

42. The system is equivalent to

$$\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & -2 & -1 \\ 0 & 0 & 1 \end{bmatrix} \mathbf{X} = \begin{bmatrix} 0 \\ 1 \\ 4 \end{bmatrix}.$$

Letting

$$\mathbf{Y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -2 & -1 \\ 0 & 0 & 1 \end{bmatrix} \mathbf{X}$$

we have

$$\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 4 \end{bmatrix}.$$

This implies $y_1 = 0$, $3y_1 + y_2 = y_2 = 1$, and $y_1 + y_2 + y_3 = 0 + 1 + y_3 = 4$ or $y_3 = 3$. Then

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & -2 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix},$$

which implies $x_3 = 3$, $-2x_2 - x_3 = -2x_2 - 3 = 1$ or $x_2 = -2$, and $x_1 + x_2 + x_3 = x_1 - 2 + 3 = 0$ or $x_1 = -1$. The solution is $\mathbf{X} = (-1, -2, 3)$.

43. Using the **Solve** function in *Mathematica* we find

(a) $x_1 = -0.0717393 - 1.43084c$, $x_2 = -0.332591 + 0.855709c$, $x_3 = c$, where c is any real number

(b) $x_1 = c/3$, $x_2 = 5c/6$, $x_3 = c$, where c is any real number

(c) $x_1 = -3.76993$, $x_2 = -1.09071$, $x_3 = -4.50461$, $x_4 = -3.12221$

(d) $x_1 = \frac{8}{3} - \frac{7}{3}b + \frac{2}{3}c$, $x_2 = \frac{2}{3} - \frac{1}{3}b - \frac{1}{3}c$, $x_3 = -3$, $x_4 = b$, $x_5 = c$, where b and c are any real numbers.

Exercises 8.3

1. $\begin{bmatrix} 3 & -1 \\ 1 & 3 \end{bmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$; The rank is 2.

2. $\begin{bmatrix} 2 & -2 \\ 0 & 0 \end{bmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$; The rank is 1.

3. $\begin{bmatrix} 2 & 1 & 3 \\ 6 & 3 & 9 \\ -1 & -\frac{1}{2} & -\frac{3}{2} \end{bmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{bmatrix} 1 & \frac{1}{2} & \frac{3}{2} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$; The rank is 1.

4. $\begin{bmatrix} 1 & 1 & 2 \\ -1 & 2 & 4 \\ -1 & 0 & 3 \end{bmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{bmatrix}$; The rank is 3.

5. $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 4 \\ 1 & 4 & 1 \end{bmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix}$; The rank is 3.

6. $\begin{bmatrix} 3 & -1 & 2 & 0 \\ 6 & 2 & 4 & 5 \end{bmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{bmatrix} 1 & -\frac{1}{3} & \frac{2}{3} & 0 \\ 0 & 1 & 0 & \frac{5}{4} \end{bmatrix}$; The rank is 2.

7. $\begin{bmatrix} 1 & -2 \\ 3 & -6 \\ 7 & -1 \\ 4 & 5 \end{bmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{bmatrix} 1 & -2 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$; The rank is 2.

8. $\begin{bmatrix} 1 & -2 & 3 & 4 \\ 1 & 4 & 6 & 8 \\ 0 & 1 & 0 & 0 \\ 2 & 5 & 6 & 8 \end{bmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{bmatrix} 1 & -2 & 3 & 4 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{4}{3} \\ 0 & 0 & 0 & 0 \end{bmatrix}$; The rank is 3.

9. $\begin{bmatrix} 0 & 2 & 4 & 2 & 2 \\ 4 & 1 & 0 & 5 & 1 \\ 2 & 1 & \frac{2}{3} & 3 & \frac{1}{3} \\ 6 & 6 & 6 & 12 & 0 \end{bmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{3} & \frac{3}{2} & \frac{1}{6} \\ 0 & 1 & \frac{4}{3} & 1 & -\frac{1}{3} \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$; The rank is 3.

10. $\begin{bmatrix} 1 & -2 & 1 & 8 & -1 & 1 & 1 & 6 \\ 0 & 0 & 1 & 3 & -1 & 1 & 1 & 5 \\ 0 & 0 & 1 & 3 & -1 & 2 & 10 & 8 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 3 \\ 1 & -2 & 1 & 8 & -1 & 1 & 2 & 6 \end{bmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{bmatrix} 1 & -2 & 1 & 8 & -1 & 1 & 1 & 6 \\ 0 & 0 & 1 & 3 & -1 & 1 & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 & 1 & 9 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$; The rank is 4.

11. $\begin{bmatrix} 1 & 2 & 3 \\ 1 & 0 & 1 \\ 1 & -1 & 5 \end{bmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$;

Since the rank of the matrix is 3 and there are 3 vectors, the vectors are linearly independent.

Exercises 8.3

$$12. \begin{bmatrix} 2 & 6 & 3 \\ 1 & -1 & 4 \\ 3 & 2 & 1 \\ 2 & 5 & 4 \end{bmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{bmatrix} 1 & -1 & 4 \\ 0 & 1 & -\frac{5}{8} \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Since the rank of the matrix is 3 and there are 4 vectors, the vectors are linearly dependent.

$$13. \begin{bmatrix} 1 & -1 & 3 & -1 \\ 1 & -1 & 4 & 2 \\ 1 & -1 & 5 & 7 \end{bmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{bmatrix} 1 & -1 & 3 & -1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Since the rank of the matrix is 3 and there are 3 vectors, the vectors are linearly independent.

$$14. \begin{bmatrix} 2 & 1 & 1 & 5 \\ 2 & 2 & 1 & 1 \\ 3 & -1 & 6 & 1 \\ 1 & 1 & 1 & -1 \end{bmatrix} \xrightarrow[\text{operations}]{\text{row}} \begin{bmatrix} 1 & 1 & 1 & -1 \\ 0 & 1 & 1 & -7 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Since the rank of the matrix is 4 and there are 4 vectors, the vectors are linearly independent.

15. Since the number of unknowns is $n = 8$ and the rank of the coefficient matrix is $r = 3$, the solution of the system has $n - r = 5$ parameters.
16. (a) The maximum possible rank of $\mathbf{A}$ is the number of rows in $\mathbf{A}$, which is 4.
 (b) The system is inconsistent if $\text{rank}(\mathbf{A}) < \text{rank}(\mathbf{A}/\mathbf{B}) = 2$ and consistent if $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}/\mathbf{B}) = 2$.
 (c) The system has $n = 6$ unknowns and the rank of $\mathbf{A}$ is $r = 3$, so the solution of the system has $n - r = 3$ parameters.
17. Since $2\mathbf{v}_1 + 3\mathbf{v}_2 - \mathbf{v}_3 = \mathbf{0}$ we conclude that $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$ are linearly dependent. Thus, the rank of $\mathbf{A}$ is at most 2.
18. Since the rank of $\mathbf{A}$ is $r = 3$ and the number of equations is $n = 6$, the solution of the system has $n - r = 3$ parameters. Thus, the solution of the system is not unique.
19. The system consists of 4 equations, so the rank of the coefficient matrix is at most 4, and the maximum number of linearly independent rows is 4. However, the maximum number of linearly independent columns is the same as the maximum number of linearly independent rows. Thus, the coefficient matrix has at most 4 linearly independent columns. Since there are 5 column vectors, they must be linearly dependent.
20. Using the **RowReduce** in *Mathematica* we find that the reduced row-echelon form of the augmented matrix is

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & \frac{834}{2215} & -\frac{261}{443} \\ 0 & 1 & 0 & 0 & 0 & \frac{1818}{2215} & \frac{282}{443} \\ 0 & 0 & 1 & 0 & 0 & \frac{13}{443} & -\frac{6}{443} \\ 0 & 0 & 0 & 1 & 0 & \frac{4214}{2215} & -\frac{130}{443} \\ 0 & 0 & 0 & 0 & 1 & -\frac{6079}{2215} & \frac{677}{443} \end{bmatrix}.$$

We conclude that the system is consistent and the solution is $x_1 = -\frac{226}{443} - \frac{834}{2215}c$, $x_2 = \frac{282}{443} - \frac{1818}{2215}c$, $x_3 = -\frac{6}{443} - \frac{13}{443}c$, $x_4 = -\frac{130}{443} - \frac{4214}{2215}c$, $x_5 = \frac{677}{443} + \frac{6079}{2215}c$, $x_6 = c$.

Exercises 8.4

$$1. M_{12} = \begin{vmatrix} 1 & 2 \\ -2 & 5 \end{vmatrix} = 9$$

$$2. M_{32} = \begin{vmatrix} 2 & 4 \\ 1 & 2 \end{vmatrix}$$

$$3. C_{13} = (-1)^{1+3} \begin{vmatrix} 1 & -1 \\ -2 & 3 \end{vmatrix} = 1$$

$$4. C_{22} = (-1)^{2+2} \begin{vmatrix} 2 & 4 \\ -2 & 5 \end{vmatrix} = 18$$

$$5. M_{33} = \begin{vmatrix} 0 & 2 & 0 \\ 1 & 2 & 3 \\ 1 & 1 & 2 \end{vmatrix} = 2$$

$$6. M_{41} = \begin{vmatrix} 2 & 4 & 0 \\ 2 & -2 & 3 \\ 1 & 0 & -1 \end{vmatrix} = 24$$

$$7. C_{34} = (-1)^{3+4} \begin{vmatrix} 0 & 2 & 4 \\ 1 & 2 & -2 \\ 1 & 1 & 1 \end{vmatrix} = 10$$

$$8. C_{23} = (-1)^{2+3} \begin{vmatrix} 0 & 2 & 0 \\ 5 & 1 & -1 \\ 1 & 1 & 2 \end{vmatrix} = 22$$

$$9. -7$$

$$10. 2$$

$$11. 17$$

$$12. -1/2$$

$$13. (1 - \lambda)(2 - \lambda) - 6 = \lambda^2 - 3\lambda - 4$$

$$14. (-3 - \lambda)(5 - \lambda) - 8 = \lambda^2 - 2\lambda - 23$$

$$15. \begin{vmatrix} 0 & 2 & 0 \\ 3 & 0 & 1 \\ 0 & 5 & 8 \end{vmatrix} = -3 \begin{vmatrix} 2 & 0 \\ 5 & 8 \end{vmatrix} = -48$$

$$16. \begin{vmatrix} 5 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 2 \end{vmatrix} = 5 \begin{vmatrix} -3 & 0 \\ 0 & 2 \end{vmatrix} = 5(-3)(2) = -30$$

$$17. \begin{vmatrix} 3 & 0 & 2 \\ 2 & 7 & 1 \\ 2 & 6 & 4 \end{vmatrix} = 3 \begin{vmatrix} 7 & 1 \\ 6 & 4 \end{vmatrix} + 2 \begin{vmatrix} 2 & 7 \\ 2 & 6 \end{vmatrix} = 3(22) + 2(-2) = 62$$

$$18. \begin{vmatrix} 1 & -1 & -1 \\ 2 & 2 & -2 \\ 1 & 1 & 9 \end{vmatrix} = \begin{vmatrix} 2 & -2 \\ 1 & 9 \end{vmatrix} - 2 \begin{vmatrix} -1 & -1 \\ 1 & 9 \end{vmatrix} + \begin{vmatrix} -1 & -1 \\ 2 & -2 \end{vmatrix} = 20 - 2(-8) + 4 = 40$$

$$19. \begin{vmatrix} 4 & 5 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{vmatrix} = 4 \begin{vmatrix} 2 & 3 \\ 2 & 3 \end{vmatrix} - 5 \begin{vmatrix} 1 & 3 \\ 1 & 3 \end{vmatrix} + 3 \begin{vmatrix} 1 & 2 \\ 1 & 2 \end{vmatrix} = 0$$

$$20. \begin{vmatrix} \frac{1}{4} & 6 & 0 \\ \frac{1}{3} & 8 & 0 \\ \frac{1}{2} & 9 & 0 \end{vmatrix} = 0, \text{ expanding along the third column.}$$

$$21. \begin{vmatrix} -2 & -1 & 4 \\ -3 & 6 & 1 \\ -3 & 4 & 8 \end{vmatrix} = -2 \begin{vmatrix} 6 & 1 \\ 4 & 8 \end{vmatrix} + 3 \begin{vmatrix} -1 & 4 \\ 4 & 8 \end{vmatrix} - 3 \begin{vmatrix} -1 & 4 \\ 6 & 1 \end{vmatrix} = -2(44) + 3(-24) - 3(-25) = -85$$

$$22. \begin{vmatrix} 3 & 5 & 1 \\ -1 & 2 & 5 \\ 7 & -4 & 10 \end{vmatrix} = 3 \begin{vmatrix} 2 & 5 \\ -4 & 10 \end{vmatrix} - 5 \begin{vmatrix} -1 & 5 \\ 7 & 10 \end{vmatrix} + \begin{vmatrix} -1 & 2 \\ 7 & -4 \end{vmatrix} = 3(40) - 5(-45) + (-10) = 335$$

$$23. \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ 2 & 3 & 4 \end{vmatrix} = \begin{vmatrix} y & z \\ 3 & 4 \end{vmatrix} - \begin{vmatrix} x & z \\ 2 & 4 \end{vmatrix} + \begin{vmatrix} x & y \\ 2 & 3 \end{vmatrix} = (4y - 3z) - (4x - 2z) + (3x - 2y) = -x + 2y - z$$

Exercises 8.4

$$\begin{aligned}
 24. \quad \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ 2+x & 3+y & 4+z \end{vmatrix} &= \begin{vmatrix} y & z \\ 3+y & 4+z \end{vmatrix} - \begin{vmatrix} x & z \\ 2+x & 4+z \end{vmatrix} + \begin{vmatrix} x & y \\ 2+x & 3+y \end{vmatrix} \\
 &= (4y + yz - 3z - yz) - (4x + xz - 2z - xz) + (3x + xy - 2y - xy) = -x + 2y - z
 \end{aligned}$$

$$25. \quad \begin{vmatrix} 1 & 1 & -3 & 0 \\ 1 & 5 & 3 & 2 \\ 1 & -2 & 1 & 0 \\ 4 & 8 & 0 & 0 \end{vmatrix} = 2 \begin{vmatrix} 1 & 1 & -3 \\ 1 & -2 & 1 \\ 4 & 8 & 0 \end{vmatrix} = 2(4) \begin{vmatrix} 1 & -3 \\ -2 & 1 \end{vmatrix} - 2(8) \begin{vmatrix} 1 & -3 \\ 1 & 1 \end{vmatrix} = 8(-5) - 16(4) = -104$$

$$26. \quad \begin{vmatrix} 2 & 1 & -2 & 1 \\ 0 & 5 & 0 & 4 \\ 1 & 6 & 1 & 0 \\ 5 & -1 & 1 & 1 \end{vmatrix} = 5 \begin{vmatrix} 2 & -2 & 1 \\ 1 & 1 & 0 \\ 5 & 1 & 1 \end{vmatrix} + 4 \begin{vmatrix} 2 & 1 & -2 \\ 1 & 6 & 1 \\ 5 & -1 & 1 \end{vmatrix} = 5(0) + 4(80) = 320$$

27. Expanding along the first column in the original matrix and each succeeding minor, we obtain $3(1)(2)(4)(2) = 48$.

28. Expanding along the bottom row we obtain

$$-1 \begin{vmatrix} 2 & 0 & 0 & -2 \\ 1 & 6 & 0 & 5 \\ 1 & 2 & -1 & 1 \\ 2 & 1 & -2 & 3 \end{vmatrix} + \begin{vmatrix} 2 & 2 & 0 & 0 \\ 1 & 1 & 6 & 0 \\ 1 & 0 & 2 & -1 \\ 2 & 0 & 1 & -2 \end{vmatrix} = -1(-48) + 0 = 48.$$

29. Solving $\lambda^2 - 2\lambda - 15 - 20 = \lambda^2 - 2\lambda - 35 = (\lambda - 7)(\lambda + 5) = 0$ we obtain $\lambda = 7$ and -5 .

30. Solving $-\lambda^3 + 3\lambda^2 - 2\lambda = -\lambda(\lambda - 2)(\lambda - 1) = 0$ we obtain $\lambda = 0, 1$, and 2 .

Exercises 8.5

1. Theorem 8.11

2. Theorem 8.14

3. Theorem 8.14

4. Theorem 8.12 and 8.11

5. Theorem 8.12 (twice)

6. Theorem 8.11 (twice)

7. Theorem 8.10

8. Theorem 8.12 and 8.9

9. Theorem 8.8

10. Theorem 8.11 (twice)

11. $\det \mathbf{A} = -5$

12. $\det \mathbf{B} = 2(3)(5) = 30$

13. $\det \mathbf{C} = -5$

14. $\det \mathbf{D} = 5$

15. $\det \mathbf{A} = 6(\frac{2}{3})(-4)(-5) = 80$

16. $\det \mathbf{B} = -a_{13}a_{22}a_{31}$

17. $\det \mathbf{C} = (-5)(7)(3) = -105$

18. $\det \mathbf{D} = 4(7)(-2) = -56$

19. $\det \mathbf{A} = 14 = \det \mathbf{A}^T$

20. $\det \mathbf{A} = 96 = \det \mathbf{A}^T$

$$21. \det \mathbf{AB} = \begin{vmatrix} 0 & -2 & 2 \\ 10 & 7 & 23 \\ 8 & 4 & 16 \end{vmatrix} = -80 = 20(-4) = \det \mathbf{A} \det \mathbf{B}$$

22. From Problem 21, $(\det \mathbf{A})^2 = \det \mathbf{A}^2 = \det \mathbf{I} = 1$, so $\det \mathbf{A} = \pm 1$.

$$23. \text{ Using Theorems 8.14, 8.12, and 8.9, } \det \mathbf{A} = \begin{vmatrix} a & 1 & 2 \\ b & 1 & 2 \\ c & 1 & 2 \end{vmatrix} = 2 \begin{vmatrix} a & 1 & 1 \\ b & 1 & 1 \\ c & 1 & 1 \end{vmatrix} = 0.$$

24. Using Theorems 8.14 and 8.9,

$$\det \mathbf{A} = \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x+y+z & x+y+z & x+y+z \end{vmatrix} = (x+y+z) \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ 1 & 1 & 1 \end{vmatrix} = 0.$$

$$25. \begin{vmatrix} 1 & 1 & 5 \\ 4 & 3 & 6 \\ 0 & -1 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 5 \\ 0 & -1 & -14 \\ 0 & -1 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 5 \\ 0 & -1 & -14 \\ 0 & 0 & 15 \end{vmatrix} = 1(-1)(15) = -15$$

$$26. \begin{vmatrix} 2 & 4 & 5 \\ 4 & 2 & 0 \\ 8 & 7 & -2 \end{vmatrix} = \begin{vmatrix} 2 & 4 & 5 \\ 0 & -6 & -10 \\ 0 & -9 & -22 \end{vmatrix} = -2 \begin{vmatrix} 2 & 4 & 5 \\ 0 & 3 & 5 \\ 0 & -9 & -22 \end{vmatrix} = -2 \begin{vmatrix} 2 & 4 & 5 \\ 0 & 3 & 5 \\ 0 & 0 & -7 \end{vmatrix} = -2(2)(3)(-7) = 84$$

$$27. \begin{vmatrix} -1 & 2 & 3 \\ 4 & -5 & -2 \\ 9 & -9 & 6 \end{vmatrix} = \begin{vmatrix} -1 & 2 & 3 \\ 0 & 3 & 10 \\ 0 & 9 & 33 \end{vmatrix} = \begin{vmatrix} -1 & 2 & 3 \\ 0 & 3 & 10 \\ 0 & 0 & 3 \end{vmatrix} = -1(3)(3) = -9$$

$$28. \begin{vmatrix} -2 & 2 & -6 \\ 5 & 0 & 1 \\ 1 & -2 & 2 \end{vmatrix} = - \begin{vmatrix} 1 & -2 & 2 \\ 5 & 0 & 1 \\ -2 & 2 & -6 \end{vmatrix} = - \begin{vmatrix} 1 & -2 & 2 \\ 0 & 10 & -9 \\ 0 & -2 & -2 \end{vmatrix} = - \begin{vmatrix} 1 & -2 & 2 \\ 0 & 10 & -9 \\ 0 & 0 & -\frac{19}{5} \end{vmatrix} = -1(10)(-\frac{19}{5}) = 38$$

$$29. \begin{vmatrix} 1 & -2 & 2 & 1 \\ 2 & 1 & -2 & 3 \\ 3 & 4 & -8 & 1 \\ 3 & -11 & 12 & 2 \end{vmatrix} = \begin{vmatrix} 1 & -2 & 2 & 1 \\ 0 & 5 & -6 & 1 \\ 0 & 10 & -14 & -2 \\ 0 & -5 & 6 & -1 \end{vmatrix} = \begin{vmatrix} 1 & -2 & 2 & 1 \\ 0 & 5 & -6 & 1 \\ 0 & 0 & -2 & -4 \\ 0 & 0 & 0 & 0 \end{vmatrix} = 1(5)(-2)(0) = 0$$

$$30. \begin{vmatrix} 0 & 1 & 4 & 5 \\ 2 & 5 & 0 & 1 \\ 1 & 2 & 2 & 0 \\ 3 & 1 & 3 & 2 \end{vmatrix} = - \begin{vmatrix} 1 & 2 & 2 & 0 \\ 2 & 5 & 0 & 1 \\ 0 & 1 & 4 & 5 \\ 3 & 1 & 3 & 2 \end{vmatrix} = - \begin{vmatrix} 1 & 2 & 2 & 0 \\ 0 & 1 & -4 & 1 \\ 0 & 1 & 4 & 5 \\ 0 & -5 & -3 & 2 \end{vmatrix} = - \begin{vmatrix} 1 & 2 & 2 & 0 \\ 0 & 1 & -4 & 1 \\ 0 & 0 & 8 & 4 \\ 0 & 0 & -23 & 7 \end{vmatrix} = - \begin{vmatrix} 1 & 2 & 2 & 0 \\ 0 & 1 & -4 & 1 \\ 0 & 0 & 8 & 4 \\ 0 & 0 & -23 & \frac{37}{2} \end{vmatrix} \\ = -(1)(1)(8)(\frac{37}{2}) = -148$$

$$31. \begin{vmatrix} 1 & 2 & 3 & 4 \\ 1 & 3 & 5 & 7 \\ 2 & 3 & 6 & 7 \\ 1 & 5 & 8 & 20 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & -1 & 0 & -1 \\ 0 & 3 & 5 & 16 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & -1 & 7 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 8 \end{vmatrix} = 1(1)(2)(8) = 16$$

Exercises 8.5

$$\begin{aligned}
 32. \quad \begin{vmatrix} 2 & 9 & 1 & 8 \\ 1 & 3 & 7 & 4 \\ 0 & 1 & 6 & 5 \\ 3 & 1 & 4 & 2 \end{vmatrix} &= - \begin{vmatrix} 1 & 3 & 7 & 4 \\ 2 & 9 & 1 & 8 \\ 0 & 1 & 6 & 5 \\ 3 & 1 & 4 & 2 \end{vmatrix} = - \begin{vmatrix} 1 & 3 & 7 & 4 \\ 0 & 3 & -13 & 0 \\ 0 & 1 & 6 & 5 \\ 0 & -8 & -17 & -10 \end{vmatrix} = \begin{vmatrix} 1 & 3 & 7 & 4 \\ 0 & 1 & 6 & 5 \\ 0 & 3 & -13 & 0 \\ 0 & -8 & -17 & -10 \end{vmatrix} \\
 &= \begin{vmatrix} 1 & 3 & 7 & 4 \\ 0 & 1 & 6 & 5 \\ 0 & 0 & -31 & -15 \\ 0 & 0 & 31 & 30 \end{vmatrix} = \begin{vmatrix} 1 & 3 & 7 & 4 \\ 0 & 1 & 6 & 5 \\ 0 & 0 & -31 & -15 \\ 0 & 0 & 0 & 15 \end{vmatrix} = 1(1)(-31)(15) = -465
 \end{aligned}$$

33. We first use the second row to reduce the third row. Then we use the first row to reduce the second row.

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ 0 & b^2 - ab & c^2 - ac \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ 0 & b - a & c - a \\ 0 & b(b - a) & c(c - a) \end{vmatrix} = (b - a)(c - a) \begin{vmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & b & c \end{vmatrix}.$$

Expanding along the first row gives $(b - a)(c - a)(c - b)$.

34. In order, we use the third row to reduce the fourth row, the second row to reduce the third row, and the first row to reduce the second row. We then pull out a common factor from each column.

$$\begin{vmatrix} 1 & 1 & 1 & 1 \\ a & b & c & d \\ a^2 & b^2 & c^2 & d^2 \\ a^3 & b^3 & c^3 & d^3 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & b - a & c - a & d - a \\ 0 & b^2 - ab & c^2 - ac & d^2 - ad \\ 0 & b^3 - ab^2 & c^3 - ac^2 & d^3 - ad^2 \end{vmatrix} = (b - a)(c - a)(d - a) \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & b & c & d \\ 0 & b^2 & c^2 & d^2 \end{vmatrix}.$$

Expanding along the first column and using Problem 33 we obtain $(b - a)(c - a)(d - a)(c - b)(d - b)(d - c)$.

35. Since $C_{11} = 4$, $C_{12} = 5$, and $C_{13} = -6$, we have $a_{21}C_{11} + a_{22}C_{12} + a_{23}C_{13} = (-1)(4) + 2(5) + 1(-6) = 0$. Since $C_{12} = 5$, $C_{22} = -7$, and $C_{23} = -3$, we have $a_{13}C_{12} + a_{23}C_{22} + a_{33}C_{32} = 2(5) + 1(-7) + 1(-3) = 0$.

36. Since $C_{11} = -7$, $C_{12} = -8$, and $C_{13} = -10$ we have $a_{21}C_{11} + a_{22}C_{12} + a_{23}C_{13} = -2(-7) + 3(-8) - 1(-10) = 0$. Since $C_{12} = -8$, $C_{22} = -19$, and $C_{32} = -7$ we have $a_{13}C_{12} + a_{23}C_{22} + a_{33}C_{32} = 5(-8) - 1(-19) - 3(-7) = 0$.

$$37. \quad \det(\mathbf{A} + \mathbf{B}) = \begin{vmatrix} 10 & 0 \\ 0 & -3 \end{vmatrix} = -30; \quad \det \mathbf{A} + \det \mathbf{B} = 10 - 31 = -21$$

$$38. \quad \det(2\mathbf{A}) = 2^5 \det \mathbf{A} = 32(-7) = -224$$

39. Factoring -1 out of each row we see that $\det(-\mathbf{A}) = (-1)^5 \det \mathbf{A} = -\det \mathbf{A}$. Then $-\det \mathbf{A} = \det(-\mathbf{A}) = \det \mathbf{A}^T = \det \mathbf{A}$ and $\det \mathbf{A} = 0$.

40. (a) Cofactors: $25! \approx 1.55(10^{25})$; Row reduction: $25^3/3 \approx 5.2(10^3)$

(b) Cofactors: about 90 billion centuries; Row reduction: about $\frac{1}{10}$ second

Exercises 8.6

$$1. \mathbf{AB} = \begin{bmatrix} 3-2 & -1+1 \\ 6-6 & -2+3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$2. \mathbf{AB} = \begin{bmatrix} 2-1 & -1+1 & -2+2 \\ 6-6 & -3+4 & 6-6 \\ 2+1-3 & -1-1+2 & 2+2-3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$3. \det \mathbf{A} = 9. \mathbf{A} \text{ is nonsingular. } \mathbf{A}^{-1} = \frac{1}{9} \begin{bmatrix} 1 & 1 \\ -4 & 5 \end{bmatrix} = \begin{bmatrix} \frac{1}{9} & \frac{1}{9} \\ -\frac{4}{9} & \frac{5}{9} \end{bmatrix}$$

$$4. \det \mathbf{A} = 5. \mathbf{A} \text{ is nonsingular. } \mathbf{A}^{-1} = \frac{1}{5} \begin{bmatrix} 3 & 1 \\ -4 & \frac{1}{3} \end{bmatrix} = \begin{bmatrix} \frac{3}{5} & \frac{1}{5} \\ -\frac{4}{5} & \frac{1}{15} \end{bmatrix}$$

$$5. \det \mathbf{A} = 12. \mathbf{A} \text{ is nonsingular. } \mathbf{A}^{-1} = \frac{1}{12} \begin{bmatrix} 2 & 0 \\ 3 & 6 \end{bmatrix} = \begin{bmatrix} \frac{1}{6} & 0 \\ \frac{1}{4} & \frac{1}{2} \end{bmatrix}$$

$$6. \det \mathbf{A} = -3\pi^2. \mathbf{A} \text{ is nonsingular. } \mathbf{A}^{-1} = -\frac{1}{3\pi^2} \begin{bmatrix} \pi & \pi \\ \pi & -2\pi \end{bmatrix} = \begin{bmatrix} -\frac{1}{3\pi} & -\frac{1}{3\pi} \\ -\frac{1}{3\pi} & \frac{2}{3\pi} \end{bmatrix}$$

$$7. \det \mathbf{A} = -16. \mathbf{A} \text{ is nonsingular. } \mathbf{A}^{-1} = -\frac{1}{16} \begin{bmatrix} 8 & -8 & -8 \\ 2 & -4 & 6 \\ -6 & 4 & -2 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{8} & \frac{1}{4} & -\frac{3}{8} \\ \frac{3}{8} & -\frac{1}{4} & \frac{1}{8} \end{bmatrix}$$

$$8. \det \mathbf{A} = 0. \mathbf{A} \text{ is singular.}$$

$$9. \det \mathbf{A} = -30. \mathbf{A} \text{ is nonsingular. } \mathbf{A}^{-1} = -\frac{1}{30} \begin{bmatrix} -14 & 13 & 16 \\ -2 & 4 & -2 \\ -4 & -7 & -4 \end{bmatrix} = \begin{bmatrix} \frac{7}{15} & -\frac{13}{30} & -\frac{8}{15} \\ \frac{1}{15} & -\frac{2}{15} & \frac{1}{15} \\ \frac{2}{15} & \frac{7}{30} & \frac{2}{15} \end{bmatrix}$$

$$10. \det \mathbf{A} = 78. \mathbf{A} \text{ is nonsingular. } \mathbf{A}^{-1} = \frac{1}{78} \begin{bmatrix} 8 & 20 & 2 \\ -2 & -5 & 19 \\ 12 & -9 & 3 \end{bmatrix} \begin{bmatrix} \frac{4}{39} & \frac{10}{39} & \frac{1}{39} \\ -\frac{1}{39} & -\frac{5}{78} & \frac{19}{78} \\ \frac{2}{13} & -\frac{3}{26} & \frac{1}{26} \end{bmatrix}$$

$$11. \det \mathbf{A} = -36. \mathbf{A} \text{ is nonsingular. } \mathbf{A}^{-1} = -\frac{1}{36} \begin{bmatrix} -12 & 0 & 0 \\ 0 & -6 & 0 \\ 0 & 0 & 18 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & 0 & 0 \\ 0 & \frac{1}{6} & 0 \\ 0 & 0 & -\frac{1}{2} \end{bmatrix}$$

$$12. \det \mathbf{A} = 16. \mathbf{A} \text{ is nonsingular. } \mathbf{A}^{-1} = \frac{1}{16} \begin{bmatrix} 0 & 0 & 2 \\ 8 & 0 & 0 \\ 0 & 16 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & \frac{1}{8} \\ \frac{1}{2} & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$13. \det \mathbf{A} = 27. \mathbf{A} \text{ is nonsingular. } \mathbf{A}^{-1} = \frac{1}{27} \begin{bmatrix} 6 & 21 & -9 & -36 \\ -1 & 1 & 6 & -3 \\ 10 & 17 & -6 & -51 \\ 4 & -4 & 3 & 12 \end{bmatrix} = \begin{bmatrix} \frac{2}{9} & \frac{7}{9} & -\frac{1}{3} & -\frac{4}{3} \\ -\frac{1}{27} & \frac{1}{27} & \frac{2}{9} & -\frac{1}{9} \\ \frac{10}{27} & \frac{17}{27} & -\frac{2}{9} & -\frac{17}{9} \\ \frac{4}{27} & -\frac{4}{27} & \frac{1}{9} & \frac{4}{9} \end{bmatrix}$$

$$14. \det \mathbf{A} = -6. \mathbf{A} \text{ is nonsingular. } \mathbf{A}^{-1} = -\frac{1}{6} \begin{bmatrix} 0 & 1 & -3 & 3 \\ 0 & 1 & 3 & -9 \\ 0 & -2 & 0 & 0 \\ -6 & -1 & -3 & 15 \end{bmatrix} = \begin{bmatrix} 0 & -\frac{1}{6} & \frac{1}{2} & -\frac{1}{2} \\ 0 & -\frac{1}{6} & -\frac{1}{2} & \frac{3}{2} \\ 0 & \frac{1}{3} & 0 & 0 \\ 1 & \frac{1}{6} & \frac{1}{2} & -\frac{5}{2} \end{bmatrix}$$

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$$15. \left[\begin{array}{cc|cc} 6 & -2 & 1 & 0 \\ 0 & 4 & 0 & 1 \end{array} \right] \xrightarrow[\frac{1}{4}R_2]{\frac{1}{6}R_1} \left[\begin{array}{cc|cc} 1 & -\frac{1}{3} & \frac{1}{6} & 0 \\ 0 & 1 & 0 & \frac{1}{4} \end{array} \right] \xrightarrow{\frac{1}{3}R_2+R_1} \left[\begin{array}{cc|cc} 1 & 0 & \frac{1}{6} & \frac{1}{12} \\ 0 & 1 & 0 & \frac{1}{4} \end{array} \right]; \quad \mathbf{A}^{-1} = \begin{bmatrix} \frac{1}{6} & \frac{1}{12} \\ 0 & \frac{1}{4} \end{bmatrix}$$

$$16. \left[\begin{array}{cc|cc} 8 & 0 & 1 & 0 \\ 0 & \frac{1}{2} & 0 & 1 \end{array} \right] \xrightarrow[2R_2]{\frac{1}{8}R_1} \left[\begin{array}{cc|cc} 1 & 0 & \frac{1}{8} & 0 \\ 0 & 1 & 0 & 2 \end{array} \right]; \quad \mathbf{A}^{-1} = \begin{bmatrix} \frac{1}{8} & 0 \\ 0 & 2 \end{bmatrix}$$

$$17. \left[\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 5 & 3 & 0 & 1 \end{array} \right] \xrightarrow{-5R_1+R_2} \left[\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 0 & -12 & -5 & 1 \end{array} \right] \xrightarrow{-\frac{1}{12}R_2} \left[\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 0 & 1 & \frac{5}{12} & -\frac{1}{12} \end{array} \right] \xrightarrow{-3R_2+R_1} \left[\begin{array}{cc|cc} 1 & 0 & -\frac{1}{4} & \frac{1}{4} \\ 0 & 1 & \frac{5}{12} & -\frac{1}{12} \end{array} \right];$$

$$\mathbf{A}^{-1} = \begin{bmatrix} -\frac{1}{4} & \frac{1}{4} \\ \frac{5}{12} & -\frac{1}{12} \end{bmatrix}$$

$$18. \left[\begin{array}{cc|cc} 2 & -3 & 1 & 0 \\ -2 & 4 & 0 & 1 \end{array} \right] \xrightarrow{\frac{1}{2}R_1} \left[\begin{array}{cc|cc} 1 & -\frac{3}{2} & \frac{1}{2} & 0 \\ -2 & 4 & 0 & 1 \end{array} \right] \xrightarrow{2R_1+R_2} \left[\begin{array}{cc|cc} 1 & -\frac{3}{2} & \frac{1}{2} & 0 \\ 0 & 1 & 1 & 1 \end{array} \right] \xrightarrow{\frac{3}{2}R_2+R_1} \left[\begin{array}{cc|cc} 1 & 0 & 2 & \frac{3}{2} \\ 0 & 1 & 1 & 1 \end{array} \right];$$

$$\mathbf{A}^{-1} = \begin{bmatrix} 2 & \frac{3}{2} \\ 1 & 1 \end{bmatrix}$$

$$19. \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 4 & 5 & 6 & 0 & 1 & 0 \\ 7 & 8 & 9 & 0 & 0 & 1 \end{array} \right] \xrightarrow[\text{operations}]{\text{row}} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 2 & \frac{4}{3} & -\frac{1}{3} & 0 \\ 0 & 0 & 0 & 1 & -2 & 1 \end{array} \right]; \quad \mathbf{A} \text{ is singular.}$$

$$20. \left[\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & -2 & 1 & 0 & 1 & 0 \\ 2 & -1 & 3 & 0 & 0 & 1 \end{array} \right] \xrightarrow[\text{operations}]{\text{row}} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{5}{9} & -\frac{1}{9} & \frac{2}{9} \\ 0 & 1 & 0 & -\frac{2}{9} & -\frac{5}{9} & \frac{1}{9} \\ 0 & 0 & 1 & -\frac{4}{9} & -\frac{1}{9} & \frac{2}{9} \end{array} \right]; \quad \mathbf{A}^{-1} = \begin{bmatrix} \frac{5}{9} & -\frac{1}{9} & \frac{2}{9} \\ -\frac{2}{9} & -\frac{5}{9} & \frac{1}{9} \\ -\frac{4}{9} & -\frac{1}{9} & \frac{2}{9} \end{bmatrix}$$

$$21. \left[\begin{array}{ccc|ccc} 4 & 2 & 3 & 1 & 0 & 0 \\ 2 & 1 & 0 & 0 & 1 & 0 \\ -1 & -2 & 0 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_{13}} \left[\begin{array}{ccc|ccc} -1 & -2 & 0 & 0 & 0 & 1 \\ 2 & 1 & 0 & 0 & 1 & 0 \\ 4 & 2 & 3 & 1 & 0 & 0 \end{array} \right] \xrightarrow[\text{operations}]{\text{row}} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & \frac{2}{3} & \frac{1}{3} \\ 0 & 1 & 0 & 0 & -\frac{1}{3} & -\frac{2}{3} \\ 0 & 0 & 1 & \frac{1}{3} & -\frac{2}{3} & 0 \end{array} \right];$$

$$\mathbf{A}^{-1} = \begin{bmatrix} 0 & \frac{2}{3} & \frac{1}{3} \\ 0 & -\frac{1}{3} & -\frac{2}{3} \\ \frac{1}{3} & -\frac{2}{3} & 0 \end{bmatrix}$$

$$22. \left[\begin{array}{ccc|ccc} 2 & 4 & -2 & 1 & 0 & 0 \\ 4 & 2 & -2 & 0 & 1 & 0 \\ 8 & 10 & -6 & 0 & 0 & 1 \end{array} \right] \xrightarrow[\text{operations}]{\text{row}} \left[\begin{array}{ccc|ccc} 1 & 2 & -1 & \frac{1}{2} & 0 & 0 \\ 0 & 1 & -\frac{1}{3} & \frac{1}{3} & -\frac{1}{6} & 0 \\ 0 & 0 & 0 & -2 & -1 & 1 \end{array} \right]; \quad \mathbf{A} \text{ is singular.}$$

$$23. \left[\begin{array}{ccc|ccc} -1 & 3 & 0 & 1 & 0 & 0 \\ 3 & -2 & 1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array} \right] \xrightarrow[\text{operations}]{\text{row}} \left[\begin{array}{ccc|ccc} 1 & -3 & 0 & -1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & -1 & -1 & 1 \end{array} \right] \xrightarrow[\text{operations}]{\text{row}} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 5 & 6 & -3 \\ 0 & 1 & 0 & 2 & 2 & -1 \\ 0 & 0 & 1 & -1 & -1 & 1 \end{array} \right];$$

$$\mathbf{A}^{-1} = \begin{bmatrix} 5 & 6 & -3 \\ 2 & 2 & -1 \\ -1 & -1 & 1 \end{bmatrix}$$

$$24. \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 4 & 0 & 1 & 0 \\ 0 & 0 & 8 & 0 & 0 & 1 \end{array} \right] \xrightarrow[\text{operations}]{\text{row}} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -2 & \frac{5}{8} \\ 0 & 1 & 0 & 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 1 & 0 & 0 & \frac{1}{8} \end{array} \right]; \quad \mathbf{A}^{-1} = \begin{bmatrix} 1 & -2 & \frac{5}{8} \\ 0 & 1 & -\frac{1}{2} \\ 0 & 0 & \frac{1}{8} \end{bmatrix}$$

$$\begin{aligned}
25. \quad & \left[\begin{array}{cccc|cccc} 1 & 2 & 3 & 1 & 1 & 0 & 0 & 0 \\ -1 & 0 & 2 & 1 & 0 & 1 & 0 & 0 \\ 2 & 1 & -3 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 2 & 1 & 0 & 0 & 0 & 1 \end{array} \right] \xrightarrow[\text{operations}]{\text{row}} \left[\begin{array}{cccc|cccc} 1 & 2 & 3 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & \frac{5}{2} & 1 & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 1 & -\frac{2}{3} & \frac{1}{3} & -1 & -\frac{2}{3} & 0 \\ 0 & 0 & 0 & 1 & -\frac{1}{2} & 1 & \frac{1}{2} & \frac{1}{2} \end{array} \right] \\
& \xrightarrow[\text{operations}]{\text{row}} \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & -\frac{1}{2} & -\frac{2}{3} & -\frac{1}{6} & \frac{7}{6} \\ 0 & 1 & 0 & 0 & 1 & \frac{1}{3} & \frac{1}{3} & -\frac{4}{3} \\ 0 & 0 & 1 & 0 & 0 & -\frac{1}{3} & -\frac{1}{3} & \frac{1}{3} \\ 0 & 0 & 0 & 1 & -\frac{1}{2} & 1 & \frac{1}{2} & \frac{1}{2} \end{array} \right]; \quad \mathbf{A}^{-1} = \left[\begin{array}{cccc} -\frac{1}{2} & -\frac{2}{3} & -\frac{1}{6} & \frac{7}{6} \\ 1 & \frac{1}{3} & \frac{1}{3} & -\frac{4}{3} \\ 0 & -\frac{1}{3} & -\frac{1}{3} & \frac{1}{3} \\ -\frac{1}{2} & 1 & \frac{1}{2} & \frac{1}{2} \end{array} \right]
\end{aligned}$$

$$26. \quad \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \end{array} \right] \xrightarrow[\text{interchange}]{\text{row}} \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \end{array} \right]; \quad \mathbf{A}^{-1} = \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

$$27. (\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1} = \begin{bmatrix} -\frac{1}{3} & \frac{1}{3} \\ -1 & \frac{10}{3} \end{bmatrix}$$

$$28. (\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1} = \begin{bmatrix} -1 & -4 & 20 \\ 2 & 6 & -30 \\ 3 & 6 & -32 \end{bmatrix}$$

$$29. \mathbf{A} = (\mathbf{A}^{-1})^{-1} = \begin{bmatrix} -2 & 3 \\ 3 & -4 \end{bmatrix}$$

$$30. \mathbf{A}^T = \begin{bmatrix} 1 & 2 \\ 4 & 10 \end{bmatrix}; \quad (\mathbf{A}^T)^{-1} = \begin{bmatrix} 5 & -1 \\ -2 & \frac{1}{2} \end{bmatrix}; \quad \mathbf{A}^{-1} = \begin{bmatrix} 5 & -2 \\ -1 & \frac{1}{2} \end{bmatrix}; \quad (\mathbf{A}^{-1})^T = \begin{bmatrix} 5 & -1 \\ -2 & \frac{1}{2} \end{bmatrix}$$

$$31. \text{ Multiplying } \begin{bmatrix} 4 & -3 \\ x & -4 \end{bmatrix} \begin{bmatrix} 4 & -3 \\ x & -4 \end{bmatrix} = \begin{bmatrix} 16-3x & 0 \\ 0 & 16-3x \end{bmatrix} \text{ we see that } x = 5.$$

$$32. \mathbf{A}^{-1} = \begin{bmatrix} \sin \theta & -\cos \theta \\ \cos \theta & \sin \theta \end{bmatrix}$$

$$33. \quad \begin{aligned} \text{(a)} \quad \mathbf{A}^T &= \begin{bmatrix} \sin \theta & -\cos \theta \\ \cos \theta & \sin \theta \end{bmatrix} = \mathbf{A}^{-1} & \quad \text{(b)} \quad \mathbf{A}^T &= \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} \end{bmatrix} = \mathbf{A}^{-1} \end{aligned}$$

34. Since $\det \mathbf{A} \cdot \det \mathbf{A}^{-1} = \det \mathbf{AA}^{-1} = \det \mathbf{I} = 1$, we see that $\det \mathbf{A}^{-1} = 1/\det \mathbf{A}$. If $\mathbf{A}$ is orthogonal, $\det \mathbf{A} = \det \mathbf{A}^T = \det \mathbf{A}^{-1} = 1/\det \mathbf{A}$ and $(\det \mathbf{A})^2 = 1$, so $\det \mathbf{A} = \pm 1$.

35. Since $\mathbf{A}$ and $\mathbf{B}$ are nonsingular, $\det \mathbf{AB} = \det \mathbf{A} \cdot \det \mathbf{B} \neq 0$, and $\mathbf{AB}$ is nonsingular.

36. Suppose $\mathbf{A}$ is singular. Then $\det \mathbf{A} = 0$, $\det \mathbf{AB} = \det \mathbf{A} \cdot \det \mathbf{B} = 0$, and $\mathbf{AB}$ is singular.

37. Since $\det \mathbf{A} \cdot \det \mathbf{A}^{-1} = \det \mathbf{AA}^{-1} = \det \mathbf{I} = 1$, $\det \mathbf{A}^{-1} = 1/\det \mathbf{A}$.

38. Suppose $\mathbf{A}^2 = \mathbf{A}$ and $\mathbf{A}$ is nonsingular. Then $\mathbf{A}^2\mathbf{A}^{-1} = \mathbf{AA}^{-1}$, and $\mathbf{A} = \mathbf{I}$. Thus, if $\mathbf{A}^2 = \mathbf{A}$, either $\mathbf{A}$ is singular or $\mathbf{A} = \mathbf{I}$.

39. If $\mathbf{A}$ is nonsingular, then $\mathbf{A}^{-1}$ exists, and $\mathbf{AB} = \mathbf{0}$ implies $\mathbf{A}^{-1}\mathbf{AB} = \mathbf{A}^{-1}\mathbf{0}$, so $\mathbf{B} = \mathbf{0}$.

40. If $\mathbf{A}$ is nonsingular, $\mathbf{A}^{-1}$ exists, and $\mathbf{AB} = \mathbf{AC}$ implies $\mathbf{A}^{-1}\mathbf{AB} = \mathbf{A}^{-1}\mathbf{AC}$, so $\mathbf{B} = \mathbf{C}$.

Exercises 8.6

41. No, consider $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$.

42. $\mathbf{A}$ is nonsingular if $a_{11}a_{22}a_{33} = 0$ or a_{11} , a_{22} , and a_{33} are all nonzero.

$$\mathbf{A}^{-1} = \begin{bmatrix} 1/a_{11} & 0 & 0 \\ 0 & 1/a_{22} & 0 \\ 0 & 0 & 1/a_{33} \end{bmatrix}$$

For any diagonal matrix, the inverse matrix is obtained by taking the reciprocals of the diagonal entries and leaving all other entries 0.

43. $\mathbf{A}^{-1} = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} \\ \frac{2}{3} & -\frac{1}{3} \end{bmatrix}$; $\mathbf{A}^{-1} \begin{bmatrix} 4 \\ 14 \end{bmatrix} = \begin{bmatrix} 6 \\ -2 \end{bmatrix}$; $x_1 = 6$, $x_2 = -2$

44. $\mathbf{A}^{-1} = \begin{bmatrix} \frac{2}{3} & \frac{1}{6} \\ -\frac{1}{3} & \frac{1}{6} \end{bmatrix}$; $\mathbf{A}^{-1} \begin{bmatrix} 2 \\ -5 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ -\frac{3}{2} \end{bmatrix}$; $x_1 = \frac{1}{2}$, $x_2 = -\frac{3}{2}$

45. $\mathbf{A}^{-1} = \begin{bmatrix} \frac{1}{16} & \frac{3}{8} \\ -\frac{1}{8} & \frac{1}{4} \end{bmatrix}$; $\mathbf{A}^{-1} \begin{bmatrix} 6 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{3}{4} \\ -\frac{1}{2} \end{bmatrix}$; $x_1 = \frac{3}{4}$, $x_2 = -\frac{1}{2}$

46. $\mathbf{A}^{-1} = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}$; $\mathbf{A}^{-1} \begin{bmatrix} 4 \\ -3 \end{bmatrix} = \begin{bmatrix} -11 \\ \frac{15}{2} \end{bmatrix}$; $x_1 = -11$, $x_2 = \frac{15}{2}$

47. $\mathbf{A}^{-1} = \begin{bmatrix} -\frac{1}{5} & \frac{1}{5} & \frac{1}{5} \\ -1 & 1 & 0 \\ \frac{6}{5} & -\frac{1}{5} & -\frac{1}{5} \end{bmatrix}$; $\mathbf{A}^{-1} \begin{bmatrix} -4 \\ 0 \\ 6 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ -6 \end{bmatrix}$; $x_1 = 2$, $x_2 = 4$, $x_3 = -6$

48. $\mathbf{A}^{-1} = \begin{bmatrix} \frac{5}{12} & -\frac{1}{12} & \frac{1}{4} \\ -\frac{2}{3} & \frac{1}{3} & 0 \\ -\frac{1}{12} & \frac{5}{12} & -\frac{1}{4} \end{bmatrix}$; $\mathbf{A}^{-1} \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} \\ 0 \\ \frac{3}{2} \end{bmatrix}$; $x_1 = -\frac{1}{2}$, $x_2 = 0$, $x_3 = \frac{3}{2}$

49. $\mathbf{A}^{-1} = \begin{bmatrix} -2 & -3 & 2 \\ \frac{1}{4} & -\frac{1}{4} & 0 \\ \frac{5}{4} & \frac{7}{4} & -1 \end{bmatrix}$; $\mathbf{A}^{-1} \begin{bmatrix} 1 \\ -3 \\ 7 \end{bmatrix} = \begin{bmatrix} 21 \\ 1 \\ -11 \end{bmatrix}$; $x_1 = 21$, $x_2 = 1$, $x_3 = -11$

50. $\mathbf{A}^{-1} = \begin{bmatrix} 2 & -1 & 1 & 1 \\ -1 & 2 & -1 & -1 \\ 1 & -1 & 1 & 1 \\ 1 & -1 & 1 & 0 \end{bmatrix}$; $\mathbf{A}^{-1} \begin{bmatrix} 2 \\ 1 \\ -5 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -1 \\ -4 \end{bmatrix}$; $x_1 = 1$, $x_2 = 2$, $x_3 = -1$, $x_4 = -4$

51. $\begin{bmatrix} 7 & -2 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$; $\mathbf{A}^{-1} = \begin{bmatrix} \frac{1}{10} & \frac{1}{10} \\ -\frac{3}{20} & \frac{7}{20} \end{bmatrix}$; $\mathbf{X} = \mathbf{A}^{-1} \begin{bmatrix} 5 \\ 4 \end{bmatrix} = \begin{bmatrix} \frac{9}{10} \\ \frac{13}{20} \end{bmatrix}$; $\mathbf{X} = \mathbf{A}^{-1} \begin{bmatrix} 10 \\ 50 \end{bmatrix} = \begin{bmatrix} 6 \\ 16 \end{bmatrix}$;

$$\mathbf{X} = \mathbf{A}^{-1} \begin{bmatrix} 0 \\ -20 \end{bmatrix} = \begin{bmatrix} -2 \\ -7 \end{bmatrix}$$

52. $\begin{bmatrix} 1 & 2 & 5 \\ 2 & 3 & 8 \\ -1 & 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$; $\mathbf{A}^{-1} = \begin{bmatrix} 2 & -1 & -1 \\ 12 & -7 & -2 \\ -5 & 3 & 1 \end{bmatrix}$; $\mathbf{X} = \mathbf{A}^{-1} \begin{bmatrix} -1 \\ 4 \\ 6 \end{bmatrix} = \begin{bmatrix} -12 \\ -52 \\ 23 \end{bmatrix}$;

$$\mathbf{X} = \mathbf{A}^{-1} \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 9 \\ -3 \end{bmatrix}; \quad \mathbf{X} = \mathbf{A}^{-1} \begin{bmatrix} 0 \\ -5 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 \\ 27 \\ -11 \end{bmatrix}$$

53. $\det \mathbf{A} = 18 \neq 0$, so the system has only the trivial solution.

54. $\det \mathbf{A} = 0$, so the system has a nontrivial solution.

55. $\det \mathbf{A} = 0$, so the system has a nontrivial solution.

56. $\det \mathbf{A} = 12 \neq 0$, so the system has only the trivial solution.

$$57. \text{ (a) } \begin{bmatrix} 1 & 1 & 1 \\ -R_1 & R_2 & 0 \\ 0 & -R_2 & R_3 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 0 \\ E_2 - E_1 \\ E_3 - E_2 \end{bmatrix}$$

(b) $\det \mathbf{A} = R_1 R_2 + R_1 R_3 + R_2 R_3 > 0$, so $\mathbf{A}$ is nonsingular.

$$\text{(c) } \mathbf{A}^{-1} = \frac{1}{R_1 R_2 + R_1 R_3 + R_2 R_3} \begin{bmatrix} R_2 R_3 & -R_2 - R_3 & -R_2 \\ R_1 R_3 & R_3 & -R_1 \\ R_1 R_2 & R_2 & R_1 + R_2 \end{bmatrix};$$

$$\mathbf{A}^{-1} \begin{bmatrix} 0 \\ E_2 - E_1 \\ E_3 - E_2 \end{bmatrix} = \frac{1}{R_1 R_2 + R_1 R_3 + R_2 R_3} \begin{bmatrix} R_2 E_1 - R_2 E_3 + R_3 E_1 - R_3 E_2 \\ R_1 E_2 - R_1 E_3 - R_3 E_1 + R_3 E_2 \\ -R_1 E_2 + R_1 E_3 - R_2 E_1 + R_2 E_3 \end{bmatrix}$$

58. (a) We write the equations in the form

$$-4u_1 + u_2 + u_4 = -200$$

$$u_1 - 4u_2 + u_3 = -300$$

$$u_2 - 4u_3 + u_4 = -300$$

$$u_1 + u_3 - 4u_4 = -200.$$

In matrix form this becomes

$$\begin{bmatrix} -4 & 1 & 0 & 1 \\ 1 & -4 & 1 & 0 \\ 0 & 1 & -4 & 1 \\ 1 & 0 & 1 & -4 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} = \begin{bmatrix} -200 \\ -300 \\ -300 \\ -200 \end{bmatrix}.$$

$$\text{(b) } \mathbf{A}^{-1} = \begin{bmatrix} -\frac{7}{24} & -\frac{1}{12} & -\frac{1}{24} & -\frac{1}{12} \\ -\frac{1}{12} & -\frac{7}{24} & -\frac{1}{12} & -\frac{1}{24} \\ -\frac{1}{24} & -\frac{1}{12} & -\frac{7}{24} & -\frac{1}{12} \\ -\frac{1}{12} & -\frac{1}{24} & -\frac{1}{12} & -\frac{7}{24} \end{bmatrix}; \quad \mathbf{A}^{-1} \begin{bmatrix} -200 \\ -300 \\ -300 \\ -200 \end{bmatrix} = \begin{bmatrix} \frac{225}{2} \\ \frac{275}{2} \\ \frac{275}{2} \\ \frac{225}{2} \end{bmatrix}; \quad u_1 = u_4 = \frac{225}{2}, \quad u_2 = u_3 = \frac{275}{2}$$

Exercises 8.7

1. $\det \mathbf{A} = 10$, $\det \mathbf{A}_1 = -6$, $\det \mathbf{A}_2 = 12$; $x_1 = \frac{-6}{10} = -\frac{3}{5}$, $x_2 = \frac{12}{10} = \frac{6}{5}$

2. $\det \mathbf{A} = -3$, $\det \mathbf{A}_1 = -6$, $\det \mathbf{A}_2 = -6$; $x_1 = \frac{-6}{-3} = 2$, $x_2 = \frac{-6}{-3} = 2$

3. $\det \mathbf{A} = 0.3$, $\det \mathbf{A}_1 = 0.03$, $\det \mathbf{A}_2 = -0.09$; $x_1 = \frac{0.03}{0.3} = 0.1$, $x_2 = \frac{-0.09}{0.3} = -0.3$

4. $\det \mathbf{A} = -0.015$, $\det \mathbf{A}_1 = -0.00315$, $\det \mathbf{A}_2 = -0.00855$; $x_1 = \frac{-0.00315}{-0.015} = 0.21$, $x_2 = \frac{-0.00855}{-0.015} = 0.57$

5. $\det \mathbf{A} = 1$, $\det \mathbf{A}_1 = 4$, $\det \mathbf{A}_2 = -7$; $x = 4$, $y = -7$

6. $\det \mathbf{A} = -70$, $\det \mathbf{A}_1 = -14$, $\det \mathbf{A}_2 = 35$; $r = \frac{-14}{-70} = \frac{1}{5}$, $s = \frac{35}{-70} = -\frac{1}{2}$

7. $\det \mathbf{A} = 11$, $\det \mathbf{A}_1 = -44$, $\det \mathbf{A}_2 = 44$, $\det \mathbf{A}_3 = -55$; $x_1 = \frac{-44}{11} = -4$, $x_2 = \frac{44}{11} = 4$, $x_3 = \frac{-55}{11} = -5$

Exercises 8.7

8. $\det \mathbf{A} = -63$, $\det \mathbf{A}_1 = 173$, $\det \mathbf{A}_2 = -136$, $\det \mathbf{A}_3 = -\frac{61}{2}$; $x_1 = -\frac{173}{63}$, $x_2 = \frac{136}{63}$, $x_3 = \frac{61}{126}$
9. $\det \mathbf{A} = -12$, $\det \mathbf{A}_1 = -48$, $\det \mathbf{A}_2 = -18$, $\det \mathbf{A}_3 = -12$; $u = \frac{48}{12} = 4$, $v = \frac{18}{12} = \frac{3}{2}$, $w = 1$
10. $\det \mathbf{A} = 1$, $\det \mathbf{A}_1 = -2$, $\det \mathbf{A}_2 = 2$, $\det \mathbf{A}_3 = 5$; $x = -2$, $y = 2$, $z = 5$
11. $\det \mathbf{A} = 6 - 5k$, $\det \mathbf{A}_1 = 12 - 7k$, $\det \mathbf{A}_2 = 6 - 7k$; $x_1 = \frac{12 - 7k}{6 - 5k}$, $x_2 = \frac{6 - 7k}{6 - 5k}$. The system is inconsistent for $k = 6/5$.
12. (a) $\det \mathbf{A} = \epsilon - 1$, $\det \mathbf{A}_1 = \epsilon - 2$, $\det \mathbf{A}_2 = 1$; $x_1 = \frac{\epsilon - 2}{\epsilon - 1} = \frac{\epsilon - 1 - 1}{\epsilon - 1} = 1 - \frac{1}{\epsilon - 1}$, $x_2 = \frac{1}{\epsilon - 1}$
 (b) When $\epsilon = 1.01$, $x_1 = -99$ and $x_2 = 100$. When $\epsilon = 0.99$, $x_1 = 101$ and $x_2 = -100$.
13. $\det \mathbf{A} \approx 0.6428$, $\det \mathbf{A}_1 \approx 289.8$, $\det \mathbf{A}_2 \approx 271.9$; $x_1 \approx \frac{289.8}{0.6428} \approx 450.8$, $x_2 \approx \frac{271.9}{0.6428} \approx 423$
14. We have $(\sin 30^\circ)F + (\sin 30^\circ)(0.5N) + N \sin 60^\circ = 400$ and $(\cos 30^\circ)F + (\cos 30^\circ)(0.5N) - N \cos 60^\circ = 0$. The system is

$$(\sin 30^\circ)F + (0.5 \sin 30^\circ + \sin 60^\circ)N = 400$$

$$(\cos 30^\circ)F + (0.5 \cos 30^\circ - \cos 60^\circ)N = 0.$$

$$\det \mathbf{A} \approx -1, \det \mathbf{A}_1 \approx -26.795, \det \mathbf{A}_2 \approx -346.41; F \approx 26.795, N \approx 346.41$$

15. The system is

$$i_1 + i_2 - i_3 = 0$$

$$r_1 i_1 - r_2 i_2 = E_1 - E_2$$

$$r_2 i_2 + R i_3 = E_2$$

$$\det \mathbf{A} = -r_1 R - r_2 R - r_1 r_2, \det \mathbf{A}_3 = -r_1 E_2, -r_2 E_1; i_3 = \frac{r_1 E_2 + r_2 E_1}{r_1 R + r_2 R + r_1 r_2}$$

Exercises 8.8

1. $\mathbf{K}_3$ since $\begin{bmatrix} 4 & 2 \\ 5 & 1 \end{bmatrix} \begin{bmatrix} -2 \\ 5 \end{bmatrix} = \begin{bmatrix} 2 \\ -5 \end{bmatrix} = (-1) \begin{bmatrix} -2 \\ 5 \end{bmatrix}$; $\lambda = -1$
2. $\mathbf{K}_1$ and $\mathbf{K}_2$ since $\begin{bmatrix} 2 & -1 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 - \sqrt{2} \end{bmatrix} = \begin{bmatrix} \sqrt{2} \\ -2 + 2\sqrt{2} \end{bmatrix} = \sqrt{2} \begin{bmatrix} 1 \\ 2 - \sqrt{2} \end{bmatrix}$, $\lambda = \sqrt{2}$
 $\begin{bmatrix} 2 & -1 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} 2 + \sqrt{2} \\ 2 \end{bmatrix} = \begin{bmatrix} 2 + 2\sqrt{2} \\ 2\sqrt{2} \end{bmatrix} = \sqrt{2} \begin{bmatrix} 2 + \sqrt{2} \\ 2 \end{bmatrix}$; $\lambda = \sqrt{2}$
3. $\mathbf{K}_3$ since $\begin{bmatrix} 6 & 3 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} -5 \\ 10 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = 0 \begin{bmatrix} -5 \\ 10 \end{bmatrix}$; $\lambda = 0$
4. $\mathbf{K}_2$ since $\begin{bmatrix} 2 & 8 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} 2 + 2i \\ -1 \end{bmatrix} = \begin{bmatrix} -4 + 4i \\ -2i \end{bmatrix} = 2i \begin{bmatrix} 2 + 2i \\ -1 \end{bmatrix}$; $\lambda = 2i$
5. $\mathbf{K}_2$ and $\mathbf{K}_3$ since $\begin{bmatrix} 1 & -2 & 2 \\ -2 & 1 & -2 \\ 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ -4 \\ 0 \end{bmatrix} = \begin{bmatrix} 12 \\ -12 \\ 0 \end{bmatrix} = 3 \begin{bmatrix} 4 \\ -4 \\ 0 \end{bmatrix}$; $\lambda = 3$
 $\begin{bmatrix} 1 & -2 & 2 \\ -2 & 1 & -2 \\ 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$; $\lambda = 1$

6. $\mathbf{K}_2$ since $\begin{bmatrix} -1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 3 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 12 \\ 9 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 4 \\ 3 \end{bmatrix}; \quad \lambda = 3$

7. We solve $\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} -1 - \lambda & 2 \\ -7 & 8 - \lambda \end{vmatrix} = (\lambda - 6)(\lambda - 1) = 0.$

For $\lambda_1 = 6$ we have $\begin{bmatrix} -7 & 2 & | & 0 \\ -7 & 2 & | & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & -2/7 & | & 0 \\ 0 & 0 & | & 0 \end{bmatrix}$

so that $k_1 = \frac{2}{7}k_2$. If $k_2 = 7$ then $\mathbf{K}_1 = \begin{bmatrix} 2 \\ 7 \end{bmatrix}$. For $\lambda_2 = 1$ we have

$$\begin{bmatrix} -2 & 2 & | & 0 \\ -7 & 7 & | & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & -1 & | & 0 \\ 0 & 0 & | & 0 \end{bmatrix}$$

so that $k_1 = k_2$. If $k_2 = 1$ then $\mathbf{K}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

8. We solve $\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} 2 - \lambda & 1 \\ 2 & 1 - \lambda \end{vmatrix} = \lambda(\lambda - 3) = 0.$

For $\lambda_1 = 0$ we have $\begin{bmatrix} 2 & 1 & | & 0 \\ 2 & 1 & | & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 1/2 & | & 0 \\ 0 & 0 & | & 0 \end{bmatrix}$

so that $k_1 = -\frac{1}{2}k_2$. If $k_2 = 2$ then $\mathbf{K}_1 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$. For $\lambda_2 = 3$ we have

$$\begin{bmatrix} -1 & 1 & | & 0 \\ 2 & -2 & | & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & -1 & | & 0 \\ 0 & 0 & | & 0 \end{bmatrix}$$

so that $k_1 = k_2$. If $k_2 = 1$ then $\mathbf{K}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

9. We solve $\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} -8 - \lambda & -1 \\ 16 & -\lambda \end{vmatrix} = (\lambda + 4)^2 = 0.$

For $\lambda_1 = \lambda_2 = -4$ we have $\begin{bmatrix} -4 & -1 & | & 0 \\ 16 & 4 & | & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 1/4 & | & 0 \\ 0 & 0 & | & 0 \end{bmatrix}$

so that $k_1 = -\frac{1}{4}k_2$. If $k_2 = 4$ then $\mathbf{K}_1 = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$.

10. We solve $\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} 1 - \lambda & 1 \\ 1/4 & 1 - \lambda \end{vmatrix} = (\lambda - 3/2)(\lambda - 1/2) = 0.$

For $\lambda_1 = 3/2$ we have $\begin{bmatrix} -1/2 & 1 & | & 0 \\ 1/4 & -1/2 & | & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & -2 & | & 0 \\ 0 & 0 & | & 0 \end{bmatrix}$

so that $k_1 = 2k_2$. If $k_2 = 1$ then $\mathbf{K}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. If $\lambda_2 = 1/2$ then

$$\begin{bmatrix} 1/2 & 1 & | & 0 \\ 1/4 & 1/2 & | & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 & | & 0 \\ 0 & 0 & | & 0 \end{bmatrix}$$

so that $k_1 = -2k_2$. If $k_2 = 1$ then $\mathbf{K}_2 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$.

Exercises 8.8

11. We solve $\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} -1 - \lambda & 2 \\ -5 & 1 - \lambda \end{vmatrix} = \lambda^2 + 9 = (\lambda - 3i)(\lambda + 3i) = 0.$

For $\lambda_1 = 3i$ we have $\left[\begin{array}{cc|c} -1 - 3i & 2 & 0 \\ -5 & 1 - 3i & 0 \end{array} \right] \Rightarrow \left[\begin{array}{cc|c} 1 & -(1/5) + (3/5)i & 0 \\ 0 & 0 & 0 \end{array} \right]$

so that $k_1 = (\frac{1}{5} - \frac{3}{5}i) k_2$. If $k_2 = 5$ then $\mathbf{K}_1 = \begin{bmatrix} 1 - 3i \\ 5 \end{bmatrix}$. For $\lambda_2 = -3i$ we have

$$\left[\begin{array}{cc|c} -1 + 3i & 2 & 0 \\ -5 & 1 + 3i & 0 \end{array} \right] \Rightarrow \left[\begin{array}{cc|c} 1 & -\frac{1}{5} - \frac{3}{5}i & 0 \\ 0 & 0 & 0 \end{array} \right]$$

so that $k_1 = (\frac{1}{5} + \frac{3}{5}i) k_2$. If $k_2 = 5$ then $\mathbf{K}_2 = \begin{bmatrix} 1 + 3i \\ 5 \end{bmatrix}$.

12. We solve $\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} 1 - \lambda & -1 \\ 1 & 1 - \lambda \end{vmatrix} = \lambda^2 - 2\lambda + 2 = 0.$

For $\lambda_1 = 1 - i$ we have $\left[\begin{array}{cc|c} i & -1 & 0 \\ 1 & i & 0 \end{array} \right] \Rightarrow \left[\begin{array}{cc|c} i & -1 & 0 \\ 0 & 0 & 0 \end{array} \right]$

so that $k_1 = -ik_2$. If $k_2 = 1$ then $\mathbf{K}_1 = \begin{bmatrix} -i \\ 1 \end{bmatrix}$ and $\mathbf{K}_2 = \overline{\mathbf{K}}_1 = \begin{bmatrix} i \\ 1 \end{bmatrix}$.

13. We solve $\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} 4 - \lambda & 8 \\ 0 & -5 - \lambda \end{vmatrix} = (\lambda - 4)(\lambda + 5) = 0.$

For $\lambda_1 = 4$ we have $\left[\begin{array}{cc|c} 0 & 8 & 0 \\ 0 & -9 & 0 \end{array} \right] \Rightarrow \left[\begin{array}{cc|c} 0 & 1 & 0 \\ 0 & 0 & 0 \end{array} \right]$

so that $k_2 = 0$. If $k_1 = 1$ then $\mathbf{K}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. For $\lambda_2 = -5$ we have

$$\left[\begin{array}{cc|c} 9 & 8 & 0 \\ 0 & 0 & 0 \end{array} \right] \Rightarrow \left[\begin{array}{cc|c} 1 & \frac{8}{9} & 0 \\ 0 & 0 & 0 \end{array} \right]$$

so that $k_1 = -\frac{8}{9} k_2$. If $k_2 = 9$ then $\mathbf{K}_2 = \begin{bmatrix} -8 \\ 9 \end{bmatrix}$.

14. We solve $\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} 7 - \lambda & 0 \\ 0 & 13 - \lambda \end{vmatrix} = (\lambda - 7)(\lambda - 13) = 0.$

For $\lambda_1 = 7$ we have $\left[\begin{array}{cc|c} 0 & 0 & 0 \\ 0 & 6 & 0 \end{array} \right] \Rightarrow \left[\begin{array}{cc|c} 0 & 1 & 0 \\ 0 & 0 & 0 \end{array} \right]$

so that $k_2 = 0$. If $k_1 = 1$ then $\mathbf{K}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. For $\lambda_2 = 13$ we have

$$\left[\begin{array}{cc|c} -6 & 0 & 0 \\ 0 & 0 & 0 \end{array} \right] \Rightarrow \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 0 & 0 \end{array} \right]$$

so that $k_1 = 0$. If $k_2 = 1$ then $\mathbf{K}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

15. We solve $\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} 5 - \lambda & -1 & 0 \\ 0 & -5 - \lambda & 9 \\ 5 & -1 & -\lambda \end{vmatrix} = \begin{vmatrix} 4 - \lambda & -1 & 0 \\ 4 - \lambda & -5 - \lambda & 9 \\ 4 - \lambda & -1 & -\lambda \end{vmatrix} = \lambda(4 - \lambda)(\lambda + 4) = 0.$

For $\lambda_1 = 0$ we have $\left[\begin{array}{ccc|c} 5 & -1 & 0 & 0 \\ 0 & -5 & 9 & 0 \\ 5 & -1 & 0 & 0 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -9/25 & 0 \\ 0 & 1 & -9/5 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$

so that $k_1 = \frac{9}{25}k_3$ and $k_2 = \frac{9}{5}k_3$. If $k_3 = 25$ then $\mathbf{K}_1 = \begin{bmatrix} 9 \\ 45 \\ 25 \end{bmatrix}$. If $\lambda_2 = 4$ then

$$\left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 0 & -9 & 9 & 0 \\ 5 & -1 & -4 & 0 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

so that $k_1 = k_3$ and $k_2 = k_3$. If $k_3 = 1$ then $\mathbf{K}_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. If $\lambda_3 = -4$ then

$$\left[\begin{array}{ccc|c} 9 & -1 & 0 & 0 \\ 0 & -1 & 9 & 0 \\ 5 & -1 & 4 & 0 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & -9 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

so that $k_1 = k_3$ and $k_2 = 9k_3$. If $k_3 = 1$ then $\mathbf{K}_3 = \begin{bmatrix} 1 \\ 9 \\ 1 \end{bmatrix}$.

16. We solve $\det(\mathbf{A} - \lambda\mathbf{I}) = \begin{vmatrix} 3-\lambda & 0 & 0 \\ 0 & 2-\lambda & 0 \\ 4 & 0 & 1-\lambda \end{vmatrix} = (3-\lambda)(2-\lambda)(1-\lambda) = 0.$

For $\lambda_1 = 1$ we have $\left[\begin{array}{ccc|c} 2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 4 & 0 & 0 & 0 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$

so that $k_1 = 0$ and $k_2 = 0$. If $k_3 = 1$ then $\mathbf{K}_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$. If $\lambda_2 = 2$ then

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 4 & 0 & -1 & 0 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

so that $k_1 = 0$ and $k_3 = 0$. If $k_2 = 1$ then $\mathbf{K}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$. If $\lambda_3 = 3$ then

$$\left[\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 4 & 0 & -2 & 0 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -1/2 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

so that $k_1 = \frac{1}{2}k_3$ and $k_2 = 0$. If $k_3 = 2$ then $\mathbf{K}_3 = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$.

Exercises 8.8

17. We solve $\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} -\lambda & 4 & 0 \\ -1 & -4-\lambda & 0 \\ 0 & 0 & -2-\lambda \end{vmatrix} = -(\lambda+2)^3 = 0.$

For $\lambda_1 = \lambda_2 = \lambda_3 = -2$ we have $\left[\begin{array}{ccc|c} 2 & 4 & 0 & 0 \\ -1 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$

so that $k_1 = -2k_2$. If $k_2 = 1$ and $k_3 = 1$ then

$$\mathbf{K}_1 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{K}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

18. We solve $\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} 1-\lambda & 6 & 0 \\ 0 & 2-\lambda & 1 \\ 0 & 1 & 2-\lambda \end{vmatrix} = \begin{vmatrix} 1-\lambda & 6 & 0 \\ 0 & 3-\lambda & 3-\lambda \\ 0 & 1 & 2-\lambda \end{vmatrix} = (3-\lambda)(1-\lambda)^2 = 0.$

For $\lambda_1 = 3$ we have $\left[\begin{array}{ccc|c} -2 & 6 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -3 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$

so that $k_1 = 3k_3$ and $k_2 = k_3$. If $k_3 = 1$ then $\mathbf{K}_1 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$. For $\lambda_2 = \lambda_3 = 1$ we have

$$\left[\begin{array}{ccc|c} 0 & 6 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

so that $k_2 = 0$ and $k_3 = 0$. If $k_1 = 1$ then $\mathbf{K}_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$.

19. We solve $\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} -\lambda & 0 & -1 \\ 0 & -\lambda & 0 \\ 1 & 1 & -1-\lambda \end{vmatrix} = -(\lambda+1)(\lambda^2+1) = 0.$

For $\lambda_1 = -1$ we have $\left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$

so that $k_1 = k_3$ and $k_2 = -k_3$. If $k_3 = 1$ then $\mathbf{K}_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$. For $\lambda_2 = i$ we have

$$\left[\begin{array}{ccc|c} -i & 0 & -1 & 0 \\ 1 & -i & 0 & 0 \\ 1 & 1 & -1-i & 0 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -i & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

so that $k_1 = ik_3$ and $k_2 = k_3$. If $k_3 = 1$ then $\mathbf{K}_2 = \begin{bmatrix} i \\ 1 \\ 1 \end{bmatrix}$ and $\mathbf{K}_3 = \overline{\mathbf{K}_2} = \begin{bmatrix} -i \\ 1 \\ 1 \end{bmatrix}$.

20. We solve

$$\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} 2 - \lambda & -1 & 0 \\ 5 & 2 - \lambda & 4 \\ 0 & 1 & 2 - \lambda \end{vmatrix} = -\lambda^3 + 6\lambda^2 - 13\lambda + 10 = (\lambda - 2)(-\lambda^2 + 4\lambda - 5) \\ = (\lambda - 2)(\lambda - (2 + i))(\lambda - (2 - i)) = 0.$$

For $\lambda_1 = 2$ we have

$$\begin{bmatrix} 0 & -1 & 0 & | & 0 \\ 5 & 0 & 4 & | & 0 \\ 0 & 1 & 0 & | & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & 4/5 & | & 0 \\ 0 & 1 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

so that $k_1 = -\frac{4}{5}k_3$ and $k_2 = 0$. If $k_3 = 5$ then $\mathbf{K}_1 = \begin{bmatrix} -4 \\ 0 \\ 5 \end{bmatrix}$. For $\lambda_2 = 2 + i$ we have

$$\begin{bmatrix} -i & -1 & 0 & | & 0 \\ 5 & -i & 4 & | & 0 \\ 0 & 1 & -i & | & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & -i & 0 & | & 0 \\ 0 & 1 & -i & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

so that $k_1 = ik_2$ and $k_2 = ik_3$. If $k_3 = i$ then $\mathbf{K}_2 = \begin{bmatrix} -i \\ -1 \\ i \end{bmatrix}$. For $\lambda_3 = 2 - i$ we have

$$\begin{bmatrix} i & -1 & 0 & | & 0 \\ 5 & i & 4 & | & 0 \\ 0 & 1 & i & | & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & i & 0 & | & 0 \\ 0 & 1 & i & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

so that $k_1 = -ik_2$ and $k_2 = -ik_3$. If $k_3 = i$ then $\mathbf{K}_3 = \begin{bmatrix} -1 \\ 1 \\ i \end{bmatrix}$.

21. We solve $\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} 1 - \lambda & 2 & 3 \\ 0 & 5 - \lambda & 6 \\ 0 & 0 & -7 - \lambda \end{vmatrix} = -(\lambda - 1)(\lambda - 5)(\lambda + 7) = 0.$

For $\lambda_1 = 1$ we have

$$\begin{bmatrix} 0 & 2 & 3 & | & 0 \\ 0 & 4 & 6 & | & 0 \\ 0 & 0 & -6 & | & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & 1 & 0 & | & 0 \\ 0 & 0 & 1 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

so that $k_2 = k_3 = 0$. If $k_1 = 1$ then $\mathbf{K}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$. For $\lambda_2 = 5$ we have

$$\begin{bmatrix} -4 & 2 & 3 & | & 0 \\ 0 & 0 & 6 & | & 0 \\ 0 & 0 & -12 & | & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & -\frac{1}{2} & 0 & | & 0 \\ 0 & 0 & 1 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

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so that $k_3 = 0$ and $k_2 = 2k_1$. If $k_1 = 1$ then $\mathbf{K}_2 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$. For $\lambda_3 = -7$ we have

$$\left[\begin{array}{ccc|c} 8 & 2 & 3 & 0 \\ 0 & 12 & 6 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & \frac{1}{4} & 0 \\ 0 & 1 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

so that $k_1 = -\frac{1}{4}k_3$ and $k_2 = -\frac{1}{2}k_3$. If $k_3 = 4$ then $\mathbf{K}_3 = \begin{bmatrix} -1 \\ -2 \\ 4 \end{bmatrix}$.

22. We solve
$$\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} -\lambda & 0 & 0 \\ 0 & -\lambda & 0 \\ 0 & 0 & 1 - \lambda \end{vmatrix} = -\lambda^2(\lambda - 1) = 0.$$

For $\lambda_1 = \lambda_2 = 0$ we have
$$\left[\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|c} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

so that $k_3 = 0$. If $k_1 = 1$ and $k_2 = 0$ then $\mathbf{K}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ and if $k_1 = 0$ and $k_2 = 1$ then $\mathbf{K}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$. For $\lambda_3 = 1$ we

have

$$\left[\begin{array}{ccc|c} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

so that $k_1 = k_2 = 0$. If $k_3 = 1$ then $\mathbf{K}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$.

23. The eigenvalues and eigenvectors of $\mathbf{A} = \begin{bmatrix} 5 & 1 \\ 1 & 5 \end{bmatrix}$ are

$$\lambda_1 = 4, \quad \lambda_2 = 6, \quad \mathbf{K}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad \mathbf{K}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

and the eigenvalues and eigenvectors of $\mathbf{A}^{-1} = \frac{1}{24} \begin{bmatrix} 5 & -1 \\ -1 & 5 \end{bmatrix}$ are

$$\lambda_1 = \frac{1}{4}, \quad \lambda_2 = \frac{1}{6}, \quad \mathbf{K}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad \mathbf{K}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

24. The eigenvalues and eigenvectors of $\mathbf{A} = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 0 & 1 \\ 4 & -4 & 5 \end{bmatrix}$ are

$$\lambda_1 = 1, \quad \lambda_2 = 2, \quad \lambda_3 = 3, \quad \mathbf{K}_1 = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}, \quad \mathbf{K}_2 = \begin{bmatrix} -2 \\ 1 \\ 4 \end{bmatrix}, \quad \mathbf{K}_3 = \begin{bmatrix} -1 \\ 1 \\ 4 \end{bmatrix}.$$

and the eigenvalues and eigenvectors of $\mathbf{A}^{-1} = \frac{1}{6} \begin{bmatrix} 4 & -6 & 2 \\ -1 & 9 & -2 \\ -4 & 12 & -2 \end{bmatrix}$ are

$$\lambda_1 = 1, \quad \lambda_2 = \frac{1}{2}, \quad \lambda_3 = \frac{1}{3}, \quad \mathbf{K}_1 = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}, \quad \mathbf{K}_2 = \begin{bmatrix} -2 \\ 1 \\ 4 \end{bmatrix}, \quad \mathbf{K}_3 = \begin{bmatrix} -1 \\ 1 \\ 4 \end{bmatrix}.$$

25. Since $\det \mathbf{A} = \begin{vmatrix} 6 & 0 \\ 3 & 0 \end{vmatrix} = 0$ the matrix is singular. Now from

$$\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} 6 - \lambda & 0 \\ 3 & -\lambda \end{vmatrix} = \lambda(\lambda - 6)$$

we see $\lambda = 0$ is an eigenvalue.

26. Since $\det \mathbf{A} = \begin{vmatrix} 1 & 0 & 1 \\ 4 & -4 & 5 \\ 7 & -4 & 8 \end{vmatrix} = 0$ the matrix is singular. Now from

$$\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} 1 - \lambda & 0 & 1 \\ 4 & -4 - \lambda & 5 \\ 7 & -4 & 8 - \lambda \end{vmatrix} = -\lambda(\lambda^2 - 5\lambda - 15)$$

we see $\lambda = 0$ is an eigenvalue.

27. (a) Since $p + 1 - p = 1$ and $q + 1 - q = 1$, the first matrix $\mathbf{A}$ is stochastic. Since $\frac{1}{2} + \frac{1}{4} + \frac{1}{4} = 1$, $\frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$, and $\frac{1}{6} + \frac{1}{3} + \frac{1}{2} = 1$, the second matrix $\mathbf{A}$ is stochastic.

(b) The matrix from part (a) is shown with its eigenvalues and corresponding eigenvectors.

$$\begin{bmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{6} & \frac{1}{3} & \frac{1}{2} \end{bmatrix}; \quad \text{eigenvalues: } 1, \frac{1}{6} - \frac{1}{12}\sqrt{2}, \frac{1}{6} + \frac{1}{12}\sqrt{2};$$

$$\text{eigenvectors: } (1, 1, 1), \left(-\frac{3(-1+\sqrt{2})}{-6+\sqrt{2}}, \frac{2(2+\sqrt{2})}{-6+\sqrt{2}}, 1\right), \left(-\frac{3(1+\sqrt{2})}{6+\sqrt{2}}, \frac{2(-2+\sqrt{2})}{6+\sqrt{2}}, 1\right)$$

Further examples indicate that 1 is always an eigenvalue with corresponding eigenvector $(1, 1, 1)$. To prove this, let $\mathbf{A}$ be a stochastic matrix and $\mathbf{K} = (1, 1, 1)$. Then

$$\mathbf{AK} = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} + \cdots + a_{1n} \\ \vdots \\ a_{n1} + \cdots + a_{nn} \end{bmatrix} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} = 1\mathbf{K},$$

and 1 is an eigenvalue of $\mathbf{A}$ with corresponding eigenvector $(1, 1, 1)$.

(c) For the 3×3 matrix in part (a) we have

$$\mathbf{A}^2 = \begin{bmatrix} \frac{3}{8} & \frac{7}{24} & \frac{1}{3} \\ \frac{1}{3} & \frac{11}{36} & \frac{13}{36} \\ \frac{5}{18} & \frac{23}{72} & \frac{29}{72} \end{bmatrix}, \quad \mathbf{A}^3 = \begin{bmatrix} \frac{49}{144} & \frac{29}{96} & \frac{103}{288} \\ \frac{71}{216} & \frac{11}{36} & \frac{79}{216} \\ \frac{5}{16} & \frac{67}{216} & \frac{163}{432} \end{bmatrix}.$$

These powers of $\mathbf{A}$ are also stochastic matrices. To prove that this is true in general for 2×2 matrices, we prove the more general theorem that any product of 2×2 stochastic matrices is stochastic. Let

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$$

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be stochastic matrices. Then

$$\mathbf{AB} = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{bmatrix}.$$

The sums of the rows are

$$\begin{aligned} a_{11}b_{11} + a_{12}b_{21} + a_{11}b_{12} + a_{12}b_{22} &= a_{11}(b_{11} + b_{12}) + a_{12}(b_{21} + b_{22}) \\ &= a_{11}(1) + a_{12}(1) = a_{11} + a_{12} = 1 \end{aligned}$$

$$\begin{aligned} a_{21}b_{11} + a_{22}b_{21} + a_{21}b_{12} + a_{22}b_{22} &= a_{21}(b_{11} + b_{12}) + a_{22}(b_{21} + b_{22}) \\ &= a_{21}(1) + a_{22}(1) = a_{21} + a_{22} = 1. \end{aligned}$$

Thus, the product matrix $\mathbf{AB}$ is stochastic. It follows that any power of a 2×2 matrix is stochastic. The proof in the case of an $n \times n$ matrix is very similar.

Exercises 8.9

1. The characteristic equation is $\lambda^2 - 6\lambda + 13 = 0$. Then

$$\mathbf{A}^2 - 6\mathbf{A} + 13\mathbf{I} = \begin{bmatrix} -7 & -12 \\ 24 & 17 \end{bmatrix} - \begin{bmatrix} 6 & -12 \\ 24 & 30 \end{bmatrix} + \begin{bmatrix} 13 & 0 \\ 0 & 13 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

2. The characteristic equation is $-\lambda^3 + \lambda^2 + 4\lambda - 1 = 0$. Then

$$-\mathbf{A}^3 + \mathbf{A}^2 + 4\mathbf{A} - \mathbf{I} = -\begin{bmatrix} 2 & 6 & 13 \\ 4 & 5 & 17 \\ 1 & 5 & 9 \end{bmatrix} + \begin{bmatrix} 1 & 2 & 5 \\ 0 & 4 & 5 \\ 1 & 1 & 4 \end{bmatrix} + 4\begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 0 & 1 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

3. The characteristic equation is $\lambda^2 - 3\lambda - 10 = 0$, with eigenvalues -2 and 5 . Substituting the eigenvalues into $\lambda^m = c_0 + c_1\lambda$ generates

$$(-2)^m = c_0 - 2c_1$$

$$5^m = c_0 + 5c_1.$$

Solving the system gives

$$c_0 = \frac{1}{7}[5(-2)^m + 2(5)^m], \quad c_1 = \frac{1}{7}[-(-2)^m + 5^m].$$

Thus

$$\mathbf{A}^m = c_0\mathbf{I} + c_1\mathbf{A} = \begin{bmatrix} \frac{1}{7}[3(-1)^m 2^{m+1} + 5^m] & \frac{3}{7}[-(-2)^m + 5^m] \\ \frac{2}{7}[-(-2)^m + 5^m] & \frac{1}{7}[(-2)^m + 6(5)^m] \end{bmatrix}$$

and

$$\mathbf{A}^3 = \begin{bmatrix} 11 & 57 \\ 38 & 106 \end{bmatrix}.$$

4. The characteristic equation is $\lambda^2 - 10\lambda + 16 = 0$, with eigenvalues 2 and 8 . Substituting the eigenvalues into $\lambda^m = c_0 + c_1\lambda$ generates

$$2^m = c_0 + 2c_1$$

$$8^m = c_0 + 8c_1.$$

Solving the system gives

$$c_0 = \frac{1}{3}(2^{m+2} - 8^m), \quad c_1 = \frac{1}{6}(-2^m + 8^m).$$

Thus

$$\mathbf{A}^m = c_0 \mathbf{I} + c_1 \mathbf{A} = \begin{bmatrix} \frac{1}{2}(2^m + 8^m) & \frac{1}{2}(2^m - 8^m) \\ \frac{1}{2}(2^m - 8^m) & \frac{1}{2}(2^m + 8^m) \end{bmatrix}$$

and

$$\mathbf{A}^4 = \begin{bmatrix} 2056 & -2040 \\ -2040 & 2056 \end{bmatrix}.$$

5. The characteristic equation is $\lambda^2 - 8\lambda - 20 = 0$, with eigenvalues -2 and 10 . Substituting the eigenvalues into $\lambda^m = c_0 + c_1 \lambda$ generates

$$\begin{aligned} (-2)^m &= c_0 - 2c_1 \\ 10^m &= c_0 + 10c_1. \end{aligned}$$

Solving the system gives

$$c_0 = \frac{1}{6}[5(-2)^m + 10^m], \quad c_1 = \frac{1}{12}[-(-2)^m + 10^m].$$

Thus

$$\mathbf{A}^m = c_0 \mathbf{I} + c_1 \mathbf{A} = \begin{bmatrix} \frac{1}{6}[(-2)^m + 2^m 5^{m+1}] & \frac{5}{12}[-(-2)^m + 10^m] \\ \frac{1}{3}[-(-2)^m + 10^m] & \frac{1}{6}[5(-2)^m + 10^m] \end{bmatrix}$$

and

$$\mathbf{A}^5 = \begin{bmatrix} 83328 & 41680 \\ 33344 & 16640 \end{bmatrix}.$$

6. The characteristic equation is $\lambda^2 + 4\lambda + 3 = 0$, with eigenvalues -3 and -1 . Substituting the eigenvalues into $\lambda^m = c_0 + c_1 \lambda$ generates

$$\begin{aligned} (-3)^m &= c_0 - 3c_1 \\ (-1)^m &= c_0 - c_1. \end{aligned}$$

Solving the system gives

$$c_0 = \frac{1}{2}[-(-3)^m + 3(-1)^m], \quad c_1 = \frac{1}{2}[-(-3)^m + (-1)^m].$$

Thus

$$\mathbf{A}^m = c_0 \mathbf{I} + c_1 \mathbf{A} = \begin{bmatrix} (-1)^m & -(-3)^m + (-1)^m \\ 0 & (-3)^m \end{bmatrix}$$

and

$$\mathbf{A}^6 = \begin{bmatrix} 1 & -728 \\ 0 & 729 \end{bmatrix}.$$

7. The characteristic equation is $-\lambda^3 + 2\lambda^2 + \lambda - 2 = 0$, with eigenvalues -1 , 1 , and 2 . Substituting the eigenvalues into $\lambda^m = c_0 + c_1 \lambda + c_2 \lambda^2$ generates

$$\begin{aligned} (-1)^m &= c_0 - c_1 + c_2 \\ 1 &= c_0 + c_1 + c_2 \\ 2^m &= c_0 + 2c_1 + 4c_2. \end{aligned}$$

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Solving the system gives

$$\begin{aligned}c_0 &= \frac{1}{3}[3 + (-1)^m - 2^m], \\c_1 &= \frac{1}{2}[1 - (-1)^m], \\c_2 &= \frac{1}{6}[-3 + (-1)^m + 2^{m+1}].\end{aligned}$$

Thus

$$\mathbf{A}^m = c_0 \mathbf{I} + c_1 \mathbf{A} + c_2 \mathbf{A}^2 = \begin{bmatrix} 1 & -1 + 2^m & -1 + 2^m \\ 0 & \frac{1}{3}[(-1)^m + 2^{m+1}] & -\frac{2}{3}[(-1)^m - 2^m] \\ 0 & \frac{1}{3}[(-1)^m + 2^m] & \frac{1}{3}[2(-1)^m + 2^m] \end{bmatrix}$$

and

$$\mathbf{A}^{10} = \begin{bmatrix} 1 & 1023 & 1023 \\ 0 & 683 & 682 \\ 0 & 341 & 342 \end{bmatrix}.$$

8. The characteristic equation is $-\lambda^3 - \lambda^2 + 2\lambda + 2 = 0$, with eigenvalues -1 , $-\sqrt{2}$, and $\sqrt{2}$. Substituting the eigenvalues into $\lambda^m = c_0 + c_1\lambda + c_2\lambda^2$ generates

$$\begin{aligned}(-1)^m &= c_0 - c_1 + c_2 \\(-\sqrt{2})^m &= c_0 - \sqrt{2}c_1 + 2c_2 \\(\sqrt{2})^m &= c_0 + \sqrt{2}c_1 + 2c_2.\end{aligned}$$

Solving the system gives

$$\begin{aligned}c_0 &= [2 - (\sqrt{2})^{m-1} - (\sqrt{2})^{m-2}](-1)^m + (\sqrt{2} - 1)(\sqrt{2})^{m-2}, \\c_1 &= \frac{1}{2}[1 - (-1)^m](\sqrt{2})^{m-1}, \\c_2 &= (-1)^{m+1} + \frac{1}{2}(1 + \sqrt{2})(-1)^m(\sqrt{2})^{m-1} + \frac{1}{2}(\sqrt{2} - 1)(\sqrt{2})^{m-1}.\end{aligned}$$

Thus $\mathbf{A}^m = c_0 \mathbf{I} + c_1 \mathbf{A} + c_2 \mathbf{A}^2$ and

$$\mathbf{A}^6 = \begin{bmatrix} 1 & 0 & 7 \\ 7 & 8 & -7 \\ 0 & 0 & 8 \end{bmatrix}.$$

9. The characteristic equation is $-\lambda^3 + 3\lambda^2 + 6\lambda - 8 = 0$, with eigenvalues -2 , 1 , and 4 . Substituting the eigenvalues into $\lambda^m = c_0 + c_1\lambda + c_2\lambda^2$ generates

$$\begin{aligned}(-2)^m &= c_0 - 2c_1 + 4c_2 \\1 &= c_0 + c_1 + c_2 \\4^m &= c_0 + 4c_1 + 16c_2.\end{aligned}$$

Solving the system gives

$$\begin{aligned}c_0 &= \frac{1}{9}[8 + (-1)^m 2^{m+1} - 4^m], \\c_1 &= \frac{1}{18}[4 - 5(-2)^m + 4^m], \\c_2 &= \frac{1}{18}[-2 + (-2)^m + 4^m].\end{aligned}$$

Thus

$$\mathbf{A}^m = c_0 \mathbf{I} + c_1 \mathbf{A} + c_2 \mathbf{A}^2 = \begin{bmatrix} \frac{1}{9}[(-2)^m + (-1)^m 2^{m+1} + 3 \cdot 2^{2m+1}] & \frac{1}{3}[(-2)^m + 4^m] & 0 \\ -\frac{2}{3}[(-2)^m - 4^m] & \frac{1}{3}[(-1)^m 2^{m+1} + 4^m] & 0 \\ \frac{1}{3}[-3 + (-2)^m + 2^{2m+1}] & \frac{1}{3}[(-2)^m + 4^m] & 1 \end{bmatrix}$$

and

$$\mathbf{A}^{10} = \begin{bmatrix} 699392 & 349184 & 0 \\ 698368 & 350208 & 0 \\ 699391 & 349184 & 1 \end{bmatrix}.$$

10. The characteristic equation is $-\lambda^3 - \frac{3}{2}\lambda^2 + \frac{3}{2}\lambda + 1 = 0$, with eigenvalues -2 , $-\frac{1}{2}$, and 1 . Substituting the eigenvalues into $\lambda^m = c_0 + c_1\lambda + c_2\lambda^2$ generates

$$\begin{aligned} (-2)^m &= c_0 - 2c_1 + 4c_2 \\ \left(-\frac{1}{2}\right)^m &= c_0 - \frac{1}{2}c_1 + \frac{1}{4}c_2 \\ 1 &= c_0 + c_1 + c_2. \end{aligned}$$

Solving the system gives

$$\begin{aligned} c_0 &= \frac{1}{9}[2^{-m}[(-4)^m + 8(-1)^m + 2^{m+1} - (-1)^m 2^{2m+1}], \\ c_1 &= -\frac{1}{9}2^{-m}[(-4)^m + 4(-1)^m - 5 \cdot 2^m], \\ c_2 &= \frac{2}{9}[1 + (-2)^m - (-1)^m 2^{m-1}]. \end{aligned}$$

Thus

$$\begin{aligned} \mathbf{A}^m &= c_0 \mathbf{I} + c_1 \mathbf{A} + c_2 \mathbf{A}^2 \\ &= \begin{bmatrix} \frac{1}{3}2^{-m}[2(-1)^m + 2^m] & \frac{1}{3}[-1 + (-\frac{1}{2})^m] & 0 \\ \frac{2}{3}[-1 + (-\frac{1}{2})^m] & \frac{1}{3}[2 + (-\frac{1}{2})^m] & 0 \\ -\frac{1}{9}2^{-m}[7(-4)^m - 6(-1)^m - 3 \cdot 2^m + (-1)^m 2^{2m+1}] & \frac{1}{3}[-1 + (-\frac{1}{2})^m] & \frac{1}{3}[(-2)^m + (-1)^m 2^{m+1}] \end{bmatrix} \end{aligned}$$

and

$$\mathbf{A}^8 = \begin{bmatrix} \frac{43}{128} & -\frac{85}{256} & 0 \\ -\frac{85}{128} & \frac{171}{256} & 0 \\ -\frac{32725}{128} & -\frac{85}{256} & 256 \end{bmatrix}.$$

11. The characteristic equation is $\lambda^2 - 8\lambda + 16 = 0$, with eigenvalues 4 and 4 . Substituting the eigenvalues into $\lambda^m = c_0 + c_1\lambda$ generates

$$\begin{aligned} 4^m &= c_0 + 4c_1 \\ 4^{m-1} &= c_1. \end{aligned}$$

Solving the system gives

$$c_0 = -4^m(m-1), \quad c_1 = 4^{m-1}m.$$

Thus

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$$\mathbf{A}^m = c_0 \mathbf{I} + c_1 \mathbf{A} = \begin{bmatrix} 4^{m-1}(3m+4) & 3 \cdot 4^{m-1}m \\ -3 \cdot 4^{m-1}m & 4^{m-1}(-3m+4) \end{bmatrix}$$

and

$$\mathbf{A}^6 = \begin{bmatrix} 22528 & 18432 \\ -18432 & -14336 \end{bmatrix}.$$

12. The characteristic equation is $-\lambda^3 - \lambda^2 + 21\lambda + 45 = 0$, with eigenvalues -3 , -3 , and 5 . Substituting the eigenvalues into $\lambda^m = c_0 + c_1\lambda + c_2\lambda^2$ generates

$$\begin{aligned} (-3)^m &= c_0 - 3c_1 + 9c_2 \\ (-3)^{m-1}m &= c_1 - 6c_2 \\ 5^m &= c_0 + 5c_1 + 25c_2. \end{aligned}$$

Solving the system gives

$$\begin{aligned} c_0 &= \frac{1}{64}[73(-3)^m - 2(-1)^m 3^{m+2} + 9 \cdot 5^m - 40(-3)^m m], \\ c_1 &= \frac{1}{96}[-(-1)^m 3^{m+2} + 9 \cdot 5^m - 8(-3)^m m], \\ c_2 &= \frac{1}{64}[-(-3)^m + 5^m - 8(-3)^{m-1}m]. \end{aligned}$$

Thus

$$\begin{aligned} \mathbf{A}^m &= c_0 \mathbf{I} + c_1 \mathbf{A} + c_2 \mathbf{A}^2 \\ &= \begin{bmatrix} \frac{1}{32}[31(-3)^m - (-1)^m 3^{m+1} + 4 \cdot 5^m] & \frac{1}{16}[-(-3)^m - (-1)^m 3^{m+1} + 4 \cdot 5^m] & \frac{1}{32}[(-3)^m + (-1)^m 3^{m+1} - 4 \cdot 5^m] \\ \frac{1}{16}[-(-3)^m - (-1)^m 3^{m+1} + 4 \cdot 5^m] & \frac{1}{8}[7(-3)^m - (-1)^m 3^{m+1} + 4 \cdot 5^m] & \frac{1}{16}[(-3)^m + (-1)^m 3^{m+1} - 4 \cdot 5^m] \\ \frac{3}{32}[(-3)^m + (-1)^m 3^{m+1} - 4 \cdot 5^m] & \frac{3}{16}[(-3)^m + (-1)^m 3^{m+1} - 4 \cdot 5^m] & \frac{1}{32}[29(-3)^m - (-1)^m 3^{m+2} + 12 \cdot 5^m] \end{bmatrix} \end{aligned}$$

and

$$\mathbf{A}^5 = \begin{bmatrix} 178 & 842 & -421 \\ 842 & 1441 & -842 \\ -1263 & -2526 & 1020 \end{bmatrix}.$$

13. (a) The characteristic equation is $\lambda^2 - 4\lambda = \lambda(\lambda - 4) = 0$, so 0 is an eigenvalue. Since the matrix satisfies the characteristic equation, $\mathbf{A}^2 = 4\mathbf{A}$, $\mathbf{A}^3 = 4\mathbf{A}^2 = 4^2\mathbf{A}$, $\mathbf{A}^4 = 4^2\mathbf{A}^2 = 4^3\mathbf{A}$, and, in general,

$$\mathbf{A}^m = 4^m \mathbf{A} = \begin{bmatrix} 4^m & 4^m \\ 3(4)^m & 3(4)^m \end{bmatrix}.$$

- (b) The characteristic equation is $\lambda^2 = 0$, so 0 is an eigenvalue. Since the matrix satisfies the characteristic equation, $\mathbf{A}^2 = \mathbf{0}$, $\mathbf{A}^3 = \mathbf{A}\mathbf{A}^2 = \mathbf{0}$, and, in general, $\mathbf{A}^m = \mathbf{0}$.
- (c) The characteristic equation is $-\lambda^3 + 5\lambda^2 - 6\lambda = 0$, with eigenvalues 0 , 2 , and 3 . Substituting $\lambda = 0$ into $\lambda^m = c_0 + c_1\lambda + c_2\lambda^2$ we find that $c_0 = 0$. Using the nonzero eigenvalues, we find

$$\begin{aligned} 2^m &= 2c_1 + 4c_2 \\ 3^m &= 3c_1 + 9c_2. \end{aligned}$$

Solving the system gives

$$c_1 = \frac{1}{6}[9(2)^m - 4(3)^m], \quad c_2 = \frac{1}{6}[-3(2)^m + 2(3)^m].$$

Thus $\mathbf{A}^m = c_1 \mathbf{A} + c_2 \mathbf{A}^2$ and

$$\mathbf{A}^m = \begin{bmatrix} 2(3)^{m-1} & 3^{m-1} & 3^{m-1} \\ \frac{1}{6}[9(2)^m - 4(3)^m] & \frac{1}{6}[3(2)^m - 2(3)^m] & \frac{1}{6}[-3(2)^m - 2(3)^m] \\ \frac{1}{6}[-9(2)^m + 8(3)^m] & \frac{1}{6}[-3(2)^m + 4(3)^m] & \frac{1}{6}[3(2)^m + 4(3)^m] \end{bmatrix}.$$

14. (a) Let

$$\mathbf{X}_{n-1} = \begin{bmatrix} x_{n-1} \\ y_{n-1} \end{bmatrix} \quad \text{and} \quad \mathbf{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

Then

$$\mathbf{X}_n = \mathbf{A}\mathbf{X}_{n-1} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_{n-1} \\ y_{n-1} \end{bmatrix} = \begin{bmatrix} x_{n-1} + y_{n-1} \\ x_{n-1} \end{bmatrix}.$$

(b) The characteristic equation of $\mathbf{A}$ is $\lambda^2 - \lambda - 1 = 0$, with eigenvalues $\lambda_1 = \frac{1}{2}(1 - \sqrt{5})$ and $\lambda_2 = \frac{1}{2}(1 + \sqrt{5})$. From $\lambda^m = c_0 + c_1 \lambda$ we get $\lambda_1^m = c_0 + c_1 \lambda_1$ and $\lambda_2^m = c_0 + c_1 \lambda_2$. Solving this system gives

$$c_0 = (\lambda_2 \lambda_1^m - \lambda_1 \lambda_2^m) / (\lambda_2 - \lambda_1) \quad \text{and} \quad c_1 = (\lambda_2^m - \lambda_1^m) / (\lambda_2 - \lambda_1).$$

Thus

$$\begin{aligned} \mathbf{A}^m &= c_0 \mathbf{I} + c_1 \mathbf{A} \\ &= \frac{1}{2^{m+1}\sqrt{5}} \begin{bmatrix} (1 + \sqrt{5})^{m+1} - (1 - \sqrt{5})^{m+1} & 2(1 + \sqrt{5})^m - 2(1 - \sqrt{5})^m \\ 2(1 + \sqrt{5})^m - 2(1 - \sqrt{5})^m & (1 + \sqrt{5})(1 - \sqrt{5})^m - (1 - \sqrt{5})(1 + \sqrt{5})^m \end{bmatrix}. \end{aligned}$$

(c) From part (a), $\mathbf{X}_2 = \mathbf{A}\mathbf{X}_1$, $\mathbf{X}_3 = \mathbf{A}\mathbf{X}_2 = \mathbf{A}^2\mathbf{X}_1$, $\mathbf{X}_4 = \mathbf{A}\mathbf{X}_3 = \mathbf{A}^3\mathbf{X}_1$, and, in general, $\mathbf{X}_n = \mathbf{A}^{n-1}\mathbf{X}_1$. With

$$\mathbf{X}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{we have} \quad \mathbf{X}_{12} = \mathbf{A}^{11}\mathbf{X}_1 = \begin{bmatrix} 144 & 89 \\ 89 & 55 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 233 \\ 144 \end{bmatrix},$$

so the number of adult pairs is 233. With

$$\mathbf{X}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{we have} \quad \mathbf{A}^{11}\mathbf{X}_1 = \begin{bmatrix} 144 & 89 \\ 89 & 55 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 144 \\ 89 \end{bmatrix},$$

so the number of baby pairs is 144. With

$$\mathbf{X}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \quad \text{we have} \quad \mathbf{A}^{11}\mathbf{X}_1 = \begin{bmatrix} 144 & 89 \\ 89 & 55 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 377 \\ 233 \end{bmatrix},$$

so the total number of pairs is 377.

15. The characteristic equation of $\mathbf{A}$ is $\lambda^2 - 5\lambda + 10 = 0$, so $\mathbf{A}^2 - 5\mathbf{A} + 10\mathbf{I} = \mathbf{0}$ and $\mathbf{I} = -\frac{1}{10}\mathbf{A}^2 + \frac{1}{2}\mathbf{A}$. Multiplying by $\mathbf{A}^{-1}$ we find

$$\mathbf{A}^{-1} = -\frac{1}{10}\mathbf{A} + \frac{1}{2}\mathbf{I} = -\frac{1}{10} \begin{bmatrix} 2 & -4 \\ 1 & 3 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{3}{10} & \frac{2}{5} \\ -\frac{1}{10} & \frac{1}{5} \end{bmatrix}.$$

16. The characteristic equation of $\mathbf{A}$ is $-\lambda^3 + 2\lambda^2 + \lambda - 2 = 0$, so $-\mathbf{A}^3 + 2\mathbf{A}^2 + \mathbf{A} - 2\mathbf{I} = \mathbf{0}$ and $\mathbf{I} = -\frac{1}{2}\mathbf{A}^3 + \mathbf{A}^2 + \frac{1}{2}\mathbf{A}$. Multiplying by $\mathbf{A}^{-1}$ we find

$$\mathbf{A}^{-1} = -\frac{1}{2}\mathbf{A}^2 + \mathbf{A} + \frac{1}{2}\mathbf{I} = \begin{bmatrix} \frac{3}{2} & \frac{1}{2} & -\frac{5}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{3}{2} \end{bmatrix}.$$

Exercises 8.9

17. (a) Since

$$\mathbf{A}^2 = \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix} \quad \text{we see that} \quad \mathbf{A}^m = \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix}$$

for all integers $m \geq 2$. Thus $\mathbf{A}$ is not nilpotent.

(b) Since $\mathbf{A}^2 = \mathbf{0}$, the matrix is nilpotent with index 2.

(c) Since $\mathbf{A}^3 = \mathbf{0}$, the matrix is nilpotent with index 3.

(d) Since $\mathbf{A}^2 = \mathbf{0}$, the matrix is nilpotent with index 2.

(e) Since $\mathbf{A}^4 = \mathbf{0}$, the matrix is nilpotent with index 4.

(f) Since $\mathbf{A}^4 = \mathbf{0}$, the matrix is nilpotent with index 4.

18. (a) If $\mathbf{A}^m = \mathbf{0}$ for some m , then $(\det \mathbf{A})^m = \det \mathbf{A}^m = \det \mathbf{0} = 0$, and $\mathbf{A}$ is a singular matrix.

(b) By (1) of Section 8.8 we have $\mathbf{A}\mathbf{K} = \lambda\mathbf{K}$, $\mathbf{A}^2\mathbf{K} = \lambda\mathbf{A}\mathbf{K} = \lambda^2\mathbf{K}$, $\mathbf{A}^3\mathbf{K} = \lambda^2\mathbf{A}\mathbf{K} = \lambda^3\mathbf{K}$, and, in general, $\mathbf{A}^m\mathbf{K} = \lambda^m\mathbf{K}$. If $\mathbf{A}$ is nilpotent with index m , then $\mathbf{A}^m = \mathbf{0}$ and $\lambda^m = 0$.

Exercises 8.10

$$1. \text{ (a)-(b)} \quad \begin{bmatrix} 0 & 0 & -4 \\ 0 & -4 & 0 \\ -4 & 0 & 15 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -4 \\ 0 \end{bmatrix} = -4 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}; \quad \lambda_1 = -4$$

$$\begin{bmatrix} 0 & 0 & -4 \\ 0 & -4 & 0 \\ -4 & 0 & 15 \end{bmatrix} \begin{bmatrix} 4 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -4 \\ 0 \\ 1 \end{bmatrix} = (-1) \begin{bmatrix} 4 \\ 0 \\ 1 \end{bmatrix}; \quad \lambda_2 = -1$$

$$\begin{bmatrix} 0 & 0 & -4 \\ 0 & -4 & 0 \\ -4 & 0 & 15 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -4 \end{bmatrix} = \begin{bmatrix} 16 \\ 0 \\ -64 \end{bmatrix} = 16 \begin{bmatrix} 1 \\ 0 \\ -4 \end{bmatrix}; \quad \lambda_3 = 16$$

$$(c) \quad \mathbf{K}_1^T \mathbf{K}_2 = [0 \quad 1 \quad 0] \begin{bmatrix} 4 \\ 0 \\ 1 \end{bmatrix} = 0; \quad \mathbf{K}_1^T \mathbf{K}_3 = [0 \quad 1 \quad 0] \begin{bmatrix} 1 \\ 0 \\ -4 \end{bmatrix} = 0; \quad \mathbf{K}_2^T \mathbf{K}_3 = [4 \quad 0 \quad 1] \begin{bmatrix} 1 \\ 0 \\ -4 \end{bmatrix} = 4 - 4 = 0$$

$$2. \text{ (a)-(b)} \quad \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -4 \\ 2 \\ 2 \end{bmatrix} = 2 \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}; \quad \lambda_1 = 2$$

$$\begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix} = 2 \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}; \quad \lambda_2 = 2$$

$$\begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ -1 \end{bmatrix} = (-1) \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}; \quad \lambda_3 = -1$$

$$(c) \quad \mathbf{K}_1^T \mathbf{K}_2 = [-2 \quad 1 \quad 1] \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} = 1 - 1 = 0; \quad \mathbf{K}_1^T \mathbf{K}_3 = [-2 \quad 1 \quad 1] \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = -2 + 1 + 1 = 0$$

$$\mathbf{K}_2^T \mathbf{K}_3 = \begin{bmatrix} 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 1 - 1 = 0$$

$$3. \text{ (a)-(b)} \quad \begin{bmatrix} 5 & 13 & 0 \\ 13 & 5 & 0 \\ 0 & 0 & -8 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \\ 0 \end{bmatrix} = \begin{bmatrix} 9\sqrt{2} \\ 9\sqrt{2} \\ 0 \end{bmatrix} = 18 \begin{bmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \\ 0 \end{bmatrix}; \quad \lambda_1 = 18$$

$$\begin{bmatrix} 5 & 13 & 0 \\ 13 & 5 & 0 \\ 0 & 0 & -8 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{3}}{3} \\ -\frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \end{bmatrix} = \begin{bmatrix} -\frac{8\sqrt{2}}{3} \\ \frac{8\sqrt{3}}{3} \\ -\frac{8\sqrt{3}}{3} \end{bmatrix} = (-8) \begin{bmatrix} \frac{\sqrt{3}}{3} \\ -\frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \end{bmatrix}; \quad \lambda_2 = -8$$

$$\begin{bmatrix} 5 & 13 & 0 \\ 13 & 5 & 0 \\ 0 & 0 & -8 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{6}}{6} \\ -\frac{\sqrt{6}}{6} \\ -\frac{\sqrt{6}}{3} \end{bmatrix} = \begin{bmatrix} -\frac{8\sqrt{6}}{6} \\ \frac{8\sqrt{6}}{6} \\ \frac{8\sqrt{6}}{3} \end{bmatrix} = (-8) \begin{bmatrix} \frac{\sqrt{6}}{6} \\ -\frac{\sqrt{6}}{6} \\ -\frac{\sqrt{6}}{3} \end{bmatrix}; \quad \lambda_3 = -8$$

$$(c) \quad \mathbf{K}_1^T \mathbf{K}_2 = \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{3}}{3} \\ -\frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \end{bmatrix} = \frac{\sqrt{6}}{6} - \frac{\sqrt{6}}{6} = 0; \quad \mathbf{K}_1^T \mathbf{K}_3 = \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{6}}{6} \\ -\frac{\sqrt{6}}{6} \\ -\frac{\sqrt{6}}{3} \end{bmatrix} = \frac{\sqrt{12}}{12} - \frac{\sqrt{12}}{12} = 0$$

$$\mathbf{K}_2^T \mathbf{K}_3 = \begin{bmatrix} \frac{\sqrt{3}}{3} & -\frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} \end{bmatrix} \begin{bmatrix} \frac{\sqrt{6}}{6} \\ -\frac{\sqrt{6}}{6} \\ -\frac{\sqrt{6}}{3} \end{bmatrix} = \frac{\sqrt{18}}{18} + \frac{\sqrt{18}}{18} - \frac{\sqrt{18}}{9} = 0$$

$$4. \text{ (a)-(b)} \quad \begin{bmatrix} 3 & 2 & 2 \\ 2 & 2 & 0 \\ 2 & 0 & 4 \end{bmatrix} \begin{bmatrix} -2 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = 0 \begin{bmatrix} -2 \\ 2 \\ 1 \end{bmatrix}; \quad \lambda_1 = 0$$

$$\begin{bmatrix} 3 & 2 & 2 \\ 2 & 2 & 0 \\ 2 & 0 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \\ -6 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix}; \quad \lambda_2 = 3$$

$$\begin{bmatrix} 3 & 2 & 2 \\ 2 & 2 & 0 \\ 2 & 0 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 12 \\ 6 \\ 12 \end{bmatrix} = 6 \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}; \quad \lambda_3 = 6$$

$$(c) \quad \mathbf{K}_1^T \mathbf{K}_2 = \begin{bmatrix} -2 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix} = -2 + 4 - 2 = 0; \quad \mathbf{K}_1^T \mathbf{K}_3 = \begin{bmatrix} -2 & 2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} = -4 + 2 + 2 = 0$$

$$\mathbf{K}_2^T \mathbf{K}_3 = \begin{bmatrix} 1 & 2 & -2 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} = 2 + 2 - 4 = 0$$

5. Orthogonal. Columns form an orthonormal set.
6. Not orthogonal. Columns one and three are not unit vectors.
7. Orthogonal. Columns form an orthonormal set.
8. Not orthogonal. The matrix is singular.

Exercises 8.10

9. Not orthogonal. Columns are not unit vectors.

10. Orthogonal. Columns form an orthogonal set.

11. $\lambda_1 = -8, \lambda_2 = 10, \mathbf{K}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \mathbf{K}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \mathbf{P} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$

12. $\lambda_1 = 7, \lambda_2 = 4, \mathbf{K}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \mathbf{K}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \mathbf{P} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

13. $\lambda_1 = 0, \lambda_2 = 10, \mathbf{K}_1 = \begin{bmatrix} 3 \\ -1 \end{bmatrix}, \mathbf{K}_2 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \mathbf{P} = \begin{bmatrix} \frac{3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \\ -\frac{1}{\sqrt{10}} & \frac{3}{\sqrt{10}} \end{bmatrix}$

14. $\lambda_1 = \frac{1}{2} + \frac{\sqrt{5}}{2}, \lambda_2 = \frac{1}{2} - \frac{\sqrt{5}}{2}, \mathbf{K}_1 = \begin{bmatrix} 1+\sqrt{5} \\ 2 \end{bmatrix}, \mathbf{K}_2 = \begin{bmatrix} 1-\sqrt{5} \\ 2 \end{bmatrix}, \mathbf{P} = \begin{bmatrix} \frac{1+\sqrt{5}}{\sqrt{10+2\sqrt{5}}} & \frac{1-\sqrt{5}}{\sqrt{10-2\sqrt{5}}} \\ \frac{2}{\sqrt{10+2\sqrt{5}}} & \frac{2}{\sqrt{10-2\sqrt{5}}} \end{bmatrix}$

15. $\lambda_1 = 0, \lambda_2 = 2, \lambda_3 = 1, \mathbf{K}_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \mathbf{K}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \mathbf{K}_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \mathbf{P} = \begin{bmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{bmatrix}$

16. $\lambda_1 = -1, \lambda_2 = 1 - \sqrt{2}, \lambda_3 = 1 + \sqrt{2}, \mathbf{K}_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \mathbf{K}_2 = \begin{bmatrix} 1 \\ -\sqrt{2} \\ 1 \end{bmatrix}, \mathbf{K}_3 = \begin{bmatrix} 1 \\ \sqrt{2} \\ 1 \end{bmatrix}, \mathbf{P} = \begin{bmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{2} & \frac{1}{2} \\ 0 & -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{1}{\sqrt{2}} & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$

17. $\lambda_1 = -11, \lambda_2 = 0, \lambda_3 = 6, \mathbf{K}_1 = \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix}, \mathbf{K}_2 = \begin{bmatrix} 1 \\ -4 \\ 7 \end{bmatrix}, \mathbf{K}_3 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \mathbf{P} = \begin{bmatrix} -\frac{3}{\sqrt{11}} & \frac{1}{\sqrt{66}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{11}} & -\frac{4}{\sqrt{66}} & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{11}} & \frac{7}{\sqrt{66}} & \frac{1}{\sqrt{6}} \end{bmatrix}$

18. $\lambda_1 = -18, \lambda_2 = 0, \lambda_3 = 9, \mathbf{K}_1 = \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix}, \mathbf{K}_2 = \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix}, \mathbf{K}_3 = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}, \mathbf{P} = \begin{bmatrix} \frac{1}{3} & -\frac{2}{3} & \frac{2}{3} \\ -\frac{2}{3} & \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{2}{3} & \frac{1}{3} \end{bmatrix}$

19. $\begin{bmatrix} \frac{3}{5} & a \\ \frac{4}{5} & b \end{bmatrix} \begin{bmatrix} \frac{3}{5} & \frac{4}{5} \\ a & b \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ implies $\frac{9}{25} + a^2 = 1$ and $\frac{16}{25} + b^2 = 1$. These equations give $a = \pm \frac{4}{5}, b = \pm \frac{3}{5}$. But

$\frac{12}{25} + ab = 0$ indicates a and b must have opposite signs. Therefore choose $a = -\frac{4}{5}, b = \frac{3}{5}$.

The matrix $\begin{bmatrix} \frac{3}{5} & -\frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{bmatrix}$ is orthogonal.

20. $\begin{bmatrix} \frac{1}{\sqrt{5}} & b \\ a & \frac{1}{\sqrt{5}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{5}} & a \\ b & \frac{1}{\sqrt{5}} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ implies $\frac{1}{5} + b^2 = 1$ and $a^2 + \frac{1}{5} = 1$. These equations give $a = \pm \frac{2}{\sqrt{5}}, b = \pm \frac{2}{\sqrt{5}}$.

But $\frac{a}{\sqrt{5}} + \frac{b}{\sqrt{5}} = 0$ indicates a and b must have opposite signs. Therefore choose $a = -\frac{2}{\sqrt{5}}, b = \frac{2}{\sqrt{5}}$.

The matrix $\begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ -\frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix}$ is orthogonal.

21. (a) $\mathbf{AK}_1 = \begin{bmatrix} 7 & 4 & -4 \\ 4 & -8 & -1 \\ -4 & -1 & -8 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 36 \\ 9 \\ -9 \end{bmatrix} = 9 \begin{bmatrix} 4 \\ 1 \\ -1 \end{bmatrix} = \lambda_1 \mathbf{K}_1$

$$\mathbf{A}\mathbf{K}_2 = \begin{bmatrix} 7 & 4 & -4 \\ 4 & -8 & -1 \\ -4 & -1 & -8 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix} = \begin{bmatrix} -9 \\ 0 \\ -36 \end{bmatrix} = -9 \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix} = \lambda_2 \mathbf{K}_2$$

$$\mathbf{A}\mathbf{K}_3 = \begin{bmatrix} 7 & 4 & -4 \\ 4 & -8 & -1 \\ -4 & -1 & -8 \end{bmatrix} \begin{bmatrix} 1 \\ -4 \\ 0 \end{bmatrix} = \begin{bmatrix} -9 \\ 36 \\ 0 \end{bmatrix} = -9 \begin{bmatrix} 1 \\ -4 \\ 0 \end{bmatrix} = \lambda_3 \mathbf{K}_3$$

(b) For $\lambda_2 = \lambda_3 = -9$ we have

$$\left[\begin{array}{ccc|c} 16 & 4 & -4 & 0 \\ 4 & 1 & -1 & 0 \\ -4 & -1 & 1 & 0 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|c} 1 & \frac{1}{4} & -\frac{1}{4} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

so that $k_1 = -\frac{1}{4}k_2 + \frac{1}{4}k_3$. The choices $k_2 = 0, k_3 = 4$ and $k_2 = -4, k_3 = 0$ give the eigenvectors in part (a). These eigenvectors are not orthogonal. However, by choosing $k_2 = k_3 = 1$ and $k_2 = -2, k_3 = 2$ we obtain, respectively,

$$\mathbf{K}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{K}_3 = \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix}.$$

Mutually orthogonal eigenvectors are

$$\begin{bmatrix} 4 \\ 1 \\ -1 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix}.$$

22. The eigenvalues and eigenvectors of $\mathbf{A}$ are

$$\lambda_1 = \lambda_2 = -1, \quad \lambda_3 = \lambda_4 = 3, \quad \mathbf{K}_1 = \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{K}_2 = \begin{bmatrix} 0 \\ 0 \\ -1 \\ 1 \end{bmatrix}, \quad \mathbf{K}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{K}_4 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}.$$

Since $\|\mathbf{K}_1\| = \|\mathbf{K}_2\| = \|\mathbf{K}_3\| = \|\mathbf{K}_4\| = \sqrt{2}$, an orthogonal matrix is

$$\mathbf{P} = \begin{bmatrix} -\frac{1}{\sqrt{2}} & 0 & 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 & 0 & \frac{1}{\sqrt{2}} \\ 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{bmatrix}.$$

23. Suppose $\mathbf{A}$ and $\mathbf{B}$ are orthogonal matrices. Then $\mathbf{A}^{-1} = \mathbf{A}^T$ and $\mathbf{B}^{-1} = \mathbf{B}^T$ and

$$(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1} = \mathbf{B}^T\mathbf{A}^T = (\mathbf{AB})^T.$$

Thus $\mathbf{AB}$ is an orthogonal matrix.

Exercises 8.11

1. Taking $\mathbf{X}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and computing $\mathbf{X}_i = \mathbf{A}\mathbf{X}_{i-1}$ for $i = 1, 2, 3, 4$ we obtain

$$\mathbf{X}_1 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \quad \mathbf{X}_2 = \begin{bmatrix} 4 \\ 4 \end{bmatrix}, \quad \mathbf{X}_3 = \begin{bmatrix} 8 \\ 8 \end{bmatrix}, \quad \mathbf{X}_4 = \begin{bmatrix} 16 \\ 16 \end{bmatrix}.$$

We conclude that a dominant eigenvector is $\mathbf{K} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ with corresponding eigenvalue $\lambda = \frac{\mathbf{A}\mathbf{K} \cdot \mathbf{K}}{\mathbf{K} \cdot \mathbf{K}} = \frac{4}{2} = 2$.

2. Taking $\mathbf{X}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and computing $\mathbf{X}_i = \mathbf{A}\mathbf{X}_{i-1}$ for $i = 1, 2, 3, 4, 5$ we obtain

$$\mathbf{X}_1 = \begin{bmatrix} -5 \\ 7 \end{bmatrix}, \quad \mathbf{X}_2 = \begin{bmatrix} 49 \\ -47 \end{bmatrix}, \quad \mathbf{X}_3 = \begin{bmatrix} -437 \\ 439 \end{bmatrix}, \quad \mathbf{X}_4 = \begin{bmatrix} 3937 \\ -3935 \end{bmatrix}, \quad \mathbf{X}_5 = \begin{bmatrix} -35429 \\ 35431 \end{bmatrix}.$$

We conclude that a dominant eigenvector is $\mathbf{K} = \frac{1}{35439} \begin{bmatrix} -35429 \\ 35431 \end{bmatrix} \approx \begin{bmatrix} -0.99994 \\ 1 \end{bmatrix}$ with corresponding eigenvalue $\lambda = \frac{\mathbf{A}\mathbf{K} \cdot \mathbf{K}}{\mathbf{K} \cdot \mathbf{K}} = -8.9998$.

3. Taking $\mathbf{X}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and computing $\mathbf{A}\mathbf{X}_0 = \begin{bmatrix} 6 \\ 16 \end{bmatrix}$, we define $\mathbf{X}_1 = \frac{1}{16} \begin{bmatrix} 6 \\ 16 \end{bmatrix} = \begin{bmatrix} 0.375 \\ 1 \end{bmatrix}$. Continuing in this manner we obtain

$$\mathbf{X}_2 = \begin{bmatrix} 0.3363 \\ 1 \end{bmatrix}, \quad \mathbf{X}_3 = \begin{bmatrix} 0.3335 \\ 1 \end{bmatrix}, \quad \mathbf{X}_4 = \begin{bmatrix} 0.3333 \\ 1 \end{bmatrix}.$$

We conclude that a dominant eigenvector is $\mathbf{K} = \begin{bmatrix} 0.3333 \\ 1 \end{bmatrix}$ with corresponding eigenvalue $\lambda = 14$.

4. Taking $\mathbf{X}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and computing $\mathbf{A}\mathbf{X}_0 = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$, we define $\mathbf{X}_1 = \frac{1}{5} \begin{bmatrix} 1 \\ 5 \end{bmatrix} = \begin{bmatrix} 0.2 \\ 1 \end{bmatrix}$. Continuing in this manner we obtain

$$\mathbf{X}_2 = \begin{bmatrix} 0.2727 \\ 1 \end{bmatrix}, \quad \mathbf{X}_3 = \begin{bmatrix} 0.2676 \\ 1 \end{bmatrix}, \quad \mathbf{X}_4 = \begin{bmatrix} 0.2680 \\ 1 \end{bmatrix}, \quad \mathbf{X}_5 = \begin{bmatrix} 0.2679 \\ 1 \end{bmatrix}.$$

We conclude that a dominant eigenvector is $\mathbf{K} = \begin{bmatrix} 0.2679 \\ 1 \end{bmatrix}$ with corresponding eigenvalue $\lambda = 6.4641$.

5. Taking $\mathbf{X}_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and computing $\mathbf{A}\mathbf{X}_0 = \begin{bmatrix} 11 \\ 11 \\ 6 \end{bmatrix}$, we define $\mathbf{X}_1 = \frac{1}{11} \begin{bmatrix} 11 \\ 11 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0.5455 \end{bmatrix}$. Continuing in this manner we obtain

$$\mathbf{X}_2 = \begin{bmatrix} 1 \\ 1 \\ 0.5045 \end{bmatrix}, \quad \mathbf{X}_3 = \begin{bmatrix} 1 \\ 1 \\ 0.5005 \end{bmatrix}, \quad \mathbf{X}_4 = \begin{bmatrix} 1 \\ 1 \\ 0.5 \end{bmatrix}.$$

We conclude that a dominant eigenvector is $\mathbf{K} = \begin{bmatrix} 1 \\ 1 \\ 0.5 \end{bmatrix}$ with corresponding eigenvalue $\lambda = 10$.

6. Taking $\mathbf{X}_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and computing $\mathbf{A}\mathbf{X}_0 = \begin{bmatrix} 5 \\ 2 \\ 2 \end{bmatrix}$, we define $\mathbf{X}_1 = \frac{1}{5} \begin{bmatrix} 5 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0.4 \\ 0.4 \end{bmatrix}$. Continuing in this manner we obtain

$$\mathbf{X}_2 = \begin{bmatrix} 1 \\ 0.2105 \\ 0.2105 \end{bmatrix}, \quad \mathbf{X}_3 = \begin{bmatrix} 1 \\ 0.1231 \\ 0.1231 \end{bmatrix}, \quad \mathbf{X}_4 = \begin{bmatrix} 1 \\ 0.0758 \\ 0.0758 \end{bmatrix}, \quad \mathbf{X}_5 = \begin{bmatrix} 1 \\ 0.0481 \\ 0.0481 \end{bmatrix}.$$

At this point if we restart with $\mathbf{X}_0 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ we see that $\mathbf{K} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ is a dominant eigenvector with corresponding eigenvalue $\lambda = 3$.

7. Taking $\mathbf{X}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and using scaling we obtain

$$\mathbf{X}_1 = \begin{bmatrix} 0.625 \\ 1 \end{bmatrix}, \quad \mathbf{X}_2 = \begin{bmatrix} 0.5345 \\ 1 \end{bmatrix}, \quad \mathbf{X}_3 = \begin{bmatrix} 0.5098 \\ 1 \end{bmatrix}, \quad \mathbf{X}_4 = \begin{bmatrix} 0.5028 \\ 1 \end{bmatrix}, \quad \mathbf{X}_5 = \begin{bmatrix} 0.5008 \\ 1 \end{bmatrix}.$$

Taking $\mathbf{K} = \begin{bmatrix} 0.5 \\ 1 \end{bmatrix}$ as the dominant eigenvector we find $\lambda_1 = 7$. Now the normalized eigenvector is

$$\mathbf{K}_1 = \begin{bmatrix} 0.4472 \\ 0.8944 \end{bmatrix} \text{ and } \mathbf{B} = \begin{bmatrix} 1.6 & -0.8 \\ -0.8 & 0.4 \end{bmatrix}. \text{ Taking } \mathbf{X}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ and using scaling again we obtain } \mathbf{X}_1 = \begin{bmatrix} 1 \\ -0.5 \end{bmatrix},$$

$$\mathbf{X}_2 = \begin{bmatrix} 1 \\ -0.5 \end{bmatrix}. \text{ Taking } \mathbf{K} = \begin{bmatrix} 1 \\ -0.5 \end{bmatrix} \text{ we find } \lambda_2 = 2. \text{ The eigenvalues are 7 and 2.}$$

8. Taking $\mathbf{X}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and using scaling we obtain $\mathbf{X}_1 = \begin{bmatrix} 0.3333 \\ 1 \end{bmatrix}$, $\mathbf{X}_2 = \begin{bmatrix} 0.3333 \\ 1 \end{bmatrix}$. Taking $\mathbf{K} = \begin{bmatrix} 1/3 \\ 1 \end{bmatrix}$ as the

dominant eigenvector we find $\lambda_1 = 10$. Now the normalized eigenvector is $\mathbf{K}_1 = \begin{bmatrix} 0.3162 \\ 0.9486 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$. An eigenvector for the zero matrix is $\lambda_2 = 0$. The eigenvalues are 10 and 0.

9. Taking $\mathbf{X}_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and using scaling we obtain

$$\mathbf{X}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{X}_2 = \begin{bmatrix} 1 \\ -0.6667 \\ 1 \end{bmatrix}, \quad \mathbf{X}_3 = \begin{bmatrix} 1 \\ -0.9091 \\ 1 \end{bmatrix}, \quad \mathbf{X}_4 = \begin{bmatrix} 1 \\ -0.9767 \\ 1 \end{bmatrix}, \quad \mathbf{X}_5 = \begin{bmatrix} 1 \\ -0.9942 \\ 1 \end{bmatrix}.$$

Taking $\mathbf{K} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ as the dominant eigenvector we find $\lambda_1 = 4$. Now the normalized eigenvector is

$$\mathbf{K}_1 = \begin{bmatrix} 0.5774 \\ -0.5774 \\ 0.5774 \end{bmatrix} \text{ and } \mathbf{B} = \begin{bmatrix} 1.6667 & 0.3333 & -1.3333 \\ 0.3333 & 0.6667 & 0.3333 \\ -1.3333 & 0.3333 & 1.6667 \end{bmatrix}. \text{ If } \mathbf{X}_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \text{ is now chosen only one more}$$

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eigenvalue is found. Thus, try $\mathbf{X}_0 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$. Using scaling we obtain

$$\mathbf{X}_1 = \begin{bmatrix} 1 \\ 0.5 \\ -0.5 \end{bmatrix}, \quad \mathbf{X}_2 = \begin{bmatrix} 1 \\ 0.2 \\ -0.8 \end{bmatrix}, \quad \mathbf{X}_3 = \begin{bmatrix} 1 \\ 0.0714 \\ -0.9286 \end{bmatrix}, \quad \mathbf{X}_4 = \begin{bmatrix} 1 \\ 0.0244 \\ -0.9756 \end{bmatrix}, \quad \mathbf{X}_5 = \begin{bmatrix} 1 \\ 0.0082 \\ -0.9918 \end{bmatrix}.$$

Taking $\mathbf{K} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$ as the eigenvector we find $\lambda_2 = 3$. The normalized eigenvector in this case is

$$\mathbf{K}_2 = \begin{bmatrix} 0.7071 \\ 0 \\ -0.7071 \end{bmatrix} \text{ and } \mathbf{C} = \begin{bmatrix} 0.1667 & 0.3333 & 0.1667 \\ 0.3333 & 0.6667 & 0.3333 \\ 0.1667 & 0.3333 & 0.1667 \end{bmatrix}. \text{ If } \mathbf{X}_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \text{ is chosen, and scaling is used we}$$

obtain $\mathbf{X}_1 = \begin{bmatrix} 0.5 \\ 1 \\ 0.5 \end{bmatrix}$, $\mathbf{X}_2 = \begin{bmatrix} 0.5 \\ 1 \\ 0.5 \end{bmatrix}$. Taking $\mathbf{K} = \begin{bmatrix} 0.5 \\ 1 \\ 0.5 \end{bmatrix}$ we find $\lambda_3 = 1$. The eigenvalues are 4, 3, and 1.

The difficulty in choosing $\mathbf{X}_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ to find the second eigenvector results from the fact that this vector is a linear combination of the eigenvectors corresponding to the other two eigenvalues, with 0 contribution from the second eigenvector. When this occurs the development of the power method, shown in the text, breaks down.

10. Taking $\mathbf{X}_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and using scaling we obtain

$$\mathbf{X}_1 = \begin{bmatrix} -0.3636 \\ -0.3636 \\ 1 \end{bmatrix}, \quad \mathbf{X}_2 = \begin{bmatrix} -0.2431 \\ 0.0884 \\ 1 \end{bmatrix}, \quad \mathbf{X}_3 = \begin{bmatrix} -0.2504 \\ -0.0221 \\ 1 \end{bmatrix}, \quad \mathbf{X}_4 = \begin{bmatrix} -0.2499 \\ -0.0055 \\ 1 \end{bmatrix}.$$

Taking $\mathbf{K} = \begin{bmatrix} -0.25 \\ 0 \\ 1 \end{bmatrix}$ as the dominant eigenvector we find $\lambda_1 = 16$. The normalized eigenvector is

$$\mathbf{K}_1 = \begin{bmatrix} -0.2425 \\ 0 \\ 0.9701 \end{bmatrix} \text{ and } \mathbf{B} = \begin{bmatrix} -0.9412 & 0 & -0.2353 \\ 0 & -4 & 0 \\ -0.2353 & 0 & -0.0588 \end{bmatrix}. \text{ Taking } \mathbf{X}_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \text{ and using scaling we obtain}$$

$$\mathbf{X}_1 = \begin{bmatrix} -0.2941 \\ -1 \\ -0.0735 \end{bmatrix}, \quad \mathbf{X}_2 = \begin{bmatrix} 0.0735 \\ 1 \\ 0.0184 \end{bmatrix}, \quad \mathbf{X}_3 = \begin{bmatrix} -0.0184 \\ -1 \\ -0.0046 \end{bmatrix}, \quad \mathbf{X}_4 = \begin{bmatrix} 0.0046 \\ 1 \\ 0.0011 \end{bmatrix}.$$

Taking $\mathbf{K} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ as the eigenvector we find $\lambda_2 = -4$. The normalized eigenvector in this case is $\mathbf{K}_2 = \mathbf{K} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ and $\mathbf{C} = \begin{bmatrix} -0.9412 & 0 & -0.2353 \\ 0 & 0 & 0 \\ -0.2353 & 0 & -0.0588 \end{bmatrix}$. Taking $\mathbf{X}_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and using scaling we obtain $\mathbf{X}_1 = \begin{bmatrix} -1 \\ 0 \\ -0.25 \end{bmatrix}$,

$\mathbf{X}_2 = \begin{bmatrix} 1 \\ 0 \\ 0.25 \end{bmatrix}$. Using $\mathbf{K} = \begin{bmatrix} 1 \\ 0 \\ 0.25 \end{bmatrix}$ we find $\lambda_3 = -1$. The eigenvalues are 16, -4, and -1.

11. The inverse matrix is $\begin{bmatrix} 4 & -1 \\ -3 & 1 \end{bmatrix}$. Taking $\mathbf{X}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and using scaling we obtain

$$\mathbf{X}_1 = \begin{bmatrix} 1 \\ -0.6667 \end{bmatrix}, \quad \mathbf{X}_2 = \begin{bmatrix} 1 \\ -0.7857 \end{bmatrix}, \quad \mathbf{X}_3 = \begin{bmatrix} 1 \\ -0.7910 \end{bmatrix}, \quad \mathbf{X}_4 = \begin{bmatrix} 1 \\ -0.7913 \end{bmatrix}.$$

Using $\mathbf{K} = \begin{bmatrix} 1 \\ -0.7913 \end{bmatrix}$ we find $\lambda = 4.7913$. The minimum eigenvalue of $\begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix}$ is $1/4.7913 \approx 0.2087$.

12. The inverse matrix is $\begin{bmatrix} 1 & 3 \\ 4 & 2 \end{bmatrix}$. Taking $\mathbf{X}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and using scaling we obtain

$$\mathbf{X}_1 = \begin{bmatrix} 0.6667 \\ 1 \end{bmatrix}, \quad \mathbf{X}_2 = \begin{bmatrix} 0.7857 \\ 1 \end{bmatrix}, \dots, \quad \mathbf{X}_{10} = \begin{bmatrix} 0.75 \\ 1 \end{bmatrix}.$$

Using $\mathbf{K} = \begin{bmatrix} 0.75 \\ 1 \end{bmatrix}$ we find $\lambda = 5$. The minimum eigenvalue of $\begin{bmatrix} -0.2 & 0.3 \\ 0.4 & -0.1 \end{bmatrix}$ is $1/5 = 0.2$.

13. (a) Replacing the second derivative with the difference expression we obtain

$$EI \frac{y_{i+1} - 2y_i + y_{i-1}}{h^2} + Py_i = 0 \quad \text{or} \quad EI(y_{i+1} - 2y_i + y_{i-1}) + Ph^2y_i = 0.$$

- (b) Expanding the difference equation for $i = 1, 2, 3$ and using $h = L/4$, $y_0 = 0$, and $y_4 = 0$ we obtain

$$\begin{aligned} EI(y_2 - 2y_1 + y_0) + \frac{PL^2}{16}y_1 &= 0 & 2y_1 - y_2 &= \frac{PL^2}{16EI}y_1 \\ EI(y_3 - 2y_2 + y_1) + \frac{PL^2}{16}y_2 &= 0 & \text{or} & \quad -y_1 + 2y_2 - y_3 &= \frac{PL^2}{16EI}y_2 \\ EI(y_4 - 2y_3 + y_2) + \frac{PL^2}{16}y_3 &= 0 & -y_2 + 2y_3 &= \frac{PL^2}{16EI}y_3. \end{aligned}$$

In matrix form this becomes

$$\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \frac{PL^2}{16EI} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}.$$

(c) $\mathbf{A}^{-1} = \begin{bmatrix} 0.75 & 0.5 & 0.25 \\ 0.5 & 1 & 0.5 \\ 0.25 & 0.5 & 0.75 \end{bmatrix}$

- (d) Taking $\mathbf{X}_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and using scaling we obtain

$$\mathbf{X}_1 = \begin{bmatrix} 0.75 \\ 1 \\ 0.75 \end{bmatrix}, \quad \mathbf{X}_2 = \begin{bmatrix} 0.7143 \\ 1 \\ 0.7143 \end{bmatrix}, \quad \mathbf{X}_3 = \begin{bmatrix} 0.7083 \\ 1 \\ 0.7083 \end{bmatrix}, \quad \mathbf{X}_4 = \begin{bmatrix} 0.7073 \\ 1 \\ 0.7073 \end{bmatrix}, \quad \mathbf{X}_5 = \begin{bmatrix} 0.7071 \\ 1 \\ 0.7071 \end{bmatrix}.$$

Using $\mathbf{K} = \begin{bmatrix} 0.7071 \\ 1 \\ 0.7071 \end{bmatrix}$ we find $\lambda = 1.7071$. Then $1/\lambda = 0.5859$ is the minimum eigenvalue of $\mathbf{A}$.

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(e) Solving $\frac{PL^2}{16EI} = 0.5859$ for P we obtain $P = 9.3726 \frac{EI}{L^2}$. In Example 3 of Section 3.9 we saw

$$P = \pi^2 \frac{EI}{L^2} \approx 9.8696 \frac{EI}{L^2}.$$

14. (a) The difference equation is

$$EI_i(y_{i+1} - 2y_i + y_{i-1}) + Ph^2y_i = 0, \quad i = 1, 2, 3,$$

where $I_0 = 0.00200$, $I_1 = 0.00175$, $I_2 = 0.00150$, $I_3 = 0.00125$, and $I_4 = 0.00100$. The system of equations is

$$\begin{aligned} 0.00175E(y_2 - 2y_1 + y_0) + \frac{PL^2}{16}y_1 &= 0 & 0.0035y_1 - 0.00175y_2 &= \frac{PL^2}{16E}y_1 \\ 0.00150E(y_3 - 2y_2 + y_1) + \frac{PL^2}{16}y_2 &= 0 & \text{or} & -0.0015y_1 + 0.003y_2 - 0.0015y_3 &= \frac{PL^2}{16E}y_2 \\ 0.00125E(y_4 - 2y_3 + y_2) + \frac{PL^2}{16}y_3 &= 0 & -0.00125y_2 + 0.0025y_3 &= \frac{PL^2}{16E}y_3. \end{aligned}$$

In matrix form this becomes

$$\begin{bmatrix} 0.0035 & -0.00175 & 0 \\ -0.0015 & 0.003 & -0.0015 \\ 0 & -0.00125 & 0.0025 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \frac{PL^2}{16E} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}.$$

(b) The inverse of $\mathbf{A}$ is

$$\mathbf{A}^{-1} = \begin{bmatrix} 428.571 & 333.333 & 200 \\ 285.714 & 666.667 & 400 \\ 142.857 & 333.333 & 600 \end{bmatrix}.$$

Taking $\mathbf{X}_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and using scaling we obtain

$$\mathbf{X}_1 = \begin{bmatrix} 0.7113 \\ 1 \\ 0.7958 \end{bmatrix}, \quad \mathbf{X}_2 = \begin{bmatrix} 0.6710 \\ 1 \\ 0.7679 \end{bmatrix}, \quad \mathbf{X}_3 = \begin{bmatrix} 0.6645 \\ 1 \\ 0.7635 \end{bmatrix}, \quad \mathbf{X}_4 = \begin{bmatrix} 0.6634 \\ 1 \\ 0.7628 \end{bmatrix}, \quad \mathbf{X}_5 = \begin{bmatrix} 0.6632 \\ 1 \\ 0.7627 \end{bmatrix}.$$

This yields the eigenvalue $\lambda = 1161.23$. The smallest eigenvalue of $\mathbf{A}$ is then $1/\lambda = 0.0008612$. The lowest critical load is

$$P = \frac{16E}{L^2}(0.0008612) - 0.01378 \frac{E}{L^2}.$$

$$15. (a) \mathbf{A}^{10} = \begin{bmatrix} 67,745,349 & -43,691,832 & 8,258,598 \\ -43,691,832 & 28,182,816 & -5,328,720 \\ 8,258,598 & -5,328,720 & 1,008,180 \end{bmatrix}$$

$$(b) \mathbf{X}_{10} = \mathbf{A}^{10} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 67,745,349 \\ -43,691,832 \\ 8,258,598 \end{bmatrix} \approx 67,745,349 \begin{bmatrix} 1 \\ -0.644942 \\ 0.121906 \end{bmatrix}$$

$$\mathbf{X}_{12} = \mathbf{A}^{12} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2,680,201,629 \\ -1,728,645,624 \\ 326,775,222 \end{bmatrix} \approx 2,680,201,629 \begin{bmatrix} 1 \\ -0.644968 \\ 0.121922 \end{bmatrix}.$$

The vectors appear to be approaching scalar multiples of $\mathbf{K} = (1, -0.644968, 0.121922)$, which approximates the dominant eigenvector.

(c) The dominant eigenvalue is $\lambda_1 = (\mathbf{AK} \cdot \mathbf{K})/(\mathbf{K} \cdot \mathbf{K}) = 6.28995$.

Exercises 8.12

1. Distinct eigenvalues $\lambda_1 = 1$, $\lambda_2 = 5$ imply $\mathbf{A}$ is diagonalizable.

$$\mathbf{P} = \begin{bmatrix} -3 & 1 \\ 1 & 1 \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix}$$

2. Distinct eigenvalues $\lambda_1 = 0$, $\lambda_2 = 6$ imply $\mathbf{A}$ is diagonalizable.

$$\mathbf{P} = \begin{bmatrix} -5 & -1 \\ 4 & 2 \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} 0 & 0 \\ 0 & 6 \end{bmatrix}$$

3. For $\lambda_1 = \lambda_2 = 1$ we obtain the single eigenvector $\mathbf{K}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Hence $\mathbf{A}$ is not diagonalizable.

4. Distinct eigenvalues $\lambda_1 = \sqrt{5}$, $\lambda_2 = -\sqrt{5}$ imply $\mathbf{A}$ is diagonalizable.

$$\mathbf{P} = \begin{bmatrix} \sqrt{5} & -\sqrt{5} \\ 1 & 1 \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} \sqrt{5} & 0 \\ 0 & -\sqrt{5} \end{bmatrix}$$

5. Distinct eigenvalues $\lambda_1 = -7$, $\lambda_2 = 4$ imply $\mathbf{A}$ is diagonalizable.

$$\mathbf{P} = \begin{bmatrix} 13 & 1 \\ 2 & 1 \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} -7 & 0 \\ 0 & 4 \end{bmatrix}$$

6. Distinct eigenvalues $\lambda_1 = -4$, $\lambda_2 = 10$ imply $\mathbf{A}$ is diagonalizable.

$$\mathbf{P} = \begin{bmatrix} -3 & 1 \\ 1 & -5 \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} -4 & 0 \\ 0 & 10 \end{bmatrix}$$

7. Distinct eigenvalues $\lambda_1 = \frac{1}{3}$, $\lambda_2 = \frac{2}{3}$ imply $\mathbf{A}$ is diagonalizable.

$$\mathbf{P} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} \frac{1}{3} & 0 \\ 0 & \frac{2}{3} \end{bmatrix}$$

8. For $\lambda_1 = \lambda_2 = -3$ we obtain the single eigenvector $\mathbf{K}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Hence $\mathbf{A}$ is not diagonalizable.

9. Distinct eigenvalues $\lambda_1 = -i$, $\lambda_2 = i$ imply $\mathbf{A}$ is diagonalizable.

$$\mathbf{P} = \begin{bmatrix} 1 & 1 \\ -i & i \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} -i & 0 \\ 0 & i \end{bmatrix}$$

10. Distinct eigenvalues $\lambda_1 = 1 + i$, $\lambda_2 = 1 - i$ imply $\mathbf{A}$ is diagonalizable.

$$\mathbf{P} = \begin{bmatrix} 2 & 2 \\ i & -i \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} 1+i & 0 \\ 0 & 1-i \end{bmatrix}$$

Exercises 8.12

11. Distinct eigenvalues $\lambda_1 = 1$, $\lambda_2 = -1$, $\lambda_3 = 2$ imply $\mathbf{A}$ is diagonalizable.

$$\mathbf{P} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

12. Distinct eigenvalues $\lambda_1 = 3$, $\lambda_2 = 4$, $\lambda_3 = 5$ imply $\mathbf{A}$ is diagonalizable.

$$\mathbf{P} = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 1 \\ 1 & 1 & -1 \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

13. Distinct eigenvalues $\lambda_1 = 0$, $\lambda_2 = 1$, $\lambda_3 = 2$ imply $\mathbf{A}$ is diagonalizable.

$$\mathbf{P} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

14. Distinct eigenvalues $\lambda_1 = 1$, $\lambda_2 = -3i$, $\lambda_3 = 3i$ imply $\mathbf{A}$ is diagonalizable.

$$\mathbf{P} = \begin{bmatrix} 0 & -3i & 3i \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -3i & 0 \\ 0 & 0 & 3i \end{bmatrix}$$

15. The eigenvalues are $\lambda_1 = \lambda_2 = 1$, $\lambda_3 = 2$. For $\lambda_1 = \lambda_2 = 1$ we obtain the single eigenvector $\mathbf{K}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$.

Hence $\mathbf{A}$ is not diagonalizable.

16. Distinct eigenvalues $\lambda_1 = 1$, $\lambda_2 = 2$, $\lambda_3 = 3$ imply $\mathbf{A}$ is diagonalizable.

$$\mathbf{P} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

17. Distinct eigenvalues $\lambda_1 = 1$, $\lambda_2 = \sqrt{5}$, $\lambda_3 = -\sqrt{5}$ imply $\mathbf{A}$ is diagonalizable.

$$\mathbf{P} = \begin{bmatrix} 0 & 1 + \sqrt{5} & 1 - \sqrt{5} \\ 0 & 2 & 2 \\ 1 & 0 & 0 \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sqrt{5} & 0 \\ 0 & 0 & -\sqrt{5} \end{bmatrix}$$

18. For $\lambda_1 = \lambda_2 = \lambda_3 = 1$ we obtain the single eigenvector $\mathbf{K}_1 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$. Hence $\mathbf{A}$ is not diagonalizable.

19. For the eigenvalues $\lambda_1 = \lambda_2 = 2$, $\lambda_3 = 1$, $\lambda_4 = -1$ we obtain four linearly independent eigenvectors. Hence $\mathbf{A}$ is diagonalizable and

$$\mathbf{P} = \begin{bmatrix} -3 & -1 & -1 & 1 \\ 0 & 1 & 0 & 0 \\ -3 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}.$$

20. The eigenvalues are $\lambda_1 = \lambda_2 = 2$, $\lambda_3 = \lambda_4 = 3$. For $\lambda_3 = \lambda_4 = 3$ we obtain the single eigenvector $\mathbf{K}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$.

Hence $\mathbf{A}$ is not diagonalizable.

21. $\lambda_1 = 0$, $\lambda_2 = 2$, $\mathbf{K}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, $\mathbf{K}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\mathbf{P} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$, $\mathbf{D} = \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix}$
22. $\lambda_1 = -1$, $\lambda_2 = 4$, $\mathbf{K}_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$, $\mathbf{K}_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $\mathbf{P} = \begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ -\frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix}$, $\mathbf{D} = \begin{bmatrix} -1 & 0 \\ 0 & 4 \end{bmatrix}$
23. $\lambda_1 = 3$, $\lambda_2 = 10$, $\mathbf{K}_1 = \begin{bmatrix} -\sqrt{10} \\ 2 \end{bmatrix}$, $\mathbf{K}_2 = \begin{bmatrix} \sqrt{10} \\ 5 \end{bmatrix}$, $\mathbf{P} = \begin{bmatrix} -\frac{\sqrt{10}}{\sqrt{14}} & \frac{\sqrt{10}}{\sqrt{35}} \\ \frac{2}{\sqrt{14}} & \frac{15}{\sqrt{35}} \end{bmatrix}$, $\mathbf{D} = \begin{bmatrix} 3 & 0 \\ 0 & 10 \end{bmatrix}$
24. $\lambda_1 = -1$, $\lambda_2 = 3$, $\mathbf{K}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\mathbf{K}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, $\mathbf{P} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$, $\mathbf{D} = \begin{bmatrix} -1 & 0 \\ 0 & 3 \end{bmatrix}$
25. $\lambda_1 = -1$, $\lambda_2 = \lambda_3 = 1$, $\mathbf{K}_1 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$, $\mathbf{K}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$, $\mathbf{K}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{P} = \begin{bmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $\mathbf{D} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
26. $\lambda_1 = \lambda_2 = -1$, $\lambda_3 = 5$, $\mathbf{K}_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{K}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$, $\mathbf{K}_3 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$, $\mathbf{P} = \begin{bmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{3}} \end{bmatrix}$,
 $\mathbf{D} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 5 \end{bmatrix}$
27. $\lambda_1 = 3$, $\lambda_2 = 6$, $\lambda_3 = 9$, $\mathbf{K}_1 = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}$, $\mathbf{K}_2 = \begin{bmatrix} 2 \\ -1 \\ -2 \end{bmatrix}$, $\mathbf{K}_3 = \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix}$, $\mathbf{P} = \begin{bmatrix} \frac{2}{3} & \frac{2}{3} & \frac{1}{3} \\ \frac{2}{3} & -\frac{1}{3} & -\frac{2}{3} \\ \frac{1}{3} & -\frac{2}{3} & \frac{2}{3} \end{bmatrix}$, $\mathbf{D} = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 9 \end{bmatrix}$
28. $\lambda_1 = 1$, $\lambda_2 = 2 - \sqrt{2}$, $\lambda_3 = 2 + \sqrt{2}$, $\mathbf{K}_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, $\mathbf{K}_2 = \begin{bmatrix} 1 - \sqrt{2} \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{K}_3 = \begin{bmatrix} 1 + \sqrt{2} \\ 0 \\ 1 \end{bmatrix}$,
 $\mathbf{P} = \begin{bmatrix} 0 & -\frac{\sqrt{2-\sqrt{2}}}{2} & \frac{\sqrt{2+\sqrt{2}}}{2} \\ 1 & 0 & 0 \\ 0 & \frac{\sqrt{2+\sqrt{2}}}{2} & \frac{\sqrt{2-\sqrt{2}}}{2} \end{bmatrix}$, $\mathbf{D} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 - 2\sqrt{2} & 0 \\ 0 & 0 & 2 + \sqrt{2} \end{bmatrix}$
29. $\lambda_1 = 1$, $\lambda_2 = -6$, $\lambda_3 = 8$, $\mathbf{K}_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, $\mathbf{K}_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$, $\mathbf{K}_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{P} = \begin{bmatrix} 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 1 & 0 & 0 \\ 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$, $\mathbf{D} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -6 & 0 \\ 0 & 0 & 8 \end{bmatrix}$
30. $\lambda_1 = \lambda_2 = 0$, $\lambda_3 = -2$, $\lambda_4 = 2$, $\mathbf{K}_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$, $\mathbf{K}_2 = \begin{bmatrix} 0 \\ -1 \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{K}_3 = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}$, $\mathbf{K}_4 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$

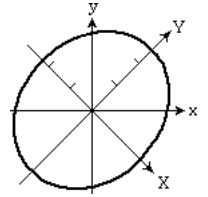
Exercises 8.12

$$\mathbf{P} = \begin{bmatrix} -\frac{1}{\sqrt{2}} & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & -\frac{1}{\sqrt{2}} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{2} & \frac{1}{2} \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

31. The given equation can be written as $\mathbf{X}^T \mathbf{A} \mathbf{X} = 24$: $\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 5 & -1 \\ -1 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 24$. Using $\lambda_1 = 6$, $\lambda_2 = 4$, $\mathbf{K}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, $\mathbf{K}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\mathbf{P} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$ and $\mathbf{X} = \mathbf{P} \mathbf{X}'$ we find

$$\begin{bmatrix} X & Y \end{bmatrix} \begin{bmatrix} 6 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix} = 24 \quad \text{or} \quad 6X^2 + 4Y^2 = 24.$$

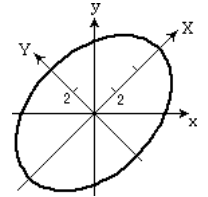
The conic section is an ellipse. Now from $\mathbf{X}' = \mathbf{P}^T \mathbf{X}$ we see that the XY -coordinates of $(1, -1)$ and $(1, 1)$ are $(\sqrt{2}, 0)$ and $(0, \sqrt{2})$, respectively. From this we conclude that the X -axis and Y -axis are as shown in the accompanying figure.



32. The given equation can be written as $\mathbf{X}^T \mathbf{A} \mathbf{X} = 288$: $\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 13 & -5 \\ -5 & 13 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 288$. Using $\lambda_1 = 8$, $\lambda_2 = 18$, $\mathbf{K}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\mathbf{K}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$, $\mathbf{P} = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$ and $\mathbf{X} = \mathbf{P} \mathbf{X}'$ we find

$$\begin{bmatrix} X & Y \end{bmatrix} \begin{bmatrix} 8 & 0 \\ 0 & 18 \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix} = 288 \quad \text{or} \quad 8X^2 + 18Y^2 = 288.$$

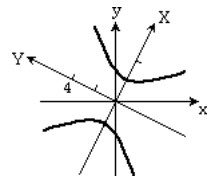
The conic section is an ellipse. Now from $\mathbf{X}' = \mathbf{P}^T \mathbf{X}$ we see that the XY -coordinates of $(1, 1)$ and $(1, -1)$ are $(\sqrt{2}, 0)$ and $(0, -\sqrt{2})$, respectively. From this we conclude that the X -axis and Y -axis are as shown in the accompanying figure.



33. The given equation can be written as $\mathbf{X}^T \mathbf{A} \mathbf{X} = 20$: $\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} -3 & 4 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 20$. Using $\lambda_1 = 5$, $\lambda_2 = -5$, $\mathbf{K}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\mathbf{K}_2 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$, $\mathbf{P} = \begin{bmatrix} \frac{1}{\sqrt{5}} & -\frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix}$ and $\mathbf{X} = \mathbf{P} \mathbf{X}'$ we find

$$\begin{bmatrix} X & Y \end{bmatrix} \begin{bmatrix} 5 & 0 \\ 0 & -5 \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix} = 20 \quad \text{or} \quad 5X^2 - 5Y^2 = 20.$$

The conic section is a hyperbola. Now from $\mathbf{X}' = \mathbf{P}^T \mathbf{X}$ we see that the XY -coordinates of $(1, 2)$ and $(-2, 1)$ are $(\sqrt{5}, 0)$ and $(0, \sqrt{5})$, respectively. From this we conclude that the X -axis and Y -axis are as shown in the accompanying figure.

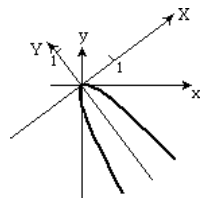


34. The given equation can be written as $\mathbf{X}^T \mathbf{A} \mathbf{X} = 288$:

$$\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 16 & 12 \\ 12 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} -3 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0.$$

Using $\lambda_1 = 25$, $\lambda_2 = 0$, $\mathbf{K}_1 = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$, $\mathbf{K}_2 = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$, $\mathbf{P} = \begin{bmatrix} \frac{4}{5} & \frac{3}{5} \\ \frac{3}{5} & -\frac{4}{5} \end{bmatrix}$ and $\mathbf{X} = \mathbf{P} \mathbf{X}'$ we find

$$\begin{bmatrix} X & Y \end{bmatrix} \begin{bmatrix} 25 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix} + \begin{bmatrix} 0 & 5 \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix} = 0 \quad \text{or} \quad 25X^2 + 5Y = 0.$$



The conic section is a parabola. Now from $\mathbf{X}' = \mathbf{P}^T \mathbf{X}$ we see that the XY -coordinates of $(4, 3)$ and $(3, -4)$ are $(5, 0)$ and $(0, -5)$, respectively. From this we conclude that the X -axis and Y -axis are as shown in the accompanying figure.

35. Since $\mathbf{D} = \mathbf{P}^{-1} \mathbf{A} \mathbf{P}$ we have $\mathbf{A} = \mathbf{P} \mathbf{D} \mathbf{P}^{-1}$. Hence

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 4 & -1 \\ 2 & 1 \end{bmatrix}.$$

36. Since eigenvectors are mutually orthogonal we use an orthogonal matrix $\mathbf{P}$ and $\mathbf{A} = \mathbf{P} \mathbf{D} \mathbf{P}^T$.

$$\mathbf{A} = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{3}} & 0 & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \end{bmatrix} = \begin{bmatrix} \frac{8}{3} & \frac{4}{3} & -\frac{1}{3} \\ \frac{4}{3} & \frac{11}{3} & \frac{4}{3} \\ -\frac{1}{3} & \frac{4}{3} & \frac{8}{3} \end{bmatrix}$$

37. Since $\mathbf{D} = \mathbf{P}^{-1} \mathbf{A} \mathbf{P}$ we have

$$\mathbf{A} = \mathbf{P} \mathbf{D} \mathbf{P}^{-1}$$

$$\mathbf{A}^2 = \mathbf{P} \mathbf{D} \mathbf{P}^{-1} \mathbf{P} \mathbf{D} \mathbf{P}^{-1} = \mathbf{P} \mathbf{D} \mathbf{D} \mathbf{P}^{-1} = \mathbf{P} \mathbf{D}^2 \mathbf{P}^{-1}$$

$$\mathbf{A}^3 = \mathbf{A}^2 \mathbf{A} = \mathbf{P} \mathbf{D}^2 \mathbf{P}^{-1} \mathbf{P} \mathbf{D} \mathbf{P}^{-1} = \mathbf{P} \mathbf{D}^2 \mathbf{D} \mathbf{P}^{-1} = \mathbf{P} \mathbf{D}^3 \mathbf{P}^{-1}$$

and so on.

$$38. \begin{bmatrix} 2^4 & 0 & 0 & 0 \\ 0 & 3^4 & 0 & 0 \\ 0 & 0 & (-1)^4 & 0 \\ 0 & 0 & 0 & (5)^4 \end{bmatrix} = \begin{bmatrix} 16 & 0 & 0 & 0 \\ 0 & 81 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 625 \end{bmatrix}$$

$$39. \lambda_1 = 2, \lambda_2 = -1, \mathbf{K}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \mathbf{K}_2 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \mathbf{P} = \begin{bmatrix} 1 & -1 \\ 1 & 2 \end{bmatrix}, \mathbf{P}^{-1} = \begin{bmatrix} \frac{2}{3} & \frac{1}{3} \\ -\frac{1}{3} & \frac{1}{3} \end{bmatrix}$$

$$\mathbf{A}^5 = \begin{bmatrix} 1 & -1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 32 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \frac{2}{3} & \frac{1}{3} \\ -\frac{1}{3} & \frac{1}{3} \end{bmatrix} = \begin{bmatrix} 21 & 11 \\ 22 & 10 \end{bmatrix}$$

$$40. \lambda_1 = 0, \lambda_2 = 1, \mathbf{K}_1 = \begin{bmatrix} 5 \\ 3 \end{bmatrix}, \mathbf{K}_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \mathbf{P} = \begin{bmatrix} 5 & 2 \\ 3 & 1 \end{bmatrix}, \mathbf{P}^{-1} = \begin{bmatrix} -1 & 2 \\ 3 & -5 \end{bmatrix}$$

$$\mathbf{A}^{10} = \begin{bmatrix} 5 & 2 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 2 \\ 3 & -5 \end{bmatrix} = \begin{bmatrix} 6 & -10 \\ 3 & -5 \end{bmatrix}$$

Exercises 8.13

1. (a) The message is $\mathbf{M} = \begin{bmatrix} 19 & 5 & 14 & 4 & 0 \\ 8 & 5 & 12 & 16 & 0 \end{bmatrix}$. The encoded message is

$$\mathbf{B} = \mathbf{A} \mathbf{M} = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 19 & 5 & 14 & 4 & 0 \\ 8 & 5 & 12 & 16 & 0 \end{bmatrix} = \begin{bmatrix} 35 & 15 & 38 & 36 & 0 \\ 27 & 10 & 26 & 20 & 0 \end{bmatrix}.$$

(b) The decoded message is

$$\mathbf{M} = \mathbf{A}^{-1} \mathbf{B} = \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 35 & 15 & 38 & 36 & 0 \\ 27 & 10 & 26 & 20 & 0 \end{bmatrix} = \begin{bmatrix} 19 & 5 & 14 & 4 & 0 \\ 8 & 5 & 12 & 16 & 0 \end{bmatrix}.$$

Exercises 8.13

2. (a) The message is $\mathbf{M} = \begin{bmatrix} 20 & 8 & 5 & 0 & 13 & 15 & 14 & 5 & 25 \\ 0 & 9 & 19 & 0 & 8 & 5 & 18 & 5 & 0 \end{bmatrix}$. The encoded message is

$$\mathbf{B} = \mathbf{A}\mathbf{M} = \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 20 & 8 & 5 & 0 & 13 & 15 & 14 & 5 & 25 \\ 0 & 9 & 19 & 0 & 8 & 5 & 18 & 5 & 0 \end{bmatrix} = \begin{bmatrix} 60 & 69 & 110 & 0 & 79 & 70 & 132 & 40 & 75 \\ 20 & 26 & 43 & 0 & 29 & 25 & 50 & 15 & 25 \end{bmatrix}.$$

- (b) The decoded message is

$$\begin{aligned} \mathbf{M} &= \mathbf{A}^{-1}\mathbf{B} = \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 60 & 69 & 110 & 0 & 79 & 70 & 132 & 40 & 75 \\ 20 & 26 & 43 & 0 & 29 & 25 & 50 & 15 & 25 \end{bmatrix} \\ &= \begin{bmatrix} 20 & 8 & 5 & 0 & 13 & 15 & 14 & 5 & 25 \\ 0 & 9 & 19 & 0 & 8 & 5 & 18 & 5 & 0 \end{bmatrix}. \end{aligned}$$

3. (a) The message is $\mathbf{M} = \begin{bmatrix} 16 & 8 & 15 & 14 & 5 \\ 0 & 8 & 15 & 13 & 5 \end{bmatrix}$. The encoded message is

$$\mathbf{B} = \mathbf{A}\mathbf{M} = \begin{bmatrix} 3 & 5 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 16 & 8 & 15 & 14 & 5 \\ 0 & 8 & 15 & 13 & 5 \end{bmatrix} = \begin{bmatrix} 48 & 64 & 120 & 107 & 40 \\ 32 & 40 & 75 & 67 & 25 \end{bmatrix}.$$

- (b) The decoded message is

$$\mathbf{M} = \mathbf{A}^{-1}\mathbf{B} = \begin{bmatrix} -3 & 5 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} 48 & 64 & 120 & 107 & 40 \\ 32 & 40 & 75 & 67 & 25 \end{bmatrix} = \begin{bmatrix} 16 & 8 & 15 & 14 & 5 \\ 0 & 8 & 15 & 13 & 5 \end{bmatrix}.$$

4. (a) The message is $\mathbf{M} = \begin{bmatrix} 7 & 15 & 0 & 14 & 15 & 18 & 20 \\ 8 & 0 & 15 & 14 & 0 & 13 & 1 \\ 9 & 14 & 0 & 19 & 20 & 0 & 0 \end{bmatrix}$. The encoded message is

$$\mathbf{B} = \mathbf{A}\mathbf{M} = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 7 & 15 & 0 & 14 & 15 & 18 & 20 \\ 8 & 0 & 15 & 14 & 0 & 13 & 1 \\ 9 & 14 & 0 & 19 & 20 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 50 & 57 & 30 & 99 & 75 & 44 & 22 \\ 33 & 43 & 15 & 66 & 55 & 31 & 21 \\ 26 & 28 & 15 & 52 & 40 & 13 & 1 \end{bmatrix}.$$

- (b) The decoded message is

$$\mathbf{M} = \mathbf{A}^{-1}\mathbf{B} = \begin{bmatrix} 0 & 1 & -1 \\ 2 & -2 & -1 \\ -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 50 & 57 & 30 & 99 & 75 & 44 & 22 \\ 33 & 43 & 15 & 66 & 55 & 31 & 21 \\ 26 & 28 & 15 & 52 & 40 & 13 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 15 & 0 & 14 & 15 & 18 & 20 \\ 8 & 0 & 15 & 14 & 0 & 13 & 1 \\ 9 & 14 & 0 & 19 & 20 & 0 & 0 \end{bmatrix}.$$

5. (a) The message is $\mathbf{M} = \begin{bmatrix} 7 & 15 & 0 & 14 & 15 & 18 & 20 \\ 8 & 0 & 15 & 14 & 0 & 13 & 1 \\ 9 & 14 & 0 & 19 & 20 & 0 & 0 \end{bmatrix}$. The encoded message is

$$\mathbf{B} = \mathbf{A}\mathbf{M} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 1 & 1 \\ -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 7 & 15 & 0 & 14 & 15 & 18 & 20 \\ 8 & 0 & 15 & 14 & 0 & 13 & 1 \\ 9 & 14 & 0 & 19 & 20 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 31 & 44 & 15 & 61 & 50 & 49 & 41 \\ 24 & 29 & 15 & 47 & 35 & 31 & 21 \\ 1 & -15 & 15 & 0 & -15 & -5 & -19 \end{bmatrix}.$$

- (b) The decoded message is

$$\mathbf{M} = \mathbf{A}^{-1}\mathbf{B} = \begin{bmatrix} 1 & -1 & 0 \\ 1 & -1 & 1 \\ -2 & 3 & -1 \end{bmatrix} \begin{bmatrix} 31 & 44 & 15 & 61 & 50 & 49 & 41 \\ 24 & 29 & 15 & 47 & 35 & 31 & 21 \\ 1 & -15 & 15 & 0 & -15 & -5 & -19 \end{bmatrix} = \begin{bmatrix} 7 & 15 & 0 & 14 & 15 & 18 & 20 \\ 8 & 0 & 15 & 14 & 0 & 13 & 1 \\ 9 & 14 & 0 & 19 & 20 & 0 & 0 \end{bmatrix}.$$

6. (a) The message is $\mathbf{M} = \begin{bmatrix} 4 & 18 & 0 & 10 & 15 & 8 \\ 14 & 0 & 9 & 19 & 0 & 20 \\ 8 & 5 & 0 & 19 & 16 & 25 \end{bmatrix}$. The encoded message is

$$\mathbf{B} = \mathbf{A}\mathbf{M} = \begin{bmatrix} 5 & 3 & 0 \\ 4 & 3 & -1 \\ 5 & 2 & 2 \end{bmatrix} \begin{bmatrix} 4 & 18 & 0 & 10 & 15 & 8 \\ 14 & 0 & 9 & 19 & 0 & 20 \\ 8 & 5 & 0 & 19 & 16 & 25 \end{bmatrix} = \begin{bmatrix} 62 & 90 & 27 & 107 & 75 & 100 \\ 50 & 67 & 27 & 78 & 44 & 67 \\ 64 & 100 & 18 & 126 & 107 & 130 \end{bmatrix}.$$

- (b) The decoded message is

$$\mathbf{M} = \mathbf{A}^{-1}\mathbf{B} = \begin{bmatrix} 8 & -6 & -3 \\ -13 & 10 & 5 \\ -7 & 5 & 3 \end{bmatrix} \begin{bmatrix} 62 & 90 & 27 & 107 & 75 & 100 \\ 50 & 67 & 27 & 78 & 44 & 67 \\ 64 & 100 & 18 & 126 & 107 & 130 \end{bmatrix} = \begin{bmatrix} 4 & 18 & 0 & 10 & 15 & 8 \\ 14 & 0 & 9 & 19 & 0 & 20 \\ 8 & 5 & 0 & 19 & 16 & 25 \end{bmatrix}.$$

7. (a) The decoded message is

$$\mathbf{M} = \mathbf{A}^{-1}\mathbf{B} = \begin{bmatrix} 2 & -3 \\ -5 & 8 \end{bmatrix} \begin{bmatrix} 152 & 184 & 171 & 86 & 212 \\ 95 & 116 & 107 & 56 & 133 \end{bmatrix} = \begin{bmatrix} 19 & 20 & 21 & 4 & 25 \\ 0 & 8 & 1 & 18 & 4 \end{bmatrix}.$$

From correspondence (1) we obtain: STUDY_HARD.

8. The decoded message is

$$\mathbf{M} = \mathbf{A}^{-1}\mathbf{B} = \begin{bmatrix} 1 & -1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 46 & -7 & -13 & 22 & -18 & 1 & 10 \\ 23 & -15 & -14 & 2 & -18 & -12 & 5 \end{bmatrix} = \begin{bmatrix} 23 & 8 & 1 & 20 & 0 & 13 & 5 \\ 0 & 23 & 15 & 18 & 18 & 25 & 0 \end{bmatrix}.$$

From correspondence (1) we obtain: WHAT_ME_WORRY_.

9. The decoded message is

$$\mathbf{M} = \mathbf{A}^{-1}\mathbf{B} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 31 & 21 & 21 & 22 & 20 & 9 \\ 19 & 0 & 9 & 13 & 16 & 15 \\ 13 & 1 & 20 & 8 & 0 & 9 \end{bmatrix} = \begin{bmatrix} 13 & 1 & 20 & 8 & 0 & 9 \\ 19 & 0 & 9 & 13 & 16 & 15 \\ 18 & 20 & 1 & 14 & 20 & 0 \end{bmatrix}.$$

From correspondence (1) we obtain: MATH_IS_IMPORTANT.

10. The decoded message is

$$\begin{aligned} \mathbf{M} = \mathbf{A}^{-1}\mathbf{B} &= \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 2 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 36 & 32 & 28 & 61 & 26 & 56 & 10 & 12 \\ -9 & -2 & -18 & -1 & -18 & -25 & 0 & 0 \\ 23 & 27 & 23 & 41 & 26 & 43 & 5 & 12 \end{bmatrix} \\ &= \begin{bmatrix} 13 & 5 & 5 & 20 & 0 & 13 & 5 & 0 \\ 1 & 20 & 0 & 20 & 8 & 5 & 0 & 12 \\ 9 & 2 & 18 & 1 & 18 & 25 & 0 & 0 \end{bmatrix}. \end{aligned}$$

From correspondence (1) we obtain: MEET_ME_AT_THE_LIBRARY__.

11. Let $\mathbf{A}^{-1} = \begin{bmatrix} u & v \\ x & y \end{bmatrix}$. Then

$$\mathbf{A}^{-1}\mathbf{B} = \begin{bmatrix} u & v \\ x & y \end{bmatrix} \begin{bmatrix} 17 & 16 & 18 & 5 & 34 & 0 & 34 & 20 & 9 & 5 & 25 \\ -30 & -31 & -32 & -10 & -59 & 0 & -54 & -35 & -13 & -6 & -50 \end{bmatrix},$$

so $17u - 30v = 4$, $16u - 31v = 1$ and $5x - 6y = 1$, $25x - 50y = 25$. Then $\mathbf{A}^{-1} = \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}$ and

$$\mathbf{A}^{-1}\mathbf{B} = \begin{bmatrix} 4 & 1 & 4 & 0 & 9 & 0 & 14 & 5 & 5 & 4 & 0 \\ 13 & 15 & 14 & 5 & 25 & 0 & 20 & 15 & 4 & 1 & 25 \end{bmatrix}.$$

Exercises 8.13

From correspondence (1) we obtain: DAD_I_NEED_MONEY_TODAY.

$$12. (a) \mathbf{M}^T = \begin{bmatrix} 22 & 8 & 19 & 27 & 21 & 3 & 3 & 27 & 21 & 18 & 21 \\ 13 & 3 & 21 & 22 & 3 & 25 & 27 & 6 & 7 & 14 & 23 \\ 2 & 27 & 21 & 7 & 27 & 5 & 21 & 17 & 2 & 25 & 7 \end{bmatrix}$$

$$(b) \mathbf{B}^T = \mathbf{M} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 37 & 38 & 61 & 56 & 51 & 33 & 51 & 50 & 30 & 57 & 51 \\ 24 & 35 & 40 & 34 & 48 & 8 & 24 & 44 & 23 & 43 & 28 \\ 11 & -24 & 0 & 15 & -24 & 20 & 6 & -11 & 5 & -11 & 16 \end{bmatrix}$$

$$(c) \mathbf{BA}^{-1} = \mathbf{B} \begin{bmatrix} -1 & 1 & 1 \\ 2 & -1 & -1 \\ 1 & 0 & -1 \end{bmatrix} = \mathbf{M}$$

$$13. (a) \mathbf{B}' = \begin{bmatrix} 15 & 22 & 20 & 8 & 23 & 6 & 21 & 22 \\ 10 & 22 & 18 & 23 & 25 & 2 & 23 & 25 \\ 3 & 26 & 26 & 14 & 23 & 16 & 26 & 12 \end{bmatrix}$$

(b) Using correspondence (1) the encoded message is: OVTHWFUVJVRWYBWYCZZNWPZL.

$$(c) \mathbf{M}'\mathbf{A}^{-1}\mathbf{B}' = \begin{bmatrix} 1 & 4 & -3 \\ 2 & 3 & -2 \\ -2 & -4 & 3 \end{bmatrix} \mathbf{B}' = \begin{bmatrix} 46 & 32 & 14 & 58 & 54 & -34 & 35 & 86 \\ 54 & 58 & 42 & 57 & 75 & -14 & 59 & 95 \\ -61 & -54 & -34 & -66 & -77 & 28 & -56 & -108 \end{bmatrix}$$

$$\mathbf{M} = \mathbf{M}'_{\text{mod } 27} = \begin{bmatrix} 19 & 5 & 14 & 4 & 0 & 20 & 8 & 5 \\ 0 & 4 & 15 & 3 & 21 & 13 & 5 & 14 \\ 20 & 0 & 20 & 15 & 4 & 1 & 25 & 0 \end{bmatrix}.$$

Using correspondence (1) the encoded message is: SEND_THE_DOCUMENT_TODAY.

Exercises 8.14

1. $[0 \ 1 \ 1 \ 0]$

2. $[1 \ 1 \ 1 \ 1]$

3. $[0 \ 0 \ 0 \ 1 \ 1]$

4. $[1 \ 0 \ 1 \ 0 \ 0]$

5. $[1 \ 0 \ 1 \ 0 \ 1 \ 0 \ 0 \ 1]$

6. $[0 \ 1 \ 1 \ 0 \ 1 \ 0 \ 1 \ 0]$

7. $[1 \ 0 \ 0]$

8. $[0 \ 0 \ 1]$

9. Parity error

10. $[1 \ 0 \ 1 \ 0]$

11. $[1 \ 0 \ 0 \ 1 \ 1]$

12. Parity error

In Problems 13-18 $\mathbf{D} = [c_1 \ c_2 \ c_3]$ and $\mathbf{P} = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$.

13. $\mathbf{D}^T = \mathbf{P}[1 \ 1 \ 1 \ 0]^T = [0 \ 0 \ 0]^T$; $\mathbf{C} = [0 \ 0 \ 1 \ 0 \ 1 \ 1 \ 0]$

14. $\mathbf{D}^T = \mathbf{P}[0 \ 0 \ 1 \ 1]^T = [1 \ 0 \ 0]^T$; $\mathbf{C} = [1 \ 0 \ 0 \ 0 \ 0 \ 1 \ 1]$

15. $\mathbf{D}^T = \mathbf{P}[0 \ 1 \ 0 \ 1]^T = [0 \ 1 \ 0]^T$; $\mathbf{C} = [0 \ 1 \ 0 \ 0 \ 1 \ 0 \ 1]$

16. $\mathbf{D}^T = \mathbf{P}[0 \ 0 \ 0 \ 1]^T = [1 \ 1 \ 1]^T$; $\mathbf{C} = [1 \ 1 \ 0 \ 1 \ 0 \ 0 \ 1]$

17. $\mathbf{D}^T = \mathbf{P}[0 \ 1 \ 1 \ 0]^T = [1 \ 1 \ 0]^T$; $\mathbf{C} = [1 \ 1 \ 0 \ 0 \ 1 \ 1 \ 0]$

18. $\mathbf{D}^T = \mathbf{P}[1 \ 1 \ 0 \ 0]^T = [0 \ 1 \ 1]^T$; $\mathbf{C} = [0 \ 1 \ 1 \ 1 \ 1 \ 0 \ 0]$

In Problems 19-28 $\mathbf{W}$ represents the correctly decoded message.

19. $\mathbf{S} = \mathbf{H}\mathbf{R}^T = \mathbf{H}[0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0] = [0 \ 0 \ 0]^T$; a code word. $\mathbf{W} = [0 \ 0 \ 0 \ 0]$

20. $\mathbf{S} = \mathbf{H}\mathbf{R}^T = \mathbf{H}[1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0] = [0 \ 1 \ 1]^T$; not a code word. The error is in the third bit.
 $\mathbf{W} = [1 \ 0 \ 0 \ 0]$

21. $\mathbf{S} = \mathbf{H}\mathbf{R}^T = \mathbf{H}[1 \ 1 \ 0 \ 1 \ 1 \ 0 \ 1] = [1 \ 0 \ 1]^T$; not a code word. The error is in the fifth bit.
 $\mathbf{W} = [0 \ 0 \ 0 \ 1]$

22. $\mathbf{S} = \mathbf{H}\mathbf{R}^T = \mathbf{H}[0 \ 1 \ 0 \ 1 \ 0 \ 1 \ 0] = [0 \ 0 \ 0]^T$; a code word. $\mathbf{W} = [0 \ 0 \ 1 \ 0]$

23. $\mathbf{S} = \mathbf{H}\mathbf{R}^T = \mathbf{H}[1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1] = [0 \ 0 \ 0]^T$; a code word. $\mathbf{W} = [1 \ 1 \ 1 \ 1]$

24. $\mathbf{S} = \mathbf{H}\mathbf{R}^T = \mathbf{H}[1 \ 1 \ 0 \ 0 \ 1 \ 1 \ 0] = [0 \ 0 \ 0]^T$; a code word. $\mathbf{W} = [0 \ 1 \ 1 \ 0]$

25. $\mathbf{S} = \mathbf{H}\mathbf{R}^T = \mathbf{H}[0 \ 1 \ 1 \ 1 \ 0 \ 0 \ 1] = [0 \ 1 \ 0]^T$; not a code word. The error is in the second bit.
 $\mathbf{W} = [1 \ 0 \ 0 \ 1]$

26. $\mathbf{S} = \mathbf{H}\mathbf{R}^T = \mathbf{H}[1 \ 0 \ 0 \ 1 \ 0 \ 0 \ 1] = [0 \ 1 \ 0]^T$; not a code word. The error is in the second bit.
 $\mathbf{W} = [0 \ 0 \ 0 \ 1]$

27. $\mathbf{S} = \mathbf{H}\mathbf{R}^T = \mathbf{H}[1 \ 0 \ 1 \ 1 \ 0 \ 1 \ 1] = [1 \ 1 \ 1]^T$; not a code word. The error is in the seventh bit.
 $\mathbf{W} = [1 \ 0 \ 1 \ 0]$

28. $\mathbf{S} = \mathbf{H}\mathbf{R}^T = \mathbf{H}[0 \ 0 \ 1 \ 0 \ 0 \ 1 \ 1] = [0 \ 1 \ 0]^T$; not a code word. The error is in the second bit.
 $\mathbf{W} = [1 \ 0 \ 1 \ 1]$

29. (a) $2^7 = 128$

(b) $2^4 = 16$

(c) $[0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0]$, $[1 \ 1 \ 0 \ 1 \ 0 \ 0 \ 1]$, $[0 \ 1 \ 0 \ 1 \ 0 \ 1 \ 0]$, $[1 \ 0 \ 0 \ 0 \ 0 \ 1 \ 1]$,
 $[1 \ 0 \ 0 \ 1 \ 1 \ 0 \ 0]$, $[0 \ 1 \ 0 \ 0 \ 1 \ 0 \ 1]$, $[1 \ 1 \ 0 \ 0 \ 1 \ 1 \ 0]$, $[0 \ 0 \ 0 \ 1 \ 1 \ 1 \ 1]$,
 $[1 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0]$, $[0 \ 0 \ 1 \ 1 \ 0 \ 0 \ 1]$, $[1 \ 0 \ 1 \ 1 \ 0 \ 1 \ 0]$, $[0 \ 1 \ 1 \ 0 \ 0 \ 1 \ 1]$,
 $[0 \ 1 \ 1 \ 1 \ 1 \ 0 \ 0]$, $[1 \ 0 \ 1 \ 0 \ 1 \ 0 \ 1]$, $[0 \ 0 \ 1 \ 0 \ 1 \ 1 \ 0]$, $[1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1]$

30. (a) $c_4 = 0, c_3 = 1, c_2 = 1, c_1 = 0$; $[0 \ 1 \ 1 \ 0 \ 0 \ 1 \ 1 \ 0]$

(b) $\mathbf{H} = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$ (c) $\mathbf{S} = \mathbf{H}\mathbf{R}^T = \mathbf{H}[0 \ 0 \ 1 \ 1 \ 1 \ 1 \ 0 \ 0]^T = [0 \ 0 \ 0 \ 0]^T$

Exercises 8.15

1. We have $\mathbf{Y}^T = [1 \ 2 \ 3 \ 2]$ and $\mathbf{A}^T = \begin{bmatrix} 2 & 3 & 4 & 5 \\ 1 & 1 & 1 & 1 \end{bmatrix}$.

Now $\mathbf{A}^T \mathbf{A} = \begin{bmatrix} 54 & 14 \\ 14 & 4 \end{bmatrix}$ and $(\mathbf{A}^T \mathbf{A})^{-1} = \frac{1}{20} \begin{bmatrix} 4 & -14 \\ -14 & 54 \end{bmatrix}$

so $\mathbf{X} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{Y} = \begin{bmatrix} \frac{2}{5} \\ \frac{3}{5} \end{bmatrix}$ and the least squares line is $y = 0.4x + 0.6$.

2. We have $\mathbf{Y}^T = [-1 \ 3 \ 5 \ 7]$ and $\mathbf{A}^T = \begin{bmatrix} 0 & 1 & 2 & 3 \\ 1 & 1 & 1 & 1 \end{bmatrix}$.

Now $\mathbf{A}^T \mathbf{A} = \begin{bmatrix} 14 & 6 \\ 6 & 4 \end{bmatrix}$ and $(\mathbf{A}^T \mathbf{A})^{-1} = \frac{1}{20} \begin{bmatrix} 4 & -6 \\ -6 & 14 \end{bmatrix}$

so $\mathbf{X} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{Y} = \begin{bmatrix} \frac{13}{5} \\ -\frac{2}{5} \end{bmatrix}$ and the least squares line is $y = 2.6x - 0.4$.

3. We have $\mathbf{Y}^T = [1 \ 1.5 \ 3 \ 4.5 \ 5]$ and $\mathbf{A}^T = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$.

Now $\mathbf{A}^T \mathbf{A} = \begin{bmatrix} 55 & 15 \\ 15 & 5 \end{bmatrix}$ and $(\mathbf{A}^T \mathbf{A})^{-1} = \frac{1}{50} \begin{bmatrix} 5 & -15 \\ -15 & 55 \end{bmatrix}$

so $\mathbf{X} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{Y} = \begin{bmatrix} 1.1 \\ -0.3 \end{bmatrix}$ and the least squares line is $y = 1.1x - 0.3$.

4. We have $\mathbf{Y}^T = [0 \ 1.5 \ 3 \ 4.5 \ 5]$ and $\mathbf{A}^T = \begin{bmatrix} 0 & 2 & 3 & 4 & 5 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$.

Now $\mathbf{A}^T \mathbf{A} = \begin{bmatrix} 54 & 14 \\ 14 & 5 \end{bmatrix}$ and $(\mathbf{A}^T \mathbf{A})^{-1} = \frac{1}{74} \begin{bmatrix} 5 & -14 \\ -14 & 54 \end{bmatrix}$

so $\mathbf{X} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{Y} = \begin{bmatrix} 1.06757 \\ -0.189189 \end{bmatrix}$ and the least squares line is $y = 1.06757x - 0.189189$.

5. We have $\mathbf{Y}^T = [2 \ 3 \ 5 \ 5 \ 9 \ 8 \ 10]$ and $\mathbf{A}^T = \begin{bmatrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$.

Now $\mathbf{A}^T \mathbf{A} = \begin{bmatrix} 91 & 21 \\ 21 & 7 \end{bmatrix}$ and $(\mathbf{A}^T \mathbf{A})^{-1} = \frac{1}{196} \begin{bmatrix} 7 & -21 \\ -21 & 91 \end{bmatrix}$

so $\mathbf{X} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{Y} = \begin{bmatrix} \frac{19}{14} \\ \frac{27}{14} \end{bmatrix}$ and the least squares line is $y = 1.35714x + 1.92857$.

6. We have $\mathbf{Y}^T = [2 \ 2.5 \ 1 \ 1.5 \ 2 \ 3.2 \ 5]$ and $\mathbf{A}^T = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$.

Now $\mathbf{A}^T \mathbf{A} = \begin{bmatrix} 140 & 28 \\ 28 & 7 \end{bmatrix}$ and $(\mathbf{A}^T \mathbf{A})^{-1} = \frac{1}{196} \begin{bmatrix} 7 & -28 \\ -28 & 140 \end{bmatrix}$

so $\mathbf{X} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{Y} = \begin{bmatrix} 0.407143 \\ 0.828571 \end{bmatrix}$ and the least squares line is $y = 0.407143x + 0.828571$.

7. We have $\mathbf{Y}^T = [220 \ 200 \ 180 \ 170 \ 150 \ 135]$ and $\mathbf{A}^T = \begin{bmatrix} 20 & 40 & 60 & 80 & 100 & 120 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$.

Now $\mathbf{A}^T \mathbf{A} = \begin{bmatrix} 36400 & 420 \\ 420 & 6 \end{bmatrix}$ and $(\mathbf{A}^T \mathbf{A})^{-1} = \frac{1}{42000} \begin{bmatrix} 6 & -420 \\ -420 & 36400 \end{bmatrix}$

so $\mathbf{X} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{Y} = \begin{bmatrix} -\frac{117}{140} \\ \frac{703}{3} \end{bmatrix}$ and the least squares line is $v = -0.835714T + 234.333$. At $T = 140$, $v \approx 117.333$ and at $T = 160$, $v \approx 100.619$.

8. We have $\mathbf{Y}^T = [0.47 \ 0.90 \ 2.0 \ 3.7 \ 7.5 \ 15]$ and $\mathbf{A}^T = \begin{bmatrix} 400 & 450 & 500 & 550 & 600 & 650 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$.

Now $\mathbf{A}^T \mathbf{A} = \begin{bmatrix} 1697500 & 3150 \\ 3150 & 6 \end{bmatrix}$ and $(\mathbf{A}^T \mathbf{A})^{-1} = \frac{1}{262500} \begin{bmatrix} 6 & -3150 \\ -3150 & 1697500 \end{bmatrix}$

so $\mathbf{X} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{Y} = \begin{bmatrix} 0.0538 \\ -23.3167 \end{bmatrix}$ and the least squares line is $R = 0.0538T - 23.3167$. At $T = 700$, $R \approx 14.3433$.

Exercises 8.16

In Problems 1-5 we use the fact that the element τ_{ij} in the transfer matrix $\mathbf{T}$ is the rate of transfer from compartment j to compartment i , and the fact that the sum of each column in $\mathbf{T}$ is 1.

1. (a) The initial state and the transfer matrix are

$$\mathbf{X}_0 = \begin{bmatrix} 90 \\ 60 \end{bmatrix} \quad \text{and} \quad \mathbf{T} = \begin{bmatrix} 0.8 & 0.4 \\ 0.2 & 0.6 \end{bmatrix}.$$

(b) We have

$$\mathbf{X}_1 = \mathbf{T}\mathbf{X}_0 = \begin{bmatrix} 0.8 & 0.4 \\ 0.2 & 0.6 \end{bmatrix} \begin{bmatrix} 90 \\ 60 \end{bmatrix} = \begin{bmatrix} 96 \\ 54 \end{bmatrix}$$

and

$$\mathbf{X}_2 = \mathbf{T}\mathbf{X}_1 = \begin{bmatrix} 0.8 & 0.4 \\ 0.2 & 0.6 \end{bmatrix} \begin{bmatrix} 96 \\ 54 \end{bmatrix} = \begin{bmatrix} 98.4 \\ 51.6 \end{bmatrix}.$$

(c) From $\mathbf{T}\hat{\mathbf{X}} - \hat{\mathbf{X}} = (\mathbf{T} - \mathbf{I})\hat{\mathbf{X}} = \mathbf{0}$ and the fact that the system is closed we obtain

$$\begin{aligned} -0.2x_1 + 0.4x_2 &= 0 \\ x_1 + x_2 &= 150. \end{aligned}$$

The solution is $x_1 = 100$, $x_2 = 50$, so the equilibrium state is $\hat{\mathbf{X}} = \begin{bmatrix} 100 \\ 50 \end{bmatrix}$.

2. (a) The initial state and the transfer matrix are

$$\mathbf{X}_0 = \begin{bmatrix} 100 \\ 200 \\ 150 \end{bmatrix} \quad \text{and} \quad \mathbf{T} = \begin{bmatrix} 0.7 & 0 & 0.5 \\ 0.3 & 0.8 & 0 \\ 0 & 0.2 & 0.5 \end{bmatrix}.$$

Exercises 8.16

(b) We have

$$\mathbf{X}_1 = \mathbf{T}\mathbf{X}_0 = \begin{bmatrix} 145 \\ 190 \\ 115 \end{bmatrix} \quad \text{and} \quad \mathbf{X}_2 = \mathbf{T}\mathbf{X}_1 = \begin{bmatrix} 159 \\ 195.5 \\ 95.5 \end{bmatrix}.$$

(c) From $\mathbf{T}\hat{\mathbf{X}} - \hat{\mathbf{X}} = (\mathbf{T} - \mathbf{I})\hat{\mathbf{X}} = \mathbf{0}$ and the fact that the system is closed we obtain

$$\begin{aligned} -0.8x_1 + 0.5x_2 &= 0 \\ 0.3x_1 - 0.9x_2 &= 0 \\ x_1 + x_2 + x_3 &= 450. \end{aligned}$$

The solution is $x_1 = 145.161$, $x_2 = 217.742$, $x_3 = 87.0968$, so the equilibrium state is $\hat{\mathbf{X}} = \begin{bmatrix} 145.161 \\ 217.742 \\ 87.097 \end{bmatrix}$.

3. (a) The initial state and the transfer matrix are

$$\mathbf{X}_0 = \begin{bmatrix} 100 \\ 0 \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{T} = \begin{bmatrix} 0.2 & 0.5 & 0 \\ 0.3 & 0.1 & 0 \\ 0.5 & 0.4 & 1 \end{bmatrix}.$$

(b) We have

$$\mathbf{X}_1 = \mathbf{T}\mathbf{X}_0 = \begin{bmatrix} 20 \\ 30 \\ 50 \end{bmatrix} \quad \text{and} \quad \mathbf{X}_2 = \mathbf{T}\mathbf{X}_1 = \begin{bmatrix} 19 \\ 9 \\ 72 \end{bmatrix}.$$

(c) From $\mathbf{T}\hat{\mathbf{X}} - \hat{\mathbf{X}} = (\mathbf{T} - \mathbf{I})\hat{\mathbf{X}} = \mathbf{0}$ and the fact that the system is closed we obtain

$$\begin{aligned} -0.8x_1 + 0.5x_2 &= 0 \\ 0.3x_1 - 0.9x_2 &= 0 \\ x_1 + x_2 + x_3 &= 100. \end{aligned}$$

The solution is $x_1 = x_2 = 0$, $x_3 = 100$, so the equilibrium state is $\hat{\mathbf{X}} = \begin{bmatrix} 0 \\ 0 \\ 100 \end{bmatrix}$.

4. (a) The transfer matrix is

$$\mathbf{T} = \begin{bmatrix} 0.7 & 0.05 & 0.15 \\ 0.3 & 0.75 & 0 \\ 0 & 0.2 & 0.85 \end{bmatrix}.$$

(b)

Year	Bare Space	Grasses	Small Shrubs
0	10.00	0.00	0.00
1	7.00	3.00	0.00
2	5.05	4.35	0.60
3	3.84	4.78	1.38
4	3.14	4.74	2.13
5	2.75	4.49	2.76
6	2.56	4.19	3.24

5. (a) The initial state and the transfer matrix are

$$\mathbf{X}_0 = \begin{bmatrix} 0 \\ 100 \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{T} = \begin{bmatrix} 0.88 & 0.02 & 0 \\ 0.06 & 0.97 & 0.05 \\ 0.06 & 0.01 & 0.95 \end{bmatrix}.$$

(b)

Year	Phytoplankton	Water	Zooplankton
0	0.00	100.00	0.00
1	2.00	97.00	1.00
2	3.70	94.26	2.04
3	5.14	91.76	3.10
4	6.36	89.47	4.17
5	7.39	87.37	5.24
6	8.25	85.46	6.30
7	8.97	83.70	7.33
8	9.56	82.10	8.34
9	10.06	80.62	9.32
10	10.46	79.28	10.26
11	10.79	78.04	11.17
12	11.06	76.90	12.04

6. From $\mathbf{T}\hat{\mathbf{X}} = 1\hat{\mathbf{X}}$ we see that the equilibrium state vector $\hat{\mathbf{X}}$ is the eigenvector of the transfer matrix $\mathbf{T}$ corresponding to the eigenvalue 1. It has the properties that its components add up to the sum of the components of the initial state vector.

Chapter 8 Review Exercises

- $\begin{bmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \\ 4 & 5 & 6 \\ 5 & 6 & 7 \end{bmatrix}$
- 4×3
- $\mathbf{AB} = \begin{bmatrix} 3 & 4 \\ 6 & 8 \end{bmatrix}; \quad \mathbf{BA} = [11]$
- $\mathbf{A}^{-1} = -\frac{1}{2} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}$
- False; consider $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
- True
- $\det(\frac{1}{2}\mathbf{A}) = (\frac{1}{2})^3(5) = \frac{5}{8}; \quad \det(-\mathbf{A}^T) = (-1)^3(5) = -5$
- $\det \mathbf{AB}^{-1} = \det \mathbf{A} / \det \mathbf{B} = 6/2 = 3$
- 0
- $\det \mathbf{C} = (-1)^3 / \det \mathbf{B} = -1/10^3(2) = -1/2000$
- False; an eigenvalue can be 0.

Chapter 8 Review Exercises

12. True
13. True
14. True, since complex roots of real polynomials occur in conjugate pairs.
15. False; if the characteristic equation of an $n \times n$ matrix has repeated roots, there may not be n linearly independent eigenvectors.
16. True
17. True
18. True
19. False; $\mathbf{A}$ is singular and thus not orthogonal.
20. True
21. $\mathbf{A} = \frac{1}{2}(\mathbf{A} + \mathbf{A}^T) + \frac{1}{2}(\mathbf{A} - \mathbf{A}^T)$ where $\frac{1}{2}(\mathbf{A} + \mathbf{A}^T)$ is symmetric and $\frac{1}{2}(\mathbf{A} - \mathbf{A}^T)$ is skew-symmetric.
22. Since $\det \mathbf{A}^2 = (\det \mathbf{A})^2 \geq 0$ and $\det \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = -1$, there is no $\mathbf{A}$ such that $\mathbf{A}^2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.
23. (a) $\begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$ is nilpotent.
 (b) Since $\det \mathbf{A}^n = (\det \mathbf{A})^n = 0$ we see that $\det \mathbf{A} = 0$ and $\mathbf{A}$ is singular.
24. (a) $\sigma_x \sigma_y = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix} = -\sigma_y \sigma_x$; $\sigma_x \sigma_z = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = -\sigma_z \sigma_x$; $\sigma_y \sigma_z = \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix} = -\sigma_z \sigma_y$
 (b) We first note that for anticommuting matrices $\mathbf{AB} = -\mathbf{BA}$, so $\mathbf{C} = 2\mathbf{AB}$. Then $\mathbf{C}_{xy} = \begin{bmatrix} 2i & 0 \\ 0 & -2i \end{bmatrix}$,

$$\mathbf{C}_{yz} = \begin{bmatrix} 0 & 2i \\ 2i & 0 \end{bmatrix}, \text{ and } \mathbf{C}_{zx} = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}.$$
25.
$$\left[\begin{array}{ccc|c} 5 & -1 & 1 & -9 \\ 2 & 4 & 0 & 27 \\ 1 & 1 & 5 & 9 \end{array} \right] \xrightarrow{R_{13}} \left[\begin{array}{ccc|c} 1 & 1 & 5 & 9 \\ 2 & 4 & 0 & 27 \\ 5 & -1 & 1 & -9 \end{array} \right] \xrightarrow[\text{operations}]{\text{row}} \left[\begin{array}{ccc|c} 1 & 1 & 5 & 9 \\ 0 & 1 & -5 & \frac{9}{2} \\ 0 & 0 & 1 & \frac{1}{2} \end{array} \right] \xrightarrow[\text{operations}]{\text{row}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & -\frac{1}{2} \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & \frac{1}{2} \end{array} \right].$$

 The solution is $\mathbf{X} = [-\frac{1}{2} \quad 7 \quad \frac{1}{2}]^T$.
26.
$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & -2 & 3 & 2 \\ 2 & 0 & -3 & 3 \end{array} \right] \xrightarrow[\text{operations}]{\text{row}} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -\frac{2}{3} & \frac{4}{3} \\ 0 & 0 & 1 & 1 \end{array} \right] \xrightarrow[\text{operations}]{\text{row}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{array} \right].$$

 The solution is $x_1 = 3, x_2 = 2, x_3 = 1$.
27. Multiplying the second row by abc we obtain the third row. Thus the determinant is 0.
28. Expanding along the first row we see that the result is an expression of the form $ay + bx^2 + cx + d = 0$, which is a parabola since, in this case $a \neq 0$ and $b \neq 0$. Letting $x = 1$ and $y = 2$ we note that the first and second rows are the same. Similarly, when $x = 2$ and $y = 3$, the first and third rows are the same; and when $x = 3$ and $y = 5$, the first and fourth rows are the same. In each case the determinant is 0 and the points lie on the parabola.
29. $4(-2)(3)(-1)(2)(5) = 240$
30. $(-3)(6)(9)(1) = -162$

31. Since $\begin{bmatrix} 1 & -1 & 1 \\ 5 & 1 & -1 \\ 1 & 2 & 1 \end{bmatrix} = 18 = 0$, the system has only the trivial solution.

32. Since $\begin{bmatrix} 1 & -1 & -1 \\ 5 & 1 & -1 \\ 1 & 2 & 1 \end{bmatrix} = 0$, the system has infinitely many solutions.

33. From $x_1\text{I}_2 + x_2\text{HNO}_3 \rightarrow x_3\text{HIO}_3 + x_4\text{NO}_2 + x_5\text{H}_2\text{O}$ we obtain the system $2x_1 = x_3$, $x_2 = x_3 + 2x_5$, $x_2 = x_4$, $3x_2 = 3x_3 + 2x_4 + x_5$. Letting $x_4 = x_2$ in the fourth equation we obtain $x_2 = 3x_3 + x_5$. Taking $x_1 = t$ we see that $x_3 = 2t$, $x_2 = 2t + 2x_5$, and $x_2 = 6t + x_5$. From the latter two equations we get $x_5 = 4t$. Taking $t = 1$ we have $x_1 = 1$, $x_2 = 10$, $x_3 = 2$, $x_4 = 10$, and $x_5 = 4$. The balanced equation is $\text{I}_2 + 10\text{HNO}_3 \rightarrow 2\text{HIO}_3 + 10\text{NO}_2 + 4\text{H}_2\text{O}$.

34. From $x_1\text{Ca} + x_2\text{H}_3\text{PO}_4 \rightarrow x_3\text{Ca}_3\text{P}_2\text{O}_8 + x_4\text{H}_2$ we obtain the system $x_1 = 3x_3$, $3x_2 = 2x_4$, $x_2 = 2x_3$, $4x_2 = 8x_3$. Letting $x_3 = t$ we see that $x_1 = 3t$, $x_2 = 2t$, and $x_4 = 3t$. Taking $t = 1$ we obtain the balanced equation $3\text{Ca} + 2\text{H}_3\text{PO}_4 \rightarrow \text{Ca}_3\text{P}_2\text{O}_8 + 3\text{H}_2$.

35. $\det \mathbf{A} = -84$, $\det \mathbf{A}_1 = 42$, $\det \mathbf{A}_2 = -21$, $\det \mathbf{A}_3 = -56$; $x_1 = \frac{42}{-84} = -\frac{1}{2}$, $x_2 = \frac{-21}{-84} = \frac{1}{4}$, $x_3 = \frac{-56}{-84} = \frac{2}{3}$

36. $\det = 4$, $\det \mathbf{A}_1 = 16$, $\det \mathbf{A}_2 = -4$, $\det \mathbf{A}_3 = 0$; $x_1 = \frac{16}{4} = 4$, $x_2 = \frac{-4}{4} = -1$, $x_3 = \frac{0}{4} = 0$

37. $\det \mathbf{A} = \cos^2 \theta + \sin^2 \theta$, $\det \mathbf{A}_1 = X \cos \theta - Y \sin \theta$, $\det \mathbf{A}_2 = Y \cos \theta + X \sin \theta$;
 $x_1 = X \cos \theta - Y \sin \theta$, $y = Y \cos \theta + X \sin \theta$

38. (a) $i_1 - i_2 - i_3 - i_4 = 0$, $i_2 R_1 = E$, $i_2 R_1 - i_3 R_2 = 0$, $i_3 R_2 - i_4 R_3 = 0$

(b) $\det \mathbf{A} = \begin{vmatrix} 1 & -1 & -1 & -1 \\ 0 & R_1 & 0 & 0 \\ 0 & R_1 & -R_2 & 0 \\ 0 & 0 & R_2 & -R_3 \end{vmatrix} = R_1 R_2 R_3$;

$\det \mathbf{A}_1 = \begin{vmatrix} 0 & -1 & -1 & -1 \\ E & R_1 & 0 & 0 \\ 0 & R_1 & -R_2 & 0 \\ 0 & 0 & R_2 & -R_3 \end{vmatrix} = -E[-R_2 R_3 - R_1(R_3 + R_2)] = E(R_2 R_3 + R_1 R_3 + R_1 R_2)$;

$i_1 = \frac{\det \mathbf{A}_1}{\det \mathbf{A}} = \frac{E(R_2 R_3 + R_1 R_3 + R_1 R_2)}{R_1 R_2 R_3} = E \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right)$

39. $\mathbf{A}\mathbf{X} = \mathbf{B}$ is $\begin{bmatrix} 2 & 3 & -1 \\ 1 & -2 & 0 \\ -2 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6 \\ -3 \\ 9 \end{bmatrix}$. Since $\mathbf{A}^{-1} = -\frac{1}{3} \begin{bmatrix} -2 & -3 & -2 \\ -1 & 0 & -1 \\ -4 & -6 & -7 \end{bmatrix}$, we have

$\mathbf{X} = \mathbf{A}^{-1}\mathbf{B} = \begin{bmatrix} 7 \\ 5 \\ 23 \end{bmatrix}$.

40. (a) $\mathbf{A}^{-1}\mathbf{B} = \begin{bmatrix} \frac{3}{2} & -\frac{1}{4} & -\frac{9}{4} \\ -1 & \frac{1}{2} & \frac{3}{2} \\ \frac{1}{2} & -\frac{1}{4} & -\frac{1}{4} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$

(b) $\mathbf{A}^{-1}\mathbf{B} = \begin{bmatrix} \frac{3}{2} & -\frac{1}{4} & -\frac{9}{4} \\ -1 & \frac{1}{2} & \frac{3}{2} \\ \frac{1}{2} & -\frac{1}{4} & -\frac{1}{4} \end{bmatrix} \begin{bmatrix} -2 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} -10 \\ 7 \\ -2 \end{bmatrix}$

Chapter 8 Review Exercises

41. From the characteristic equation $\lambda^2 - 4\lambda - 5 = 0$ we see that the eigenvalues are $\lambda_1 = -1$ and $\lambda_2 = 5$. For $\lambda_1 = -1$ we have $2k_1 + 2k_2 = 0$, $4k_1 - 4k_2 = 0$ and $\mathbf{K}_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$. For $\lambda_2 = 5$ we have $-4k_1 + 2k_2 = 0$, $4k_1 - 2k_2 = 0$

and $\mathbf{K}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

42. From the characteristic equation $\lambda^2 = 0$ we see that the eigenvalues are $\lambda_1 = \lambda_2 = 0$. For $\lambda_1 = \lambda_2 = 0$ we have $4k_1 = 0$ and $\mathbf{K}_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ is a single eigenvector.

43. From the characteristic equation $-\lambda^3 + 6\lambda^2 + 15\lambda + 8 = -(\lambda + 1)^2(\lambda - 8) = 0$ we see that the eigenvalues are $\lambda_1 = \lambda_2 = -1$ and $\lambda_3 = 8$. For $\lambda_1 = \lambda_2 = -1$ we have

$$\left[\begin{array}{ccc|c} 4 & 2 & 4 & 0 \\ 2 & 1 & 2 & 0 \\ 4 & 2 & 4 & 0 \end{array} \right] \xrightarrow[\text{operations}]{\text{row}} \left[\begin{array}{ccc|c} 1 & \frac{1}{2} & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

Thus $\mathbf{K}_1 = [1 \ -2 \ 0]^T$ and $\mathbf{K}_2 = [1 \ 0 \ -1]^T$. For $\lambda_3 = 8$ we have

$$\left[\begin{array}{ccc|c} -5 & 2 & 4 & 0 \\ 2 & -8 & 2 & 0 \\ 4 & 2 & -5 & 0 \end{array} \right] \xrightarrow[\text{operations}]{\text{row}} \left[\begin{array}{ccc|c} 1 & -\frac{2}{5} & -\frac{4}{5} & 0 \\ 0 & 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

Thus $\mathbf{K}_3 = [2 \ 1 \ 2]^T$.

44. From the characteristic equation $-\lambda^3 + 18\lambda^2 - 99\lambda + 162 = -(\lambda - 9)(\lambda - 6)(\lambda - 3) = 0$ we see that the eigenvalues are $\lambda_1 = 9$, $\lambda_2 = 6$, and $\lambda_3 = 3$. For $\lambda_1 = 9$ we have

$$\left[\begin{array}{ccc|c} -2 & -2 & 0 & 0 \\ -2 & -3 & 2 & 0 \\ 0 & 2 & -4 & 0 \end{array} \right] \xrightarrow[\text{operations}]{\text{row}} \left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

Thus $\mathbf{K}_1 = [-2 \ 2 \ 1]^T$. For $\lambda_2 = 6$ we have

$$\left[\begin{array}{ccc|c} 1 & -2 & 0 & 0 \\ -2 & 0 & 2 & 0 \\ 0 & 2 & -1 & 0 \end{array} \right] \xrightarrow[\text{operations}]{\text{row}} \left[\begin{array}{ccc|c} 1 & -2 & 0 & 0 \\ 0 & 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

Thus $\mathbf{K}_2 = [2 \ 1 \ 2]^T$. For $\lambda_3 = 3$ we have

$$\left[\begin{array}{ccc|c} 4 & -2 & 0 & 0 \\ -2 & 3 & 2 & 0 \\ 0 & 2 & 2 & 0 \end{array} \right] \xrightarrow[\text{operations}]{\text{row}} \left[\begin{array}{ccc|c} 1 & -\frac{1}{2} & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

Thus $\mathbf{K}_3 = [1 \ 2 \ -2]^T$.

45. From the characteristic equation $-\lambda^3 - \lambda^2 + 21\lambda + 45 = -(\lambda + 3)^2(\lambda - 5) = 0$ we see that the eigenvalues are $\lambda_1 = \lambda_2 = -3$ and $\lambda_3 = 5$. For $\lambda_1 = \lambda_2 = -3$ we have

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & 0 \\ 2 & 4 & -6 & 0 \\ -1 & -2 & 3 & 0 \end{array} \right] \xrightarrow[\text{operations}]{\text{row}} \left[\begin{array}{ccc|c} 1 & 2 & -3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

Thus $\mathbf{K}_1 = [-2 \ 1 \ 0]^T$ and $\mathbf{K}_2 = [3 \ 0 \ 1]^T$. For $\lambda_3 = 5$ we have

$$\left[\begin{array}{ccc|c} -7 & 2 & -3 & 0 \\ 2 & -4 & -6 & 0 \\ -1 & -2 & -5 & 0 \end{array} \right] \xrightarrow[\text{operations}]{\text{row}} \left[\begin{array}{ccc|c} 1 & -\frac{2}{7} & \frac{3}{7} & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

Thus $\mathbf{K}_3 = [-1 \ -2 \ 1]^T$.

46. From the characteristic equation $-\lambda^3 + \lambda^2 + 2\lambda = -\lambda(\lambda + 1)(\lambda - 2) = 0$ we see that the eigenvalues are $\lambda_1 = 0$, $\lambda_2 = -1$, and $\lambda_3 = 2$. For $\lambda_1 = 0$ we have $k_3 = 0$, $2k_1 + 2k_2 + k_3 = 0$ and $\mathbf{K}_1 = [1 \ -1 \ 0]^T$. For $\lambda_2 = -1$ we have

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 \end{array} \right] \xrightarrow[\text{operations}]{\text{row}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

Thus $\mathbf{K}_2 = [0 \ 1 \ -1]^T$. For $\lambda_3 = 2$ we have

$$\left[\begin{array}{ccc|c} -2 & 0 & 0 & 0 \\ 0 & -2 & 1 & 0 \\ 2 & 2 & -1 & 0 \end{array} \right] \xrightarrow[\text{operations}]{\text{row}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

Thus $\mathbf{K}_3 = [0 \ 1 \ 2]^T$.

47. Let $\mathbf{X}_1 = [a \ b \ c]^T$ be the first column of the matrix. Then $\mathbf{X}_1^T [-\frac{1}{\sqrt{2}} \ 0 \ \frac{1}{\sqrt{2}}]^T = \frac{1}{\sqrt{2}}(c - a) = 0$ and $\mathbf{X}_1^T [\frac{1}{\sqrt{3}} \ \frac{1}{\sqrt{3}} \ \frac{1}{\sqrt{3}}]^T = \frac{1}{\sqrt{3}}(a + b + c) = 0$. Also $\mathbf{X}_1^T \mathbf{X}_1 = a^2 + b^2 + c^2 = 1$. We see that $c = a$ and $b = -2a$ from the first two equations. Then $a^2 + 4a^2 + a^2 = 6a^2 = 1$ and $a = \frac{1}{\sqrt{6}}$. Thus $\mathbf{X}_1 = [\frac{1}{\sqrt{6}} \ -\frac{2}{\sqrt{6}} \ \frac{1}{\sqrt{6}}]^T$.
48. (a) Eigenvalues are $\lambda_1 = \lambda_2 = 0$ and $\lambda_3 = 5$ with corresponding eigenvectors $\mathbf{K}_1 = [0 \ 1 \ 0]^T$, $\mathbf{K}_2 = [2 \ 0 \ 1]^T$, and $\mathbf{K}_3 = [-1 \ 0 \ 2]^T$. Since $\|\mathbf{K}_1\| = 1$, $\|\mathbf{K}_2\| = \sqrt{5}$, and $\|\mathbf{K}_3\| = \sqrt{5}$, we have

$$\mathbf{P} = \begin{bmatrix} 0 & \frac{2}{\sqrt{5}} & -\frac{1}{\sqrt{5}} \\ 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{bmatrix} \quad \text{and} \quad \mathbf{P}^{-1} = \mathbf{P}^T = \begin{bmatrix} 0 & 1 & 0 \\ \frac{2}{\sqrt{5}} & 0 & \frac{1}{\sqrt{5}} \\ -\frac{1}{\sqrt{5}} & 0 & \frac{2}{\sqrt{5}} \end{bmatrix}.$$

$$(b) \mathbf{P}^{-1} \mathbf{A} \mathbf{P} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

49. We identify $\mathbf{A} = \begin{bmatrix} 1 & \frac{3}{2} \\ \frac{3}{2} & 1 \end{bmatrix}$. Eigenvalues are $\lambda_1 = -\frac{1}{2}$ and $\lambda_2 = \frac{5}{2}$ so $\mathbf{D} = \begin{bmatrix} -\frac{1}{2} & 0 \\ 0 & \frac{5}{2} \end{bmatrix}$ and the equation becomes $[X \ Y] \mathbf{D} \begin{bmatrix} X \\ Y \end{bmatrix} = -\frac{1}{2}X^2 + \frac{5}{2}Y^2 = 1$. The graph is a hyperbola.

50. We measure years in units of 10, with 0 corresponding to 1890. Then $\mathbf{Y} = [63 \ 76 \ 92 \ 106 \ 123]^T$ and $\mathbf{A} = \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}^T$, so $\mathbf{A}^T \mathbf{A} = \begin{bmatrix} 30 & 10 \\ 10 & 5 \end{bmatrix}$. Thus

$$\mathbf{X} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{Y} = \frac{1}{50} \begin{bmatrix} 5 & -10 \\ -10 & 30 \end{bmatrix} \mathbf{A}^T \mathbf{Y} = \begin{bmatrix} 15 \\ 62 \end{bmatrix},$$

and the least squares line is $y = 15t + 62$. At $t = 5$ (corresponding to 1940) we have $y = 137$. The error in the predicted population is 5 million or 3.7%.

51. The encoded message is

$$\begin{aligned} \mathbf{B} = \mathbf{A} \mathbf{M} &= \begin{bmatrix} 10 & 1 \\ 9 & 1 \end{bmatrix} \begin{bmatrix} 19 & 1 & 20 & 5 & 12 & 12 & 9 & 20 & 5 & 0 & 12 & 1 & 21 \\ 14 & 3 & 8 & 5 & 4 & 0 & 15 & 14 & 0 & 6 & 18 & 9 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 204 & 13 & 208 & 55 & 124 & 120 & 105 & 214 & 50 & 6 & 138 & 19 & 210 \\ 185 & 12 & 188 & 50 & 112 & 108 & 96 & 194 & 45 & 6 & 126 & 18 & 189 \end{bmatrix}. \end{aligned}$$

Chapter 8 Review Exercises

52. The encoded message is

$$\begin{aligned}\mathbf{B} = \mathbf{A}\mathbf{M} &= \begin{bmatrix} 10 & 1 \\ 9 & 1 \end{bmatrix} \begin{bmatrix} 19 & 5 & 3 & 0 & 1 & 7 & 14 & 20 & 0 & 1 & 18 \\ 18 & 22 & 19 & 0 & 20 & 21 & 5 & 19 & 0 & 1 & 13 \end{bmatrix} \\ &= \begin{bmatrix} 208 & 72 & 49 & 0 & 30 & 91 & 145 & 219 & 0 & 11 & 193 \\ 189 & 67 & 46 & 0 & 29 & 84 & 131 & 199 & 0 & 10 & 175 \end{bmatrix}.\end{aligned}$$

53. The decoded message is

$$\mathbf{M} = \mathbf{A}^{-1}\mathbf{B} = \begin{bmatrix} -3 & 2 & -1 \\ 1 & 0 & 0 \\ 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} 19 & 0 & 15 & 14 & 0 & 20 \\ 35 & 10 & 27 & 53 & 1 & 54 \\ 5 & 15 & -3 & 48 & 2 & 39 \end{bmatrix} = \begin{bmatrix} 8 & 5 & 12 & 16 & 0 & 9 \\ 19 & 0 & 15 & 14 & 0 & 20 \\ 8 & 5 & 0 & 23 & 1 & 25 \end{bmatrix}.$$

From correspondence (1) we obtain: HELP_IS_ON_THE_WAY.

54. The decoded message is

$$\mathbf{M} = \mathbf{A}^{-1}\mathbf{B} = \begin{bmatrix} -3 & 2 & -1 \\ 1 & 0 & 0 \\ 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} 5 & 2 & 21 \\ 27 & 17 & 40 \\ 21 & 13 & -2 \end{bmatrix} = \begin{bmatrix} 18 & 15 & 19 \\ 5 & 2 & 21 \\ 4 & 0 & 0 \end{bmatrix}.$$

From correspondence (1) we obtain: ROSEBUD__.

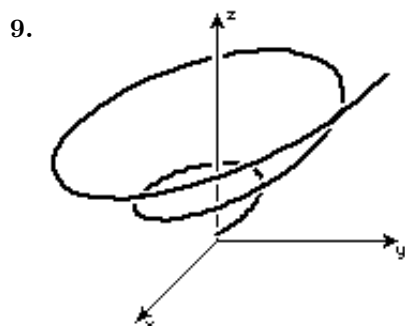
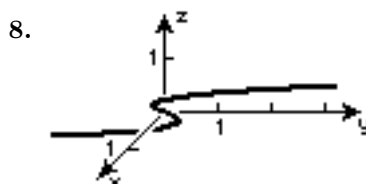
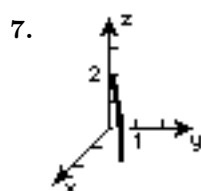
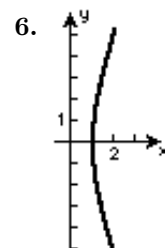
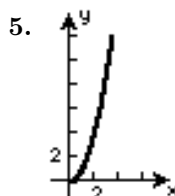
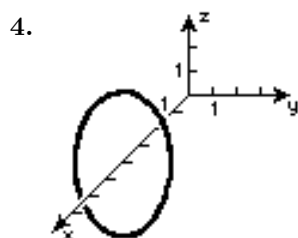
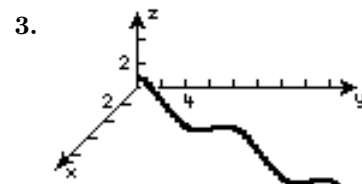
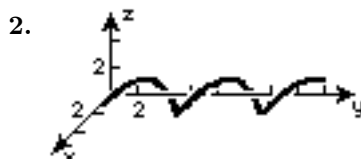
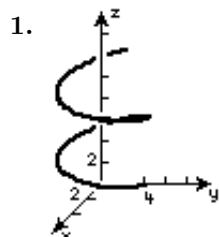
55. (a) The parity is even so the decoded message is $[1 \ 1 \ 0 \ 0 \ 1]$

(b) The parity is odd; there is a parity error.

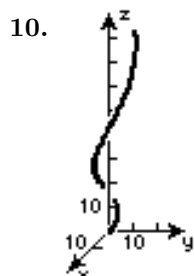
56. From $\begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ we obtain the codeword $[0 \ 0 \ 1 \ 1 \ 0 \ 0 \ 1]$.

9 Vector Calculus

Exercises 9.1

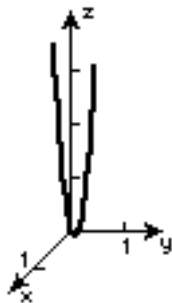


Note: the scale is distorted in this graph. For $t = 0$, the graph starts at $(1,0,1)$. The upper loop shown intersects the xz -plane at about $(286751, 0, 286751)$.

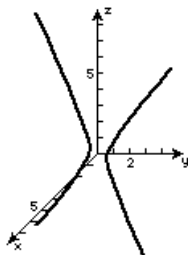


Exercises 9.1

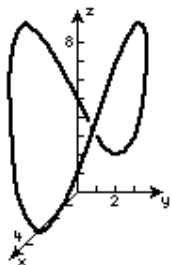
11. $x = t, y = t, z = t^2 + t^2 = 2t^2; \mathbf{r}(t) = t\mathbf{i} + t\mathbf{j} + 2t^2\mathbf{k}$



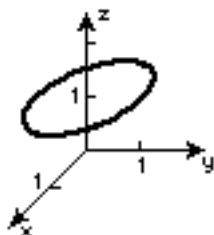
12. $x = t, y = 2t, z = \pm\sqrt{t^2 + 4t^2 + 1} = \pm\sqrt{5t^2 + 1}; \mathbf{r}(t) = t\mathbf{i} + 2t\mathbf{j} \pm \sqrt{5t^2 + 1}\mathbf{k}$



13. $x = 3\cos t, z = 9 - 9\cos^2 t = 9\sin^2 t, y = 3\sin t; \mathbf{r}(t) = 3\cos t\mathbf{i} + 3\sin t\mathbf{j} + 9\sin^2 t\mathbf{k}$



14. $x = \sin t, z = 1, y = \cos t; \mathbf{r}(t) = \sin t\mathbf{i} + \cos t\mathbf{j} + \mathbf{k}$



15. $\mathbf{r}(t) = \frac{\sin 2t}{t}\mathbf{i} + (t-2)^5\mathbf{j} + \frac{\ln t}{1/t}\mathbf{k}$. Using L'Hôpital's Rule,

$$\lim_{t \rightarrow 0^+} \mathbf{r}(t) = \left[\frac{2\cos 2t}{1}\mathbf{i} + (t-2)^5\mathbf{j} + \frac{1/t}{-1/t^2}\mathbf{k} \right] = 2\mathbf{i} - 32\mathbf{j}.$$

16. (a) $\lim_{t \rightarrow \alpha} [-4\mathbf{r}_1(t) + 3\mathbf{r}_2(t)] = -4(\mathbf{i} - 2\mathbf{j} + \mathbf{k}) + 3(2\mathbf{i} + 5\mathbf{j} + 7\mathbf{k}) = 2\mathbf{i} + 23\mathbf{j} + 17\mathbf{k}$

(b) $\lim_{t \rightarrow \alpha} \mathbf{r}_1(t) \cdot \mathbf{r}_2(t) = (\mathbf{i} - 2\mathbf{j} + \mathbf{k}) \cdot (2\mathbf{i} + 5\mathbf{j} + 7\mathbf{k}) = -1$

17. $\mathbf{r}'(t) = \frac{1}{t}\mathbf{i} - \frac{1}{t^2}\mathbf{j}; \mathbf{r}''(t) = -\frac{1}{t^2}\mathbf{i} + \frac{2}{t^3}\mathbf{j}$

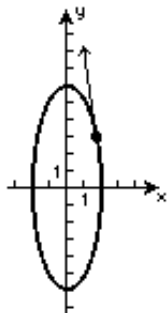
18. $\mathbf{r}'(t) = \langle -t \sin t, 1 - \sin t \rangle; \mathbf{r}''(t) = \langle -t \cos t - \sin t, -\cos t \rangle$

19. $\mathbf{r}'(t) = \langle 2te^{2t} + e^{2t}, 3t^2, 8t - 1 \rangle$; $\mathbf{r}''(t) = \langle 4te^{2t} + 4e^{2t}, 6t, 8 \rangle$

20. $\mathbf{r}'(t) = 2t\mathbf{i} + 3t^2\mathbf{j} + \frac{1}{1+t^2}\mathbf{k}$; $\mathbf{r}''(t) = 2\mathbf{i} + 6t\mathbf{j} - \frac{2t}{(1+t^2)^2}\mathbf{k}$

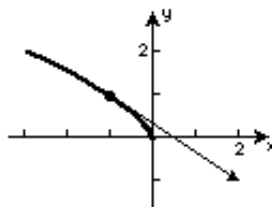
21. $\mathbf{r}'(t) = -2\sin t\mathbf{i} + 6\cos t\mathbf{j}$

$\mathbf{r}'(\pi/6) = -\mathbf{i} + 3\sqrt{3}\mathbf{j}$



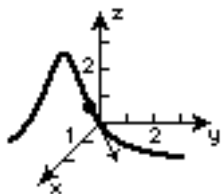
22. $\mathbf{r}'(t) = 3t^2\mathbf{i} + 2t\mathbf{j}$

$\mathbf{r}'(-1) = 3\mathbf{i} - 2\mathbf{j}$



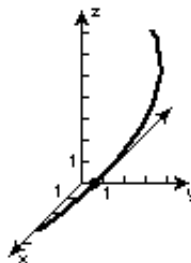
23. $\mathbf{r}'(t) = \mathbf{j} - \frac{8t}{(1+t^2)^2}\mathbf{k}$

$\mathbf{r}'(1) = \mathbf{j} - 2\mathbf{k}$



24. $\mathbf{r}'(t) = -3\sin t\mathbf{i} + 3\cos t\mathbf{j} + 2\mathbf{k}$

$\mathbf{r}'(\pi/4) = \frac{-3\sqrt{2}}{2}\mathbf{i} + \frac{3\sqrt{2}}{2}\mathbf{j} + 2\mathbf{k}$



25. $\mathbf{r}(t) = t\mathbf{i} + \frac{1}{2}t^2\mathbf{j} + \frac{1}{3}t^3\mathbf{k}$; $\mathbf{r}(2) = 2\mathbf{i} + 2\mathbf{j} + \frac{8}{3}\mathbf{k}$; $\mathbf{r}'(t) = \mathbf{i} + t\mathbf{j} + t^2\mathbf{k}$; $\mathbf{r}'(2) = \mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$

Using the point $(2, 2, 8/3)$ and the direction vector $\mathbf{r}'(2)$, we have $x = 2 + t$, $y = 2 + 2t$, $z = 8/3 + 4t$.

26. $\mathbf{r}(t) = (t^3 - t)\mathbf{i} + \frac{6t}{t+1}\mathbf{j} + (2t+1)^2\mathbf{k}$; $\mathbf{r}(1) = 3\mathbf{j} + 9\mathbf{k}$; $\mathbf{r}'(t) = (3t^2 - 1)\mathbf{i} + \frac{6}{(t+1)^2}\mathbf{j} + (8t+4)\mathbf{k}$; $\mathbf{r}'(1) = 2\mathbf{i} + \frac{3}{2}\mathbf{j} + 12\mathbf{k}$.

Using the point $(0, 3, 9)$ and the direction vector $\mathbf{r}'(1)$, we have $x = 2t$, $y = 3 + \frac{3}{2}t$, $z = 9 + 12t$.

27. $\frac{d}{dt}[\mathbf{r}(t) \times \mathbf{r}'(t)] = \mathbf{r}(t) \times \mathbf{r}''(t) + \mathbf{r}'(t) \times \mathbf{r}'(t) = \mathbf{r}(t) \times \mathbf{r}''(t)$

28. $\frac{d}{dt}[\mathbf{r}(t) \cdot (t\mathbf{r}(t))] = \mathbf{r}(t) \cdot \frac{d}{dt}(t\mathbf{r}(t)) + \mathbf{r}'(t) \cdot (t\mathbf{r}(t)) = \mathbf{r}(t) \cdot (t\mathbf{r}'(t) + \mathbf{r}(t)) + \mathbf{r}'(t) \cdot (t\mathbf{r}(t))$
 $= \mathbf{r}(t) \cdot (t\mathbf{r}'(t)) + \mathbf{r}(t) \cdot \mathbf{r}(t) + \mathbf{r}'(t) \cdot (t\mathbf{r}(t)) = 2t(\mathbf{r}(t) \cdot \mathbf{r}'(t)) + \mathbf{r}(t) \cdot \mathbf{r}(t)$

29. $\frac{d}{dt}[\mathbf{r}(t) \cdot (\mathbf{r}'(t) \times \mathbf{r}''(t))] = \mathbf{r}(t) \cdot \frac{d}{dt}(\mathbf{r}'(t) \times \mathbf{r}''(t)) + \mathbf{r}'(t) \cdot (\mathbf{r}'(t) \times \mathbf{r}''(t))$
 $= \mathbf{r}(t) \cdot (\mathbf{r}'(t) \times \mathbf{r}'''(t) + \mathbf{r}''(t) \times \mathbf{r}''(t)) + \mathbf{r}'(t) \cdot (\mathbf{r}'(t) \times \mathbf{r}''(t))$
 $= \mathbf{r}(t) \cdot (\mathbf{r}'(t) \times \mathbf{r}'''(t))$

30. $\frac{d}{dt}[\mathbf{r}_1(t) \times (\mathbf{r}_2(t) \times \mathbf{r}_3(t))] = \mathbf{r}_1(t) \times \frac{d}{dt}(\mathbf{r}_2(t) \times \mathbf{r}_3(t)) + \mathbf{r}_1'(t) \times (\mathbf{r}_2(t) \times \mathbf{r}_3(t))$
 $= \mathbf{r}_1(t) \times (\mathbf{r}_2(t) \times \mathbf{r}_3'(t) + \mathbf{r}_2'(t) \times \mathbf{r}_3(t)) + \mathbf{r}_1'(t) \times (\mathbf{r}_2(t) \times \mathbf{r}_3(t))$
 $= \mathbf{r}_1(t) \times (\mathbf{r}_2(t) \times \mathbf{r}_3'(t)) + \mathbf{r}_1(t) \times (\mathbf{r}_2'(t) \times \mathbf{r}_3(t)) + \mathbf{r}_1'(t) \times (\mathbf{r}_2(t) \times \mathbf{r}_3(t))$

31. $\frac{d}{dt}\left[\mathbf{r}_1(2t) + \mathbf{r}_2\left(\frac{1}{t}\right)\right] = 2\mathbf{r}_1'(2t) - \frac{1}{t^2}\mathbf{r}_2'\left(\frac{1}{t}\right)$

Exercises 9.1

$$32. \frac{d}{dt}[t^3 \mathbf{r}(t^2)] = t^3(2t)\mathbf{r}'(t^2) + 3t^2\mathbf{r}(t^2) = 2t^4\mathbf{r}'(t^2) + 3t^2\mathbf{r}(t^2)$$

$$33. \int_{-1}^2 \mathbf{r}(t) dt = \left[\int_{-1}^2 t dt \right] \mathbf{i} + \left[\int_{-1}^2 3t^2 dt \right] \mathbf{j} + \left[\int_{-1}^2 4t^3 dt \right] \mathbf{k} = \frac{1}{2}t^2 \Big|_{-1}^2 \mathbf{i} + t^3 \Big|_{-1}^2 \mathbf{j} + t^4 \Big|_{-1}^2 \mathbf{k} = \frac{3}{2}\mathbf{i} + 9\mathbf{j} + 15\mathbf{k}$$

$$34. \int_0^4 \mathbf{r}(t) dt = \left[\int_0^4 \sqrt{2t+1} dt \right] \mathbf{i} + \left[\int_0^4 -\sqrt{t} dt \right] \mathbf{j} + \left[\int_0^4 \sin \pi t dt \right] \mathbf{k} \\ = \frac{1}{3}(2t+1)^{3/2} \Big|_0^4 \mathbf{i} - \frac{2}{3}t^{3/2} \Big|_0^4 \mathbf{j} - \frac{1}{\pi} \cos \pi t \Big|_0^4 \mathbf{k} = \frac{26}{3}\mathbf{i} - \frac{16}{3}\mathbf{j}$$

$$35. \int \mathbf{r}(t) dt = \left[\int te^t dt \right] \mathbf{i} + \left[\int -e^{-2t} dt \right] \mathbf{j} + \left[\int te^{t^2} dt \right] \mathbf{k} \\ = [te^t - e^t + c_1]\mathbf{i} + \left[\frac{1}{2}e^{-2t} + c_2 \right] \mathbf{j} + \left[\frac{1}{2}e^{t^2} + c_3 \right] \mathbf{k} = e^t(t-1)\mathbf{i} + \frac{1}{2}e^{-2t}\mathbf{j} + \frac{1}{2}e^{t^2}\mathbf{k} + \mathbf{c},$$

where $\mathbf{c} = c_1\mathbf{i} + c_2\mathbf{j} + c_3\mathbf{k}$.

$$36. \int \mathbf{r}(t) dt = \left[\int \frac{1}{1+t^2} dt \right] \mathbf{i} + \left[\int \frac{t}{1+t^2} dt \right] \mathbf{j} + \left[\int \frac{t^2}{1+t^2} dt \right] \mathbf{k} \\ = [\tan^{-1} t + c_1]\mathbf{i} + \left[\frac{1}{2} \ln(1+t^2) + c_2 \right] \mathbf{j} + \left[\int \left(1 - \frac{1}{1+t^2} \right) dt \right] \mathbf{k} \\ = [\tan^{-1} t + c_1]\mathbf{i} + \left[\frac{1}{2} \ln(1+t^2) + c_2 \right] \mathbf{j} + [t - \tan^{-1} t + c_3]\mathbf{k} \\ = \tan^{-1} t \mathbf{i} + \frac{1}{2} \ln(1+t^2)\mathbf{j} + (t - \tan^{-1} t)\mathbf{k} + \mathbf{c},$$

where $\mathbf{c} = c_1\mathbf{i} + c_2\mathbf{j} + c_3\mathbf{k}$.

$$37. \mathbf{r}(t) = \int \mathbf{r}'(t) dt = \left[\int 6 dt \right] \mathbf{i} + \left[\int 6t dt \right] \mathbf{j} + \left[\int 3t^2 dt \right] \mathbf{k} = [6t + c_1]\mathbf{i} + [3t^2 + c_2]\mathbf{j} + [t^3 + c_3]\mathbf{k}$$

Since $\mathbf{r}(0) = \mathbf{i} - 2\mathbf{j} + \mathbf{k} = c_1\mathbf{i} + c_2\mathbf{j} + c_3\mathbf{k}$, $c_1 = 1$, $c_2 = -2$, and $c_3 = 1$. Thus,

$$\mathbf{r}(t) = (6t + 1)\mathbf{i} + (3t^2 - 2)\mathbf{j} + (t^3 + 1)\mathbf{k}.$$

$$38. \mathbf{r}(t) = \int \mathbf{r}'(t) dt = \left[\int t \sin t^2 dt \right] \mathbf{i} + \left[\int -\cos 2t dt \right] \mathbf{j} = -\left[\frac{1}{2} \cos t^2 + c_1 \right] \mathbf{i} + \left[-\frac{1}{2} \sin 2t + c_2 \right] \mathbf{j}$$

Since $\mathbf{r}(0) = \frac{3}{2}\mathbf{i} = (-\frac{1}{2} + c_1)\mathbf{i} + c_2\mathbf{j}$, $c_1 = 2$ and $c_2 = 0$. Thus,

$$\mathbf{r}(t) = \left(-\frac{1}{2} \cos t^2 + 2 \right) \mathbf{i} - \frac{1}{2} \sin 2t \mathbf{j}.$$

$$39. \mathbf{r}'(t) = \int \mathbf{r}''(t) dt = \left[\int 12t dt \right] \mathbf{i} + \left[\int -3t^{-1/2} dt \right] \mathbf{j} + \left[\int 2 dt \right] \mathbf{k} = [6t^2 + c_1]\mathbf{i} + [-6t^{1/2} + c_2]\mathbf{j} + [2t + c_3]\mathbf{k}$$

Since $\mathbf{r}'(1) = \mathbf{j} = (6 + c_1)\mathbf{i} + (-6 + c_2)\mathbf{j} + (2 + c_3)\mathbf{k}$, $c_1 = -6$, $c_2 = 7$, and $c_3 = -2$. Thus,

$$\mathbf{r}'(t) = (6t^2 - 6)\mathbf{i} + (-6t^{1/2} + 7)\mathbf{j} + (2t - 2)\mathbf{k}.$$

$$\mathbf{r}(t) = \int \mathbf{r}'(t) dt = \left[\int (6t^2 - 6) dt \right] \mathbf{i} + \left[\int (-6t^{1/2} + 7) dt \right] \mathbf{j} + \left[\int (2t - 2) dt \right] \mathbf{k} \\ = [2t^3 - 6t + c_4]\mathbf{i} + [-4t^{3/2} + 7t + c_5]\mathbf{j} + [t^2 - 2t + c_6]\mathbf{k}.$$

Since

$$\mathbf{r}(1) = 2\mathbf{i} - \mathbf{k} = (-4 + c_4)\mathbf{i} + (3 + c_5)\mathbf{j} + (-1 + c_6)\mathbf{k},$$

$c_4 = 6$, $c_5 = -3$, and $c_6 = 0$. Thus,

$$\mathbf{r}(t) = (2t^3 - 6t + 6)\mathbf{i} + (-4t^{3/2} + 7t - 3)\mathbf{j} + (t^2 - 2t)\mathbf{k}.$$

$$\begin{aligned}
 40. \quad \mathbf{r}'(t) &= \int \mathbf{r}''(t) dt = \left[\int \sec^2 t dt \right] \mathbf{i} + \left[\int \cos t dt \right] \mathbf{j} + \left[\int -\sin t dt \right] \mathbf{k} \\
 &= [\tan t + c_1] \mathbf{i} + [\sin t + c_2] \mathbf{j} + [\cos t + c_3] \mathbf{k}
 \end{aligned}$$

Since $\mathbf{r}'(0) = \mathbf{i} + \mathbf{j} + \mathbf{k} = c_1 \mathbf{i} + c_2 \mathbf{j} + (1 + c_3) \mathbf{k}$, $c_1 = 1$, $c_2 = 1$, and $c_3 = 0$. Thus,

$$\mathbf{r}'(t) = (\tan t + 1) \mathbf{i} + (\sin t + 1) \mathbf{j} + \cos t \mathbf{k}.$$

$$\begin{aligned}
 \mathbf{r}(t) &= \int \mathbf{r}'(t) dt = \left[\int (\tan t + 1) dt \right] \mathbf{i} + \left[\int (\sin t + 1) dt \right] \mathbf{j} + \left[\int \cos t dt \right] \mathbf{k} \\
 &= [\ln |\sec t| + t + c_4] \mathbf{i} + [-\cos t + t + c_5] \mathbf{j} + [\sin t + c_6] \mathbf{k}.
 \end{aligned}$$

Since $\mathbf{r}(0) = -\mathbf{j} + 5\mathbf{k} = c_4 \mathbf{i} + (-1 + c_5) \mathbf{j} + c_6 \mathbf{k}$, $c_4 = 0$, $c_5 = 0$, and $c_6 = 5$. Thus,

$$\mathbf{r}(t) = (\ln |\sec t| + t) \mathbf{i} + (-\cos t + t) \mathbf{j} + (\sin t + 5) \mathbf{k}.$$

$$41. \quad \mathbf{r}'(t) = -a \sin t \mathbf{i} + a \cos t \mathbf{j} + c \mathbf{k}; \quad \|\mathbf{r}'(t)\| = \sqrt{(-a \sin t)^2 + (a \cos t)^2 + c^2} = \sqrt{a^2 + c^2}$$

$$s = \int_0^{2\pi} \sqrt{a^2 + c^2} dt = \sqrt{a^2 + c^2} t \Big|_0^{2\pi} = 2\pi \sqrt{a^2 + c^2}$$

$$42. \quad \mathbf{r}'(t) = \mathbf{i} + (\cos t - t \sin t) \mathbf{j} + (\sin t + t \cos t) \mathbf{k}$$

$$\|\mathbf{r}'(t)\| = \sqrt{1^2 + (\cos t - t \sin t)^2 + (\sin t + t \cos t)^2} = \sqrt{2 + t^2}$$

$$s = \int_0^\pi \sqrt{2 + t^2} dt = \left(\frac{t}{2} \sqrt{2 + t^2} + \ln |t + \sqrt{2 + t^2}| \right) \Big|_0^\pi = \frac{\pi}{2} \sqrt{2 + \pi^2} + \ln(\pi + \sqrt{2 + \pi^2}) - \ln \sqrt{2}$$

$$43. \quad \mathbf{r}'(t) = (-2e^t \sin 2t + e^t \cos 2t) \mathbf{i} + (2e^t \cos 2t + e^t \sin 2t) \mathbf{j} + e^t \mathbf{k}$$

$$\|\mathbf{r}'(t)\| = \sqrt{5e^{2t} \cos^2 2t + 5e^{2t} \sin^2 2t + e^{2t}} = \sqrt{6e^{2t}} = \sqrt{6} e^t$$

$$s = \int_0^{3\pi} \sqrt{6} e^t dt = \sqrt{6} e^t \Big|_0^{3\pi} = \sqrt{6} (e^{3\pi} - 1)$$

$$44. \quad \mathbf{r}'(t) = 3\mathbf{i} + 2\sqrt{3}t \mathbf{j} + 2t^2 \mathbf{k}; \quad \|\mathbf{r}'(t)\| = \sqrt{3^2 + (2\sqrt{3}t)^2 + (2t^2)^2} = \sqrt{9 + 12t^2 + 4t^4} = 3 + 2t^2$$

$$s = \int_0^1 (3 + 2t^2) dt = \left(3t + \frac{2}{3}t^3 \right) \Big|_0^1 = 3 + \frac{2}{3} = \frac{11}{3}$$

$$45. \quad \mathbf{r}'(t) = -a \sin t \mathbf{i} + a \cos t \mathbf{j}; \quad \|\mathbf{r}'(t)\| = \sqrt{a^2 \sin^2 t + a^2 \cos^2 t} = a, \quad a > 0; \quad s = \int_0^t a du = at$$

$$\mathbf{r}(s) = a \cos(s/a) \mathbf{i} + a \sin(s/a) \mathbf{j}; \quad \mathbf{r}'(s) = -\sin(s/a) \mathbf{i} + \cos(s/a) \mathbf{j}$$

$$\|\mathbf{r}'(s)\| = \sqrt{\sin^2(s/a) + \cos^2(s/a)} = 1$$

$$46. \quad \mathbf{r}'(s) = -\frac{2}{\sqrt{5}} \sin(s/\sqrt{5}) \mathbf{i} + \frac{2}{\sqrt{5}} \cos(s/\sqrt{5}) \mathbf{j} + \frac{1}{\sqrt{5}} \mathbf{k}$$

$$\|\mathbf{r}'(s)\| = \sqrt{\frac{4}{5} \sin^2(s/\sqrt{5}) + \frac{4}{5} \cos^2(s/\sqrt{5}) + \frac{1}{5}} = \sqrt{\frac{4}{5} + \frac{1}{5}} = 1$$

$$47. \quad \text{Since } \frac{d}{dt}(\mathbf{r} \cdot \mathbf{r}) = \frac{d}{dt} \|\mathbf{r}\|^2 = \frac{d}{dt} c^2 = 0 \text{ and } \frac{d}{dt}(\mathbf{r} \cdot \mathbf{r}) = \mathbf{r} \cdot \mathbf{r}' + \mathbf{r}' \cdot \mathbf{r} = 2\mathbf{r} \cdot \mathbf{r}', \text{ we have } \mathbf{r} \cdot \mathbf{r}' = 0. \text{ Thus,}$$

$\mathbf{r}'$ is perpendicular to $\mathbf{r}$.

$$48. \quad \text{Since } \|\mathbf{r}(t)\| \text{ is the length of } \mathbf{r}(t), \|\mathbf{r}(t)\| = c \text{ represents a curve lying on a sphere of radius } c \text{ centered at the origin.}$$

Exercises 9.1

49. Let $\mathbf{r}_1(t) = x(t)\mathbf{i} + y(t)\mathbf{j}$. Then

$$\begin{aligned}\frac{d}{dt}[u(t)\mathbf{r}_1(t)] &= \frac{d}{dt}[u(t)x(t)\mathbf{i} + u(t)y(t)\mathbf{j}] = [u(t)x'(t) + u'(t)x(t)]\mathbf{i} + [u(t)y'(t) + u'(t)y(t)]\mathbf{j} \\ &= u(t)[x'(t)\mathbf{i} + y'(t)\mathbf{j}] + u'(t)[x(t)\mathbf{i} + y(t)\mathbf{j}] = u(t)\mathbf{r}_1'(t) + u'(t)\mathbf{r}_1(t).\end{aligned}$$

50. Let $\mathbf{r}_1(t) = x_1(t)\mathbf{i} + y_1(t)\mathbf{j}$ and $\mathbf{r}_2(t) = x_2(t)\mathbf{i} + y_2(t)\mathbf{j}$. Then

$$\begin{aligned}\frac{d}{dt}[\mathbf{r}_1(t) \cdot \mathbf{r}_2(t)] &= \frac{d}{dt}[x_1(t)x_2(t) + y_1(t)y_2(t)] = x_1(t)x_2'(t) + x_1'(t)x_2(t) + y_1(t)y_2'(t) + y_1'(t)y_2(t) \\ &= [x_1(t)x_2'(t) + y_1(t)y_2'(t)] + [x_1'(t)x_2(t) + y_1'(t)y_2(t)] = \mathbf{r}_1(t) \cdot \mathbf{r}_2'(t) + \mathbf{r}_1'(t) \cdot \mathbf{r}_2(t).\end{aligned}$$

$$\begin{aligned}51. \quad \frac{d}{dt}[\mathbf{r}_1(t) \times \mathbf{r}_2(t)] &= \lim_{h \rightarrow 0} \frac{\mathbf{r}_1(t+h) \times \mathbf{r}_2(t+h) - \mathbf{r}_1(t) \times \mathbf{r}_2(t)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\mathbf{r}_1(t+h) \times \mathbf{r}_2(t+h) - \mathbf{r}_1(t+h) \times \mathbf{r}_2(t) + \mathbf{r}_1(t+h) \times \mathbf{r}_2(t) - \mathbf{r}_1(t) \times \mathbf{r}_2(t)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\mathbf{r}_1(t+h) \times [\mathbf{r}_2(t+h) - \mathbf{r}_2(t)]}{h} + \lim_{h \rightarrow 0} \frac{[\mathbf{r}_1(t+h) - \mathbf{r}_1(t)] \times \mathbf{r}_2(t)}{h} \\ &= \mathbf{r}_1(t) \times \left(\lim_{h \rightarrow 0} \frac{\mathbf{r}_2(t+h) - \mathbf{r}_2(t)}{h} \right) + \left(\lim_{h \rightarrow 0} \frac{\mathbf{r}_1(t+h) - \mathbf{r}_1(t)}{h} \right) \times \mathbf{r}_2(t) \\ &= \mathbf{r}_1(t) \times \mathbf{r}_2'(t) + \mathbf{r}_1'(t) \times \mathbf{r}_2(t)\end{aligned}$$

52. Let $\mathbf{v} = a\mathbf{i} + b\mathbf{j}$ and $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}$. Then

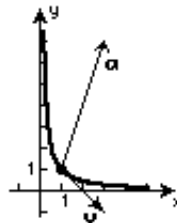
$$\int_a^b \mathbf{v} \cdot \mathbf{r}(t) dt = \int_a^b [ax(t) + by(t)] dt = a \int_a^b x(t) dt + b \int_a^b y(t) dt = \mathbf{v} \cdot \int_a^b \mathbf{r}(t) dt.$$

Exercises 9.2

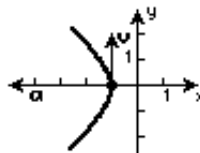
1. $\mathbf{v}(t) = 2t\mathbf{i} + t^3\mathbf{j}$; $\mathbf{v}(1) = 2\mathbf{i} + \mathbf{j}$; $\|\mathbf{v}(1)\| = \sqrt{4+1} = \sqrt{5}$;
 $\mathbf{a}(t) = 2\mathbf{i} + 3t^2\mathbf{j}$; $\mathbf{a}(1) = 2\mathbf{i} + 3\mathbf{j}$



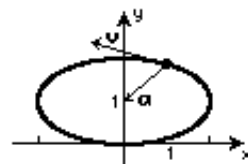
2. $\mathbf{v}(t) = 2t\mathbf{i} - \frac{2}{t^3}\mathbf{j}$; $\mathbf{v}(1) = 2\mathbf{i} - 2\mathbf{j}$; $\|\mathbf{v}(1)\| = \sqrt{4+4} = 2\sqrt{2}$;
 $\mathbf{a}(t) = 2\mathbf{i} + \frac{6}{t^4}\mathbf{j}$; $\mathbf{a}(1) = 2\mathbf{i} + 6\mathbf{j}$



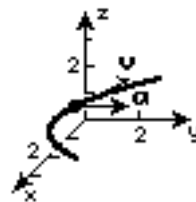
3. $\mathbf{v}(t) = -2\sinh 2t\mathbf{i} + 2\cosh 2t\mathbf{j}$; $\mathbf{v}(0) = 2\mathbf{j}$; $\|\mathbf{v}(0)\| = 2$;
 $\mathbf{a}(t) = -4\cosh 2t\mathbf{i} + 4\sinh 2t\mathbf{j}$; $\mathbf{a}(0) = -4\mathbf{i}$



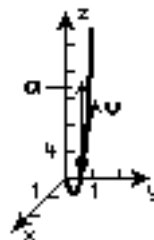
4. $\mathbf{v}(t) = -2\sin t\mathbf{i} + \cos t\mathbf{j}$; $\mathbf{v}(\pi/3) = -\sqrt{3}\mathbf{i} + \frac{1}{2}\mathbf{j}$; $\|\mathbf{v}(\pi/3)\| = \sqrt{3+1/4} = \sqrt{13}/2$;
 $\mathbf{a}(t) = -2\cos t\mathbf{i} - \sin t\mathbf{j}$; $\mathbf{a}(\pi/3) = -\mathbf{i} - \frac{\sqrt{3}}{2}\mathbf{j}$



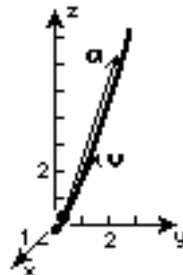
5. $\mathbf{v}(t) = (2t - 2)\mathbf{j} + \mathbf{k}$; $\mathbf{v}(2) = 2\mathbf{j} + \mathbf{k}$ $\|\mathbf{v}(2)\| = \sqrt{4+1} = \sqrt{5}$;
 $\mathbf{a}(t) = 2\mathbf{j}$; $\mathbf{a}(2) = 2\mathbf{j}$



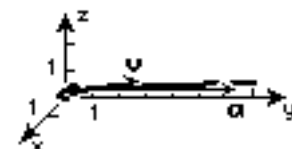
6. $\mathbf{v}(t) = \mathbf{i} + \mathbf{j} + 3t^2\mathbf{k}$; $\mathbf{v}(2) = \mathbf{i} + \mathbf{j} + 12\mathbf{k}$; $\|\mathbf{v}(2)\| = \sqrt{1+1+144} = \sqrt{146}$; $\mathbf{a}(t) = 6t\mathbf{k}$;
 $\mathbf{a}(2) = 12\mathbf{k}$



7. $\mathbf{v}(t) = \mathbf{i} + 2t\mathbf{j} + 3t^2\mathbf{k}$; $\mathbf{v}(1) = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$; $\|\mathbf{v}(1)\| = \sqrt{1+4+9} = \sqrt{14}$;
 $\mathbf{a}(t) = 2\mathbf{j} + 6t\mathbf{k}$; $\mathbf{a}(1) = 2\mathbf{j} + 6\mathbf{k}$



8. $\mathbf{v}(t) = \mathbf{i} + 3t^2\mathbf{j} + \mathbf{k}$; $\mathbf{v}(1) = \mathbf{i} + 3\mathbf{j} + \mathbf{k}$; $\|\mathbf{v}(1)\| = \sqrt{1+9+1} = \sqrt{11}$;
 $\mathbf{a}(t) = 6t\mathbf{j}$; $\mathbf{a}(1) = 6\mathbf{j}$



9. The particle passes through the xy -plane when $z(t) = t^2 - 5t = 0$ or $t = 0, 5$ which gives us the points $(0, 0, 0)$ and $(25, 115, 0)$. $\mathbf{v}(t) = 2t\mathbf{i} + (3t^2 - 2)\mathbf{j} + (2t - 5)\mathbf{k}$; $\mathbf{v}(0) = -2\mathbf{j} - 5\mathbf{k}$, $\mathbf{v}(5) = 10\mathbf{i} + 73\mathbf{j} + 5\mathbf{k}$; $\mathbf{a}(t) = 2\mathbf{i} + 6t\mathbf{j} + 2\mathbf{k}$;
 $\mathbf{a}(0) = 2\mathbf{i} + 2\mathbf{k}$, $\mathbf{a}(5) = 2\mathbf{i} + 30\mathbf{j} + 2\mathbf{k}$
10. If $\mathbf{a}(t) = \mathbf{0}$, then $\mathbf{v}(t) = \mathbf{c}_1$ and $\mathbf{r}(t) = \mathbf{c}_1 t + \mathbf{c}_2$. The graph of this equation is a straight line.
11. Initially we are given $\mathbf{s}_0 = \mathbf{0}$ and $\mathbf{v}_0 = (480 \cos 30^\circ)\mathbf{i} + (480 \sin 30^\circ)\mathbf{j} = 240\sqrt{3}\mathbf{i} + 240\mathbf{j}$. Using $\mathbf{a}(t) = -32\mathbf{j}$ we find

$$\mathbf{v}(t) = \int \mathbf{a}(t) dt = -32t\mathbf{j} + \mathbf{c}$$

$$240\sqrt{3}\mathbf{i} + 240\mathbf{j} = \mathbf{v}(0) = \mathbf{c}$$

$$\mathbf{v}(t) = -32t\mathbf{j} + 240\sqrt{3}\mathbf{i} + 240\mathbf{j} = 240\sqrt{3}\mathbf{i} + (240 - 32t)\mathbf{j}$$

$$\mathbf{r}(t) = \int \mathbf{v}(t) dt = 240\sqrt{3}t\mathbf{i} + (240t - 16t^2)\mathbf{j} + \mathbf{b}$$

$$\mathbf{0} = \mathbf{r}(0) = \mathbf{b}.$$

- (a) The shell's trajectory is given by $\mathbf{r}(t) = 240\sqrt{3}t\mathbf{i} + (240t - 16t^2)\mathbf{j}$ or $x = 240\sqrt{3}t$, $y = 240t - 16t^2$.
- (b) Solving $dy/dt = 240 - 32t = 0$, we see that y is maximum when $t = 15/2$. The maximum altitude is $y(15/2) = 900$ ft.
- (c) Solving $y(t) = 240t - 16t^2 = 16t(15 - t) = 0$, we see that the shell is at ground level when $t = 0$ and $t = 15$. The range of the shell is $x(15) = 3600\sqrt{3} \approx 6235$ ft.

Exercises 9.2

(d) From (c), impact is when $t = 15$. The speed at impact is

$$\|\mathbf{v}(15)\| = |240\sqrt{3}\mathbf{i} + (240 - 32 \cdot 15)\mathbf{j}| = \sqrt{240^2 \cdot 3 + (-240)^2} = 480 \text{ ft/s.}$$

12. Initially we are given $\mathbf{s}_0 = 1600\mathbf{j}$ and $\mathbf{v}_0 = (480 \cos 30^\circ)\mathbf{i} + (480 \sin 30^\circ)\mathbf{j} = 240\sqrt{3}\mathbf{i} + 240\mathbf{j}$. Using $\mathbf{a}(t) = -32\mathbf{j}$ we find

$$\begin{aligned}\mathbf{v}(t) &= \int \mathbf{a}(t) dt = -32t\mathbf{j} + \mathbf{c} \\ 240\sqrt{3}\mathbf{i} + 240\mathbf{j} &= \mathbf{v}(0) = \mathbf{c} \\ \mathbf{v}(t) &= -32t\mathbf{j} + 240\sqrt{3}\mathbf{i} + 240\mathbf{j} = 240\sqrt{3}\mathbf{i} + (240 - 32t)\mathbf{j} \\ \mathbf{r}(t) &= \int \mathbf{v}(t) dt = 240\sqrt{3}t\mathbf{i} + (240t - 16t^2)\mathbf{j} + \mathbf{b} \\ 1600\mathbf{j} &= \mathbf{r}(0) = \mathbf{b}.\end{aligned}$$

- (a) The shell's trajectory is given by $\mathbf{r}(t) = 240\sqrt{3}t\mathbf{i} + (240t - 16t^2 + 1600)\mathbf{j}$ or $x = 240\sqrt{3}t$, $y = 240t - 16t^2 + 1600$.
- (b) Solving $dy/dt = 240 - 32t = 0$, we see that y is maximum when $t = 15/2$. The maximum altitude is $y(15/2) = 2500$ ft.
- (c) Solving $y(t) = -16t^2 + 240t + 1600 = -16(t - 20)(t + 5) = 0$, we see that the shell hits the ground when $t = 20$. The range of the shell is $x(20) = 4800\sqrt{3} \approx 8314$ ft.
- (d) From (c), impact is when $t = 20$. The speed at impact is

$$\|\mathbf{v}(20)\| = |240\sqrt{3}\mathbf{i} + (240 - 32 \cdot 20)\mathbf{j}| = \sqrt{240^2 \cdot 3 + (-400)^2} = 160\sqrt{13} \approx 577 \text{ ft/s.}$$

13. We are given $\mathbf{s}_0 = 81\mathbf{j}$ and $\mathbf{v}_0 = 4\mathbf{i}$. Using $\mathbf{a}(t) = -32\mathbf{j}$, we have

$$\begin{aligned}\mathbf{v}(t) &= \int \mathbf{a}(t) dt = -32t\mathbf{j} + \mathbf{c} \\ 4\mathbf{i} &= \mathbf{v}(0) = \mathbf{c} \\ \mathbf{v}(t) &= 4\mathbf{i} - 32t\mathbf{j} \\ \mathbf{r}(t) &= \int \mathbf{v}(t) dt = 4t\mathbf{i} - 16t^2\mathbf{j} + \mathbf{b} \\ 81\mathbf{j} &= \mathbf{r}(0) = \mathbf{b} \\ \mathbf{r}(t) &= 4t\mathbf{i} + (81 - 16t^2)\mathbf{j}.\end{aligned}$$

Solving $y(t) = 81 - 16t^2 = 0$, we see that the car hits the water when $t = 9/4$. Then

$$\|\mathbf{v}(9/4)\| = |4\mathbf{i} - 32(9/4)\mathbf{j}| = \sqrt{4^2 + 72^2} = 20\sqrt{13} \approx 72.11 \text{ ft/s.}$$

14. Let θ be the angle of elevation. Then $\mathbf{v}(0) = 98 \cos \theta \mathbf{i} + 98 \sin \theta \mathbf{j}$. Using $\mathbf{a}(t) = -9.8\mathbf{j}$, we have

$$\begin{aligned}\mathbf{v}(t) &= \int \mathbf{a}(t) dt = -9.8t\mathbf{j} + \mathbf{c} \\ 98 \cos \theta \mathbf{i} + 98 \sin \theta \mathbf{j} &= \mathbf{v}(0) = \mathbf{c} \\ \mathbf{v}(t) &= 98 \cos \theta \mathbf{i} + (98 \sin \theta - 9.8t)\mathbf{j} \\ \mathbf{r}(t) &= 98t \cos \theta \mathbf{i} + (98t \sin \theta - 4.9t^2)\mathbf{j} + \mathbf{b}.\end{aligned}$$

Since $\mathbf{r}(0) = \mathbf{0}$, $\mathbf{b} = \mathbf{0}$ and $\mathbf{r}(t) = 98t \cos \theta \mathbf{i} + (98t \sin \theta - 4.9t^2)\mathbf{j}$. Setting $y(t) = 98t \sin \theta - 4.9t^2 = t(98 \sin \theta - 4.9t) = 0$, we see that the projectile hits the ground when $t = 20 \sin \theta$. Thus, using $x(t) = 98t \cos \theta$, $490 = x(t) = 98(20 \sin \theta) \cos \theta$ or $\sin 2\theta = 0.5$. Then $2\theta = 30^\circ$ or 150° . The angles of elevation are 15° and 75° .

15. Let s be the initial speed. Then $\mathbf{v}(0) = s \cos 45^\circ \mathbf{i} + s \sin 45^\circ \mathbf{j} = \frac{s\sqrt{2}}{2} \mathbf{i} + \frac{s\sqrt{2}}{2} \mathbf{j}$. Using $\mathbf{a}(t) = -32\mathbf{j}$, we have

$$\begin{aligned}\mathbf{v}(t) &= \int \mathbf{a}(t) dt = -32t\mathbf{j} + \mathbf{c} \\ \frac{s\sqrt{2}}{2} \mathbf{i} + \frac{s\sqrt{2}}{2} \mathbf{j} &= \mathbf{v}(0) = \mathbf{c} \\ \mathbf{v}(t) &= \frac{s\sqrt{2}}{2} \mathbf{i} + \left(\frac{s\sqrt{2}}{2} - 32t \right) \mathbf{j} \\ \mathbf{r}(t) &= \frac{s\sqrt{2}}{2} t \mathbf{i} + \left(\frac{s\sqrt{2}}{2} t - 16t^2 \right) \mathbf{j} + \mathbf{b}.\end{aligned}$$

Since $\mathbf{r}(0) = \mathbf{0}$, $\mathbf{b} = \mathbf{0}$ and

$$\mathbf{r}(t) = \frac{s\sqrt{2}}{2} t \mathbf{i} + \left(\frac{s\sqrt{2}}{2} t - 16t^2 \right) \mathbf{j}.$$

Setting $y(t) = s\sqrt{2}t/2 - 16t^2 = t(s\sqrt{2}/2 - 16t) = 0$ we see that the ball hits the ground when $t = \sqrt{2}s/32$. Thus, using $x(t) = s\sqrt{2}t/2$ and the fact that 100 yd = 300 ft, $300 = x(t) = \frac{s\sqrt{2}}{2}(\sqrt{2}s/32) = \frac{s^2}{32}$ and $s = \sqrt{9600} \approx 97.98$ ft/s.

16. Let s be the initial speed and θ the initial angle. Then $\mathbf{v}(0) = s \cos \theta \mathbf{i} + s \sin \theta \mathbf{j}$. Using $\mathbf{a}(t) = -32\mathbf{j}$, we have

$$\begin{aligned}\mathbf{v}(t) &= \int \mathbf{a}(t) dt = -32t\mathbf{j} + \mathbf{c} \\ s \cos \theta \mathbf{i} + s \sin \theta \mathbf{j} &= \mathbf{v}(0) = \mathbf{c} \\ \mathbf{v}(t) &= s \cos \theta \mathbf{i} + (s \sin \theta - 32t)\mathbf{j} \\ \mathbf{r}(t) &= st \cos \theta \mathbf{i} + (st \sin \theta - 16t^2)\mathbf{j} + \mathbf{b}.\end{aligned}$$

Since $\mathbf{r}(0) = \mathbf{0}$, $\mathbf{b} = \mathbf{0}$ and $\mathbf{r}(t) = st \cos \theta \mathbf{i} + (st \sin \theta - 16t^2)\mathbf{j}$. Setting $y(t) = st \sin \theta - 16t^2 = t(s \sin \theta - 16t) = 0$, we see that the ball hits the ground when $t = (s \sin \theta)/16$. Using $x(t) = st \cos \theta \mathbf{i}$, we see that the range of the ball is

$$x\left(\frac{s \sin \theta}{16}\right) = \frac{s^2 \sin \theta \cos \theta}{16} = \frac{s^2 \sin 2\theta}{32}.$$

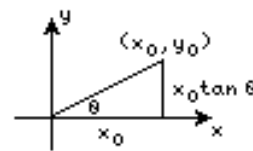
For $\theta = 30^\circ$, the range is $s^2 \sin 60^\circ/32 = \sqrt{3}s^2/64$ and for $\theta = 60^\circ$ the range is $s^2 \sin 120^\circ/32 = \sqrt{3}s^2/64$. In general, when the angle is $90^\circ - \theta$ the range is

$$[s^2 \sin 2(90^\circ - \theta)]/32 = s^2[\sin(180^\circ - 2\theta)]/32 = s^2(\sin 2\theta)/32.$$

Thus, for angles θ and $90^\circ - \theta$, the range is the same.

Exercises 9.2

17. Let the initial speed of the projectile be s and let the target be at (x_0, y_0) . Then $\mathbf{v}_p(0) = s \cos \theta \mathbf{i} + s \sin \theta \mathbf{j}$ and $\mathbf{v}_t(0) = \mathbf{0}$. Using $\mathbf{a}(t) = -32\mathbf{j}$, we have



$$\begin{aligned}\mathbf{v}_p(t) &= \int \mathbf{a}(t) dt = -32t\mathbf{j} + \mathbf{c} \\ s \cos \theta \mathbf{i} + s \sin \theta \mathbf{j} &= \mathbf{v}_p(0) = \mathbf{c} \\ \mathbf{v}_p(t) &= s \cos \theta \mathbf{i} + (s \sin \theta - 32t)\mathbf{j} \\ \mathbf{r}_p(t) &= st \cos \theta \mathbf{i} + (st \sin \theta - 16t^2)\mathbf{j} + \mathbf{b}.\end{aligned}$$

Since $\mathbf{r}_p(0) = \mathbf{0}$, $\mathbf{b} = \mathbf{0}$ and $\mathbf{r}_p(t) = st \cos \theta \mathbf{i} + (st \sin \theta - 16t^2)\mathbf{j}$. Also, $\mathbf{v}_t(t) = -32t\mathbf{j} + \mathbf{c}$ and since $\mathbf{v}_t(0) = \mathbf{0}$, $\mathbf{c} = \mathbf{0}$ and $\mathbf{v}_t(t) = -32t\mathbf{j}$. Then $\mathbf{r}_t(t) = -16t^2\mathbf{j} + \mathbf{b}$. Since $\mathbf{r}_t(0) = x_0\mathbf{i} + y_0\mathbf{j}$, $\mathbf{b} = x_0\mathbf{i} + y_0\mathbf{j}$ and $\mathbf{r}_t(t) = x_0\mathbf{i} + (y_0 - 16t^2)\mathbf{j}$. Now, the horizontal component of $\mathbf{r}_p(t)$ will be x_0 when $t = x_0/s \cos \theta$ at which time the vertical component of $\mathbf{r}_p(t)$ will be

$$(sx_0/s \cos \theta) \sin \theta - 16(x_0/s \cos \theta)^2 = x_0 \tan \theta - 16(x_0/s \cos \theta)^2 = y_0 - 16(x_0/s \cos \theta)^2.$$

Thus, $\mathbf{r}_p(x_0/s \cos \theta) = \mathbf{r}_t(x_0/s \cos \theta)$ and the projectile will strike the target as it falls.

18. The initial angle is $\theta = 0$, the initial height is 1024 ft, and the initial speed is $s = 180(5280)/3600 = 264$ ft/s. Then $x(t) = 264t$ and $y(t) = -16t^2 + 1024$. Solving $y(t) = 0$ we see that the pack hits the ground at $t = 8$ seconds. The horizontal distance travelled is $x(8) = 2112$ feet. From the figure in the text, $\tan \alpha = 1024/2112 = 16/33$ and $\alpha \approx 0.45$ radian or 25.87° .

19. $\mathbf{r}'(t) = \mathbf{v}(t) = -r_0\omega \sin \omega t \mathbf{i} + r_0\omega \cos \omega t \mathbf{j}$; $v = \|\mathbf{v}(t)\| = \sqrt{r_0^2\omega^2 \sin^2 \omega t + r_0^2\omega^2 \cos^2 \omega t} = r_0\omega$
 $\omega = v/r_0$; $\mathbf{a}(t) = \mathbf{r}''(t) = -r_0\omega^2 \cos \omega t \mathbf{i} - r_0\omega^2 \sin \omega t \mathbf{j}$
 $a = \|\mathbf{a}(t)\| = \sqrt{r_0^2\omega^4 \cos^2 \omega t + r_0^2\omega^4 \sin^2 \omega t} = r_0\omega^2 = r_0(v/r_0)^2 = v^2/r_0.$

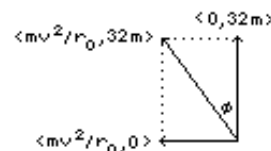
20. (a) $\mathbf{v}(t) = -b \sin t \mathbf{i} + b \cos t \mathbf{j} + c\mathbf{k}$; $\|\mathbf{v}(t)\| = \sqrt{b^2 \sin^2 t + b^2 \cos^2 t + c^2} = \sqrt{b^2 + c^2}$

(b) $s = \int_0^t \|\mathbf{v}(u)\| du = \int_0^t \sqrt{b^2 + c^2} du = t\sqrt{b^2 + c^2}$; $\frac{ds}{dt} = \sqrt{b^2 + c^2}$

(c) $\frac{d^2s}{dt^2} = 0$; $\mathbf{a}(t) = -b \cos t \mathbf{i} - b \sin t \mathbf{j}$; $\|\mathbf{a}(t)\| = \sqrt{b^2 \cos^2 t + b^2 \sin^2 t} = |b|$. Thus, $d^2s/dt^2 \neq \|\mathbf{a}(t)\|$.

21. By Problem 19, $a = v^2/r_0 = 1530^2/(4000 \cdot 5280) \approx 0.1108$. We are given $mg = 192$, so $m = 192/32$ and $w_e = 192 - (192/32)(0.1108) \approx 191.33$ lb.

22. By Problem 19, the centripetal acceleration is v^2/r_0 . Then the horizontal force is mv^2/r_0 . The vertical force is $32m$. The resultant force is $\mathbf{U} = (mv^2/r_0)\mathbf{i} + 32m\mathbf{j}$. From the figure, we see that $\tan \phi = (mv^2/r_0)/32m = v^2/32r_0$. Using $r_0 = 60$ and $v = 44$ we obtain $\tan \phi = 44^2/32(60) \approx 1.0083$ and $\phi \approx 45.24^\circ$.



23. Solving $x(t) = (v_0 \cos \theta)t$ for t and substituting into $y(t) = -\frac{1}{2}gt^2 + (v_0 \sin \theta)t + s_0$ we obtain

$$y = -\frac{1}{2}g \left(\frac{x}{v_0 \cos \theta} \right)^2 + (v_0 \sin \theta) \frac{x}{v_0 \cos \theta} + s_0 = -\frac{g}{2v_0^2 \cos^2 \theta} x^2 + (\tan \theta)x + s_0,$$

which is the equation of a parabola.

24. Since the projectile is launched from ground level, $s_0 = 0$. To find the maximum height we maximize $y(t) = -\frac{1}{2}gt^2 + (v_0 \sin \theta)t$. Solving $y'(t) = -gt + v_0 \sin \theta = 0$, we see that $t = (v_0/g) \sin \theta$ is a critical point.

Since $y''(t) = -g < 0$,

$$H = y\left(\frac{v_0 \sin \theta}{g}\right) = -\frac{1}{2}g \frac{v_0^2 \sin^2 \theta}{g^2} + v_0 \sin \theta \frac{v_0 \sin \theta}{g} = \frac{v_0^2 \sin^2 \theta}{2g}$$

is the maximum height. To find the range we solve $y(t) = -\frac{1}{2}gt^2 + (v_0 \sin \theta)t = t(v_0 \sin \theta - \frac{1}{2}gt) = 0$. The positive solution of this equation is $t = (2v_0 \sin \theta)/g$. The range is thus

$$x(t) = (v_0 \cos \theta) \frac{2v_0 \sin \theta}{g} = \frac{v_0^2 \sin 2\theta}{g}.$$

25. Letting $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$, the equation $d\mathbf{r}/dt = \mathbf{v}$ is equivalent to $dx/dt = 6t^2x$, $dy/dt = -4ty^2$, $dz/dt = 2t(z+1)$. Separating variables and integrating, we obtain $dx/x = 6t^2 dt$, $dy/y^2 = -4t dt$, $dz/(z+1) = 2t dt$, and $\ln x = 2t^3 + c_1$, $-1/y = -2t^2 + c_2$, $\ln(z+1) = t^2 + c_3$. Thus,

$$\mathbf{r}(t) = k_1 e^{2t^3} \mathbf{i} + \frac{1}{2t^2 + k_2} \mathbf{j} + (k_3 e^{t^2} - 1) \mathbf{k}.$$

26. We require the fact that $d\mathbf{r}/dt = \mathbf{v}$. Then

$$\frac{d\mathbf{L}}{dt} = \frac{d}{dt}(\mathbf{r} \times \mathbf{p}) = \mathbf{r} \times \frac{d\mathbf{p}}{dt} + \frac{d\mathbf{r}}{dt} \times \mathbf{p} = \boldsymbol{\tau} + \mathbf{v} \times \mathbf{p} = \boldsymbol{\tau} + \mathbf{v} \times m\mathbf{v} = \boldsymbol{\tau} + m(\mathbf{v} \times \mathbf{v}) = \boldsymbol{\tau} + \mathbf{0} = \boldsymbol{\tau}.$$

27. (a) Since $\mathbf{F}$ is directed along $\mathbf{r}$ we have $\mathbf{F} = c\mathbf{r}$ for some constant c . Then

$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F} = \mathbf{r} \times (c\mathbf{r}) = c(\mathbf{r} \times \mathbf{r}) = \mathbf{0}.$$

(b) If $\boldsymbol{\tau} = \mathbf{0}$ then $d\mathbf{L}/dt = \mathbf{0}$ and $\mathbf{L}$ is constant.

28. (a) Using Problem 27, $\mathbf{F} = -k(Mm/r^2)\mathbf{u} = m\mathbf{a}$. Then $\mathbf{a} = d^2\mathbf{r}/dt^2 = -k(M/r^2)\mathbf{u}$.

(b) Using $\mathbf{u} = \mathbf{r}/r$ we have

$$\mathbf{r} \times \mathbf{r}'' = \mathbf{r} \times \left(-k \frac{M}{r^2} \mathbf{u}\right) = -\frac{kM}{r^2} \left[\mathbf{r} \times \left(\frac{1}{r} \mathbf{r}\right)\right] = -\frac{kM}{r^3}(\mathbf{r} \times \mathbf{r}) = \mathbf{0}.$$

(c) From Theorem 9.4 (iv) we have

$$\frac{d}{dt}(\mathbf{r} \times \mathbf{v}) = \mathbf{r} \times \frac{d\mathbf{v}}{dt} + \frac{d\mathbf{r}}{dt} \times \mathbf{v} = \mathbf{r} \times \mathbf{r}'' + \mathbf{v} \times \mathbf{v} = \mathbf{0} + \mathbf{0} = \mathbf{0}.$$

(d) Since $\mathbf{r} = r\mathbf{u}$ we have $\mathbf{c} = \mathbf{r} \times \mathbf{v} = r\mathbf{u} \times r\mathbf{u}' = r^2(\mathbf{u} \times \mathbf{u}')$.

(e) Since $\mathbf{u} = (1/r)\mathbf{r}$ is a unit vector, $\mathbf{u} \cdot \mathbf{u} = 1$ and

$$\frac{d}{dt}(\mathbf{u} \cdot \mathbf{u}) = \mathbf{u} \cdot \frac{d\mathbf{u}}{dt} + \frac{d\mathbf{u}}{dt} \cdot \mathbf{u} = 2\mathbf{u} \cdot \frac{d\mathbf{u}}{dt} = \frac{d}{dt}(1) = 0.$$

Thus, $\mathbf{u} \cdot \mathbf{u}' = 0$.

$$\begin{aligned} \text{(f)} \quad \frac{d}{dt}(\mathbf{v} \times \mathbf{c}) &= \mathbf{v} \times \frac{d\mathbf{c}}{dt} + \frac{d\mathbf{v}}{dt} \times \mathbf{c} = \mathbf{v} \times \mathbf{0} + \mathbf{a} \times \mathbf{c} = -\frac{kM}{r^2} \mathbf{u} \times \mathbf{c} = -\frac{kM}{r^2} \mathbf{u} \times [r^2(\mathbf{u} \times \mathbf{u}')] \\ &= -kM[\mathbf{u} \times (\mathbf{u} \times \mathbf{u}')] = -kM = -kM[(\mathbf{u} \cdot \mathbf{u}')\mathbf{u} - (\mathbf{u} \cdot \mathbf{u})\mathbf{u}'] \quad \boxed{\text{by (10) of 7.4}} \\ &= -kM[\mathbf{0} - \mathbf{u}'] = kM\mathbf{u}' = kM \frac{d\mathbf{u}}{dt} \end{aligned}$$

(g) Since

$$\begin{aligned} \mathbf{r} \cdot (\mathbf{v} \times \mathbf{c}) &= (\mathbf{r} \times \mathbf{v}) \cdot \mathbf{c} \quad \boxed{\text{by Problem 61 in 7.4}} \\ &= \mathbf{c} \cdot \mathbf{c} = c^2 \quad \boxed{\text{where } c = \|\mathbf{c}\|} \end{aligned}$$

Exercises 9.2

and

$$\begin{aligned}(kM\mathbf{u} + \mathbf{d}) \cdot \mathbf{r} &= (kM\mathbf{u} + \mathbf{d}) \cdot r\mathbf{u} = kMr\mathbf{u} \cdot \mathbf{u} + r\mathbf{d} \cdot \mathbf{u} \\ &= kMr + rd \cos \theta \quad \boxed{\text{where } d = \|\mathbf{d}\|}\end{aligned}$$

$$\text{we have } c^2 = kMr + rd \cos \theta \text{ or } r = \frac{c^2}{kM + d \cos \theta} = \frac{c^2/kM}{1 + (d/kM) \cos \theta}.$$

(h) First note that $c > 0$ (otherwise there is no orbit) and $d > 0$ (since the orbit is not a circle). We recognize the equation in (g) to be that of a conic section with eccentricity $e = d/kM$. Since the orbit of the planet is closed it must be an ellipse.

(i) At perihelion $c = \|\mathbf{c}\| = \|\mathbf{r} \times \mathbf{v}\| = r_0 v_0 \sin(\pi/r) = r_0 v_0$. Since r is minimum at this point, we want the denominator in the equation $r_0 = [c^2/kM]/[1 + (d/kM) \cos \theta]$ to be maximum. This occurs when $\theta = 0$. In this case

$$r_0 = \frac{r_0^2 v_0^2 / kM}{1 + d/kM} \quad \text{and} \quad d = r_0 v_0^2 - kM.$$

Exercises 9.3

1. $\mathbf{r}'(t) = -t \sin t \mathbf{i} + t \cos t \mathbf{j} + 2t \mathbf{k}$; $|\mathbf{r}'(t)| = \sqrt{t^2 \sin^2 t + t^2 \cos^2 t + 4t^2} = \sqrt{5}t$;

$$\mathbf{T}(t) = -\frac{\sin t}{\sqrt{5}} \mathbf{i} + \frac{\cos t}{\sqrt{5}} \mathbf{j} + \frac{2}{\sqrt{5}} \mathbf{k}$$

2. $\mathbf{r}'(t) = e^t(-\sin t + \cos t)\mathbf{i} + e^t(\cos t + \sin t)\mathbf{j} + \sqrt{2}e^t\mathbf{k}$,

$$|\mathbf{r}'(t)| = [e^{2t}(\sin^2 t - 2 \sin t \cos t + \cos^2 t) + e^{2t}(\cos^2 t + 2 \sin t \cos t + \sin^2 t) + 2e^{2t}]^{1/2} = \sqrt{4e^{2t}} = 2e^t;$$

$$\mathbf{T}(t) = \frac{1}{2}(-\sin t + \cos t)\mathbf{i} + \frac{1}{2}(\cos t + \sin t)\mathbf{j} + \frac{\sqrt{2}}{2}\mathbf{k}$$

3. We assume $a > 0$. $\mathbf{r}'(t) = -a \sin t \mathbf{i} + a \cos t \mathbf{j} + c \mathbf{k}$; $|\mathbf{r}'(t)| = \sqrt{a^2 \sin^2 t + a^2 \cos^2 t + c^2} = \sqrt{a^2 + c^2}$;

$$\mathbf{T}(t) = -\frac{a \sin t}{\sqrt{a^2 + c^2}} \mathbf{i} + \frac{a \cos t}{\sqrt{a^2 + c^2}} \mathbf{j} + \frac{c}{\sqrt{a^2 + c^2}} \mathbf{k}; \quad \frac{d\mathbf{T}}{dt} = -\frac{a \cos t}{\sqrt{a^2 + c^2}} \mathbf{i} - \frac{a \sin t}{\sqrt{a^2 + c^2}} \mathbf{j},$$

$$\left| \frac{d\mathbf{T}}{dt} \right| = \sqrt{\frac{a^2 \cos^2 t}{a^2 + c^2} + \frac{a^2 \sin^2 t}{a^2 + c^2}} = \frac{a}{\sqrt{a^2 + c^2}}; \quad \mathbf{N} = -\cos t \mathbf{i} - \sin t \mathbf{j};$$

$$\mathbf{B} = \mathbf{T} \times \mathbf{N} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -\frac{a \sin t}{\sqrt{a^2 + c^2}} & \frac{a \cos t}{\sqrt{a^2 + c^2}} & \frac{c}{\sqrt{a^2 + c^2}} \\ -\cos t & -\sin t & 0 \end{vmatrix} = \frac{c \sin t}{\sqrt{a^2 + c^2}} \mathbf{i} - \frac{c \cos t}{\sqrt{a^2 + c^2}} \mathbf{j} + \frac{a}{\sqrt{a^2 + c^2}} \mathbf{k};$$

$$\kappa = \frac{|d\mathbf{T}/dt|}{|\mathbf{r}'(t)|} = \frac{a/\sqrt{a^2 + c^2}}{\sqrt{a^2 + c^2}} = \frac{a}{a^2 + c^2}$$

4. $\mathbf{r}'(t) = \mathbf{i} + t\mathbf{j} + t^2\mathbf{k}$, $\mathbf{r}'(1) = \mathbf{i} + \mathbf{j} + \mathbf{k}$; $|\mathbf{r}'(t)| = \sqrt{1 + t^2 + t^4}$, $|\mathbf{r}'(1)| = \sqrt{3}$;

$$\mathbf{T}(t) = (1 + t^2 + t^4)^{-1/2}(\mathbf{i} + t\mathbf{j} + t^2\mathbf{k}), \quad \mathbf{T}(1) = \frac{1}{\sqrt{3}}(\mathbf{i} + \mathbf{j} + \mathbf{k});$$

$$\begin{aligned} \frac{d\mathbf{T}}{dt} &= -\frac{1}{2}(1 + t^2 + t^4)^{-3/2}(2t + 4t^3)\mathbf{i} + [(1 + t^2 + t^4)^{-1/2} - \frac{t}{2}(1 + t^2 + t^4)^{-3/2}(2t + 4t^3)]\mathbf{j} \\ &\quad + [2t(1 + t^2 + t^4)^{-1/2} - \frac{t^2}{2}(1 + t^2 + t^4)^{-3/2}(2t + 4t^3)]\mathbf{k}; \end{aligned}$$

$$\frac{d}{dt} \mathbf{T}(1) = -\frac{1}{\sqrt{3}} \mathbf{i} + \frac{1}{\sqrt{3}} \mathbf{k}, \quad \left| \frac{d}{dt} \mathbf{T}(1) \right| = \sqrt{\frac{1}{3} + \frac{1}{3}} = \frac{\sqrt{2}}{\sqrt{3}}; \quad \mathbf{N}(1) = -\frac{1}{\sqrt{2}}(\mathbf{i} - \mathbf{k});$$

$$\mathbf{B}(1) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ -1/\sqrt{2} & 0 & 1/\sqrt{2} \end{vmatrix} = \frac{1}{\sqrt{6}}(\mathbf{i} - 2\mathbf{j} + \mathbf{k}); \quad \kappa = \left| \frac{d}{dt} \mathbf{T}(1) \right| / |\mathbf{r}'(1)| = \frac{\sqrt{2}/\sqrt{3}}{\sqrt{3}} = \frac{\sqrt{2}}{3}$$

5. From Example 2 in the text, a normal to the osculating plane is $\mathbf{B}(\pi/4) = \frac{1}{\sqrt{26}}(3\mathbf{i} - 3\mathbf{j} + 2\sqrt{2}\mathbf{k})$. The point on the curve when $t = \pi/4$ is $(\sqrt{2}, \sqrt{2}, 3\pi/4)$. An equation of the plane is $3(x - \sqrt{2}) - 3(y - \sqrt{2}) + 2\sqrt{2}(z - 3\pi/4) = 0$, $3x - 3y + 2\sqrt{2}z = 3\sqrt{2}\pi/2$, or $3\sqrt{2}x - 3\sqrt{2}y + 4z = 3\pi$.

6. From Problem 4, a normal to the osculating plane is $\mathbf{B}(1) = \frac{1}{\sqrt{6}}(\mathbf{i} - 2\mathbf{j} + \mathbf{k})$. The point on the curve when $t = 1$ is $(1, 1/2, 1/3)$. An equation of the plane is $(x - 1) - 2(y - 1/2) + (z - 1/3) = 0$ or $x - 2y + z = 1/3$.

7. $\mathbf{v}(t) = \mathbf{j} + 2t\mathbf{k}$, $|\mathbf{v}(t)| = \sqrt{1 + 4t^2}$; $\mathbf{a}(t) = 2\mathbf{k}$; $\mathbf{v} \cdot \mathbf{a} = 4t$, $\mathbf{v} \times \mathbf{a} = 2\mathbf{i}$, $|\mathbf{v} \times \mathbf{a}| = 2$;

$$a_T = \frac{4t}{\sqrt{1 + 4t^2}}, \quad a_N = \frac{2}{\sqrt{1 + 4t^2}}$$

8. $\mathbf{v}(t) = -3\sin t\mathbf{i} + 2\cos t\mathbf{j} + \mathbf{k}$,

$$|\mathbf{v}(t)| = \sqrt{9\sin^2 t + 4\cos^2 t + 1} = \sqrt{5\sin^2 t + 4\sin^2 t + 4\cos^2 t + 1} = \sqrt{5}\sqrt{\sin^2 t + 1};$$

$$\mathbf{a}(t) = -3\cos t\mathbf{i} - 2\sin t\mathbf{j}; \quad \mathbf{v} \cdot \mathbf{a} = 9\sin t \cos t - 4\sin t \cos t = 5\sin t \cos t,$$

$$\mathbf{v} \times \mathbf{a} = 2\sin t\mathbf{i} - 3\cos t\mathbf{j} + 6\mathbf{k}, \quad |\mathbf{v} \times \mathbf{a}| = \sqrt{4\sin^2 t + 9\cos^2 t + 36} = \sqrt{5}\sqrt{\cos^2 t + 8};$$

$$a_T = \frac{\sqrt{5}\sin t \cos t}{\sqrt{\sin^2 t + 1}}, \quad a_N = \sqrt{\frac{\cos^2 t + 8}{\sin^2 t + 1}}$$

9. $\mathbf{v}(t) = 2t\mathbf{i} + 2t\mathbf{j} + 4t\mathbf{k}$, $|\mathbf{v}(t)| = 2\sqrt{6}t$, $t > 0$; $\mathbf{a}(t) = 2\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$; $\mathbf{v} \cdot \mathbf{a} = 24t$, $\mathbf{v} \times \mathbf{a} = \mathbf{0}$;

$$a_T = \frac{24t}{2\sqrt{6}t} = 2\sqrt{6}, \quad a_N = 0, \quad t > 0$$

10. $\mathbf{v}(t) = 2t\mathbf{i} - 3t^2\mathbf{j} + 4t^3\mathbf{k}$, $|\mathbf{v}(t)| = t\sqrt{4 + 9t^2 + 16t^4}$, $t > 0$; $\mathbf{a}(t) = 2\mathbf{i} - 6t\mathbf{j} + 12t^2\mathbf{k}$;

$$\mathbf{v} \cdot \mathbf{a} = 4t + 18t^3 + 48t^5; \quad \mathbf{v} \times \mathbf{a} = -12t^4\mathbf{i} - 16t^3\mathbf{j} - 6t^2\mathbf{k}, \quad |\mathbf{v} \times \mathbf{a}| = 2t^2\sqrt{36t^4 + 64t^2 + 9};$$

$$a_T = \frac{4 + 18t^2 + 48t^4}{\sqrt{4 + 9t^2 + 16t^4}}, \quad a_N = \frac{2t\sqrt{36t^4 + 64t^2 + 9}}{\sqrt{4 + 9t^2 + 16t^4}}, \quad t > 0$$

11. $\mathbf{v}(t) = 2\mathbf{i} + 2t\mathbf{j}$, $|\mathbf{v}(t)| = 2\sqrt{1 + t^2}$; $\mathbf{a}(t) = 2\mathbf{j}$; $\mathbf{v} \cdot \mathbf{a} = 4t$; $\mathbf{v} \times \mathbf{a} = 4\mathbf{k}$, $|\mathbf{v} \times \mathbf{a}| = 4$;

$$a_T = \frac{2t}{\sqrt{1 + t^2}}, \quad a_N = \frac{2}{\sqrt{1 + t^2}}$$

12. $\mathbf{v}(t) = \frac{1}{1 + t^2}\mathbf{i} + \frac{t}{1 + t^2}\mathbf{j}$, $|\mathbf{v}(t)| = \frac{\sqrt{1 + t^2}}{1 + t^2}$; $\mathbf{a}(t) = -\frac{2t}{(1 + t^2)^2}\mathbf{i} + \frac{1 - t^2}{(1 + t^2)^2}\mathbf{j}$;

$$\mathbf{v} \cdot \mathbf{a} = -\frac{2t}{(1 + t^2)^3} + \frac{t - t^3}{(1 + t^2)^3} = -\frac{t}{(1 + t^2)^2}; \quad \mathbf{v} \times \mathbf{a} = \frac{1}{(1 + t^2)^2}\mathbf{k}, \quad |\mathbf{v} \times \mathbf{a}| = \frac{1}{(1 + t^2)^2};$$

$$a_T = -\frac{t/(1 + t^2)^2}{\sqrt{1 + t^2}/(1 + t^2)} = -\frac{t}{(1 + t^2)^{3/2}}, \quad a_N = \frac{1/(1 + t^2)^2}{\sqrt{1 + t^2}/(1 + t^2)} = \frac{1}{(1 + t^2)^{3/2}}$$

13. $\mathbf{v}(t) = -5\sin t\mathbf{i} + 5\cos t\mathbf{j}$, $|\mathbf{v}(t)| = 5$; $\mathbf{a}(t) = -5\cos t\mathbf{i} - 5\sin t\mathbf{j}$; $\mathbf{v} \cdot \mathbf{a} = 0$, $\mathbf{v} \times \mathbf{a} = 25\mathbf{k}$, $|\mathbf{v} \times \mathbf{a}| = 25$;
 $a_T = 0$, $a_N = 5$

14. $\mathbf{v}(t) = \sinh t\mathbf{i} + \cosh t\mathbf{j}$, $|\mathbf{v}(t)| = \sqrt{\sinh^2 t + \cosh^2 t}$; $\mathbf{a}(t) = \cosh t\mathbf{i} + \sinh t\mathbf{j}$; $\mathbf{v} \cdot \mathbf{a} = 2\sinh t \cosh t$;

$$\mathbf{v} \times \mathbf{a} = (\sinh^2 t - \cosh^2 t)\mathbf{k} = -\mathbf{k}, \quad |\mathbf{v} \times \mathbf{a}| = 1; \quad a_T = \frac{2\sinh t \cosh t}{\sqrt{\sinh^2 t + \cosh^2 t}}, \quad a_N = \frac{1}{\sqrt{\sinh^2 t + \cosh^2 t}}$$

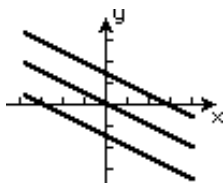
15. $\mathbf{v}(t) = -e^{-t}(\mathbf{i} + \mathbf{j} + \mathbf{k})$, $|\mathbf{v}(t)| = \sqrt{3}e^{-t}$; $\mathbf{a}(t) = e^{-t}(\mathbf{i} + \mathbf{j} + \mathbf{k})$; $\mathbf{v} \cdot \mathbf{a} = -3e^{-2t}$; $\mathbf{v} \times \mathbf{a} = \mathbf{0}$, $|\mathbf{v} \times \mathbf{a}| = 0$;
 $a_T = -\sqrt{3}e^{-t}$, $a_N = 0$

Exercises 9.3

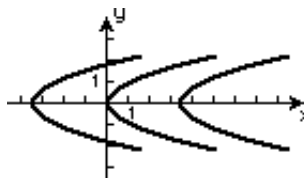
16. $\mathbf{v}(t) = \mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$, $|\mathbf{v}(t)| = \sqrt{21}$; $\mathbf{a}(t) = \mathbf{0}$; $\mathbf{v} \cdot \mathbf{a} = 0$, $\mathbf{v} \times \mathbf{a} = \mathbf{0}$, $|\mathbf{v} \times \mathbf{a}| = 0$; $a_T = 0$, $a_N = 0$
17. $\mathbf{v}(t) = -a \sin t \mathbf{i} + b \cos t \mathbf{j} + c \mathbf{k}$, $|\mathbf{v}(t)| = \sqrt{a^2 \sin^2 t + b^2 \cos^2 t + c^2}$; $\mathbf{a}(t) = -a \cos t \mathbf{i} - b \sin t \mathbf{j}$;
 $\mathbf{v} \times \mathbf{a} = bc \sin t \mathbf{i} - ac \cos t \mathbf{j} + ab \mathbf{k}$, $|\mathbf{v} \times \mathbf{a}| = \sqrt{b^2 c^2 \sin^2 t + a^2 c^2 \cos^2 t + a^2 b^2}$
 $\kappa = \frac{|\mathbf{v} \times \mathbf{a}|}{|\mathbf{v}|^3} = \frac{\sqrt{b^2 c^2 \sin^2 t + a^2 c^2 \cos^2 t + a^2 b^2}}{(a^2 \sin^2 t + b^2 \cos^2 t + c^2)^{3/2}}$;
18. (a) $\mathbf{v}(t) = -a \sin t \mathbf{i} + b \cos t \mathbf{j}$, $|\mathbf{v}(t)| = \sqrt{a^2 \sin^2 t + b^2 \cos^2 t}$; $\mathbf{a}(t) = -a \cos t \mathbf{i} - b \sin t \mathbf{j}$;
 $\mathbf{v} \times \mathbf{a} = ab \mathbf{k}$; $|\mathbf{v} \times \mathbf{a}| = ab$; $\kappa = \frac{ab}{(a^2 \sin^2 t + b^2 \cos^2 t)^{3/2}}$
 (b) When $a = b$, $|\mathbf{v}(t)| = a$, $|\mathbf{v} \times \mathbf{a}| = a^2$, and $\kappa = a^2/a^3 = 1/a$.
19. The equation of a line is $\mathbf{r}(t) = \mathbf{b} + t\mathbf{c}$, where $\mathbf{b}$ and $\mathbf{c}$ are constant vectors.
 $\mathbf{v}(t) = \mathbf{c}$, $|\mathbf{v}(t)| = |\mathbf{c}|$; $\mathbf{a}(t) = \mathbf{0}$; $\mathbf{v} \times \mathbf{a} = \mathbf{0}$, $|\mathbf{v} \times \mathbf{a}| = 0$; $\kappa = |\mathbf{v} \times \mathbf{a}|/|\mathbf{v}|^3 = 0$
20. $\mathbf{v}(t) = a(1 - \cos t)\mathbf{i} + a \sin t \mathbf{j}$; $\mathbf{v}(\pi) = 2a\mathbf{i}$, $|\mathbf{v}(\pi)| = 2a$; $\mathbf{a}(t) = a \sin t \mathbf{i} + a \cos t \mathbf{j}$, $\mathbf{a}(\pi) = -a\mathbf{j}$;
 $|\mathbf{v} \times \mathbf{a}| = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2a & 0 & 0 \\ 0 & -a & 0 \end{vmatrix} = -2a^2 \mathbf{k}$; $|\mathbf{v} \times \mathbf{a}| = 2a^2$; $\kappa = \frac{|\mathbf{v} \times \mathbf{a}|}{|\mathbf{v}|^3} = \frac{2a^2}{8a^3} = \frac{1}{4a}$
21. $\mathbf{v}(t) = f'(t)\mathbf{i} + g'(t)\mathbf{j}$, $|\mathbf{v}(t)| = \sqrt{[f'(t)]^2 + [g'(t)]^2}$; $\mathbf{a}(t) = f''(t)\mathbf{i} + g''(t)\mathbf{j}$;
 $\mathbf{v} \times \mathbf{a} = [f'(t)g''(t) - g'(t)f''(t)]\mathbf{k}$, $|\mathbf{v} \times \mathbf{a}| = |f'(t)g''(t) - g'(t)f''(t)|$;
 $\kappa = \frac{|\mathbf{v} \times \mathbf{a}|}{|\mathbf{v}|^3} = \frac{|f'(t)g''(t) - g'(t)f''(t)|}{([f'(t)]^2 + [g'(t)]^2)^{3/2}}$
22. For $y = F(x)$, $\mathbf{r}(x) = x\mathbf{i} + F(x)\mathbf{j}$. We identify $f(x) = x$ and $g(x) = F(x)$ in Problem 21. Then $f'(x) = 1$, $f''(x) = 0$, $g'(x) = F'(x)$, $g''(x) = F''(x)$, and $\kappa = |F''(x)|/(1 + [F'(x)]^2)^{3/2}$.
23. $F(x) = x^2$, $F(0) = 0$, $F(1) = 1$; $F'(x) = 2x$, $F'(0) = 0$, $F'(1) = 2$; $F''(x) = 2$, $F''(0) = 2$, $F''(1) = 2$;
 $\kappa(0) = \frac{2}{(1+0^2)^{3/2}} = 2$; $\rho(0) = \frac{1}{2}$; $\kappa(1) = \frac{2}{(1+2^2)^{3/2}} = \frac{2}{5\sqrt{5}} \approx 0.18$;
 $\rho(1) = \frac{5\sqrt{5}}{2} \approx 5.59$; Since $2 > 2/5\sqrt{5}$, the curve is “sharper” at $(0, 0)$.
24. $F(x) = x^3$, $F(-1) = -1$, $F(1/2) = 1/8$; $F'(x) = 3x^2$, $F'(-1) = 3$, $F'(1/2) = 3/4$; $F''(x) = 6x$,
 $F''(-1) = -6$, $F''(1/2) = 3$; $\kappa(-1) = \frac{|-6|}{(1+3^2)^{3/2}} = \frac{6}{10\sqrt{10}} = \frac{3}{5\sqrt{10}} \approx 0.19$;
 $\rho(-1) = \frac{5\sqrt{10}}{3} \approx 5.27$; $\kappa(\frac{1}{2}) = \frac{3}{[1+(3/4)^2]^{3/2}} = \frac{3}{125/64} = \frac{192}{125} \approx 1.54$; $\rho(\frac{1}{2}) = \frac{125}{192} \approx 0.65$
 Since $1.54 > 0.19$, the curve is “sharper” at $(1/2, 1/8)$.
25. At a point of inflection $(x_0, F(x_0))$, if $F''(x_0)$ exists then $F''(x_0) = 0$. Thus, assuming that $\lim_{x \rightarrow x_0} F''(x)$ exists, $F''(x)$ and hence κ is near 0 for x near x_0 .
26. We use the fact that $\mathbf{T} \cdot \mathbf{N} = 0$ and $\mathbf{T} \cdot \mathbf{T} = \mathbf{N} \cdot \mathbf{N} = 1$. Then
 $|\mathbf{a}(t)|^2 = \mathbf{a} \cdot \mathbf{a} = (a_N \mathbf{N} + a_T \mathbf{T}) \cdot (a_N \mathbf{N} + a_T \mathbf{T}) = a_N^2 \mathbf{N} \cdot \mathbf{N} + 2a_N a_T \mathbf{N} \cdot \mathbf{T} + a_T^2 \mathbf{T} \cdot \mathbf{T} = a_N^2 + a_T^2$.

Exercises 9.4

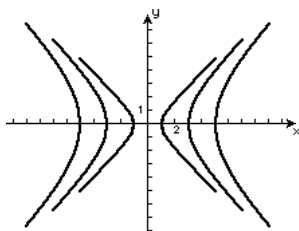
1. $y = -\frac{1}{2}x + C$



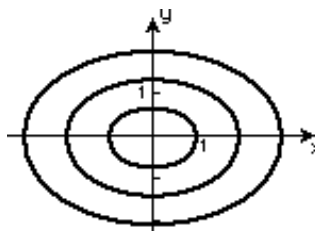
2. $x = y^2 - c$



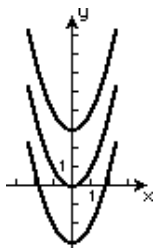
3. $x^2 - y^2 = 1 + c^2$



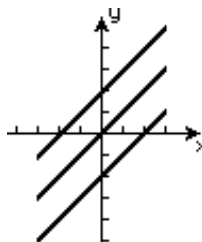
4. $4x^2 + 9y^2 = 36 - c^2, -6 \leq c \leq 6$



5. $y = x^2 + \ln c, c > 0$



6. $y = x + \tan c, -\pi/2 < c < \pi/2$



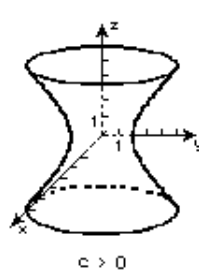
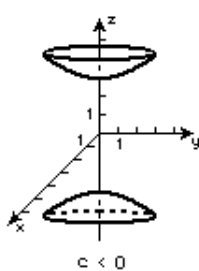
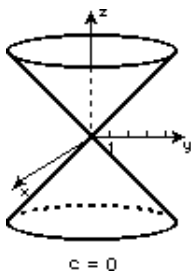
7. $x^2/9 + z^2/4 = c$; elliptical cylinder

8. $x^2 + y^2 + z^2 = c$; sphere

9. $x^2 + 3y^2 + 6z^2 = c$; ellipsoid

10. $4y - 2z + 1 = c$; plane

11.



12. Setting $x = -4$, $y = 2$, and $z = -3$ in $x^2/16 + y^2/4 + z^2/9 = c$ we obtain $c = 3$. The equation of the surface is $x^2/16 + y^2/4 + z^2/9 = 3$. Setting $y = z = 0$ we find the x -intercepts are $\pm 4\sqrt{3}$. Similarly, the y -intercepts are $\pm 2\sqrt{3}$ and the z -intercepts are $\pm 3\sqrt{3}$.

13. $z_x = 2x - y^2$; $z_y = -2xy + 20y^4$

14. $z_x = -3x^2 + 12xy^3$; $z_y = 18x^2y^2 + 10y$

15. $z_x = 20x^3y^3 - 2xy^6 + 30x^4$; $z_y = 15x^4y^2 - 6x^2y^5 - 4$

16. $z_x = 3x^2y^2 \sec^2(x^3y^2)$; $z_y = 2x^3y \sec^2(x^3y^2)$

Exercises 9.4

17. $z_x = \frac{2}{\sqrt{x}(3y^2+1)}$; $z_y = -\frac{24y\sqrt{x}}{(3y^2+1)^2}$
18. $z_x = 12x^2 - 10x + 8$; $z_y = 0$
19. $z_x = -(x^3 - y^2)^{-2}(3x^2) = -3x^2(x^3 - y^2)^{-2}$; $z_y = -(x^3 - y^2)^{-2}(-2y) = 2y(x^3 - y^2)^{-2}$
20. $z_x = 6(-x^4 + 7y^2 + 3y)^5(-4x^3) = -24x^3(-x^4 + 7y^2 + 3y)^5$; $z_y = 6(-x^4 + 7y^2 + 3y)^5(14y + 3)$
21. $z_x = 2(\cos 5x)(-\sin 5x)(5) = -10 \sin 5x \cos 5x$; $z_y = 2(\sin 5y)(\cos 5y)(5) = 10 \sin 5y \cos 5y$
22. $z_x = (2x \tan^{-1} y^2)e^{x^2 \tan^{-1} y^2}$; $z_y = \frac{2x^2 y}{1+y^4} e^{x^2 \tan^{-1} y^2}$
23. $f_x = x(3x^2 y e^{x^3 y} + e^{x^3 y}) = (3x^3 y + 1)e^{x^3 y}$; $f_y = x^4 e^{x^3 y}$
24. $f_\theta = \phi^2 \left(\cos \frac{\theta}{\phi} \right) \left(\frac{1}{\phi} \right) = \phi \cos \frac{\theta}{\phi}$; $f_\phi = \phi^2 \left(\cos \frac{\theta}{\phi} \right) \left(-\frac{\theta}{\phi^2} \right) + 2\phi \sin \frac{\theta}{\phi} = -\theta \cos \frac{\theta}{\phi} + 2\phi \sin \frac{\theta}{\phi}$
25. $f_x = \frac{(x+2y)3 - (3x-y)}{(x+2y)^2} = \frac{7y}{(x+2y)^2}$; $f_y = \frac{(x+2y)(-1) - (3x-y)(2)}{(x+2y)^2} = \frac{-7x}{(x+2y)^2}$
26. $f_x = \frac{(x^2 - y^2)^2 y - xy[2(x^2 - y^2)2x]}{(x^2 - y^2)^4} = \frac{-3x^2 y - y^3}{(x^2 - y^2)^3}$;
 $f_y = \frac{(x^2 - y^2)^2 x - xy[2(x^2 - y^2)(-2y)]}{(x^2 - y^2)^4} = \frac{3xy^2 + x^3}{(x^2 - y^2)^3}$
27. $g_u = \frac{8u}{4u^2 + 5v^3}$; $g_v = \frac{15v^2}{4u^2 + 5v^3}$
28. $h_r = \frac{1}{2s\sqrt{r}} + \frac{\sqrt{s}}{r^2}$; $h_s = -\frac{\sqrt{r}}{s^2} - \frac{1}{2r\sqrt{s}}$
29. $w_x = \frac{y}{\sqrt{x}}$; $w_y = 2\sqrt{x} - y \left(\frac{1}{z} e^{y/z} \right) - e^{y/z} = 2\sqrt{x} - \left(\frac{y}{z} + 1 \right) e^{y/z}$; $w_z = -y e^{y/z} \left(-\frac{y}{z^2} \right) = \frac{y^2}{z^2} e^{y/z}$
30. $w_x = xy \left(\frac{1}{x} \right) + (\ln xz)y = y + y \ln xz$; $w_y = x \ln xz$; $w_z = \frac{xy}{z}$
31. $F_u = 2uw^2 - v^3 - vwt^2 \sin(ut^2)$; $F_v = -3uv^2 + w \cos(ut^2)$;
 $F_x = 4(2x^2 t)^3(4xt) = 16xt(2x^2 t)^3 = 128x^7 t^4$; $F_t = -2uvw t \sin(ut^2) + 64x^8 t^3$
32. $G_p = r^4 s^5 (p^2 q^3)^{r^4 s^5 - 1} (2pq^3) = 2pq^3 r^4 s^5 (p^2 q^3)^{r^4 s^5 - 1}$;
 $G_q = r^4 s^5 (p^2 q^3)^{r^4 s^5 - 1} (3p^2 q^2) = 3p^2 q^2 r^4 s^5 (p^2 q^3)^{r^4 s^5 - 1}$; $G_r = (p^2 q^3)^{r^4 s^5} (4r^3 s^5) \ln(p^2 q^3)$;
 $G_s = (p^2 q^3)^{r^4 s^5} (5r^4 s^4) \ln(p^2 q^3)$
33. $\frac{\partial z}{\partial x} = \frac{2x}{x^2 + y^2}$, $\frac{\partial^2 z}{\partial x^2} = \frac{(x^2 + y^2)2 - 2x(2x)}{(x^2 + y^2)^2} = \frac{2y^2 - 2x^2}{(x^2 + y^2)^2}$; $\frac{\partial z}{\partial y} = \frac{2y}{x^2 + y^2}$,
 $\frac{\partial^2 z}{\partial y^2} = \frac{(x^2 + y^2)2 - 2y(2y)}{(x^2 + y^2)^2} = \frac{2x^2 - 2y^2}{(x^2 + y^2)^2}$; $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = \frac{2y^2 - 2x^2 + 2x^2 - 2y^2}{(x^2 + y^2)^2} = 0$
34. $\frac{\partial z}{\partial x} = e^{x^2 - y^2}(-2y \sin 2xy) + 2xe^{x^2 - y^2} \cos 2xy$
 $\frac{\partial^2 z}{\partial x^2} = e^{x^2 - y^2}(-4y^2 \cos 2xy - 8xy \sin 2xy + 4x^2 \cos 2xy + 2 \cos 2xy)$
 $\frac{\partial z}{\partial y} = e^{x^2 - y^2}(-2x \sin 2xy) - 2ye^{x^2 - y^2} \cos 2xy$
 $\frac{\partial^2 z}{\partial y^2} = e^{x^2 - y^2}(-4x^2 \cos 2xy + 8xy \sin 2xy + 4y^2 \cos 2xy - 2 \cos 2xy)$

Adding the second partial derivatives gives

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = [-4(y^2 + x^2) \cos 2xy + 4(x^2 + y^2) \cos 2xy] = 0.$$

$$35. \quad \frac{\partial u}{\partial x} = \cos at \cos x, \quad \frac{\partial^2 u}{\partial x^2} = -\cos at \sin x; \quad \frac{\partial u}{\partial t} = -a \sin at \sin x, \quad \frac{\partial^2 u}{\partial t^2} = -a^2 \cos at \sin x;$$

$$a^2 \frac{\partial^2 u}{\partial x^2} = a^2(-\cos at \sin x) = \frac{\partial^2 u}{\partial t^2}$$

$$36. \quad \frac{\partial u}{\partial x} = -\sin(x+at) + \cos(x-at), \quad \frac{\partial^2 u}{\partial x^2} = -\cos(x+at) - \sin(x-at);$$

$$\frac{\partial u}{\partial t} = -a \sin(x+at) - a \cos(x-at), \quad \frac{\partial^2 u}{\partial t^2} = -a^2 \cos(x+at) - a^2 \sin(x-at);$$

$$a^2 \frac{\partial^2 u}{\partial x^2} = -a^2 \cos(x+at) - a^2 \sin(x-at) = \frac{\partial^2 u}{\partial t^2}$$

$$37. \quad \frac{\partial C}{\partial x} = -\frac{2x}{kt} t^{-1/2} e^{-x^2/kt}, \quad \frac{\partial^2 C}{\partial x^2} = \frac{4x^2}{k^2 t^2} t^{-1/2} e^{-x^2/kt} - \frac{2}{kt} t^{-1/2} e^{-x^2/kt};$$

$$\frac{\partial C}{\partial t} = t^{-1/2} \frac{x^2}{kt^2} e^{-x^2/kt} - \frac{t^{-3/2}}{2} e^{-x^2/kt}; \quad \frac{k}{4} \frac{\partial^2 C}{\partial x^2} = \frac{x^2}{kt^2} t^{-1/2} e^{-x^2/kt} - \frac{t^{-1/2}}{2t} e^{-x^2/kt} = \frac{\partial C}{\partial t}$$

$$38. \quad (a) \quad P_v = -k(T/V^2)$$

$$(b) \quad PV = kt, \quad PV_T = k, \quad V_T = k/P$$

$$(c) \quad PV = kT, \quad V = kT_p, \quad T_p = V/k$$

$$39. \quad z_x = v^2 e^{uv^2} (3x^2) + 2uve^{uv^2} (1) = 3x^2 v^2 e^{uv^2} + 2uve^{uv^2}; \quad z_y = v^2 e^{uv^2} (0) + 2uve^{uv^2} (-2y) = -4yuve^{uv^2}$$

$$40. \quad z_x = (2u \cos 4v)(2xy^3) - (4u^2 \sin 4v)(3x^2) = 4xy^3 u \cos 4v - 12x^2 u^2 \sin 4v$$

$$z_y = (2u \cos 4v)(3x^2 y^2) - (4v^2 \sin 4v)(3y^2) = 6x^2 y^2 u \cos 4v - 12y^2 u^2 \sin 4v$$

$$41. \quad z_u = 4(4u^3) - 10y[2(2u-v)(2)] = 16u^3 - 40(2u-v)y$$

$$z_v = 4(-24v^2) - 10y[2(2u-v)(-1)] = -96v^2 + 20(2u-v)y$$

$$42. \quad z_u = \frac{2y}{(x+y)^2} \left(\frac{1}{v} \right) + \frac{-2x}{(x+y)^2} \left(-\frac{v^2}{u^2} \right) = \frac{2y}{v(x+y)^2} + \frac{2xv^2}{u^2(x+y)^2}$$

$$z_v = \frac{2y}{(x+y)^2} \left(-\frac{u}{v^2} \right) + \frac{-2x}{(x+y)^2} \left(\frac{2v}{u} \right) = -\frac{2yu}{v^2(x+y)^2} - \frac{4xv}{u(x+y)^2}$$

$$43. \quad w_t = \frac{3}{2}(u^2 + v^2)^{1/2}(2u)(-e^{-t} \sin \theta) + \frac{3}{2}(u^2 + v^2)^{1/2}(2v)(-e^{-t} \cos \theta)$$

$$= -3u(u^2 + v^2)^{1/2} e^{-t} \sin \theta - 3v(u^2 + v^2)^{1/2} e^{-t} \cos \theta$$

$$w_\theta = \frac{3}{2}(u^2 + v^2)^{1/2}(2u)e^{-t} \cos \theta + \frac{3}{2}(u^2 + v^2)^{1/2}(2v)(-e^{-t} \sin \theta)$$

$$= 3u(u^2 + v^2)^{1/2} e^{-t} \cos \theta - 3v(u^2 + v^2)^{1/2} e^{-t} \sin \theta$$

$$44. \quad w_r = \frac{v/2\sqrt{uv}}{1+uv}(2r) + \frac{u/2\sqrt{uv}}{1+uv}(2rs^2) = \frac{rv}{\sqrt{uv}(1+uv)} + \frac{rs^2u}{\sqrt{uv}(1+uv)}$$

$$w_s = \frac{v/2\sqrt{uv}}{1+uv}(-2s) + \frac{u/2\sqrt{uv}}{1+uv}(2r^2s) = \frac{-sv}{\sqrt{uv}(1+uv)} + \frac{r^2su}{\sqrt{uv}(1+uv)}$$

$$45. \quad R_u = s^2 t^4 (e^{v^2}) + 2rst^4 (-2uve^{-u^2}) + 4rs^2 t^3 (2uv^2 e^{u^2 v^2}) = s^2 t^4 e^{v^2} - 4uvrst^4 e^{-u^2} + 8uv^2 rs^2 t^3 e^{u^2 v^2}$$

$$R_v = s^2 t^4 (2uve^{v^2}) + 2rst^4 (e^{-u^2}) + 4rs^2 t^3 (2u^2 ve^{u^2 v^2}) = 2s^2 t^4 uve^{v^2} + 2rst^4 e^{-u^2} + 8rs^2 t^3 u^2 ve^{u^2 v^2}$$

$$46. \quad Q_x = \frac{1}{P} \left(\frac{t^2}{\sqrt{1-x^2}} \right) + \frac{1}{q} \left(\frac{1}{t^2} \right) + \frac{1}{r} \left(\frac{1/t}{1+(x/t)^2} \right) = \frac{t^2}{p\sqrt{1-x^2}} + \frac{1}{qt^2} + \frac{t}{r(t^2+x^2)}$$

Exercises 9.4

$$Q_t = \frac{1}{p}(2t \sin^{-1} x) + \frac{1}{q} \left(-\frac{2x}{t^3} \right) + \frac{1}{r} \left(\frac{-x/t^2}{1 + (x/t)^2} \right) = \frac{2t \sin^{-1} x}{p} - \frac{2x}{qt^3} - \frac{x}{r(t^2 + x^2)}$$

$$\begin{aligned} 47. \quad w_t &= \frac{2x}{2\sqrt{x^2 + y^2}} \frac{u}{rs + tu} + \frac{2y}{2\sqrt{x^2 + y^2}} \frac{\cosh rs}{u} = \frac{xu}{\sqrt{x^2 + y^2}(rs + tu)} + \frac{y \cosh rs}{u\sqrt{x^2 + y^2}} \\ w_r &= \frac{2x}{2\sqrt{x^2 + y^2}} \frac{s}{rs + tu} + \frac{2y}{2\sqrt{x^2 + y^2}} \frac{st \sinh rs}{u} = \frac{xs}{\sqrt{x^2 + y^2}(rs + tu)} + \frac{yst \sinh rs}{u\sqrt{x^2 + y^2}} \\ w_u &= \frac{2x}{2\sqrt{x^2 + y^2}} \frac{t}{rs + tu} + \frac{2y}{2\sqrt{x^2 + y^2}} \frac{-t \cosh rs}{u^2} = \frac{xt}{\sqrt{x^2 + y^2}(rs + tu)} - \frac{yt \cosh rs}{u^2\sqrt{x^2 + y^2}} \end{aligned}$$

$$\begin{aligned} 48. \quad s_\phi &= 2pe^{3\theta} + 2q[-\sin(\phi + \theta)] - 2r\theta^2 + 4(2) = 2pe^{3\theta} - 2q\sin(\phi + \theta) - 2r\theta^2 + 8 \\ s_\theta &= 2p(3\phi e^{3\theta}) + 2q[-\sin(\phi + \theta)] - 2r(2\phi\theta) + 4(8) = 6p\phi e^{3\theta} - 2q\sin(\phi + \theta) - 4r\phi\theta + 32 \end{aligned}$$

$$49. \quad \frac{dz}{dt} = \frac{2u}{u^2 + v^2}(2t) + \frac{2v}{u^2 + v^2}(-2t^{-3}) = \frac{4ut - 4vt^{-3}}{u^2 + v^2}$$

$$50. \quad \frac{dz}{dt} = (3u^2v - v^4)(-5e^{-5t}) + (u^3 - 4uv^3)(5 \sec 5t \tan 5t) = -5(3u^2v - v^4)e^{-5t} + 5(u^3 - 4uv^3) \sec 5t \tan 5t$$

$$\begin{aligned} 51. \quad \frac{dw}{dt} &= -3\sin(3u + 4v)(2) - 4\sin(3u + 4v)(-1); \quad u(\pi) = 5\pi/2, \quad v(\pi) = -5\pi/4 \\ \frac{dw}{dt} \Big|_\pi &= -6\sin\left(\frac{15\pi}{2} - 5\pi\right) + 4\sin\left(\frac{15\pi}{2} - 5\pi\right) = -2\sin\frac{5\pi}{2} = -2 \end{aligned}$$

$$52. \quad \frac{dw}{dt} = ye^{xy} \left[\frac{-8}{(2t+1)^2} \right] + xe^{xy}(3); \quad x(0) = 4, \quad y(0) = 5; \quad \frac{dw}{dt} \Big|_0 = 5e^{20}(-8) + 4e^{20}(3) = -28e^{20}$$

$$53. \quad \text{With } x = r \cos \theta \text{ and } y = r \sin \theta$$

$$\begin{aligned} \frac{\partial u}{\partial r} &= \frac{\partial u}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial u}{\partial y} \frac{\partial y}{\partial r} = \frac{\partial u}{\partial x} \cos \theta + \frac{\partial u}{\partial y} \sin \theta \\ \frac{\partial^2 u}{\partial r^2} &= \frac{\partial^2 u}{\partial x^2} \frac{\partial x}{\partial r} \cos \theta + \frac{\partial^2 u}{\partial y^2} \frac{\partial y}{\partial r} \sin \theta = \frac{\partial^2 u}{\partial x^2} \cos^2 \theta + \frac{\partial^2 u}{\partial y^2} \sin^2 \theta \\ \frac{\partial u}{\partial \theta} &= \frac{\partial u}{\partial x} \frac{\partial x}{\partial \theta} + \frac{\partial u}{\partial y} \frac{\partial y}{\partial \theta} = \frac{\partial u}{\partial x}(-r \sin \theta) + \frac{\partial u}{\partial y}(r \cos \theta) \\ \frac{\partial^2 u}{\partial \theta^2} &= \frac{\partial u}{\partial x}(-r \cos \theta) + \frac{\partial^2 u}{\partial x^2} \frac{\partial x}{\partial \theta}(-r \sin \theta) + \frac{\partial u}{\partial y}(-r \sin \theta) + \frac{\partial^2 u}{\partial y^2} \frac{\partial y}{\partial \theta}(r \cos \theta) \\ &= -r \frac{\partial u}{\partial x} \cos \theta + r^2 \frac{\partial^2 u}{\partial x^2} \sin^2 \theta - r \frac{\partial u}{\partial y} \sin \theta + r^2 \frac{\partial^2 u}{\partial y^2} \cos^2 \theta. \end{aligned}$$

Using $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$, we have

$$\begin{aligned} \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} &= \frac{\partial^2 u}{\partial x^2} \cos^2 \theta + \frac{\partial^2 u}{\partial y^2} \sin^2 \theta + \frac{1}{r} \left(\frac{\partial u}{\partial x} \cos \theta + \frac{\partial u}{\partial y} \sin \theta \right) \\ &\quad + \frac{1}{r^2} \left(-r \frac{\partial u}{\partial x} \cos \theta + r^2 \frac{\partial^2 u}{\partial x^2} \sin^2 \theta - r \frac{\partial u}{\partial y} \sin \theta + r^2 \frac{\partial^2 u}{\partial y^2} \cos^2 \theta \right) \\ &= \frac{\partial^2 u}{\partial x^2} (\cos^2 \theta + \sin^2 \theta) + \frac{\partial^2 u}{\partial y^2} (\sin^2 \theta + \cos^2 \theta) + \frac{\partial u}{\partial x} \left(\frac{1}{r} \cos \theta - \frac{1}{r} \cos \theta \right) \\ &\quad + \frac{\partial u}{\partial y} \left(\frac{1}{r} \sin \theta - \frac{1}{r} \sin \theta \right) \\ &= \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0. \end{aligned}$$

$$\begin{aligned}
 54. \quad \frac{dP}{dt} &= \frac{(V - 0.0427)(0.08)dT/dt}{(V - 0.0427)^2} - \frac{0.08T(dV/dt)}{(V - 0.0427)^2} + \frac{3.6}{V^3} \frac{dV}{dt} \\
 &= \frac{0.08}{V - 0.0427} \frac{dT}{dt} + \left(\frac{3.6}{V^3} - \frac{0.08T}{(V - 0.0427)^2} \right) \frac{dV}{dt}
 \end{aligned}$$

55. Since $dT/dT = 1$ and $\partial P/\partial T = 0$,

$$0 = F_T = \frac{\partial F}{\partial P} \frac{\partial P}{\partial T} + \frac{\partial F}{\partial V} \frac{\partial V}{\partial T} + \frac{\partial F}{\partial T} \frac{dT}{dT} \implies \frac{\partial V}{\partial T} = -\frac{\partial F/\partial T}{\partial F/\partial V} = -\frac{1}{\partial T/\partial V}.$$

56. We are given $dE/dt = 2$ and $dR/dt = -1$. Then $\frac{dI}{dt} = \frac{\partial I}{\partial E} \frac{dE}{dt} + \frac{\partial I}{\partial R} \frac{dR}{dt} = \frac{1}{R}(2) - \frac{E}{R^2}(-1)$, and

$$\text{when } E = 60 \text{ and } R = 50, \quad \frac{dI}{dt} = \frac{2}{50} + \frac{60}{50^2} = \frac{1}{25} + \frac{3/5}{25} = \frac{8}{125} \text{ amp/min.}$$

57. Since the height of the triangle is $x \sin \theta$, the area is given by $A = \frac{1}{2}xy \sin \theta$. Then

$$\frac{dA}{dt} \frac{\partial A}{\partial x} \frac{dx}{dt} + \frac{\partial A}{\partial y} \frac{dy}{dt} + \frac{\partial A}{\partial \theta} \frac{d\theta}{dt} = \frac{1}{2}y \sin \theta \frac{dx}{dt} + \frac{1}{2}x \sin \theta \frac{dy}{dt} + \frac{1}{2}xy \cos \theta \frac{d\theta}{dt}.$$

When $x = 10$, $y = 8$, $\theta = \pi/6$, $dx/dt = 0.3$, $dy/dt = 0.5$, and $d\theta/dt = 0.1$,

$$\begin{aligned}
 \frac{dA}{dt} &= \frac{1}{2}(8) \left(\frac{1}{2} \right) (0.3) + \frac{1}{2}(10) \left(\frac{1}{2} \right) (0.5) + \frac{1}{2}(10)(8) \left(\frac{\sqrt{3}}{2} \right) (0.1) \\
 &= 0.6 + 1.25 + 2\sqrt{3} = 1.85 + 2\sqrt{3} \approx 5.31 \text{ cm}^2/\text{s}.
 \end{aligned}$$

$$\begin{aligned}
 58. \quad \frac{dw}{dt} &= \frac{\partial w}{\partial x} \frac{dx}{dt} + \frac{\partial w}{\partial y} \frac{dy}{dt} + \frac{\partial w}{\partial z} \frac{dz}{dt} = \frac{x dx/dt + y dy/dt + z dz/dt}{\sqrt{x^2 + y^2 + z^2}} = \frac{-4x \sin t + 4y \cos t + 5z}{\sqrt{16 \cos^2 t + 16 \sin^2 t + 25t^2}} \\
 &= \frac{-16 \sin t \cos t + 16 \sin t \cos t + 25t}{\sqrt{16 + 25t^2}} = \frac{25t}{\sqrt{16 + 25t^2}}
 \end{aligned}$$

$$\left. \frac{dw}{dt} \right|_{t=5\pi/2} = \frac{125\pi/2}{\sqrt{16 + 625\pi^2/4}} = \frac{125\pi}{\sqrt{64 + 625\pi^2}} \approx 4.9743$$

Exercises 9.5

1. $\nabla f = (2x - 3x^2y^2)\mathbf{i} + (4y^3 - 2x^3y)\mathbf{j}$

2. $\nabla f = 4xye^{-2x^2y}\mathbf{i} + (1 + 2x^2e^{-2x^2y})\mathbf{j}$

3. $\nabla F = \frac{y^2}{z^3}\mathbf{i} + \frac{2xy}{z^3}\mathbf{j} - \frac{3xy^2}{z^4}\mathbf{k}$

4. $\nabla F = y \cos yz \mathbf{i} + (x \cos yz - xyz \sin yz) \mathbf{j} - xy^2 \sin yz \mathbf{k}$

5. $\nabla f = 2x\mathbf{i} - 8y\mathbf{j}$; $\nabla f(2, 4) = 4\mathbf{i} - 32\mathbf{j}$

6. $\nabla f = \frac{3x^2}{2\sqrt{x^3y - y^4}}\mathbf{i} + \frac{x^3 - 4y^3}{2\sqrt{x^3y - y^4}}\mathbf{j}$; $\nabla f(3, 2) = \frac{27}{\sqrt{38}}\mathbf{i} - \frac{5}{2\sqrt{38}}\mathbf{j}$

7. $\nabla F = 2xz^2 \sin 4y\mathbf{i} + 4x^2z^2 \cos 4y\mathbf{j} + 2x^2z \sin 4y\mathbf{k}$

$$\nabla F(-2, \pi/3, 1) = -4 \sin \frac{4\pi}{3} \mathbf{i} + 16 \cos \frac{4\pi}{3} \mathbf{j} + 8 \sin \frac{4\pi}{3} \mathbf{k} = 2\sqrt{3}\mathbf{i} - 8\mathbf{j} - 4\sqrt{3}\mathbf{k}$$

8. $\nabla F = \frac{2x}{x^2 + y^2 + z^2}\mathbf{i} + \frac{2y}{x^2 + y^2 + z^2}\mathbf{j} + \frac{2z}{x^2 + y^2 + z^2}\mathbf{k}$; $\nabla F(-4, 3, 5) = -\frac{4}{25}\mathbf{i} + \frac{3}{25}\mathbf{j} + \frac{1}{5}\mathbf{k}$

Exercises 9.5

9. $D_{\mathbf{u}}f(x, y) = \lim_{h \rightarrow 0} \frac{f(x + h\sqrt{3}/2, y + h/2) - f(x, y)}{h} = \lim_{h \rightarrow 0} \frac{(x + h\sqrt{3}/2)^2 + (y + h/2)^2 - x^2 - y^2}{h}$
 $= \lim_{h \rightarrow 0} \frac{h\sqrt{3}x + 3h^2/4 + hy + h^2/4}{h} = \lim_{h \rightarrow 0} (\sqrt{3}x + 3h/4 + y + h/4) = \sqrt{3}x + y$
10. $D_{\mathbf{u}}f(x, y) = \lim_{h \rightarrow 0} \frac{f(x + h\sqrt{2}/2, y + h\sqrt{2}/2) - f(x, y)}{h} = \lim_{h \rightarrow 0} \frac{3x + 3h\sqrt{2}/2 - (y + h\sqrt{2}/2)^2 - 3x + y^2}{h}$
 $= \lim_{h \rightarrow 0} \frac{3h\sqrt{2}/2 - h\sqrt{2}y - h^2/2}{h} = \lim_{h \rightarrow 0} (3\sqrt{2}/2 - \sqrt{2}y - h/2) = 3\sqrt{2}/2 - \sqrt{2}y$
11. $\mathbf{u} = \frac{\sqrt{3}}{2}\mathbf{i} + \frac{1}{2}\mathbf{j}$; $\nabla f = 15x^2y^6\mathbf{i} + 30x^3y^5\mathbf{j}$; $\nabla f(-1, 1) = 15\mathbf{i} - 30\mathbf{j}$;
 $D_{\mathbf{u}}f(-1, 1) = \frac{15\sqrt{3}}{2} - 15 = \frac{15}{2}(\sqrt{3} - 2)$
12. $\mathbf{u} = \frac{\sqrt{2}}{2}\mathbf{i} + \frac{\sqrt{2}}{2}\mathbf{j}$; $\nabla f = (4 + y^2)\mathbf{i} + (2xy - 5)\mathbf{j}$; $\nabla f(3, -1) = 5\mathbf{i} - 11\mathbf{j}$;
 $D_{\mathbf{u}}f(3, -1) = \frac{5\sqrt{2}}{2} - \frac{11\sqrt{2}}{2} = -3\sqrt{2}$
13. $\mathbf{u} = \frac{\sqrt{10}}{10}\mathbf{i} - \frac{3\sqrt{10}}{10}\mathbf{j}$; $\nabla f = \frac{-y}{x^2 + y^2}\mathbf{i} + \frac{x}{x^2 + y^2}\mathbf{j}$; $\nabla f(2, -2) = \frac{1}{4}\mathbf{i} + \frac{1}{4}\mathbf{j}$
 $D_{\mathbf{u}}f(2, -2) = \frac{\sqrt{10}}{40} - \frac{3\sqrt{10}}{40} = -\frac{\sqrt{10}}{20}$
14. $\mathbf{u} = \frac{6}{10}\mathbf{i} + \frac{8}{10}\mathbf{j} = \frac{3}{5}\mathbf{i} + \frac{4}{5}\mathbf{j}$; $\nabla f = \frac{y^2}{(x + y)^2}\mathbf{i} + \frac{x^2}{(x + y)^2}\mathbf{j}$; $\nabla f(2, -1) = \mathbf{i} + 4\mathbf{j}$
 $D_{\mathbf{u}}f(2, -1) = \frac{3}{5} + \frac{16}{5} = \frac{19}{5}$
15. $\mathbf{u} = (2\mathbf{i} + \mathbf{j})/\sqrt{5}$; $\nabla f = 2y(xy + 1)\mathbf{i} + 2x(xy + 1)\mathbf{j}$; $\nabla f(3, 2) = 28\mathbf{i} + 42\mathbf{j}$
 $D_{\mathbf{u}}f(3, 2) = \frac{2(28)}{\sqrt{5}} + \frac{42}{\sqrt{5}} = \frac{98}{\sqrt{5}}$
16. $\mathbf{u} = -\mathbf{i}$; $\nabla f = 2x \tan y \mathbf{i} + x^2 \sec^2 y \mathbf{j}$; $\nabla f(1/2, \pi/3) = \sqrt{3}\mathbf{i} + \mathbf{j}$; $D_{\mathbf{u}}f(1/2, \pi/3) = -\sqrt{3}$
17. $\mathbf{u} = \frac{1}{\sqrt{2}}\mathbf{j} + \frac{1}{\sqrt{2}}\mathbf{k}$; $\nabla F = 2xy^2(2z + 1)^2\mathbf{i} + 2x^2y(2z + 1)^2\mathbf{j} + 4x^2y^2(2z + 1)\mathbf{k}$
 $\nabla F(1, -1, 1) = 18\mathbf{i} - 18\mathbf{j} + 12\mathbf{k}$; $D_{\mathbf{u}}F(1, -1, 1) = -\frac{18}{\sqrt{2}} + \frac{12}{\sqrt{2}} = -\frac{6}{\sqrt{2}} = -3\sqrt{2}$
18. $\mathbf{u} = \frac{1}{\sqrt{6}}\mathbf{i} - \frac{2}{\sqrt{6}}\mathbf{j} + \frac{1}{\sqrt{6}}\mathbf{k}$; $\nabla F = \frac{2x}{z^2}\mathbf{i} - \frac{2y}{z^2}\mathbf{j} + \frac{2y^2 - 2x^2}{z^3}\mathbf{k}$; $\nabla F(2, 4, -1) = 4\mathbf{i} - 8\mathbf{j} - 24\mathbf{k}$
 $D_{\mathbf{u}}F(2, 4, -1) = \frac{4}{\sqrt{6}} - \frac{16}{\sqrt{6}} - \frac{24}{\sqrt{6}} = -6\sqrt{6}$
19. $\mathbf{u} = -\mathbf{k}$; $\nabla F = \frac{xy}{\sqrt{x^2y + 2y^2z}}\mathbf{i} + \frac{x^2 + 4z}{2\sqrt{x^2y + 2y^2z}}\mathbf{j} + \frac{y^2}{\sqrt{x^2y + 2y^2z}}\mathbf{k}$
 $\nabla F(-2, 2, 1) = -\mathbf{i} + \mathbf{j} + \mathbf{k}$; $D_{\mathbf{u}}F(-2, 2, 1) = -1$
20. $\mathbf{u} = -(4\mathbf{i} - 4\mathbf{j} + 2\mathbf{k})/\sqrt{36} = -\frac{2}{3}\mathbf{i} + \frac{2}{3}\mathbf{j} - \frac{1}{3}\mathbf{k}$; $\nabla F = 2\mathbf{i} - 2y\mathbf{j} + 2z\mathbf{k}$; $\nabla F(4, -4, 2) = 2\mathbf{i} + 8\mathbf{j} + 4\mathbf{k}$
 $D_{\mathbf{u}}F(4, -4, 2) = -\frac{4}{3} + \frac{16}{3} - \frac{4}{3} = \frac{8}{3}$
21. $\mathbf{u} = (-4\mathbf{i} - \mathbf{j})/\sqrt{17}$; $\nabla f = 2(x - y)\mathbf{i} - 2(x - y)\mathbf{j}$; $\nabla f(4, 2) = 4\mathbf{i} - 4\mathbf{j}$; $D_{\mathbf{u}}F(4, 2) = -\frac{16}{\sqrt{17}} \frac{4}{\sqrt{17}} = -\frac{12}{\sqrt{17}}$

22. $\mathbf{u} = (-2\mathbf{i} + 5\mathbf{j})/\sqrt{29}$; $\nabla f = (3x^2 - 5y)\mathbf{i} - (5x - 2y)\mathbf{j}$; $\nabla f(1, 1) = -2\mathbf{i} - 3\mathbf{j}$;
 $D_{\mathbf{u}}f(1, 1) = \frac{4}{\sqrt{29}} - \frac{15}{\sqrt{29}} = -\frac{11}{\sqrt{29}}$
23. $\nabla f = 2e^{2x} \sin y \mathbf{i} + e^{2x} \cos y \mathbf{j}$; $\nabla f(0, \pi/4) = \sqrt{2}\mathbf{i} + \frac{\sqrt{2}}{2}\mathbf{j}$
 The maximum $D_{\mathbf{u}}$ is $[(\sqrt{2})^2 + (\sqrt{2}/2)^2]^{1/2} = \sqrt{5/2}$ in the direction $\sqrt{2}\mathbf{i} + (\sqrt{2}/2)\mathbf{j}$.
24. $\nabla f = (xye^{x-y} + ye^{x-y})\mathbf{i} + (-xye^{x-y} + xe^{x-y})\mathbf{j}$; $\nabla f(5, 5) = 30\mathbf{i} - 20\mathbf{j}$
 The maximum $D_{\mathbf{u}}$ is $[30^2 + (-20)^2]^{1/2} = 10\sqrt{13}$ in the direction $30\mathbf{i} - 20\mathbf{j}$.
25. $\nabla F = (2x + 4z)\mathbf{i} + 2z^2\mathbf{j} + (4x + 4yz)\mathbf{k}$; $\nabla F(1, 2, -1) = -2\mathbf{i} + 2\mathbf{j} - 4\mathbf{k}$
 The maximum $D_{\mathbf{u}}$ is $[(-2)^2 + 2^2 + (-4)^2]^{1/2} = 2\sqrt{6}$ in the direction $-2\mathbf{i} + 2\mathbf{j} - 4\mathbf{k}$.
26. $\nabla F = yz\mathbf{i} + xz\mathbf{j} + xy\mathbf{k}$; $\nabla F(3, 1, -5) = -5\mathbf{i} - 15\mathbf{j} + 3\mathbf{k}$
 The maximum $D_{\mathbf{u}}$ is $[(-5)^2 + (-15)^2 + 3^2]^{1/2} = \sqrt{259}$ in the direction $-5\mathbf{i} - 15\mathbf{j} + 3\mathbf{k}$.
27. $\nabla f = 2x \sec^2(x^2 + y^2)\mathbf{i} + 2y \sec^2(x^2 + y^2)\mathbf{j}$;
 $\nabla f(\sqrt{\pi/6}, \sqrt{\pi/6}) = 2\sqrt{\pi/6} \sec^2(\pi/3)(\mathbf{i} + \mathbf{j}) = 8\sqrt{\pi/6}(\mathbf{i} + \mathbf{j})$
 The minimum $D_{\mathbf{u}}$ is $-8\sqrt{\pi/6}(1^2 + 1^2)^{1/2} = -8\sqrt{\pi/3}$ in the direction $-(\mathbf{i} + \mathbf{j})$.
28. $\nabla f = 3x^2\mathbf{i} - 3y^2\mathbf{j}$; $\nabla f(2, -2) = 12\mathbf{i} - 12\mathbf{j} = 12(\mathbf{i} - \mathbf{j})$
 The minimum $D_{\mathbf{u}}$ is $-12[1^2 + (-1)^2]^{1/2} = -12\sqrt{2}$ in the direction $-(\mathbf{i} - \mathbf{j}) = -\mathbf{i} + \mathbf{j}$.
29. $\nabla F = \frac{\sqrt{z}e^y}{2\sqrt{x}}\mathbf{i} + \sqrt{xz}e^y\mathbf{j} + \frac{\sqrt{x}}{2\sqrt{z}}\mathbf{k}$; $\nabla F(16, 0, 9) = \frac{3}{8}\mathbf{i} + 12\mathbf{j} + \frac{2}{3}\mathbf{k}$. The minimum $D_{\mathbf{u}}$ is
 $-[(3/8)^2 + 12^2 + (2/3)^2]^{1/2} = -\sqrt{83,281}/24$ in the direction $-\frac{3}{8}\mathbf{i} - 12\mathbf{j} - \frac{2}{3}\mathbf{k}$.
30. $\nabla F = \frac{1}{x}\mathbf{i} + \frac{1}{y}\mathbf{j} - \frac{1}{z}\mathbf{k}$; $\nabla F(1/2, 1/6, 1/3) = 2\mathbf{i} + 6\mathbf{j} - 3\mathbf{k}$
 The minimum $D_{\mathbf{u}}$ is $-[2^2 + 6^2 + (-3)^2]^{1/2} = -7$ in the direction $-2\mathbf{i} - 6\mathbf{j} + 3\mathbf{k}$.
31. Using implicit differentiation on $2x^2 + y^2 = 9$ we find $y' = -2x/y$. At $(2, 1)$ the slope of the tangent line is $-2(2)/1 = -4$. Thus, $\mathbf{u} = \pm(\mathbf{i} - 4\mathbf{j})/\sqrt{17}$. Now, $\nabla f = \mathbf{i} + 2y\mathbf{j}$ and $\nabla f(3, 4) = \mathbf{i} + 8\mathbf{j}$. Thus,
 $D_{\mathbf{u}} = \pm(1/\sqrt{17} - 32\sqrt{17}) = \pm 31/\sqrt{17}$.
32. $\nabla f = (2x + y - 1)\mathbf{i} + (x + 2y)\mathbf{j}$; $D_{\mathbf{u}}f(x, y) = \frac{2x + y - 1}{\sqrt{2}} + \frac{x + 2y}{\sqrt{2}} = \frac{3x + 3y - 1}{\sqrt{2}}$
 Solving $(3x + 3y - 1)/\sqrt{2} = 0$ we see that $D_{\mathbf{u}}$ is 0 for all points on the line $3x + 3y = 1$.
33. (a) Vectors perpendicular to $4\mathbf{i} + 3\mathbf{j}$ are $\pm(3\mathbf{i} - 4\mathbf{j})$. Take $\mathbf{u} = \pm\left(\frac{3}{5}\mathbf{i} - \frac{4}{5}\mathbf{j}\right)$.
 (b) $\mathbf{u} = (4\mathbf{i} + 3\mathbf{j})/\sqrt{16 + 9} = \frac{4}{5}\mathbf{i} + \frac{3}{5}\mathbf{j}$
 (c) $\mathbf{u} = -\frac{4}{5}\mathbf{i} - \frac{3}{5}\mathbf{j}$
34. $D_{-\mathbf{u}}f(a, b) = \nabla f(a, b) \cdot (-\mathbf{u}) = -\nabla f(a, b) \cdot \mathbf{u} = -D_{\mathbf{u}}f(a, b) = -6$
35. (a) $\nabla f = (3x^2 - 6xy^2)\mathbf{i} + (-6x^2y + 3y^2)\mathbf{j}$
 $D_{\mathbf{u}}f(x, y) = \frac{3(3x^2 - 6xy^2)}{\sqrt{10}} + \frac{-6x^2y + 3y^2}{\sqrt{10}} = \frac{9x^2 - 18xy^2 - 6x^2y + 3y^2}{\sqrt{10}}$
 (b) $F(x, y) = \frac{3}{\sqrt{10}}(3x^2 - 6xy^2 - 2x^2y + y^2)$; $\nabla F = \frac{3}{\sqrt{10}}[(6x - 6y^2 - 4xy)\mathbf{i} + (-12xy - 2x^2 + 2y)\mathbf{j}]$

Exercises 9.5

$$\begin{aligned} D_{\mathbf{u}}F(x, y) &= \left(\frac{3}{\sqrt{10}}\right) \left(\frac{3}{\sqrt{10}}\right) (6x - 6y^2 - 4xy) + \left(\frac{1}{\sqrt{10}}\right) \left(\frac{3}{\sqrt{10}}\right) (-12xy - 2x^2 + 2y) \\ &= \frac{9}{5}(3x - 3y^2 - 2xy) + \frac{3}{5}(-6xy - x^2 + y) = \frac{1}{5}(27x - 27y^2 - 36xy - 3x^2 + 3y) \end{aligned}$$

$$36. \nabla U = \frac{Gmx}{(x^2 + y^2)^{3/2}} \mathbf{i} + \frac{Gmy}{(x^2 + y^2)^{3/2}} \mathbf{j} = \frac{Gm}{(x^2 + y^2)^{3/2}} (x\mathbf{i} + y\mathbf{j})$$

The maximum and minimum values of $D_{\mathbf{u}}U(x, y)$ are obtained when $\mathbf{u}$ is in the directions ∇U and $-\nabla U$, respectively. Thus, at a point (x, y) , not $(0, 0)$, the directions of maximum and minimum increase in U are $x\mathbf{i} + y\mathbf{j}$ and $-x\mathbf{i} - y\mathbf{j}$, respectively. A vector at (x, y) in the direction $\pm(x\mathbf{i} + y\mathbf{j})$ lies on a line through the origin.

37. $\nabla f = (3x^2 - 12)\mathbf{i} + (2y - 10)\mathbf{j}$. Setting $|\nabla f| = [(3x^2 - 12)^2 + (2y - 10)^2]^{1/2} = 0$, we obtain $3x^2 - 12 = 0$ and $2y - 10 = 0$. The points where $|\nabla f| = 0$ are $(2, 5)$ and $(-2, 5)$.

38. Let $\nabla f(a, b) = \alpha\mathbf{i} + \beta\mathbf{j}$. Then

$$D_{\mathbf{u}}f(a, b) = \nabla f(a, b) \cdot \mathbf{u} = \frac{5}{13}\alpha - \frac{12}{13}\beta = 7 \quad \text{and} \quad D_{\mathbf{v}}f(a, b) = \nabla f(a, b) \cdot \mathbf{v} = \frac{5}{13}\alpha - \frac{12}{13}\beta = 3.$$

Solving for α and β , we obtain $\alpha = 13$ and $\beta = -13/6$. Thus, $\nabla f(a, b) = 13\mathbf{i} - (13/6)\mathbf{j}$.

39. $\nabla T = 4x\mathbf{i} + 2y\mathbf{j}$; $\nabla T(4, 2) = 16\mathbf{i} + 4\mathbf{j}$. The minimum change in temperature (that is, the maximum decrease in temperature) is in the direction $-\nabla T(4, 3) = -16\mathbf{i} - 4\mathbf{j}$.

40. Let $x(t)\mathbf{i} + y(t)\mathbf{j}$ be the vector equation of the path. At (x, y) on this curve, the direction of a tangent vector is $x'(t)\mathbf{i} + y'(t)\mathbf{j}$. Since we want the direction of motion to be $-\nabla T(x, y)$, we have $x'(t)\mathbf{i} + y'(t)\mathbf{j} = -\nabla T(x, y) = 4x\mathbf{i} + 2y\mathbf{j}$. Separating variables in $dx/dt = 4x$, we obtain $dx/x = 4 dt$, $\ln x = 4t + c_1$, and $x = C_1 e^{4t}$. Separating variables in $dy/dt = 2y$, we obtain $dy/y = 2 dt$, $\ln y = 2t + c_2$, and $y = C_2 e^{2t}$. Since $x(0) = 4$ and $y(0) = 2$, we have $x = 4e^{4t}$ and $y = 2e^{2t}$. The equation of the path is $4e^{4t}\mathbf{i} + 2e^{2t}\mathbf{j}$ for $t \geq 0$, or eliminating the parameter, $x = y^2$, $y \geq 0$.

41. Let $x(t)\mathbf{i} + y(t)\mathbf{j}$ be the vector equation of the path. At (x, y) on this curve, the direction of a tangent vector is $x'(t)\mathbf{i} + y'(t)\mathbf{j}$. Since we want the direction of motion to be $\nabla T(x, y)$, we have $x'(t)\mathbf{i} + y'(t)\mathbf{j} = \nabla T(x, y) = -4x\mathbf{i} - 2y\mathbf{j}$. Separating variables in $dx/dt = -4x$ we obtain $dx/x = -4 dt$, $\ln x = -4t + c_1$ and $x = C_1 e^{-4t}$. Separating variables in $dy/dt = -2y$ we obtain $dy/y = -2 dt$, $\ln y = -2t + c_2$ and $y = C_2 e^{-2t}$. Since $x(0) = 3$ and $y(0) = 4$, we have $x = 3e^{-4t}$ and $y = 4e^{-2t}$. The equation of the path is $3e^{-4t}\mathbf{i} + 4e^{-2t}\mathbf{j}$, or eliminating the parameter, $16x = 3y^2$, $y \geq 0$.

42. Substituting $x = 0$, $y = 0$, $z = 1$, and $T = 500$ into $T = k/(x^2 + y^2 + z^2)$ we see that $k = 500$ and $T(x, y, z) = 500/(x^2 + y^2 + z^2)$.

$$(a) \mathbf{u} = \frac{1}{3}\langle 1, -2, -2 \rangle = \frac{1}{3}\mathbf{i} - \frac{2}{3}\mathbf{j} - \frac{2}{3}\mathbf{k}$$

$$\nabla T = -\frac{1000x}{(x^2 + y^2 + z^2)^2} \mathbf{i} - \frac{1000y}{(x^2 + y^2 + z^2)^2} \mathbf{j} - \frac{1000z}{(x^2 + y^2 + z^2)^2} \mathbf{k}$$

$$\nabla T(2, 3, 3) = -\frac{500}{121}\mathbf{i} - \frac{750}{121}\mathbf{j} - \frac{750}{121}\mathbf{k}$$

$$D_{\mathbf{u}}T(2, 3, 3) = \frac{1}{3} \left(-\frac{500}{121} \right) - \frac{2}{3} \left(-\frac{750}{121} \right) - \frac{2}{3} \left(-\frac{750}{121} \right) = \frac{2500}{363}$$

(b) The direction of maximum increase is $\nabla T(2, 3, 3) = -\frac{500}{121}\mathbf{i} - \frac{750}{121}\mathbf{j} - \frac{750}{121}\mathbf{k} = \frac{250}{121}(-2\mathbf{i} - 3\mathbf{j} - 3\mathbf{k})$.

(c) The maximum rate of change of T is $|\nabla T(2, 3, 3)| = \frac{250}{121}\sqrt{4 + 9 + 9} = \frac{250\sqrt{22}}{121}$.

43. Since $\nabla f = f_x(x, y)\mathbf{i} + f_y(x, y)\mathbf{j}$, we have $\partial f/\partial x = 3x^2 + y^3 + ye^{xy}$. Integrating, we obtain $f(x, y) = x^3 + xy^3 + e^{xy} + g(y)$. Then $f_y = 3xy^2 + xe^{xy} + g'(y) = -2y^2 + 3xy^2 + xe^{xy}$. Thus, $g'(y) = -2y^2$, $g(y) = -\frac{2}{3}y^3 + c$, and $f(x, y) = x^3 + xy^3 + e^{xy} - \frac{2}{3}y^3 + C$.

44. Let $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j}$ and $\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j}$. $D_{\mathbf{v}}f = (f_x\mathbf{i} + f_y\mathbf{j}) \cdot \mathbf{v} = v_1f_x + v_2f_y$

$$D_{\mathbf{u}}D_{\mathbf{v}}f = \left[\frac{\partial}{\partial x}(v_1f_x + v_2f_y)\mathbf{i} + \frac{\partial}{\partial y}(v_1f_x + v_2f_y)\mathbf{j} \right] \cdot \mathbf{u} = [(v_1f_{xx} + v_2f_{yx})\mathbf{i} + (v_1f_{xy} + v_2f_{yy})\mathbf{j}] \cdot \mathbf{u}$$

$$= u_1v_1f_{xx} + u_1v_2f_{yx} + u_2v_1f_{xy} + u_2v_2f_{yy}$$

$$D_{\mathbf{u}}f = (f_x\mathbf{i} + f_y\mathbf{j}) \cdot \mathbf{u} = u_1f_x + u_2f_y$$

$$D_{\mathbf{v}}D_{\mathbf{u}}f = \left[\frac{\partial}{\partial x}(u_1f_x + u_2f_y)\mathbf{i} + \frac{\partial}{\partial y}(u_1f_x + u_2f_y)\mathbf{j} \right] \cdot \mathbf{v} = [(u_1f_{xx} + u_2f_{xy})\mathbf{i} + (u_1f_{xy} + u_2f_{yy})\mathbf{j}] \cdot \mathbf{v}$$

$$= u_1v_1f_{xx} + u_2v_1f_{xy} + u_1v_2f_{xy} + u_2v_2f_{yy}$$

Since the second partial derivatives are continuous, $f_{xy} = f_{yx}$ and $D_{\mathbf{u}}D_{\mathbf{v}}f = D_{\mathbf{v}}D_{\mathbf{u}}f$. [Note that this result is a generalization of $f_{xy} = f_{yx}$ since $D_{\mathbf{i}}D_{\mathbf{j}}f = f_{yx}$ and $D_{\mathbf{j}}D_{\mathbf{i}}f = f_{xy}$.]

45. $\nabla(cf) = \frac{\partial}{\partial x}(cf)\mathbf{i} + \frac{\partial}{\partial y}(cf)\mathbf{j} = cf_x\mathbf{i} + cf_y\mathbf{j} = c(f_x\mathbf{i} + f_y\mathbf{j}) = c\nabla f$

46. $\nabla(f + g) = (f_x + g_x)\mathbf{i} + (f_y + g_y)\mathbf{j} = (f_x\mathbf{i} + f_y\mathbf{j}) + (g_x\mathbf{i} + g_y\mathbf{j}) = \nabla f + \nabla g$

47. $\nabla(fg) = (fg_x + f_xg)\mathbf{i} + (fg_y + f_yg)\mathbf{j} = f(g_x\mathbf{i} + g_y\mathbf{j}) + g(f_x\mathbf{i} + f_y\mathbf{j}) = f\nabla g + g\nabla f$

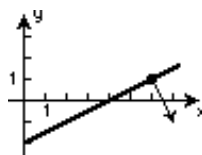
48. $\nabla(f/g) = [(gf_x - fg_x)/g^2]\mathbf{i} + [(gf_y - fg_y)/g^2]\mathbf{j} = g(f_x\mathbf{i} + f_y\mathbf{j})/g^2 - f(g_x\mathbf{i} + g_y\mathbf{j})/g^2$
 $= g\nabla f/g^2 - f\nabla g/g^2 = (g\nabla f - f\nabla g)/g^2$

49. $\nabla \times F = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ f_1 & f_2 & f_3 \end{vmatrix} = \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z} \right) \mathbf{i} + \left(\frac{\partial f_1}{\partial z} - \frac{\partial f_3}{\partial x} \right) \mathbf{j} + \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right) \mathbf{k}$

Exercises 9.6

1. Since $f(6, 1) = 4$, the level curve is $x - 2y = 4$.

$$\nabla f = \mathbf{i} - 2\mathbf{j}; \quad \nabla f(6, 1) = \mathbf{i} - 2\mathbf{j}$$



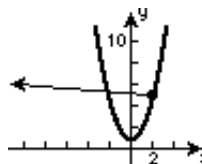
2. Since $f(1, 3) = 5$, the level curve is $y + 2x = 5$ or $y = 5 - 2x$, $x \neq 0$.

$$\nabla f = -\frac{y}{x^2}\mathbf{i} + \frac{1}{x}\mathbf{j}; \quad \nabla f(1, 3) = -3\mathbf{i} + \mathbf{j}$$



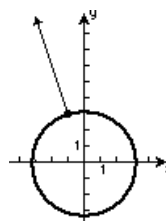
3. Since $f(2, 5) = 1$, the level curve is $y = x^2 + 1$.

$$\nabla f = -2x\mathbf{i} + \mathbf{j}; \quad \nabla f(2, 5) = -10\mathbf{i} + \mathbf{j}$$



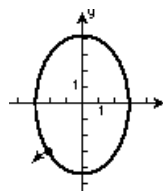
4. Since $f(-1, 3) = 10$, the level curve is $x^2 + y^2 = 10$.

$$\nabla f = 2x\mathbf{i} + 2y\mathbf{j}; \quad \nabla f(-1, 3) = -2\mathbf{i} + 6\mathbf{j}$$



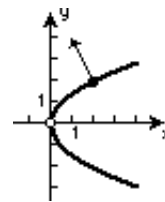
5. Since $f(-2, -3) = 2$, the level curve is $x^2/4 + y^2/9 = 2$

or $x^2/8 + y^2/18 = 1$. $\nabla f = \frac{x}{2}\mathbf{i} + \frac{2y}{9}\mathbf{j}; \quad \nabla f(-2, -3) = -\mathbf{i} - \frac{2}{3}\mathbf{j}$

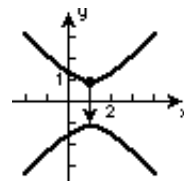


Exercises 9.6

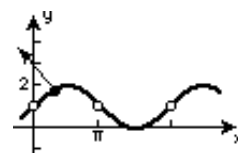
6. Since $f(2, 2) = 2$, the level curve is $y^2 = 2x$, $x \neq 0$. $\nabla f = -\frac{y^2}{x^2}\mathbf{i} + \frac{2y}{x}\mathbf{j}$;
 $\nabla f(2, 2) = -\mathbf{i} + 2\mathbf{j}$



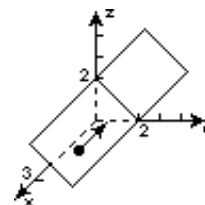
7. Since $f(1, 1) = -1$, the level curve is $(x - 1)^2 - y^2 = -1$ or $y^2 - (x - 1)^2 = 1$.
 $\nabla f = 2(x - 1)\mathbf{i} - 2y\mathbf{j}$; $\nabla f(1, 1) = -2\mathbf{j}$



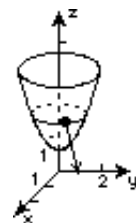
8. Since $f(\pi/6, 3/2) = 1$, the level curve is $y - 1 = \sin x$ or $y = 1 + \sin x$, $\sin x \neq 0$.
 $\nabla f = \frac{-(y - 1) \cos x}{\sin^2 x}\mathbf{i} + \frac{1}{\sin x}\mathbf{j}$; $\nabla f(\pi/6, 3/2) = -\sqrt{3}\mathbf{i} + 2\mathbf{j}$



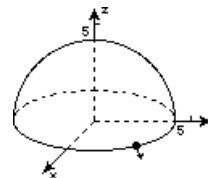
9. Since $F(3, 1, 1) = 2$, the level surface is $y + z = 2$. $\nabla F = \mathbf{j} + \mathbf{k}$;
 $\nabla F(3, 1, 1) = \mathbf{j} + \mathbf{k}$



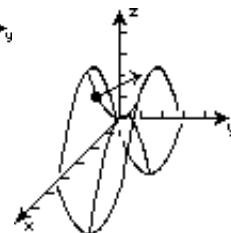
10. Since $F(1, 1, 3) = -1$, the level surface is $x^2 + y^2 - z = -1$ or $z = 1 + x^2 + y^2$.
 $\nabla F = 2x\mathbf{i} + 2y\mathbf{j} - \mathbf{k}$; $\nabla F(1, 1, 3) = 2\mathbf{i} + 2\mathbf{j} - \mathbf{k}$



11. Since $F(3, 4, 0) = 5$, the level surface is $x^2 + y^2 + z^2 = 25$.
 $\nabla F = \frac{x}{\sqrt{x^2 + y^2 + z^2}}\mathbf{i} + \frac{y}{\sqrt{x^2 + y^2 + z^2}}\mathbf{j} + \frac{z}{\sqrt{x^2 + y^2 + z^2}}\mathbf{k}$;
 $\nabla F(3, 4, 0) = \frac{3}{5}\mathbf{i} + \frac{4}{5}\mathbf{j}$



12. Since $F(0, -1, 1) = 0$, the level surface is $x^2 - y^2 + z = 0$ or $z = y^2 - x^2$.
 $\nabla F = 2x\mathbf{i} - 2y\mathbf{j} + \mathbf{k}$; $\nabla F(0, -1, 1) = 2\mathbf{j} + \mathbf{k}$



13. $F(x, y, z) = x^2 + y^2 - z$; $\nabla F = 2x\mathbf{i} + 2y\mathbf{j} - \mathbf{k}$. We want $\nabla F = c(4\mathbf{i} + \mathbf{j} + \frac{1}{2}\mathbf{k})$ or $2x = 4c$, $2y = c$, $-1 = c/2$. From the third equation $c = -2$. Thus, $x = -4$ and $y = -1$. Since $z = x^2 + y^2 = 16 + 1 = 17$, the point on the surface is $(-4, -1, 17)$.

14. $F(x, y, z) = x^3 + y^2 + z$; $\nabla F = 3x^2\mathbf{i} + 2y\mathbf{j} + \mathbf{k}$. We want $\nabla F = c(27\mathbf{i} + 8\mathbf{j} + \mathbf{k})$ or $3x^2 = 27c$, $2y = 8c$, $1 = c$. From $c = 1$ we obtain $x = \pm 3$ and $y = 4$. Since $z = 15 - x^3 - y^2 = 15 - (\pm 3)^3 - 16 = -1 \mp 27$, the points on the surface are $(3, 4, -28)$ and $(-3, 4, 26)$.

15. $F(x, y, z) = x^2 + y^2 + z^2$; $\nabla F = 2x\mathbf{i} + 2y\mathbf{j} + 2z\mathbf{k}$. $\nabla F(-2, 2, 1) = -4\mathbf{i} + 4\mathbf{j} + 2\mathbf{k}$. The equation of the tangent plane is $-4(x + 2) + 4(y - 2) + 2(z - 1) = 0$ or $-2x + 2y + z = 9$.
16. $F(x, y, z) = 5x^2 - y^2 + 4z^2$; $\nabla F = 10x\mathbf{i} - 2y\mathbf{j} + 8z\mathbf{k}$; $\nabla F(2, 4, 1) = 20\mathbf{i} - 8\mathbf{j} + 8\mathbf{k}$. The equation of the tangent plane is $20(x - 2) - 8(y - 4) + 8(z - 1) = 0$ or $5x - 2y + 2z = 4$.
17. $F(x, y, z) = x^2 - y^2 - 3z^2$; $\nabla F = 2x\mathbf{i} - 2y\mathbf{j} - 6z\mathbf{k}$; $\nabla F(6, 2, 3) = 12\mathbf{i} - 4\mathbf{j} - 18\mathbf{k}$. The equation of the tangent plane is $12(x - 6) - 4(y - 2) - 18(z - 3) = 0$ or $6x - 2y - 9z = 5$.
18. $F(x, y, z) = xy + yz + zx$; $\nabla F = (y + z)\mathbf{i} + (x + z)\mathbf{j} + (y + x)\mathbf{k}$; $\nabla F(1, -3, -5) = -8\mathbf{i} - 4\mathbf{j} - 2\mathbf{k}$. The equation of the tangent plane is $-8(x - 1) - 4(y + 3) - 2(z + 5) = 0$ or $4x + 2y + z = -7$.
19. $F(x, y, z) = x^2 + y^2 + z$; $\nabla F = 2x\mathbf{i} + 2y\mathbf{j} + \mathbf{k}$; $\nabla F(3, -4, 0) = 6\mathbf{i} - 8\mathbf{j} + \mathbf{k}$. The equation of the tangent plane is $6(x - 3) - 8(y + 4) + z = 0$ or $6x - 8y + z = 50$.
20. $F(x, y, z) = xz$; $\nabla F = z\mathbf{i} + x\mathbf{k}$; $\nabla F(2, 0, 3) = 3\mathbf{i} + 2\mathbf{k}$. The equation of the tangent plane is $3(x - 2) + 2(z - 3) = 0$ or $3x + 2z = 12$.

21. $F(x, y, z) = \cos(2x + y) - z$; $\nabla F = -2\sin(2x + y)\mathbf{i} - \sin(2x + y)\mathbf{j} - \mathbf{k}$; $\nabla F(\pi/2, \pi/4, -1/\sqrt{2}) = \sqrt{2}\mathbf{i} + \frac{\sqrt{2}}{2}\mathbf{j} - \mathbf{k}$.

The equation of the tangent plane is $\sqrt{2}\left(x - \frac{\pi}{2}\right) + \frac{\sqrt{2}}{2}\left(y - \frac{\pi}{4}\right) - \left(z + \frac{1}{\sqrt{2}}\right) = 0$,

$$2\left(x - \frac{\pi}{2}\right) + \left(y - \frac{\pi}{4}\right) - \sqrt{2}\left(z + \frac{1}{\sqrt{2}}\right) = 0, \text{ or } 2x + y - \sqrt{2}z = \frac{5\pi}{4} + 1.$$

22. $F(x, y, z) = x^2y^3 + 6z$; $\nabla F = 2xy^3\mathbf{i} + 3x^2y^2\mathbf{j} + 6\mathbf{k}$; $\nabla F(2, 1, 1) = 4\mathbf{i} + 12\mathbf{j} + 6\mathbf{k}$. The equation of the tangent plane is $4(x - 2) + 12(y - 1) + 6(z - 1) = 0$ or $2x + 6y + 3z = 13$.

23. $F(x, y, z) = \ln(x^2 + y^2) - z$; $\nabla F = \frac{2x}{x^2 + y^2}\mathbf{i} + \frac{2y}{x^2 + y^2}\mathbf{j} - \mathbf{k}$; $\nabla F(1/\sqrt{2}, 1/\sqrt{2}, 0) = \sqrt{2}\mathbf{i} + \sqrt{2}\mathbf{j} - \mathbf{k}$.

The equation of the tangent plane is $\sqrt{2}\left(x - \frac{1}{\sqrt{2}}\right) + \sqrt{2}\left(y - \frac{1}{\sqrt{2}}\right) - (z - 0) = 0$,

$$2\left(x - \frac{1}{\sqrt{2}}\right) + 2\left(y - \frac{1}{\sqrt{2}}\right) - \sqrt{2}z = 0, \text{ or } 2x + 2y - \sqrt{2}z = 2\sqrt{2}.$$

24. $F(x, y, z) = 8e^{-2y} \sin 4x - z$; $\nabla F = 32e^{-2y} \cos 4x\mathbf{i} - 16e^{-2y} \sin 4x\mathbf{j} - \mathbf{k}$; $\nabla F(\pi/24, 0, 4) = 16\sqrt{3}\mathbf{i} - 8\mathbf{j} - \mathbf{k}$.

The equation of the tangent plane is

$$16\sqrt{3}(x - \pi/24) - 8(y - 0) - (z - 4) = 0 \quad \text{or} \quad 16\sqrt{3}x - 8y - z = \frac{2\sqrt{3}\pi}{3} - 4.$$

25. The gradient of $F(x, y, z) = x^2 + y^2 + z^2$ is $\nabla F = 2x\mathbf{i} + 2y\mathbf{j} + 2z\mathbf{k}$, so the normal vector to the surface at (x_0, y_0, z_0) is $2x_0\mathbf{i} + 2y_0\mathbf{j} + 2z_0\mathbf{k}$. A normal vector to the plane $2x + 4y + 6z = 1$ is $2\mathbf{i} + 4\mathbf{j} + 6\mathbf{k}$. Since we want the tangent plane to be parallel to the given plane, we find c so that $2x_0 = 2c$, $2y_0 = 4c$, $2z_0 = 6c$ or $x_0 = c$, $y_0 = 2c$, $z_0 = 3c$. Now, (x_0, y_0, z_0) is on the surface, so $c^2 + (2c)^2 + (3c)^2 = 14c^2 = 7$ and $c = \pm 1/\sqrt{2}$. Thus, the points on the surface are $(\sqrt{2}/2, \sqrt{2}, 3\sqrt{2}/2)$ and $(-\sqrt{2}/2, -\sqrt{2}, -3\sqrt{2}/2)$.
26. The gradient of $F(x, y, z) = x^2 - 2y^2 - 3z^2$ is $\nabla F(x, y, z) = 2x\mathbf{i} - 4y\mathbf{j} - 6z\mathbf{k}$, so a normal vector to the surface at (x_0, y_0, z_0) is $\nabla F(x_0, y_0, z_0) = 2x_0\mathbf{i} - 4y_0\mathbf{j} - 6z_0\mathbf{k}$. A normal vector to the plane $8x + 4y + 6z = 5$ is $8\mathbf{i} + 4\mathbf{j} + 6\mathbf{k}$. Since we want the tangent plane to be parallel to the given plane, we find c so that $2x_0 = 8c$, $-4y_0 = 4c$, $-6z_0 = 6c$ or $x_0 = 4c$, $y_0 = -c$, $z_0 = -c$. Now, (x_0, y_0, z_0) is on the surface, so $(4c)^2 - 2(-c)^2 - 3(-c)^2 = 11c^2 = 33$ and $c = \pm\sqrt{3}$. Thus, the points on the surface are $(4\sqrt{3}, -\sqrt{3}, -\sqrt{3})$ and $(-4\sqrt{3}, \sqrt{3}, \sqrt{3})$.

Exercises 9.6

27. The gradient of $F(x, y, z) = x^2 + 4x + y^2 + z^2 - 2z$ is $\nabla F = (2x + 4)\mathbf{i} + 2y\mathbf{j} + (2z - 2)\mathbf{k}$, so a normal to the surface at (x_0, y_0, z_0) is $(2x_0 + 4)\mathbf{i} + 2y_0\mathbf{j} + (2z_0 - 2)\mathbf{k}$. A horizontal plane has normal $c\mathbf{k}$ for $c \neq 0$. Thus, we want $2x_0 + 4 = 0$, $2y_0 = 0$, $2z_0 - 2 = c$ or $x_0 = -2$, $y_0 = 0$, $z_0 = c + 1$. Since (x_0, y_0, z_0) is on the surface, $(-2)^2 + 4(-2) + (c + 1)^2 - 2(c + 1) = c^2 - 5 = 11$ and $c = \pm 4$. The points on the surface are $(-2, 0, 5)$ and $(-2, 0, -3)$.
28. The gradient of $F(x, y, z) = x^2 + 3y^2 + 4z^2 - 2xy$ is $\nabla F = (2x - 2y)\mathbf{i} + (6y - 2x)\mathbf{j} + 8z\mathbf{k}$, so a normal to the surface at (x_0, y_0, z_0) is $2(x_0 - y_0)\mathbf{i} + 2(3y_0 - x_0)\mathbf{j} + 8z_0\mathbf{k}$.
- (a) A normal to the xz plane is $c\mathbf{j}$ for $c \neq 0$. Thus, we want $2(x_0 - y_0) = 0$, $2(3y_0 - x_0) = c$, $8z_0 = 0$ or $x_0 = y_0$, $3y_0 - x_0 = c/2$, $z_0 = 0$. Solving the first two equations, we obtain $x_0 = y_0 = c/4$. Since (x_0, y_0, z_0) is on the surface, $(c/4)^2 + 3(c/4)^2 + 4(0)^2 - 2(c/4)(c/4) = 2c^2/16 = 16$ and $c = \pm 16/\sqrt{2}$. Thus, the points on the surface are $(4/\sqrt{2}, 4/\sqrt{2}, 0)$ and $(-4/\sqrt{2}, -4/\sqrt{2}, 0)$.
- (b) A normal to the yz -plane is $c\mathbf{i}$ for $c \neq 0$. Thus, we want $2(x_0 - y_0) = c$, $2(3y_0 - x_0) = 0$, $8z_0 = 0$ or $x_0 - y_0 = c/2$, $x_0 = 3y_0$, $z_0 = 0$. Solving the first two equations, we obtain $x_0 = 3c/4$ and $y_0 = c/4$. Since (x_0, y_0, z_0) is on the surface, $(3c/4)^2 + 3(c/4)^2 + 4(0)^2 - 2(3c/4)(c/4) = 6c^2/16 = 16$ and $c = \pm 16\sqrt{6}$. Thus, the points on the surface are $(12/\sqrt{6}, 4/\sqrt{6}, 0)$ on the surface are $(12/\sqrt{6}, 4/\sqrt{6}, 0)$ and $(-12/\sqrt{6}, -4/\sqrt{6}, 0)$.
- (c) A normal to the xy -plane is $c\mathbf{k}$ for $c \neq 0$. Thus, we want $2(x_0 - y_0) = 0$, $2(3y_0 - x_0) = 0$, $8z_0 = c$ or $x_0 = y_0$, $3y_0 - x_0 = 0$, $z_0 = c/8$. Solving the first two equations, we obtain $x_0 = y_0 = 0$. Since (x_0, y_0, z_0) is on the surface, $0^2 + 3(0)^2 + 4(c/8)^2 - 2(0)(0) = c^2/16 = 16$ and $c = \pm 16$. Thus, the points on the surface are $(0, 0, 2)$ and $(0, 0, -2)$.
29. If (x_0, y_0, z_0) is on $x^2/a^2 + y^2/b^2 + z^2/c^2 = 1$, then $x_0^2/a^2 + y_0^2/b^2 + z_0^2/c^2 = 1$ and (x_0, y_0, z_0) is on the plane $xx_0/a^2 + yy_0/b^2 + zz_0/c^2 = 1$. A normal to the surface at (x_0, y_0, z_0) is

$$\nabla F(x_0, y_0, z_0) = (2x_0/a^2)\mathbf{i} + (2y_0/b^2)\mathbf{j} + (2z_0/c^2)\mathbf{k}.$$

A normal to the plane is $(x_0/a^2)\mathbf{i} + (y_0/b^2)\mathbf{j} + (z_0/c^2)\mathbf{k}$. Since the normal to the surface is a multiple of the normal to the plane, the normal vectors are parallel and the plane is tangent to the surface.

30. If (x_0, y_0, z_0) is on $x^2/a^2 - y^2/b^2 + z^2/c^2 = 1$, then $x_0^2/a^2 - y_0^2/b^2 + z_0^2/c^2 = 1$ and (x_0, y_0, z_0) is on the plane $xx_0/a^2 - yy_0/b^2 + zz_0/c^2 = 1$. A normal to the surface at (x_0, y_0, z_0) is

$$\nabla F(x_0, y_0, z_0) = (2x_0/a^2)\mathbf{i} - (2y_0/b^2)\mathbf{j} + (2z_0/c^2)\mathbf{k}.$$

A normal to the plane is $(x_0/a^2)\mathbf{i} - (y_0/b^2)\mathbf{j} + (z_0/c^2)\mathbf{k}$. Since the normal to the surface is a multiple of the normal to the plane, the normal vectors are parallel, and the plane is tangent to the surface.

31. Let $F(x, y, z) = x^2 + y^2 - z^2$. Then $\nabla F = 2x\mathbf{i} + 2y\mathbf{j} - 2z\mathbf{k}$ and a normal to the surface at (x_0, y_0, z_0) is $x_0\mathbf{i} + y_0\mathbf{j} - z_0\mathbf{k}$. An equation of the tangent plane at (x_0, y_0, z_0) is $x_0(x - x_0) + y_0(y - y_0) - z_0(z - z_0) = 0$ or $x_0x + y_0y - z_0z = x_0^2 + y_0^2 - z_0^2$. Since (x_0, y_0, z_0) is on the surface, $z_0^2 = x_0^2 + y_0^2$ and $x_0^2 + y_0^2 - z_0^2 = 0$. Thus, the equation of the tangent plane is $x_0x + y_0y - z_0z = 0$, which passes through the origin.
32. Let $F(x, y, z) = \sqrt{x} + \sqrt{y} + \sqrt{z}$. Then $\nabla F = \frac{1}{2\sqrt{x}}\mathbf{i} + \frac{1}{2\sqrt{y}}\mathbf{j} + \frac{1}{2\sqrt{z}}\mathbf{k}$ and a normal to the surface at (x_0, y_0, z_0) is $\frac{1}{2\sqrt{x_0}}\mathbf{i} + \frac{1}{2\sqrt{y_0}}\mathbf{j} + \frac{1}{2\sqrt{z_0}}\mathbf{k}$. An equation of the tangent plane at (x_0, y_0, z_0) is

$$\frac{1}{2\sqrt{x_0}}(x - x_0) + \frac{1}{2\sqrt{y_0}}(y - y_0) + \frac{1}{2\sqrt{z_0}}(z - z_0) = 0$$

or

$$\frac{1}{\sqrt{x_0}}x + \frac{1}{\sqrt{y_0}}y + \frac{1}{\sqrt{z_0}}z = \sqrt{x_0} + \sqrt{y_0} + \sqrt{z_0} = \sqrt{a}.$$

The sum of the intercepts is $\sqrt{x_0}\sqrt{a} + \sqrt{y_0}\sqrt{a} + \sqrt{z_0}\sqrt{a} = (\sqrt{x_0} + \sqrt{y_0} + \sqrt{z_0})\sqrt{a} = \sqrt{a} \cdot \sqrt{a} = a$.

33. $F(x, y, z) = x^2 + 2y^2 + z^2$; $\nabla F = 2x\mathbf{i} + 4y\mathbf{j} + 2z\mathbf{k}$; $\nabla F(1, -1, 1) = 2\mathbf{i} - 4\mathbf{j} + 2\mathbf{k}$. Parametric equations of the line are $x = 1 + 2t$, $y = -1 - 4t$, $z = 1 + 2t$.
34. $F(x, y, z) = 2x^2 - 4y^2 - z$; $\nabla F = 4x\mathbf{i} - 8y\mathbf{j} - \mathbf{k}$; $\nabla F(3, -2, 2) = 12\mathbf{i} + 16\mathbf{j} - \mathbf{k}$. Parametric equations of the line are $x = 3 + 12t$, $y = -2 + 16t$, $z = 2 - t$.
35. $F(x, y, z) = 4x^2 + 9y^2 - z$; $\nabla F = 8x\mathbf{i} + 18y\mathbf{j} - \mathbf{k}$; $\nabla F(1/2, 1/3, 3) = 4\mathbf{i} + 6\mathbf{j} - \mathbf{k}$. Symmetric equations of the line are $\frac{x - 1/2}{4} = \frac{y - 1/3}{6} = \frac{z - 3}{-1}$.
36. $F(x, y, z) = x^2 + y^2 - z^2$; $\nabla F = 2x\mathbf{i} + 2y\mathbf{j} - 2z\mathbf{k}$; $\nabla F(3, 4, 5) = 6\mathbf{i} + 8\mathbf{j} - 10\mathbf{k}$. Symmetric equations of the line are $\frac{x - 3}{6} = \frac{y - 4}{8} = \frac{z - 5}{-10}$.
37. A normal to the surface at (x_0, y_0, z_0) is $\nabla F(x_0, y_0, z_0) = 2x_0\mathbf{i} + 2y_0\mathbf{j} + 2z_0\mathbf{k}$. Parametric equations of the normal line are $x = x_0 + 2x_0t$, $y = y_0 + 2y_0t$, $z = z_0 + 2z_0t$. Letting $t = -1/2$, we see that the normal line passes through the origin.
38. The normal lines to $F(x, y, z) = 0$ and $G(x, y, z) = 0$ are $F_x\mathbf{i} + F_y\mathbf{j} + F_z\mathbf{k}$ and $G_x\mathbf{i} + G_y\mathbf{j} + G_z\mathbf{k}$, respectively. These vectors are orthogonal if and only if their dot product is 0. Thus, the surfaces are orthogonal at P if and only if $F_xG_x + F_yG_y + F_zG_z = 0$.
39. Let $F(x, y, z) = x^2 + y^2 + z^2 - 25$ and $G(x, y, z) = -x^2 + y^2 + z^2$. Then

$$F_xG_x + F_yG_y + F_zG_z = (2x)(-2x) + (2y)(2y) + (2z)(2z) = 4(-x^2 + y^2 + z^2).$$

For (x, y, z) on both surfaces, $F(x, y, z) = G(x, y, z) = 0$. Thus, $F_xG_x + F_yG_y + F_zG_z = 4(0) = 0$ and the surfaces are orthogonal at points of intersection.

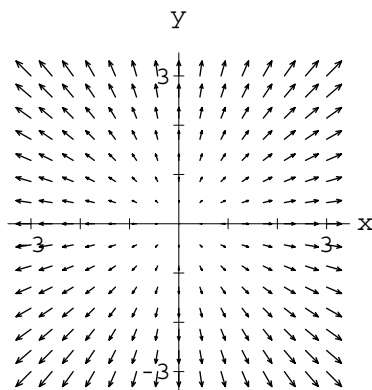
40. Let $F(x, y, z) = x^2 - y^2 + z^2 - 4$ and $G(x, y, z) = 1/xy^2 - z$. Then

$$\begin{aligned} F_xG_x + F_yG_y + F_zG_z &= (2x)(-1/x^2y^2) + (-2y)(-2/xy^3) + (2z)(-1) \\ &= -2/xy^2 + 4/xy^2 - 2z = 2(1/xy^2 - z). \end{aligned}$$

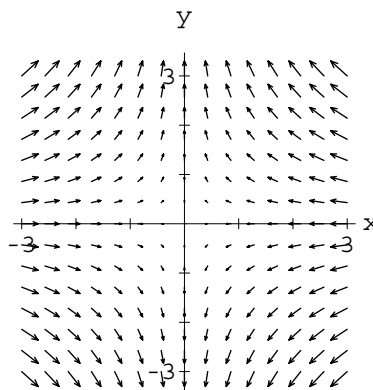
For (x, y, z) on both surfaces, $F(x, y, z) = G(x, y, z) = 0$. Thus, $F_xG_x + F_yG_y + F_zG_z = 2(0)$ and the surfaces are orthogonal at points of intersection.

Exercises 9.7

1.

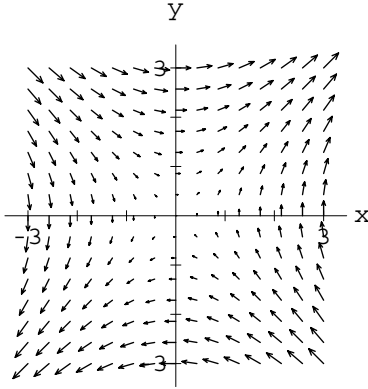


2.

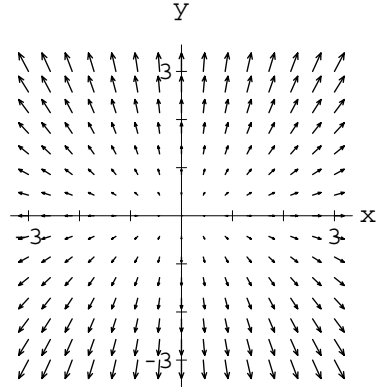


Exercises 9.7

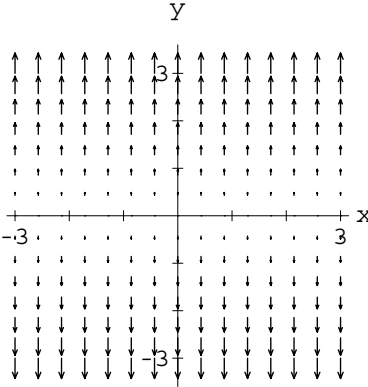
3.



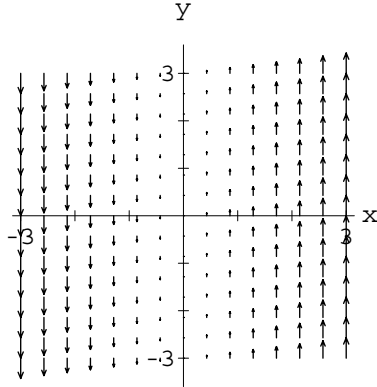
4.



5.



6.



7. $\text{curl } \mathbf{F} = (x - y)\mathbf{i} + (x - y)\mathbf{j}; \text{ div } \mathbf{F} = 2z$

8. $\text{curl } \mathbf{F} = -2x^2\mathbf{i} + (10y - 18x^2)\mathbf{j} + (4xz - 10z)\mathbf{k}; \text{ div } \mathbf{F} = 0$

9. $\text{curl } \mathbf{F} = \mathbf{0}; \text{ div } \mathbf{F} = 4y + 8z$

10. $\text{curl } \mathbf{F} = (xe^{2y} + ye^{-yz} + 2xye^{2y})\mathbf{i} - ye^{2y}\mathbf{j} + 3(x - y)^2\mathbf{k}; \text{ div } \mathbf{F} = 3(x - y)^2 - ze^{-yz}$

11. $\text{curl } \mathbf{F} = (4y^3 - 6xz^2)\mathbf{i} + (2z^3 - 3x^2)\mathbf{k}; \text{ div } \mathbf{F} = 6xy$

12. $\text{curl } \mathbf{F} = -x^3z\mathbf{i} + (3x^2yz - z)\mathbf{j} + (\frac{3}{2}x^2y^2 - y - 15y^2)\mathbf{k}; \text{ div } \mathbf{F} = (x^3y - x) - (x^3y - x) = 0$

13. $\text{curl } \mathbf{F} = (3e^{-z} - 8yz)\mathbf{i} - xe^{-z}\mathbf{j}; \text{ div } \mathbf{F} = e^{-z} + 4z^2 - 3ye^{-z}$

14. $\text{curl } \mathbf{F} = (2xyz^3 + 3y)\mathbf{i} + (y \ln x - y^2z^3)\mathbf{j} + (2 - z \ln x)\mathbf{k}; \text{ div } \mathbf{F} = \frac{yz}{x} - 3z + 3xy^2z^2$

15. $\text{curl } \mathbf{F} = (xy^2e^y + 2xye^y + x^3ye^z + x^3yze^z)\mathbf{i} - y^2e^y\mathbf{j} + (-3x^2yze^z - xe^x)\mathbf{k}; \text{ div } \mathbf{F} = xye^x + ye^x - x^3ze^z$

16. $\text{curl } \mathbf{F} = (5xye^{5xy} + e^{5xy} + 3xz^3 \sin xz^3 - \cos xz^3)\mathbf{i} + (x^2y \cos yz - 5y^2e^{5xy})\mathbf{j}$
 $+ (-z^4 \sin xz^3 - x^2z \cos yz)\mathbf{k}; \text{ div } \mathbf{F} = 2x \sin yz$

17. $\text{div } \mathbf{r} = 1 + 1 + 1 = 3$

18. $\text{curl } \mathbf{r} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x & y & z \end{vmatrix} = 0\mathbf{i} - 0\mathbf{j} + 0\mathbf{k} = \mathbf{0}$

19. $\mathbf{a} \times \nabla = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \end{vmatrix} = \left(a_2 \frac{\partial}{\partial z} - a_3 \frac{\partial}{\partial y}\right)\mathbf{i} + \left(a_3 \frac{\partial}{\partial x} - a_1 \frac{\partial}{\partial z}\right)\mathbf{j} + \left(a_1 \frac{\partial}{\partial y} - a_2 \frac{\partial}{\partial x}\right)\mathbf{k}$

$$\begin{aligned}
(\mathbf{a} \times \nabla) \times \mathbf{r} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_2 \frac{\partial}{\partial z} - a_3 & a_3 \frac{\partial}{\partial x} - a_1 \frac{\partial}{\partial z} & a_1 \frac{\partial}{\partial y} - a_2 \frac{\partial}{\partial x} \\ x & y & z \end{vmatrix} \\
&= (-a_1 - a_1)\mathbf{i} - (a_2 + a_2)\mathbf{j} + (-a_3 - a_3)\mathbf{k} = -2\mathbf{a}
\end{aligned}$$

$$20. \nabla \times (\mathbf{a} \times \mathbf{r}) = (\nabla \cdot \mathbf{r})\mathbf{a} - (\nabla \cdot \mathbf{a})\mathbf{r} = (1 + 1 + 1)\mathbf{a} - \left(a_1 \frac{\partial}{\partial x} + a_2 \frac{\partial}{\partial y} + a_3 \frac{\partial}{\partial z}\right)\mathbf{r} = 3\mathbf{a} - (a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}) = 2\mathbf{a}$$

$$21. \nabla \cdot (\mathbf{a} \times \mathbf{r}) = \begin{vmatrix} \partial/\partial x & \partial/\partial y & \partial/\partial z \\ a_1 & a_2 & a_3 \\ x & y & z \end{vmatrix} = \frac{\partial}{\partial x}(a_2z - a_3y) - \frac{\partial}{\partial y}(a_1z - a_3x) + \frac{\partial}{\partial z}(a_1y - a_2x) = 0$$

$$22. \nabla \times \mathbf{r} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x & y & z \end{vmatrix} = \mathbf{0}; \quad \mathbf{a} \times (\nabla \times \mathbf{r}) = \mathbf{a} \times \mathbf{0} = \mathbf{0}$$

$$23. \mathbf{r} \times \mathbf{a} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x & y & z \\ a_1 & a_2 & a_3 \end{vmatrix} = (a_3y - a_2z)\mathbf{i} - (a_3x - a_1z)\mathbf{j} + (a_2x - a_1y)\mathbf{k}; \quad \mathbf{r} \cdot \mathbf{r} = x^2 + y^2 + z^2$$

$$\begin{aligned}
\nabla \times [(\mathbf{r} \cdot \mathbf{r})\mathbf{a}] &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ (\mathbf{r} \cdot \mathbf{r})a_1 & (\mathbf{r} \cdot \mathbf{r})a_2 & (\mathbf{r} \cdot \mathbf{r})a_3 \end{vmatrix} \\
&= (2ya_3 - 2za_2)\mathbf{i} - (2xa_3 - 2za_1)\mathbf{j} + (2xa_2 - 2ya_1)\mathbf{k} = 2(\mathbf{r} \times \mathbf{a})
\end{aligned}$$

$$24. \mathbf{r} \cdot \mathbf{a} = a_1x + a_2y + a_3z; \quad \mathbf{r} \cdot \mathbf{r} = x^2 + y^2 + z^2; \quad \nabla \cdot [(\mathbf{r} \cdot \mathbf{r})\mathbf{a}] = 2xa_1 + 2ya_2 + 2za_3 = 2(\mathbf{r} \cdot \mathbf{a})$$

$$25. \text{ Let } \mathbf{F} = P(x, y, z)\mathbf{i} + Q(x, y, z)\mathbf{j} + R(x, y, z)\mathbf{k} \text{ and } \mathbf{G} = S(x, y, z)\mathbf{i} + T(x, y, z)\mathbf{j} + U(x, y, z)\mathbf{k}.$$

$$\begin{aligned}
\nabla \cdot (\mathbf{F} + \mathbf{G}) &= \nabla \cdot [(P + S)\mathbf{i} + (Q + T)\mathbf{j} + (R + U)\mathbf{k}] = P_x + S_x + Q_y + T_y + R_z + U_z \\
&= (P_x + Q_y + R_z) + (S_x + T_y + U_z) = \nabla \cdot \mathbf{F} + \nabla \cdot \mathbf{G}
\end{aligned}$$

$$26. \text{ Let } \mathbf{F} = P(x, y, z)\mathbf{i} + Q(x, y, z)\mathbf{j} + R(x, y, z)\mathbf{k} \text{ and } \mathbf{G} = S(x, y, z)\mathbf{i} + T(x, y, z)\mathbf{j} + U(x, y, z)\mathbf{k}.$$

$$\begin{aligned}
\nabla \times (\mathbf{F} + \mathbf{G}) &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ P + S & Q + T & R + U \end{vmatrix} \\
&= (R_y + U_y - Q_z - T_z)\mathbf{i} - (R_x + U_x - P_z - S_z)\mathbf{j} + (Q_x + T_x - P_y - S_y)\mathbf{k} \\
&= (R_y - Q_z)\mathbf{i} - (R_x - P_z)\mathbf{j} + (Q_x - P_y)\mathbf{k} + (U_y - T_z)\mathbf{i} - (U_x - S_z)\mathbf{j} + (T_x - S_y)\mathbf{k} \\
&= \nabla \times \mathbf{F} + \nabla \times \mathbf{G}
\end{aligned}$$

$$\begin{aligned}
27. \nabla \cdot (f\mathbf{F}) &= \nabla \cdot (fP\mathbf{i} + fQ\mathbf{j} + fR\mathbf{k}) = fP_x + Pf_x + fQ_y + Qf_y + fR_z + Rf_z \\
&= f(P_x + Q_y + R_z) + (Pf_x + Qf_y + Rf_z) = f(\nabla \cdot \mathbf{F}) + \mathbf{F} \cdot (\nabla f)
\end{aligned}$$

Exercises 9.7

$$\begin{aligned}
 28. \quad \nabla \times (f\mathbf{F}) &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ fP & fQ & fR \end{vmatrix} \\
 &= (fR_y + Rf_y - fQ_z - Qf_z)\mathbf{i} - (fR_x + Rf_x - fP_z - Pf_z)\mathbf{j} + (fQ_x + Qf_x - fP_y - Pf_y)\mathbf{k} \\
 &= (fR_y - fQ_z)\mathbf{i} - (fR_x - fP_z)\mathbf{j} + (fQ_x - fP_y)\mathbf{k} + (Rf_y - Qf_z)\mathbf{i} \\
 &\quad - (Rf_x - Pf_z)\mathbf{j} + (Qf_x - Pf_y)\mathbf{k} \\
 &= f[(R_y - Q_z)\mathbf{i} - (R_x - P_z)\mathbf{j} + (Q_x - P_y)\mathbf{k}] + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ f_x & f_y & f_z \\ P & Q & R \end{vmatrix} = f(\nabla \times \mathbf{F}) + (\nabla f) \times \mathbf{F}
 \end{aligned}$$

29. Assuming continuous second partial derivatives,

$$\begin{aligned}
 \text{curl}(\text{grad } f) &= \nabla \times (f_x\mathbf{i} + f_y\mathbf{j} + f_z\mathbf{k}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ f_x & f_y & f_z \end{vmatrix} \\
 &= (f_{zy} - f_{yz})\mathbf{i} - (f_{zx} - f_{xz})\mathbf{j} + (f_{yx} - f_{xy})\mathbf{k} = \mathbf{0}.
 \end{aligned}$$

30. Assuming continuous second partial derivatives,

$$\begin{aligned}
 \text{div}(\text{curl } \mathbf{F}) &= \nabla \cdot [(R_y - Q_z)\mathbf{i} - (R_x - P_z)\mathbf{j} + (Q_x - P_y)\mathbf{k}] \\
 &= (R_{yx} - Q_{zx} - (R_{xy} - P_{zy}) + (Q_{xz} - P_{yz})) = 0.
 \end{aligned}$$

31. Let $\mathbf{F} = P(x, y, z)\mathbf{i} + Q(x, y, z)\mathbf{j} + R(x, y, z)\mathbf{k}$ and $\mathbf{G} = S(x, y, z)\mathbf{i} + T(x, y, z)\mathbf{j} + U(x, y, z)\mathbf{k}$.

$$\begin{aligned}
 \mathbf{F} \times \mathbf{G} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ P & Q & R \\ S & T & U \end{vmatrix} = (QU - RT)\mathbf{i} - (PU - RS)\mathbf{j} + (PT - QS)\mathbf{k} \\
 \text{div}(\mathbf{F} \times \mathbf{G}) &= (QU_x + Q_xU - RT_x - R_xT) - (PU_y + P_yU - RS_y - R_yS) + (PT_z + P_zT - QS_z - Q_zS) \\
 &= S(R_y - Q_z) + T(P_z - R_x) + U(Q_x - P_y) - P(U_y - T_z) - Q(S_z - U_x) - R(T_x - S_y) \\
 &= \mathbf{G} \cdot (\text{curl } \mathbf{F}) - \mathbf{F} \cdot (\text{curl } \mathbf{G})
 \end{aligned}$$

32. Using Problems 26 and 29,

$$\begin{aligned}
 \text{curl}(\text{curl } \mathbf{F} + \text{grad } f) &= \nabla \times (\text{curl } \mathbf{F} + \text{grad } f) = \nabla \times (\text{curl } \mathbf{F}) + \nabla \times (\text{grad } f) \\
 &= \text{curl}(\text{curl } \mathbf{F}) + \text{curl}(\text{grad } f) = \text{curl}(\text{curl } \mathbf{F}) + \mathbf{0} = \text{curl}(\text{curl } \mathbf{F}).
 \end{aligned}$$

$$33. \quad \nabla \cdot \nabla f = \nabla \cdot (f_x\mathbf{i} + f_y\mathbf{j} + f_z\mathbf{k}) = f_{xx} + f_{yy} + f_{zz}$$

$$34. \quad \text{Using Problem 27, } \nabla \cdot (f\nabla f) = f(\nabla \cdot \nabla f) + \nabla f \cdot \nabla f = f(\nabla^2 f) + |\nabla f|^2.$$

$$35. \quad \text{curl } \mathbf{F} = -8yz\mathbf{i} - 2z\mathbf{j} - x\mathbf{k}; \quad \text{curl}(\text{curl } \mathbf{F}) = 2\mathbf{i} - (8y - 1)\mathbf{j} + 8z\mathbf{k}$$

36. (a) For $\mathbf{F} = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$,

$$\begin{aligned}
 \text{curl}(\text{curl } \mathbf{F}) &= (Q_{xy} - P_{yy} - P_{zz} + R_{xz})\mathbf{i} + (R_{yz} - Q_{zz} - Q_{xx} + P_{yx})\mathbf{j} \\
 &\quad + (P_{zx} - R_{xx} - R_{yy} + Q_{zy})\mathbf{k}
 \end{aligned}$$

and

$$\begin{aligned}
-\nabla^2 \mathbf{F} + \text{grad} (\text{div } \mathbf{F}) &= -(P_{xx} + P_{yy} + P_{zz})\mathbf{i} - (Q_{xx} + Q_{yy} + Q_{zz})\mathbf{j} - (R_{xx} + R_{yy} + R_{zz})\mathbf{k} \\
&\quad + \text{grad} (P_x + Q_y + R_z) \\
&= -P_{xx}\mathbf{i} - Q_{yy}\mathbf{j} - R_{zz}\mathbf{k} + (-P_{yy} - P_{zz})\mathbf{i} + (-Q_{xx} - Q_{zz})\mathbf{j} \\
&\quad + (-R_{xx} - R_{yy})\mathbf{k} + (P_{xx} + Q_{yx} + R_{zx})\mathbf{i} + (P_{xy} + Q_{yy} + R_{zy})\mathbf{j} \\
&\quad + (P_{xz} + Q_{yz} + R_{zz})\mathbf{k} \\
&= (-P_{yy} - P_{zz} + Q_{yx} + R_{zx})\mathbf{i} + (-Q_{xx} - Q_{zz} + P_{xy} + R_{zy})\mathbf{j} \\
&\quad + (-R_{xx} - R_{yy} + P_{xz} + Q_{yz})\mathbf{k}.
\end{aligned}$$

Thus, $\text{curl} (\text{curl } \mathbf{F}) = -\nabla^2 \mathbf{F} + \text{grad} (\text{div } \mathbf{F})$.

(b) For $\mathbf{F} = xy\mathbf{i} + 4yz^2\mathbf{j} + 2xz\mathbf{k}$, $\nabla^2 \mathbf{F} = 0\mathbf{i} + 8y\mathbf{j} + 0\mathbf{k}$, $\text{div } \mathbf{F} = y + 4z^2 + 2x$, and $\text{grad} (\text{div } \mathbf{F}) = 2\mathbf{i} + \mathbf{j} + 8z\mathbf{k}$.

Then $\text{curl} (\text{curl } \mathbf{F}) = -8y\mathbf{j} + 2\mathbf{i} + \mathbf{j} + 8z\mathbf{k} = 2\mathbf{i} + (1 - 8y)\mathbf{j} + 8z\mathbf{k}$.

$$\begin{aligned}
37. \quad \frac{\partial f}{\partial x} &= -x(x^2 + y^2 + z^2)^{-3/2} \\
\frac{\partial f}{\partial y} &= -y(x^2 + y^2 + z^2)^{-3/2} \\
\frac{\partial f}{\partial z} &= -z(x^2 + y^2 + z^2)^{-3/2} \\
\frac{\partial^2 f}{\partial x^2} &= 3x^2(x^2 + y^2 + z^2)^{-5/2} - (x^2 + y^2 + z^2)^{-3/2} \\
\frac{\partial^2 f}{\partial y^2} &= 3y^2(x^2 + y^2 + z^2)^{-5/2} - (x^2 + y^2 + z^2)^{-3/2} \\
\frac{\partial^2 f}{\partial z^2} &= 3z^2(x^2 + y^2 + z^2)^{-5/2} - (x^2 + y^2 + z^2)^{-3/2}
\end{aligned}$$

Adding the second partial derivatives gives

$$\begin{aligned}
\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} &= \frac{3(x^2 + y^2 + z^2)}{(x^2 + y^2 + z^2)^{5/2}} - 3(x^2 + y^2 + z^2)^{-3/2} \\
&= 3(x^2 + y^2 + z^2)^{-3/2} - 3(x^2 + y^2 + z^2)^{-3/2} = 0
\end{aligned}$$

except when $x = y = z = 0$.

$$\begin{aligned}
38. \quad f_x &= \frac{1}{1 + \frac{4y^2}{(x^2 + y^2 - 1)^2}} \left(-\frac{4xy}{(x^2 + y^2 - 1)^2} \right) = -\frac{4xy}{(x^2 + y^2 - 1)^2 + 4y^2} \\
f_{xx} &= -\frac{[(x^2 + y^2 - 1)^2 + 4y^2]4y - 4xy[4x(x^2 + y^2 - 1)]}{[(x^2 + y^2 - 1)^2 + 4y^2]^2} = \frac{12x^4y - 4y^5 + 8x^2y^3 - 8x^2y - 8y^3 - 4y}{[(x^2 + y^2 - 1)^2 + 4y^2]^2} \\
f_y &= \frac{1}{1 + \frac{4y^2}{(x^2 + y^2 - 1)^2}} \left[\frac{2(x^2 + y^2 - 1) - 4y^2}{(x^2 + y^2 - 1)^2} \right] = \frac{2(x^2 - y^2 - 1)}{(x^2 + y^2 - 1)^2 + 4y^2} \\
f_{yy} &= \frac{[(x^2 + y^2 - 1)^2 + 4y^2](-4y) - 2(x^2 - y^2 - 1)[4y(x^2 + y^2 - 1) + 8y]}{[(x^2 + y^2 - 1)^2 + 4y^2]^2} \\
&= \frac{-12x^4y + 4y^5 - 8x^2y^3 + 8x^2y + 8y^3 + 4y}{[(x^2 + y^2 - 1)^2 + 4y^2]^2} \\
\nabla^2 f &= f_{xx} + f_{yy} = 0
\end{aligned}$$

Exercises 9.7

$$\begin{aligned}
 39. \quad \operatorname{curl} \mathbf{F} &= -Gm_1m_2 \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x/|\mathbf{r}|^3 & y/|\mathbf{r}|^3 & z/|\mathbf{r}|^3 \end{vmatrix} \\
 &= -Gm_1m_2 [(-3yz/|\mathbf{r}|^5 + 3yz/|\mathbf{r}|^5)\mathbf{i} - (-3xz/|\mathbf{r}|^5 + 3xz/|\mathbf{r}|^5)\mathbf{j} + (-3xy/|\mathbf{r}|^5 + 3xy/|\mathbf{r}|^5)\mathbf{k}] \\
 &= \mathbf{0}
 \end{aligned}$$

$$\operatorname{div} \mathbf{F} = -Gm_1m_2 \left[\frac{-2x^2 + y^2 + z^2}{|\mathbf{r}|^{5/2}} + \frac{x^2 - 2y^2 + z^2}{|\mathbf{r}|^{5/2}} + \frac{x^2 + y^2 - 2z^2}{|\mathbf{r}|^{5/2}} \right] = 0$$

$$\begin{aligned}
 40. \quad \frac{1}{2} \operatorname{curl} \mathbf{v} &= \frac{1}{2} \operatorname{curl} (\boldsymbol{\omega} \times \mathbf{r}) = \frac{1}{2} \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \omega_2 z - \omega_3 y & \omega_3 x - \omega_1 z & \omega_1 y - \omega_2 x \end{vmatrix} \\
 &= \frac{1}{2} [(\omega_1 + \omega_1)\mathbf{i} - (-\omega_2 - \omega_2)\mathbf{j} + (\omega_3 + \omega_3)\mathbf{k}] = \omega_1 \mathbf{i} + \omega_2 \mathbf{j} + \omega_3 \mathbf{k} = \boldsymbol{\omega}
 \end{aligned}$$

41. Using Problems 31 and 29,

$$\nabla \cdot \mathbf{F} = \operatorname{div} (\nabla f \times \nabla g) = \nabla g \cdot (\operatorname{curl} \nabla f) - \nabla f \cdot (\operatorname{curl} \nabla g) = \nabla g \cdot \mathbf{0} - \nabla f \cdot \mathbf{0} = 0.$$

42. Recall that $\mathbf{a} \cdot (\mathbf{a} \times \mathbf{b}) = 0$. Then, using Problems 31, 29, and 28,

$$\begin{aligned}
 \nabla \cdot \mathbf{F} &= \operatorname{div} (\nabla f \times f \nabla g) = f \nabla g \cdot (\operatorname{curl} \nabla f) - \nabla f \cdot (\operatorname{curl} f \nabla g) = f \nabla g \cdot \mathbf{0} - \nabla f \cdot (\nabla \times f \nabla g) \\
 &= -\nabla f \cdot [f(\nabla \times \nabla g) + (\nabla f \times \nabla g)] = -\nabla f \cdot [f \operatorname{curl} \nabla g + (\nabla f \times \nabla g)] \\
 &= -\nabla f \cdot [f \mathbf{0} + (\nabla f \times \nabla g)] = -\nabla f \cdot (\nabla f \times \nabla g) = 0.
 \end{aligned}$$

43. (a) Expressing the vertical component of $\mathbf{V}$ in polar coordinates, we have

$$\frac{2xy}{(x^2 + y^2)^2} = \frac{2r^2 \sin \theta \cos \theta}{r^4} = \frac{\sin 2\theta}{r^2}.$$

Similarly,

$$\frac{x^2 - y^2}{(x^2 + y^2)^2} = \frac{r^2(\cos^2 \theta - \sin^2 \theta)}{r^4} = \frac{\cos 2\theta}{r^2}.$$

Since $\lim_{r \rightarrow \infty} (\sin 2\theta)/r^2 = \lim_{r \rightarrow \infty} (\cos 2\theta)/r^2 = 0$, $\mathbf{V} \approx A\mathbf{i}$ for r large or (x, y) far from the origin.

(b) Identifying $P(x, y) = A \left[1 - \frac{x^2 - y^2}{(x^2 + y^2)^2} \right]$, $Q(x, y) = -\frac{2Axy}{(x^2 + y^2)^2}$, and $R(x, y) = 0$, we have

$$P_y = \frac{2Ay(3x^2 - y^2)}{(x^2 + y^2)^3}, \quad Q_x = \frac{2Ay(3x^2 - y^2)}{(x^2 + y^2)^3}, \quad \text{and} \quad P_z = Q_z = R_x = R_y = 0.$$

Thus, $\operatorname{curl} \mathbf{V} = (R_y - Q_z)\mathbf{i} + (P_z - R_x)\mathbf{j} + (Q_x - P_y)\mathbf{k} = \mathbf{0}$ and $\mathbf{V}$ is irrotational.

(c) Since $P_x = \frac{2Ax(x^2 - 3y^2)}{(x^2 + y^2)^3}$, $Q_y = \frac{2Ax(3y^2 - x^2)}{(x^2 + y^2)^3}$, and $R_z = 0$, $\nabla \cdot \mathbf{F} = P_x + Q_y + R_z = 0$ and $\mathbf{V}$ is incompressible.

44. We first note that $\operatorname{curl} (\partial \mathbf{H} / \partial t) = \partial (\operatorname{curl} \mathbf{H}) / \partial t$ and $\operatorname{curl} (\partial \mathbf{E} / \partial t) = \partial (\operatorname{curl} \mathbf{E}) / \partial t$. Then, from Problem 30,

$$\begin{aligned}
 -\nabla^2 \mathbf{E} &= -\nabla^2 \mathbf{E} + \mathbf{0} = -\nabla^2 \mathbf{E} + \operatorname{grad} 0 = -\nabla^2 \mathbf{E} + \operatorname{grad} (\operatorname{div} \mathbf{E}) = \operatorname{curl} (\operatorname{curl} \mathbf{E}) \\
 &= \operatorname{curl} \left(-\frac{1}{c} \frac{\partial \mathbf{H}}{\partial t} \right) = -\frac{1}{c} \frac{\partial}{\partial t} \operatorname{curl} \mathbf{H} = -\frac{1}{c} \frac{\partial}{\partial t} \left(\frac{1}{c} \frac{\partial \mathbf{E}}{\partial t} \right) = -\frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2}
 \end{aligned}$$

and $\nabla^2 \mathbf{E} = \frac{1}{c^2} \partial^2 \mathbf{E} / \partial t^2$. Similarly,

$$\begin{aligned} -\nabla^2 \mathbf{H} &= -\nabla^2 \mathbf{H} + \text{grad} (\text{div } \mathbf{H}) = \text{curl} (\text{curl } \mathbf{H}) = \text{curl} \left(\frac{1}{c} \frac{\partial \mathbf{E}}{\partial t} \right) = \frac{1}{c} \frac{\partial}{\partial t} \text{curl } \mathbf{E} \\ &= \frac{1}{c} \frac{\partial}{\partial t} \left(-\frac{1}{c} \frac{\partial \mathbf{H}}{\partial t} \right) = -\frac{1}{c^2} \frac{\partial^2 \mathbf{H}}{\partial t^2} \end{aligned}$$

and $\nabla^2 \mathbf{H} = \frac{1}{c^2} \partial^2 \mathbf{H} / \partial t^2$.

45. We note that $\text{div } \mathbf{F} = 2xyz - 2xyz + 1 = 1 \neq 0$. If $\mathbf{F} = \text{curl } \mathbf{G}$, then $\text{div} (\text{curl } \mathbf{G}) = \text{div } \mathbf{F} = 1$. But, by Problem 30, for any vector field $\mathbf{G}$, $\text{div} (\text{curl } \mathbf{G}) = 0$. Thus, $\mathbf{F}$ cannot be the curl of $\mathbf{G}$.

Exercises 9.8

- $$\begin{aligned} \int_C 2xy \, dx &= \int_0^{\pi/4} 2(5 \cos t)(5 \sin t)(-5 \sin t) \, dt = -250 \int_0^{\pi/4} \sin^2 t \cos t \, dt = -250 \left(\frac{1}{3} \sin^3 t \right) \Big|_0^{\pi/4} = -\frac{125\sqrt{2}}{6} \\ \int_C 2xy \, dy &= \int_0^{\pi/4} 2(5 \cos t)(5 \sin t)(5 \cos t) \, dt = 250 \int_0^{\pi/4} \cos^2 t \sin t \, dt = 250 \left(-\frac{1}{3} \cos^3 t \right) \Big|_0^{\pi/4} \\ &= \frac{250}{3} \left(1 - \frac{\sqrt{2}}{4} \right) = \frac{125}{6} (4 - \sqrt{2}) \\ \int_C 2xy \, ds &= \int_0^{\pi/4} 2(5 \cos t)(5 \sin t) \sqrt{25 \sin^2 t + 25 \cos^2 t} \, dt = 250 \int_0^{\pi/4} \sin t \cos t \, dt \\ &= 250 \left(\frac{1}{2} \sin^2 t \right) \Big|_0^{\pi/4} = \frac{125}{2} \end{aligned}$$
- $$\begin{aligned} \int_C (x^3 + 2xy^2 + 2x) \, dx &= \int_0^1 [8t^3 + 2(2t)(t^4) + 2(2t)] 2 \, dt = 2 \int_0^1 (8t^3 + 4t^5 + 4t) \, dt \\ &= 2 \left(2t^4 + \frac{2}{3}t^6 + 2t^2 \right) \Big|_0^1 = \frac{28}{3} \\ \int_C (x^3 + 2xy^2 + 2x) \, dy &= \int_0^1 [8t^3 + 2(2t)(t^4) + 2(2t)] 2t \, dt = 2 \int_0^1 (8t^4 + 4t^6 + 4t^2) \, dt \\ &= 2 \left(\frac{8}{5}t^5 + \frac{4}{7}t^7 + \frac{4}{3}t^3 \right) \Big|_0^1 = \frac{736}{105} \\ \int_C (x^3 + 2xy^2 + 2x) \, ds &= \int_0^1 [8t^3 + 2(2t)(t^4) + 2(2t)] \sqrt{4 + 4t^2} \, dt = 8 \int_0^1 t(1 + t^2)^{5/2} \, dt \\ &= 8 \left(\frac{1}{7}(1 + t^2)^{7/2} \right) \Big|_0^1 = \frac{8}{7}(2^{7/2} - 1) \end{aligned}$$
- $$\begin{aligned} \int_C (3x^2 + 6y^2) \, dx &= \int_{-1}^0 [3x^2 + 6(2x + 1)^2] \, dx = \int_{-1}^0 (27x^2 + 24x + 6) \, dx = (9x^3 + 12x^2 + 6x) \Big|_{-1}^0 \\ &= -(-9 + 12 - 6) = 3 \\ \int_C (3x^2 + 6y^2) \, dy &= \int_{-1}^0 [3x^2 + 6(2x + 1)^2] 2 \, dx = 6 \\ \int_C (3x^2 + 6y^2) \, ds &= \int_{-1}^0 [3x^2 + 6(2x + 1)^2] \sqrt{1 + 4} \, dx = 3\sqrt{5} \end{aligned}$$

Exercises 9.8

$$\begin{aligned}
 4. \quad \int_C \frac{x^2}{y^3} dx &= \int_1^8 \frac{x^2}{27x^2/8} dx = \frac{8}{27} \int_1^8 dx = \frac{56}{27} \\
 \int_C \frac{x^2}{y^3} dy &= \int_1^8 \frac{x^2}{27x^2/8} x^{-1/3} dx = \frac{8}{27} \int_1^8 x^{-1/3} dx = \frac{4}{9} x^{2/3} \Big|_1^8 = \frac{4}{3} \\
 \int_C \frac{x^2}{y^3} ds &= \int_1^8 \frac{x^2}{27x^2/8} \sqrt{1+x^{-2/3}} dx = \frac{8}{27} \int_1^8 x^{-1/3} \sqrt{1+x^{2/3}} dx = \frac{8}{27} (1+x^{2/3})^{3/2} \Big|_1^8 = \frac{8}{27} (5^{3/2} - 2^{3/2})
 \end{aligned}$$

$$\begin{aligned}
 5. \quad \int_C z dx &= \int_0^{\pi/2} t(-\sin t) dt \quad \boxed{\text{Integration by parts}} \\
 &= (t \cos t - \sin t) \Big|_0^{\pi/2} = -1
 \end{aligned}$$

$$\begin{aligned}
 \int_C z dy &= \int_0^{\pi/2} t \cos t dt \quad \boxed{\text{Integration by parts}} \\
 &= (t \sin t + \cos t) \Big|_0^{\pi/2} = \frac{\pi}{2} - 1
 \end{aligned}$$

$$\int_C z dz = \int_0^{\pi/2} t dt = \frac{1}{2} t^2 \Big|_0^{\pi/2} = \frac{\pi^2}{8}$$

$$\int_C z ds = \int_0^{\pi/2} t \sqrt{\sin^2 t + \cos^2 t + 1} dt = \sqrt{2} \int_0^{\pi/2} t dt = \frac{\pi^2 \sqrt{2}}{8}$$

$$\begin{aligned}
 6. \quad \int_C 4xyz dx &= \int_0^1 4 \left(\frac{1}{3} t^3 \right) (t^2)(2t)t^2 dt = \frac{8}{3} \int_0^1 t^8 dt = \frac{8}{27} t^9 \Big|_0^1 = \frac{8}{27} \\
 \int_C 4xyz dy &= \int_0^1 4 \left(\frac{1}{3} t^3 \right) (t^2)(2t)2t dt = \frac{16}{3} \int_0^1 t^7 dt = \frac{2}{3} t^8 \Big|_0^1 = \frac{2}{3} \\
 \int_C 4xyz dz &= \int_0^1 4 \left(\frac{1}{3} t^3 \right) (t^2)(2t)2 dt = \frac{16}{3} \int_0^1 t^6 dt = \frac{16}{21} t^7 \Big|_0^1 = \frac{16}{21} \\
 \int_C 4xyz ds &= \int_0^1 4 \left(\frac{1}{3} t^3 \right) (t^2)(2t) \sqrt{t^4 + 4t^2 + 4} dt = \frac{8}{3} \int_0^1 t^6 (t^2 + 2) dt = \frac{8}{3} \left(\frac{1}{9} t^9 + \frac{2}{7} t^7 \right) \Big|_0^1 = \frac{200}{189}
 \end{aligned}$$

7. Using x as the parameter, $dy = dx$ and

$$\begin{aligned}
 \int_C (2x + y) dx + xy dy &= \int_{-1}^2 (2x + x + 3 + x^2 + 3x) dx = \int_{-1}^2 (x^2 + 6x + 3) dx \\
 &= \left(\frac{1}{3} x^3 + 3x^2 + 3x \right) \Big|_{-1}^2 = 21.
 \end{aligned}$$

8. Using x as the parameter, $dy = 2x dx$ and

$$\begin{aligned}
 \int_C (2x + y) dx + xy dy &= \int_{-1}^2 (2x + x^2 + 1) dx + \int_{-1}^2 x(x^2 + 1) 2x dx = \int_{-1}^2 (2x^4 + 3x^2 + 2x + 1) dx \\
 &= \left(\frac{2}{5} x^5 + x^3 + x^2 + x \right) \Big|_{-1}^2 = \frac{141}{5}.
 \end{aligned}$$

9. From $(-1, 2)$ to $(2, 2)$ we use x as a parameter with $y = 2$ and $dy = 0$. From $(2, 2)$ to $(2, 5)$ we use y as a parameter with $x = 2$ and $dx = 0$.

$$\int_C (2x + y) dx + xy dy = \int_{-1}^2 (2x + 2) dx + \int_2^5 2y dy = (x^2 + 2x) \Big|_{-1}^2 + y^2 \Big|_2^5 = 9 + 21 = 30$$

10. From $(-1, 2)$ to $(-1, 0)$ we use y as a parameter with $x = -1$ and $dx = 0$. From $(-1, 0)$ to $(2, 0)$ we use x as a parameter with $y = dy = 0$. From $(2, 0)$ to $(2, 5)$ we use y as a parameter with $x = 2$ and $dx = 0$.

$$\begin{aligned}\int_C (2x + y) dx + xy dy &= \int_2^0 (-1) y dy + \int_{-1}^2 2x dx + \int_0^5 2y dy = -\frac{1}{2} y^2 \Big|_2^0 + x^2 \Big|_{-1}^2 + y^2 \Big|_0^5 \\ &= 2 + 3 + 25 = 30\end{aligned}$$

11. Using x as the parameter, $dy = 2x dx$.

$$\int_C y dx + x dy = \int_0^1 x^2 dx + \int_0^1 x(2x) dx = \int_0^1 3x^2 dx = x^3 \Big|_0^1 = 1$$

12. Using x as the parameter, $dy = dx$.

$$\int_C y dx + x dy = \int_0^1 x dx + \int_0^1 x dx = \int_0^1 2x dx = x^2 \Big|_0^1 = 1$$

13. From $(0, 0)$ to $(0, 1)$ we use y as a parameter with $x = dx = 0$. From $(0, 1)$ to $(1, 1)$ we use x as a parameter with $y = 1$ and $dy = 0$.

$$\int_C y dx + x dy = 0 + \int_0^1 1 dx = 1$$

14. From $(0, 0)$ to $(1, 0)$ we use x as a parameter with $y = dy = 0$. From $(1, 0)$ to $(1, 1)$ we use y as a parameter with $x = 1$ and $dx = 0$.

$$\int_C y dx + x dy = 0 + \int_0^1 1 dy = 1$$

15.
$$\begin{aligned}\int_C (6x^2 + 2y^2) dx + 4xy dy &= \int_4^9 (6t + 2t^2) \frac{1}{2} t^{-1/2} dt + \int_4^9 4\sqrt{t} t dt = \int_4^9 (3t^{1/2} + 5t^{3/2}) dt \\ &= (2t^{3/2} + 2t^{5/2}) \Big|_4^9 = 460\end{aligned}$$

16.
$$\int_C (-y^2) dx + xy dy = \int_0^2 (-t^6) 2 dt + \int_0^2 (2t)(t^3) 3t^2 dt = \int_0^2 4t^6 dt = \frac{4}{7} t^7 \Big|_0^2 = \frac{512}{7}$$

17.
$$\begin{aligned}\int_C 2x^3 y dx + (3x + y) dy &= \int_{-1}^1 2(y^6) y 2y dy + \int_{-1}^1 (3y^2 + y) dy = \int_{-1}^1 (4y^8 + 3y^2 + y) dy \\ &= \left(\frac{4}{9} y^9 + y^3 + \frac{1}{2} y^2 \right) \Big|_{-1}^1 = \frac{26}{9}\end{aligned}$$

18.
$$\begin{aligned}\int_C 4x dx + 2y dy &= \int_{-1}^2 4(y^3 + 1) 3y^2 dy + \int_{-1}^2 2y dy = \int_{-1}^2 (12y^5 + 12y^2 + 2y) dy \\ &= (2y^6 + 4y^3 + y^2) \Big|_{-1}^2 = 165\end{aligned}$$

19. From $(-2, 0)$ to $(2, 0)$ we use x as a parameter with $y = dy = 0$. From $(2, 0)$ to $(-2, 0)$ we parameterize the semicircle as $x = 2 \cos \theta$ and $y = 2 \sin \theta$ for $0 \leq \theta \leq \pi$.

$$\begin{aligned}\oint_C (x^2 + y^2) dx - 2xy dy &= \int_{-2}^2 x^2 dx + \int_0^\pi 4(-2 \sin \theta d\theta) - \int_0^\pi 8 \cos \theta \sin \theta (2 \cos \theta d\theta) \\ &= \frac{1}{3} x^3 \Big|_{-2}^2 - 8 \int_0^\pi (\sin \theta + 2 \cos^2 \theta \sin \theta) d\theta \\ &= \frac{16}{3} - 8 \left(-\cos \theta - \frac{2}{3} \cos^3 \theta \right) \Big|_0^\pi = \frac{16}{3} - \frac{80}{3} = -\frac{64}{3}\end{aligned}$$

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20. We start at $(0, 0)$ and use x as a parameter.

$$\begin{aligned}\oint_C (x^2 + y^2) dx - 2xy dy &= \int_0^1 (x^2 + x^4) dx - 2 \int_0^1 xx^2(2x dx) + \int_1^0 (x^2 + x) dx \\ &\quad - 2 \int_1^0 x\sqrt{x} \left(\frac{1}{2}x^{-1/2}\right) dx \\ &= \int_0^1 (x^2 - 3x^4) dx + \int_1^0 x^2 dx = \int_0^1 (-3x^4) dx = -\frac{3}{5}x^5 \Big|_0^1 = -\frac{3}{5}\end{aligned}$$

21. From $(1, 1)$ to $(-1, 1)$ and $(-1, -1)$ to $(1, -1)$ we use x as a parameter with $y = 1$ and $y = -1$, respectively, and $dy = 0$. From $(-1, 1)$ to $(-1, -1)$ and $(1, -1)$ to $(1, 1)$ we use y as a parameter with $x = -1$ and $x = 1$, respectively, and $dx = 0$.

$$\begin{aligned}\oint_C x^2y^3 dx - xy^2 dy &= \int_1^{-1} x^2(1) dx + \int_1^{-1} -(-1)y^2 dy + \int_{-1}^1 x^2(-1)^3 dx + \int_{-1}^1 -(1)y^2 dy \\ &= \frac{1}{3}x^3 \Big|_1^{-1} + \frac{1}{3}y^3 \Big|_1^{-1} - \frac{1}{3}x^3 \Big|_{-1}^1 - \frac{1}{3}y^3 \Big|_{-1}^1 = -\frac{8}{3}\end{aligned}$$

22. From $(2, 4)$ to $(0, 4)$ we use x as a parameter with $y = 4$ and $dy = 0$. From $(0, 4)$ to $(0, 0)$ we use y as a parameter with $x = 0$ and $dx = 0$. From $(0, 0)$ to $(2, 0)$ we use x as a parameter with $y = 0$ and $dy = 0$. From $(2, 0)$ to $(2, 4)$ we use y as a parameter with $x = 2$ and $dx = 0$.

$$\begin{aligned}\oint_C x^2y^3 dx - xy^2 dy &= \int_2^0 x^2(64) dx - \int_4^0 0 dy + \int_0^2 x^2(8x^3) dx - \int_0^2 x(4x^2)2 dx \\ &= \frac{64}{3}x^3 \Big|_2^0 + \frac{4}{3}x^6 \Big|_0^2 - 2x^4 \Big|_0^2 = -\frac{512}{3} + \frac{256}{3} - 32 = -\frac{352}{3}\end{aligned}$$

23.
$$\begin{aligned}\oint_C (x^2 - y^2) ds &= \int_0^{2\pi} (25 \cos^2 \theta - 25 \sin^2 \theta) \sqrt{25 \sin^2 \theta + 25 \cos^2 \theta} d\theta = 125 \int_0^{2\pi} (\cos^2 \theta - \sin^2 \theta) d\theta \\ &= 125 \int_0^{2\pi} \cos 2\theta d\theta = \frac{125}{2} \sin 2\theta \Big|_0^{2\pi} = 0\end{aligned}$$

24.
$$\oint_C y dx - x dy = \int_0^\pi 3 \sin t(-2 \sin t) dt - \int_0^\pi 2 \cos t(3 \cos t) dt = -6 \int_0^\pi (\sin^2 t + \cos^2 t) dt = -6 \int_0^\pi dt = -6\pi$$

Thus, $\int_{-C} y dx - x dy = 6\pi$.

25. We parameterize the line segment from $(0, 0, 0)$ to $(2, 3, 4)$ by $x = 2t$, $y = 3t$, $z = 4t$ for $0 \leq t \leq 1$. We parameterize the line segment from $(2, 3, 4)$ to $(6, 8, 5)$ by $x = 2 + 4t$, $y = 3 + 5t$, $z = 4 + t$, $0 \leq t \leq 1$.

$$\begin{aligned}\oint_C y dx + z dy + x dz &= \int_0^1 3t(2 dt) + \int_0^1 4t(3 dt) + \int_0^1 2t(4 dt) + \int_0^1 (3 + 5t)(4 dt) \\ &\quad + \int_0^1 (4 + t)(5 dt) + \int_0^1 (2 + 4t) dt \\ &= \int_0^1 (55t + 34) dt = \left(\frac{55}{2}t^2 + 34t\right) \Big|_0^1 = \frac{123}{2}\end{aligned}$$

26.
$$\begin{aligned}\int_C y dx + z dy + x dz &= \int_0^2 t^3(3 dt) + \int_0^2 \left(\frac{5}{4}t^2\right)(3t^2 dt) + \int_0^2 (3t)\left(\frac{5}{2}t dt\right) \\ &= \int_0^2 \left(3t^3 + \frac{15}{4}t^4 + \frac{15}{2}t^2\right) dt = \left(\frac{3}{4}t^4 + \frac{3}{4}t^5 + \frac{5}{2}t^3\right) \Big|_0^2 = 56\end{aligned}$$

27. From $(0, 0, 0)$ to $(6, 0, 0)$ we use x as a parameter with $y = dy = 0$ and $z = dz = 0$. From $(6, 0, 0)$ to $(6, 0, 5)$ we use z as a parameter with $x = 6$ and $dx = 0$ and $y = dy = 0$. From $(6, 0, 5)$ to $(6, 8, 5)$ we use y as a parameter with $x = 6$ and $dx = 0$ and $z = 5$ and $dz = 0$.

$$\int_C y \, dx + z \, dy + x \, dz = \int_0^6 0 + \int_0^5 6 \, dz + \int_0^8 5 \, dy = 70$$

28. We parametrize the line segment from $(0, 0, 0)$ to $(6, 8, 0)$ by $x = 6t$, $y = 8t$, $z = 0$ for $0 \leq t \leq 1$. From $(6, 8, 0)$ to $(6, 8, 5)$ we use z as a parameter with $x = 6$, $dx = 0$, and $y = 8$, $dy = 0$.

$$\int_C y \, dx + z \, dy + x \, dz = \int_0^1 8t(6 \, dt) + \int_0^5 6 \, dz = 24t^2 \Big|_0^1 + 30 = 54$$

29. $\mathbf{F} = e^{3t}\mathbf{i} - (e^{-4t})e^t\mathbf{j} = e^{3t}\mathbf{i} - e^{-3t}\mathbf{j}$; $d\mathbf{r} = (-2e^{-2t}\mathbf{i} + e^t\mathbf{j}) \, dt$; $\mathbf{F} \cdot d\mathbf{r} = (-2e^t - e^{-2t}) \, dt$;

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{\ln 2} (-2e^t - e^{-2t}) \, dt = \left(-2e^t + \frac{1}{2}e^{-2t} \right) \Big|_0^{\ln 2} = -\frac{31}{8} - \left(-\frac{3}{2} \right) = -\frac{19}{8}$$

30. $\mathbf{F} = e^t\mathbf{i} + te^{t^3}\mathbf{j} + t^3e^{t^6}\mathbf{k}$; $d\mathbf{r} = (\mathbf{i} + 2t\mathbf{j} + 3t^2\mathbf{k}) \, dt$;

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^1 (e^t + 2t^2e^{t^3} + 3t^5e^{t^6}) \, dt = \left(e^t + \frac{2}{3}e^{t^3} + \frac{1}{2}e^{t^6} \right) \Big|_0^1 = \frac{13}{6}(e - 1)$$

31. Using x as a parameter, $\mathbf{r}(x) = x\mathbf{i} + \ln x\mathbf{j}$. Then $\mathbf{F} = \ln x\mathbf{i} + x\mathbf{j}$, $d\mathbf{r} = (\mathbf{i} + \frac{1}{x}\mathbf{j}) \, dx$, and

$$W = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_1^e (\ln x + 1) \, dx = (x \ln x) \Big|_1^e = e.$$

32. Let $\mathbf{r}_1 = (-2 + 2t)\mathbf{i} + (2 - 2t)\mathbf{j}$ and $\mathbf{r}_2 = 2t\mathbf{i} + 3t\mathbf{j}$ for $0 \leq t \leq 1$. Then

$$d\mathbf{r}_1 = 2\mathbf{i} - 2\mathbf{j}, \quad d\mathbf{r}_2 = 2\mathbf{i} + 3\mathbf{j},$$

$$\mathbf{F}_1 = 2(-2 + 2t)(2 - 2t)\mathbf{i} + 4(2 - 2t)^2\mathbf{j} = (-8t^2 + 16t - 8)\mathbf{i} + (16t^2 - 32t + 16)\mathbf{j},$$

$$\mathbf{F}_2 = 2(2t)(3t)\mathbf{i} + 4(3t)^2\mathbf{j} = 12t^2\mathbf{i} + 36t^2\mathbf{j},$$

and

$$\begin{aligned} W &= \int_{C_1} \mathbf{F}_1 \cdot d\mathbf{r}_1 + \int_{C_2} \mathbf{F}_2 \cdot d\mathbf{r}_2 = \int_0^1 (-16t^2 + 32t - 16 - 32t^2 + 64t - 32) \, dt + \int_0^1 (24t^2 + 108t^2) \, dt \\ &= \int_0^1 (84t^2 + 96t - 48) \, dt = (28t^3 + 48t^2 - 48t) \Big|_0^1 = 28. \end{aligned}$$

33. Let $\mathbf{r}_1 = (1 + 2t)\mathbf{i} + \mathbf{j}$, $\mathbf{r}_2 = 3\mathbf{i} + (1 + t)\mathbf{j}$, and $\mathbf{r}_3 = (3 - 2t)\mathbf{i} + (2 - t)\mathbf{j}$ for $0 \leq t \leq 1$. Then

$$d\mathbf{r}_1 = 2\mathbf{i}, \quad d\mathbf{r}_2 = \mathbf{j}, \quad d\mathbf{r}_3 = -2\mathbf{i} - \mathbf{j},$$

$$\mathbf{F}_1 = (1 + 2t + 2)\mathbf{i} + (6 - 2 - 4t)\mathbf{j} = (3 + 2t)\mathbf{i} + (4 - 4t)\mathbf{j},$$

$$\mathbf{F}_2 = (3 + 2 + 2t)\mathbf{i} + (6 + 6t - 6)\mathbf{j} = (5 + 2t)\mathbf{i} + 6t\mathbf{j},$$

$$\mathbf{F}_3 = (3 - 2t + 4 - 2t)\mathbf{i} + (12 - 6t - 6 + 4t)\mathbf{j} = (7 - 4t)\mathbf{i} + (6 - 2t)\mathbf{j},$$

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and

$$\begin{aligned} W &= \int_{C_1} \mathbf{F}_1 \cdot d\mathbf{r}_1 + \int_{C_2} \mathbf{F}_2 \cdot d\mathbf{r}_2 + \int_{C_3} \mathbf{F}_3 \cdot d\mathbf{r}_3 \\ &= \int_0^1 (6 + 4t) dt + \int_0^1 6t dt + \int_0^1 (-14 + 8t - 6 + 2t) dt \\ &= \int_0^1 (-14 + 20t) dt = (-14t + 10t^2) \Big|_0^1 = -4. \end{aligned}$$

34. $\mathbf{F} = t^3\mathbf{i} + t^4\mathbf{j} + t^5\mathbf{k}$; $d\mathbf{r} = 3t^2\mathbf{i} + 2t\mathbf{j} + \mathbf{k}$; $W = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_1^3 (3t^5 + 2t^5 + t^5) dt = \int_1^3 6t^5 dt = t^6 \Big|_1^3 = 728$

35. $\mathbf{r} = 3 \cos t \mathbf{i} + 3 \sin t \mathbf{j}$, $0 \leq t \leq 2\pi$; $d\mathbf{r} = -3 \sin t \mathbf{i} + 3 \cos t \mathbf{j}$; $\mathbf{F} = a\mathbf{i} + b\mathbf{j}$;

$$W = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} (-3a \sin t + 3b \cos t) dt = (3a \cos t + 3b \sin t) \Big|_0^{2\pi} = 0$$

36. Let $\mathbf{r} = t\mathbf{i} + t\mathbf{j} + t\mathbf{k}$ for $1 \leq t \leq 3$. Then $d\mathbf{r} = \mathbf{i} + \mathbf{j} + \mathbf{k}$, and

$$\begin{aligned} \mathbf{F} &= \frac{c}{|\mathbf{r}|^3} (t\mathbf{i} + t\mathbf{j} + t\mathbf{k}) = \frac{ct}{(\sqrt{3}t^2)^3} (\mathbf{i} + \mathbf{j} + \mathbf{k}) = \frac{c}{3\sqrt{3}t^2} (\mathbf{i} + \mathbf{j} + \mathbf{k}), \\ W &= \int_C \mathbf{F} \cdot d\mathbf{r} = \int_1^3 \frac{c}{3\sqrt{3}t^2} (1 + 1 + 1) dt = \frac{c}{\sqrt{3}} \int_1^3 \frac{1}{t^2} dt = \frac{c}{\sqrt{3}} \left(-\frac{1}{t} \right) \Big|_1^3 = \frac{c}{\sqrt{3}} \left(-\frac{1}{3} + 1 \right) = \frac{2c}{3\sqrt{3}}. \end{aligned}$$

37. $\int_{C_1} y^2 dx + xy dy = \int_0^1 (4t + 2)^2 2 dt + \int_0^1 (2t + 1)(4t + 2) 4 dt = \int_0^1 (64t^2 + 64t + 16) dt$

$$= \left(\frac{64}{3} t^3 + 32t^2 + 16t \right) \Big|_0^1 = \frac{64}{3} + 32 + 16 = \frac{208}{3}$$

$$\int_{C_2} y^2 dx + xy dy = \int_1^{\sqrt{3}} 4t^4 (2t) dt + \int_1^{\sqrt{3}} 2t^4 (4t) dt = \int_1^{\sqrt{3}} 16t^5 dt = \frac{8}{3} t^6 \Big|_1^{\sqrt{3}} = 72 - \frac{8}{3} = \frac{208}{3}$$

$$\int_{C_3} y^2 dx + xy dy = \int_e^{e^3} 4(\ln t)^2 \frac{1}{t} dt + \int_e^{e^3} 2(\ln t)^2 \frac{2}{t} dt = \int_e^{e^3} \frac{8}{t} (\ln t)^2 dt = \frac{8}{3} (\ln t)^3 \Big|_e^{e^3} = \frac{8}{3} (27 - 1) = \frac{208}{3}$$

38. $\int_{C_1} xy ds = \int_0^2 t(2t)\sqrt{1+4} dt = 2\sqrt{5} \int_0^2 t^2 dt = 2\sqrt{5} \left(\frac{1}{3} t^3 \right) \Big|_0^2 = \frac{16\sqrt{5}}{3}$

$$\begin{aligned} \int_{C_2} xy ds &= \int_0^2 t(t^2)\sqrt{1+4t^2} dt = \int_0^2 t^3 \sqrt{1+4t^2} dt \quad \boxed{u = 1 + 4t^2, \quad du = 8t dt, \quad t^2 = \frac{1}{4}(u - 1)} \\ &= \int_1^{17} \frac{1}{4}(u - 1)u^{1/2} \frac{1}{8} du = \frac{1}{32} \int_1^{17} (u^{3/2} - u^{1/2}) du = \frac{1}{32} \left(\frac{2}{5} u^{5/2} - \frac{2}{3} u^{3/2} \right) \Big|_1^{17} = \frac{391\sqrt{17} + 1}{120} \end{aligned}$$

$$\int_{C_3} xy ds = \int_2^3 (2t - 4)(4t - 8)\sqrt{4 + 16} dt = 16\sqrt{5} \int_2^3 (t - 2)^2 dt = 16\sqrt{5} \left[\frac{1}{3} (t - 2)^3 \right] \Big|_2^3 = \frac{16\sqrt{5}}{3}$$

C_1 and C_3 are different parameterizations of the same curve, while C_1 and C_2 are different curves.

39. Since $\mathbf{v} \cdot \mathbf{v} = v^2$, $\frac{d}{dt} v^2 = \frac{d}{dt} (\mathbf{v} \cdot \mathbf{v}) = \mathbf{v} \cdot \frac{d\mathbf{v}}{dt} + \frac{d\mathbf{v}}{dt} \cdot \mathbf{v} = 2 \frac{d\mathbf{v}}{dt} \cdot \mathbf{v}$. Then

$$\begin{aligned} W &= \int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b m\mathbf{a} \cdot \left(\frac{d\mathbf{r}}{dt} dt \right) = m \int_a^b \frac{d\mathbf{v}}{dt} \cdot \mathbf{v} dt = m \int_a^b \frac{1}{2} \left(\frac{d}{dt} v^2 \right) dt \\ &= \frac{1}{2} m(v^2) \Big|_a^b = \frac{1}{2} m[v(b)]^2 - \frac{1}{2} m[v(a)]^2. \end{aligned}$$

40. We are given $\rho = kx$. Then

$$\begin{aligned} m &= \int_C \rho \, ds = \int_0^\pi kx \, ds = k \int_0^\pi (1 + \cos t) \sqrt{\sin^2 t + \cos^2 t} \, dt = k \int_0^\pi (1 + \cos t) \, dt \\ &= k(t + \sin t) \Big|_0^\pi = k\pi. \end{aligned}$$

41. From Problem 40, $m = k\pi$ and $ds = dt$.

$$M_x = \int_C y \rho \, ds = \int_C kxy \, ds = k \int_0^\pi (1 + \cos t) \sin t \, dt = k \left(-\cos t + \frac{1}{2} \sin^2 t \right) \Big|_0^\pi = 2k$$

$$\begin{aligned} M_y &= \int_C x \rho \, ds = \int_C kx^2 \, ds = k \int_0^\pi (1 + \cos t)^2 \, dt = k \int_0^\pi (1 + 2\cos t + \cos^2 t) \, dt \\ &= k \left(t + 2\sin t + \frac{1}{2}t + \frac{1}{4}\sin 2t \right) \Big|_0^\pi = \frac{3}{2}k\pi \end{aligned}$$

$$\bar{x} = M_y/m = \frac{3k\pi/2}{k\pi} = \frac{3}{2}; \quad \bar{y} = M_x/m = \frac{2k}{k\pi} = \frac{2}{\pi}. \quad \text{The center of mass is } (3/2, 2/\pi).$$

42. On C_1 , $\mathbf{T} = \mathbf{i}$ and $\mathbf{F} \cdot \mathbf{T} = \text{comp}_{\mathbf{T}} \mathbf{F} \approx 1$. On C_2 , $\mathbf{T} = -\mathbf{j}$ and $\mathbf{F} \cdot \mathbf{T} = \text{comp}_{\mathbf{T}} \mathbf{F} \approx 2$. On C_3 , $\mathbf{T} = -\mathbf{i}$ and $\mathbf{F} \cdot \mathbf{T} = \text{comp}_{\mathbf{T}} \mathbf{F} \approx 1.5$. Using the fact that the lengths of C_1 , C_2 , and C_3 are 4, 5, and 5, respectively, we have

$$W = \int_C \mathbf{F} \cdot \mathbf{T} \, ds = \int_{C_1} \mathbf{F} \cdot \mathbf{T} \, ds + \int_{C_2} \mathbf{F} \cdot \mathbf{T} \, ds + \int_{C_3} \mathbf{F} \cdot \mathbf{T} \, ds \approx 1(4) + 2(5) + 1.5(5) = 21.5 \text{ ft-lb.}$$

Exercises 9.9

1. (a) $P_y = 0 = Q_x$ and the integral is independent of path. $\phi_x = x^2$, $\phi = \frac{1}{3}x^3 + g(y)$,

$$\phi_y = g'(y) = y^2, \quad g(y) = \frac{1}{3}y^3, \quad \phi = \frac{1}{3}x^3 + \frac{1}{3}y^3, \quad \int_{(0,0)}^{(2,2)} x^2 \, dx + y^2 \, dy = \frac{1}{3}(x^3 + y^3) \Big|_{(0,0)}^{(2,2)} = \frac{16}{3}$$

(b) Use $y = x$ for $0 \leq x \leq 2$. $\int_{(0,0)}^{(2,2)} x^2 \, dx + y^2 \, dy = \int_0^2 (x^2 + x^2) \, dx = \frac{2}{3}x^3 \Big|_0^2 = \frac{16}{3}$

2. (a) $P_y = 2x = Q_x$ and the integral is independent of path. $\phi_x = 2xy$, $\phi = x^2y + g(y)$,

$$\phi_y = x^2 + g'(y) = x^2, \quad g(y) = 0, \quad \phi = x^2y, \quad \int_{(1,1)}^{(2,4)} 2xy \, dx + x^2 \, dy = x^2y \Big|_{(1,1)}^{(2,4)} = 16 - 1 = 15$$

(b) Use $y = 3x - 2$ for $1 \leq x \leq 2$.

$$\int_{(1,1)}^{(2,4)} 2xy \, dx + x^2 \, dy = \int_1^2 [2x(3x - 2) + x^2(3)] \, dx = \int_1^2 (9x^2 - 4x) \, dx = (3x^3 - 2x^2) \Big|_1^2 = 15$$

3. (a) $P_y = 2 = Q_x$ and the integral is independent of path. $\phi_x = x + 2y$, $\phi = \frac{1}{2}x^2 + 2xy + g(y)$,

$$\phi_y = 2x + g'(y) = 2x - y, \quad g(y) = -\frac{1}{2}y^2, \quad \phi = \frac{1}{2}x^2 + 2xy - \frac{1}{2}y^2,$$

$$\int_{(1,0)}^{(3,2)} (x + 2y) \, dx + (2x - y) \, dy = \left(\frac{1}{2}x^2 + 2xy - \frac{1}{2}y^2 \right) \Big|_{(1,0)}^{(3,2)} = 14$$

Exercises 9.9

- (b) Use $y = x - 1$ for $1 \leq x \leq 3$.

$$\begin{aligned} \int_{(1,0)}^{(3,2)} (x+2y) dx + (2x-y) dy &= \int_1^3 [x+2(x-1)+2x-(x-1)] dx \\ &= \int_1^3 (4x-1) dx = (2x^2-x) \Big|_1^3 = 14 \end{aligned}$$

4. (a) $P_y = -\cos x \sin y = Q_x$ and the integral is independent of path. $\phi_x = \cos x \cos y$,
 $\phi = \sin x \cos y + g(y)$, $\phi_y = -\sin x \sin y + g'(y) = 1 - \sin x \sin y$, $g(y) = y$, $\phi = \sin x \cos y + y$,

$$\int_{(0,0)}^{(\pi/2,0)} \cos x \cos y dx + (1 - \sin x \sin y) dy = (\sin x \cos y + y) \Big|_{(0,0)}^{(\pi/2,0)} = 1$$

- (b) Use $y = 0$ for $0 \leq x \leq \pi/2$.

$$\int_{(0,0)}^{(\pi/2,0)} \cos x \cos y dx + (1 - \sin x \sin y) dy = \int_0^{\pi/2} \cos x dx = \sin x \Big|_0^{\pi/2} = 1$$

5. (a) $P_y = 1/y^2 = Q_x$ and the integral is independent of path. $\phi_x = -\frac{1}{y}$, $\phi = -\frac{x}{y} + g(y)$, $\phi_y = \frac{x}{y^2} + g'(y) = \frac{x}{y^2}$,

$$g(y) = 0, \quad \phi = -\frac{x}{y}, \quad \int_{(4,1)}^{(4,4)} -\frac{1}{y} dx + \frac{x}{y^2} dy = \left(-\frac{x}{y}\right) \Big|_{(4,1)}^{(4,4)} = 3$$

- (b) Use $x = 4$ for $1 \leq y \leq 4$.

$$\int_{(4,1)}^{(4,4)} -\frac{1}{y} dx + \frac{x}{y^2} dy = \int_1^4 \frac{4}{y^2} dy = -\frac{4}{y} \Big|_1^4 = 3$$

6. (a) $P_y = -xy(x^2+y^2)^{-3/2} = Q_x$ and the integral is independent of path. $\phi_x = \frac{x}{\sqrt{x^2+y^2}}$,

$$\phi = \sqrt{x^2+y^2} + g(y), \quad \phi_y = \frac{y}{\sqrt{x^2+y^2}} + g'(y) = \frac{y}{\sqrt{x^2+y^2}}, \quad g(y) = 0, \quad \phi = \sqrt{x^2+y^2},$$

$$\int_{(1,0)}^{(3,4)} \frac{x dx + y dy}{\sqrt{x^2+y^2}} = \sqrt{x^2+y^2} \Big|_{(1,0)}^{(3,4)} = 4$$

- (b) Use $y = 2x - 2$ for $1 \leq x \leq 3$.

$$\int_{(1,0)}^{(3,4)} \frac{x dx + y dy}{\sqrt{x^2+y^2}} = \int_1^3 \frac{x + (2x-2)2}{\sqrt{x^2+(2x-2)^2}} dx = \int_1^3 \frac{5x-4}{\sqrt{5x^2-8x+4}} = \sqrt{5x^2-8x+4} \Big|_1^3 = 4$$

7. (a) $P_y = 4xy = Q_x$ and the integral is independent of path. $\phi_x = 2y^2x - 3$, $\phi = x^2y^2 - 3x + g(y)$,
 $\phi_y = 2x^2y + g'(y) = 2x^2y + 4$, $g(y) = 4y$, $\phi = x^2y^2 - 3x + 4y$,

$$\int_{(1,2)}^{(3,6)} (2y^2x - 3) dx + (2yx^2 + 4) dy = (x^2y^2 - 3x + 4y) \Big|_{(1,2)}^{(3,6)} = 330$$

- (b) Use $y = 2x$ for $1 \leq x \leq 3$.

$$\begin{aligned} \int_{(1,2)}^{(3,6)} (2y^2x - 3) dx + (2yx^2 + 4) dy &= \int_1^3 ([2(2x)^2x - 3] + [2(2x)x^2 + 4]2) dx \\ &= \int_1^3 (16x^3 + 5) dx = (4x^4 + 5x) \Big|_1^3 = 330 \end{aligned}$$

8. (a) $P_y = 4 = Q_x$ and the integral is independent of path. $\phi_x = 5x + 4y$, $\phi = \frac{5}{2}x^2 + 4xy + g(y)$,
 $\phi_y = 4x + g'(y) = 4x - 8y^3$, $g(y) = -2y^4$, $\phi = \frac{5}{2}x^2 + 4xy - 2y^4$,

$$\int_{(-1,1)}^{(0,0)} (5x + 4y) dx + (4x - 8y^3) dy = \left(\frac{5}{2}x^2 + 4xy - 2y^4 \right) \Big|_{(-1,1)}^{(0,0)} = \frac{7}{2}$$

(b) Use $y = -x$ for $-1 \leq x \leq 0$.

$$\begin{aligned} \int_{(-1,1)}^{(0,0)} (5x + 4y) dx + (4x - 8y^3) dy &= \int_{-1}^0 [(5x - 4x) + (4x + 8x^3)(-1)] dx \\ &= \int_{-1}^0 (-3x - 8x^3) dx = \left(-\frac{3}{2}x^2 - 2x^4 \right) \Big|_{-1}^0 = \frac{7}{2} \end{aligned}$$

9. (a) $P_y = 3y^2 + 3x^2 = Q_x$ and the integral is independent of path. $\phi_x = y^3 + 3x^2y$,
 $\phi = xy^3 + x^3y + g(y)$, $\phi_y = 3xy^2 + x^3 + g'(y) = x^3 + 3y^2x + 1$, $g(y) = y$, $\phi = xy^3 + x^3y + y$,
 $\int_{(0,0)}^{(2,8)} (y^3 + 3x^2y) dx + (x^3 + 3y^2x + 1) dy = (xy^3 + x^3y + y) \Big|_{(0,0)}^{(2,8)} = 1096$

(b) Use $y = 4x$ for $0 \leq x \leq 2$.

$$\begin{aligned} \int_{(0,0)}^{(2,8)} (y^3 + 3x^2y) dx + (x^3 + 3y^2x + 1) dy &= \int_0^2 [(64x^3 + 12x^3) + (x^3 + 48x^3 + 1)(4)] dx \\ &= \int_0^2 (272x^3 + 4) dx = (68x^4 + 4x) \Big|_0^2 = 1096 \end{aligned}$$

10. (a) $P_y = -xy \cos xy - \sin xy - 20y^3 = Q_x$ and the integral is independent of path.

$$\begin{aligned} \phi_x &= 2x - y \sin xy - 5y^4, \quad \phi = x^2 + \cos xy - 5xy^4 + g(y), \\ \phi_y &= -x \sin xy - 20xy^3 + g'(y) = -20xy^3 - x \sin xy, \quad g(y) = 0, \quad \phi = x^2 + \cos xy - 5xy^4, \\ \int_{(-2,0)}^{(1,0)} (2x - y \sin xy - 5y^4) dx - (20xy^3 + x \sin xy) dy &= (x^2 + \cos xy - 5xy^4) \Big|_{(-2,0)}^{(1,0)} = -3 \end{aligned}$$

(b) Use $y = 0$ for $-2 \leq x \leq 1$.

$$\int_{(-2,0)}^{(1,0)} (2x - y \sin xy - 5y^4) dx - (20xy^3 + x \sin xy) dy = \int_{-2}^1 2x dx = x^2 \Big|_{-2}^1 = -3$$

11. $P_y = 12x^3y^2 = Q_x$ and the vector field is a gradient field. $\phi_x = 4x^3y^3 + 3$, $\phi = x^4y^3 + 3x + g(y)$,
 $\phi_y = 3x^4y^2 + g'(y) = 3x^4y^2 + 1$, $g(y) = y$, $\phi = x^4y^3 + 3x + y$
12. $P_y = 6xy^2 = Q_x$ and the vector field is a gradient field. $\phi_x = 2xy^3$, $\phi = x^2y^3 + g(y)$,
 $\phi_y = 3x^2y^2 + g'(y) = 3x^2y^2 + 3y^2$, $g(y) = y^3$, $\phi = x^2y^3 + y^3$
13. $P_y = -2xy^3 \sin xy^2 + 2y \cos xy^2$, $Q_x = -2xy^3 \cos xy^2 - 2y \sin xy^2$ and the vector field is not a gradient field.
14. $P_y = -4xy(x^2 + y^2 + 1)^{-3} = Q_x$ and the vector field is a gradient field.
 $\phi_x = x(x^2 + y^2 + 1)^{-2}$, $\phi = -\frac{1}{2}(x^2 + y^2 + 1)^{-1} + g(y)$, $\phi_y = y(x^2 + y^2 + 1)^{-2} + g'(y) = y(x^2 + y^2 + 1)^{-2}$,
 $g(y) = 0$, $\phi = -\frac{1}{2}(x^2 + y^2 + 1)^{-1}$
15. $P_y = 1 = Q_x$ and the vector field is a gradient field. $\phi_x = x^3 + y$, $\phi = \frac{1}{4}x^4 + xy + g(y)$, $\phi_y = x + g'(y) = x + y^3$,
 $g(y) = \frac{1}{4}y^4$, $\phi = \frac{1}{4}x^4 + xy + \frac{1}{4}y^4$
16. $P_y = 4e^{2y}$, $Q_x = e^{2y}$ and the vector field is not a gradient field.

Exercises 9.9

17. Since $P_y = -e^{-y} = Q_x$, $\mathbf{F}$ is conservative and $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of the path. Thus, instead of the given curve we may use the simpler curve $C_1: y = x, 0 \leq x \leq 1$. Then

$$\begin{aligned} W &= \int_{C_1} (2x + e^{-y}) dx + (4y - xe^{-y}) dy \\ &= \int_0^1 (2x + e^{-x}) dx + \int_0^1 (4x - xe^{-x}) dx \quad \boxed{\text{Integration by parts}} \\ &= (x^2 - e^{-x}) \Big|_0^1 + (2x^2 + xe^{-x} + e^{-x}) \Big|_0^1 \\ &= [(1 - e^{-1}) - (-1)] + [(2 + e^{-1} + e^{-1}) - (1)] = 3 + e^{-1}. \end{aligned}$$

18. Since $P_y = -e^{-y} = Q_x$, $\mathbf{F}$ is conservative and $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of the path. Thus, instead of the given curve we may use the simpler curve $C_1: y = 0, -2 \leq -x \leq 2$. Then $dy = 0$ and

$$W = \int_{C_1} (2x + e^{-y}) dx + (4y - xe^{-y}) dy = \int_2^{-2} (2x + 1) dx = (x^2 + x) \Big|_2^{-2} = (4 - 2) - (4 + 2) = -4.$$

19. $P_y = z = Q_x$, $Q_z = x = R_y$, $R_x = y = P_z$, and the integral is independent of path. Parameterize the line segment between the points by $x = 1 + t$, $y = 1 + 3t$, $z = 1 + 7t$, for $0 \leq t \leq 1$. Then $dx = dt$, $dy = 3 dt$, $dz = 7 dt$ and

$$\begin{aligned} \int_{(1,1,1)}^{(2,4,8)} yz dx + xz dy + xy dz &= \int_0^1 [(1 + 3t)(1 + 7t) + (1 + t)(1 + 7t)(3) + (1 + t)(1 + 3t)(7)] dt \\ &= \int_0^1 (11 + 62t + 63t^2) dt = (11t + 31t^2 + 21t^3) \Big|_0^1 = 63. \end{aligned}$$

20. $P_y = 0 = Q_x$, $Q_z = 0 = R_y$, $R_x = 0 = P_z$ and the integral is independent of path. Parameterize the line segment between the points by $x = t$, $y = t$, $z = t$, for $0 \leq t \leq 1$. Then $dx = dy = dz = dt$ and

$$\int_{(0,0,0)}^{(1,1,1)} 2x dx + 3y^2 dy + 4z^3 dz = \int_0^1 (2t + 3t^2 + 4t^3) dt = (t^2 + t^3 + t^4) \Big|_0^1 = 3.$$

21. $P_y = 2x \cos y = Q_x$, $Q_z = 0 = R_y$, $R_x = 3e^{3z} = P_z$, and the integral is independent of path. Integrating $\phi_x = 2x \sin y + e^{3z}$ we find $\phi = x^2 \sin y + xe^{3z} + g(y, z)$. Then $\phi_y = x^2 \cos y + g_y = Q = x^2 \cos y$, so $g_y = 0$, $g(y, z) = h(z)$, and $\phi = x^2 \sin y + xe^{3z} + h(z)$. Now $\phi_z = 3xe^{3z} + h'(z) = R = 3xe^{3z} + 5$, so $h'(z) = 5$ and $h(z) = 5z$. Thus $\phi = x^2 \sin y + xe^{3z} + 5z$ and

$$\begin{aligned} \int_{(1,0,0)}^{(2,\pi/2,1)} (2x \sin y + e^{3z}) dx + x^2 \cos y dy + (3xe^{3z} + 5) dz \\ = (x^2 \sin y + xe^{3z} + 5z) \Big|_{(1,0,0)}^{(2,\pi/2,1)} = [4(1) + 2e^3 + 5] - [0 + 1 + 0] = 8 + 2e^3. \end{aligned}$$

22. $P_y = 0 = Q_x$, $Q_z = 0 = R_y$, $R_x = 0 = P_z$, and the integral is independent of path. Parameterize the line segment between the points by $x = 1 + 2t$, $y = 2 + 2t$, $z = 1$, for $0 \leq t \leq 1$. Then $dx = 2 dt$, $dz = 0$ and

$$\begin{aligned} \int_{(1,2,1)}^{(3,4,1)} (2x + 1) dx + 3y^2 dy + \frac{1}{z} dz &= \int_0^1 [(2 + 4t + 1)2 + 3(2 + 2t)^2 2] dt \\ &= \int_0^1 (24t^2 + 56t + 30) dt = (8t^3 + 28t^2 + 30t) \Big|_0^1 = 66. \end{aligned}$$

23. $P_y = 0 = Q_x$, $Q_z = 0 = R_y$, $R_x = 2e^{2z} = P_z$ and the integral is independent of path. Parameterize the line segment between the points by $x = 1 + t$, $y = 1 + t$, $z = \ln 3$, for $0 \leq t \leq 1$. Then $dx = dy = dt$, $dz = 0$ and

$$\int_{(1,1,\ln 3)}^{(2,2,\ln 3)} e^{2z} dx + 3y^2 dy + 2xe^{2z} dz = \int_0^1 [e^{2\ln 3} + 3(1+t)^2] dt = [9t + (1+t)^3] \Big|_0^1 = 16.$$

24. $P_y = 0 = Q_x$, $Q_z = 2y = R_y$, $R_x = 2x = P_z$ and the integral is independent of path. Parameterize the line segment between the points by $x = -2(1-t)$, $y = 3(1-t)$, $z = 1-t$, for $0 \leq t \leq 1$. Then $dx = 2dt$, $dy = -3dt$, $dz = -dt$, and

$$\begin{aligned} \int_{(-2,3,1)}^{(0,0,0)} 2xz dx + 2yz dy + (x^2 + y^2) dz \\ &= \int_0^1 [-4(1-t)^2(2) + 6(1-t)^2(-3) + 4(1-t)^2(-1) + 9(1-t)^2(-1)] dt \\ &= \int_0^1 -39(1-t)^2 dt = 13(1-t)^3 \Big|_0^1 = -13. \end{aligned}$$

25. $P_y = 1 - z \sin x = Q_x$, $Q_z = \cos x = R_y$, $R_x = -y \sin x = P_z$ and the integral is independent of path. Integrating $\phi_x = y - yz \sin x$ we find $\phi = xy + yz \cos x + g(y, z)$. Then $\phi_y = x + z \cos x + g_y(y, z) = Q = x + z \cos x$, so $g_y = 0$, $g(y, z) = h(z)$, and $\phi = xy + yz \cos x + h(z)$. Now $\phi_z = y \cos x + h'(z) = R = y \cos x$, so $h'(z) = 0$ and $\phi = xy + yz \cos x$. Since $\mathbf{r}(0) = 4\mathbf{j}$ and $\mathbf{r}(\pi/2) = \pi\mathbf{i} + \mathbf{j} + 4\mathbf{k}$,

$$\int_C \mathbf{F} \cdot d\mathbf{r} = (xy + yz \cos x) \Big|_{(0,4,0)}^{(\pi,1,4)} = (\pi - 4) - (0 + 0) = \pi - 4.$$

26. $P_y = 0 = Q_x$, $Q_z = 0 = R_y$, $R_x = -e^z = P_z$ and the integral is independent of path. Integrating $\phi_x = 2 - e^z$ we find $\phi = 2x - xe^z + g(y, z)$. Then $\phi_y = g_y = 2y - 1$, so $g(y, z) = y^2 - y + h(z)$ and $\phi = 2x - xe^z + y^2 - y + h(z)$. Now $\phi_z = -xe^z + h'(z) = R = 2 - xe^z$, so $h'(z) = 2$, $h(z) = 2z$, and $\phi = 2x - xe^z + y^2 - y + 2z$. Thus

$$\int_C \mathbf{F} \cdot d\mathbf{r} = (2x - xe^z + y^2 - y + 2z) \Big|_{(-1,1,-1)}^{(2,2,2)} = (4 - 2e^2 + 4 - 2 + 4) - (-2 + e^{-1} + 1 - 1 - 2) = 14 - 2e^2 - e^{-1}.$$

27. Since $P_y = Gm_1m_2(2xy/|\mathbf{r}|^5) = Q_x$, $Q_z = Gm_1m_2(2yz/|\mathbf{r}|^5) = R_y$, and $R_x = Gm_1m_2(2xz/|\mathbf{r}|^5) = P_z$, the force field is conservative.

$$\phi_x = -Gm_1m_2 \frac{x}{(x^2 + y^2 + z^2)^{3/2}}, \quad \phi = Gm_1m_2(x^2 + y^2 + z^2)^{-1/2} + g(y, z),$$

$$\phi_y = -Gm_1m_2 \frac{y}{(x^2 + y^2 + z^2)^{3/2}} + g_y(y, z) = -Gm_1m_2 \frac{y}{(x^2 + y^2 + z^2)^{3/2}}, \quad g(y, z) = h(z),$$

$$\phi = Gm_1m_2(x^2 + y^2 + z^2)^{-1/2} + h(z),$$

$$\phi_z = -Gm_1m_2 \frac{z}{(x^2 + y^2 + z^2)^{3/2}} + h'(z) = -Gm_1m_2 \frac{z}{(x^2 + y^2 + z^2)^{3/2}},$$

$$h(z) = 0, \quad \phi = \frac{Gm_1m_2}{\sqrt{x^2 + y^2 + z^2}} = \frac{Gm_1m_2}{|\mathbf{r}|}$$

28. Since $P_y = 24xy^2z = Q_x$, $Q_z = 12x^2y^2 = R_y$, and $R_x = 8xy^3 = P_z$, $\mathbf{F}$ is conservative. Thus, the work done between two points is independent of the path. From $\phi_x = 8xy^3z$ we obtain $\phi = 4x^2y^3z$ which is a potential function for $\mathbf{F}$. Then

$$W = \int_{(2,0,0)}^{(1,\sqrt{3},\pi/3)} \mathbf{F} \cdot d\mathbf{r} = 4x^2y^3z \Big|_{(2,0,0)}^{(1,\sqrt{3},\pi/3)} = 4\sqrt{3}\pi \quad \text{and} \quad W = \int_{(2,0,0)}^{(0,2,\pi/2)} \mathbf{F} \cdot d\mathbf{r} = 0.$$

Exercises 9.9

29. Since $\mathbf{F}$ is conservative, $\int_{C_1} \mathbf{F} \cdot d\mathbf{r} = \int_{-C_2} \mathbf{F} \cdot d\mathbf{r}$. Then, since the simply closed curve C is composed of C_1 and C_2 ,

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \int_{C_1} \mathbf{F} \cdot d\mathbf{r} + \int_{C_2} \mathbf{F} \cdot d\mathbf{r} = \int_{C_1} \mathbf{F} \cdot d\mathbf{r} - \int_{-C_2} \mathbf{F} \cdot d\mathbf{r} = 0.$$



30. From $\mathbf{F} = (x^2 + y^2)^{n/2}(\mathbf{x}\mathbf{i} + \mathbf{y}\mathbf{j})$ we obtain $P_y = nxy(x^2 + y^2)^{n/2-1} = Q_x$, so that $\mathbf{F}$ is conservative. From $\phi_x = x(x^2 + y^2)^{n/2}$ we obtain the potential function $\phi = (x^2 + y^2)^{(n+2)/2}/(n+2)$. Then

$$W = \int_{(x_1, y_1)}^{(x_2, y_2)} \mathbf{F} \cdot d\mathbf{r} = \left(\frac{(x^2 + y^2)^{(n+2)/2}}{n+2} \right) \Big|_{(x_1, y_1)}^{(x_2, y_2)} = \frac{1}{n+2} \left[(x_2^2 + y_2^2)^{(n+2)/2} - (x_1^2 + y_1^2)^{(n+2)/2} \right].$$

31. From Problem 39 in Exercises 9.8, $\frac{d\mathbf{v}}{dt} \cdot \frac{d\mathbf{r}}{dt} = \frac{d\mathbf{v}}{dt} \cdot \mathbf{v} = \frac{1}{2} \frac{d}{dt} v^2$. Then, using $\frac{dp}{dt} = \frac{\partial p}{\partial x} \frac{dx}{dt} + \frac{\partial p}{\partial y} \frac{dy}{dt} = \nabla p \cdot \frac{d\mathbf{r}}{dt}$, we have

$$\begin{aligned} \int m \frac{d\mathbf{v}}{dt} \cdot \frac{d\mathbf{r}}{dt} dt + \int \nabla p \cdot \frac{d\mathbf{r}}{dt} dt &= \int 0 dt \\ \frac{1}{2} m \int \frac{d}{dt} v^2 dt + \int \frac{dp}{dt} dt &= \text{constant} \\ \frac{1}{2} m v^2 + p &= \text{constant}. \end{aligned}$$

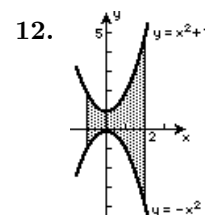
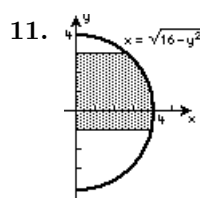
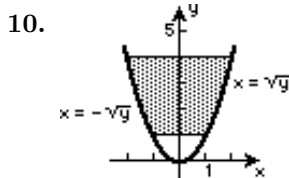
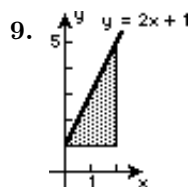
32. By Problem 31, the sum of kinetic and potential energies in a conservative force field is constant. That is, it is independent of points A and B , so $p(B) + K(B) = p(A) + K(A)$.

Exercises 9.10

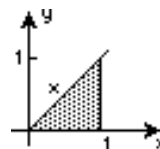
- $\int_{-1}^3 (6xy - 5e^y) dx = (3x^2y - 5xe^y) \Big|_{-1}^3 = (27y - 15e^y) - (3y + 5e^y) = 24y - 20e^y$
- $\int_1^2 \tan xy dy = \frac{1}{x} \ln |\sec xy| \Big|_1^2 = \frac{1}{x} \ln |\sec 2x - \sec x|$
- $\int_1^{3x} x^3 e^{xy} dy = x^2 e^{xy} \Big|_1^{3x} = x^2 (e^{3x^2} - e^x)$
- $\int_{\sqrt{y}}^{y^3} (8x^3y - 4xy^2) dx = (2x^4y - 2x^2y^2) \Big|_{\sqrt{y}}^{y^3} = (2y^{13} - 2y^8) - (2y^3 - 2y^3) = 2y^{13} - 2y^8$
- $\int_0^{2x} \frac{xy}{x^2 + y^2} dy = \frac{x}{2} \ln(x^2 + y^2) \Big|_0^{2x} = \frac{x}{2} [\ln(x^2 + 4x^2) - \ln x^2] = \frac{x}{2} \ln 5$
- $\int_{x^3}^x e^{2y/x} dy = \frac{x}{2} e^{2y/x} \Big|_{x^3}^x = \frac{x}{2} (e^{2x/x} - e^{2x^3/x}) = \frac{x}{2} (e^2 - e^{2x^2})$

$$\begin{aligned}
 7. \quad \int_{\tan y}^{\sec y} (2x + \cos y) dx &= (x^2 + x \cos y) \Big|_{\tan y}^{\sec y} = \sec^2 y + \sec y \cos y - \tan^2 y - \tan y \cos y \\
 &= \sec^2 y + 1 - \tan^2 y - \sin y = 2 - \sin y
 \end{aligned}$$

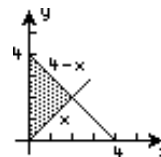
$$\begin{aligned}
 8. \quad \int_{\sqrt{y}}^1 y \ln x dx &\quad \boxed{\text{Integration by parts}} \\
 &= y(x \ln x - x) \Big|_{\sqrt{y}}^1 = y(0 - 1) - y(\sqrt{y} \ln \sqrt{y} - \sqrt{y}) = -y - y\sqrt{y} \left(\frac{1}{2} \ln y - 1 \right)
 \end{aligned}$$



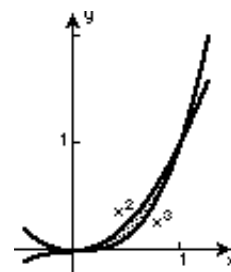
$$13. \quad \iint_R x^3 y^2 dA = \int_0^1 \int_0^x x^3 y^2 dy dx = \int_0^1 \frac{1}{3} x^3 y^3 \Big|_0^x dx = \frac{1}{3} \int_0^1 x^6 dx = \frac{1}{21} x^7 \Big|_0^1 = \frac{1}{21}$$



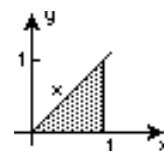
$$\begin{aligned}
 14. \quad \iint_R (x+1) dA &= \int_0^2 \int_x^{4-x} (x+1) dy dx = \int_0^2 (xy + y) \Big|_x^{4-x} dx \\
 &= \int_0^2 [(4x - x^2 + 4 - x) - (x^2 + x)] dx = \int_0^2 (2x - 2x^2 + 4) dx \\
 &= \left(x^2 - \frac{2}{3} x^3 + 4x \right) \Big|_0^2 = \frac{20}{3}
 \end{aligned}$$



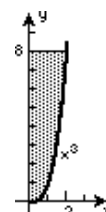
$$\begin{aligned}
 15. \quad \iint_R (2x + 4y + 1) dA &= \int_0^1 \int_{x^3}^{x^2} (2x + 4y + 1) dy dx = \int_0^1 (2xy + 2y^2 + y) \Big|_{x^3}^{x^2} dx \\
 &= \int_0^1 [(2x^3 + 2x^4 + x^2) - (2x^4 + 2x^6 + x^3)] dx \\
 &= \int_0^1 (x^3 + x^2 - 2x^6) dx = \left(\frac{1}{4} x^4 + \frac{1}{3} x^3 - \frac{2}{7} x^7 \right) \Big|_0^1 \\
 &= \frac{1}{4} + \frac{1}{3} - \frac{2}{7} = \frac{25}{84}
 \end{aligned}$$



$$\begin{aligned}
 16. \quad \iint_R x e^y dA &= \int_0^1 \int_0^x x e^y dy dx = \int_0^1 x e^y \Big|_0^x dx = \int_0^1 (x e^x - x) dx \\
 &\quad \boxed{\text{Integration by parts}} \\
 &= \left(x e^x - e^x - \frac{1}{2} x^2 \right) \Big|_0^1 = \left(e - e - \frac{1}{2} \right) - (-1) = \frac{1}{2}
 \end{aligned}$$

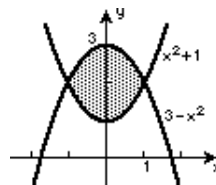


$$17. \quad \iint_R 2xy dA = \int_0^2 \int_{x^3}^8 2xy dy dx = \int_0^2 x y^2 \Big|_{x^3}^8 dx = \int_0^2 (64x - x^7) dx = \left(32x^2 - \frac{1}{8} x^8 \right) \Big|_0^2 = 96$$

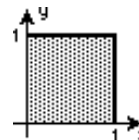


Exercises 9.10

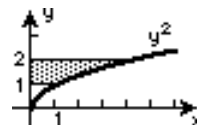
$$\begin{aligned}
 18. \quad \iint_R \frac{x}{\sqrt{y}} dA &= \int_{-1}^1 \int_{x^2+1}^{3-x^2} xy^{-1/2} dy dx = \int_{-1}^1 2x\sqrt{y} \Big|_{x^2+1}^{3-x^2} dx \\
 &= 2 \int_{-1}^1 (x\sqrt{3-x^2} - x\sqrt{x^2+1}) dx \\
 &= 2 \left[-\frac{1}{3}(3-x^2)^{3/2} - \frac{1}{3}(x^2+1)^{3/2} \right] \Big|_{-1}^1 \\
 &= -\frac{2}{3}[(2^{3/2} + 2^{3/2}) - (2^{3/2} + 2^{3/2})] = 0
 \end{aligned}$$



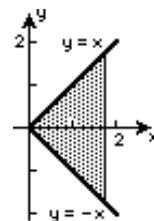
$$\begin{aligned}
 19. \quad \iint_R \frac{y}{1+xy} dA &= \int_0^1 \int_0^1 \frac{y}{1+xy} dx dy = \int_0^1 \ln(1+xy) \Big|_0^1 dy = \int_0^1 \ln(1+y) dy \\
 &= [(1+y)\ln(1+y) - (1+y)] \Big|_0^1 = (2\ln 2 - 2) - (-1) = 2\ln 2 - 1
 \end{aligned}$$



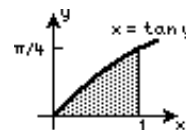
$$\begin{aligned}
 20. \quad \iint_R \sin \frac{\pi x}{y} dA &= \int_1^2 \int_0^{y^2} \sin \frac{\pi x}{y} dx dy = \int_1^2 \left(-\frac{y}{\pi} \cos \frac{\pi x}{y} \right) \Big|_0^{y^2} dy \\
 &= \int_1^2 \left(-\frac{y}{\pi} \cos \pi y + \frac{y}{\pi} \right) dy \quad \boxed{\text{Integration by parts}} \\
 &= \left(-\frac{y}{\pi^2} \sin \pi y - \frac{1}{\pi^3} \cos \pi y + \frac{y^2}{2\pi} \right) \Big|_1^2 = \left(-\frac{1}{\pi^3} + \frac{2}{\pi} \right) - \left(\frac{1}{\pi^3} + \frac{1}{2\pi} \right) \\
 &= \frac{3\pi^2 - 4}{2\pi^3}
 \end{aligned}$$



$$\begin{aligned}
 21. \quad \iint_R \sqrt{x^2+1} dA &= \int_0^{\sqrt{3}} \int_{-x}^x \sqrt{x^2+1} dy dx = \int_0^{\sqrt{3}} y\sqrt{x^2+1} \Big|_{-x}^x dx \\
 &= \int_0^{\sqrt{3}} (x\sqrt{x^2+1} + x\sqrt{x^2+1}) dx = \int_0^{\sqrt{3}} 2x\sqrt{x^2+1} dx \\
 &= \frac{2}{3}(x^2+1)^{3/2} \Big|_0^{\sqrt{3}} = \frac{2}{3}(4^{3/2} - 1^{3/2}) = \frac{14}{3}
 \end{aligned}$$



$$\begin{aligned}
 22. \quad \iint_R x dA &= \int_0^{\pi/4} \int_{\tan y}^1 x dx dy = \int_0^{\pi/4} \frac{1}{2}x^2 \Big|_{\tan y}^1 dy = \frac{1}{2} \int_0^{\pi/4} (1 - \tan^2 y) dy \\
 &= \frac{1}{2} \int_0^{\pi/4} (2 - \sec^2 y) dy = \frac{1}{2}(2y - \tan y) \Big|_0^{\pi/4} = \frac{1}{2} \left(\frac{\pi}{2} - 1 \right) = \frac{\pi}{4} - \frac{1}{2}
 \end{aligned}$$



23. The correct integral is (c).

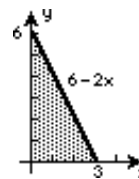
$$\begin{aligned}
 V &= 2 \int_{-2}^2 \int_0^{\sqrt{4-y^2}} (4-y) dx dy = 2 \int_{-2}^2 (4-y)x \Big|_0^{\sqrt{4-y^2}} dy = 2 \int_{-2}^2 (4-y)\sqrt{4-y^2} dy \\
 &= 2 \left[2y\sqrt{4-y^2} + 8 \sin^{-1} \frac{y}{2} + \frac{1}{3}(4-y^2)^{3/2} \right] \Big|_{-2}^2 = 2(4\pi - (-4\pi)) = 16\pi
 \end{aligned}$$

24. The correct integral is (b).

$$\begin{aligned}
 A &= 8 \int_0^2 \int_0^{\sqrt{4-y^2}} (4-y^2)^{1/2} dx dy = 8 \int_0^2 (4-y^2)^{1/2} x \Big|_0^{\sqrt{4-y^2}} dy = 8 \int_0^2 (4-y^2) dy \\
 &= 8 \left(4y - \frac{1}{3}y^3 \right) \Big|_0^2 = \frac{128}{3}
 \end{aligned}$$

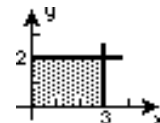
25. Setting $z = 0$ we have $y = 6 - 2x$.

$$\begin{aligned} V &= \int_0^3 \int_0^{6-2x} (6-2x-y) dy dx = \int_0^3 \left(6y - 2xy - \frac{1}{2}y^2 \right) \Big|_0^{6-2x} dx \\ &= \int_0^3 [6(6-2x) - 2x(6-2x) - \frac{1}{2}(6-2x)^2] dx = \int_0^3 (18 - 12x + 2x^2) dx \\ &= \left(18x - 6x^2 + \frac{2}{3}x^3 \right) \Big|_0^3 = 18 \end{aligned}$$



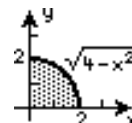
26. Setting $z = 0$ we have $y = \pm 2$.

$$V = \int_0^3 \int_0^2 (4-y^2) dy dx = \int_0^3 \left(4y - \frac{1}{3}y^3 \right) \Big|_0^2 dx = \int_0^3 \frac{16}{3} dx = 16$$



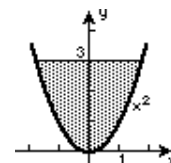
27. Solving for z , we have $x = 2 - \frac{1}{2}x + \frac{1}{2}y$. Setting $z = 0$, we see that this surface (plane) intersects the xy -plane in the line $y = x - 4$. since $z(0, 0) = 2 > 0$, the surface lies above the xy -plane over the quarter-circular region.

$$\begin{aligned} V &= \int_0^2 \int_0^{\sqrt{4-x^2}} \left(2 - \frac{1}{2}x + \frac{1}{2}y \right) dy dx = \int_0^2 \left(2y - \frac{1}{2}xy + \frac{1}{4}y^2 \right) \Big|_0^{\sqrt{4-x^2}} dx \\ &= \int_0^2 \left(2\sqrt{4-x^2} - \frac{1}{2}x\sqrt{4-x^2} + 1 - \frac{1}{4}x^2 \right) dx = \left[x\sqrt{4-x^2} + 4\sin^{-1} \frac{x}{2} + \frac{1}{6}(4-x^2)^{3/2} + x - \frac{1}{12}x^3 \right] \Big|_0^2 \\ &= \left(2\pi + 2 - \frac{2}{3} \right) - \frac{4}{3} = 2\pi \end{aligned}$$



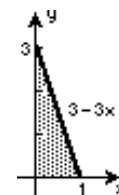
28. Setting $z = 0$ we have $y = 3$. Using symmetry,

$$\begin{aligned} V &= 2 \int_0^{\sqrt{3}} \int_{x^2}^3 (3-y) dy dx = 2 \int_0^{\sqrt{3}} \left(3y - \frac{1}{2}y^2 \right) \Big|_{x^2}^3 dx = 2 \int_0^{\sqrt{3}} \left(\frac{9}{2} - 3x^2 + \frac{1}{2}x^4 \right) dx \\ &= 2 \left(\frac{9}{2}x - x^3 + \frac{1}{10}x^5 \right) \Big|_0^{\sqrt{3}} = 2 \left(\frac{9}{2}\sqrt{3} - 3\sqrt{3} + \frac{9}{10}\sqrt{3} \right) = \frac{24\sqrt{3}}{5}. \end{aligned}$$



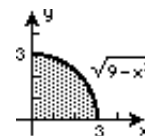
29. Note that $z = 1 + x^2 + y^2$ is always positive. Then

$$\begin{aligned} V &= \int_0^1 \int_0^{3-3x} (1+x^2+y^2) dy dx = \int_0^1 \left(y + x^2y + \frac{1}{3}y^3 \right) \Big|_0^{3-3x} dx \\ &= \int_0^1 [(3-3x) + x^2(3-3x) + 9(1-x)^3] dx = \int_0^1 (12-30x+30x^2-12x^3) dx \\ &= (12x-15x^2+10x^3-3x^4) \Big|_0^1 = 4. \end{aligned}$$



30. In the first octant, $z = x + y$ is nonnegative. Then

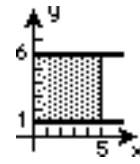
$$\begin{aligned} V &= \int_0^3 \int_0^{\sqrt{9-x^2}} (x+y) dy dx = \int_0^3 \left(xy + \frac{1}{2}y^2 \right) \Big|_0^{\sqrt{9-x^2}} dx \\ &= \int_0^3 \left(x\sqrt{9-x^2} + \frac{9}{2} - \frac{1}{2}x^2 \right) dx = \left[-\frac{1}{3}(9-x^2)^{3/2} + \frac{9}{2}x - \frac{1}{6}x^3 \right] \Big|_0^3 = \left(\frac{27}{2} - \frac{9}{2} \right) - (-9) = 18. \end{aligned}$$



Exercises 9.10

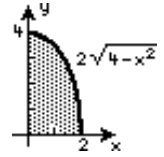
31. In the first octant $z = 6/y$ is positive. Then

$$V = \int_1^6 \int_0^5 \frac{6}{y} dx dy = \int_1^6 \frac{6x}{y} \Big|_0^5 dy = 30 \int_1^6 \frac{dy}{y} = 30 \ln y \Big|_1^6 = 30 \ln 6.$$



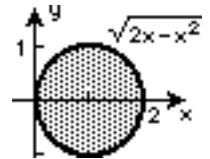
32. Setting $z = 0$, we have $x^2/4 + y^2/16 = 1$. Using symmetry,

$$\begin{aligned} V &= 4 \int_0^2 \int_0^{2\sqrt{4-x^2}} \left(4 - x^2 - \frac{1}{4}y^2\right) dy dx = 4 \int_0^2 \left(4y - x^2y - \frac{1}{12}y^3\right) \Big|_0^{2\sqrt{4-x^2}} dx \\ &= 4 \int_0^2 \left[8\sqrt{4-x^2} - 2x^2\sqrt{4-x^2} - \frac{2}{3}(4-x^2)^{3/2}\right] dx \quad \boxed{\text{Trig substitution}} \\ &= 4 \left[4x\sqrt{4-x^2} + 16\sin^{-1} \frac{x}{2} - \frac{1}{4}x(2x^2-4)\sqrt{4-x^2} - 4\sin^{-1} \frac{x}{2} + \frac{1}{12}x(2x^2-20)\sqrt{4-x^2} - 4\sin \frac{x}{2}\right] \Big|_0^2 \\ &= 4 \left(\frac{16\pi}{2} - \frac{4\pi}{2} - \frac{4\pi}{2}\right) - (0) = 16\pi. \end{aligned}$$



33. Note that $z = 4 - y^2$ is positive for $|y| \leq 1$. Using symmetry,

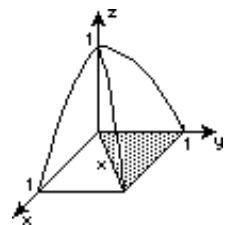
$$\begin{aligned} V &= 2 \int_0^2 \int_0^{\sqrt{2x-x^2}} (4 - y^2) dy dx = 2 \int_0^2 \left(4y - \frac{1}{3}y^3\right) \Big|_0^{\sqrt{2x-x^2}} dx \\ &= 2 \int_0^2 \left[4\sqrt{2x-x^2} - \frac{1}{3}(2x-x^2)\sqrt{2x-x^2}\right] dx \\ &= 2 \int_0^2 \left(4\sqrt{1-(x-1)^2} - \frac{1}{3}[1-(x-1)^2]\sqrt{1-(x-1)^2}\right) dx \quad \boxed{u = x-1, du = dx} \\ &= 2 \int_{-1}^1 \left[4\sqrt{1-u^2} - \frac{1}{3}(1-u^2)\sqrt{1-u^2}\right] du = 2 \int_{-1}^1 \left(\frac{11}{3}\sqrt{1-u^2} + \frac{1}{3}u^2\sqrt{1-u^2}\right) du \\ &\quad \boxed{\text{Trig substitution}} \\ &= 2 \left[\frac{11}{6}u\sqrt{1-u^2} + \frac{11}{6}\sin u + \frac{1}{24}x(2x^2-1)\sqrt{1-u^2} + \frac{1}{24}\sin^{-1} u\right] \Big|_{-1}^1 \\ &= 2 \left[\left(\frac{11}{6}\frac{\pi}{2} + \frac{1}{24}\frac{\pi}{2}\right) - \left(-\frac{11}{6}\frac{\pi}{2} - \frac{1}{24}\frac{\pi}{2}\right)\right] = \frac{15\pi}{4}. \end{aligned}$$



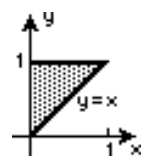
34. From $z = 1 - x^2$ and $z = 1 - y^2$ we have $1 - x^2 = 1 - y^2$ or $y = x$ (in the first octant).

Thus, the surfaces intersect in the plane $y = x$. Using symmetry,

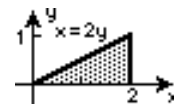
$$\begin{aligned} V &= 2 \int_0^1 \int_x^1 (1 - y^2) dy dx = 2 \int_0^1 \left(y - \frac{1}{3}y^3\right) \Big|_x^1 dx = 2 \int_0^1 \left(\frac{2}{3} - x + \frac{1}{3}x^3\right) dx \\ &= 2 \left(\frac{2}{3}x - \frac{1}{2}x^2 + \frac{1}{12}x^4\right) \Big|_0^1 = \frac{1}{2}. \end{aligned}$$



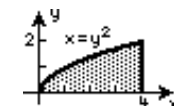
$$\begin{aligned} 35. \int_0^1 \int_x^1 x^2 \sqrt{1+y^4} dy dx &= \int_0^1 \int_0^y x^2 \sqrt{1+y^4} dx dy = \int_0^1 \frac{1}{3}x^3 \sqrt{1+y^4} \Big|_0^y dy \\ &= \frac{1}{3} \int_0^1 y^3 \sqrt{1+y^4} dy = \frac{1}{3} \left[\frac{1}{6}(1+y^4)^{3/2}\right] \Big|_0^1 = \frac{1}{18}(2\sqrt{2}-1) \end{aligned}$$



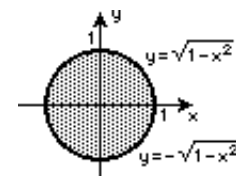
$$\begin{aligned}
 36. \quad \int_0^1 \int_{2y}^2 e^{-y/x} dx dy &= \int_0^2 \int_0^{x/2} e^{-y/x} dy dx = \int_0^2 -xe^{-y/x} \Big|_0^{x/2} = \int_0^2 (-xe^{-1/2} + x) dx \\
 &= \int_0^2 (1 - e^{-1/2})x dx = \frac{1}{2}(1 - e^{-1/2})x^2 \Big|_0^2 = 2(1 - e^{-1/2})
 \end{aligned}$$



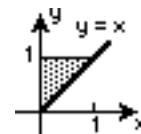
$$\begin{aligned}
 37. \quad \int_0^2 \int_{y^2}^4 \cos x^{3/2} dx dy &= \int_0^4 \int_0^{\sqrt{x}} \cos x^{3/2} dy dx = \int_0^4 y \cos x^{3/2} \Big|_0^{\sqrt{x}} dx \\
 &= \int_0^4 \sqrt{x} \cos x^{3/2} dx = \frac{2}{3} \sin x^{3/2} \Big|_0^4 = \frac{2}{3} \sin 8
 \end{aligned}$$



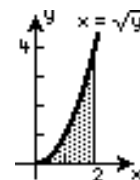
$$\begin{aligned}
 38. \quad \int_{-1}^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} x \sqrt{1-x^2-y^2} dy dx &= \int_{-1}^1 \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} x \sqrt{1-x^2-y^2} dx dy \\
 &= \int_{-1}^1 \left[-\frac{1}{3}(1-x^2-y^2)^{3/2} \right] \Big|_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} dy = -\frac{1}{3} \int_{-1}^1 (0-0) dy = 0
 \end{aligned}$$



$$\begin{aligned}
 39. \quad \int_0^1 \int_x^1 \frac{1}{1+y^4} dy dx &= \int_0^1 \int_0^y \frac{1}{1+y^4} dx dy = \int_0^1 \frac{x}{1+y^4} \Big|_0^y dy = \int_0^1 \frac{y}{1+y^4} dy \\
 &= \frac{1}{2} \tan^{-1} y^2 \Big|_0^1 = \frac{\pi}{8}
 \end{aligned}$$



$$\begin{aligned}
 40. \quad \int_0^4 \int_{\sqrt{y}}^2 \sqrt{x^3+1} dx dy &= \int_0^2 \int_0^{x^2} \sqrt{x^3+1} dy dx = \int_0^2 y \sqrt{x^3+1} \Big|_0^{x^2} dx \\
 &= \int_0^2 x^2 \sqrt{x^3+1} dx = \frac{2}{9} (x^3+1)^{3/2} \Big|_0^2 = \frac{2}{9} (9^{3/2} - 1^{3/2}) = \frac{52}{9}
 \end{aligned}$$

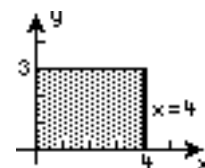


$$41. \quad m = \int_0^3 \int_0^4 xy dx dy = \int_0^3 \frac{1}{2} x^2 y \Big|_0^4 dy = \int_0^3 8y dy = 4y^2 \Big|_0^3 = 36$$

$$M_y = \int_0^3 \int_0^4 x^2 y dx dy = \int_0^3 \frac{1}{3} x^3 y \Big|_0^4 dy = \int_0^3 \frac{64}{3} y dy = \frac{32}{3} y^2 \Big|_0^3 = 96$$

$$M_x = \int_0^3 \int_0^4 xy^2 dx dy = \int_0^3 \frac{1}{2} x^2 y^2 \Big|_0^4 dy = \int_0^3 8y^2 dy = \frac{8}{3} y^3 \Big|_0^3 = 72$$

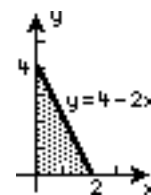
$\bar{x} = M_y/m = 96/36 = 8/3$; $\bar{y} = M_x/m = 72/36 = 2$. The center of mass is $(8/3, 2)$.



$$\begin{aligned}
 42. \quad m &= \int_0^2 \int_0^{4-2x} x^2 dy dx = \int_0^2 x^2 y \Big|_0^{4-2x} dx = \int_0^2 x^2 (4-2x) dx \\
 &= \int_0^2 (4x^2 - 2x^3) dx = \left(\frac{4}{3} x^3 - \frac{1}{2} x^4 \right) \Big|_0^2 = \frac{32}{3} - 8 = \frac{8}{3}
 \end{aligned}$$

$$\begin{aligned}
 M_y &= \int_0^2 \int_0^{4-2x} x^3 dy dx = \int_0^2 x^3 y \Big|_0^{4-2x} dx = \int_0^2 x^3 (4-2x) dx = \int_0^2 (4x^3 - 2x^4) dx \\
 &= \left(x^4 - \frac{2}{5} x^5 \right) \Big|_0^2 = 16 - \frac{64}{5} = \frac{16}{5}
 \end{aligned}$$

$$\begin{aligned}
 M_x &= \int_0^2 \int_0^{4-2x} x^2 y dy dx = \int_0^2 \frac{1}{2} x^2 y^2 \Big|_0^{4-2x} dx = \frac{1}{2} \int_0^2 x^2 (4-2x)^2 dx = \frac{1}{2} \int_0^2 (16x^2 - 16x^3 + 4x^4) dx \\
 &= 2 \int_0^2 (4x^2 - 4x^3 + x^4) dx = 2 \left(\frac{4}{3} x^3 - x^4 + \frac{1}{5} x^5 \right) \Big|_0^2 = 2 \left(\frac{32}{3} - 16 + \frac{32}{5} \right) = \frac{32}{15}
 \end{aligned}$$



Exercises 9.10

$\bar{x} = M_y/m = \frac{16/5}{8/3} = 6/5$; $\bar{y} = M_x/m = \frac{32/15}{8/3} = 4/5$. The center of mass is $(6/5, 4/5)$.

43. Since both the region and ρ are symmetric with respect to the line $x = 3$, $\bar{x} = 3$.

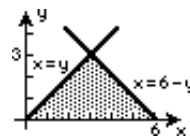
$$m = \int_0^3 \int_y^{6-y} 2y \, dx \, dy = \int_0^3 2xy \Big|_y^{6-y} \, dy = \int_0^3 2y(6-y-y) \, dy = \int_0^3 (12y - 4y^2) \, dy$$

$$= \left(6y^2 - \frac{4}{3}y^3 \right) \Big|_0^3 = 18$$

$$M_x = \int_0^3 \int_y^{6-y} 2y^2 \, dx \, dy = \int_0^3 2xy^2 \Big|_y^{6-y} \, dy = \int_0^3 2y^2(6-y-y) \, dy = \int_0^3 (12y^2 - 4y^3) \, dy$$

$$= (4y^3 - y^4) \Big|_0^3 = 27$$

$\bar{y} = M_x/m = 27/18 = 3/2$. The center of mass is $(3, 3/2)$.



44. Since both the region and ρ are symmetric with respect to the y -axis, $\bar{x} = 0$. Using symmetry,

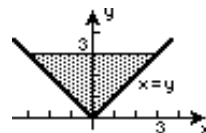
$$m = \int_0^3 \int_0^y (x^2 + y^2) \, dx \, dy = \int_0^3 \left(\frac{1}{3}x^3 + xy^2 \right) \Big|_0^y \, dy = \int_0^3 \left(\frac{1}{3}y^3 + y^3 \right) \, dy$$

$$= \frac{4}{3} \int_0^3 y^3 \, dy = \frac{1}{3}y^4 \Big|_0^3 = 27.$$

$$M_x = \int_0^3 \int_0^y (x^2 y + y^3) \, dx \, dy = \int_0^3 \left(\frac{1}{3}x^3 y + xy^3 \right) \Big|_0^y \, dy = \int_0^3 \left(\frac{1}{3}y^4 + y^4 \right) \, dy = \frac{4}{3} \int_0^3 y^4 \, dy$$

$$= \frac{4}{15}y^5 \Big|_0^3 = \frac{324}{5}$$

$\bar{y} = M_x/m = \frac{324/5}{27} = 12/5$. The center of mass is $(0, 12/5)$.



45. $m = \int_0^1 \int_0^{x^2} (x+y) \, dy \, dx = \int_0^1 \left(xy + \frac{1}{2}y^2 \right) \Big|_0^{x^2} \, dx = \int_0^1 \left(x^3 + \frac{1}{2}x^4 \right) \, dx$

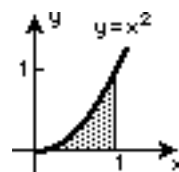
$$= \left(\frac{1}{4}x^4 + \frac{1}{10}x^5 \right) \Big|_0^1 = \frac{7}{20}$$

$$M_y = \int_0^1 \int_0^{x^2} (x^2 + xy) \, dy \, dx = \int_0^1 \left(x^2 y + \frac{1}{2}xy^2 \right) \Big|_0^{x^2} \, dx = \int_0^1 \left(x^4 + \frac{1}{2}x^5 \right) \, dx$$

$$= \left(\frac{1}{5}x^5 + \frac{1}{12}x^6 \right) \Big|_0^1 = \frac{17}{60}$$

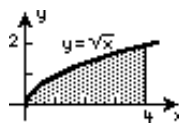
$$M_x = \int_0^1 \int_0^{x^2} (xy + y^2) \, dy \, dx = \int_0^1 \left(\frac{1}{2}xy^2 + \frac{1}{3}y^3 \right) \Big|_0^{x^2} \, dx = \int_0^1 \left(\frac{1}{2}x^5 + \frac{1}{3}x^6 \right) \, dx = \left(\frac{1}{12}x^6 + \frac{1}{21}x^7 \right) \Big|_0^1 = \frac{11}{84}$$

$\bar{x} = M_y/m = \frac{17/60}{7/20} = 17/21$; $\bar{y} = M_x/m = \frac{11/84}{7/20} = 55/147$. The center of mass is $(17/21, 55/147)$.



46. $m = \int_0^4 \int_0^{\sqrt{x}} (y+5) \, dy \, dx = \int_0^4 \left(\frac{1}{2}y^2 + 5y \right) \Big|_0^{\sqrt{x}} \, dx = \int_0^4 \left(\frac{1}{2}x + 5\sqrt{x} \right) \, dx$

$$= \left(\frac{1}{4}x^2 + \frac{10}{3}x^{3/2} \right) \Big|_0^4 = \frac{92}{3}$$



$$M_y = \int_0^4 \int_0^{\sqrt{x}} (xy + 5x) dy dx = \int_0^4 \left(\frac{1}{2}xy^2 + 5xy \right) \Big|_0^{\sqrt{x}} dx = \int_0^4 \left(\frac{1}{2}x^2 + 5x^{3/2} \right) dx$$

$$= \left(\frac{1}{6}x^3 + 2x^{5/2} \right) \Big|_0^4 = \frac{224}{3}$$

$$M_x = \int_0^4 \int_0^{\sqrt{x}} (y^2 + 5y) dy dx = \int_0^4 \left(\frac{1}{3}y^3 + \frac{5}{2}y^2 \right) \Big|_0^{\sqrt{x}} dx = \int_0^4 \left(\frac{1}{3}x^{3/2} + \frac{5}{2}x \right) dx$$

$$= \left(\frac{2}{15}x^{5/2} + \frac{5}{4}x^2 \right) \Big|_0^4 = \frac{364}{15}$$

$$\bar{x} = M_y/m = \frac{224/3}{92/3} = 56/23; \quad \bar{y} = M_x/m = \frac{364/15}{92/3} = 91/115. \text{ The center of mass is } (56/23, 91/115).$$

47. The density is $\rho = ky$. Since both the region and ρ are symmetric with respect to the y -axis, $\bar{x} = 0$. Using symmetry,

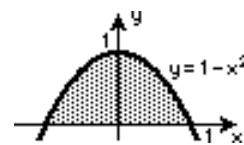
$$m = 2 \int_0^1 \int_0^{1-x^2} ky dy dx = 2k \int_0^1 \frac{1}{2}y^2 \Big|_0^{1-x^2} dx = k \int_0^1 (1-x^2)^2 dx$$

$$= k \int_0^1 (1 - 2x^2 + x^4) dx = k \left(x - \frac{2}{3}x^3 + \frac{1}{5}x^5 \right) \Big|_0^1 = k \left(1 - \frac{2}{3} + \frac{1}{5} \right) = \frac{8}{15}k$$

$$M_x = 2 \int_0^1 \int_0^{1-x^2} ky^2 dy dx = 2k \int_0^1 \frac{1}{3}y^3 \Big|_0^{1-x^2} dx = \frac{2}{3}k \int_0^1 (1-x^2)^3 dx = \frac{2}{3}k \int_0^1 (1 - 3x^2 + 3x^4 - x^6) dx$$

$$= \frac{2}{3}k \left(x - x^3 + \frac{3}{5}x^5 - \frac{1}{7}x^7 \right) \Big|_0^1 = \frac{2}{3}k \left(1 - 1 + \frac{3}{5} - \frac{1}{7} \right) = \frac{32}{105}k$$

$$\bar{y} = M_x/m = \frac{32k/105}{8k/15} = 4/7. \text{ The center of mass is } (0, 4/7).$$



48. The density is $\rho = kx$.

$$m = \int_0^\pi \int_0^{\sin x} kx dy dx = \int_0^\pi kxy \Big|_0^{\sin x} dx = \int_0^\pi kx \sin x dx$$

Integration by parts

$$= k(\sin x - x \cos x) \Big|_0^\pi = k\pi$$

$$M_y = \int_0^\pi \int_0^{\sin x} kx^2 dy dx = \int_0^\pi kx^2 y \Big|_0^{\sin x} dx = \int_0^\pi kx^2 \sin x dx$$

Integration by parts

$$= k(-x^2 \cos x + 2 \cos x + 2x \sin x) \Big|_0^\pi = k[(\pi^2 - 2) - 2] = k(\pi^2 - 4)$$

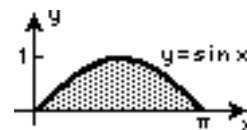
$$M_x = \int_0^\pi \int_0^{\sin x} kxy dy dx = \int_0^\pi \frac{1}{2}kxy^2 \Big|_0^{\sin x} dx = \int_0^\pi \frac{1}{2}kx \sin^2 x dx = \int_0^\pi \frac{1}{4}kx(1 - \cos 2x) dx$$

$$= \frac{1}{4}k \left[\int_0^\pi x dx - \int_0^\pi x \cos 2x dx \right]$$

Integration by parts

$$= \frac{1}{4}k \left[\frac{1}{2}x^2 \Big|_0^\pi - \frac{1}{4}(\cos 2x + 2x \sin 2x) \Big|_0^\pi \right] = \frac{1}{4}k \left(\frac{1}{2}\pi^2 \right) = \frac{1}{8}k\pi^2$$

$$\bar{x} = M_y/m = \frac{k(\pi^2 - 4)}{k\pi} = \pi - 4/\pi; \quad \bar{y} = M_x/m = \frac{k\pi^2/8}{k\pi} = \pi/8. \text{ The center of mass is } (\pi - 4/\pi, \pi/8).$$



Exercises 9.10

$$49. m = \int_0^1 \int_0^{e^x} y^3 dy dx = \int_0^1 \frac{1}{4} y^4 \Big|_0^{e^x} dx = \int_0^1 \frac{1}{4} e^{4x} dx = \frac{1}{16} e^{4x} \Big|_0^1 = \frac{1}{16} (e^4 - 1)$$

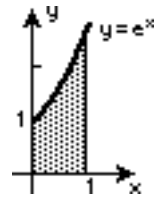
$$M_y = \int_0^1 \int_0^{e^x} xy^3 dy dx = \int_0^1 \frac{1}{4} xy^4 \Big|_0^{e^x} dx = \int_0^1 \frac{1}{4} x e^{4x} dx \quad \boxed{\text{Integration by parts}}$$

$$= \frac{1}{4} \left(\frac{1}{4} x e^{4x} - \frac{1}{16} e^{4x} \right) \Big|_0^1 = \frac{1}{4} \left(\frac{3}{16} e^4 + \frac{1}{16} \right) = \frac{1}{64} (3e^4 + 1)$$

$$M_x = \int_0^1 \int_0^{e^x} y^4 dy dx = \int_0^1 \frac{1}{5} y^5 \Big|_0^{e^x} dx = \int_0^1 \frac{1}{5} e^{5x} dx = \frac{1}{25} e^{5x} \Big|_0^1 = \frac{1}{25} (e^5 - 1)$$

$$\bar{x} = M_y/m = \frac{(3e^4 + 1)/64}{(e^4 - 1)/16} = \frac{3e^4 + 1}{4(e^4 - 1)}; \quad \bar{y} = M_x/m = \frac{(e^5 - 1)/25}{(e^4 - 1)/16} = \frac{16(e^5 - 1)}{25(e^4 - 1)}$$

The center of mass is $\left(\frac{3e^4 + 1}{4(e^4 - 1)}, \frac{16(e^5 - 1)}{25(e^4 - 1)} \right) \approx (0.77, 1.76)$.



50. Since both the region and ρ are symmetric with respect to the y -axis, $\bar{x} = 0$. Using symmetry,

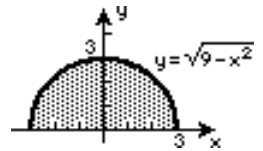
$$m = 2 \int_0^3 \int_0^{\sqrt{9-x^2}} x^2 dy dx = 2 \int_0^3 x^2 y \Big|_0^{\sqrt{9-x^2}} dx = 2 \int_0^3 x^2 \sqrt{9-x^2} dx$$

Trig substitution

$$= 2 \left[\frac{x}{8} (2x^2 - 9) \sqrt{9-x^2} + \frac{81}{8} \sin^{-1} \frac{x}{3} \right] \Big|_0^3 = \frac{81}{4} \frac{\pi}{2} = \frac{81\pi}{2}.$$

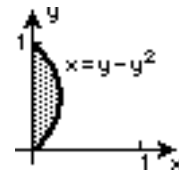
$$M_x = 2 \int_0^3 \int_0^{\sqrt{9-x^2}} x^2 y dy dx = 2 \int_0^3 \frac{1}{2} x^2 y^2 \Big|_0^{\sqrt{9-x^2}} dy dx = \int_0^3 x^2 (9-x^2) dx = \left(3x^2 - \frac{1}{5} x^5 \right) \Big|_0^3 = \frac{162}{5}$$

$$\bar{y} = M_x/m = \frac{162/5}{81\pi/8} = 16/5\pi. \text{ The center of mass is } (0, 16/5\pi).$$



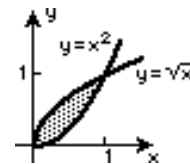
$$51. I_x = \int_0^1 \int_0^{y-y^2} 2xy^2 dx dy = \int_0^1 x^2 y^2 \Big|_0^{y-y^2} dy = \int_0^1 (y-y^2)^2 y^2 dy$$

$$= \int_0^1 (y^4 - 2y^5 + y^6) dy = \left(\frac{1}{5} y^5 - \frac{2}{6} y^6 + \frac{1}{7} y^7 \right) \Big|_0^1 = \frac{1}{105}$$



$$52. I_x = \int_0^1 \int_{x^2}^{\sqrt{x}} x^2 y^2 dy dx = \int_0^1 \frac{1}{3} x^2 y^3 \Big|_{x^2}^{\sqrt{x}} dx = \frac{1}{3} \int_0^1 (x^{7/2} - x^8) dx$$

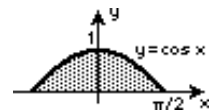
$$= \frac{1}{3} \left(\frac{2}{9} x^{9/2} - \frac{1}{9} x^9 \right) \Big|_0^1 = \frac{1}{27}$$



53. Using symmetry,

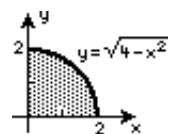
$$I_x = 2 \int_0^{\pi/2} \int_0^{\cos x} ky^2 dy dx = 2k \int_0^{\pi/2} \frac{1}{3} y^3 \Big|_0^{\cos x} dx = \frac{2}{3} k \int_0^{\pi/2} \cos^3 x dx$$

$$= \frac{2}{3} k \int_0^{\pi/2} \cos x (1 - \sin^2 x) dx = \frac{2}{3} k \left(\sin x - \frac{1}{3} \sin^3 x \right) \Big|_0^{\pi/2} = \frac{4}{9} k.$$



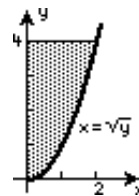
$$54. I_x = \int_0^2 \int_0^{\sqrt{4-x^2}} y^3 dy dx = \int_0^2 \frac{1}{4} y^4 \Big|_0^{\sqrt{4-x^2}} dx = \frac{1}{4} \int_0^2 (4-x^2)^2 dx$$

$$= \frac{1}{4} \int_0^2 (16 - 8x^2 + x^4) dx = \frac{1}{4} \left(16x - \frac{8}{3} x^3 + \frac{1}{5} x^5 \right) \Big|_0^2 = \frac{1}{4} \left(32 - \frac{64}{3} + \frac{32}{5} \right)$$

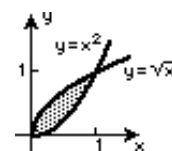


$$= 8 \left(1 - \frac{2}{3} + \frac{1}{5} \right) = \frac{64}{15}$$

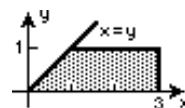
$$\begin{aligned} 55. \quad I_y &= \int_0^4 \int_0^{\sqrt{y}} x^2 y \, dx \, dy = \int_0^4 \frac{1}{3} x^3 y \Big|_0^{\sqrt{y}} dy = \frac{1}{3} \int_0^4 y^{3/2} y \, dy = \frac{1}{3} \int_0^4 y^{5/2} dy \\ &= \frac{1}{3} \left(\frac{2}{7} y^{7/2} \right) \Big|_0^4 = \frac{2}{21} (4^{7/2}) = \frac{256}{21} \end{aligned}$$



$$56. \quad I_y = \int_0^1 \int_{x^2}^{\sqrt{x}} x^4 \, dy \, dx = \int_0^1 x^4 y \Big|_{x^2}^{\sqrt{x}} dx = \int_0^1 (x^{9/2} - x^6) dx = \left(\frac{2}{11} x^{11/2} - \frac{1}{7} x^7 \right) \Big|_0^1 = \frac{3}{77}$$

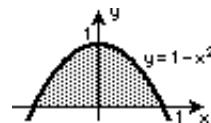


$$\begin{aligned} 57. \quad I_y &= \int_0^1 \int_y^3 (4x^3 + 3x^2 y) \, dx \, dy = \int_0^1 (x^4 + x^3 y) \Big|_y^3 dy = \int_0^1 (81 + 27y - 2y^4) dy \\ &= \left(81y + \frac{27}{2} y^2 - \frac{2}{5} y^5 \right) \Big|_0^1 = \frac{941}{10} \end{aligned}$$



58. The density is $\rho = ky$. Using symmetry,

$$\begin{aligned} I_y &= 2 \int_0^1 \int_0^{1-x^2} kx^2 y \, dy \, dx = 2 \int_0^1 \frac{1}{2} kx^2 y^2 \Big|_0^{1-x^2} dx = k \int_0^1 x^2 (1-x^2)^2 dx \\ &= k \int_0^1 (x^2 - 2x^4 + x^6) dx = k \left(\frac{1}{3} x^3 - \frac{2}{5} x^5 + \frac{1}{7} x^7 \right) \Big|_0^1 = \frac{8k}{105}. \end{aligned}$$

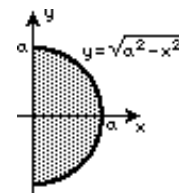


59. Using symmetry,

$$\begin{aligned} m &= 2 \int_0^a \int_0^{\sqrt{a^2-y^2}} x \, dx \, dy = 2 \int_0^a \frac{1}{2} x^2 \Big|_0^{\sqrt{a^2-y^2}} dy = \int_0^a (a^2 - y^2) dy \\ &= \left(a^2 y - \frac{1}{3} y^3 \right) \Big|_0^a = \frac{2}{3} a^3. \end{aligned}$$

$$\begin{aligned} I_y &= 2 \int_0^a \int_0^{\sqrt{a^2-y^2}} x^3 \, dx \, dy = 2 \int_0^a \frac{1}{4} x^4 \Big|_0^{\sqrt{a^2-y^2}} dy = \frac{1}{2} \int_0^a (a^2 - y^2)^2 dy \\ &= \frac{1}{2} \int_0^a (a^4 - 2a^2 y^2 + y^4) dy = \frac{1}{2} \left(a^4 y - \frac{2}{3} a^2 y^3 + \frac{1}{5} y^5 \right) \Big|_0^a = \frac{4}{15} a^5 \end{aligned}$$

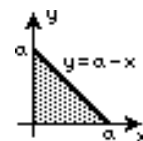
$$R_g = \sqrt{\frac{I_y}{m}} = \sqrt{\frac{4a^5/15}{2a^3/3}} = \sqrt{\frac{2}{5}} a$$



$$60. \quad m = \int_0^a \int_0^{a-x} k \, dy \, dx = \int_0^a ky \Big|_0^{a-x} dx = k \int_0^a (a-x) dx = k \left(ax - \frac{1}{2} x^2 \right) \Big|_0^a = \frac{1}{2} ka^2$$

$$\begin{aligned} I_x &= \int_0^a \int_0^{a-x} ky^2 \, dy \, dx = \int_0^a \frac{1}{3} ky^3 \Big|_0^{a-x} dx = \frac{1}{3} k \int_0^a (a-x)^3 dx \\ &= \frac{1}{3} k \int_0^a (a^3 - 3a^2 x + 3ax^2 - x^3) dx = \frac{1}{3} k \left(a^3 x - \frac{3}{2} a^2 x^2 + ax^3 - \frac{1}{4} x^4 \right) \Big|_0^a = \frac{1}{12} ka^4 \end{aligned}$$

$$R_g = \sqrt{\frac{I_x}{m}} = \sqrt{\frac{ka^4/12}{ka^2/2}} = \sqrt{\frac{1}{6}} a$$



Exercises 9.10

61. (a) Using symmetry,

$$\begin{aligned} I_x &= 4 \int_0^a \int_0^{b\sqrt{a^2-x^2}/a} y^2 dy dx = \frac{4b^3}{3a^3} \int_0^a (a^2 - x^2)^{3/2} dx && \boxed{x = a \sin \theta, \quad dx = a \cos \theta d\theta} \\ &= \frac{4}{3}ab^3 \int_0^{\pi/2} \cos^4 \theta d\theta = \frac{4}{3}ab^3 \int_0^{\pi/2} \frac{1}{4}(1 + \cos 2\theta)^2 d\theta = \frac{1}{3}ab^3 \int_0^{\pi/2} \left(1 + \cos 2\theta + \frac{1}{2} + \frac{1}{2} \cos 4\theta\right) d\theta \\ &= \frac{1}{3}ab^3 \left(\frac{3}{2}\theta + \frac{1}{2} \sin 2\theta + \frac{1}{8} \sin 4\theta\right) \Big|_0^{\pi/2} = \frac{ab^3\pi}{4}. \end{aligned}$$

(b) Using symmetry,

$$\begin{aligned} I_y &= 4 \int_0^a \int_0^{b\sqrt{a^2-x^2}/a} x^2 dy dx = \frac{4b}{a} \int_0^a x^2 \sqrt{a^2 - x^2} dx && \boxed{x = a \sin \theta, \quad dx = a \cos \theta d\theta} \\ &= 4a^3b \int_0^{\pi/2} \sin^2 \theta \cos^2 \theta d\theta = 4a^3b \int_0^{\pi/2} \frac{1}{4}(1 - \cos^2 2\theta) d\theta \\ &= a^3b \int_0^{\pi/2} \left(1 - \frac{1}{2} - \frac{1}{2} \cos 4\theta\right) d\theta = a^3b \left(\frac{1}{2}\theta - \frac{1}{8} \sin 4\theta\right) \Big|_0^{\pi/2} = \frac{a^3b\pi}{4}. \end{aligned}$$

(c) Using $m = \pi ab$, $R_g = \sqrt{I_x/m} = \frac{1}{2} \sqrt{ab^3\pi/\pi ab} = \frac{1}{2}b$.

(d) $R_g = \sqrt{I_y/m} = \frac{1}{2} \sqrt{a^3b\pi/\pi ab} = \frac{1}{2}a$

62. The equation of the ellipse is $9x^2/a^2 + 4y^2/b^2 = 1$ and the equation of the parabola is $y = \pm(9bx^2/8a^2 - b/2)$. Letting I_e and I_p represent the moments of inertia of the ellipse and parabola, respectively, about the x -axis, we have

$$\begin{aligned} I_e &= 2 \int_{-a/3}^0 \int_0^{b\sqrt{a^2-9x^2}/2a} y^2 dy dx = \frac{b^3}{12a^3} \int_{-a/3}^0 (a^2 - 9x^2)^{3/2} dx && \boxed{x = \frac{a}{3} \sin \theta, \quad dx = \frac{a}{3} \cos \theta d\theta} \\ &= \frac{b^3}{12a^3} \frac{a^4}{3} \int_{-\pi/3}^0 \cos^4 \theta d\theta = \frac{b^3a}{36} \frac{3\pi}{16} = \frac{ab^3\pi}{192} \end{aligned}$$

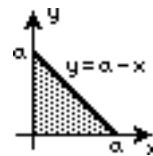
and

$$\begin{aligned} I_p &= 2 \int_0^{2a/3} \int_0^{b/2-9bx^2/8a^2} y^2 dy dx = \frac{2}{3} \int_0^{2a/3} \left(\frac{b}{2} - \frac{9b}{8a^2}x^2\right)^3 dx = \frac{2}{3} \frac{b^3}{8} \int_0^{2a/3} \left(1 - \frac{9}{4a^2}x^2\right)^3 dx \\ &= \frac{b^3}{12} \int_0^{2a/3} \left(1 - \frac{27}{4a^2}x^2 + \frac{243}{16a^4}x^4 - \frac{729}{64a^6}x^6\right) dx = \frac{b^3}{12} \left(x - \frac{9}{4a^2}x^3 + \frac{243}{80a^4}x^5 - \frac{729}{64a^6}x^7\right) \Big|_0^{2a/3} \\ &= \frac{b^3}{12} \frac{32a}{105} = \frac{8ab^3}{315}. \end{aligned}$$

Then $I_x = I_e + I_p = \frac{ab^3\pi}{192} + \frac{8ab^3}{315}$.

63. From Problem 60, $m = \frac{1}{2}ka^2$ and $I_x = \frac{1}{12}ka^4$.

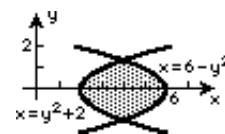
$$\begin{aligned} I_y &= \int_0^a \int_0^{a-x} kx^2 dy dx = \int_0^a kx^2 y \Big|_0^{a-x} dx = k \int_0^a x^2(a-x) dx \\ &= k \left(\frac{1}{3}ax^3 - \frac{1}{4}x^4\right) \Big|_0^a = \frac{1}{12}ka^4 \\ I_0 &= I_x + I_y = \frac{1}{12}ka^4 + \frac{1}{12}ka^4 = \frac{1}{6}ka^4 \end{aligned}$$



64. From Problem 52, $I_x = \frac{1}{27}$, and from Problem 56, $I_y = \frac{3}{77}$. Thus, $I_0 = I_x + I_y = \frac{1}{27} + \frac{3}{77} = \frac{158}{2079}$.

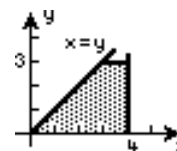
65. The density is $\rho = k/(x^2 + y^2)$. Using symmetry,

$$\begin{aligned} I_0 &= 2 \int_0^{\sqrt{2}} \int_{y^2+2}^{6-y^2} (x^2 + y^2) \frac{k}{x^2 + y^2} dx dy = 2 \int_0^{\sqrt{2}} kx \Big|_{y^2+2}^{6-y^2} dy \\ &= 2k \int_0^{\sqrt{2}} (6 - y^2 - y^2 - 2) dy = 2k \left(4y - \frac{2}{3}y^3 \right) \Big|_0^{\sqrt{2}} = 2k \left(\frac{8}{3}\sqrt{2} \right) = \frac{16\sqrt{2}}{3} k. \end{aligned}$$



66.
$$I_0 = \int_0^3 \int_y^4 k(x^2 + y^2) dx dy = k \int_0^3 \left(\frac{1}{3}x^3 + xy^2 \right) \Big|_y^4 dy$$

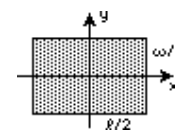
$$= k \int_0^3 \left(\frac{64}{3} + 4y^2 - \frac{1}{3}y^3 - y^3 \right) dy = k \left(\frac{64}{3}y + \frac{4}{3}y^3 - \frac{1}{3}y^4 \right) \Big|_0^3 = 73k$$



67. From Problem 60, $m = \frac{1}{2}ka^2$, and from Problem 61, $I_0 = \frac{1}{6}ka^4$. Then $R_g = \sqrt{I_0/m} = \sqrt{\frac{ka^4/6}{ka^2/2}} = \sqrt{\frac{1}{3}} a$.

68. Since the plate is homogeneous, the density is $\rho = m/\ell\omega$. Using symmetry,

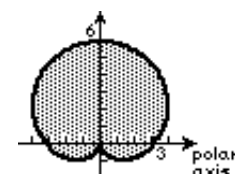
$$\begin{aligned} I_0 &= 4 \int_0^{\ell/2} \int_0^{\omega/2} \frac{m}{\ell\omega} (x^2 + y^2) dy dx = \frac{4m}{\ell\omega} \int_0^{\ell/2} \left(x^2y + \frac{1}{3}y^3 \right) \Big|_0^{\omega/2} dx \\ &= \frac{4m}{\ell\omega} \int_0^{\ell/2} \left(\frac{\omega}{2}x^2 + \frac{\omega^3}{24} \right) dx = \frac{4m}{\ell\omega} \left(\frac{\omega}{6}x^3 + \frac{\omega^3}{24}x \right) \Big|_0^{\ell/2} = \frac{4m}{\ell\omega} \left(\frac{\omega\ell^3}{48} + \frac{\ell\omega^3}{48} \right) = m \frac{\ell^2 + \omega^2}{12}. \end{aligned}$$



Exercises 9.11

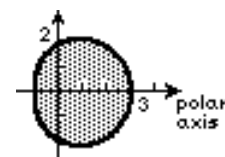
1. Using symmetry,

$$\begin{aligned} A &= 2 \int_{-\pi/2}^{\pi/2} \int_0^{3+3\sin\theta} r dr d\theta = 2 \int_{-\pi/2}^{\pi/2} \frac{1}{2}r^2 \Big|_0^{3+3\sin\theta} d\theta = \int_{-\pi/2}^{\pi/2} 9(1 + \sin\theta)^2 d\theta \\ &= 9 \int_{-\pi/2}^{\pi/2} (1 + 2\sin\theta + \sin^2\theta) d\theta = 9 \left(\theta - 2\cos\theta + \frac{1}{2}\theta - \frac{1}{4}\sin 2\theta \right) \Big|_{-\pi/2}^{\pi/2} \\ &= 9 \left[\frac{3}{2}\frac{\pi}{2} - \frac{3}{2}\left(-\frac{\pi}{2}\right) \right] = \frac{27\pi}{2} \end{aligned}$$



2. Using symmetry,

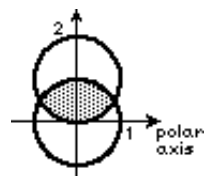
$$\begin{aligned} A &= 2 \int_0^{\pi} \int_0^{2+\cos\theta} r dr d\theta = 2 \int_0^{\pi} \frac{1}{2}r^2 \Big|_0^{2+\cos\theta} d\theta = \int_0^{\pi} (2 + \cos\theta)^2 d\theta \\ &= \int_0^{\pi} (4 + 4\cos\theta + \cos^2\theta) d\theta = \left(4\theta + 4\sin\theta + \frac{1}{2}\theta + \frac{1}{4}\cos 2\theta \right) \Big|_0^{\pi} \\ &= \left(4\pi + \frac{\pi}{2} + \frac{1}{4} \right) - \left(\frac{1}{4} \right) = \frac{9\pi}{2}. \end{aligned}$$



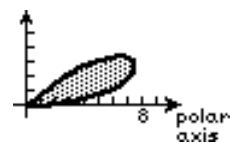
Exercises 9.11

3. Solving $r = 2 \sin \theta$ and $r = 1$, we obtain $\sin \theta = 1/2$ or $\theta = \pi/6$. Using symmetry,

$$\begin{aligned} A &= 2 \int_0^{\pi/6} \int_0^{2 \sin \theta} r \, dr \, d\theta + 2 \int_{\pi/6}^{\pi/2} \int_0^1 r \, dr \, d\theta \\ &= 2 \int_0^{\pi/6} \frac{1}{2} r^2 \Big|_0^{2 \sin \theta} d\theta + 2 \int_{\pi/6}^{\pi/2} \frac{1}{2} r^2 \Big|_0^1 d\theta = \int_0^{\pi/6} 4 \sin^2 \theta \, d\theta + \int_{\pi/6}^{\pi/2} d\theta \\ &= (2\theta - \sin 2\theta) \Big|_0^{\pi/6} + \left(\frac{\pi}{2} - \frac{\pi}{6} \right) = \frac{\pi}{3} - \frac{\sqrt{3}}{2} + \frac{\pi}{3} = \frac{4\pi - 3\sqrt{3}}{6} \end{aligned}$$



$$\begin{aligned} 4. \quad A &= \int_0^{\pi/4} \int_0^{8 \sin 4\theta} r \, dr \, d\theta = \int_0^{\pi/4} \frac{1}{2} r^2 \Big|_0^{8 \sin 4\theta} d\theta = \frac{1}{2} \int_0^{\pi/4} 64 \sin^2 4\theta \, d\theta \\ &= 32 \left(\frac{1}{2} \theta - \frac{1}{16} \sin 8\theta \right) \Big|_0^{\pi/4} = 4\pi \end{aligned}$$

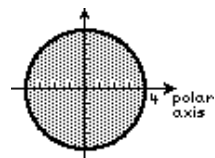


5. Using symmetry,

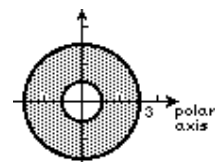
$$\begin{aligned} V &= 2 \int_0^{\pi/6} \int_0^{5 \cos 3\theta} 4r \, dr \, d\theta = 4 \int_0^{\pi/6} r^2 \Big|_0^{5 \cos 3\theta} d\theta = 4 \int_0^{\pi/6} 25 \cos^2 3\theta \, d\theta \\ &= 100 \left(\frac{1}{2} \theta + \frac{1}{12} \sin 6\theta \right) \Big|_0^{\pi/6} = \frac{25\pi}{3} \end{aligned}$$



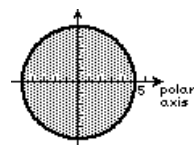
$$\begin{aligned} 6. \quad V &= \int_0^{2\pi} \int_0^2 \sqrt{9 - r^2} \, r \, dr \, d\theta = \int_0^{2\pi} -\frac{1}{3} (9 - r^2)^{3/2} \Big|_0^2 d\theta \\ &= -\frac{1}{3} \int_0^{2\pi} (5^{3/2} - 27) \, d\theta = \frac{1}{3} (27 - 5^{3/2}) 2\pi = \frac{2\pi(27 - 5\sqrt{5})}{3} \end{aligned}$$



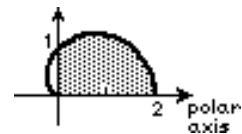
$$\begin{aligned} 7. \quad V &= \int_0^{2\pi} \int_1^3 \sqrt{16 - r^2} \, r \, dr \, d\theta = \int_0^{2\pi} -\frac{1}{3} (16 - r^2)^{3/2} \Big|_1^3 d\theta \\ &= -\frac{1}{3} \int_0^{2\pi} (7^{3/2} - 15^{3/2}) \, d\theta = \frac{1}{3} (15^{3/2} - 7^{3/2}) 2\pi = \frac{2\pi(15\sqrt{15} - 7\sqrt{7})}{3} \end{aligned}$$



$$8. \quad V = \int_0^{2\pi} \int_0^5 \sqrt{r^2} \, r \, dr \, d\theta = \int_0^{2\pi} \frac{1}{3} r^3 \Big|_0^5 d\theta = \int_0^{2\pi} \frac{125}{3} \, d\theta = \frac{250\pi}{3}$$

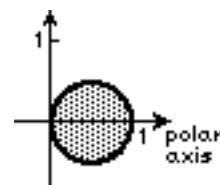


$$\begin{aligned} 9. \quad V &= \int_0^{\pi/2} \int_0^{1+\cos \theta} (r \sin \theta) r \, dr \, d\theta = \int_0^{\pi/2} \frac{1}{3} r^3 \sin \theta \Big|_0^{1+\cos \theta} d\theta \\ &= \frac{1}{3} \int_0^{\pi/2} (1 + \cos \theta)^3 \sin \theta \, d\theta = \frac{1}{3} \left[-\frac{1}{4} (1 + \cos \theta)^4 \right] \Big|_0^{\pi/2} = -\frac{1}{12} (1 - 2^4) = \frac{5}{4} \end{aligned}$$



10. Using symmetry,

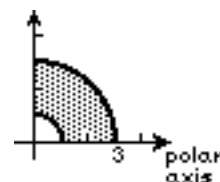
$$\begin{aligned}
 V &= 2 \int_0^{\pi/2} \int_0^{\cos \theta} (2 + r^2) r \, dr \, d\theta = \int_0^{\pi/2} \left(r^2 + \frac{1}{4} r^4 \right) \Big|_0^{\cos \theta} d\theta \\
 &= 2 \int_0^{\pi/2} \left(\cos^2 \theta + \frac{1}{4} \cos^4 \theta \right) d\theta = 2 \int_0^{\pi/2} \left[\cos^2 \theta + \frac{1}{4} \left(\frac{1 + \cos 2\theta}{2} \right)^2 \right] d\theta \\
 &= \int_0^{\pi/2} \left(2 \cos^2 \theta + \frac{1}{8} + \frac{1}{4} \cos 2\theta + \frac{1}{8} \cos^2 2\theta \right) d\theta \\
 &= \left(\theta + \frac{1}{2} \sin 2\theta + \frac{1}{8} \theta + \frac{1}{8} \sin 2\theta + \frac{1}{16} \theta + \frac{1}{64} \sin 4\theta \right) \Big|_0^{\pi/2} = \frac{19\pi}{32}.
 \end{aligned}$$



$$11. \quad m = \int_0^{\pi/2} \int_1^3 kr \, dr \, d\theta = k \int_0^{\pi/2} \frac{1}{2} r^2 \Big|_1^3 d\theta = \frac{1}{2} k \int_0^{\pi/2} 8 \, d\theta = 2k\pi$$

$$\begin{aligned}
 M_y &= \int_0^{\pi/2} \int_1^3 kxr \, dr \, d\theta = k \int_0^{\pi/2} \int_1^3 r^2 \cos \theta \, dr \, d\theta = k \int_0^{\pi/2} \frac{1}{3} r^3 \cos \theta \Big|_1^3 d\theta \\
 &= \frac{1}{3} k \int_0^{\pi/2} 26 \cos \theta \, d\theta = \frac{26}{3} k \sin \theta \Big|_0^{\pi/2} = \frac{26}{3} k
 \end{aligned}$$

$\bar{x} = M_y/m = \frac{26k/3}{2k\pi} = \frac{13}{3\pi}$. Since the region and density function are symmetric about the ray $\theta = \pi/4$, $\bar{y} = \bar{x} = 13/3\pi$ and the center of mass is $(13/3\pi, 13/3\pi)$.



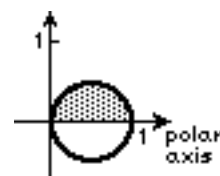
12. The interior of the upper-half circle is traced from
- $\theta = 0$
- to
- $\pi/2$
- . The density is
- kr
- .

Since both the region and the density are symmetric about the polar axis, $\bar{y} = 0$.

$$\begin{aligned}
 m &= \int_0^{\pi/2} \int_0^{\cos \theta} kr^2 \, dr \, d\theta = k \int_0^{\pi/2} \frac{1}{3} r^3 \Big|_0^{\cos \theta} d\theta = \frac{k}{3} \int_0^{\pi/2} \cos^3 \theta \, d\theta \\
 &= \frac{k}{3} \left(\frac{2}{3} + \frac{1}{3} \cos^2 \theta \right) \sin \theta \Big|_0^{\pi/2} = \frac{2k}{9}
 \end{aligned}$$

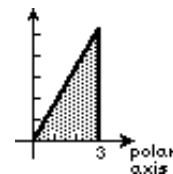
$$\begin{aligned}
 M_y &= k \int_0^{\pi/2} \int_0^{\cos \theta} (r \cos \theta)(r)(r \, dr \, d\theta) = k \int_0^{\pi/2} \int_0^{\cos \theta} r^3 \cos \theta \, dr \, d\theta = k \int_0^{\pi/2} \frac{1}{4} r^4 \cos \theta \Big|_0^{\cos \theta} d\theta \\
 &= \frac{k}{4} \int_0^{\pi/2} \cos^5 \theta \, d\theta = \frac{k}{4} \left(\sin \theta - \frac{2}{3} \sin^3 \theta + \frac{1}{5} \sin^5 \theta \right) \Big|_0^{\pi/2} = \frac{2k}{15}
 \end{aligned}$$

Thus, $\bar{x} = \frac{2k/15}{2k/9} = 3/5$ and the center of mass is $(3/5, 0)$.



13. In polar coordinates the line
- $x = 3$
- becomes
- $r \cos \theta = 3$
- or
- $r = 3 \sec \theta$
- . The angle of inclination of the line
- $y = \sqrt{3}x$
- is
- $\pi/3$
- .

$$\begin{aligned}
 m &= \int_0^{\pi/3} \int_0^{3 \sec \theta} r^2 r \, dr \, d\theta = \int_0^{\pi/3} \frac{1}{4} r^4 \Big|_0^{3 \sec \theta} d\theta = \frac{81}{4} \int_0^{\pi/3} \sec^4 \theta \, d\theta \\
 &= \frac{81}{4} \int_0^{\pi/3} (1 + \tan^2 \theta) \sec^2 \theta \, d\theta = \frac{81}{4} \left(\tan \theta + \frac{1}{3} \tan^3 \theta \right) \Big|_0^{\pi/3} = \frac{81}{4} (\sqrt{3} + \sqrt{3}) = \frac{81}{2} \sqrt{3}
 \end{aligned}$$

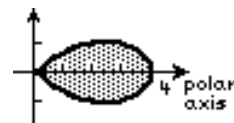


Exercises 9.11

$$\begin{aligned}
 M_y &= \int_0^{\pi/3} \int_0^{3 \sec \theta} x r^2 dr d\theta = \int_0^{\pi/3} \int_0^{3 \sec \theta} r^4 \cos \theta dr d\theta = \int_0^{\pi/3} \frac{1}{5} r^5 \cos \theta \Big|_0^{3 \sec \theta} d\theta \\
 &= \frac{243}{5} \int_0^{\pi/3} \sec^5 \theta \cos \theta d\theta = \frac{243}{5} \int_0^{\pi/3} \sec^4 \theta d\theta = \frac{243}{5} (2\sqrt{3}) = \frac{486}{5} \sqrt{3} \\
 M_x &= \int_0^{\pi/3} \int_0^{3 \sec \theta} y r^2 dr d\theta = \int_0^{\pi/3} \int_0^{3 \sec \theta} r^4 \sin \theta d\theta = \int_0^{\pi/3} \frac{1}{5} r^5 \sin \theta \Big|_0^{3 \sec \theta} d\theta \\
 &= \frac{243}{5} \int_0^{\pi/3} \sec^5 \theta \sin \theta d\theta = \frac{243}{5} \int_0^{\pi/3} \tan \theta \sec^4 \theta d\theta = \frac{243}{5} \int_0^{\pi/3} \tan \theta (1 + \tan^2 \theta) \sec^2 \theta d\theta \\
 &= \frac{243}{5} \int_0^{\pi/3} (\tan \theta + \tan^3 \theta) \sec^2 \theta d\theta = \frac{243}{5} \left(\frac{1}{2} \tan^2 \theta + \frac{1}{4} \tan^4 \theta \right) \Big|_0^{\pi/3} = \frac{243}{5} \left(\frac{3}{2} + \frac{9}{4} \right) = \frac{729}{4} \\
 \bar{x} = M_y/m &= \frac{486\sqrt{3}/5}{81\sqrt{3}/2} = 12/5; \quad \bar{y} = M_x/m = \frac{729/4}{81\sqrt{3}/2} = 3\sqrt{3}/2. \text{ The center of mass is } (12/5, 3\sqrt{3}/2).
 \end{aligned}$$

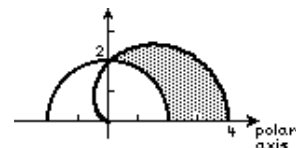
14. Since both the region and the density are symmetric about the x -axis, $\bar{y} = 0$. Using symmetry,

$$\begin{aligned}
 m &= 2 \int_0^{\pi/4} \int_0^{4 \cos 2\theta} k r dr d\theta = 2k \int_0^{\pi/4} \frac{1}{2} r^2 \Big|_0^{4 \cos 2\theta} d\theta = 16k \int_0^{\pi/4} \cos^2 2\theta d\theta \\
 &= 16k \left(\frac{1}{2} \theta + \frac{1}{8} \sin 4\theta \right) \Big|_0^{\pi/4} = 2k\pi \\
 M_y &= 2 \int_0^{\pi/4} \int_0^{4 \cos 2\theta} k x r dr d\theta = 2k \int_0^{\pi/4} \int_0^{4 \cos 2\theta} r^2 \cos \theta dr d\theta = 2k \int_0^{\pi/4} \frac{1}{3} r^3 \cos \theta \Big|_0^{4 \cos 2\theta} d\theta \\
 &= \frac{128}{3} k \int_0^{\pi/4} \cos^3 2\theta \cos \theta d\theta = \frac{128}{3} k \int_0^{\pi/4} (1 - 2 \sin^2 \theta)^3 \cos \theta d\theta \\
 &= \frac{128}{3} k \int_0^{\pi/4} (1 - 6 \sin^2 \theta + 12 \sin^4 \theta - 8 \sin^6 \theta) \cos \theta d\theta = \frac{128}{3} k \left(\sin \theta - 2 \sin^3 \theta + \frac{12}{5} \sin^5 \theta - \frac{8}{7} \sin^7 \theta \right) \Big|_0^{\pi/4} \\
 &= \frac{128}{3} k \left(\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} + \frac{3\sqrt{2}}{10} - \frac{\sqrt{2}}{14} \right) = \frac{1024}{105} \sqrt{2} k \\
 \bar{x} = M_y/m &= \frac{1024\sqrt{2}k/105}{2k\pi} = \frac{512\sqrt{2}}{105\pi}. \text{ The center of mass is } (512\sqrt{2}/105\pi, 0) \text{ or approximately } (2.20, 0).
 \end{aligned}$$



15. The density is $\rho = k/r$.

$$\begin{aligned}
 m &= \int_0^{\pi/2} \int_2^{2+2 \cos \theta} \frac{k}{r} r dr d\theta = k \int_0^{\pi/2} \int_2^{2+2 \cos \theta} dr d\theta \\
 &= k \int_0^{\pi/2} 2 \cos \theta d\theta = 2k(\sin \theta) \Big|_0^{\pi/2} = 2k \\
 M_y &= \int_0^{\pi/2} \int_2^{2+2 \cos \theta} x \frac{k}{r} r dr d\theta = k \int_0^{\pi/2} \int_2^{2+2 \cos \theta} r \cos \theta dr d\theta = k \int_0^{\pi/2} \frac{1}{2} r^2 \Big|_2^{2+2 \cos \theta} \cos \theta d\theta \\
 &= \frac{1}{2} k \int_0^{\pi/2} (8 \cos \theta + 4 \cos^2 \theta) \cos \theta d\theta = 2k \int_0^{\pi/2} (2 \cos^2 \theta + \cos \theta - \sin^2 \theta \cos \theta) d\theta \\
 &= 2k \left(\theta + \frac{1}{2} \sin 2\theta + \sin \theta - \frac{1}{3} \sin^3 \theta \right) \Big|_0^{\pi/2} = 2k \left(\frac{\pi}{2} + \frac{2}{3} \right) = \frac{3\pi + 4}{3} k
 \end{aligned}$$



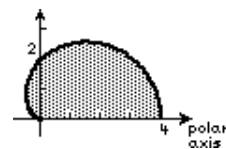
$$M_x = \int_0^{\pi/2} \int_2^{2+2\cos\theta} y \frac{k}{r} r dr d\theta = k \int_0^{\pi/2} \int_2^{2+2\cos\theta} r \sin\theta dr d\theta = k \int_0^{\pi/2} \frac{1}{2} r^2 \Big|_2^{2+2\cos\theta} \sin\theta d\theta$$

$$= \frac{1}{2} k \int_0^{\pi/2} (8\cos\theta + 4\cos^2\theta) \sin\theta d\theta = \frac{1}{2} k \left(-4\cos^2\theta - \frac{4}{3}\cos^3\theta \right) \Big|_0^{\pi/2} = \frac{1}{2} k \left[-\left(-4 - \frac{4}{3}\right) \right] = \frac{8}{3} k$$

$$\bar{x} = M_y/m = \frac{(3\pi+4)k/3}{2k} = \frac{3\pi+4}{6}; \quad \bar{y} = M_x/m = \frac{8k/3}{2k} = \frac{4}{3}. \text{ The center of mass is } ((3\pi+4)/6, 4/3).$$

$$16. \quad m = \int_0^\pi \int_0^{2+2\cos\theta} kr dr d\theta = k \int_0^\pi \frac{1}{2} r^2 \Big|_0^{2+2\cos\theta} d\theta = 2k \int_0^\pi (1+\cos\theta)^2 d\theta$$

$$= 2k \int_0^\pi (1+2\cos\theta+\cos^2\theta) d\theta = 2k \left(\theta + 2\sin\theta + \frac{1}{2}\theta + \frac{1}{4}\sin 2\theta \right) \Big|_0^\pi = 3\pi k$$



$$M_y = \int_0^\pi \int_0^{2+2\cos\theta} kxr dr d\theta = k \int_0^\pi \int_0^{2+2\cos\theta} r^2 \cos\theta dr d\theta = k \int_0^\pi \frac{1}{3} r^3 \Big|_0^{2+2\cos\theta} \cos\theta d\theta$$

$$= \frac{8}{3} k \int_0^\pi (1+\cos\theta)^3 \cos\theta d\theta = \frac{8}{3} k \int_0^\pi (\cos\theta + 3\cos^2\theta + 3\cos^3\theta + \cos^4\theta) d\theta$$

$$= \frac{8}{3} k \left[\sin\theta + \left(\frac{3}{2}\theta + \frac{3}{4}\sin 2\theta \right) + (3\sin\theta - \sin^3\theta) + \left(\frac{3}{8}\theta + \frac{1}{4}\sin 2\theta + \frac{1}{32}\sin 4\theta \right) \right] \Big|_0^\pi = \frac{8}{3} k \left(\frac{15}{8}\pi \right) = 5\pi k$$

$$M_x = \int_0^\pi \int_0^{2+2\cos\theta} kyr dr d\theta = k \int_0^\pi \int_0^{2+2\cos\theta} r^2 \sin\theta dr d\theta = k \int_0^\pi \frac{1}{3} r^3 \Big|_0^{2+2\cos\theta} \sin\theta d\theta$$

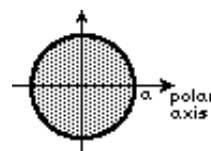
$$= \frac{8}{3} k \int_0^\pi (1+\cos\theta)^3 \sin\theta d\theta = \frac{8}{3} k \int_0^\pi (1+3\cos\theta+3\cos^2\theta+\cos^3\theta) \sin\theta d\theta$$

$$= \frac{8}{3} k \left(-\cos\theta - \frac{3}{2}\cos^2\theta - \cos^3\theta - \frac{1}{4}\cos^4\theta \right) \Big|_0^\pi = \frac{8}{3} k \left[\frac{1}{4} - \left(-\frac{15}{4}\right) \right] = \frac{32}{3} k$$

$$\bar{x} = M_y/m = \frac{5\pi k}{3\pi k} = 5/3; \quad \bar{y} = M_x/m = \frac{32k/3}{3\pi k} = 32/9\pi. \text{ The center of mass is } (5/3, 32/9\pi).$$

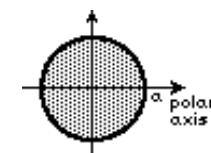
$$17. \quad I_x = \int_0^{2\pi} \int_0^a y^2 kr dr d\theta = k \int_0^{2\pi} \int_0^a r^3 \sin^2\theta dr d\theta = k \int_0^{2\pi} \frac{1}{4} r^4 \sin^2\theta \Big|_0^a d\theta$$

$$= \frac{ka^4}{4} \int_0^{2\pi} \sin^2\theta d\theta = \frac{ka^4}{4} \left(\frac{1}{2}\theta - \frac{1}{4}\sin 2\theta \right) \Big|_0^{2\pi} = \frac{k\pi a^4}{4}$$



$$18. \quad I_x = \int_0^{2\pi} \int_0^a y^2 \frac{1}{1+r^4} r dr d\theta = \int_0^{2\pi} \int_0^a \frac{r^3}{1+r^4} \sin^2\theta dr d\theta$$

$$= \int_0^{2\pi} \frac{1}{4} \ln(1+r^4) \Big|_0^a \sin^2\theta d\theta = \frac{1}{4} \ln(1+a^4) \left(\frac{1}{2}\theta - \frac{1}{4}\sin 2\theta \right) \Big|_0^{2\pi} = \frac{\pi}{4} \ln(1+a^4)$$

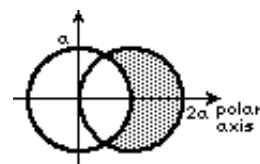


19. Solving $a = 2a \cos\theta$, $\cos\theta = 1/2$ or $\theta = \pi/3$. The density is k/r^3 . Using symmetry,

$$I_y = 2 \int_0^{\pi/3} \int_a^{2a\cos\theta} x^2 \frac{k}{r^3} r dr d\theta = 2k \int_0^{\pi/3} \int_a^{2a\cos\theta} \cos^2\theta dr d\theta$$

$$= 2k \int_0^{\pi/3} (2a\cos^3\theta - a\cos^2\theta) d\theta = 2ak \left(2\sin\theta - \frac{2}{3}\sin^3\theta - \frac{1}{2}\theta - \frac{1}{4}\sin 2\theta \right) \Big|_0^{\pi/3}$$

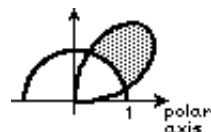
$$= 2ak \left(\sqrt{3} - \frac{\sqrt{3}}{4} - \frac{\pi}{6} - \frac{\sqrt{3}}{8} \right) = \frac{5ak\sqrt{3}}{4} - \frac{ak\pi}{3}$$



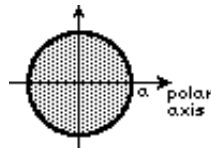
Exercises 9.11

20. Solving $1 = 2 \sin 2\theta$, we obtain $\sin 2\theta = 1/2$ or $\theta = \pi/12$ and $\theta = 5\pi/12$.

$$\begin{aligned} I_y &= \int_{\pi/12}^{5\pi/12} \int_1^{2\sin 2\theta} x^2 \sec^2 \theta r dr d\theta = \int_{\pi/12}^{5\pi/12} \int_1^{2\sin 2\theta} r^3 dr d\theta \\ &= \int_{\pi/12}^{5\pi/12} \frac{1}{4} r^4 \Big|_1^{2\sin 2\theta} d\theta = 4 \int_{\pi/12}^{5\pi/12} \sin^4 2\theta d\theta = 2 \left(\frac{3}{4} \theta - \frac{1}{4} \sin 4\theta + \frac{1}{32} \sin 8\theta \right) \Big|_{\pi/12}^{5\pi/12} \\ &= 2 \left[\left(\frac{5\pi}{16} + \frac{\sqrt{3}}{8} - \frac{\sqrt{3}}{64} \right) - \left(\frac{\pi}{16} - \frac{\sqrt{3}}{8} + \frac{\sqrt{3}}{64} \right) \right] = \frac{8\pi + 7\sqrt{3}}{16} \end{aligned}$$

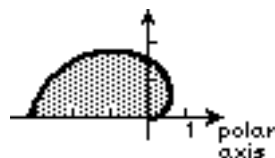


21. From Problem 17, $I_x = k\pi a^4/4$. By symmetry, $I_y = I_x$. Thus $I_0 = k\pi a^4/2$.



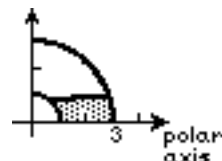
22. The density is $\rho = kr$.

$$\begin{aligned} I_0 &= \int_0^\pi \int_0^\theta r^2 (kr) r dr d\theta = k \int_0^\pi \int_0^\theta r^4 dr d\theta = k \int_0^\pi \frac{1}{5} r^5 \Big|_0^\theta d\theta \\ &= \frac{1}{5} k \int_0^\pi \theta^5 d\theta = \frac{1}{5} k \left(\frac{1}{6} \theta^6 \right) \Big|_0^\pi = \frac{k\pi^6}{30} \end{aligned}$$

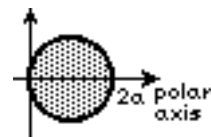


23. The density is $\rho = k/r$.

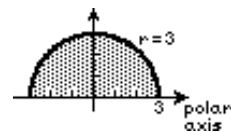
$$I_0 = \int_1^3 \int_0^{1/r} r^2 \frac{k}{r} r d\theta dr = k \int_1^3 \int_0^{1/r} r^2 d\theta dr = k \int_1^3 r^2 \left(\frac{1}{r} \right) dr = k \left(\frac{1}{2} r^2 \right) \Big|_1^3 = 4k$$



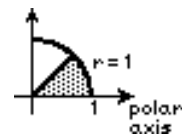
$$\begin{aligned} 24. \quad I_0 &= \int_0^\pi \int_0^{2a \cos \theta} r^2 k r dr d\theta = k \int_0^\pi \frac{1}{4} r^4 \Big|_0^{2a \cos \theta} d\theta = 4ka^4 \int_0^\pi \cos^4 \theta d\theta \\ &= 4ka^4 \left(\frac{3}{8} \theta + \frac{1}{4} \sin 2\theta + \frac{1}{32} \sin 4\theta \right) \Big|_0^\pi = 4ka^4 \left(\frac{3\pi}{8} \right) = \frac{3k\pi a^4}{2} \end{aligned}$$



$$25. \quad \int_{-3}^3 \int_0^{\sqrt{9-x^2}} \sqrt{x^2+y^2} dy dx = \int_0^\pi \int_0^3 |r| r dr d\theta = \int_0^\pi \frac{1}{3} r^3 \Big|_0^3 d\theta = 9 \int_0^\pi d\theta = 9\pi$$



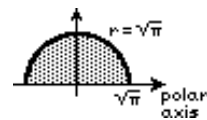
$$\begin{aligned} 26. \quad \int_0^{\sqrt{2}/2} \int_y^{\sqrt{1-y^2}} \frac{y^2}{\sqrt{x^2+y^2}} dx dy &= \int_0^{\pi/4} \int_0^1 \frac{r^2 \sin^2 \theta}{|r|} r dr d\theta \\ &= \int_0^{\pi/4} \int_0^1 r^2 \sin^2 \theta dr d\theta = \int_0^{\pi/4} \frac{1}{3} r^3 \sin^2 \theta \Big|_0^1 d\theta = \frac{1}{3} \int_0^{\pi/4} \sin^2 \theta d\theta \\ &= \frac{1}{3} \left(\frac{1}{2} \theta - \frac{1}{4} \sin 2\theta \right) \Big|_0^{\pi/4} = \frac{\pi-2}{24} \end{aligned}$$



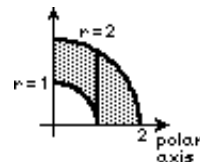
$$\begin{aligned} 27. \quad \int_0^1 \int_0^{\sqrt{1-y^2}} e^{x^2+y^2} dx dy &= \int_0^{\pi/2} \int_0^1 e^{r^2} r dr d\theta = \int_0^{\pi/2} \frac{1}{2} e^{r^2} \Big|_0^1 d\theta \\ &= \frac{1}{2} \int_0^{\pi/2} (e-1) d\theta = \frac{\pi(e-1)}{4} \end{aligned}$$



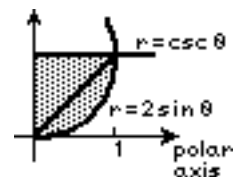
$$\begin{aligned}
 28. \quad \int_{-\sqrt{\pi}}^{\sqrt{\pi}} \int_0^{\sqrt{\pi-x^2}} \sin(x^2+y^2) \, dy \, dx &= \int_0^{\pi} \int_0^{\sqrt{\pi}} (\sin r^2) r \, dr \, d\theta = \int_0^{\pi} \left. -\frac{1}{2} \cos r^2 \right|_0^{\sqrt{\pi}} d\theta \\
 &= -\frac{1}{2} \int_0^{\pi} (-1 - 1) \, d\theta = \pi
 \end{aligned}$$



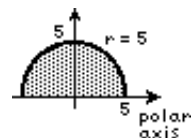
$$\begin{aligned}
 29. \quad \int_0^1 \int_{\sqrt{1-x^2}}^{\sqrt{4-x^2}} \frac{x^2}{x^2+y^2} \, dy \, dx + \int_1^2 \int_0^{\sqrt{4-x^2}} \frac{x^2}{x^2+y^2} \, dy \, dx \\
 = \int_0^{\pi/2} \int_1^2 \frac{r^2 \cos^2 \theta}{r^2} r \, dr \, d\theta = \int_0^{\pi/2} \int_1^2 r \cos^2 \theta \, dr \, d\theta \\
 = \int_0^{\pi/2} \left. \frac{1}{2} r^2 \right|_1^2 \cos^2 \theta \, d\theta = \frac{3}{2} \int_0^{\pi/2} \cos^2 \theta \, d\theta = \frac{3}{2} \left(\frac{1}{2} \theta + \frac{1}{4} \sin 2\theta \right) \Big|_0^{\pi/2} = \frac{3\pi}{8}
 \end{aligned}$$



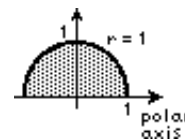
$$\begin{aligned}
 30. \quad \int_0^1 \int_0^{\sqrt{2y-y^2}} (1-x^2-y^2) \, dx \, dy \\
 = \int_0^{\pi/4} \int_0^{2\sin\theta} (1-r^2) r \, dr \, d\theta + \int_{\pi/4}^{\pi/2} \int_0^{\csc\theta} (1-r^2) r \, dr \, d\theta \\
 = \int_0^{\pi/4} \left(\frac{1}{2} r^2 - \frac{1}{4} r^4 \right) \Big|_0^{2\sin\theta} d\theta + \int_{\pi/4}^{\pi/2} \left(\frac{1}{2} r^2 - \frac{1}{4} r^4 \right) \Big|_0^{\csc\theta} d\theta \\
 = \int_0^{\pi/4} (2\sin^2\theta - 4\sin^4\theta) \, d\theta + \int_{\pi/4}^{\pi/2} \left(\frac{1}{2} \csc^2\theta - \frac{1}{4} \csc^4\theta \right) d\theta \\
 = \left[\theta - \frac{1}{2} \sin 2\theta - \left(\frac{3}{2} \theta - \sin 2\theta + \frac{1}{8} \sin 4\theta \right) \right] + \left[-\frac{1}{2} \cot \theta - \frac{1}{4} \left(-\cot \theta - \frac{1}{3} \cot^3 \theta \right) \right] \Big|_{\pi/4}^{\pi/2} \\
 = \left(-\frac{\pi}{8} + \frac{1}{2} \right) + \left[0 - \left(-\frac{1}{4} + \frac{1}{12} \right) \right] = \frac{16-3\pi}{24}
 \end{aligned}$$



$$\begin{aligned}
 31. \quad \int_{-5}^5 \int_0^{\sqrt{25-x^2}} (4x+3y) \, dy \, dx = \int_0^{\pi} \int_0^5 (4r \cos \theta + 3r \sin \theta) r \, dr \, d\theta \\
 = \int_0^{\pi} \int_0^5 (4r^2 \cos \theta + 3r^2 \sin \theta) \, dr \, d\theta = \int_0^{\pi} \left(\frac{4}{3} r^3 \cos \theta + r^3 \sin \theta \right) \Big|_0^5 d\theta \\
 = \int_0^{\pi} \left(\frac{500}{3} \cos \theta + 125 \sin \theta \right) d\theta = \left(\frac{500}{3} \sin \theta - 125 \cos \theta \right) \Big|_0^{\pi} = 250
 \end{aligned}$$



$$\begin{aligned}
 32. \quad \int_0^1 \int_0^{\sqrt{1-y^2}} \frac{1}{1+\sqrt{x^2+y^2}} \, dx \, dy = \int_0^{\pi/2} \int_0^1 \frac{1}{1+r} r \, dr \, d\theta \\
 = \int_0^{\pi/2} \int_0^1 \left(1 - \frac{1}{1+r} \right) dr \, d\theta = \int_0^{\pi/2} [r - \ln(1+r)] \Big|_0^1 d\theta \\
 = \int_0^{\pi/2} (1 - \ln 2) d\theta = \frac{\pi}{2} (1 - \ln 2)
 \end{aligned}$$



33. The volume of the cylindrical portion of the tank is $V_c = \pi(4.2)^2 19.3 \approx 1069.56 \text{ m}^3$. We take the equation of the ellipsoid to be

$$\frac{x^2}{(4.2)^2} + \frac{z^2}{(5.15)^2} = 1 \quad \text{or} \quad z = \pm \frac{5.15}{4.2} \sqrt{(4.2)^2 - x^2 - y^2}.$$

Exercises 9.11

The volume of the ellipsoid is

$$\begin{aligned} V_e &= 2 \left(\frac{5.15}{4.2} \right) \iint_R \sqrt{(4.2)^2 - x^2 - y^2} \, dx \, dy = \frac{10.3}{4.2} \int_0^{2\pi} \int_0^{4.2} [(4.2)^2 - r^2]^{1/2} r \, dr \, d\theta \\ &= \frac{10.3}{4.2} \int_0^{2\pi} \left[\left(-\frac{1}{2} \right) \frac{2}{3} [(4.2)^2 - r^2]^{3/2} \Big|_0^{4.2} \right] d\theta = \frac{10.3}{4.2} \frac{1}{3} \int_0^{2\pi} (4.2)^3 \, d\theta \\ &= \frac{2\pi}{3} \frac{10.3}{4.2} (4.2)^3 \approx 380.53. \end{aligned}$$

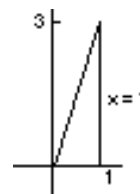
The volume of the tank is approximately $1069.56 + 380.53 = 1450.09 \text{ m}^3$.

$$\begin{aligned} 34. \quad \iint_R (x+y) \, dA &= \int_0^{\pi/2} \int_{2\sin\theta}^2 (r\cos\theta + r\sin\theta) r \, dr \, d\theta = \int_0^{\pi/2} \int_{2\sin\theta}^2 r^2 (\cos\theta + \sin\theta) \, dr \, d\theta \\ &= \int_0^{\pi/2} \frac{1}{3} r^3 (\cos\theta + \sin\theta) \Big|_{2\sin\theta}^2 d\theta = \frac{8}{3} \int_0^{\pi/2} (\cos\theta + \sin\theta - \sin^3\theta \cos\theta - \sin^4\theta) \, d\theta \\ &= \frac{8}{3} \left(\sin\theta - \cos\theta - \frac{1}{4} \sin^4\theta + \frac{1}{4} \sin^3\theta \cos\theta - \frac{3}{8}\theta + \frac{3}{16} \sin 2\theta \right) \Big|_0^{\pi/2} \\ &= \frac{8}{3} \left[\left(1 - \frac{1}{4} - \frac{3\pi}{16} \right) - (-1) \right] = \frac{28-3\pi}{6} \\ 35. \quad I^2 &= \int_0^\infty \int_0^\infty e^{-(x^2+y^2)} \, dx \, dy = \int_0^{\pi/2} \int_0^\infty e^{-r^2} r \, dr \, d\theta = \int_0^{\pi/2} \lim_{t \rightarrow \infty} \left(-\frac{1}{2} e^{-r^2} \right) \Big|_0^t d\theta \\ &= \int_0^{\pi/2} \lim_{t \rightarrow \infty} \left(-\frac{1}{2} e^{-t^2} + \frac{1}{2} \right) d\theta = \int_0^{\pi/2} \frac{1}{2} \, d\theta = \frac{\pi}{4}; \quad I = \frac{\sqrt{\pi}}{2} \end{aligned}$$

Exercises 9.12

1. The sides of the triangle are C_1 : $y = 0$, $0 \leq x \leq 1$; C_2 : $x = 1$, $0 \leq y \leq 3$; C_3 : $y = 3x$, $0 \leq -x \leq 1$.

$$\begin{aligned} \oint_C (x-y) \, dx + xy \, dy &= \int_0^1 x \, dx + \int_0^3 y \, dy + \int_1^0 (x-3x) \, dx + \int_1^0 x(3x) 3 \, dx \\ &= \left(\frac{1}{2} x^2 \right) \Big|_0^1 + \left(\frac{1}{2} y^2 \right) \Big|_0^3 + (-x^2) \Big|_0^1 + (3x^2) \Big|_1^0 = \frac{1}{2} + \frac{9}{2} + 1 - 3 = 3 \end{aligned}$$



$$\begin{aligned} \iint_R (y+1) \, dA &= \int_0^1 \int_0^{3x} (y+1) \, dy \, dx = \int_0^1 \left(\frac{1}{2} y^2 + y \right) \Big|_0^{3x} dx = \int_0^1 \left(\frac{9}{2} x^2 + 3x \right) dx \\ &= \left(\frac{3}{2} x^3 + \frac{3}{2} x^2 \right) \Big|_0^1 = 3 \end{aligned}$$

2. The sides of the rectangle are C_1 : $y = 0$, $-1 \leq x \leq 1$; C_2 : $x = 1$, $0 \leq y \leq 1$; C_3 : $y = 1$, $1 \geq x \geq -1$; C_4 : $x = -1$, $1 \geq y \geq 0$.

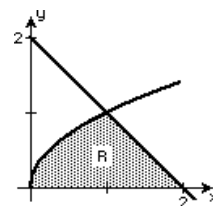
$$\begin{aligned} \oint_C 3x^2 y \, dx + (x^2 - 5y) \, dy &= \int_{-1}^1 0 \, dx + \int_0^1 (1 - 5y) \, dy = \int_{-1}^1 0 \, dx + \int_0^1 (1 - 5y) \, dy \\ &= \int_{-1}^{-1} 3x^2 \, dx + \int_1^0 (1 - 5y) \, dy = \left(y - \frac{5}{2} y^2 \right) \Big|_1^0 + x^3 \Big|_{-1}^{-1} + \left(y - \frac{5}{2} y^2 \right) \Big|_1^0 = -2 \\ \iint_R (2x - 3x^2) \, dA &= \int_0^1 \int_{-1}^1 (2x - 3x^2) \, dx \, dy = \int_0^1 (x^2 - x^3) \Big|_{-1}^1 dy = \int_0^1 (-2) \, dy = -2 \end{aligned}$$

$$\begin{aligned}
3. \quad \oint_C -y^2 dx + x^2 dy &= \int_0^{2\pi} (-9 \sin^2 t)(-3 \sin t) dt + \int_0^{2\pi} 9 \cos^2 t(3 \cos t) dt \\
&= 27 \int_0^{2\pi} [(1 - \cos^2 t) \sin t + (1 - \sin^2 t) \cos t] dt \\
&= 27 \left(-\cos t + \frac{1}{3} \cos^3 t + \sin t - \frac{1}{3} \sin^3 t \right) \Big|_0^{2\pi} = 27(0) = 0 \\
\iint_R (2x + 2y) dA &= 2 \int_0^{2\pi} \int_0^3 (r \cos \theta + r \sin \theta) r dr d\theta = 2 \int_0^{2\pi} \int_0^3 r^2 (\cos \theta + \sin \theta) dr d\theta \\
&= 2 \int_0^{2\pi} \left[\frac{1}{3} r^3 (\cos \theta + \sin \theta) \right] \Big|_0^3 d\theta = 18 \int_0^{2\pi} (\cos \theta + \sin \theta) d\theta \\
&= 18(\sin \theta - \cos \theta) \Big|_0^{2\pi} = 18(0) = 0
\end{aligned}$$

4. The sides of the region are C_1 : $y = 0$, $0 \leq x \leq 2$; C_2 : $y = -x + 2$, $2 \geq x \geq 1$; C_3 : $y = \sqrt{x}$, $1 \geq x \geq 0$.

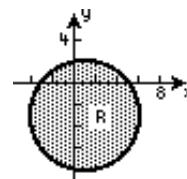
$$\begin{aligned}
\oint_C -2y^2 dx + 4xy dy &= \int_0^2 0 dx + \int_2^1 -2(-x+2)^2 dx + \int_2^1 4x(-x+2)(-dx) \\
&\quad + \int_1^0 -2x dx + \int_1^0 4x\sqrt{x} \left(\frac{1}{2\sqrt{x}} \right) dx \\
&= 0 + \frac{2}{3} + \frac{8}{3} + 1 - 1 = \frac{10}{3}
\end{aligned}$$

$$\iint_R 8y dA = \int_0^1 \int_{y^2}^{2-y} 8y dx dy = \int_0^1 8y(2-y-y^2) dy = \left(8y^2 - \frac{8}{3}y^3 - 2y^4 \right) \Big|_0^1 = \frac{10}{3}$$



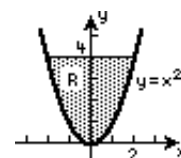
5. $P = 2y$, $P_y = 2$, $Q = 5x$, $Q_x = 5$

$$\oint_C 2y dx + 5x dy = \iint_R (5 - 2) dA = 3 \iint_R dA = 3(25\pi) = 75\pi$$



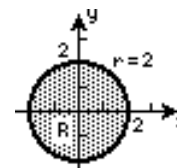
6. $P = x + y^2$, $P_y = 2y$, $Q = 2x^2 - y$, $Q_x = 4x$

$$\begin{aligned}
\oint_C (x + y^2) dx + (2x^2 - y) dy &= \iint_R (4x - 2y) dA = \int_{-2}^2 \int_{x^2}^4 (4x - 2y) dy dx \\
&= \int_{-2}^2 (4xy - y^2) \Big|_{x^2}^4 dx = \int_{-2}^2 (16x - 16 - 4x^3 + x^4) dx \\
&= \left(8x^2 - 16x - x^4 + \frac{1}{5}x^5 \right) \Big|_{-2}^2 = -\frac{96}{5}
\end{aligned}$$



7. $P = x^4 - 2y^3$, $P_y = -6y^2$, $Q = 2x^3 - y^4$, $Q_x = 6x^2$. Using polar coordinates,

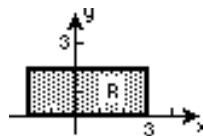
$$\begin{aligned}
\oint_C (x^4 - 2y^3) dx + (2x^3 - y^4) dy &= \iint_R (6x^2 + 6y^2) dA = \int_0^{2\pi} \int_0^2 6r^2 r dr d\theta \\
&= \int_0^{2\pi} \left(\frac{3}{2} r^4 \right) \Big|_0^2 d\theta = \int_0^{2\pi} 24 d\theta = 48\pi.
\end{aligned}$$



Exercises 9.12

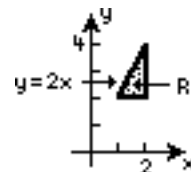
8. $P = x - 3y$, $P_y = -3$, $Q = 4x + y$, $Q_x = 4$

$$\oint_C (x - 3y) dx + 4(x + y) dy = \iint_R (4 + 3) dA = 7(10) = 70$$



9. $P = 2xy$, $P_y = 2x$, $Q = 3xy^2$, $Q_x = 3y^2$

$$\begin{aligned} \oint_C 2xy dx + 3xy^2 dy &= \iint_R (3y^2 - 2x) dA = \int_1^2 \int_2^{2x} (3y^2 - 2x) dy dx \\ &= \int_1^2 (y^3 - 2xy) \Big|_2^{2x} dx = \int_1^2 (8x^3 - 4x^2 - 8 + 4x) dx \\ &= \left(2x^4 - \frac{4}{3}x^3 - 8x + 2x^2 \right) \Big|_1^2 = \frac{40}{3} - \left(-\frac{16}{3} \right) = \frac{56}{3} \end{aligned}$$

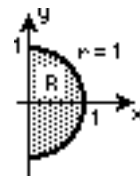


10. $P = e^{2x} \sin 2y$, $P_y = 2e^{2x} \cos 2y$, $Q = e^{2x} \cos 2y$, $Q_x = 2e^{2x} \cos 2y$

$$\oint_C = e^{2x} \sin 2y dx + e^{2x} \cos 2y dy = \iint_R 0 dA = 0$$

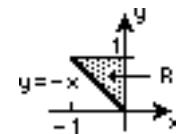
11. $P = xy$, $P_y = x$, $Q = x^2$, $Q_x = 2x$. Using polar coordinates,

$$\begin{aligned} \oint_C xy dx + x^2 dy &= \iint_R (2x - x) dA = \int_{-\pi/2}^{\pi/2} \int_0^1 r \cos \theta r dr d\theta \\ &= \int_{-\pi/2}^{\pi/2} \left(\frac{1}{3} r^3 \cos \theta \right) \Big|_0^1 d\theta = \int_{-\pi/2}^{\pi/2} \frac{1}{3} \cos \theta d\theta = \frac{1}{3} \sin \theta \Big|_{-\pi/2}^{\pi/2} = \frac{2}{3} \end{aligned}$$



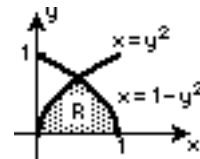
12. $P = e^{x^2}$, $P_y = 0$, $Q = 2 \tan^{-1} x$, $Q_x = \frac{2}{1+x^2}$

$$\begin{aligned} \oint_C e^{x^2} dx + 2 \tan^{-1} x dy &= \iint_R \frac{2}{1+x^2} dA = \int_{-1}^0 \int_{-x}^1 \frac{2}{1+x^2} dy dx \\ &= \int_{-1}^0 \left(\frac{2y}{1+x^2} \right) \Big|_{-x}^1 dx = \int_{-1}^0 \left(\frac{2}{1+x^2} + \frac{2x}{1+x^2} \right) dx \\ &= [2 \tan^{-1} x + \ln(1+x^2)] \Big|_{-1}^0 = 0 - \left(-\frac{\pi}{2} + \ln 2 \right) = \frac{\pi}{2} - \ln 2 \end{aligned}$$



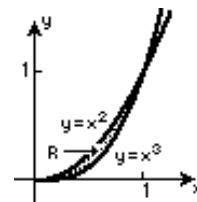
13. $P = \frac{1}{3}y^3$, $P_y = y^2$, $Q = xy + xy^2$, $Q_x = y + y^2$

$$\begin{aligned} \oint_C \frac{1}{3}y^3 dx + (xy + xy^2) dy &= \iint_R y dA = \int_0^{1/\sqrt{2}} \int_{y^2}^{1-y^2} y dx dy \\ &= \int_0^{1/\sqrt{2}} (xy) \Big|_{y^2}^{1-y^2} dy = \int_0^{1/\sqrt{2}} (y - y^3 - y^3) dy \\ &= \left(\frac{1}{2}y^2 - \frac{1}{2}y^4 \right) \Big|_0^{1/\sqrt{2}} = \frac{1}{4} - \frac{1}{8} = \frac{1}{8} \end{aligned}$$



14. $P = xy^2$, $P_y = 2xy$, $Q = 3 \cos y$, $Q_x = 0$

$$\begin{aligned}\oint_C xy^2 dx + 3 \cos y dy &= \iint_R (-2xy) dA = - \int_0^1 \int_{x^3}^{x^2} 2xy dy dx \\ &= - \int_0^1 (xy) \Big|_{x^3}^{x^2} dx = - \int_0^1 (x^3 - x^4) dx = \left(\frac{1}{4}x^4 - \frac{1}{5}x^5 \right) \Big|_0^1 = -\frac{1}{20}\end{aligned}$$



15. $P = ay$, $P_y = a$, $Q = bx$, $Q_x = b$. $\oint_C ay dx + bx dy = \iint_R (b - a) dA = (b - a) \times (\text{area bounded by } C)$

16. $P = P(x)$, $P_y = 0$, $Q = Q(y)$, $Q_x = 0$. $\oint_C P(x) dx + Q(y) dy = \iint_R 0 dA = 0$

17. For the first integral: $P = 0$, $P_y = 0$, $Q = x$, $Q_x = 1$; $\oint_C x dy = \iint_R 1 dA = \text{area of } R$.

For the second integral: $P = y$, $P_y = 1$, $Q = 0$, $Q_x = 0$; $-\oint_C y dx = -\iint_R -1 dA = \text{area of } R$.

Thus, $\oint_C x dy = -\oint_C y dx$.

18. $P = -y$, $P_y = -1$, $Q = x$, $Q_x = 1$. $\frac{1}{2} \oint_C -y dx + x dy = \frac{1}{2} \iint_R 2 dA = \iint_R dA = \text{area of } R$

19. $A = \iint_R dA = \oint_C x dy = \int_0^{2\pi} a \cos^3 t (3a \sin^2 t \cos t dt) = 3a^2 \int_0^{2\pi} \sin^2 t \cos^4 t dt$
 $= 3a^2 \left(\frac{1}{16}t - \frac{1}{64} \sin 4t + \frac{1}{48} \sin^3 2t \right) \Big|_0^{2\pi} = \frac{3}{8} \pi a^2$

20. $A = \iint_R dA = \oint_C x dy = \int_0^{2\pi} a \cos t (b \cos t dt) = ab \int_0^{2\pi} \cos^2 t dt = ab \left(\frac{1}{2}t + \frac{1}{4} \sin 2t \right) \Big|_0^{2\pi} = \pi ab$

21. (a) Parameterize C by $x = x_1 + (x_2 - x_1)t$ and $y = y_1 + (y_2 - y_1)t$ for $0 \leq t \leq 1$. Then

$$\begin{aligned}\int_C -y dx + x dy &= \int_0^1 -[y_1 + (y_2 - y_1)t](x_2 - x_1) dt + \int_0^1 [x_1 + (x_2 - x_1)t](y_2 - y_1) dt \\ &= -(x_2 - x_1) \left[y_1 t + \frac{1}{2}(y_2 - y_1)t^2 \right] \Big|_0^1 + (y_2 - y_1) \left[x_1 t + \frac{1}{2}(x_2 - x_1)t^2 \right] \Big|_0^1 \\ &= -(x_2 - x_1) \left[y_1 + \frac{1}{2}(y_2 - y_1) \right] + (y_2 - y_1) \left[x_1 + \frac{1}{2}(x_2 - x_1) \right] = x_1 y_2 - x_2 y_1.\end{aligned}$$

- (b) Let C_i be the line segment from (x_i, y_i) to (x_{i+1}, y_{i+1}) for $i = 1, 2, \dots, n-1$, and C_2 the line segment from (x_n, y_n) to (x_1, y_1) . Then

$$\begin{aligned}A &= \frac{1}{2} \oint_C -y dx + x dy \quad \boxed{\text{Using Problem 18}} \\ &= \frac{1}{2} \left[\int_{C_1} -y dx + x dy + \int_{C_2} -y dx + x dy + \dots + \int_{C_{n-1}} -y dx + x dy + \int_{C_n} -y dx + x dy \right] \\ &= \frac{1}{2} (x_1 y_2 - x_2 y_1) + \frac{1}{2} (x_2 y_3 - x_3 y_2) + \frac{1}{2} (x_{n-1} y_n - x_n y_{n-1}) + \frac{1}{2} (x_n y_1 - x_1 y_n).\end{aligned}$$

22. From part (b) of Problem 21

$$\begin{aligned}A &= \frac{1}{2} [(-1)(1) - (1)(3)] + \frac{1}{2} [(1)(2) - (4)(1)] + \frac{1}{2} [(4)(5) - (3)(2)] + \frac{1}{2} [(3)(3) - (-1)(5)] \\ &= \frac{1}{2} (-4 - 2 + 14 + 14) = 11.\end{aligned}$$

Exercises 9.12

23. $P = 4x^2 - y^3$, $P_y = -3y^2$; $Q = x^3 + y^2$, $Q_x = 3x^2$.

$$\begin{aligned}\oint_C (4x^2 - y^3) dx + (x^3 + y^2) dy &= \iint_R (3x^2 + 3y^2) dA = \int_0^{2\pi} \int_1^2 3r^2 (r dr d\theta) = \int_0^{2\pi} \left(\frac{3}{4} r^4 \right) \Big|_1^2 d\theta \\ &= \int_0^{2\pi} \frac{45}{4} d\theta = \frac{45\pi}{2}\end{aligned}$$

24. $P = \cos x^2 - y$, $P_y = -1$; $Q = \sqrt{y^3 + 1}$, $Q_x = 0$

$$\oint_C (\cos x^2 - y) dx + \sqrt{y^3 + 1} dy = \iint_R (0 + 1) dA = \iint_R dA = (6\sqrt{2})^2 - \pi(2)(4) = 72 - 8\pi$$

25. We first observe that $P_y = (y^4 - 3x^2y^2)/(x^2 + y^2)^3 = Q_x$. Letting C' be the circle $x^2 + y^2 = \frac{1}{4}$ we have

$$\begin{aligned}\oint_C \frac{-y^3 dx + xy^2 dy}{(x^2 + y^2)^2} &= \oint_{C'} \frac{-y^3 dx + xy^2 dy}{(x^2 + y^2)^2} \\ &\quad \boxed{x = \frac{1}{4} \cos t, \quad dx = -\frac{1}{4} \sin t dt, \quad y = \frac{1}{4} \sin t, \quad dy = \frac{1}{4} \cos t dt} \\ &= \int_0^{2\pi} \frac{-\frac{1}{64} \sin^3 t (-\frac{1}{4} \sin t dt) + \frac{1}{4} \cos t (\frac{1}{16} \sin^2 t) (\frac{1}{4} \cos t dt)}{1/256} \\ &= \int_0^{2\pi} (\sin^4 t + \sin^2 t \cos^2 t) dt = \int_0^{2\pi} (\sin^4 t + (\sin^2 t - \sin^4 t) dt \\ &= \int_0^{2\pi} \sin^2 t dt = \left(\frac{1}{2} t - \frac{1}{4} \sin 2t \right) \Big|_0^{2\pi} = \pi\end{aligned}$$

26. We first observe that $P_y = [4y^2 - (x+1)^2]/[(x+1)^2 + 4y^2]^2 = Q_x$. Letting C' be the ellipse $(x+1)^2 + 4y^2 = 4$ we have

$$\begin{aligned}\oint_C \frac{-y}{(x+1)^2 + 4y^2} dx + \frac{x+1}{(x+1)^2 + 4y^2} dy &= \oint_{C'} \frac{-y}{(x+1)^2 + 4y^2} dx + \frac{x+1}{(x+1)^2 + 4y^2} dy \\ &\quad \boxed{x+1 = 2 \cos t, \quad dx = -2 \sin t dt, \quad y = \sin t, \quad dy = \cos t dt} \\ &= \int_0^{2\pi} \left[\frac{-\sin t}{4} (-2 \sin t) + \frac{2 \cos t}{4} \cos t \right] dt = \frac{1}{2} \int_0^{2\pi} (\sin^2 t + \cos^2 t) dt = \pi.\end{aligned}$$

27. Writing $\iint_R x^2 dA = \iint_R (Q_x - P_y) dA$ we identify $Q = 0$ and $P = -x^2y$. Then, with C : $x = 3 \cos t$, $y = 2 \sin t$, $0 \leq t \leq 2\pi$, we have

$$\begin{aligned}\iint_R x^2 dA &= \oint_C P dx + Q dy = \oint_C -x^2y dx = - \int_0^{2\pi} 9 \cos^2 t (2 \sin t) (-3 \sin t) dt \\ &= \frac{54}{4} \int_0^{2\pi} 4 \sin^2 t \cos^2 t dt = \frac{27}{2} \int_0^{2\pi} \sin^2 2t dt = \frac{27}{4} \int_0^{2\pi} (1 - \cos 4t) dt \\ &= \frac{27}{4} \left(t - \frac{1}{4} \sin 4t \right) \Big|_0^{2\pi} = \frac{27\pi}{2}.\end{aligned}$$

28. Writing $\iint_R [1 - 2(y-1)] dA = \iint_R (Q_x - P_y) dA$ we identify $Q = x$ and $P = (y-1)^2$. Then, with

C_1 : $x = \cos t$, $y - 1 = \sin t$, $-\pi/2 \leq t \leq \pi/2$, and C_2 : $x = 0$, $2 \geq y \geq 0$,

$$\begin{aligned} \iint_R [1 - 2(y - 1)] dA &= \int_{C_1} P dx + Q dy + \int_{C_2} P dx + Q dy = \int_{C_1} (y - 1)^2 dx + x dy + \int_{C_2} 0 dy \\ &= \int_{-\pi/2}^{\pi/2} [\sin^2 t(-\sin t) + \cos t \cos t] dt = \int_{-\pi/2}^{\pi/2} [\cos^2 t - (1 - \cos^2 t) \sin t] dt \\ &= \int_{-\pi/2}^{\pi/2} \left[\frac{1}{2}(1 + \cos 2t) - \sin t + \cos^2 t \sin t \right] dt \\ &= \left(\frac{1}{2}t + \frac{1}{4} \sin 2t + \cos t - \frac{1}{3} \cos^3 t \right) \Big|_{-\pi/2}^{\pi/2} = \frac{\pi}{4} - \left(-\frac{\pi}{4} \right) = \frac{\pi}{2}. \end{aligned}$$

29. $P = x - y$, $P_y = -1$, $Q = x + y$, $Q_x = 1$; $W = \oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_R 2 dA = 2 \times \text{area} = 2 \left(\frac{3\pi}{4} \right) = \frac{3}{2}\pi$

30. $P = -xy^2$, $P_y = -2xy$, $Q = x^2y$, $Q_x = 2xy$. Using polar coordinates,

$$\begin{aligned} W &= \oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_R 4xy dA = \int_0^{\pi/2} \int_1^2 4(r \cos \theta)(r \sin \theta) r dr d\theta = \int_0^{\pi/2} (r^4 \cos \theta \sin \theta) \Big|_1^2 d\theta \\ &= 15 \int_0^{\pi/2} \sin \theta \cos \theta d\theta = \frac{15}{2} \sin^2 \theta \Big|_0^{\pi/2} = \frac{15}{2}. \end{aligned}$$

31. Since $\int_A^B P dx + Q dy$ is independent of path, $P_y = Q_x$ by Theorem 9.9. Then, by Green's Theorem

$$\oint_C P dx + Q dy = \iint_R (Q_x - P_y) dA = \iint_R 0 dA = 0.$$

32. Let $P = 0$ and $Q = x^2$. Then $Q_x - P_y = 2x$ and

$$\frac{1}{2A} \oint_C x^2 dy = \frac{1}{2A} \iint_R 2x dA = \frac{\iint_R x dA}{A} = \bar{x}.$$

Let $P = y^2$ and $Q = 0$. Then $Q_x - P_y = -2y$ and

$$-\frac{1}{2A} \oint_C y^2 dx = -\frac{1}{2A} \iint_R -2y dA = \frac{\iint_R y dA}{A} = \bar{y}.$$

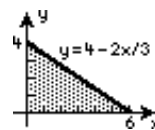
33. Using Green's Theorem,

$$\begin{aligned} W &= \oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C -y dx + x dy = \iint_R 2 dA = 2 \int_0^{2\pi} \int_0^{1+\cos \theta} r dr d\theta \\ &= 2 \int_0^{2\pi} \left(\frac{1}{2} r^2 \right) \Big|_0^{1+\cos \theta} d\theta = \int_0^{2\pi} (1 + 2 \cos \theta + \cos^2 \theta) d\theta \\ &= \left(\theta + 2 \sin \theta + \frac{1}{2} \theta + \frac{1}{4} \sin 2\theta \right) \Big|_0^{2\pi} = 3\pi. \end{aligned}$$

Exercises 9.13

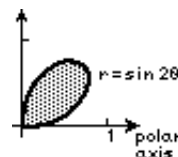
1. Letting $z = 0$, we have $2x + 3y = 12$. Using $f(x, y) = z = 3 - \frac{1}{2}x - \frac{3}{4}y$ we have $f_x = -\frac{1}{2}$, $f_y = -\frac{3}{4}$, $1 + f_x^2 + f_y^2 = \frac{29}{16}$. Then

$$\begin{aligned} A &= \int_0^6 \int_0^{4-2x/3} \sqrt{29/16} \, dy \, dx = \frac{\sqrt{29}}{4} \int_0^6 \left(4 - \frac{2}{3}x\right) dx = \frac{\sqrt{29}}{4} \left(4x - \frac{1}{3}x^2\right) \Big|_0^6 \\ &= \frac{\sqrt{29}}{4}(24 - 12) = 3\sqrt{29}. \end{aligned}$$



2. We see from the graph in Problem 1 that the plane is entirely above the region bounded by $r = \sin 2\theta$ in the first octant. Using $f(x, y) = z = 3 - \frac{1}{2}x - \frac{3}{4}y$ we have $f_x = -\frac{1}{2}$, $f_y = -\frac{3}{4}$, $1 + f_x^2 + f_y^2 = \frac{29}{16}$. Then

$$\begin{aligned} A &= \int_0^{\pi/2} \int_0^{\sin 2\theta} \sqrt{29/16} \, r \, dr \, d\theta = \frac{\sqrt{29}}{4} \int_0^{\pi/2} \frac{1}{2} r^2 \Big|_0^{\sin 2\theta} d\theta = \frac{\sqrt{29}}{8} \int_0^{\pi/2} \sin^2 2\theta \, d\theta \\ &= \frac{\sqrt{29}}{8} \left(\frac{1}{2}\theta - \frac{1}{8}\sin 4\theta \right) \Big|_0^{\pi/2} = \frac{\sqrt{29}\pi}{32}. \end{aligned}$$

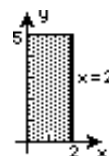


3. Using $f(x, y) = z = \sqrt{16 - x^2}$ we see that for $0 \leq x \leq 2$ and $0 \leq y \leq 5$, $z > 0$.

Thus, the surface is entirely above the region. Now $f_x = -\frac{x}{\sqrt{16 - x^2}}$, $f_y = 0$,

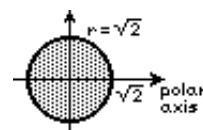
$$1 + f_x^2 + f_y^2 = 1 + \frac{x^2}{16 - x^2} = \frac{16}{16 - x^2} \text{ and}$$

$$A = \int_0^5 \int_0^2 \frac{4}{\sqrt{16 - x^2}} \, dx \, dy = 4 \int_0^5 \sin^{-1} \frac{x}{4} \Big|_0^2 dy = 4 \int_0^5 \frac{\pi}{6} \, dy = \frac{10\pi}{3}.$$



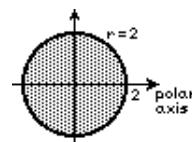
4. The region in the xy -plane beneath the surface is bounded by the graph of $x^2 + y^2 = 2$. Using $f(x, y) = z = x^2 + y^2$ we have $f_x = 2x$, $f_y = 2y$, $1 + f_x^2 + f_y^2 = 1 + 4(x^2 + y^2)$. Then,

$$A = \int_0^{2\pi} \int_0^{\sqrt{2}} \sqrt{1 + 4r^2} \, r \, dr \, d\theta = \int_0^{2\pi} \frac{1}{12} (1 + 4r^2)^{3/2} \Big|_0^{\sqrt{2}} d\theta = \frac{1}{12} \int_0^{2\pi} (27 - 1) d\theta = \frac{13\pi}{3}.$$



5. Letting $z = 0$ we have $x^2 + y^2 = 4$. Using $f(x, y) = z = 4 - (x^2 + y^2)$ we have $f_x = -2x$, $f_y = -2y$, $1 + f_x^2 + f_y^2 = 1 + 4(x^2 + y^2)$. Then

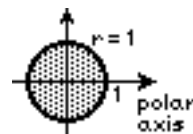
$$\begin{aligned} A &= \int_0^{2\pi} \int_0^2 \sqrt{1 + 4r^2} \, r \, dr \, d\theta = \int_0^{2\pi} \frac{1}{3} (1 + 4r^2)^{3/2} \Big|_0^2 d\theta \\ &= \frac{1}{12} \int_0^{2\pi} (17^{3/2} - 1) d\theta = \frac{\pi}{6} (17^{3/2} - 1). \end{aligned}$$



6. The surfaces $x^2 + y^2 + z^2 = 2$ and $z^2 = x^2 + y^2$ intersect on the cylinder $2x^2 + 2y^2 = 2$ or $x^2 + y^2 = 1$. There are portions of the sphere within the cone both above and below the xy -plane. Using $f(x, y) = \sqrt{2 - x^2 - y^2}$ we have $f_x = -\frac{x}{\sqrt{2 - x^2 - y^2}}$,

$$f_y = -\frac{y}{\sqrt{2 - x^2 - y^2}}, \quad 1 + f_x^2 + f_y^2 = \frac{2}{2 - x^2 - y^2}. \quad \text{Then}$$

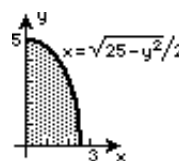
$$\begin{aligned} A &= 2 \left[\int_0^{2\pi} \int_0^1 \frac{\sqrt{2}}{\sqrt{2 - r^2}} r \, dr \, d\theta \right] = 2\sqrt{2} \int_0^{2\pi} -\sqrt{2 - r^2} \Big|_0^1 d\theta \\ &= 2\sqrt{2} \int_0^{2\pi} (\sqrt{2} - 1) d\theta = 4\pi\sqrt{2}(\sqrt{2} - 1). \end{aligned}$$



7. Using $f(x, y) = z = \sqrt{25 - x^2 - y^2}$ we have $f_x = -\frac{x}{\sqrt{25 - x^2 - y^2}}$,

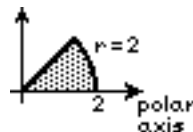
$$f_y = -\frac{y}{\sqrt{25 - x^2 - y^2}}, \quad 1 + f_x^2 + f_y^2 = \frac{25}{25 - x^2 - y^2}. \quad \text{Then}$$

$$\begin{aligned} A &= \int_0^5 \int_0^{\sqrt{25-y^2}/2} \frac{5}{\sqrt{25 - x^2 - y^2}} dx \, dy = 5 \int_0^5 \sin^{-1} \frac{x}{\sqrt{25 - y^2}} \Big|_0^{\sqrt{25-y^2}/2} dy \\ &= 5 \int_0^5 \frac{\pi}{6} dy = \frac{25\pi}{6}. \end{aligned}$$



8. In the first octant, the graph of $z = x^2 - y^2$ intersects the xy -plane in the line $y = x$. The surface is in the first octant for $x > y$. Using $f(x, y) = z = x^2 - y^2$ we have $f_x = 2x$, $f_y = -2y$, $1 + f_x^2 + f_y^2 = 1 + 4x^2 + 4y^2$. Then

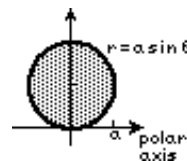
$$\begin{aligned} A &= \int_0^{\pi/4} \int_0^2 \sqrt{1 + 4r^2} r \, dr \, d\theta = \int_0^{\pi/4} \frac{1}{12} (1 + 4r^2)^{3/2} \Big|_0^2 d\theta \\ &= \frac{1}{12} \int_0^{\pi/4} (17^{3/2} - 1) d\theta = \frac{\pi}{48} (17^{3/2} - 1). \end{aligned}$$



9. There are portions of the sphere within the cylinder both above and below the xy -plane. Using $f(x, y) = z = \sqrt{a^2 - x^2 - y^2}$ we have $f_x = -\frac{x}{\sqrt{a^2 - x^2 - y^2}}$, $f_y = -\frac{y}{\sqrt{a^2 - x^2 - y^2}}$,

$$1 + f_x^2 + f_y^2 = \frac{a^2}{a^2 - x^2 - y^2}. \quad \text{Then, using symmetry,}$$

$$\begin{aligned} A &= 2 \left[2 \int_0^{\pi/2} \int_0^{a \sin \theta} \frac{a}{\sqrt{a^2 - r^2}} r \, dr \, d\theta \right] = 4a \int_0^{\pi/2} -\sqrt{a^2 - r^2} \Big|_0^{a \sin \theta} d\theta \\ &= 4a \int_0^{\pi/2} (a - a\sqrt{1 - \sin^2 \theta}) d\theta = 4a^2 \int_0^{\pi/2} (1 - \cos \theta) d\theta \\ &= 4a^2 (\theta - \sin \theta) \Big|_0^{\pi/2} = 4a^2 \left(\frac{\pi}{2} - 1 \right) = 2a^2(\pi - 2). \end{aligned}$$

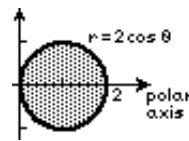


Exercises 9.13

10. There are portions of the cone within the cylinder both above and below the xy -plane. Using $f(x, y) = \frac{1}{2}\sqrt{x^2 + y^2}$, we have $f_x = \frac{x}{2\sqrt{x^2 + y^2}}$, $f_y = \frac{y}{2\sqrt{x^2 + y^2}}$, $1 + f_x^2 + f_y^2 = \frac{5}{4}$.

Then, using symmetry,

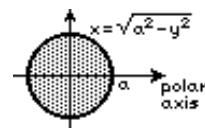
$$\begin{aligned} A &= 2 \left[2 \int_0^{\pi/2} \int_0^{2\cos\theta} \sqrt{\frac{5}{4}} r \, dr \, d\theta \right] = 2\sqrt{5} \int_0^{\pi/2} \frac{1}{2} r^2 \Big|_0^{2\cos\theta} d\theta \\ &= 4\sqrt{5} \int_0^{\pi/2} \cos^2\theta \, d\theta = 4\sqrt{5} \left(\frac{1}{2}\theta + \frac{1}{4}\sin 2\theta \right) \Big|_0^{\pi/2} = \sqrt{5}\pi. \end{aligned}$$



11. There are portions of the surface in each octant with areas equal to the area of the portion in the first octant. Using $f(x, y) = z = \sqrt{a^2 - y^2}$ we have $f_x = 0$, $f_y = \frac{y}{\sqrt{a^2 - y^2}}$,

$$1 + f_x^2 + f_y^2 = \frac{a^2}{a^2 - y^2}. \text{ Then}$$

$$A = 8 \int_0^a \int_0^{\sqrt{a^2 - y^2}} \frac{a}{\sqrt{a^2 - y^2}} \, dx \, dy = 8a \int_0^a \frac{x}{\sqrt{a^2 - y^2}} \Big|_0^{\sqrt{a^2 - y^2}} dy = 8a \int_0^a dy = 8a^2.$$

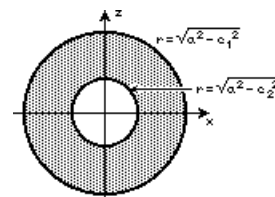


12. From Example 1, the area of the portion of the hemisphere within $x^2 + y^2 = b^2$ is $2\pi a(a - \sqrt{a^2 - b^2})$. Thus, the area of the sphere is $A = 2 \lim_{b \rightarrow a} 2\pi a(a - \sqrt{a^2 - b^2}) = 2(2\pi a^2) = 4\pi a^2$.

13. The projection of the surface onto the xz -plane is shown in the graph. Using $f(x, z) = y = \sqrt{a^2 - x^2 - z^2}$ we have $f_x = -\frac{x}{\sqrt{a^2 - x^2 - z^2}}$,

$$f_z = -\frac{z}{\sqrt{a^2 - x^2 - z^2}}, \quad 1 + f_x^2 + f_z^2 = \frac{a^2}{a^2 - x^2 - z^2}. \text{ Then}$$

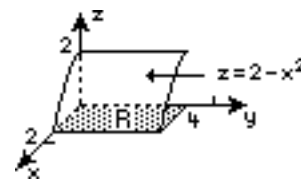
$$\begin{aligned} A &= \int_0^{2\pi} \int_{\sqrt{a^2 - c_2^2}}^{\sqrt{a^2 - c_1^2}} \frac{a}{\sqrt{a^2 - r^2}} r \, dr \, d\theta = a \int_0^{2\pi} -\sqrt{a^2 - r^2} \Big|_{\sqrt{a^2 - c_2^2}}^{\sqrt{a^2 - c_1^2}} d\theta \\ &= a \int_0^{2\pi} (c_2 - c_1) \, d\theta = 2\pi a(c_2 - c_1). \end{aligned}$$



14. The surface area of the cylinder $x^2 + z^2 = a^2$ from $y = c_1$ to $y = c_2$ is the area of a cylinder of radius a and height $c_2 - c_1$. This is $2\pi a(c_2 - c_1)$.

15. $z_x = -2x$, $z_y = 0$; $dS = \sqrt{1 + 4x^2} \, dA$

$$\begin{aligned} \iint_S x \, dS &= \int_0^4 \int_0^{\sqrt{2}} x \sqrt{1 + 4x^2} \, dx \, dy = \int_0^4 \frac{1}{12} (1 + 4x^2)^{3/2} \Big|_0^{\sqrt{2}} dy \\ &= \int_0^4 \frac{13}{6} \, dy = \frac{26}{3} \end{aligned}$$



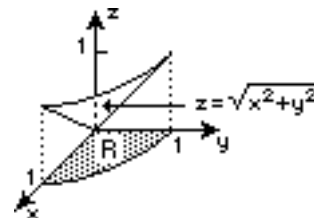
16. See Problem 15.

$$\begin{aligned} \iint_S xy(9 - 4z) \, dS &= \iint_S xy(1 + 4x^2) \, dS = \int_0^4 \int_0^{\sqrt{2}} xy(1 + 4x^2)^{3/2} \, dx \, dy \\ &= \int_0^4 \frac{y}{20} (1 + 4x^2)^{5/2} \Big|_0^{\sqrt{2}} dy = \int_0^4 \frac{242}{20} y \, dy = \frac{121}{10} \int_0^4 y \, dy = \frac{121}{10} \left(\frac{1}{2} y^2 \right) \Big|_0^4 = \frac{484}{5} \end{aligned}$$

17. $z_x = \frac{x}{\sqrt{x^2 + y^2}}, z_y = \frac{y}{\sqrt{x^2 + y^2}}; dS = \sqrt{2} dA.$

Using polar coordinates,

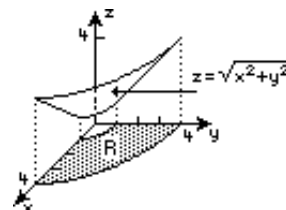
$$\begin{aligned}\iint_S xz^3 dS &= \iint_R x(x^2 + y^2)^{3/2} \sqrt{2} dA = \sqrt{2} \int_0^{2\pi} \int_0^1 (r \cos \theta) r^{3/2} r dr d\theta \\ &= \sqrt{2} \int_0^{2\pi} \int_0^1 r^{7/2} \cos \theta dr d\theta = \sqrt{2} \int_0^{2\pi} \frac{2}{9} r^{9/2} \cos \theta \Big|_0^1 d\theta \\ &= \sqrt{2} \int_0^{2\pi} \frac{2}{9} \cos \theta d\theta = \frac{2\sqrt{2}}{9} \sin \theta \Big|_0^{2\pi} = 0.\end{aligned}$$



18. $z_x = \frac{x}{\sqrt{x^2 + y^2}}, z_y = \frac{y}{\sqrt{x^2 + y^2}}; dS = \sqrt{2} dA.$

Using polar coordinates,

$$\begin{aligned}\iint_S (x + y + z) dS &= \iint_R (x + y + \sqrt{x^2 + y^2}) \sqrt{2} dA \\ &= \sqrt{2} \int_0^{2\pi} \int_1^4 (r \cos \theta + r \sin \theta + r) r dr d\theta \\ &= \sqrt{2} \int_0^{2\pi} \int_1^4 r^2 (1 + \cos \theta + \sin \theta) dr d\theta = \sqrt{2} \int_0^{2\pi} \frac{1}{3} r^3 (1 + \cos \theta + \sin \theta) \Big|_1^4 d\theta \\ &= \frac{63\sqrt{2}}{3} \int_0^{2\pi} (1 + \cos \theta + \sin \theta) d\theta = 21\sqrt{2} (\theta + \sin \theta - \cos \theta) \Big|_0^{2\pi} = 42\sqrt{2} \pi.\end{aligned}$$

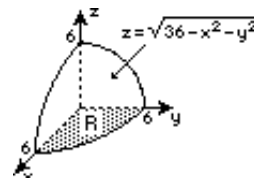


19. $z = \sqrt{36 - x^2 - y^2}, z_x = -\frac{x}{\sqrt{36 - x^2 - y^2}}, z_y = -\frac{y}{\sqrt{36 - x^2 - y^2}};$

$$dS = \sqrt{1 + \frac{x^2}{36 - x^2 - y^2} + \frac{y^2}{36 - x^2 - y^2}} dA = \frac{6}{\sqrt{36 - x^2 - y^2}} dA.$$

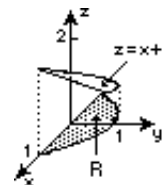
Using polar coordinates,

$$\begin{aligned}\iint_S (x^2 + y^2) z dS &= \iint_R (x^2 + y^2) \sqrt{36 - x^2 - y^2} \frac{6}{\sqrt{36 - x^2 - y^2}} dA \\ &= 6 \int_0^{2\pi} \int_0^6 r^2 r dr d\theta = 6 \int_0^{2\pi} \frac{1}{4} r^4 \Big|_0^6 d\theta = 6 \int_0^{2\pi} 324 d\theta = 972\pi.\end{aligned}$$



20. $z_x = 1, z_y = 0; dS = \sqrt{2} dA$

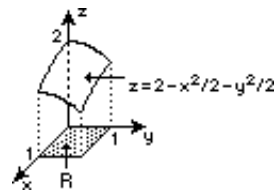
$$\begin{aligned}\iint_S z^2 dS &= \int_{-1}^1 \int_0^{1-x^2} (x+1)^2 \sqrt{2} dy dx = \sqrt{2} \int_{-1}^1 y(x+1)^2 \Big|_0^{1-x^2} dx \\ &= \sqrt{2} \int_{-1}^1 (1-x^2)(x+1)^2 dx = \sqrt{2} \int_{-1}^1 (1+2x-2x^3-x^4) dx \\ &= \sqrt{2} \left(x + x^2 - \frac{1}{2} x^4 - \frac{1}{5} x^5 \right) \Big|_{-1}^1 = \frac{8\sqrt{2}}{5}\end{aligned}$$



Exercises 9.13

21. $z_x = -x$, $z_y = -y$; $dS = \sqrt{1+x^2+y^2} dA$

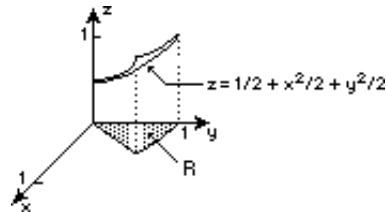
$$\begin{aligned}\iint_S xy dS &= \int_0^1 \int_0^1 xy \sqrt{1+x^2+y^2} dx dy = \int_0^1 \frac{1}{3} y(1+x^2+y^2)^{3/2} \Big|_0^1 dy \\ &= \int_0^1 \left[\frac{1}{3} y(2+y^2)^{3/2} - \frac{1}{3} y(1+y^2)^{3/2} \right] dy \\ &= \left[\frac{1}{15} (2+y^2)^{5/2} - \frac{1}{15} (1+y^2)^{5/2} \right] \Big|_0^1 = \frac{1}{15} (3^{5/2} - 2^{7/2} + 1)\end{aligned}$$



22. $z = \frac{1}{2} + \frac{1}{2}x^2 + \frac{1}{2}y^2$, $z_x = x$, $z_y = y$; $dS = \sqrt{1+x^2+y^2} dA$.

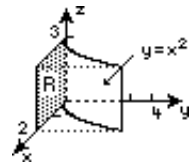
Using polar coordinates,

$$\begin{aligned}\iint_S 2z dS &= \iint_R (1+x^2+y^2) \sqrt{1+x^2+y^2} dA \\ &= \int_{\pi/3}^{\pi/2} \int_0^1 (1+r^2) \sqrt{1+r^2} r dr d\theta \\ &= \int_{\pi/3}^{\pi/2} \int_0^1 (1+r^2)^{3/2} r dr d\theta = \int_{\pi/3}^{\pi/2} \frac{1}{5} (1+r^2)^{5/2} \Big|_0^1 d\theta = \frac{1}{5} \int_{\pi/3}^{\pi/2} (2^{5/2} - 1) d\theta \\ &= \frac{4\sqrt{2}-1}{5} \left(\frac{\pi}{2} - \frac{\pi}{3} \right) = \frac{(4\sqrt{2}-1)\pi}{30}.\end{aligned}$$



23. $y_x = 2x$, $y_z = 0$; $dS = \sqrt{1+4x^2} dA$

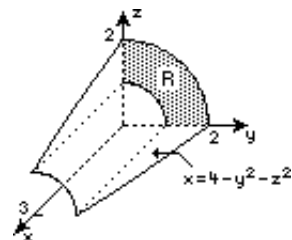
$$\begin{aligned}\iint_S 24\sqrt{y} z dS &= \int_0^3 \int_0^2 24xz \sqrt{1+4x^2} dx dz = \int_0^3 2z(1+4x^2)^{3/2} \Big|_0^2 dz \\ &= 2(17^{3/2} - 1) \int_0^3 z dz = 2(17^{3/2} - 1) \left(\frac{1}{2} z^2 \right) \Big|_0^3 = 9(17^{3/2} - 1)\end{aligned}$$



24. $x_y = -2y$, $x_z = -2z$; $dS = \sqrt{1+4y^2+4z^2} dA$.

Using polar coordinates,

$$\begin{aligned}\iint_S (1+4y^2+4z^2)^{1/2} dS &= \int_0^{\pi/2} \int_1^2 (1+4r^2) r dr d\theta \\ &= \int_0^{\pi/2} \frac{1}{16} (1+4r^2)^2 \Big|_1^2 d\theta = \frac{1}{16} \int_0^{\pi/2} 12 d\theta = \frac{3\pi}{8}.\end{aligned}$$



25. Write the equation of the surface as $y = \frac{1}{2}(6-x-3z)$.

$$y_x = -\frac{1}{2}, y_z = -\frac{3}{2}; dS = \sqrt{1+1/4+9/4} = \frac{\sqrt{14}}{2}.$$

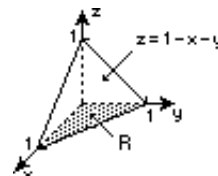
$$\begin{aligned}\iint_S (3z^2 + 4yz) dS &= \int_0^2 \int_0^{6-3z} \left[3z^2 + 4z \frac{1}{2}(6-x-3z) \right] \frac{\sqrt{14}}{2} dx dz \\ &= \frac{\sqrt{14}}{2} \int_0^2 [3z^2 x - z(6-x-3z)^2] \Big|_0^{6-3z} dz \\ &= \frac{\sqrt{14}}{2} \int_0^2 ([3z^2(6-3z) - 0] - [0 - z(6-3z)^2]) dz \\ &= \frac{\sqrt{14}}{2} \int_0^2 (36z - 18z^2) dz = \frac{\sqrt{14}}{2} (18z^2 - 6z^3) \Big|_0^2 = \frac{\sqrt{14}}{2} (72 - 48) = 12\sqrt{14}\end{aligned}$$

26. Write the equation of the surface as $x = 6 - 2y - 3z$. Then $x_y = -2$, $x_z = -3$; $dS = \sqrt{1 + 4 + 9} = \sqrt{14}$.

$$\begin{aligned}\iint_S (3z^2 + 4yz) dS &= \int_0^2 \int_0^{3-3z/2} (3z^2 + 4yz) \sqrt{14} dy dz = \sqrt{14} \int_0^2 (3yz + 2y^2 z) \Big|_0^{3-3z/2} dz \\ &= \sqrt{14} \int_0^2 \left[9z \left(1 - \frac{z}{2} \right) + 18z \left(1 - \frac{z}{2} \right)^2 \right] dz = \sqrt{14} \int_0^2 \left(27z - \frac{45}{2} z^2 + \frac{9}{2} z^3 \right) dz \\ &= \sqrt{14} \left(\frac{27}{2} z^2 - \frac{15}{2} z^3 + \frac{9}{8} z^4 \right) \Big|_0^2 = \sqrt{14} (54 - 60 + 18) = 2\sqrt{14}\end{aligned}$$

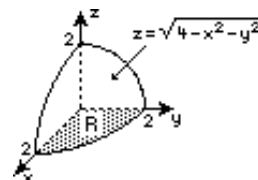
27. The density is $\rho = kx^2$. The surface is $z = 1 - x - y$. Then $z_x = -1$, $z_y = -1$; $dS = \sqrt{3} dA$.

$$\begin{aligned}m &= \iint_S kx^2 dS = k \int_0^1 \int_0^{1-x} x^2 \sqrt{3} dy dx = \sqrt{3} k \int_0^1 \frac{1}{3} x^3 \Big|_0^{1-x} dx \\ &= \frac{\sqrt{3}}{3} k \int_0^1 (1-x)^3 dx = \frac{\sqrt{3}}{3} k \left[-\frac{1}{4} (1-x)^4 \right] \Big|_0^1 = \frac{\sqrt{3}}{12} k\end{aligned}$$



28. $z_x = -\frac{x}{\sqrt{4-x^2-y^2}}$, $z_y = -\frac{y}{\sqrt{4-x^2-y^2}}$;

$$dS = \sqrt{1 + \frac{x^2}{4-x^2-y^2} + \frac{y^2}{4-x^2-y^2}} dA = \frac{2}{\sqrt{4-x^2-y^2}} dA.$$



Using symmetry and polar coordinates,

$$\begin{aligned}m &= 4 \iint_S |xy| dS = 4 \int_0^{\pi/2} \int_0^2 (r^2 \cos \theta \sin \theta) \frac{2}{\sqrt{4-r^2}} r dr d\theta \\ &= 4 \int_0^{\pi/2} \int_0^2 r^2 (4-r^2)^{-1/2} \sin 2\theta (r dr) d\theta \quad \boxed{u = 4 - r^2, du = -2r dr, r^2 = 4 - u} \\ &= 4 \int_0^{\pi/2} \int_4^0 (4-u) u^{-1/2} \sin 2\theta \left(-\frac{1}{2} du \right) d\theta = -2 \int_0^{\pi/2} \int_4^0 (4u^{-1/2} - u^{1/2}) \sin 2\theta du d\theta \\ &= -2 \int_0^{\pi/2} \left(8u^{1/2} - \frac{2}{3} u^{3/2} \right) \Big|_4^0 \sin 2\theta d\theta = -2 \int_0^{\pi/2} \left(-\frac{32}{3} \sin 2\theta \right) d\theta = \frac{64}{3} \left(-\frac{1}{2} \cos 2\theta \right) \Big|_0^{\pi/2} = \frac{64}{3}.\end{aligned}$$

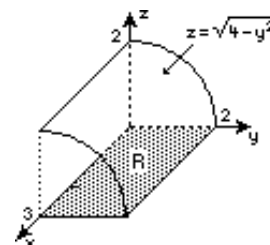
29. The surface is $g(x, y, z) = y^2 + z^2 - 4 = 0$. $\nabla g = 2y\mathbf{j} + 2z\mathbf{k}$,

$$|\nabla g| = 2\sqrt{y^2 + z^2}; \quad \mathbf{n} = \frac{y\mathbf{j} + z\mathbf{k}}{\sqrt{y^2 + z^2}};$$

$$\mathbf{F} \cdot \mathbf{n} = \frac{2yz}{\sqrt{y^2 + z^2}} + \frac{yz}{\sqrt{y^2 + z^2}} = \frac{3yz}{\sqrt{y^2 + z^2}}; \quad z = \sqrt{4 - y^2}, \quad z_x = 0,$$

$$z_y = -\frac{y}{\sqrt{4 - y^2}}; \quad dS = \sqrt{1 + \frac{y^2}{4 - y^2}} dA = \frac{2}{\sqrt{4 - y^2}} dA$$

$$\begin{aligned}\text{Flux} &= \iint_S \mathbf{F} \cdot \mathbf{n} dS = \iint_R \frac{3yz}{\sqrt{y^2 + z^2}} \frac{2}{\sqrt{4 - y^2}} dA = \iint_R \frac{3y\sqrt{4 - y^2}}{\sqrt{y^2 + 4 - y^2}} \frac{2}{\sqrt{4 - y^2}} dA \\ &= \int_0^3 \int_0^2 3y dy dx = \int_0^3 \frac{3}{2} y^2 \Big|_0^2 dx = \int_0^3 6 dx = 18\end{aligned}$$



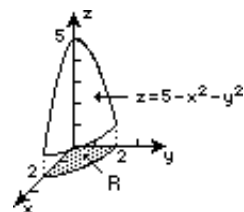
Exercises 9.13

30. The surface is $g(x, y, z) = x^2 + y^2 + z - 5 = 0$. $\nabla g = 2x\mathbf{i} + 2y\mathbf{j} + \mathbf{k}$,

$$|\nabla g| = \sqrt{1 + 4x^2 + 4y^2}; \quad \mathbf{n} = \frac{2x\mathbf{i} + 2y\mathbf{j} + \mathbf{k}}{\sqrt{1 + 4x^2 + 4y^2}}; \quad \mathbf{F} \cdot \mathbf{n} = \frac{z}{\sqrt{1 + 4x^2 + 4y^2}};$$

$z_x = -2x$, $z_y = -2y$, $dS = \sqrt{1 + 4x^2 + 4y^2} dA$. Using polar coordinates,

$$\begin{aligned} \text{Flux} &= \iint_S \mathbf{F} \cdot \mathbf{n} dS = \iint_R \frac{z}{\sqrt{1 + 4x^2 + 4y^2}} \sqrt{1 + 4x^2 + 4y^2} dA = \iint_R (5 - x^2 - y^2) dA \\ &= \int_0^{2\pi} \int_0^2 (5 - r^2) r dr d\theta = \int_0^{2\pi} \left(\frac{5}{2} r^2 - \frac{1}{4} r^4 \right) \Big|_0^2 d\theta = \int_0^{2\pi} 6 d\theta = 12\pi. \end{aligned}$$



31. From Problem 30, $\mathbf{n} = \frac{2x\mathbf{i} + 2y\mathbf{j} + \mathbf{k}}{\sqrt{1 + 4x^2 + 4y^2}}$. Then $\mathbf{F} \cdot \mathbf{n} = \frac{2x^2 + 2y^2 + z}{\sqrt{1 + 4x^2 + 4y^2}}$. Also, from Problem 30,

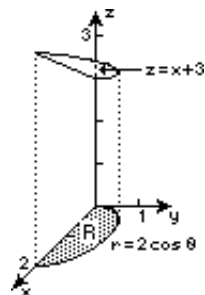
$dS = \sqrt{1 + 4x^2 + 4y^2} dA$. Using polar coordinates,

$$\begin{aligned} \text{Flux} &= \iint_S \mathbf{F} \cdot \mathbf{n} dS = \iint_R \frac{2x^2 + 2y^2 + z}{\sqrt{1 + 4x^2 + 4y^2}} \sqrt{1 + 4x^2 + 4y^2} dA = \iint_R (2x^2 + 2y^2 + 5 - x^2 - y^2) dA \\ &= \int_0^{2\pi} \int_0^2 (r^2 + 5) r dr d\theta = \int_0^{2\pi} \left(\frac{1}{4} r^4 + \frac{5}{2} r^2 \right) \Big|_0^2 d\theta = \int_0^{2\pi} 14 d\theta = 28\pi. \end{aligned}$$

32. The surface is $g(x, y, z) = z - x - 3 = 0$. $\nabla g = -\mathbf{i} + \mathbf{k}$, $|\nabla g| = \sqrt{2}$; $\mathbf{n} = \frac{-\mathbf{i} + \mathbf{k}}{\sqrt{2}}$;

$\mathbf{F} \cdot \mathbf{n} = \frac{1}{\sqrt{2}} x^3 y + \frac{1}{\sqrt{2}} x y^3$; $z_x = 1$, $z_y = 0$, $dS = \sqrt{2} dA$. Using polar coordinates,

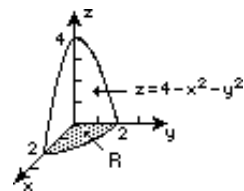
$$\begin{aligned} \text{Flux} &= \iint_S \mathbf{F} \cdot \mathbf{n} dS = \iint_R \frac{1}{\sqrt{2}} (x^3 y + x y^3) \sqrt{2} dA = \iint_R x y (x^2 + y^2) dA \\ &= \int_0^{\pi/2} \int_0^{2 \cos \theta} (r^2 \cos \theta \sin \theta) r^2 r dr d\theta = \int_0^{\pi/2} \int_0^{2 \cos \theta} r^5 \cos \theta \sin \theta dr d\theta \\ &= \int_0^{\pi/2} \frac{1}{6} r^6 \cos \theta \sin \theta \Big|_0^{2 \cos \theta} d\theta = \frac{1}{6} \int_0^{\pi/2} 64 \cos^7 \theta \sin \theta d\theta = \frac{32}{3} \left(-\frac{1}{8} \cos^8 \theta \right) \Big|_0^{\pi/2} = \frac{4}{3}. \end{aligned}$$



33. The surface is $g(x, y, z) = x^2 + y^2 + z - 4 = 0$. $\nabla g = 2x\mathbf{i} + 2y\mathbf{j} + \mathbf{k}$,

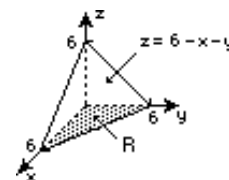
$$|\nabla g| = \sqrt{4x^2 + 4y^2 + 1}; \quad \mathbf{n} = \frac{2x\mathbf{i} + 2y\mathbf{j} + \mathbf{k}}{\sqrt{4x^2 + 4y^2 + 1}}; \quad \mathbf{F} \cdot \mathbf{n} = \frac{x^3 + y^3 + z}{\sqrt{4x^2 + 4y^2 + 1}};$$

$z_x = -2x$, $z_y = -2y$, $dS = \sqrt{1 + 4x^2 + 4y^2} dA$. Using polar coordinates,



$$\begin{aligned} \text{Flux} &= \iint_S \mathbf{F} \cdot \mathbf{n} dS = \iint_R (x^3 + y^3 + z) dA = \iint_R (4 - x^2 - y^2 + x^3 + y^3) dA \\ &= \int_0^{2\pi} \int_0^2 (4 - r^2 + r^3 \cos^3 \theta + r^3 \sin^3 \theta) r dr d\theta \\ &= \int_0^{2\pi} \left(2r^2 - \frac{1}{4} r^4 + \frac{1}{5} r^5 \cos^3 \theta + \frac{1}{5} r^5 \sin^3 \theta \right) \Big|_0^2 d\theta \\ &= \int_0^{2\pi} \left(4 + \frac{32}{5} \cos^3 \theta + \frac{32}{5} \sin^3 \theta \right) d\theta = 4\theta \Big|_0^{2\pi} + 0 + 0 = 8\pi. \end{aligned}$$

34. The surface is $g(x, y, z) = x + y + z - 6$. $\nabla g = \mathbf{i} + \mathbf{j} + \mathbf{k}$, $|\nabla g| = \sqrt{3}$; $\mathbf{n} = (\mathbf{i} + \mathbf{j} + \mathbf{k})/\sqrt{3}$;
 $\mathbf{F} \cdot \mathbf{n} = (e^y + e^x + 18y)/\sqrt{3}$; $z_x = -1$, $z_y = -1$, $dS = \sqrt{1+1+1} dA = \sqrt{3} dA$.



$$\begin{aligned} \text{Flux} &= \iint_S \mathbf{F} \cdot \mathbf{n} dS = \iint_r (e^y + e^x + 18y) dA = \int_0^6 \int_0^{6-x} (e^y + e^x + 18y) dy dx \\ &= \int_0^6 (e^y + ye^x + 9y^2) \Big|_0^{6-x} dx = \int_0^6 [e^{6-x} + (6-x)e^x + 9(6-x)^2 - 1] dx \\ &= [-e^{6-x} + 6e^x - xe^x + e^x - 3(6-x)^3 - x] \Big|_0^6 \\ &= (-1 + 6e^6 - 6e^6 + e^6 - 6) - (-e^6 + 6 + 1 - 648) = 2e^6 + 634 \approx 1440.86 \end{aligned}$$

35. For S_1 : $g(x, y, z) = x^2 + y^2 - z$, $\nabla g = 2x\mathbf{i} + 2y\mathbf{j} - \mathbf{k}$, $|\nabla g| = \sqrt{4x^2 + 4y^2 + 1}$; $\mathbf{n}_1 = \frac{2x\mathbf{i} + 2y\mathbf{j} - \mathbf{k}}{\sqrt{4x^2 + 4y^2 + 1}}$;

$$\mathbf{F} \cdot \mathbf{n}_1 = \frac{2xy^2 + 2x^2y - 5z}{\sqrt{4x^2 + 4y^2 + 1}}; \quad z_x = 2x, \quad z_y = 2y, \quad dS_1 = \sqrt{1 + 4x^2 + 4y^2} dA. \quad \text{For } S_2: g(x, y, z) = z - 1,$$

$\nabla g = \mathbf{k}$, $|\nabla g| = 1$; $\mathbf{n}_2 = \mathbf{k}$; $\mathbf{F} \cdot \mathbf{n}_2 = 5z$; $z_x = 0$, $z_y = 0$, $dS_2 = dA$. Using polar coordinates and $R: x^2 + y^2 \leq 1$ we have

$$\begin{aligned} \text{Flux} &= \iint_{S_1} \mathbf{F} \cdot \mathbf{n}_1 dS_1 + \iint_{S_2} \mathbf{F} \cdot \mathbf{n}_2 dS_2 = \iint_R (2xy^2 + 2x^2y - 5z) dA + \iint_R 5z dA \\ &= \iint_R [2xy^2 + 2x^2y - 5(x^2 + y^2) + 5(1)] dA \\ &= \int_0^{2\pi} \int_0^1 (2r^3 \cos \theta \sin^2 \theta + 2r^3 \cos^2 \theta \sin \theta - 5r^2 + 5)r dr d\theta \\ &= \int_0^{2\pi} \left(\frac{2}{5} r^5 \cos \theta \sin^2 \theta + \frac{2}{5} r^5 \cos^2 \theta \sin \theta - \frac{5}{4} r^4 + \frac{5}{2} r^2 \right) \Big|_0^1 d\theta \\ &= \int_0^{2\pi} \left[\frac{2}{5} (\cos \theta \sin^2 \theta + \cos^2 \theta \sin \theta) + \frac{5}{4} \right] d\theta = \frac{2}{5} \left(\frac{1}{3} \sin^3 \theta - \frac{1}{3} \cos^3 \theta \right) \Big|_0^{2\pi} + \frac{5}{4} \theta \Big|_0^{2\pi} \\ &= \frac{2}{5} \left[-\frac{1}{3} - \left(-\frac{1}{3} \right) \right] + \frac{5}{2} \pi = \frac{5}{2} \pi. \end{aligned}$$

36. For S_1 : $g(x, y, z) = x^2 + y^2 + z - 4$, $\nabla g = 2x\mathbf{i} + 2y\mathbf{j} + \mathbf{k}$, $|\nabla g| = \sqrt{4x^2 + 4y^2 + 1}$;

$$\mathbf{n}_1 = \frac{2x\mathbf{i} + 2y\mathbf{j} + \mathbf{k}}{\sqrt{4x^2 + 4y^2 + 1}}; \quad \mathbf{F} \cdot \mathbf{n}_1 = 6z^2/\sqrt{4x^2 + 4y^2 + 1}; \quad z_x = -2x, \quad z_y = -2y,$$

$$dS_1 = \sqrt{1 + 4x^2 + 4y^2} dA. \quad \text{For } S_2: g(x, y, z) = x^2 + y^2 - z, \quad \nabla g = 2x\mathbf{i} + 2y\mathbf{j} - \mathbf{k},$$

$$|\nabla g| = \sqrt{4x^2 + 4y^2 + 1}; \quad \mathbf{n}_2 = \frac{2x\mathbf{i} + 2y\mathbf{j} - \mathbf{k}}{\sqrt{4x^2 + 4y^2 + 1}}; \quad \mathbf{F} \cdot \mathbf{n}_2 = -6z^2/\sqrt{4x^2 + 4y^2 + 1}; \quad z_x = 2x, \quad z_y = 2y,$$

$dS_2 = \sqrt{1 + 4x^2 + 4y^2} dA$. Using polar coordinates and $R: x^2 + y^2 \leq 2$ we have

$$\begin{aligned} \text{Flux} &= \iint_{S_1} \mathbf{F} \cdot \mathbf{n}_1 dS_1 + \iint_{S_2} \mathbf{F} \cdot \mathbf{n}_2 dS_2 = \iint_R 6z^2 dA + \iint_R -6z^2 dA \\ &= \iint_R [6(4 - x^2 - y^2)^2 - 6(x^2 + y^2)^2] dA = 6 \int_0^{2\pi} \int_0^{\sqrt{2}} [(4 - r^2)^2 - r^4] r dr d\theta \\ &= 6 \int_0^{2\pi} \left[-\frac{1}{6} (4 - r^2)^3 - \frac{1}{6} r^6 \right] \Big|_0^{\sqrt{2}} d\theta = - \int_0^{2\pi} [(2^3 - 4^3) + (\sqrt{2})^6] d\theta = \int_0^{2\pi} 48 d\theta = 96\pi. \end{aligned}$$

Exercises 9.13

37. The surface is $g(x, y, z) = x^2 + y^2 + z^2 - a^2 = 0$. $\nabla g = 2x\mathbf{i} + 2y\mathbf{j} + 2z\mathbf{k}$,

$$|\nabla g| = 2\sqrt{x^2 + y^2 + z^2}; \quad \mathbf{n} = \frac{x\mathbf{i} + y\mathbf{j} + z\mathbf{k}}{\sqrt{x^2 + y^2 + z^2}};$$

$$\mathbf{F} \cdot \mathbf{n} = -(2x\mathbf{i} + 2y\mathbf{j} + 2z\mathbf{k}) \cdot \frac{x\mathbf{i} + y\mathbf{j} + z\mathbf{k}}{\sqrt{x^2 + y^2 + z^2}} = -\frac{2x^2 + 2y^2 + 2z^2}{\sqrt{x^2 + y^2 + z^2}} = -2\sqrt{x^2 + y^2 + z^2} = -2a.$$

$$\text{Flux} = \iint_S -2a \, dS = -2a \times \text{area} = -2a(4\pi a^2) = -8\pi a^3$$

38. $\mathbf{n}_1 = \mathbf{k}$, $\mathbf{n}_2 = -\mathbf{i}$, $\mathbf{n}_3 = \mathbf{j}$, $\mathbf{n}_4 = -\mathbf{k}$, $\mathbf{n}_5 = \mathbf{i}$, $\mathbf{n}_6 = -\mathbf{j}$; $\mathbf{F} \cdot \mathbf{n}_1 = z = 1$, $\mathbf{F} \cdot \mathbf{n}_2 = -x = 0$, $\mathbf{F} \cdot \mathbf{n}_3 = y = 1$,

$$\mathbf{F} \cdot \mathbf{n}_4 = -z = 0, \mathbf{F} \cdot \mathbf{n}_5 = x = 1, \mathbf{F} \cdot \mathbf{n}_6 = -y = 0; \quad \text{Flux} = \iint_{S_1} 1 \, dS + \iint_{S_3} 1 \, dS + \iint_{S_5} 1 \, dS = 3$$

39. Referring to the solution to Problem 37, we find $\mathbf{n} = \frac{x\mathbf{i} + y\mathbf{j} + z\mathbf{k}}{\sqrt{x^2 + y^2 + z^2}}$ and $dS = \frac{a}{\sqrt{a^2 - x^2 - y^2}} \, dA$.

Now

$$\mathbf{F} \cdot \mathbf{n} = kq \frac{\mathbf{r}}{|\mathbf{r}|^3} \cdot \frac{\mathbf{r}}{|\mathbf{r}|} = \frac{kq}{|\mathbf{r}|^4} |\mathbf{r}|^2 = \frac{kq}{|\mathbf{r}|^2} = \frac{kq}{x^2 + y^2 + z^2} = \frac{kq}{a^2}$$

and

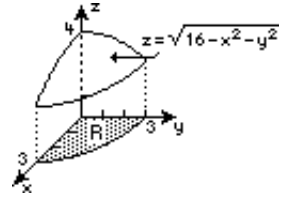
$$\text{Flux} = \iint_S \mathbf{F} \cdot \mathbf{n} \, dS = \iint_S \frac{kq}{a^2} \, dS = \frac{kq}{a^2} \times \text{area} = \frac{kq}{a^2} (4\pi a^2) = 4\pi kq.$$

40. We are given $\sigma = kz$. Now $z_x = \frac{x}{\sqrt{16 - x^2 - y^2}}$, $z_y = -\frac{y}{\sqrt{16 - x^2 - y^2}}$;

$$dS = \sqrt{1 + \frac{x^2}{16 - x^2 - y^2} + \frac{y^2}{16 - x^2 - y^2}} \, dA = \frac{4}{\sqrt{16 - x^2 - y^2}} \, dA$$

Using polar coordinates,

$$\begin{aligned} Q &= \iint_S kz \, dS = k \iint_R \sqrt{16 - x^2 - y^2} \frac{4}{\sqrt{16 - x^2 - y^2}} \, dA = 4k \int_0^{2\pi} \int_0^3 r \, dr \, d\theta \\ &= 4k \int_0^{2\pi} \frac{1}{2} r^2 \Big|_0^3 \, d\theta = 4k \int_0^{2\pi} \frac{9}{2} \, d\theta = 36\pi k. \end{aligned}$$



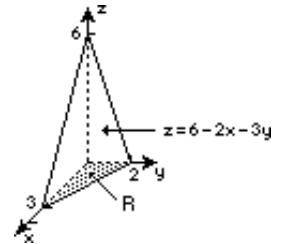
41. The surface is $z = 6 - 2x - 3y$. Then $z_x = -2$, $z_y = -3$, $dS = \sqrt{1 + 4 + 9} = \sqrt{14} \, dA$.

The area of the surface is

$$\begin{aligned} A(s) &= \iint_S dS = \int_0^3 \int_0^{2-2x/3} \sqrt{14} \, dy \, dx = \sqrt{14} \int_0^3 \left(2 - \frac{2}{3}x\right) \, dx \\ &= \sqrt{14} \left(2x - \frac{1}{3}x^2\right) \Big|_0^3 = 3\sqrt{14}. \end{aligned}$$

$$\begin{aligned} \bar{x} &= \frac{1}{3\sqrt{14}} \iint_S x \, dS = \frac{1}{3\sqrt{14}} \int_0^3 \int_0^{2-2x/3} \sqrt{14} x \, dy \, dx = \frac{1}{3} \int_0^3 xy \Big|_0^{2-2x/3} \, dx \\ &= \frac{1}{3} \int_0^3 \left(2x - \frac{2}{3}x^2\right) \, dx = \frac{1}{3} \left(x^2 - \frac{2}{9}x^3\right) \Big|_0^3 = 1 \end{aligned}$$

$$\begin{aligned} \bar{y} &= \frac{1}{3\sqrt{14}} \iint_S y \, dS = \frac{1}{3\sqrt{14}} \int_0^3 \int_0^{2-2x/3} \sqrt{14} y \, dy \, dx = \frac{1}{3} \int_0^3 \frac{1}{2} y^2 \Big|_0^{2-2x/3} \, dx \\ &= \frac{1}{6} \int_0^3 \left(2 - \frac{2}{3}x\right)^2 \, dx = \frac{1}{6} \left[-\frac{1}{2} \left(2 - \frac{2}{3}x\right)^3\right] \Big|_0^3 = \frac{2}{3} \end{aligned}$$



$$\begin{aligned}\bar{z} &= \frac{1}{3\sqrt{14}} \iint_S z \, dS = \frac{1}{3\sqrt{14}} \int_0^3 \int_0^{2-2x/3} (6-2x-3y)\sqrt{14} \, dy \, dx \\ &= \frac{1}{3} \int_0^3 \left(6y - 2xy - \frac{3}{2}y^2 \right) \Big|_0^{2-2x/3} dx = \frac{1}{3} \int_0^3 \left(6 - 4x + \frac{2}{3}x^2 \right) dx = \frac{1}{3} \left(6x - 2x^2 + \frac{2}{9}x^3 \right) \Big|_0^3 = 2\end{aligned}$$

The centroid is $(1, 2/3, 2)$.

42. The area of the hemisphere is $A(s) = 2\pi a^2$. By symmetry, $\bar{x} = \bar{y} = 0$.

$$z_x = -\frac{x}{\sqrt{a^2 - x^2 - y^2}}, \quad z_y = -\frac{y}{\sqrt{a^2 - x^2 - y^2}};$$

$$dS = \sqrt{1 + \frac{x^2}{a^2 - x^2 - y^2} + \frac{y^2}{a^2 - x^2 - y^2}} \, dA = \frac{a}{\sqrt{a^2 - x^2 - y^2}} \, dA$$

Using polar coordinates,

$$\begin{aligned}z &= \iint_S \frac{z \, dS}{2\pi a^2} = \frac{1}{2\pi a^2} \iint_R \sqrt{a^2 - x^2 - y^2} \frac{a}{\sqrt{a^2 - x^2 - y^2}} \, dA = \frac{1}{2\pi a} \int_0^{2\pi} \int_0^a r \, dr \, d\theta \\ &= \frac{1}{2\pi a} \int_0^{2\pi} \frac{1}{2} r^2 \Big|_0^a d\theta = \frac{1}{2\pi a} \int_0^{2\pi} \frac{1}{2} a^2 d\theta = \frac{a}{2}.\end{aligned}$$

The centroid is $(0, 0, a/2)$.

43. The surface is $g(x, y, z) = z - f(x, y) = 0$. $\nabla g = -f_x \mathbf{i} - f_y \mathbf{j} + \mathbf{k}$, $|\nabla g| = \sqrt{f_x^2 + f_y^2 + 1}$;

$$\mathbf{n} = \frac{-f_x \mathbf{i} - f_y \mathbf{j} + \mathbf{k}}{\sqrt{1 + f_x^2 + f_y^2}}; \quad \mathbf{F} \cdot \mathbf{n} = \frac{-Pf_x - Qf_y + R}{\sqrt{1 + f_x^2 + f_y^2}}; \quad dS = \sqrt{1 + f_x^2 + f_y^2} \, dA$$

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, dS = \iint_R \frac{-Pf_x - Qf_y + R}{\sqrt{1 + f_x^2 + f_y^2}} \sqrt{1 + f_x^2 + f_y^2} \, dA = \iint_R (-Pf_x - Qf_y + R) \, dA$$

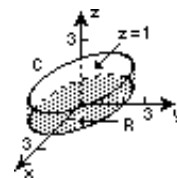
Exercises 9.14

1. **Surface Integral:** $\text{curl } \mathbf{F} = -10\mathbf{k}$. Letting $g(x, y, z) = z - 1$, we have $\nabla g = \mathbf{k}$ and $\mathbf{n} = \mathbf{k}$. Then

$$\iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS = \iint_S (-10) \, dS = -10 \times (\text{area of } S) = -10(4\pi) = -40\pi.$$

Line Integral: Parameterize the curve C by $x = 2 \cos t$, $y = 2 \sin t$, $z = 1$, for $0 \leq t \leq 2\pi$. Then

$$\begin{aligned}\oint_C \mathbf{F} \cdot d\mathbf{r} &= \oint_C 5y \, dx - 5x \, dy + 3 \, dz = \int_0^{2\pi} [10 \sin t(-2 \sin t) - 10 \cos t(2 \cos t)] \, dt \\ &= \int_0^{2\pi} (-20 \sin^2 t - 20 \cos^2 t) \, dt = \int_0^{2\pi} -20 \, dt = -40\pi.\end{aligned}$$

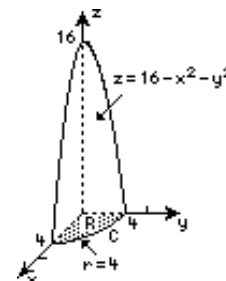


2. **Surface Integral:** $\text{curl } \mathbf{F} = 4\mathbf{i} - 2\mathbf{j} - 3\mathbf{k}$. Letting $g(x, y, z) = x^2 + y^2 + z - 16$, $\nabla g = 2x\mathbf{i} + 2y\mathbf{j} + \mathbf{k}$, and $\mathbf{n} = (2x\mathbf{i} + 2y\mathbf{j} + \mathbf{k})/\sqrt{4x^2 + 4y^2 + 1}$. Thus,

$$\iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS = \iint_S \frac{8x - 4y - 3}{\sqrt{4x^2 + 4y^2 + 1}} \, dS.$$

Letting the surface be $z = 16 - x^2 - y^2$, we have $z_x = -2x$, $z_y = -2y$, and

$dS = \sqrt{1 + 4x^2 + 4y^2} \, dA$. Then, using polar coordinates,



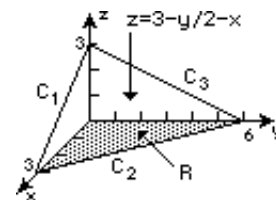
Exercises 9.14

$$\begin{aligned}
 \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS &= \iint_R (8x - 4y - 3) \, dA = \int_0^{2\pi} \int_0^4 (8r \cos \theta - 4r \sin \theta - 3) r \, dr \, d\theta \\
 &= \int_0^{2\pi} \left(\frac{8}{3} r^3 \cos \theta - \frac{4}{3} r^3 \sin \theta - \frac{3}{2} r^2 \right) \Big|_0^4 \, d\theta = \int_0^{2\pi} \left(\frac{512}{3} \cos \theta - \frac{256}{3} \sin \theta - 24 \right) d\theta \\
 &= \left(\frac{512}{3} \sin \theta + \frac{256}{3} \cos \theta - 24\theta \right) \Big|_0^{2\pi} = -48\pi.
 \end{aligned}$$

Line Integral: Parameterize the curve C by $x = 4 \cos t$, $y = 4 \sin t$, $z = 0$, for $0 \leq t \leq 2\pi$. Then,

$$\begin{aligned}
 \oint_C \mathbf{F} \cdot d\mathbf{r} &= \oint_C 2z \, dx - 3x \, dy + 4y \, dz = \int_0^{2\pi} [-12 \cos t (4 \cos t)] \, dt \\
 &= \int_0^{2\pi} -48 \cos^2 t \, dt = (-24t - 12 \sin 2t) \Big|_0^{2\pi} = -48\pi.
 \end{aligned}$$

- 3. Surface Integral:** $\text{curl } \mathbf{F} = \mathbf{i} + \mathbf{j} + \mathbf{k}$. Letting $g(x, y, z) = 2x + y + 2z - 6$, we have $\nabla g = 2\mathbf{i} + \mathbf{j} + 2\mathbf{k}$ and $\mathbf{n} = (2\mathbf{i} + \mathbf{j} + 2\mathbf{k})/3$. Then $\iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS = \iint_S \frac{5}{3} \, dS$. Letting the surface be $z = 3 - \frac{1}{2}y - x$ we have $z_x = -1$, $z_y = -\frac{1}{2}$, and $dS = \sqrt{1 + (-1)^2 + (-\frac{1}{2})^2} \, dA = \frac{3}{2} \, dA$. Then



$$\iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS = \iint_R \frac{5}{3} \left(\frac{3}{2} \right) \, dA = \frac{5}{2} \times (\text{area of } R) = \frac{5}{2}(9) = \frac{45}{2}.$$

Line Integral: C_1 : $z = 3 - x$, $0 \leq x \leq 3$, $y = 0$; C_2 : $y = 6 - 2x$, $3 \geq x \geq 0$, $z = 0$; C_3 : $z = 3 - y/2$, $6 \geq y \geq 0$, $x = 0$.

$$\begin{aligned}
 \oint_C z \, dx + x \, dy + y \, dz &= \iint_{C_1} z \, dx + \int_{C_2} x \, dy + \int_{C_3} y \, dz \\
 &= \int_0^3 (3 - x) \, dx + \int_3^0 x(-2 \, dx) + \int_6^0 y(-dy/2) \\
 &= \left(3x - \frac{1}{2}x^2 \right) \Big|_0^3 - x^2 \Big|_3^0 - \frac{1}{4}y^2 \Big|_6^0 = \frac{9}{2} - (0 - 9) - \frac{1}{4}(0 - 36) = \frac{45}{2}
 \end{aligned}$$

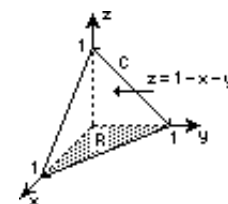
- 4. Surface Integral:** $\text{curl } \mathbf{F} = \mathbf{0}$ and $\iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS = 0$.

Line Integral: the curve is $x = \cos t$, $y = \sin t$, $z = 0$, $0 \leq t \leq 2\pi$.

$$\oint_C x \, dx + y \, dy + z \, dz = \int_0^{2\pi} [\cos t(-\sin t) + \sin t(\cos t)] \, dt = 0.$$

- 5.** $\text{curl } \mathbf{F} = 2\mathbf{i} + \mathbf{j}$. A unit vector normal to the plane is $\mathbf{n} = (\mathbf{i} + \mathbf{j} + \mathbf{k})/\sqrt{3}$. Taking the equation of the plane to be $z = 1 - x - y$, we have $z_x = z_y = -1$. Thus, $dS = \sqrt{1 + 1 + 1} \, dA = \sqrt{3} \, dA$ and

$$\begin{aligned}
 \oint_C \mathbf{F} \cdot d\mathbf{r} &= \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS = \iint_S \sqrt{3} \, dS = \sqrt{3} \iint_R \sqrt{3} \, dA \\
 &= 3 \times (\text{area of } R) = 3(1/2) = 3/2.
 \end{aligned}$$



- 6.** $\text{curl } \mathbf{F} = -2xz\mathbf{i} + z^2\mathbf{k}$. A unit vector normal to the plane is $\mathbf{n} = (\mathbf{j} + \mathbf{k})/\sqrt{2}$. From $z = 1 - y$, we have $z_x = 0$ and $z_y = -1$. Thus, $dS = \sqrt{1 + 1} \, dA = \sqrt{2} \, dA$ and

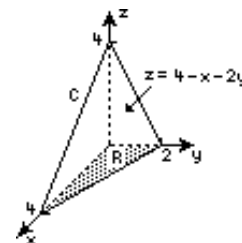
$$\begin{aligned}\oint_C \mathbf{F} \cdot d\mathbf{r} &= \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} dS = \iint_R \frac{1}{\sqrt{2}} z^2 \sqrt{2} dA = \iint_R (1-y)^2 dA \\ &= \int_0^2 \int_0^1 (1-y)^2 dy dx = \int_0^2 \left. -\frac{1}{3}(1-y)^3 \right|_0^1 dx = \int_0^2 \frac{1}{3} dx = \frac{2}{3}.\end{aligned}$$

7. $\text{curl } \mathbf{F} = -2y\mathbf{i} - z\mathbf{j} - x\mathbf{k}$. A unit vector normal to the plane is $\mathbf{n} = (\mathbf{j} + \mathbf{k})/\sqrt{2}$. From $z = 1 - y$ we have $z_x = 0$ and $z_y = -1$. Then $dS = \sqrt{1+1} dA = \sqrt{2} dA$ and

$$\begin{aligned}\oint_C \mathbf{F} \cdot d\mathbf{r} &= \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} dS = \iint_R \left[-\frac{1}{\sqrt{2}}(z+x) \right] \sqrt{2} dA = \iint_R (y-x-1) dA \\ &= \int_0^2 \int_0^1 (y-x-1) dy dx = \int_0^2 \left(\frac{1}{2}y^2 - xy - y \right) \Big|_0^1 dx = \int_0^2 \left(-x - \frac{1}{2} \right) dx \\ &= \left(-\frac{1}{2}x^2 - \frac{1}{2}x \right) \Big|_0^2 = -3.\end{aligned}$$

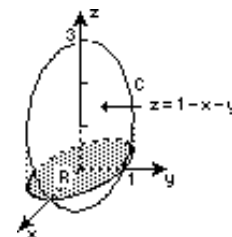
8. $\text{curl } \mathbf{F} = 2\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$. Letting $g(x, y, z) = x + 2y + z - 4$, we have $\nabla g = \mathbf{i} + 2\mathbf{j} + \mathbf{k}$ and $\mathbf{n} = (\mathbf{i} + 2\mathbf{j} + \mathbf{k})/\sqrt{6}$. From $z = 4 - x - 2y$ we have $z_x = -1$ and $z_y = -2$. Then $dS = \sqrt{6} dA$ and

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} dS = \iint_R \frac{1}{\sqrt{6}} (9) \sqrt{6} dA = \iint_R 9 dA = 9(4) = 36.$$



9. $\text{curl } \mathbf{F} = (-3x^2 - 3y^2)\mathbf{k}$. A unit vector normal to the plane is $\mathbf{n} = (\mathbf{i} + \mathbf{j} + \mathbf{k})/\sqrt{3}$. From $z = 1 - x - y$, we have $z_x = z_y = -1$ and $dS = \sqrt{3} dA$. Then, using polar coordinates,

$$\begin{aligned}\oint_C \mathbf{F} \cdot d\mathbf{r} &= \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} dS = \iint_R (-\sqrt{3}x^2 - \sqrt{3}y^2) \sqrt{3} dA \\ &= 3 \iint_R (-x^2 - y^2) dA = 3 \int_0^{2\pi} \int_0^1 (-r^2) r dr d\theta \\ &= 3 \int_0^{2\pi} \left. -\frac{1}{4}r^4 \right|_0^1 d\theta = 3 \int_0^{2\pi} -\frac{1}{4} d\theta = -\frac{3\pi}{2}.\end{aligned}$$



10. $\text{curl } \mathbf{F} = 2xyzi - y^2zj + (1-x^2)k$. A unit vector normal to the surface is $\mathbf{n} = \frac{2y\mathbf{j} + \mathbf{k}}{\sqrt{4y^2 + 1}}$. From $z = 9 - y^2$ we have $z_x = 0$, $z_y = -2y$ and $dS = \sqrt{1+4y^2} dA$. Then

$$\begin{aligned}\oint_C \mathbf{F} \cdot d\mathbf{r} &= \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} dS = \iint_R (-2y^3z + 1 - x^2) dA = \int_0^3 \int_0^{y/2} [-2y^3(9-y^2) + 1 - x^2] dx dy \\ &= \int_0^3 \left(-18y^3x + 2y^5x + x - \frac{1}{3}x^3 \right) \Big|_0^{y/2} dy = \int_0^3 \left(-9y^4 + y^6 + \frac{1}{2}y - \frac{1}{24}y^3 \right) dy \\ &= \left(-\frac{9}{5}y^5 + \frac{1}{7}y^7 + \frac{1}{4}y^2 - \frac{1}{96}y^4 \right) \Big|_0^3 \approx 123.57.\end{aligned}$$

11. $\text{curl } \mathbf{F} = 3x^2y^2\mathbf{k}$. A unit vector normal to the surface is

$$\mathbf{n} = \frac{8x\mathbf{i} + 2y\mathbf{j} + 2z\mathbf{k}}{\sqrt{64x^2 + 4y^2 + 4z^2}} = \frac{4x\mathbf{i} + y\mathbf{j} + z\mathbf{k}}{\sqrt{16x^2 + y^2 + z^2}}.$$

Exercises 9.14

From $z_x = -\frac{4x}{\sqrt{4-4x^2-y^2}}$, $z_y = -\frac{y}{\sqrt{4-4x^2-y^2}}$ we obtain $dS = 2\sqrt{\frac{1+3x^2}{4-4x^2-y^2}} dA$. Then

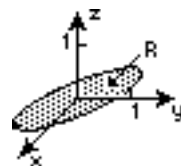
$$\begin{aligned}\oint_C \mathbf{F} \cdot d\mathbf{r} &= \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} dS = \iint_R \frac{3x^2 y^2 z}{\sqrt{16x^2 + y^2 + z^2}} \left(2\sqrt{\frac{1+3x^2}{4-4x^2-y^2}} \right) dA \\ &= \iint_R 3x^2 y^2 dA \quad \boxed{\text{Using symmetry}} \\ &= 12 \int_0^1 \int_0^{2\sqrt{1-x^2}} x^2 y^2 dy dx = 12 \int_0^1 \left(\frac{1}{3} x^2 y^3 \right) \Big|_0^{2\sqrt{1-x^2}} dx \\ &= 32 \int_0^1 x^2 (1-x^2)^{3/2} dx \quad \boxed{x = \sin t, dx = \cos t dt} \\ &= 32 \int_0^{\pi/2} \sin^2 t \cos^4 t dt = \pi.\end{aligned}$$

12. $\text{curl } \mathbf{F} = \mathbf{i} + \mathbf{j} + \mathbf{k}$. Taking the surface S bounded by C to be the portion of the plane $x + y + z = 0$ inside C , we have $\mathbf{n} = (\mathbf{i} + \mathbf{j} + \mathbf{k})/\sqrt{3}$ and $dS = \sqrt{3} dA$.

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} dS = \iint_S \sqrt{3} dS = \sqrt{3} \iint_R \sqrt{3} dA = 3 \times (\text{area of } R)$$

The region R is obtained by eliminating z from the equations of the plane and the sphere. This gives $x^2 + xy + y^2 = \frac{1}{2}$. Rotating axes, we see that R is enclosed by the ellipse $X^2/(1/3) + Y^2/1 = 1$ in a rotated coordinate system. Thus,

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = 3 \times (\text{area of } R) = 3 \left(\pi \frac{1}{\sqrt{3}} 1 \right) = \sqrt{3} \pi.$$

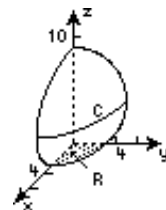


13. Parameterize C by $x = 4 \cos t$, $y = 2 \sin t$, $z = 4$, for $0 \leq t \leq 2\pi$. Then

$$\begin{aligned}\iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} dS &= \oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C 6yz dx + 5x dy + yze^{x^2} dz \\ &= \int_0^{2\pi} [6(2 \sin t)(4)(-4 \sin t) + 5(4 \cos t)(2 \cos t) + 0] dt \\ &= 8 \int_0^{2\pi} (-24 \sin^2 t + 5 \cos^2 t) dt = 8 \int_0^{2\pi} (5 - 29 \sin^2 t) dt = -152\pi.\end{aligned}$$

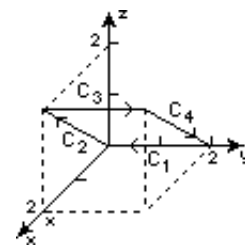
14. Parameterize C by $x = 5 \cos t$, $y = 5 \sin t$, $z = 4$, for $0 \leq t \leq 2\pi$. Then,

$$\begin{aligned}\iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} dS &= \oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C y dx + (y-x) dy + z^2 dz \\ &= \int_0^{2\pi} [(5 \sin t)(-5 \sin t) + (5 \sin t - 5 \cos t)(5 \cos t)] dt \\ &= \int_0^{2\pi} (25 \sin t \cos t - 25) dt = \left(\frac{25}{2} \sin^2 t - 25t \right) \Big|_0^{2\pi} = -50\pi.\end{aligned}$$



15. Parameterize C by $C_1: x = 0, z = 0, 2 \geq y \geq 0$; $C_2: z = x, y = 0, 0 \leq x \leq 2$; $C_3: x = 2, z = 2, 0 \leq y \leq 2$; $C_4: z = x, y = 2, 2 \geq x \geq 0$. Then

$$\begin{aligned} \iint_S (\operatorname{curl} \mathbf{F}) \cdot \mathbf{n} dS &= \oint_C \mathbf{F} \cdot \mathbf{r} = \oint_C 3x^2 dx + 8x^3 y dy + 3x^2 y dz \\ &= \int_{C_1} 0 dx + 0 dy + 0 dz + \int_{C_2} 3x^2 dx + \int_{C_3} 64 dy + \int_{C_4} 3x^2 dx + 6x^2 dx \\ &= \int_0^2 3x^2 dx + \int_0^2 64 dy + \int_2^0 9x^2 dx = x^3 \Big|_0^2 + 64y \Big|_0^2 + 3x^3 \Big|_2^0 = 112. \end{aligned}$$



16. Parameterize C by $x = \cos t, y = \sin t, z = \sin t, 0 \leq t \leq 2\pi$. Then

$$\begin{aligned} \iint_S (\operatorname{curl} \mathbf{F}) \cdot \mathbf{n} dS &= \oint_C \mathbf{F} \cdot \mathbf{r} = \oint_C 2xy^2 z dx + 2x^2 yz dy + (x^2 y^2 - 6x) dz \\ &= \int_0^{2\pi} [2 \cos t \sin^2 t \sin t (-\sin t) + 2 \cos^2 t \sin t \sin t \cos t \\ &\quad + (\cos^2 t \sin^2 t - 6 \cos t) \cos t] dt \\ &= \int_0^{2\pi} (-2 \cos t \sin^4 t + 3 \cos^3 t \sin^2 t - 6 \cos^2 t) dt = -6\pi. \end{aligned}$$

17. We take the surface to be $z = 0$. Then $\mathbf{n} = \mathbf{k}$ and $dS = dA$. Since $\operatorname{curl} \mathbf{F} = \frac{1}{1+y^2} \mathbf{i} + 2ze^{x^2} \mathbf{j} + y^2 \mathbf{k}$,

$$\begin{aligned} \oint_C z^2 e^{x^2} dx + xy dy + \tan^{-1} y dz &= \iint_S (\operatorname{curl} \mathbf{F}) \cdot \mathbf{n} dS = \iint_S y^2 dS = \iint_R y^2 dA \\ &= \int_0^{2\pi} \int_0^3 r^2 \sin^2 \theta r dr d\theta = \int_0^{2\pi} \frac{1}{4} r^4 \sin^2 \theta \Big|_0^3 d\theta \\ &= \frac{81}{4} \int_0^{2\pi} \sin^2 \theta d\theta = \frac{81\pi}{4}. \end{aligned}$$

18. (a) $\operatorname{curl} \mathbf{F} = xz\mathbf{i} - yz\mathbf{j}$. A unit vector normal to the surface is $\mathbf{n} = \frac{2x\mathbf{i} + 2y\mathbf{j} + \mathbf{k}}{\sqrt{4x^2 + 4y^2 + 1}}$ and

$dS = \sqrt{1 + 4x^2 + 4y^2} dA$. Then, using $x = \cos t, y = \sin t, 0 \leq t \leq 2\pi$, we have

$$\begin{aligned} \iint_S (\operatorname{curl} \mathbf{F}) \cdot \mathbf{n} dS &= \iint_R (2x^2 z - 2y^2 z) dA = \iint_R (2x^2 - 2y^2)(1 - x^2 - y^2) dA \\ &= \iint_R (2x^2 - 2y^2 - 2x^4 + 2y^4) dA \\ &= \int_0^{2\pi} \int_0^1 (2r^2 \cos^2 \theta - 2r^2 \sin^2 \theta - 2r^4 \cos^4 \theta + 2r^4 \sin^4 \theta) r dr d\theta \\ &= 2 \int_0^{2\pi} \int_0^1 [r^3 \cos 2\theta - r^5 (\cos^2 \theta - \sin^2 \theta)(\cos^2 \theta + \sin^2 \theta)] dr d\theta \\ &= 2 \int_0^{2\pi} \int_0^1 (r^3 \cos 2\theta - r^5 \cos 2\theta) dr d\theta = 2 \int_0^{2\pi} \cos 2\theta \left(\frac{1}{4} r^4 - \frac{1}{6} r^6 \right) \Big|_0^1 d\theta \\ &= \frac{1}{6} \int_0^{2\pi} \cos 2\theta d\theta = 0. \end{aligned}$$

- (b) We take the surface to be $z = 0$. Then $\mathbf{n} = \mathbf{k}$, $\operatorname{curl} \mathbf{F} \cdot \mathbf{n} = \operatorname{curl} \mathbf{F} \cdot \mathbf{k} = 0$ and $\iint_S (\operatorname{curl} \mathbf{F}) \cdot \mathbf{n} dS = 0$.

Exercises 9.14

(c) By Stoke's Theorem, using $z = 0$, we have

$$\iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS = \oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C xyz \, dz = \oint_C xy(0) \, dz = 0.$$

Exercises 9.15

1.
$$\begin{aligned} \int_2^4 \int_{-2}^2 \int_{-1}^1 (x + y + z) \, dx \, dy \, dz &= \int_2^4 \int_{-2}^2 \left(\frac{1}{2}x^2 + xy + xz \right) \Big|_{-1}^1 \, dy \, dz \\ &= \int_2^4 \int_{-2}^2 (2y + 2z) \, dy \, dz = \int_2^4 (y^2 + 2yz) \Big|_{-2}^2 \, dz = \int_2^4 8z \, dz = 4z^2 \Big|_2^4 = 48 \end{aligned}$$
2.
$$\begin{aligned} \int_1^3 \int_1^x \int_2^{xy} 24xyz \, dz \, dy \, dx &= \int_1^3 \int_1^x 24xyz \Big|_2^{xy} \, dy \, dx = \int_1^3 \int_1^x (24x^2y^2 - 48xy) \, dy \, dx \\ &= \int_1^3 (8x^2y^3 - 24xy^2) \Big|_1^x \, dx = \int_1^3 (8x^5 - 24x^3 - 8x^2 + 24x) \, dx \\ &= \left(\frac{4}{3}x^6 - 6x^4 - \frac{8}{3}x^3 + 12x^2 \right) \Big|_1^3 = 522 - \frac{14}{3} = \frac{1552}{3} \end{aligned}$$
3.
$$\begin{aligned} \int_0^6 \int_0^{6-x} \int_0^{6-x-z} dy \, dz \, dx &= \int_0^6 \int_0^{6-x} (6 - x - z) \, dz \, dx = \int_0^6 \left(6z - xz - \frac{1}{2}z^2 \right) \Big|_0^{6-x} \, dx \\ &= \int_0^6 \left[6(6-x) - x(6-x) - \frac{1}{2}(6-x)^2 \right] \, dx = \int_0^6 \left(18 - 6x + \frac{1}{2}x^2 \right) \, dx \\ &= \left(18x - 3x^2 + \frac{1}{6}x^3 \right) \Big|_0^6 = 36 \end{aligned}$$
4.
$$\begin{aligned} \int_0^1 \int_0^{1-x} \int_0^{\sqrt{y}} 4x^2z^3 \, dz \, dy \, dx &= \int_0^1 \int_0^{1-x} x^2z^4 \Big|_0^{\sqrt{y}} \, dy \, dx = \int_0^1 \int_0^{1-x} x^2y^2 \, dy \, dx \\ &= \int_0^1 \frac{1}{3}x^2y^3 \Big|_0^{1-x} \, dx = \frac{1}{3} \int_0^1 x^2(1-x)^3 \, dx = \frac{1}{3} \int_0^1 (x^2 - 3x^3 + 3x^4 - x^5) \, dx \\ &= \frac{1}{3} \left(\frac{1}{3}x^3 - \frac{3}{4}x^4 + \frac{3}{5}x^5 - \frac{1}{6}x^6 \right) \Big|_0^1 = \frac{1}{180} \end{aligned}$$
5.
$$\begin{aligned} \int_0^{\pi/2} \int_0^{y^2} \int_0^y \cos \frac{x}{y} \, dz \, dx \, dy &= \int_0^{\pi/2} \int_0^{y^2} y \cos \frac{x}{y} \, dx \, dy = \int_0^{\pi/2} y^2 \sin \frac{x}{y} \Big|_0^y \, dy \\ &= \int_0^{\pi/2} y^2 \sin y \, dy \quad \boxed{\text{Integration by parts}} \\ &= (-y^2 \cos y + 2 \cos y + 2y \sin y) \Big|_0^{\pi/2} = \pi - 2 \end{aligned}$$
6.
$$\begin{aligned} \int_0^{\sqrt{2}} \int_{\sqrt{y}}^2 \int_0^{e^{x^2}} x \, dz \, dx \, dy &= \int_0^{\sqrt{2}} \int_{\sqrt{y}}^2 xe^{x^2} \, dx \, dy = \int_0^{\sqrt{2}} \frac{1}{2}e^{x^2} \Big|_{\sqrt{y}}^2 \, dy = \frac{1}{2} \int_0^{\sqrt{2}} (e^4 - e^y) \, dy \\ &= \frac{1}{2} (ye^4 - e^y) \Big|_0^{\sqrt{2}} = \frac{1}{2} [(e^4\sqrt{2} - e^{\sqrt{2}}) - (-1)] = \frac{1}{2} (1 + e^4\sqrt{2} - e^{\sqrt{2}}) \end{aligned}$$

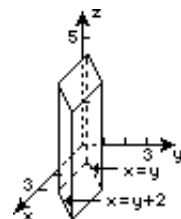
$$\begin{aligned}
 7. \quad \int_0^1 \int_0^1 \int_0^{2-x^2-y^2} xye^z dz dx dy &= \int_0^1 \int_0^1 xye^z \Big|_0^{2-x^2-y^2} dx dy = \int_0^1 \int_0^1 (xye^{2-x^2-y^2} - xy) dx dy \\
 &= \int_0^1 \left(-\frac{1}{2}ye^{2-x^2-y^2} - \frac{1}{2}x^2y \right) \Big|_0^1 dy = \int_0^1 \left(-\frac{1}{2}ye^{1-y^2} - \frac{1}{2}y + \frac{1}{2}ye^{2-y^2} \right) dy \\
 &= \left(\frac{1}{4}e^{1-y^2} - \frac{1}{4}y^2 - \frac{1}{4}e^{2-y^2} \right) \Big|_0^1 = \left(\frac{1}{4} - \frac{1}{4} - \frac{1}{4}e \right) - \left(\frac{1}{4}e - \frac{1}{4}e^2 \right) = \frac{1}{4}e^2 - \frac{1}{2}e
 \end{aligned}$$

$$8. \quad \int_0^4 \int_0^{1/2} \int_0^{x^2} \frac{1}{\sqrt{x^2-y^2}} dy dx dz = \int_0^4 \int_0^{1/2} \sin^{-1} \frac{y}{x} \Big|_0^{x^2} dx dz = \int_0^4 \int_0^{1/2} \sin^{-1} x dx dz$$

Integration by parts

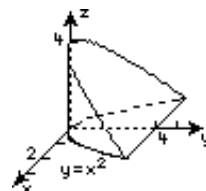
$$= \int_0^4 (x \sin^{-1} x + \sqrt{1-x^2}) \Big|_0^{1/2} dz = \int_0^4 \left(\frac{1}{2} \frac{\pi}{6} + \frac{\sqrt{3}}{2} - 1 \right) dz = \frac{\pi}{3} + 2\sqrt{3} - 4$$

$$\begin{aligned}
 9. \quad \iiint_D z dV &= \int_0^5 \int_1^3 \int_y^{y+2} z dx dy dz = \int_0^5 \int_1^3 xz \Big|_y^{y+2} dy dz = \int_0^5 \int_1^3 2z dy dz \\
 &= \int_0^5 2yz \Big|_1^3 dz = \int_0^5 4z dz = 2z^2 \Big|_0^5 = 50
 \end{aligned}$$



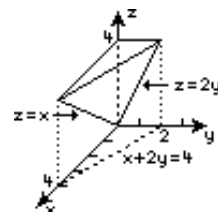
10. Using symmetry,

$$\begin{aligned}
 \iiint_D (x^2 + y^2) dV &= 2 \int_0^2 \int_{x^2}^4 \int_0^{4-y} (x^2 + y^2) dz dy dx = 2 \int_0^2 \int_{x^2}^4 (x^2 + y^2)z \Big|_0^{4-y} dy dx \\
 &= 2 \int_0^2 \int_{x^2}^4 (4x^2 - x^2y + 4y^2 - y^3) dy dx \\
 &= 2 \int_0^2 \left(4x^2y - \frac{1}{2}x^2y^2 + \frac{4}{3}y^3 - \frac{1}{4}y^4 \right) \Big|_{x^2}^4 dx \\
 &= 2 \int_0^2 \left[\left(8x^2 + \frac{64}{3} \right) - \left(4x^4 + \frac{5}{6}x^6 - \frac{1}{4}x^8 \right) \right] dx \\
 &= 2 \left(\frac{8}{3}x^3 + \frac{64}{3}x - \frac{4}{5}x^5 - \frac{5}{42}x^7 + \frac{1}{36}x^9 \right) \Big|_0^2 = \frac{23,552}{315}.
 \end{aligned}$$



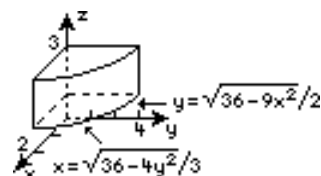
11. The other five integrals are $\int_0^4 \int_0^{2-x/2} \int_{x+2y}^4 F(x, y, z) dz dy dx$,

$$\begin{aligned}
 &\int_0^4 \int_0^z \int_0^{(z-x)/2} F(x, y, z) dy dx dz, \quad \int_0^4 \int_x^4 \int_0^{(z-x)/2} F(x, y, z) dy dz dx, \\
 &\int_0^4 \int_0^{z/2} \int_0^{z-2y} F(x, y, z) dx dy dz, \quad \int_0^2 \int_{2y}^4 \int_0^{z-2y} F(x, y, z) dx dz dy.
 \end{aligned}$$



12. The other five integrals are $\int_0^3 \int_0^{\sqrt{36-4y^2}/3} \int_1^3 F(x, y, z) dz dx dy$,

$$\begin{aligned}
 &\int_1^3 \int_0^2 \int_0^{\sqrt{36-9x^2}/2} F(x, y, z) dy dx dz, \quad \int_1^3 \int_0^3 \int_0^{\sqrt{36-4y^2}/3} F(x, y, z) dx dy dz, \\
 &\int_0^3 \int_1^3 \int_0^{\sqrt{36-4y^2}/3} F(x, y, z) dx dz dy, \quad \int_0^2 \int_1^3 \int_0^{\sqrt{36-9x^2}/2} F(x, y, z) dy dz dx.
 \end{aligned}$$



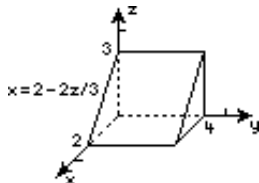
Exercises 9.15

13. (a) $V = \int_0^2 \int_{x^3}^8 \int_0^4 dz dy dx$ (b) $V = \int_0^8 \int_0^4 \int_0^{y^{1/3}} dx dz dy$ (c) $V = \int_0^4 \int_0^2 \int_{x^2}^8 dy dx dz$

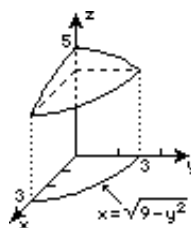
14. Solving $z = \sqrt{x}$ and $x + z = 2$, we obtain $x = 1$, $z = 1$. (a) $V = \int_0^3 \int_0^1 \int_{z^2}^{2-z} dx dz dy$

(b) $V = \int_0^1 \int_{z^2}^{2-z} \int_0^3 dy dx dz$ (c) $V = \int_0^3 \int_0^1 \int_0^{\sqrt{x}} dz dx dy + \int_0^3 \int_1^2 \int_0^{2-x} dz dx dy$

15.

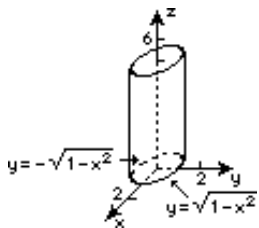


16.

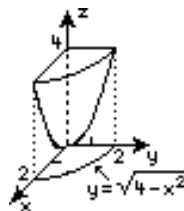


The region in the first octant is shown.

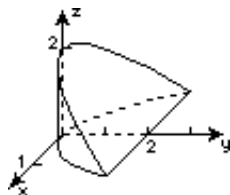
17.



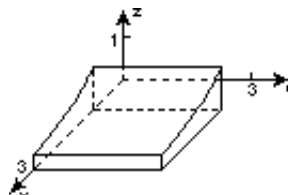
18.



19.

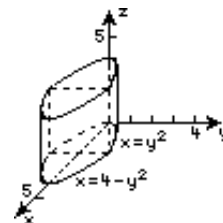


20.

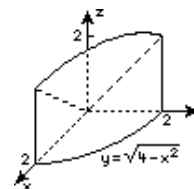


21. Solving $x = y^2$ and $4 - x = y^2$, we obtain $x = 2$, $y = \pm\sqrt{2}$. Using symmetry,

$$\begin{aligned} V &= 2 \int_0^3 \int_0^{\sqrt{2}} \int_{y^2}^{4-y^2} dx dy dz = 2 \int_0^3 \int_0^{\sqrt{2}} (4 - 2y^2) dy dz \\ &= 2 \int_0^3 \left(4y - \frac{2}{3}y^3 \right) \Big|_0^{\sqrt{2}} dz = 2 \int_0^3 \frac{8\sqrt{2}}{3} dz = 16\sqrt{2}. \end{aligned}$$

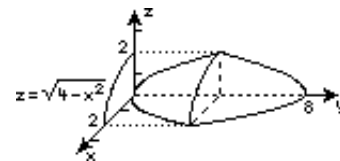


$$\begin{aligned} 22. \quad V &= \int_0^2 \int_0^{\sqrt{4-x^2}} \int_0^{x+y} dz dy dx = \int_0^2 \int_0^{\sqrt{4-x^2}} z \Big|_0^{x+y} dy dx \\ &= \int_0^2 \int_0^{\sqrt{4-x^2}} (x+y) dy dx = \int_0^2 \left(xy + \frac{1}{2}y^2 \right) \Big|_0^{\sqrt{4-x^2}} dx \\ &= \int_0^2 \left[x\sqrt{4-x^2} + \frac{1}{2}(4-x^2) \right] dx = \left[-\frac{1}{3}(4-x^2)^{3/2} + 2x - \frac{1}{6}x^3 \right] \Big|_0^2 \\ &= \left(4 - \frac{4}{3} \right) - \left(-\frac{8}{3} \right) = \frac{16}{3} \end{aligned}$$



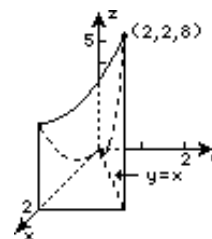
23. Adding the two equations, we obtain $2y = 8$. Thus, the paraboloids intersect in the plane $y = 4$. Their intersection is a circle of radius 2. Using symmetry,

$$\begin{aligned} V &= 4 \int_0^2 \int_0^{\sqrt{4-x^2}} \int_{x^2+z^2}^{8-x^2-z^2} dy \, dz \, dx = 4 \int_0^2 \int_0^{\sqrt{4-x^2}} (8-2x^2-2z^2) \, dz \, dx \\ &= 4 \int_0^2 \left[2(4-x^2)z - \frac{2}{3}z^3 \right] \Big|_0^{\sqrt{4-x^2}} dx = 4 \int_0^2 \frac{4}{3}(4-x^2)^{3/2} dx \quad \boxed{\text{Trig substitution}} \\ &= \frac{16}{3} \left[-\frac{x}{8}(2x^2-20)\sqrt{4-x^2} + 6 \sin^{-1} \frac{x}{2} \right] \Big|_0^2 = 16\pi. \end{aligned}$$



24. Solving $x = 2$, $y = x$, and $z = x^2 + y^2$, we obtain the point $(2, 2, 8)$.

$$\begin{aligned} V &= \int_0^2 \int_0^x \int_0^{x^2+y^2} dz \, dy \, dx = \int_0^2 \int_0^x (x^2 + y^2) \, dy \, dx = \int_0^2 \left(x^2 y + \frac{1}{3} y^3 \right) \Big|_0^x dx \\ &= \int_0^2 \frac{4}{3} x^3 \, dx = \frac{1}{3} x^4 \Big|_0^2 = \frac{16}{3}. \end{aligned}$$



25. We are given $\rho(x, y, z) = kz$.

$$\begin{aligned} m &= \int_0^8 \int_0^4 \int_0^{y^{1/3}} kz \, dx \, dz \, dy = k \int_0^8 \int_0^4 xz \Big|_0^{y^{1/3}} dz \, dy = k \int_0^8 \int_0^4 y^{1/3} z \, dz \, dy \\ &= k \int_0^8 \frac{1}{2} y^{1/3} z^2 \Big|_0^4 dy = 8k \int_0^8 y^{1/3} dy = 8k \left(\frac{3}{4} y^{4/3} \right) \Big|_0^8 = 96k \\ M_{xy} &= \int_0^8 \int_0^4 \int_0^{y^{1/3}} kz^2 \, dx \, dz \, dy = k \int_0^8 \int_0^4 xz^2 \Big|_0^{y^{1/3}} dz \, dy = k \int_0^8 \int_0^4 y^{1/3} z^2 \, dz \, dy \\ &= k \int_0^8 \frac{1}{3} y^{1/3} z^3 \Big|_0^4 dy = \frac{64}{3} k \int_0^8 y^{1/3} dy = \frac{64}{3} k \left(\frac{3}{4} y^{4/3} \right) \Big|_0^8 = 256k \\ M_{xz} &= \int_0^8 \int_0^4 \int_0^{y^{1/3}} kyz \, dx \, dz \, dy = k \int_0^8 \int_0^4 xyz \Big|_0^{y^{1/3}} dz \, dy = k \int_0^8 \int_0^4 y^{4/3} z \, dz \, dy \\ &= k \int_0^8 \frac{1}{2} y^{4/3} z^2 \Big|_0^4 dy = 8k \int_0^8 y^{4/3} dy = 8k \left(\frac{3}{7} y^{7/3} \right) \Big|_0^8 = \frac{3072}{7} k \\ M_{yz} &= \int_0^8 \int_0^4 \int_0^{y^{1/3}} kxz \, dx \, dz \, dy = k \int_0^8 \int_0^4 \frac{1}{2} x^2 z \Big|_0^{y^{1/3}} dz \, dy = \frac{1}{2} k \int_0^8 \int_0^4 y^{2/3} z \, dz \, dy \\ &= \frac{1}{2} k \int_0^8 \frac{1}{2} y^{2/3} z^2 \Big|_0^4 dy = 4k \int_0^8 y^{2/3} dy = 4k \left(\frac{3}{5} y^{5/3} \right) \Big|_0^8 = \frac{384}{5} k \\ \bar{x} = M_{yz}/m &= \frac{384k/5}{96k} = 4/5; \quad \bar{y} = M_{xz}/m = \frac{3072k/7}{96k} = 32/7; \quad \bar{z} = M_{xy}/m = \frac{256k}{96k} = 8/3 \end{aligned}$$

The center of mass is $(4/5, 32/7, 8/3)$.

26. We use the form of the integral in Problem 14(b) of this section. Without loss of generality, we take $\rho = 1$.

$$\begin{aligned} m &= \int_0^1 \int_{z^2}^{2-z} \int_0^3 dy \, dx \, dz = \int_0^1 \int_{z^2}^{2-z} 3 \, dx \, dz = 3 \int_0^1 (2-z-z^2) \, dz \\ &= 3 \left(2z - \frac{1}{2} z^2 - \frac{1}{3} z^3 \right) \Big|_0^1 = \frac{7}{2} \end{aligned}$$

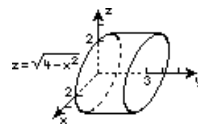
Exercises 9.15

$$\begin{aligned}
 M_{xy} &= \int_0^1 \int_{z^2}^{2-z} \int_0^3 z \, dy \, dx \, dz = \int_0^1 \int_{z^2}^{2-z} yz \Big|_0^3 \, dx \, dz = \int_0^1 \int_{z^2}^{2-z} 3z \, dx \, dz \\
 &= 3 \int_0^1 xz \Big|_{z^2}^{2-z} \, dz = 3 \int_0^1 (2z - z^2 - z^3) \, dz = 3 \left(z^2 - \frac{1}{3}z^3 - \frac{1}{4}z^4 \right) \Big|_0^1 = \frac{5}{4} \\
 M_{xz} &= \int_0^1 \int_{z^2}^{2-z} \int_0^3 y \, dy \, dx \, dz = \int_0^1 \int_{z^2}^{2-z} \frac{1}{2}y^2 \Big|_0^3 \, dx \, dz = \frac{9}{2} \int_0^1 \int_{z^2}^{2-z} dx \, dz \\
 &= \frac{9}{2} \int_0^1 (2 - z - z^2) \, dz = \frac{9}{2} \left(2z - \frac{1}{2}z^2 - \frac{1}{3}z^3 \right) \Big|_0^1 = \frac{21}{4} \\
 M_{yz} &= \int_0^1 \int_{z^2}^{2-z} \int_0^3 x \, dy \, dx \, dz = \int_0^1 \int_{z^2}^{2-z} xy \Big|_0^3 \, dx \, dz = \int_0^1 \int_{z^2}^{2-z} 3x \, dx \, dz \\
 &= 3 \int_0^1 \frac{1}{2}x^2 \Big|_{z^2}^{2-z} \, dz = \frac{3}{2} \int_0^1 (4 - 4z + z^2 - z^4) \, dz = \frac{3}{2} \left(4z - 2z^2 + \frac{1}{3}z^3 - \frac{1}{5}z^5 \right) \Big|_0^1 = \frac{16}{5}
 \end{aligned}$$

$$\bar{x} = M_{yz}/m = \frac{16/5}{7/2} = 32/35, \quad \bar{y} = M_{xz}/m = \frac{21/4}{7/2} = 3/2, \quad \bar{z} = M_{xy}/m = \frac{5/4}{7/2} = 5/14.$$

The centroid is $(32/35, 3/2, 5/14)$.

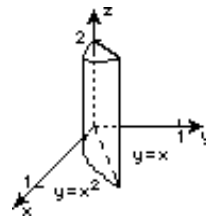
- 27.** The density is $\rho(x, y, z) = ky$. Since both the region and the density function are symmetric with respect to the xy - and yz -planes, $\bar{x} = \bar{z} = 0$. Using symmetry,



$$\begin{aligned}
 m &= 4 \int_0^3 \int_0^2 \int_0^{\sqrt{4-x^2}} ky \, dz \, dx \, dy = 4k \int_0^3 \int_0^2 yz \Big|_0^{\sqrt{4-x^2}} \, dx \, dy = 4k \int_0^3 \int_0^2 y\sqrt{4-x^2} \, dx \, dy \\
 &= 4k \int_0^3 y \left(\frac{x}{2}\sqrt{4-x^2} + 2\sin^{-1}\frac{x}{2} \right) \Big|_0^2 \, dy = 4k \int_0^3 \pi y \, dy = 4\pi k \left(\frac{1}{2}y^2 \right) \Big|_0^3 = 18\pi k \\
 M_{xz} &= 4 \int_0^3 \int_0^2 \int_0^{\sqrt{4-x^2}} ky^2 \, dz \, dx \, dy = 4k \int_0^3 \int_0^2 y^2 z \Big|_0^{\sqrt{4-x^2}} \, dx \, dy = 4k \int_0^3 \int_0^2 y^2\sqrt{4-x^2} \, dx \, dy \\
 &= 4k \int_0^3 y^2 \left(\frac{x}{2}\sqrt{4-x^2} + 2\sin^{-1}\frac{x}{2} \right) \Big|_0^2 \, dy = 4k \int_0^3 \pi y^2 \, dy = 4\pi k \left(\frac{1}{3}y^3 \right) \Big|_0^3 = 36\pi k. \\
 \bar{y} &= M_{xz}/m = \frac{36\pi k}{18\pi k} = 2. \text{ The center of mass is } (0, 2, 0).
 \end{aligned}$$

- 28.** The density is $\rho(x, y, z) = kz$.

$$\begin{aligned}
 m &= \int_0^1 \int_{x^2}^x \int_0^{y+2} kz \, dz \, dy \, dx = k \int_0^1 \int_{x^2}^x \frac{1}{2}z^2 \Big|_0^{y+2} \, dy \, dx \\
 &= \frac{1}{2}k \int_0^1 \int_{x^2}^x (y+2)^2 \, dy \, dx = \frac{1}{2}k \int_0^1 \frac{1}{3}(y+2)^3 \Big|_{x^2}^x \, dx \\
 &= \frac{1}{6}k \int_0^1 [(x+2)^3 - (x^2+2)^3] \, dx = \frac{1}{6}k \int_0^1 [(x+2)^3 - (x^6 + 6x^4 + 12x^2 + 8)] \, dx \\
 &= \frac{1}{6}k \left[\frac{1}{4}(x+2)^4 - \frac{1}{7}x^7 - \frac{6}{5}x^5 - 4x^3 - 8x \right] \Big|_0^1 = \frac{407}{840}k
 \end{aligned}$$

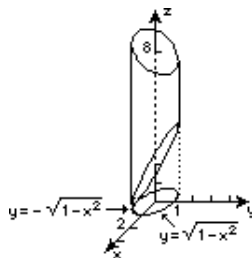


$$\begin{aligned}
M_{xy} &= \int_0^1 \int_{x^2}^x \int_0^{y+2} kz^2 dz dy dx = k \int_0^1 \int_{x^2}^x \frac{1}{3} z^3 \Big|_0^{y+2} dy dx = \frac{1}{3} k \int_0^1 \int_{x^2}^x (y+2)^3 dy dx \\
&= \frac{1}{3} k \int_0^1 \frac{1}{4} (y+2)^4 \Big|_{x^2}^x dx = \frac{1}{12} k \int_0^1 [(x+2)^4 - (x^2+2)^4] dx \\
&= \frac{1}{12} k \int_0^1 [(x+2)^4 - (x^8 + 8x^6 + 24x^4 + 32x^2 + 16)] dx \\
&= \frac{1}{12} k \left[\frac{1}{5} (x+2)^5 - \frac{1}{9} x^9 - \frac{8}{7} x^7 - \frac{24}{5} x^5 - \frac{32}{3} x^3 - 16x \right] \Big|_0^1 = \frac{1493}{1890} k \\
M_{xz} &= \int_0^1 \int_{x^2}^x \int_0^{y+2} kyz dz dy dx = k \int_0^1 \int_{x^2}^x \frac{1}{2} yz^2 \Big|_0^{y+2} dy dx = \frac{1}{2} k \int_0^1 \int_{x^2}^x y(y+2)^2 dy dx \\
&= \frac{1}{2} k \int_0^1 \int_{x^2}^x (y^3 + 4y^2 + 4y) dy dx = \frac{1}{2} k \int_0^1 \left(\frac{1}{4} y^4 + \frac{4}{3} y^3 + 2y^2 \right) \Big|_{x^2}^x dx \\
&= \frac{1}{2} k \int_0^1 \left(-\frac{1}{4} x^8 - \frac{4}{3} x^6 - 74x^4 + \frac{4}{3} x^3 + 2x^2 \right) dx \\
&= \frac{1}{2} k \left(-\frac{1}{36} x^9 - \frac{4}{21} x^7 - \frac{7}{20} x^5 + \frac{1}{3} x^4 + \frac{2}{3} x^3 \right) \Big|_0^1 = \frac{68}{315} k \\
M_{yz} &= \int_0^1 \int_{x^2}^x \int_0^{y+2} kxz dz dy dx = k \int_0^1 \int_{x^2}^x \frac{1}{2} xz^2 \Big|_0^{y+2} dy dx = \frac{1}{2} k \int_0^1 \int_{x^2}^x x(y+2)^2 dy dx \\
&= \frac{1}{2} k \int_0^1 \frac{1}{3} x(y+2)^3 \Big|_{x^2}^x dx = \frac{1}{6} k \int_0^1 [x(x+2)^3 - x(x^2+2)^3] dx \\
&= \frac{1}{6} k \int_0^1 [x^4 + 6x^3 + 12x^2 + 8x - x(x^2+2)^3] dx \\
&= \frac{1}{6} k \left[\frac{1}{5} x^5 + \frac{3}{2} x^4 + 4x^3 + 4x^2 - \frac{1}{8} (x^2+2)^4 \right] \Big|_0^1 = \frac{21}{80} k
\end{aligned}$$

$$\bar{x} = M_{yz}/m = \frac{21k/80}{407k/840} = 441/814, \quad \bar{y} = M_{xz}/m = \frac{68k/315}{407k/840} = 544/1221,$$

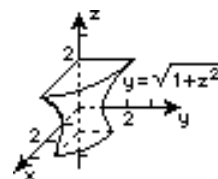
$$\bar{z} = M_{xy}/m = \frac{1493k/1890}{407k/840} = 5972/3663. \text{ The center of mass is } (441/814, 544/1221, 5972/3663).$$

$$29. \quad m = \int_{-1}^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_{2+2y}^{8-y} (x+y+4) dz dy dx$$



30. Both the region and the density function are symmetric with respect to the xz - and

$$yz\text{-planes. Thus, } m = 4 \int_{-1}^2 \int_0^{\sqrt{1+z^2}} \int_0^{\sqrt{1+z^2-y^2}} z^2 dx dy dz.$$



Exercises 9.15

31. We are given $\rho(x, y, z) = kz$.

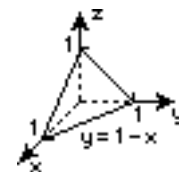
$$\begin{aligned} I_y &= \int_0^8 \int_0^4 \int_0^{y^{1/3}} kz(x^2 + z^2) dx dz dy = k \int_0^8 \int_0^4 \left(\frac{1}{3}x^3 z + xz^3 \right) \Big|_0^{y^{1/3}} dz dy \\ &= k \int_0^8 \int_0^4 \left(\frac{1}{3}yz + y^{1/3}z^3 \right) dz dy = k \int_0^8 \left(\frac{1}{6}yz^2 + \frac{1}{4}y^{1/3}z^4 \right) \Big|_0^4 dy \\ &= k \int_0^8 \left(\frac{8}{3}y + 64y^{1/3} \right) dy = k \left(\frac{4}{3}y^2 + 48y^{4/3} \right) \Big|_0^8 = \frac{2560}{3}k \end{aligned}$$

From Problem 25, $m = 96k$. Thus, $R_g = \sqrt{I_y/m} = \sqrt{\frac{2560k/3}{96k}} = \frac{4\sqrt{5}}{3}$.

32. We are given $\rho(x, y, z) = k$.

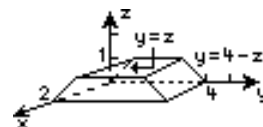
$$\begin{aligned} I_x &= \int_0^1 \int_{z^2}^{2-z} \int_0^3 k(y^2 + z^2) dy dx dz = k \int_0^1 \int_{z^2}^{2-z} \left(\frac{1}{3}y^3 + yz^2 \right) \Big|_0^3 dx dz = k \int_0^1 \int_{z^2}^{2-z} (9 + 3z^2) dx dz \\ &= k \int_0^1 (9x + 3xz^2) \Big|_{z^2}^{2-z} dz = k \int_0^1 (18 - 9z - 3z^2 - 3z^3 - 3z^4) dz \\ &= k \left(18z - \frac{9}{2}z^2 - z^3 - \frac{3}{4}z^4 - \frac{3}{5}z^5 \right) \Big|_0^1 = \frac{223}{20}k \\ m &= \int_0^1 \int_{z^2}^{2-z} \int_0^3 k dy dx dz = k \int_0^1 \int_{z^2}^{2-z} 3 dx dz = 3k \int_0^1 (2 - z - z^2) dz = 3k \left(2z - \frac{1}{2}z^2 - \frac{1}{3}z^3 \right) \Big|_0^1 = \frac{7}{2}k \\ R_g &= \sqrt{\frac{I_x}{m}} = \sqrt{\frac{223k/20}{7k/2}} = \sqrt{\frac{223}{70}} \end{aligned}$$

33.
$$\begin{aligned} I_z &= k \int_0^1 \int_0^{1-x} \int_0^{1-x-y} (x^2 + y^2) dz dy dx = k \int_0^1 \int_0^{1-x} (x^2 + y^2)(1 - x - y) dy dx \\ &= k \int_0^1 \int_0^{1-x} (x^2 - x^3 - x^2y + y^2 - xy^2 - y^3) dy dx \\ &= k \int_0^1 \left[(x^2 - x^3)y - \frac{1}{2}x^2y^2 + \frac{1}{3}(1-x)y^3 - \frac{1}{4}y^4 \right] \Big|_0^{1-x} dx \\ &= k \int_0^1 \left[\frac{1}{2}x^2 - x^3 + \frac{1}{2}x^4 + \frac{1}{12}(1-x)^4 \right] dx = k \left[\frac{1}{6}x^6 - \frac{1}{4}x^4 + \frac{1}{10}x^5 - \frac{1}{60}(1-x)^5 \right] \Big|_0^1 = \frac{k}{30} \end{aligned}$$



34. We are given $\rho(x, y, z) = kx$.

$$\begin{aligned} I_y &= \int_0^1 \int_0^2 \int_z^{4-z} kx(x^2 + z^2) dy dx dz = k \int_0^1 \int_0^2 (x^3 + xz^2)y \Big|_z^{4-z} dx dz \\ &= k \int_0^1 \int_0^2 (x^3 + xz^2)(4 - 2z) dx dz = k \int_0^1 \left(\frac{1}{4}x^4 + \frac{1}{2}x^2z^2 \right) (4 - 2z) \Big|_0^2 dz \\ &= k \int_0^1 (4 + 2z^2)(4 - 2z) dz = 4k \int_0^1 (4 - 2z + 2z^2 - z^3) dz = 4k \left(4z - z^2 + \frac{2}{3}z^3 - \frac{1}{4}z^4 \right) \Big|_0^1 = \frac{41}{3}k \end{aligned}$$



35. $x = 10 \cos 3\pi/4 = -5\sqrt{2}$; $y = 10 \sin 3\pi/4 = 5\sqrt{2}$; $(-5\sqrt{2}, 5\sqrt{2}, 5)$

36. $x = 2 \cos 5\pi/6 = -\sqrt{3}$; $y = 2 \sin 5\pi/6 = 1$; $(-\sqrt{3}, 1, -3)$

37. $x = \sqrt{3} \cos \pi/3 = \sqrt{3}/2$; $y = \sqrt{3} \sin \pi/3 = 3/2$; $(\sqrt{3}/2, 3/2, -4)$

38. $x = 4 \cos 7\pi/4 = 2\sqrt{2}$; $y = 4 \sin 7\pi/4 = -2\sqrt{2}$; $(2\sqrt{2}, -2\sqrt{2}, 0)$

39. With $x = 1$ and $y = -1$ we have $r^2 = 2$ and $\tan \theta = -1$. The point is $(\sqrt{2}, -\pi/4, -9)$.

40. With $x = 2\sqrt{3}$ and $y = 2$ we have $r^2 = 16$ and $\tan \theta = 1/\sqrt{3}$. The point is $(4, \pi/6, 17)$.

41. With $x = -\sqrt{2}$ and $y = \sqrt{6}$ we have $r^2 = 8$ and $\tan \theta = -\sqrt{3}$. The point is $(2\sqrt{2}, 2\pi/3, 2)$.

42. With $x = 1$ and $y = 2$ we have $r^2 = 5$ and $\tan \theta = 2$. The point is $(\sqrt{5}, \tan^{-1} 2, 7)$.

43. $r^2 + z^2 = 25$

44. $r \cos \theta + r \sin \theta - z = 1$

45. $r^2 - z^2 = 1$

46. $r^2 \cos^2 \theta + z^2 = 16$

47. $z = x^2 + y^2$

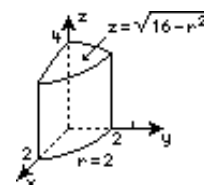
48. $z = 2y$

49. $r \cos \theta = 5, x = 5$

50. $\tan \theta = 1/\sqrt{3}, y/x = 1/\sqrt{3}, x = \sqrt{3}y, x > 0$

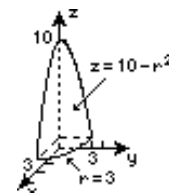
51. The equations are $r^2 = 4$, $r^2 + z^2 = 16$, and $z = 0$.

$$\begin{aligned} V &= \int_0^{2\pi} \int_0^2 \int_0^{\sqrt{16-r^2}} r \, dz \, dr \, d\theta = \int_0^{2\pi} \int_0^2 r \sqrt{16-r^2} \, dr \, d\theta \\ &= \int_0^{2\pi} -\frac{1}{3}(16-r^2)^{3/2} \Big|_0^2 d\theta = \int_0^{2\pi} \frac{1}{3}(64-24\sqrt{3}) \, d\theta = \frac{2\pi}{3}(64-24\sqrt{3}) \end{aligned}$$



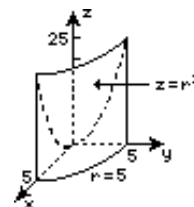
52. The equation is $z = 10 - r^2$.

$$\begin{aligned} V &= \int_0^{2\pi} \int_0^3 \int_1^{10-r^2} r \, dz \, dr \, d\theta = \int_0^{2\pi} \int_0^3 r(9-r^2) \, dr \, d\theta = \int_0^{2\pi} \left(\frac{9}{2}r^2 - \frac{1}{4}r^4 \right) \Big|_0^3 d\theta \\ &= \int_0^{2\pi} \frac{81}{4} \, d\theta = \frac{81\pi}{2}. \end{aligned}$$



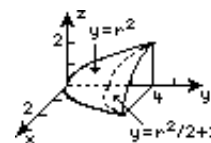
53. The equations are $z = r^2$, $r = 5$, and $z = 0$.

$$\begin{aligned} V &= \int_0^{2\pi} \int_0^5 \int_0^{r^2} r \, dz \, dr \, d\theta = \int_0^{2\pi} \int_0^5 r^3 \, dr \, d\theta = \int_0^{2\pi} \frac{1}{4}r^4 \Big|_0^5 d\theta \\ &= \int_0^{2\pi} \frac{625}{4} \, d\theta = \frac{625\pi}{2} \end{aligned}$$



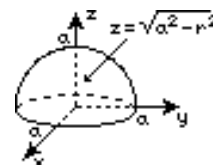
54. Substituting the first equation into the second, we see that the surfaces intersect in the plane $y = 4$. Using polar coordinates in the xz -plane, the equations of the surfaces become $y = r^2$ and $y = \frac{1}{2}r^2 + 2$.

$$\begin{aligned} V &= \int_0^{2\pi} \int_0^2 \int_{r^2}^{r^2/2+2} r \, dy \, dr \, d\theta = \int_0^{2\pi} \int_0^2 r \left(\frac{r^2}{2} + 2 - r^2 \right) \, dr \, d\theta \\ &= \int_0^{2\pi} \int_0^2 \left(2r - \frac{1}{2}r^3 \right) \, dr \, d\theta = \int_0^{2\pi} \left(r^2 - \frac{1}{8}r^4 \right) \Big|_0^2 d\theta = \int_0^{2\pi} 2 \, d\theta = 4\pi \end{aligned}$$



55. The equation is $z = \sqrt{a^2 - r^2}$. By symmetry, $\bar{x} = \bar{y} = 0$.

$$\begin{aligned} m &= \int_0^{2\pi} \int_0^a \int_0^{\sqrt{a^2-r^2}} r \, dz \, dr \, d\theta = \int_0^{2\pi} \int_0^a r \sqrt{a^2-r^2} \, dr \, d\theta \\ &= \int_0^{2\pi} -\frac{1}{3}(a^2-r^2)^{3/2} \Big|_0^a d\theta = \int_0^{2\pi} \frac{1}{3}a^3 \, d\theta = \frac{2}{3}\pi a^3 \end{aligned}$$

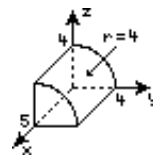


Exercises 9.15

$$\begin{aligned}
 M_{xy} &= \int_0^{2\pi} \int_0^a \int_0^{\sqrt{a^2-r^2}} zr \, dz \, dr \, d\theta = \int_0^{2\pi} \int_0^a \frac{1}{2}rz^2 \Big|_0^{\sqrt{a^2-r^2}} dr \, d\theta = \frac{1}{2} \int_0^{2\pi} \int_0^a r(a^2 - r^2) \, dr \, d\theta \\
 &= \frac{1}{2} \int_0^{2\pi} \left(\frac{1}{2}a^2r^2 - \frac{1}{4}r^4 \right) \Big|_0^a d\theta = \frac{1}{2} \int_0^{2\pi} \frac{1}{4}a^4 \, d\theta = \frac{1}{4}\pi a^4 \\
 \bar{z} &= M_{xy}/m = \frac{\pi a^4/4}{2\pi a^3/3} = 3a/8. \text{ The centroid is } (0, 0, 3a/8).
 \end{aligned}$$

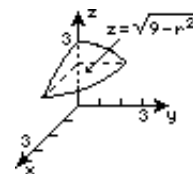
56. We use polar coordinates in the yz -plane. The density is $\rho(x, y, z) = kz$. By symmetry, $\bar{y} = \bar{z} = 0$.

$$\begin{aligned}
 m &= \int_0^{2\pi} \int_0^4 \int_0^5 kxr \, dx \, dr \, d\theta = k \int_0^{2\pi} \int_0^4 \frac{1}{2}rz^2 \Big|_0^5 dr \, d\theta = \frac{k}{2} \int_0^{2\pi} \int_0^4 25r \, dr \, d\theta \\
 &= \frac{25k}{2} \int_0^{2\pi} \frac{1}{2}r^2 \Big|_0^4 d\theta = \frac{25k}{2} \int_0^{2\pi} 8 \, d\theta = 200k\pi \\
 M_{yz} &= \int_0^{2\pi} \int_0^4 \int_0^5 kx^2r \, dx \, dr \, d\theta = k \int_0^{2\pi} \int_0^4 \frac{1}{3}rx^3 \Big|_0^5 dr \, d\theta = \frac{1}{3}k \int_0^{2\pi} \int_0^4 125r \, dr \, d\theta \\
 &= \frac{1}{3}k \int_0^{2\pi} \frac{125}{2}r^2 \Big|_0^4 d\theta = \frac{1}{3}k \int_0^{2\pi} 1000 \, d\theta = \frac{2000}{3}k\pi \\
 \bar{x} &= M_{yz}/m = \frac{2000k\pi/3}{200k\pi} = 10/3. \text{ The center of mass of the given solid is } (10/3, 0, 0).
 \end{aligned}$$



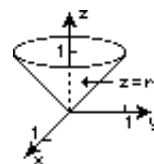
57. The equation is $z = \sqrt{9-r^2}$ and the density is $\rho = k/r^2$. When $z = 2$, $r = \sqrt{5}$.

$$\begin{aligned}
 I_z &= \int_0^{2\pi} \int_0^{\sqrt{5}} \int_2^{\sqrt{9-r^2}} r^2(k/r^2)r \, dz \, dr \, d\theta = k \int_0^{2\pi} \int_0^{\sqrt{5}} rz \Big|_2^{\sqrt{9-r^2}} dr \, d\theta \\
 &= k \int_0^{2\pi} \int_0^{\sqrt{5}} (r\sqrt{9-r^2} - 2r) \, dr \, d\theta = k \int_0^{2\pi} \left[-\frac{1}{3}(9-r^2)^{3/2} - r^2 \right] \Big|_0^{\sqrt{5}} d\theta \\
 &= k \int_0^{2\pi} \frac{4}{3} \, d\theta = \frac{8}{3}\pi k
 \end{aligned}$$



58. The equation is $z = r$ and the density is $\rho = kr$.

$$\begin{aligned}
 I_x &= \int_0^{2\pi} \int_0^1 \int_r^1 (y^2 + z^2)(kr)r \, dz \, dr \, d\theta = k \int_0^{2\pi} \int_0^1 \int_r^1 (r^4 \sin^2 \theta + r^2 z^2) \, dz \, dr \, d\theta \\
 &= k \int_0^{2\pi} \int_0^1 \left[(r^4 \sin^2 \theta)z + \frac{1}{3}r^2 z^3 \right] \Big|_r^1 dr \, d\theta \\
 &= k \int_0^{2\pi} \int_0^1 \left(r^4 \sin^2 \theta + \frac{1}{3}r^2 - r^5 \sin^2 \theta - \frac{1}{3}r^5 \right) dr \, d\theta \\
 &= k \int_0^{2\pi} \left(\frac{1}{5}r^5 \sin^2 \theta + \frac{1}{9}r^3 - \frac{1}{6}r^6 \sin^2 \theta - \frac{1}{18}r^6 \right) \Big|_0^1 d\theta = k \int_0^{2\pi} \left(\frac{1}{30} \sin^2 \theta + \frac{1}{18} \right) d\theta \\
 &= k \left(\frac{1}{60}\theta - \frac{1}{120} \sin 2\theta + \frac{1}{18}\theta \right) \Big|_0^{2\pi} = \frac{13}{90}\pi k
 \end{aligned}$$



59. (a) $x = (2/3) \sin(\pi/2) \cos(\pi/6) = \sqrt{3}/3$; $y = (2/3) \sin(\pi/2) \sin(\pi/6) = 1/3$;
 $z = (2/3) \cos(\pi/2) = 0$; $(\sqrt{3}/3, 1/3, 0)$

(b) With $x = \sqrt{3}/3$ and $y = 1/3$ we have $r^2 = 4/9$ and $\tan \theta = \sqrt{3}/3$. The point is $(2/3, \pi/6, 0)$.

60. (a) $x = 5 \sin(5\pi/4) \cos(2\pi/3) = 5\sqrt{2}/4$; $y = 5 \sin(5\pi/4) \sin(2\pi/3) = -5\sqrt{6}/4$;
 $z = 5 \cos(5\pi/4) = -5\sqrt{2}/2$; $(5\sqrt{2}/4, -5\sqrt{6}/4, -5\sqrt{2}/2)$

(b) With $x = 5\sqrt{2}/4$ and $y = -5\sqrt{6}/4$ we have $r^2 = 25/2$ and $\tan \theta = -\sqrt{3}$.

The point is $(5/\sqrt{2}, 2\pi/3, -5\sqrt{2}/2)$.

61. (a) $x = 8 \sin(\pi/4) \cos(3\pi/4) = -4$; $y = 8 \sin(\pi/4) \sin(3\pi/4) = 4$; $z = 8 \cos(\pi/4) = 4\sqrt{2}$;
 $(-4, 4, 4\sqrt{2})$

(b) With $x = -4$ and $y = 4$ we have $r^2 = 32$ and $\tan \theta = -1$. The point is $(4\sqrt{2}, 3\pi/4, 4\sqrt{2})$.

62. (a) $x = (1/3) \sin(5\pi/3) \cos(\pi/6) = -1/4$; $y = (1/3) \sin(5\pi/3) \sin(\pi/6) = -\sqrt{3}/12$;
 $z = (1/3) \cos(5\pi/3) = 1/6$; $(-1/4, -\sqrt{3}/12, 1/6)$

(b) With $x = -1/4$ and $y = -\sqrt{3}/12$ we have $r^2 = 1/12$ and $\tan \theta = \sqrt{3}/3$.

The point is $(1/2\sqrt{3}, \pi/6, 1/6)$.

63. With $x = -5$, $y = -5$, and $z = 0$, we have $\rho^2 = 50$, $\tan \theta = 1$, and $\cos \phi = 0$. The point is $(5\sqrt{2}, \pi/2, 5\pi/4)$.

64. With $x = 1$, $y = -\sqrt{3}$, and $z = 1$, we have $\rho^2 = 5$, $\tan \theta = -\sqrt{3}$, and $\cos \phi = 1/\sqrt{5}$.

The point is $(\sqrt{5}, \cos^{-1} 1/\sqrt{5}, -\pi/3)$.

65. With $x = \sqrt{3}/2$, $y = 1/2$, and $z = 1$, we have $\rho^2 = 2$, $\tan \theta = 1/\sqrt{3}$, and $\cos \phi = 1/\sqrt{2}$.

The point is $(\sqrt{2}, \pi/4, \pi/6)$.

66. With $x = -\sqrt{3}/2$, $y = 0$, and $z = -1/2$, we have $\rho^2 = 1$, $\tan \theta = 0$, and $\cos \phi = -1/2$.

The point is $(1, 2\pi/3, 0)$.

67. $\rho = 8$

68. $\rho^2 = 4\rho \cos \phi$; $\rho = 4 \cos \phi$

69. $4z^2 = 3x^2 + 3y^2 + 3z^2$; $4\rho^2 \cos^2 \phi = 3\rho^2$; $\cos \phi = \pm\sqrt{3}/2$; $\phi = \pi/6, 5\pi/6$

70. $-x^2 - y^2 - z^2 = 1 - 2z^2$; $-\rho^2 = 1 - 2\rho^2 \cos^2 \phi$; $\rho^2(2 \cos^2 \phi - 1) = 1$

71. $x^2 + y^2 + z^2 = 100$

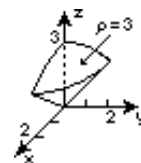
72. $\cos \phi = 1/2$; $\rho^2 \cos^2 \phi = \rho^2/4$; $4z^2 = x^2 + y^2 + z^2$; $x^2 + y^2 = 3z^2$

73. $\rho \cos \phi = 2$; $z = 2$

74. $\rho(1 - \cos^2 \phi) = \cos \phi$; $\rho^2 - \rho^2 \cos^2 \phi = \rho \cos \phi$; $x^2 + y^2 + z^2 - z^2 = z$; $z = x^2 + y^2$

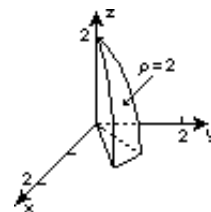
75. The equations are $\phi = \pi/4$ and $\rho = 3$.

$$\begin{aligned} V &= \int_0^{2\pi} \int_0^{\pi/4} \int_0^3 \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = \int_0^{2\pi} \int_0^{\pi/4} \frac{1}{3} \rho^3 \sin \phi \Big|_0^3 \, d\phi \, d\theta = \int_0^{2\pi} \int_0^{\pi/4} 9 \sin \phi \, d\phi \, d\theta \\ &= \int_0^{2\pi} -9 \cos \phi \Big|_0^{\pi/4} \, d\theta = -9 \int_0^{2\pi} \left(\frac{\sqrt{2}}{2} - 1 \right) d\theta = 9\pi(2 - \sqrt{2}) \end{aligned}$$



76. The equations are $\rho = 2$, $\theta = \pi/4$, and $\theta = \pi/3$.

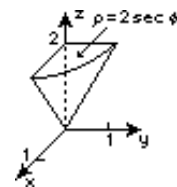
$$\begin{aligned} \int_{\pi/4}^{\pi/3} \int_0^{\pi/2} \int_0^2 \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta &= \int_{\pi/4}^{\pi/3} \int_0^{\pi/2} \frac{1}{3} \rho^3 \sin \phi \Big|_0^2 \, d\phi \, d\theta \\ &= \int_{\pi/4}^{\pi/3} \int_0^{\pi/2} \frac{8}{3} \sin \phi \, d\phi \, d\theta = \frac{8}{3} \int_{\pi/4}^{\pi/3} -\cos \phi \Big|_0^{\pi/2} \, d\theta \\ &= \frac{8}{3} \int_{\pi/4}^{\pi/3} (0 + 1) \, d\theta = \frac{2\pi}{9} \end{aligned}$$



Exercises 9.15

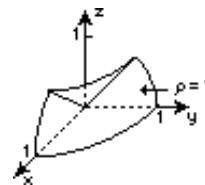
77. Using Problem 69, the equations are $\phi = \pi/6$, $\theta = \pi/2$, $\theta = 0$, and $\rho \cos \phi = 2$.

$$\begin{aligned} V &= \int_0^{\pi/2} \int_0^{\pi/6} \int_0^{2 \sec \phi} \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = \int_0^{\pi/2} \int_0^{\pi/6} \frac{1}{3} \rho^3 \sin \phi \Big|_0^{2 \sec \phi} d\phi \, d\theta \\ &= \frac{8}{3} \int_0^{\pi/2} \int_0^{\pi/6} \sec^3 \phi \sin \phi \, d\phi \, d\theta = \frac{8}{3} \int_0^{\pi/2} \int_0^{\pi/6} \sec^2 \phi \tan \phi \, d\phi \, d\theta \\ &= \frac{8}{3} \int_0^{\pi/2} \frac{1}{2} \tan^2 \phi \Big|_0^{\pi/6} d\theta = \frac{4}{3} \int_0^{\pi/2} \frac{1}{3} d\theta = \frac{2}{9} \pi \end{aligned}$$



78. The equations are $\rho = 1$ and $\phi = \pi/4$. We find the volume above the xy -plane and double.

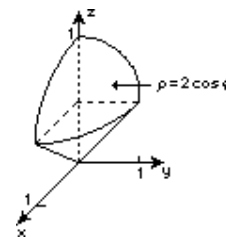
$$\begin{aligned} V &= 2 \int_0^{2\pi} \int_{\pi/4}^{\pi/2} \int_0^1 \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = 2 \int_0^{2\pi} \int_{\pi/4}^{\pi/2} \frac{1}{3} \rho^3 \sin \phi \Big|_0^1 d\phi \, d\theta \\ &= \frac{2}{3} \int_0^{2\pi} \int_{\pi/4}^{\pi/2} \sin \phi \, d\phi \, d\theta = \frac{2}{3} \int_0^{2\pi} -\cos \phi \Big|_{\pi/4}^{\pi/2} d\theta = \frac{2}{3} \int_0^{2\pi} \frac{\sqrt{2}}{2} d\theta = \frac{2\pi\sqrt{2}}{3} \end{aligned}$$



79. By symmetry, $\bar{x} = \bar{y} = 0$. The equations are $\phi = \pi/4$ and $\rho = 2 \cos \phi$.

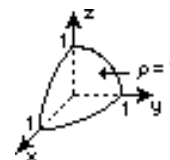
$$\begin{aligned} m &= \int_0^{2\pi} \int_0^{\pi/4} \int_0^{2 \cos \phi} \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = \int_0^{2\pi} \int_0^{\pi/4} \frac{1}{3} \rho^3 \sin \phi \Big|_0^{2 \cos \phi} d\phi \, d\theta \\ &= \frac{8}{3} \int_0^{2\pi} \int_0^{\pi/4} \sin \phi \cos^3 \phi \, d\phi \, d\theta = \frac{8}{3} \int_0^{2\pi} -\frac{1}{4} \cos^4 \phi \Big|_0^{\pi/4} d\theta \\ &= -\frac{2}{3} \int_0^{2\pi} \left(\frac{1}{4} - 1 \right) d\theta = \pi \\ M_{xy} &= \int_0^{2\pi} \int_0^{\pi/4} \int_0^{2 \cos \phi} z \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = \int_0^{2\pi} \int_0^{\pi/4} \int_0^{2 \cos \phi} \rho^3 \sin \phi \cos \phi \, d\rho \, d\phi \, d\theta \\ &= \int_0^{2\pi} \int_0^{\pi/4} \frac{1}{4} \rho^4 \sin \phi \cos \phi \Big|_0^{2 \cos \phi} d\phi \, d\theta = 4 \int_0^{2\pi} \int_0^{\pi/4} \cos^5 \phi \sin \phi \, d\phi \, d\theta \\ &= 4 \int_0^{2\pi} -\frac{1}{6} \cos^6 \phi \Big|_0^{\pi/4} d\theta = -\frac{2}{3} \int_0^{2\pi} \left(\frac{1}{8} - 1 \right) d\theta = \frac{7}{6} \pi \end{aligned}$$

$$\bar{z} = M_{xy}/m = \frac{7\pi/6}{\pi} = 7/6. \text{ The centroid is } (0, 0, 7/6).$$



80. We are given density $= kz$. By symmetry, $\bar{x} = \bar{y} = 0$. The equation is $\rho = 1$.

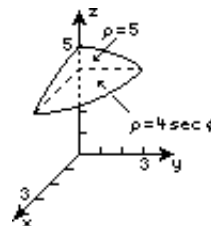
$$\begin{aligned} m &= \int_0^{2\pi} \int_0^{\pi/2} \int_0^1 kz \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = k \int_0^{2\pi} \int_0^{\pi/2} \int_0^1 \rho^3 \sin \phi \cos \phi \, d\rho \, d\phi \, d\theta \\ &= k \int_0^{2\pi} \int_0^{\pi/2} \frac{1}{4} \rho^4 \sin \phi \cos \phi \Big|_0^1 d\phi \, d\theta = \frac{1}{4} k \int_0^{2\pi} \int_0^{\pi/2} \sin \phi \cos \phi \, d\phi \, d\theta \\ &= \frac{1}{4} k \int_0^{2\pi} \frac{1}{2} \sin^2 \phi \Big|_0^{\pi/2} d\theta = \frac{1}{8} k \int_0^{2\pi} d\theta = \frac{k\pi}{4} \\ M_{xy} &= \int_0^{2\pi} \int_0^{\pi/2} \int_0^1 kz^2 \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = k \int_0^{2\pi} \int_0^{\pi/2} \int_0^1 \rho^4 \cos^2 \phi \sin \phi \, d\rho \, d\phi \, d\theta \\ &= k \int_0^{2\pi} \int_0^{\pi/2} \frac{1}{5} \rho^5 \cos^2 \phi \sin \phi \Big|_0^1 d\phi \, d\theta = \frac{1}{5} k \int_0^{2\pi} \int_0^{\pi/2} \cos^2 \phi \sin \phi \, d\phi \, d\theta \\ &= \frac{1}{5} k \int_0^{2\pi} -\frac{1}{3} \cos^3 \phi \Big|_0^{\pi/2} d\theta = -\frac{1}{15} k \int_0^{2\pi} (0 - 1) d\theta = \frac{2}{15} k\pi \end{aligned}$$



$\bar{z} = M_{xy}/m = \frac{2k\pi/15}{k\pi/4} = 8/15$. The center of mass is $(0, 0, 8/15)$.

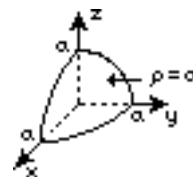
81. We are given density $= k/\rho$.

$$\begin{aligned} m &= \int_0^{2\pi} \int_0^{\cos^{-1} 4/5} \int_{4 \sec \phi}^5 \frac{k}{\rho} \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = k \int_0^{2\pi} \int_0^{\cos^{-1} 4/5} \frac{1}{2} \rho^2 \sin \phi \bigg|_{4 \sec \phi}^5 \, d\phi \, d\theta \\ &= \frac{1}{2} k \int_0^{2\pi} \int_0^{\cos^{-1} 4/5} (25 \sin \phi - 16 \tan \phi \sec \phi) \, d\phi \, d\theta \\ &= \frac{1}{2} k \int_0^{2\pi} (-25 \cos \phi - 16 \sec \phi) \bigg|_0^{\cos^{-1} 4/5} \, d\theta = \frac{1}{2} k \int_0^{2\pi} [-25(4/5) - 16(5/4) - (-25 - 16)] \, d\theta \\ &= \frac{1}{2} k \int_0^{2\pi} d\theta = k\pi \end{aligned}$$



82. We are given density $= k\rho$.

$$\begin{aligned} I_z &= \int_0^{2\pi} \int_0^\pi \int_0^a (x^2 + y^2)(k\rho) \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta \\ &= k \int_0^{2\pi} \int_0^\pi \int_0^a (\rho^2 \sin^2 \phi \cos^2 \theta + \rho^2 \sin^2 \phi \sin^2 \theta) \rho^3 \sin \phi \, d\rho \, d\phi \, d\theta \\ &= k \int_0^{2\pi} \int_0^\pi \int_0^a \rho^5 \sin^3 \phi \, d\rho \, d\phi \, d\theta = k \int_0^{2\pi} \int_0^\pi \frac{1}{6} \rho^6 \sin^3 \phi \bigg|_0^a \, d\phi \, d\theta = \frac{1}{6} k a^6 \int_0^{2\pi} \int_0^\pi \sin^3 \phi \, d\phi \, d\theta \\ &= \frac{1}{6} k a^3 \int_0^{2\pi} \int_0^\pi (1 - \cos^2 \phi) \sin \phi \, d\phi \, d\theta = \frac{1}{6} k a^3 \int_0^{2\pi} \left(-\cos \phi + \frac{1}{3} \cos^3 \phi \right) \bigg|_0^\pi \, d\theta = \frac{1}{6} k a^3 \int_0^{2\pi} \frac{4}{3} \, d\theta = \frac{4\pi}{9} k a^6 \end{aligned}$$

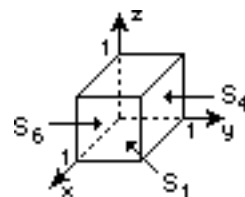


Exercises 9.16

1. $\text{div } \mathbf{F} = y + z + x$

The Triple Integral:

$$\begin{aligned} \iiint_D \text{div } \mathbf{F} \, dV &= \int_0^1 \int_0^1 \int_0^1 (x + y + z) \, dx \, dy \, dz \\ &= \int_0^1 \int_0^1 \left(\frac{1}{2} x^2 + xy + xz \right) \bigg|_0^1 \, dy \, dz \\ &= \int_0^1 \int_0^1 \left(\frac{1}{2} + y + z \right) \, dy \, dz = \int_0^1 \left(\frac{1}{2} y + \frac{1}{2} y^2 + yz \right) \bigg|_0^1 \, dz \\ &= \int_0^1 (1 + z) \, dz = \frac{1}{2} (1 + z^2) \bigg|_0^1 = 2 - \frac{1}{2} = \frac{3}{2} \end{aligned}$$



The Surface Integral: Let the surfaces be S_1 in $z = 0$, S_2 in $z = 1$, S_3 in $y = 0$, S_4 in $y = 1$, S_5 in $x = 0$, and S_6 in $x = 1$. The unit outward normal vectors are $-\mathbf{k}$, $\mathbf{k}$, $-\mathbf{j}$, $\mathbf{j}$, $-\mathbf{i}$ and $\mathbf{i}$, respectively. Then

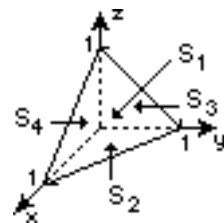
$$\begin{aligned} \iint_S \mathbf{F} \cdot \mathbf{n} \, dS &= \iint_{S_1} \mathbf{F} \cdot (-\mathbf{k}) \, dS_1 + \iint_{S_2} \mathbf{F} \cdot \mathbf{k} \, dS_2 + \iint_{S_3} \mathbf{F} \cdot (-\mathbf{j}) \, dS_3 + \iint_{S_4} \mathbf{F} \cdot \mathbf{j} \, dS_4 \\ &\quad + \iint_{S_5} \mathbf{F} \cdot (-\mathbf{i}) \, dS_5 + \iint_{S_6} \mathbf{F} \cdot \mathbf{i} \, dS_6 \end{aligned}$$

$$\begin{aligned}
&= \iint_{S_1} (-xz) dS_1 + \iint_{S_2} xz dS_2 + \iint_{S_3} (-yz) dS_3 + \iint_{S_4} yz dS_4 \\
&\quad + \iint_{S_5} (-xy) dS_5 + \iint_{S_6} xy dS_6 \\
&= \iint_{S_2} x dS_2 + \iint_{S_4} z dS_4 + \iint_{S_6} y dS_6 \\
&= \int_0^1 \int_0^1 x dx dy + \int_0^1 \int_0^1 z dz dx + \int_0^1 \int_0^1 y dy dz \\
&= \int_0^1 \frac{1}{2} dy + \int_0^1 \frac{1}{2} dx + \int_0^1 \frac{1}{2} dz = \frac{3}{2}.
\end{aligned}$$

2. $\operatorname{div} \mathbf{F} = 6y + 4z$

The Triple Integral:

$$\begin{aligned}
\iiint_D \operatorname{div} \mathbf{F} dV &= \int_0^1 \int_0^{1-x} \int_0^{1-x-y} (6y + 4z) dz dy dx \\
&= \int_0^1 \int_0^{1-x} (6yz + 2z^2) \Big|_0^{1-x-y} dy dx \\
&= \int_0^1 \int_0^{1-x} (-4y^2 + 2y - 2xy + 2x^2 - 4x + 2) dy dx \\
&= \int_0^1 \left(-\frac{4}{3}y^3 + y^2 - xy^2 + 2x^2y - 4xy + 2y \right) \Big|_0^{1-x} dx \\
&= \int_0^1 \left(-\frac{5}{3}x^3 + 5x^2 - 5x + \frac{5}{3} \right) dx = \left(-\frac{5}{12}x^4 + \frac{5}{3}x^3 - \frac{5}{2}x^2 + \frac{5}{3}x \right) \Big|_0^1 = \frac{5}{12}
\end{aligned}$$



The Surface Integral: Let the surfaces be S_1 in the plane $x + y + z = 1$, S_2 in $z = 0$, S_3 in $x = 0$, and S_4 in $y = 0$. The unit outward normal vectors are $\mathbf{n}_1 = (\mathbf{i} + \mathbf{j} + \mathbf{k})/\sqrt{3}$, $\mathbf{n}_2 = -\mathbf{k}$, $\mathbf{n}_3 = -\mathbf{i}$, and $\mathbf{n}_4 = -\mathbf{j}$, respectively. Now on S_1 , $dS_1 = \sqrt{3} dA_1$, on S_3 , $x = 0$, and on S_4 , $y = 0$, so

$$\begin{aligned}
\iint_S \mathbf{F} \cdot \mathbf{n} dS &= \iint_{S_1} \mathbf{F} \cdot \mathbf{n}_1 dS_1 + \iint_{S_2} \mathbf{F} \cdot (-\mathbf{k}) dS_2 + \iint_{S_3} \mathbf{F} \cdot (-\mathbf{i}) dS_3 + \iint_{S_4} \mathbf{F} \cdot (-\mathbf{j}) dS_4 \\
&= \int_0^1 \int_0^{1-x} (6xy + 4y(1-x-y) + xe^{-y}) dy dx + \int_0^1 \int_0^{1-x} (-xe^{-y}) dy dx \\
&\quad + \iint_{S_3} (-6xy) dS_3 + \iint_{S_4} (-4yz) dS_4 \\
&= \int_0^1 \left(xy^2 + 2y^2 - \frac{4}{3}y^3 - xe^{-y} \right) \Big|_0^{1-x} dx + \int_0^1 xe^{-y} \Big|_0^{1-x} dx + 0 + 0 \\
&= \int_0^1 [x(1-x)^2 + 2(1-x)^2 - \frac{4}{3}(1-x)^3 - xe^{x-1} + x] dx + \int_0^1 (xe^{x-1} - x) dx \\
&= \left[\frac{1}{2}x^2 - \frac{2}{3}x^3 + \frac{1}{4}x^4 - \frac{2}{3}(1-x)^3 + \frac{1}{3}(1-x)^4 \right] \Big|_0^1 = \frac{5}{12}.
\end{aligned}$$

3. $\operatorname{div} \mathbf{F} = 3x^2 + 3y^2 + 3z^2$. Using spherical coordinates,

$$\begin{aligned}\iint_S \mathbf{F} \cdot \mathbf{n} dS &= \iiint_D 3(x^2 + y^2 + z^2) dV = \int_0^{2\pi} \int_0^\pi \int_0^a 3\rho^2 \rho^2 \sin \phi d\rho d\phi d\theta \\ &= \int_0^{2\pi} \int_0^\pi \left. \frac{3}{5} \rho^5 \sin \phi \right|_0^a d\phi d\theta = \frac{3a^5}{5} \int_0^{2\pi} \int_0^\pi \sin \phi d\phi d\theta \\ &= \frac{3a^5}{5} \int_0^{2\pi} -\cos \phi \Big|_0^\pi d\theta = \frac{6a^5}{5} \int_0^{2\pi} d\theta = \frac{12\pi a^5}{5}.\end{aligned}$$

4. $\operatorname{div} \mathbf{F} = 4 + 1 + 4 = 9$. Using the formula for the volume of a sphere,

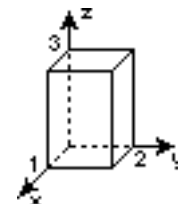
$$\iint_S \mathbf{F} \cdot \mathbf{n} dS = \iiint_D 9 dV = 9 \left(\frac{4}{3} \pi 2^3 \right) = 96\pi.$$

5. $\operatorname{div} \mathbf{F} = 2(z - 1)$. Using cylindrical coordinates,

$$\begin{aligned}\iint_S \mathbf{F} \cdot \mathbf{n} dS &= \iiint_D 2(z - 1) dV = \int_0^{2\pi} \int_0^4 \int_1^5 2(z - 1) dz r dr d\theta = \int_0^{2\pi} \int_0^4 (z - 1)^2 \Big|_1^5 r dr d\theta \\ &= \int_0^{2\pi} \int_0^4 16r dr d\theta = \int_0^{2\pi} 8r^2 \Big|_0^4 d\theta = 128 \int_0^{2\pi} d\theta = 256\pi.\end{aligned}$$

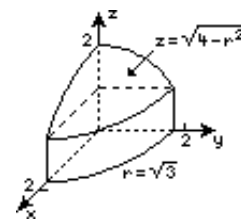
6. $\operatorname{div} \mathbf{F} = 2x + 2z + 12z^2$.

$$\begin{aligned}\iint_S \mathbf{F} \cdot \mathbf{n} dS &= \iiint_D \operatorname{div} \mathbf{F} dV = \int_0^3 \int_0^2 \int_0^1 (2x + 2z + 12xz^2) dx dy dz \\ &= \int_0^3 \int_0^2 (x^2 + 2xz + 12xz^2) \Big|_0^1 dy dz = \int_0^3 \int_0^2 (1 + 2z + 12xz^2) dy dz \\ &= \int_0^3 2(1 + 2z + 12z^2) dz = (2z + 2z^2 + 8z^3) \Big|_0^3 = 240\end{aligned}$$



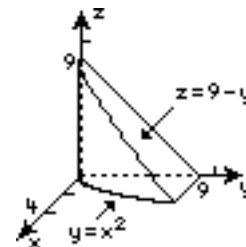
7. $\operatorname{div} \mathbf{F} = 3z^2$. Using cylindrical coordinates,

$$\begin{aligned}\iint_S \mathbf{F} \cdot \mathbf{n} dS &= \iiint_D \operatorname{div} \mathbf{F} dV = \int_0^{2\pi} \int_0^{\sqrt{3}} \int_0^{\sqrt{4-r^2}} 3z^2 r dz dr d\theta \\ &= \int_0^{2\pi} \int_0^{\sqrt{3}} r z^3 \Big|_0^{\sqrt{4-r^2}} dr d\theta = \int_0^{2\pi} \int_0^{\sqrt{3}} r(4 - r^2)^{3/2} dr d\theta \\ &= \int_0^{2\pi} -\frac{1}{5}(4 - r^2)^{5/2} \Big|_0^{\sqrt{3}} d\theta = \int_0^{2\pi} -\frac{1}{5}(1 - 32) d\theta = \int_0^{2\pi} \frac{31}{5} d\theta = \frac{62\pi}{5}.\end{aligned}$$



8. $\operatorname{div} \mathbf{F} = 2x$.

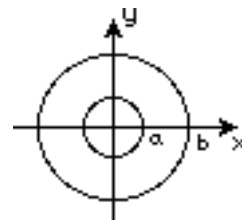
$$\begin{aligned}\iint_S \mathbf{F} \cdot \mathbf{n} dS &= \iiint_D \operatorname{div} \mathbf{F} dV = \int_0^3 \int_{x^2}^9 \int_0^{9-y} 2x dz dy dx \\ &= \int_0^3 \int_{x^2}^9 2x(9 - y) dy dx = \int_0^3 -x(9 - y)^2 \Big|_{x^2}^9 dx = \int_0^3 x(9 - x)^2 dx \\ &= \int_0^3 (x^3 - 18x^2 + 81x) dx = \left(\frac{1}{4}x^4 - 6x^3 + \frac{81}{2}x^2 \right) \Big|_0^3 = \frac{891}{4}\end{aligned}$$



Exercises 9.16

9. $\operatorname{div} \mathbf{F} = \frac{1}{x^2 + y^2 + z^2}$. Using spherical coordinates,

$$\begin{aligned}\iint_S \mathbf{F} \cdot \mathbf{n} dS &= \iiint_D \operatorname{div} \mathbf{F} dV = \int_0^{2\pi} \int_0^\pi \int_a^b \frac{1}{\rho^2} \rho^2 \sin \phi d\rho d\phi d\theta \\ &= \int_0^{2\pi} \int_0^\pi (b-a) \sin \phi d\phi d\theta = (b-a) \int_0^{2\pi} -\cos \phi \Big|_0^\pi d\theta \\ &= (b-a) \int_0^{2\pi} 2 d\theta = 4\pi(b-a).\end{aligned}$$



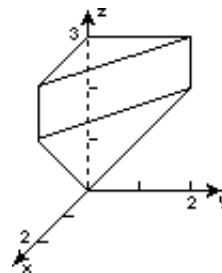
10. Since $\operatorname{div} \mathbf{F} = 0$, $\iint_S \mathbf{F} \cdot \mathbf{n} dS = \iiint_D 0 dV = 0$.

11. $\operatorname{div} \mathbf{F} = 2z + 10y - 2z = 10y$.

$$\begin{aligned}\iint_S \mathbf{F} \cdot \mathbf{n} dS &= \iiint_D 10y dV = \int_0^2 \int_0^{2-x^2/2} \int_z^{4-z} 10y dy dz dx = \int_0^2 \int_0^{2-x^2/2} 5y^2 \Big|_z^{4-z} dz dx \\ &= \int_0^2 \int_0^{2-x^2/2} (80 - 40z) dz dx = \int_0^2 (80z - 20z^2) \Big|_0^{2-x^2/2} dx = \int_0^2 (80 - 5x^4) dx = (80x - x^5) \Big|_0^2 = 128\end{aligned}$$

12. $\operatorname{div} \mathbf{F} = 30xy$.

$$\begin{aligned}\iint_S \mathbf{F} \cdot \mathbf{n} dS &= \iiint_D 30xy dV = \int_0^2 \int_0^{2-x} \int_{x+y}^3 30xy dz dy dx \\ &= \int_0^2 \int_0^{2-x} 30xyz \Big|_{x+y}^3 dy dx \\ &= \int_0^2 \int_0^{2-x} (90xy - 30x^2y - 30xy^2) dy dx \\ &= \int_0^2 (45xy^2 - 15x^2y^2 - 10xy^3) \Big|_0^{2-x} dx \\ &= \int_0^2 (-5x^4 + 45x^3 - 120x^2 + 100x) dx = \left(-x^5 + \frac{45}{4}x^4 - 40x^3 + 50x^2 \right) \Big|_0^2 = 28\end{aligned}$$



13. $\operatorname{div} \mathbf{F} = 6xy^2 + 1 - 6xy^2 = 1$. Using cylindrical coordinates,

$$\begin{aligned}\iint_S \mathbf{F} \cdot \mathbf{n} dS &= \iiint_D dV = \int_0^\pi \int_0^{2\sin\theta} \int_{r^2}^{2r\sin\theta} dz r dr d\theta = \int_0^\pi \int_0^{2\sin\theta} (2r\sin\theta - r^2) r dr d\theta \\ &= \int_0^\pi \left(\frac{2}{3}r^3 \sin\theta - \frac{1}{4}r^4 \right) \Big|_0^{2\sin\theta} d\theta = \int_0^\pi \left(\frac{16}{3} \sin^4\theta - 4 \sin^4\theta \right) d\theta \\ &= \frac{4}{3} \int_0^\pi \sin^4\theta d\theta = \frac{4}{3} \left(\frac{3}{8}\theta - \frac{1}{4}\sin 2\theta + \frac{1}{32}\sin 4\theta \right) \Big|_0^\pi = \frac{\pi}{2}\end{aligned}$$

14. $\operatorname{div} \mathbf{F} = y^2 + x^2$. Using spherical coordinates, we have $x^2 + y^2 = \rho^2 \sin^2 \phi$ and $z = \rho \cos \phi$ or $\rho = z \sec \phi$. Then

$$\begin{aligned}\iint_S \mathbf{F} \cdot \mathbf{n} dS &= \iiint_D (x^2 + y^2) dS = \int_0^{2\pi} \int_0^{\pi/4} \int_{2\sec\phi}^{4\sec\phi} \rho^2 \sin^2 \phi \rho^2 \sin \phi d\rho d\phi d\theta \\ &= \int_0^{2\pi} \int_0^{\pi/4} \frac{1}{5} \rho^5 \sin^3 \phi \Big|_{2\sec\phi}^{4\sec\phi} d\phi d\theta = \int_0^{2\pi} \int_0^{\pi/4} \frac{992}{5} \sec^5 \phi \sin^3 \phi d\phi d\theta \\ &= \int_0^{2\pi} \int_0^{\pi/4} \frac{992}{5} \tan^3 \phi \sec^2 \phi d\phi d\theta = \frac{992}{5} \int_0^{2\pi} \frac{1}{4} \tan^4 \phi \Big|_0^{\pi/4} d\theta = \frac{992}{5} \int_0^{2\pi} \frac{1}{4} d\theta = \frac{496\pi}{5}.\end{aligned}$$

15. (a) $\operatorname{div} \mathbf{E} = q \left[\frac{-2x^2 + y^2 + z^2}{(x^2 + y^2 + z^2)^{5/2}} + \frac{x^2 - 2y^2 + z^2}{(x^2 + y^2 + z^2)^{5/2}} + \frac{x^2 + y^2 - 2z^2}{(x^2 + y^2 + z^2)^{5/2}} \right] = 0$

$$\iint_{S \cup S_a} (\mathbf{E} \cdot \mathbf{n}) dS = \iiint_D \operatorname{div} \mathbf{E} dV = \iiint_D 0 dV = 0$$

(b) From (a), $\iint_S (\mathbf{E} \cdot \mathbf{n}) dS + \iint_{S_a} (\mathbf{E} \cdot \mathbf{n}) dS = 0$ and $\iint_S (\mathbf{E} \cdot \mathbf{n}) dS = -\iint_{S_a} (\mathbf{E} \cdot \mathbf{n}) dS$. On S_a ,

$|\mathbf{r}| = a$, $\mathbf{n} = -(\mathbf{x}\mathbf{i} + \mathbf{y}\mathbf{j} + \mathbf{z}\mathbf{k})/a = -\mathbf{r}/a$ and $\mathbf{E} \cdot \mathbf{n} = (q\mathbf{r}/a^3) \cdot (-\mathbf{r}/a) = -qa^2/a^4 = -q/a^2$. Thus

$$\iint_S (\mathbf{E} \cdot \mathbf{n}) dS = -\iint_{S_a} \left(-\frac{q}{a^2}\right) dS = \frac{q}{a^2} \iint_{S_a} dS = \frac{q}{a^2} \times (\text{area of } S_a) = \frac{q}{a^2} (4\pi a^2) = 4\pi q.$$

16. (a) By Gauss' Law $\iint_S (\mathbf{E} \cdot \mathbf{n}) dS = \iiint_D 4\pi\rho dV$, and by the Divergence Theorem

$$\iint_S (\mathbf{E} \cdot \mathbf{n}) dS = \iiint_D \operatorname{div} \mathbf{E} dV. \text{ Thus } \iiint_D 4\pi\rho dV = \iiint_D \operatorname{div} \mathbf{E} dV \text{ and } \iiint_D (4\pi\rho - \operatorname{div} \mathbf{E}) dV = 0.$$

Since this holds for all regions D , $4\pi\rho - \operatorname{div} \mathbf{E} = 0$ and $\operatorname{div} \mathbf{E} = 4\pi\rho$.

(b) Since $\mathbf{E}$ is irrotational, $\mathbf{E} = \nabla\phi$ and $\nabla^2\phi = \nabla \cdot \nabla\phi = \nabla\mathbf{E} = \operatorname{div} \mathbf{E} = 4\pi\rho$.

17. Since $\operatorname{div} \mathbf{a} = 0$, by the Divergence Theorem

$$\iint_S (\mathbf{a} \cdot \mathbf{n}) dS = \iiint_D \operatorname{div} \mathbf{a} dV = \iiint_D 0 dV = 0.$$

18. By the Divergence Theorem and Problem 30 in Section 9.7,

$$\iint_S (\operatorname{curl} \mathbf{F} \cdot \mathbf{n}) dS = \iiint_D \operatorname{div} (\operatorname{curl} \mathbf{F}) dV = \iiint_D 0 dV = 0.$$

19. By the Divergence Theorem and Problem 27 in Section 9.7,

$$\begin{aligned} \iint_S (f\nabla g) \cdot \mathbf{n} dS &= \iiint_D \operatorname{div} (f\nabla g) dV = \iiint_D \nabla \cdot (f\nabla g) dV = \iiint_D [f(\nabla \cdot \nabla g) + \nabla g \cdot \nabla f] dV \\ &= \iiint_D (f\nabla^2 g + \nabla g \cdot \nabla f) dV. \end{aligned}$$

20. By the Divergence Theorem and Problems 25 and 27 in Section 8.7,

$$\begin{aligned} \iint_S (f\nabla g - g\nabla f) \cdot \mathbf{n} dS &= \iiint_D \operatorname{div} (f\nabla g - g\nabla f) dV = \iiint_D \nabla \cdot (f\nabla g - g\nabla f) dV \\ &= \iiint_D [f(\nabla \cdot \nabla g) + \nabla g \cdot \nabla f - g(\nabla \cdot \nabla f) - \nabla f \cdot \nabla g] dV \\ &= \iiint_D (f\nabla^2 g - g\nabla^2 f) dV. \end{aligned}$$

21. If $G(x, y, z)$ is a vector valued function then we define surface integrals and triple integrals of $\mathbf{G}$ component-wise.

In this case, if $\mathbf{a}$ is a constant vector it is easily shown that

$$\iint_S \mathbf{a} \cdot \mathbf{G} dS = \mathbf{a} \cdot \iint_S \mathbf{G} dS \quad \text{and} \quad \iiint_D \mathbf{a} \cdot \mathbf{G} dV = \mathbf{a} \cdot \iiint_D \mathbf{G} dV.$$

Now let $\mathbf{F} = f\mathbf{a}$. Then

$$\iint_S \mathbf{F} \cdot \mathbf{n} dS = \iint_S (f\mathbf{a}) \cdot \mathbf{n} dS = \iint_S \mathbf{a} \cdot (f\mathbf{n}) dS$$

and, using Problem 27 in Section 8.7 and the fact that $\nabla \cdot \mathbf{a} = 0$, we have

$$\iiint_D \operatorname{div} \mathbf{F} dV = \iiint_D \nabla \cdot (f\mathbf{a}) dV = \iiint_D [f(\nabla \cdot \mathbf{a}) + \mathbf{a} \cdot \nabla f] dV = \iiint_D \mathbf{a} \cdot \nabla f dV.$$

Exercises 9.16

By the Divergence Theorem,

$$\iint_S \mathbf{a} \cdot (f\mathbf{n}) dS = \iint_S \mathbf{F} \cdot \mathbf{n} dS = \iiint_D \operatorname{div} \mathbf{F} dV = \iiint_D \mathbf{a} \cdot \nabla f dV$$

and

$$\mathbf{a} \cdot \left(\iint_S f\mathbf{n} dS \right) = \mathbf{a} \cdot \left(\iiint_D \nabla f dV \right) \quad \text{or} \quad \mathbf{a} \cdot \left(\iint_S f\mathbf{n} dS - \iiint_D \nabla f dV \right) = 0.$$

Since $\mathbf{a}$ is arbitrary,

$$\iint_S f\mathbf{n} dS - \iiint_D \nabla f dV = 0 \quad \text{and} \quad \iint_S f\mathbf{n} dS = \iiint_D \nabla f dV.$$

$$\begin{aligned} 22. \quad \mathbf{B} + \mathbf{W} &= - \iint_S p\mathbf{n} dS + m\mathbf{g} = m\mathbf{g} - \iiint_D \nabla p dV = m\mathbf{g} - \iiint_D \rho\mathbf{g} dV \\ &= m\mathbf{g} - \left(\iiint_D \rho dV \right) \mathbf{g} = m\mathbf{g} - m\mathbf{g} = \mathbf{0} \end{aligned}$$

Exercises 9.17

1. $T: (0,0) \rightarrow (0,0); (0,2) \rightarrow (-2,8); (4,0) \rightarrow (16,20); (4,2) \rightarrow (14,28)$
2. Writing $x^2 = v - u$ and $y = v + u$ and solving for u and v , we obtain $u = (y - x^2)/2$ and $v = (x^2 + y)/2$. Then the images under T^{-1} are $(1,1) \rightarrow (0,1); (1,3) \rightarrow (1,2); (\sqrt{2},2) \rightarrow (0,2)$.

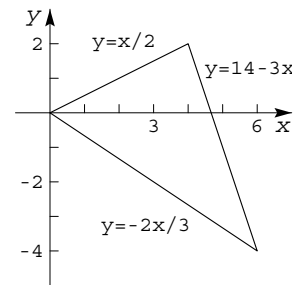
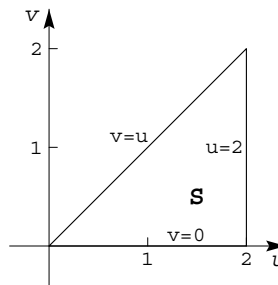
3. The uv -corner points $(0,0), (2,0), (2,2)$ correspond to xy -points $(0,0), (4,2), (6,-4)$.

$$v = 0: x = 2u, y = u \implies y = x/2$$

$$u = 2: x = 4 + v, y = 2 - 3v \implies$$

$$y = 2 - 3(x - 4) = -3x + 14$$

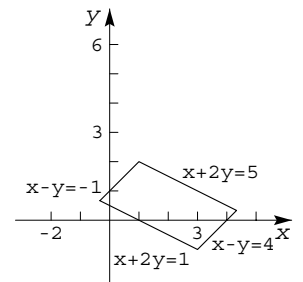
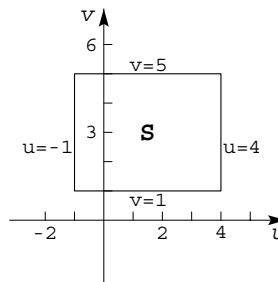
$$v = u: x = 3u, y = -2u \implies y = -2x/3$$



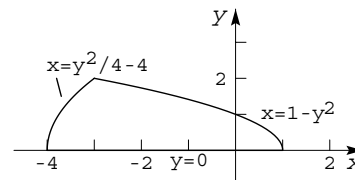
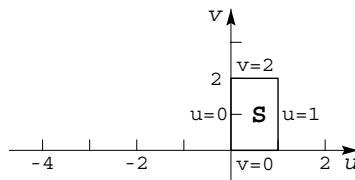
4. Solving for x and y we see that the transformation is $x = 2u/3 + v/3, y = -u/3 + v/3$. The uv -corner points $(-1,1), (4,1), (4,5), (-1,5)$ correspond to the xy -points $(-1/3, 2/3), (3, -1), (13/3, 1/3), (1, 2)$.

$$v = 1: x + 2y = 1; \quad v = 5: x + 2y = 5;$$

$$u = -1: x - y = -1; \quad u = 4: x - y = 4$$



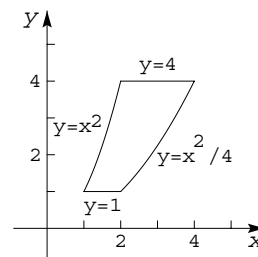
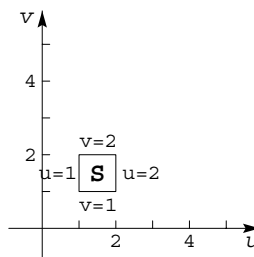
5. The uv -corner points $(0, 0)$, $(1, 0)$, $(1, 2)$, $(0, 2)$ correspond to the xy -points $(0, 0)$, $(1, 0)$, $(-3, 2)$, $(-4, 0)$.



$$\begin{aligned} v = 0: x = u^2, y = 0 &\implies y = 0 \text{ and } 0 \leq x \leq 1 \\ u = 1: x = 1 - v^2, y = v &\implies x = 1 - y^2 \\ v = 2: x = u^2 - 4, y = 2u &\implies x = y^2/4 - 4 \\ u = 0: x = -v^2, y = 0 &\implies y = 0 \text{ and } -4 \leq x \leq 0 \end{aligned}$$

6. The uv -corner points $(1, 1)$, $(2, 1)$, $(2, 2)$, $(1, 2)$ correspond to the xy -points $(1, 1)$, $(2, 1)$, $(4, 4)$, $(2, 4)$.

$$\begin{aligned} v = 1: x = u, y = 1 &\implies y = 1, 1 \leq x \leq 2 \\ u = 2: x = 2v, y = v^2 &\implies y = x^2/4 \\ v = 2: x = 2u, y = 4 &\implies y = 4, 2 \leq x \leq 4 \\ u = 1: x = v, y = v^2 &\implies y = x^2 \end{aligned}$$



7. $\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} -ve^{-u} & e^{-u} \\ ve^u & e^u \end{vmatrix} = -2v$

8. $\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} 3e^{3u} \sin v & e^{3u} \cos v \\ 3e^{3u} \cos v & -e^{3u} \sin v \end{vmatrix} = -3e^{6u}$

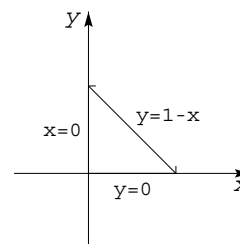
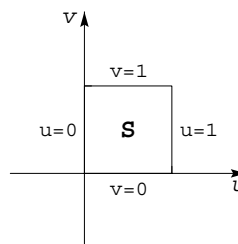
9. $\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} -2y/x^3 & 1/x^2 \\ -y^2/x^2 & 2y/x \end{vmatrix} = -\frac{3y^2}{x^4} = -3\left(\frac{y}{x^2}\right)^2 = -3u^2; \quad \frac{\partial(x, y)}{\partial(u, v)} = \frac{1}{-3u^2} = -\frac{1}{3u^2}$

10. $\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{2(y^2 - x^2)}{(x^2 + y^2)^2} & \frac{-4xy}{(x^2 + y^2)^2} \\ \frac{4xy}{(x^2 + y^2)^2} & \frac{2(y^2 - x^2)}{(x^2 + y^2)^2} \end{vmatrix} = \frac{4}{(x^2 + y^2)^2}$

From $u = 2x/(x^2 + y^2)$ and $v = -2y/(x^2 + y^2)$ we obtain $u^2 + v^2 = 4/(x^2 + y^2)$. Then $x^2 + y^2 = 4/(u^2 + v^2)$ and $\partial(x, y)/\partial(u, v) = (x^2 + y^2)^2/4 = 4/(u^2 + v^2)^2$.

11. (a) The uv -corner points $(0, 0)$, $(1, 0)$, $(1, 1)$, $(0, 1)$ correspond to the xy -points $(0, 0)$, $(1, 0)$, $(0, 1)$, $(0, 0)$.

$$\begin{aligned} v = 0: x = u, y = 0 &\implies y = 0, 0 \leq x \leq 1 \\ u = 1: x = 1 - v, y = v &\implies y = 1 - x \\ v = 1: x = 0, y = u &\implies x = 0, 0 \leq y \leq 1 \\ u = 0: x = 0, y = 0 &\end{aligned}$$



- (b) Since the segment $u = 0, 0 \leq v \leq 1$ in the uv -plane maps to the origin in the xy -plane, the transformation is not one-to-one.

12. $\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} 1 - v & v \\ -u & u \end{vmatrix} = u$. The transformation is 0 when u is 0, for $0 \leq v \leq 1$.

Exercises 9.17

13. $R1: x + y = -1 \implies v = -1$

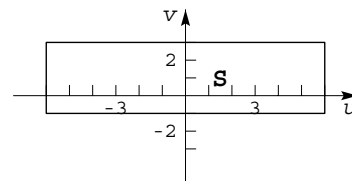
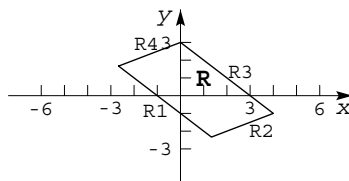
$R2: x - 2y = 6 \implies u = 3$

$R3: x + y = 3 \implies v = 3$

$R4: x - 2y = -6 \implies u = -6$

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} 1 & -2 \\ 1 & 1 \end{vmatrix} = 3 \implies \frac{\partial(x, y)}{\partial(u, v)} = \frac{1}{3}$$

$$\iint_R (x + y) dA = \iint_S v \left(\frac{1}{3} \right) dA' = \frac{1}{3} \int_{-1}^3 \int_{-6}^6 v du dv = \frac{1}{3} (12) \int_{-1}^3 v dv = 4 \left(\frac{1}{2} \right) v^2 \Big|_{-1}^3 = 16$$



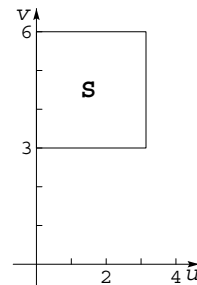
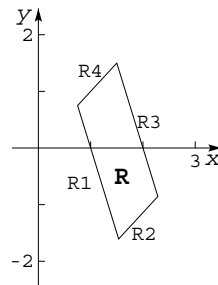
14. $R1: y = -3x + 3 \implies v = 3$

$R2: y = x - \pi \implies u = \pi$

$R3: y = -3x + 6 \implies v = 6$

$R4: y = x \implies u = 0$

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} 1 & -1 \\ 3 & 1 \end{vmatrix} = 4 \implies \frac{\partial(x, y)}{\partial(u, v)} = \frac{1}{4}$$



$$\begin{aligned} \iint_R \frac{\cos \frac{1}{2}(x - y)}{3x + y} dA &= \iint_S \frac{\cos u/2}{v} \left(\frac{1}{4} \right) dA' = \frac{1}{4} \int_3^6 \int_0^\pi \frac{\cos u/2}{v} du dv = \frac{1}{4} \int_3^6 \frac{2 \sin u/2}{v} \Big|_0^\pi dv \\ &= \frac{1}{2} \int_3^6 \frac{dv}{v} = \frac{1}{2} \ln v \Big|_3^6 = \frac{1}{2} \ln 2 \end{aligned}$$

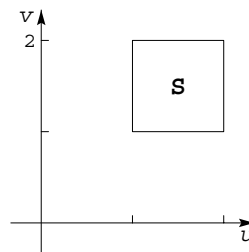
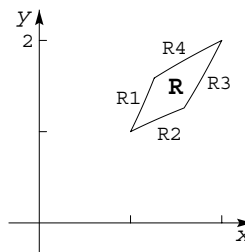
15. $R1: y = x^2 \implies u = 1$

$R2: x = y^2 \implies v = 1$

$R3: y = \frac{1}{2}x^2 \implies u = 2$

$R4: x = \frac{1}{2}y^2 \implies v = 2$

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} 2x/y & -x^2/y^2 \\ -y^2/x^2 & 2y/x \end{vmatrix} = 3 \implies \frac{\partial(x, y)}{\partial(u, v)} = \frac{1}{3}$$



$$\iint_R \frac{y^2}{x} dA = \iint_S v \left(\frac{1}{3} \right) dA' = \frac{1}{3} \int_1^2 \int_1^2 v du dv = \frac{1}{3} \int_1^2 v dv = \frac{1}{6} v^2 \Big|_1^2 = \frac{1}{2}$$

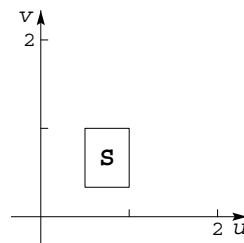
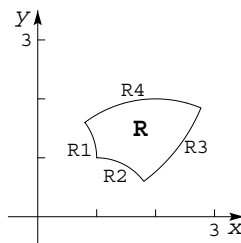
16. $R1: x^2 + y^2 = 2y \implies v = 1$

$R2: x^2 + y^2 = 2x \implies u = 1$

$R3: x^2 + y^2 = 6y \implies v = 1/3$

$R4: x^2 + y^2 = 4x \implies u = 1/2$

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{2(y^2 - x^2)}{(x^2 + y^2)^2} & \frac{-4xy}{(x^2 + y^2)^2} \\ \frac{-4xy}{(x^2 + y^2)^2} & \frac{2(x^2 - y^2)}{(x^2 + y^2)^2} \end{vmatrix} = \frac{-4}{(x^2 + y^2)^2}$$



Using $u^2 + v^2 = 4/(x^2 + y^2)$ we see that $\partial(x, y)/\partial(u, v) = -4/(u^2 + v^2)^2$.

$$\iint_R (x^2 + y^2)^{-3} dA = \iint_S \left(\frac{4}{u^2 + v^2} \right)^{-3} \left| \frac{-4}{(u^2 + v^2)^2} \right| dA' = \frac{1}{16} \int_{1/3}^1 \int_{1/2}^1 (u^2 + v^2) du dv = \frac{115}{5184}$$

17. $R1: 2xy = c \implies v = c$

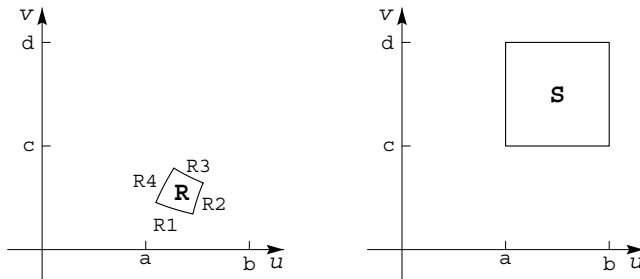
$R2: x^2 - y^2 = b \implies u = b$

$R3: 2xy = d \implies v = d$

$R4: x^2 - y^2 = a \implies u = a$

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} 2x & -2y \\ 2y & 2x \end{vmatrix} = 4(x^2 + y^2)$$

$$\implies \frac{\partial(x, y)}{\partial(u, v)} = \frac{1}{4(x^2 + y^2)}$$



$$\iint_R (x^2 + y^2) dA = \iint_S (x^2 + y^2) \frac{1}{4(x^2 + y^2)} dA' = \frac{1}{4} \int_c^d \int_a^b du dv = \frac{1}{4}(b-a)(d-c)$$

18. $R1: xy = -2 \implies v = -2$

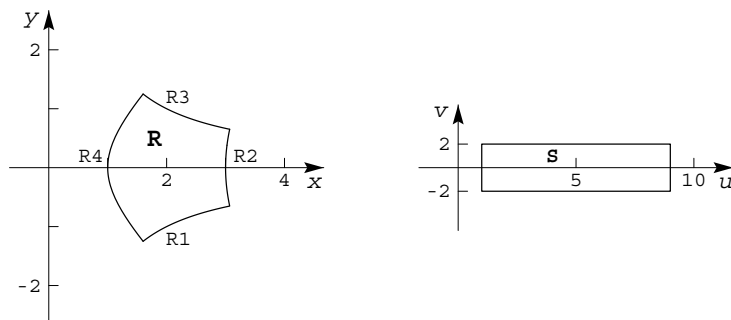
$R2: x^2 - y^2 = 9 \implies u = 9$

$R3: xy = 2 \implies v = 2$

$R4: x^2 - y^2 = 1 \implies u = 1$

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} 2x & -2y \\ y & x \end{vmatrix} = 2(x^2 + y^2)$$

$$\implies \frac{\partial(x, y)}{\partial(u, v)} = \frac{1}{2(x^2 + y^2)}$$



$$\iint_R (x^2 + y^2) \sin xy dA = \iint_S (x^2 + y^2) \sin v \left(\frac{1}{2(x^2 + y^2)} \right) dA' = \frac{1}{2} \int_{-2}^2 \int_1^9 \sin v du dv = \frac{1}{2} \int_{-2}^2 8 \sin v dv = 0$$

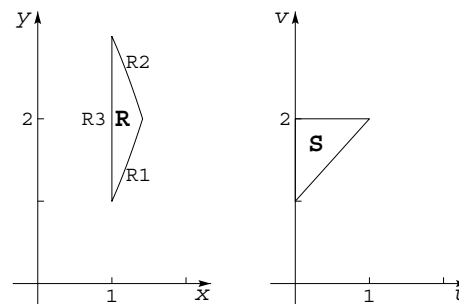
19. $R1: y = x^2 \implies v + u = v - u \implies u = 0$

$R2: y = 4 - x^2 \implies v + u = 4 - (v - u)$

$$\implies v + u = 4 - v + u \implies v = 2$$

$R3: x = 1 \implies v - u = 1 \implies v = 1 + u$

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} -\frac{1}{2\sqrt{v-u}} & \frac{1}{2\sqrt{v-u}} \\ 1 & 1 \end{vmatrix} = -\frac{1}{\sqrt{v-u}}$$



$$\begin{aligned} \iint_R \frac{x}{y + x^2} dA &= \iint_S \frac{\sqrt{v-u}}{2v} \left| -\frac{1}{\sqrt{v-u}} \right| dA' = \frac{1}{2} \int_0^1 \int_{1+u}^2 \frac{1}{v} dv du = \frac{1}{2} \int_0^1 [\ln 2 - \ln(1+u)] du \\ &= \frac{1}{2} \ln 2 - \frac{1}{2} [(1+u) \ln(1+u) - (1+u)] \Big|_0^1 = \frac{1}{2} \ln 2 - \frac{1}{2} [2 \ln 2 - 2 - (0 - 1)] = \frac{1}{2} - \frac{1}{2} \ln 2 \end{aligned}$$

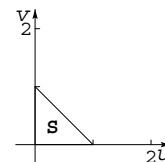
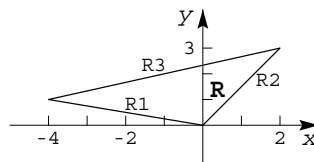
Exercises 9.17

20. Solving $x = 2u - 4v$, $y = 3u + v$ for

u and v we obtain $u = \frac{1}{14}x + \frac{2}{7}y$,

$v = -\frac{3}{14}x + \frac{1}{7}y$. The xy -corner points $(-4, 1)$,

$(0, 0)$, $(2, 3)$ correspond to the uv -points $(0, 1)$, $(0, 0)$, $(1, 0)$.



$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} 2 & -4 \\ 3 & 1 \end{vmatrix} = 14$$

$$\begin{aligned} \iint_R y \, dA &= \iint_S (3u + v)(14) \, dA' = 14 \int_0^1 \int_0^{1-u} (3u + v) \, dv \, du = 14 \int_0^1 \left(3uv + \frac{1}{2}v^2 \right) \Big|_0^{1-u} du \\ &= 14 \int_0^1 \left(\frac{1}{2} + 2u - \frac{5}{2}u^2 \right) du = \left(7u + 14u^2 - \frac{35}{3}u^3 \right) \Big|_0^1 = \frac{28}{3} \end{aligned}$$

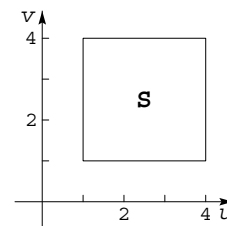
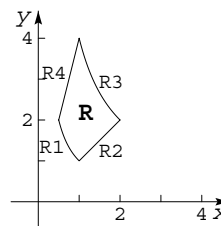
21. $R1: y = 1/x \implies u = 1$

$R2: y = x \implies v = 1$

$R3: y = 4/x \implies u = 4$

$R4: y = 4x \implies v = 4$

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} y & x \\ -y/x^2 & 1/x \end{vmatrix} = \frac{2y}{x} \implies \frac{\partial(x, y)}{\partial(u, v)} = \frac{x}{2y}$$



$$8 \iint_R y^4 \, dA = \iint_S u^2 v^2 \left(\frac{1}{2v} \right) du \, dv = \frac{1}{2} \int_1^4 u^2 v \, du \, dv = \frac{1}{2} \int_1^4 \frac{1}{3} u^3 v \Big|_1^4 dv = \frac{1}{6} \int_1^4 63v \, dv = \frac{21}{4} v^2 \Big|_1^4 = \frac{315}{4}$$

22. Under the transformation $u = y + z$, $v = -y + z$, $w = x - y$ the parallelepiped D is mapped to the parallelepiped $E: 1 \leq u \leq 3, -1 \leq v \leq 1, 0 \leq w \leq 3$.

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = \begin{vmatrix} 0 & 1 & 1 \\ 0 & -1 & 1 \\ 1 & -1 & 0 \end{vmatrix} = 2 \implies \frac{\partial(x, y, z)}{\partial(u, v, w)} = \frac{1}{2}$$

$$\begin{aligned} \iiint_D (4z + 2x - 2y) \, dV &= \iiint_E (2u + 2v + 2w) \frac{1}{2} \, dV' = \frac{1}{2} \int_0^3 \int_{-1}^1 \int_1^3 (2u + 2v + 2w) \, du \, dv \, dw \\ &= \frac{1}{2} \int_0^3 \int_{-1}^1 (u^2 + 2uv + 2uw) \Big|_1^3 dv \, dw = \frac{1}{2} \int_0^3 \int_{-1}^1 (8 + 4v + 4w) \, dv \, dw \\ &= \int_0^3 (4v + v^2 + 2vw) \Big|_{-1}^1 dw = \int_0^3 (8 + 4w) \, dw = (8w + 2w^2) \Big|_0^3 = 42 \end{aligned}$$

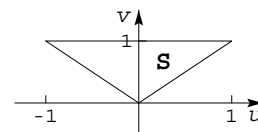
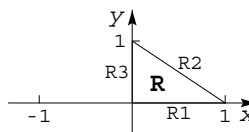
23. We let $u = y - x$ and $v = y + x$.

$R1: y = 0 \implies u = -x, v = x \implies v = -u$

$R2: x + y = 1 \implies v = 1$

$R3: x = 0 \implies u = y, v = y \implies v = u$

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} -1 & 1 \\ 1 & 1 \end{vmatrix} = -2 \implies \frac{\partial(x, y)}{\partial(u, v)} = -\frac{1}{2}$$



$$\begin{aligned}\iint_R e^{(y-x)/(y+x)} dA &= \iint_S e^{u/v} \left| -\frac{1}{2} \right| dA' = \frac{1}{2} \int_0^1 \int_{-v}^v e^{u/v} du dv = \frac{1}{2} \int_0^1 v e^{u/v} \Big|_{-v}^v dv \\ &= \frac{1}{2} \int_0^1 v(e - e^{-1}) dv = \frac{1}{2}(e - e^{-1}) \frac{1}{2} v^2 \Big|_0^1 = \frac{1}{4}(e - e^{-1})\end{aligned}$$

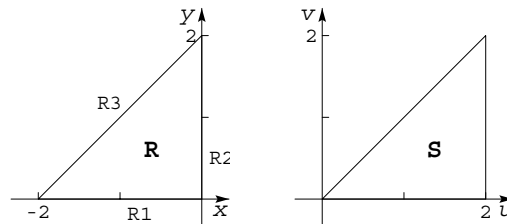
24. We let $u = y - x$ and $v = y$.

$$R1: y = 0 \implies v = 0, u = -x \implies v = 0, 0 \leq u \leq 2$$

$$R2: x = 0 \implies v = u$$

$$R3: y = x + 2 \implies u = 2$$

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} -1 & 1 \\ 0 & 1 \end{vmatrix} = -1 \implies \frac{\partial(x, y)}{\partial(u, v)} = -1$$



$$\iint_R e^{y^2-2xy+x^2} dA = \iint_S e^{u^2} |-1| dA' = \int_0^2 \int_0^u e^{u^2} dv du = \int_0^2 u e^{u^2} du = \frac{1}{2} e^{u^2} \Big|_0^2 = \frac{1}{2}(e^4 - 1)$$

25. Noting that $R2$, $R3$, and $R4$ have equations $y+2x=8$, $y-2x=0$,

and $y+2x=2$, we let $u = y/x$ and $v = y+2x$.

$$R1: y = 0 \implies u = 0, v = 2x \implies u = 0, 2 \leq v \leq 8$$

$$R2: y+2x=8 \implies v=8$$

$$R3: y-2x=0 \implies u=2$$

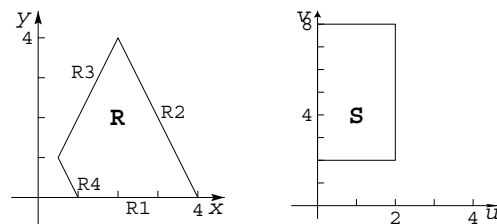
$$R4: y+2x=2 \implies v=2$$

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} -y/x^2 & 1/x \\ 2 & 1 \end{vmatrix} = -\frac{y+2x}{x^2} \implies \frac{\partial(x, y)}{\partial(u, v)} = -\frac{x^2}{y+2x}$$

$$\iint_R (6x+3y) dA = 3 \iint_S (y+2x) \left| -\frac{x^2}{y+2x} \right| dA' = 3 \iint_S x^2 dA'$$

From $y = ux$ we see that $v = ux + 2x$ and $x = v/(u+2)$. Then

$$3 \iint_S x^2 dA' = 3 \int_0^2 \int_2^8 v^2 (u+2)^2 dv du = \int_0^2 \frac{v^3}{(u+2)^2} \Big|_2^8 du = 504 \int_0^2 \frac{du}{(u+2)^2} = -\frac{504}{u+2} \Big|_0^2 = 126.$$



26. We let $u = x + y$ and $v = x - y$.

$$R1: x + y = 1 \implies u = 1$$

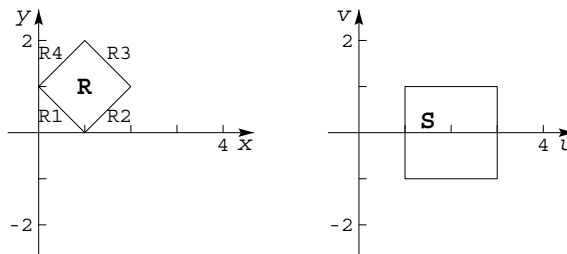
$$R2: x - y = 1 \implies v = 1$$

$$R3: x + y = 3 \implies u = 3$$

$$R4: x - y = -1 \implies v = -1$$

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = -2 \implies \frac{\partial(x, y)}{\partial(u, v)} = -\frac{1}{2}$$

$$\begin{aligned}\iint_R (x+y)^4 e^{x-y} dA &= \iint_S u^4 e^v \left| -\frac{1}{2} \right| dA' = \frac{1}{2} \int_1^3 \int_{-1}^1 u^4 e^v dv du = \frac{1}{2} \int_1^3 u^4 e^v \Big|_{-1}^1 du \\ &= \frac{e - e^{-1}}{2} \int_1^3 u^4 du = \frac{e - e^{-1}}{10} u^5 \Big|_1^3 = \frac{242(e - e^{-1})}{10} = \frac{121}{5}(e - e^{-1})\end{aligned}$$



27. Let $u = xy$ and $v = xy^{1.4}$. Then $xy^{1.4} = c \implies v = c$; $xy = b \implies u = b$; $xy^{1.4} = d \implies v = d$; $xy = a \implies u = a$.

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} y & x \\ y^{1.4} & 1.4xy^{0.4} \end{vmatrix} = 0.4xy^{1.4} = 0.4v \implies \frac{\partial(x, y)}{\partial(u, v)} = \frac{5}{2v}$$

Exercises 9.17

$$\iint_R dA = \iint_S \frac{5}{2v} dA' = \int_c^d \int_a^b \frac{5}{2v} du dv = \frac{5}{2}(b-a) \int_c^d \frac{dv}{v} = \frac{5}{2}(b-a)(\ln d - \ln c)$$

28. The image of the ellipsoid $x^2/a^2 + y^2/b^2 + z^2/c^2 = 1$ under the transformation $u = x/a$, $v = y/b$, $w = z/c$, is the unit sphere $u^2 + v^2 + w^2 = 1$. The volume of this sphere is $\frac{4}{3}\pi$. Now

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{vmatrix} = abc$$

and

$$\iiint_D dV = \iiint_E abc dV' = abc \iiint_E dV' = abc \left(\frac{4}{3}\pi \right) = \frac{4}{3}\pi abc.$$

29. The image of the ellipse is the unit circle $x^2 + y^2 = 1$. From $\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} 5 & 0 \\ 0 & 3 \end{vmatrix} = 15$ we obtain

$$\begin{aligned} \iint_R \left(\frac{x^2}{25} + \frac{y^2}{9} \right) dA &= \iint_S (u^2 + v^2) 15 dA' = 15 \int_0^{2\pi} \int_0^1 r^2 r dr d\theta = \frac{15}{4} \int_0^{2\pi} r^4 \Big|_0^1 d\theta \\ &= \frac{15}{4} \int_0^{2\pi} d\theta = \frac{15\pi}{2}. \end{aligned}$$

$$\begin{aligned} 30. \quad \frac{\partial(x, y, z)}{\partial(\rho, \phi, \theta)} &= \begin{vmatrix} \sin \phi \cos \theta & \rho \cos \phi \cos \theta & -\rho \sin \phi \sin \theta \\ \sin \phi \sin \theta & \rho \cos \phi \sin \theta & \rho \sin \phi \cos \theta \\ \cos \phi & -\rho \sin \phi & 0 \end{vmatrix} \\ &= \cos \phi (\rho^2 \sin \phi \cos \phi \cos^2 \theta + \rho^2 \sin \phi \cos \phi \sin^2 \theta) + \rho \sin \phi (\rho \sin^2 \phi \cos^2 \theta + \rho \sin^2 \phi \sin^2 \theta) \\ &= \rho^2 \sin \phi \cos^2 \phi (\cos^2 \theta + \sin^2 \theta) + \rho^2 \sin^3 \phi (\cos^2 \theta + \sin^2 \theta) = \rho^2 \sin \phi (\cos^2 \phi + \sin^2 \phi) = \rho^2 \sin \phi \end{aligned}$$

Chapter 9 Review Exercises

- True; $|\mathbf{v}(t)| = \sqrt{2}$
- True; for all t , $y = 4$.
- True
- False; consider $\mathbf{r}(t) = t^2 \mathbf{i}$. In this case, $\mathbf{v}(t) = 2t \mathbf{i}$ and $\mathbf{a}(t) = 2 \mathbf{i}$. Since $\mathbf{v} \cdot \mathbf{a} = 4t$, the velocity and acceleration vectors are not orthogonal for $t \neq 0$.
- False; ∇f is perpendicular to the level curve $f(x, y) = c$.
- False; consider $f(x, y) = xy$ at $(0, 0)$.
- True; the value is $4/3$.
- True; since $2xy dx - x^2 dy$ is not exact.
- False; $\int_C x dx + x^2 dy = 0$ from $(-1, 0)$ to $(1, 0)$ along the x -axis and along the semicircle $y = \sqrt{1 - x^2}$, but since $x dx + x^2 dy$ is not exact, the integral is not independent of path.
- True
- False; unless the first partial derivatives are continuous.
- True

13. True

14. True; since $\text{curl } \mathbf{F} = \mathbf{0}$ when $\mathbf{F}$ is a conservative vector field.

15. True

16. True

17. True

18. True

19. $\mathbf{F} = \nabla\phi = -x(x^2 + y^2)^{-3/2}\mathbf{i} - y(x^2 + y^2)^{-3/2}\mathbf{j}$

$$20. \text{curl } \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ f(x) & g(y) & h(z) \end{vmatrix} = \mathbf{0}$$

21. $\mathbf{v}(t) = 6\mathbf{i} + \mathbf{j} + 2t\mathbf{k}$; $\mathbf{a}(t) = 2\mathbf{k}$. To find when the particle passes through the plane, we solve $-6t + t + t^2 = -4$ or $t^2 - 5t + 4 = 0$. This gives $t = 1$ and $t = 4$. $\mathbf{v}(1) = 6\mathbf{i} + \mathbf{j} + 2\mathbf{k}$, $\mathbf{a}(1) = 2\mathbf{k}$; $\mathbf{v}(4) = 6\mathbf{i} + \mathbf{j} + 8\mathbf{k}$, $\mathbf{a}(4) = 2\mathbf{k}$

22. We are given $\mathbf{r}(0) = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$.

$$\mathbf{r}(t) = \int \mathbf{v}(t) dt = \int (-10t\mathbf{i} + (3t^2 - 4t)\mathbf{j} + \mathbf{k}) dt = -5t^2\mathbf{i} + (t^3 - 2t^2)\mathbf{j} + t\mathbf{k} + \mathbf{c}$$

$$\mathbf{i} + 2\mathbf{j} + 3\mathbf{k} = \mathbf{r}(0) = \mathbf{c}$$

$$\mathbf{r}(t) = (1 - 5t^2)\mathbf{i} + (t^3 - 2t^2 + 2)\mathbf{j} + (t + 3)\mathbf{k}$$

$$\mathbf{r}(2) = -19\mathbf{i} + 2\mathbf{j} + 5\mathbf{k}$$

$$23. \mathbf{v}(t) = \int \mathbf{a}(t) dt = \int (\sqrt{2} \sin t \mathbf{i} + \sqrt{2} \cos t \mathbf{j}) dt = -\sqrt{2} \cos t \mathbf{i} + \sqrt{2} \sin t \mathbf{j} + \mathbf{c};$$

$$-\mathbf{i} + \mathbf{j} + \mathbf{k} = \mathbf{v}(\pi/4) = -\sqrt{2} \cos t \mathbf{i} + \sqrt{2} \sin t \mathbf{j} + \mathbf{c}; \quad \mathbf{c} = \mathbf{k}; \quad \mathbf{v}(t) = -\sqrt{2} \cos t \mathbf{i} + \sqrt{2} \sin t \mathbf{j} + \mathbf{k};$$

$$\mathbf{r}(t) = -\sqrt{2} \sin t \mathbf{i} - \sqrt{2} \cos t \mathbf{j} + t\mathbf{k} + \mathbf{b}; \quad \mathbf{i} + 2\mathbf{j} + (\pi/4)\mathbf{k} = \mathbf{r}(\pi/4) = -\sqrt{2} \sin t \mathbf{i} - \sqrt{2} \cos t \mathbf{j} + t\mathbf{k} + \mathbf{b}, \quad \mathbf{b} = 2\mathbf{i} + 3\mathbf{j};$$

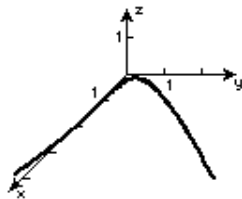
$$\mathbf{r}(t) = (2 - \sqrt{2} \sin t)\mathbf{i} + (3 - \sqrt{2} \cos t)\mathbf{j} + t\mathbf{k}; \quad \mathbf{r}(3\pi/4) = \mathbf{i} + 4\mathbf{j} + (3\pi/4)\mathbf{k}$$

$$24. \mathbf{v}(t) = t\mathbf{i} + t^2\mathbf{j} - t\mathbf{k}; \quad |\mathbf{v}| = t\sqrt{t^2 + 2}, \quad t > 0; \quad \mathbf{a}(t) = \mathbf{i} + 2t\mathbf{j} - \mathbf{k}; \quad \mathbf{v} \cdot \mathbf{a} = t + 2t^3 + t = 2t + 2t^3;$$

$$\mathbf{v} \times \mathbf{a} = t^2\mathbf{i} + t^2\mathbf{k}, \quad |\mathbf{v} \times \mathbf{a}| = t^2\sqrt{2}; \quad a_T = \frac{2t + 2t^3}{t\sqrt{t^2 + 2}} = \frac{2 + 2t^2}{\sqrt{t^2 + 2}}, \quad a_N = \frac{t^2\sqrt{2}}{t\sqrt{t^2 + 2}} = \frac{\sqrt{2}t}{\sqrt{t^2 + 2}};$$

$$\kappa = \frac{t^2\sqrt{2}}{t^3(t^2 + 2)^{3/2}} = \frac{\sqrt{2}}{t(t^2 + 2)^{3/2}}$$

25.



$$26. \mathbf{r}'(t) = \sinh t \mathbf{i} + \cosh t \mathbf{j} + \mathbf{k}, \quad \mathbf{r}'(1) = \sinh 1 \mathbf{i} + \cosh 1 \mathbf{j} + \mathbf{k};$$

$$|\mathbf{r}'(t)| = \sqrt{\sinh^2 t + \cosh^2 t + 1} = \sqrt{2 \cosh^2 t} = \sqrt{2} \cosh t; \quad |\mathbf{r}'(1)| = \sqrt{2} \cosh 1;$$

$$\mathbf{T}(t) = \frac{1}{\sqrt{2}} \tanh t \mathbf{i} + \frac{1}{\sqrt{2}} \mathbf{j} + \frac{1}{\sqrt{2}} \operatorname{sech} t \mathbf{k}, \quad \mathbf{T}(1) = \frac{1}{\sqrt{2}} (\tanh 1 \mathbf{i} + \mathbf{j} + \operatorname{sech} 1 \mathbf{k});$$

$$\frac{d\mathbf{T}}{dt} = \frac{1}{\sqrt{2}} \operatorname{sech}^2 t \mathbf{i} - \frac{1}{\sqrt{2}} \operatorname{sech} t \tanh t \mathbf{k}; \quad \frac{d}{dt} \mathbf{T}(1) = \frac{1}{\sqrt{2}} \operatorname{sech}^2 1 \mathbf{i} - \frac{1}{\sqrt{2}} \operatorname{sech} 1 \tanh 1 \mathbf{k},$$

$$\left| \frac{d}{dt} \mathbf{T}(1) \right| = \frac{\operatorname{sech} 1}{\sqrt{2}} \sqrt{\operatorname{sech}^2 1 + \tanh^2 1} = \frac{1}{\sqrt{2}} \operatorname{sech} 1; \quad \mathbf{N}(1) = \operatorname{sech} 1 \mathbf{i} - \tanh 1 \mathbf{k};$$

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$$\begin{aligned}\mathbf{B}(1) &= \mathbf{T}(1) \times \mathbf{N}(1) = -\frac{1}{\sqrt{2}} \tanh 1 \mathbf{i} + \frac{1}{\sqrt{2}} (\tanh^2 1 + \operatorname{sech}^2 1) \mathbf{j} - \frac{1}{\sqrt{2}} \operatorname{sech} 1 \mathbf{k} \\ &= \frac{1}{\sqrt{2}} (-\tanh 1 \mathbf{i} + \mathbf{j} - \operatorname{sech} 1 \mathbf{k})\end{aligned}$$

$$\kappa = \left| \frac{d}{dt} \mathbf{T}(1) \right| / |\mathbf{r}'(1)| = \frac{(\operatorname{sech} 1)/\sqrt{2}}{\sqrt{2} \cosh 1} = \frac{1}{2} \operatorname{sech}^2 1$$

$$27. \nabla f = (2xy - y^2) \mathbf{i} + (x^2 - 2xy) \mathbf{j}; \quad \mathbf{u} = \frac{2}{\sqrt{40}} \mathbf{i} + \frac{6}{\sqrt{40}} \mathbf{j} = \frac{1}{\sqrt{10}} (\mathbf{i} + 3\mathbf{j});$$

$$D_{\mathbf{u}} f = \frac{1}{\sqrt{10}} (2xy - y^2 + 3x^2 - 6xy) = \frac{1}{\sqrt{10}} (3x^2 - 4xy - y^2)$$

$$28. \nabla F = \frac{2x}{x^2 + y^2 + z^2} \mathbf{i} + \frac{2y}{x^2 + y^2 + z^2} \mathbf{j} + \frac{2z}{x^2 + y^2 + z^2} \mathbf{k}; \quad \mathbf{u} = -\frac{2}{3} \mathbf{i} + \frac{1}{3} \mathbf{j} + \frac{2}{3} \mathbf{k}; \quad D_{\mathbf{u}} F = \frac{-4x + 2y + 4z}{3(x^2 + y^2 + z^2)}$$

$$29. f_x = 2xy^4, \quad f_y = 4x^2y^3.$$

$$(a) \quad \mathbf{u} = \mathbf{i}, \quad D_{\mathbf{u}}(1, 1) = f_x(1, 1) = 2$$

$$(b) \quad \mathbf{u} = (\mathbf{i} - \mathbf{j})/\sqrt{2}, \quad D_{\mathbf{u}}(1, 1) = (2 - 4)/\sqrt{2} = -2/\sqrt{2}$$

$$(c) \quad \mathbf{u} = \mathbf{j}, \quad D_{\mathbf{u}}(1, 1) = f_y(1, 1) = 4$$

$$30. (a) \quad \frac{dw}{dt} = \frac{\partial w}{\partial x} \frac{dx}{dt} + \frac{\partial w}{\partial y} \frac{dy}{dt} + \frac{\partial w}{\partial z} \frac{dz}{dt}$$

$$\begin{aligned}&= \frac{x}{\sqrt{x^2 + y^2 + z^2}} 6 \cos 2t + \frac{y}{\sqrt{x^2 + y^2 + z^2}} (-8 \sin 2t) + \frac{z}{\sqrt{x^2 + y^2 + z^2}} 15t^2 \\ &= \frac{(6x \cos 2t - 8y \sin 2t + 15zt^2)}{\sqrt{x^2 + y^2 + z^2}}\end{aligned}$$

$$(b) \quad \frac{\partial w}{\partial t} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial t} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial t}$$

$$\begin{aligned}&= \frac{x}{\sqrt{x^2 + y^2 + z^2}} \frac{6}{r} \cos \frac{2t}{r} + \frac{y}{\sqrt{x^2 + y^2 + z^2}} \left(\frac{8r}{t^2} \sin \frac{2r}{t} \right) + \frac{z}{\sqrt{x^2 + y^2 + z^2}} 15t^2 r^3 \\ &= \frac{\left(\frac{6x}{r} \cos \frac{2t}{r} + \frac{8yr}{t^2} \sin \frac{2r}{t} + 15zt^2 r^3 \right)}{\sqrt{x^2 + y^2 + z^2}}\end{aligned}$$

$$31. F(x, y, z) = \sin xy - z; \quad \nabla F = y \cos xy \mathbf{i} + x \cos xy \mathbf{j} - \mathbf{k}; \quad \nabla F(1/2, 2\pi/3, \sqrt{3}/2) = \frac{\pi}{3} \mathbf{i} + \frac{1}{4} \mathbf{j} - \mathbf{k}.$$

$$\text{The equation of the tangent plane is } \frac{\pi}{3} \left(x - \frac{1}{2} \right) + \frac{1}{4} \left(y - \frac{2\pi}{3} \right) - \left(z - \frac{\sqrt{3}}{2} \right) = 0 \text{ or } 4\pi x + 3y - 12z = 4\pi - 6\sqrt{3}.$$

$$32. \text{ We want to find a normal to the surface that is parallel to } \mathbf{k}. \quad \nabla F = (y-2)\mathbf{i} + (x-2y)\mathbf{j} + 2z\mathbf{k}. \text{ We need } y-2=0 \text{ and } x-2y=0. \text{ The tangent plane is parallel to } z=2 \text{ when } y=2 \text{ and } x=4. \text{ In this case } z^2=5. \text{ The points are } (4, 2, \sqrt{5}) \text{ and } (4, 2, -\sqrt{5}).$$

$$33. (a) \quad V = \int_0^1 \int_x^{2x} \sqrt{1-x^2} dy dx = \int_0^1 y \sqrt{1-x^2} \Big|_x^{2x} dx = \int_0^1 x \sqrt{1-x^2} dx = -\frac{1}{3} (1-x^2)^{3/2} \Big|_0^1 = \frac{1}{3}$$

$$(b) \quad V = \int_0^1 \int_{y/2}^y \sqrt{1-x^2} dx dy + \int_1^2 \int_{y/2}^1 \sqrt{1-x^2} dx dy$$

34. We are given $\rho = k(x^2 + y^2)$.

$$\begin{aligned}
 m &= \int_0^1 \int_{x^3}^{x^2} k(x^2 + y^2) dy dx = k \int_0^1 \left(x^2 y + \frac{1}{3} y^3 \right) \Big|_{x^3}^{x^2} dx \\
 &= k \int_0^1 \left(x^4 + \frac{1}{3} x^6 - x^5 - \frac{1}{3} x^9 \right) dx = k \left(\frac{1}{5} x^5 + \frac{1}{21} x^7 - \frac{1}{6} x^6 - \frac{1}{30} x^{10} \right) \Big|_0^1 = \frac{k}{21} \\
 M_y &= \int_0^1 \int_{x^3}^{x^2} k(x^3 + xy^2) dy dx = k \int_0^1 \left(x^3 y + \frac{1}{3} xy^3 \right) \Big|_{x^3}^{x^2} dx = k \int_0^1 \left(x^5 + \frac{1}{3} x^7 - x^6 - \frac{1}{3} x^{10} \right) dx \\
 &= k \left(\frac{1}{6} x^6 + \frac{1}{24} x^8 - \frac{1}{7} x^7 - \frac{1}{33} x^{11} \right) \Big|_0^1 = \frac{65k}{1848} \\
 M_x &= \int_0^1 \int_{x^3}^{x^2} k(x^2 y + y^3) dy dx = k \int_0^1 \left(\frac{1}{2} x^2 y^2 + \frac{1}{4} y^4 \right) \Big|_{x^3}^{x^2} dx = k \int_0^1 \left(\frac{1}{2} x^6 + \frac{1}{4} x^8 - \frac{1}{2} x^8 - \frac{1}{4} x^{12} \right) dx \\
 &= k \left(\frac{1}{14} x^7 - \frac{1}{36} x^9 - \frac{1}{52} x^{13} \right) \Big|_0^1 = \frac{20k}{819} \\
 \bar{x} = M_y/m &= \frac{65k/1848}{k/21} = 65/88; \quad \bar{y} = M_x/m = \frac{20k/819}{k/21} = 20/39 \quad \text{The center of mass is } (65/88, 20/39).
 \end{aligned}$$

$$\begin{aligned}
 35. \quad I_y &= \int_0^1 \int_{x^3}^{x^2} k(x^4 + x^2 y^2) dy dx = k \int_0^1 \left(x^4 y + \frac{1}{3} x^2 y^3 \right) \Big|_{x^3}^{x^2} dx = k \int_0^1 \left(x^6 + \frac{1}{3} x^8 - x^7 - \frac{1}{3} x^{11} \right) dx \\
 &= k \left(\frac{1}{7} x^7 + \frac{1}{27} x^9 - \frac{1}{8} x^8 - \frac{1}{36} x^{12} \right) \Big|_0^1 = \frac{41}{1512} k
 \end{aligned}$$

36. (a) Using symmetry,

$$\begin{aligned}
 V &= 8 \int_0^a \int_0^{\sqrt{a^2-x^2}} \int_0^{\sqrt{a^2-x^2-y^2}} dz dy dx = 8 \int_0^a \int_0^{\sqrt{a^2-x^2}} \sqrt{a^2-x^2-y^2} dy dx \\
 &\quad \boxed{\text{Trig substitution}} \\
 &= 8 \int_0^a \left(\frac{y}{2} \sqrt{a^2-x^2-y^2} + \frac{a^2-x^2}{2} \sin^{-1} \frac{y}{\sqrt{a^2-x^2}} \right) \Big|_0^{\sqrt{a^2-x^2}} dx = 8 \int_0^a \frac{\pi}{2} \frac{a^2-x^2}{2} dx \\
 &= 2\pi \left(a^2 x - \frac{1}{3} x^3 \right) \Big|_0^a = \frac{4}{3} \pi a^3
 \end{aligned}$$

(b) Using symmetry,

$$\begin{aligned}
 V &= 2 \int_0^{2\pi} \int_0^a \int_0^{\sqrt{a^2-r^2}} r dz dr d\theta = 2 \int_0^{2\pi} \int_0^a r \sqrt{a^2-r^2} dr d\theta \\
 &= 2 \int_0^{2\pi} \left(-\frac{1}{3} (a^2-r^2)^{3/2} \right) \Big|_0^a d\theta = \frac{2}{3} \int_0^{2\pi} a^3 d\theta = \frac{4}{3} \pi a^3
 \end{aligned}$$

$$\begin{aligned}
 (c) \quad V &= \int_0^{2\pi} \int_0^\pi \int_0^a \rho^2 \sin \phi d\rho d\phi d\theta = \int_0^{2\pi} \int_0^\pi \frac{1}{3} \rho^3 \sin \phi \Big|_0^a d\phi d\theta \\
 &= \frac{1}{3} \int_0^{2\pi} \int_0^\pi a^3 \sin \phi d\phi d\theta = \frac{1}{3} \int_0^{2\pi} -a^3 \cos \phi \Big|_0^\pi d\theta = \frac{1}{3} \int_0^{2\pi} 2a^3 d\theta = \frac{4}{3} \pi a^3
 \end{aligned}$$

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37. We use spherical coordinates.

$$\begin{aligned} V &= \int_0^{2\pi} \int_{\tan^{-1} 1/3}^{\pi/4} \int_0^{3 \sec \phi} \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = \int_0^{2\pi} \int_{\tan^{-1} 1/3}^{\pi/4} \frac{1}{3} \rho^3 \sin \phi \Big|_0^{3 \sec \phi} d\phi \, d\theta \\ &= \frac{1}{3} \int_0^{2\pi} \int_{\tan^{-1} 1/3}^{\pi/4} 27 \sec^3 \phi \sin \phi \, d\phi \, d\theta = 9 \int_0^{2\pi} \int_{\tan^{-1} 1/3}^{\pi/4} \tan \phi \sec^2 \phi \, d\phi \, d\theta \\ &= 9 \int_0^{2\pi} \frac{1}{2} \tan^2 \phi \Big|_{\tan^{-1} 1/3}^{\pi/4} d\theta = \frac{9}{2} \int_0^{2\pi} \left(1 - \frac{1}{9}\right) d\theta = 8\pi \end{aligned}$$

$$\begin{aligned} 38. \quad V &= \int_0^{2\pi} \int_0^{\pi/6} \int_1^2 \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = \int_0^{2\pi} \int_0^{\pi/6} \frac{1}{3} \rho^3 \sin \phi \Big|_1^2 d\phi \, d\theta \\ &= \int_0^{2\pi} \int_0^{\pi/6} \left(\frac{8}{3} \sin \phi - \frac{1}{3} \sin \phi\right) d\phi \, d\theta = \frac{7}{3} \int_0^{2\pi} \int_0^{\pi/6} \sin \phi \, d\phi \, d\theta = \frac{7}{3} \int_0^{2\pi} -\cos \phi \Big|_0^{\pi/6} d\theta \\ &= \frac{7}{3} \int_0^{2\pi} \left[-\frac{\sqrt{3}}{2} - (-1)\right] d\theta = \frac{7}{3} \left(1 - \frac{\sqrt{3}}{2}\right) 2\pi = \frac{7\pi}{3}(2 - \sqrt{3}) \end{aligned}$$

39. $2xy + 2xy + 2xy = 6xy$

$$40. \quad \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x^2y & xy^2 & 2xyz \end{vmatrix} = 2xz\mathbf{i} - 2yz\mathbf{j} + (y^2 - x^2)\mathbf{k}$$

$$41. \quad \frac{\partial}{\partial x}(2xz) - \frac{\partial}{\partial y}(2yz) + \frac{\partial}{\partial z}(y^2 - x^2) = 0$$

$$42. \quad \nabla(6xy) = 6y\mathbf{i} + 6x\mathbf{j}$$

$$43. \quad \int_C \frac{z^2}{x^2 + y^2} ds = \int_{\pi}^{2\pi} \frac{4t^2}{\cos^2 2t + \sin^2 2t} \sqrt{4\sin^2 2t + 4\cos^2 2t + 4} dt = \int_{\pi}^{2\pi} 8\sqrt{2} t^2 dt = \frac{8\sqrt{2}}{3} t^3 \Big|_{\pi}^{2\pi} = \frac{56\sqrt{2}\pi^3}{3}$$

$$44. \quad \int_C (xy + 4x) ds = \int_1^0 [x(2 - 2x) + 4x] \sqrt{1 + 4} dx = \sqrt{5} \int_1^0 (6x - 2x^2) dx = \sqrt{5} \left(3x^2 - \frac{2}{3}x^3\right) \Big|_1^0 = -\frac{7\sqrt{5}}{3}$$

45. Since $P_y = 6x^2y = Q_x$, the integral is independent of path.

$$\phi_x = 3x^2y^2, \quad \phi = x^3y^2 + g(y), \quad \phi_y = 2x^3y + g'(y) = 2x^3y - 3y^2; \quad g(y) = -y^3; \quad \phi = x^3y^2 - y^3;$$

$$\int_{(0,0)}^{(1,-2)} 3x^2y^2 dx + (2x^3y - 3y^2) dy = (x^3y^2 - y^3) \Big|_{(0,0)}^{(1,-2)} = 12$$

46. Let $x = a \cos t$, $y = a \sin t$, $0 \leq t \leq 2\pi$. Then using $dx = -a \sin t dt$, $dy = a \cos t dt$, $x^2 + y^2 = a^2$ we have

$$\oint_C \frac{-y dx + x dy}{x^2 + y^2} = \int_0^{2\pi} \frac{1}{a^2} [-a \sin t(-a \sin t) + a \cos t(a \cos t)] dt = \int_0^{2\pi} (\sin^2 t + \cos^2 t) dt = \int_0^{2\pi} dt = 2\pi.$$

$$47. \quad \int_C y \sin \pi z dx + x^2 e^y dy + 3xyz dz$$

$$\begin{aligned} &= \int_0^1 [t^2 \sin \pi t^3 + t^2 e^{t^2} (2t) + 3t t^2 t^3 (3t^2)] dt = \int_0^1 (t^2 \sin \pi t^3 + 2t^3 e^{t^2} + 9t^8) dt \\ &= \left(-\frac{1}{3\pi} \cos \pi t^3 + t^9\right) \Big|_0^1 + 2 \int_0^1 t^3 e^{t^2} dt \quad \boxed{\text{Integration by parts}} \\ &= \frac{2}{3\pi} + 1 + (t^2 e^{t^2} - e^{t^2}) \Big|_0^1 = \frac{2}{3\pi} + 2 \end{aligned}$$

48. Parameterize C by $x = \cos t$, $y = \sin t$; $0 \leq t \leq 2\pi$. Then

$$\begin{aligned}\oint_C \mathbf{F} \cdot d\mathbf{r} &= \int_0^{2\pi} [4 \sin t(-\sin t dt) + 6 \cos t(\cos t) dt] = \int_0^{2\pi} (6 \cos^2 t - 4 \sin^2 t) dt \\ &= \int_0^{2\pi} (10 \cos^2 t - 4) dt = \left(5t + \frac{5}{2} \sin 2t - 4t \right) \Big|_0^{2\pi} = 2\pi.\end{aligned}$$

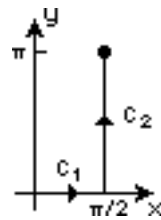
Using Green's Theorem, $Q_x - P_y = 6 - 4 = 2$ and $\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_R 2 dA = 2(\pi \cdot 1^2) = 2\pi$.

49. Let $\mathbf{r}_1 = \frac{\pi}{2} t \mathbf{i}$ and $\mathbf{r}_2 = \frac{\pi}{2} \mathbf{i} + \pi t \mathbf{j}$ for $0 \leq t \leq 1$. Then $d\mathbf{r}_1 = \frac{\pi}{2} \mathbf{i}$, $d\mathbf{r}_2 = \pi \mathbf{j}$, $\mathbf{F}_1 = \mathbf{0}$,

$$\mathbf{F}_2 = \frac{\pi}{2} \sin \pi t \mathbf{i} + \pi t \sin \frac{\pi}{2} \mathbf{j} = \frac{\pi}{2} \sin \pi t \mathbf{i} + \pi t \mathbf{j},$$

and

$$W = \int_{C_1} \mathbf{F}_1 \cdot d\mathbf{r}_1 + \int_{C_2} \mathbf{F}_2 \cdot d\mathbf{r}_2 = \int_0^1 \pi^2 t dt = \frac{1}{2} \pi^2 t^2 \Big|_0^1 = \frac{\pi^2}{2}.$$

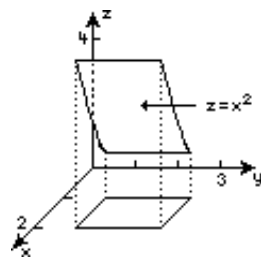


50. Parameterize the line segment from $(-1/2, 1/2)$ to $(-1, 1)$ using $y = -x$ as x goes from $-1/2$ to -1 . Parameterize the line segment from $(-1, 1)$ to $(1, 1)$ using $y = 1$ as x goes from -1 to 1 . Parameterize the line segment from $(1, 1)$ to $(1, \sqrt{3})$ using $x = 1$ as y goes from 1 to $\sqrt{3}$. Then

$$\begin{aligned}W &= \int_C \mathbf{F} \cdot d\mathbf{r} = \int_{-1/2}^{-1} \mathbf{F} \cdot (dx \mathbf{i} - dx \mathbf{j}) + \int_{-1}^1 \mathbf{F} \cdot (dx \mathbf{i}) + \int_1^{\sqrt{3}} \mathbf{F} \cdot (dy \mathbf{j}) \\ &= \int_{-1/2}^{-1} \left(\frac{2}{x^2 + (-x)^2} - \frac{1}{x^2 + (-x)^2} \right) dx + \int_{-1}^1 \frac{2}{x^2 + 1} dx + \int_1^{\sqrt{3}} \frac{1}{1 + y^2} dy \\ &= \int_{-1/2}^{-1} \frac{1}{2x^2} dx + \int_{-1}^1 \frac{2}{1 + x^2} dx + \int_1^{\sqrt{3}} \frac{1}{1 + y^2} dy \\ &= -\frac{1}{2x} \Big|_{-1/2}^{-1} + 2 \tan^{-1} x \Big|_{-1}^1 + \tan^{-1} y \Big|_1^{\sqrt{3}} = -\frac{1}{2} + 2 \left(\frac{\pi}{2} \right) + \frac{\pi}{12} = \frac{13\pi - 6}{12}.\end{aligned}$$

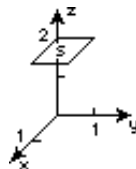
51. $z_x = 2x$, $z_y = 0$; $dS = \sqrt{1 + 4x^2} dA$

$$\begin{aligned}\iint_S \frac{z}{xy} dS &= \int_1^3 \int_1^2 \frac{x^2}{xy} \sqrt{1 + 4x^2} dx dy = \int_1^3 \frac{1}{y} \left[\frac{1}{12} (1 + 4x^2)^{3/2} \right] \Big|_1^2 dy \\ &= \frac{1}{12} \int_1^3 \frac{17^{3/2} - 5^{3/2}}{y} dy = \frac{17\sqrt{17} - 5\sqrt{5}}{12} \ln y \Big|_1^3 \\ &= \frac{17\sqrt{17} - 5\sqrt{5}}{12} \ln 3\end{aligned}$$



52. $\mathbf{n} = \mathbf{k}$, $\mathbf{F} \cdot \mathbf{n} = 3$;

$$\text{flux} = \iint_S \mathbf{F} \cdot \mathbf{n} dS = 3 \iint_S dS = 3 \times (\text{area of } S) = 3(1) = 3$$



53. The surface is $g(x, y, z) = x^2 + y^2 + z^2 - a^2 = 0$. $\nabla g = 2(x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) = 2\mathbf{r}$, $\mathbf{n} = \mathbf{r}/|\mathbf{r}|$,

$$\mathbf{F} = c \nabla(1/|\mathbf{r}|) + c \nabla(x^2 + y^2 + z^2)^{-1/2} = c \frac{-x\mathbf{i} - y\mathbf{j} - z\mathbf{k}}{(x^2 + y^2 + z^2)^{3/2}} = -c\mathbf{r}/|\mathbf{r}|^3$$

$$\mathbf{F} \cdot \mathbf{n} = -\frac{\mathbf{r}}{|\mathbf{r}|^3} \cdot \frac{\mathbf{r}}{|\mathbf{r}|} = -c \frac{\mathbf{r} \cdot \mathbf{r}}{|\mathbf{r}|^4} = -c \frac{|\mathbf{r}|^2}{|\mathbf{r}|^4} = -\frac{c}{|\mathbf{r}|^2} = -\frac{c}{a^2}$$

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$$\text{flux} = \iint_S \mathbf{F} \cdot \mathbf{n} dS = -\frac{c}{a^2} \iint_S dS = -\frac{c}{a^2} \times (\text{area of } S) = -\frac{c}{a^2}(4\pi a^2) = -4\pi c$$

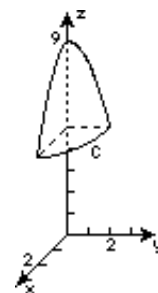
54. In Problem 53, $\mathbf{F}$ is not continuous at $(0, 0, 0)$ which is in any acceptable region containing the sphere.

55. Since $\mathbf{F} = c\nabla(1/r)$, $\text{div } \mathbf{F} = \nabla \cdot (c\nabla(1/r)) = c\nabla^2(1/r) = c\nabla^2[(x^2 + y^2 + z^2)^{-1/2}] = 0$ by Problem 37 in Section 9.7. Then, by the Divergence Theorem,

$$\text{flux } \mathbf{F} = \iint_S \mathbf{F} \cdot \mathbf{n} dS = \iiint_D \text{div } \mathbf{F} dV = \iiint_D 0 dV = 0.$$

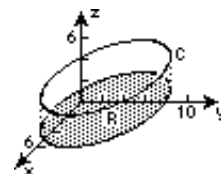
56. Parameterize C by $x = 2 \cos t$, $y = 2 \sin t$, $z = 5$, for $0 \leq t \leq 2\pi$. Then

$$\begin{aligned} \iint_S (\text{curl } \mathbf{F} \cdot \mathbf{n}) dS &= \oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C 6x dx + 7z dy + 8y dz \\ &= \int_0^{2\pi} [12 \cos t(-2 \sin t) + 35(2 \cos t)] dt \\ &= \int_0^{2\pi} (70 \cos t - 24 \sin t \cos t) dt = (70 \sin t - 12 \sin^2 t) \Big|_0^{2\pi} = 0. \end{aligned}$$



57. Identify $\mathbf{F} = -2y\mathbf{i} + 3xz\mathbf{j} + 10z\mathbf{k}$. Then $\text{curl } \mathbf{F} = 5\mathbf{k}$. The curve C lies in the plane $z = 3$, so $\mathbf{n} = \mathbf{k}$ and $dS = dA$. Thus,

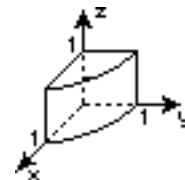
$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} dS = \iint_R 5 dA = 5 \times (\text{area of } R) = 5(25\pi) = 125\pi.$$



58. Since $\text{curl } \mathbf{F} = \mathbf{0}$, $\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\text{curl } \mathbf{F} \cdot \mathbf{n}) dS = \iint_S 0 dS = 0$.

59. $\text{div } \mathbf{f} = 1 + 1 = 2$;

$$\iint_S \mathbf{F} \cdot \mathbf{n} dS = \iiint_D \text{div } \mathbf{F} dV = \iiint_D 3 dV = 3 \times (\text{volume of } D) = 3\pi$$

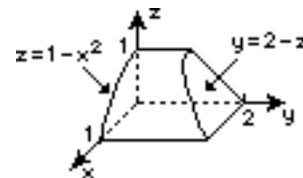


60. $\text{div } \mathbf{F} = x^2 + y^2 + z^2$. Using cylindrical coordinates,

$$\begin{aligned} \iint_S \mathbf{F} \cdot \mathbf{n} dS &= \iiint_D \text{div } \mathbf{F} dV = \iiint_D (x^2 + y^2 + z^2) dV = \int_0^{2\pi} \int_0^1 \int_0^1 (r^2 + z^2) r dz dr d\theta \\ &= \int_0^{2\pi} \int_0^1 \left(r^3 z + \frac{1}{3} r z^3 \right) \Big|_0^1 dr d\theta = \int_0^{2\pi} \int_0^1 \left(r^3 + \frac{1}{3} r \right) dr d\theta \\ &= \int_0^{2\pi} \left(\frac{1}{4} r^4 + \frac{1}{6} r^2 \right) \Big|_0^1 d\theta = \int_0^{2\pi} \frac{5}{12} d\theta = \frac{5\pi}{6}. \end{aligned}$$

61. $\text{div } \mathbf{F} = 2x + 2(x + y) - 2y = 4x$

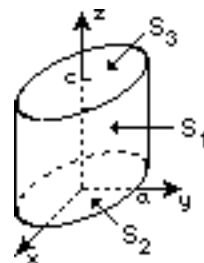
$$\begin{aligned} \iint_S \mathbf{F} \cdot \mathbf{n} dS &= \iiint_D \text{div } \mathbf{F} dV = \iiint_D 4x dV = \int_0^1 \int_0^{1-x^2} \int_0^{2-z} 4x dy dz dx \\ &= \int_0^1 \int_0^{1-x^2} 4x(2-z) dz dx = \int_0^1 \int_0^{1-x^2} (8x - 4xz) dz dx \\ &= \int_0^1 (8xz - 2xz^2) \Big|_0^{1-x^2} dx = \int_0^1 [8x(1-x^2) - 2x(1-x^2)^2] dx \\ &= \left[-2(1-x^2)^2 + \frac{1}{3}(1-x^2)^3 \right] \Big|_0^1 = \frac{5}{3} \end{aligned}$$



62. For S_1 , $\mathbf{n} = (x\mathbf{i} + y\mathbf{j})/\sqrt{x^2 + y^2}$; for S_2 , $\mathbf{n}_2 = -\mathbf{k}$ and $z = 0$; and for S_3 , $\mathbf{n}_3 = \mathbf{k}$ and $z = c$.

Then

$$\begin{aligned}
 \iint_S \mathbf{F} \cdot \mathbf{n} \, dS &= \iint_{S_1} \mathbf{F} \cdot \mathbf{n}_1 \, dS_1 + \iint_{S_2} \mathbf{F} \cdot \mathbf{n}_2 \, dS_2 + \iint_{S_3} \mathbf{F} \cdot \mathbf{n}_3 \, dS_3 \\
 &= \iint_{S_1} \frac{x^2 + y^2}{\sqrt{x^2 + y^2}} \, dS_1 + \iint_{S_2} (-z^2 - 1) \, dS_2 + \iint_{S_3} (z^2 + 1) \, dS_3 \\
 &= \iint_{S_1} \sqrt{x^2 + y^2} \, dS_1 + \iint_{S_2} (-1) \, dS_2 + \iint_{S_3} (c^2 + 1) \, dS_3 \\
 &= a \iint_{S_1} dS_1 - \iint_{S_2} dS_2 + (c^2 + 1) \iint_{S_3} dS_3 \\
 &= a(2\pi ac) - \pi a^2 + (c^2 + 1)\pi a^2 = 2\pi a^2 c + \pi a^2 c^2.
 \end{aligned}$$



63. $x = 0 \implies u = 0, v = -y^2 \implies u = 0, -1 \leq v \leq 0$

$$x = 1 \implies u = 2y, v = 1 - y^2 = 1 - u^2/4$$

$$y = 0 \implies u = 0, v = x^2 \implies u = 0, 0 \leq v \leq 1$$

$$y = 1 \implies u = 2x, v = x^2 - 1 = u^2/4 - 1$$

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} 2y & 2x \\ 2x & -2y \end{vmatrix} = -4(x^2 + y^2) \implies \frac{\partial(x, y)}{\partial(u, v)} = -\frac{1}{4(x^2 + y^2)}$$

$$\begin{aligned}
 \iint_R (x^2 + y^2) \sqrt[3]{x^2 + y^2} \, dA &= \iint_S (x^2 + y^2) \sqrt[3]{v} \left| -\frac{1}{4(x^2 + y^2)} \right| dA' = \frac{1}{4} \int_0^2 \int_{u^2/4-1}^{1-u^2/4} v^{1/3} \, dv \, du \\
 &= \frac{1}{4} \int_0^2 \frac{3}{4} v^{4/3} \Big|_{u^2/4-1}^{1-u^2/4} du = \frac{3}{16} \int_0^2 \left[(1 - u^2/4)^{4/3} - (u^2/4 - 1)^{4/3} \right] du \\
 &= \frac{3}{16} \int_0^2 \left[(1 - u^2/4)^{4/3} - (1 - u^2/4)^{4/3} \right] du = 0
 \end{aligned}$$

64. $y = x \implies u + uv = v + uv \implies v = u$

$$x = 2 \implies u + uv = 2 \implies v = (2 - u)/u$$

$$y = 0 \implies v + uv = 0 \implies v = 0 \text{ or } u = -1$$

(we take $v = 0$)

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} 1 + v & u \\ v & 1 + u \end{vmatrix} = 1 + u + v$$

Using $x = u + uv$ and $y = v + uv$ we find

$$(x - y)^2 = (u + uv - v - uv)^2 = (u - v)^2 = u^2 - 2uv + v^2$$

$$x + y = u + uv + v + uv = u + v + 2uv$$

$$(x - y)^2 + 2(x + y) + 1 = u^2 + 2uv + v^2 + 2(u + v) + 1 = (u + v)^2 + 2(u + v) + 1 = (u + v + 1)^2.$$

Then

$$\begin{aligned}
 \iint_R \frac{1}{\sqrt{(x - y)^2 + 2(x + y) + 1}} \, dA &= \iint_S \frac{1}{u + v + 1} (u + v + 1) \, dA' = \int_0^1 \int_v^{2/(1+v)} du \, dv \\
 &= \int_0^1 \left(\frac{2}{1+v} - v \right) dv = \left[2 \ln(1+v) - \frac{1}{2} v^2 \right] \Big|_0^1 = 2 \ln 2 - \frac{1}{2}.
 \end{aligned}$$

65. The equations of the spheres are $x^2 + y^2 + z^2 = a^2$ and $x^2 + y^2 + (z - a)^2 = 1$. Subtracting these equations, we

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obtain $(z - a)^2 - z^2 = 1 - a^2$ or $-2az + a^2 = 1 - a^2$. Thus, the spheres intersect on the plane $z = a - 1/2a$. The region of integration is $x^2 + y^2 + (a - 1/2a)^2 = a^2$ or $r^2 = 1 - 1/4a^2$. The area is

$$\begin{aligned} A &= a \int_0^{2\pi} \int_0^{\sqrt{1-1/4a^2}} (a^2 - r^2)^{-1/2} r \, dr \, d\theta = 2\pi a \left[-(a^2 - r^2)^{1/2} \right]_0^{\sqrt{1-1/4a^2}} \\ &= 2\pi a \left(a - \left[a^2 - \left(1 - \frac{1}{4a^2} \right) \right]^{1/2} \right) = 2\pi a \left(a - \left[\left(a - \frac{1}{2a} \right)^2 \right]^{1/2} \right) = \pi. \end{aligned}$$

66. (a) Both states span 7 degrees of longitude and 4 degrees of latitude, but Colorado is larger because it lies to the south of Wyoming. Lines of longitude converge as they go north, so the east-west dimensions of Wyoming are shorter than those of Colorado.
- (b) We use the function $f(x, y) = \sqrt{R^2 - x^2 - y^2}$ to describe the northern hemisphere, where $R \approx 3960$ miles is the radius of the Earth. We need to compute the surface area over a polar rectangle P of the form $\theta_1 \leq \theta \leq \theta_2$, $R \cos \phi_2 \leq r \leq R \cos \phi_1$. We have

$$f_x = \frac{-x}{\sqrt{R^2 - x^2 - y^2}} \quad \text{and} \quad f_y = \frac{-y}{\sqrt{R^2 - x^2 - y^2}}$$

so that

$$\sqrt{1 + f_x^2 + f_y^2} = \sqrt{1 + \frac{x^2 + y^2}{R^2 - x^2 - y^2}} = \frac{R}{\sqrt{R^2 - r^2}}.$$

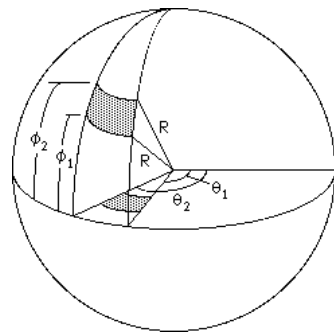
Thus

$$\begin{aligned} A &= \iint_P \sqrt{1 + f_x^2 + f_y^2} \, dA = \int_{\theta_1}^{\theta_2} \int_{R \cos \phi_2}^{R \cos \phi_1} \frac{R}{\sqrt{R^2 - r^2}} r \, dr \, d\theta \\ &= (\theta_2 - \theta_1) R \sqrt{R^2 - r^2} \Big|_{R \cos \phi_1}^{R \cos \phi_2} = (\theta_2 - \theta_1) R^2 (\sin \phi_2 - \sin \phi_1). \end{aligned}$$

The ratio of Wyoming to Colorado is then $\frac{\sin 45^\circ - \sin 41^\circ}{\sin 41^\circ - \sin 37^\circ} \approx 0.941$. Thus Wyoming is about 6% smaller

than Colorado.

- (c) $97,914/104,247 \approx 0.939$, which is close to the theoretical value of 0.941. (Our formula for the area says that the area of Colorado is approximately 103,924 square miles, while the area of Wyoming is approximately 97,801 square miles.)



10 Systems of Linear Differential Equations

Exercises 10.1

1. Let $\mathbf{X} = \begin{pmatrix} x \\ y \end{pmatrix}$. Then

$$\mathbf{X}' = \begin{pmatrix} 3 & -5 \\ 4 & 8 \end{pmatrix} \mathbf{X}.$$

2. Let $\mathbf{X} = \begin{pmatrix} x \\ y \end{pmatrix}$. Then

$$\mathbf{X}' = \begin{pmatrix} 4 & -7 \\ 5 & 0 \end{pmatrix} \mathbf{X}.$$

3. Let $\mathbf{X} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$. Then

$$\mathbf{X}' = \begin{pmatrix} -3 & 4 & -9 \\ 6 & -1 & 0 \\ 10 & 4 & 3 \end{pmatrix} \mathbf{X}.$$

4. Let $\mathbf{X} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$. Then

$$\mathbf{X}' = \begin{pmatrix} 1 & -1 & 0 \\ 1 & 0 & 2 \\ -1 & 0 & 1 \end{pmatrix} \mathbf{X}.$$

5. Let $\mathbf{X} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$. Then

$$\mathbf{X}' = \begin{pmatrix} 1 & -1 & 1 \\ 2 & 1 & -1 \\ 1 & 1 & 1 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 0 \\ -3t^2 \\ t^2 \end{pmatrix} + \begin{pmatrix} t \\ 0 \\ -t \end{pmatrix} + \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix}.$$

6. Let $\mathbf{X} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$. Then

$$\mathbf{X}' = \begin{pmatrix} -3 & 4 & 0 \\ 5 & 0 & 9 \\ 0 & 1 & 6 \end{pmatrix} \mathbf{X} + \begin{pmatrix} e^{-t} \sin 2t \\ 4e^{-t} \cos 2t \\ -e^{-t} \end{pmatrix}$$

7. $\frac{dx}{dt} = 4x + 2y + e^t$; $\frac{dy}{dt} = -x + 3y - e^t$

8. $\frac{dx}{dt} = 7x + 5y - 9z - 8e^{-2t}$; $\frac{dy}{dt} = 4x + y + z + 2e^{5t}$; $\frac{dz}{dt} = -2y + 3z + e^{5t} - 3e^{-2t}$

Exercises 10.1

9. $\frac{dx}{dt} = x - y + 2z + e^{-t} - 3t$; $\frac{dy}{dt} = 3x - 4y + z + 2e^{-t} + t$; $\frac{dz}{dt} = -2x + 5y + 6z + 2e^{-t} - t$

10. $\frac{dx}{dt} = 3x - 7y + 4\sin t + (t-4)e^{4t}$; $\frac{dy}{dt} = x + y + 8\sin t + (2t+1)e^{4t}$

11. Since

$$\mathbf{X}' = \begin{pmatrix} -5 \\ -10 \end{pmatrix} e^{-5t} \quad \text{and} \quad \begin{pmatrix} 3 & -4 \\ 4 & -7 \end{pmatrix} \mathbf{X} = \begin{pmatrix} -5 \\ -10 \end{pmatrix} e^{-5t}$$

we see that

$$\mathbf{X}' = \begin{pmatrix} 3 & -4 \\ 4 & -7 \end{pmatrix} \mathbf{X}.$$

12. Since

$$\mathbf{X}' = \begin{pmatrix} 5\cos t - 5\sin t \\ 2\cos t - 4\sin t \end{pmatrix} e^t \quad \text{and} \quad \begin{pmatrix} -2 & 5 \\ -2 & 4 \end{pmatrix} \mathbf{X} = \begin{pmatrix} 5\cos t - 5\sin t \\ 2\cos t - 4\sin t \end{pmatrix} e^t$$

we see that

$$\mathbf{X}' = \begin{pmatrix} -2 & 5 \\ -2 & 4 \end{pmatrix} \mathbf{X}.$$

13. Since

$$\mathbf{X}' = \begin{pmatrix} 3/2 \\ -3 \end{pmatrix} e^{-3t/2} \quad \text{and} \quad \begin{pmatrix} -1 & 1/4 \\ 1 & -1 \end{pmatrix} \mathbf{X} = \begin{pmatrix} 3/2 \\ -3 \end{pmatrix} e^{-3t/2}$$

we see that

$$\mathbf{X}' = \begin{pmatrix} -1 & 1/4 \\ 1 & -1 \end{pmatrix} \mathbf{X}.$$

14. Since

$$\mathbf{X}' = \begin{pmatrix} 5 \\ -1 \end{pmatrix} e^t + \begin{pmatrix} 4 \\ -4 \end{pmatrix} te^t \quad \text{and} \quad \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix} \mathbf{X} = \begin{pmatrix} 5 \\ -1 \end{pmatrix} e^t + \begin{pmatrix} 4 \\ -4 \end{pmatrix} te^t$$

we see that

$$\mathbf{X}' = \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix} \mathbf{X}.$$

15. Since

$$\mathbf{X}' = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & 2 & 1 \\ 6 & -1 & 0 \\ -1 & -2 & -1 \end{pmatrix} \mathbf{X} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

we see that

$$\mathbf{X}' = \begin{pmatrix} 1 & 2 & 1 \\ 6 & -1 & 0 \\ -1 & -2 & -1 \end{pmatrix} \mathbf{X}.$$

16. Since

$$\mathbf{X}' = \begin{pmatrix} \cos t \\ \frac{1}{2}\sin t - \frac{1}{2}\cos t \\ -\cos t - \sin t \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ -2 & 0 & -1 \end{pmatrix} \mathbf{X} = \begin{pmatrix} \cos t \\ \frac{1}{2}\sin t - \frac{1}{2}\cos t \\ -\cos t - \sin t \end{pmatrix}$$

we see that

$$\mathbf{X}' = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ -2 & 0 & -1 \end{pmatrix} \mathbf{X}.$$

17. Yes, since $W(\mathbf{X}_1, \mathbf{X}_2) = -2e^{-8t} \neq 0$ and $\mathbf{X}_1$ and $\mathbf{X}_2$ are linearly independent on $-\infty < t < \infty$.

18. Yes, since $W(\mathbf{X}_1, \mathbf{X}_2) = 8e^{2t} \neq 0$ and $\mathbf{X}_1$ and $\mathbf{X}_2$ are linearly independent on $-\infty < t < \infty$.

19. No, since $W(\mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3) = 0$ and $\mathbf{X}_1, \mathbf{X}_2$, and $\mathbf{X}_3$ are linearly dependent on $-\infty < t < \infty$.

20. Yes, since $W(\mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3) = -84e^{-t} \neq 0$ and $\mathbf{X}_1, \mathbf{X}_2$, and $\mathbf{X}_3$ are linearly independent on $-\infty < t < \infty$.

21. Since

$$\mathbf{X}'_p = \begin{pmatrix} 2 \\ -1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & 4 \\ 3 & 2 \end{pmatrix} \mathbf{X}_p + \begin{pmatrix} 2 \\ -4 \end{pmatrix} t + \begin{pmatrix} -7 \\ -18 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$$

we see that

$$\mathbf{X}'_p = \begin{pmatrix} 1 & 4 \\ 3 & 2 \end{pmatrix} \mathbf{X}_p + \begin{pmatrix} 2 \\ -4 \end{pmatrix} t + \begin{pmatrix} -7 \\ -18 \end{pmatrix}.$$

22. Since

$$\mathbf{X}'_p = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix} \mathbf{X}_p + \begin{pmatrix} -5 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

we see that

$$\mathbf{X}'_p = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix} \mathbf{X}_p + \begin{pmatrix} -5 \\ 2 \end{pmatrix}.$$

23. Since

$$\mathbf{X}'_p = \begin{pmatrix} 2 \\ 0 \end{pmatrix} e^t + \begin{pmatrix} 1 \\ -1 \end{pmatrix} te^t \quad \text{and} \quad \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix} \mathbf{X}_p - \begin{pmatrix} 1 \\ 7 \end{pmatrix} e^t = \begin{pmatrix} 2 \\ 0 \end{pmatrix} e^t + \begin{pmatrix} 1 \\ -1 \end{pmatrix} te^t$$

we see that

$$\mathbf{X}'_p = \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix} \mathbf{X}_p - \begin{pmatrix} 1 \\ 7 \end{pmatrix} e^t.$$

24. Since

$$\mathbf{X}'_p = \begin{pmatrix} 3 \cos 3t \\ 0 \\ -3 \sin 3t \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & 2 & 3 \\ -4 & 2 & 0 \\ -6 & 1 & 0 \end{pmatrix} \mathbf{X}_p + \begin{pmatrix} -1 \\ 4 \\ 3 \end{pmatrix} \sin 3t = \begin{pmatrix} 3 \cos 3t \\ 0 \\ -3 \sin 3t \end{pmatrix}$$

we see that

$$\mathbf{X}'_p = \begin{pmatrix} 1 & 2 & 3 \\ -4 & 2 & 0 \\ -6 & 1 & 0 \end{pmatrix} \mathbf{X}_p + \begin{pmatrix} -1 \\ 4 \\ 3 \end{pmatrix} \sin 3t.$$

25. Let

$$\mathbf{X}_1 = \begin{pmatrix} 6 \\ -1 \\ -5 \end{pmatrix} e^{-t}, \quad \mathbf{X}_2 = \begin{pmatrix} -3 \\ 1 \\ 1 \end{pmatrix} e^{-2t}, \quad \mathbf{X}_3 = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} e^{3t}, \quad \text{and} \quad \mathbf{A} = \begin{pmatrix} 0 & 6 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$$

Then

$$\mathbf{X}'_1 = \begin{pmatrix} -6 \\ 1 \\ 5 \end{pmatrix} e^{-t} = \mathbf{A}\mathbf{X}_1,$$

$$\mathbf{X}'_2 = \begin{pmatrix} 6 \\ -2 \\ -2 \end{pmatrix} e^{-2t} = \mathbf{A}\mathbf{X}_2,$$

$$\mathbf{X}'_3 = \begin{pmatrix} 6 \\ 3 \\ 3 \end{pmatrix} e^{3t} = \mathbf{A}\mathbf{X}_3,$$

and $W(\mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3) = 20 \neq 0$ so that $\mathbf{X}_1, \mathbf{X}_2$, and $\mathbf{X}_3$ form a fundamental set for $\mathbf{X}' = \mathbf{A}\mathbf{X}$ on $-\infty < t < \infty$.

Exercises 10.1

26. Let

$$\mathbf{X}_1 = \begin{pmatrix} 1 \\ -1 - \sqrt{2} \end{pmatrix} e^{\sqrt{2}t},$$

$$\mathbf{X}_2 = \begin{pmatrix} 1 \\ -1 + \sqrt{2} \end{pmatrix} e^{-\sqrt{2}t},$$

$$\mathbf{X}_p = \begin{pmatrix} 1 \\ 0 \end{pmatrix} t^2 + \begin{pmatrix} -2 \\ 4 \end{pmatrix} t + \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

and

$$\mathbf{A} = \begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix}.$$

Then

$$\mathbf{X}'_1 = \begin{pmatrix} \sqrt{2} \\ -2 - \sqrt{2} \end{pmatrix} e^{\sqrt{2}t} = \mathbf{A}\mathbf{X}_1,$$

$$\mathbf{X}'_2 = \begin{pmatrix} -\sqrt{2} \\ -2 + \sqrt{2} \end{pmatrix} e^{-\sqrt{2}t} = \mathbf{A}\mathbf{X}_2,$$

$$\mathbf{X}'_p = \begin{pmatrix} 2 \\ 0 \end{pmatrix} t + \begin{pmatrix} -2 \\ 4 \end{pmatrix} = \mathbf{A}\mathbf{X}_p + \begin{pmatrix} 1 \\ 1 \end{pmatrix} t^2 + \begin{pmatrix} 4 \\ -6 \end{pmatrix} t + \begin{pmatrix} -1 \\ 5 \end{pmatrix},$$

and $W(\mathbf{X}_1, \mathbf{X}_2) = 2\sqrt{2} \neq 0$ so that $\mathbf{X}_p$ is a particular solution and $\mathbf{X}_1$ and $\mathbf{X}_2$ form a fundamental set on $-\infty < t < \infty$.

Exercises 10.2

1. The system is

$$\mathbf{X}' = \begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix} \mathbf{X}$$

and $\det(\mathbf{A} - \lambda\mathbf{I}) = (\lambda - 5)(\lambda + 1) = 0$. For $\lambda_1 = 5$ we obtain

$$\left(\begin{array}{cc|c} -4 & 2 & 0 \\ 4 & -2 & 0 \end{array} \right) \Rightarrow \left(\begin{array}{cc|c} 1 & -1/2 & 0 \\ 0 & 0 & 0 \end{array} \right) \quad \text{so that} \quad \mathbf{K}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

For $\lambda_2 = -1$ we obtain

$$\left(\begin{array}{cc|c} 2 & 2 & 0 \\ 4 & 4 & 0 \end{array} \right) \Rightarrow \left(\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 0 & 0 \end{array} \right) \quad \text{so that} \quad \mathbf{K}_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

Then

$$\mathbf{X} = c_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{5t} + c_2 \begin{pmatrix} -1 \\ 1 \end{pmatrix} e^{-t}.$$

2. The system is

$$\mathbf{X}' = \begin{pmatrix} 2 & 2 \\ 1 & 3 \end{pmatrix} \mathbf{X}$$

and $\det(\mathbf{A} - \lambda\mathbf{I}) = (\lambda - 1)(\lambda - 4) = 0$. For $\lambda_1 = 1$ we obtain

$$\left(\begin{array}{cc|c} 1 & 2 & 0 \\ 1 & 2 & 0 \end{array} \right) \Rightarrow \left(\begin{array}{cc|c} 1 & 2 & 0 \\ 0 & 0 & 0 \end{array} \right) \quad \text{so that} \quad \mathbf{K}_1 = \begin{pmatrix} -2 \\ 1 \end{pmatrix}.$$

For $\lambda_2 = 4$ we obtain

$$\left(\begin{array}{cc|c} -2 & 2 & 0 \\ 1 & -1 & 0 \end{array} \right) \Rightarrow \left(\begin{array}{cc|c} -1 & 1 & 0 \\ 0 & 0 & 0 \end{array} \right) \quad \text{so that} \quad \mathbf{K}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Then

$$\mathbf{X} = c_1 \begin{pmatrix} -2 \\ 1 \end{pmatrix} e^t + c_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{4t}.$$

3. The system is

$$\mathbf{X}' = \begin{pmatrix} -4 & 2 \\ -5/2 & 2 \end{pmatrix} \mathbf{X}$$

and $\det(\mathbf{A} - \lambda \mathbf{I}) = (\lambda - 1)(\lambda + 3) = 0$. For $\lambda_1 = 1$ we obtain

$$\left(\begin{array}{cc|c} -5 & 2 & 0 \\ -5/2 & 1 & 0 \end{array} \right) \Rightarrow \left(\begin{array}{cc|c} -5 & 2 & 0 \\ 0 & 0 & 0 \end{array} \right) \quad \text{so that} \quad \mathbf{K}_1 = \begin{pmatrix} 2 \\ 5 \end{pmatrix}.$$

For $\lambda_2 = -3$ we obtain

$$\left(\begin{array}{cc|c} -1 & 2 & 0 \\ -5/2 & 5 & 0 \end{array} \right) \Rightarrow \left(\begin{array}{cc|c} -1 & 2 & 0 \\ 0 & 0 & 0 \end{array} \right) \quad \text{so that} \quad \mathbf{K}_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Then

$$\mathbf{X} = c_1 \begin{pmatrix} 2 \\ 5 \end{pmatrix} e^t + c_2 \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{-3t}.$$

4. The system is

$$\mathbf{X}' = \begin{pmatrix} -5/2 & 2 \\ 3/4 & -2 \end{pmatrix} \mathbf{X}$$

and $\det(\mathbf{A} - \lambda \mathbf{I}) = (\lambda + 1)(2\lambda + 7) = 0$. For $\lambda_1 = -7/2$ we obtain

$$\left(\begin{array}{cc|c} 1 & 2 & 0 \\ 3/4 & 3/2 & 0 \end{array} \right) \Rightarrow \left(\begin{array}{cc|c} -1 & 2 & 0 \\ 0 & 0 & 0 \end{array} \right) \quad \text{so that} \quad \mathbf{K}_1 = \begin{pmatrix} -2 \\ 1 \end{pmatrix}.$$

For $\lambda_2 = -1$ we obtain

$$\left(\begin{array}{cc|c} -3/2 & 2 & 0 \\ 3/4 & -1 & 0 \end{array} \right) \Rightarrow \left(\begin{array}{cc|c} -3 & 4 & 0 \\ 0 & 0 & 0 \end{array} \right) \quad \text{so that} \quad \mathbf{K}_2 = \begin{pmatrix} 4 \\ 3 \end{pmatrix}.$$

Then

$$\mathbf{X} = c_1 \begin{pmatrix} -2 \\ 1 \end{pmatrix} e^{-7t/2} + c_2 \begin{pmatrix} 4 \\ 3 \end{pmatrix} e^{-t}.$$

5. The system is

$$\mathbf{X}' = \begin{pmatrix} 10 & -5 \\ 8 & -12 \end{pmatrix} \mathbf{X}$$

and $\det(\mathbf{A} - \lambda \mathbf{I}) = (\lambda - 8)(\lambda + 10) = 0$. For $\lambda_1 = 8$ we obtain

$$\left(\begin{array}{cc|c} 2 & -5 & 0 \\ 8 & -20 & 0 \end{array} \right) \Rightarrow \left(\begin{array}{cc|c} 1 & -5/2 & 0 \\ 0 & 0 & 0 \end{array} \right) \quad \text{so that} \quad \mathbf{K}_1 = \begin{pmatrix} 5 \\ 2 \end{pmatrix}.$$

For $\lambda_2 = -10$ we obtain

$$\left(\begin{array}{cc|c} 20 & -5 & 0 \\ 8 & -2 & 0 \end{array} \right) \Rightarrow \left(\begin{array}{cc|c} 1 & -1/4 & 0 \\ 0 & 0 & 0 \end{array} \right) \quad \text{so that} \quad \mathbf{K}_2 = \begin{pmatrix} 1 \\ 4 \end{pmatrix}.$$

Then

$$\mathbf{X} = c_1 \begin{pmatrix} 5 \\ 2 \end{pmatrix} e^{8t} + c_2 \begin{pmatrix} 1 \\ 4 \end{pmatrix} e^{-10t}.$$

6. The system is

$$\mathbf{X}' = \begin{pmatrix} -6 & 2 \\ -3 & 1 \end{pmatrix} \mathbf{X}$$

Exercises 10.2

and $\det(\mathbf{A} - \lambda\mathbf{I}) = \lambda(\lambda + 5) = 0$. For $\lambda_1 = 0$ we obtain

$$\left(\begin{array}{cc|c} -6 & 2 & 0 \\ -3 & 1 & 0 \end{array}\right) \Rightarrow \left(\begin{array}{cc|c} 1 & -1/3 & 0 \\ 0 & 0 & 0 \end{array}\right) \quad \text{so that} \quad \mathbf{K}_1 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}.$$

For $\lambda_2 = -5$ we obtain

$$\left(\begin{array}{cc|c} -1 & 2 & 0 \\ -3 & 6 & 0 \end{array}\right) \Rightarrow \left(\begin{array}{cc|c} 1 & -2 & 0 \\ 0 & 0 & 0 \end{array}\right) \quad \text{so that} \quad \mathbf{K}_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Then

$$\mathbf{X} = c_1 \begin{pmatrix} 1 \\ 3 \end{pmatrix} + c_2 \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{-5t}.$$

7. The system is

$$\mathbf{X}' = \begin{pmatrix} 1 & 1 & -1 \\ 0 & 2 & 0 \\ 0 & 1 & -1 \end{pmatrix} \mathbf{X}$$

and $\det(\mathbf{A} - \lambda\mathbf{I}) = (\lambda - 1)(2 - \lambda)(\lambda + 1) = 0$. For $\lambda_1 = 1$, $\lambda_2 = 2$, and $\lambda_3 = -1$ we obtain

$$\mathbf{K}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{K}_2 = \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}, \quad \text{and} \quad \mathbf{K}_3 = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix},$$

so that

$$\mathbf{X} = c_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} e^t + c_2 \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} e^{2t} + c_3 \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} e^{-t}.$$

8. The system is

$$\mathbf{X}' = \begin{pmatrix} 2 & -7 & 0 \\ 5 & 10 & 4 \\ 0 & 5 & 2 \end{pmatrix} \mathbf{X}$$

and $\det(\mathbf{A} - \lambda\mathbf{I}) = (2 - \lambda)(\lambda - 5)(\lambda - 7) = 0$. For $\lambda_1 = 2$, $\lambda_2 = 5$, and $\lambda_3 = 7$ we obtain

$$\mathbf{K}_1 = \begin{pmatrix} 4 \\ 0 \\ -5 \end{pmatrix}, \quad \mathbf{K}_2 = \begin{pmatrix} -7 \\ 3 \\ 5 \end{pmatrix}, \quad \text{and} \quad \mathbf{K}_3 = \begin{pmatrix} -7 \\ 5 \\ 5 \end{pmatrix},$$

so that

$$\mathbf{X} = c_1 \begin{pmatrix} 4 \\ 0 \\ -5 \end{pmatrix} e^{2t} + c_2 \begin{pmatrix} -7 \\ 3 \\ 5 \end{pmatrix} e^{5t} + c_3 \begin{pmatrix} -7 \\ 5 \\ 5 \end{pmatrix} e^{7t}.$$

9. We have $\det(\mathbf{A} - \lambda\mathbf{I}) = -(\lambda + 1)(\lambda - 3)(\lambda + 2) = 0$. For $\lambda_1 = -1$, $\lambda_2 = 3$, and $\lambda_3 = -2$ we obtain

$$\mathbf{K}_1 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \quad \mathbf{K}_2 = \begin{pmatrix} 1 \\ 4 \\ 3 \end{pmatrix}, \quad \text{and} \quad \mathbf{K}_3 = \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix},$$

so that

$$\mathbf{X} = c_1 \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} e^{-t} + c_2 \begin{pmatrix} 1 \\ 4 \\ 3 \end{pmatrix} e^{3t} + c_3 \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix} e^{-2t}.$$

10. We have $\det(\mathbf{A} - \lambda\mathbf{I}) = -\lambda(\lambda - 1)(\lambda - 2) = 0$. For $\lambda_1 = 0$, $\lambda_2 = 1$, and $\lambda_3 = 2$ we obtain

$$\mathbf{K}_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \quad \mathbf{K}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \text{and} \quad \mathbf{K}_3 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix},$$

so that

$$\mathbf{X} = c_1 \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} e^t + c_3 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} e^{2t}.$$

11. We have $\det(\mathbf{A} - \lambda\mathbf{I}) = -(\lambda + 1)(\lambda + 1/2)(\lambda + 3/2) = 0$. For $\lambda_1 = -1$, $\lambda_2 = -1/2$, and $\lambda_3 = -3/2$ we obtain

$$\mathbf{K}_1 = \begin{pmatrix} 4 \\ 0 \\ -1 \end{pmatrix}, \quad \mathbf{K}_2 = \begin{pmatrix} -12 \\ 6 \\ 5 \end{pmatrix}, \quad \text{and} \quad \mathbf{K}_3 = \begin{pmatrix} 4 \\ 2 \\ -1 \end{pmatrix},$$

so that

$$\mathbf{X} = c_1 \begin{pmatrix} 4 \\ 0 \\ -1 \end{pmatrix} e^{-t} + c_2 \begin{pmatrix} -12 \\ 6 \\ 5 \end{pmatrix} e^{-t/2} + c_3 \begin{pmatrix} 4 \\ 2 \\ -1 \end{pmatrix} e^{-3t/2}.$$

12. We have $\det(\mathbf{A} - \lambda\mathbf{I}) = (\lambda - 3)(\lambda + 5)(6 - \lambda) = 0$. For $\lambda_1 = 3$, $\lambda_2 = -5$, and $\lambda_3 = 6$ we obtain

$$\mathbf{K}_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{K}_2 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \quad \text{and} \quad \mathbf{K}_3 = \begin{pmatrix} 2 \\ -2 \\ 11 \end{pmatrix},$$

so that

$$\mathbf{X} = c_1 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} e^{3t} + c_2 \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} e^{-5t} + c_3 \begin{pmatrix} 2 \\ -2 \\ 11 \end{pmatrix} e^{6t}.$$

13. We have $\det(\mathbf{A} - \lambda\mathbf{I}) = (\lambda + 1/2)(\lambda - 1/2) = 0$. For $\lambda_1 = -1/2$ and $\lambda_2 = 1/2$ we obtain

$$\mathbf{K}_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad \text{and} \quad \mathbf{K}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix},$$

so that

$$\mathbf{X} = c_1 \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{-t/2} + c_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{t/2}.$$

If

$$\mathbf{X}(0) = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$$

then $c_1 = 2$ and $c_2 = 3$.

14. We have $\det(\mathbf{A} - \lambda\mathbf{I}) = (2 - \lambda)(\lambda - 3)(\lambda + 1) = 0$. For $\lambda_1 = 2$, $\lambda_2 = 3$, and $\lambda_3 = -1$ we obtain

$$\mathbf{K}_1 = \begin{pmatrix} 5 \\ -3 \\ 2 \end{pmatrix}, \quad \mathbf{K}_2 = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}, \quad \text{and} \quad \mathbf{K}_3 = \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix},$$

so that

$$\mathbf{X} = c_1 \begin{pmatrix} 5 \\ -3 \\ 2 \end{pmatrix} e^{2t} + c_2 \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} e^{3t} + c_3 \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix} e^{-t}.$$

Exercises 10.2

If

$$\mathbf{X}(0) = \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix}$$

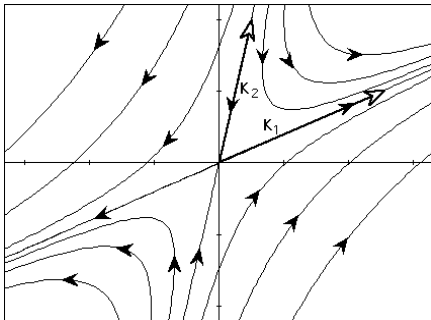
then $c_1 = -1$, $c_2 = 5/2$, and $c_3 = -1/2$.

$$15. \mathbf{X} = c_1 \begin{pmatrix} 0.382175 \\ 0.851161 \\ 0.359815 \end{pmatrix} e^{8.58979t} + c_2 \begin{pmatrix} 0.405188 \\ -0.676043 \\ 0.615458 \end{pmatrix} e^{2.25684t} + c_3 \begin{pmatrix} -0.923562 \\ -0.132174 \\ 0.35995 \end{pmatrix} e^{-0.0466321t}$$

$$16. \mathbf{X} = c_1 \begin{pmatrix} 0.0312209 \\ 0.949058 \\ 0.239535 \\ 0.195825 \\ 0.0508861 \end{pmatrix} e^{5.05452t} + c_2 \begin{pmatrix} -0.280232 \\ -0.836611 \\ -0.275304 \\ 0.176045 \\ 0.338775 \end{pmatrix} e^{4.09561t} + c_3 \begin{pmatrix} 0.262219 \\ -0.162664 \\ -0.826218 \\ -0.346439 \\ 0.31957 \end{pmatrix} e^{-2.92362t}$$

$$+ c_4 \begin{pmatrix} 0.313235 \\ 0.64181 \\ 0.31754 \\ 0.173787 \\ -0.599108 \end{pmatrix} e^{2.02882t} + c_5 \begin{pmatrix} -0.301294 \\ 0.466599 \\ 0.222136 \\ 0.0534311 \\ -0.799567 \end{pmatrix} e^{-0.155338t}$$

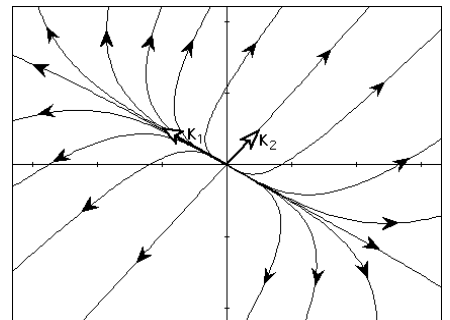
17. (a)



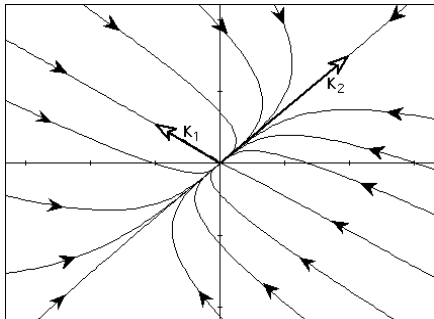
(b) Letting $c_1 = 1$ and $c_2 = 0$ we get $x = 5e^{8t}$, $y = 2e^{8t}$. Eliminating the parameter we find $y = \frac{2}{5}x$, $x > 0$. When $c_1 = -1$ and $c_2 = 0$ we find $y = \frac{2}{5}x$, $x < 0$. Letting $c_1 = 0$ and $c_2 = 1$ we get $x = e^{-10t}$, $y = 4e^{-10t}$. Eliminating the parameter we find $y = 4x$, $x > 0$. Letting $c_1 = 0$ and $c_2 = -1$ we find $y = 4x$, $x < 0$.

(c) The eigenvectors $\mathbf{K}_1 = (5, 2)$ and $\mathbf{K}_2 = (1, 4)$ are shown in the figure in part (a).

18. (a) Letting $c_1 = 1$ and $c_2 = 0$ we get $x = -2e^t$, $y = e^t$. Eliminating the parameter we find $y = -\frac{1}{2}x$, $x < 0$. When $c_1 = -1$ and $c_2 = 0$ we find $y = -\frac{1}{2}x$, $x > 0$. Letting $c_1 = 0$ and $c_2 = 1$ we get $x = e^{4t}$, $y = e^{4t}$. Eliminating the parameter we find $y = x$, $x > 0$. When $c_1 = 0$ and $c_2 = -1$ we find $y = x$, $x < 0$.



Letting $c_1 = 1$ and $c_2 = 0$ we get $x = -2e^{-7t/2}$, $y = e^{-7t/2}$. Eliminating the parameter we find $y = -\frac{1}{2}x$, $x < 0$. When $c_1 = -1$ and $c_2 = 0$ we find $y = -\frac{1}{2}x$, $x > 0$. Letting $c_1 = 0$ and $c_2 = 1$ we get $x = 4e^{-t}$, $y = 3e^{-t}$. Eliminating the parameter we find $y = \frac{3}{4}x$, $x > 0$. When $c_1 = 0$ and $c_2 = -1$ we find $y = \frac{3}{4}x$, $x < 0$.



19. We have $\det(\mathbf{A} - \lambda\mathbf{I}) = \lambda^2 = 0$. For $\lambda_1 = 0$ we obtain

$$\mathbf{K} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}.$$

A solution of $(\mathbf{A} - \lambda_1\mathbf{I})\mathbf{P} = \mathbf{K}$ is

$$\mathbf{P} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

so that

$$\mathbf{X} = c_1 \begin{pmatrix} 1 \\ 3 \end{pmatrix} + c_2 \left[\begin{pmatrix} 1 \\ 3 \end{pmatrix} t + \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right].$$

20. We have $\det(\mathbf{A} - \lambda\mathbf{I}) = (\lambda + 1)^2 = 0$. For $\lambda_1 = -1$ we obtain

$$\mathbf{K} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

A solution of $(\mathbf{A} - \lambda_1\mathbf{I})\mathbf{P} = \mathbf{K}$ is

$$\mathbf{P} = \begin{pmatrix} 0 \\ 1/5 \end{pmatrix}$$

so that

$$\mathbf{X} = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{-t} + c_2 \left[\begin{pmatrix} 1 \\ 1 \end{pmatrix} t e^{-t} + \begin{pmatrix} 0 \\ 1/5 \end{pmatrix} e^{-t} \right].$$

21. We have $\det(\mathbf{A} - \lambda\mathbf{I}) = (\lambda - 2)^2 = 0$. For $\lambda_1 = 2$ we obtain

$$\mathbf{K} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

A solution of $(\mathbf{A} - \lambda_1\mathbf{I})\mathbf{P} = \mathbf{K}$ is

$$\mathbf{P} = \begin{pmatrix} -1/3 \\ 0 \end{pmatrix}$$

so that

$$\mathbf{X} = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{2t} + c_2 \left[\begin{pmatrix} 1 \\ 1 \end{pmatrix} t e^{2t} + \begin{pmatrix} -1/3 \\ 0 \end{pmatrix} e^{2t} \right].$$

22. We have $\det(\mathbf{A} - \lambda\mathbf{I}) = (\lambda - 6)^2 = 0$. For $\lambda_1 = 6$ we obtain

$$\mathbf{K} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}.$$

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A solution of $(\mathbf{A} - \lambda_1 \mathbf{I})\mathbf{P} = \mathbf{K}$ is

$$\mathbf{P} = \begin{pmatrix} 1/2 \\ 0 \end{pmatrix}$$

so that

$$\mathbf{X} = c_1 \begin{pmatrix} 3 \\ 2 \end{pmatrix} e^{6t} + c_2 \left[\begin{pmatrix} 3 \\ 2 \end{pmatrix} t e^{6t} + \begin{pmatrix} 1/2 \\ 0 \end{pmatrix} e^{6t} \right].$$

23. We have $\det(\mathbf{A} - \lambda \mathbf{I}) = (1 - \lambda)(\lambda - 2)^2 = 0$. For $\lambda_1 = 1$ we obtain

$$\mathbf{K}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

For $\lambda_2 = 2$ we obtain

$$\mathbf{K}_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \quad \text{and} \quad \mathbf{K}_3 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}.$$

Then

$$\mathbf{X} = c_1 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} e^t + c_2 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} e^{2t} + c_3 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} e^{2t}.$$

24. We have $\det(\mathbf{A} - \lambda \mathbf{I}) = (\lambda - 8)(\lambda + 1)^2 = 0$. For $\lambda_1 = 8$ we obtain

$$\mathbf{K}_1 = \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}.$$

For $\lambda_2 = -1$ we obtain

$$\mathbf{K}_2 = \begin{pmatrix} 0 \\ -2 \\ 1 \end{pmatrix} \quad \text{and} \quad \mathbf{K}_3 = \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix}.$$

Then

$$\mathbf{X} = c_1 \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} e^{8t} + c_2 \begin{pmatrix} 0 \\ -2 \\ 1 \end{pmatrix} e^{-t} + c_3 \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix} e^{-t}.$$

25. We have $\det(\mathbf{A} - \lambda \mathbf{I}) = -\lambda(5 - \lambda)^2 = 0$. For $\lambda_1 = 0$ we obtain

$$\mathbf{K}_1 = \begin{pmatrix} -4 \\ -5 \\ 2 \end{pmatrix}.$$

For $\lambda_2 = 5$ we obtain

$$\mathbf{K} = \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix}.$$

A solution of $(\mathbf{A} - \lambda_1 \mathbf{I})\mathbf{P} = \mathbf{K}$ is

$$\mathbf{P} = \begin{pmatrix} 5/2 \\ 1/2 \\ 0 \end{pmatrix}$$

so that

$$\mathbf{X} = c_1 \begin{pmatrix} -4 \\ -5 \\ 2 \end{pmatrix} + c_2 \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix} e^{5t} + c_3 \left[\begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix} t e^{5t} + \begin{pmatrix} 5/2 \\ 1/2 \\ 0 \end{pmatrix} e^{5t} \right].$$

26. We have $\det(\mathbf{A} - \lambda \mathbf{I}) = (1 - \lambda)(\lambda - 2)^2 = 0$. For $\lambda_1 = 1$ we obtain

$$\mathbf{K}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.$$

For $\lambda_2 = 2$ we obtain

$$\mathbf{K} = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}.$$

A solution of $(\mathbf{A} - \lambda_2 \mathbf{I})\mathbf{P} = \mathbf{K}$ is

$$\mathbf{P} = \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix}$$

so that

$$\mathbf{X} = c_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} e^t + c_2 \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} e^{2t} + c_3 \left[\begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} t e^{2t} + \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix} e^{2t} \right].$$

27. We have $\det(\mathbf{A} - \lambda \mathbf{I}) = -(\lambda - 1)^3 = 0$. For $\lambda_1 = 1$ we obtain

$$\mathbf{K} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}.$$

Solutions of $(\mathbf{A} - \lambda_1 \mathbf{I})\mathbf{P} = \mathbf{K}$ and $(\mathbf{A} - \lambda_1 \mathbf{I})\mathbf{Q} = \mathbf{P}$ are

$$\mathbf{P} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad \text{and} \quad \mathbf{Q} = \begin{pmatrix} 1/2 \\ 0 \\ 0 \end{pmatrix}$$

so that

$$\mathbf{X} = c_1 \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + c_2 \left[\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} t e^t + \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} e^t \right] + c_3 \left[\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \frac{t^2}{2} e^t + \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} t e^t + \begin{pmatrix} 1/2 \\ 0 \\ 0 \end{pmatrix} e^t \right].$$

28. We have $\det(\mathbf{A} - \lambda \mathbf{I}) = (\lambda - 4)^3 = 0$. For $\lambda_1 = 4$ we obtain

$$\mathbf{K} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.$$

Solutions of $(\mathbf{A} - \lambda_1 \mathbf{I})\mathbf{P} = \mathbf{K}$ and $(\mathbf{A} - \lambda_1 \mathbf{I})\mathbf{Q} = \mathbf{P}$ are

$$\mathbf{P} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad \text{and} \quad \mathbf{Q} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

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so that

$$\mathbf{X} = c_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} e^{4t} + c_2 \left[\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} t e^{4t} + \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} e^{4t} \right] + c_3 \left[\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \frac{t^2}{2} e^{4t} + \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} t e^{4t} + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} e^{4t} \right].$$

29. We have $\det(\mathbf{A} - \lambda \mathbf{I}) = (\lambda - 4)^2 = 0$. For $\lambda_1 = 4$ we obtain

$$\mathbf{K} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

A solution of $(\mathbf{A} - \lambda_1 \mathbf{I})\mathbf{P} = \mathbf{K}$ is

$$\mathbf{P} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

so that

$$\mathbf{X} = c_1 \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{4t} + c_2 \left[\begin{pmatrix} 2 \\ 1 \end{pmatrix} t e^{4t} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{4t} \right].$$

If

$$\mathbf{X}(0) = \begin{pmatrix} -1 \\ 6 \end{pmatrix}$$

then $c_1 = -7$ and $c_2 = 13$.

30. We have $\det(\mathbf{A} - \lambda \mathbf{I}) = -(\lambda + 1)(\lambda - 1)^2 = 0$. For $\lambda_1 = -1$ we obtain

$$\mathbf{K}_1 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}.$$

For $\lambda_2 = 1$ we obtain

$$\mathbf{K}_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \quad \text{and} \quad \mathbf{K}_3 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

so that

$$\mathbf{X} = c_1 \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} e^{-t} + c_2 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} e^t + c_3 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} e^t.$$

If

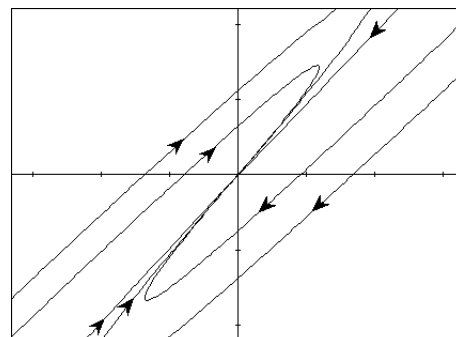
$$\mathbf{X}(0) = \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix}$$

then $c_1 = 2$, $c_2 = 3$, and $c_3 = 2$.

31. In this case $\det(\mathbf{A} - \lambda \mathbf{I}) = (2 - \lambda)^5$, and $\lambda_1 = 2$ is an eigenvalue of multiplicity 5. Linearly independent eigenvectors are

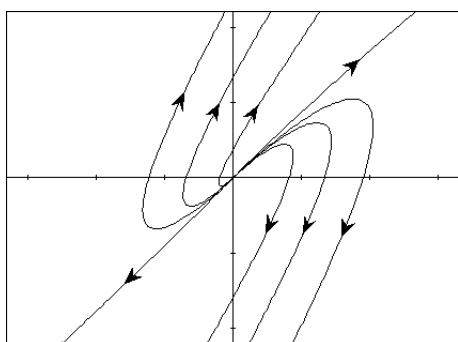
$$\mathbf{K}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{K}_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad \text{and} \quad \mathbf{K}_3 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}.$$

32. Letting $c_1 = 1$ and $c_2 = 0$ we get $x = e^t$, $y = e^t$. Eliminating the parameter we find $y = x$, $x > 0$. When $c_1 = -1$ and $c_2 = 0$ we find $y = x$, $x < 0$.



Phase portrait for the system in Problem 20

Letting $c_1 = 1$ and $c_2 = 0$ we get $x = e^{2t}$, $y = e^{2t}$. Eliminating the parameter we find $y = x$, $x > 0$. When $c_1 = -1$ and $c_2 = 0$ we find $y = x$, $x < 0$.



Phase portrait for the system in Problem 21

In Problems 33-46 the form of the answer will vary according to the choice of eigenvector. For example, in Problem 33, if $\mathbf{K}_1$ is chosen to be $\begin{pmatrix} 1 \\ 2-i \end{pmatrix}$ the solution has the form

$$\mathbf{X} = c_1 \begin{pmatrix} \cos t \\ 2 \cos t + \sin t \end{pmatrix} e^{4t} + c_2 \begin{pmatrix} \sin t \\ 2 \sin t - \cos t \end{pmatrix} e^{4t}.$$

33. We have $\det(\mathbf{A} - \lambda \mathbf{I}) = \lambda^2 - 8\lambda + 17 = 0$. For $\lambda_1 = 4 + i$ we obtain

$$\mathbf{K}_1 = \begin{pmatrix} 2+i \\ 5 \end{pmatrix}$$

so that

$$\mathbf{X}_1 = \begin{pmatrix} 2+i \\ 5 \end{pmatrix} e^{(4+i)t} = \begin{pmatrix} 2 \cos t - \sin t \\ 5 \cos t \end{pmatrix} e^{4t} + i \begin{pmatrix} \cos t + 2 \sin t \\ 5 \sin t \end{pmatrix} e^{4t}.$$

Then

$$\mathbf{X} = c_1 \begin{pmatrix} 2 \cos t - \sin t \\ 5 \cos t \end{pmatrix} e^{4t} + c_2 \begin{pmatrix} 2 \sin t + \cos t \\ 5 \sin t \end{pmatrix} e^{4t}.$$

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34. We have $\det(\mathbf{A} - \lambda\mathbf{I}) = \lambda^2 + 1 = 0$. For $\lambda_1 = i$ we obtain

$$\mathbf{K}_1 = \begin{pmatrix} -1 - i \\ 2 \end{pmatrix}$$

so that

$$\mathbf{X}_1 = \begin{pmatrix} -1 - i \\ 2 \end{pmatrix} e^{it} = \begin{pmatrix} \sin t - \cos t \\ 2 \cos t \end{pmatrix} + i \begin{pmatrix} -\cos t - \sin t \\ 2 \sin t \end{pmatrix}.$$

Then

$$\mathbf{X} = c_1 \begin{pmatrix} \sin t - \cos t \\ 2 \cos t \end{pmatrix} + c_2 \begin{pmatrix} -\cos t - \sin t \\ 2 \sin t \end{pmatrix}.$$

35. We have $\det(\mathbf{A} - \lambda\mathbf{I}) = \lambda^2 - 8\lambda + 17 = 0$. For $\lambda_1 = 4 + i$ we obtain

$$\mathbf{K}_1 = \begin{pmatrix} -1 - i \\ 2 \end{pmatrix}$$

so that

$$\mathbf{X}_1 = \begin{pmatrix} -1 - i \\ 2 \end{pmatrix} e^{(4+i)t} = \begin{pmatrix} \sin t - \cos t \\ 2 \cos t \end{pmatrix} e^{4t} + i \begin{pmatrix} -\sin t - \cos t \\ 2 \sin t \end{pmatrix} e^{4t}.$$

Then

$$\mathbf{X} = c_1 \begin{pmatrix} \sin t - \cos t \\ 2 \cos t \end{pmatrix} e^{4t} + c_2 \begin{pmatrix} -\sin t - \cos t \\ 2 \sin t \end{pmatrix} e^{4t}.$$

36. We have $\det(\mathbf{A} - \lambda\mathbf{I}) = \lambda^2 - 10\lambda + 34 = 0$. For $\lambda_1 = 5 + 3i$ we obtain

$$\mathbf{K}_1 = \begin{pmatrix} 1 - 3i \\ 2 \end{pmatrix}$$

so that

$$\mathbf{X}_1 = \begin{pmatrix} 1 - 3i \\ 2 \end{pmatrix} e^{(5+3i)t} = \begin{pmatrix} \cos 3t + 3 \sin 3t \\ 2 \cos 3t \end{pmatrix} e^{5t} + i \begin{pmatrix} \sin 3t - 3 \cos 3t \\ 2 \cos 3t \end{pmatrix} e^{5t}.$$

Then

$$\mathbf{X} = c_1 \begin{pmatrix} \cos 3t + 3 \sin 3t \\ 2 \cos 3t \end{pmatrix} e^{5t} + c_2 \begin{pmatrix} \sin 3t - 3 \cos 3t \\ 2 \cos 3t \end{pmatrix} e^{5t}.$$

37. We have $\det(\mathbf{A} - \lambda\mathbf{I}) = \lambda^2 + 9 = 0$. For $\lambda_1 = 3i$ we obtain

$$\mathbf{K}_1 = \begin{pmatrix} 4 + 3i \\ 5 \end{pmatrix}$$

so that

$$\mathbf{X}_1 = \begin{pmatrix} 4 + 3i \\ 5 \end{pmatrix} e^{3it} = \begin{pmatrix} 4 \cos 3t - 3 \sin 3t \\ 5 \cos 3t \end{pmatrix} + i \begin{pmatrix} 4 \sin 3t + 3 \cos 3t \\ 5 \sin 3t \end{pmatrix}.$$

Then

$$\mathbf{X} = c_1 \begin{pmatrix} 4 \cos 3t - 3 \sin 3t \\ 5 \cos 3t \end{pmatrix} + c_2 \begin{pmatrix} 4 \sin 3t + 3 \cos 3t \\ 5 \sin 3t \end{pmatrix}.$$

38. We have $\det(\mathbf{A} - \lambda\mathbf{I}) = \lambda^2 + 2\lambda + 5 = 0$. For $\lambda_1 = -1 + 2i$ we obtain

$$\mathbf{K}_1 = \begin{pmatrix} 2 + 2i \\ 1 \end{pmatrix}$$

so that

$$\begin{aligned} \mathbf{X}_1 &= \begin{pmatrix} 2 + 2i \\ 1 \end{pmatrix} e^{(-1+2i)t} \\ &= (2 \cos 2t - 2 \sin 2t \cos 2t) e^{-t} + i \begin{pmatrix} 2 \cos 2t + 2 \sin 2t \\ \sin 2t \end{pmatrix} e^{-t}. \end{aligned}$$

Then

$$\mathbf{X} = c_1 \begin{pmatrix} 2 \cos 2t - 2 \sin 2t \\ \cos 2t \end{pmatrix} e^{-t} + c_2 \begin{pmatrix} 2 \cos 2t + 2 \sin 2t \\ \sin 2t \end{pmatrix} e^{-t}.$$

39. We have $\det(\mathbf{A} - \lambda \mathbf{I}) = -\lambda(\lambda^2 + 1) = 0$. For $\lambda_1 = 0$ we obtain

$$\mathbf{K}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.$$

For $\lambda_2 = i$ we obtain

$$\mathbf{K}_2 = \begin{pmatrix} -i \\ i \\ 1 \end{pmatrix}$$

so that

$$\mathbf{X}_2 = \begin{pmatrix} -i \\ i \\ 1 \end{pmatrix} e^{it} = \begin{pmatrix} \sin t \\ -\sin t \\ \cos t \end{pmatrix} + i \begin{pmatrix} -\cos t \\ \cos t \\ \sin t \end{pmatrix}.$$

Then

$$\mathbf{X} = c_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} \sin t \\ -\sin t \\ \cos t \end{pmatrix} + c_3 \begin{pmatrix} -\cos t \\ \cos t \\ \sin t \end{pmatrix}.$$

40. We have $\det(\mathbf{A} - \lambda \mathbf{I}) = -(\lambda + 3)(\lambda^2 - 2\lambda + 5) = 0$. For $\lambda_1 = -3$ we obtain

$$\mathbf{K}_1 = \begin{pmatrix} 0 \\ -2 \\ 1 \end{pmatrix}.$$

For $\lambda_2 = 1 + 2i$ we obtain

$$\mathbf{K}_2 = \begin{pmatrix} -2 - i \\ -3i \\ 2 \end{pmatrix}$$

so that

$$\mathbf{X}_2 = \begin{pmatrix} -2 \cos 2t + \sin 2t \\ 3 \sin 2t \\ 2 \cos 2t \end{pmatrix} e^t + i \begin{pmatrix} -\cos 2t - 2 \sin 2t \\ -3 \cos 2t \\ 2 \sin 2t \end{pmatrix} e^t.$$

Then

$$\mathbf{X} = c_1 \begin{pmatrix} 0 \\ -2 \\ 1 \end{pmatrix} e^{-3t} + c_2 \begin{pmatrix} -2 \cos 2t + \sin 2t \\ 3 \sin 2t \\ 2 \cos 2t \end{pmatrix} e^t + c_3 \begin{pmatrix} -\cos 2t - 2 \sin 2t \\ -3 \cos 2t \\ 2 \sin 2t \end{pmatrix} e^t.$$

41. We have $\det(\mathbf{A} - \lambda \mathbf{I}) = (1 - \lambda)(\lambda^2 - 2\lambda + 2) = 0$. For $\lambda_1 = 1$ we obtain

$$\mathbf{K}_1 = \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix}.$$

For $\lambda_2 = 1 + i$ we obtain

$$\mathbf{K}_2 = \begin{pmatrix} 1 \\ i \\ i \end{pmatrix}$$

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so that

$$\mathbf{X}_2 = \begin{pmatrix} 1 \\ i \\ i \end{pmatrix} e^{(1+i)t} = \begin{pmatrix} \cos t \\ -\sin t \\ -\sin t \end{pmatrix} e^t + i \begin{pmatrix} \sin t \\ \cos t \\ \cos t \end{pmatrix} e^t.$$

Then

$$\mathbf{X} = c_1 \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} e^t + c_2 \begin{pmatrix} \cos t \\ -\sin t \\ -\sin t \end{pmatrix} e^t + c_3 \begin{pmatrix} \sin t \\ \cos t \\ \cos t \end{pmatrix} e^t.$$

42. We have $\det(\mathbf{A} - \lambda \mathbf{I}) = -(\lambda - 6)(\lambda^2 - 8\lambda + 20) = 0$. For $\lambda_1 = 6$ we obtain

$$\mathbf{K}_1 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

For $\lambda_2 = 4 + 2i$ we obtain

$$\mathbf{K}_2 = \begin{pmatrix} -i \\ 0 \\ 2 \end{pmatrix}$$

so that

$$\mathbf{X}_2 = \begin{pmatrix} -i \\ 0 \\ 2 \end{pmatrix} e^{(4+2i)t} = \begin{pmatrix} \sin 2t \\ 0 \\ 2 \cos 2t \end{pmatrix} e^{4t} + i \begin{pmatrix} -\cos 2t \\ 0 \\ 2 \sin 2t \end{pmatrix} e^{4t}.$$

Then

$$\mathbf{X} = c_1 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} e^{6t} + c_2 \begin{pmatrix} \sin 2t \\ 0 \\ 2 \cos 2t \end{pmatrix} e^{4t} + c_3 \begin{pmatrix} -\cos 2t \\ 0 \\ 2 \sin 2t \end{pmatrix} e^{4t}.$$

43. We have $\det(\mathbf{A} - \lambda \mathbf{I}) = (2 - \lambda)(\lambda^2 + 4\lambda + 13) = 0$. For $\lambda_1 = 2$ we obtain

$$\mathbf{K}_1 = \begin{pmatrix} 28 \\ -5 \\ 25 \end{pmatrix}.$$

For $\lambda_2 = -2 + 3i$ we obtain

$$\mathbf{K}_2 = \begin{pmatrix} 4 + 3i \\ -5 \\ 0 \end{pmatrix}$$

so that

$$\mathbf{X}_2 = \begin{pmatrix} 4 + 3i \\ -5 \\ 0 \end{pmatrix} e^{(-2+3i)t} = \begin{pmatrix} 4 \cos 3t - 3 \sin 3t \\ -5 \cos 3t \\ 0 \end{pmatrix} e^{-2t} + i \begin{pmatrix} 4 \sin 3t + 3 \cos 3t \\ -5 \sin 3t \\ 0 \end{pmatrix} e^{-2t}.$$

Then

$$\mathbf{X} = c_1 \begin{pmatrix} 28 \\ -5 \\ 25 \end{pmatrix} e^{2t} + c_2 \begin{pmatrix} 4 \cos 3t - 3 \sin 3t \\ -5 \cos 3t \\ 0 \end{pmatrix} e^{-2t} + c_3 \begin{pmatrix} 4 \sin 3t + 3 \cos 3t \\ -5 \sin 3t \\ 0 \end{pmatrix} e^{-2t}.$$

44. We have $\det(\mathbf{A} - \lambda\mathbf{I}) = -(\lambda + 2)(\lambda^2 + 4) = 0$. For $\lambda_1 = -2$ we obtain

$$\mathbf{K}_1 = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}.$$

For $\lambda_2 = 2i$ we obtain

$$\mathbf{K}_2 = \begin{pmatrix} -2 - 2i \\ 1 \\ 1 \end{pmatrix}$$

so that

$$\mathbf{X}_2 = \begin{pmatrix} -2 - 2i \\ 1 \\ 1 \end{pmatrix} e^{2it} = \begin{pmatrix} -2 \cos 2t + 2 \sin 2t \\ \cos 2t \\ \cos 2t \end{pmatrix} + i \begin{pmatrix} -2 \cos 2t - 2 \sin 2t \\ \sin 2t \\ \sin 2t \end{pmatrix}.$$

Then

$$\mathbf{X} = c_1 \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} e^{-2t} + c_2 \begin{pmatrix} -2 \cos 2t + 2 \sin 2t \\ \cos 2t \\ \cos 2t \end{pmatrix} + c_3 \begin{pmatrix} -2 \cos 2t - 2 \sin 2t \\ \sin 2t \\ \sin 2t \end{pmatrix}.$$

45. We have $\det(\mathbf{A} - \lambda\mathbf{I}) = (1 - \lambda)(\lambda^2 + 25) = 0$. For $\lambda_1 = 1$ we obtain

$$\mathbf{K}_1 = \begin{pmatrix} 25 \\ -7 \\ 6 \end{pmatrix}.$$

For $\lambda_2 = 5i$ we obtain

$$\mathbf{K}_2 = \begin{pmatrix} 1 + 5i \\ 1 \\ 1 \end{pmatrix}$$

so that

$$\mathbf{X}_2 = \begin{pmatrix} 1 + 5i \\ 1 \\ 1 \end{pmatrix} e^{5it} = \begin{pmatrix} \cos 5t - 5 \sin 5t \\ \cos 5t \\ \cos 5t \end{pmatrix} + i \begin{pmatrix} \sin 5t + 5 \cos 5t \\ \sin 5t \\ \sin 5t \end{pmatrix}.$$

Then

$$\mathbf{X} = c_1 \begin{pmatrix} 25 \\ -7 \\ 6 \end{pmatrix} e^t + c_2 \begin{pmatrix} \cos 5t - 5 \sin 5t \\ \cos 5t \\ \cos 5t \end{pmatrix} + c_3 \begin{pmatrix} \sin 5t + 5 \cos 5t \\ \sin 5t \\ \sin 5t \end{pmatrix}.$$

If

$$\mathbf{X}(0) = \begin{pmatrix} 4 \\ 6 \\ -7 \end{pmatrix}$$

then $c_1 = c_2 = -1$ and $c_3 = 6$.

46. We have $\det(\mathbf{A} - \lambda\mathbf{I}) = \lambda^2 - 10\lambda + 29 = 0$. For $\lambda_1 = 5 + 2i$ we obtain

$$\mathbf{K}_1 = \begin{pmatrix} 1 \\ 1 - 2i \end{pmatrix}$$

so that

$$\mathbf{X}_1 = \begin{pmatrix} 1 \\ 1 - 2i \end{pmatrix} e^{(5+2i)t} = \begin{pmatrix} \cos 2t \\ \cos 2t + 2 \sin 2t \end{pmatrix} e^{5t} + i \begin{pmatrix} \sin 2t \\ \sin 2t - 2 \cos 2t \end{pmatrix} e^{5t}.$$

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and

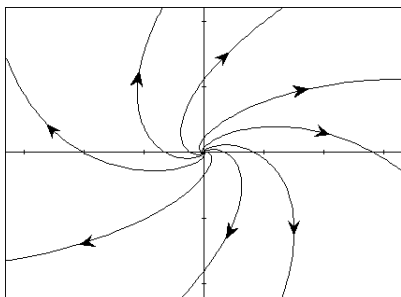
$$\mathbf{X} = c_1 \begin{pmatrix} \cos 2t \\ \cos 2t + 2 \sin 2t \end{pmatrix} e^{5t} + c_3 \begin{pmatrix} \sin 2t \\ \sin 2t - 2 \cos 2t \end{pmatrix} e^{5t}.$$

If

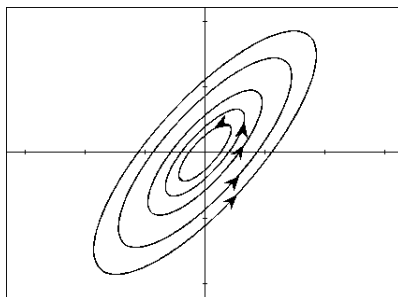
$$\mathbf{X}(0) = \begin{pmatrix} -2 \\ 8 \end{pmatrix}$$

then $c_1 = -2$ and $c_2 = 5$.

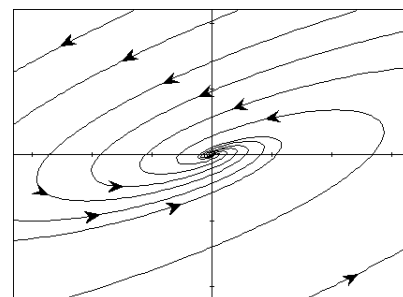
47.



Phase portrait for the system in Problem 36



Phase portrait for the system in Problem 37



Phase portrait for the system in Problem 38

48. (a) From $\det(\mathbf{A} - \lambda \mathbf{I}) = \lambda(\lambda - 2) = 0$ we get $\lambda_1 = 0$ and $\lambda_2 = 2$. For $\lambda_1 = 0$ we obtain

$$\begin{pmatrix} 1 & 1 & | & 0 \\ 1 & 1 & | & 0 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & 1 & | & 0 \\ 0 & 0 & | & 0 \end{pmatrix} \quad \text{so that} \quad \mathbf{K}_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

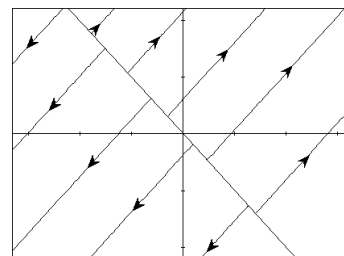
For $\lambda_2 = 2$ we obtain

$$\begin{pmatrix} -1 & 1 & | & 0 \\ 1 & -1 & | & 0 \end{pmatrix} \Rightarrow \begin{pmatrix} -1 & 1 & | & 0 \\ 0 & 0 & | & 0 \end{pmatrix} \quad \text{so that} \quad \mathbf{K}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Then

$$\mathbf{X} = c_1 \begin{pmatrix} -1 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{2t}.$$

The line $y = -x$ is not a trajectory of the system. Trajectories are $x = -c_1 + c_2 e^{2t}$, $y = c_1 + c_2 e^{2t}$ or $y = x + 2c_1$. This is a family of lines perpendicular to the line $y = -x$. All of the constant solutions of the system do, however, lie on the line $y = -x$.



(b) From $\det(\mathbf{A} - \lambda \mathbf{I}) = \lambda^2 = 0$ we get $\lambda_1 = 0$ and

$$\mathbf{K} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

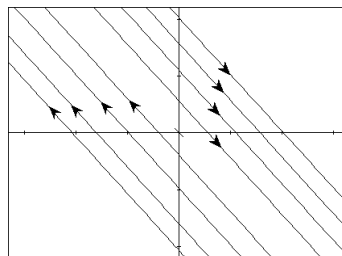
A solution of $(\mathbf{A} - \lambda_1 \mathbf{I})\mathbf{P} = \mathbf{K}$ is

$$\mathbf{P} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$$

so that

$$\mathbf{X} = c_1 \begin{pmatrix} -1 \\ 1 \end{pmatrix} + c_2 \left[\begin{pmatrix} -1 \\ 1 \end{pmatrix} t + \begin{pmatrix} -1 \\ 0 \end{pmatrix} \right].$$

All trajectories are parallel to $y = -x$, but $y = -x$ is not a trajectory. There are constant solutions of the system, however, that do lie on the line $y = -x$.



49. The system of differential equations is

$$x_1' = 2x_1 + x_2$$

$$x_2' = 2x_2$$

$$x_3' = 2x_3$$

$$x_4' = 2x_4 + x_5$$

$$x_5' = 2x_5.$$

We see immediately that $x_2 = c_2 e^{2t}$, $x_3 = c_3 e^{2t}$, and $x_5 = c_5 e^{2t}$. Then

$$x_1' = 2x_1 + c_2 e^{2t} \implies x_1 = c_2 t e^{2t} + c_1 e^{2t}$$

and

$$x_4' = 2x_4 + c_5 e^{2t} \implies x_4 = c_5 t e^{2t} + c_4 e^{2t}.$$

The general solution of the system is

$$\begin{aligned} \mathbf{X} &= \begin{pmatrix} c_2 t e^{2t} + c_1 e^{2t} \\ c_2 e^{2t} \\ c_3 e^{2t} \\ c_5 t e^{2t} + c_4 e^{2t} \\ c_5 e^{2t} \end{pmatrix} \\ &= c_1 \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} e^{2t} + c_2 \left[\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} t e^{2t} + \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} e^{2t} \right] \\ &\quad + c_3 \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} e^{2t} + c_4 \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} e^{2t} + c_5 \left[\begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} t e^{2t} + \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} e^{2t} \right] \end{aligned}$$

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$$= c_1 \mathbf{K}_1 e^{2t} + c_2 \left[\mathbf{K}_1 t e^{2t} + \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} e^{2t} \right] \\ + c_3 \mathbf{K}_2 e^{2t} + c_4 \mathbf{K}_3 e^{2t} + c_5 \left[\mathbf{K}_3 t e^{2t} + \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} e^{2t} \right].$$

There are three solutions of the form $\mathbf{X} = \mathbf{K}e^{2t}$, where $\mathbf{K}$ is an eigenvector, and two solutions of the form $\mathbf{X} = \mathbf{K}te^{2t} + \mathbf{P}e^{2t}$. See (12) in the text. From (13) and (14) in the text

$$(\mathbf{A} - 2\mathbf{I})\mathbf{K}_1 = \mathbf{0}$$

and

$$(\mathbf{A} - 2\mathbf{I})\mathbf{K}_2 = \mathbf{K}_1.$$

This implies

$$\begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} p_1 \\ p_2 \\ p_3 \\ p_4 \\ p_5 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix},$$

so $p_2 = 1$ and $p_5 = 0$, while p_1 , p_3 , and p_4 are arbitrary. Choosing $p_1 = p_3 = p_4 = 0$ we have

$$\mathbf{P} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

Therefore a solution is

$$\mathbf{X} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} t e^{2t} + \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} e^{2t}.$$

Repeating for $\mathbf{K}_3$ we find

$$\mathbf{P} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix},$$

so another solution is

$$\mathbf{X} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} te^{2t} + \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} e^{2t}.$$

50. From $x = 2 \cos 2t - 2 \sin 2t$, $y = -\cos 2t$ we find $x + 2y = -2 \sin 2t$. Then

$$(x + 2y)^2 = 4 \sin^2 2t = 4(1 - \cos^2 2t) = 4 - 4 \cos^2 2t = 4 - 4y^2$$

and

$$x^2 + 4xy + 4y^2 = 4 - 4y^2 \quad \text{or} \quad x^2 + 4xy + 8y^2 = 4.$$

This is a rotated conic section and, from the discriminant $b^2 - 4ac = 16 - 32 < 0$, we see that the curve is an ellipse.

51. Suppose the eigenvalues are $\alpha \pm i\beta$, $\beta > 0$. In Problem 36 the eigenvalues are $5 \pm 3i$, in Problem 37 they are $\pm 3i$, and in Problem 38 they are $-1 \pm 2i$. From Problem 47 we deduce that the phase portrait will consist of a family of closed curves when $\alpha = 0$ and spirals when $\alpha \neq 0$. The origin will be a repeller when $\alpha > 0$, and an attractor when $\alpha < 0$.
52. (a) The given system can be written as

$$x_1'' = -\frac{k_1 + k_2}{m_1} x_1 + \frac{k_2}{m_1} x_2, \quad x_2'' = \frac{k_2}{m_1} x_1 - \frac{k_2}{m_1} x_2.$$

In terms of matrices this is $\mathbf{X}'' = \mathbf{A}\mathbf{X}$ where

$$\mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \text{and} \quad \mathbf{A} = \begin{bmatrix} -\frac{k_1 + k_2}{m_1} & \frac{k_2}{m_1} \\ \frac{k_2}{m_1} & -\frac{k_2}{m_1} \end{bmatrix}.$$

- (b) If $\mathbf{X} = \mathbf{K}e^{\omega t}$ then $\mathbf{X}'' = \omega^2 \mathbf{K}e^{\omega t}$ and $\mathbf{A}\mathbf{X} = \mathbf{A}\mathbf{K}e^{\omega t}$ so that $\mathbf{X}'' = \mathbf{A}\mathbf{X}$ becomes $\omega^2 \mathbf{K}e^{\omega t} = \mathbf{A}\mathbf{K}e^{\omega t}$ or $(\mathbf{A} - \omega^2 \mathbf{I})\mathbf{K} = 0$. Now let $\omega^2 = \lambda$.

- (c) When $m_1 = 1$, $m_2 = 1$, $k_1 = 3$, and $k_2 = 2$ we obtain $\begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix}$. The eigenvalues and corresponding eigenvectors of $\mathbf{A}$ are $\lambda_1 = -1$, $\lambda_2 = -6$, $\mathbf{K}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\mathbf{K}_2 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$. Since $\omega_1 = i$, $\omega_2 = -i$, $\omega_3 = \sqrt{6}i$, and $\omega_4 = -\sqrt{6}i$ a solution is

$$\mathbf{X} = c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{it} + c_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{-it} + c_3 \begin{bmatrix} -2 \\ 1 \end{bmatrix} e^{\sqrt{6}it} + c_4 \begin{bmatrix} -2 \\ 1 \end{bmatrix} e^{-\sqrt{6}it}.$$

Using $e^{it} = \cos t + i \sin t$ and $e^{\sqrt{6}it} = \cos \sqrt{6}t + i \sin \sqrt{6}t$ the preceding solution can be rewritten as

$$\mathbf{X} = b_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} \cos t + b_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} \sin t + b_3 \begin{bmatrix} -2 \\ 1 \end{bmatrix} \cos \sqrt{6}t + b_4 \begin{bmatrix} -2 \\ 1 \end{bmatrix} \sin \sqrt{6}t$$

where b_1 , b_2 , b_3 , and b_4 are constants.

Exercises 10.3

1. $\lambda_1 = 7, \lambda_2 = -4, \mathbf{K}_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \mathbf{K}_2 = \begin{bmatrix} -2 \\ 3 \end{bmatrix}, \mathbf{P} = \begin{bmatrix} 3 & -2 \\ 1 & 3 \end{bmatrix};$

$$\mathbf{X} = \mathbf{P}\mathbf{Y} = \begin{bmatrix} 3 & -2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} c_1 e^{7t} \\ c_2 e^{-4t} \end{bmatrix} = \begin{bmatrix} 3c_1 e^{7t} - 2c_2 e^{-4t} \\ c_1 e^{7t} + 3c_2 e^{-4t} \end{bmatrix} = c_1 \begin{bmatrix} 3 \\ 1 \end{bmatrix} e^{7t} + c_2 \begin{bmatrix} -2 \\ 3 \end{bmatrix} e^{-4t}$$

2. $\lambda_1 = 0, \lambda_2 = 1, \mathbf{K}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \mathbf{K}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \mathbf{P} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix};$

$$\mathbf{X} = \mathbf{P}\mathbf{Y} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 e^t \end{bmatrix} = \begin{bmatrix} c_1 + c_2 e^t \\ -c_1 + c_2 e^t \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^t$$

3. $\lambda_1 = \frac{1}{2}, \lambda_2 = \frac{3}{2}, \mathbf{K}_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \mathbf{K}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \mathbf{P} = \begin{bmatrix} 1 & 1 \\ -2 & 2 \end{bmatrix};$

$$\mathbf{X} = \mathbf{P}\mathbf{Y} = \begin{bmatrix} 1 & 1 \\ -2 & 2 \end{bmatrix} \begin{bmatrix} c_1 e^{t/2} \\ c_2 e^{3t/2} \end{bmatrix} = \begin{bmatrix} c_1 e^{t/2} + c_2 e^{3t/2} \\ -2c_1 e^{t/2} + 2c_2 e^{3t/2} \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ -2 \end{bmatrix} e^{t/2} + c_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{3t/2}$$

4. $\lambda_1 = -\sqrt{2}, \lambda_2 = \sqrt{2}, \mathbf{K}_1 = \begin{bmatrix} -1 \\ 1 + \sqrt{2} \end{bmatrix}, \mathbf{K}_2 = \begin{bmatrix} -1 \\ 1 - \sqrt{2} \end{bmatrix}, \mathbf{P} = \begin{bmatrix} -1 & -1 \\ 1 + \sqrt{2} & 1 - \sqrt{2} \end{bmatrix};$

$$\begin{aligned} \mathbf{X} = \mathbf{P}\mathbf{Y} &= \begin{bmatrix} -1 & -1 \\ 1 + \sqrt{2} & 1 - \sqrt{2} \end{bmatrix} \begin{bmatrix} c_1 e^{-\sqrt{2}t} \\ c_2 e^{\sqrt{2}t} \end{bmatrix} = \begin{bmatrix} -c_1 e^{-\sqrt{2}t} - c_2 e^{\sqrt{2}t} \\ (1 + \sqrt{2})c_1 e^{-\sqrt{2}t} + (1 - \sqrt{2})c_2 e^{\sqrt{2}t} \end{bmatrix} \\ &= c_1 \begin{bmatrix} -1 \\ 1 + \sqrt{2} \end{bmatrix} e^{-\sqrt{2}t} + c_2 \begin{bmatrix} -1 \\ 1 - \sqrt{2} \end{bmatrix} e^{\sqrt{2}t} \end{aligned}$$

5. $\lambda_1 = -4, \lambda_2 = 2, \lambda_3 = 6, \mathbf{K}_1 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \mathbf{K}_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \mathbf{K}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \mathbf{P} = \begin{bmatrix} -1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix};$

$$\mathbf{X} = \mathbf{P}\mathbf{Y} = \begin{bmatrix} -1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} c_1 e^{-4t} \\ c_2 e^{2t} \\ c_3 e^{6t} \end{bmatrix} = \begin{bmatrix} -c_1 e^{-4t} + c_2 e^{2t} \\ c_1 e^{-4t} + c_2 e^{2t} \\ c_2 e^{2t} + c_3 e^{6t} \end{bmatrix} = c_1 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} e^{-4t} + c_2 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^{2t} + c_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{6t}$$

6. $\lambda_1 = -1, \lambda_2 = 1, \lambda_3 = 4, \mathbf{K}_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \mathbf{K}_2 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}, \mathbf{K}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \mathbf{P} = \begin{bmatrix} -1 & 1 & 1 \\ 0 & -2 & 1 \\ 1 & 1 & 1 \end{bmatrix};$

$$\mathbf{X} = \mathbf{P}\mathbf{Y} = \begin{bmatrix} -1 & 1 & 1 \\ 0 & -2 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} c_1 e^{-t} \\ c_2 e^t \\ c_3 e^{4t} \end{bmatrix} = \begin{bmatrix} -c_1 + c_2 e^t + c_3 e^{4t} \\ -2c_2 e^t + c_3 e^{4t} \\ c_1 e^{-t} + c_2 e^t + c_3 e^{4t} \end{bmatrix} = c_1 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} e^t + c_3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^{4t}$$

7. $\lambda_1 = -1, \lambda_2 = 2, \lambda_3 = 2, \mathbf{K}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \mathbf{K}_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \mathbf{K}_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \mathbf{P} = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix};$

$$\mathbf{X} = \mathbf{P}\mathbf{Y} = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_1 e^{-t} \\ c_2 e^{2t} \\ c_3 e^{2t} \end{bmatrix} = \begin{bmatrix} c_1 e^{-t} - c_2 e^{2t} - c_3 e^{2t} \\ c_1 e^{-t} + c_2 e^{2t} \\ c_1 e^{-t} + c_3 e^{2t} \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} e^{2t} + c_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} e^{2t}$$

$$8. \lambda_1 = 0, \lambda_2 = 0, \lambda_3 = 0, \lambda_4 = 4, \mathbf{K}_1 = \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \mathbf{K}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \mathbf{K}_3 = \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \mathbf{K}_4 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix},$$

$$\mathbf{P} = \begin{bmatrix} -1 & -1 & -1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix};$$

$$\begin{aligned} \mathbf{X} = \mathbf{P}\mathbf{Y} &= \begin{bmatrix} -1 & -1 & -1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 e^{4t} \end{bmatrix} = \begin{bmatrix} -c_1 - c_2 - c_3 + c_4 e^{4t} \\ c_1 \\ c_2 \\ c_3 + c_4 e^{4t} \end{bmatrix} \\ &= c_1 \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix} + c_4 \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} e^{4t} \end{aligned}$$

$$9. \lambda_1 = 1, \lambda_2 = 2, \lambda_3 = 3, \mathbf{K}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \mathbf{K}_2 = \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}, \mathbf{K}_3 = \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}, \mathbf{P} = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 4 \\ 1 & 3 & 5 \end{bmatrix};$$

$$\mathbf{X} = \mathbf{P}\mathbf{Y} = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 4 \\ 1 & 3 & 5 \end{bmatrix} \begin{bmatrix} c_1 e^t \\ c_2 e^{2t} \\ c_3 e^{3t} \end{bmatrix} = \begin{bmatrix} c_1 e^t + 2c_2 e^{2t} + 3c_3 e^{3t} \\ c_1 e^t + 2c_2 e^{2t} + 4c_3 e^{3t} \\ c_1 e^t + 3c_2 e^{2t} + 5c_3 e^{3t} \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^t + c_2 \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix} e^{2t} + c_3 \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} e^{3t}$$

$$10. \lambda_1 = 0, \lambda_2 = -2\sqrt{2}, \lambda_3 = 2\sqrt{2}, \mathbf{K}_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \mathbf{K}_2 = \begin{bmatrix} 1 \\ -\sqrt{2} \\ 1 \end{bmatrix}, \mathbf{K}_3 = \begin{bmatrix} 1 \\ \sqrt{2} \\ 1 \end{bmatrix}, \mathbf{P} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -\sqrt{2} & \sqrt{2} \\ -1 & 1 & 1 \end{bmatrix};$$

$$\begin{aligned} \mathbf{X} = \mathbf{P}\mathbf{Y} &= \begin{bmatrix} 1 & 1 & 1 \\ 0 & -\sqrt{2} & \sqrt{2} \\ -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 e^{-2\sqrt{2}t} \\ c_3 e^{2\sqrt{2}t} \end{bmatrix} = \begin{bmatrix} c_1 + c_2 e^{-2\sqrt{2}t} + c_3 e^{2\sqrt{2}t} \\ -\sqrt{2} c_2 e^{-2\sqrt{2}t} + \sqrt{2} c_3 e^{2\sqrt{2}t} \\ -c_1 + c_2 e^{-2\sqrt{2}t} + c_3 e^{2\sqrt{2}t} \end{bmatrix} \\ &= c_1 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ -\sqrt{2} \\ 1 \end{bmatrix} e^{-2\sqrt{2}t} + c_3 \begin{bmatrix} 1 \\ \sqrt{2} \\ 1 \end{bmatrix} e^{2\sqrt{2}t} \end{aligned}$$

Exercises 10.4

1. From

$$\mathbf{X}' = \begin{pmatrix} 3 & -3 \\ 2 & -2 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 4 \\ -1 \end{pmatrix}$$

we obtain

$$\mathbf{X}_c = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 3 \\ 2 \end{pmatrix} e^t.$$

Then

$$\Phi = \begin{pmatrix} 1 & 3e^t \\ 1 & 2e^t \end{pmatrix} \quad \text{and} \quad \Phi^{-1} = \begin{pmatrix} -2 & 3 \\ e^{-t} & -e^{-t} \end{pmatrix}$$

Exercises 10.4

so that

$$\mathbf{U} = \int \Phi^{-1} \mathbf{F} dt = \int \begin{pmatrix} -11 \\ 5e^{-t} \end{pmatrix} dt = \begin{pmatrix} -11t \\ -5e^{-t} \end{pmatrix}$$

and

$$\mathbf{X}_p = \Phi \mathbf{U} = \begin{pmatrix} -11 \\ -11 \end{pmatrix} t + \begin{pmatrix} -15 \\ -10 \end{pmatrix}.$$

2. From

$$\mathbf{X}' = \begin{pmatrix} 2 & -1 \\ 3 & -2 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 0 \\ 4 \end{pmatrix} t$$

we obtain

$$\mathbf{X}_c = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^t + c_2 \begin{pmatrix} 1 \\ 3 \end{pmatrix} e^{-t}.$$

Then

$$\Phi = \begin{pmatrix} e^t & e^{-t} \\ e^t & 3e^{-t} \end{pmatrix} \quad \text{and} \quad \Phi^{-1} = \begin{pmatrix} \frac{3}{2}e^{-t} & -\frac{1}{2}e^{-t} \\ -\frac{1}{2}e^t & \frac{1}{2}e^t \end{pmatrix}$$

so that

$$\mathbf{U} = \int \Phi^{-1} \mathbf{F} dt = \int \begin{pmatrix} -2te^{-t} \\ 2te^t \end{pmatrix} dt = \begin{pmatrix} 2te^{-t} + 2e^{-t} \\ 2te^t - 2e^t \end{pmatrix}$$

and

$$\mathbf{X}_p = \Phi \mathbf{U} = \begin{pmatrix} 4 \\ 8 \end{pmatrix} t + \begin{pmatrix} 0 \\ -4 \end{pmatrix}.$$

3. From

$$\mathbf{X}' = \begin{pmatrix} 3 & -5 \\ 3/4 & -1 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{t/2}$$

we obtain

$$\mathbf{X}_c = c_1 \begin{pmatrix} 10 \\ 3 \end{pmatrix} e^{3t/2} + c_2 \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{t/2}.$$

Then

$$\Phi = \begin{pmatrix} 10e^{3t/2} & 2e^{t/2} \\ 3e^{3t/2} & e^{t/2} \end{pmatrix} \quad \text{and} \quad \Phi^{-1} = \begin{pmatrix} \frac{1}{4}e^{-3t/2} & -\frac{1}{2}e^{-3t/2} \\ -\frac{3}{4}e^{-t/2} & \frac{5}{2}e^{-t/2} \end{pmatrix}$$

so that

$$\mathbf{U} = \int \Phi^{-1} \mathbf{F} dt = \int \begin{pmatrix} \frac{3}{4}e^{-t} \\ -\frac{13}{4}t \end{pmatrix} dt = \begin{pmatrix} -\frac{3}{4}e^{-t} \\ -\frac{13}{4}t \end{pmatrix}$$

and

$$\mathbf{X}_p = \Phi \mathbf{U} = \begin{pmatrix} -13/2 \\ -13/4 \end{pmatrix} te^{t/2} + \begin{pmatrix} -15/2 \\ -9/4 \end{pmatrix} e^{t/2}.$$

4. From

$$\mathbf{X}' = \begin{pmatrix} 2 & -1 \\ 4 & 2 \end{pmatrix} \mathbf{X} + \begin{pmatrix} \sin 2t \\ 2 \cos 2t \end{pmatrix}$$

we obtain

$$\mathbf{X}_c = c_1 \begin{pmatrix} -\sin 2t \\ 2 \cos 2t \end{pmatrix} e^{2t} + c_2 \begin{pmatrix} \cos 2t \\ 2 \sin 2t \end{pmatrix} e^{2t}.$$

Then

$$\Phi = \begin{pmatrix} -e^{2t} \sin 2t & e^{2t} \cos 2t \\ 2e^{2t} \cos 2t & 2e^{2t} \sin 2t \end{pmatrix} \quad \text{and} \quad \Phi^{-1} = \begin{pmatrix} -\frac{1}{2}e^{-2t} \sin 2t & \frac{1}{4}e^{-2t} \cos 2t \\ \frac{1}{2}e^{-2t} \cos 2t & \frac{1}{4}e^{-2t} \sin 2t \end{pmatrix}$$

so that

$$\mathbf{U} = \int \Phi^{-1} \mathbf{F} dt = \int \begin{pmatrix} \frac{1}{2} \cos 4t \\ \frac{1}{2} \sin 4t \end{pmatrix} dt = \begin{pmatrix} \frac{1}{8} \sin 4t \\ -\frac{1}{8} \cos 4t \end{pmatrix}$$

and

$$\mathbf{X}_p = \Phi \mathbf{U} = \begin{pmatrix} -\frac{1}{8} \sin 2t \cos 4t - \frac{1}{8} \cos 2t \cos 4t \\ \frac{1}{4} \cos 2t \sin 4t - \frac{1}{4} \sin 2t \cos 4t \end{pmatrix} e^{2t}.$$

5. From

$$\mathbf{X}' = \begin{pmatrix} 0 & 2 \\ -1 & 3 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^t$$

we obtain

$$\mathbf{X}_c = c_1 \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^t + c_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{2t}.$$

Then

$$\Phi = \begin{pmatrix} 2e^t & e^{2t} \\ e^t & e^{2t} \end{pmatrix} \quad \text{and} \quad \Phi^{-1} = \begin{pmatrix} e^{-t} & -e^{-t} \\ -e^{-2t} & 2e^{-2t} \end{pmatrix}$$

so that

$$\mathbf{U} = \int \Phi^{-1} \mathbf{F} dt = \int \begin{pmatrix} 2 \\ -3e^{-t} \end{pmatrix} dt = \begin{pmatrix} 2t \\ 3e^{-t} \end{pmatrix}$$

and

$$\mathbf{X}_p = \Phi \mathbf{U} = \begin{pmatrix} 4 \\ 2 \end{pmatrix} te^t + \begin{pmatrix} 3 \\ 3 \end{pmatrix} e^t.$$

6. From

$$\mathbf{X}' = \begin{pmatrix} 0 & 2 \\ -1 & 3 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 2 \\ e^{-3t} \end{pmatrix}$$

we obtain

$$\mathbf{X}_c = c_1 \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^t + c_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{2t}.$$

Then

$$\Phi = \begin{pmatrix} 2e^t & e^{2t} \\ e^t & e^{2t} \end{pmatrix} \quad \text{and} \quad \Phi^{-1} = \begin{pmatrix} e^{-t} & -e^{-t} \\ -e^{-2t} & 2e^{-2t} \end{pmatrix}$$

so that

$$\mathbf{U} = \int \Phi^{-1} \mathbf{F} dt = \int \begin{pmatrix} 2e^{-t} - e^{-4t} \\ -2e^{-2t} + 2e^{-5t} \end{pmatrix} dt = \begin{pmatrix} -2e^{-t} + \frac{1}{4}e^{-4t} \\ e^{-2t} - \frac{2}{5}e^{-5t} \end{pmatrix}$$

and

$$\mathbf{X}_p = \Phi \mathbf{U} = \begin{pmatrix} \frac{1}{10}e^{-3t} - 3 \\ -\frac{3}{20}e^{-3t} - 1 \end{pmatrix}.$$

7. From

$$\mathbf{X}' = \begin{pmatrix} 1 & 8 \\ 1 & -1 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 12 \\ 12 \end{pmatrix} t$$

we obtain

$$\mathbf{X}_c = c_1 \begin{pmatrix} 4 \\ 1 \end{pmatrix} e^{3t} + c_2 \begin{pmatrix} -2 \\ 1 \end{pmatrix} e^{-3t}.$$

Then

$$\Phi = \begin{pmatrix} 4e^{3t} & -2e^{-3t} \\ e^{3t} & e^{-3t} \end{pmatrix} \quad \text{and} \quad \Phi^{-1} = \begin{pmatrix} \frac{1}{6}e^{-3t} & \frac{1}{3}e^{-3t} \\ -\frac{1}{6}e^{3t} & \frac{2}{3}e^{3t} \end{pmatrix}$$

so that

$$\mathbf{U} = \int \Phi^{-1} \mathbf{F} dt = \int \begin{pmatrix} 6te^{-3t} \\ 6te^{3t} \end{pmatrix} dt = \begin{pmatrix} -2te^{-3t} - \frac{2}{3}e^{-3t} \\ 2te^{3t} - \frac{2}{3}e^{3t} \end{pmatrix}$$

and

$$\mathbf{X}_p = \Phi \mathbf{U} = \begin{pmatrix} -12 \\ 0 \end{pmatrix} t + \begin{pmatrix} -4/3 \\ -4/3 \end{pmatrix}.$$

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8. From

$$\mathbf{X}' = \begin{pmatrix} 1 & 8 \\ 1 & -1 \end{pmatrix} \mathbf{X} + \begin{pmatrix} e^{-t} \\ te^t \end{pmatrix}$$

we obtain

$$\mathbf{X}_c = c_1 \begin{pmatrix} 4 \\ 1 \end{pmatrix} e^{3t} + c_2 \begin{pmatrix} -2 \\ 1 \end{pmatrix} e^{-3t}.$$

Then

$$\Phi = \begin{pmatrix} 4e^{3t} & -2e^{3t} \\ e^{3t} & e^{-3t} \end{pmatrix} \quad \text{and} \quad \Phi^{-1} = \begin{pmatrix} \frac{1}{6}e^{-3t} & \frac{1}{3}e^{-3t} \\ -\frac{1}{6}e^{3t} & \frac{2}{3}e^{3t} \end{pmatrix}$$

so that

$$\mathbf{U} = \int \Phi^{-1} \mathbf{F} dt = \int \begin{pmatrix} \frac{1}{6}e^{-4t} + \frac{1}{3}te^{-2t} \\ -\frac{1}{6}e^{2t} + \frac{2}{3}te^{4t} \end{pmatrix} dt = \begin{pmatrix} -\frac{1}{24}e^{-4t} - \frac{1}{6}te^{-2t} - \frac{1}{12}e^{-2t} \\ -\frac{1}{12}e^{2t} + \frac{1}{6}te^{4t} - \frac{1}{24}e^{4t} \end{pmatrix}$$

and

$$\mathbf{X}_p = \Phi \mathbf{U} = \begin{pmatrix} -te^t - \frac{1}{4}e^t \\ -\frac{1}{8}e^{-t} - \frac{1}{8}e^t \end{pmatrix}.$$

9. From

$$\mathbf{X}' = \begin{pmatrix} 3 & 2 \\ -2 & -1 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{-t}$$

we obtain

$$\mathbf{X}_c = c_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^t + c_2 \left[\begin{pmatrix} 1 \\ -1 \end{pmatrix} te^t + \begin{pmatrix} 0 \\ 1/2 \end{pmatrix} e^t \right].$$

Then

$$\Phi = \begin{pmatrix} e^t & te^t \\ -e^t & \frac{1}{2}e^t - te^t \end{pmatrix} \quad \text{and} \quad \Phi^{-1} = \begin{pmatrix} e^{-t} - 2te^{-t} & -2te^{-t} \\ 2e^{-t} & 2e^{-t} \end{pmatrix}$$

so that

$$\mathbf{U} = \int \Phi^{-1} \mathbf{F} dt = \int \begin{pmatrix} 2e^{-2t} - 6te^{-2t} \\ 6e^{-2t} \end{pmatrix} dt = \begin{pmatrix} \frac{1}{2}e^{-2t} + 3te^{-2t} \\ -3e^{-2t} \end{pmatrix}$$

and

$$\mathbf{X}_p = \Phi \mathbf{U} = \begin{pmatrix} 1/2 \\ -2 \end{pmatrix} e^{-t}.$$

10. From

$$\mathbf{X}' = \begin{pmatrix} 3 & 2 \\ -2 & -1 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

we obtain

$$\mathbf{X}_c = c_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^t + c_2 \left[\begin{pmatrix} 1 \\ -1 \end{pmatrix} te^t + \begin{pmatrix} 0 \\ 1/2 \end{pmatrix} e^t \right].$$

Then

$$\Phi = \begin{pmatrix} e^t & te^t \\ -e^t & \frac{1}{2}e^t - te^t \end{pmatrix} \quad \text{and} \quad \Phi^{-1} = \begin{pmatrix} e^{-t} - 2te^{-t} & -2te^{-t} \\ 2e^{-t} & 2e^{-t} \end{pmatrix}$$

so that

$$\mathbf{U} = \int \Phi^{-1} \mathbf{F} dt = \int \begin{pmatrix} e^{-t} - 4te^{-t} \\ 2e^{-t} \end{pmatrix} dt = \begin{pmatrix} 3e^{-t} + 4te^{-t} \\ -2e^{-t} \end{pmatrix}$$

and

$$\mathbf{X}_p = \Phi \mathbf{U} = \begin{pmatrix} 3 \\ -5 \end{pmatrix}.$$

11. From

$$\mathbf{X}' = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \mathbf{X} + \begin{pmatrix} \sec t \\ 0 \end{pmatrix}$$

we obtain

$$\mathbf{X}_c = c_1 \begin{pmatrix} \cos t \\ \sin t \end{pmatrix} + c_2 \begin{pmatrix} \sin t \\ -\cos t \end{pmatrix}.$$

Then

$$\Phi = \begin{pmatrix} \cos t & \sin t \\ \sin t & -\cos t \end{pmatrix} \quad \text{and} \quad \Phi^{-1} = \begin{pmatrix} \cos t & \sin t \\ \sin t & -\cos t \end{pmatrix}$$

so that

$$\mathbf{U} = \int \Phi^{-1} \mathbf{F} dt = \int \begin{pmatrix} 1 \\ \tan t \end{pmatrix} dt = \begin{pmatrix} t \\ \ln |\sec t| \end{pmatrix}$$

and

$$\mathbf{X}_p = \Phi \mathbf{U} = \begin{pmatrix} t \cos t + \sin t \ln |\sec t| \\ t \sin t - \cos t \ln |\sec t| \end{pmatrix}.$$

12. From

$$\mathbf{X}' = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 3 \\ 3 \end{pmatrix} e^t$$

we obtain

$$\mathbf{X}_c = c_1 \begin{pmatrix} -\sin t \\ \cos t \end{pmatrix} e^t + c_2 \begin{pmatrix} \cos t \\ \sin t \end{pmatrix} e^t.$$

Then

$$\Phi = \begin{pmatrix} -\sin t & \cos t \\ \cos t & \sin t \end{pmatrix} e^t \quad \text{and} \quad \Phi^{-1} = \begin{pmatrix} -\sin t & \cos t \\ \cos t & \sin t \end{pmatrix} e^{-t}$$

so that

$$\mathbf{U} = \int \Phi^{-1} \mathbf{F} dt = \int \begin{pmatrix} -3 \sin t + 3 \cos t \\ 3 \cos t + 3 \sin t \end{pmatrix} dt = \begin{pmatrix} 3 \cos t + 3 \sin t \\ 3 \sin t - 3 \cos t \end{pmatrix}$$

and

$$\mathbf{X}_p = \Phi \mathbf{U} = \begin{pmatrix} -3 \\ 3 \end{pmatrix} e^t.$$

13. From

$$\mathbf{X}' = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \mathbf{X} + \begin{pmatrix} \cos t \\ \sin t \end{pmatrix} e^t$$

we obtain

$$\mathbf{X}_c = c_1 \begin{pmatrix} -\sin t \\ \cos t \end{pmatrix} e^t + c_2 \begin{pmatrix} \cos t \\ \sin t \end{pmatrix} e^t.$$

Then

$$\Phi = \begin{pmatrix} -\sin t & \cos t \\ \cos t & \sin t \end{pmatrix} e^t \quad \text{and} \quad \Phi^{-1} = \begin{pmatrix} -\sin t & \cos t \\ \cos t & \sin t \end{pmatrix} e^{-t}$$

so that

$$\mathbf{U} = \int \Phi^{-1} \mathbf{F} dt = \int \begin{pmatrix} 0 \\ 1 \end{pmatrix} dt = \begin{pmatrix} 0 \\ t \end{pmatrix}$$

and

$$\mathbf{X}_p = \Phi \mathbf{U} = \begin{pmatrix} \cos t \\ \sin t \end{pmatrix} t e^t.$$

14. From

$$\mathbf{X}' = \begin{pmatrix} 2 & -2 \\ 8 & -6 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 1 \\ 3 \end{pmatrix} \frac{1}{t} e^{-2t}$$

we obtain

$$\mathbf{X}_c = c_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{-2t} + c_2 \left[\begin{pmatrix} 1 \\ 2 \end{pmatrix} t e^{-2t} + \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix} e^{-2t} \right].$$

Then

$$\Phi = \begin{pmatrix} 1 & 2t+1 \\ 2 & 4t+1 \end{pmatrix} e^{-2t} \quad \text{and} \quad \Phi^{-1} = \begin{pmatrix} -(4t+1) & 2t+1 \\ 2 & -1 \end{pmatrix} e^{2t}$$

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so that

$$\mathbf{U} = \int \Phi^{-1} \mathbf{F} dt = \int \begin{pmatrix} 2t + 2 \ln t \\ -\ln t \end{pmatrix} dt$$

and

$$\mathbf{X}_p = \Phi \mathbf{U} = \begin{pmatrix} 2t + \ln t - 2t \ln t \\ 4t + 3 \ln t - 4t \ln t \end{pmatrix} e^{-2t}.$$

15. From

$$\mathbf{X}' = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 0 \\ \sec t \tan t \end{pmatrix}$$

we obtain

$$\mathbf{X}_c = c_1 \begin{pmatrix} \cos t \\ -\sin t \end{pmatrix} + c_2 \begin{pmatrix} \sin t \\ \cos t \end{pmatrix}.$$

Then

$$\Phi = \begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix} t \quad \text{and} \quad \Phi^{-1} = \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}$$

so that

$$\mathbf{U} = \int \Phi^{-1} \mathbf{F} dt = \int \begin{pmatrix} -\tan^2 t \\ \tan t \end{pmatrix} dt = \begin{pmatrix} t - \tan t \\ \ln |\sec t| \end{pmatrix}$$

and

$$\mathbf{X}_p = \Phi \mathbf{U} = \begin{pmatrix} \cos t \\ -\sin t \end{pmatrix} t + \begin{pmatrix} -\sin t \\ \sin t \tan t \end{pmatrix} + \begin{pmatrix} \sin t \\ \cos t \end{pmatrix} \ln |\sec t|.$$

16. From

$$\mathbf{X}' = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 1 \\ \cot t \end{pmatrix}$$

we obtain

$$\mathbf{X}_c = c_1 \begin{pmatrix} \cos t \\ -\sin t \end{pmatrix} + c_2 \begin{pmatrix} \sin t \\ \cos t \end{pmatrix}.$$

Then

$$\Phi = \begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix} \quad \text{and} \quad \Phi^{-1} = \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}$$

so that

$$\mathbf{U} = \int \Phi^{-1} \mathbf{F} dt = \int \begin{pmatrix} 0 \\ \csc t \end{pmatrix} dt = \begin{pmatrix} 0 \\ \ln |\csc t - \cot t| \end{pmatrix}$$

and

$$\mathbf{X}_p = \Phi \mathbf{U} = \begin{pmatrix} \sin t \ln |\csc t - \cot t| \\ \cos t \ln |\csc t - \cot t| \end{pmatrix}.$$

17. From

$$\mathbf{X}' = \begin{pmatrix} 1 & 2 \\ -1/2 & 1 \end{pmatrix} \mathbf{X} + \begin{pmatrix} \csc t \\ \sec t \end{pmatrix} e^t$$

we obtain

$$\mathbf{X}_c = c_1 \begin{pmatrix} 2 \sin t \\ \cos t \end{pmatrix} e^t + c_2 \begin{pmatrix} 2 \cos t \\ -\sin t \end{pmatrix} e^t.$$

Then

$$\Phi = \begin{pmatrix} 2 \sin t & 2 \cos t \\ \cos t & -\sin t \end{pmatrix} e^t \quad \text{and} \quad \Phi^{-1} = \begin{pmatrix} \frac{1}{2} \sin t & \cos t \\ \frac{1}{2} \cos t & -\sin t \end{pmatrix} e^{-t}$$

so that

$$\mathbf{U} = \int \Phi^{-1} \mathbf{F} dt = \int \begin{pmatrix} \frac{3}{2} \\ \frac{1}{2} \cos t - \tan t \end{pmatrix} dt = \begin{pmatrix} \frac{3}{2} t \\ \frac{1}{2} \ln |\sin t| - \ln |\sec t| \end{pmatrix}$$

and

$$\mathbf{X}_p = \Phi \mathbf{U} = \begin{pmatrix} 3 \sin t \\ \frac{3}{2} \cos t \end{pmatrix} t e^t + \begin{pmatrix} \cos t \\ -\frac{1}{2} \sin t \end{pmatrix} \ln |\sin t| + \begin{pmatrix} -2 \cos t \\ \sin t \end{pmatrix} \ln |\sec t|.$$

18. From

$$\mathbf{X}' = \begin{pmatrix} 1 & -2 \\ 1 & -1 \end{pmatrix} \mathbf{X} + \begin{pmatrix} \tan t \\ 1 \end{pmatrix}$$

we obtain

$$\mathbf{X}_c = c_1 \begin{pmatrix} \cos t - \sin t \\ \cos t \end{pmatrix} + c_2 \begin{pmatrix} \cos t + \sin t \\ \sin t \end{pmatrix}.$$

Then

$$\Phi = \begin{pmatrix} \cos t - \sin t & \cos t + \sin t \\ \cos t & \sin t \end{pmatrix} \quad \text{and} \quad \Phi^{-1} = \begin{pmatrix} -\sin t & \cos t + \sin t \\ \cos t & \sin t - \cos t \end{pmatrix}$$

so that

$$\mathbf{U} = \int \Phi^{-1} \mathbf{F} dt = \int \begin{pmatrix} 2 \cos t + \sin t - \sec t \\ 2 \sin t - \cos t \end{pmatrix} dt = \begin{pmatrix} 2 \sin t - \cos t - \ln |\sec t + \tan t| \\ -2 \cos t - \sin t \end{pmatrix}$$

and

$$\mathbf{X}_p = \Phi \mathbf{U} = \begin{pmatrix} 3 \sin t \cos t - \cos^2 t - 2 \sin^2 t + (\sin t - \cos t) \ln |\sec t + \tan t| \\ \sin^2 t - \cos^2 t - \cos t (\ln |\sec t + \tan t|) \end{pmatrix}.$$

19. From

$$\mathbf{X}' = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix} \mathbf{X} + \begin{pmatrix} e^t \\ e^{2t} \\ te^{3t} \end{pmatrix}$$

we obtain

$$\mathbf{X}_c = c_1 \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} e^{2t} + c_3 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} e^{3t}.$$

Then

$$\Phi = \begin{pmatrix} 1 & e^{2t} & 0 \\ -1 & e^{2t} & 0 \\ 0 & 0 & e^{3t} \end{pmatrix} \quad \text{and} \quad \Phi^{-1} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} & 0 \\ \frac{1}{2}e^{-2t} & \frac{1}{2}e^{-2t} & 0 \\ 0 & 0 & e^{-3t} \end{pmatrix}$$

so that

$$\mathbf{U} = \int \Phi^{-1} \mathbf{F} dt = \int \begin{pmatrix} \frac{1}{2}e^t - \frac{1}{2}e^{2t} \\ \frac{1}{2}e^{-t} + \frac{1}{2} \\ t \end{pmatrix} dt = \begin{pmatrix} \frac{1}{2}e^t - \frac{1}{4}e^{2t} \\ -\frac{1}{2}e^{-t} + \frac{1}{2}t \\ \frac{1}{2}t^2 \end{pmatrix}$$

and

$$\mathbf{X}_p = \Phi \mathbf{U} = \begin{pmatrix} -\frac{1}{4}e^{2t} + \frac{1}{2}te^{2t} \\ -e^t + \frac{1}{4}e^{2t} + \frac{1}{2}te^{2t} \\ \frac{1}{2}t^2e^{3t} \end{pmatrix}.$$

20. From

$$\mathbf{X}' = \begin{pmatrix} 3 & -1 & -1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 0 \\ t \\ 2e^t \end{pmatrix}$$

we obtain

$$\mathbf{X}_c = c_1 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} e^t + c_2 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} e^{2t} + c_3 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} e^{2t}.$$

Then

$$\Phi = \begin{pmatrix} e^t & e^{2t} & e^{2t} \\ e^t & e^{2t} & 0 \\ e^t & 0 & e^{2t} \end{pmatrix} \quad \text{and} \quad \Phi^{-1} = \begin{pmatrix} -e^{-t} & e^{-t} & e^{-t} \\ e^{-2t} & 0 & -e^{-2t} \\ e^{-2t} & -e^{-2t} & 0 \end{pmatrix}$$

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so that

$$\mathbf{U} = \int \Phi^{-1} \mathbf{F} dt = \int \begin{pmatrix} te^{-t} + 2 \\ -2e^{-t} \\ -te^{-2t} \end{pmatrix} dt = \begin{pmatrix} -te^{-t} - e^{-t} + 2t \\ 2e^{-t} \\ \frac{1}{2}te^{-2t} + \frac{1}{4}e^{-2t} \end{pmatrix}$$

and

$$\mathbf{X}_p = \Phi \mathbf{U} = \begin{pmatrix} -1/2 \\ -1 \\ -1/2 \end{pmatrix} t + \begin{pmatrix} -3/4 \\ -1 \\ -3/4 \end{pmatrix} + \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix} e^t + \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix} te^t.$$

21. From

$$\mathbf{X}' = \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 4e^{2t} \\ 4e^{4t} \end{pmatrix}$$

we obtain

$$\Phi = \begin{pmatrix} -e^{4t} & e^{2t} \\ e^{4t} & e^{2t} \end{pmatrix}, \quad \Phi^{-1} = \begin{pmatrix} -\frac{1}{2}e^{-4t} & \frac{1}{2}e^{-4t} \\ \frac{1}{2}e^{-2t} & \frac{1}{2}e^{-2t} \end{pmatrix},$$

and

$$\begin{aligned} \mathbf{X} &= \Phi \Phi^{-1}(0) \mathbf{X}(0) + \Phi \int_0^t \Phi^{-1} \mathbf{F} ds = \Phi \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} + \Phi \cdot \begin{pmatrix} e^{-2t} + 2t - 1 \\ e^{2t} + 2t - 1 \end{pmatrix} \\ &= \begin{pmatrix} 2 \\ 2 \end{pmatrix} te^{2t} + \begin{pmatrix} -1 \\ 1 \end{pmatrix} e^{2t} + \begin{pmatrix} -2 \\ 2 \end{pmatrix} te^{4t} + \begin{pmatrix} 2 \\ 0 \end{pmatrix} e^{4t}. \end{aligned}$$

22. From

$$\mathbf{X}' = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 1/t \\ 1/t \end{pmatrix}$$

we obtain

$$\Phi = \begin{pmatrix} 1 & 1+t \\ 1 & t \end{pmatrix}, \quad \Phi^{-1} = \begin{pmatrix} -t & 1+t \\ 1 & -1 \end{pmatrix},$$

and

$$\mathbf{X} = \Phi \Phi^{-1}(1) \mathbf{X}(1) + \Phi \int_1^t \Phi^{-1} \mathbf{F} ds = \Phi \cdot \begin{pmatrix} -4 \\ 3 \end{pmatrix} + \Phi \cdot \begin{pmatrix} \ln t \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} t - \begin{pmatrix} 1 \\ 4 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} \ln t.$$

23. $\lambda_1 = -1, \lambda_2 = -2, \mathbf{K}_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \mathbf{K}_2 = \begin{bmatrix} 2 \\ 7 \end{bmatrix}, \mathbf{P} = \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}, \mathbf{P}^{-1} = \begin{bmatrix} 7 & -2 \\ -3 & 1 \end{bmatrix}, \mathbf{P}^{-1} \mathbf{F} = \begin{bmatrix} 34 \\ -14 \end{bmatrix};$

$$\mathbf{Y}' = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \mathbf{Y} + \begin{bmatrix} 34 \\ -14 \end{bmatrix}$$

$$y_1 = 34 + c_1 e^{-t}, \quad y_2 = -7 + c_2 e^{-2t}$$

$$\mathbf{X} = \mathbf{P} \mathbf{Y} = \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix} \begin{bmatrix} 34 + c_1 e^{-t} \\ -7 + c_2 e^{-2t} \end{bmatrix} = \begin{bmatrix} 20 + c_1 e^{-t} + 2c_2 e^{-2t} \\ 53 + 3c_1 e^{-t} + 7c_2 e^{-2t} \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 2 \\ 7 \end{bmatrix} e^{-2t} + \begin{bmatrix} 20 \\ 53 \end{bmatrix}$$

24. $\lambda_1 = -1, \lambda_2 = 4, \mathbf{K}_1 = \begin{bmatrix} 3 \\ -2 \end{bmatrix}, \mathbf{K}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \mathbf{P} = \begin{bmatrix} 3 & 1 \\ -2 & 1 \end{bmatrix}, \mathbf{P}^{-1} = \frac{1}{5} \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}, \mathbf{P}^{-1} \mathbf{F} = \begin{bmatrix} 0 \\ e^t \end{bmatrix};$

$$\mathbf{Y}' = \begin{bmatrix} -1 & 0 \\ 0 & 4 \end{bmatrix} \mathbf{Y} + \begin{bmatrix} 0 \\ e^t \end{bmatrix}$$

$$y_1 = c_1 e^{-t}, \quad y_2 = -\frac{1}{3} e^t + c_2 e^{4t}$$

$$\mathbf{X} = \mathbf{P} \mathbf{Y} = \begin{bmatrix} 3 & 1 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} c_1 e^{-t} \\ -\frac{1}{3} e^t + c_2 e^{4t} \end{bmatrix} = \begin{bmatrix} -\frac{1}{3} e^t + 3c_1 e^{-t} + c_2 e^{4t} \\ -\frac{1}{3} e^t - 2c_1 e^{-t} + c_2 e^{4t} \end{bmatrix} = c_1 \begin{bmatrix} 3 \\ -2 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{4t} - \frac{1}{3} \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^t$$

25. $\lambda_1 = 0, \lambda_2 = 10, \mathbf{K}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \mathbf{K}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \mathbf{P} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}, \mathbf{P}^{-1} = \frac{1}{2} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}, \mathbf{P}^{-1}\mathbf{F} = \begin{bmatrix} t-4 \\ t+4 \end{bmatrix};$

$$\mathbf{Y}' = \begin{bmatrix} 0 & 0 \\ 0 & 10 \end{bmatrix} \mathbf{Y} + \begin{bmatrix} t-4 \\ t+4 \end{bmatrix}$$

$$y_1 = \frac{1}{2}t^2 - 4t + c_1, \quad y_2 = -\frac{1}{10}t - \frac{41}{100} + c_2e^{10t}$$

$$\begin{aligned} \mathbf{X} = \mathbf{P}\mathbf{Y} &= \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{2}t^2 - 4t + c_1 \\ -\frac{1}{10}t - \frac{41}{100} + c_2e^{10t} \end{bmatrix} = \begin{bmatrix} \frac{1}{2}t^2 - \frac{41}{10}t - \frac{41}{100} + c_1 + c_2e^{10t} \\ -\frac{1}{2}t^2 + \frac{39}{10}t - \frac{41}{100} - c_1 + c_2e^{10t} \end{bmatrix} \\ &= c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{10t} + \frac{1}{2} \begin{bmatrix} 1 \\ -1 \end{bmatrix} t^2 + \frac{1}{10} \begin{bmatrix} -41 \\ 39 \end{bmatrix} t - \frac{41}{100} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \end{aligned}$$

26. $\lambda_1 = -1, \lambda_2 = 1, \mathbf{K}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \mathbf{K}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \mathbf{P} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}, \mathbf{P}^{-1} = \frac{1}{2} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}, \mathbf{P}^{-1}\mathbf{F} = \begin{bmatrix} 2-4e^{-2t} \\ 2+4e^{-2t} \end{bmatrix};$

$$\mathbf{Y}' = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \mathbf{Y} + \begin{bmatrix} 2-4e^{-2t} \\ 2+4e^{-2t} \end{bmatrix}$$

$$y_1 = 2 + 4e^{-2t} + c_1e^{-t}, \quad y_2 = -2 - \frac{4}{3}e^{-2t} + c_2e^t$$

$$\begin{aligned} \mathbf{X} = \mathbf{P}\mathbf{Y} &= \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 2 + 4e^{-2t} + c_1e^{-t} \\ -2 - \frac{4}{3}e^{-2t} + c_2e^t \end{bmatrix} = \begin{bmatrix} \frac{8}{3}e^{-2t} + c_1e^{-t} + c_2e^t \\ -4 - \frac{16}{3}e^{-2t} - c_1e^{-t} + c_2e^t \end{bmatrix} \\ &= c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^t + \frac{8}{3} \begin{bmatrix} 1 \\ -2 \end{bmatrix} e^{-2t} + \begin{bmatrix} 0 \\ -4 \end{bmatrix} \end{aligned}$$

27. Let $\mathbf{I} = \begin{pmatrix} i_1 \\ i_2 \end{pmatrix}$ so that

$$\mathbf{I}' = \begin{pmatrix} -11 & 3 \\ 3 & -3 \end{pmatrix} \mathbf{I} + \begin{pmatrix} 100 \sin t \\ 0 \end{pmatrix}$$

and

$$\mathbf{X}_c = c_1 \begin{pmatrix} 1 \\ 3 \end{pmatrix} e^{-2t} + c_2 \begin{pmatrix} 3 \\ -1 \end{pmatrix} e^{-12t}.$$

Then

$$\Phi = \begin{pmatrix} e^{-2t} & 3e^{-12t} \\ 3e^{-2t} & -e^{-12t} \end{pmatrix}, \quad \Phi^{-1} = \begin{pmatrix} \frac{1}{10}e^{2t} & \frac{3}{10}e^{2t} \\ \frac{3}{10}e^{12t} & -\frac{1}{10}e^{12t} \end{pmatrix},$$

$$\mathbf{U} = \int \Phi^{-1}\mathbf{F} dt = \int \begin{pmatrix} 10e^{2t} \sin t \\ 30e^{12t} \sin t \end{pmatrix} dt = \begin{pmatrix} 2e^{2t}(2 \sin t - \cos t) \\ \frac{6}{29}e^{12t}(12 \sin t - \cos t) \end{pmatrix},$$

and

$$\mathbf{I}_p = \Phi \mathbf{U} = \begin{pmatrix} \frac{332}{29} \sin t - \frac{76}{29} \cos t \\ \frac{276}{29} \sin t - \frac{168}{29} \cos t \end{pmatrix}$$

so that

$$\mathbf{I} = c_1 \begin{pmatrix} 1 \\ 3 \end{pmatrix} e^{-2t} + c_2 \begin{pmatrix} 3 \\ -1 \end{pmatrix} e^{-12t} + \mathbf{I}_p.$$

If $\mathbf{I}(0) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ then $c_1 = 2$ and $c_2 = \frac{6}{29}$.

28. For the system

$$\frac{d}{dt} \begin{bmatrix} i_1 \\ q \end{bmatrix} = \begin{bmatrix} -\frac{20}{3} & -\frac{50}{3} \\ \frac{1}{3} & -\frac{5}{3} \end{bmatrix} \begin{bmatrix} i_1 \\ q \end{bmatrix} + \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

Exercises 10.4

we find $\lambda_1 = -5$, $\lambda_2 = -\frac{10}{3}$, $\mathbf{K}_1 = \begin{bmatrix} 10 \\ -1 \end{bmatrix}$, $\mathbf{K}_2 = \begin{bmatrix} 5 \\ -1 \end{bmatrix}$, $\mathbf{X}_p = \begin{bmatrix} \frac{1}{5} \\ \frac{1}{25} \end{bmatrix}$ and so

$$\mathbf{X} = \begin{bmatrix} i_1 \\ q \end{bmatrix} = c_1 \begin{bmatrix} 10 \\ -1 \end{bmatrix} e^{-5t} + c_2 \begin{bmatrix} 5 \\ -1 \end{bmatrix} e^{-10t/3} + \frac{1}{25} \begin{bmatrix} 5 \\ 1 \end{bmatrix}.$$

29. Let $\mathbf{I} = \begin{bmatrix} i_2 \\ i_3 \end{bmatrix}$ so that

$$\mathbf{I}' = \begin{bmatrix} -2 & -2 \\ -2 & -5 \end{bmatrix} \mathbf{I} + \begin{bmatrix} 60 \\ 60 \end{bmatrix} \quad \text{and} \quad \mathbf{X}_c = c_1 \begin{bmatrix} 2 \\ -1 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{-6t}.$$

If $\mathbf{X}_p = \begin{bmatrix} a_1 \\ b_1 \end{bmatrix}$ then $\mathbf{X}_p = \begin{bmatrix} 30 \\ 0 \end{bmatrix}$ so that $\mathbf{X} = c_1 \begin{bmatrix} 2 \\ -1 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{-6t} + \begin{bmatrix} 30 \\ 0 \end{bmatrix}$. For $\mathbf{I}(0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

we find $c_1 = -12$ and $c_2 = -6$. Thus

$$i_1(t) = i_2(t) + i_3(t) = -12e^{-t} - 18e^{-6t} + 30.$$

30. (a) The eigenvalues are 0, 1, 3, and 4, with corresponding eigenvectors

$$\begin{pmatrix} -6 \\ -4 \\ 1 \\ 2 \end{pmatrix}, \quad \begin{pmatrix} 2 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 3 \\ 1 \\ 2 \\ 1 \end{pmatrix}, \quad \text{and} \quad \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix}.$$

$$(b) \quad \Phi = \begin{pmatrix} -6 & 2e^t & 3e^{3t} & -e^{4t} \\ -4 & e^t & e^{3t} & e^{4t} \\ 1 & 0 & 2e^{3t} & 0 \\ 2 & 0 & e^{3t} & 0 \end{pmatrix}, \quad \Phi^{-1} = \begin{pmatrix} 0 & 0 & -\frac{1}{3} & \frac{2}{3} \\ \frac{1}{3}e^{-t} & \frac{1}{3}e^{-t} & -2e^{-t} & \frac{8}{3}e^{-t} \\ 0 & 0 & \frac{2}{3}e^{-3t} & -\frac{1}{3}e^{-3t} \\ -\frac{1}{3}e^{-4t} & \frac{2}{3}e^{-4t} & 0 & \frac{1}{3}e^{-4t} \end{pmatrix}$$

$$(c) \quad \Phi^{-1}(t)\mathbf{F}(t) = \begin{pmatrix} \frac{2}{3} - \frac{1}{3}e^{2t} \\ \frac{1}{3}e^{-2t} + \frac{8}{3}e^{-t} - 2e^t + \frac{1}{3}t \\ -\frac{1}{3}e^{-3t} + \frac{2}{3}e^{-t} \\ \frac{2}{3}e^{-5t} + \frac{1}{3}e^{-4t} - \frac{1}{3}te^{-3t} \end{pmatrix},$$

$$\int \Phi^{-1}(t)\mathbf{F}(t)dt = \begin{pmatrix} -\frac{1}{6}e^{2t} + \frac{2}{3}t \\ -\frac{1}{6}e^{-2t} - \frac{8}{3}e^{-t} - 2e^t + \frac{1}{6}t^2 \\ \frac{1}{9}e^{-3t} - \frac{2}{3}e^{-t} \\ -\frac{2}{15}e^{-5t} - \frac{1}{12}e^{-4t} + \frac{1}{27}e^{-3t} + \frac{1}{9}te^{-3t} \end{pmatrix},$$

$$\Phi(t) \int \Phi^{-1}(t)\mathbf{F}(t)dt = \begin{pmatrix} -5e^{2t} - \frac{1}{5}e^{-t} - \frac{1}{27}e^t - \frac{1}{9}te^t + \frac{1}{3}t^2e^t - 4t - \frac{59}{12} \\ -2e^{2t} - \frac{3}{10}e^{-t} + \frac{1}{27}e^t + \frac{1}{9}te^t + \frac{1}{6}t^2e^t - \frac{8}{3}t - \frac{95}{36} \\ -\frac{3}{2}e^{2t} + \frac{2}{3}t + \frac{2}{9} \\ -e^{2t} + \frac{4}{3}t - \frac{1}{9} \end{pmatrix},$$

$$\Phi(t)\mathbf{C} = \begin{pmatrix} -6c_1 + 2c_2e^t + 3c_3e^{3t} - c_4e^{4t} \\ -4c_1 + c_2e^t + c_3e^{3t} + c_4e^{4t} \\ c_1 + 2c_3e^{3t} \\ 2c_1 + c_3e^{3t} \end{pmatrix},$$

$$\begin{aligned}
& \Phi(t)\mathbf{C} + \Phi(t) \int \Phi^{-1}(t)\mathbf{F}(t)dt \\
&= \begin{pmatrix} -6c_1 + 2c_2e^t + 3c_3e^{3t} - c_4e^{4t} \\ -4c_1 + c_2e^t + c_3e^{3t} + c_4e^{4t} \\ c_1 + 2c_3e^{3t} \\ 2c_1 + c_3e^{3t} \end{pmatrix} + \begin{pmatrix} -5e^{2t} - \frac{1}{5}e^{-t} - \frac{1}{27}e^t - \frac{1}{9}te^t + \frac{1}{3}t^2e^t - 4t - \frac{59}{12} \\ -2e^{2t} - \frac{3}{10}e^{-t} + \frac{1}{27}e^t + \frac{1}{9}te^t + \frac{1}{6}t^2e^t - \frac{8}{3}t - \frac{95}{36} \\ -\frac{3}{2}e^{2t} + \frac{2}{3}t + \frac{2}{9} \\ -e^{2t} + \frac{4}{3}t - \frac{1}{9} \end{pmatrix} \\
\text{(d) } \mathbf{X}(t) &= c_1 \begin{pmatrix} -6 \\ -4 \\ 1 \\ 2 \end{pmatrix} + c_2 \begin{pmatrix} 2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + c_3 \begin{pmatrix} 3 \\ 1 \\ 2 \\ 1 \end{pmatrix} + c_4 \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} \\
&+ \begin{pmatrix} -5e^{2t} - \frac{1}{5}e^{-t} - \frac{1}{27}e^t - \frac{1}{9}te^t + \frac{1}{3}t^2e^t - 4t - \frac{59}{12} \\ -2e^{2t} - \frac{3}{10}e^{-t} + \frac{1}{27}e^t + \frac{1}{9}te^t + \frac{1}{6}t^2e^t - \frac{8}{3}t - \frac{95}{36} \\ -\frac{3}{2}e^{2t} + \frac{2}{3}t + \frac{2}{9} \\ -e^{2t} + \frac{4}{3}t - \frac{1}{9} \end{pmatrix}
\end{aligned}$$

Exercises 10.5

1. For $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$ we have

$$\mathbf{A}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix},$$

$$\mathbf{A}^3 = \mathbf{A}\mathbf{A}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 8 \end{pmatrix},$$

$$\mathbf{A}^4 = \mathbf{A}\mathbf{A}^3 = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 8 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 16 \end{pmatrix},$$

and so on. In general

$$\mathbf{A}^k = \begin{pmatrix} 1 & 0 \\ 0 & 2^k \end{pmatrix} \quad \text{for } k = 1, 2, 3, \dots$$

Thus

$$\begin{aligned}
e^{\mathbf{A}t} &= \mathbf{I} + \frac{\mathbf{A}}{1!}t + \frac{\mathbf{A}^2}{2!}t^2 + \frac{\mathbf{A}^3}{3!}t^3 + \dots \\
&= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \frac{1}{1!} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} t + \frac{1}{2!} \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix} t^2 + \frac{1}{3!} \begin{pmatrix} 1 & 0 \\ 0 & 8 \end{pmatrix} t^3 + \dots \\
&= \begin{pmatrix} 1 + t + \frac{t^2}{2!} + \frac{t^3}{3!} + \dots & 0 \\ 0 & 1 + t + \frac{(2t)^2}{2!} + \frac{(2t)^3}{3!} + \dots \end{pmatrix} = \begin{pmatrix} e^t & 0 \\ 0 & e^{2t} \end{pmatrix}
\end{aligned}$$

and

$$e^{-\mathbf{A}t} = \begin{pmatrix} e^{-t} & 0 \\ 0 & e^{-2t} \end{pmatrix}.$$

Exercises 10.5

2. For $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ we have

$$\mathbf{A}^2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \mathbf{I}$$

$$\mathbf{A}^3 = \mathbf{A}\mathbf{A}^2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \mathbf{I} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \mathbf{A}$$

$$\mathbf{A}^4 = (\mathbf{A}^2)^2 = \mathbf{I}$$

$$\mathbf{A}^5 = \mathbf{A}\mathbf{A}^4 = \mathbf{A}\mathbf{I} = \mathbf{A}$$

and so on. In general

$$\mathbf{A}^k = \begin{cases} \mathbf{A}, & k = 1, 3, 5, \dots \\ \mathbf{I}, & k = 2, 4, 6, \dots \end{cases}$$

Thus

$$\begin{aligned} e^{\mathbf{A}t} &= \mathbf{I} + \frac{\mathbf{A}}{1!}t + \frac{\mathbf{A}^2}{2!}t^2 + \frac{\mathbf{A}^3}{3!}t^3 + \dots \\ &= \mathbf{I} + \mathbf{A}t + \frac{1}{2!}\mathbf{I}t^2 + \frac{1}{3!}\mathbf{A}t^3 + \dots \\ &= \mathbf{I} \left(1 + \frac{1}{2!}t^2 + \frac{1}{4!}t^4 + \dots \right) + \mathbf{A} \left(t + \frac{1}{3!}t^3 + \frac{1}{5!}t^5 + \dots \right) \\ &= \mathbf{I} \cosh t + \mathbf{A} \sinh t = \begin{pmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{pmatrix} \end{aligned}$$

and

$$e^{-\mathbf{A}t} = \begin{pmatrix} \cosh(-t) & \sinh(-t) \\ \sinh(-t) & \cosh(-t) \end{pmatrix} = \begin{pmatrix} \cosh t & -\sinh t \\ -\sinh t & \cosh t \end{pmatrix}.$$

3. For

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ -2 & -2 & -2 \end{pmatrix}$$

we have

$$\mathbf{A}^2 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ -2 & -2 & -2 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ -2 & -2 & -2 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Thus, $\mathbf{A}^3 = \mathbf{A}^4 = \mathbf{A}^5 = \dots = \mathbf{0}$ and

$$e^{\mathbf{A}t} = \mathbf{I} + \mathbf{A}t = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} t & t & t \\ t & t & t \\ -2t & -2t & -2t \end{pmatrix} = \begin{pmatrix} t+1 & t & t \\ t & t+1 & t \\ -2t & -2t & -2t+1 \end{pmatrix}.$$

4. For

$$\mathbf{A} = \begin{pmatrix} 0 & 0 & 0 \\ 3 & 0 & 0 \\ 5 & 1 & 0 \end{pmatrix}$$

we have

$$\mathbf{A}^2 = \begin{pmatrix} 0 & 0 & 0 \\ 3 & 0 & 0 \\ 5 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 3 & 0 & 0 \\ 5 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 3 & 0 & 0 \end{pmatrix}$$

$$\mathbf{A}^3 = \mathbf{A}\mathbf{A}^2 = \begin{pmatrix} 0 & 0 & 0 \\ 3 & 0 & 0 \\ 5 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 3 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Thus, $\mathbf{A}^4 = \mathbf{A}^5 = \mathbf{A}^6 = \cdots = \mathbf{0}$ and

$$\begin{aligned} e^{\mathbf{A}t} &= \mathbf{I} + \mathbf{A}t + \frac{1}{2}\mathbf{A}^2t^2 \\ &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ 3t & 0 & 0 \\ 5t & t & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ \frac{3}{2}t^2 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 3t & 1 & 0 \\ \frac{3}{2}t^2 + 5t & t & 1 \end{pmatrix}. \end{aligned}$$

5. Using the result of Problem 1

$$\mathbf{X} = \begin{pmatrix} e^t & 0 \\ 0 & e^{2t} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = c_1 \begin{pmatrix} e^t \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ e^t \end{pmatrix}.$$

6. Using the result of Problem 2

$$\mathbf{X} = \begin{pmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = c_1 \begin{pmatrix} \cosh t \\ \sinh t \end{pmatrix} + c_2 \begin{pmatrix} \sinh t \\ \cosh t \end{pmatrix}.$$

7. Using the result of Problem 3

$$\mathbf{X} = \begin{pmatrix} t+1 & t & t \\ t & t+1 & t \\ -2t & -2t & -2t+1 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = c_1 \begin{pmatrix} t+1 \\ t \\ -2t \end{pmatrix} + c_2 \begin{pmatrix} t \\ t+1 \\ -2t \end{pmatrix} + c_3 \begin{pmatrix} t \\ t \\ -2t+1 \end{pmatrix}.$$

8. Using the result of Problem 4

$$\mathbf{X} = \begin{pmatrix} 1 & 0 & 0 \\ 3t & 1 & 0 \\ \frac{3}{2}t^2 + 5t & t & 1 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 3t \\ \frac{3}{2}t^2 + 5t \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 1 \\ t \end{pmatrix} + c_3 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

9. To solve

$$\mathbf{X}' = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 3 \\ -1 \end{pmatrix}$$

we identify $t_0 = 0$, $\mathbf{F}(s) = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$, and use the results of Problem 1 and equation (6) in the text.

$$\begin{aligned} \mathbf{X}(t) &= e^{\mathbf{A}t}\mathbf{C} + e^{\mathbf{A}t} \int_{t_0}^t e^{-\mathbf{A}s}\mathbf{F}(s) ds \\ &= \begin{pmatrix} e^t & 0 \\ 0 & e^{2t} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} e^t & 0 \\ 0 & e^{2t} \end{pmatrix} \int_0^t \begin{pmatrix} e^{-s} & 0 \\ 0 & e^{-2s} \end{pmatrix} \begin{pmatrix} 3 \\ -1 \end{pmatrix} ds \\ &= \begin{pmatrix} c_1 e^t \\ c_2 e^{2t} \end{pmatrix} + \begin{pmatrix} e^t & 0 \\ 0 & e^{2t} \end{pmatrix} \int_0^t \begin{pmatrix} 3e^{-s} \\ -e^{-2s} \end{pmatrix} ds \end{aligned}$$

Exercises 10.5

$$\begin{aligned}
&= \begin{pmatrix} c_1 e^t \\ c_2 e^{2t} \end{pmatrix} + \begin{pmatrix} e^t & 0 \\ 0 & e^{2t} \end{pmatrix} \begin{pmatrix} -3e^{-s} \\ \frac{1}{2}e^{-2s} \end{pmatrix} \Big|_0^t \\
&= \begin{pmatrix} c_1 e^t \\ c_2 e^{2t} \end{pmatrix} + \begin{pmatrix} e^t & 0 \\ 0 & e^{2t} \end{pmatrix} \begin{pmatrix} -3e^{-t} - 3 \\ \frac{1}{2}e^{-2t} - \frac{1}{2} \end{pmatrix} \\
&= \begin{pmatrix} c_1 e^t \\ c_2 e^{2t} \end{pmatrix} + \begin{pmatrix} -3 - 3e^t \\ \frac{1}{2} - \frac{1}{2}e^{2t} \end{pmatrix} = c_3 \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^t + c_4 \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{2t} + \begin{pmatrix} -3 \\ \frac{1}{2} \end{pmatrix}.
\end{aligned}$$

10. To solve

$$\mathbf{X}' = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \mathbf{X} + \begin{pmatrix} t \\ e^{4t} \end{pmatrix}$$

we identify $t_0 = 0$, $\mathbf{F}(s) = \begin{pmatrix} s \\ e^{4s} \end{pmatrix}$, and use the results of Problem 1 and equation (6) in the text.

$$\begin{aligned}
\mathbf{X}(t) &= e^{\mathbf{A}t} \mathbf{C} + e^{\mathbf{A}t} \int_{t_0}^t e^{-\mathbf{A}s} \mathbf{F}(s) ds \\
&= \begin{pmatrix} e^t & 0 \\ 0 & e^{2t} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} e^t & 0 \\ 0 & e^{2t} \end{pmatrix} \int_0^t \begin{pmatrix} e^{-s} & 0 \\ 0 & e^{-2s} \end{pmatrix} \begin{pmatrix} s \\ e^{4s} \end{pmatrix} ds \\
&= \begin{pmatrix} c_1 e^t \\ c_2 e^{2t} \end{pmatrix} + \begin{pmatrix} e^t & 0 \\ 0 & e^{2t} \end{pmatrix} \int_0^t \begin{pmatrix} s e^{-s} \\ e^{2s} \end{pmatrix} ds \\
&= \begin{pmatrix} c_1 e^t \\ c_2 e^{2t} \end{pmatrix} + \begin{pmatrix} e^t & 0 \\ 0 & e^{2t} \end{pmatrix} \begin{pmatrix} -s e^{-s} - e^{-s} \\ \frac{1}{2} e^{2s} \end{pmatrix} \Big|_0^t \\
&= \begin{pmatrix} c_1 e^t \\ c_2 e^{2t} \end{pmatrix} + \begin{pmatrix} e^t & 0 \\ 0 & e^{2t} \end{pmatrix} \begin{pmatrix} -t e^{-t} - e^{-t} + 1 \\ \frac{1}{2} e^{2t} - \frac{1}{2} \end{pmatrix} \\
&= \begin{pmatrix} c_1 e^t \\ c_2 e^{2t} \end{pmatrix} + \begin{pmatrix} -t - 1 + e^t \\ \frac{1}{2} e^{4t} - \frac{1}{2} e^{2t} \end{pmatrix} = c_3 \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^t + c_4 \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{2t} + \begin{pmatrix} -t - 1 \\ \frac{1}{2} e^{4t} \end{pmatrix}.
\end{aligned}$$

11. To solve

$$\mathbf{X}' = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

we identify $t_0 = 0$, $\mathbf{F}(s) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, and use the results of Problem 2 and equation (6) in the text.

$$\begin{aligned}
\mathbf{X}(t) &= e^{\mathbf{A}t} \mathbf{C} + e^{\mathbf{A}t} \int_{t_0}^t e^{-\mathbf{A}s} \mathbf{F}(s) ds \\
&= \begin{pmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{pmatrix} \int_0^t \begin{pmatrix} \cosh s & -\sinh s \\ -\sinh s & \cosh s \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} ds \\
&= \begin{pmatrix} c_1 \cosh t + c_2 \sinh t \\ c_1 \sinh t + c_2 \cosh t \end{pmatrix} + \begin{pmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{pmatrix} \int_0^t \begin{pmatrix} \cosh s & -\sinh s \\ -\sinh s & \cosh s \end{pmatrix} ds \\
&= \begin{pmatrix} c_1 \cosh t + c_2 \sinh t \\ c_1 \sinh t + c_2 \cosh t \end{pmatrix} + \begin{pmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{pmatrix} \begin{pmatrix} \sinh s - \cosh s \\ -\cosh s + \sinh s \end{pmatrix} \Big|_0^t
\end{aligned}$$

$$\begin{aligned}
&= \begin{pmatrix} c_1 \cosh t + c_2 \sinh t \\ c_1 \sinh t + c_2 \cosh t \end{pmatrix} + \begin{pmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{pmatrix} \begin{pmatrix} \sinh t - \cosh t \\ -\cosh t + \sinh t \end{pmatrix} \\
&= \begin{pmatrix} c_1 \cosh t + c_2 \sinh t \\ c_1 \sinh t + c_2 \cosh t \end{pmatrix} + \begin{pmatrix} \sinh^2 t - \cosh^2 t \\ \sinh^2 t - \cosh^2 t \end{pmatrix} = c_1 \begin{pmatrix} \cosh t \\ \sinh t \end{pmatrix} + c_2 \begin{pmatrix} \sinh t \\ \cosh t \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \end{pmatrix}.
\end{aligned}$$

12. To solve

$$\mathbf{X}' = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \mathbf{X} + \begin{pmatrix} \cosh t \\ \sinh t \end{pmatrix}$$

we identify $t_0 = 0$, $\mathbf{F}(s) = \begin{pmatrix} \cosh s \\ \sinh s \end{pmatrix}$, and use the results of Problem 2 and equation (6) in the text.

$$\begin{aligned}
\mathbf{X}(t) &= e^{\mathbf{A}t} \mathbf{C} + e^{\mathbf{A}t} \int_{t_0}^t e^{-\mathbf{A}s} \mathbf{F}(s) ds \\
&= \begin{pmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{pmatrix} \int_0^t \begin{pmatrix} \cosh s & -\sinh s \\ -\sinh s & \cosh s \end{pmatrix} \begin{pmatrix} \cosh s \\ \sinh s \end{pmatrix} ds \\
&= \begin{pmatrix} c_1 \cosh t + c_2 \sinh t \\ c_1 \sinh t + c_2 \cosh t \end{pmatrix} + \begin{pmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{pmatrix} \int_0^t \begin{pmatrix} 1 \\ 0 \end{pmatrix} ds \\
&= \begin{pmatrix} c_1 \cosh t + c_2 \sinh t \\ c_1 \sinh t + c_2 \cosh t \end{pmatrix} + \begin{pmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{pmatrix} \begin{pmatrix} s \\ 0 \end{pmatrix} \Big|_0^t \\
&= \begin{pmatrix} c_1 \cosh t + c_2 \sinh t \\ c_1 \sinh t + c_2 \cosh t \end{pmatrix} + \begin{pmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{pmatrix} \begin{pmatrix} t \\ 0 \end{pmatrix} \\
&= \begin{pmatrix} c_1 \cosh t + c_2 \sinh t \\ c_1 \sinh t + c_2 \cosh t \end{pmatrix} + \begin{pmatrix} t \cosh t \\ t \sinh t \end{pmatrix} = c_1 \begin{pmatrix} \cosh t \\ \sinh t \end{pmatrix} + c_2 \begin{pmatrix} \sinh t \\ \cosh t \end{pmatrix} + t \begin{pmatrix} \cosh t \\ \sinh t \end{pmatrix}.
\end{aligned}$$

13. We have

$$\mathbf{X}(0) = c_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + c_3 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} 1 \\ -4 \\ 6 \end{pmatrix}.$$

Thus, the solution of the initial-value problem is

$$\mathbf{X} = \begin{pmatrix} t+1 \\ t \\ -2t \end{pmatrix} - 4 \begin{pmatrix} t \\ t+1 \\ -2t \end{pmatrix} + 6 \begin{pmatrix} t \\ t \\ -2t+1 \end{pmatrix}.$$

14. We have

$$\mathbf{X}(0) = c_3 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + c_4 \begin{pmatrix} 0 \\ 1 \end{pmatrix} + \begin{pmatrix} -3 \\ \frac{1}{2} \end{pmatrix} = \begin{pmatrix} c_3 - 3 \\ c_4 + \frac{1}{2} \end{pmatrix} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}.$$

Thus, $c_3 = 7$ and $c_4 = \frac{5}{2}$, so

$$\mathbf{X} = 7 \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^t + \frac{5}{2} \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{2t} + \begin{pmatrix} -3 \\ \frac{1}{2} \end{pmatrix}.$$

15. From $s\mathbf{I} - \mathbf{A} = \begin{pmatrix} s-4 & -3 \\ 4 & s+4 \end{pmatrix}$ we find

$$(s\mathbf{I} - \mathbf{A})^{-1} = \begin{pmatrix} \frac{3/2}{s-2} - \frac{1/2}{s+2} & \frac{3/4}{s-2} - \frac{3/4}{s+2} \\ \frac{-1}{s-2} + \frac{1}{s+2} & \frac{-1/2}{s-2} + \frac{3/2}{s+2} \end{pmatrix}$$

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and

$$e^{\mathbf{A}t} = \begin{pmatrix} \frac{3}{2}e^{2t} - \frac{1}{2}e^{-2t} & \frac{3}{4}e^{2t} - \frac{3}{4}e^{-2t} \\ -e^{2t} + e^{-2t} & -\frac{1}{2}e^{2t} + \frac{3}{2}e^{-2t} \end{pmatrix}.$$

The general solution of the system is then

$$\begin{aligned} \mathbf{X} &= e^{\mathbf{A}t} \mathbf{C} = \begin{pmatrix} \frac{3}{2}e^{2t} - \frac{1}{2}e^{-2t} & \frac{3}{4}e^{2t} - \frac{3}{4}e^{-2t} \\ -e^{2t} + e^{-2t} & -\frac{1}{2}e^{2t} + \frac{3}{2}e^{-2t} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \\ &= c_1 \begin{pmatrix} 3/2 \\ -1 \end{pmatrix} e^{2t} + c_1 \begin{pmatrix} -1/2 \\ 1 \end{pmatrix} e^{-2t} + c_2 \begin{pmatrix} 3/4 \\ -1/2 \end{pmatrix} e^{2t} + c_2 \begin{pmatrix} -3/4 \\ 3/2 \end{pmatrix} e^{-2t} \\ &= \left(\frac{1}{2}c_1 + \frac{1}{4}c_2 \right) \begin{pmatrix} 3 \\ -2 \end{pmatrix} e^{2t} + \left(-\frac{1}{2}c_1 - \frac{3}{4}c_2 \right) \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{-2t} \\ &= c_3 \begin{pmatrix} 3 \\ -2 \end{pmatrix} e^{2t} + c_4 \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{-2t}. \end{aligned}$$

16. From $s\mathbf{I} - \mathbf{A} = \begin{pmatrix} s-4 & 2 \\ -1 & s-1 \end{pmatrix}$ we find

$$(s\mathbf{I} - \mathbf{A})^{-1} = \begin{pmatrix} \frac{2}{s-3} - \frac{1}{s-2} & -\frac{2}{s-3} + \frac{2}{s-2} \\ \frac{1}{s-3} - \frac{1}{s-2} & \frac{-1}{s-3} + \frac{2}{s-2} \end{pmatrix}$$

and

$$e^{\mathbf{A}t} = \begin{pmatrix} 2e^{3t} - e^{2t} & -2e^{3t} + 2e^{2t} \\ e^{3t} - e^{2t} & -e^{3t} + 2e^{2t} \end{pmatrix}.$$

The general solution of the system is then

$$\begin{aligned} \mathbf{X} &= e^{\mathbf{A}t} \mathbf{C} = \begin{pmatrix} 2e^{3t} - e^{2t} & -2e^{3t} + 2e^{2t} \\ e^{3t} - e^{2t} & -e^{3t} + 2e^{2t} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \\ &= c_1 \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{3t} + c_1 \begin{pmatrix} -1 \\ -1 \end{pmatrix} e^{2t} + c_2 \begin{pmatrix} -2 \\ -1 \end{pmatrix} e^{3t} + c_2 \begin{pmatrix} 2 \\ 2 \end{pmatrix} e^{2t} \\ &= (c_1 - c_2) \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{3t} + (-c_1 + 2c_2) \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{2t} \\ &= c_3 \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{3t} + c_4 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{2t}. \end{aligned}$$

17. From $s\mathbf{I} - \mathbf{A} = \begin{pmatrix} s-5 & 9 \\ -1 & s+1 \end{pmatrix}$ we find

$$(s\mathbf{I} - \mathbf{A})^{-1} = \begin{pmatrix} \frac{1}{s-2} + \frac{3}{(s-2)^2} & -\frac{9}{(s-2)^2} \\ \frac{1}{(s-2)^2} & \frac{1}{s-2} - \frac{3}{(s-2)^2} \end{pmatrix}$$

and

$$e^{\mathbf{A}t} = \begin{pmatrix} e^{2t} + 3te^{2t} & -9te^{2t} \\ te^{2t} & e^{2t} - 3te^{2t} \end{pmatrix}.$$

The general solution of the system is then

$$\begin{aligned}
\mathbf{X} &= e^{\mathbf{A}t} \mathbf{C} = \begin{pmatrix} e^{2t} + 3te^{2t} & -9te^{2t} \\ te^{2t} & e^{2t} - 3te^{2t} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \\
&= c_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{2t} + c_1 \begin{pmatrix} 3 \\ 1 \end{pmatrix} te^{2t} + c_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{2t} + c_2 \begin{pmatrix} -9 \\ -3 \end{pmatrix} te^{2t} \\
&= c_1 \begin{pmatrix} 1 + 3t \\ t \end{pmatrix} e^{2t} + c_2 \begin{pmatrix} -9t \\ 1 - 3t \end{pmatrix} e^{2t}.
\end{aligned}$$

18. From $s\mathbf{I} - \mathbf{A} = \begin{pmatrix} s & -1 \\ 2 & s+2 \end{pmatrix}$ we find

$$(s\mathbf{I} - \mathbf{A})^{-1} = \begin{pmatrix} \frac{s+1+1}{(s+1)^2+1} & \frac{1}{(s+1)^2+1} \\ \frac{-2}{(s+1)^2+1} & \frac{s+1+1}{(s+1)^2+1} \end{pmatrix}$$

and

$$e^{\mathbf{A}t} = \begin{pmatrix} e^{-t} \cos t + e^{-t} \sin t & e^{-t} \sin t \\ -2e^{-t} \sin t & e^{-t} \cos t - e^{-t} \sin t \end{pmatrix}.$$

The general solution of the system is then

$$\begin{aligned}
\mathbf{X} &= e^{\mathbf{A}t} \mathbf{C} = \begin{pmatrix} e^{-t} \cos t + e^{-t} \sin t & e^{-t} \sin t \\ -2e^{-t} \sin t & e^{-t} \cos t - e^{-t} \sin t \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \\
&= c_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{-t} \cos t + c_1 \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{-t} \sin t + c_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{-t} \cos t + c_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-t} \sin t \\
&= c_1 \begin{pmatrix} \cos t + \sin t \\ -2 \sin t \end{pmatrix} e^{-t} + c_2 \begin{pmatrix} \sin t \\ \cos t - \sin t \end{pmatrix} e^{-t}.
\end{aligned}$$

19. The eigenvalues are $\lambda_1 = 1$ and $\lambda_2 = 6$. This leads to the system

$$e^t = b_0 + b_1$$

$$e^{6t} = b_0 + 6b_1,$$

which has the solution $b_0 = \frac{6}{5}e^t - \frac{1}{5}e^{6t}$ and $b_1 = -\frac{1}{5}e^t + \frac{1}{5}e^{6t}$. Then

$$e^{\mathbf{A}t} = b_0 \mathbf{I} + b_1 \mathbf{A} = \begin{pmatrix} \frac{4}{5}e^t + \frac{1}{5}e^{6t} & \frac{2}{5}e^t - \frac{2}{5}e^{6t} \\ \frac{2}{5}e^t - \frac{2}{5}e^{6t} & \frac{1}{5}e^t + \frac{4}{5}e^{6t} \end{pmatrix}.$$

The general solution of the system is then

$$\begin{aligned}
\mathbf{X} &= e^{\mathbf{A}t} \mathbf{C} = \begin{pmatrix} \frac{4}{5}e^t + \frac{1}{5}e^{6t} & \frac{2}{5}e^t - \frac{2}{5}e^{6t} \\ \frac{2}{5}e^t - \frac{2}{5}e^{6t} & \frac{1}{5}e^t + \frac{4}{5}e^{6t} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \\
&= c_1 \begin{pmatrix} 4/5 \\ 2/5 \end{pmatrix} e^t + c_1 \begin{pmatrix} 1/5 \\ -2/5 \end{pmatrix} e^{6t} + c_2 \begin{pmatrix} 2/5 \\ 1/5 \end{pmatrix} e^t + c_2 \begin{pmatrix} -2/5 \\ 4/5 \end{pmatrix} e^{6t} \\
&= \left(\frac{2}{5}c_1 + \frac{1}{5}c_2 \right) \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^t + \left(\frac{1}{5}c_1 - \frac{2}{5}c_2 \right) \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{6t} \\
&= c_3 \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^t + c_4 \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{6t}.
\end{aligned}$$

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20. The eigenvalues are $\lambda_1 = 2$ and $\lambda_2 = 3$. This leads to the system

$$e^{2t} = b_0 + 2b_1$$

$$e^{3t} = b_0 + 3b_1,$$

which has the solution $b_0 = 3e^{2t} - 2e^{3t}$ and $b_1 = -e^{2t} + e^{3t}$. Then

$$e^{\mathbf{A}t} = b_0 \mathbf{I} + b_1 \mathbf{A} = \begin{pmatrix} 2e^{2t} - e^{3t} & -2e^{2t} + 2e^{3t} \\ e^{2t} - e^{3t} & -e^{2t} + 2e^{3t} \end{pmatrix}.$$

The general solution of the system is then

$$\begin{aligned} \mathbf{X} &= e^{\mathbf{A}t} \mathbf{C} = \begin{pmatrix} 2e^{2t} - e^{3t} & -2e^{2t} + 2e^{3t} \\ e^{2t} - e^{3t} & -e^{2t} + 2e^{3t} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \\ &= c_1 \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{2t} + c_1 \begin{pmatrix} -1 \\ -1 \end{pmatrix} e^{3t} + c_2 \begin{pmatrix} -2 \\ -1 \end{pmatrix} e^{2t} + c_2 \begin{pmatrix} 2 \\ 2 \end{pmatrix} e^{3t} \\ &= (c_1 - c_2) \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{2t} + (-c_1 + 2c_2) \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{3t} \\ &= c_3 \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{2t} + c_4 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{3t}. \end{aligned}$$

21. The eigenvalues are $\lambda_1 = -1$ and $\lambda_2 = 3$. This leads to the system

$$e^{-t} = b_0 - b_1$$

$$e^{3t} = b_0 + 3b_1,$$

which has the solution $b_0 = \frac{3}{4}e^{-t} + \frac{1}{4}e^{3t}$ and $b_1 = -\frac{1}{4}e^{-t} + \frac{1}{4}e^{3t}$. Then

$$e^{\mathbf{A}t} = b_0 \mathbf{I} + b_1 \mathbf{A} = \begin{pmatrix} e^{3t} & -2e^{-t} + 2e^{3t} \\ 0 & e^{-t} \end{pmatrix}.$$

The general solution of the system is then

$$\begin{aligned} \mathbf{X} &= e^{\mathbf{A}t} \mathbf{C} = \begin{pmatrix} e^{3t} & -2e^{-t} + 2e^{3t} \\ 0 & e^{-t} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \\ &= c_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{3t} + c_2 \begin{pmatrix} -2 \\ 1 \end{pmatrix} e^{-t} + c_2 \begin{pmatrix} 2 \\ 0 \end{pmatrix} e^{3t} \\ &= c_2 \begin{pmatrix} -2 \\ 1 \end{pmatrix} e^{-t} + (c_1 + 2c_2) \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{3t} \\ &= c_3 \begin{pmatrix} -2 \\ 1 \end{pmatrix} e^{-t} + c_4 \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{3t}. \end{aligned}$$

22. The eigenvalues are $\lambda_1 = \frac{1}{4}$ and $\lambda_2 = \frac{1}{2}$. This leads to the system

$$e^{t/4} = b_0 + \frac{1}{4}b_1$$

$$e^{t/2} = b_0 + \frac{1}{2}b_1,$$

which has the solution $b_0 = 2e^{t/4} + e^{t/2}$ and $b_1 = -4e^{t/4} + 4e^{t/2}$. Then

$$e^{\mathbf{A}t} = b_0 \mathbf{I} + b_1 \mathbf{A} = \begin{pmatrix} -2e^{t/4} + 3e^{t/2} & 6e^{t/4} - 6e^{t/2} \\ -e^{t/4} + e^{t/2} & 3e^{t/4} - 2e^{t/2} \end{pmatrix}.$$

The general solution of the system is then

$$\begin{aligned} \mathbf{X} &= e^{\mathbf{A}t} \mathbf{C} = \begin{pmatrix} -2e^{t/4} + 3e^{t/2} & 6e^{t/4} - 6e^{t/2} \\ -e^{t/4} + e^{t/2} & 3e^{t/4} - 2e^{t/2} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \\ &= c_1 \begin{pmatrix} -2 \\ -1 \end{pmatrix} e^{t/4} + c_1 \begin{pmatrix} 3 \\ 1 \end{pmatrix} e^{t/2} + c_2 \begin{pmatrix} 6 \\ 3 \end{pmatrix} e^{t/4} + c_2 \begin{pmatrix} -6 \\ -2 \end{pmatrix} e^{t/2} \\ &= (-c_1 + 3c_2) \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{t/4} + (c_1 - 2c_2) \begin{pmatrix} 3 \\ 1 \end{pmatrix} e^{t/2} \\ &= c_3 \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{t/4} + c_4 \begin{pmatrix} 3 \\ 1 \end{pmatrix} e^{t/2}. \end{aligned}$$

23. From equation (2) in the text we have $e^{\mathbf{D}t} = \mathbf{I} + t\mathbf{D} + \frac{t^2}{2!}\mathbf{D}^2 + \frac{t^3}{3!}\mathbf{D}^3 + \cdots$ so that

$$\mathbf{P}e^{\mathbf{D}t}\mathbf{P}^{-1} = \mathbf{P}\mathbf{P}^{-1} + t(\mathbf{P}\mathbf{D}\mathbf{P}^{-1}) + \frac{t^2}{2!}(\mathbf{P}\mathbf{D}^2\mathbf{P}^{-1}) + \frac{t^3}{3!}(\mathbf{P}\mathbf{D}^3\mathbf{P}^{-1}) + \cdots.$$

But $\mathbf{P}\mathbf{P}^{-1} = \mathbf{I}$, $\mathbf{P}\mathbf{D}\mathbf{P}^{-1} = \mathbf{A}$ and $\mathbf{P}\mathbf{D}^n\mathbf{P}^{-1} = \mathbf{A}^n$ (see Problem 37, Exercises 8.12). Thus,

$$\mathbf{P}e^{\mathbf{D}t}\mathbf{P}^{-1} = \mathbf{I} + t\mathbf{A} + \frac{t^2}{2!}\mathbf{A}^2 + \frac{t^3}{3!}\mathbf{A}^3 + \cdots = e^{\mathbf{A}t}.$$

24. From equation (2) in the text

$$\begin{aligned} e^{\mathbf{D}t} &= \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} + \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix} + \frac{1}{2!}t^2 \begin{bmatrix} \lambda_1^2 & 0 & \cdots & 0 \\ 0 & \lambda_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n^2 \end{bmatrix} \\ &\quad + \frac{1}{3!}t^3 \begin{bmatrix} \lambda_1^3 & 0 & \cdots & 0 \\ 0 & \lambda_2^3 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n^3 \end{bmatrix} + \cdots \\ &= \begin{bmatrix} 1 + \lambda_1 t + \frac{1}{2!}(\lambda_1 t)^2 + \cdots & 0 & \cdots & 0 \\ 0 & 1 + \lambda_2 t + \frac{1}{2!}(\lambda_2 t)^2 + \cdots & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 + \lambda_n t + \frac{1}{2!}(\lambda_n t)^2 + \cdots \end{bmatrix} \\ &= \begin{bmatrix} e^{\lambda_1 t} & 0 & \cdots & 0 \\ 0 & e^{\lambda_2 t} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & e^{\lambda_n t} \end{bmatrix} \end{aligned}$$

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25. From Problems 23 and 24 and equation (1) in the text

$$\begin{aligned}\mathbf{X} &= e^{\mathbf{A}t}\mathbf{C} = \mathbf{P}e^{\mathbf{D}t}\mathbf{P}^{-1}\mathbf{C} = \begin{bmatrix} e^{3t} & e^{5t} \\ e^{3t} & 3e^{5t} \end{bmatrix} \begin{bmatrix} e^{3t} & 0 \\ 0 & e^{5t} \end{bmatrix} \begin{bmatrix} \frac{3}{2}e^{-3t} & -\frac{1}{2}e^{-3t} \\ -\frac{1}{2}e^{-5t} & \frac{1}{2}e^{-5t} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} \\ &= \begin{bmatrix} \frac{3}{2}e^{3t} - \frac{1}{2}e^{5t} & -\frac{1}{2}e^{3t} + \frac{1}{2}e^{5t} \\ \frac{3}{2}e^{3t} - \frac{3}{2}e^{5t} & -\frac{1}{2}e^{3t} + \frac{3}{2}e^{5t} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}.\end{aligned}$$

26. From Problems 23 and 24 and equation (1) in the text

$$\begin{aligned}\mathbf{X} &= e^{\mathbf{A}t}\mathbf{C} = \mathbf{P}e^{\mathbf{D}t}\mathbf{P}^{-1}\mathbf{C} = \begin{bmatrix} -e^t & e^{3t} \\ e^t & e^{3t} \end{bmatrix} \begin{bmatrix} e^t & 0 \\ 0 & e^{3t} \end{bmatrix} \begin{bmatrix} -\frac{1}{2}e^{-t} & \frac{1}{2}e^{-t} \\ \frac{1}{2}e^{3t} & \frac{1}{2}e^{-3t} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{2}e^t + \frac{1}{2}e^{9t} & -\frac{1}{2}e^t + \frac{1}{2}e^{3t} \\ -\frac{1}{2}e^t + \frac{1}{2}e^{9t} & \frac{1}{2}e^t + \frac{1}{2}e^{3t} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}.\end{aligned}$$

27. (a) The following commands can be used in *Mathematica*:

```
A={{4, 2},{3, 3}};
c={c1, c2};
m=MatrixExp[A t];
sol=Expand[m.c]
Collect[sol, {c1, c2}] // MatrixForm
```

The output gives

$$\begin{aligned}x(t) &= c_1 \left(\frac{2}{5}e^t + \frac{3}{5}e^{6t} \right) + c_2 \left(-\frac{2}{5}e^t + \frac{2}{5}e^{6t} \right) \\ y(t) &= c_1 \left(-\frac{3}{5}e^t + \frac{3}{5}e^{6t} \right) + c_2 \left(\frac{3}{5}e^t + \frac{2}{5}e^{6t} \right).\end{aligned}$$

The eigenvalues are 1 and 6 with corresponding eigenvectors

$$\begin{pmatrix} -2 \\ 3 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 \\ 1 \end{pmatrix},$$

so the solution of the system is

$$\mathbf{X}(t) = b_1 \begin{pmatrix} -2 \\ 3 \end{pmatrix} e^t + b_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{6t}$$

or

$$\begin{aligned}x(t) &= -2b_1e^t + b_2e^{6t} \\ y(t) &= 3b_1e^t + b_2e^{6t}.\end{aligned}$$

If we replace b_1 with $-\frac{1}{5}c_1 + \frac{1}{5}c_2$ and b_2 with $\frac{3}{5}c_1 + \frac{2}{5}c_2$, we obtain the solution found using the matrix exponential.

(b)
$$\begin{aligned}x(t) &= c_1e^{-2t}\cos t - (c_1 + c_2)e^{-2t}\sin t \\ y(t) &= c_2e^{-2t}\cos t + (2c_1 + c_2)e^{-2t}\sin t\end{aligned}$$

28.
$$\begin{aligned}x(t) &= c_1(3e^{-2t} - 2e^{-t}) + c_3(-6e^{-2t} + 6e^{-t}) \\ y(t) &= c_2(4e^{-2t} - 3e^{-t}) + c_4(4e^{-2t} - 4e^{-t}) \\ z(t) &= c_1(e^{-2t} - e^{-t}) + c_3(-2e^{-2t} + 3e^{-t}) \\ w(t) &= c_2(-3e^{-2t} + 3e^{-t}) + c_4(-3e^{-2t} + 4e^{-t})\end{aligned}$$

Chapter 10 Review Exercises

1. If $\mathbf{X} = k \begin{pmatrix} 4 \\ 5 \end{pmatrix}$, then $\mathbf{X}' = \mathbf{0}$ and

$$k \begin{pmatrix} 1 & 4 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 4 \\ 5 \end{pmatrix} - \begin{pmatrix} 8 \\ 1 \end{pmatrix} = k \begin{pmatrix} 24 \\ 3 \end{pmatrix} - \begin{pmatrix} 8 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

We see that $k = \frac{1}{3}$.

2. Solving for c_1 and c_2 we find $c_1 = -\frac{3}{4}$ and $c_2 = \frac{1}{4}$.
3. Since

$$\begin{pmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 12 \\ 4 \\ -4 \end{pmatrix} = 4 \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix},$$

we see that $\lambda = 4$ is an eigenvalue with eigenvector $\mathbf{K}_3$. The corresponding solution is $\mathbf{X} = \mathbf{K}_3 e^{4t}$.

4. The other eigenvalue is $\lambda_2 = 1 - 2i$ with corresponding eigenvector $\mathbf{K}_2 = \begin{pmatrix} 1 \\ -i \end{pmatrix}$. The general solution is

$$\mathbf{X}(t) = c_1 \begin{pmatrix} \cos 2t \\ -\sin 2t \end{pmatrix} e^t + c_2 \begin{pmatrix} \sin 2t \\ \cos 2t \end{pmatrix} e^t.$$

5. We have $\det(\mathbf{A} - \lambda \mathbf{I}) = (\lambda - 1)^2 = 0$ and $\mathbf{K} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$. A solution to $(\mathbf{A} - \lambda \mathbf{I})\mathbf{P} = \mathbf{K}$ is $\mathbf{P} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ so that

$$\mathbf{X} = c_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^t + c_2 \left[\begin{pmatrix} 1 \\ -1 \end{pmatrix} t e^t + \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^t \right].$$

6. We have $\det(\mathbf{A} - \lambda \mathbf{I}) = (\lambda + 6)(\lambda + 2) = 0$ so that

$$\mathbf{X} = c_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-6t} + c_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{-2t}.$$

7. We have $\det(\mathbf{A} - \lambda \mathbf{I}) = \lambda^2 - 2\lambda + 5 = 0$. For $\lambda = 1 + 2i$ we obtain $\mathbf{K}_1 = \begin{pmatrix} 1 \\ i \end{pmatrix}$ and

$$\mathbf{X}_1 = \begin{pmatrix} 1 \\ i \end{pmatrix} e^{(1+2i)t} = \begin{pmatrix} \cos 2t \\ -\sin 2t \end{pmatrix} e^t + i \begin{pmatrix} \sin 2t \\ \cos 2t \end{pmatrix} e^t.$$

Then

$$\mathbf{X} = c_1 \begin{pmatrix} \cos 2t \\ -\sin 2t \end{pmatrix} e^t + c_2 \begin{pmatrix} \sin 2t \\ \cos 2t \end{pmatrix} e^t.$$

8. We have $\det(\mathbf{A} - \lambda \mathbf{I}) = \lambda^2 - 2\lambda + 2 = 0$. For $\lambda = 1 + i$ we obtain $\mathbf{K}_1 = \begin{pmatrix} 3 - i \\ 2 \end{pmatrix}$ and

$$\mathbf{X}_1 = \begin{pmatrix} 3 - i \\ 2 \end{pmatrix} e^{(1+i)t} = \begin{pmatrix} 3 \cos t + \sin t \\ 2 \cos t \end{pmatrix} e^t + i \begin{pmatrix} -\cos t + 3 \sin t \\ 2 \sin t \end{pmatrix} e^t.$$

Then

$$\mathbf{X} = c_1 \begin{pmatrix} 3 \cos t + \sin t \\ 2 \cos t \end{pmatrix} e^t + c_2 \begin{pmatrix} -\cos t + 3 \sin t \\ 2 \sin t \end{pmatrix} e^t.$$

Chapter 10 Review Exercises

9. We have $\det(\mathbf{A} - \lambda\mathbf{I}) = -(\lambda - 2)(\lambda - 4)(\lambda + 3) = 0$ so that

$$\mathbf{X} = c_1 \begin{pmatrix} -2 \\ 3 \\ 1 \end{pmatrix} e^{2t} + c_2 \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} e^{4t} + c_3 \begin{pmatrix} 7 \\ 12 \\ -16 \end{pmatrix} e^{-3t}.$$

10. We have $\det(\mathbf{A} - \lambda\mathbf{I}) = -(s+2)(s^2 - 2s + 3) = 0$. The eigenvalues are $\lambda_1 = -2$, $\lambda_2 = 1 + \sqrt{2}i$, and $\lambda_3 = 1 - \sqrt{2}i$, with eigenvectors

$$\mathbf{K}_1 = \begin{pmatrix} -7 \\ 5 \\ 4 \end{pmatrix}, \quad \mathbf{K}_2 = \begin{pmatrix} 1 \\ \sqrt{2}i/2 \\ 1 \end{pmatrix}, \quad \text{and} \quad \mathbf{K}_3 = \begin{pmatrix} 1 \\ -\sqrt{2}i/2 \\ 1 \end{pmatrix}.$$

Thus

$$\begin{aligned} \mathbf{X} &= c_1 \begin{pmatrix} -7 \\ 5 \\ 4 \end{pmatrix} e^{-2t} + c_2 \left[\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \cos \sqrt{2}t - \begin{pmatrix} 0 \\ \sqrt{2}/2 \\ 0 \end{pmatrix} \sin \sqrt{2}t \right] e^t \\ &\quad + c_3 \left[\begin{pmatrix} 0 \\ \sqrt{2}/2 \\ 0 \end{pmatrix} \cos \sqrt{2}t + \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \sin \sqrt{2}t \right] e^t \\ &= c_1 \begin{pmatrix} -7 \\ 5 \\ 4 \end{pmatrix} e^{-2t} + c_2 \begin{pmatrix} \cos \sqrt{2}t \\ -\sqrt{2} \sin \sqrt{2}t/2 \\ \cos \sqrt{2}t \end{pmatrix} e^t + c_3 \begin{pmatrix} \sin \sqrt{2}t \\ \sqrt{2} \cos \sqrt{2}t/2 \\ \sin \sqrt{2}t \end{pmatrix} e^t. \end{aligned}$$

11. We have

$$\mathbf{X}_c = c_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{2t} + c_2 \begin{pmatrix} 4 \\ 1 \end{pmatrix} e^{4t}.$$

Then

$$\Phi = \begin{pmatrix} e^{2t} & 4e^{4t} \\ 0 & e^{4t} \end{pmatrix}, \quad \Phi^{-1} = \begin{pmatrix} e^{-2t} & -4e^{-2t} \\ 0 & e^{-4t} \end{pmatrix},$$

and

$$\mathbf{U} = \int \Phi^{-1} \mathbf{F} dt = \int \begin{pmatrix} 2e^{-2t} - 64te^{-2t} \\ 16te^{-4t} \end{pmatrix} dt = \begin{pmatrix} 15e^{-2t} + 32te^{-2t} \\ -e^{-4t} - 4te^{-4t} \end{pmatrix},$$

so that

$$\mathbf{X}_p = \Phi \mathbf{U} = \begin{pmatrix} 11 + 16t \\ -1 - 4t \end{pmatrix}.$$

12. We have

$$\mathbf{X}_c = c_1 \begin{pmatrix} 2 \cos t \\ -\sin t \end{pmatrix} e^t + c_2 \begin{pmatrix} 2 \sin t \\ \cos t \end{pmatrix} e^t.$$

Then

$$\Phi = \begin{pmatrix} 2 \cos t & 2 \sin t \\ -\sin t & \cos t \end{pmatrix} e^t, \quad \Phi^{-1} = \begin{pmatrix} \frac{1}{2} \cos t & -\sin t \\ \frac{1}{2} \sin t & \cos t \end{pmatrix} e^{-t},$$

and

$$\mathbf{U} = \int \Phi^{-1} \mathbf{F} dt = \int \begin{pmatrix} \cos t - \sec t \\ \sin t \end{pmatrix} dt = \begin{pmatrix} \sin t - \ln |\sec t + \tan t| \\ -\cos t \end{pmatrix},$$

so that

$$\mathbf{X}_p = \Phi \mathbf{U} = \begin{pmatrix} -2 \cos t \ln |\sec t + \tan t| \\ -1 + \sin t \ln |\sec t + \tan t| \end{pmatrix}$$

13. We have

$$\mathbf{X}_c = c_1 \begin{pmatrix} \cos t + \sin t \\ 2 \cos t \end{pmatrix} + c_2 \begin{pmatrix} \sin t - \cos t \\ 2 \sin t \end{pmatrix}.$$

Then

$$\Phi = \begin{pmatrix} \cos t + \sin t & \sin t - \cos t \\ 2 \cos t & 2 \sin t \end{pmatrix}, \quad \Phi^{-1} = \begin{pmatrix} \sin t & \frac{1}{2} \cos t - \frac{1}{2} \sin t \\ -\cos t & \frac{1}{2} \cos t + \frac{1}{2} \sin t \end{pmatrix},$$

and

$$\begin{aligned} \mathbf{U} &= \int \Phi^{-1} \mathbf{F} dt = \int \begin{pmatrix} \frac{1}{2} \sin t - \frac{1}{2} \cos t + \frac{1}{2} \csc t \\ -\frac{1}{2} \sin t - \frac{1}{2} \cos t + \frac{1}{2} \csc t \end{pmatrix} dt \\ &= \begin{pmatrix} -\frac{1}{2} \cos t - \frac{1}{2} \sin t + \frac{1}{2} \ln |\csc t - \cot t| \\ \frac{1}{2} \cos t - \frac{1}{2} \sin t + \frac{1}{2} \ln |\csc t - \cot t| \end{pmatrix}, \end{aligned}$$

so that

$$\mathbf{X}_p = \Phi \mathbf{U} = \begin{pmatrix} -1 \\ -1 \end{pmatrix} + \begin{pmatrix} \sin t \\ \sin t + \cos t \end{pmatrix} \ln |\csc t - \cot t|.$$

14. We have

$$\mathbf{X}_c = c_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{2t} + c_2 \left[\begin{pmatrix} 1 \\ -1 \end{pmatrix} t e^{2t} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{2t} \right].$$

Then

$$\Phi = \begin{pmatrix} e^{2t} & t e^{2t} + e^{2t} \\ -e^{2t} & -t e^{2t} \end{pmatrix}, \quad \Phi^{-1} = \begin{pmatrix} -t e^{-2t} & -t e^{-2t} - e^{-2t} \\ e^{-2t} & e^{-2t} \end{pmatrix},$$

and

$$\mathbf{U} = \int \Phi^{-1} \mathbf{F} dt = \int \begin{pmatrix} t-1 \\ -1 \end{pmatrix} dt = \begin{pmatrix} \frac{1}{2} t^2 - t \\ -t \end{pmatrix},$$

so that

$$\mathbf{X}_p = \Phi \mathbf{U} = \begin{pmatrix} -1/2 \\ 1/2 \end{pmatrix} t^2 e^{2t} + \begin{pmatrix} -2 \\ 1 \end{pmatrix} t e^{2t}.$$

15. (a) Letting

$$\mathbf{K} = \begin{pmatrix} k_1 \\ k_2 \\ k_3 \end{pmatrix}$$

we note that $(\mathbf{A} - 2\mathbf{I})\mathbf{K} = \mathbf{0}$ implies that $3k_1 + 3k_2 + 3k_3 = 0$, so $k_1 = -(k_2 + k_3)$. Choosing $k_2 = 0$, $k_3 = 1$ and then $k_2 = 1$, $k_3 = 0$ we get

$$\mathbf{K}_1 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \quad \text{and} \quad \mathbf{K}_2 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix},$$

respectively. Thus,

$$\mathbf{X}_1 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} e^{2t} \quad \text{and} \quad \mathbf{X}_2 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} e^{2t}$$

Chapter 10 Review Exercises

are two solutions.

(b) From $\det(\mathbf{A} - \lambda \mathbf{I}) = \lambda^2(3 - \lambda) = 0$ we see that $\lambda_1 = 3$, and 0 is an eigenvalue of multiplicity two. Letting

$$\mathbf{K} = \begin{pmatrix} k_1 \\ k_2 \\ k_3 \end{pmatrix},$$

as in part (a), we note that $(\mathbf{A} - 0\mathbf{I})\mathbf{K} = \mathbf{A}\mathbf{K} = \mathbf{0}$ implies that $k_1 + k_2 + k_3 = 0$, so $k_1 = -(k_2 + k_3)$. Choosing $k_2 = 0$, $k_3 = 1$, and then $k_2 = 1$, $k_3 = 0$ we get

$$\mathbf{K}_2 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \quad \text{and} \quad \mathbf{K}_3 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix},$$

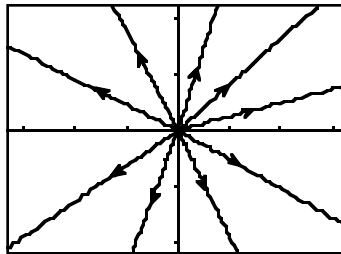
respectively. Since the eigenvector corresponding to $\lambda_1 = 3$ is

$$\mathbf{K}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix},$$

the general solution of the system is

$$\mathbf{X} = c_1 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} e^{3t} + c_2 \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} + c_3 \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}.$$

16. For $\mathbf{X} = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} e^t$ we have $\mathbf{X}' = \mathbf{X} = \mathbf{I}\mathbf{X}$.



11 Systems of Nonlinear Differential Equations

Exercises 11.1

1. The corresponding plane autonomous system is

$$x' = y, \quad y' = -9 \sin x.$$

If (x, y) is a critical point, $y = 0$ and $-9 \sin x = 0$. Therefore $x = \pm n\pi$ and so the critical points are $(\pm n\pi, 0)$ for $n = 0, 1, 2, \dots$.

2. The corresponding plane autonomous system is

$$x' = y, \quad y' = -2x - y^2.$$

If (x, y) is a critical point, then $y = 0$ and so $-2x - y^2 = -2x = 0$. Therefore $(0, 0)$ is the sole critical point.

3. The corresponding plane autonomous system is

$$x' = y, \quad y' = x^2 - y(1 - x^3).$$

If (x, y) is a critical point, $y = 0$ and so $x^2 - y(1 - x^3) = x^2 = 0$. Therefore $(0, 0)$ is the sole critical point.

4. The corresponding plane autonomous system is

$$x' = y, \quad y' = -4 \frac{x}{1+x^2} - 2y.$$

If (x, y) is a critical point, $y = 0$ and so $-4 \frac{x}{1+x^2} - 2(0) = 0$. Therefore $x = 0$ and so $(0, 0)$ is the sole critical point.

5. The corresponding plane autonomous system is

$$x' = y, \quad y' = -x + \epsilon x^3.$$

If (x, y) is a critical point, $y = 0$ and $-x + \epsilon x^3 = 0$. Hence $x(-1 + \epsilon x^2) = 0$ and so $x = 0, \sqrt{1/\epsilon}, -\sqrt{1/\epsilon}$. The critical points are $(0, 0), (\sqrt{1/\epsilon}, 0)$ and $(-\sqrt{1/\epsilon}, 0)$.

6. The corresponding plane autonomous system is

$$x' = y, \quad y' = -x + \epsilon x|x|.$$

If (x, y) is a critical point, $y = 0$ and $-x + \epsilon x|x| = x(-1 + \epsilon|x|) = 0$. Hence $x = 0, 1/\epsilon, -1/\epsilon$. The critical points are $(0, 0), (1/\epsilon, 0)$ and $(-1/\epsilon, 0)$.

7. From $x + xy = 0$ we have $x(1+y) = 0$. Therefore $x = 0$ or $y = -1$. If $x = 0$, then, substituting into $-y - xy = 0$, we obtain $y = 0$. Likewise, if $y = -1$, $1 + x = 0$ or $x = -1$. We may conclude that $(0, 0)$ and $(-1, -1)$ are critical points of the system.

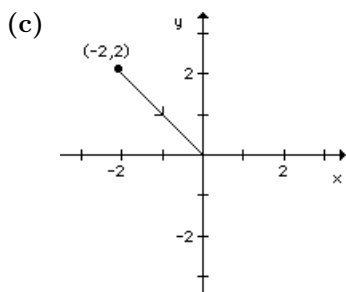
8. From $y^2 - x = 0$ we have $x = y^2$. Substituting into $x^2 - y = 0$, we obtain $y^4 - y = 0$ or $y(y^3 - 1) = 0$. It follows that $y = 0, 1$ and so $(0, 0)$ and $(1, 1)$ are the critical points of the system.

9. From $x - y = 0$ we have $y = x$. Substituting into $3x^2 - 4y = 0$ we obtain $3x^2 - 4x = x(3x - 4) = 0$. It follows that $(0, 0)$ and $(4/3, 4/3)$ are the critical points of the system.

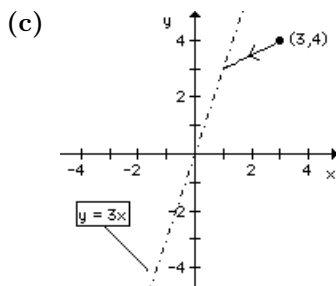
10. From $x^3 - y = 0$ we have $y = x^3$. Substituting into $x - y^3 = 0$ we obtain $x - x^9 = 0$ or $x(1 - x^8) = 0$. Therefore $x = 0, 1, -1$ and so the critical points of the system are $(0, 0), (1, 1)$, and $(-1, -1)$.

Exercises 11.1

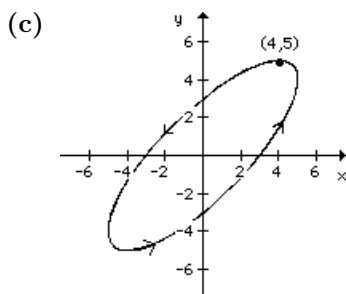
11. From $x(10 - x - \frac{1}{2}y) = 0$ we obtain $x = 0$ or $x + \frac{1}{2}y = 10$. Likewise $y(16 - y - x) = 0$ implies that $y = 0$ or $x + y = 16$. We therefore have four cases. If $x = 0$, $y = 0$ or $y = 16$. If $x + \frac{1}{2}y = 10$, we may conclude that $y(-\frac{1}{2}y + 6) = 0$ and so $y = 0, 12$. Therefore the critical points of the system are $(0, 0)$, $(0, 16)$, $(10, 0)$, and $(4, 12)$.
12. Adding the two equations we obtain $10 - 15 \frac{y}{y+5} = 0$. It follows that $y = 10$, and from $-2x + y + 10 = 0$ we may conclude that $x = 10$. Therefore $(10, 10)$ is the sole critical point of the system.
13. From $x^2 e^y = 0$ we have $x = 0$. Since $e^x - 1 = e^0 - 1 = 0$, the second equation is satisfied for an arbitrary value of y . Therefore any point of the form $(0, y)$ is a critical point.
14. From $\sin y = 0$ we have $y = \pm n\pi$. From $e^{x-y} = 1$, we may conclude that $x - y = 0$ or $x = y$. The critical points of the system are therefore $(\pm n\pi, \pm n\pi)$ for $n = 0, 1, 2, \dots$.
15. From $x(1 - x^2 - 3y^2) = 0$ we have $x = 0$ or $x^2 + 3y^2 = 1$. If $x = 0$, then substituting into $y(3 - x^2 - 3y^2)$ gives $y(3 - 3y^2) = 0$. Therefore $y = 0, 1, -1$. Likewise $x^2 = 1 - 3y^2$ yields $2y = 0$ so that $y = 0$ and $x^2 = 1 - 3(0)^2 = 1$. The critical points of the system are therefore $(0, 0)$, $(0, 1)$, $(0, -1)$, $(1, 0)$, and $(-1, 0)$.
16. From $-x(4 - y^2) = 0$ we obtain $x = 0$, $y = 2$, or $y = -2$. If $x = 0$, then substituting into $4y(1 - x^2)$ yields $y = 0$. Likewise $y = 2$ gives $8(1 - x^2) = 0$ or $x = 1, -1$. Finally $y = -2$ yields $-8(1 - x^2) = 0$ or $x = 1, -1$. The critical points of the system are therefore $(0, 0)$, $(1, 2)$, $(-1, 2)$, $(1, -2)$, and $(-1, -2)$.
17. (a) From Exercises 10.2, Problem 1, $x = c_1 e^{5t} - c_2 e^{-t}$ and $y = 2c_1 e^{5t} + c_2 e^{-t}$.
 (b) From $\mathbf{X}(0) = (2, -1)$ it follows that $c_1 = 0$ and $c_2 = 2$. Therefore $x = -2e^{-t}$ and $y = 2e^{-t}$.



18. (a) From Exercises 10.2, Problem 6, $x = c_1 + 2c_2 e^{-5t}$ and $y = 3c_1 + c_2 e^{-5t}$.
 (b) From $\mathbf{X}(0) = (3, 4)$ it follows that $c_1 = c_2 = 1$. Therefore $x = 1 + 2e^{-5t}$ and $y = 3 + e^{-5t}$.

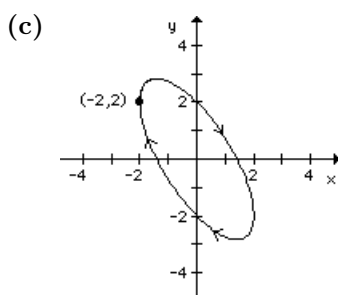


19. (a) From Exercises 10.2, Problem 37, $x = c_1(4 \cos 3t - 3 \sin 3t) + c_2(4 \sin 3t + 3 \cos 3t)$ and $y = c_1(5 \cos 3t) + c_2(5 \sin 3t)$. All solutions are one periodic with $p = 2\pi/3$.
 (b) From $\mathbf{X}(0) = (4, 5)$ it follows that $c_1 = 1$ and $c_2 = 0$. Therefore $x = 4 \cos 3t - 3 \sin 3t$ and $y = 5 \cos 3t$.



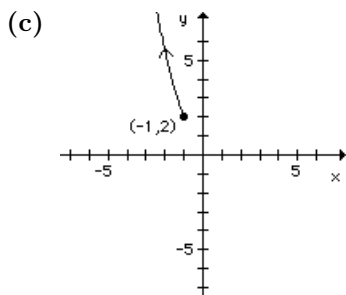
20. (a) From Exercises 10.2, Problem 34, $x = c_1(\sin t - \cos t) + c_2(-\cos t - \sin t)$ and $y = 2c_1 \cos t + 2c_2 \sin t$. All solutions are periodic with $p = 2\pi$.

(b) From $\mathbf{X}(0) = (-2, 2)$ it follows that $c_1 = c_2 = 1$. Therefore $x = -2 \cos t$ and $y = 2 \cos t + 2 \sin t$.



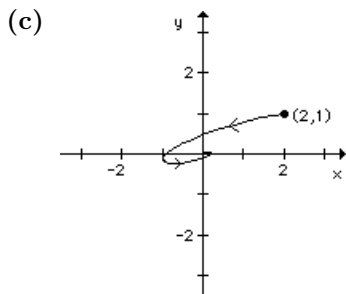
21. (a) From Exercises 10.2, Problem 35, $x = c_1(\sin t - \cos t)e^{4t} + c_2(-\sin t - \cos t)e^{4t}$ and $y = 2c_1(\cos t)e^{4t} + 2c_2(\sin t)e^{4t}$. Because of the presence of e^{4t} , there are no periodic solutions.

(b) From $\mathbf{X}(0) = (-1, 2)$ it follows that $c_1 = 1$ and $c_2 = 0$. Therefore $x = (\sin t - \cos t)e^{4t}$ and $y = 2(\cos t)e^{4t}$.



22. (a) From Exercises 10.2, Problem 38, $x = c_1 e^{-t}(2 \cos 2t - 2 \sin 2t) + c_2 e^{-t}(2 \cos 2t + 2 \sin 2t)$ and $y = c_1 e^{-t} \cos 2t + c_2 e^{-t} \sin 2t$. Because of the presence of e^{-t} , there are no periodic solutions.

(b) From $\mathbf{X}(0) = (2, 1)$ it follows that $c_1 = 1$ and $c_2 = 0$. Therefore $x = e^{-t}(2 \cos 2t - 2 \sin 2t)$ and $y = e^{-t} \cos 2t$.



Exercises 11.1

23. Switching to polar coordinates,

$$\frac{dr}{dt} = \frac{1}{r} \left(x \frac{dx}{dt} + y \frac{dy}{dt} \right) = \frac{1}{r} (-xy - x^2 r^4 + xy - y^2 r^4) = -r^5$$

$$\frac{d\theta}{dt} = \frac{1}{r^2} \left(-y \frac{dx}{dt} + x \frac{dy}{dt} \right) = \frac{1}{r^2} (y^2 + xyr^4 + x^2 - xyr^4) = 1.$$

If we use separation of variables on $\frac{dr}{dt} = -r^5$ we obtain

$$r = \left(\frac{1}{4t + c_1} \right)^{1/4} \quad \text{and} \quad \theta = t + c_2.$$

Since $\mathbf{X}(0) = (4, 0)$, $r = 4$ and $\theta = 0$ when $t = 0$. It follows that $c_2 = 0$ and $c_1 = \frac{1}{256}$. The final solution may be written as

$$r = \frac{4}{\sqrt[4]{1024t + 1}}, \quad \theta = t$$

and so the solution spirals toward the origin as t increases.

24. Switching to polar coordinates,

$$\frac{dr}{dt} = \frac{1}{r} \left(x \frac{dx}{dt} + y \frac{dy}{dt} \right) = \frac{1}{r} (xy - x^2 r^2 - xy + y^2 r^2) = r^3$$

$$\frac{d\theta}{dt} = \frac{1}{r^2} \left(-y \frac{dx}{dt} + x \frac{dy}{dt} \right) = \frac{1}{r^2} (-y^2 - xyr^2 - x^2 + xyr^2) = -1.$$

If we use separation of variables, it follows that

$$r = \frac{1}{\sqrt{-2t + c_1}} \quad \text{and} \quad \theta = -t + c_2.$$

Since $\mathbf{X}(0) = (4, 0)$, $r = 4$ and $\theta = 0$ when $t = 0$. It follows that $c_2 = 0$ and $c_1 = \frac{1}{16}$. The final solution may be written as

$$r = \frac{4}{\sqrt{1 - 32t}}, \quad \theta = -t.$$

Note that $r \rightarrow \infty$ as $t \rightarrow \left(\frac{1}{32}\right)^-$. Because $0 \leq t \leq \frac{1}{32}$, the curve is not a spiral.

25. Switching to polar coordinates,

$$\frac{dr}{dt} = \frac{1}{r} \left(x \frac{dx}{dt} + y \frac{dy}{dt} \right) = \frac{1}{r} [-xy + x^2(1 - r^2) + xy + y^2(1 - r^2)] = r(1 - r^2)$$

$$\frac{d\theta}{dt} = \frac{1}{r^2} \left(-y \frac{dx}{dt} + x \frac{dy}{dt} \right) = \frac{1}{r^2} [y^2 - xy(1 - r^2) + x^2 + xy(1 - r^2)] = 1.$$

Now $\frac{dr}{dt} = r - r^3$ or $\frac{dr}{dt} - r = -r^3$ is a Bernoulli differential equation. Following the procedure in Section 2.5 of the text, we let $w = r^{-2}$ so that $w' = -2r^{-3} \frac{dr}{dt}$. Therefore $w' + 2w = 2$, a linear first order differential equation. It follows that $w = 1 + c_1 e^{-2t}$ and so $r^2 = \frac{1}{1 + c_1 e^{-2t}}$. The general solution may be written as

$$r = \frac{1}{\sqrt{1 + c_1 e^{-2t}}}, \quad \theta = t + c_2.$$

If $\mathbf{X}(0) = (1, 0)$, $r = 1$ and $\theta = 0$ when $t = 0$. Therefore $c_1 = 0 = c_2$ and so $x = r \cos t = \cos t$ and $y = r \sin t = \sin t$. This solution generates the circle $r = 1$. If $\mathbf{X}(0) = (2, 0)$, $r = 2$ and $\theta = 0$ when $t = 0$.

Therefore $c_1 = -3/4$, $c_2 = 0$ and so

$$r = \frac{1}{\sqrt{1 - \frac{3}{4}e^{-2t}}}, \quad \theta = t.$$

This solution spirals toward the circle $r = 1$ as t increases.

26. Switching to polar coordinates,

$$\begin{aligned} \frac{dr}{dt} &= \frac{1}{r} \left(x \frac{dx}{dt} + y \frac{dy}{dt} \right) = \frac{1}{r} \left[xy - \frac{x^2}{r}(4 - r^2) - xy - \frac{y^2}{r}(4 - r^2) \right] = r^2 - 4 \\ \frac{d\theta}{dt} &= \frac{1}{r^2} \left(-y \frac{dx}{dt} + x \frac{dy}{dt} \right) = \frac{1}{r^2} \left[-y^2 + \frac{xy}{r}(4 - r^2) - x^2 - \frac{xy}{r}(4 - r^2) \right] = -1. \end{aligned}$$

From Example 3, Section 2.2,

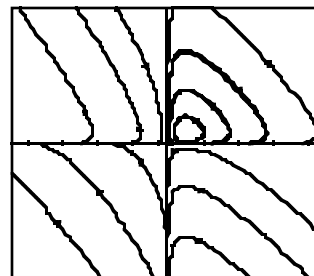
$$r = 2 \frac{1 + c_1 e^{4t}}{1 - c_1 e^{4t}} \quad \text{and} \quad \theta = -t + c_2.$$

If $\mathbf{X}(0) = (1, 0)$, $r = 1$ and $\theta = 0$ when $t = 0$. It follows that $c_2 = 0$ and $c_1 = -\frac{1}{3}$. Therefore

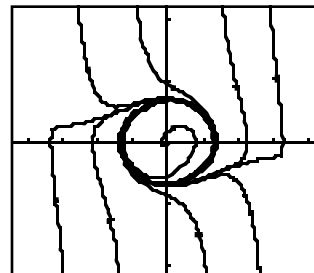
$$r = 2 \frac{1 - \frac{1}{3}e^{4t}}{1 + \frac{1}{3}e^{4t}} \quad \text{and} \quad \theta = -t.$$

Note that $r = 0$ when $e^{4t} = 3$ or $t = \frac{\ln 3}{4}$ and $r \rightarrow -2$ as $t \rightarrow \infty$. The solution therefore approaches the circle $r = 2$. If $\mathbf{X}(0) = (2, 0)$, it follows that $c_1 = c_2 = 0$. Therefore $r = 2$ and $\theta = -t$ so that the solution generates the circle $r = 2$ traversed in the clockwise direction. Note also that the original system is not defined at $(0, 0)$ but the corresponding polar system is defined for $r = 0$. If the Runge-Kutta method is applied to the original system, the solution corresponding to $\mathbf{X}(0) = (1, 0)$ will stall at the origin.

27. The system has no critical points, so there are no periodic solutions.
28. From $x(6y - 1) = 0$ and $y(2 - 8x) = 0$ we see that $(0, 0)$ and $(1/4, 1/6)$ are critical points. From the graph we see that there are periodic solutions around $(1/4, 1/6)$.



29. The only critical point is $(0, 0)$. There appears to be a single periodic solution around $(0, 0)$.



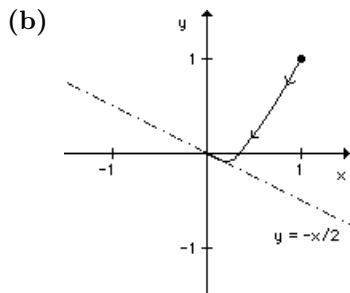
30. The system has no critical points, so there are no periodic solutions.
31. If $\mathbf{X}(t) = (x(t), y(t))$ is a solution,

$$\frac{d}{dt} f(x(t), y(t)) = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} = QP - PQ = 0,$$

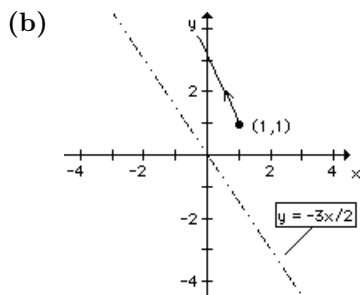
using the chain rule. Therefore $f(x(t), y(t)) = c$ for some constant c , and the solution lies on a level curve of f .

Exercises 11.2

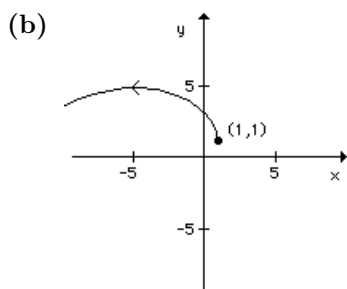
1. (a) If $\mathbf{X}(0) = \mathbf{X}_0$ lies on the line $y = 2x$, then $\mathbf{X}(t)$ approaches $(0, 0)$ along this line. For all other initial conditions, $\mathbf{X}(t)$ approaches $(0, 0)$ from the direction determined by the line $y = -x/2$.



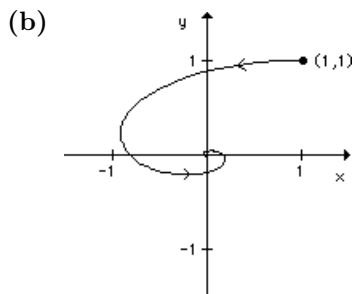
2. (a) If $\mathbf{X}(0) = \mathbf{X}_0$ lies on the line $y = -x$, then $\mathbf{X}(t)$ becomes unbounded along this line. For all other initial conditions, $\mathbf{X}(t)$ becomes unbounded and $y = -3x/2$ serves as an asymptote.



3. (a) All solutions are unstable spirals which become unbounded as t increases.

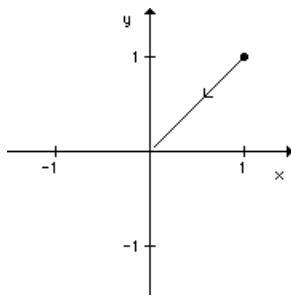


4. (a) All solutions are spirals which approach the origin.



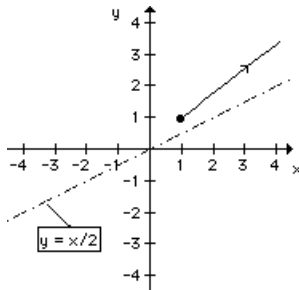
5. (a) All solutions approach $(0, 0)$ from the direction specified by the line $y = x$.

(b)



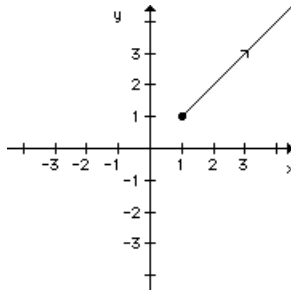
6. (a) All solutions become unbounded and $y = x/2$ serves as the asymptote.

(b)



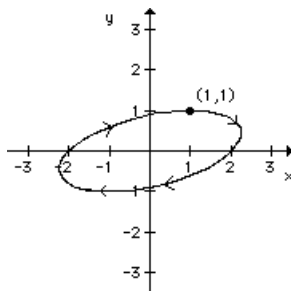
7. (a) If $\mathbf{X}(0) = \mathbf{X}_0$ lies on the line $y = 3x$, then $\mathbf{X}(t)$ approaches $(0,0)$ along this line. For all other initial conditions, $\mathbf{X}(t)$ becomes unbounded and $y = x$ serves as the asymptote.

(b)



8. (a) The solutions are ellipses which encircle the origin.

(b)

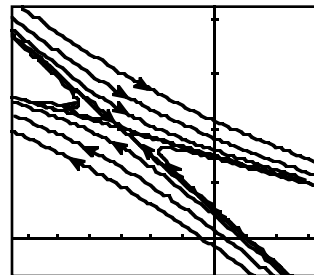


9. Since $\Delta = -41 < 0$, we may conclude from Figure 11.18 that $(0,0)$ is a saddle point.
10. Since $\Delta = 29$ and $\tau = -12$, $\tau^2 - 4\Delta > 0$ and so from Figure 11.18, $(0,0)$ is a stable node.
11. Since $\Delta = -19 < 0$, we may conclude from Figure 11.18 that $(0,0)$ is a saddle point.
12. Since $\Delta = 1$ and $\tau = -1$, $\tau^2 - 4\Delta = -3$ and so from Figure 11.18, $(0,0)$ is a stable spiral point.
13. Since $\Delta = 1$ and $\tau = -2$, $\tau^2 - 4\Delta = 0$ and so from Figure 11.18, $(0,0)$ is a degenerate stable node.
14. Since $\Delta = 1$ and $\tau = 2$, $\tau^2 - 4\Delta = 0$ and so from Figure 11.18, $(0,0)$ is a degenerate unstable node.

Exercises 11.2

15. Since $\Delta = 0.01$ and $\tau = -0.03$, $\tau^2 - 4\Delta < 0$ and so from Figure 11.18, $(0, 0)$ is a stable spiral point.
16. Since $\Delta = 0.0016$ and $\tau = 0.08$, $\tau^2 - 4\Delta = 0$ and so from Figure 11.18, $(0, 0)$ is a degenerate unstable node.
17. $\Delta = 1 - \mu^2$, $\tau = 0$, and so we need $\Delta = 1 - \mu^2 > 0$ for $(0, 0)$ to be a center. Therefore $|\mu| < 1$.
18. Note that $\Delta = 1$ and $\tau = \mu$. Therefore we need both $\tau = \mu < 0$ and $\tau^2 - 4\Delta = \mu^2 - 4 < 0$ for $(0, 0)$ to be a stable spiral point. These two conditions may be written as $-2 < \mu < 0$.
19. Note that $\Delta = \mu + 1$ and $\tau = \mu + 1$ and so $\tau^2 - 4\Delta = (\mu + 1)^2 - 4(\mu + 1) = (\mu + 1)(\mu - 3)$. It follows that $\tau^2 - 4\Delta < 0$ if and only if $-1 < \mu < 3$. We may conclude that $(0, 0)$ will be a saddle point when $\mu < -1$. Likewise $(0, 0)$ will be an unstable spiral point when $\tau = \mu + 1 > 0$ and $\tau^2 - 4\Delta < 0$. This condition reduces to $-1 < \mu < 3$.
20. $\tau = 2\alpha$, $\Delta = \alpha^2 + \beta^2 > 0$, and $\tau^2 - 4\Delta = -4\beta < 0$. If $\alpha < 0$, $(0, 0)$ is a stable spiral point. If $\alpha > 0$, $(0, 0)$ is an unstable spiral point. Therefore $(0, 0)$ cannot be a node or saddle point.
21. $\mathbf{A}\mathbf{X}_1 + \mathbf{F} = \mathbf{0}$ implies that $\mathbf{A}\mathbf{X}_1 = -\mathbf{F}$ or $\mathbf{X}_1 = -\mathbf{A}^{-1}\mathbf{F}$. Since $\mathbf{X}_p(t) = -\mathbf{A}^{-1}\mathbf{F}$ is a particular solution, it follows from Theorem 10.6 that $\mathbf{X}(t) = \mathbf{X}_c(t) + \mathbf{X}_1$ is the general solution to $\mathbf{X}' = \mathbf{A}\mathbf{X} + \mathbf{F}$. If $\tau < 0$ and $\Delta > 0$ then $\mathbf{X}_c(t)$ approaches $(0, 0)$ by Theorem 11.1(a). It follows that $\mathbf{X}(t)$ approaches $\mathbf{X}_1$ as $t \rightarrow \infty$.
22. If $bc < 1$, $\Delta = ad\hat{x}\hat{y}(1 - bc) > 0$ and $\tau^2 - 4\Delta = (a\hat{x} - d\hat{y})^2 + 4abcd\hat{x}\hat{y} > 0$. Therefore $(0, 0)$ is a stable node.
23. (a) The critical point is $\mathbf{X}_1 = (-3, 4)$.

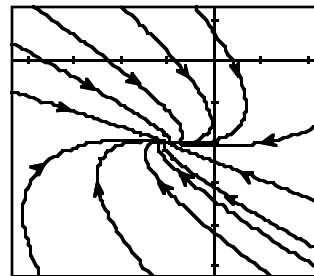
(b) From the graph, $\mathbf{X}_1$ appears to be an unstable node or a saddle point.



(c) Since $\Delta = -1$, $(0, 0)$ is a saddle point.

24. (a) The critical point is $\mathbf{X}_1 = (-1, -2)$.

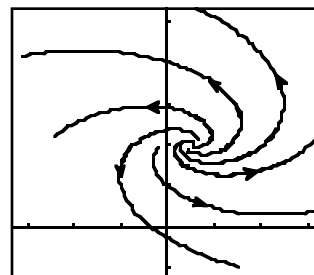
(b) From the graph, $\mathbf{X}_1$ appears to be a stable node or a degenerate stable node.



(c) Since $\tau = -16$, $\Delta = 64$, and $\tau^2 - 4\Delta = 0$, $(0, 0)$ is a degenerate stable node.

25. (a) The critical point is $\mathbf{X}_1 = (0.5, 2)$.

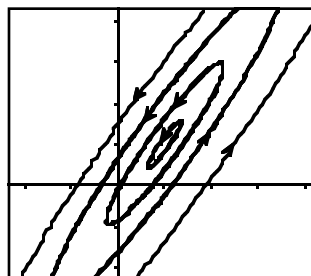
(b) From the graph, $\mathbf{X}_1$ appears to be an unstable spiral point.



(c) Since $\tau = 0.2$, $\Delta = 0.03$, and $\tau^2 - 4\Delta = -0.08$, $(0, 0)$ is an unstable spiral point.

26. (a) The critical point is $\mathbf{X}_1 = (1, 1)$.

(b) From the graph, $\mathbf{X}_1$ appears to be a center.



(c) Since $\tau = 0$ and $\Delta = 1$, $(0, 0)$ is a center.

Exercises 11.3

1. Switching to polar coordinates,

$$\frac{dr}{dt} = \frac{1}{r} \left(x \frac{dx}{dt} + y \frac{dy}{dt} \right) = \frac{1}{r} (\alpha x^2 - \beta xy + xy^2 + \beta xy + \alpha y^2 - xy^2) = \frac{1}{r} \alpha r^2 = \alpha r.$$

Therefore $r = ce^{\alpha t}$ and so $r \rightarrow 0$ if and only if $\alpha < 0$.

2. The differential equation $\frac{dr}{dt} = \alpha r(5 - r)$ is a logistic differential equation. [See Section 2.8, (4) and (5).] It follows that

$$r = \frac{5}{1 + c_1 e^{-5\alpha t}} \quad \text{and} \quad \theta = -t + c_2.$$

If $\alpha > 0$, $r \rightarrow 5$ as $t \rightarrow +\infty$ and so the critical point $(0, 0)$ is unstable. If $\alpha < 0$, $r \rightarrow 0$ as $t \rightarrow +\infty$ and so $(0, 0)$ is asymptotically stable.

3. The critical points are $x = 0$ and $x = n + 1$. Since $g'(x) = k(n + 1) - 2kx$, $g'(0) = k(n + 1) > 0$ and $g'(n + 1) = -k(n + 1) < 0$. Therefore $x = 0$ is unstable while $x = n + 1$ is asymptotically stable. See Theorem 11.2.

4. Note that $x = k$ is the only critical point since $\ln(x/k)$ is not defined at $x = 0$. Since $g'(x) = -k - k \ln(x/k)$, $g'(k) = -k < 0$. Therefore $x = k$ is an asymptotically stable critical point by Theorem 11.2.

5. The only critical point is $T = T_0$. Since $g'(T) = k$, $g'(T_0) = k > 0$. Therefore $T = T_0$ is unstable by Theorem 11.2.

6. The only critical point is $v = mg/k$. Now $g(v) = g - (k/m)v$ and so $g'(v) = -k/m < 0$. Therefore $v = mg/k$ is an asymptotically stable critical point by Theorem 11.2.

Exercises 11.3

7. Critical points occur at $x = \alpha, \beta$. Since $g'(x) = k(-\alpha - \beta + 2x)$, $g'(\alpha) = k(\alpha - \beta)$ and $g'(\beta) = k(\beta - \alpha)$. Since $\alpha > \beta$, $g'(\alpha) > 0$ and so $x = \alpha$ is unstable. Likewise $x = \beta$ is asymptotically stable.
8. Critical points occur at $x = \alpha, \beta, \gamma$. Since

$$g'(x) = k(\alpha - x)(-\beta - \gamma - 2x) + k(\beta - x)(\gamma - x)(-1),$$

$g'(\alpha) = -k(\beta - \alpha)(\gamma - \alpha) < 0$ since $\alpha > \beta > \gamma$. Therefore $x = \alpha$ is asymptotically stable. Similarly $g'(\beta) > 0$ and $g'(\gamma) < 0$. Therefore $x = \beta$ is unstable while $x = \gamma$ is asymptotically stable.

9. Critical points occur at $P = a/b, c$ but not at $P = 0$. Since $g'(P) = (a - bP) + (P - c)(-b)$,

$$g'(a/b) = (a/b - c)(-b) = -a + bc \quad \text{and} \quad g'(c) = a - bc.$$

Since $a < bc$, $-a + bc > 0$ and $a - bc < 0$. Therefore $P = a/b$ is unstable while $P = c$ is asymptotically stable.

10. Since $A > 0$, the only critical point is $A = K^2$. Since $g'(A) = \frac{1}{2}kKA^{-1/2} - k$, $g'(K^2) = -k/2 < 0$. Therefore $A = K^2$ is asymptotically stable.
11. The sole critical point is $(1/2, 1)$ and

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} -2y & -2x \\ 2y & 2x - 1 \end{pmatrix}.$$

Computing $\mathbf{g}'((1/2, 1))$ we find that $\tau = -2$ and $\Delta = 2$ so that $\tau^2 - 4\Delta = -4 < 0$. Therefore $(1/2, 1)$ is a stable spiral point.

12. Critical points are $(1, 0)$ and $(-1, 0)$, and

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 2x & -2y \\ 0 & 2 \end{pmatrix}.$$

At $\mathbf{X} = (1, 0)$, $\tau = 4$, $\Delta = 4$, and so $\tau^2 - 4\Delta = 0$. We may conclude that $(1, 0)$ is unstable but we are unable to classify this critical point any further. At $\mathbf{X} = (-1, 0)$, $\Delta = -4 < 0$ and so $(-1, 0)$ is a saddle point.

13. $y' = 2xy - y = y(2x - 1)$. Therefore if (x, y) is a critical point, either $x = 1/2$ or $y = 0$. The case $x = 1/2$ and $y - x^2 + 2 = 0$ implies that $(x, y) = (1/2, -7/4)$. The case $y = 0$ leads to the critical points $(\sqrt{2}, 0)$ and $(-\sqrt{2}, 0)$. We next use the Jacobian matrix

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} -2x & 1 \\ 2y & 2x - 1 \end{pmatrix}$$

to classify these three critical points. For $\mathbf{X} = (\sqrt{2}, 0)$ or $(-\sqrt{2}, 0)$, $\tau = -1$ and $\Delta < 0$. Therefore both critical points are saddle points. For $\mathbf{X} = (1/2, -7/4)$, $\tau = -1$, $\Delta = 7/2$ and so $\tau^2 - 4\Delta = -13 < 0$. Therefore $(1/2, -7/4)$ is a stable spiral point.

14. $y' = -y + xy = y(-1 + x)$. Therefore if (x, y) is a critical point, either $y = 0$ or $x = 1$. The case $y = 0$ and $2x - y^2 = 0$ implies that $(x, y) = (0, 0)$. The case $x = 1$ leads to the critical points $(1, \sqrt{2})$ and $(1, -\sqrt{2})$. We next use the Jacobian matrix

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 2 & -2y \\ y & x - 1 \end{pmatrix}$$

to classify these critical points. For $\mathbf{X} = (0, 0)$, $\Delta = -2 < 0$ and so $(0, 0)$ is a saddle point. For either $(1, \sqrt{2})$ or $(1, -\sqrt{2})$, $\tau = 2$, $\Delta = 4$, and so $\tau^2 - 4\Delta = -12$. Therefore $(1, \sqrt{2})$ and $(1, -\sqrt{2})$ are unstable spiral points.

15. Since $x^2 - y^2 = 0$, $y^2 = x^2$ and so $x^2 - 3x + 2 = (x - 1)(x - 2) = 0$. It follows that the critical points are $(1, 1)$, $(1, -1)$, $(2, 2)$, and $(2, -2)$. We next use the Jacobian

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} -3 & 2y \\ 2x & -2y \end{pmatrix}$$

to classify these four critical points. For $\mathbf{X} = (1, 1)$, $\tau = -5$, $\Delta = 2$, and so $\tau^2 - 4\Delta = 17 > 0$. Therefore $(1, 1)$ is a stable node. For $\mathbf{X} = (1, -1)$, $\Delta = -2 < 0$ and so $(1, -1)$ is a saddle point. For $\mathbf{X} = (2, 2)$, $\Delta = -4 < 0$ and so we have another saddle point. Finally, if $\mathbf{X} = (2, -2)$, $\tau = 1$, $\Delta = 4$, and so $\tau^2 - 4\Delta = -15 < 0$. Therefore $(2, -2)$ is an unstable spiral point.

16. From $y^2 - x^2 = 0$, $y = x$ or $y = -x$. The case $y = x$ leads to $(4, 4)$ and $(-1, 1)$ but the case $y = -x$ leads to $x^2 - 3x + 4 = 0$ which has no real solutions. Therefore $(4, 4)$ and $(-1, 1)$ are the only critical points. We next use the Jacobian matrix

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} y & x-3 \\ -2x & 2y \end{pmatrix}$$

to classify these two critical points. For $\mathbf{X} = (4, 4)$, $\tau = 12$, $\Delta = 40$, and so $\tau^2 - 4\Delta < 0$. Therefore $(4, 4)$ is an unstable spiral point. For $\mathbf{X} = (-1, 1)$, $\tau = -3$, $\Delta = 10$, and so $\tau^2 - 4\Delta < 0$. It follows that $(-1, 1)$ is a stable spiral point.

17. Since $x' = -2xy = 0$, either $x = 0$ or $y = 0$. If $x = 0$, $y(1 - y^2) = 0$ and so $(0, 0)$, $(0, 1)$, and $(0, -1)$ are critical points. The case $y = 0$ leads to $x = 0$. We next use the Jacobian matrix

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} -2y & -2x \\ -1+y & 1+x-3y^2 \end{pmatrix}$$

to classify these three critical points. For $\mathbf{X} = (0, 0)$, $\tau = 1$ and $\Delta = 0$ and so the test is inconclusive. For $\mathbf{X} = (0, 1)$, $\tau = -4$, $\Delta = 4$ and so $\tau^2 - 4\Delta = 0$. We can conclude that $(0, 1)$ is a stable critical point but we are unable to classify this critical point further in this borderline case. For $\mathbf{X} = (0, -1)$, $\Delta = -4 < 0$ and so $(0, -1)$ is a saddle point.

18. We found that $(0, 0)$, $(0, 1)$, $(0, -1)$, $(1, 0)$ and $(-1, 0)$ were the critical points in Exercise 15, Section 11.1. The Jacobian is

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 1 - 3x^2 - 3y^2 & -6xy \\ -2xy & 3 - x^2 - 9y^2 \end{pmatrix}.$$

For $\mathbf{X} = (0, 0)$, $\tau = 4$, $\Delta = 3$ and so $\tau^2 - 4\Delta = 4 > 0$. Therefore $(0, 0)$ is an unstable node. Both $(0, 1)$ and $(0, -1)$ give $\tau = -8$, $\Delta = 12$, and $\tau^2 - 4\Delta = 16 > 0$. These two critical points are therefore stable nodes. For $\mathbf{X} = (1, 0)$ or $(-1, 0)$, $\Delta = -4 < 0$ and so saddle points occur.

19. We found the critical points $(0, 0)$, $(10, 0)$, $(0, 16)$ and $(4, 12)$ in Exercise 11, Section 11.1. Since the Jacobian is

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 10 - 2x - \frac{1}{2}y & -\frac{1}{2}x \\ -y & 16 - 2y - x \end{pmatrix}$$

we may classify the critical points as follows:

$\mathbf{X}$	τ	Δ	$\tau^2 - 4\Delta$	Conclusion
$(0, 0)$	26	160	36	unstable node
$(10, 0)$	-4	-60	-	saddle point
$(0, 16)$	-14	-32	-	saddle point
$(4, 12)$	-16	24	160	stable node

20. We found the sole critical point $(10, 10)$ in Exercise 12, Section 11.1. The Jacobian is

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} -2 & 1 \\ 2 & -1 - \frac{15}{(y+5)^2} \end{pmatrix},$$

$\mathbf{g}'((10, 10))$ has trace $\tau = -46/15$, $\Delta = 2/15$, and $\tau^2 - 4\Delta > 0$. Therefore $(0, 0)$ is a stable node.

Exercises 11.3

21. The corresponding plane autonomous system is

$$\theta' = y, \quad y' = \left(\cos \theta - \frac{1}{2}\right) \sin \theta.$$

Since $|\theta| < \pi$, it follows that critical points are $(0, 0)$, $(\pi/3, 0)$ and $(-\pi/3, 0)$. The Jacobian matrix is

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 0 & 1 \\ \cos 2\theta - \frac{1}{2} \cos \theta & 0 \end{pmatrix}$$

and so at $(0, 0)$, $\tau = 0$ and $\Delta = -1/2$. Therefore $(0, 0)$ is a saddle point. For $\mathbf{X} = (\pm\pi/3, 0)$, $\tau = 0$ and $\Delta = 3/4$. It is not possible to classify either critical point in this borderline case.

22. The corresponding plane autonomous system is

$$x' = y, \quad y' = -x + \left(\frac{1}{2} - 3y^2\right)y - x^2.$$

If (x, y) is a critical point, $y = 0$ and so $-x - x^2 = -x(1 + x) = 0$. Therefore $(0, 0)$ and $(-1, 0)$ are the only two critical points. We next use the Jacobian matrix

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 0 & 1 \\ -1 - 2x & \frac{1}{2} - 9y^2 \end{pmatrix}$$

to classify these critical points. For $\mathbf{X} = (0, 0)$, $\tau = 1/2$, $\Delta = 1$, and $\tau^2 - 4\Delta < 0$. Therefore $(0, 0)$ is an unstable spiral point. For $\mathbf{X} = (-1, 0)$, $\tau = 1/2$, $\Delta = -1$ and so $(-1, 0)$ is a saddle point.

23. The corresponding plane autonomous system is

$$x' = y, \quad y' = x^2 - y(1 - x^3)$$

and the only critical point is $(0, 0)$. Since the Jacobian matrix is

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 0 & 1 \\ 2x + 3x^2y & x^3 - 1 \end{pmatrix},$$

$\tau = -1$ and $\Delta = 0$, and we are unable to classify the critical point in this borderline case.

24. The corresponding plane autonomous system is

$$x' = y, \quad y' = -\frac{4x}{1+x^2} - 2y$$

and the only critical point is $(0, 0)$. Since the Jacobian matrix is

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 0 & 1 \\ -4\frac{1-x^2}{(1+x^2)^2} & -2 \end{pmatrix},$$

$\tau = -2$, $\Delta = 4$, $\tau^2 - 4\Delta = -12$, and so $(0, 0)$ is a stable spiral point.

25. In Exercise 5, Section 11.1, we showed that $(0, 0)$, $(\sqrt{1/\epsilon}, 0)$ and $(-\sqrt{1/\epsilon}, 0)$ are the critical points. We will use the Jacobian matrix

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 0 & 1 \\ -1 + 3\epsilon x^2 & 0 \end{pmatrix}$$

to classify these three critical points. For $\mathbf{X} = (0, 0)$, $\tau = 0$ and $\Delta = 1$ and we are unable to classify this critical point. For $(\pm\sqrt{1/\epsilon}, 0)$, $\tau = 0$ and $\Delta = -2$ and so both of these critical points are saddle points.

26. In Exercise 6, Section 11.1, we showed that $(0, 0)$, $(1/\epsilon, 0)$, and $(-1/\epsilon, 0)$ are the critical points. Since $D_x x|x| = 2|x|$, the Jacobian matrix is

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 0 & 1 \\ 2\epsilon|x| - 1 & 0 \end{pmatrix}.$$

For $\mathbf{X} = (0, 0)$, $\tau = 0$, $\Delta = 1$ and we are unable to classify this critical point. For $(\pm 1/\epsilon, 0)$, $\tau = 0$, $\Delta = -1$, and so both of these critical points are saddle points.

27. The corresponding plane autonomous system is

$$x' = y, \quad y' = -\frac{(\beta + \alpha^2 y^2)x}{1 + \alpha^2 x^2}$$

and the Jacobian matrix is

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 0 & 1 \\ \frac{(\beta + \alpha^2 y^2)(\alpha^2 x^2 - 1)}{(1 + \alpha^2 x^2)^2} & \frac{-2\alpha^2 yx}{1 + \alpha^2 x^2} \end{pmatrix}.$$

For $\mathbf{X} = (0, 0)$, $\tau = 0$ and $\Delta = \beta$. Since $\beta < 0$, we may conclude that $(0, 0)$ is a saddle point.

28. From $x' = -\alpha x + xy = x(-\alpha + y) = 0$, either $x = 0$ or $y = \alpha$. If $x = 0$, then $1 - \beta y = 0$ and so $y = 1/\beta$. The case $y = \alpha$ implies that $1 - \beta\alpha - x^2 = 0$ or $x^2 = 1 - \alpha\beta$. Since $\alpha\beta > 1$, this equation has no real solutions. It follows that $(0, 1/\beta)$ is the unique critical point. Since the Jacobian matrix is

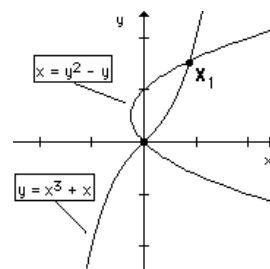
$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} -\alpha + y & x \\ -2x & -\beta \end{pmatrix},$$

$\tau = -\alpha - \beta + \frac{1}{\beta} = -\beta + \frac{1 - \alpha\beta}{\beta} < 0$ and $\Delta = \alpha\beta - 1 > 0$. Therefore $(0, 1/\beta)$ is a stable critical point.

29. (a) The graphs of $-x + y - x^3 = 0$ and $-x - y + y^2 = 0$ are shown in the figure. The Jacobian matrix is

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} -1 - 3x^2 & 1 \\ -1 & -1 + 2y \end{pmatrix}.$$

For $\mathbf{X} = (0, 0)$, $\tau = -2$, $\Delta = 2$, $\tau^2 - 4\Delta = -4$, and so $(0, 0)$ is a stable spiral point.



- (b) For $\mathbf{X}_1$, $\Delta = -6.07 < 0$ and so a saddle point occurs at $\mathbf{X}_1$.

30. (a) The corresponding plane autonomous system is

$$x' = y, \quad y' = \epsilon(y - \frac{1}{3}y^3) - x$$

and so the only critical point is $(0, 0)$. Since the Jacobian matrix is

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 0 & 1 \\ -1 & \epsilon(1 - y^2) \end{pmatrix},$$

$\tau = \epsilon$, $\Delta = 1$, and so $\tau^2 - 4\Delta = \epsilon^2 - 4$ at the critical point $(0, 0)$.

- (b) When $\tau = \epsilon > 0$, $(0, 0)$ is an unstable critical point.
(c) When $\epsilon < 0$ and $\tau^2 - 4\Delta = \epsilon^2 - 4 < 0$, $(0, 0)$ is a stable spiral point. These two requirements can be written as $-2 < \epsilon < 0$.
(d) When $\epsilon = 0$, $x'' + x = 0$ and so $x = c_1 \cos t + c_2 \sin t$. Therefore all solutions are periodic (with period 2π) and so $(0, 0)$ is a center.
31. $\frac{dy}{dx} = \frac{y'}{x'} = \frac{-2x^3}{y}$ may be solved by separating variables. It follows that $y^2 + x^4 = c$. If $\mathbf{X}(0) = (x_0, 0)$ where $x_0 > 0$, then $c = x_0^4$ so that $y^2 = x_0^4 - x^4$. Therefore if $-x_0 < x < x_0$, $y^2 > 0$ and so there are two values of y corresponding to each value of x . Therefore the solution $\mathbf{X}(t)$ with $\mathbf{X}(0) = (x_0, 0)$ is periodic and so $(0, 0)$ is a center.

Exercises 11.3

32. $\frac{dy}{dx} = \frac{y'}{x'} = \frac{x^2 - 2x}{y}$ may be solved by separating variables. It follows that $\frac{y^2}{2} = \frac{x^3}{3} - x^2 + c$ and since $\mathbf{X}(0) = (x(0), x'(0)) = (1, 0)$, $c = \frac{2}{3}$. Therefore

$$\frac{y^2}{2} = \frac{x^3 - 3x^2 + 2}{3} = \frac{(x-1)(x^2 - 2x - 2)}{3}.$$

But $(x-1)(x^2 - 2x - 2) > 0$ for $1 - \sqrt{3} < x < 1$ and so each x in this interval has 2 corresponding values of y . therefore $\mathbf{X}(t)$ is a periodic solution.

33. (a) $x' = 2xy = 0$ implies that either $x = 0$ or $y = 0$. If $x = 0$, then from $1 - x^2 + y^2 = 0$, $y^2 = -1$ and there are no real solutions. If $y = 0$, $1 - x^2 = 0$ and so $(1, 0)$ and $(-1, 0)$ are critical points. The Jacobian matrix is

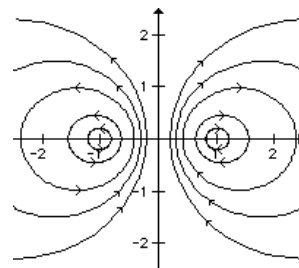
$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 2y & 2x \\ -2x & 2y \end{pmatrix}$$

and so $\tau = 0$ and $\Delta = 4$ at either $\mathbf{X} = (1, 0)$ or $(-1, 0)$. We obtain no information about these critical points in this borderline case.

- (b) $\frac{dy}{dx} = \frac{y'}{x'} = \frac{1 - x^2 + y^2}{2xy}$ or $2xy \frac{dy}{dx} = 1 - x^2 + y^2$. Letting $\mu = \frac{y^2}{x}$, it follows that $\frac{d\mu}{dx} = \frac{1}{x^2} - 1$ and so $\mu = -\frac{1}{x} - x + 2c$. Therefore $\frac{y^2}{x} = -\frac{1}{x} - x + 2c$ which can be put in the form

$$(x - c)^2 + y^2 = c^2 - 1.$$

The solution curves are shown and so both $(1, 0)$ and $(-1, 0)$ are centers.



34. (a) $\frac{dy}{dx} = \frac{y'}{x'} = \frac{-x - y^2}{y} = -\frac{x}{y} - y$ and so $\frac{dy}{dx} + y = -xy^{-1}$.
 (b) Let $w = y^{1-n} = y^2$. It follows that $\frac{dw}{dx} + 2w = -2x$, a linear first order differential equation whose solution is

$$y^2 = w = ce^{-2x} + \left(\frac{1}{2} - x\right).$$

Since $x(0) = \frac{1}{2}$ and $y(0) = x'(0) = 0$, $0 = c$ and so

$$y^2 = \frac{1}{2} - x,$$

a parabola with vertex at $(1/2, 0)$. Therefore the solution $\mathbf{X}(t)$ with $\mathbf{X}(0) = (1/2, 0)$ is not periodic.

35. $\frac{dy}{dx} = \frac{y'}{x'} = \frac{x^3 - x}{y}$ and so $\frac{y^2}{2} = \frac{x^4}{4} - \frac{x^2}{2} + c$ or $y^2 = \frac{x^4}{2} - x^2 + c_1$. Since $x(0) = 0$ and $y(0) = x'(0) = v_0$, it follows that $c_1 = v_0^2$ and so

$$y^2 = \frac{1}{2}x^4 - x^2 + v_0^2 = \frac{(x^2 - 1)^2 + 2v_0^2 - 1}{2}.$$

The x -intercepts on this graph satisfy

$$x^2 = 1 \pm \sqrt{1 - 2v_0^2}$$

and so we must require that $1 - 2v_0^2 \geq 0$ (or $|v_0| \leq \frac{1}{\sqrt{2}}$) for real solutions to exist. If $x_0^2 = 1 - \sqrt{1 - 2v_0^2}$ and $-x_0 < x < x_0$, then $(x^2 - 1)^2 + 2v_0^2 - 1 > 0$ and so there are two corresponding values of y . Therefore $\mathbf{X}(t)$ with $\mathbf{X}(0) = (0, v_0)$ is periodic provided that $|v_0| \leq \frac{1}{\sqrt{2}}$.

36. The corresponding plane autonomous system is

$$x' = y, \quad y' = \epsilon x^2 - x + 1$$

and so the critical points must satisfy $y = 0$ and

$$x = \frac{1 \pm \sqrt{1-4\epsilon}}{2\epsilon}.$$

Therefore we must require that $\epsilon \leq \frac{1}{4}$ for real solutions to exist. We will use the Jacobian matrix

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 0 & 1 \\ 2\epsilon x - 1 & 0 \end{pmatrix}$$

to attempt to classify $((1 \pm \sqrt{1-4\epsilon})/2\epsilon, 0)$ when $\epsilon \leq 1/4$. Note that $\tau = 0$ and $\Delta = \mp\sqrt{1-4\epsilon}$. For $\mathbf{X} = ((1 + \sqrt{1-4\epsilon})/2\epsilon, 0)$ and $\epsilon < 1/4$, $\Delta < 0$ and so a saddle point occurs. For $\mathbf{X} = ((1 - \sqrt{1-4\epsilon})/2\epsilon, 0)$ $\Delta \geq 0$ and we are not able to classify this critical point using linearization.

37. The corresponding plane autonomous system is

$$x' = y, \quad y' = -\frac{\alpha}{L}x - \frac{\beta}{L}x^3 - \frac{R}{L}y$$

where $x = q$ and $y = q'$. If $\mathbf{X} = (x, y)$ is a critical point, $y = 0$ and $-\alpha x - \beta x^3 = -x(\alpha + \beta x^2) = 0$. If $\beta > 0$, $\alpha + \beta x^2 = 0$ has no real solutions and so $(0, 0)$ is the only critical point. Since

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 0 & 1 \\ \frac{-\alpha - 3\beta x^2}{L} & -\frac{R}{L} \end{pmatrix},$$

$\tau = -R/L < 0$ and $\Delta = \alpha/L > 0$. Therefore $(0, 0)$ is a stable critical point. If $\beta < 0$, $(0, 0)$ and $(\pm\hat{x}, 0)$, where $\hat{x}^2 = -\alpha/\beta$ are critical points. At $\mathbf{X}(\pm\hat{x}, 0)$, $\tau = -R/L < 0$ and $\Delta = -2\alpha/L < 0$. Therefore both critical points are saddles.

38. If we let $dx/dt = y$, then $dy/dt = -x^3 - x$. From this we obtain the first-order differential equation

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = -\frac{x^3 + x}{y}.$$

Separating variables and integrating we obtain

$$\int y \, dy = - \int (x^3 + x) \, dx$$

and

$$\frac{1}{2}y^2 = -\frac{1}{4}x^4 - \frac{1}{2}x^2 + c_1.$$

Completing the square we can write the solution as $y^2 = -\frac{1}{2}(x^2+1)^2 + c_2$. If $\mathbf{X}(0) = (x_0, 0)$, then $c_2 = \frac{1}{2}(x_0^2+1)^2$ and so

$$\begin{aligned} y^2 &= -\frac{1}{2}(x^2+1)^2 + \frac{1}{2}(x_0^2+1)^2 = \frac{x_0^4 + 2x_0^2 + 1 - x^4 - 2x^2 - 1}{2} \\ &= \frac{(x_0^2 + x^2)(x_0^2 - x^2) + 2(x_0^2 - x^2)}{2} = \frac{(x_0^2 + x^2 + 2)(x_0^2 - x^2)}{2}. \end{aligned}$$

Note that $y = 0$ when $x = -x_0$. In addition, the right-hand side is positive for $-x_0 < x < x_0$, and so there are two corresponding values of y for each x between $-x_0$ and x_0 . The solution $\mathbf{X} = \mathbf{X}(t)$ that satisfies $\mathbf{X}(0) = (x_0, 0)$ is therefore periodic, and so $(0, 0)$ is a center.

39. (a) Letting $x = \theta$ and $y = x'$ we obtain the system $x' = y$ and $y' = 1/2 - \sin x$. Since $\sin \pi/6 = \sin 5\pi/6 = 1/2$ we see that $(\pi/6, 0)$ and $(5\pi/6, 0)$ are critical points of the system.

Exercises 11.3

(b) The Jacobian matrix is

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 0 & 1 \\ -\cos x & 0 \end{pmatrix}$$

and so

$$\mathbf{A}_1 = \mathbf{g}'((\pi/6, 0)) = \begin{pmatrix} 0 & 1 \\ -\sqrt{3}/2 & 0 \end{pmatrix} \quad \text{and} \quad \mathbf{A}_2 = \mathbf{g}'((5\pi/6, 0)) = \begin{pmatrix} 0 & 1 \\ \sqrt{3}/2 & 0 \end{pmatrix}.$$

Since $\det \mathbf{A}_1 > 0$ and the trace of $\mathbf{A}_1$ is 0, no conclusion can be drawn regarding the critical point $(\pi/6, 0)$. Since $\det \mathbf{A}_2 < 0$, we see that $(5\pi/6, 0)$ is a saddle point.

(c) From the system in part (a) we obtain the first-order differential equation

$$\frac{dy}{dx} = \frac{1/2 - \sin x}{y}.$$

Separating variables and integrating we obtain

$$\int y \, dy = \int \left(\frac{1}{2} - \sin x \right) dx$$

and

$$\frac{1}{2}y^2 = \frac{1}{2}x + \cos x + c_1$$

or

$$y^2 = x + 2 \cos x + c_2.$$

For x_0 near $\pi/6$, if $\mathbf{X}(0) = (x_0, 0)$ then $c_2 = -x_0 - 2 \cos x_0$ and $y^2 = x + 2 \cos x - x_0 - 2 \cos x_0$. Thus, there are two values of y for each x in a sufficiently small interval around $\pi/6$. Therefore $(\pi/6, 0)$ is a center.

40. (a) Writing the system as $x' = x(x^3 - 2y^3)$ and $y' = y(2x^3 - y^3)$ we see that $(0, 0)$ is a critical point. Setting $x^3 - 2y^3 = 0$ we have $x^3 = 2y^3$ and $2x^3 - y^3 = 4y^3 - y^3 = 3y^3$. Thus, $(0, 0)$ is the only critical point of the system.

(b) From the system we obtain the first-order differential equation

$$\frac{dy}{dx} = \frac{2x^3y - y^4}{x^4 - 2xy^3}$$

or

$$(2x^3y - y^4) \, dx + (2xy^3 - x^4) \, dy = 0$$

which is homogeneous. If we let $y = ux$ it follows that

$$(2x^4u - x^4u^4) \, dx + (2x^4u^3 - x^4)(u \, dx + x \, du) = 0$$

$$x^4u(1 + u^3) \, dx + x^5(2u^3 - 1) \, du = 0$$

$$\frac{1}{x} \, dx + \frac{2u^3 - 1}{u(u^3 + 1)} \, du = 0$$

$$\frac{1}{x} \, dx + \left(\frac{1}{u+1} - \frac{1}{u} + \frac{2u-1}{u^2-u+1} \right) du = 0.$$

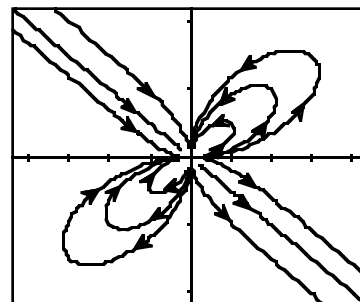
Integrating gives

$$\ln |x| + \ln |u+1| - \ln |u| + \ln |u^2 - u + 1| = c_1$$

or

$$\begin{aligned}
 x \left(\frac{u+1}{u} \right) (u^2 - u + 1) &= c_2 \\
 x \left(\frac{y+x}{y} \right) \left(\frac{y^2}{x^2} - \frac{y}{x} + 1 \right) &= c_2 \\
 (xy + x^2)(y^2 - xy + x^2) &= c_2 x^2 y \\
 xy^3 + x^4 &= c_2 x^2 y \\
 x^3 + y^2 &= 3c_3 xy.
 \end{aligned}$$

- (c) We see from the graph that $(0,0)$ is unstable. It is not possible to classify the critical point as a node, saddle, center, or spiral point.



Exercises 11.4

1. We are given that $x(0) = \theta(0) = \frac{\pi}{3}$ and $y(0) = \theta'(0) = w_0$. Since $y^2 = \frac{2g}{l} \cos x + c$, $w_0^2 = \frac{2g}{l} \cos \frac{\pi}{3} + c = \frac{g}{l} + c$ and so $c = w_0^2 - \frac{g}{l}$. Therefore

$$y^2 = \frac{2g}{l} \left(\cos x - \frac{1}{2} + \frac{l}{2g} w_0^2 \right)$$

and the x -intercepts occur where $\cos x = \frac{1}{2} - \frac{l}{2g} w_0^2$ and so $\frac{1}{2} - \frac{l}{2g} w_0^2$ must be greater than -1 for solutions to exist. This condition is equivalent to $|w_0| < \sqrt{\frac{3g}{l}}$.

2. (a) Since $y^2 = \frac{2g}{l} \cos x + c$, $x(0) = \theta(0) = \theta_0$ and $y(0) = \theta'(0) = 0$, $c = -\frac{2g}{l} \cos \theta_0$ and so $y^2 = \frac{2g}{l} (\cos \theta - \cos \theta_0)$.
When $\theta = -\theta_0$, $y^2 = \frac{2g}{l} (\cos(-\theta_0) - \cos(\theta_0)) = 0$. Therefore $y = \frac{d\theta}{dt} = 0$ when $\theta = \theta_0$.

- (b) Since $y = \frac{d\theta}{dt}$ and θ is decreasing between the time when $\theta = \theta_0$, $t = 0$, and $\theta = -\theta_0$, that is, $t = T$,

$$\frac{d\theta}{dt} = -\sqrt{\frac{2g}{l}} \sqrt{\cos \theta - \cos \theta_0}.$$

Therefore $\frac{dt}{d\theta} = -\sqrt{\frac{l}{2g}} \frac{1}{\sqrt{\cos \theta - \cos \theta_0}}$ and so

$$T = -\sqrt{\frac{l}{2g}} \int_{\theta=\theta_0}^{\theta=-\theta_0} \frac{1}{\sqrt{\cos \theta - \cos \theta_0}} d\theta = \sqrt{\frac{l}{2g}} \int_{-\theta_0}^{\theta_0} \frac{1}{\sqrt{\cos \theta - \cos \theta_0}} d\theta.$$

3. The corresponding plane autonomous system is

$$x' = y, \quad y' = -g \frac{f'(x)}{1 + [f'(x)]^2} - \frac{\beta}{m} y$$

Exercises 11.4

and

$$\frac{\partial}{\partial x} \left(-g \frac{f'(x)}{1 + [f'(x)]^2} - \frac{\beta}{m} y \right) = -g \frac{(1 + [f'(x)]^2)f''(x) - f'(x)2f'(x)f''(x)}{(1 + [f'(x)]^2)^2}.$$

If $\mathbf{X}_1 = (x_1, y_1)$ is a critical point, $y_1 = 0$ and $f'(x_1) = 0$. The Jacobian at this critical point is therefore

$$\mathbf{g}'(\mathbf{X}_1) = \begin{pmatrix} 0 & 1 \\ -gf''(x_1) & -\frac{\beta}{m} \end{pmatrix}.$$

4. When $\beta = 0$ the Jacobian matrix is

$$\begin{pmatrix} 0 & 1 \\ -gf''(x_1) & 0 \end{pmatrix}$$

which has complex eigenvalues $\lambda = \pm \sqrt{gf''(x_1)} i$. The approximating linear system with $x'(0) = 0$ has solution

$$x(t) = x(0) \cos \sqrt{gf''(x_1)} t$$

and period $2\pi/\sqrt{gf''(x_1)}$. Therefore $p \approx 2\pi/\sqrt{gf''(x_1)}$ for the actual solution.

5. (a) If $f(x) = \frac{x^2}{2}$, $f'(x) = x$ and so

$$\frac{dy}{dx} = \frac{y'}{x'} = -g \frac{x}{1+x^2} \frac{1}{y}.$$

We may separate variables to show that $y^2 = -g \ln(1+x^2) + c$. But $x(0) = x_0$ and $y(0) = x'(0) = v_0$. Therefore $c = v_0^2 + g \ln(1+x_0^2)$ and so

$$y^2 = v_0^2 - g \ln \left(\frac{1+x^2}{1+x_0^2} \right).$$

Now

$$v_0^2 - g \ln \left(\frac{1+x^2}{1+x_0^2} \right) \geq 0 \quad \text{if and only if} \quad x^2 \leq e^{v_0^2/g}(1+x_0^2) - 1.$$

Therefore, if $|x| \leq [e^{v_0^2/g}(1+x_0^2) - 1]^{1/2}$, there are two values of y for a given value of x and so the solution is periodic.

- (b) Since $z = \frac{x^2}{2}$, the maximum height occurs at the largest value of x on the cycle. From (a), $x_{\max} = [e^{v_0^2/g}(1+x_0^2) - 1]^{1/2}$ and so

$$z_{\max} = \frac{x_{\max}^2}{2} = \frac{1}{2}[e^{v_0^2/g}(1+x_0^2) - 1].$$

6. (a) If $f(x) = \cosh x$, $f'(x) = \sinh x$ and $[f'(x)]^2 + 1 = \sinh^2 x + 1 = \cosh^2 x$. Therefore

$$\frac{dy}{dx} = \frac{y'}{x'} = -g \frac{\sinh x}{\cosh^2 x} \frac{1}{y}.$$

We may separate variables to show that $y^2 = \frac{2g}{\cosh x} + c$. But $x(0) = x_0$ and $y(0) = x'(0) = v_0$. Therefore

$$c = v_0^2 - \frac{2g}{\cosh x_0} \quad \text{and so}$$

$$y^2 = \frac{2g}{\cosh x} - \frac{2g}{\cosh x_0} + v_0^2.$$

Now

$$\frac{2g}{\cosh x} - \frac{2g}{\cosh x_0} + v_0^2 \geq 0 \quad \text{if and only if} \quad \cosh x \leq \frac{2g \cosh x_0}{2g - v_0^2 \cosh x_0}$$

and the solution to this inequality is an interval $[-a, a]$. Therefore each x in $(-a, a)$ has two corresponding values of y and so the solution is periodic.

- (b) Since $z = \cosh x$, the maximum height occurs at the largest value of x on the cycle. From (a), $x_{\max} = a$ where $\cosh a = \frac{2g \cosh x_0}{2g - v_0^2 \cosh x_0}$. Therefore

$$z_{\max} = \frac{2g \cosh x_0}{2g - v_0^2 \cosh x_0}.$$

7. If $x_m < x_1 < x_n$, then $F(x_1) > F(x_m) = F(x_n)$. Letting $x = x_1$,

$$G(y) = \frac{c_0}{F(x_1)} = \frac{F(x_m)G(a/b)}{F(x_1)} < G(a/b).$$

Therefore from Figure 11.35(b), $G(y) = \frac{c_0}{F(x_1)}$ has two solutions y_1 and y_2 that satisfy $y_1 < a/b < y_2$.

8. From (i) when $y = a/b$, x_n is taken on at some time t . From (iii) if $x > x_n$ there is no corresponding value of y . Therefore the maximum number of predators is x_n and x_n occurs when $y = a/b$.

9. (a) In the Lotka-Volterra Model the average number of predators is d/c and the average number of prey is a/b . But

$$x' = -ax + bxy - \epsilon_1 x = -(a + \epsilon_1)x + bxy$$

$$y' = -cxy + dy - \epsilon_2 y = -cxy + (d - \epsilon_2)y$$

and so the new critical point in the first quadrant is $(d/c - \epsilon_2/c, a/b + \epsilon_1/b)$.

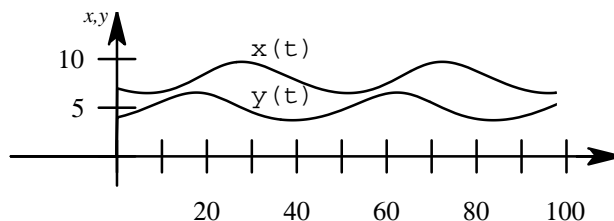
- (b) The average number of predators $d/c - \epsilon_2/c$ has decreased while the average number of prey $a/b + \epsilon_1/b$ has increased. The fishery science model is consistent with Volterra's principle.

10. (a) Solving

$$x(-0.1 + 0.02y) = 0$$

$$y(0.2 - 0.025x) = 0$$

in the first quadrant we obtain the critical point $(8, 5)$. The graphs are plotted using $x(0) = 7$ and $y(0) = 4$.



- (b) The graph in part (a) was obtained using **NDSolve** in *Mathematica*. We see that the period is around 40. Since $x(0) = 7$, we use the **FindRoot** equation solver in *Mathematica* to approximate the solution of $x(t) = 7$ for t near 40. From this we see that the period is more closely approximated by $t = 44.65$.

11. Solving

$$x(20 - 0.4x - 0.3y) = 0$$

$$y(10 - 0.1y - 0.3x) = 0$$

we see that critical points are $(0, 0)$, $(0, 100)$, $(50, 0)$, and $(20, 40)$. The Jacobian matrix is

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 0.08(20 - 0.8x - 0.3y) & -0.024x \\ -0.018y & 0.06(10 - 0.2y - 0.3x) \end{pmatrix}$$

and so

$$\mathbf{A}_1 = \mathbf{g}'((0, 0)) = \begin{pmatrix} 1.6 & 0 \\ 0 & 0.6 \end{pmatrix}$$

$$\mathbf{A}_2 = \mathbf{g}'((0, 100)) = \begin{pmatrix} -0.8 & 0 \\ -1.8 & -0.6 \end{pmatrix}$$

$$\mathbf{A}_3 = \mathbf{g}'((50, 0)) = \begin{pmatrix} -1.6 & -1.2 \\ 0 & -0.3 \end{pmatrix}$$

$$\mathbf{A}_4 = \mathbf{g}'((20, 40)) = \begin{pmatrix} -0.64 & -0.48 \\ -0.72 & -0.24 \end{pmatrix}.$$

Exercises 11.4

Since $\det(\mathbf{A}_1) = \Delta_1 = 0.96 > 0$, $\tau = 2.2 > 0$, and $\tau_1^2 - 4\Delta_1 = 1 > 0$, we see that $(0, 0)$ is an unstable node. Since $\det(\mathbf{A}_2) = \Delta_2 = 0.48 > 0$, $\tau = -1.4 < 0$, and $\tau_2^2 - 4\Delta_2 = 0.04 > 0$, we see that $(0, 100)$ is a stable node. Since $\det(\mathbf{A}_3) = \Delta_3 = 0.48 > 0$, $\tau = -1.9 < 0$, and $\tau_3^2 - 4\Delta_3 = 1.69 > 0$, we see that $(50, 0)$ is a stable node. Since $\det(\mathbf{A}_4) = -0.192 < 0$ we see that $(20, 40)$ is a saddle point.

12. $\Delta = r_1 r_2$, $\tau = r_1 + r_2$ and $\tau^2 - 4\Delta = (r_1 + r_2)^2 - 4r_1 r_2 = (r_1 - r_2)^2$. Therefore when $r_1 \neq r_2$, $(0, 0)$ is an unstable node.
13. For $\mathbf{X} = (K_1, 0)$, $\tau = -r_1 + r_2 \left(1 - \frac{K_1}{K_2} \alpha_{21}\right)$ and $\Delta = -r_1 r_2 \left(1 - \frac{K_1}{K_2} \alpha_{21}\right)$. If we let $c = 1 - \frac{K_1}{K_2} \alpha_{21}$, $\tau^2 - 4\Delta = (cr_2 + r_1)^2 > 0$. Now if $k_1 > \frac{K_2}{\alpha_{21}}$, $c < 0$ and so $\tau < 0$, $\Delta > 0$. Therefore $(K_1, 0)$ is a stable node. If $K_1 < \frac{K_2}{\alpha_{21}}$, $c > 0$ and so $\Delta < 0$. In this case $(K_1, 0)$ is a saddle point.
14. $(\hat{x}, \hat{y})$ is a stable node if and only if $\frac{K_1}{\alpha_{12}} > K_2$ and $\frac{K_2}{\alpha_{21}} > K_1$. [See Figure 11.38(a) in the text.] From Problem 12, $(0, 0)$ is an unstable node and from Problem 13, since $K_1 < \frac{K_2}{\alpha_{21}}$, $(K_1, 0)$ is a saddle point. Finally, when $K_2 < \frac{K_1}{\alpha_{12}}$, $(0, K_2)$ is a saddle point. This is Problem 12 with the roles of 1 and 2 interchanged. Therefore $(0, 0)$, $(K_1, 0)$, and $(0, K_2)$ are unstable.
15. $\frac{K_1}{\alpha_{12}} < K_2 < K_1 \alpha_{21}$ and so $\alpha_{12} \alpha_{21} > 1$. Therefore $\Delta = (1 - \alpha_{12} \alpha_{21}) \hat{x} \hat{y} \frac{r_1 r_2}{K_1 K_2} < 0$ and so $(\hat{x}, \hat{y})$ is a saddle point.
16. (a) The corresponding plane autonomous system is

$$x' = y, \quad y' = \frac{-g}{l} \sin x - \frac{\beta}{ml} y$$

and so critical points must satisfy both $y = 0$ and $\sin x = 0$. Therefore $(\pm n\pi, 0)$ are critical points.

- (b) The Jacobian matrix

$$\begin{pmatrix} 0 & 1 \\ -\frac{g}{l} \cos x & -\frac{\beta}{ml} \end{pmatrix}$$

has trace $\tau = -\frac{\beta}{ml}$ and determinant $\Delta = \frac{g}{l} > 0$ at $(0, 0)$. Therefore

$$\tau^2 - 4\Delta = \frac{\beta^2}{m^2 l^2} - 4 \frac{g}{l} = \frac{\beta^2 - 4g l m^2}{m^2 l^2}.$$

We may conclude that $(0, 0)$ is a stable spiral point provided $\beta^2 - 4g l m^2 < 0$ or $\beta < 2m\sqrt{gl}$.

17. (a) The corresponding plane autonomous system is

$$x = y, \quad y' = -\frac{\beta}{m} y|y| - \frac{k}{m} x$$

and so a critical point must satisfy both $y = 0$ and $x = 0$. Therefore $(0, 0)$ is the unique critical point.

- (b) The Jacobian matrix is

$$\begin{pmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{\beta}{m} 2|y| \end{pmatrix}$$

and so $\tau = 0$ and $\Delta = \frac{k}{m} > 0$. Therefore $(0, 0)$ is a center, stable spiral point, or an unstable spiral point. Physical considerations suggest that $(0, 0)$ must be asymptotically stable and so $(0, 0)$ must be a stable spiral point.

18. (a) The magnitude of the frictional force between the bead and the wire is $\mu(mg \cos \theta)$ for some $\mu > 0$. The component of this frictional force in the x -direction is

$$(\mu mg \cos \theta) \cos \theta = \mu mg \cos^2 \theta.$$

But

$$\cos \theta = \frac{1}{\sqrt{1 + [f'(x)]^2}} \quad \text{and so} \quad \mu mg \cos^2 \theta = \frac{\mu mg}{1 + [f'(x)]^2}.$$

It follows from Newton's Second Law that

$$mx'' = -mg \frac{f'(x)}{1 + [f'(x)]^2} - \beta x' + mg \frac{\mu}{1 + [f'(x)]^2}$$

and so

$$x'' = g \frac{\mu - f'(x)}{1 + [f'(x)]^2} - \frac{\beta}{m} x'.$$

- (b) A critical point (x, y) must satisfy $y = 0$ and $f'(x) = \mu$. Therefore critical points occur at $(x_1, 0)$ where $f'(x_1) = \mu$. The Jacobian matrix of the plane autonomous system is

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 0 & 1 \\ g \frac{(1 + [f'(x)]^2)(-f''(x)) - (\mu - f'(x))2f'(x)f''(x)}{(1 + [f'(x)]^2)^2} & -\frac{\beta}{m} \end{pmatrix}$$

and so at a critical point $\mathbf{X}_1$,

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 0 & 1 \\ \frac{-gf''(x_1)}{1 + \mu^2} & -\frac{\beta}{m} \end{pmatrix}.$$

Therefore $\tau = -\frac{\beta}{m} < 0$ and $\Delta = \frac{gf''(x_1)}{1 + \mu^2}$. When $f''(x_1) < 0$, $\Delta < 0$ and so a saddle point occurs. When $f''(x_1) > 0$ and

$$\tau^2 - 4\Delta = \frac{\beta^2}{m^2} - 4g \frac{f''(x_1)}{1 + \mu^2} < 0,$$

$(x_1, 0)$ is a stable spiral point. This condition may also be written as

$$\beta^2 < 4gm^2 \frac{f''(x_1)}{1 + \mu^2}.$$

19. $\frac{dy}{dx} = \frac{y'}{x'} = -\frac{f(x)}{y}$ and so using separation of variables, $\frac{y^2}{2} = -\int_0^x f(\mu) d\mu + c$ or $y^2 + 2F(x) = c$. We may conclude that for a given value of x there are at most two corresponding values of y . If $(0, 0)$ were a stable spiral point there would exist an x with more than two corresponding values of y . Note that the condition $f(0) = 0$ is required for $(0, 0)$ to be a critical point of the corresponding plane autonomous system $x' = y$, $y' = -f(x)$.
20. (a) $x' = x(-a + by) = 0$ implies that $x = 0$ or $y = a/b$. If $x = 0$, then, from

$$-cxy + \frac{r}{K} y(K - y) = 0,$$

$y = 0$ or K . Therefore $(0, 0)$ and $(0, K)$ are critical points. If $\hat{y} = a/b$, then

$$\hat{y} \left[-cx + \frac{r}{K}(K - \hat{y}) \right] = 0.$$

Exercises 11.4

The corresponding value of x , $x = \hat{x}$, therefore satisfies the equation $c\hat{x} = \frac{r}{K}(K - \hat{y})$.

(b) The Jacobian matrix is

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} -a + by & bx \\ -cy & -cx + \frac{r}{K}(K - 2y) \end{pmatrix}$$

and so at $\mathbf{X}_1 = (0, 0)$, $\Delta = -ar < 0$. For $\mathbf{X}_1 = (0, K)$, $\Delta = n(Kb - a) = -rb\left(K - \frac{a}{b}\right)$. Since we are given that $K > \frac{a}{b}$, $\Delta < 0$ in this case. Therefore $(0, 0)$ and $(0, K)$ are each saddle points. For $\mathbf{X}_1 = (\hat{x}, \hat{y})$ where $\hat{y} = \frac{a}{b}$ and $c\hat{x} = \frac{r}{K}(K - \hat{y})$, we may write the Jacobian matrix as

$$\mathbf{g}'((\hat{x}, \hat{y})) = \begin{pmatrix} 0 & b\hat{x} \\ -c\hat{y} & -\frac{r}{K}\hat{y} \end{pmatrix}$$

and so $\tau = -\frac{r}{K}\hat{y} < 0$ and $\Delta = bc\hat{x}\hat{y} > 0$. Therefore $(\hat{x}, \hat{y})$ is a stable critical point and so it is either a stable node (perhaps degenerate) or a stable spiral point.

(c) Write

$$\tau^2 - 4\Delta = \frac{r^2}{K^2}\hat{y}^2 - 4bc\hat{x}\hat{y} = \hat{y} \left[\frac{r^2}{K^2}\hat{y} - 4bc\hat{x} \right] = \hat{y} \left[\frac{r^2}{K^2}\hat{y} - 4b\frac{r}{K}(K - \hat{y}) \right]$$

using

$$c\hat{x} = \frac{r}{K}(K - \hat{y}) = \frac{r}{K}\hat{y} \left[\left(\frac{r}{K} + 4b \right) \hat{y} - 4bK \right].$$

Therefore $\tau^2 - 4\Delta < 0$ if and only if

$$\hat{y} < \frac{4bK}{\frac{r}{K} + 4b} = \frac{4bK^2}{r + 4bK}.$$

Note that

$$\frac{4bK^2}{r + 4bK} = \frac{4bK}{r + 4bK} \cdot K \approx K$$

where K is large, and $\hat{y} = \frac{a}{b} < K$. Therefore $\tau^2 - 4\Delta < 0$ when K is large and a stable spiral point will result.

21. The equation

$$x' = \alpha \frac{y}{1+y} x - x = x \left(\frac{\alpha y}{1+y} - 1 \right) = 0$$

implies that $x = 0$ or $y = \frac{1}{\alpha - 1}$. When $\alpha > 0$, $\hat{y} = \frac{1}{\alpha - 1} > 0$. If $x = 0$, then from the differential equation for y' , $y = \beta$. On the other hand, if $\hat{y} = \frac{1}{\alpha - 1}$, $\frac{\hat{y}}{1 + \hat{y}} = \frac{1}{\alpha}$ and so $\frac{1}{\alpha}\hat{x} - \frac{1}{\alpha - 1} + \beta = 0$. It follows that

$$\hat{x} = \alpha \left(\beta - \frac{1}{\alpha - 1} \right) = \frac{\alpha}{\alpha - 1} [(\alpha - 1)\beta - 1]$$

and if $\beta(\alpha - 1) > 1$, $\hat{x} > 0$. Therefore $(\hat{x}, \hat{y})$ is the unique critical point in the first quadrant. The Jacobian matrix is

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} \alpha \frac{y}{y+1} - 1 & \frac{\alpha x}{(1+y)^2} \\ -\frac{y}{1+y} & \frac{-x}{(1+y)^2} - 1 \end{pmatrix}$$

and for $\mathbf{X} = (\hat{x}, \hat{y})$, the Jacobian can be written in the form

$$\mathbf{g}'((\hat{x}, \hat{y})) = \begin{pmatrix} 0 & \frac{(\alpha-1)^2}{\alpha} \hat{x} \\ -\frac{1}{\alpha} & -\frac{(\alpha-1)^2}{\alpha^2} - 1 \end{pmatrix}.$$

It follows that

$$\tau = -\left[\frac{(\alpha-1)^2}{\alpha^2} \hat{x} + 1\right] < 0, \quad \Delta = \frac{(\alpha-1)^2}{\alpha^2} \hat{x}$$

and so $\tau = -(\Delta + 1)$. Therefore $\tau^2 - 4\Delta = (\Delta + 1)^2 - 4\Delta = (\Delta - 1)^2 > 0$. Therefore $(\hat{x}, \hat{y})$ is a stable node.

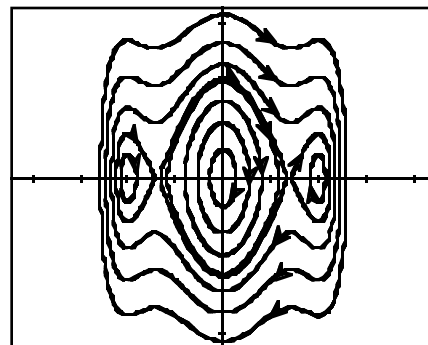
22. Letting $y = x'$ we obtain the plane autonomous system

$$\begin{aligned} x' &= y \\ y' &= -8x + 6x^3 - x^5. \end{aligned}$$

Solving $x^5 - 6x^3 + 8x = x(x^2 - 4)(x^2 - 2) = 0$ we see that critical points are $(0, 0)$, $(0, -2)$, $(0, 2)$, $(0, -\sqrt{2})$, and $(0, \sqrt{2})$. The Jacobian matrix is

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 0 & 1 \\ -8 + 18x^2 - 5x^4 & 0 \end{pmatrix}$$

and we see that $\det(\mathbf{g}'(\mathbf{X})) = 5x^4 - 18x^2 + 8$ and the trace of $\mathbf{g}'(\mathbf{X})$ is 0. Since $\det(\mathbf{g}'((\pm\sqrt{2}, 0))) = -8 < 0$, $(\pm\sqrt{2}, 0)$ are saddle points. For the other critical points the determinant is positive and linearization discloses no information. The graph of the phase plane suggests that $(0, 0)$ and $(\pm 2, 0)$ are centers.



Exercises 11.5

1. $y' = x - y = 0$ implies that $y = x$ and so $2 + xy = 2 + x^2 > 0$. Therefore the system has no critical points, and so, by the corollary to Theorem 11.4, there are no periodic solutions.
2. $x' = 2x - xy = x(2 - y) = 0$ implies that $x = 0$ or $y = 2$. If $x = 0$, then from $-1 - x^2 + 2x - y^2 = 0$, $y^2 = -1$ and there are no real solutions. If $y = 2$, $-x^2 + 2x - 5 = 0$ which has no real solutions. Therefore the system has no critical points and so, by the corollary to Theorem 11.4, there are no periodic solutions.
3. For $P = -x + y^2$ and $Q = x - y$, $\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = -2 < 0$. Therefore there are no periodic solutions by Theorem 11.5.
4. For $P = xy^2 - x^2y$ and $Q = x^2y - 1$, $\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = y^2 - 2xy + x^2 = (y - x)^2$. Therefore $\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}$ does not change signs and so there are no periodic solutions by Theorem 11.5.
5. For $P = -\mu x - y$ and $Q = x + y^3$, $\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = -\mu + 9y^2 > 0$ since $\mu < 0$. Therefore there are no periodic solutions by Theorem 11.5.
6. From $y' = xy - y = y(x - 1) = 0$ either $y = 0$ or $x = 1$. If $y = 0$, then from $2x + y^2 = 0$, $x = 0$. Likewise $x = 1$ implies that $2 + y^2 = 0$, which has no real solutions. Therefore $(0, 0)$ is the only critical point. But $\mathbf{g}'((0, 0))$ has determinant $\Delta = -2$. The single critical point is a saddle point and so, by the corollary to Theorem 11.4, there are no periodic solutions.

Exercises 11.5

7. The corresponding plane autonomous system is $x' = y$, $y' = 2x - y^4$. Therefore $(0, 0)$ is the only critical point. But $\mathbf{g}'((0, 0))$ has determinant $\Delta = -2 < 0$. The single critical point is a saddle point and so, by the corollary to Theorem 11.4, there are no periodic solutions.

8. The corresponding plane autonomous system is

$$x' = y, \quad y' = -x + \left(\frac{1}{2} + 3y^2\right)y - x^2$$

and so $\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = 0 + \frac{1}{2} + 9y^2 > 0$. Therefore there are no periodic solutions by Theorem 11.5.

9. For $\delta(x, y) = e^{ax+by}$, $\frac{\partial}{\partial x}(\delta P) + \frac{\partial}{\partial y}(\delta Q)$ can be simplified to

$$e^{ax+by}[-bx^2 - 2ax + axy + (2b + 1)y].$$

Setting $a = 0$ and $b = -1/2$,

$$\frac{\partial}{\partial x}(\delta P) + \frac{\partial}{\partial y}(\delta Q) = \frac{1}{2}x^2e^{-\frac{1}{2}y}$$

which does not change signs. Therefore by Theorem 11.6 there are no periodic solutions.

10. For $\delta(x, y) = ax^2 + by^2$, $\frac{\partial}{\partial x}(\delta P) + \frac{\partial}{\partial y}(\delta Q)$ can be simplified to

$$-5ax^4 - 3bx^2y^2 + 10(a - b)x^2y.$$

Setting $a = b = 1$,

$$\frac{\partial}{\partial x}(\delta P) + \frac{\partial}{\partial y}(\delta Q) = -5x^4 - 3x^2y^2$$

which does not change signs. Therefore by Theorem 11.6 there are no periodic solutions.

11. For $P = x(1 - x^2 - 3y^2)$ and $Q = y(3 - x^2 - 3y^2)$,

$$\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = 4(1 - x^2 - 3y^2) \quad \text{and so} \quad \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} > 0$$

for $x^2 + 3y^2 < 1$. Therefore there are no periodic solutions in the elliptical region $x^2 + 3y^2 < 1$.

12. The corresponding plane autonomous system is $x' = y$, $y' = g(x, y)$ and so

$$\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = \frac{\partial g}{\partial y} = \frac{\partial g}{\partial x'} \neq 0$$

in the region R . Therefore $\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}$ cannot change signs and so there are no periodic solutions by Theorem 11.5.

13. For $\delta(x, y) = \frac{1}{xy}$, $\delta P = -\frac{a}{y} + b$, $\delta Q = -c + \frac{r}{K} \frac{1}{x}(K - y)$ and so

$$\frac{\partial(\delta P)}{\partial x} + \frac{\partial(\delta Q)}{\partial y} = -\frac{r}{Kx} < 0$$

in the first quadrant. By Theorem 11.6, there are no periodic solutions in the first quadrant.

14. If $\mathbf{n} = (-2x, -2y)$,

$$\mathbf{V} \cdot \mathbf{n} = 2xy + 2x^2e^{x+y} - 2xy + 2y^2e^{x+y} = 2(x^2 + y^2)e^{x+y} \geq 0.$$

Therefore any circular region of the form $x^2 + y^2 \leq r^2$ is an invariant region by Theorem 11.7.

15. If $\mathbf{n} = (-2x, -2y)$,

$$\mathbf{V} \cdot \mathbf{n} = 2x^2 - 4xy + 2y^2 + 2y^4 = 2(x - y)^2 + 2y^4 \geq 0.$$

Therefore $x^2 + y^2 \leq r$ serves as an invariant region for any $r > 0$ by Theorem 11.7.

16. $\mathbf{n} = -\nabla t = (-6x^5, -6y)$. Since the corresponding plane autonomous system is $x' = y$, $y' = -y - y^3 - x^5$,

$$\mathbf{n} \cdot \mathbf{V} = -6x^5y + 6y^2 + 6y^4 + 6x^5y = 6y^2 + 6y^4.$$

Therefore $\mathbf{n} \cdot \mathbf{V} \geq 0$ and so by Theorem 11.7, the region $x^6 + 3y^2 \leq 1$ serves as an invariant region.

17. We showed in Example 8 that $\frac{1}{16} \leq x^2 + y^2 \leq 1$ is an invariant region for the plane autonomous system. If the only critical point is $(0, 0)$, this critical point lies outside the invariant region and so Theorem 11.8(ii) is applicable. There is at least one periodic solution in R .
18. The corresponding plane autonomous system is

$$x' = y, \quad y' = y(1 - 3x^2 - 2y^2) - x$$

and it is easy to see that $(0, 0)$ is the only critical point. If $\mathbf{n} = (-2x, -2y)$ then

$$\mathbf{V} \cdot \mathbf{n} = -2xy - 2y^2(1 - 3x^2 - 2y^2) + 2xy = -2y^2(1 - 2r^2 - x^2).$$

If $r = \frac{1}{2}\sqrt{2}$, $2r^2 = 1$ and so $\mathbf{V} \cdot \mathbf{n} = 2x^2y^2 \geq 0$. Therefore $\frac{1}{4} \leq x^2 + y^2 \leq \frac{1}{2}$ serves as an invariant region. By Theorem 11.8(ii) there is at least one periodic solution.

19. If $r < 1$ and $\mathbf{n} = (-2x, -2y)$ then

$$\mathbf{V} \cdot \mathbf{n} = -2xy + 2xy + 2y^2(1 - x^2) = 2y^2(1 - x^2) \geq 0$$

since $x^2 < 1$. Therefore $x^2 + y^2 \leq r^2$ serves as an invariant region. Now $(0, 0)$ is the only critical point and, since $\tau = -1$ and $\Delta = 1$, $\tau^2 - 4\Delta < 0$. Therefore $(0, 0)$ is a stable spiral point and so, by Theorem 11.9(ii), $\lim_{t \rightarrow \infty} \mathbf{X}(t) = (0, 0)$.

20. Since $\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = -1 - 3y^2 < 0$, there are no periodic solutions. If $\mathbf{n} = (-2x, -2y)$,

$$\mathbf{V} \cdot \mathbf{n} = -2xy + 2x^2 + 2xy + 2y^4 = 2(x^2 + y^4) \geq 0.$$

Therefore the circular region $x^2 + y^2 \leq r^2$ serves as an invariant region for any $r > 0$. If (x, y) is a critical point, $y - x = 0$ or $y = x$. From $-x - y^3 = 0$ we have $-y(1 + y^2) = 0$. Therefore $y = 0$ and so $(0, 0)$ is the only critical point. It is easy to check that $\tau = -1$, $\Delta = 1$, $\tau^2 - 4\Delta = -3$ and so $(0, 0)$ is a stable spiral point. By Theorem 11.9(ii), $(0, 0)$ is globally stable. For any initial condition, $\lim_{t \rightarrow \infty} \mathbf{X}(t) = (0, 0)$.

21. (a) $\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = 2xy - 1 - x^2 \leq 2x - 1 - x^2 = -(x - 1)^2 \leq 0$. Therefore there are no periodic solutions.
- (b) If (x, y) is a critical point, $x^2y = \frac{1}{2}$ and so from $x' = x^2y - x + 1$, $\frac{1}{2} - x + 1 = 0$. Therefore $x = 3/2$ and so $y = 2/9$. For this critical point, $\tau = -31/12 < 0$, $\Delta = 9/4 > 0$, and $\tau^2 - 4\Delta < 0$. Therefore $(3/2, 2/9)$ is a stable spiral point and so, from Theorem 11.9(ii), $\lim_{x \rightarrow \infty} \mathbf{X}(t) = (3/2, 2/9)$.

22. (a) From $x \left(\frac{2y}{y+2} - 1 \right) = 0$, either $x = 0$ or $y = 2$. For the case $x = 0$, from $y \left(1 - \frac{2x}{y+2} - \frac{y}{8} \right) = 0$, $y \left(1 - \frac{y}{8} \right) = 0$. Therefore $(0, 0)$ and $(0, 8)$ are critical points. If $y = 2$, $1 - \frac{2x}{4} - \frac{1}{4} = 0$ and so $x = 3/2$.

Therefore $(3/2, 2)$ is the additional critical point. We may classify these critical points as follows:

$\mathbf{X}$	τ	Δ	$\tau^2 - 4\Delta$	Conclusion
$(0, 0)$	$-$	-1	$-$	saddle point
$(0, 8)$	$-$	$-\frac{3}{5}$	$-$	saddle point
$(\frac{3}{2}, 2)$	$\frac{1}{8}$	$\frac{3}{8}$	$-\frac{95}{64}$	unstable spiral point

Exercises 11.5

- (b) By Theorem 11.8(i), since there is a unique unstable critical point inside the invariant region, there is at least one periodic solution.

Chapter 11 Review Exercises

1. True
2. True
3. A center or a saddle point
4. Complex with negative real parts
5. False; there are initial conditions for which $\lim_{t \rightarrow \infty} \mathbf{X}(t) = (0, 0)$.
6. True
7. False; this is a borderline case. See Figure 11.25 in the text.
8. False; see Figure 11.29 in the text.
9. True
10. False; we also need to have no critical points on the boundary of R .
11. Switching to polar coordinates,

$$\begin{aligned}\frac{dr}{dt} &= \frac{1}{r} \left(x \frac{dx}{dt} + y \frac{dy}{dt} \right) = \frac{1}{r} (-xy - x^2 r^3 + xy - y^2 r^3) = -r^4 \\ \frac{d\theta}{dt} &= \frac{1}{r^2} \left(-y \frac{dx}{dt} + x \frac{dy}{dt} \right) = \frac{1}{r^2} (y^2 + xyr^3 + x^2 - xyr^3) = 1.\end{aligned}$$

Using separation of variables it follows that $r = \frac{1}{\sqrt[3]{3t + c_1}}$ and $\theta = t + c_2$. Since $\mathbf{X}(0) = (1, 0)$, $r = 1$ and $\theta = 0$. It follows that $c_1 = 1$, $c_2 = 0$, and so

$$r = \frac{1}{\sqrt[3]{3t + 1}}, \quad \theta = t.$$

As $t \rightarrow \infty$, $r \rightarrow 0$ and the solution spirals toward the origin.

12. (a) If $\mathbf{X}(0) = \mathbf{X}_0$ lies on the line $y = -2x$, then $\mathbf{X}(t)$ approaches $(0, 0)$ along this line. For all other initial conditions, $\mathbf{X}(t)$ approaches $(0, 0)$ from the direction determined by the line $y = x$.
- (b) If $\mathbf{X}(0) = \mathbf{X}_0$ lies on the line $y = -x$, then $\mathbf{X}(t)$ approaches $(0, 0)$ along this line. For all other initial conditions, $\mathbf{X}(t)$ becomes unbounded and $y = 2x$ serves as an asymptote.
13. (a) $\tau = 0$, $\Delta = 11 > 0$ and so $(0, 0)$ is a center.
- (b) $\tau = -2$, $\Delta = 1$, $\tau^2 - 4\Delta = 0$ and so $(0, 0)$ is a degenerate stable node.
14. From $x' = x(1 + y - 3x) = 0$, either $x = 0$ or $1 + y - 3x = 0$. If $x = 0$, then, from $y(4 - 2x - y) = 0$ we obtain $y(4 - y) = 0$. It follows that $(0, 0)$ and $(0, 4)$ are critical points. If $1 + y - 3x = 0$, then $y(5 - 5x) = 0$. Therefore $(1/3, 0)$ and $(1, 2)$ are the remaining critical points. We will use the Jacobian matrix

$$\mathbf{g}'(\mathbf{X}) = \begin{bmatrix} 1 + y - 6x & x \\ -2y & 4 - 2x - 2y \end{bmatrix}$$

to classify these four critical points. The results are as follows:

$\mathbf{X}$	τ	Δ	$\tau^2 - 4\Delta$	Conclusion
$(0, 0)$	5	4	9	unstable node
$(0, 4)$	—	—20	—	saddle point
$(\frac{1}{3}, 0)$	—	— $\frac{10}{3}$	—	saddle point
$(1, 2)$	—5	10	—15	stable spiral point

If $\delta(x, y) = \frac{1}{xy}$, $\delta P = \frac{1}{y} + 1 - 3\frac{x}{y}$ and $\delta Q = \frac{4}{x} - 2 - \frac{y}{x}$. It follows that

$$\frac{\partial}{\partial x}(\delta P) + \frac{\partial}{\partial y}(\delta Q) = -\frac{3}{y} - \frac{1}{x} < 0$$

in quadrant one. Therefore there are no periodic solutions in the first quadrant.

15. The corresponding plane autonomous system is $x' = y$, $y' = \mu(1 - x^2) - x$ and so the Jacobian at the critical point $(0, 0)$ is

$$\mathbf{g}'((0, 0)) = \begin{bmatrix} 0 & 1 \\ -1 & \mu \end{bmatrix}.$$

Therefore $\tau = \mu$, $\Delta = 1$ and $\tau^2 - 4\Delta = \mu^2 - 4$. Now $\mu^2 - 4 < 0$ if and only if $-2 < \mu < 2$. We may therefore conclude that $(0, 0)$ is a stable node for $\mu < -2$, a stable spiral point for $-2 < \mu < 0$, an unstable spiral point for $0 < \mu < 2$, and an unstable node for $\mu > 2$.

16. Critical points occur at $x = \pm 1$. Since

$$g'(x) = -\frac{1}{2}e^{-x/2}(x^2 - 4x - 1),$$

$g'(1) > 0$ and $g'(-1) < 0$. Therefore $x = 1$ is unstable and $x = -1$ is asymptotically stable.

17. $\frac{dy}{dx} = \frac{y'}{x'} = \frac{-2x\sqrt{y^2+1}}{y}$. We may separate variables to show that $\sqrt{y^2+1} = -x^2 + c$. But $x(0) = x_0$ and $y(0) = x'(0) = 0$. It follows that $c = 1 + x_0^2$ so that

$$y^2 = (1 + x_0^2 - x^2)^2 - 1.$$

Note that $1 + x_0^2 - x^2 > 1$ for $-x_0 < x < x_0$ and $y = 0$ for $x = \pm x_0$. Each x with $-x_0 < x < x_0$ has two corresponding values of y and so the solution $\mathbf{X}(t)$ with $\mathbf{X}(0) = (x_0, 0)$ is periodic.

18. The corresponding plane autonomous system is

$$x' = y, \quad y' = -\frac{\beta}{m}y - \frac{k}{m}(s+x)^3 + g$$

and so the Jacobian is

$$\mathbf{g}'(\mathbf{X}) = \begin{bmatrix} 0 & 1 \\ -\frac{3k}{m}(s+x)^2 & -\frac{\beta}{m} \end{bmatrix}.$$

For $\mathbf{X} = (0, 0)$, $\tau = -\frac{\beta}{m} < 0$, $\Delta = \frac{3k}{m}s^2 > 0$. Therefore

$$\tau^2 - 4\Delta = \frac{\beta^2}{m^2} - \frac{12k}{m}s^2 = \frac{1}{m^2}(\beta^2 - 12kms^2).$$

Therefore $(0, 0)$ is a stable node if $\beta^2 > 12kms^2$ and a stable spiral point provided $\beta^2 < 12kms^2$, where $ks^3 = mg$.

Chapter 11 Review Exercises

19. For $P = 4x + 2y - 2x^2$ and $Q = 4x - 3y + 4xy$.

$$\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = 4 - 4x - 3 + 4x = 1 > 0.$$

Therefore there are no periodic solutions by Theorem 11.5.

20. If (x, y) is a critical point,

$$P = \epsilon x + y - x(x^2 + y^2) = 0$$

$$Q = -x + \epsilon y - y(x^2 + y^2) = 0.$$

But $yP - xQ = y^2 + x^2$. Therefore $x^2 + y^2 = 0$ and so $x = y = 0$. It follows that $(0, 0)$ is the only critical point and

$$\mathbf{g}'((0, 0)) = \begin{bmatrix} \epsilon & 1 \\ -1 & \epsilon \end{bmatrix}$$

so that $\tau = 2\epsilon$, $\Delta = \epsilon^2 + 1 > 0$, and so $\tau^2 - 4\Delta = -4$. Note that if $\mathbf{n} = (-2x, -2y)$, $\mathbf{n} \cdot \mathbf{V} = 2r^2(r^2 - \epsilon)$. If $\epsilon > 0$, then $\tau > 0$ and so $(0, 0)$ is an unstable spiral point. If $r = 2\epsilon$, $\mathbf{n} \cdot \mathbf{V} \geq 0$ and so $x^2 + y^2 \leq 4\epsilon^2$ is an invariant region. It follows from Theorem 11.8(i) that there is at least one periodic solution in this region. If $\epsilon < 0$, $\tau < 0$ and so $(0, 0)$ is a stable spiral point. Note that $\mathbf{n} \cdot \mathbf{V} = 2r^2(r^2 - \epsilon) \geq 0$ for any r and so $x^2 + y^2 \leq r^2$ is an invariant region. It follows from Theorem 11.9(ii) that $\lim_{t \rightarrow +\infty} \mathbf{X}(t) = (0, 0)$ for any choice of initial position $\mathbf{X}_0$.

21. (a) If $x = \theta$ and $y = x' = \theta'$, the corresponding plane autonomous system is

$$x' = y, \quad y' = \omega^2 \sin x \cos x - \frac{g}{l} \sin x - \frac{\beta}{ml} y.$$

Therefore $\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = -\frac{\beta}{ml} < 0$ and so there are no periodic solutions.

- (b) If (x, y) is a critical point, $y = 0$ and so $\sin x(\omega^2 \cos x - g/l) = 0$. Either $\sin x = 0$ (in which case $x = 0$) or $\cos x = g/\omega^2 l$. But if $\omega^2 < g/l$, $g/\omega^2 l > 1$ and so the latter equation has no real solutions. Therefore $(0, 0)$ is the only critical point if $\omega^2 < g/l$. The Jacobian matrix is

$$\mathbf{g}'(\mathbf{X}) = \begin{bmatrix} 0 & 1 \\ \omega^2 \cos 2x - \frac{g}{l} \cos x & -\frac{\beta}{ml} \end{bmatrix}$$

and so $\tau = -\beta/ml < 0$ and $\Delta = g/l - \omega^2 > 0$ for $\mathbf{X} = (0, 0)$. It follows that $(0, 0)$ is asymptotically stable and so after a small displacement, the pendulum will return to $\theta = 0$, $\theta' = 0$.

- (c) If $\omega^2 > g/l$, $\cos x = g/\omega^2 l$ will have two solutions $x = \pm \hat{x}$ that satisfy $-\pi < x < \pi$. Therefore $(\pm \hat{x}, 0)$ are two additional critical points. If $\mathbf{X}_1 = (0, 0)$, $\Delta = g/l - \omega^2 < 0$ and so $(0, 0)$ is a saddle point. If $\mathbf{X}_1 = (\pm \hat{x}, 0)$, $\tau = -\beta/ml < 0$ and

$$\Delta = \frac{g}{l} \cos \hat{x} - \omega^2 \cos 2\hat{x} = \frac{g^2}{\omega^2 l^2} - \omega^2 \left(2 \frac{g^2}{\omega^4 l^2} - 1 \right) = \omega^2 - \frac{g^2}{\omega^2 l^2} > 0.$$

Therefore $(\hat{x}, 0)$ and $(-\hat{x}, 0)$ are each stable. When $\theta(0) = \theta_0$, $\theta'(0) = 0$ and θ_0 is small we expect the pendulum to reach one of these two stable equilibrium positions.

- (d) In (b), $(0, 0)$ is a stable spiral point provided

$$\tau^2 - 4\Delta = \frac{\beta^2}{m^2 l^2} - 4 \left(\frac{g}{l} - \omega^2 \right) < 0.$$

This condition is equivalent to $\beta < 2ml\sqrt{g/l - \omega^2}$. In (c), $(\pm \hat{x}, 0)$ are stable spiral points provided that

$$\tau^2 - 4\Delta = \frac{\beta^2}{m^2 l^2} - 4 \left(\omega^2 - \frac{g^2}{\omega^2 l^2} \right) < 0.$$

This condition is equivalent to $\beta < 2ml\sqrt{\omega^2 - g^2/(\omega^2 l^2)}$.

22. The corresponding plane autonomous system is

$$x' = y, \quad y' = 2ky - cy^3 - \omega^2 x$$

and so $(0, 0)$ is the only critical point. Since

$$\mathbf{g}'(\mathbf{X}) = \begin{bmatrix} 0 & 1 \\ -\omega^2 & 2k - 3cy^2 \end{bmatrix}.$$

$\tau = 2k$ and $\Delta = \omega^2 > 0$. Since $k > 0$, $(0, 0)$ is an unstable critical point. Assuming that a Type I invariant region exists that contains $(0, 0)$ in its interior, we may apply Theorem 11.8(i) to conclude that there is at least one periodic solution.

12 Orthogonal Functions and Fourier Series

Exercises 12.1

1. $\int_{-2}^2 x x^2 dx = \frac{1}{4} x^4 \Big|_{-2}^2 = 0$
2. $\int_{-1}^1 x^3(x^2 + 1) dx = \frac{1}{6} x^6 \Big|_{-1}^1 + \frac{1}{4} x^4 \Big|_{-1}^1 = 0$
3. $\int_0^2 e^x(xe^{-x} - e^{-x}) dx = \int_0^2 (x - 1) dx = \left(\frac{1}{2} x^2 - x \right) \Big|_0^2 = 0$
4. $\int_0^\pi \cos x \sin^2 x dx = \frac{1}{3} \sin^3 x \Big|_0^\pi = 0$
5. $\int_{-\pi/2}^{\pi/2} x \cos 2x dx = \frac{1}{2} \left(\frac{1}{2} \cos 2x + x \sin 2x \right) \Big|_{-\pi/2}^{\pi/2} = 0$
6. $\int_{\pi/4}^{5\pi/4} e^x \sin x dx = \left(\frac{1}{2} e^x \sin x - \frac{1}{2} e^x \cos x \right) \Big|_{\pi/4}^{5\pi/4} = 0$
7. For $m \neq n$

$$\begin{aligned}
 & \int_0^{\pi/2} \sin(2n+1)x \sin(2m+1)x dx \\
 &= \frac{1}{2} \int_0^{\pi/2} [\cos 2(n-m)x - \cos 2(n+m+1)x] dx \\
 &= \frac{1}{4(n-m)} \sin 2(n-m)x \Big|_0^{\pi/2} - \frac{1}{4(n+m+1)} \sin 2(n+m+1)x \Big|_0^{\pi/2} \\
 &= 0.
 \end{aligned}$$

For $m = n$

$$\begin{aligned}
 \int_0^{\pi/2} \sin^2(2n+1)x dx &= \int_0^{\pi/2} \left(\frac{1}{2} - \frac{1}{2} \cos 2(2n+1)x \right) dx \\
 &= \frac{1}{2} x \Big|_0^{\pi/2} - \frac{1}{4(2n+1)} \sin 2(2n+1)x \Big|_0^{\pi/2} \\
 &= \frac{\pi}{4}
 \end{aligned}$$

so that

$$\| \sin(2n+1)x \| = \frac{1}{2} \sqrt{\pi}.$$

8. For $m \neq n$

$$\begin{aligned}
 \int_0^{\pi/2} \cos(2n+1)x \cos(2m+1)x \, dx &= \frac{1}{2} \int_0^{\pi/2} [\cos 2(n-m)x + \cos 2(n+m+1)x] \, dx \\
 &= \frac{1}{4(n-m)} \sin 2(n-m)x \Big|_0^{\pi/2} + \frac{1}{4(n+m+1)} \sin 2(n+m+1)x \Big|_0^{\pi/2} \\
 &= 0.
 \end{aligned}$$

For $m = n$

$$\begin{aligned}
 \int_0^{\pi/2} \cos^2(2n+1)x \, dx &= \int_0^{\pi/2} \left(\frac{1}{2} + \frac{1}{2} \cos 2(2n+1)x \right) \, dx \\
 &= \frac{1}{2}x \Big|_0^{\pi/2} + \frac{1}{4(2n+1)} \sin 2(2n+1)x \Big|_0^{\pi/2} \\
 &= \frac{\pi}{4}
 \end{aligned}$$

so that

$$\|\cos(2n+1)x\| = \frac{1}{2}\sqrt{\pi}.$$

9. For $m \neq n$

$$\begin{aligned}
 \int_0^{\pi} \sin nx \sin mx \, dx &= \frac{1}{2} \int_0^{\pi} [\cos(n-m)x - \cos(n+m)x] \, dx \\
 &= \frac{1}{2(n-m)} \sin(n-m)x \Big|_0^{\pi} - \frac{1}{2(n+m)} \sin 2(n+m)x \Big|_0^{\pi} \\
 &= 0.
 \end{aligned}$$

For $m = n$

$$\int_0^{\pi} \sin^2 nx \, dx = \int_0^{\pi} \left[\frac{1}{2} - \frac{1}{2} \cos 2nx \right] \, dx = \frac{1}{2}x \Big|_0^{\pi} - \frac{1}{4n} \sin 2nx \Big|_0^{\pi} = \frac{\pi}{2}$$

so that

$$\|\sin nx\| = \sqrt{\frac{\pi}{2}}.$$

10. For $m \neq n$

$$\begin{aligned}
 \int_0^p \sin \frac{n\pi}{p}x \sin \frac{m\pi}{p}x \, dx &= \frac{1}{2} \int_0^p \left(\cos \frac{(n-m)\pi}{p}x - \cos \frac{(n+m)\pi}{p}x \right) \, dx \\
 &= \frac{p}{2(n-m)\pi} \sin \frac{(n-m)\pi}{p}x \Big|_0^p - \frac{p}{2(n+m)\pi} \sin \frac{(n+m)\pi}{p}x \Big|_0^p \\
 &= 0.
 \end{aligned}$$

For $m = n$

$$\int_0^p \sin^2 \frac{n\pi}{p}x \, dx = \int_0^p \left[\frac{1}{2} - \frac{1}{2} \cos \frac{2n\pi}{p}x \right] \, dx = \frac{1}{2}x \Big|_0^p - \frac{p}{4n\pi} \sin \frac{2n\pi}{p}x \Big|_0^p = \frac{p}{2}$$

so that

$$\left\| \sin \frac{n\pi}{p}x \right\| = \sqrt{\frac{p}{2}}.$$

Exercises 12.1

11. For $m \neq n$

$$\begin{aligned} \int_0^p \cos \frac{n\pi}{p}x \cos \frac{m\pi}{p}x dx &= \frac{1}{2} \int_0^p \left(\cos \frac{(n-m)\pi}{p}x + \cos \frac{(n+m)\pi}{p}x \right) dx \\ &= \frac{p}{2(n-m)\pi} \sin \frac{(n-m)\pi}{p}x \Big|_0^p + \frac{p}{2(n+m)\pi} \sin \frac{(n+m)\pi}{p}x \Big|_0^p \\ &= 0. \end{aligned}$$

For $m = n$

$$\int_0^p \cos^2 \frac{n\pi}{p}x dx = \int_0^p \left(\frac{1}{2} + \frac{1}{2} \cos \frac{2n\pi}{p}x \right) dx = \frac{1}{2}x \Big|_0^p + \frac{p}{4n\pi} \sin \frac{2n\pi}{p}x \Big|_0^p = \frac{p}{2}.$$

Also

$$\int_0^p 1 \cdot \cos \frac{n\pi}{p}x dx = \frac{p}{n\pi} \sin \frac{n\pi}{p}x \Big|_0^p = 0 \quad \text{and} \quad \int_0^p 1^2 dx = p$$

so that

$$\|1\| = \sqrt{p} \quad \text{and} \quad \left\| \cos \frac{n\pi}{p}x \right\| = \sqrt{\frac{p}{2}}.$$

12. For $m \neq n$, we use Problems 11 and 10:

$$\begin{aligned} \int_{-p}^p \cos \frac{n\pi}{p}x \cos \frac{m\pi}{p}x dx &= 2 \int_0^p \cos \frac{n\pi}{p}x \cos \frac{m\pi}{p}x dx = 0 \\ \int_{-p}^p \sin \frac{n\pi}{p}x \sin \frac{m\pi}{p}x dx &= 2 \int_0^p \sin \frac{n\pi}{p}x \sin \frac{m\pi}{p}x dx = 0. \end{aligned}$$

Also

$$\begin{aligned} \int_{-p}^p \sin \frac{n\pi}{p}x \cos \frac{m\pi}{p}x dx &= \frac{1}{2} \int_{-p}^p \left(\sin \frac{(n-m)\pi}{p}x + \sin \frac{(n+m)\pi}{p}x \right) dx = 0, \\ \int_{-p}^p 1 \cdot \cos \frac{n\pi}{p}x dx &= \frac{p}{n\pi} \sin \frac{n\pi}{p}x \Big|_{-p}^p = 0, \\ \int_{-p}^p 1 \cdot \sin \frac{n\pi}{p}x dx &= -\frac{p}{n\pi} \cos \frac{n\pi}{p}x \Big|_{-p}^p = 0, \end{aligned}$$

and

$$\int_{-p}^p \sin \frac{n\pi}{p}x \cos \frac{n\pi}{p}x dx = \int_{-p}^p \frac{1}{2} \sin \frac{2n\pi}{p}x dx = -\frac{p}{4n\pi} \cos \frac{2n\pi}{p}x \Big|_{-p}^p = 0.$$

For $m = n$

$$\begin{aligned} \int_{-p}^p \cos^2 \frac{n\pi}{p}x dx &= \int_{-p}^p \left(\frac{1}{2} + \frac{1}{2} \cos \frac{2n\pi}{p}x \right) dx = p, \\ \int_{-p}^p \sin^2 \frac{n\pi}{p}x dx &= \int_{-p}^p \left(\frac{1}{2} - \frac{1}{2} \cos \frac{2n\pi}{p}x \right) dx = p, \end{aligned}$$

and

$$\int_{-p}^p 1^2 dx = 2p$$

so that

$$\|1\| = \sqrt{2p}, \quad \left\| \cos \frac{n\pi}{p}x \right\| = \sqrt{p}, \quad \text{and} \quad \left\| \sin \frac{n\pi}{p}x \right\| = \sqrt{p}.$$

13. Since

$$\int_{-\infty}^{\infty} e^{-x^2} \cdot 1 \cdot 2x \, dx = -e^{-x^2} \Big|_{-\infty}^0 - e^{-x^2} \Big|_0^{\infty} = 0,$$

$$\begin{aligned} \int_{-\infty}^{\infty} e^{-x^2} \cdot 1 \cdot (4x^2 - 2) \, dx &= 2 \int_{-\infty}^{\infty} x (2xe^{-x^2}) \, dx - 2 \int_{-\infty}^{\infty} e^{-x^2} \, dx \\ &= 2 \left(-xe^{-x^2} \Big|_{-\infty}^{\infty} + \int_{-\infty}^{\infty} e^{-x^2} \, dx \right) - 2 \int_{-\infty}^{\infty} e^{-x^2} \, dx \\ &= 2 \left(-xe^{-x^2} \Big|_{-\infty}^0 - xe^{-x^2} \Big|_0^{\infty} \right) = 0, \end{aligned}$$

and

$$\begin{aligned} \int_{-\infty}^{\infty} e^{-x^2} \cdot 2x \cdot (4x^2 - 2) \, dx &= 4 \int_{-\infty}^{\infty} x^2 (2xe^{-x^2}) \, dx - 4 \int_{-\infty}^{\infty} xe^{-x^2} \, dx \\ &= 4 \left(-x^2 e^{-x^2} \Big|_{-\infty}^{\infty} + 2 \int_{-\infty}^{\infty} xe^{-x^2} \, dx \right) - 4 \int_{-\infty}^{\infty} xe^{-x^2} \, dx \\ &= 4 \left(-x^2 e^{-x^2} \Big|_{-\infty}^0 - x^2 e^{-x^2} \Big|_0^{\infty} \right) + 2 \int_{-\infty}^{\infty} 2xe^{-x^2} \, dx = 0, \end{aligned}$$

the functions are orthogonal.

14. Since

$$\int_0^{\infty} e^{-x^2} \cdot 1(1-x) \, dx = (x-1)e^{-x} \Big|_0^{\infty} - \int_0^{\infty} e^{-x} \, dx = 0,$$

$$\begin{aligned} \int_0^{\infty} e^{-x} \cdot 1 \cdot \left(\frac{1}{2}x^2 - 2x + 1 \right) \, dx &= \left(2x - 1 - \frac{1}{2}x^2 \right) e^{-x} \Big|_0^{\infty} + \int_0^{\infty} e^{-x}(x-2) \, dx \\ &= 1 + (2-x)e^{-x} \Big|_0^{\infty} + \int_0^{\infty} e^{-x} \, dx = 0, \end{aligned}$$

and

$$\begin{aligned} \int_0^{\infty} e^{-x} \cdot (1-x) \left(\frac{1}{2}x^2 - 2x + 1 \right) \, dx &= \int_0^{\infty} e^{-x} \left(-\frac{1}{2}x^3 + \frac{5}{2}x^2 - 3x + 1 \right) \, dx \\ &= e^{-x} \left(\frac{1}{2}x^3 - \frac{5}{2}x^2 + 3x - 1 \right) \Big|_0^{\infty} + \int_0^{\infty} e^{-x} \left(-\frac{3}{2}x^2 + 5x - 3 \right) \, dx \\ &= 1 + e^{-x} \left(\frac{3}{2}x^2 - 5x + 3 \right) \Big|_0^{\infty} + \int_0^{\infty} e^{-x}(5-3x) \, dx \\ &= 1 - 3 + e^{-x}(3x-5) \Big|_0^{\infty} - 3 \int_0^{\infty} e^{-x} \, dx = 0, \end{aligned}$$

the functions are orthogonal.

15. By orthogonality $\int_a^b \phi_0(x)\phi_n(x)dx = 0$ for $n = 1, 2, 3, \dots$; that is, $\int_a^b \phi_n(x)dx = 0$ for $n = 1, 2, 3, \dots$.

Exercises 12.1

16. Using the facts that ϕ_0 and ϕ_1 are orthogonal to ϕ_n for $n > 1$, we have

$$\begin{aligned}\int_a^b (\alpha x + \beta) \phi_n(x) dx &= \alpha \int_a^b x \phi_n(x) dx + \beta \int_a^b 1 \cdot \phi_n(x) dx \\ &= \alpha \int_a^b \phi_1(x) \phi_n(x) dx + \beta \int_a^b \phi_0(x) \phi_n(x) dx \\ &= \alpha \cdot 0 + \beta \cdot 0 = 0\end{aligned}$$

for $n = 2, 3, 4, \dots$.

17. Using the fact that ϕ_n and ϕ_m are orthogonal for $n \neq m$ we have

$$\begin{aligned}\|\phi_m(x) + \phi_n(x)\|^2 &= \int_a^b [\phi_m(x) + \phi_n(x)]^2 dx = \int_a^b [\phi_m^2(x) + 2\phi_m(x)\phi_n(x) + \phi_n^2(x)] dx \\ &= \int_a^b \phi_m^2(x) dx + 2 \int_a^b \phi_m(x)\phi_n(x) dx + \int_a^b \phi_n^2(x) dx \\ &= \|\phi_m(x)\|^2 + \|\phi_n(x)\|^2.\end{aligned}$$

18. Setting

$$0 = \int_{-2}^2 f_3(x) f_1(x) dx = \int_{-2}^2 (x^2 + c_1 x^3 + c_2 x^4) dx = \frac{16}{3} + \frac{64}{5} c_2$$

and

$$0 = \int_{-2}^2 f_3(x) f_2(x) dx = \int_{-2}^2 (x^3 + c_1 x^4 + c_2 x^5) dx = \frac{64}{5} c_1$$

we obtain $c_1 = 0$ and $c_2 = -5/12$.

19. Since $\sin nx$ is an odd function on $[-\pi, \pi]$,

$$(1, \sin nx) = \int_{-\pi}^{\pi} \sin nx dx = 0$$

and $f(x) = 1$ is orthogonal to every member of $\{\sin nx\}$. Thus $\{\sin nx\}$ is not complete.

$$20. (f_1 + f_2, f_3) = \int_a^b [f_1(x) + f_2(x)] f_3(x) dx = \int_a^b f_1(x) f_3(x) dx + \int_a^b f_2(x) f_3(x) dx = (f_1, f_3) + (f_2, f_3)$$

21. (a) The fundamental period is $2\pi/2\pi = 1$.

(b) The fundamental period is $2\pi/(4/L) = \frac{1}{2}\pi L$.

(c) The fundamental period of $\sin x + \sin 2x$ is 2π .

(d) The fundamental period of $\sin 2x + \cos 4x$ is $2\pi/2 = \pi$.

(e) The fundamental period of $\sin 3x + \cos 4x$ is 2π since the smallest integer multiples of $2\pi/3$ and $2\pi/4 = \pi/2$ that are equal are 3 and 4, respectively.

(f) The fundamental period of $f(x)$ is $2\pi/(n\pi/p) = 2p/n$.

Exercises 12.2

1. $a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_0^{\pi} 1 dx = 1$
 $a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos \frac{n\pi}{\pi} x dx = \frac{1}{\pi} \int_0^{\pi} \cos nx dx = 0$
 $b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin \frac{n\pi}{\pi} x dx = \frac{1}{\pi} \int_0^{\pi} \sin nx dx = \frac{1}{n\pi} (1 - \cos n\pi) = \frac{1}{n\pi} [1 - (-1)^n]$
 $f(x) = \frac{1}{2} + \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n} \sin nx$

2. $a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_{-\pi}^0 -1 dx + \frac{1}{\pi} \int_0^{\pi} 2 dx = 1$
 $a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{1}{\pi} \int_{-\pi}^0 -\cos nx dx + \frac{1}{\pi} \int_0^{\pi} 2 \cos nx dx = 0$
 $b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = \frac{1}{\pi} \int_{-\pi}^0 -\sin nx dx + \frac{1}{\pi} \int_0^{\pi} 2 \sin nx dx = \frac{3}{n\pi} [1 - (-1)^n]$
 $f(x) = \frac{1}{2} + \frac{3}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n} \sin nx$

3. $a_0 = \int_{-1}^1 f(x) dx = \int_{-1}^0 1 dx + \int_0^1 x dx = \frac{3}{2}$
 $a_n = \int_{-1}^1 f(x) \cos n\pi x dx = \int_{-1}^0 \cos n\pi x dx + \int_0^1 x \cos n\pi x dx = \frac{1}{n^2\pi^2} [(-1)^n - 1]$
 $b_n = \int_{-1}^1 f(x) \sin n\pi x dx = \int_{-1}^0 \sin n\pi x dx + \int_0^1 x \sin n\pi x dx = -\frac{1}{n\pi}$
 $f(x) = \frac{3}{4} + \sum_{n=1}^{\infty} \left[\frac{(-1)^n - 1}{n^2\pi^2} \cos n\pi x - \frac{1}{n\pi} \sin n\pi x \right]$

4. $a_0 = \int_{-1}^1 f(x) dx = \int_0^1 x dx = \frac{1}{2}$
 $a_n = \int_{-1}^1 f(x) \cos n\pi x dx = \int_0^1 x \cos n\pi x dx = \frac{1}{n^2\pi^2} [(-1)^n - 1]$
 $b_n = \int_{-1}^1 f(x) \sin n\pi x dx = \int_0^1 x \sin n\pi x dx = \frac{(-1)^{n+1}}{n\pi}$
 $f(x) = \frac{3}{4} + \sum_{n=1}^{\infty} \left[\frac{(-1)^n - 1}{n^2\pi^2} \cos n\pi x + \frac{(-1)^{n+1}}{n\pi} \sin n\pi x \right]$

5. $a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_0^{\pi} x^2 dx = \frac{1}{3}\pi^2$
 $a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{1}{\pi} \int_0^{\pi} x^2 \cos nx dx = \frac{1}{\pi} \left(\frac{x^2}{\pi} \sin nx \Big|_0^{\pi} - \frac{2}{n} \int_0^{\pi} x \sin nx dx \right) = \frac{2(-1)^n}{n^2}$
 $b_n = \frac{1}{\pi} \int_0^{\pi} x^2 \sin nx dx = \frac{1}{\pi} \left(-\frac{x^2}{n} \cos nx \Big|_0^{\pi} + \frac{2}{n} \int_0^{\pi} x \cos nx dx \right) = \frac{\pi}{n} (-1)^{n+1} + \frac{2}{n^3\pi} [(-1)^n - 1]$

Exercises 12.2

$$f(x) = \frac{\pi^2}{6} + \sum_{n=1}^{\infty} \left[\frac{2(-1)^n}{n^2} \cos nx + \left(\frac{\pi}{n}(-1)^{n+1} + \frac{2[(-1)^n - 1]}{n^3\pi} \right) \sin nx \right]$$

$$6. \quad a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_{-\pi}^0 \pi^2 dx + \frac{1}{\pi} \int_0^{\pi} (\pi^2 - x^2) dx = \frac{5}{3}\pi^2$$

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{1}{\pi} \int_{-\pi}^0 \pi^2 \cos nx dx + \frac{1}{\pi} \int_0^{\pi} (\pi^2 - x^2) \cos nx dx \\ &= \frac{1}{\pi} \left(\frac{\pi^2 - x^2}{n} \sin nx \Big|_0^{\pi} + \frac{2}{n} \int_0^{\pi} x \sin nx dx \right) = \frac{2}{n^2}(-1)^{n+1} \end{aligned}$$

$$\begin{aligned} b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = \frac{1}{\pi} \int_{-\pi}^0 \pi^2 \sin nx dx + \frac{1}{\pi} \int_0^{\pi} (\pi^2 - x^2) \sin nx dx \\ &= \frac{\pi}{n} [(-1)^n - 1] + \frac{1}{\pi} \left(\frac{x^2 - \pi^2}{n} \cos nx \Big|_0^{\pi} - \frac{2}{n} \int_0^{\pi} x \cos nx dx \right) = \frac{\pi}{n}(-1)^n + \frac{2}{n^3\pi} [1 - (-1)^n] \end{aligned}$$

$$f(x) = \frac{5\pi^2}{6} + \sum_{n=1}^{\infty} \left[\frac{2}{n^2}(-1)^{n+1} \cos nx + \left(\frac{\pi}{n}(-1)^n + \frac{2[1 - (-1)^n]}{n^3\pi} \right) \sin nx \right]$$

$$7. \quad a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_{-\pi}^{\pi} (x + \pi) dx = 2\pi$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{1}{\pi} \int_{-\pi}^{\pi} (x + \pi) \cos nx dx = 0$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = \frac{2}{n}(-1)^{n+1}$$

$$f(x) = \pi + \sum_{n=1}^{\infty} \frac{2}{n}(-1)^{n+1} \sin nx$$

$$8. \quad a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_{-\pi}^{\pi} (3 - 2x) dx = 6$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{1}{\pi} \int_{-\pi}^{\pi} (3 - 2x) \cos nx dx = 0$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} (3 - 2x) \sin nx dx = \frac{4}{n}(-1)^n$$

$$f(x) = 3 + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \sin nx$$

$$9. \quad a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_0^{\pi} \sin x dx = \frac{2}{\pi}$$

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{1}{\pi} \int_0^{\pi} \sin x \cos nx dx = \frac{1}{2\pi} \int_0^{\pi} [\sin(n+1)x + \sin(1-n)x] dx \\ &= \frac{1 + (-1)^n}{\pi(1 - n^2)} \quad \text{for } n = 2, 3, 4, \dots \end{aligned}$$

$$a_1 = \frac{1}{2\pi} \int_0^{\pi} \sin 2x dx = 0$$

$$\begin{aligned} b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = \frac{1}{\pi} \int_0^{\pi} \sin x \sin nx dx \\ &= \frac{1}{2\pi} \int_0^{\pi} [\cos(1-n)x - \cos(1+n)x] dx = 0 \quad \text{for } n = 2, 3, 4, \dots \end{aligned}$$

$$b_1 = \frac{1}{2\pi} \int_0^\pi (1 - \cos 2x) dx = \frac{1}{2}$$

$$f(x) = \frac{1}{\pi} + \frac{1}{2} \sin x + \sum_{n=2}^{\infty} \frac{1 + (-1)^n}{\pi(1 - n^2)} \cos nx$$

$$10. a_0 = \frac{2}{\pi} \int_{-\pi/2}^{\pi/2} f(x) dx = \frac{2}{\pi} \int_0^{\pi/2} \cos x dx = \frac{2}{\pi}$$

$$\begin{aligned} a_n &= \frac{2}{\pi} \int_{-\pi/2}^{\pi/2} f(x) \cos 2nx dx = \frac{2}{\pi} \int_0^{\pi/2} \cos x \cos 2nx dx = \frac{1}{\pi} \int_0^{\pi/2} [\cos(2n-1)x + \cos(2n+1)x] dx \\ &= \frac{2(-1)^{n+1}}{\pi(4n^2-1)} \end{aligned}$$

$$\begin{aligned} b_n &= \frac{2}{\pi} \int_{-\pi/2}^{\pi/2} f(x) \sin 2nx dx = \frac{2}{\pi} \int_0^{\pi/2} \cos x \sin 2nx dx = \frac{1}{\pi} \int_0^{\pi/2} [\sin(2n-1)x + \sin(2n+1)x] dx \\ &= \frac{4n}{\pi(4n^2-1)} \end{aligned}$$

$$f(x) = \frac{1}{\pi} + \sum_{n=1}^{\infty} \left[\frac{2(-1)^{n+1}}{\pi(4n^2-1)} \cos 2nx + \frac{4n}{\pi(4n^2-1)} \sin 2nx \right]$$

$$11. a_0 = \frac{1}{2} \int_{-2}^2 f(x) dx = \frac{1}{2} \left(\int_{-1}^0 -2 dx + \int_0^1 1 dx \right) = -\frac{1}{2}$$

$$a_n = \frac{1}{2} \int_{-2}^2 f(x) \cos \frac{n\pi}{2} x dx = \frac{1}{2} \left(\int_{-1}^0 \left[-2 \cos \frac{n\pi}{2} x \right] dx + \int_0^1 \cos \frac{n\pi}{2} x dx \right) = -\frac{1}{n\pi} \sin \frac{n\pi}{2}$$

$$b_n = \frac{1}{2} \int_{-2}^2 f(x) \sin \frac{n\pi}{2} x dx = \frac{1}{2} \left(\int_{-1}^0 \left[-2 \sin \frac{n\pi}{2} x \right] dx + \int_0^1 \sin \frac{n\pi}{2} x dx \right) = \frac{3}{n\pi} \left(1 - \cos \frac{n\pi}{2} \right)$$

$$f(x) = -\frac{1}{4} + \sum_{n=1}^{\infty} \left[-\frac{1}{n\pi} \sin \frac{n\pi}{2} \cos \frac{n\pi}{2} x + \frac{3}{n\pi} \left(1 - \cos \frac{n\pi}{2} \right) \sin \frac{n\pi}{2} x \right]$$

$$12. a_0 = \frac{1}{2} \int_{-2}^2 f(x) dx = \frac{1}{2} \left(\int_0^1 x dx + \int_1^2 1 dx \right) = \frac{3}{4}$$

$$a_n = \frac{1}{2} \int_{-2}^2 f(x) \cos \frac{n\pi}{2} x dx = \frac{1}{2} \left(\int_0^1 x \cos \frac{n\pi}{2} x dx + \int_1^2 \cos \frac{n\pi}{2} x dx \right) = \frac{2}{n^2\pi^2} \left(\cos \frac{n\pi}{2} - 1 \right)$$

$$b_n = \frac{1}{2} \int_{-2}^2 f(x) \sin \frac{n\pi}{2} x dx = \frac{1}{2} \left(\int_0^1 x \sin \frac{n\pi}{2} x dx + \int_1^2 \sin \frac{n\pi}{2} x dx \right) = \frac{2}{n^2\pi^2} \left(\sin \frac{n\pi}{2} + \frac{n\pi}{2} (-1)^{n+1} \right)$$

$$f(x) = \frac{3}{8} + \sum_{n=1}^{\infty} \left[\frac{2}{n^2\pi^2} \left(\cos \frac{n\pi}{2} - 1 \right) \cos \frac{n\pi}{2} x + \frac{2}{n^2\pi^2} \left(\sin \frac{n\pi}{2} + \frac{n\pi}{2} (-1)^{n+1} \right) \sin \frac{n\pi}{2} x \right]$$

$$13. a_0 = \frac{1}{5} \int_{-5}^5 f(x) dx = \frac{1}{5} \left(\int_{-5}^0 1 dx + \int_0^5 (1+x) dx \right) = \frac{9}{2}$$

$$a_n = \frac{1}{5} \int_{-5}^5 f(x) \cos \frac{n\pi}{5} x dx = \frac{1}{5} \left(\int_{-5}^0 \cos \frac{n\pi}{5} x dx + \int_0^5 (1+x) \cos \frac{n\pi}{5} x dx \right) = \frac{5}{n^2\pi^2} [(-1)^n - 1]$$

$$b_n = \frac{1}{5} \int_{-5}^5 f(x) \sin \frac{n\pi}{5} x dx = \frac{1}{5} \left(\int_{-5}^0 \sin \frac{n\pi}{5} x dx + \int_0^5 (1+x) \sin \frac{n\pi}{5} x dx \right) = \frac{5}{n\pi} (-1)^{n+1}$$

$$f(x) = \frac{9}{4} + \sum_{n=1}^{\infty} \left[\frac{5}{n^2\pi^2} [(-1)^n - 1] \cos \frac{n\pi}{5} x + \frac{5}{n\pi} (-1)^{n+1} \sin \frac{n\pi}{5} x \right]$$

Exercises 12.2

$$\begin{aligned}
 14. \quad a_0 &= \frac{1}{2} \int_{-2}^2 f(x) dx = \frac{1}{2} \left(\int_{-2}^0 (2+x) dx + \int_0^2 2 dx \right) = 3 \\
 a_n &= \frac{1}{2} \int_{-2}^2 f(x) \cos \frac{n\pi}{2} x dx = \frac{1}{2} \left(\int_{-2}^0 (2+x) \cos \frac{n\pi}{2} x dx + \int_0^2 2 \cos \frac{n\pi}{2} x dx \right) = \frac{2}{n^2 \pi^2} [1 - (-1)^n] \\
 b_n &= \frac{1}{2} \int_{-2}^2 f(x) \sin \frac{n\pi}{2} x dx = \frac{1}{2} \left(\int_{-2}^0 (2+x) \sin \frac{n\pi}{2} x dx + \int_0^2 2 \sin \frac{n\pi}{2} x dx \right) = \frac{2}{n\pi} (-1)^{n+1} \\
 f(x) &= \frac{3}{2} + \sum_{n=1}^{\infty} \left[\frac{2}{n^2 \pi^2} [1 - (-1)^n] \cos \frac{n\pi}{2} x + \frac{2}{n\pi} (-1)^{n+1} \sin \frac{n\pi}{2} x \right]
 \end{aligned}$$

$$\begin{aligned}
 15. \quad a_0 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_{-\pi}^{\pi} e^x dx = \frac{1}{\pi} (e^{\pi} - e^{-\pi}) \\
 a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{(-1)^n (e^{\pi} - e^{-\pi})}{\pi(1+n^2)} \\
 b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = \frac{1}{\pi} \int_{-\pi}^{\pi} e^x \sin nx dx = \frac{(-1)^n n (e^{-\pi} - e^{\pi})}{\pi(1+n^2)} \\
 f(x) &= \frac{e^{\pi} - e^{-\pi}}{2\pi} + \sum_{n=1}^{\infty} \left[\frac{(-1)^n (e^{\pi} - e^{-\pi})}{\pi(1+n^2)} \cos nx + \frac{(-1)^n n (e^{-\pi} - e^{\pi})}{\pi(1+n^2)} \sin nx \right]
 \end{aligned}$$

$$\begin{aligned}
 16. \quad a_0 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_0^{\pi} (e^x - 1) dx = \frac{1}{\pi} (e^{\pi} - \pi - 1) \\
 a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{1}{\pi} \int_0^{\pi} (e^x - 1) \cos nx dx = \frac{[e^{\pi}(-1)^n - 1]}{\pi(1+n^2)} \\
 b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = \frac{1}{\pi} \int_0^{\pi} (e^x - 1) \sin nx dx = \frac{1}{\pi} \left(\frac{n e^{\pi} (-1)^{n+1}}{1+n^2} + \frac{n}{1+n^2} + \frac{(-1)^n}{n} - \frac{1}{n} \right) \\
 f(x) &= \frac{e^{\pi} - \pi - 1}{2\pi} + \sum_{n=1}^{\infty} \left[\frac{e^{\pi}(-1)^n - 1}{\pi(1+n^2)} \cos nx + \left(\frac{n}{1+n^2} [e^{\pi}(-1)^{n+1} + 1] + \frac{(-1)^n - 1}{n} \right) \sin nx \right]
 \end{aligned}$$

17. The function in Problem 5 is discontinuous at $x = \pi$, so the corresponding Fourier series converges to $\pi^2/2$ at $x = \pi$. That is,

$$\begin{aligned}
 \frac{\pi^2}{2} &= \frac{\pi^2}{6} + \sum_{n=1}^{\infty} \left[\frac{2(-1)^n}{n^2} \cos n\pi + \left(\frac{\pi}{n} (-1)^{n+1} + \frac{2[(-1)^n - 1]}{n^3 \pi} \right) \sin n\pi \right] \\
 &= \frac{\pi^2}{6} + \sum_{n=1}^{\infty} \frac{2(-1)^n}{n^2} (-1)^n = \frac{\pi^2}{6} + \sum_{n=1}^{\infty} \frac{2}{n^2} = \frac{\pi^2}{6} + 2 \left(1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots \right)
 \end{aligned}$$

and

$$\frac{\pi^2}{6} = \frac{1}{2} \left(\frac{\pi^2}{2} - \frac{\pi^2}{6} \right) = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots$$

At $x = 0$ the series converges to 0 and

$$0 = \frac{\pi^2}{6} + \sum_{n=1}^{\infty} \frac{2(-1)^n}{n^2} = \frac{\pi^2}{6} + 2 \left(-1 + \frac{1}{2^2} - \frac{1}{3^2} + \frac{1}{4^2} - \cdots \right)$$

so

$$\frac{\pi^2}{12} = 1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots$$

18. From Problem 17

$$\frac{\pi^2}{8} = \frac{1}{2} \left(\frac{\pi^2}{6} + \frac{\pi^2}{12} \right) = \frac{1}{2} \left(2 + \frac{2}{3^2} + \frac{2}{5^2} + \cdots \right) = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \cdots$$

19. The function in Problem 7 is continuous at $x = \pi/2$ so

$$\frac{3\pi}{2} = f\left(\frac{\pi}{2}\right) = \pi + \sum_{n=1}^{\infty} \frac{2}{n} (-1)^{n+1} \sin \frac{n\pi}{2} = \pi + 2 \left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots\right)$$

and

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots.$$

20. The function in Problem 9 is continuous at $x = \pi/2$ so

$$\begin{aligned} 1 &= f\left(\frac{\pi}{2}\right) = \frac{1}{\pi} + \frac{1}{2} + \sum_{n=2}^{\infty} \frac{1 + (-1)^n}{\pi(1 - n^2)} \cos \frac{n\pi}{2} \\ &= \frac{1}{\pi} + \frac{1}{2} + \frac{2}{3\pi} - \frac{2}{3 \cdot 5\pi} + \frac{2}{5 \cdot 7\pi} - \cdots \end{aligned}$$

and

$$\pi = 1 + \frac{\pi}{2} + \frac{2}{3} - \frac{2}{3 \cdot 5} + \frac{2}{5 \cdot 7} - \cdots$$

or

$$\frac{\pi}{4} = \frac{1}{2} + \frac{1}{1 \cdot 3} - \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} - \cdots.$$

21. Writing

$$f(x) = \frac{a_0}{2} + a_1 \cos \frac{\pi}{p} x + \cdots + a_n \cos \frac{n\pi}{p} x + \cdots + b_1 \sin \frac{\pi}{p} x + \cdots + b_n \sin \frac{n\pi}{p} x + \cdots$$

we see that $f^2(x)$ consists exclusively of squared terms of the form

$$\frac{a_0^2}{4}, \quad a_n^2 \cos^2 \frac{n\pi}{p} x, \quad b_n^2 \sin^2 \frac{n\pi}{p} x$$

and cross-product terms, with $m \neq n$, of the form

$$\begin{aligned} a_0 a_n \cos \frac{n\pi}{p} x, \quad a_0 b_n \sin \frac{n\pi}{p} x, \quad 2a_m a_n \cos \frac{m\pi}{p} x \cos \frac{n\pi}{p} x, \\ 2a_m b_n \cos \frac{m\pi}{p} x \sin \frac{n\pi}{p} x, \quad 2b_m b_n \sin \frac{m\pi}{p} x \sin \frac{n\pi}{p} x. \end{aligned}$$

The integral of each cross-product term taken over the interval $(-p, p)$ is zero by orthogonality. For the squared terms we have

$$\frac{a_0^2}{4} \int_{-p}^p dx = \frac{a_0^2 p}{2}, \quad a_n^2 \int_{-p}^p \cos^2 \frac{n\pi}{p} x dx = a_n^2 p, \quad b_n^2 \int_{-p}^p \sin^2 \frac{n\pi}{p} x dx = b_n^2 p.$$

Thus

$$RMS(f) = \sqrt{\frac{1}{4} a_0^2 + \frac{1}{2} \sum_{n=1}^{\infty} (a_n^2 + b_n^2)}.$$

Exercises 12.3

1. Since $f(-x) = \sin(-3x) = -\sin 3x = -f(x)$, $f(x)$ is an odd function.
2. Since $f(-x) = -x \cos(-x) = -x \cos x = -f(x)$, $f(x)$ is an odd function.
3. Since $f(-x) = (-x)^2 - x = x^2 - x$, $f(x)$ is neither even nor odd.
4. Since $f(-x) = (-x)^3 + 4x = -(x^3 - 4x) = -f(x)$, $f(x)$ is an odd function.
5. Since $f(-x) = e^{|-x|} = e^{|x|} = f(x)$, $f(x)$ is an even function.
6. Since $f(-x) = e^{-x} - e^x = -f(x)$, $f(x)$ is an odd function.
7. For $0 < x < 1$, $f(-x) = (-x)^2 = x^2 = -f(x)$, $f(x)$ is an odd function.
8. For $0 \leq x < 2$, $f(-x) = -x + 5 = f(x)$, $f(x)$ is an even function.
9. Since $f(x)$ is not defined for $x < 0$, it is neither even nor odd.
10. Since $f(-x) = |(-x)^5| = |x^5| = f(x)$, $f(x)$ is an even function.
11. Since $f(x)$ is an odd function, we expand in a sine series:

$$b_n = \frac{2}{\pi} \int_0^\pi 1 \cdot \sin nx \, dx = \frac{2}{n\pi} [1 - (-1)^n].$$

Thus

$$f(x) = \sum_{n=1}^{\infty} \frac{2}{n\pi} [1 - (-1)^n] \sin nx.$$

12. Since $f(x)$ is an even function, we expand in a cosine series:

$$a_0 = \int_1^2 1 \, dx = 1$$

$$a_n = \int_1^2 \cos \frac{n\pi}{2} x \, dx = -\frac{2}{n\pi} \sin \frac{n\pi}{2}.$$

Thus

$$f(x) = \frac{1}{2} + \sum_{n=1}^{\infty} \frac{-2}{n\pi} \sin \frac{n\pi}{2} \cos \frac{n\pi}{2} x.$$

13. Since $f(x)$ is an even function, we expand in a cosine series:

$$a_0 = \frac{2}{\pi} \int_0^\pi x \, dx = \pi$$

$$a_n = \frac{2}{\pi} \int_0^\pi x \cos nx \, dx = \frac{2}{n^2\pi} [(-1)^n - 1].$$

Thus

$$f(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{n^2\pi} [(-1)^n - 1] \cos nx.$$

14. Since $f(x)$ is an odd function, we expand in a sine series:

$$b_n = \frac{2}{\pi} \int_0^\pi x \sin nx \, dx = \frac{2}{n} (-1)^{n+1}.$$

Thus

$$f(x) = \sum_{n=1}^{\infty} \frac{2}{n} (-1)^{n+1} \sin nx.$$

15. Since $f(x)$ is an even function, we expand in a cosine series:

$$a_0 = 2 \int_0^1 x^2 dx = \frac{2}{3}$$

$$a_n = 2 \int_0^1 x^2 \cos n\pi x dx = 2 \left(\frac{x^2}{n\pi} \sin n\pi x \Big|_0^1 - \frac{2}{n\pi} \int_0^1 x \sin n\pi x dx \right) = \frac{4}{n^2 \pi^2} (-1)^n.$$

Thus

$$f(x) = \frac{1}{3} + \sum_{n=1}^{\infty} \frac{4}{n^2 \pi^2} (-1)^n \cos n\pi x.$$

16. Since $f(x)$ is an odd function, we expand in a sine series:

$$b_n = 2 \int_0^1 x^2 \sin n\pi x dx = 2 \left(-\frac{x^2}{n\pi} \cos n\pi x \Big|_0^1 + \frac{2}{n\pi} \int_0^1 x \cos n\pi x dx \right)$$

$$= \frac{2(-1)^{n+1}}{n\pi} + \frac{4}{n^3 \pi^3} [(-1)^n - 1].$$

Thus

$$f(x) = \sum_{n=1}^{\infty} \left(\frac{2(-1)^{n+1}}{n\pi} + \frac{4}{n^3 \pi^3} [(-1)^n - 1] \right) \sin n\pi x.$$

17. Since $f(x)$ is an even function, we expand in a cosine series:

$$a_0 = \frac{2}{\pi} \int_0^{\pi} (\pi^2 - x^2) dx = \frac{4}{3} \pi^2$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} (\pi^2 - x^2) \cos nx dx = \frac{2}{\pi} \left(\frac{\pi^2 - x^2}{n} \sin nx \Big|_0^{\pi} + \frac{2}{n} \int_0^{\pi} x \sin nx dx \right) = \frac{4}{n^2} (-1)^{n+1}.$$

Thus

$$f(x) = \frac{2}{3} \pi^2 + \sum_{n=1}^{\infty} \frac{4}{n^2} (-1)^{n+1} \cos nx dx.$$

18. Since $f(x)$ is an odd function, we expand in a sine series:

$$b_n = \frac{2}{\pi} \int_0^{\pi} x^3 \sin nx dx = \frac{2}{\pi} \left(-\frac{x^3}{n} \cos nx \Big|_0^{\pi} + \frac{3}{n} \int_0^{\pi} x^2 \cos nx dx \right)$$

$$= \frac{2\pi^2}{n} (-1)^{n+1} - \frac{12}{n^2 \pi} \int_0^{\pi} x \sin nx dx$$

$$= \frac{2\pi^2}{n} (-1)^{n+1} - \frac{12}{n^2 \pi} \left(-\frac{x}{n} \cos nx \Big|_0^{\pi} + \frac{1}{n} \int_0^{\pi} \cos nx dx \right) = \frac{2\pi^2}{n} (-1)^{n+1} + \frac{12}{n^3} (-1)^n.$$

Thus

$$f(x) = \sum_{n=1}^{\infty} \left(\frac{2\pi^2}{n} (-1)^{n+1} + \frac{12}{n^3} (-1)^n \right) \sin nx.$$

19. Since $f(x)$ is an odd function, we expand in a sine series:

$$b_n = \frac{2}{\pi} \int_0^{\pi} (x+1) \sin nx dx = \frac{2(\pi+1)}{n\pi} (-1)^{n+1} + \frac{2}{n\pi}.$$

Thus

$$f(x) = \sum_{n=1}^{\infty} \left(\frac{2(\pi+1)}{n\pi} (-1)^{n+1} + \frac{2}{n\pi} \right) \sin nx.$$

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20. Since $f(x)$ is an odd function, we expand in a sine series:

$$\begin{aligned} b_n &= 2 \int_0^1 (x-1) \sin n\pi x \, dx = 2 \left[\int_0^1 x \sin n\pi x \, dx - \int_0^1 \sin n\pi x \, dx \right] \\ &= 2 \left[\frac{1}{n^2\pi^2} \sin n\pi x - \frac{x}{n\pi} \cos n\pi x + \frac{1}{n\pi} \cos n\pi x \right]_0^1 = -\frac{2}{n\pi}. \end{aligned}$$

Thus

$$f(x) = -\sum_{n=1}^{\infty} \frac{2}{n\pi} \sin n\pi x.$$

21. Since $f(x)$ is an even function, we expand in a cosine series:

$$\begin{aligned} a_0 &= \int_0^1 x \, dx + \int_1^2 1 \, dx = \frac{3}{2} \\ a_n &= \int_0^1 x \cos \frac{n\pi}{2} x \, dx + \int_1^2 \cos \frac{n\pi}{2} x \, dx = \frac{4}{n^2\pi^2} \left(\cos \frac{n\pi}{2} - 1 \right). \end{aligned}$$

Thus

$$f(x) = \frac{3}{4} + \sum_{n=1}^{\infty} \frac{4}{n^2\pi^2} \left(\cos \frac{n\pi}{2} - 1 \right) \cos \frac{n\pi}{2} x.$$

22. Since $f(x)$ is an odd function, we expand in a sine series:

$$b_n = \frac{1}{\pi} \int_0^{\pi} x \sin \frac{n}{2} x \, dx + \int_{\pi}^{2\pi} \pi \sin \frac{n}{2} x \, dx = \frac{4}{n^2\pi} \sin \frac{n\pi}{2} + \frac{2}{n} (-1)^{n+1}.$$

Thus

$$f(x) = \sum_{n=1}^{\infty} \left(\frac{4}{n^2\pi} \sin \frac{n\pi}{2} + \frac{2}{n} (-1)^{n+1} \right) \sin \frac{n}{2} x.$$

23. Since $f(x)$ is an even function, we expand in a cosine series:

$$\begin{aligned} a_0 &= \frac{2}{\pi} \int_0^{\pi} \sin x \, dx = \frac{4}{\pi} \\ a_n &= \frac{2}{\pi} \int_0^{\pi} \sin x \cos nx \, dx = \frac{1}{\pi} \int_0^{\pi} [\sin(n+1)x + \sin(1-n)x] \, dx \\ &= \frac{2}{\pi(1-n^2)} [1 + (-1)^n] \quad \text{for } n = 2, 3, 4, \dots \\ a_1 &= \frac{1}{\pi} \int_0^{\pi} \sin 2x \, dx = 0. \end{aligned}$$

Thus

$$f(x) = \frac{2}{\pi} + \sum_{n=2}^{\infty} \frac{2[1 + (-1)^n]}{\pi(1-n^2)} \cos nx.$$

24. Since $f(x)$ is an even function, we expand in a cosine series. [See the solution of Problem 10 in Exercise 12.2 for the computation of the integrals.]

$$\begin{aligned} a_0 &= \frac{2}{\pi/2} \int_0^{\pi/2} \cos x \, dx = \frac{4}{\pi} \\ a_n &= \frac{2}{\pi/2} \int_0^{\pi/2} \cos x \cos \frac{n\pi}{\pi/2} x \, dx = \frac{4(-1)^{n+1}}{\pi(4n^2-1)} \end{aligned}$$

Thus

$$f(x) = \frac{2}{\pi} + \sum_{n=1}^{\infty} \frac{4(-1)^{n+1}}{\pi(4n^2 - 1)} \cos 2nx.$$

$$25. \quad a_0 = 2 \int_0^{1/2} 1 \, dx = 1$$

$$a_n = 2 \int_0^{1/2} 1 \cdot \cos n\pi x \, dx = \frac{2}{n\pi} \sin \frac{n\pi}{2}$$

$$b_n = 2 \int_0^{1/2} 1 \cdot \sin n\pi x \, dx = \frac{2}{n\pi} \left(1 - \cos \frac{n\pi}{2}\right)$$

$$f(x) = \frac{1}{2} + \sum_{n=1}^{\infty} \frac{2}{n\pi} \sin \frac{n\pi}{2} \cos n\pi x$$

$$f(x) = \sum_{n=1}^{\infty} \frac{2}{n\pi} \left(1 - \cos \frac{n\pi}{2}\right) \sin n\pi x$$

$$26. \quad a_0 = 2 \int_{1/2}^1 1 \, dx = 1$$

$$a_n = 2 \int_{1/2}^1 1 \cdot \cos n\pi x \, dx = -\frac{2}{n\pi} \sin \frac{n\pi}{2}$$

$$b_n = 2 \int_{1/2}^1 1 \cdot \sin n\pi x \, dx = \frac{2}{n\pi} \left(\cos \frac{n\pi}{2} + (-1)^{n+1}\right)$$

$$f(x) = \frac{1}{2} + \sum_{n=1}^{\infty} \left(-\frac{2}{n\pi} \sin \frac{n\pi}{2} \cos n\pi x\right)$$

$$f(x) = \sum_{n=1}^{\infty} \frac{2}{n\pi} \left(\cos \frac{n\pi}{2} + (-1)^{n+1}\right) \sin n\pi x$$

$$27. \quad a_0 = \frac{4}{\pi} \int_0^{\pi/2} \cos x \, dx = \frac{4}{\pi}$$

$$a_n = \frac{4}{\pi} \int_0^{\pi/2} \cos x \cos 2nx \, dx = \frac{2}{\pi} \int_0^{\pi/2} [\cos(2n+1)x + \cos(2n-1)x] \, dx = \frac{4(-1)^n}{\pi(1-4n^2)}$$

$$b_n = \frac{4}{\pi} \int_0^{\pi/2} \cos x \sin 2nx \, dx = \frac{2}{\pi} \int_0^{\pi/2} [\sin(2n+1)x + \sin(2n-1)x] \, dx = \frac{8n}{\pi(4n^2-1)}$$

$$f(x) = \frac{2}{\pi} + \sum_{n=1}^{\infty} \frac{4(-1)^n}{\pi(1-4n^2)} \cos 2nx$$

$$f(x) = \sum_{n=1}^{\infty} \frac{8n}{\pi(4n^2-1)} \sin 2nx$$

$$28. \quad a_0 = \frac{2}{\pi} \int_0^{\pi} \sin x \, dx = \frac{4}{\pi}$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} \sin x \cos nx \, dx = \frac{1}{\pi} \int_0^{\pi} [\sin(n+1)x - \sin(n-1)x] \, dx = \frac{2[(-1)^n + 1]}{\pi(1-n^2)} \quad \text{for } n = 2, 3, 4, \dots$$

$$b_n = \frac{2}{\pi} \int_0^{\pi} \sin x \sin nx \, dx = \frac{1}{\pi} \int_0^{\pi} [\cos(n-1)x - \cos(n+1)x] \, dx = 0 \quad \text{for } n = 2, 3, 4, \dots$$

$$a_1 = \frac{1}{\pi} \int_0^{\pi} \sin 2x \, dx = 0$$

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$$b_1 = \frac{2}{\pi} \int_0^\pi \sin^2 x \, dx = 1$$

$$f(x) = \sin x$$

$$f(x) = \frac{2}{\pi} + \frac{2}{\pi} \sum_{n=2}^{\infty} \frac{(-1)^n + 1}{1 - n^2} \cos nx$$

$$29. \quad a_0 = \frac{2}{\pi} \left(\int_0^{\pi/2} x \, dx + \int_{\pi/2}^{\pi} (\pi - x) \, dx \right) = \frac{\pi}{2}$$

$$a_n = \frac{2}{\pi} \left(\int_0^{\pi/2} x \cos nx \, dx + \int_{\pi/2}^{\pi} (\pi - x) \cos nx \, dx \right) = \frac{2}{n^2 \pi} \left(2 \cos \frac{n\pi}{2} + (-1)^{n+1} - 1 \right)$$

$$b_n = \frac{2}{\pi} \left(\int_0^{\pi/2} x \sin nx \, dx + \int_{\pi/2}^{\pi} (\pi - x) \sin nx \, dx \right) = \frac{4}{n^2 \pi} \sin \frac{n\pi}{2}$$

$$f(x) = \frac{\pi}{4} + \sum_{n=1}^{\infty} \frac{2}{n^2 \pi} \left(2 \cos \frac{n\pi}{2} + (-1)^{n+1} - 1 \right) \cos nx$$

$$f(x) = \sum_{n=1}^{\infty} \frac{4}{n^2 \pi} \sin \frac{n\pi}{2} \sin nx$$

$$30. \quad a_0 = \frac{1}{\pi} \int_{\pi}^{2\pi} (x - \pi) \, dx = \frac{\pi}{2}$$

$$a_n = \frac{1}{\pi} \int_{\pi}^{2\pi} (x - \pi) \cos \frac{n}{2} x \, dx = \frac{4}{n^2 \pi} \left[(-1)^n - \cos \frac{n\pi}{2} \right]$$

$$b_n = \frac{1}{\pi} \int_{\pi}^{2\pi} (x - \pi) \sin \frac{n}{2} x \, dx = \frac{2}{n} (-1)^{n+1} - \frac{4}{n^2 \pi} \sin \frac{n\pi}{2}$$

$$f(x) = \frac{\pi}{4} + \sum_{n=1}^{\infty} \frac{4}{n^2 \pi} \left[(-1)^n - \cos \frac{n\pi}{2} \right] \cos \frac{n}{2} x$$

$$f(x) = \sum_{n=1}^{\infty} \left(\frac{2}{n} (-1)^{n+1} - \frac{4}{n^2 \pi} \sin \frac{n\pi}{2} \right) \sin \frac{n}{2} x$$

$$31. \quad a_0 = \int_0^1 x \, dx + \int_1^2 1 \, dx = \frac{3}{2}$$

$$a_n = \int_0^1 x \cos \frac{n\pi}{2} x \, dx = \frac{4}{n^2 \pi^2} \left(\cos \frac{n\pi}{2} - 1 \right)$$

$$b_n = \int_0^1 x \sin \frac{n\pi}{2} x \, dx + \int_1^2 1 \cdot \sin \frac{n\pi}{2} x \, dx = \frac{4}{n^2 \pi^2} \sin \frac{n\pi}{2} + \frac{2}{n\pi} (-1)^{n+1}$$

$$f(x) = \frac{3}{4} + \sum_{n=1}^{\infty} \frac{4}{n^2 \pi^2} \left(\cos \frac{n\pi}{2} - 1 \right) \cos \frac{n\pi}{2} x$$

$$f(x) = \sum_{n=1}^{\infty} \left(\frac{4}{n^2 \pi^2} \sin \frac{n\pi}{2} + \frac{2}{n\pi} (-1)^{n+1} \right) \sin \frac{n\pi}{2} x$$

$$32. a_0 = \int_0^1 1 \, dx + \int_1^2 (2-x) \, dx = \frac{3}{2}$$

$$a_n = \int_0^1 1 \cdot \cos \frac{n\pi}{2} x \, dx + \int_1^2 (2-x) \cos \frac{n\pi}{2} x \, dx = \frac{4}{n^2 \pi^2} \left(\cos \frac{n\pi}{2} + (-1)^{n+1} \right)$$

$$b_n = \int_0^1 1 \cdot \sin \frac{n\pi}{2} x \, dx + \int_1^2 (2-x) \sin \frac{n\pi}{2} x \, dx = \frac{2}{n\pi} + \frac{4}{n^2 \pi^2} \sin \frac{n\pi}{2}$$

$$f(x) = \frac{3}{4} + \sum_{n=1}^{\infty} \frac{4}{n^2 \pi^2} \left(\cos \frac{n\pi}{2} + (-1)^{n+1} \right) \cos \frac{n\pi}{2} x$$

$$f(x) = \sum_{n=1}^{\infty} \left(\frac{2}{n\pi} + \frac{4}{n^2 \pi^2} \sin \frac{n\pi}{2} \right) \sin \frac{n\pi}{2} x$$

$$33. a_0 = 2 \int_0^1 (x^2 + x) \, dx = \frac{5}{3}$$

$$a_n = 2 \int_0^1 (x^2 + x) \cos n\pi x \, dx = \frac{2(x^2 + x)}{n\pi} \sin n\pi x \Big|_0^1 - \frac{2}{n\pi} \int_0^1 (2x + 1) \sin n\pi x \, dx = \frac{2}{n^2 \pi^2} [3(-1)^n - 1]$$

$$b_n = 2 \int_0^1 (x^2 + x) \sin n\pi x \, dx = -\frac{2(x^2 + x)}{n\pi} \cos n\pi x \Big|_0^1 + \frac{2}{n\pi} \int_0^1 (2x + 1) \cos n\pi x \, dx$$

$$= \frac{4}{n\pi} (-1)^{n+1} + \frac{4}{n^3 \pi^3} [(-1)^n - 1]$$

$$f(x) = \frac{5}{6} + \sum_{n=1}^{\infty} \frac{2}{n^2 \pi^2} [3(-1)^n - 1] \cos n\pi x$$

$$f(x) = \sum_{n=1}^{\infty} \left(\frac{4}{n\pi} (-1)^{n+1} + \frac{4}{n^3 \pi^3} [(-1)^n - 1] \right) \sin n\pi x$$

$$34. a_0 = \int_0^2 (2x - x^2) \, dx = \frac{4}{3}$$

$$a_n = \int_0^2 (2x - x^2) \cos \frac{n\pi}{2} x \, dx = \frac{8}{n^2 \pi^2} [(-1)^{n+1} - 1]$$

$$b_n = \int_0^2 (2x - x^2) \sin \frac{n\pi}{2} x \, dx = \frac{16}{n^3 \pi^3} [1 - (-1)^n]$$

$$f(x) = \frac{2}{3} + \sum_{n=1}^{\infty} \frac{8}{n^2 \pi^2} [(-1)^{n+1} - 1] \cos \frac{n\pi}{2} x$$

$$f(x) = \sum_{n=1}^{\infty} \frac{16}{n^3 \pi^3} [1 - (-1)^n] \sin \frac{n\pi}{2} x$$

$$35. a_0 = \frac{1}{\pi} \int_0^{2\pi} x^2 \, dx = \frac{8}{3} \pi^2$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} x^2 \cos nx \, dx = \frac{4}{n^2}$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} x^2 \sin nx \, dx = -\frac{4\pi}{n}$$

$$f(x) = \frac{4}{3} \pi^2 + \sum_{n=1}^{\infty} \left(\frac{4}{n^2} \cos nx - \frac{4\pi}{n} \sin nx \right)$$

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$$36. a_0 = \frac{2}{\pi} \int_0^\pi x \, dx = \pi$$

$$a_n = \frac{2}{\pi} \int_0^\pi x \cos 2nx \, dx = 0$$

$$b_n = \frac{2}{\pi} \int_0^\pi x \sin 2nx \, dx = -\frac{1}{n}$$

$$f(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \left(-\frac{1}{n} \sin 2nx \right)$$

$$37. a_0 = 2 \int_0^1 (x+1) \, dx = 3$$

$$a_n = 2 \int_0^1 (x+1) \cos 2n\pi x \, dx = 0$$

$$b_n = 2 \int_0^1 (x+1) \sin 2n\pi x \, dx = -\frac{1}{n\pi}$$

$$f(x) = \frac{3}{2} - \sum_{n=1}^{\infty} \frac{1}{n\pi} \sin 2n\pi x$$

$$38. a_0 = 2 \int_0^2 (2-x) \, dx = 2$$

$$a_n = 2 \int_0^2 (2-x) \cos n\pi x \, dx = 0$$

$$b_n = 2 \int_0^2 (2-x) \sin n\pi x \, dx = \frac{2}{n\pi}$$

$$f(x) = 1 + \sum_{n=1}^{\infty} \frac{2}{n\pi} \sin n\pi x$$

39. We have

$$b_n = \frac{2}{\pi} \int_0^\pi 5 \sin nt \, dt = \frac{10}{n\pi} [1 - (-1)^n]$$

so that

$$f(t) = \sum_{n=1}^{\infty} \frac{10[1 - (-1)^n]}{n\pi} \sin nt.$$

Substituting the assumption $x_p(t) = \sum_{n=1}^{\infty} B_n \sin nt$ into the differential equation then gives

$$x_p'' + 10x_p = \sum_{n=1}^{\infty} B_n(10 - n^2) \sin nt = \sum_{n=1}^{\infty} \frac{10[1 - (-1)^n]}{n\pi} \sin nt$$

and so $B_n = \frac{10[1 - (-1)^n]}{n\pi(10 - n^2)}$. Thus

$$x_p(t) = \frac{10}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n(10 - n^2)} \sin nt.$$

40. We have

$$b_n = \frac{2}{\pi} \int_0^1 (1-t) \sin n\pi t \, dt = \frac{2}{n\pi}$$

so that

$$f(t) = \sum_{n=1}^{\infty} \frac{2}{n\pi} \sin n\pi t.$$

Substituting the assumption $x_p(t) = \sum_{n=1}^{\infty} B_n \sin n\pi t$ into the differential equation then gives

$$x_p'' + 10x_p = \sum_{n=1}^{\infty} B_n(10 - n^2\pi^2) \sin n\pi t = \sum_{n=1}^{\infty} \frac{2}{n\pi} \sin n\pi t$$

and so $B_n = \frac{2}{n\pi(10 - n^2\pi^2)}$. Thus

$$x_p(t) = \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{n(10 - n^2\pi^2)} \sin n\pi t.$$

41. We have

$$a_0 = \frac{2}{\pi} \int_0^{\pi} (2\pi t - t^2) dt = \frac{4}{3}\pi^2$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} (2\pi t - t^2) \cos nt dt = -4n^2$$

so that

$$f(t) = \frac{2\pi^2}{3} + \sum_{n=1}^{\infty} \frac{(-4)}{n^2} \cos nt.$$

Substituting the assumption

$$x_p(t) = \frac{A_0}{2} + \sum_{n=1}^{\infty} A_n \cos nt$$

into the differential equation then gives

$$\frac{1}{4} x_p'' + 12x_p = 6A_0 + \sum_{n=1}^{\infty} A_n \left(-\frac{1}{4}n^2 + 12 \right) \cos nt = \frac{2\pi^2}{3} + \sum_{n=1}^{\infty} \frac{(-4)}{n^2} \cos nt$$

and $A_0 = \frac{\pi^2}{9}$, $A_n = \frac{16}{n^2(n^2 - 48)}$. Thus

$$x_p(t) = \frac{\pi^2}{18} + 16 \sum_{n=1}^{\infty} \frac{1}{n^2(n^2 - 48)} \cos nt.$$

42. We have

$$a_0 = \frac{2}{(1/2)} \int_0^{1/2} t dt = \frac{1}{2}$$

$$a_n = \frac{2}{(1/2)} \int_0^{1/2} t \cos 2n\pi t dt = \frac{1}{n^2\pi^2} [(-1)^n - 1]$$

so that

$$f(t) = \frac{1}{4} + \sum_{n=1}^{\infty} \frac{(-1)^n - 1}{n^2\pi^2} \cos 2n\pi t.$$

Substituting the assumption

$$x_p(t) = \frac{A_0}{2} + \sum_{n=1}^{\infty} A_n \cos 2n\pi t$$

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into the differential equation then gives

$$\frac{1}{4}x_p'' + 12x_p = 6A_0 + \sum_{n=1}^{\infty} A_n(12 - n^2\pi^2) \cos 2n\pi t = \frac{1}{4} + \sum_{n=1}^{\infty} \frac{(-1)^n - 1}{n^2\pi^2} \cos 2n\pi t$$

and $A_0 = \frac{1}{24}$, $A_n = \frac{(-1)^n - 1}{n^2\pi^2(12 - n^2\pi^2)}$. Thus

$$x_p(t) = \frac{1}{48} + \frac{1}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n - 1}{n^2(12 - n^2\pi^2)} \cos 2n\pi t.$$

- 43. (a)** The general solution is $x(t) = c_1 \cos \sqrt{10}t + c_2 \sin \sqrt{10}t + x_p(t)$, where

$$x_p(t) = \frac{10}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n(10 - n^2)} \sin nt.$$

The initial condition $x(0) = 0$ implies $c_1 + x_p(0) = 0$. Since $x_p(0) = 0$, we have $c_1 = 0$ and $x(t) = c_2 \sin \sqrt{10}t + x_p(t)$. Then $x'(t) = c_2 \sqrt{10} \cos \sqrt{10}t + x_p'(t)$ and $x'(0) = 0$ implies

$$c_2 \sqrt{10} + \frac{10}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{10 - n^2} \cos 0 = 0.$$

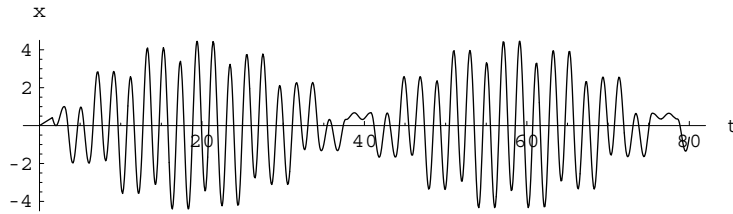
Thus

$$c_2 = -\frac{\sqrt{10}}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{10 - n^2}$$

and

$$x(t) = \frac{10}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{10 - n^2} \left[\frac{1}{n} \sin nt - \frac{1}{\sqrt{10}} \sin \sqrt{10}t \right].$$

- (b)** The graph is plotted using eight nonzero terms in the series expansion of $x(t)$.



- 44. (a)** The general solution is $x(t) = c_1 \cos 4\sqrt{3}t + c_2 \sin 4\sqrt{3}t + x_p(t)$, where

$$x_p(t) = \frac{\pi^2}{18} + 16 \sum_{n=1}^{\infty} \frac{1}{n^2(n^2 - 48)} \cos nt.$$

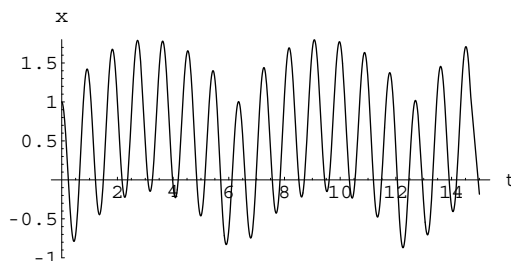
The initial condition $x(0) = 0$ implies $c_1 + x_p(0) = 1$ or

$$c_1 = 1 - x_p(0) = 1 - \frac{\pi^2}{18} - 16 \sum_{n=1}^{\infty} \frac{1}{n^2(n^2 - 48)}.$$

Now $x'(t) = -4\sqrt{3}c_1 \sin 4\sqrt{3}t + 4c_2 \cos 4\sqrt{3}t + x_p'(t)$, so $x'(0) = 0$ implies $4c_2 + x_p'(0) = 0$. Since $x_p'(0) = 0$, we have $c_2 = 0$ and

$$\begin{aligned}
 x(t) &= \left(1 - \frac{\pi^2}{18} - 16 \sum_{n=1}^{\infty} \frac{1}{n^2(n^2 - 48)}\right) \cos 4\sqrt{3}t + \frac{\pi^2}{18} + 16 \sum_{n=1}^{\infty} \frac{1}{n^2(n^2 - 48)} \cos nt \\
 &= \frac{\pi^2}{18} + \left(1 - \frac{\pi^2}{18}\right) \cos 4\sqrt{3}t + 16 \sum_{n=1}^{\infty} \frac{1}{n^2(n^2 - 48)} [\cos nt - \cos 4\sqrt{3}t].
 \end{aligned}$$

(b) The graph is plotted using five nonzero terms in the series expansion of $x(t)$.



45. We have

$$b_n = \frac{2}{L} \int_0^L \frac{w_0 x}{L} \sin \frac{n\pi}{L} x \, dx = \frac{2w_0}{n\pi} (-1)^{n+1}$$

so that

$$w(x) = \sum_{n=1}^{\infty} \frac{2w_0}{n\pi} (-1)^{n+1} \sin \frac{n\pi}{L} x.$$

If we assume $y(x) = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi}{L} x$ then

$$y^{(4)} = \sum_{n=1}^{\infty} \frac{n^4 \pi^4}{L^4} B_n \sin \frac{n\pi}{L} x$$

and so the differential equation $EIy^{(4)} = w(x)$ gives

$$B_n = \frac{2w_0(-1)^{n+1}L^4}{EI\pi^5 n^5}.$$

Thus

$$y(x) = \frac{2w_0L^4}{EI\pi^5} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^5} \sin \frac{n\pi}{L} x.$$

46. We have

$$b_n = \frac{2}{L} \int_{L/3}^{2L/3} w_0 \sin \frac{n\pi}{L} x \, dx = \frac{2w_0}{n\pi} \left[\cos \frac{n\pi}{3} - \cos \frac{2n\pi}{3} \right]$$

so that

$$w(x) = \sum_{n=1}^{\infty} \frac{2w_0}{n\pi} \left[\cos \frac{n\pi}{3} - \cos \frac{2n\pi}{3} \right] \sin \frac{n\pi}{L} x.$$

If we assume $y(x) = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi}{L} x$ then

$$y^{(4)} = \sum_{n=1}^{\infty} \frac{n^4 \pi^4}{L^4} B_n \sin \frac{n\pi}{L} x$$

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and so the differential equation $EIy^{(4)} = w(x)$ gives

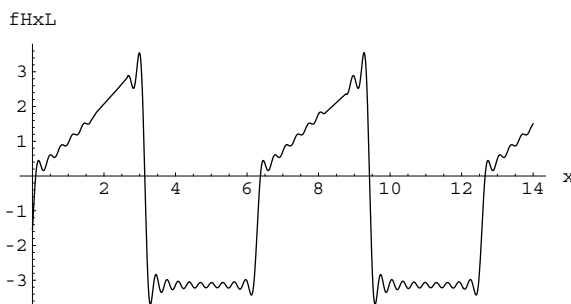
$$B_n = 2w_0L^4 \frac{\cos \frac{n\pi}{3} - \cos \frac{2n\pi}{3}}{EI n^5 \pi^5}.$$

Thus

$$y(x) = \frac{2w_0L^4}{EI\pi^5} \sum_{n=1}^{\infty} \frac{\cos \frac{n\pi}{3} - \cos \frac{2n\pi}{3}}{n^5} \sin \frac{n\pi}{L} x.$$

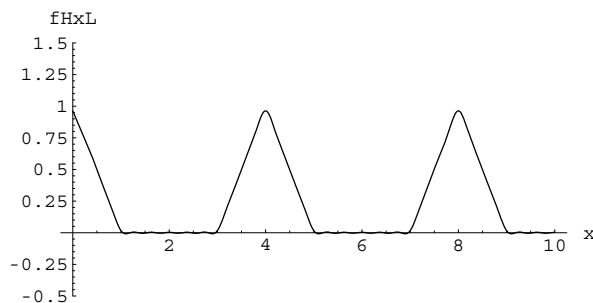
47. The graph is obtained by summing the series from $n = 1$ to 20. It appears that

$$f(x) = \begin{cases} x, & 0 < x < \pi \\ -\pi, & \pi < x < 2\pi. \end{cases}$$



48. The graph is obtained by summing the series from $n = 1$ to 10. It appears that

$$f(x) = \begin{cases} 1 - x, & 0 < x < 1 \\ 0, & 1 < x < 2. \end{cases}$$



49. (a) If f and g are even and $h(x) = f(x)g(x)$ then

$$h(-x) = f(-x)g(-x) = f(x)g(x) = h(x)$$

and h is even.

(c) If f is even and g is odd and $h(x) = f(x)g(x)$ then

$$h(-x) = f(-x)g(-x) = f(x)[-g(x)] = -h(x)$$

and h is odd.

(d) Let $h(x) = f(x) \pm g(x)$ where f and g are even. Then

$$h(-x) = f(-x) \pm g(-x) = f(x) \pm g(x),$$

and h is an even function.

(f) If f is even then

$$\int_{-a}^a f(x) dx = -\int_a^0 f(-u) du + \int_0^a f(x) dx = \int_0^a f(u) du + \int_0^a f(x) dx = 2 \int_0^a f(x) dx.$$

(g) If f is odd then

$$\begin{aligned} \int_{-a}^a f(x) dx &= -\int_{-a}^0 f(-x) dx + \int_0^a f(x) dx = \int_a^0 f(u) du + \int_0^a f(x) dx \\ &= -\int_0^a f(u) du + \int_0^a f(x) dx = 0. \end{aligned}$$

50. The answers are not unique. The function in Problem 47 could also be

$$f(x) = \begin{cases} x, & 0 < x < \pi \\ 1, & x = \pi \\ -\pi, & \pi < x < 2\pi. \end{cases}$$

In Problem 48 the function could also be

$$f(x) = \begin{cases} 0, & -2 < x < -1 \\ x+1, & -1 < x < 0 \\ -x+1, & 0 < x < 1 \\ 0, & 1 < x < 2. \end{cases}$$

Exercises 12.4

In this section we make use of the following identities due to Euler's formula:

$$e^{in\pi} = e^{-in\pi} = (-1)^n, \quad e^{-2in\pi} = 1, \quad e^{-in\pi/2} = (-i)^n.$$

1. Identifying $p = 2$ we have

$$\begin{aligned} c_n &= \frac{1}{4} \int_{-2}^2 f(x) e^{-in\pi x/2} dx = \frac{1}{4} \left[\int_{-2}^0 (-1) e^{-in\pi x/2} dx + \int_0^2 e^{-in\pi x/2} dx \right] \\ &= \frac{i}{2n\pi} [-1 + e^{in\pi} + e^{-in\pi} - 1] = \frac{i}{2n\pi} [-1 + (-1)^n + (-1)^n - 1] = \frac{1 - (-1)^n}{n\pi i} \end{aligned}$$

and

$$c_0 = \frac{1}{4} \int_{-2}^2 f(x) dx = 0.$$

Thus

$$f(x) = \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} \frac{1 - (-1)^n}{in\pi} e^{in\pi x/2}.$$

Exercises 12.4

2. Identifying $2p = 2$ or $p = 1$ we have

$$\begin{aligned} c_n &= \frac{1}{2} \int_0^2 f(x) e^{-in\pi x} dx = \frac{1}{2} \int_1^2 e^{-in\pi x} dx = -\frac{1}{2in\pi} e^{-in\pi x} \Big|_1^2 \\ &= -\frac{1}{2in\pi} (e^{-2in\pi} - e^{-in\pi}) = -\frac{1}{2in\pi} [1 - (-1)^n] = \frac{i}{2n\pi} [1 - (-1)^n] \end{aligned}$$

and

$$c_0 = \frac{1}{2} \int_0^2 f(x) dx = \frac{1}{2} \int_1^2 dx = \frac{1}{2}.$$

Thus

$$f(x) = \frac{1}{2} + \frac{i}{2\pi} \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} \frac{1 - (-1)^n}{n} e^{in\pi x}.$$

3. Identifying $p = 1/2$ we have

$$\begin{aligned} c_n &= \int_{-1/2}^{1/2} f(x) e^{-2in\pi x} dx = \int_0^{1/4} e^{-2in\pi x} dx = -\frac{1}{2in\pi} e^{-2in\pi x} \Big|_0^{1/4} \\ &= -\frac{1}{2in\pi} [e^{-in\pi/2} - 1] = -\frac{1}{2in\pi} [(-i)^n - 1] = \frac{i}{2n\pi} [(-i)^n - 1] \end{aligned}$$

and

$$c_0 = \int_0^{1/4} dx = \frac{1}{4}.$$

Thus

$$f(x) = \frac{1}{4} + \frac{i}{2\pi} \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} \frac{(-i)^n - 1}{n} e^{2in\pi x}.$$

4. Identifying $p = \pi$ we have

$$\begin{aligned} c_n &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx/\pi} dx = \frac{1}{2\pi} \int_0^{\pi} x e^{-inx/\pi} dx \\ &= \frac{1}{2} \left(\frac{\pi}{n^2} + \frac{ix}{n} \right) e^{-inx/\pi} \Big|_0^{\pi} = \frac{\pi(1+in)}{2n^2} e^{-in} - \frac{\pi}{2n^2} \end{aligned}$$

and

$$c_0 = \frac{1}{2\pi} \int_0^{\pi} x dx = \frac{\pi}{4}.$$

Thus

$$f(x) = \frac{\pi}{4} + \frac{\pi}{2} \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} \frac{1}{n^2} [(1+in)e^{-in} - 1] e^{inx}.$$

5. Identifying $2p = 2\pi$ or $p = \pi$ we have

$$\begin{aligned} c_n &= \frac{1}{2\pi} \int_0^{2\pi} f(x) e^{-inx} dx = \frac{1}{2\pi} \int_0^{2\pi} x e^{-inx} dx \\ &= \frac{1}{2\pi} \left(\frac{1}{n^2} + \frac{ix}{n} \right) e^{-inx} \Big|_0^{2\pi} = \frac{1+2in\pi}{2n^2\pi} - \frac{1}{2n^2\pi} = \frac{i}{n} \end{aligned}$$

and

$$c_0 = \frac{1}{2\pi} \int_0^{2\pi} x \, dx = \pi.$$

Thus

$$f(x) = \pi + \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} \frac{i}{n} e^{inx}.$$

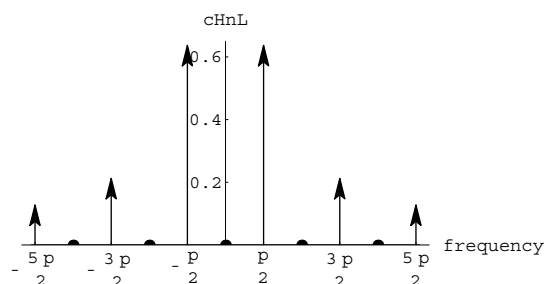
6. Identifying $p = 1$ we have

$$\begin{aligned} c_n &= \frac{1}{2} \int_{-1}^1 f(x) e^{-in\pi x} dx = \frac{1}{2} \left[\int_{-1}^0 e^x e^{-in\pi x} dx + \int_0^1 e^{-x} e^{-in\pi x} dx \right] \\ &= \frac{1}{2} \left[-\frac{1}{1-in\pi} e^{(1-in\pi)x} \Big|_{-1}^0 - \frac{1}{1+in\pi} e^{-(1+in\pi)x} \Big|_0^1 \right] \\ &= \frac{e - (-1)^n}{e(1-in\pi)} + \frac{1 - e^{-1}(-1)^n}{1+in\pi} = \frac{2[e - (-1)^n]}{e(1+n^2\pi^2)}. \end{aligned}$$

Thus

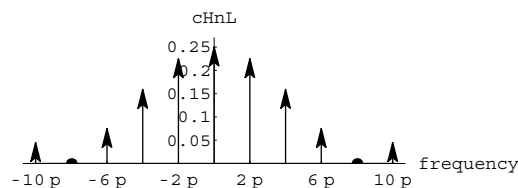
$$f(x) = \sum_{n=-\infty}^{\infty} \frac{2[e - (-1)^n]}{e(1+n^2\pi^2)} e^{in\pi x}.$$

7. The fundamental period is $T = 4$, so $\omega = 2\pi/4 = \pi/2$ and the values of $n\omega$ are $0, \pm\pi/2, \pm\pi, \pm3\pi/2, \dots$. From Problem 1, $c_0 = 0$ and $|c_n| = (1 - (-1)^n)/n\pi$. The table shows some values of n with corresponding values of $|c_n|$. The graph is a portion of the frequency spectrum.



n	-5	-4	-3	-2	-1	0	1	2	3	4	5
c _n	0.1273	0.0000	0.2122	0.0000	0.6366	0.0000	0.6366	0.0000	0.2122	0.0000	0.1273

8. The fundamental period is $T = 1$, so $\omega = 2\pi$ and the values of $n\omega$ are $0, \pm2\pi, \pm4\pi, \pm6\pi, \dots$. From Problem 3, $c_0 = \frac{1}{4}$ and $|c_n| = |(-i)^n - 1|/2n\pi$, or $c_1 = c_{-1} = \sqrt{2}/2\pi$, $c_2 = c_{-2} = 1/2\pi$, $c_3 = c_{-3} = \sqrt{2}/6\pi$, $c_4 = c_{-4} = 0$, $c_5 = c_{-5} = \sqrt{2}/10\pi$, $c_6 = c_{-6} = 1/6\pi$, $c_7 = c_{-7} = \sqrt{2}/14\pi$, $c_8 = c_{-8} = 0, \dots$. The table shows some values of n with corresponding values of $|c_n|$. The graph is a portion of the frequency spectrum.

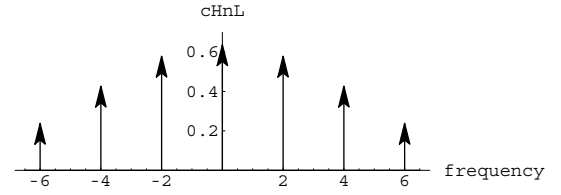
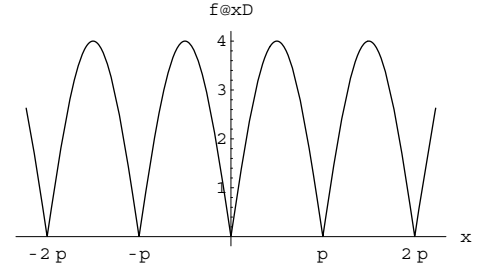


n	-5	-4	-3	-2	-1	0	1	2	3	4	5
c _n	0.0450	0.0000	0.0750	0.1592	0.2251	0.2500	0.2251	0.1592	0.0750	0.0000	0.0450

Exercises 12.4

9. Identifying $2p = \pi$ or $p = \pi/2$, and using $\sin x = (e^{ix} - e^{-ix})/2i$, we have

$$\begin{aligned}
 c_n &= \frac{1}{\pi} \int_0^\pi f(x) e^{-2inx/\pi} dx \\
 &= \frac{1}{\pi} \int_0^\pi (\sin x) e^{-2inx/\pi} dx \\
 &= \frac{1}{\pi} \int_0^\pi \frac{1}{2i} (e^{ix} - e^{-ix}) e^{-2inx/\pi} dx \\
 &= \frac{1}{2\pi i} \int_0^\pi \left(e^{(1-2n/\pi)ix} - e^{-(1+2n/\pi)ix} \right) dx \\
 &= \frac{1}{2\pi i} \left[\frac{1}{i(1-2n/\pi)} e^{(1-2n/\pi)ix} \right. \\
 &\quad \left. + \frac{1}{i(1+2n/\pi)} e^{-(1+2n/\pi)ix} \right]_0^\pi \\
 &= \frac{\pi(1 + e^{-2in})}{\pi^2 - 4n^2}.
 \end{aligned}$$

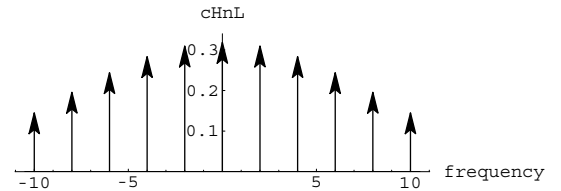
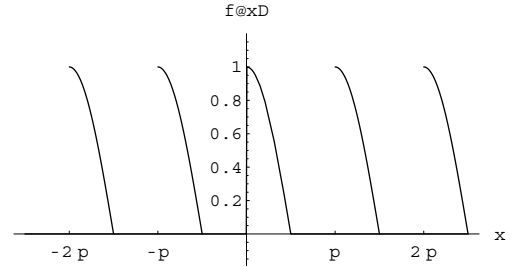


The fundamental period is $T = \pi$, so $\omega = 2\pi/\pi = 2$ and the values of $n\omega$ are $0, \pm 2, \pm 4, \pm 6, \dots$. Values of $|c_n|$ for $n = 0, \pm 1, \pm 2, \pm 3, \pm 4$, and ± 5 are shown in the table. The bottom graph is a portion of the frequency spectrum.

n	-5	-4	-3	-2	-1	0	1	2	3	4	5
c_n	0.0198	0.0759	0.2380	0.4265	0.5784	0.6366	0.5784	0.4265	0.2380	0.0759	0.0198

10. Identifying $2p = \pi$ or $p = \pi/2$, and using $\cos x = (e^{ix} + e^{-ix})/2$, we have

$$\begin{aligned}
 c_n &= \frac{1}{\pi} \int_0^\pi f(x) e^{-2inx/\pi} dx \\
 &= \frac{1}{\pi} \int_0^{\pi/2} (\cos x) e^{-2inx/\pi} dx \\
 &= \frac{1}{\pi} \int_0^{\pi/2} \frac{1}{2} (e^{ix} + e^{-ix}) e^{-2inx/\pi} dx \\
 &= \frac{1}{2\pi} \int_0^{\pi/2} \left(e^{(1-2n/\pi)ix} + e^{-(1+2n/\pi)ix} \right) dx \\
 &= \frac{1}{2\pi} \left[\frac{1}{i(1-2n/\pi)} e^{(1-2n/\pi)ix} \right. \\
 &\quad \left. + \frac{1}{i(1+2n/\pi)} e^{-(1+2n/\pi)ix} \right]_0^{\pi/2} \\
 &= \frac{2ne^{-in} + i\pi}{\pi^2 - 4n^2}.
 \end{aligned}$$



The fundamental period is $T = \pi$, so $\omega = 2\pi/\pi = 2$ and the values of $n\omega$ are $0, \pm 2, \pm 4, \pm 6, \dots$. Values of $|c_n|$ for $n = 0, \pm 1, \pm 2, \pm 3, \pm 4$, and ± 5 are shown in the table. The bottom graph is a portion of the frequency spectrum.

n	-5	-4	-3	-2	-1	0	1	2	3	4	5
c_n	0.1447	0.1954	0.2437	0.2833	0.3093	0.3183	0.3093	0.2833	0.2437	0.1954	0.1447

11. (a) Adding $c_n = \frac{1}{2}(a_n - ib_n)$ and $c_{-n} = \frac{1}{2}(a_n + ib_n)$ we get $c_n + c_{-n} = a_n$. Subtracting, we get $c_n - c_{-n} = -ib_n$. Multiplying both sides by i we obtain $i(c_n - c_{-n}) = b_n$.

(b) From

$$a_n = c_n + c_{-n} = (-1)^n \frac{\sinh \pi}{\pi} \left[\frac{1 - in}{n^2 + 1} + \frac{1 + in}{n^2 + 1} \right] = \frac{2(-1)^n \sinh \pi}{\pi(n^2 + 1)}, \quad n = 0, 1, 2, \dots$$

and

$$b_n = i(c_n - c_{-n}) = i(-1)^n \frac{\sinh \pi}{\pi} \left[\frac{1 - in}{n^2 + 1} - \frac{1 + in}{n^2 + 1} \right] = i(-1)^n \frac{\sinh \pi}{\pi} \left[-\frac{2in}{n^2 + 1} \right] = \frac{2(-1)^n n \sinh \pi}{\pi(n^2 + 1)},$$

the Fourier series of f is

$$f(x) = \frac{\sinh \pi}{\pi} + \frac{2 \sinh \pi}{\pi} \sum_{n=1}^{\infty} \left[\frac{(-1)^n}{n^2 + 1} \cos nx + \frac{n(-1)^n}{n^2 + 1} \sin nx \right].$$

12. From Problem 11 and the fact that f is odd, $c_n + c_{-n} = a_n = 0$, so $c_{-n} = -c_n$. Then $b_n = i(c_n - c_{-n}) = 2ic_n$. From Problem 1, $b_n = 2i[1 - (-1)^n]/n\pi = 2[1 - (-1)^n]/n\pi$, and the Fourier sine series of f is

$$f(x) = \sum_{i=1}^{\infty} \frac{2[1 - (-1)^n]}{n\pi} \sin \frac{n\pi x}{2}.$$

Exercises 12.5

1. For $\lambda \leq 0$ the only solution of the boundary-value problem is $y = 0$. For $\lambda > 0$ we have

$$y = c_1 \cos \sqrt{\lambda} x + c_2 \sin \sqrt{\lambda} x.$$

Now

$$y'(x) = -c_1 \sqrt{\lambda} \sin \sqrt{\lambda} x + c_2 \sqrt{\lambda} \cos \sqrt{\lambda} x$$

and $y'(0) = 0$ implies $c_2 = 0$, so

$$y(1) + y'(1) = c_1(\cos \sqrt{\lambda} - \sqrt{\lambda} \sin \sqrt{\lambda}) = 0 \quad \text{or} \quad \cot \sqrt{\lambda} = \sqrt{\lambda}.$$

The eigenvalues are $\lambda_n = x_n^2$ where $x_1, x_2, x_3, \dots$ are the consecutive positive solutions of $\cot \sqrt{\lambda} = \sqrt{\lambda}$. The corresponding eigenfunctions are $\cos \sqrt{\lambda_n} x = \cos x_n x$ for $n = 1, 2, 3, \dots$. Using a CAS we find that the first four eigenvalues are 0.7402, 11.7349, 41.4388, and 90.8082 with corresponding eigenfunctions $\cos 0.8603x$, $\cos 3.4256x$, $\cos 6.4373x$, and $\cos 9.5293x$.

2. For $\lambda < 0$ the only solution of the boundary-value problem is $y = 0$. For $\lambda = 0$ we have $y = c_1 x + c_2$. Now $y' = c_1$ and the boundary conditions both imply $c_1 + c_2 = 0$. Thus, $\lambda = 0$ is an eigenvalue with corresponding eigenfunction $y_0 = x - 1$.

Exercises 12.5

For $\lambda > 0$ we have

$$y = c_1 \cos \sqrt{\lambda} x + c_2 \sin \sqrt{\lambda} x$$

and

$$y' = -c_1 \sqrt{\lambda} \sin \sqrt{\lambda} x + c_2 \sqrt{\lambda} \cos \sqrt{\lambda} x.$$

The boundary conditions imply

$$\begin{aligned} c_1 + c_2 \sqrt{\lambda} &= 0 \\ c_1 \cos \sqrt{\lambda} + c_2 \sin \sqrt{\lambda} &= 0 \end{aligned}$$

which gives

$$-c_2 \sqrt{\lambda} \cos \sqrt{\lambda} + c_2 \sin \sqrt{\lambda} = 0 \quad \text{or} \quad \tan \sqrt{\lambda} = \sqrt{\lambda}.$$

The eigenvalues are $\lambda_n = x_n^2$ where $x_1, x_2, x_3, \dots$ are the consecutive positive solutions of $\tan \sqrt{\lambda} = \sqrt{\lambda}$. The corresponding eigenfunctions are $\sqrt{\lambda_n} \cos \sqrt{\lambda_n} x - \sin \sqrt{\lambda_n} x$ (obtained by taking $c_2 = -1$ in the first equation of the system.) Using a CAS we find that the first four positive eigenvalues are 20.1907, 59.6795, 118.9000, and 197.858 with corresponding eigenfunctions $4.4934 \cos 4.4934x - \sin 4.4934x$, $7.7253 \cos 7.7253x - \sin 7.7253x$, $10.9041 \cos 10.9041x - \sin 10.9041x$, and $14.0662 \cos 14.0662x - \sin 14.0662x$.

3. For $\lambda = 0$ the solution of $y'' = 0$ is $y = c_1 x + c_2$. The condition $y'(0) = 0$ implies $c_1 = 0$, so $\lambda = 0$ is an eigenvalue with corresponding eigenfunction 1. For $\lambda < 0$ we have $y = c_1 \cosh \sqrt{-\lambda} x + c_2 \sinh \sqrt{-\lambda} x$ and $y' = c_1 \sqrt{-\lambda} \sinh \sqrt{-\lambda} x + c_2 \sqrt{-\lambda} \cosh \sqrt{-\lambda} x$. The condition $y'(0) = 0$ implies $c_2 = 0$ and so $y = c_1 \cosh \sqrt{-\lambda} x$. Now the condition $y'(L) = 0$ implies $c_1 = 0$. Thus $y = 0$ and there are no negative eigenvalues. For $\lambda > 0$ we have $y = c_1 \cos \sqrt{\lambda} x + c_2 \sin \sqrt{\lambda} x$ and $y' = -c_1 \sqrt{\lambda} \sin \sqrt{\lambda} x + c_2 \sqrt{\lambda} \cos \sqrt{\lambda} x$. The condition $y'(0) = 0$ implies $c_2 = 0$ and so $y = c_1 \cos \sqrt{\lambda} x$. Now the condition $y'(L) = 0$ implies $-c_1 \sqrt{\lambda} \sin \sqrt{\lambda} L = 0$. For $c_1 \neq 0$ this condition will hold when $\sqrt{\lambda} L = n\pi$ or $\lambda = n^2 \pi^2 / L^2$, where $n = 1, 2, 3, \dots$. These are the positive eigenvalues with corresponding eigenfunctions $\cos(n\pi/L)x$, $n = 1, 2, 3, \dots$.

4. For $\lambda < 0$ we have

$$\begin{aligned} y &= c_1 \cosh \sqrt{-\lambda} x + c_2 \sinh \sqrt{-\lambda} x \\ y' &= c_1 \sqrt{-\lambda} \sinh \sqrt{-\lambda} x + c_2 \sqrt{-\lambda} \cosh \sqrt{-\lambda} x. \end{aligned}$$

Using the fact that $\cosh x$ is an even function and $\sinh x$ is odd we have

$$\begin{aligned} y(-L) &= c_1 \cosh(-\sqrt{-\lambda} L) + c_2 \sinh(-\sqrt{-\lambda} L) \\ &= c_1 \cosh \sqrt{-\lambda} L - c_2 \sinh \sqrt{-\lambda} L \end{aligned}$$

and

$$\begin{aligned} y'(-L) &= c_1 \sqrt{-\lambda} \sinh(-\sqrt{-\lambda} L) + c_2 \sqrt{-\lambda} \cosh(-\sqrt{-\lambda} L) \\ &= -c_1 \sqrt{-\lambda} \sinh \sqrt{-\lambda} L + c_2 \sqrt{-\lambda} \cosh \sqrt{-\lambda} L. \end{aligned}$$

The boundary conditions imply

$$c_1 \cosh \sqrt{-\lambda} L - c_2 \sinh \sqrt{-\lambda} L = c_1 \cosh \sqrt{-\lambda} L + c_2 \sinh \sqrt{-\lambda} L$$

or

$$2c_2 \sinh \sqrt{-\lambda} L = 0$$

and

$$-c_1 \sqrt{-\lambda} \sinh \sqrt{-\lambda} L + c_2 \sqrt{-\lambda} \cosh \sqrt{-\lambda} L = c_1 \sqrt{-\lambda} \sinh \sqrt{-\lambda} L + c_2 \sqrt{-\lambda} \cosh \sqrt{-\lambda} L$$

or

$$2c_1 \sqrt{-\lambda} \sinh \sqrt{-\lambda} L = 0.$$

Since $\sqrt{-\lambda}L \neq 0$, $c_1 = c_2 = 0$ and the only solution of the boundary-value problem in this case is $y = 0$.

For $\lambda = 0$ we have

$$\begin{aligned}y &= c_1 x + c_2 \\y' &= c_1.\end{aligned}$$

From $y(-L) = y(L)$ we obtain

$$-c_1 L + c_2 = c_1 L + c_2.$$

Then $c_1 = 0$ and $y = 1$ is an eigenfunction corresponding to the eigenvalue $\lambda = 0$.

For $\lambda > 0$ we have

$$\begin{aligned}y &= c_1 \cos \sqrt{\lambda} x + c_2 \sin \sqrt{\lambda} x \\y' &= -c_1 \sqrt{\lambda} \sin \sqrt{\lambda} x + c_2 \sqrt{\lambda} \cos \sqrt{\lambda} x.\end{aligned}$$

The first boundary condition implies

$$c_1 \cos \sqrt{\lambda} L - c_2 \sin \sqrt{\lambda} L = c_1 \cos \sqrt{\lambda} L + c_2 \sin \sqrt{\lambda} L$$

or

$$2c_2 \sin \sqrt{\lambda} L = 0.$$

Thus, if $c_1 = 0$ and $c_2 \neq 0$,

$$\sqrt{\lambda} L = n\pi \quad \text{or} \quad \lambda = \frac{n^2 \pi^2}{L^2}, \quad n = 1, 2, 3, \dots$$

The corresponding eigenfunctions are $\sin \frac{n\pi}{L} x$, for $n = 1, 2, 3, \dots$. Similarly, the second boundary condition implies

$$2c_1 \sqrt{\lambda} \sin \sqrt{\lambda} L = 0.$$

If $c_2 = 0$ and $c_1 \neq 0$,

$$\sqrt{\lambda} L = n\pi \quad \text{or} \quad \lambda = \frac{n^2 \pi^2}{L^2}, \quad n = 1, 2, 3, \dots,$$

and the corresponding eigenfunctions are $\cos \frac{n\pi}{L} x$, for $n = 1, 2, 3, \dots$.

5. The eigenfunctions are $\cos \sqrt{\lambda_n} x$ where $\cot \sqrt{\lambda_n} = \sqrt{\lambda_n}$. Thus

$$\begin{aligned}\|\cos \sqrt{\lambda_n} x\|^2 &= \int_0^1 \cos^2 \sqrt{\lambda_n} x \, dx = \frac{1}{2} \int_0^1 (1 + \cos 2\sqrt{\lambda_n} x) \, dx \\&= \frac{1}{2} \left(x + \frac{1}{2\sqrt{\lambda_n}} \sin 2\sqrt{\lambda_n} x \right) \Big|_0^1 = \frac{1}{2} \left(1 + \frac{1}{2\sqrt{\lambda_n}} \sin 2\sqrt{\lambda_n} \right) \\&= \frac{1}{2} \left[1 + \frac{1}{2\sqrt{\lambda_n}} (2 \sin \sqrt{\lambda_n} \cos \sqrt{\lambda_n}) \right] \\&= \frac{1}{2} \left[1 + \frac{1}{\sqrt{\lambda_n}} \sin \sqrt{\lambda_n} \cot \sqrt{\lambda_n} \sin \sqrt{\lambda_n} \right] \\&= \frac{1}{2} \left[1 + \frac{1}{\sqrt{\lambda_n}} (\sin \sqrt{\lambda_n}) \sqrt{\lambda_n} (\sin \sqrt{\lambda_n}) \right] = \frac{1}{2} (1 + \sin^2 \sqrt{\lambda_n}).\end{aligned}$$

Exercises 12.5

6. The eigenfunctions are $\sin \sqrt{\lambda_n} x$ where $\tan \sqrt{\lambda_n} = -\lambda_n$. Thus

$$\begin{aligned}\|\sin \sqrt{\lambda_n} x\|^2 &= \int_0^1 \sin^2 \sqrt{\lambda_n} x \, dx = \frac{1}{2} \int_0^1 (1 - \cos 2\sqrt{\lambda_n} x) \, dx \\ &= \frac{1}{2} \left(x - \frac{1}{2\sqrt{\lambda_n}} \sin 2\sqrt{\lambda_n} x \right) \Big|_0^1 = \frac{1}{2} \left(1 - \frac{1}{2\sqrt{\lambda_n}} \sin 2\sqrt{\lambda_n} \right) \\ &= \frac{1}{2} \left[1 - \frac{1}{2\sqrt{\lambda_n}} (2 \sin \sqrt{\lambda_n} \cos \sqrt{\lambda_n}) \right] \\ &= \frac{1}{2} \left[1 - \frac{1}{\sqrt{\lambda_n}} \tan \sqrt{\lambda_n} \cos \sqrt{\lambda_n} \cos \sqrt{\lambda_n} \right] \\ &= \frac{1}{2} \left[1 - \frac{1}{\sqrt{\lambda_n}} (-\sqrt{\lambda_n} \cos^2 \sqrt{\lambda_n}) \right] = \frac{1}{2} (1 + \cos^2 \sqrt{\lambda_n}).\end{aligned}$$

7. (a) If $\lambda \leq 0$ the initial conditions imply $y = 0$. For $\lambda > 0$ the general solution of the Cauchy-Euler differential equation is $y = c_1 \cos(\sqrt{\lambda} \ln x) + c_2 \sin(\sqrt{\lambda} \ln x)$. The condition $y(1) = 0$ implies $c_1 = 0$, so that $y = c_2 \sin(\sqrt{\lambda} \ln x)$. The condition $y(5) = 0$ implies $\sqrt{\lambda} \ln 5 = n\pi$, $n = 1, 2, 3, \dots$. Thus, the eigenvalues are $n^2\pi^2/(\ln 5)^2$ for $n = 1, 2, 3, \dots$, with corresponding eigenfunctions $\sin[(n\pi/\ln 5) \ln x]$.

- (b) The self-adjoint form is

$$\frac{d}{dx}[xy'] + \frac{\lambda}{x}y = 0.$$

- (c) An orthogonality relation is

$$\int_1^5 \sin\left(\frac{m\pi}{\ln 5} \ln x\right) \sin\left(\frac{n\pi}{\ln 5} \ln x\right) dx = 0.$$

8. (a) The roots of the auxiliary equation $m^2 + m + \lambda = 0$ are $\frac{1}{2}(-1 \pm \sqrt{1-4\lambda})$. When $\lambda = 0$ the general solution of the differential equation is $c_1 + c_2 e^{-x}$. The initial conditions imply $c_1 + c_2 = 0$ and $c_1 + c_2 e^{-2} = 0$. Since the determinant of the coefficients is not 0, the only solution of this homogeneous system is $c_1 = c_2 = 0$, in which case $y = 0$. Similarly, if $0 < \lambda < \frac{1}{4}$, the general solution is

$$y = c_1 e^{\frac{1}{2}(-1+\sqrt{1-4\lambda})x} + c_2 e^{\frac{1}{2}(-1-\sqrt{1-4\lambda})x}.$$

In this case the initial conditions again imply $c_1 = c_2 = 0$, and so $y = 0$. Now, for $\lambda > \frac{1}{4}$, the general solution of the differential equation is

$$y = c_1 e^{-x/2} \cos \sqrt{4\lambda-1} x + c_2 e^{-x/2} \sin \sqrt{4\lambda-1} x.$$

The condition $y(0) = 0$ implies $c_1 = 0$ so $y = c_2 e^{-x/2} \sin \sqrt{4\lambda-1} x$. From

$$y(2) = c_2 e^{-1} \sin 2\sqrt{4\lambda-1} = 0$$

we see that the eigenvalues are determined by $2\sqrt{4\lambda-1} = n\pi$ for $n = 1, 2, 3, \dots$. Thus, the eigenvalues are $n^2\pi^2/4^2 + 1/4$ for $n = 1, 2, 3, \dots$, with corresponding eigenfunctions $e^{-x/2} \sin \frac{n\pi}{2} x$.

- (b) The self-adjoint form is

$$\frac{d}{dx}[e^x y'] + \lambda e^x y = 0.$$

- (c) An orthogonality relation is

$$\int_0^2 e^x \left(e^{-x/2} \sin \frac{m\pi}{2} x \right) \left(e^{-x/2} \cos \frac{n\pi}{2} x \right) dx = \int_0^2 \left(\sin \frac{m\pi}{2} x \right) \left(\cos \frac{n\pi}{2} x \right) dx = 0.$$

9. (a) An orthogonality relation is

$$\int_0^1 \cos x_m x \cos x_n x = 0$$

where $x_m \neq x_n$ are positive solutions of $\cot x = x$.

- (b) Referring to Problem 1 we use a CAS to compute

$$\int_0^1 (\cos 0.8603x)(\cos 3.4256x) dx = -1.8771 \times 10^{-6}.$$

10. (a) An orthogonality relation is

$$\int_0^1 (x_m \cos x_m x - \sin x_m x)(x_n \cos x_n x - \sin x_n x) dx = 0$$

where $x_m \neq x_n$ are positive solutions of $\tan x = x$.

- (b) Referring to Problem 2 we use a CAS to compute

$$\int_0^1 (4.4934 \cos 4.4934x - \sin 4.4934x)(7.7253 \cos 7.7253x - \sin 7.7253x) dx = -2.5650 \times 10^{-4}.$$

11. To obtain the self-adjoint form we note that an integrating factor is $(1/x)e^{\int (1-x)dx/x} = e^{-x}$. Thus, the differential equation is

$$xe^{-x}y'' + (1-x)e^{-x}y' + ne^{-x}y = 0$$

and the self-adjoint form is

$$\frac{d}{dx} [xe^{-x}y'] + ne^{-x}y = 0.$$

Identifying the weight function $p(x) = e^{-x}$ and noting that since $r(x) = xe^{-x}$, $r(0) = 0$ and $\lim_{x \rightarrow \infty} r(x) = 0$, we have the orthogonality relation

$$\int_0^{\infty} e^{-x} L_n(x) L_m(x) dx = 0, \quad m \neq n.$$

12. To obtain the self-adjoint form we note that an integrating factor is $e^{\int -2x dx} = e^{-x^2}$. Thus, the differential equation is

$$e^{-x^2}y'' - 2xe^{-x^2}y' + 2ne^{-x^2}y = 0$$

and the self-adjoint form is

$$\frac{d}{dx} [e^{-x^2}y'] + 2ne^{-x^2}y = 0.$$

Identifying the weight function $p(x) = 2e^{-x^2}$ and noting that since $r(x) = e^{-x^2}$, $\lim_{x \rightarrow -\infty} r(x) = \lim_{x \rightarrow \infty} r(x) = 0$, we have the orthogonality relation

$$\int_{-\infty}^{\infty} 2e^{-x^2} H_n(x) H_m(x) dx = 0, \quad m \neq n.$$

13. (a) The differential equation is

$$(1+x^2)y'' + 2xy' + \frac{\lambda}{1+x^2}y = 0.$$

Exercises 12.5

Letting $x = \tan \theta$ we have $\theta = \tan^{-1} x$ and

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{d\theta} \frac{d\theta}{dx} = \frac{1}{1+x^2} \frac{dy}{d\theta} \\ \frac{d^2y}{dx^2} &= \frac{d}{dx} \left[\frac{1}{1+x^2} \frac{dy}{d\theta} \right] = \frac{1}{1+x^2} \left(\frac{d^2y}{d\theta^2} \frac{d\theta}{dx} \right) - \frac{2x}{(1+x^2)^2} \frac{dy}{d\theta} \\ &= \frac{1}{(1+x^2)^2} \frac{d^2y}{d\theta^2} - \frac{2x}{(1+x^2)^2} \frac{dy}{d\theta}.\end{aligned}$$

The differential equation can then be written in terms of $y(\theta)$ as

$$\begin{aligned}(1+x^2) \left[\frac{1}{(1+x^2)^2} \frac{d^2y}{d\theta^2} - \frac{2x}{(1+x^2)^2} \frac{dy}{d\theta} \right] + 2x \left[\frac{1}{1+x^2} \frac{dy}{d\theta} \right] + \frac{\lambda}{1+x^2} y \\ = \frac{1}{1+x^2} \frac{d^2y}{d\theta^2} + \frac{\lambda}{1+x^2} y = 0\end{aligned}$$

or

$$\frac{d^2y}{d\theta^2} + \lambda y = 0.$$

The boundary conditions become $y(0) = y(\pi/4) = 0$. For $\lambda \leq 0$ the only solution of the boundary-value problem is $y = 0$. For $\lambda > 0$ the general solution of the differential equation is $y = c_1 \cos \sqrt{\lambda} \theta + c_2 \sin \sqrt{\lambda} \theta$. The condition $y(0) = 0$ implies $c_1 = 0$ so $y = c_2 \sin \sqrt{\lambda} \theta$. Now the condition $y(\pi/4) = 0$ implies $c_2 \sin \sqrt{\lambda} \pi/4 = 0$. For $c_2 \neq 0$ this condition will hold when $\sqrt{\lambda} \pi/4 = n\pi$ or $\lambda = 16n^2$, where $n = 1, 2, 3, \dots$. These are the eigenvalues with corresponding eigenfunctions $\sin 4n\theta = \sin(4n \tan^{-1} x)$, for $n = 1, 2, 3, \dots$.

(b) An orthogonality relation is

$$\int_0^1 \frac{1}{x^2+1} \sin(4m \tan^{-1} x) \sin(4n \tan^{-1} x) dx = 0.$$

14. (a) This is the parametric Bessel equation with $\nu = 1$. The general solution is

$$y = c_1 J_1(\lambda x) + c_2 Y_1(\lambda x).$$

Since Y is unbounded at 0 we must have $c_2 = 0$, so that $y = c_1 J_1(\lambda x)$. The condition $J_1(3\lambda) = 0$ defines the eigenvalues $\lambda_1, \lambda_2, \lambda_3, \dots$. (When $\lambda = 0$ the differential equation is Cauchy-Euler and the only solution satisfying the boundary condition is $y = 0$, so $\lambda = 0$ is not an eigenvalue.)

(b) From Table 5.2 in the text we see that eigenvalues are determined by $3\lambda_1 = 3.832$, $3\lambda_2 = 7.016$, $3\lambda_3 = 10.173$, and $3\lambda_4 = 13.323$. The first four eigenvalues are thus $\lambda_1 = 1.2773$, $\lambda_2 = 2.3387$, $\lambda_3 = 3.391$, and $\lambda_4 = 4.441$.

Exercises 12.6

1. Identifying $b = 3$, the first four eigenvalues are

$$\lambda_1 = \frac{3.832}{3} \approx 1.277$$

$$\lambda_2 = \frac{7.016}{3} \approx 2.339$$

$$\lambda_3 = \frac{10.173}{3} = 3.391$$

$$\lambda_4 = \frac{13.323}{3} \approx 4.441.$$

2. We first note from Case III in the text that 0 is an eigenvalue. Now, since $J'_0(2\lambda) = 0$ is equivalent to $J_1(2\lambda) = 0$, the next three eigenvalues are

$$\lambda_2 = \frac{3.832}{2} = 1.916$$

$$\lambda_3 = \frac{7.016}{2} = 3.508$$

$$\lambda_4 = \frac{10.173}{2} = 5.087.$$

3. The boundary condition indicates that we use (15) and (16) of Section 12.6. With $b = 2$ we obtain

$$\begin{aligned} c_i &= \frac{2}{4J_1^2(2\lambda_i)} \int_0^2 x J_0(\lambda_i x) dx && \boxed{t = \lambda_i x \quad dt = \lambda_i dx} \\ &= \frac{1}{2J_1^2(2\lambda_i)} \cdot \frac{1}{\lambda_i^2} \int_0^{2\lambda_i} t J_0(t) dt \\ &= \frac{1}{2\lambda_i^2 J_1^2(2\lambda_i)} \int_0^{2\lambda_i} \frac{d}{dt} [t J_1(t)] dt && [\text{From (4) in the text}] \\ &= \frac{1}{2\lambda_i^2 J_1^2(2\lambda_i)} t J_1(t) \Big|_0^{2\lambda_i} \\ &= \frac{1}{\lambda_i J_1(2\lambda_i)}. \end{aligned}$$

Thus

$$f(x) = \sum_{i=1}^{\infty} \frac{1}{\lambda_i J_1(2\lambda_i)} J_0(\lambda_i x).$$

4. The boundary condition indicates that we use (19) and (20) of Section 12.6. With $b = 2$ we obtain

$$\begin{aligned} c_1 &= \frac{2}{4} \int_0^2 x dx = \frac{2}{4} \frac{x^2}{2} \Big|_0^2 = 1, \\ c_i &= \frac{2}{4J_0^2(2\lambda_i)} \int_0^2 x J_0(\lambda_i x) dx && \boxed{t = \lambda_i x \quad dt = \lambda_i dx} \\ &= \frac{1}{2J_0^2(2\lambda_i)} \cdot \frac{1}{\lambda_i^2} \int_0^{2\lambda_i} t J_0(t) dt \end{aligned}$$

Exercises 12.6

$$\begin{aligned}
 &= \frac{1}{2\lambda_i^2 J_0^2(2\lambda_i)} \int_0^{2\lambda_i} \frac{d}{dt} [tJ_1(t)] dt && \text{[From (4) in the text]} \\
 &= \frac{1}{2\lambda_i^2 J_0^2(2\lambda_i)} tJ_1(t) \Big|_0^{2\lambda_i} \\
 &= \frac{J_1(2\lambda_i)}{\lambda_i J_0^2(2\lambda_i)}.
 \end{aligned}$$

Now since $J_0'(2\lambda_i) = 0$ is equivalent to $J_1(2\lambda_i) = 0$ we conclude $c_i = 0$ for $i = 2, 3, 4, \dots$. Thus the expansion of f on $0 < x < 2$ consists of a series with one nontrivial term:

$$f(x) = c_1 = 1.$$

5. The boundary condition indicates that we use (17) and (18) of Section 12.6. With $b = 2$ and $h = 1$ we obtain

$$\begin{aligned}
 c_i &= \frac{2\lambda_i^2}{(4\lambda_i^2 + 1)J_0^2(2\lambda_i)} \int_0^2 xJ_0(\lambda_i x) dx && \boxed{t = \lambda_i x \quad dt = \lambda_i dx} \\
 &= \frac{2\lambda_i^2}{(4\lambda_i^2 + 1)J_0^2(2\lambda_i)} \cdot \frac{1}{\lambda_i^2} \int_0^{2\lambda_i} tJ_0(t) dt \\
 &= \frac{2}{(4\lambda_i^2 + 1)J_0^2(2\lambda_i)} \int_0^{2\lambda_i} \frac{d}{dt} [tJ_1(t)] dt && \text{[From (4) in the text]} \\
 &= \frac{2}{(4\lambda_i^2 + 1)J_0^2(2\lambda_i)} tJ_1(t) \Big|_0^{2\lambda_i} \\
 &= \frac{4\lambda_i J_1(2\lambda_i)}{(4\lambda_i^2 + 1)J_0^2(2\lambda_i)}.
 \end{aligned}$$

Thus

$$f(x) = 4 \sum_{i=1}^{\infty} \frac{\lambda_i J_1(2\lambda_i)}{(4\lambda_i^2 + 1)J_0^2(2\lambda_i)} J_0(\lambda_i x).$$

6. Writing the boundary condition in the form

$$2J_0(2\lambda) + 2\lambda J_0'(2\lambda) = 0$$

we identify $b = 2$ and $h = 2$. Using (17) and (18) of Section 12.6 we obtain

$$\begin{aligned}
 c_i &= \frac{2\lambda_i^2}{(4\lambda_i^2 + 4)J_0^2(2\lambda_i)} \int_0^2 xJ_0(\lambda_i x) dx && \boxed{t = \lambda_i x \quad dt = \lambda_i dx} \\
 &= \frac{\lambda_i^2}{2(\lambda_i^2 + 1)J_0^2(2\lambda_i)} \cdot \frac{1}{\lambda_i^2} \int_0^{2\lambda_i} tJ_0(t) dt \\
 &= \frac{1}{2(\lambda_i^2 + 1)J_0^2(2\lambda_i)} \int_0^{2\lambda_i} \frac{d}{dt} [tJ_1(t)] dt && \text{[From (4) in the text]} \\
 &= \frac{1}{2(\lambda_i^2 + 1)J_0^2(2\lambda_i)} tJ_1(t) \Big|_0^{2\lambda_i} \\
 &= \frac{\lambda_i J_1(2\lambda_i)}{(\lambda_i^2 + 1)J_0^2(2\lambda_i)}.
 \end{aligned}$$

Thus

$$f(x) = \sum_{i=1}^{\infty} \frac{\lambda_i J_1(2\lambda_i)}{(\lambda_i^2 + 1)J_0^2(2\lambda_i)} J_0(\lambda_i x).$$

7. The boundary condition indicates that we use (17) and (18) of Section 12.6. With $n = 1$, $b = 4$, and $h = 3$ we obtain

$$\begin{aligned} c_i &= \frac{2\lambda_i^2}{(16\lambda_i^2 - 1 + 9)J_1^2(4\lambda_i)} \int_0^4 x J_1(\lambda_i x) 5x \, dx && \boxed{t = \lambda_i x \quad dt = \lambda_i \, dx} \\ &= \frac{5\lambda_i^2}{4(2\lambda_i^2 + 1)J_1^2(4\lambda_i)} \cdot \frac{1}{\lambda_i^3} \int_0^{4\lambda_i} t^2 J_1(t) \, dt \\ &= \frac{5}{4\lambda_i(2\lambda_i^2 + 1)J_1^2(4\lambda_i)} \int_0^{4\lambda_i} \frac{d}{dt} [t^2 J_2(t)] \, dt && [\text{From (4) in the text}] \\ &= \frac{5}{4\lambda_i(2\lambda_i^2 + 1)J_1^2(4\lambda_i)} t^2 J_2(t) \Big|_0^{4\lambda_i} \\ &= \frac{20\lambda_i J_2(4\lambda_i)}{(2\lambda_i^2 + 1)J_1^2(4\lambda_i)}. \end{aligned}$$

Thus

$$f(x) = 20 \sum_{i=1}^{\infty} \frac{\lambda_i J_2(4\lambda_i)}{(2\lambda_i^2 + 1)J_1^2(4\lambda_i)} J_1(\lambda_i x).$$

8. The boundary condition indicates that we use (15) and (16) of Section 12.6. With $n = 2$ and $b = 1$ we obtain

$$\begin{aligned} c_1 &= \frac{2}{J_3^2(\lambda_i)} \int_0^1 x J_2(\lambda_i x) x^2 \, dx && \boxed{t = \lambda_i x \quad dt = \lambda_i \, dx} \\ &= \frac{2}{J_3^2(\lambda_i)} \cdot \frac{1}{\lambda_i^4} \int_0^{\lambda_i} t^3 J_2(t) \, dt \\ &= \frac{2}{\lambda_i^4 J_3^2(\lambda_i)} \int_0^{\lambda_i} \frac{d}{dt} [t^3 J_3(t)] \, dt && [\text{From (4) in the text}] \\ &= \frac{2}{\lambda_i^4 J_3^2(\lambda_i)} t^3 J_3(t) \Big|_0^{\lambda_i} \\ &= \frac{2}{\lambda_i J_3(\lambda_i)}. \end{aligned}$$

Thus

$$f(x) = 2 \sum_{i=1}^{\infty} \frac{1}{\lambda_i J_3(\lambda_i)} J_2(\lambda_i x).$$

9. The boundary condition indicates that we use (19) and (20) of Section 12.6. With $b = 3$ we obtain

$$\begin{aligned} c_1 &= \frac{2}{9} \int_0^3 x x^2 \, dx = \frac{2}{9} \frac{x^4}{4} \Big|_0^3 = \frac{9}{2}, \\ c_i &= \frac{2}{9J_0^2(3\lambda_i)} \int_0^3 x J_0(\lambda_i x) x^2 \, dx && \boxed{t = \lambda_i x \quad dt = \lambda_i \, dx} \end{aligned}$$

Exercises 12.6

$$\begin{aligned}
 &= \frac{2}{9J_0^2(3\lambda_i)} \cdot \frac{1}{\lambda_i^4} \int_0^{3\lambda_i} t^3 J_0(t) dt \\
 &= \frac{2}{9\lambda_i^4 J_0^2(3\lambda_i)} \int_0^{3\lambda_i} t^2 \frac{d}{dt} [tJ_1(t)] dt
 \end{aligned}$$

$u = t^2$	$dv = \frac{d}{dt} [tJ_1(t)] dt$
$du = 2t dt$	$v = tJ_1(t)$

$$= \frac{2}{9\lambda_i^4 J_0^2(3\lambda_i)} \left(t^3 J_1(t) \Big|_0^{3\lambda_i} - 2 \int_0^{3\lambda_i} t^2 J_1(t) dt \right)$$

With $n = 0$ in equation (5) in Section 12.6 in the text we have $J_0'(x) = -J_1(x)$, so the boundary condition $J_0'(3\lambda_i) = 0$ implies $J_1(3\lambda_i) = 0$. Then

$$\begin{aligned}
 c_i &= \frac{2}{9\lambda_i^4 J_0^2(3\lambda_i)} \left(-2 \int_0^{3\lambda_i} \frac{d}{dt} [t^2 J_2(t)] dt \right) = \frac{2}{9\lambda_i^4 J_0^2(3\lambda_i)} \left(-2t^2 J_2(t) \Big|_0^{3\lambda_i} \right) \\
 &= \frac{2}{9\lambda_i^4 J_0^2(3\lambda_i)} [-18\lambda_i^2 J_2(3\lambda_i)] = \frac{-4J_2(3\lambda_i)}{\lambda_i^2 J_0^2(3\lambda_i)}.
 \end{aligned}$$

Thus

$$f(x) = \frac{9}{2} - 4 \sum_{i=1}^{\infty} \frac{J_2(3\lambda_i)}{\lambda_i^2 J_0^2(3\lambda_i)} J_0(\lambda_i x).$$

10. The boundary condition indicates that we use (15) and (16) of Section 12.6. With $b = 1$ it follows that

$$\begin{aligned}
 c_i &= \frac{2}{J_1^2(\lambda_i)} \int_0^1 x(1-x^2) J_0(\lambda_i x) dx \\
 &= \frac{2}{J_1^2(\lambda_i)} \left[\int_0^1 x J_0(\lambda_i x) dx - \int_0^1 x^3 J_0(\lambda_i x) dx \right]
 \end{aligned}$$

$t = \lambda_i x$	$dt = \lambda_i dx$
-------------------	---------------------

$$\begin{aligned}
 &= \frac{2}{J_1^2(\lambda_i)} \left[\frac{1}{\lambda_i^2} \int_0^{\lambda_i} t J_0(t) dt - \frac{1}{\lambda_i^4} \int_0^{\lambda_i} t^3 J_0(t) dt \right] \\
 &= \frac{2}{J_1^2(\lambda_i)} \left[\frac{1}{\lambda_i^2} \int_0^{\lambda_i} \frac{d}{dt} (tJ_1(t)) dt - \frac{1}{\lambda_i^4} \int_0^{\lambda_i} t^2 \frac{d}{dt} [tJ_1(t)] dt \right]
 \end{aligned}$$

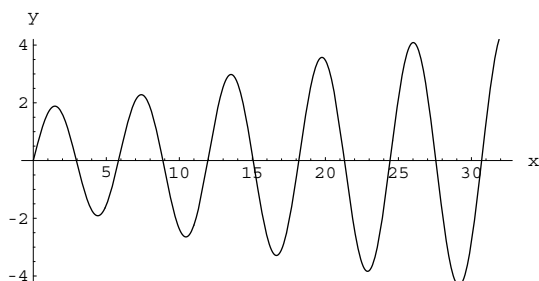
$u = t^2$	$dv = \frac{d}{dt} [tJ_1(t)] dt$
$du = 2t dt$	$v = tJ_1(t)$

$$\begin{aligned}
 &= \frac{2}{J_1^2(\lambda_i)} \left[\frac{1}{\lambda_i^2} tJ_1(t) \Big|_0^{\lambda_i} - \frac{1}{\lambda_i^4} \left(t^3 J_1(t) \Big|_0^{\lambda_i} - 2 \int_0^{\lambda_i} t^2 J_1(t) dt \right) \right] \\
 &= \frac{2}{J_1^2(\lambda_i)} \left[\frac{J_1(\lambda_i)}{\lambda_i} - \frac{J_1(\lambda_i)}{\lambda_i} + \frac{2}{\lambda_i^4} \int_0^{\lambda_i} \frac{d}{dt} [t^2 J_2(t)] dt \right] \\
 &= \frac{2}{J_1^2(\lambda_i)} \left[\frac{2}{\lambda_i^4} t^2 J_2(t) \Big|_0^{\lambda_i} \right] = \frac{4J_2(\lambda_i)}{\lambda_i^2 J_1^2(\lambda_i)}.
 \end{aligned}$$

Thus

$$f(x) = 4 \sum_{i=1}^{\infty} \frac{J_2(\lambda_i)}{\lambda_i^2 J_1^2(\lambda_i)} j_0(\lambda_i x).$$

11. (a)



(b) Using **FindRoot** in *Mathematica* we find the roots $x_1 = 2.9496$, $x_2 = 5.84113$, $x_3 = 8.87273$, $x_4 = 11.9561$, and $x_5 = 15.0624$.

(c) Dividing the roots in part (b) by 4 we find the eigenvalues $\lambda_1 = 0.7374$, $\lambda_2 = 1.46028$, $\lambda_3 = 2.21818$, $\lambda_4 = 2.98904$, and $\lambda_5 = 3.76559$.

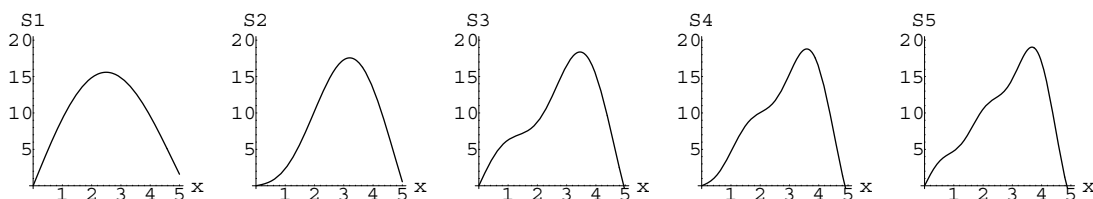
(d) The next five eigenvalues are $\lambda_6 = 4.54508$, $\lambda_7 = 5.32626$, $\lambda_8 = 6.1085$, $\lambda_9 = 6.89145$, and $\lambda_{10} = 7.6749$.

12. (a) From Problem 7, the coefficients of the Fourier-Bessel series are

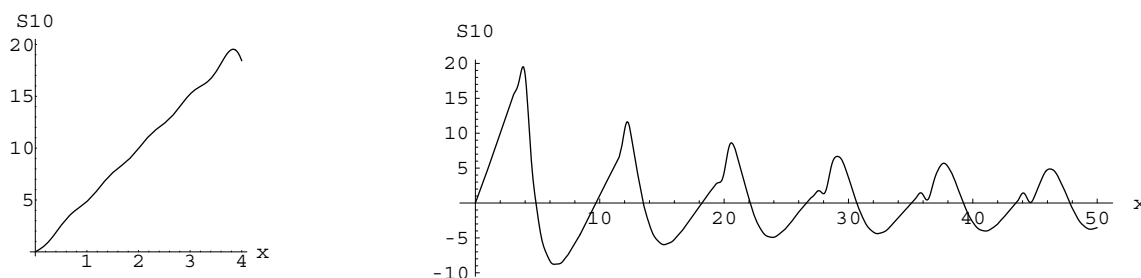
$$c_1 = \frac{20\lambda_i J_2(4\lambda_i)}{(2\lambda_i^2 + 1)J_1^2(4\lambda_i)}.$$

Using a CAS we find $c_1 = 26.7896$, $c_2 = -12.4624$, $c_3 = 7.1404$, $c_4 = -4.68705$, and $c_5 = 3.35619$.

(b)



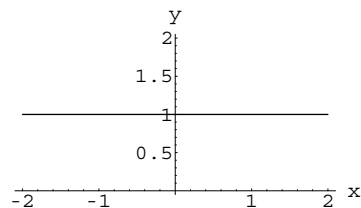
(c)



13. Since f is expanded as a series of Bessel functions, $J_1(\lambda_i x)$ and J_1 is an odd function, the series should represent an odd function.

Exercises 12.6

14. (a) Since J_0 is an even function, a series expansion of a function defined on $(0, 2)$ would converge to the even extension of the function on $(-2, 0)$.



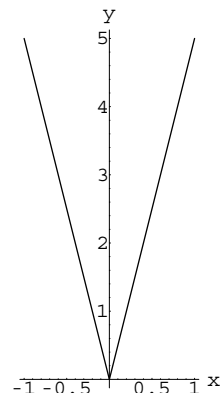
- (b) In Section 5.3 we saw that $J'_2(x) = 2J_2(x)/x - J_3(x)$. Since J_2 is even and J_3 is odd we see that

$$\begin{aligned} J'_2(-x) &= 2J_2(-x)/(-x) - J_3(-x) \\ &= -2J_2(x)/x + J_3(x) = -J'_2(x), \end{aligned}$$

so that J'_2 is an odd function. Now, if $f(x) = 3J_2(x) + 2xJ'_2(x)$, we see that

$$\begin{aligned} f(-x) &= 3J_2(-x) - 2xJ'_2(-x) \\ &= 3J_2(x) + 2xJ'_2(x) = f(x), \end{aligned}$$

so that f is an even function. Thus, a series expansion of a function defined on $(0, 1)$ would converge to the even extension of the function on $(-1, 0)$.



15. We compute

$$c_0 = \frac{1}{2} \int_0^1 x P_0(x) dx = \frac{1}{2} \int_0^1 x dx = \frac{1}{4}$$

$$c_1 = \frac{3}{2} \int_0^1 x P_1(x) dx = \frac{3}{2} \int_0^1 x^2 dx = \frac{1}{2}$$

$$c_2 = \frac{5}{2} \int_0^1 x P_2(x) dx = \frac{5}{2} \int_0^1 \frac{1}{2}(3x^3 - x) dx = \frac{5}{16}$$

$$c_3 = \frac{7}{2} \int_0^1 x P_3(x) dx = \frac{7}{2} \int_0^1 \frac{1}{2}(5x^4 - 3x^2) dx = 0$$

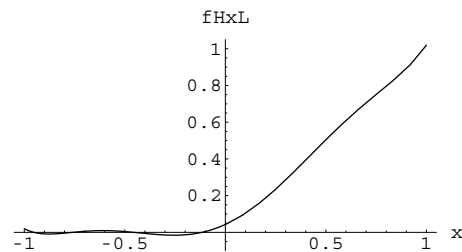
$$c_4 = \frac{9}{2} \int_0^1 x P_4(x) dx = \frac{9}{2} \int_0^1 \frac{1}{8}(35x^5 - 30x^3 + 3x) dx = -\frac{3}{32}$$

$$c_5 = \frac{11}{2} \int_0^1 x P_5(x) dx = \frac{11}{2} \int_0^1 \frac{1}{8}(63x^6 - 70x^4 + 15x^2) dx = 0$$

$$c_6 = \frac{13}{2} \int_0^1 x P_6(x) dx = \frac{13}{2} \int_0^1 \frac{1}{16}(231x^7 - 315x^5 + 105x^3 - 5x) dx = \frac{13}{256}.$$

Thus

$$f(x) = \frac{1}{4}P_0(x) + \frac{1}{2}P_1(x) + \frac{5}{16}P_2(x) - \frac{3}{32}P_4(x) + \frac{13}{256}P_6(x) + \cdots.$$



16. We compute

$$c_0 = \frac{1}{2} \int_{-1}^1 e^x P_0(x) dx = \frac{1}{2} \int_{-1}^1 e^x dx = \frac{1}{2}(e - e^{-1})$$

$$c_1 = \frac{3}{2} \int_{-1}^1 e^x P_1(x) dx = \frac{3}{2} \int_{-1}^1 x e^x dx = 3e^{-1}$$

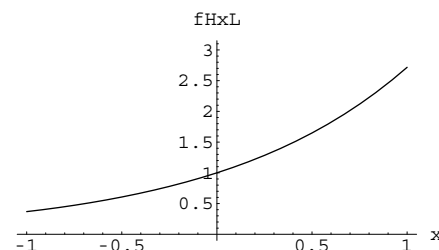
$$c_2 = \frac{5}{2} \int_{-1}^1 e^x P_2(x) dx = \frac{5}{2} \int_{-1}^1 \frac{1}{2}(3x^2 e^x - e^x) dx = \frac{5}{2}(e - 7e^{-1})$$

$$c_3 = \frac{7}{2} \int_{-1}^1 e^x P_3(x) dx = \frac{7}{2} \int_{-1}^1 \frac{1}{2}(5x^3 e^x - 3x e^x) dx = \frac{7}{2}(-5e + 37e^{-1})$$

$$c_4 = \frac{9}{2} \int_{-1}^1 e^x P_4(x) dx = \frac{9}{2} \int_{-1}^1 \frac{1}{8}(35x^4 e^x - 30x^2 e^x + 3e^x) dx = \frac{9}{2}(36e - 266e^{-1}).$$

Thus

$$\begin{aligned} f(x) &= \frac{1}{2}(e - e^{-1})P_0(x) + 3e^{-1}P_1(x) + \frac{5}{2}(e - 7e^{-1})P_2(x) \\ &\quad + \frac{7}{2}(-5e + 37e^{-1})P_3(x) + \frac{9}{2}(36e - 266e^{-1})P_4(x) + \cdots \end{aligned}$$



17. Using $\cos^2 \theta = \frac{1}{2}(\cos 2\theta + 1)$ we have

$$\begin{aligned} P_2(\cos \theta) &= \frac{1}{2}(3 \cos^2 \theta - 1) = \frac{3}{2} \cos^2 \theta - \frac{1}{2} \\ &= \frac{3}{4}(\cos 2\theta + 1) - \frac{1}{2} = \frac{3}{4} \cos 2\theta + \frac{1}{4} = \frac{1}{4}(3 \cos 2\theta + 1). \end{aligned}$$

18. From Problem 17 we have

$$P_2(\cos 2\theta) = \frac{1}{4}(3 \cos 2\theta + 1)$$

or

$$\cos 2\theta = \frac{4}{3}P_2(\cos \theta) - \frac{1}{3}.$$

Then, using $P_0(\cos \theta) = 1$,

$$\begin{aligned} F(\theta) &= 1 - \cos 2\theta = 1 - \left[\frac{4}{3}P_2(\cos \theta) - \frac{1}{3} \right] \\ &= \frac{4}{3} - \frac{4}{3}P_2(\cos \theta) = \frac{4}{3}P_0(\cos \theta) - \frac{4}{3}P_2(\cos \theta). \end{aligned}$$

19. If f is an even function on $(-1, 1)$ then

$$\int_{-1}^1 f(x) P_{2n}(x) dx = 2 \int_0^1 f(x) P_{2n}(x) dx$$

and

$$\int_{-1}^1 f(x) P_{2n+1}(x) dx = 0.$$

Thus

$$\begin{aligned} c_{2n} &= \frac{2(2n) + 1}{2} \int_{-1}^1 f(x) P_{2n}(x) dx = \frac{4n + 1}{2} \left(2 \int_0^1 f(x) P_{2n}(x) dx \right) \\ &= (4n + 1) \int_0^1 f(x) P_{2n}(x) dx, \end{aligned}$$

Exercises 12.6

$c_{2n+1} = 0$, and

$$f(x) = \sum_{n=0}^{\infty} c_{2n} P_{2n}(x).$$

20. If f is an odd function on $(-1, 1)$ then

$$\int_{-1}^1 f(x) P_{2n}(x) dx = 0$$

and

$$\int_{-1}^1 f(x) P_{2n+1}(x) dx = 2 \int_0^1 f(x) P_{2n+1}(x) dx.$$

Thus

$$\begin{aligned} c_{2n+1} &= \frac{2(2n+1)+1}{2} \int_{-1}^1 f(x) P_{2n+1}(x) dx = \frac{4n+3}{2} \left(2 \int_0^1 f(x) P_{2n+1}(x) dx \right) \\ &= (4n+1) \int_0^1 f(x) P_{2n+1}(x) dx, \end{aligned}$$

$c_{2n} = 0$, and

$$f(x) = \sum_{n=0}^{\infty} c_{2n+1} P_{2n+1}(x).$$

21. From (26) in Problem 19 in the text we find

$$c_0 = \int_0^1 x P_0(x) dx = \int_0^1 x dx = \frac{1}{2},$$

$$c_2 = 5 \int_0^1 x P_2(x) dx = 5 \int_0^1 \frac{1}{2} (3x^3 - x) dx = \frac{5}{8},$$

$$c_4 = 9 \int_0^1 x P_4(x) dx = 9 \int_0^1 \frac{1}{8} (35x^5 - 30x^3 + 3x) dx = -\frac{3}{16},$$

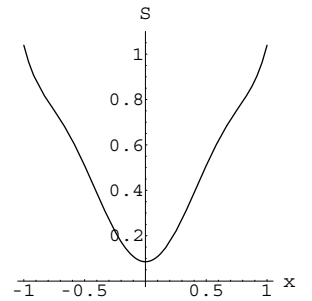
and

$$c_6 = 13 \int_0^1 x P_6(x) dx = 13 \int_0^1 \frac{1}{16} (231x^7 - 315x^5 + 105x^3 - 5x) dx = \frac{13}{128}.$$

Hence, from (25) in the text,

$$f(x) = \frac{1}{2} P_0(x) + \frac{5}{8} P_2(x) - \frac{3}{16} P_4(x) + \frac{13}{128} P_6(x) + \cdots.$$

On the interval $-1 < x < 1$ this series represents the function $f(x) = |x|$.



22. From (28) in Problem 20 in the text we find

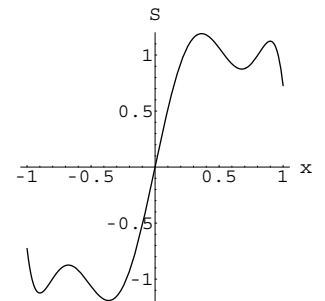
$$c_1 = 3 \int_0^1 P_1(x) dx = 3 \int_0^1 x dx = \frac{3}{2},$$

$$c_3 = 7 \int_0^1 P_3(x) dx = 7 \int_0^1 \frac{1}{2} (5x^3 - 3x) dx = -\frac{7}{8},$$

$$c_5 = 11 \int_0^1 P_5(x) dx = 11 \int_0^1 \frac{1}{8} (63x^5 - 70x^3 + 15x) dx = \frac{11}{16}$$

and

$$c_7 = 15 \int_0^1 P_7(x) dx = 15 \int_0^1 \frac{1}{16} (429x^7 - 693x^5 + 315x^3 - 35x) dx = -\frac{75}{128}.$$



Hence, from (27) in the text,

$$f(x) = \frac{3}{2}P_1(x) - \frac{7}{8}P_3(x) + \frac{11}{16}P_5(x) - \frac{75}{128}P_7(x) + \cdots$$

On the interval $-1 < x < 1$ this series represents the odd function

$$f(x) = \begin{cases} -1, & -1 < x < 0 \\ 1, & 0 < x < 1. \end{cases}$$

Chapter 12 Review Exercises

1. True, since $\int_{-\pi}^{\pi} (x^2 - 1)x^5 dx = 0$
2. Even, since if f and g are odd then $h(-x) = f(-x)g(-x) = -f(x)[-g(x)] = f(x)g(x) = h(x)$
3. Cosine, since f is even
4. True
5. False; the Sturm-Liouville problem,

$$\frac{d}{dx}[r(x)y'] + \lambda p(x)y = 0, \quad y'(a) = 0, \quad y'(b) = 0,$$

on the interval $[a, b]$, has eigenvalue $\lambda = 0$.

6. $3/2$, the average of 3 and 0
7. The Fourier series will converge to 1, the cosine series to 1, and the sine series to 0 at $x = 0$. Respectively, this is because the rule $(x^2 + 1)$ defining $f(x)$ determines a continuous function on $(-3, 3)$, the even extension of f to $(-3, 0)$ is continuous at 0, and the odd extension of f to $(-3, 0)$ approaches -1 as x approaches 0 from the left.
8. $\cos 5x$, since the general solution is $y = c_1 \cos \sqrt{\lambda}x + c_2 \sin \sqrt{\lambda}x$ and $y'(0) = 0$ implies $c_2 = 0$.
9. Since the coefficient of y in the differential equation is n^2 , the weight function is the integrating factor

$$\frac{1}{a(x)} e^{\int (b/a) dx} = \frac{1}{1-x^2} e^{\int -\frac{x}{1-x^2} dx} = \frac{1}{1-x^2} e^{\frac{1}{2} \ln(1-x^2)} = \frac{\sqrt{1-x^2}}{1-x^2} = \frac{1}{\sqrt{1-x^2}}$$

on the interval $[-1, 1]$. The orthogonality relation is

$$\int_{-1}^1 \frac{1}{\sqrt{1-x^2}} T_m(x) T_n(x) dx = 0, \quad m \neq n.$$

10. Since $P_n(x)$ is orthogonal to $P_0(x) = 1$ for $n > 0$,

$$\int_{-1}^1 P_n(x) dx = \int_{-1}^1 P_0(x) P_n(x) dx = 0.$$

11. We know from a half-angle formula in trigonometry that $\cos^2 x = \frac{1}{2} + \frac{1}{2} \cos 2x$, which is a cosine series.
12. (a) For $m \neq n$

$$\int_0^L \sin \frac{(2n+1)\pi}{2L} x \sin \frac{(2m+1)\pi}{2L} x dx = \frac{1}{2} \int_0^L \left(\cos \frac{n-m}{L} \pi x - \cos \frac{n+m+\pi}{L} \pi x \right) dx = 0.$$

- (b) From

$$\int_0^L \sin^2 \frac{(2n+1)\pi}{2L} x dx = \int_0^L \left(\frac{1}{2} - \frac{1}{2} \cos \frac{(2n+1)\pi}{2L} x \right) dx = \frac{L}{2}$$

Chapter 12 Review Exercises

we see that

$$\left\| \sin \frac{(2n+1)\pi}{2L} x \right\| = \sqrt{\frac{L}{2}}.$$

13. Since

$$A_0 = \int_{-1}^0 (-2x) dx = 1,$$

$$A_n = \int_{-1}^0 (-2x) \cos n\pi x dx = \frac{2}{n^2\pi^2} [(-1)^n - 1],$$

and

$$B_n = \int_{-1}^0 (-2x) \sin n\pi x dx = \frac{4}{n\pi} (-1)^n$$

for $n = 1, 2, 3, \dots$ we have

$$f(x) = \frac{1}{2} + \sum_{n=1}^{\infty} \left(\frac{2}{n^2\pi^2} [(-1)^n - 1] \cos n\pi x + \frac{4}{n\pi} (-1)^n \sin n\pi x \right).$$

14. Since

$$A_0 = \int_{-1}^1 (2x^2 - 1) dx = -\frac{2}{3},$$

$$A_n = \int_{-1}^1 (2x^2 - 1) \cos n\pi x dx = \frac{8}{n^2\pi^2} (-1)^n,$$

and

$$B_n = \int_{-1}^1 (2x^2 - 1) \sin n\pi x dx = 0$$

for $n = 1, 2, 3, \dots$ we have

$$f(x) = -\frac{1}{3} + \sum_{n=1}^{\infty} \frac{8}{n^2\pi^2} (-1)^n \cos n\pi x.$$

15. Since

$$A_0 = 2 \int_0^1 e^{-x} dx$$

and

$$A_n = 2 \int_{-1}^1 e^{-x} \cos n\pi x dx = \frac{2}{1 + n^2\pi^2} [(1 - (-1)^n e^{-1})]$$

for $n = 1, 2, 3, \dots$ we have

$$f(x) = 1 - e^{-1} + 2 \sum_{n=1}^{\infty} \frac{1 - (-1)^n e^{-1}}{1 + n^2\pi^2} \cos n\pi x.$$

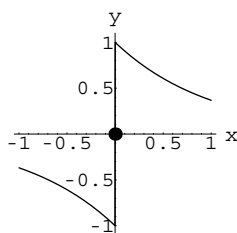
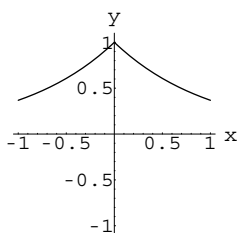
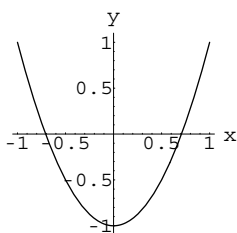
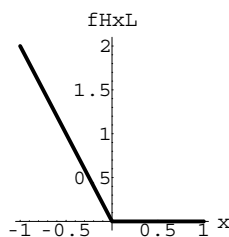
Since

$$B_n = 2 \int_0^1 e^{-x} \sin n\pi x dx = \frac{2n\pi}{1 + n^2\pi^2} [(1 - (-1)^n e^{-1})]$$

for $n = 1, 2, 3, \dots$ we have

$$f(x) = \sum_{n=1}^{\infty} \frac{2n\pi}{1 + n^2\pi^2} [(1 - (-1)^n e^{-1})] \sin n\pi x.$$

16.



$$f(x) = |x| - 1 \quad f(x) = 2x^2 - 1 \quad f(x) = e^{-|x|} \quad f(x) = \begin{cases} e^{-x}, & 0 < x < 1 \\ 0, & x = 0 \\ -e^x, & -1 < x < 0 \end{cases}$$

17. For $\lambda > 0$ a general solution of the given differential equation is

$$y = c_1 \cos(3\sqrt{\lambda} \ln x) + c_2 \sin(3\sqrt{\lambda} \ln x)$$

and

$$y' = -\frac{3c_1\sqrt{\lambda}}{x} \sin(3\sqrt{\lambda} \ln x) + \frac{3c_2\sqrt{\lambda}}{x} \cos(3\sqrt{\lambda} \ln x).$$

Since $\ln 1 = 0$, the boundary condition $y'(1) = 0$ implies $c_2 = 0$. Therefore

$$y = c_1 \cos(3\sqrt{\lambda} \ln x).$$

Using $\ln e = 1$ we find that $y(e) = 0$ implies $c_1 \cos 3\sqrt{\lambda} = 0$ or $3\sqrt{\lambda} = \frac{2n-1}{2}\pi$, for $n = 1, 2, 3, \dots$. The eigenvalues are $\lambda = (2n-1)^2\pi^2/36$ with corresponding eigenfunctions $\cos\left(\frac{2n-1}{2}\pi \ln x\right)$ for $n = 1, 2, 3, \dots$.

18. To obtain the self-adjoint form of the differential equation in Problem 17 we note that an integrating factor is $(1/x^2)e^{\int dx/x} = 1/x$. Thus the weight function is $9/x$ and an orthogonality relation is

$$\int_1^e \frac{9}{x} \cos\left(\frac{2n-1}{2}\pi \ln x\right) \cos\left(\frac{2m-1}{2}\pi \ln x\right) dx = 0, \quad m \neq n.$$

19. The boundary condition indicates that we use (15) and (16) of Section 12.6. With $b = 4$ we obtain

$$\begin{aligned} c_i &= \frac{2}{16J_1^2(4\lambda_i)} \int_0^4 x J_0(\lambda_i x) f(x) dx \\ &= \frac{1}{8J_1^2(4\lambda_i)} \int_0^2 x J_0(\lambda_i x) dx \\ &= \frac{1}{8J_1^2(4\lambda_i)} \cdot \frac{1}{\lambda_i^2} \int_0^{2\lambda_i} t J_0(t) dt \\ &= \frac{1}{8J_1^2(4\lambda_i)} \int_0^{2\lambda_i} \frac{d}{dt} [t J_1(t)] dt \\ &= \frac{1}{8J_1^2(4\lambda_i)} t J_1(t) \Big|_0^{2\lambda_i} = \frac{J_1(2\lambda_i)}{4\lambda_i J_1^2(4\lambda_i)}. \end{aligned}$$

$t = \lambda_i x \quad dt = \lambda_i dx$

[From (4) in 12.6 in the text]

Thus

$$f(x) = \frac{1}{4} \sum_{i=1}^{\infty} \frac{J_1(2\lambda_i)}{\lambda_i J_1^2(4\lambda_i)} J_0(\lambda_i x).$$

20. Since $f(x) = x^4$ is a polynomial in x , an expansion of f in polynomials in x must terminate with the term having the same degree as f . Using the fact that $x^4 P_1(x)$ and $x^4 P_3(x)$ are odd functions, we see immediately

Chapter 12 Review Exercises

that $c_1 = c_3 = 0$. Now

$$c_0 = \frac{1}{2} \int_{-1}^1 x^4 P_0(x) dx = \frac{1}{2} \int_{-1}^1 x^4 dx = \frac{1}{5}$$

$$c_2 = \frac{5}{2} \int_{-1}^1 x^4 P_2(x) dx = \frac{5}{2} \int_{-1}^1 \frac{1}{2}(3x^6 - x^4) dx = \frac{4}{7}$$

$$c_4 = \frac{9}{2} \int_{-1}^1 x^4 P_4(x) dx = \frac{9}{2} \int_{-1}^1 \frac{1}{8}(35x^8 - 30x^6 + 3x^4) dx = \frac{8}{35}.$$

Thus

$$f(x) = \frac{1}{5}P_0(x) + \frac{4}{7}P_2(x) + \frac{8}{35}P_4(x).$$

13 Boundary-Value Problems in Rectangular Coordinates

Exercises 13.1

1. If $u = XY$ then

$$u_x = X'Y,$$

$$u_y = XY',$$

$$X'Y = XY',$$

and

$$\frac{X'}{X} = \frac{Y'}{Y} = \pm\lambda^2.$$

Then

$$X' \mp \lambda^2 X = 0 \quad \text{and} \quad Y' \mp \lambda^2 Y = 0$$

so that

$$X = A_1 e^{\pm\lambda^2 x},$$

$$Y = A_2 e^{\pm\lambda^2 y},$$

and

$$u = XY = c_1 e^{c_2(x+y)}.$$

2. If $u = XY$ then

$$u_x = X'Y,$$

$$u_y = XY',$$

$$X'Y = -3XY',$$

and

$$\frac{X'}{-3X} = \frac{Y'}{Y} = \pm\lambda^2.$$

Then

$$X' \pm 3\lambda^2 X = 0 \quad \text{and} \quad Y' \mp \lambda^2 Y = 0$$

so that

$$X = A_1 e^{\mp 3\lambda^2 x},$$

$$Y = A_2 e^{\pm\lambda^2 y},$$

and

$$u = XY = c_1 e^{c_2(y-3x)}.$$

3. If $u = XY$ then

$$u_x = X'Y,$$

$$u_y = XY',$$

$$X'Y = X(Y - Y'),$$

and

Exercises 13.1

$$\frac{X'}{X} = \frac{Y - Y'}{Y} = \pm\lambda^2.$$

Then

$$X' \mp \lambda^2 X = 0 \quad \text{and} \quad Y' - (1 \mp \lambda^2)Y = 0$$

so that

$$X = A_1 e^{\pm\lambda^2 x},$$

$$Y = A_2 e^{(1 \mp \lambda^2)y},$$

and

$$u = XY = c_1 e^{y+c_2(x-y)}.$$

4. If $u = XY$ then

$$u_x = X'Y,$$

$$u_y = XY',$$

$$X'Y = X(Y + Y'),$$

and

$$\frac{X'}{X} = \frac{Y + Y'}{Y} = \pm\lambda^2.$$

Then

$$X' \mp \lambda^2 X = 0 \quad \text{and} \quad Y' - (-1 \pm \lambda^2)Y = 0$$

so that

$$X = A_1 e^{\pm\lambda^2 x},$$

$$Y = A_2 e^{(-1 \pm \lambda^2)y},$$

and

$$u = XY = c_1 e^{-y+c_2(x+y)}.$$

5. If $u = XY$ then

$$u_x = X'Y,$$

$$u_y = XY',$$

$$xX'Y = yXY',$$

and

$$\frac{xX'}{X} = \frac{yY'}{Y} = \pm\lambda^2.$$

Then

$$X \mp \frac{1}{x}\lambda^2 X = 0 \quad \text{and} \quad Y' \mp \frac{1}{y}\lambda^2 Y = 0$$

so that

$$X = A_1 x^{\pm\lambda^2},$$

$$Y = A_2 y^{\pm\lambda^2},$$

and

$$u = XY = c_1 (xy)^{c_2}.$$

6. If $u = XY$ then

$$\begin{aligned}u_x &= X'Y, \\u_y &= XY', \\yX'Y &= xXY',\end{aligned}$$

and

$$\frac{X'}{xX} = \frac{Y'}{-yY} = \pm\lambda^2.$$

Then

$$X \mp \lambda^2 xX = 0 \quad \text{and} \quad Y' \pm \lambda^2 yY = 0$$

so that

$$\begin{aligned}X &= A_1 e^{\pm\lambda^2 x^2/2}, \\Y &= A_2 e^{\mp\lambda^2 y^2/2},\end{aligned}$$

and

$$u = XY = c_1 e^{c_2(x^2 - y^2)}.$$

7. If $u = XY$ then

$$u_{xx} = X''Y, \quad u_{yy} = XY'', \quad u_{yx} = X'Y',$$

and

$$X''Y + X'Y' + XY'' = 0,$$

which is not separable.

8. If $u = XY$ then

$$\begin{aligned}u_{yx} &= X'Y', \\yX'Y' + XY &= 0,\end{aligned}$$

and

$$\frac{X'}{-X} = \frac{Y}{yY'} = \pm\lambda^2.$$

Then

$$X' \mp \lambda^2 X = 0 \quad \text{and} \quad \pm\lambda^2 yY' - Y = 0$$

so that

$$\begin{aligned}X &= A_1 e^{\mp\lambda^2 x}, \\Y &= A_2 y^{\pm 1/\lambda^2},\end{aligned}$$

and

$$u = XY = c_1 e^{-c_2 x} y^{1/c_2}.$$

9. If $u = XT$ then

$$\begin{aligned}u_t &= XT', \\u_{xx} &= X''T, \\kX''T - XT &= XT',\end{aligned}$$

and we choose

$$\frac{T'}{T} = \frac{kX'' - X}{X} = -1 \pm k\lambda^2$$

Exercises 13.1

so that

$$T' - (-1 \pm k\lambda^2)T = 0 \quad \text{and} \quad X'' - (\pm\lambda^2)X = 0.$$

For $\lambda^2 > 0$ we obtain

$$X = A_1 \cosh \lambda x + A_2 \sinh \lambda x \quad \text{and} \quad T = A_3 e^{(-1+k\lambda^2)t}$$

so that

$$u = XT = e^{(-1+k\lambda^2)t} (c_1 \cosh \lambda x + c_2 \sinh \lambda x).$$

For $-\lambda^2 < 0$ we obtain

$$X = A_1 \cos \lambda x + A_2 \sin \lambda x \quad \text{and} \quad T = A_3 e^{(-1-k\lambda^2)t}$$

so that

$$u = XT = e^{(-1-k\lambda^2)t} (c_3 \cos \lambda x + c_4 \sin \lambda x).$$

If $\lambda^2 = 0$ then

$$X'' = 0 \quad \text{and} \quad T' + T = 0,$$

and we obtain

$$X = A_1 x + A_2 \quad \text{and} \quad T = A_3 e^{-t}.$$

In this case

$$u = XT = e^{-t} (c_5 x + c_6)$$

10. If $u = XT$ then

$$u_t = XT',$$

$$u_{xx} = X''T,$$

$$kX''T = XT',$$

and

$$\frac{X''}{X} = \frac{T'}{kT} = \pm\lambda^2$$

so that

$$X'' \mp \lambda^2 X = 0 \quad \text{and} \quad T' \mp \lambda^2 kT = 0.$$

For $\lambda^2 > 0$ we obtain

$$X = A_1 \cos \lambda x + A_2 \sin \lambda x \quad \text{and} \quad T = A_3 e^{\lambda^2 kt},$$

so that

$$u = XT = e^{-c_3^2 kt} (c_1 \cos c_3 x + c_2 \sin c_3 x).$$

For $-\lambda^2 < 0$ we obtain

$$X = A_1 e^{-\lambda x} + A_2 e^{\lambda x},$$

$$T = A_3 e^{\lambda^2 kt},$$

and

$$u = XT = e^{c_3^2 kt} (c_1 e^{-c_3 x} + c_2 e^{c_3 x}).$$

For $\lambda^2 = 0$ we obtain

$$T = A_3, \quad X = A_1 x + A_2, \quad \text{and} \quad u = XT = c_1 x + c_2.$$

11. If $u = XT$ then

$$\begin{aligned}u_{xx} &= X''T, \\u_{tt} &= XT'', \\a^2X''T &= XT'',\end{aligned}$$

and

$$\frac{X''}{X} = \frac{T''}{a^2T} = \pm\lambda^2$$

so that

$$X'' \mp \lambda^2 X = 0 \quad \text{and} \quad T'' \mp a^2\lambda^2 T = 0.$$

For $\lambda^2 > 0$ we obtain

$$\begin{aligned}X &= A_1 \cos \lambda x + A_2 \sin \lambda x, \\T &= A_3 \cos a\lambda t + A_4 \sin a\lambda t,\end{aligned}$$

and

$$u = XT = (A_1 \cos \lambda x + A_2 \sin \lambda x)(A_3 \cos a\lambda t + A_4 \sin a\lambda t).$$

For $-\lambda^2 < 0$ we obtain

$$\begin{aligned}X &= A_1 e^{\lambda x} + A_2 e^{-\lambda x}, \\T &= A_3 e^{a\lambda t} + A_4 e^{-a\lambda t},\end{aligned}$$

and

$$u = XT = (A_1 e^{\lambda x} + A_2 e^{-\lambda x})(A_3 e^{a\lambda t} + A_4 e^{-a\lambda t}).$$

For $\lambda^2 = 0$ we obtain

$$\begin{aligned}X &= A_1 x + A_2, \\T &= A_3 t + A_4,\end{aligned}$$

and

$$u = XT = (A_1 x + A_2)(A_3 t + A_4).$$

12. If $u = XT$ then

$$\begin{aligned}u_t &= XT', \\u_{tt} &= XT'', \\u_{xx} &= X''T, \\a^2X''T &= XT'' + 2kXT',\end{aligned}$$

and

$$\frac{X''}{X} = \frac{T'' + 2kT'}{a^2T} = \pm\lambda^2$$

so that

$$X'' \mp \lambda^2 X = 0 \quad \text{and} \quad T'' + 2kT' \mp a^2\lambda^2 T = 0.$$

For $\lambda^2 > 0$ we obtain

$$\begin{aligned}X &= A_1 e^{\lambda x} + A_2 e^{-\lambda x}, \\T &= A_3 e^{(-k + \sqrt{k^2 + a^2\lambda^2})t} + A_4 e^{(-k - \sqrt{k^2 + a^2\lambda^2})t},\end{aligned}$$

and

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$$u = XT = (A_1 e^{\lambda x} + A_2 e^{-\lambda x}) \left(A_3 e^{(-k + \sqrt{k^2 - a^2 \lambda^2})t} + A_4 e^{(-k - \sqrt{k^2 - a^2 \lambda^2})t} \right).$$

For $-\lambda^2 < 0$ we obtain

$$X = A_1 \cos \lambda x + A_2 \sin \lambda x.$$

If $k^2 - a^2 \lambda^2 > 0$ then

$$T = A_3 e^{(-k + \sqrt{k^2 - a^2 \lambda^2})t} + A_4 e^{(-k - \sqrt{k^2 - a^2 \lambda^2})t}.$$

If $k^2 - a^2 \lambda^2 < 0$ then

$$T = e^{-kt} \left(A_3 \cos \sqrt{a^2 \lambda^2 - k^2} t + A_4 \sin \sqrt{a^2 \lambda^2 - k^2} t \right).$$

If $k^2 - a^2 \lambda^2 = 0$ then

$$T = A_3 e^{-kt} + A_4 t e^{-kt}$$

so that

$$\begin{aligned} u = XT &= (A_1 \cos \lambda x + A_2 \sin \lambda x) \left(A_3 e^{(-k + \sqrt{k^2 - a^2 \lambda^2})t} + A_4 e^{(-k - \sqrt{k^2 - a^2 \lambda^2})t} \right) \\ &= (A_1 \cos \lambda x + A_2 \sin \lambda x) e^{-kt} \left(A_3 \cos \sqrt{a^2 \lambda^2 - k^2} t + A_4 \sin \sqrt{a^2 \lambda^2 - k^2} t \right) \\ &= \left(A_1 \cos \frac{k}{a} x + A_2 \sin \frac{k}{a} x \right) (A_3 e^{-kt} + A_4 t e^{-kt}). \end{aligned}$$

For $\lambda^2 = 0$ we obtain

$$\begin{aligned} X &= A_1 x + A_2, \\ T &= A_3 + A_4 e^{-2kt}, \end{aligned}$$

and

$$u = XT = (A_1 x + A_2)(A_3 + A_4 e^{-2kt}).$$

13. If $u = XY$ then

$$\begin{aligned} u_{xx} &= X''Y, \\ u_{yy} &= XY'', \\ X''Y + xY'' &= 0, \end{aligned}$$

and

$$\frac{X''}{-X} = \frac{Y''}{Y} = \pm \lambda^2$$

so that

$$X'' \pm \lambda^2 X = 0 \quad \text{and} \quad Y'' \mp \lambda^2 Y = 0.$$

For $\lambda^2 > 0$ we obtain

$$\begin{aligned} X &= A_1 \cos \lambda x + A_2 \sin \lambda x, \\ Y &= A_3 e^{\lambda y} + A_4 e^{-\lambda y}, \end{aligned}$$

and

$$u = XY = (A_1 \cos \lambda x + A_2 \sin \lambda x)(A_3 e^{\lambda y} + A_4 e^{-\lambda y}).$$

For $-\lambda^2 < 0$ we obtain

$$\begin{aligned} X &= A_1 e^{\lambda x} + A_2 e^{-\lambda x}, \\ Y &= A_3 \cos \lambda y + A_4 \sin \lambda y, \end{aligned}$$

and

$$u = XY = (A_1 e^{\lambda x} + A_2 e^{-\lambda x})(A_3 \cos \lambda y + A_4 \sin \lambda y).$$

For $\lambda^2 = 0$ we obtain

$$u = XY = (A_1 x + A_2)(A_3 y + A_4).$$

14. If $u = XY$ then

$$\begin{aligned} u_{xx} &= X''Y, \\ u_{yy} &= XY'', \\ x^2 X''Y + xY'' &= 0, \end{aligned}$$

and

$$\frac{x^2 X''}{-X} = \frac{Y''}{Y} = \pm \lambda^2$$

so that

$$x^2 X'' \pm \lambda^2 X = 0 \quad \text{and} \quad Y'' \mp \lambda^2 Y = 0.$$

For $\lambda^2 > 0$ we obtain

$$Y = A_3 e^{\lambda y} + A_4 e^{-\lambda y}.$$

If $1 - 4\lambda^2 > 0$ then

$$X = A_1 x^{(1/2 + \sqrt{1-4\lambda^2}/2)} + A_2 x^{(1/2 - \sqrt{1-4\lambda^2}/2)}.$$

If $1 - 4\lambda^2 < 0$ then

$$X = x^{1/2} \left(A_1 \cos \frac{1}{2} \sqrt{4\lambda^2 - 1} \ln x + A_2 \sin \frac{1}{2} \sqrt{4\lambda^2 - 1} \ln x \right).$$

If $1 - 4\lambda^2 = 0$ then

$$X = A_1 x^{1/2} + A_2 x^{1/2} \ln x$$

so that

$$\begin{aligned} u = XY &= \left(A_1 x^{(1/2 + \sqrt{1-4\lambda^2}/2)} + A_2 x^{(1/2 - \sqrt{1-4\lambda^2}/2)} \right) (A_3 e^{\lambda y} + A_4 e^{-\lambda y}) \\ &= x^{1/2} \left(A_1 \cos \frac{1}{2} \sqrt{4\lambda^2 - 1} \ln x + A_2 \sin \frac{1}{2} \sqrt{4\lambda^2 - 1} \ln x \right) (A_3 e^{y/2} + A_4 e^{-y/2}) \\ &= (A_1 x^{1/2} + A_2 x^{1/2} \ln x) (A_3 e^{y/2} + A_4 e^{-y/2}). \end{aligned}$$

For $-\lambda^2 < 0$ we obtain

$$X = A_1 e^{(1/2 + \sqrt{1+4\lambda^2}/2)x} + A_2 e^{(1/2 - \sqrt{1+4\lambda^2}/2)x},$$

$$Y = A_3 \cos \lambda y + A_4 \sin \lambda y,$$

and

$$u = XY = \left(A_1 e^{(1/2 + \sqrt{1+4\lambda^2}/2)x} + A_2 e^{(1/2 - \sqrt{1+4\lambda^2}/2)x} \right) (A_3 \cos \lambda y + A_4 \sin \lambda y).$$

For $\lambda^2 = 0$ we obtain

$$X = A_1 x + A_2,$$

$$Y = A_3 y + A_4,$$

and

$$u = XY = (A_1 x + A_2)(A_3 y + A_4).$$

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15. If $u = XY$ then

$$u_{xx} = X''Y,$$

$$u_{yy} = XY'',$$

$$X''Y + XY'' = XY,$$

and

$$\frac{X''}{X} = \frac{Y - Y''}{Y} = \pm\lambda^2$$

so that

$$X'' \mp \lambda^2 X = 0 \quad \text{and} \quad Y'' + (\pm\lambda^2 - 1)Y = 0.$$

For $\lambda^2 > 0$ we obtain

$$X = A_1 e^{\lambda x} + A_2 e^{-\lambda x}.$$

If $\lambda^2 - 1 > 0$ then

$$Y = A_3 \cos \sqrt{\lambda^2 - 1} y + A_4 \sin \sqrt{\lambda^2 - 1} y.$$

If $\lambda^2 - 1 < 0$ then

$$Y = A_3 e^{\sqrt{1-\lambda^2} y} + A_4 e^{-\sqrt{1-\lambda^2} y}.$$

If $\lambda^2 - 1 = 0$ then $Y = A_3 y + A_4$ so that

$$\begin{aligned} u = XY &= (A_1 e^{\lambda x} + A_2 e^{-\lambda x}) \left(A_3 \cos \sqrt{\lambda^2 - 1} y + A_4 \sin \sqrt{\lambda^2 - 1} y \right), \\ &= (A_1 e^{\lambda x} + A_2 e^{-\lambda x}) \left(A_3 e^{\sqrt{1-\lambda^2} y} + A_4 e^{-\sqrt{1-\lambda^2} y} \right) \\ &= (A_1 e^x + A_2 e^{-x})(A_3 y + A_4). \end{aligned}$$

For $-\lambda^2 < 0$ we obtain

$$\begin{aligned} X &= A_1 \cos \lambda x + A_2 \sin \lambda x, \\ Y &= A_3 e^{\sqrt{1+\lambda^2} y} + A_4 e^{-\sqrt{1+\lambda^2} y}, \end{aligned}$$

and

$$u = XY = (A_1 \cos \lambda x + A_2 \sin \lambda x) \left(A_3 e^{\sqrt{1+\lambda^2} y} + A_4 e^{-\sqrt{1+\lambda^2} y} \right).$$

For $\lambda^2 = 0$ we obtain

$$\begin{aligned} X &= A_1 x + A_2, \\ Y &= A_3 e^y + A_4 e^{-y}, \end{aligned}$$

and

$$u = XY = (A_1 x + A_2)(A_3 e^y + A_4 e^{-y}).$$

16. If $u = XT$ then

$$u_{tt} = XT'', \quad u_{xx} = X''T, \quad \text{and} \quad a^2 X''T - g = XT'',$$

which is not separable.

17. Identifying $A = B = C = 1$, we compute $B^2 - 4AC = -3 < 0$. The equation is elliptic.

18. Identifying $A = 3$, $B = 5$, and $C = 1$, we compute $B^2 - 4AC = 13 > 0$. The equation is hyperbolic.

19. Identifying $A = 1$, $B = 6$, and $C = 9$, we compute $B^2 - 4AC = 0$. The equation is parabolic.

20. Identifying $A = 1$, $B = -1$, and $C = -3$, we compute $B^2 - 4AC = 13 > 0$. The equation is hyperbolic.

21. Identifying $A = 1$, $B = -9$, and $C = 0$, we compute $B^2 - 4AC = 81 > 0$. The equation is hyperbolic.
22. Identifying $A = 0$, $B = 1$, and $C = 0$, we compute $B^2 - 4AC = 1 > 0$. The equation is hyperbolic.
23. Identifying $A = 1$, $B = 2$, and $C = 1$, we compute $B^2 - 4AC = 0$. The equation is parabolic.
24. Identifying $A = 1$, $B = 0$, and $C = 1$, we compute $B^2 - 4AC = -4 < 0$. The equation is elliptic.
25. Identifying $A = a^2$, $B = 0$, and $C = -1$, we compute $B^2 - 4AC = 4a^2 > 0$. The equation is hyperbolic.
26. Identifying $A = k > 0$, $B = 0$, and $C = 0$, we compute $B^2 - 4AC = -4k < 0$. The equation is elliptic.
27. If $u = RT$ then

$$\begin{aligned}u_r &= R'T, \\u_{rr} &= R''T, \\u_t &= RT', \\RT' &= k \left(R''T + \frac{1}{r}R'T \right),\end{aligned}$$

and

$$\frac{r^2 R'' + rR'}{r^2 R} = \frac{T'}{kT} = \pm \lambda^2.$$

If we use $-\lambda^2 < 0$ then

$$r^2 R'' + rR' + \lambda^2 r^2 R = 0 \quad \text{and} \quad T'' \mp \lambda^2 kT = 0$$

so that

$$\begin{aligned}R &= A_2 J_0(\lambda r) + A_3 Y_0(\lambda r), \\T &= A_1 e^{-k\lambda^2 t},\end{aligned}$$

and

$$u = RT = e^{-k\lambda^2 t} [c_1 J_0(\lambda r) + c_2 Y_0(\lambda r)]$$

28. If $u = R\Theta$ then

$$\begin{aligned}u_r &= R'\Theta, \\u_{rr} &= R''\Theta, \\u_t &= R\Theta', \\u_{tt} &= R\Theta'', \\R''\Theta + \frac{1}{r}R'\Theta + \frac{1}{r^2}R\Theta'' &= 0,\end{aligned}$$

and

$$\frac{r^2 R'' + rR'}{R} = -\frac{\Theta''}{\Theta} = \lambda^2.$$

Then

$$r^2 R'' + rR' - \lambda^2 R = 0 \quad \text{and} \quad \Theta'' + \lambda^2 \Theta = 0$$

so that

$$\begin{aligned}R &= c_3 r^\lambda + c_4 r^{-\lambda} \\ \Theta &= c_1 \cos \lambda \theta + c_2 \sin \lambda \theta,\end{aligned}$$

and

$$u = R\Theta = (c_1 \cos \lambda \theta + c_2 \sin \lambda \theta)(c_3 r^\lambda + c_4 r^{-\lambda}).$$

Exercises 13.1

29. For $u = A_1 e^{\lambda^2 y} \cosh 2\lambda x + B_1 e^{\lambda^2 y} \sinh 2\lambda x$ we compute

$$\frac{\partial^2 u}{\partial x^2} = 4\lambda^2 A_1 e^{\lambda^2 y} \cosh 2\lambda x + 4\lambda^2 B_1 e^{\lambda^2 y} \sinh 2\lambda x$$

and

$$\frac{\partial u}{\partial y} = \lambda^2 A_1 e^{\lambda^2 y} \cosh 2\lambda x + \lambda^2 B_1 e^{\lambda^2 y} \sinh 2\lambda x.$$

Then $\partial^2 u / \partial x^2 = 4 \partial u / \partial y$.

For $u = A_2 e^{-\lambda^2 y} \cos 2\lambda x + B_2 e^{-\lambda^2 y} \sin 2\lambda x$ we compute

$$\frac{\partial^2 u}{\partial x^2} = -4\lambda^2 A_2 e^{-\lambda^2 y} \cos 2\lambda x - 4\lambda^2 B_2 e^{-\lambda^2 y} \sin 2\lambda x$$

and

$$\frac{\partial u}{\partial y} = -\lambda^2 A_2 \cos 2\lambda x - \lambda^2 B_2 \sin 2\lambda x.$$

Then $\partial^2 u / \partial x^2 = 4 \partial u / \partial y$.

For $u = A_3 x + B_3$ we compute $\partial^2 u / \partial x^2 = \partial u / \partial y = 0$. Then $\partial^2 u / \partial x^2 = 4 \partial u / \partial y$.

30. We identify $A = xy + 1$, $B = x + 2y$, and $C = 1$. Then $B^2 - 4AC = x^2 + 4y^2 - 4$. The equation $x^2 + 4y^2 = 4$ defines an ellipse. The partial differential equation is hyperbolic outside the ellipse, parabolic on the ellipse, and elliptic inside the ellipse.

Exercises 13.2

1. $k \frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}, \quad 0 < x < L, \quad t > 0$

$$u(0, t) = 0, \quad \left. \frac{\partial u}{\partial x} \right|_{x=L} = 0, \quad t > 0$$

$$u(x, 0) = f(x), \quad 0 < x < L$$

2. $k \frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}, \quad 0 < x < L, \quad t > 0$

$$u(0, t) = u_0, \quad u(L, t) = u_1, \quad t > 0$$

$$u(x, 0) = 0, \quad 0 < x < L$$

3. $k \frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}, \quad 0 < x < L, \quad t > 0$

$$u(0, t) = 100, \quad \left. \frac{\partial u}{\partial x} \right|_{x=L} = -hu(L, t), \quad t > 0$$

$$u(x, 0) = f(x), \quad 0 < x < L$$

4. $k \frac{\partial^2 u}{\partial x^2} + h(u - 50) = \frac{\partial u}{\partial t}, \quad 0 < x < L, \quad t > 0$

$$\left. \frac{\partial u}{\partial x} \right|_{x=0} = 0, \quad \left. \frac{\partial u}{\partial x} \right|_{x=L} = 0, \quad t > 0$$

$$u(x, 0) = 100, \quad 0 < x < L$$

5. $a^2 \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2}, \quad 0 < x < L, \quad t > 0$

$$u(0, t) = 0, \quad u(L, t) = 0, \quad t > 0$$

$$u(x, 0) = x(L - x), \quad \left. \frac{\partial u}{\partial x} \right|_{t=0} = 0, \quad 0 < x < L$$

6. $a^2 \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2}$, $0 < x < L$, $t > 0$
 $u(0, t) = 0$, $u(L, t) = 0$, $t > 0$
 $u(x, 0) = 0$, $\left. \frac{\partial u}{\partial x} \right|_{t=0} = \sin \frac{\pi x}{L}$, $0 < x < L$
7. $a^2 \frac{\partial^2 u}{\partial x^2} - 2\beta \frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial t^2}$, $0 < x < L$, $t > 0$
 $u(0, t) = 0$, $u(L, t) = \sin \pi t$, $t > 0$
 $u(x, 0) = f(x)$, $\left. \frac{\partial u}{\partial t} \right|_{t=0} = 0$, $0 < x < L$
8. $a^2 \frac{\partial^2 u}{\partial x^2} + Ax = \frac{\partial^2 u}{\partial t^2}$, $0 < x < L$, $t > 0$, A a constant
 $u(0, t) = 0$, $u(L, t) = 0$, $t > 0$
 $u(x, 0) = 0$, $\left. \frac{\partial u}{\partial x} \right|_{t=0} = 0$, $0 < x < L$
9. $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$, $0 < x < 4$, $0 < y < 2$
 $\left. \frac{\partial u}{\partial x} \right|_{x=0} = 0$, $u(4, y) = f(y)$, $0 < y < 2$
 $\left. \frac{\partial u}{\partial y} \right|_{y=0} = 0$, $u(x, 2) = 0$, $0 < x < 4$
10. $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$, $0 < x < \pi$, $y > 0$
 $u(0, y) = e^{-y}$, $u(\pi, y) = \begin{cases} 100, & 0 < y \leq 1 \\ 0, & y > 1 \end{cases}$
 $u(x, 0) = f(x)$, $0 < x < \pi$

Exercises 13.3

1. Using $u = XT$ and $-\lambda^2$ as a separation constant leads to

$$X'' + \lambda^2 X = 0,$$

$$X(0) = 0,$$

$$X(L) = 0,$$

and

$$T' + k\lambda^2 T = 0.$$

Then

$$X = c_1 \sin \frac{n\pi}{L} x \quad \text{and} \quad T = c_2 e^{-\frac{kn^2\pi^2}{L^2} t}$$

for $n = 1, 2, 3, \dots$ so that

$$u = \sum_{n=1}^{\infty} A_n e^{-\frac{kn^2\pi^2}{L^2} t} \sin \frac{n\pi}{L} x.$$

Exercises 13.3

Imposing

$$u(x, 0) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{L} x$$

gives

$$A_n = \frac{2}{L} \int_0^{L/2} \sin \frac{n\pi}{L} x \, dx = \frac{2}{n\pi} \left(1 - \cos \frac{n\pi}{2} \right)$$

for $n = 1, 2, 3, \dots$ so that

$$u(x, t) = \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1 - \cos \frac{n\pi}{2}}{n} e^{-\frac{kn^2\pi^2}{L^2}t} \sin \frac{n\pi}{L} x.$$

2. Using $u = XT$ and $-\lambda^2$ as a separation constant leads to

$$X'' + \lambda^2 X = 0,$$

$$X(0) = 0,$$

$$X(L) = 0,$$

and

$$T' + k\lambda^2 T = 0.$$

Then

$$X = c_1 \sin \frac{n\pi}{L} x \quad \text{and} \quad T = c_2 e^{-\frac{kn^2\pi^2}{L^2}t}$$

for $n = 1, 2, 3, \dots$ so that

$$u = \sum_{n=1}^{\infty} A_n e^{-\frac{kn^2\pi^2}{L^2}t} \sin \frac{n\pi}{L} x.$$

Imposing

$$u(x, 0) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{L} x$$

gives

$$A_n = \frac{2}{L} \int_0^L x(L-x) \sin \frac{n\pi}{L} x \, dx = \frac{4L^2}{n^3\pi^3} [1 - (-1)^n]$$

for $n = 1, 2, 3, \dots$ so that

$$u(x, t) = \frac{4L^2}{\pi^3} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n^3} e^{-\frac{kn^2\pi^2}{L^2}t} \sin \frac{n\pi}{L} x.$$

3. Using $u = XT$ and $-\lambda^2$ as a separation constant leads to

$$X'' + \lambda^2 X = 0,$$

$$X'(0) = 0,$$

$$X'(L) = 0,$$

and

$$T' + k\lambda^2 T = 0.$$

Then

$$X = c_1 \cos \frac{n\pi}{L} x \quad \text{and} \quad T = c_2 e^{-\frac{kn^2\pi^2}{L^2}t}$$

for $n = 0, 1, 2, \dots$ so that

$$u = \sum_{n=0}^{\infty} A_n e^{-\frac{kn^2\pi^2}{L^2}t} \cos \frac{n\pi}{L} x.$$

Imposing

$$u(x, 0) = f(x) = \sum_{n=0}^{\infty} A_n \cos \frac{n\pi}{L} x$$

gives

$$u(x, t) = \frac{1}{L} \int_0^L f(x) dx + \frac{2}{L} \sum_{n=1}^{\infty} \left(\int_0^L f(x) \cos \frac{n\pi}{L} x dx \right) e^{-\frac{kn^2\pi^2}{L^2}t} \cos \frac{n\pi}{L} x.$$

4. If $L = 2$ and $f(x)$ is x for $0 < x < 1$ and $f(x)$ is 0 for $1 < x < 2$ then

$$u(x, t) = \frac{1}{4} + 4 \sum_{n=1}^{\infty} \left[\frac{1}{2n\pi} \sin \frac{n\pi}{2} + \frac{1}{n^2\pi^2} \left(\cos \frac{n\pi}{2} - 1 \right) \right] e^{-\frac{kn^2\pi^2}{4}t} \cos \frac{n\pi}{2} x.$$

5. Using $u = XT$ and $-\lambda^2$ as a separation constant leads to

$$X'' + \lambda^2 X = 0,$$

$$X'(0) = 0,$$

$$X'(L) = 0,$$

and

$$T' + (h + k\lambda^2)T = 0.$$

Then

$$X = c_1 \cos \frac{n\pi}{L} x \quad \text{and} \quad T = c_2 e^{-\left(h + \frac{kn^2\pi^2}{L^2}\right)t}$$

for $n = 0, 1, 2, \dots$ so that

$$u = \sum_{n=0}^{\infty} A_n e^{-\left(h + \frac{kn^2\pi^2}{L^2}\right)t} \cos \frac{n\pi}{L} x.$$

Imposing

$$u(x, 0) = f(x) = \sum_{n=0}^{\infty} A_n \cos \frac{n\pi}{L} x$$

gives

$$u(x, t) = \frac{e^{-ht}}{L} \int_0^L f(x) dx + \frac{2}{L} \sum_{n=1}^{\infty} \left(\int_0^L f(x) \cos \frac{n\pi}{L} x dx \right) e^{-\left(h + \frac{kn^2\pi^2}{L^2}\right)t} \cos \frac{n\pi}{L} x.$$

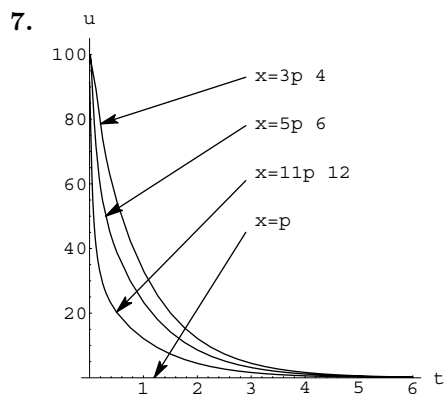
6. In Problem 5 we instead find that $X(0) = 0$ and $X(L) = 0$ so that

$$X = c_1 \sin \frac{n\pi}{L} x$$

and

$$u = \frac{2}{L} \sum_{n=1}^{\infty} \left(\int_0^L f(x) \sin \frac{n\pi}{L} x dx \right) e^{-\left(h + \frac{kn^2\pi^2}{L^2}\right)t} \sin \frac{n\pi}{L} x.$$

Exercises 13.3



8. (a) The solution is

$$u(x, t) = \sum_{n=1}^{\infty} A_n e^{-k(n^2 \pi^2 / 100^2)t} \sin \frac{n\pi}{100} x,$$

where

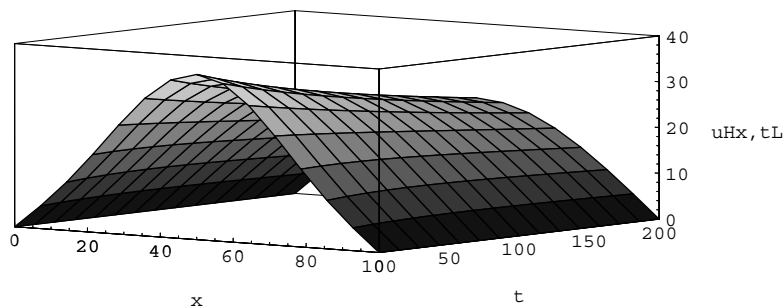
$$A_n = \frac{2}{100} \left[\int_0^{50} 0.8x \sin \frac{n\pi}{100} x \, dx + \int_{50}^{100} 0.8(100 - x) \sin \frac{n\pi}{100} x \, dx \right] = \frac{320}{n^2 \pi^2} \sin \frac{n\pi}{2}.$$

Thus,

$$u(x, t) = \frac{320}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \left(\sin \frac{n\pi}{2} \right) e^{-k(n^2 \pi^2 / 100^2)t} \sin \frac{n\pi}{100} x.$$

- (b) Since $A_n = 0$ for n even, the first five nonzero terms correspond to $n = 1, 3, 5, 7, 9$. In this case $\sin(n\pi/2) = \sin(2p-1)\pi/2 = (-1)^{p+1}$ for $p = 1, 2, 3, 4, 5$, and

$$u(x, t) = \frac{320}{\pi^2} \sum_{p=1}^{\infty} \frac{(-1)^{p+1}}{(2p-1)^2} e^{(-1.6352(2p-1)^2 \pi^2 / 100^2)t} \sin \frac{(2p-1)\pi}{100} x.$$



Exercises 13.4

1. Using $u = XT$ and $-\lambda^2$ as a separation constant leads to

$$X'' + \lambda^2 X = 0,$$

$$X(0) = 0,$$

$$X(L) = 0,$$

and

$$T'' + \lambda^2 a^2 T = 0.$$

Then

$$X = c_1 \sin \frac{n\pi}{L}x \quad \text{and} \quad T = c_2 \cos \frac{n\pi a}{L}t + c_3 \sin \frac{n\pi a}{L}t$$

for $n = 1, 2, 3, \dots$ so that

$$u = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi a}{L}t + B_n \sin \frac{n\pi a}{L}t \right) \sin \frac{n\pi}{L}x.$$

Imposing

$$u(x, 0) = \frac{1}{4}x(L - x) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{L}x$$

and

$$u_t(x, 0) = 0 = \sum_{n=1}^{\infty} B_n \frac{n\pi a}{L} \sin \frac{n\pi}{L}x$$

gives

$$A_n = \frac{L}{n^3\pi^3}[1 - (-1)^n] \quad \text{and} \quad B_n = 0$$

for $n = 1, 2, 3, \dots$ so that

$$u(x, t) = \frac{L^2}{\pi^3} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n^3} \cos \frac{n\pi a}{L}t \sin \frac{n\pi}{L}x.$$

2. Using $u = XT$ and $-\lambda^2$ as a separation constant leads to

$$X'' + \lambda^2 X = 0,$$

$$X(0) = 0,$$

$$X(L) = 0,$$

and

$$T'' + \lambda^2 a^2 T = 0.$$

Then

$$X = c_1 \sin \frac{n\pi}{L}x \quad \text{and} \quad T = c_2 \cos \frac{n\pi a}{L}t + c_3 \sin \frac{n\pi a}{L}t$$

for $n = 1, 2, 3, \dots$ so that

$$u = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi a}{L}t + B_n \sin \frac{n\pi a}{L}t \right) \sin \frac{n\pi}{L}x.$$

Imposing

$$u(x, 0) = 0 = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{L}x$$

and

$$u_t(x, 0) = x(L - x) = \sum_{n=1}^{\infty} B_n \frac{n\pi a}{L} \sin \frac{n\pi}{L}x$$

gives

$$A_n = 0 \quad \text{and} \quad B_n \frac{n\pi a}{L} = \frac{4L^2}{n^3\pi^3}[1 - (-1)^n]$$

for $n = 1, 2, 3, \dots$ so that

$$u(x, t) = \frac{4L^3}{a\pi^4} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n^4} \sin \frac{n\pi a}{L}t \sin \frac{n\pi}{L}x.$$

Exercises 13.4

3. Using $u = XT$ and $-\lambda^2$ as a separation constant leads to

$$X'' + \lambda^2 X = 0,$$

$$X(0) = 0,$$

$$X(L) = 0,$$

and

$$T'' + \lambda^2 a^2 T = 0.$$

Then

$$X = c_1 \sin \frac{n\pi}{L} x \quad \text{and} \quad T = c_2 \cos \frac{n\pi a}{L} t + c_3 \sin \frac{n\pi a}{L} t$$

for $n = 1, 2, 3, \dots$ so that

$$u = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi a}{L} t + B_n \sin \frac{n\pi a}{L} t \right) \sin \frac{n\pi}{L} x.$$

Imposing

$$u(x, 0) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{L} x$$

gives

$$A_n = \frac{2}{L} \left(\int_0^{L/3} \frac{3}{L} x \sin \frac{n\pi}{L} x \, dx + \int_{L/3}^{2L/3} \sin \frac{n\pi}{L} x \, dx + \int_{2L/3}^L \left(3 - \frac{3}{L} x \right) \sin \frac{n\pi}{L} x \, dx \right)$$

so that

$$A_1 = \frac{6\sqrt{3}}{\pi^2},$$

$$A_2 = A_3 = A_4 = 0,$$

$$A_5 = -\frac{6\sqrt{3}}{5^2 \pi^2},$$

$$A_6 = 0,$$

$$A_7 = \frac{6\sqrt{3}}{7^2 \pi^2} \dots$$

Imposing

$$u_t(x, 0) = 0 = \sum_{n=1}^{\infty} B_n \frac{n\pi a}{L} \sin \frac{n\pi}{L} x$$

gives $B_n = 0$ for $n = 1, 2, 3, \dots$ so that

$$u(x, t) = \frac{6\sqrt{3}}{\pi^2} \left(\cos \frac{\pi a}{L} t \sin \frac{\pi}{L} x - \frac{1}{5^2} \cos \frac{5\pi a}{L} t \sin \frac{5\pi}{L} x + \frac{1}{7^2} \cos \frac{7\pi a}{L} t \sin \frac{7\pi}{L} x - \dots \right).$$

4. Using $u = XT$ and $-\lambda^2$ as a separation constant leads to

$$X'' + \lambda^2 X = 0,$$

$$X(0) = 0,$$

$$X(\pi) = 0,$$

and

$$T'' + \lambda^2 a^2 T = 0.$$

Then

$$X = c_1 \sin nx \quad \text{and} \quad T = c_2 \cos nat + c_3 \sin nat$$

for $n = 1, 2, 3, \dots$ so that

$$u = \sum_{n=1}^{\infty} (A_n \cos nat + B_n \sin nat) \sin nx.$$

Imposing

$$u(x, 0) = \frac{1}{6}x(\pi^2 - x^2) = \sum_{n=1}^{\infty} A_n \sin nx \quad \text{and} \quad u_t(x, 0) = 0$$

gives

$$B_n = 0 \quad \text{and} \quad A_n = \frac{2}{n^3}(-1)^{n+1}$$

for $n = 1, 2, 3, \dots$ so that

$$u(x, t) = 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^3} \cos nat \sin nx.$$

5. Using $u = XT$ and $-\lambda^2$ as a separation constant leads to

$$X'' + \lambda^2 X = 0,$$

$$X(0) = 0,$$

$$X(\pi) = 0,$$

and

$$T'' + \lambda^2 a^2 T = 0.$$

Then

$$X = c_1 \sin nx \quad \text{and} \quad T = c_2 \cos nat + c_3 \sin nat$$

for $n = 1, 2, 3, \dots$ so that

$$u = \sum_{n=1}^{\infty} (A_n \cos nat + B_n \sin nat) \sin nx.$$

Imposing

$$u(x, 0) = 0 = \sum_{n=1}^{\infty} A_n \sin nx \quad \text{and} \quad u_t(x, 0) = \sin x = \sum_{n=1}^{\infty} B_n na \sin nx$$

gives

$$A_n = 0, \quad B_1 = \frac{1}{a^2}, \quad \text{and} \quad B_n = 0$$

for $n = 2, 3, 4, \dots$ so that

$$u(x, t) = \frac{1}{a} \sin at \sin x.$$

6. Using $u = XT$ and $-\lambda^2$ as a separation constant leads to

$$X'' + \lambda^2 X = 0,$$

$$X(0) = 0,$$

$$X(1) = 0,$$

and

$$T'' + \lambda^2 a^2 T = 0.$$

Then

$$X = c_1 \sin n\pi x \quad \text{and} \quad T = c_2 \cos n\pi at + c_3 \sin n\pi at$$

Exercises 13.4

for $n = 1, 2, 3, \dots$ so that

$$u = \sum_{n=1}^{\infty} (A_n \cos n\pi at + B_n \sin n\pi at) \sin n\pi x.$$

Imposing

$$u(x, 0) = 0.01 \sin 3\pi x = \sum_{n=1}^{\infty} A_n \sin n\pi x$$

and

$$u_t(x, 0) = 0 = \sum_{n=1}^{\infty} B_n n\pi a \sin n\pi x$$

gives $B_n = 0$ for $n = 1, 2, 3, \dots$, $A_3 = 0.01$, and $A_n = 0$ for $n = 1, 2, 4, 5, 6, \dots$ so that

$$u(x, t) = 0.01 \sin 3\pi x \cos 3\pi at.$$

7. Using $u = XT$ and $-\lambda^2$ as a separation constant leads to

$$X'' + \lambda^2 X = 0,$$

$$X(0) = 0,$$

$$X(L) = 0,$$

and

$$T'' + \lambda^2 a^2 T = 0.$$

Then

$$X = c_1 \sin \frac{n\pi}{L} x \quad \text{and} \quad T = c_2 \cos \frac{n\pi a}{L} t + c_3 \sin \frac{n\pi a}{L} t$$

for $n = 1, 2, 3, \dots$ so that

$$u = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi a}{L} t + B_n \sin \frac{n\pi a}{L} t \right) \sin \frac{n\pi}{L} x.$$

Imposing

$$u(x, 0) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{L} x$$

and

$$u_t(x, 0) = 0 = \sum_{n=1}^{\infty} B_n \frac{n\pi a}{L} \sin \frac{n\pi}{L} x$$

gives

$$B_n = 0 \quad \text{and} \quad A_n = \frac{8h}{n^2 \pi^2} \sin \frac{n\pi}{2}$$

for $n = 1, 2, 3, \dots$ so that

$$u(x, t) = \frac{8h}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin \frac{n\pi}{2} \cos \frac{n\pi a}{L} t \sin \frac{n\pi}{L} x.$$

8. Using $u = XT$ and $-\lambda^2$ as a separation constant leads to

$$X'' + \lambda^2 X = 0,$$

$$X'(0) = 0,$$

$$X'(L) = 0,$$

and

$$T'' + a^2 \lambda^2 T = 0.$$

Then

$$X = c_1 \cos \frac{n\pi}{L}x \quad \text{and} \quad T = c_2 \cos \frac{n\pi a}{L}t$$

for $n = 1, 2, 3, \dots$. Since $\lambda = 0$ is also an eigenvalue with eigenfunction $X(x) = 1$ we have

$$u = A_0 + \sum_{n=1}^{\infty} A_n \cos \frac{n\pi a}{L}t \cos \frac{n\pi}{L}x.$$

Imposing

$$u(x, 0) = x = \sum_{n=0}^{\infty} A_n \cos \frac{n\pi}{L}x$$

gives

$$A_0 = \frac{1}{L} \int_0^L x \, dx = \frac{L}{2}$$

and

$$A_n = \frac{2}{L} \int_0^L x \cos \frac{n\pi}{L}x \, dx = \frac{2L}{n^2\pi^2} [(-1)^n - 1]$$

for $n = 1, 2, 3, \dots$, so that

$$u(x, t) = \frac{L}{2} + \frac{2L}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n - 1}{n^2} \cos \frac{n\pi a}{L}t \cos \frac{n\pi}{L}x.$$

9. Using $u = XT$ and $-\lambda^2$ as a separation constant leads to

$$X'' + \lambda^2 X = 0,$$

$$X(0) = 0,$$

$$X(\pi) = 0,$$

and

$$T'' + 2\beta T' + \lambda^2 T = 0.$$

Then

$$X = c_1 \sin nx \quad \text{and} \quad T = e^{-\beta t} \left(c_2 \cos \sqrt{n^2 - \beta^2} t + c_3 \sin \sqrt{n^2 - \beta^2} t \right)$$

so that

$$u = \sum_{n=1}^{\infty} e^{-\beta t} \left(A_n \cos \sqrt{n^2 - \beta^2} t + B_n \sin \sqrt{n^2 - \beta^2} t \right) \sin nx.$$

Imposing

$$u(x, 0) = f(x) = \sum_{n=1}^{\infty} A_n \sin nx$$

and

$$u_t(x, 0) = 0 = \sum_{n=1}^{\infty} \left(B_n \sqrt{n^2 - \beta^2} - \beta A_n \right) \sin nx$$

gives

$$u(x, t) = e^{-\beta t} \sum_{n=1}^{\infty} A_n \left(\cos \sqrt{n^2 - \beta^2} t + \frac{\beta}{\sqrt{n^2 - \beta^2}} \sin \sqrt{n^2 - \beta^2} t \right) \sin nx,$$

where

$$A_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx.$$

10. Using $u = XT$ and $-\lambda^2$ as a separation constant leads to $X'' + \lambda^2 X = 0$, $X(0) = 0$, $X(\pi) = 0$ and

Exercises 13.4

$T'' + (1 + \lambda^2)T = 0$, $T'(0) = 0$. Then $X = c_2 \sin nx$ and $T = c_3 \cos \sqrt{n^2 + 1} t$ for $n = 1, 2, 3, \dots$ so that

$$u = \sum_{n=1}^{\infty} B_n \cos \sqrt{n^2 + 1} t \sin nx.$$

Imposing $u(x, 0) = \sum_{n=1}^{\infty} B_n \sin nx$ gives

$$\begin{aligned} B_n &= \frac{2}{\pi} \int_0^{\pi/2} x \sin nx \, dx + \frac{2}{\pi} \int_{\pi/2}^{\pi} (\pi - x) \sin nx \, dx = \frac{4}{\pi n^2} \sin \frac{n\pi}{2} \\ &= \begin{cases} 0, & n \text{ even} \\ \frac{4}{\pi n^2} (-1)^{(n+3)/2}, & n = 2k - 1, k = 1, 2, 3, \dots \end{cases} \end{aligned}$$

Thus with $n = 2k - 1$,

$$u(x, t) = \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{\sin \frac{n\pi}{2}}{n^2} \cos \sqrt{n^2 + 1} t \sin nx = \frac{4}{\pi} \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{(2k-1)^2} \cos \sqrt{(2k-1)^2 + 1} t \sin(2k-1)x.$$

11. Separating variables in the partial differential equation gives

$$\frac{X^{(4)}}{X} = -\frac{T''}{a^2 T} = \lambda^4$$

so that

$$X^{(4)} - \lambda^4 X = 0$$

$$T'' + a^2 \lambda^4 T = 0$$

and

$$X = c_1 \cosh \lambda x + c_2 \sinh \lambda x + c_3 \cos \lambda x + c_4 \sin \lambda x$$

$$T = c_5 \cos a \lambda^2 t + c_6 \sin a \lambda^2 t.$$

The boundary conditions translate into $X(0) = X(L) = 0$ and $X''(0) = X''(L) = 0$. From $X(0) = X''(0) = 0$ we find $c_1 = c_3 = 0$. From

$$X(L) = c_2 \sinh \lambda L + c_4 \sin \lambda L = 0$$

$$X''(L) = \lambda^2 c_2 \sinh \lambda L - \lambda^2 c_4 \sin \lambda L = 0$$

we see by subtraction that $c_4 \sin \lambda L = 0$. This equation yields the eigenvalues $\lambda = n\pi/L$ for $n = 1, 2, 3, \dots$.

The corresponding eigenfunctions are

$$X = c_4 \sin \frac{n\pi}{L} x.$$

Thus

$$u(x, t) = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n^2 \pi^2}{L^2} at + B_n \sin \frac{n^2 \pi^2}{L^2} at \right) \sin \frac{n\pi}{L} x.$$

From

$$u(x, 0) = f(x) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{L} x$$

we obtain

$$A_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi}{L} x \, dx.$$

From

$$\frac{\partial u}{\partial t} = \sum_{n=1}^{\infty} \left(-A_n \frac{n^2 \pi^2 a}{L^2} \sin \frac{n^2 \pi^2}{L^2} at + B_n \frac{n^2 \pi^2 a}{L^2} \cos \frac{n^2 \pi^2}{L^2} at \right) \sin \frac{n\pi}{L} x$$

and

$$\left. \frac{\partial u}{\partial t} \right|_{t=0} = g(x) = \sum_{n=1}^{\infty} B_n \frac{n^2 \pi^2 a}{L^2} \sin \frac{n\pi}{L} x$$

we obtain

$$B_n \frac{n^2 \pi^2 a}{L^2} = \frac{2}{L} \int_0^L g(x) \sin \frac{n\pi}{L} x dx$$

and

$$B_n = \frac{2L}{n^2 \pi^2 a} \int_0^L g(x) \sin \frac{n\pi}{L} x dx.$$

12. (a) Using

$$X = c_1 \cosh \lambda x + c_2 \sinh \lambda x + c_3 \cos \lambda x + c_4 \sin \lambda x$$

and $X(0) = 0$, $X'(0) = 0$ we find, in turn, $c_3 = -c_1$ and $c_4 = -c_2$. The conditions $X(L) = 0$ and $X'(L) = 0$ then yield the system of equations for c_1 and c_2 :

$$c_1(\cosh \lambda L - \cos \lambda L) + c_2(\sinh \lambda L - \sin \lambda L) = 0$$

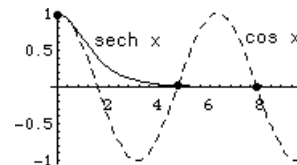
$$c_1(\lambda \sinh \lambda L + \lambda \sin \lambda L) + c_2(\lambda \cosh \lambda L - \lambda \cos \lambda L) = 0.$$

In order that this system have nontrivial solutions the determinant of the coefficients must be zero:

$$\lambda(\cosh \lambda L - \cos \lambda L)^2 - \lambda(\sinh^2 \lambda L - \sin^2 \lambda L) = 0.$$

$\lambda = 0$ is not an eigenvalue since this leads to $X = 0$. Thus the last equation simplifies to $\cosh \lambda L \cos \lambda L = 1$ or $\cosh x \cos x = 1$, where $x = \lambda L$.

(b) The equation $\cosh x \cos x = 1$ is the same as $\cos x = \operatorname{sech} x$. The figure indicates that the equation has an infinite number of roots.



(c) Using a CAS we find the first four positive roots to be $x_1 = 4.7300$, $x_2 = 7.8532$, $x_3 = 10.9956$, and $x_4 = 14.1372$. Thus the first four eigenvalues are $\lambda_1 = x_1/L = 4.7300/L$, $\lambda_2 = x_2/L = 7.8532/L$, $\lambda_3 = x_3/L = 10.9956/L$, and $\lambda_4 = 14.1372/L$.

13. From (5) in the text we have

$$u(x, t) = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi a}{L} t + B_n \sin \frac{n\pi a}{L} t \right) \sin \frac{n\pi}{L} x.$$

Since $u_t(x, 0) = g(x) = 0$ we have $B_n = 0$ and

$$\begin{aligned} u(x, t) &= \sum_{n=1}^{\infty} A_n \cos \frac{n\pi a}{L} t \sin \frac{n\pi}{L} x \\ &= \sum_{n=1}^{\infty} A_n \frac{1}{2} \left[\sin \left(\frac{n\pi}{L} x + \frac{n\pi a}{L} t \right) + \sin \left(\frac{n\pi}{L} x - \frac{n\pi a}{L} t \right) \right] \\ &= \frac{1}{2} \sum_{n=1}^{\infty} A_n \left[\sin \frac{n\pi}{L} (x + at) + \sin \frac{n\pi}{L} (x - at) \right]. \end{aligned}$$

From

$$u(x, 0) = f(x) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{L} x$$

Exercises 13.4

we identify

$$f(x + at) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{L}(x + at)$$

and

$$f(x - at) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{L}(x - at),$$

so that

$$u(x, t) = \frac{1}{2}[f(x + at) + f(x - at)].$$

14. (a) We note that $\xi_x = \eta_x = 1$, $\xi_t = a$, and $\eta_t = -a$. Then

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial \xi} \frac{\partial \xi}{\partial x} + \frac{\partial u}{\partial \eta} \frac{\partial \eta}{\partial x} = u_\xi + u_\eta$$

and

$$\begin{aligned} \frac{\partial^2 u}{\partial x^2} &= \frac{\partial}{\partial x}(u_\xi + u_\eta) = \frac{\partial u_\xi}{\partial \xi} \frac{\partial \xi}{\partial x} + \frac{\partial u_\xi}{\partial \eta} \frac{\partial \eta}{\partial x} + \frac{\partial u_\eta}{\partial \xi} \frac{\partial \xi}{\partial x} + \frac{\partial u_\eta}{\partial \eta} \frac{\partial \eta}{\partial x} \\ &= u_{\xi\xi} + 2u_{\xi\eta} + u_{\eta\eta}. \end{aligned}$$

Similarly

$$\frac{\partial^2 u}{\partial t^2} = a^2(u_{\xi\xi} - 2u_{\xi\eta} + u_{\eta\eta}).$$

Thus

$$a^2 \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2} \quad \text{becomes} \quad \frac{\partial^2 u}{\partial \xi \partial \eta} = 0.$$

(b) Integrating

$$\frac{\partial^2 u}{\partial \xi \partial \eta} = \frac{\partial}{\partial \eta} u_\xi = 0$$

we obtain

$$\begin{aligned} \int \frac{\partial}{\partial \eta} u_\xi d\eta &= \int 0 d\eta \\ u_\xi &= f(\xi). \end{aligned}$$

Integrating this result with respect to ξ we obtain

$$\begin{aligned} \int \frac{\partial u}{\partial \xi} d\xi &= \int f(\xi) d\xi \\ u &= F(\xi) + G(\eta). \end{aligned}$$

Since $\xi = x + at$ and $\eta = x - at$, we then have

$$u = F(\xi) + G(\eta) = F(x + at) + G(x - at).$$

Next, we have

$$\begin{aligned} u(x, t) &= F(x + at) + G(x - at) \\ u(x, 0) &= F(x) + G(x) = f(x) \\ u_t(x, 0) &= aF'(x) - aG'(x) = g(x) \end{aligned}$$

Integrating the last equation with respect to x gives

$$F(x) - G(x) = \frac{1}{a} \int_{x_0}^x g(s) ds + c_1.$$

Substituting $G(x) = f(x) - F(x)$ we obtain

$$F(x) = \frac{1}{2} f(x) + \frac{1}{2a} \int_{x_0}^x g(s) ds + C$$

where $c = c_1/2$. Thus

$$G(x) = \frac{1}{2} f(x) - \frac{1}{2a} \int_{x_0}^x g(s) ds - c.$$

(c) From the expressions for F and G ,

$$F(x+at) = \frac{1}{2} f(x+at) + \frac{1}{2a} \int_{x_0}^{x+at} g(s) ds + c$$

$$G(x-at) = \frac{1}{2} f(x-at) - \frac{1}{2a} \int_{x_0}^{x-at} g(s) ds - c.$$

Thus,

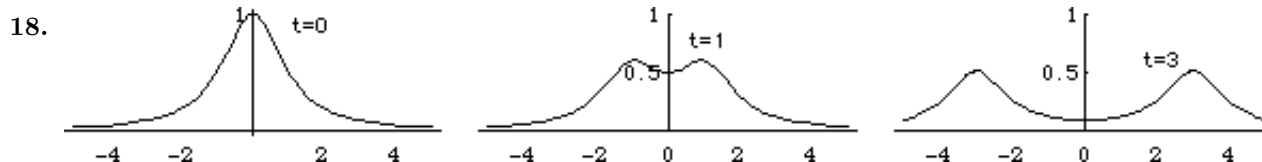
$$u(x, t) = F(x+at) + G(x-at) = \frac{1}{2} [f(x+at) + f(x-at)] + \frac{1}{2a} \int_{x-at}^{x+at} g(s) ds.$$

Here we have used $-\int_{x_0}^{x-at} g(s) ds = \int_{x-at}^{x_0} g(s) ds$.

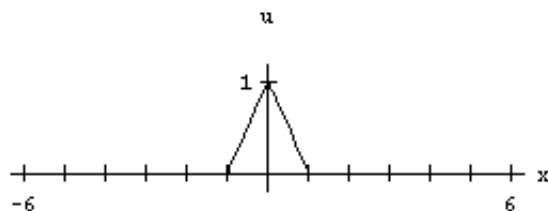
$$\begin{aligned} 15. \quad u(x, t) &= \frac{1}{2} [\sin(x+at) + \sin(x-at)] + \frac{1}{2a} \int_{x-at}^{x+at} ds \\ &= \frac{1}{2} [\sin x \cos at + \cos x \sin at + \sin x \cos at - \cos x \sin at] + \frac{1}{2a} s \Big|_{x-at}^{x+at} = \sin x \cos at + t \end{aligned}$$

$$\begin{aligned} 16. \quad u(x, t) &= \frac{1}{2} \sin(x+at) + \sin(x-at) + \frac{1}{2a} \int_{x-at}^{x+at} \cos s ds \\ &= \sin x \cos at + \frac{1}{2a} [\sin(x+at) - \sin(x-at)] = \sin x \cos at + \frac{1}{a} \cos x \sin at \end{aligned}$$

$$\begin{aligned} 17. \quad u(x, t) &= 0 + \frac{1}{2a} \int_{x-at}^{x+at} \sin 2s ds = \frac{1}{2a} \left[\frac{-\cos(2x+2at) + \cos(2x-2at)}{2} \right] \\ &= \frac{1}{4a} [-\cos 2x \cos 2at + \sin 2x \sin 2at + \cos 2x \cos 2at + \sin 2x \sin 2at] = \frac{1}{2a} \sin 2x \sin 2at \end{aligned}$$



19. (a)

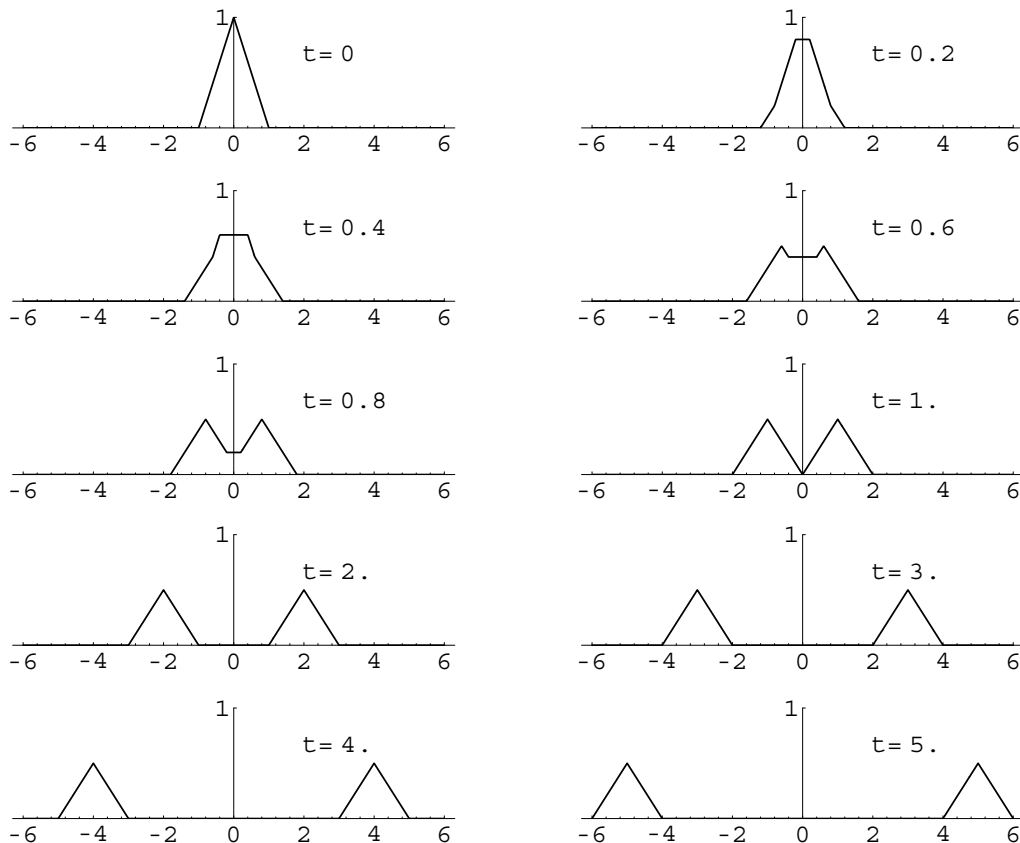


(b) Since $g(x) = 0$, d'Alembert's solution with $a = 1$ is

$$u(x, t) = \frac{1}{2} [f(x+t) + f(x-t)].$$

Exercises 13.4

Sample plots are shown below.

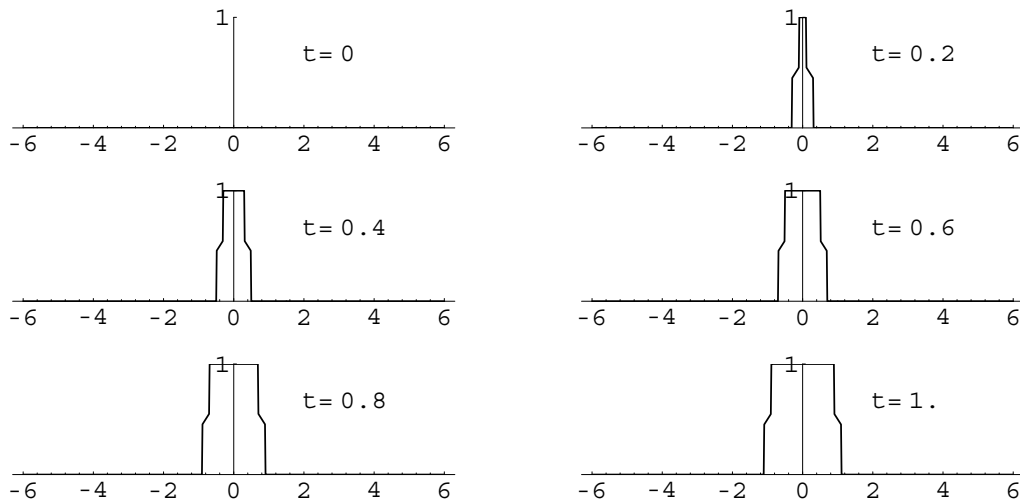


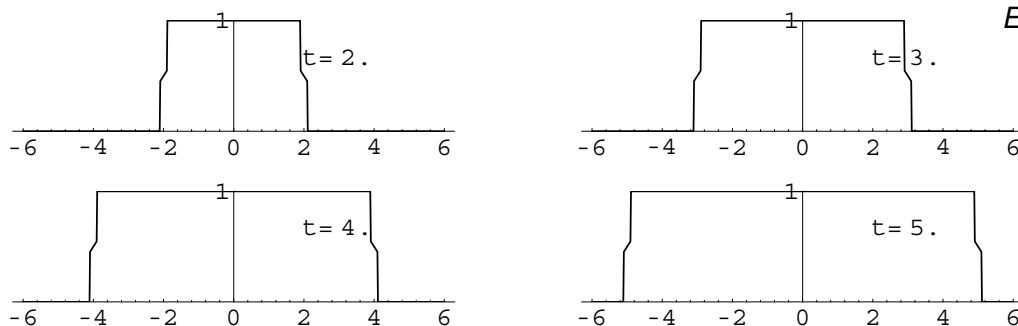
(c) The single peaked wave dissolves into two peaks moving outward.

20. (a) With $a = 1$, d'Alembert's solution is

$$u(x, t) = \frac{1}{2} \int_{x-t}^{x+t} g(s) ds \quad \text{where} \quad g(s) = \begin{cases} 1, & |s| \leq 0.1 \\ 0, & |s| > 0.1 \end{cases}.$$

Sample plots are shown below.





(b) Some frames of the movie are shown in part (a). The string has a roughly rectangular shape with the base on the x -axis increasing in length.

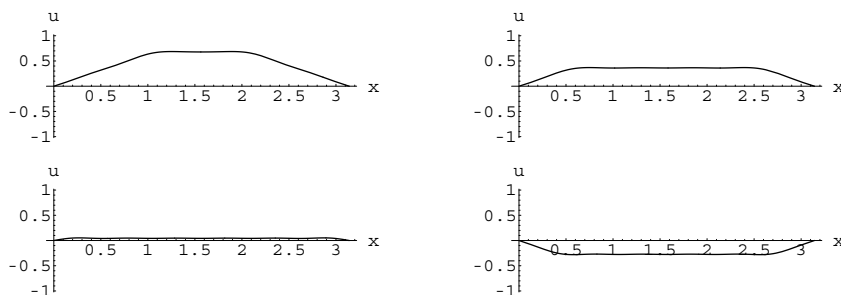
21. (a) and (b) With the given parameters, the solution is

$$u(x, t) = \frac{8}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin \frac{n\pi}{2} \cos nt \sin nx.$$

For n even, $\sin(n\pi/2) = 0$, so the first six nonzero terms correspond to $n = 1, 3, 5, 7, 9, 11$. In this case $\sin(n\pi/2) = \sin(2p-1)/2 = (-1)^{p+1}$ for $p = 1, 2, 3, 4, 5, 6$, and

$$u(x, t) = \frac{8}{\pi^2} \sum_{p=1}^{\infty} \frac{(-1)^{p+1}}{(2p-1)^2} \cos(2p-1)t \sin(2p-1)x.$$

Frames of the movie corresponding to $t = 0.5, 1, 1.5$, and 2 are shown.



Exercises 13.5

1. Using $u = XY$ and $-\lambda^2$ as a separation constant leads to

$$X'' + \lambda^2 X = 0,$$

$$X(0) = 0,$$

$$X(a) = 0,$$

and

$$Y'' - \lambda^2 Y = 0,$$

$$Y(0) = 0.$$

Then

$$X = c_1 \sin \frac{n\pi}{a} x \quad \text{and} \quad Y = c_2 \sinh \frac{n\pi}{a} y$$

for $n = 1, 2, 3, \dots$ so that

$$u = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{a} x \sinh \frac{n\pi}{a} y.$$

Exercises 13.5

Imposing

$$u(x, b) = f(x) = \sum_{n=1}^{\infty} A_n \sinh \frac{n\pi b}{a} \sin \frac{n\pi}{a} x$$

gives

$$A_n \sinh \frac{n\pi b}{a} = \frac{2}{a} \int_0^a f(x) \sin \frac{n\pi}{a} x dx$$

so that

$$u(x, y) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{a} x \sinh \frac{n\pi}{a} y$$

where

$$A_n = \frac{2}{a} \operatorname{csch} \frac{n\pi b}{a} \int_0^a f(x) \sin \frac{n\pi}{a} x dx.$$

2. Using $u = XY$ and $-\lambda^2$ as a separation constant leads to

$$X'' + \lambda^2 X = 0,$$

$$X(0) = 0,$$

$$X(a) = 0,$$

and

$$Y'' - \lambda^2 Y = 0,$$

$$Y'(0) = 0.$$

Then

$$X = c_1 \sin \frac{n\pi}{a} x \quad \text{and} \quad Y = c_2 \cosh \frac{n\pi}{a} y$$

for $n = 1, 2, 3, \dots$ so that

$$u = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{a} x \cosh \frac{n\pi}{a} y.$$

Imposing

$$u(x, b) = f(x) = \sum_{n=1}^{\infty} A_n \cosh \frac{n\pi b}{a} \sin \frac{n\pi}{a} x$$

gives

$$A_n \cosh \frac{n\pi b}{a} = \frac{2}{a} \int_0^a f(x) \sin \frac{n\pi}{a} x dx$$

so that

$$u(x, y) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{a} x \cosh \frac{n\pi}{a} y$$

where

$$A_n = \frac{2}{a} \operatorname{sech} \frac{n\pi b}{a} \int_0^a f(x) \sin \frac{n\pi}{a} x dx.$$

3. Using $u = XY$ and $-\lambda^2$ as a separation constant leads to

$$X'' + \lambda^2 X = 0,$$

$$X(0) = 0,$$

$$X(a) = 0,$$

and

$$Y'' - \lambda^2 Y = 0,$$

$$Y(b) = 0.$$

Then

$$X = c_1 \sin \frac{n\pi}{a} x \quad \text{and} \quad Y = c_2 \cosh \frac{n\pi}{a} y - c_2 \frac{\cosh \frac{n\pi b}{a}}{\sinh \frac{n\pi b}{a}} \sinh \frac{n\pi}{a} y$$

for $n = 1, 2, 3, \dots$ so that

$$u = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{a} x \left(\cosh \frac{n\pi}{a} y - \frac{\cosh \frac{n\pi b}{a}}{\sinh \frac{n\pi b}{a}} \sinh \frac{n\pi}{a} y \right).$$

Imposing

$$u(x, 0) = f(x) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{a} x$$

gives

$$A_n = \frac{2}{a} \int_0^a f(x) \sin \frac{n\pi}{a} x \, dx$$

so that

$$u(x, y) = \frac{2}{a} \sum_{n=1}^{\infty} \left(\int_0^a f(x) \sin \frac{n\pi}{a} x \, dx \right) \sin \frac{n\pi}{a} x \left(\cosh \frac{n\pi}{a} y - \frac{\cosh \frac{n\pi b}{a}}{\sinh \frac{n\pi b}{a}} \sinh \frac{n\pi}{a} y \right).$$

4. Using $u = XY$ and $-\lambda^2$ as a separation constant leads to

$$X'' + \lambda^2 X = 0,$$

$$X'(0) = 0,$$

$$X'(a) = 0,$$

and

$$Y'' - \lambda^2 Y = 0,$$

$$Y(b) = 0.$$

Then

$$X = c_1 \cos \frac{n\pi}{a} x$$

for $n = 0, 1, 2, \dots$ and

$$Y = c_2(y - b) \quad \text{or} \quad Y = c_2 \cosh \frac{n\pi}{a} y - c_2 \frac{\cosh \frac{n\pi b}{a}}{\sinh \frac{n\pi b}{a}} \sinh \frac{n\pi}{a} y$$

so that

$$u = A_0(y - b) + \sum_{n=1}^{\infty} A_n \cos \frac{n\pi}{a} x \left(\cosh \frac{n\pi}{a} y - \frac{\cosh \frac{n\pi b}{a}}{\sinh \frac{n\pi b}{a}} \sinh \frac{n\pi}{a} y \right).$$

Imposing

$$u(x, 0) = x = -A_0 b + \sum_{n=1}^{\infty} A_n \cos \frac{n\pi}{a} x$$

gives

$$-A_0 b = \frac{1}{a} \int_0^a x \, dx = \frac{1}{2} a$$

and

$$A_n = \frac{2}{a} \int_0^a x \cos \frac{n\pi}{a} x \, dx = \frac{2a}{n^2 \pi^2} [(-1)^n - 1]$$

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so that

$$u(x, y) = \frac{a}{2b}(b - y) + \frac{2a}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n - 1}{n^2} \cos \frac{n\pi}{a} x \left(\cosh \frac{n\pi}{a} y - \frac{\cosh \frac{n\pi b}{a}}{\sinh \frac{n\pi b}{a}} \sinh \frac{n\pi}{a} y \right).$$

5. Using $u = XY$ and λ^2 as a separation constant leads to

$$X'' + -\lambda^2 X = 0,$$

$$X(0) = 0,$$

and

$$Y'' + \lambda^2 Y = 0,$$

$$Y'(0) = 0,$$

$$Y'(1) = 0.$$

Then

$$Y = c_1 \cos n\pi y$$

for $n = 0, 1, 2, \dots$ and

$$X = c_2 x \quad \text{or} \quad X = c_2 \sinh n\pi x$$

for $n = 1, 2, 3, \dots$ so that

$$u = A_0 x + \sum_{n=1}^{\infty} A_n \sinh n\pi x \cos n\pi y.$$

Imposing

$$u(1, y) = 1 - y = A_0 + \sum_{n=1}^{\infty} A_n \sinh n\pi x \cos n\pi y$$

gives

$$A_0 = \int_0^1 (1 - y) dy$$

and

$$A_n \sinh n\pi = 2 \int_0^1 (1 - y) \cos n\pi y = \frac{2[1 - (-1)^n]}{n^2 \pi^2 \sinh n\pi}$$

for $n = 1, 2, 3, \dots$ so that

$$u(x, y) = \frac{1}{2}x + \frac{2}{\pi^2} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n^2 \sinh n\pi} \sinh n\pi x \cos n\pi y.$$

6. Using $u = XY$ and λ^2 as a separation constant leads to

$$X'' - \lambda^2 X = 0,$$

$$X'(1) = 0,$$

and

$$Y'' + \lambda^2 Y = 0,$$

$$Y'(0) = 0,$$

$$Y'(\pi) = 0.$$

Then

$$Y = c_1 \cos ny$$

for $n = 0, 1, 2, \dots$ and

$$X = c_2 \cosh nx - c_2 \frac{\sinh n}{\cosh n} \sinh nx$$

for $n = 0, 1, 2, \dots$ so that

$$u = A_0 + \sum_{n=1}^{\infty} A_n \left(\cosh nx - \frac{\sinh n}{\cosh n} \sinh nx \right) \cos ny.$$

Imposing

$$u(0, y) = g(y) = A_0 + \sum_{n=1}^{\infty} A_n \cos ny$$

gives

$$A_0 = \frac{1}{\pi} \int_0^{\pi} g(y) dy \quad \text{and} \quad A_n = \frac{2}{\pi} \int_0^{\pi} g(y) \cos ny dy$$

for $n = 1, 2, 3, \dots$ so that

$$u(x, y) = \frac{1}{\pi} \int_0^{\pi} g(y) dy + \sum_{n=1}^{\infty} \left(\frac{2}{\pi} \int_0^{\pi} g(y) \cos ny dy \right) \left(\cosh nx - \frac{\sinh n}{\cosh n} \sinh nx \right) \cos ny.$$

7. Using $u = XY$ and λ^2 as a separation constant leads to

$$X'' - \lambda^2 X = 0,$$

$$X'(0) = X(0),$$

and

$$Y'' + \lambda^2 Y = 0,$$

$$Y(0) = 0,$$

$$Y(\pi) = 0.$$

Then

$$Y = c_1 \sin ny \quad \text{and} \quad X = c_2 (n \cosh nx + \sinh nx)$$

for $n = 1, 2, 3, \dots$ so that

$$u = \sum_{n=1}^{\infty} A_n (n \cosh nx + \sinh nx) \sin ny.$$

Imposing

$$u(\pi, y) = 1 = \sum_{n=1}^{\infty} A_n (n \cosh n\pi + \sinh n\pi) \sin ny$$

gives

$$A_n (n \cosh n\pi + \sinh n\pi) = \frac{2}{\pi} \int_0^{\pi} \sin ny dy = \frac{2[1 - (-1)^n]}{n\pi}$$

for $n = 1, 2, 3, \dots$ so that

$$u(x, y) = \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n} \frac{n \cosh nx + \sinh nx}{n \cosh n\pi + \sinh n\pi} \sin ny.$$

8. Using $u = XY$ and $-\lambda^2$ as a separation constant leads to

$$X'' + \lambda^2 X = 0,$$

$$X(0) = 0,$$

$$X(1) = 0$$

and

Exercises 13.5

$$Y'' - \lambda^2 Y = 0,$$

$$Y'(0) = Y(0).$$

Then

$$X = c_1 \sin n\pi x \quad \text{and} \quad Y = c_2(n \cosh n\pi y + \sinh n\pi y)$$

for $n = 1, 2, 3, \dots$ so that

$$u = \sum_{n=1}^{\infty} A_n(n \cosh n\pi y + \sinh n\pi y) \sin n\pi x.$$

Imposing

$$u(x, 1) = f(x) = \sum_{n=1}^{\infty} A_n(n \cosh n\pi + \sinh n\pi) \sin n\pi x$$

gives

$$A_n(n \cosh n\pi + \sinh n\pi) = \frac{2}{\pi} \int_0^{\pi} f(x) \sin n\pi x \, dx$$

for $n = 1, 2, 3, \dots$ so that

$$u(x, y) = \sum_{n=1}^{\infty} A_n(n \cosh n\pi y + \sinh n\pi y) \sin n\pi x$$

where

$$A_n = \frac{2}{n\pi \cosh n\pi + \pi \sinh n\pi} \int_0^1 f(x) \sin n\pi x \, dx.$$

9. This boundary-value problem has the form of Problem 1 in this section, with $a = b = 1$, $f(x) = 100$, and $g(x) = 200$. The solution, then, is

$$u(x, y) = \sum_{n=1}^{\infty} (A_n \cosh n\pi y + B_n \sinh n\pi y) \sin n\pi x,$$

where

$$A_n = 2 \int_0^1 100 \sin n\pi x \, dx = 200 \left(\frac{1 - (-1)^n}{n\pi} \right)$$

and

$$\begin{aligned} B_n &= \frac{1}{\sinh n\pi} \left[2 \int_0^1 200 \sin n\pi x \, dx - A_n \cosh n\pi \right] \\ &= \frac{1}{\sinh n\pi} \left[400 \left(\frac{1 - (-1)^n}{n\pi} \right) - 200 \left(\frac{1 - (-1)^n}{n\pi} \right) \cosh n\pi \right] \\ &= 200 \left[\frac{1 - (-1)^n}{n\pi} \right] [2 \operatorname{csch} n\pi - \coth n\pi]. \end{aligned}$$

10. This boundary-value problem has the form of Problem 2 in this section, with $a = 1$ and $b = 1$. Thus, the solution has the form

$$u(x, y) = \sum_{n=1}^{\infty} (A_n \cosh n\pi x + B_n \sinh n\pi x) \sin n\pi y.$$

The boundary condition $u(0, y) = 10y$ implies

$$10y = \sum_{n=1}^{\infty} A_n \sin n\pi y$$

and

$$A_n = \frac{2}{1} \int_0^1 10y \sin n\pi y \, dy = \frac{20}{n\pi} (-1)^{n+1}.$$

The boundary condition $u_x(1, y) = -1$ implies

$$-1 = \sum_{n=1}^{\infty} (n\pi A_n \sinh n\pi + n\pi B_n \cosh n\pi) \sin n\pi y$$

and

$$\begin{aligned} n\pi A_n \sinh n\pi + n\pi B_n \cosh n\pi &= \frac{2}{1} \int_0^1 (-\sin n\pi y) dy \\ A_n \sinh n\pi + B_n \cosh n\pi &= -\frac{2}{n\pi} [1 - (-1)^n] \\ B_n &= \frac{2}{n\pi} [(-1)^n - 1] \operatorname{sech} n\pi - \frac{20}{n\pi} (-1)^{n+1} \tanh n\pi. \end{aligned}$$

11. Using $u = XY$ and $-\lambda^2$ as a separation constant leads to

$$\begin{aligned} X'' + \lambda^2 X &= 0, \\ X(0) &= 0, \\ X(\pi) &= 0, \end{aligned}$$

and

$$Y'' - \lambda^2 Y = 0.$$

Then the boundedness of u as $y \rightarrow \infty$ gives $Y = c_1 e^{-ny}$ and $X = c_2 \sin nx$ for $n = 1, 2, 3, \dots$ so that

$$u = \sum_{n=1}^{\infty} A_n e^{-ny} \sin nx.$$

Imposing

$$u(x, 0) = f(x) = \sum_{n=1}^{\infty} A_n \sin nx$$

gives

$$A_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx$$

so that

$$u(x, y) = \sum_{n=1}^{\infty} \left(\frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx \right) e^{-ny} \sin nx.$$

12. Using $u = XY$ and $-\lambda^2$ as a separation constant leads to

$$\begin{aligned} X'' + \lambda^2 X &= 0, \\ X'(0) &= 0, \\ X'(\pi) &= 0, \end{aligned}$$

and

$$Y'' - \lambda^2 Y = 0.$$

By the boundedness of u as $y \rightarrow \infty$ we obtain $Y = c_1 e^{-ny}$ for $n = 1, 2, 3, \dots$ or $Y = c_1$ and $X = c_2 \cos nx$ for $n = 0, 1, 2, \dots$ so that

$$u = A_0 + \sum_{n=1}^{\infty} A_n e^{-ny} \cos nx.$$

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Imposing

$$u(x, 0) = f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos nx$$

gives

$$A_0 = \frac{1}{\pi} \int_0^{\pi} f(x) dx \quad \text{and} \quad A_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx$$

so that

$$u(x, y) = \frac{1}{\pi} \int_0^{\pi} f(x) dx + \sum_{n=1}^{\infty} \left(\frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx \right) e^{-ny} \cos nx.$$

13. Since the boundary conditions at $y = 0$ and $y = b$ are functions of x we choose to separate Laplace's equation as

$$\frac{X''}{X} = -\frac{Y''}{Y} = -\lambda^2$$

so that

$$X'' + \lambda^2 X = 0$$

$$Y'' - \lambda^2 Y = 0$$

and

$$X(x) = c_1 \cos \lambda x + c_2 \sin \lambda x$$

$$Y(y) = c_3 \cosh \lambda y + c_2 \sinh \lambda y.$$

Now $X(0) = 0$ gives $c_1 = 0$ and $X(a) = 0$ implies $\sin \lambda a = 0$ or $\lambda = n\pi/a$ for $n = 1, 2, 3, \dots$. Thus

$$u_n(x, y) = XY = \left(A_n \cosh \frac{n\pi}{a} y + B_n \sinh \frac{n\pi}{a} y \right) \sin \frac{n\pi}{a} x$$

and

$$u(x, y) = \sum_{n=1}^{\infty} \left(A_n \cosh \frac{n\pi}{a} y + B_n \sinh \frac{n\pi}{a} y \right) \sin \frac{n\pi}{a} x. \quad (1)$$

At $y = 0$ we then have

$$f(x) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{a} x$$

and consequently

$$A_n = \frac{2}{a} \int_0^a f(x) \sin \frac{n\pi}{a} x dx. \quad (2)$$

At $y = b$,

$$g(y) = \sum_{n=1}^{\infty} \left(A_n \cosh \frac{n\pi}{a} b + B_n \sinh \frac{n\pi}{a} b \right) \sin \frac{n\pi}{a} x$$

indicates that the entire expression in the parentheses is given by

$$A_n \cosh \frac{n\pi}{a} b + B_n \sinh \frac{n\pi}{a} b = \frac{2}{a} \int_0^a g(x) \sin \frac{n\pi}{a} x dx.$$

We can now solve for B_n :

$$\begin{aligned} B_n \sinh \frac{n\pi}{a} b &= \frac{2}{a} \int_0^a g(x) \sin \frac{n\pi}{a} x dx - A_n \cosh \frac{n\pi}{a} b \\ B_n &= \frac{1}{\sinh \frac{n\pi}{a} b} \left(\frac{2}{a} \int_0^a g(x) \sin \frac{n\pi}{a} x dx - A_n \cosh \frac{n\pi}{a} b \right). \end{aligned} \quad (3)$$

A solution to the given boundary-value problem consists of the series (1) with coefficients A_n and B_n given in (2) and (3), respectively.

14. Since the boundary conditions at $x = 0$ and $x = a$ are functions of y we choose to separate Laplace's equation as

$$\frac{X''}{X} = -\frac{Y''}{Y} = \lambda^2$$

so that

$$X'' - \lambda^2 X = 0$$

$$Y'' + \lambda^2 Y = 0$$

and

$$X(x) = c_1 \cosh \lambda x + c_2 \sinh \lambda x$$

$$Y(y) = c_3 \cos \lambda y + c_2 \sin \lambda y.$$

Now $Y(0) = 0$ gives $c_3 = 0$ and $Y(b) = 0$ implies $\sin \lambda b = 0$ or $\lambda = n\pi/b$ for $n = 1, 2, 3, \dots$. Thus

$$u_n(x, y) = XY = \left(A_n \cosh \frac{n\pi}{b} x + B_n \sinh \frac{n\pi}{b} x \right) \sin \frac{n\pi}{b} y$$

and

$$u(x, y) = \sum_{n=1}^{\infty} \left(A_n \cosh \frac{n\pi}{b} x + B_n \sinh \frac{n\pi}{b} x \right) \sin \frac{n\pi}{b} y. \quad (4)$$

At $x = 0$ we then have

$$F(y) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{b} y$$

and consequently

$$A_n = \frac{2}{b} \int_0^b F(y) \sin \frac{n\pi}{b} y \, dy. \quad (5)$$

At $x = a$,

$$G(y) = \sum_{n=1}^{\infty} \left(A_n \cosh \frac{n\pi}{b} a + B_n \sinh \frac{n\pi}{b} a \right) \sin \frac{n\pi}{b} y$$

indicates that the entire expression in the parentheses is given by

$$A_n \cosh \frac{n\pi}{b} a + B_n \sinh \frac{n\pi}{b} a = \frac{2}{b} \int_0^b G(y) \sin \frac{n\pi}{b} y \, dy.$$

We can now solve for B_n :

$$\begin{aligned} B_n \sinh \frac{n\pi}{b} a &= \frac{2}{b} \int_0^b G(y) \sin \frac{n\pi}{b} y \, dy - A_n \cosh \frac{n\pi}{b} a \\ B_n &= \frac{1}{\sinh \frac{n\pi}{b} a} \left(\frac{2}{b} \int_0^b G(y) \sin \frac{n\pi}{b} y \, dy - A_n \cosh \frac{n\pi}{b} a \right). \end{aligned} \quad (6)$$

A solution to the given boundary-value problem consists of the series (4) with coefficients A_n and B_n given in (5) and (6), respectively.

Exercises 13.5

In Problems 15 and 16 we refer to the discussion in the text under the heading **Superposition Principle**.

15. We identify $a = b = \pi$, $f(x) = 0$, $g(x) = 1$, $F(y) = 1$, and $G(y) = 1$. Then $A_n = 0$ and

$$u_1(x, y) = \sum_{n=1}^{\infty} B_n \sinh ny \sin nx$$

where

$$B_n = \frac{2}{\pi \sinh n\pi} \int_0^{\pi} \sin nx \, dx = \frac{2[1 - (-1)^n]}{n\pi \sinh n\pi}.$$

Next

$$u_2(x, y) = \sum_{n=1}^{\infty} (A_n \cosh nx + B_n \sinh nx) \sin ny$$

where

$$A_n = \frac{2}{\pi} \int_0^{\pi} \sin ny \, dy = \frac{2[1 - (-1)^n]}{n\pi}$$

and

$$\begin{aligned} B_n &= \frac{1}{\sinh n\pi} \left(\frac{2}{\pi} \int_0^{\pi} \sin ny \, dy - A_n \cosh n\pi \right) \\ &= \frac{1}{\sinh n\pi} \left(\frac{2[1 - (-1)^n]}{n\pi} - \frac{2[1 - (-1)^n]}{n\pi} \cosh n\pi \right) \\ &= \frac{2[1 - (-1)^n]}{n\pi \sinh n\pi} (1 - \cosh n\pi). \end{aligned}$$

Now

$$\begin{aligned} A_n \cosh nx + B_n \sinh nx &= \frac{2[1 - (-1)^n]}{n\pi} \left[\cosh nx + \frac{\sinh nx}{\sinh n\pi} (1 - \cosh n\pi) \right] \\ &= \frac{2[1 - (-1)^n]}{n\pi \sinh n\pi} [\cosh nx \sinh n\pi + \sinh nx - \sinh nx \cosh n\pi] \\ &= \frac{2[1 - (-1)^n]}{n\pi \sinh n\pi} [\sinh nx + \sinh n(\pi - x)] \end{aligned}$$

and

$$\begin{aligned} u(x, y) = u_1 + u_2 &= \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n \sinh n\pi} \sinh ny \sin nx \\ &\quad + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{[1 - (-1)^n][\sinh nx + \sinh n(\pi - x)]}{n \sinh n\pi} \sin ny. \end{aligned}$$

16. We identify $a = b = 2$, $f(x) = 0$, $g(x) = \begin{cases} x, & 0 < x < 1 \\ 2 - x, & 1 < x < 2 \end{cases}$, $F(y) = 0$, and $G(y) = y(2 - y)$. Then $A_n = 0$ and

$$u_1(x, y) = \sum_{n=1}^{\infty} B_n \sinh \frac{n\pi}{2} y \sin \frac{n\pi}{2} x$$

where

$$\begin{aligned} B_n &= \frac{1}{\sinh n\pi} \int_0^2 g(x) \sin \frac{n\pi}{2} x \, dx \\ &= \frac{1}{\sinh n\pi} \left(\int_0^1 x \sin \frac{n\pi}{2} x \, dx + \int_1^2 (2 - x) \sin \frac{n\pi}{2} x \, dx \right) \\ &= \frac{8 \sin \frac{n\pi}{2}}{n^2 \pi^2 \sinh n\pi}. \end{aligned}$$

Next, since $A_n = 0$ in u_2 , we have

$$u_2(x, y) = \sum_{n=1}^{\infty} B_n \sinh \frac{n\pi}{2} x \sin \frac{n\pi}{2} y$$

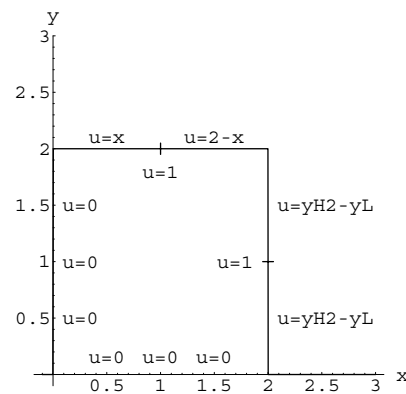
where

$$B_n = \frac{1}{\sinh n\pi} \int_0^b y(2-y) \sin \frac{n\pi}{2} y dy = \frac{16[1 - (-1)^n]}{n^3 \pi^3 \sinh n\pi}.$$

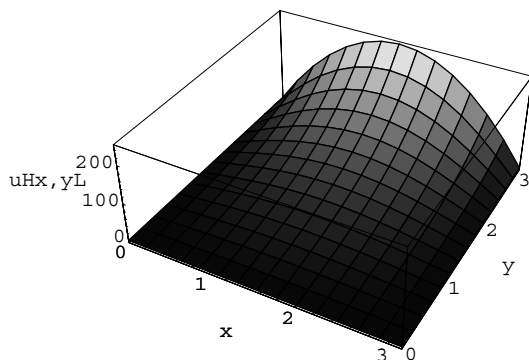
Thus

$$\begin{aligned} u(x, y) = u_1 + u_2 &= \frac{8}{\pi^2} \sum_{n=1}^{\infty} \frac{\sin \frac{n\pi}{2}}{n^2 \sinh n\pi} \sinh \frac{n\pi}{2} y \sin \frac{n\pi}{2} x \\ &+ \frac{16}{\pi^3} \sum_{n=1}^{\infty} \frac{[1 - (-1)^n]}{n^3 \sinh n\pi} \sinh \frac{n\pi}{2} x \sin \frac{n\pi}{2} y. \end{aligned}$$

17. From the figure showing the boundary conditions we see that the maximum value of the temperature is 1 at $(1, 2)$ and $(2, 1)$.



18. (a)

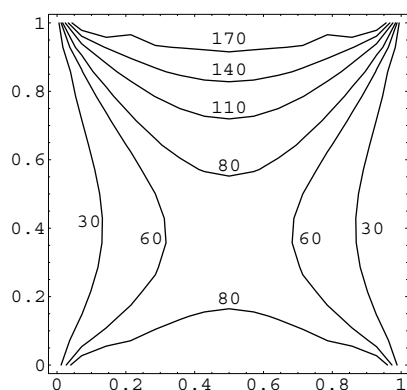


- (b) The maximum value occurs at $(\pi/2, \pi)$ and is $f(\pi/2) = 25\pi^2$.
(c) The coefficients are

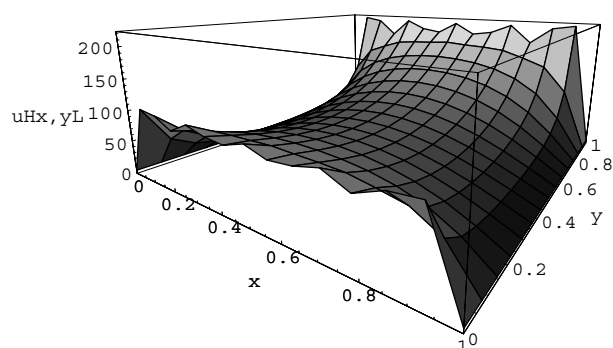
$$\begin{aligned} A_n &= \frac{2}{\pi} \operatorname{csch} n\pi \int_0^\pi 100x(\pi - x) \sin nx dx \\ &= \frac{200 \operatorname{csch} n\pi}{\pi} \left[\frac{200}{n^3} (1 - (-1)^n) \right] = \frac{400}{n^3 \pi} [1 - (-1)^n] \operatorname{csch} n\pi. \end{aligned}$$

Exercises 13.5

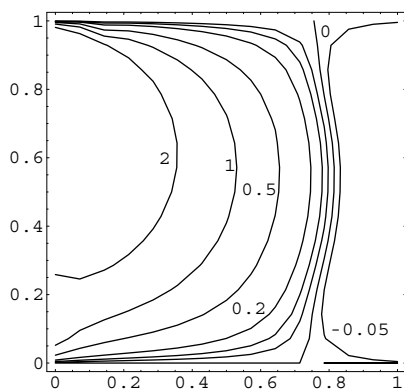
19. (a)



(b)



20.



Exercises 13.6

- Using $v(x, t) = u(x, t) - 100$ we wish to solve $kv_{xx} = v_t$ subject to $v(0, t) = 0$, $v(1, t) = 0$, and $v(x, 0) = -100$. Let $v = XT$ and use $-\lambda^2$ as a separation constant so that

$$X'' + \lambda^2 X = 0,$$

$$X(0) = 0,$$

$$X(1) = 0,$$

and

$$T' + \lambda^2 kT = 0.$$

Then

$$X = c_1 \sin n\pi x \quad \text{and} \quad T = c_2 e^{-kn^2\pi^2 t}$$

for $n = 1, 2, 3, \dots$ so that

$$v = \sum_{n=1}^{\infty} A_n e^{-kn^2\pi^2 t} \sin n\pi x.$$

Imposing

$$v(x, 0) = -100 = \sum_{n=1}^{\infty} A_n \sin n\pi x$$

gives

$$A_n = 2 \int_0^1 (-100) \sin n\pi x \, dx = \frac{-200}{n\pi} [1 - (-1)^n]$$

so that

$$u(x, t) = v(x, t) + 100 = 100 + \frac{200}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n - 1}{n} e^{-kn^2\pi^2 t} \sin n\pi x.$$

2. Letting $u(x, t) = v(x, t) + \psi(x)$ and proceeding as in Example 1 in the text we find $\psi(x) = u_0 - u_0x$. Then $v(x, t) = u(x, t) + u_0x - u_0$ and we wish to solve $kv_{xx} = v_t$ subject to $v(0, t) = 0$, $v(1, t) = 0$, and $v(x, 0) = f(x) + u_0x - u_0$. Let $v = XT$ and use $-\lambda^2$ as a separation constant so that

$$X'' + \lambda^2 X = 0,$$

$$X(0) = 0,$$

$$X(1) = 0,$$

and

$$T' + \lambda^2 kT = 0.$$

Then

$$X = c_1 \sin n\pi x \quad \text{and} \quad T = c_2 e^{-kn^2\pi^2 t}$$

for $n = 1, 2, 3, \dots$ so that

$$v = \sum_{n=1}^{\infty} A_n e^{-kn^2\pi^2 t} \sin n\pi x.$$

Imposing

$$v(x, 0) = f(x) + u_0x - u_0 = \sum_{n=1}^{\infty} A_n \sin n\pi x$$

gives

$$A_n = 2 \int_0^1 (f(x) + u_0x - u_0) \sin n\pi x \, dx$$

so that

$$u(x, t) = v(x, t) + u_0 - u_0x = u_0 - u_0x + \sum_{n=1}^{\infty} A_n e^{-kn^2\pi^2 t} \sin n\pi x.$$

3. If we let $u(x, t) = v(x, t) + \psi(x)$, then we obtain as in Example 1 in the text

$$k\psi'' + r = 0$$

or

$$\psi(x) = -\frac{r}{2k}x^2 + c_1x + c_2.$$

Exercises 13.6

The boundary conditions become

$$u(0, t) = v(0, t) + \psi(0) = u_0$$

$$u(1, t) = v(1, t) + \psi(1) = u_0.$$

Letting $\psi(0) = \psi(1) = u_0$ we obtain homogeneous boundary conditions in v :

$$v(0, t) = 0 \quad \text{and} \quad v(1, t) = 0.$$

Now $\psi(0) = \psi(1) = u_0$ implies $c_2 = u_0$ and $c_1 = r/2k$. Thus

$$\psi(x) = -\frac{r}{2k}x^2 + \frac{r}{2k}x + u_0 = u_0 - \frac{r}{2k}x(x-1).$$

To determine $v(x, t)$ we solve

$$k \frac{\partial^2 v}{\partial x^2} = \frac{\partial v}{\partial t}, \quad 0 < x < 1, \quad t > 0$$

$$v(0, t) = 0, \quad v(1, t) = 0,$$

$$v(x, 0) = \frac{r}{2k}x(x-1) - u_0.$$

Separating variables, we find

$$v(x, t) = \sum_{n=1}^{\infty} A_n e^{-kn^2\pi^2 t} \sin n\pi x,$$

where

$$A_n = 2 \int_0^1 \left[\frac{r}{2k}x(x-1) - u_0 \right] \sin n\pi x \, dx = 2 \left[\frac{u_0}{n\pi} + \frac{r}{kn^3\pi^3} \right] [(-1)^n - 1]. \quad (1)$$

Hence, a solution of the original problem is

$$u(x, t) = \psi(x) + v(x, t)$$

$$= u_0 - \frac{r}{2k}x(x-1) + \sum_{n=1}^{\infty} A_n e^{-kn^2\pi^2 t} \sin n\pi x,$$

where A_n is defined in (1).

4. If we let $u(x, t) = v(x, t) + \psi(x)$, then we obtain as in Example 1 in the text

$$k\psi'' + r = 0$$

or

$$\psi(x) = -\frac{r}{2k}x^2 + c_1x + c_2.$$

The boundary conditions become

$$u(0, t) = v(0, t) + \psi(0) = u_0$$

$$u(1, t) = v(1, t) + \psi(1) = u_1.$$

Letting $\psi(0) = u_0$ and $\psi(1) = u_1$ we obtain homogeneous boundary conditions in v :

$$v(0, t) = 0 \quad \text{and} \quad v(1, t) = 0.$$

Now $\psi(0) = u_0$ and $\psi(1) = u_1$ imply $c_2 = u_0$ and $c_1 = u_1 - u_0 + r/2k$. Thus

$$\psi(x) = -\frac{r}{2k}x^2 + \left(u_1 - u_0 + \frac{r}{2k} \right) x + u_0.$$

To determine $v(x, t)$ we solve

$$\begin{aligned} k \frac{\partial^2 v}{\partial x^2} &= \frac{\partial v}{\partial t}, \quad 0 < x < 1, \quad t > 0 \\ v(0, t) &= 0, \quad v(1, t) = 0, \\ v(x, 0) &= f(x) - \psi(x). \end{aligned}$$

Separating variables, we find

$$v(x, t) = \sum_{n=1}^{\infty} A_n e^{-kn^2\pi^2 t} \sin n\pi x,$$

where

$$A_n = 2 \int_0^1 [f(x) - \psi(x)] \sin n\pi x \, dx. \quad (2)$$

Hence, a solution of the original problem is

$$\begin{aligned} u(x, t) &= \psi(x) + v(x, t) \\ &= -\frac{r}{2k}x^2 + \left(u_1 - u_0 + \frac{r}{2k}\right)x + u_0 + \sum_{n=1}^{\infty} A_n e^{-kn^2\pi^2 t} \sin n\pi x, \end{aligned}$$

where A_n is defined in (2).

5. Substituting $u(x, t) = v(x, t) + \psi(x)$ into the partial differential equation gives

$$k \frac{\partial^2 v}{\partial x^2} + k\psi'' + Ae^{-\beta x} = \frac{\partial v}{\partial t}.$$

This equation will be homogeneous provided ψ satisfies

$$k\psi'' + Ae^{-\beta x} = 0.$$

The general solution of this differential equation is

$$\psi(x) = -\frac{A}{\beta^2 k} e^{-\beta x} + c_1 x + c_2.$$

From $\psi(0) = 0$ and $\psi(1) = 0$ we find

$$c_1 = \frac{A}{\beta^2 k} (e^{-\beta} - 1) \quad \text{and} \quad c_2 = \frac{A}{\beta^2 k}.$$

Hence

$$\begin{aligned} \psi(x) &= -\frac{A}{\beta^2 k} e^{-\beta x} + \frac{A}{\beta^2 k} (e^{-\beta} - 1)x + \frac{A}{\beta^2 k} \\ &= \frac{A}{\beta^2 k} [1 - e^{-\beta x} + (e^{-\beta} - 1)x]. \end{aligned}$$

Now the new problem is

$$\begin{aligned} k \frac{\partial^2 v}{\partial x^2} &= \frac{\partial v}{\partial t}, \quad 0 < x < 1, \quad t > 0, \\ v(0, t) &= 0, \quad v(1, t) = 0, \quad t > 0, \\ v(x, 0) &= f(x) - \psi(x), \quad 0 < x < 1. \end{aligned}$$

Identifying this as the heat equation solved in Section 13.3 in the text with $L = 1$ we obtain

$$v(x, t) = \sum_{n=1}^{\infty} A_n e^{-kn^2\pi^2 t} \sin n\pi x$$

Exercises 13.6

where

$$A_n = 2 \int_0^1 [f(x) - \psi(x)] \sin n\pi x \, dx.$$

Thus

$$u(x, t) = \frac{A}{\beta^2 k} [1 - e^{-\beta x} + (e^{-\beta} - 1)x] + \sum_{n=1}^{\infty} A_n e^{-kn^2 \pi^2 t} \sin n\pi x.$$

6. Substituting $u(x, t) = v(x, t) + \psi(x)$ into the partial differential equation gives

$$k \frac{\partial^2 v}{\partial x^2} + k\psi'' - hv - h\psi = \frac{\partial v}{\partial t}.$$

This equation will be homogeneous provided ψ satisfies

$$k\psi'' - h\psi = 0.$$

Since k and h are positive, the general solution of this latter equation is

$$\psi(x) = c_1 \cosh \sqrt{\frac{h}{k}} x + c_2 \sinh \sqrt{\frac{h}{k}} x.$$

From $\psi(0) = 0$ and $\psi(\pi) = u_0$ we find $c_1 = 0$ and $c_2 = u_0 / \sinh \sqrt{h/k} \pi$. Hence

$$\psi(x) = u_0 \frac{\sinh \sqrt{h/k} x}{\sinh \sqrt{h/k} \pi}.$$

Now the new problem is

$$\begin{aligned} k \frac{\partial^2 v}{\partial x^2} - hv &= \frac{\partial v}{\partial t}, & 0 < x < \pi, \, t > 0 \\ v(0, t) &= 0, \quad v(\pi, t) = 0, & t > 0 \\ v(x, 0) &= -\psi(x), & 0 < x < \pi. \end{aligned}$$

If we let $v = XT$ then

$$\frac{X''}{X} = \frac{T' + hT}{kT} = -\lambda^2$$

gives the separated differential equations

$$X'' + \lambda^2 X = 0 \quad \text{and} \quad T' + (h + k\lambda^2) T = 0.$$

The respective solutions are

$$\begin{aligned} X(x) &= c_3 \cos \lambda x + c_4 \sin \lambda x \\ T(t) &= c_5 e^{-(h+k\lambda^2)t}. \end{aligned}$$

From $X(0) = 0$ we get $c_3 = 0$ and from $X(\pi) = 0$ we find $\lambda = n$ for $n = 1, 2, 3, \dots$. Consequently, it follows that

$$v(x, t) = \sum_{n=1}^{\infty} A_n e^{-(h+kn^2)t} \sin nx$$

where

$$A_n = -\frac{2}{\pi} \int_0^{\pi} \psi(x) \sin nx \, dx.$$

Hence a solution of the original problem is

$$u(x, t) = u_0 \frac{\sinh \sqrt{h/k} x}{\sinh \sqrt{h/k} \pi} + e^{-ht} \sum_{n=1}^{\infty} A_n e^{-kn^2 t} \sin nx$$

where

$$A_n = -\frac{2}{\pi} \int_0^\pi u_0 \frac{\sinh \sqrt{h/k} x}{\sinh \sqrt{h/k} \pi} \sin nx \, dx.$$

Using the exponential definition of the hyperbolic sine and integration by parts we find

$$A_n = \frac{2u_0nk(-1)^n}{\pi(h + kn^2)}.$$

7. Substituting $u(x, t) = v(x, t) + \psi(x)$ into the partial differential equation gives

$$k \frac{\partial^2 v}{\partial x^2} + k\psi'' - hv - h\psi + hu_0 = \frac{\partial v}{\partial t}.$$

This equation will be homogeneous provided ψ satisfies

$$k\psi'' - h\psi + hu_0 = 0 \quad \text{or} \quad k\psi'' - h\psi = -hu_0.$$

This second-order, linear, non-homogeneous differential equation has solution

$$\psi(x) = c_1 \cosh \sqrt{\frac{h}{k}} x + c_2 \sinh \sqrt{\frac{h}{k}} x + u_0,$$

where we assume $h > 0$ and $k > 0$. From $\psi(0) = u_0$ and $\psi(1) = 0$ we find $c_1 = 0$ and $c_2 = -u_0/\sinh \sqrt{h/k}$. Thus, the steady-state solution is

$$\psi(x) = -\frac{u_0}{\sinh \sqrt{\frac{h}{k}}} \sinh \sqrt{\frac{h}{k}} x + u_0 = u_0 \left(1 - \frac{\sinh \sqrt{\frac{h}{k}} x}{\sinh \sqrt{\frac{h}{k}}} \right).$$

8. The partial differential equation is

$$k \frac{\partial^2 u}{\partial x^2} - hu = \frac{\partial u}{\partial t}.$$

Substituting $u(x, t) = v(x, t) + \psi(x)$ gives

$$k \frac{\partial^2 v}{\partial x^2} + k\psi'' - hv - h\psi = \frac{\partial v}{\partial t}.$$

This equation will be homogeneous provided ψ satisfies

$$k\psi'' - h\psi = 0.$$

Assuming $h > 0$ and $k > 0$, we have

$$\psi = c_1 e^{\sqrt{h/k} x} + c_2 e^{-\sqrt{h/k} x},$$

where we have used the exponential form of the solution since the rod is infinite. Now, in order that the steady-state temperature $\psi(x)$ be bounded as $x \rightarrow \infty$, we require $c_1 = 0$. Then

$$\psi(x) = c_2 e^{-\sqrt{h/k} x}$$

and $\psi(0) = u_0$ implies $c_2 = u_0$. Thus

$$\psi(x) = u_0 e^{-\sqrt{h/k} x}.$$

9. Substituting $u(x, t) = v(x, t) + \psi(x)$ into the partial differential equation gives

$$a^2 \frac{\partial^2 v}{\partial x^2} + a^2 \psi'' + Ax = \frac{\partial^2 v}{\partial t^2}.$$

This equation will be homogeneous provided ψ satisfies

$$a^2 \psi'' + Ax = 0.$$

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The general solution of this differential equation is

$$\psi(x) = -\frac{A}{6a^2}x^3 + c_1x + c_2.$$

From $\psi(0) = 0$ we obtain $c_2 = 0$, and from $\psi(1) = 0$ we obtain $c_1 = A/6a^2$. Hence

$$\psi(x) = \frac{A}{6a^2}(x - x^3).$$

Now the new problem is

$$\begin{aligned} a^2 \frac{\partial^2 v}{\partial x^2} &= \frac{\partial^2 v}{\partial t^2} \\ v(0, t) &= 0, \quad v(1, t) = 0, \quad t > 0, \\ v(x, 0) &= -\psi(x), \quad v_t(x, 0) = 0, \quad 0 < x < 1. \end{aligned}$$

Identifying this as the wave equation solved in Section 13.4 in the text with $L = 1$, $f(x) = -\psi(x)$, and $g(x) = 0$ we obtain

$$v(x, t) = \sum_{n=1}^{\infty} A_n \cos n\pi at \sin n\pi x$$

where

$$A_n = 2 \int_0^1 [-\psi(x)] \sin n\pi x \, dx = \frac{A}{3a^2} \int_0^1 (x^3 - x) \sin n\pi x \, dx = \frac{2A(-1)^n}{a^2\pi^3 n^3}.$$

Thus

$$u(x, t) = \frac{A}{6a^2}(x - x^3) + \frac{2A}{a^2\pi^3} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^3} \cos n\pi at \sin n\pi x.$$

10. We solve

$$\begin{aligned} a^2 \frac{\partial^2 u}{\partial x^2} - g &= \frac{\partial^2 u}{\partial t^2}, \quad 0 < x < 1, \quad t > 0 \\ u(0, t) &= 0, \quad u(1, t) = 0, \quad t > 0 \\ u(x, 0) &= 0, \quad \left. \frac{\partial u}{\partial t} \right|_{t=0} = 0, \quad 0 < x < 1. \end{aligned}$$

The partial differential equation is nonhomogeneous. The substitution $u(x, t) = v(x, t) + \psi(x)$ yields a homogeneous partial differential equation provided ψ satisfies

$$a^2 \psi'' - g = 0.$$

By integrating twice we find

$$\psi(x) = \frac{g}{2a^2}x^2 + c_1x + c_2.$$

The imposed conditions $\psi(0) = 0$ and $\psi(1) = 0$ then lead to $c_2 = 0$ and $c_1 = -g/2a^2$. Hence

$$\psi(x) = \frac{g}{2a^2}(x^2 - x).$$

The new problem is now

$$\begin{aligned} a^2 \frac{\partial^2 v}{\partial x^2} &= \frac{\partial^2 v}{\partial t^2}, \quad 0 < x < 1, \quad t > 0 \\ v(0, t) &= 0, \quad v(1, t) = 0 \\ v(x, 0) &= \frac{g}{2a^2}(x - x^2), \quad \left. \frac{\partial v}{\partial t} \right|_{t=0} = 0. \end{aligned}$$

Substituting $v = XT$ we find in the usual manner

$$\begin{aligned} X'' + \lambda^2 X &= 0 \\ T'' + a^2 \lambda^2 T &= 0 \end{aligned}$$

with solutions

$$\begin{aligned} X(x) &= c_3 \cos \lambda x + c_4 \sin \lambda x \\ T(t) &= c_5 \cos a\lambda t + c_6 \sin a\lambda t. \end{aligned}$$

The conditions $X(0) = 0$ and $X(1) = 0$ imply in turn that $c_3 = 0$ and $\lambda = n\pi$ for $n = 1, 2, 3, \dots$. The condition $T'(0) = 0$ implies $c_6 = 0$. Hence, by the superposition principle

$$v(x, t) = \sum_{n=1}^{\infty} A_n \cos(an\pi t) \sin(n\pi x).$$

At $t = 0$,

$$\frac{g}{2a^2} (x - x^2) = \sum_{n=1}^{\infty} A_n \sin n\pi x$$

and so

$$A_n = \frac{g}{a^2} \int_0^1 (x - x^2) \sin n\pi x \, dx = \frac{2g}{a^2 n^3 \pi^3} [1 - (-1)^n].$$

Thus the solution to the original problem is

$$u(x, t) = \psi(x) + v(x, t) = \frac{g}{2a^2} (x^2 - x) + \frac{2g}{a^2 \pi^3} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n^3} \cos(an\pi t) \sin(n\pi x).$$

11. Substituting $u(x, y) = v(x, y) + \psi(y)$ into Laplace's equation we obtain

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \psi''(y) = 0.$$

This equation will be homogeneous provided ψ satisfies $\psi(y) = c_1 y + c_2$. Considering

$$\begin{aligned} u(x, 0) &= v(x, 0) + \psi(0) = u_1 \\ u(x, 1) &= v(x, 1) + \psi(1) = u_0 \\ u(0, y) &= v(0, y) + \psi(y) = 0 \end{aligned}$$

we require that $\psi(0) = u_1$, $\psi(1) = u_0$ and $v(0, y) = -\psi(y)$. Then $c_1 = u_0 - u_1$ and $c_2 = u_1$. The new boundary-value problem is

$$\begin{aligned} \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} &= 0 \\ v(x, 0) &= 0, \quad v(x, 1) = 0, \\ v(0, y) &= -\psi(y), \quad 0 < y < 1, \end{aligned}$$

where $v(x, y)$ is bounded at $x \rightarrow \infty$. This problem is similar to Problem 11 in Section 13.5. The solution is

$$\begin{aligned} v(x, y) &= \sum_{n=1}^{\infty} \left(2 \int_0^1 [-\psi(y) \sin n\pi y] \, dy \right) e^{-n\pi x} \sin n\pi y \\ &= 2 \sum_{n=1}^{\infty} \left[(u_1 - u_0) \int_0^1 y \sin n\pi y \, dy - u_1 \int_0^1 \sin n\pi y \, dy \right] e^{-n\pi x} \sin n\pi y \\ &= \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{u_0(-1)^n - u_1}{n} e^{-n\pi x} \sin n\pi y. \end{aligned}$$

Exercises 13.6

Thus

$$\begin{aligned} u(x, y) &= v(x, y) + \psi(y) \\ &= (u_0 - u_1)y + u_1 + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{u_0(-1)^n - u_1}{n} e^{-n\pi x} \sin n\pi y. \end{aligned}$$

12. Substituting $u(x, y) = v(x, y) + \psi(x)$ into Poisson's equation we obtain

$$\frac{\partial^2 v}{\partial x^2} + \psi''(x) + h + \frac{\partial^2 v}{\partial y^2} = 0.$$

The equation will be homogeneous provided ψ satisfies $\psi''(x) + h = 0$ or $\psi(x) = -\frac{h}{2}x^2 + c_1x + c_2$. From $\psi(0) = 0$ we obtain $c_2 = 0$. From $\psi(\pi) = 1$ we obtain

$$c_1 = \frac{1}{\pi} + \frac{h\pi}{2}.$$

Then

$$\psi(x) = \left(\frac{1}{\pi} + \frac{h\pi}{2}\right)x - \frac{h}{2}x^2.$$

The new boundary-value problem is

$$\begin{aligned} \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} &= 0 \\ v(0, y) &= 0, \quad v(\pi, y) = 0, \\ v(x, 0) &= -\psi(x), \quad 0 < x < \pi. \end{aligned}$$

This is Problem 11 in Section 13.5. The solution is

$$v(x, y) = \sum_{n=1}^{\infty} A_n e^{-ny} \sin nx$$

where

$$\begin{aligned} A_n &= \frac{2}{\pi} \int_0^{\pi} [-\psi(x) \sin nx] dx \\ &= \frac{2(-1)^n}{n} \left(\frac{1}{\pi} + \frac{h\pi}{2}\right) - h(-1)^n \left(\frac{\pi}{n} + \frac{2}{n^2}\right). \end{aligned}$$

Thus

$$u(x, y) = v(x, y) + \psi(x) = \left(\frac{1}{\pi} + \frac{h\pi}{2}\right)x - \frac{h}{2}x^2 + \sum_{n=1}^{\infty} A_n e^{-ny} \sin nx.$$

Exercises 13.7

1. Referring to Example 1 in the text we have

$$X(x) = c_1 \cos \lambda x + c_2 \sin \lambda x$$

and

$$T(t) = c_3 e^{-k\lambda^2 t}.$$

From $X'(0) = 0$ (since the left end of the rod is insulated), we find $c_2 = 0$. Then $X(x) = c_1 \cos \lambda x$ and the other boundary condition $X'(1) = -hX(1)$ implies

$$-\lambda \sin \lambda + h \cos \lambda = 0 \quad \text{or} \quad \cot \lambda = \frac{\lambda}{h}.$$

Denoting the consecutive positive roots of this latter equation by λ_n for $n = 1, 2, 3, \dots$, we have

$$u(x, t) = \sum_{n=1}^{\infty} A_n e^{-k\lambda_n^2 t} \cos \lambda_n x.$$

From the initial condition $u(x, 0) = 1$ we obtain

$$1 = \sum_{n=1}^{\infty} A_n \cos \lambda_n x$$

and

$$\begin{aligned} A_n &= \frac{\int_0^1 \cos \lambda_n x \, dx}{\int_0^1 \cos^2 \lambda_n x \, dx} = \frac{\sin \lambda_n / \lambda_n}{\frac{1}{2} \left[1 + \frac{1}{2\lambda_n} \sin 2\lambda_n \right]} \\ &= \frac{2 \sin \lambda_n}{\lambda_n \left[1 + \frac{1}{\lambda_n} \sin \lambda_n \cos \lambda_n \right]} = \frac{2 \sin \lambda_n}{\lambda_n \left[1 + \frac{1}{h\lambda_n} \sin \lambda_n (\lambda_n \sin \lambda_n) \right]} \\ &= \frac{2h \sin \lambda_n}{\lambda_n [h + \sin^2 \lambda_n]}. \end{aligned}$$

The solution is

$$u(x, t) = 2h \sum_{n=1}^{\infty} \frac{\sin \lambda_n}{\lambda_n (h + \sin^2 \lambda_n)} e^{-k\lambda_n^2 t} \cos \lambda_n x.$$

2. Substituting $u(x, t) = v(x, t) + \psi(x)$ into the partial differential equation gives

$$k \frac{\partial^2 v}{\partial x^2} + k\psi'' = \frac{\partial v}{\partial t}.$$

This equation will be homogeneous if $\psi''(x) = 0$ or $\psi(x) = c_1 x + c_2$. The boundary condition $u(0, t) = 0$ implies $\psi(0) = 0$ which implies $c_2 = 0$. Thus $\psi(x) = c_1 x$. Using the second boundary condition we obtain

$$-\left(\frac{\partial v}{\partial x} + \psi' \right) \Big|_{x=1} = -h[v(1, t) + \psi(1) - u_0],$$

which will be homogeneous when

$$-\psi'(1) = -h\psi(1) + hu_0.$$

Since $\psi(1) = \psi'(1) = c_1$ we have $-c_1 = -hc_1 + hu_0$ and $c_1 = hu_0/(h-1)$. Thus

$$\psi(x) = \frac{hu_0}{h-1} x.$$

The new boundary-value problem is

$$\begin{aligned} k \frac{\partial^2 v}{\partial x^2} &= \frac{\partial v}{\partial t}, \quad 0 < x < 1, \quad t > 0 \\ v(0, t) &= 0, \quad \frac{\partial v}{\partial x} \Big|_{x=1} = -hv(1, t), \quad h > 0, \quad t > 0 \\ v(x, 0) &= f(x) - \frac{hu_0}{h-1} x, \quad 0 < x < 1. \end{aligned}$$

Referring to Example 1 in the text we see that

$$v(x, t) = \sum_{n=1}^{\infty} A_n e^{-k\lambda_n^2 t} \sin \lambda_n x$$

Exercises 13.7

and

$$u(x, t) = v(x, t) + \psi(x) = \frac{hu_0}{h-1}x + \sum_{n=1}^{\infty} A_n e^{-k\lambda_n^2 t} \sin \lambda_n x$$

where

$$f(x) - \frac{hu_0}{h-1}x = \sum_{n=1}^{\infty} A_n \sin \lambda_n x$$

and λ_n is a solution of $\lambda_n \cos \lambda_n = -h \sin \lambda_n$. The coefficients are

$$\begin{aligned} A_n &= \frac{\int_0^1 [f(x) - hu_0x/(h-1)] \sin \lambda_n x \, dx}{\int_0^1 \sin^2 \lambda_n x \, dx} \\ &= \frac{\int_0^1 [f(x) - hu_0x/(h-1)] \sin \lambda_n x \, dx}{\frac{1}{2} \left[1 - \frac{1}{2\lambda_n} \sin 2\lambda_n \right]} \\ &= \frac{2 \int_0^1 [f(x) - hu_0x/(h-1)] \sin \lambda_n x \, dx}{1 - \frac{1}{\lambda_n} \sin \lambda_n \cos \lambda_n} \\ &= \frac{2 \int_0^1 [f(x) - hu_0x/(h-1)] \sin \lambda_n x \, dx}{1 - \frac{1}{h\lambda_n} (h \sin \lambda_n) \cos \lambda_n} \\ &= \frac{2 \int_0^1 [f(x) - hu_0x/(h-1)] \sin \lambda_n x \, dx}{1 - \frac{1}{h\lambda_n} (-\lambda_n \cos \lambda_n) \cos \lambda_n} \\ &= \frac{2h}{h + \cos^2 \lambda_n} \int_0^1 \left[f(x) - \frac{hu_0}{h-1}x \right] \sin \lambda_n x \, dx. \end{aligned}$$

3. Separating variables in Laplace's equation gives

$$X'' + \lambda^2 X = 0$$

$$Y'' - \lambda^2 Y = 0$$

and

$$X(x) = c_1 \cos \lambda x + c_2 \sin \lambda x$$

$$Y(y) = c_3 \cosh \lambda y + c_4 \sinh \lambda y.$$

From $u(0, y) = 0$ we obtain $X(0) = 0$ and $c_1 = 0$. From $u_x(a, y) = -hu(a, y)$ we obtain $X'(a) = -hX(a)$ and

$$\lambda \cos \lambda a = -h \sin \lambda a \quad \text{or} \quad \tan \lambda a = -\frac{\lambda}{h}.$$

Let λ_n , where $n = 1, 2, 3, \dots$, be the consecutive positive roots of this equation. From $u(x, 0) = 0$ we obtain $Y(0) = 0$ and $c_3 = 0$. Thus

$$u(x, y) = \sum_{n=1}^{\infty} A_n \sinh \lambda_n y \sin \lambda_n x.$$

Now

$$f(x) = \sum_{n=1}^{\infty} A_n \sinh \lambda_n b \sin \lambda_n x$$

and

$$A_n \sinh \lambda_n b = \frac{\int_0^a f(x) \sin \lambda_n x \, dx}{\int_0^a \sin^2 \lambda_n x \, dx}.$$

Since

$$\begin{aligned}\int_0^a \sin^2 \lambda_n x \, dx &= \frac{1}{2} \left[a - \frac{1}{2\lambda_n} \sin 2\lambda_n a \right] = \frac{1}{2} \left[a - \frac{1}{\lambda_n} \sin \lambda_n a \cos \lambda_n a \right] \\ &= \frac{1}{2} \left[a - \frac{1}{h\lambda_n} (h \sin \lambda_n a) \cos \lambda_n a \right] \\ &= \frac{1}{2} \left[a - \frac{1}{h\lambda_n} (-\lambda_n \cos \lambda_n a) \cos \lambda_n a \right] = \frac{1}{2h} [ah + \cos^2 \lambda_n a],\end{aligned}$$

we have

$$A_n = \frac{2h}{\sinh \lambda_n b [ah + \cos^2 \lambda_n a]} \int_0^a f(x) \sin \lambda_n x \, dx.$$

4. Letting $u(x, y) = X(x)Y(y)$ and separating variables gives

$$X''Y + XY'' = 0.$$

The boundary conditions

$$\left. \frac{\partial u}{\partial y} \right|_{y=0} = 0 \quad \text{and} \quad \left. \frac{\partial u}{\partial y} \right|_{y=1} = -hu(x, 1)$$

correspond to

$$X(x)Y'(0) = 0 \quad \text{and} \quad X(x)Y'(1) = -hX(x)Y(1)$$

or

$$Y'(0) = 0 \quad \text{and} \quad Y'(1) = -hY(1).$$

Since these homogeneous boundary conditions are in terms of Y , we separate the differential equation as

$$\frac{X''}{X} = -\frac{Y''}{Y} = \lambda^2.$$

Then

$$Y'' + \lambda^2 Y = 0$$

and

$$X'' - \lambda^2 X = 0$$

have solutions

$$Y(y) = c_1 \cos \lambda y + c_2 \sin \lambda y$$

and

$$X(x) = c_3 e^{-\lambda x} + c_4 e^{\lambda x}.$$

We use exponential functions in the solution of $X(x)$ since $\cosh \lambda x$ and $\sinh \lambda x$ are both unbounded as $x \rightarrow \infty$. Now, $Y'(0) = 0$ implies $c_2 = 0$, so $Y(y) = c_1 \cos \lambda y$. Since $Y'(y) = -c_1 \lambda \sin \lambda y$, the boundary condition $Y'(1) = -hY(1)$ implies

$$-c_1 \lambda \sin \lambda = -hc_1 \cos \lambda \quad \text{or} \quad \cot \lambda = \frac{\lambda}{h}.$$

Consideration of the graphs of $f(\lambda) = \cot \lambda$ and $g(\lambda) = \lambda/h$ show that $\cot \lambda = \lambda/h$ has an infinite number of roots. The consecutive positive roots λ_n for $n = 1, 2, 3, \dots$, are the eigenvalues of the problem. The corresponding eigenfunctions are $Y_n(y) = c_1 \cos \lambda_n y$. The condition $\lim_{x \rightarrow \infty} u(x, y) = 0$ is equivalent to $\lim_{x \rightarrow \infty} X(x) = 0$. Thus $c_4 = 0$ and $X(x) = c_3 e^{-\lambda x}$. Therefore

$$u_n(x, y) = X_n(x)Y_n(y) = A_n e^{-\lambda_n x} \cos \lambda_n y$$

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and by the superposition principle

$$u(x, y) = \sum_{n=1}^{\infty} A_n e^{-\lambda_n x} \cos \lambda_n y.$$

[It is easily shown that there are no eigenvalues corresponding to $\lambda = 0$.] Finally, the condition $u(0, y) = u_0$ implies

$$u_0 = \sum_{n=1}^{\infty} A_n \cos \lambda_n y.$$

This is not a Fourier cosine series since the coefficients λ_n of y are not integer multiples of π/p , where $p = 1$ in this problem. The functions $\cos \lambda_n y$ are however orthogonal since they are eigenfunctions of the Sturm-Liouville problem

$$Y'' + \lambda^2 Y = 0,$$

$$Y'(0) = 0$$

$$Y'(1) + hY(1) = 0,$$

with weight function $p(x) = 1$. Thus we find

$$A_n = \frac{\int_0^1 u_0 \cos \lambda_n y \, dy}{\int_0^1 \cos^2 \lambda_n y \, dy}.$$

Now

$$\int_0^1 u_0 \cos \lambda_n y \, dy = \frac{u_0}{\lambda_n} \sin \lambda_n y \Big|_0^1 = \frac{u_0}{\lambda_n} \sin \lambda_n$$

and

$$\begin{aligned} \int_0^1 \cos^2 \lambda_n y \, dy &= \frac{1}{2} \int_0^1 (1 + \cos 2\lambda_n y) \, dy = \frac{1}{2} \left[y + \frac{1}{2\lambda_n} \sin 2\lambda_n y \right]_0^1 \\ &= \frac{1}{2} \left[1 + \frac{1}{2\lambda_n} \sin 2\lambda_n \right] = \frac{1}{2} \left[1 + \frac{1}{\lambda_n} \sin \lambda_n \cos \lambda_n \right]. \end{aligned}$$

Since $\cot \lambda = \lambda/h$,

$$\frac{\cos \lambda}{\lambda} = \frac{\sin \lambda}{h}$$

and

$$\int_0^1 \cos^2 \lambda_n y \, dy = \frac{1}{2} \left[1 + \frac{\sin^2 \lambda_n}{h} \right].$$

Then

$$A_n = \frac{\frac{u_0}{\lambda_n} \sin \lambda_n}{\frac{1}{2} \left[1 + \frac{1}{h} \sin^2 \lambda_n \right]} = \frac{2hu_0 \sin \lambda_n}{\lambda_n (h + \sin^2 \lambda_n)}$$

and

$$u(x, y) = 2hu_0 \sum_{n=1}^{\infty} \frac{\sin \lambda_n}{\lambda_n (h + \sin^2 \lambda_n)} e^{-\lambda_n x} \cos \lambda_n y$$

where λ_n for $n = 1, 2, 3, \dots$ are the consecutive positive roots of $\cot \lambda = \lambda/h$.

5. The boundary-value problem is

$$k \frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}, \quad 0 < x < L, \quad t > 0,$$

$$u(0, t) = 0, \quad \frac{\partial u}{\partial x} \Big|_{x=L} = 0, \quad t > 0,$$

$$u(x, 0) = f(x), \quad 0 < x < L.$$

Separation of variables leads to

$$X'' + \lambda^2 X = 0$$

$$T' + k\lambda^2 T = 0$$

and

$$X(x) = c_1 \cos \lambda x + c_2 \sin \lambda x$$

$$T(t) = c_3 e^{-k\lambda^2 t}.$$

From $X(0) = 0$ we find $c_1 = 0$. From $X'(L) = 0$ we obtain $\cos \lambda L = 0$ and

$$\lambda = \frac{\pi(2n-1)}{2L}, \quad n = 1, 2, 3, \dots$$

Thus

$$u(x, t) = \sum_{n=1}^{\infty} A_n e^{-k(2n-1)^2 \pi^2 t / 4L^2} \sin \left(\frac{2n-1}{2L} \right) \pi x$$

where

$$A_n = \frac{\int_0^L f(x) \sin \left(\frac{2n-1}{2L} \right) \pi x \, dx}{\int_0^L \sin^2 \left(\frac{2n-1}{2L} \right) \pi x \, dx} = \frac{2}{L} \int_0^L f(x) \sin \left(\frac{2n-1}{2L} \right) \pi x \, dx.$$

6. Substituting $u(x, t) = v(x, t) + \psi(x)$ into the partial differential equation gives

$$a^2 \frac{\partial^2 v}{\partial x^2} + \psi''(x) = \frac{\partial^2 v}{\partial t^2}.$$

This equation will be homogeneous if $\psi''(x) = 0$ or $\psi(x) = c_1 x + c_2$. The boundary condition $u(0, t) = 0$ implies $\psi(0) = 0$ which implies $c_2 = 0$. Thus $\psi(x) = c_1 x$. Using the second boundary condition, we obtain

$$E \left(\frac{\partial v}{\partial x} + \psi' \right) \Big|_{x=L} = F_0,$$

which will be homogeneous when

$$E\psi'(L) = F_0.$$

Since $\psi'(x) = c_1$ we conclude that $c_1 = F_0/E$ and

$$\psi(x) = \frac{F_0}{E} x.$$

The new boundary-value problem is

$$a^2 \frac{\partial^2 v}{\partial x^2} = \frac{\partial^2 v}{\partial t^2}, \quad 0 < x < L, \quad t > 0$$

$$v(0, t) = 0, \quad \frac{\partial v}{\partial x} \Big|_{x=L} = 0, \quad t > 0,$$

$$v(x, 0) = -\frac{F_0}{E} x, \quad \frac{\partial v}{\partial t} \Big|_{t=0} = 0, \quad 0 < x < L.$$

Referring to Example 2 in the text we see that

$$v(x, t) = \sum_{n=1}^{\infty} A_n \cos a \left(\frac{2n-1}{2L} \right) \pi t \sin \left(\frac{2n-1}{2L} \right) \pi x$$

where

$$-\frac{F_0}{E} x = \sum_{n=1}^{\infty} A_n \sin \left(\frac{2n-1}{2L} \right) \pi x$$

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and

$$A_n = \frac{-F_0 \int_0^L x \sin\left(\frac{2n-1}{2L}\right) \pi x dx}{E \int_0^L \sin^2\left(\frac{2n-1}{2L}\right) \pi x dx} = \frac{8F_0 L (-1)^n}{E \pi^2 (2n-1)^2}.$$

Thus

$$\begin{aligned} u(x, t) &= v(x, t) + \psi(x) \\ &= \frac{F_0}{E} x + \frac{8F_0 L}{E \pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n-1)^2} \cos a\left(\frac{2n-1}{2L}\right) \pi t \sin\left(\frac{2n-1}{2L}\right) \pi x. \end{aligned}$$

7. Separation of variables leads to

$$Y'' + \lambda^2 Y = 0$$

$$X'' - \lambda^2 X = 0$$

and

$$Y(y) = c_1 \cos \lambda y + c_2 \sin \lambda y$$

$$X(x) = c_3 \cosh \lambda x + c_4 \sinh \lambda x.$$

From $Y(0) = 0$ we find $c_1 = 0$. From $Y'(1) = 0$ we obtain $\cos \lambda = 0$ and

$$\lambda = \frac{\pi(2n-1)}{2}, \quad n = 1, 2, 3, \dots$$

Thus

$$Y(y) = c_2 \sin\left(\frac{2n-1}{2}\right) \pi y.$$

From $X'(0) = 0$ we find $c_4 = 0$. Then

$$u(x, y) = \sum_{n=1}^{\infty} A_n \cosh\left(\frac{2n-1}{2}\right) \pi x \sin\left(\frac{2n-1}{2}\right) \pi y$$

where

$$u_0 = u(1, y) = \sum_{n=1}^{\infty} A_n \cosh\left(\frac{2n-1}{2}\right) \pi \sin\left(\frac{2n-1}{2}\right) \pi y$$

and

$$A_n \cosh\left(\frac{2n-1}{2}\right) \pi = \frac{\int_0^1 u_0 \sin\left(\frac{2n-1}{2}\right) \pi y dy}{\int_0^1 \sin^2\left(\frac{2n-1}{2}\right) \pi y dy} = \frac{4u_0}{(2n-1)\pi}.$$

Thus

$$u(x, y) = \frac{4u_0}{\pi} \sum_{n=1}^{\infty} \frac{1}{(2n-1) \cosh\left(\frac{2n-1}{2}\right) \pi} \cosh\left(\frac{2n-1}{2}\right) \pi x \sin\left(\frac{2n-1}{2}\right) \pi y.$$

8. The boundary-value problem is

$$k \frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}, \quad 0 < x < 1, \quad t > 0$$

$$\left. \frac{\partial u}{\partial x} \right|_{x=0} = hu(0, t), \quad \left. \frac{\partial u}{\partial x} \right|_{x=1} = -hu(1, t), \quad h > 0, \quad t > 0,$$

$$u(x, 0) = f(x), \quad 0 < x < 1.$$

Referring to Example 1 in the text we have

$$X(x) = c_1 \cos \lambda x + c_2 \sin \lambda x$$

and

$$T(t) = c_3 e^{-k\lambda^2 t}.$$

Applying the boundary conditions, we obtain

$$X'(0) = hX(0)$$

$$X'(1) = -hX(1)$$

or

$$\lambda c_2 = hc_1$$

$$-\lambda c_1 \sin \lambda + \lambda c_2 \cos \lambda = -hc_1 \cos \lambda - hc_2 \sin \lambda.$$

Choosing $c_1 = \lambda$ and $c_2 = h$ (to satisfy the first equation above) we obtain

$$-\lambda^2 \sin \lambda + h\lambda \cos \lambda = -h\lambda \cos \lambda - h^2 \sin \lambda$$

$$2h\lambda \cos \lambda = (\lambda^2 - h^2) \sin \lambda.$$

The eigenvalues λ_n are the consecutive positive roots of

$$\tan \lambda = \frac{2h\lambda}{\lambda^2 - h^2}.$$

Then

$$u(x, t) = \sum_{n=1}^{\infty} A_n e^{-k\lambda_n^2 t} (\lambda_n \cos \lambda_n x + h \sin \lambda_n x)$$

where

$$f(x) = u(x, 0) = \sum_{n=1}^{\infty} A_n (\lambda_n \cos \lambda_n x + h \sin \lambda_n x)$$

and

$$\begin{aligned} A_n &= \frac{\int_0^1 f(x) (\lambda_n \cos \lambda_n x + h \sin \lambda_n x) dx}{\int_0^1 (\lambda_n \cos \lambda_n x + h \sin \lambda_n x)^2 dx} \\ &= \frac{2}{\lambda_n^2 + 2h + h^2} \int_0^1 f(x) (\lambda_n \cos \lambda_n x + h \sin \lambda_n x) dx. \end{aligned}$$

[Note: the evaluation and simplification of the integral in the denominator requires the use of the relationship $(\lambda^2 - h^2) \sin \lambda = 2h\lambda \cos \lambda$.]

9. (a) Using $u = XT$ and separation constant λ^4 we find

$$X^{(4)} - \lambda^4 X = 0$$

and

$$X(x) = c_1 \cos \lambda x + c_2 \sin \lambda x + c_3 \cosh \lambda x + c_4 \sinh \lambda x.$$

Since $u = XT$ the boundary conditions become

$$X(0) = 0, \quad X'(0) = 0, \quad X''(1) = 0, \quad X'''(1) = 0.$$

Now $X(0) = 0$ implies $c_1 + c_3 = 0$, while $X'(0) = 0$ implies $c_2 + c_4 = 0$. Thus

$$X(x) = c_1 \cos \lambda x + c_2 \sin \lambda x - c_1 \cosh \lambda x - c_2 \sinh \lambda x.$$

The boundary condition $X''(1) = 0$ implies

$$-c_1 \cos \lambda - c_2 \sin \lambda - c_1 \cosh \lambda - c_2 \sinh \lambda = 0$$

while the boundary condition $X'''(1) = 0$ implies

$$c_1 \sin \lambda - c_2 \cos \lambda - c_1 \sinh \lambda - c_2 \cosh \lambda = 0.$$

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We then have the system of two equations in two unknowns

$$(\cos \lambda + \cosh \lambda)c_1 + (\sin \lambda + \sinh \lambda)c_2 = 0$$

$$(\sin \lambda - \sinh \lambda)c_1 - (\cos \lambda + \cosh \lambda)c_2 = 0.$$

This homogeneous system will have nontrivial solutions for c_1 and c_2 provided

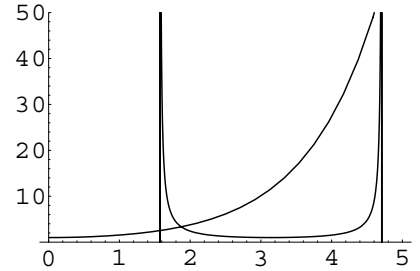
$$\begin{vmatrix} \cos \lambda + \cosh \lambda & \sin \lambda + \sinh \lambda \\ \sin \lambda - \sinh \lambda & -\cos \lambda - \cosh \lambda \end{vmatrix} = 0$$

or

$$-2 - 2 \cos \lambda \cosh \lambda = 0.$$

Thus, the eigenvalues are determined by the equation $\cos \lambda \cosh \lambda = -1$.

- (b) Using a computer to graph $\cosh \lambda$ and $-1/\cos \lambda = -\sec \lambda$ we see that the first two positive eigenvalues occur near 1.9 and 4.7. Applying Newton's method with these initial values we find that the eigenvalues are $\lambda_1 = 1.8751$ and $\lambda_2 = 4.6941$.



10. (a) In this case the boundary conditions are

$$\begin{aligned} u(0, t) &= 0, & \frac{\partial u}{\partial x} \Big|_{x=0} &= 0 \\ u(1, t) &= 0, & \frac{\partial u}{\partial x} \Big|_{x=1} &= 0. \end{aligned}$$

Separating variables leads to

$$X(x) = c_1 \cos \lambda x + c_2 \sin \lambda x + c_3 \cosh \lambda x + c_4 \sinh \lambda x$$

subject to

$$X(0) = 0, \quad X'(0) = 0, \quad X(1) = 0, \quad \text{and} \quad X'(1) = 0.$$

Now $X(0) = 0$ implies $c_1 + c_3 = 0$ while $X'(0) = 0$ implies $c_2 + c_4 = 0$. Thus

$$X(x) = c_1 \cos \lambda x + c_2 \sin \lambda x - c_1 \cosh \lambda x - c_2 \sinh \lambda x.$$

The boundary condition $X(1) = 0$ implies

$$c_1 \cos \lambda + c_2 \sin \lambda - c_1 \cosh \lambda - c_2 \sinh \lambda = 0$$

while the boundary condition $X'(1) = 0$ implies

$$-c_1 \sin \lambda + c_2 \cos \lambda - c_1 \sinh \lambda - c_2 \cosh \lambda = 0.$$

We then have the system of two equations in two unknowns

$$(\cos \lambda - \cosh \lambda)c_1 + (\sin \lambda - \sinh \lambda)c_2 = 0$$

$$-(\sin \lambda + \sinh \lambda)c_1 + (\cos \lambda - \cosh \lambda)c_2 = 0.$$

This homogeneous system will have nontrivial solutions for c_1 and c_2 provided

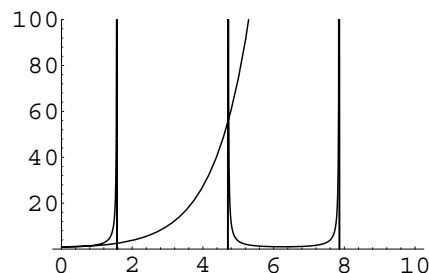
$$\begin{vmatrix} \cos \lambda - \cosh \lambda & \sin \lambda - \sinh \lambda \\ -\sin \lambda - \sinh \lambda & \cos \lambda - \cosh \lambda \end{vmatrix} = 0$$

or

$$2 - 2 \cos \lambda \cosh \lambda = 0.$$

Thus, the eigenvalues are determined by the equation $\cos \lambda \cosh \lambda = 1$.

- (b) Using a computer to graph $\cosh \lambda$ and $1/\cos \lambda = \sec \lambda$ we see that the first two positive eigenvalues occur near the vertical asymptotes of $\sec \lambda$, at $3\pi/2$ and $5\pi/2$. Applying Newton's method with these initial values we find that the eigenvalues are $\lambda_1 = 4.7300$ and $\lambda_2 = 7.8532$.



Exercises 13.8

1. This boundary-value problem was solved in Example 1 in the text. Identifying $b = c = \pi$ and $f(x, y) = u_0$ we have

$$u(x, y, t) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} A_{mn} e^{-k(m^2+n^2)t} \sin mx \sin ny$$

where

$$\begin{aligned} A_{mn} &= \frac{4}{\pi^2} \int_0^{\pi} \int_0^{\pi} u_0 \sin mx \sin ny \, dx \, dy \\ &= \frac{4u_0}{\pi^2} \int_0^{\pi} \sin mx \, dx \int_0^{\pi} \sin ny \, dy \\ &= \frac{4u_0}{mn\pi^2} [1 - (-1)^m][1 - (-1)^n]. \end{aligned}$$

2. This boundary-value problem was solved in Example 1 in the text. Identifying $b = c = 1$ and $f(x, y) = xy(x-1)(y-1)$ we have

$$u(x, y, t) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} A_{mn} e^{-k\pi(m^2+n^2)t} \sin m\pi x \sin n\pi y$$

where

$$\begin{aligned} A_{mn} &= 4 \int_0^1 \int_0^1 xy(x-1)(y-1) \sin m\pi x \sin n\pi y \, dx \, dy \\ &= 4 \left(\int_0^1 x(x-1) \sin m\pi x \, dx \right) \left(\int_0^1 y(y-1) \sin n\pi y \, dy \right) \\ &= 4 \left(\frac{2}{m^3\pi^3} [(-1)^m - 1] \right) \left(\frac{2}{n^3\pi^3} [(-1)^n - 1] \right) \\ &= \frac{16}{m^3n^3\pi^6} [(-1)^m - 1] [(-1)^n - 1]. \end{aligned}$$

In Problems 3 and 4 we need to solve the partial differential equation

$$a^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) = \frac{\partial^2 u}{\partial t^2}.$$

Exercises 13.8

To separate this equation we try $u(x, y, t) = X(x)Y(y)T(t)$:

$$a^2(X''YT + XY''T) = XYT''$$

$$\frac{X''}{X} = -\frac{Y''}{Y} + \frac{T''}{a^2T} = -\lambda^2.$$

Then

$$X'' + \lambda^2 X = 0 \tag{1}$$

$$\frac{Y''}{Y} = \frac{T''}{a^2T} + \lambda^2 = -\mu^2$$

$$Y'' + \mu^2 Y = 0 \tag{2}$$

$$T'' + a^2(\lambda^2 + \mu^2)T = 0. \tag{3}$$

The general solutions of equations (1), (2), and (3) are, respectively,

$$X(x) = c_1 \cos \lambda x + c_2 \sin \lambda x$$

$$Y(y) = c_3 \cos \mu y + c_4 \sin \mu y$$

$$T(t) = c_5 \cos a\sqrt{\lambda^2 + \mu^2}t + c_6 \sin a\sqrt{\lambda^2 + \mu^2}t.$$

3. The conditions $X(0) = 0$ and $Y(0) = 0$ give $c_1 = 0$ and $c_3 = 0$. The conditions $X(\pi) = 0$ and $Y(\pi) = 0$ yield two sets of eigenvalues:

$$\lambda = m, \quad m = 1, 2, 3, \dots \quad \text{and} \quad \mu = n, \quad n = 1, 2, 3, \dots$$

A product solution of the partial differential equation that satisfies the boundary conditions is

$$u_{mn}(x, y, t) = \left(A_{mn} \cos a\sqrt{m^2 + n^2}t + B_{mn} \sin a\sqrt{m^2 + n^2}t \right) \sin mx \sin ny.$$

To satisfy the initial conditions we use the superposition principle:

$$u(x, y, t) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \left(A_{mn} \cos a\sqrt{m^2 + n^2}t + B_{mn} \sin a\sqrt{m^2 + n^2}t \right) \sin mx \sin ny.$$

The initial condition $u_t(x, y, 0) = 0$ implies $B_{mn} = 0$ and

$$u(x, y, t) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} A_{mn} \cos a\sqrt{m^2 + n^2}t \sin mx \sin ny.$$

At $t = 0$ we have

$$xy(x - \pi)(y - \pi) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} A_{mn} \sin mx \sin ny.$$

Using (11) and (12) in the text, it follows that

$$\begin{aligned} A_{mn} &= \frac{4}{\pi^2} \int_0^\pi \int_0^\pi xy(x - \pi)(y - \pi) \sin mx \sin ny \, dx \, dy \\ &= \frac{4}{\pi^2} \int_0^\pi x(x - \pi) \sin mx \, dx \int_0^\pi y(y - \pi) \sin ny \, dy \\ &= \frac{16}{m^3 n^3 \pi^2} [(-1)^m - 1][(-1)^n - 1]. \end{aligned}$$

4. The conditions $X(0) = 0$ and $Y(0) = 0$ give $c_1 = 0$ and $c_3 = 0$. The conditions $X(b) = 0$ and $Y(c) = 0$ yield two sets of eigenvalues

$$\lambda = m\pi/b, \quad m = 1, 2, 3, \dots \quad \text{and} \quad \mu = n\pi/c, \quad n = 1, 2, 3, \dots$$

A product solution of the partial differential that satisfies the boundary conditions is

$$u_{mn}(x, y, t) = (A_{mn} \cos a\omega_{mn}t + B_{mn} \sin a\omega_{mn}t) \sin\left(\frac{m\pi}{b}x\right) \sin\left(\frac{n\pi}{c}y\right),$$

where $\omega_{mn} = \sqrt{(m\pi/b)^2 + (n\pi/c)^2}$. To satisfy the initial conditions we use the superposition principle:

$$u(x, y, t) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} (A_{mn} \cos a\omega_{mn}t + B_{mn} \sin a\omega_{mn}t) \sin\left(\frac{m\pi}{b}x\right) \sin\left(\frac{n\pi}{c}y\right).$$

At $t = 0$ we have

$$f(x, y) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} A_{mn} \sin\left(\frac{m\pi}{b}x\right) \sin\left(\frac{n\pi}{c}y\right)$$

and

$$g(x, y) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} B_{mn} a\omega_{mn} \sin\left(\frac{m\pi}{b}x\right) \sin\left(\frac{n\pi}{c}y\right).$$

Using (11) and (12) in the text, it follows that

$$A_{mn} = \frac{4}{bc} \int_0^c \int_0^b f(x, y) \sin\left(\frac{m\pi}{b}x\right) \sin\left(\frac{n\pi}{c}y\right) dx dy$$

$$B_{mn} = \frac{4}{abc\omega_{mn}} \int_0^c \int_0^b g(x, y) \sin\left(\frac{m\pi}{b}x\right) \sin\left(\frac{n\pi}{c}y\right) dx dy.$$

To separate Laplace's equation in three dimensions we try $u(x, y, z) = X(x)Y(y)Z(z)$:

$$X''YZ + XY''Z + XYZ'' = 0$$

$$\frac{X''}{X} = -\frac{Y''}{Y} - \frac{Z''}{Z} = -\lambda^2.$$

Then

$$X'' + \lambda^2 X = 0 \tag{4}$$

$$\frac{Y''}{Y} = -\frac{Z''}{Z} + \lambda^2 = -\mu^2$$

$$Y'' + \mu^2 Y = 0 \tag{5}$$

$$Z'' - (\lambda^2 + \mu^2)Z = 0. \tag{6}$$

The general solutions of equations (4), (5), and (6) are, respectively

$$X(x) = c_1 \cos \lambda x + c_2 \sin \lambda x$$

$$Y(y) = c_3 \cos \mu y + c_4 \sin \mu y$$

$$Z(z) = c_5 \cosh \sqrt{\lambda^2 + \mu^2} z + c_6 \sinh \sqrt{\lambda^2 + \mu^2} z.$$

5. The boundary and initial conditions are

$$u(0, y, z) = 0, \quad u(a, y, z) = 0$$

$$u(x, 0, z) = 0, \quad u(x, b, z) = 0$$

$$u(x, y, 0) = 0, \quad u(x, y, c) = f(x, y).$$

Exercises 13.8

The conditions $X(0) = Y(0) = Z(0) = 0$ give $c_1 = c_3 = c_5 = 0$. The conditions $X(a) = 0$ and $Y(b) = 0$ yield two sets of eigenvalues:

$$\lambda = \frac{m\pi}{a}, \quad m = 1, 2, 3, \dots \quad \text{and} \quad \mu = \frac{n\pi}{b}, \quad n = 1, 2, 3, \dots$$

By the superposition principle

$$u(x, y, t) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} A_{mn} \sinh \omega_{mn} z \sin \frac{m\pi}{a} x \sin \frac{n\pi}{b} y$$

where

$$\omega_{mn}^2 = \frac{m^2 \pi^2}{a^2} + \frac{n^2 \pi^2}{b^2}$$

and

$$A_{mn} = \frac{4}{ab \sinh \omega_{mn} c} \int_0^b \int_0^a f(x, y) \sin \frac{m\pi}{a} x \sin \frac{n\pi}{b} y \, dx \, dy.$$

6. The boundary and initial conditions are

$$\begin{aligned} u(0, y, z) &= 0, & u(a, y, z) &= 0, \\ u(x, 0, z) &= 0, & u(x, b, z) &= 0, \\ u(x, y, 0) &= f(x, y), & u(x, y, c) &= 0. \end{aligned}$$

The conditions $X(0) = Y(0) = 0$ give $c_1 = c_3 = 0$. The conditions $X(a) = Y(b) = 0$ yield two sets of eigenvalues:

$$\lambda = \frac{m\pi}{a}, \quad m = 1, 2, 3, \dots \quad \text{and} \quad \mu = \frac{n\pi}{b}, \quad n = 1, 2, 3, \dots$$

Let

$$\omega_{mn}^2 = \frac{m^2 \pi^2}{a^2} + \frac{n^2 \pi^2}{b^2}.$$

Then the boundary condition $Z(c) = 0$ gives

$$c_5 \cosh c \omega_{mn} + c_6 \sinh c \omega_{mn} = 0$$

from which we obtain

$$\begin{aligned} Z(z) &= c_5 \left(\cosh \omega_{mn} z - \frac{\cosh c \omega_{mn}}{\sinh c \omega_{mn}} \sinh \omega_{mn} z \right) \\ &= \frac{c_5}{\sinh c \omega_{mn}} (\sinh c \omega_{mn} \cosh \omega_{mn} z - \cosh c \omega_{mn} \sinh \omega_{mn} z) \\ &= c_{mn} \sinh \omega_{mn} (c - z). \end{aligned}$$

By the superposition principle

$$u(x, y, t) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} A_{mn} \sinh \omega_{mn} (c - z) \sin \frac{m\pi}{a} x \sin \frac{n\pi}{b} y$$

where

$$A_{mn} = \frac{4}{ab \sinh c \omega_{mn}} \int_0^b \int_0^a f(x, y) \sin \frac{m\pi}{a} x \sin \frac{n\pi}{b} y \, dx \, dy.$$

7. The boundary and initial conditions are

$$\begin{aligned} u(0, y, z) &= 0, & u(1, y, z) &= 0, \\ u(x, 0, z) &= 0, & u(x, 1, z) &= 0, \\ u(x, y, 0) &= -u_0, & u(x, y, 1) &= u_0. \end{aligned}$$

Applying the superposition principle to the solutions in Problems 5 and 6, with $a = b = c = 1$ and $f(x, y) = u_0$ in Problem 5 and $f(x, y) = -u_0$ in Problem 6, we get

$$u(x, y, t) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} A_{mn} [\sinh \omega_{mn} z - \sinh \omega_m (1 - z)] \sin m\pi x \sin n\pi y$$

where

$$\omega_{mn}^2 = (m^2 + n^2)\pi^2$$

and

$$\begin{aligned} A_{mn} &= \frac{4}{\sinh \omega_{mn}} \int_0^1 \int_0^1 u_0 \sin m\pi x \sin n\pi y \, dx \, dy = \frac{4u_0}{\sinh \omega_{mn}} \left(\int_0^1 \sin m\pi x \, dx \right) \left(\int_0^1 \sin n\pi y \, dy \right) \\ &= \frac{4u_0}{\sinh \omega_{mn}} \left(\frac{1}{m\pi} [1 - (-1)^n] \right) \left(\frac{1}{n\pi} [1 - (-1)^m] \right) = \frac{4u_0}{mn\pi^2 \sinh \omega_{mn}} [1 - (-1)^m] [1 - (-1)^n]. \end{aligned}$$

Chapter 13 Review Exercises

1. Let $u = XY$ so that

$$u_{yx} = X'Y',$$

$$X'Y' = XY,$$

and

$$\frac{X'}{X} = \frac{Y'}{Y} = \pm \lambda^2.$$

Then

$$X' \mp \lambda^2 X = 0 \quad \text{and} \quad Y \mp \lambda^2 Y' = 0.$$

For $\lambda^2 > 0$ we obtain

$$X = c_1 e^{\lambda^2 x} \quad \text{and} \quad Y = c_2 e^{y/\lambda^2}.$$

For $-\lambda^2 < 0$ we obtain

$$X = c_1 e^{-\lambda^2 x} \quad \text{and} \quad Y = c_2 e^{-y/\lambda^2}.$$

For $\lambda^2 = 0$ we obtain $X = c_1$ and $Y = 0$. The general form is

$$u = XY = A_1 e^{A_2 x + y/A_2}.$$

2. If $u = XY$ then

$$u_x = X'Y,$$

$$u_y = XY',$$

$$u_{xx} = X''Y,$$

$$u_{yy} = XY'',$$

and

$$X''Y + XY'' + 2X'Y + 2XY' = 0$$

Chapter 13 Review Exercises

so that

$$(X'' + 2X')Y + X(Y'' + 2Y') = 0,$$

$$\frac{X'' + 2X'}{-X} = \frac{Y'' + 2Y'}{Y} = \pm\lambda^2,$$

$$X'' + 2X' \pm \lambda^2 X = 0,$$

and

$$Y'' + 2Y' \mp \lambda^2 Y = 0.$$

Using λ^2 as a separation constant we obtain

$$Y = c_1 e^{(-1+\sqrt{1+\lambda^2})y} + c_2 e^{(-1-\sqrt{1+\lambda^2})y}.$$

If $1 - \lambda^2 < 0$ then

$$X = e^{-x} \left(c_3 \cos \sqrt{\lambda^2 - 1} x + c_4 \sin \sqrt{\lambda^2 - 1} x \right).$$

If $1 - \lambda^2 > 0$ then

$$X = c_3 e^{(-1+\sqrt{1-\lambda^2})x} + c_4 e^{(-1-\sqrt{1-\lambda^2})x}.$$

If $1 - \lambda^2 = 0$ then

$$X = c_3 e^{-x} + c_4 x e^{-x}$$

so that

$$\begin{aligned} u = XY &= \left(A_1 e^{(-1+\sqrt{1+A_5})y} + A_2 e^{(-1-\sqrt{1+A_5})y} \right) e^{-x} \left(A_3 \cos \sqrt{A_5 - 1} x + A_4 \sin \sqrt{A_5 - 1} x \right) \\ &= \left(A_1 e^{(-1+\sqrt{1+A_5})y} + A_2 e^{(-1-\sqrt{1+A_5})y} \right) \left(A_3 e^{(-1+\sqrt{1-A_5})x} + A_4 e^{(-1-\sqrt{1-A_5})x} \right) \\ &= \left(A_1 e^{(-1+\sqrt{2})y} + A_2 e^{(-1-\sqrt{2})y} \right) (A_3 e^{-x} + A_4 x e^{-x}). \end{aligned}$$

Using $-\lambda^2$ we obtain the same three solutions except that x and y are interchanged. Using $\lambda^2 = 0$ we obtain

$$u = XY = (A_1 + A_2 e^{-2x}) (A_3 + A_4 e^{-2y}).$$

3. Substituting $u(x, t) = v(x, t) + \psi(x)$ into the partial differential equation we obtain

$$k \frac{\partial^2 v}{\partial x^2} + k \psi''(x) = \frac{\partial v}{\partial t}.$$

This equation will be homogeneous provided ψ satisfies

$$k\psi'' = 0 \quad \text{or} \quad \psi = c_1 x + c_2.$$

Considering

$$u(0, t) = v(0, t) + \psi(0) = u_0$$

we set $\psi(0) = u_0$ so that $\psi(x) = c_1 x + u_0$. Now

$$-\frac{\partial u}{\partial x} \Big|_{x=\pi} = -\frac{\partial v}{\partial x} \Big|_{x=\pi} - \psi'(x) = v(\pi, t) + \psi(\pi) - u_1$$

is equivalent to

$$\frac{\partial v}{\partial x} \Big|_{x=\pi} + v(\pi, t) = u_1 - \psi'(x) - \psi(\pi) = u_1 - c_1 - (c_1 \pi + u_0),$$

which will be homogeneous when

$$u_1 - c_1 - c_1 \pi - u_0 = 0 \quad \text{or} \quad c_1 = \frac{u_1 - u_0}{1 + \pi}.$$

The steady-state solution is

$$\psi(x) = \left(\frac{u_1 - u_0}{1 + \pi} \right) x + u_0.$$

4. The solution of the problem represents the heat of a thin rod of length π . The left boundary $x = 0$ is kept at constant temperature u_0 for $t > 0$. Heat is lost from the right end of the rod by being in contact with a medium that is held at constant temperature u_1 .
5. The boundary-value problem is

$$\begin{aligned} a^2 \frac{\partial^2 u}{\partial x^2} &= \frac{\partial^2 u}{\partial t^2}, \quad 0 < x < 1, \quad t > 0, \\ u(0, t) &= 0, \quad u(1, t) = 0, \quad t > 0, \\ u(x, 0) &= 0, \quad \left. \frac{\partial u}{\partial t} \right|_{t=0} = g(x), \quad 0 < x < 1. \end{aligned}$$

From Section 13.4 in the text we see that $A_n = 0$,

$$\begin{aligned} B_n &= \frac{2}{n\pi a} \int_0^1 g(x) \sin n\pi x \, dx = \frac{2}{n\pi a} \int_{1/4}^{3/4} h \sin n\pi x \, dx \\ &= \frac{2h}{n\pi a} \left(-\frac{1}{n\pi} \cos n\pi x \right) \Big|_{1/4}^{3/4} = \frac{2h}{n^2 \pi^2 a} \left(\cos \frac{n\pi}{4} - \cos \frac{3n\pi}{4} \right) \end{aligned}$$

and

$$u(x, t) = \sum_{n=1}^{\infty} B_n \sin n\pi a t \sin n\pi x.$$

6. The boundary-value problem is

$$\begin{aligned} \frac{\partial^2 u}{\partial x^2} + x^2 &= \frac{\partial^2 u}{\partial t^2}, \quad 0 < x < 1, \quad t > 0, \\ u(0, t) &= 1, \quad u(1, t) = 0, \quad t > 0, \\ u(x, 0) &= f(x), \quad u_t(x, 0) = 0, \quad 0 < x < 1. \end{aligned}$$

Substituting $u(x, t) = v(x, t) + \psi(x)$ into the partial differential equation gives

$$\frac{\partial^2 v}{\partial x^2} + \psi''(x) + x^2 = \frac{\partial^2 v}{\partial t^2}.$$

This equation will be homogeneous provided $\psi''(x) + x^2 = 0$ or

$$\psi(x) = -\frac{1}{12}x^4 + c_1x + c_2.$$

From $\psi(0) = 1$ and $\psi(1) = 0$ we obtain $c_1 = -11/12$ and $c_2 = 1$. The new problem is

$$\begin{aligned} \frac{\partial^2 v}{\partial x^2} &= \frac{\partial^2 v}{\partial t^2}, \quad 0 < x < 1, \quad t > 0, \\ v(0, t) &= 0, \quad v(1, t) = 0, \quad t > 0, \\ v(x, 0) &= f(x) - \psi(x), \quad v_t(x, 0) = 0, \quad 0 < x < 1. \end{aligned}$$

From Section 13.4 in the text we see that $B_n = 0$,

$$A_n = 2 \int_0^1 [f(x) - \psi(x)] \sin n\pi x \, dx = 2 \int_0^1 \left[f(x) + \frac{1}{12}x^4 + \frac{11}{12}x - 1 \right] \sin n\pi x \, dx,$$

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and

$$v(x, t) = \sum_{n=1}^{\infty} A_n \cos n\pi t \sin n\pi x.$$

Thus

$$u(x, t) = v(x, t) + \psi(x) = -\frac{1}{12}x^4 - \frac{11}{12}x + 1 + \sum_{n=1}^{\infty} A_n \cos n\pi t \sin n\pi x.$$

7. Using $u = XY$ and λ^2 as a separation constant leads to

$$X'' - \lambda^2 X = 0,$$

$$X(0) = 0,$$

and

$$Y'' + \lambda^2 Y = 0,$$

$$Y(0) = 0,$$

$$Y(\pi) = 0.$$

Then

$$Y = c_1 \sin ny \quad \text{and} \quad X = c_2 \sinh nx$$

for $n = 1, 2, 3, \dots$ so that

$$u = \sum_{n=1}^{\infty} A_n \sinh nx \sin ny.$$

Imposing

$$u(\pi, y) = 50 = \sum_{n=1}^{\infty} A_n \sinh n\pi \sin ny$$

gives

$$A_n \sinh n\pi = \frac{100}{n\pi} \frac{1 - (-1)^n}{\sinh n\pi}$$

so that

$$u(x, y) = \frac{100}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n \sinh n\pi} \sinh nx \sin ny.$$

8. Using $u = XY$ and λ^2 as a separation constant leads to

$$X'' - \lambda^2 X = 0,$$

and

$$Y'' + \lambda^2 Y = 0,$$

$$Y'(0) = 0,$$

$$Y'(\pi) = 0.$$

Then

$$Y = c_1 \cos ny$$

for $n = 0, 1, 2, \dots$ and

$$X = c_2 \quad \text{or} \quad X = c_2 e^{-nx}$$

for $n = 1, 2, 3, \dots$ (since u must be bounded as $x \rightarrow \infty$) so that

$$u = A_0 + \sum_{n=1}^{\infty} A_n e^{-nx} \cos ny.$$

Imposing

$$u(0, y) = 50 = A_0 + \sum_{n=1}^{\infty} A_n \cos ny$$

gives

$$A_0 = \frac{1}{\pi} \int_0^{\pi} 50 \, dy = 50$$

and

$$A_n = \frac{2}{\pi} \int_0^{\pi} 50 \cos ny \, dy = 0$$

for $n = 1, 2, 3, \dots$ so that

$$u(x, y) = 50.$$

9. Using $u = XY$ and λ^2 as a separation constant leads to

$$X'' - \lambda^2 X = 0,$$

and

$$Y'' + \lambda^2 Y = 0,$$

$$Y(0) = 0,$$

$$Y(\pi) = 0.$$

Then

$$Y = c_1 \sin ny \quad \text{and} \quad X = c_2 e^{-nx}$$

for $n = 1, 2, 3, \dots$ (since u must be bounded as $x \rightarrow \infty$) so that

$$u = \sum_{n=1}^{\infty} A_n e^{-nx} \sin ny.$$

Imposing

$$u(0, y) = 50 = \sum_{n=1}^{\infty} A_n \sin ny$$

gives

$$A_n = \frac{2}{\pi} \int_0^{\pi} 50 \sin ny \, dy = \frac{100}{n\pi} [1 - (-1)^n]$$

so that

$$u(x, y) = \sum_{n=1}^{\infty} \frac{100}{n\pi} [1 - (-1)^n] e^{-nx} \sin ny.$$

10. The boundary-value problem is

$$k \frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}, \quad -L < x < L, \quad t > 0,$$

$$u(-L, t) = 0, \quad u(L, t) = 0, \quad t > 0,$$

$$u(x, 0) = u_0, \quad -L < x < L.$$

Referring to Section 13.3 in the text we have

$$X(x) = c_1 \cos \lambda x + c_2 \sin \lambda x$$

and

$$T(t) = c_3 e^{-k\lambda^2 t}.$$

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Using the boundary conditions $u(-L, 0) = X(-L)T(0) = 0$ and $u(L, 0) = X(L)T(0) = 0$ we obtain

$$c_1 \cos(-\lambda L) + c_2 \sin(-\lambda L) = 0$$

$$c_1 \cos \lambda L + c_2 \sin \lambda L = 0$$

or

$$c_1 \cos \lambda L - c_2 \sin \lambda L = 0$$

$$c_1 \cos \lambda L + c_2 \sin \lambda L = 0.$$

Adding, we find $\cos \lambda L = 0$ which gives the eigenvalues

$$\lambda = \frac{2n-1}{2L}\pi, \quad n = 1, 2, 3, \dots$$

Thus

$$u(x, t) = \sum_{n=1}^{\infty} A_n e^{-\left(\frac{2n-1}{2L}\pi\right)^2 kt} \cos\left(\frac{2n-1}{2L}\pi x\right).$$

From

$$u(x, 0) = u_0 = \sum_{n=1}^{\infty} A_n \cos\left(\frac{2n-1}{2L}\pi x\right)$$

we find

$$A_n = \frac{2 \int_0^L u_0 \cos\left(\frac{2n-1}{2L}\pi x\right) \pi x \, dx}{2 \int_0^L \cos^2\left(\frac{2n-1}{2L}\pi x\right) \pi x \, dx} = \frac{u_0(-1)^{n+1} 2L/\pi(2n-1)}{L/2} = \frac{4u_0(-1)^{n+1}}{\pi(2n-1)}.$$

11. The coefficients of the series

$$u(x, 0) = \sum_{n=1}^{\infty} B_n \sin nx$$

are

$$\begin{aligned} B_n &= \frac{2}{\pi} \int_0^{\pi} \sin x \sin nx \, dx = \frac{2}{\pi} \int_0^{\pi} \frac{1}{2} [\cos(1-n)x - \cos(1+n)x] \, dx \\ &= \frac{1}{\pi} \left[\frac{\sin(1-n)x}{1-n} \Big|_0^{\pi} - \frac{\sin(1+n)x}{1+n} \Big|_0^{\pi} \right] = 0 \text{ for } n \neq 1. \end{aligned}$$

For $n = 1$,

$$B_1 = \frac{2}{\pi} \int_0^{\pi} \sin^2 x \, dx = \frac{1}{\pi} \int_0^{\pi} (1 - \cos 2x) \, dx = 1.$$

Thus

$$u(x, t) = \sum_{n=1}^{\infty} B_n e^{-n^2 t} \sin nx$$

reduces to $u(x, t) = e^{-t} \sin x$ for $n = 1$.

12. Substituting $u(x, t) = v(x, t) + \psi(x)$ into the partial differential equation gives

$$k \frac{\partial^2 v}{\partial x^2} + k \psi'' + \sin 2\pi x = \frac{\partial v}{\partial t}.$$

This equation will be homogeneous provided ψ satisfies

$$k \psi'' + \sin 2\pi x = 0.$$

The general solution of this equation is

$$\psi(x) = \frac{1}{4k\pi^2} \sin 2\pi x + c_1 x + c_2.$$

From $\psi(0) = \psi(1) = 0$ we find that $c_1 = c_2 = 0$ and

$$\psi(x) = \frac{1}{4k\pi^2} \sin 2\pi x.$$

Now the new problem is

$$\begin{aligned} k \frac{\partial^2 v}{\partial x^2} &= \frac{\partial v}{\partial t}, & 0 < x < 1, & \quad t > 0 \\ v(0, t) &= 0, & v(1, t) &= 0, & \quad t > 0 \\ v(x, 0) &= \sin \pi x - \psi(x), & 0 < x < 1. \end{aligned}$$

If we let $v = XT$ then

$$\frac{X''}{X} = \frac{T'}{kT} = -\lambda^2$$

gives the separated differential equations

$$X'' + \lambda^2 X = 0 \quad \text{and} \quad T' + k\lambda^2 T = 0.$$

The respective solutions are

$$\begin{aligned} X(x) &= c_3 \cos \lambda x + c_4 \sin \lambda x \\ T(t) &= c_5 e^{-k\lambda^2 t}. \end{aligned}$$

From $X(0) = 0$ we get $c_3 = 0$ and from $X(1) = 0$ we find $\lambda = n\pi$ for $n = 1, 2, 3, \dots$. Consequently, it follows that

$$v(x, t) = \sum_{n=1}^{\infty} A_n e^{-kn^2\pi^2 t} \sin n\pi x$$

where

$$v(x, 0) = \sin \pi x - \frac{1}{4k\pi^2} \sin 2\pi x = 0$$

implies

$$A_n = 2 \int_0^1 \left(\sin \pi x - \frac{1}{4k\pi^2} \sin 2\pi x \right) \sin n\pi x \, dx.$$

By orthogonality $A_n = 0$ for $n = 3, 4, 5, \dots$, and only A_1 and A_2 can be nonzero. We have

$$A_1 = 2 \left[\int_0^1 \sin^2 \pi x \, dx - \frac{1}{4k\pi^2} \int_0^1 \sin 2\pi x \sin \pi x \, dx \right] = 2 \int_0^1 \frac{1}{2} (1 - \cos 2\pi x) \, dx = 1$$

and

$$\begin{aligned} A_2 &= 2 \left[\int_0^1 \sin \pi x \sin 2\pi x \, dx - \frac{1}{4k\pi^2} \int_0^1 \sin^2 2\pi x \, dx \right] \\ &= -\frac{1}{2k\pi^2} \int_0^1 \frac{1}{2} (1 - \cos 4\pi x) \, dx = -\frac{1}{4k\pi^2}. \end{aligned}$$

Therefore

$$\begin{aligned} v(x, t) &= A_1 e^{-k\pi^2 t} \sin \pi x + A_2 e^{-4k\pi^2 t} \sin 2\pi x \\ &= e^{-k\pi^2 t} \sin \pi x - \frac{1}{4k\pi^2} e^{-4k\pi^2 t} \sin 2\pi x \end{aligned}$$

and

$$u(x, t) = v(x, t) + \psi(x) = e^{-k\pi^2 t} \sin \pi x + \frac{1}{4k\pi^2} (1 - e^{-4k\pi^2 t}) \sin 2\pi x.$$

13. Using $u = XT$ and $-\lambda^2$ as a separation constant we find

$$X'' + 2X' + \lambda^2 X = 0 \quad \text{and} \quad T'' + 2T' + (1 + \lambda^2)T = 0.$$

Chapter 13 Review Exercises

Thus for $\lambda > 1$

$$\begin{aligned} X &= c_1 e^{-x} \cos \sqrt{\lambda^2 - 1} x + c_2 e^{-x} \sin \sqrt{\lambda^2 - 1} x \\ T &= c_3 e^{-t} \cos \lambda t + c_4 e^{-t} \sin \lambda t. \end{aligned}$$

For $0 \leq \lambda \leq 1$ we only obtain $X = 0$. Now the boundary conditions $X(0) = 0$ and $X(\pi) = 0$ give, in turn, $c_1 = 0$ and $\sqrt{\lambda^2 - 1} \pi = n\pi$ or $\lambda^2 = n^2 + 1$, $n = 1, 2, 3, \dots$. The corresponding solutions are $X = c_2 e^{-x} \sin nx$. The initial condition $T'(0) = 0$ implies $c_3 = \lambda c_4$ and so

$$T = c_4 e^{-t} \left[\sqrt{n^2 + 1} \cos \sqrt{n^2 + 1} t + \sin \sqrt{n^2 + 1} t \right].$$

Using $u = XT$ and the superposition principle, a formal series solution is

$$u(x, t) = e^{-(x+t)} \sum_{n=1}^{\infty} A_n \left[\sqrt{n^2 + 1} \cos \sqrt{n^2 + 1} t + \sin \sqrt{n^2 + 1} t \right] \sin nx.$$

14. Letting $c = XT$ and separating variables we obtain

$$\frac{kX'' - hX'}{X} = \frac{T'}{T} \quad \text{or} \quad \frac{X'' - \alpha X'}{X} = \frac{T'}{kT} = -\lambda^2$$

where $\alpha = h/k$. This leads to the separated differential equations

$$X'' - \alpha X' + \lambda^2 X = 0 \quad \text{and} \quad T' + k\lambda^2 T = 0.$$

The solution of the second equation is

$$T(t) = c_3 e^{-k\lambda^2 t}.$$

For the first equation we have $m = \frac{1}{2}(\alpha \pm \sqrt{\alpha^2 - 4\lambda^2})$, and we consider three cases using the boundary conditions $X(0) = X(1) = 0$:

$\boxed{\alpha^2 > 4\lambda^2}$ The solution is $X = c_1 e^{m_1 x} + c_2 e^{m_2 x}$, where the boundary conditions imply $c_1 = c_2 = 0$, so $X = 0$. (Note in this case that if $\lambda = 0$, the solution is $X = c_1 + c_2 e^{\alpha x}$ and the boundary conditions again imply $c_1 = c_2 = 0$, so $X = 0$.)

$\boxed{\alpha^2 = 4\lambda^2}$ The solution is $X = c_1 e^{m_1 x} + c_2 x e^{m_1 x}$, where the boundary conditions imply $c_1 = c_2 = 0$, so $X = 0$.

$\boxed{\alpha^2 < 4\lambda^2}$ The solution is

$$X(x) = c_1 e^{\alpha x/2} \cos \frac{\sqrt{4\lambda^2 - \alpha^2}}{2} x + c_2 e^{\alpha x/2} \sin \frac{\sqrt{4\lambda^2 - \alpha^2}}{2} x.$$

From $X(0) = 0$ we see that $c_1 = 0$. From $X(1) = 0$ we find

$$\frac{1}{2} \sqrt{4\lambda^2 - \alpha^2} = n\pi \quad \text{or} \quad \lambda^2 = \frac{1}{4}(4n^2\pi^2 + \alpha^2).$$

Thus

$$X(x) = c_2 e^{\alpha x/2} \sin n\pi x,$$

and

$$c(x, t) = \sum_{n=1}^{\infty} A_n e^{\alpha x/2} e^{-k(4n^2\pi^2 + \alpha^2)t/4} \sin 2n\pi x.$$

The initial condition $c(x, 0) = c_0$ implies

$$c_0 = \sum_{n=1}^{\infty} A_n e^{\alpha x/2} \sin n\pi x. \quad (1)$$

From the self-adjoint form

$$\frac{d}{dx} [e^{-\alpha x} X'] + \lambda^2 e^{-\alpha x} X = 0$$

the eigenfunctions are orthogonal on $[0, 1]$ with weight function $e^{-\alpha x}$. That is

$$\int_0^1 e^{-\alpha x} (e^{\alpha x/2} \sin n\pi x) (e^{\alpha x/2} \sin m\pi x) dx = 0, \quad n \neq m.$$

Multiplying (1) by $e^{-\alpha x} e^{\alpha x/2} \sin m\pi x$ and integrating we obtain

$$\int_0^1 c_0 e^{-\alpha x} e^{\alpha x/2} \sin m\pi x dx = \sum_{n=1}^{\infty} A_n e^{-\alpha x} e^{\alpha x/2} (\sin m\pi x) e^{\alpha x/2} \sin n\pi x dx$$

$$c_0 \int_0^1 e^{-\alpha x/2} \sin n\pi x dx = A_n \int_0^1 \sin^2 n\pi x dx = \frac{1}{2} A_n$$

and

$$A_n = 2c_0 \int_0^1 e^{-\alpha x/2} \sin n\pi x dx = \frac{4c_0 [2e^{\alpha/2} n\pi - 2n\pi (-1)^n]}{e^{\alpha/2} (\alpha^2 + 4n^2\pi^2)} = \frac{8n\pi c_0 [e^{\alpha/2} - (-1)^n]}{e^{\alpha/2} (\alpha^2 + 4n^2\pi^2)}.$$

14 Boundary-Value Problems in Other Coordinate Systems

Exercises 14.1

1. We have

$$A_0 = \frac{1}{2\pi} \int_0^\pi u_0 d\theta = \frac{u_0}{2}$$

$$A_n = \frac{1}{\pi} \int_0^\pi u_0 \cos n\theta d\theta = 0$$

$$B_n = \frac{1}{\pi} \int_0^\pi u_0 \sin n\theta d\theta = \frac{u_0}{n\pi} [1 - (-1)^n]$$

and so

$$u(r, \theta) = \frac{u_0}{2} + \frac{u_0}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n} r^n \sin n\theta.$$

2. We have

$$A_0 = \frac{1}{2\pi} \int_0^\pi \theta d\theta + \frac{1}{2\pi} \int_\pi^{2\pi} (\pi - \theta) d\theta = 0$$

$$A_n = \frac{1}{\pi} \int_0^\pi \theta \cos n\theta d\theta + \frac{1}{\pi} \int_\pi^{2\pi} (\pi - \theta) \cos n\theta d\theta = \frac{2}{n^2\pi} [(-1)^n - 1]$$

$$B_n = \frac{1}{\pi} \int_0^\pi \theta \sin n\theta d\theta + \frac{1}{\pi} \int_\pi^{2\pi} (\pi - \theta) \sin n\theta d\theta = \frac{1}{n} [1 - (-1)^n]$$

and so

$$u(r, \theta) = \sum_{n=1}^{\infty} r^n \left[\frac{(-1)^n - 1}{n^2\pi} \cos n\theta + \frac{1 - (-1)^n}{n} \sin n\theta \right].$$

3. We have

$$A_0 = \frac{1}{2\pi} \int_0^{2\pi} (2\pi\theta - \theta^2) d\theta = \frac{2\pi^2}{3}$$

$$A_n = \frac{1}{\pi} \int_0^{2\pi} (2\pi\theta - \theta^2) \cos n\theta d\theta = -\frac{4}{n^2}$$

$$B_n = \frac{1}{\pi} \int_0^{2\pi} (2\pi\theta - \theta^2) \sin n\theta d\theta = 0$$

and so

$$u(r, \theta) = \frac{2\pi^2}{3} - 4 \sum_{n=1}^{\infty} \frac{r^n}{n^2} \cos n\theta.$$

4. We have

$$A_0 = \frac{1}{2\pi} \int_0^{2\pi} \theta d\theta = \pi$$

$$A_n = \frac{1}{\pi} \int_0^{2\pi} \theta \cos n\theta d\theta = 0$$

$$B_n = \frac{1}{\pi} \int_0^{2\pi} \theta \sin n\theta d\theta = -\frac{2}{n}$$

and so

$$u(r, \theta) = \pi - 2 \sum_{n=1}^{\infty} \frac{r^n}{n} \sin n\theta.$$

5. As in Example 1 in the text we have $R(r) = c_3 r^n + c_4 r^{-n}$. In order that the solution be bounded as $r \rightarrow \infty$ we must define $c_3 = 0$. Hence

$$u(r, \theta) = A_0 + \sum_{n=1}^{\infty} r^{-n} (A_n \cos n\theta + B_n \sin n\theta)$$

where

$$\begin{aligned} A_0 &= \frac{1}{2\pi} \int_0^{2\pi} f(\theta) d\theta \\ A_n &= \frac{c^n}{\pi} \int_0^{2\pi} f(\theta) \cos n\theta d\theta \\ B_n &= \frac{c^n}{\pi} \int_0^{2\pi} f(\theta) \sin n\theta d\theta. \end{aligned}$$

6. We solve

$$\begin{aligned} \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} &= 0, \quad 0 < \theta < \frac{\pi}{2}, \quad 0 < r < c, \\ u(c, \theta) &= f(\theta), \quad 0 < \theta < \frac{\pi}{2}, \\ u(r, 0) &= 0, \quad u(r, \pi/2) = 0, \quad 0 < r < c. \end{aligned}$$

Proceeding as in Example 1 in the text we obtain the separated differential equations

$$\begin{aligned} r^2 R'' + rR' - \lambda^2 R &= 0 \\ \Theta'' + \lambda^2 \Theta &= 0 \end{aligned}$$

with solutions

$$\begin{aligned} \Theta(\theta) &= c_1 \cos \lambda\theta + c_2 \sin \lambda\theta \\ R(r) &= c_3 r^\lambda + c_4 r^{-\lambda}. \end{aligned}$$

Since we want $R(r)$ to be bounded as $r \rightarrow 0$ we require $c_4 = 0$. Applying the boundary conditions $\Theta(0) = 0$ and $\Theta(\pi/2) = 0$ we find that $c_1 = 0$ and $\lambda = 2n$ for $n = 1, 2, 3, \dots$. Therefore

$$u(r, \theta) = \sum_{n=1}^{\infty} A_n r^{2n} \sin 2n\theta.$$

From

$$u(c, \theta) = f(\theta) = \sum_{n=1}^{\infty} A_n c^{2n} \sin 2n\theta$$

we find

$$A_n = \frac{4}{\pi c^{2n}} \int_0^{\pi/2} f(\theta) \sin 2n\theta d\theta.$$

7. Referring to the solution of Problem 6 above we have

$$\begin{aligned} \Theta(\theta) &= c_1 \cos \lambda\theta + c_2 \sin \lambda\theta \\ R(r) &= c_3 r^n. \end{aligned}$$

Applying the boundary conditions $\Theta'(0) = 0$ and $\Theta'(\pi/2) = 0$ we find that $c_2 = 0$ and $\lambda = 2n$ for

Exercises 14.1

$n = 0, 1, 2, \dots$. Therefore

$$u(r, \theta) = A_0 + \sum_{n=1}^{\infty} A_n r^{2n} \cos 2n\theta.$$

From

$$u(c, \theta) = \begin{cases} 1, & 0 < \theta < \pi/4 \\ 0, & \pi/4 < \theta < \pi/2 \end{cases} = A_0 + \sum_{n=1}^{\infty} A_n c^{2n} \cos 2n\theta$$

we find

$$A_0 = \frac{1}{\pi/2} \int_0^{\pi/4} d\theta = \frac{1}{2}$$

and

$$c^{2n} A_n = \frac{2}{\pi/2} \int_0^{\pi/4} \cos 2n\theta d\theta = \frac{2}{n\pi} \sin \frac{n\pi}{2}.$$

Thus

$$u(r, \theta) = \frac{1}{2} + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi}{2} \left(\frac{r}{c}\right)^{2n} \cos 2n\theta.$$

8. We solve

$$\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0, \quad 0 < \theta < \pi/4, \quad r > 0$$

$$u(r, 0) = 0, \quad r > 0$$

$$u(r, \pi/4) = 30, \quad r > 0.$$

Proceeding as in Example 1 in the text we find the separated ordinary differential equations to be

$$r^2 R'' + rR' - \lambda^2 R = 0$$

$$\Theta'' + \lambda^2 \Theta = 0.$$

The corresponding general solutions are

$$R(r) = c_1 r^\lambda + c_2 r^{-\lambda}$$

$$\Theta(\theta) = c_3 \cos \lambda\theta + c_4 \sin \lambda\theta.$$

The condition $\Theta(0) = 0$ implies $c_3 = 0$ so that $\Theta = c_4 \sin \lambda\theta$. Now, in order that the temperature be bounded as $r \rightarrow \infty$ we define $c_1 = 0$. Similarly, in order that the temperature be bounded as $r \rightarrow 0$ we are forced to define $c_2 = 0$. Thus $R(r) = 0$ and so no nontrivial solution exists for $\lambda > 0$. For $\lambda = 0$ the separated differential equations are

$$r^2 R'' + rR' = 0 \quad \text{and} \quad \Theta'' = 0.$$

Solutions of these latter equations are

$$R(r) = c_1 + c_2 \ln r \quad \text{and} \quad \Theta(\theta) = c_2 \theta + c_3.$$

$\Theta(0) = 0$ still implies $c_3 = 0$, whereas boundedness as $r \rightarrow 0$ demands $c_2 = 0$. Thus, a product solution is

$$u = c_1 c_2 \theta = A\theta.$$

From $u(r, \pi/4) = 0$ we obtain $A = 120/\pi$. Thus, a solution to the problem is

$$u(r, \theta) = \frac{120}{\pi} \theta.$$

9. Proceeding as in Example 1 in the text and again using the periodicity of $u(r, \theta)$, we have

$$\Theta(\theta) = c_1 \cos \lambda \theta + c_2 \sin \lambda \theta$$

where $\lambda = n$ for $n = 0, 1, 2, \dots$. Then

$$R(r) = c_3 r^n + c_4 r^{-n}.$$

[We do not have $c_4 = 0$ in this case since $0 < a \leq r$.] Since $u(b, \theta) = 0$ we have

$$u(r, \theta) = A_0 \ln \frac{r}{b} + \sum_{n=1}^{\infty} \left[\left(\frac{b}{r} \right)^n - \left(\frac{r}{b} \right)^n \right] [A_n \cos n\theta + B_n \sin n\theta].$$

From

$$u(a, \theta) = f(\theta) = A_0 \ln \frac{a}{b} + \sum_{n=1}^{\infty} \left[\left(\frac{b}{a} \right)^n - \left(\frac{a}{b} \right)^n \right] [A_n \cos n\theta + B_n \sin n\theta]$$

we find

$$A_0 \ln \frac{a}{b} = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) d\theta,$$

$$\left[\left(\frac{b}{a} \right)^n - \left(\frac{a}{b} \right)^n \right] A_n = \frac{1}{\pi} \int_0^{2\pi} f(\theta) \cos n\theta d\theta,$$

and

$$\left[\left(\frac{b}{a} \right)^n - \left(\frac{a}{b} \right)^n \right] B_n = \frac{1}{\pi} \int_0^{2\pi} f(\theta) \sin n\theta d\theta.$$

10. Substituting $u(r, \theta) = v(r, \theta) + \psi(r)$ into the partial differential equation we obtain

$$\frac{\partial^2 v}{\partial r^2} + \psi''(r) + \frac{1}{r} \left[\frac{\partial v}{\partial r} + \psi'(r) \right] + \frac{1}{r^2} \frac{\partial^2 v}{\partial \theta^2} = 0.$$

This equation will be homogeneous provided

$$\psi''(r) + \frac{1}{r} \psi'(r) = 0 \quad \text{or} \quad r^2 \psi''(r) + r \psi'(r) = 0.$$

The general solution of this Cauchy-Euler differential equation is

$$\psi(r) = c_1 + c_2 \ln r.$$

From

$$u_0 = u(a, \theta) = v(a, \theta) + \psi(a) \quad \text{and} \quad u_1 = u(b, \theta) = v(b, \theta) + \psi(b)$$

we see that in order for the boundary values $v(a, \theta)$ and $v(b, \theta)$ to be 0 we need $\psi(a) = u_0$ and $\psi(b) = u_1$. From this we have

$$\psi(a) = c_1 + c_2 \ln a = u_0$$

$$\psi(b) = c_1 + c_2 \ln b = u_1.$$

Solving for c_1 and c_2 we obtain

$$c_1 = \frac{u_1 \ln a - u_0 \ln b}{\ln a/b} \quad \text{and} \quad c_2 = \frac{u_0 - u_1}{\ln a/b}.$$

Then

$$\psi(r) = \frac{u_1 \ln a - u_0 \ln b}{\ln a/b} + \frac{u_0 - u_1}{\ln a/b} \ln r = \frac{u_0 \ln r/b - u_1 \ln r/a}{\ln a/b}.$$

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From Problem 9 with $f(\theta) = 0$ we see that the solution of

$$\frac{\partial^2 v}{\partial r^2} + \frac{1}{r} \frac{\partial v}{\partial r} + \frac{1}{r^2} \frac{\partial^2 v}{\partial \theta^2} = 0, \quad 0 < \theta < 2\pi, \quad a < r < b,$$

$$v(a, \theta) = 0, \quad v(b, \theta) = 0, \quad 0 < \theta < 2\pi$$

is $v(r, \theta) = 0$. Thus the steady-state temperature of the ring is

$$u(r, \theta) = v(r, \theta) + \psi(r) = \frac{u_0 \ln r/b - u_1 \ln r/a}{\ln a/b}.$$

11. We solve

$$\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0, \quad 0 < \theta < \pi, \quad a < r < b,$$

$$u(a, \theta) = \theta(\pi - \theta), \quad u(b, \theta) = 0, \quad 0 < \theta < \pi,$$

$$u(r, 0) = 0, \quad u(r, \pi) = 0, \quad a < r < b.$$

Proceeding as in Example 1 in the text we obtain the separated differential equations

$$r^2 R'' + rR' - \lambda^2 R = 0$$

$$\Theta'' + \lambda^2 \Theta = 0$$

with solutions

$$\Theta(\theta) = c_1 \cos \lambda \theta + c_2 \sin \lambda \theta$$

$$R(r) = c_3 r^\lambda + c_4 r^{-\lambda}.$$

Applying the boundary conditions $\Theta(0) = 0$ and $\Theta(\pi) = 0$ we find that $c_1 = 0$ and $\lambda = n$ for $n = 1, 2, 3, \dots$. The boundary condition $R(b) = 0$ gives

$$c_3 b^n + c_4 b^{-n} = 0 \quad \text{and} \quad c_4 = -c_3 b^{2n}.$$

Then

$$R(r) = c_3 \left(r^n - \frac{b^{2n}}{r^n} \right) = c_3 \left(\frac{r^{2n} - b^{2n}}{r^n} \right)$$

and

$$u(r, \theta) = \sum_{n=1}^{\infty} A_n \left(\frac{r^{2n} - b^{2n}}{r^n} \right) \sin n\theta.$$

From

$$u(a, \theta) = \theta(\pi - \theta) = \sum_{n=1}^{\infty} A_n \left(\frac{a^{2n} - b^{2n}}{a^n} \right) \sin n\theta$$

we find

$$A_n \left(\frac{a^{2n} - b^{2n}}{a^n} \right) = \frac{2}{\pi} \int_0^\pi (\theta\pi - \theta^2) \sin n\theta \, d\theta = \frac{4}{n^3 \pi} [1 - (-1)^n].$$

Thus

$$u(r, \theta) = \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n^3} \frac{r^{2n} - b^{2n}}{a^{2n} - b^{2n}} \left(\frac{a}{r} \right)^n \sin n\theta.$$

12. Letting $u(r, \theta) = v(r, \theta) + \psi(\theta)$ we obtain $\psi''(\theta) = 0$ and so $\psi(\theta) = c_1 \theta + c_2$. From $\psi(0) = 0$ and $\psi(\pi) = u_0$ we find, in turn, $c_2 = 0$ and $c_1 = u_0/\pi$. Therefore $\psi(\theta) = \frac{u_0}{\pi} \theta$.

Now $u(1, \theta) = v(1, \theta) + \psi(\theta)$ so that $v(1, \theta) = u_0 - \frac{u_0}{\pi} \theta$. From

$$v(r, \theta) = \sum_{n=1}^{\infty} A_n r^n \sin n\theta \quad \text{and} \quad v(1, \theta) = \sum_{n=1}^{\infty} A_n \sin n\theta$$

we obtain

$$A_n = \frac{2}{\pi} \int_0^\pi \left(u_0 - \frac{u_0}{\pi} \theta \right) \sin n\theta \, d\theta = \frac{2u_0}{\pi n}.$$

Thus

$$u(r, \theta) = \frac{u_0}{\pi} \theta + \frac{2u_0}{\pi} \sum_{n=1}^{\infty} \frac{r^n}{n} \sin n\theta.$$

13. We solve

$$\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0, \quad 0 < \theta < \pi, \quad 0 < r < 2,$$

$$u(2, \theta) = \begin{cases} u_0, & 0 < \theta < \pi/2 \\ 0, & \pi/2 < \theta < \pi \end{cases}$$

$$\frac{\partial u}{\partial \theta} \Big|_{\theta=0} = 0, \quad \frac{\partial u}{\partial \theta} \Big|_{\theta=\pi} = 0, \quad 0 < r < 2.$$

Proceeding as in Example 1 in the text we obtain the separated differential equations

$$r^2 R'' + rR' - \lambda^2 R = 0$$

$$\Theta'' + \lambda^2 \Theta = 0$$

with solutions

$$\Theta(\theta) = c_1 \cos \lambda \theta + c_2 \sin \lambda \theta$$

$$R(r) = c_3 r^\lambda + c_4 r^{-\lambda}.$$

Applying the boundary conditions $\Theta'(0) = 0$ and $\Theta'(\pi) = 0$ we find that $c_2 = 0$ and $\lambda = n$ for $n = 0, 1, 2, \dots$. Since we want $R(r)$ to be bounded as $r \rightarrow 0$ we require $c_4 = 0$. Thus

$$u(r, \theta) = A_0 + \sum_{n=1}^{\infty} A_n r^n \cos n\theta.$$

From

$$u(2, \theta) = \begin{cases} u_0, & 0 < \theta < \pi/2 \\ 0, & \pi/2 < \theta < \pi \end{cases} = A_0 + \sum_{n=1}^{\infty} A_n 2^n \cos n\theta$$

we find

$$A_0 = \frac{1}{2} \frac{2}{\pi} \int_0^{\pi/2} u_0 \, d\theta = \frac{u_0}{2}$$

and

$$2^n A_n = \frac{2u_0}{\pi} \int_0^{\pi/2} \cos n\theta \, d\theta = \frac{2u_0}{\pi} \frac{\sin n\pi/2}{n}.$$

Therefore

$$u(r, \theta) = \frac{u_0}{2} + \frac{2u_0}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \left(\sin \frac{n\pi}{2} \right) \left(\frac{r}{2} \right)^n \cos n\theta.$$

14. Let u_1 be the solution of the boundary-value problem

$$\frac{\partial^2 u_1}{\partial r^2} + \frac{1}{r} \frac{\partial u_1}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u_1}{\partial \theta^2} = 0, \quad 0 < \theta < \pi, \quad a < r < b$$

$$u_1(a, \theta) = f(\theta), \quad 0 < \theta < \pi$$

$$u_1(b, \theta) = 0, \quad 0 < \theta < \pi,$$

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and let u^2 be the solution of the boundary-value problem

$$\frac{\partial^2 u_2}{\partial r^2} + \frac{1}{r} \frac{\partial u_2}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u_1}{\partial \theta^2} = 0, \quad 0 < \theta < \pi, \quad a < r < b$$

$$u_2(a, \theta) = 0, \quad 0 < \theta < \pi$$

$$u_2(b, \theta) = g(\theta), \quad 0 < \theta < \pi.$$

Each of these problems can be solved using the methods shown in Problem 9 of this section. Now if $u(r, \theta) = u_1(r, \theta) + u_2(r, \theta)$, then

$$u(a, \theta) = u_1(a, \theta) + u_2(a, \theta) = f(\theta)$$

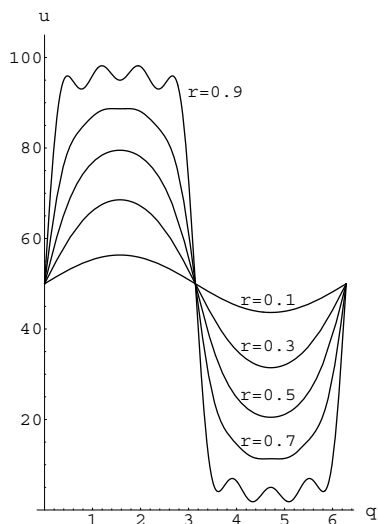
$$u(b, \theta) = u_1(b, \theta) + u_2(b, \theta) = g(\theta)$$

and $u(r, \theta)$ will be the steady-state temperature of the circular ring with boundary conditions $u(a, \theta) = f(\theta)$ and $u(b, \theta) = g(\theta)$.

15. (a) From Problem 1 in this section, with $u_0 = 100$,

$$u(r, \theta) = 50 + \frac{100}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n} r^n \sin n\theta.$$

(b)



- (c) Using a partial sum including the term with r^{99} we find

$u(0.9, 1.3) \approx 96.5268$	$u(0.9, 2\pi - 1.3) \approx 3.4731$
$u(0.7, 2) \approx 87.871$	$u(0.7, 2\pi - 2) \approx 12.129$
$u(0.5, 3.5) \approx 36.0744$	$u(0.5, 2\pi - 3.5) \approx 63.9256$
$u(0.3, 4) \approx 35.2674$	$u(0.3, 2\pi - 4) \approx 64.7326$
$u(0.1, 5.5) \approx 45.4934$	$u(0.1, 2\pi - 5.5) \approx 54.5066$

- (d) At the center of the plate $u(0, 0) = 50$. From the graphs in part (b) we observe that the solution curves are symmetric about the point $(\pi, 50)$. In part (c) we observe that the horizontal pairs add up to 100, and hence average 50. This is consistent with the observation about part (b), so it is appropriate to say the average temperature in the plate is 50° .

Exercises 14.2

1. Referring to the solution of Example 1 in the text we have

$$R(r) = c_1 J_0(\lambda_n r) \quad \text{and} \quad T(t) = c_3 \cos a\lambda_n t + c_4 \sin a\lambda_n t$$

where the eigenvalues λ_n are the positive roots of $J_0(\lambda c) = 0$. Now, the initial condition $u(r, 0) = R(r)T(0) = 0$ implies $c_3 = 0$. Thus

$$u(r, t) = \sum_{n=1}^{\infty} A_n \sin a\lambda_n t J_0(\lambda_n r) \frac{\partial u}{\partial t} = \sum_{n=1}^{\infty} a\lambda_n A_n \cos a\lambda_n t J_0(\lambda_n r).$$

From

$$\left. \frac{\partial u}{\partial t} \right|_{t=0} = 1 = \sum_{n=1}^{\infty} a\lambda_n A_n J_0 \lambda_n r$$

we find

$$\begin{aligned} a\lambda_n A_n &= \frac{2}{c^2 J_1^2(\lambda_n c)} \int_0^c r J_0(\lambda_n r) dr && \boxed{x = \lambda_n r, \quad dx = \lambda_n dr} \\ &= \frac{2}{c^2 J_1^2(\lambda_n c)} \int_0^{\lambda_n c} \frac{1}{\lambda_n^2} x J_0(x) dx \\ &= \frac{2}{c^2 J_1^2(\lambda_n c)} \int_0^{\lambda_n c} \frac{1}{\lambda_n^2} \frac{d}{dx} [x J_1(x)] dx && \boxed{\text{see (4) of Section 11.5 in text}} \\ &= \frac{2}{c^2 \lambda_n^2 J_1^2(\lambda_n c)} (x J_1(x)) \Big|_0^{\lambda_n c} = \frac{2}{c \lambda_n J_1(\lambda_n c)}. \end{aligned}$$

Then

$$A_n = \frac{2}{ac \lambda_n^2 J_1(\lambda_n c)}$$

and

$$u(r, t) = \frac{2}{ac} \sum_{n=1}^{\infty} \frac{J_0(\lambda_n r)}{\lambda_n^2 J_1(\lambda_n c)} \sin a\lambda_n t.$$

2. From Example 1 in the text we have $B_n = 0$ and

$$A_n = \frac{2}{J_1^2(\lambda_n)} \int_0^1 r(1-r^2) J_0(\lambda_n r) dr.$$

From Problem 10, Exercises 12.6 we obtained $A_n = \frac{4J_2(\lambda_n)}{\lambda_n^2 J_1^2(\lambda_n)}$. Thus

$$u(r, t) = 4 \sum_{n=1}^{\infty} \frac{J_2(\lambda_n)}{J_1^2(\lambda_n)} \cos a\lambda_n t J_0(\lambda_n r).$$

3. Referring to Example 2 in the text we have

$$R(r) = c_1 J_0(\lambda r) + c_2 Y_0(\lambda r)$$

$$Z(z) = c_3 \cosh \lambda z + c_4 \sinh \lambda z$$

where $c_2 = 0$ and $J_0(2\lambda) = 0$ defines the positive eigenvalues λ_n . From $Z(4) = 0$ we obtain

$$c_3 \cosh 4\lambda_n + c_4 \sinh 4\lambda_n = 0 \quad \text{or} \quad c_4 = -c_3 \frac{\cosh 4\lambda_n}{\sinh 4\lambda_n}.$$

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Then

$$\begin{aligned} Z(z) &= c_3 \left[\cosh \lambda_n z - \frac{\cosh 4\lambda_n}{\sinh 4\lambda_n} \sinh \lambda_n z \right] = c_3 \frac{\sinh 4\lambda_n \cosh \lambda_n z - \cosh 4\lambda_n \sinh \lambda_n z}{\sinh 4\lambda_n} \\ &= c_3 \frac{\sinh \lambda_n (4 - z)}{\sinh 4\lambda_n} \end{aligned}$$

and

$$u(r, z) = \sum_{n=1}^{\infty} A_n \frac{\sinh \lambda_n (4 - z)}{\sinh 4\lambda_n} J_0(\lambda_n r).$$

From

$$u(r, 0) = u_0 = \sum_{n=1}^{\infty} A_n J_0(\lambda_n r)$$

we obtain

$$A_n = \frac{2u_0}{4J_1^2(2\lambda_n)} \int_0^2 r J_0(\lambda_n r) dr = \frac{u_0}{\lambda_n J_1(2\lambda_n)}.$$

Thus the temperature in the cylinder is

$$u(r, z) = u_0 \sum_{n=1}^{\infty} \frac{\sinh \lambda_n (4 - z) J_0(\lambda_n r)}{\lambda_n \sinh 4\lambda_n J_1(2\lambda_n)}.$$

4. (a) The boundary condition $u_r(2, z) = 0$ implies $R'(2) = 0$ or $J'_0(2\lambda) = 0$. Thus $\lambda = 0$ is also an eigenvalue and the separated equations are in this case $rR'' + R' = 0$ and $z'' = 0$. The solutions of these equations are then

$$R(r) = c_1 + c_2 \ln r, \quad Z(z) = c_3 z + c_4.$$

Now $Z(0) = 0$ yields $c_4 = 0$ and the implicit condition that the temperature is bounded as $r \rightarrow 0$ demands that we define $c_2 = 0$. Using the superposition principle then gives

$$u(r, z) = A_1 z + \sum_{n=2}^{\infty} A_n \sinh \lambda_n z J_0(\lambda_n r). \quad (1)$$

At $z = 4$ we obtain

$$f(r) = 4A_1 + \sum_{n=2}^{\infty} A_n \sinh 4\lambda_n J_0(\lambda_n r).$$

Thus from (19) and (20) of Section 12.6 in the text we can write with $b = 2$,

$$A_1 = \frac{1}{8} \int_0^2 r f(r) dr \quad (2)$$

$$A_n = \frac{1}{2 \sinh 4\lambda_n J_0^2(2\lambda_n)} \int_0^2 r f(r) J_0(\lambda_n r) dr. \quad (3)$$

A solution of the problem consists of the series (1) with coefficients A_1 and A_n defined in (2) and (3), respectively.

- (b) When $f(r) = u_0$ we get $A_1 = u_0/4$ and

$$A_n = \frac{u_0 J_1(2\lambda_n)}{\lambda_n \sinh 4\lambda_n J_0^2(2\lambda_n)} = 0$$

since $J'_0(2\lambda) = 0$ is equivalent to $J_1(2\lambda) = 0$. A solution of the problem is then $u(r, z) = \frac{u_0}{4} z$.

5. Letting $u(r, t) = R(r)T(t)$ and separating variables we obtain

$$\frac{R'' + \frac{1}{r}R'}{R} = \frac{T'}{kT} = \mu \quad \text{and} \quad R'' + \frac{1}{r}R' - \mu R = 0, \quad T' - \mu kT = 0.$$

From the second equation we find $T(t) = e^{\mu kt}$. If $\mu > 0$, $T(t)$ increases without bound as $t \rightarrow \infty$. Thus we assume $\mu = -\lambda^2 \leq 0$. Now

$$R'' + \frac{1}{r}R' + \lambda^2 R = 0$$

is a parametric Bessel equation with solution

$$R(r) = c_1 J_0(\lambda r) + c_2 Y_0(\lambda r).$$

Since Y_0 is unbounded as $r \rightarrow 0$ we take $c_2 = 0$. Then $R(r) = c_1 J_0(\lambda r)$ and the boundary condition $u(c, t) = R(c)T(t) = 0$ implies $J_0(\lambda c) = 0$. This latter equation defines the positive eigenvalues λ_n . Thus

$$u(r, t) = \sum_{n=1}^{\infty} A_n J_0(\lambda_n r) e^{-\lambda_n^2 kt}.$$

From

$$u(r, 0) = f(r) = \sum_{n=1}^{\infty} A_n J_0(\lambda_n r)$$

we find

$$A_n = \frac{2}{c^2 J_1^2(\lambda_n c)} \int_0^c r J_0(\lambda_n r) f(r) dr, \quad n = 1, 2, 3, \dots$$

6. If the edge $r = c$ is insulated we have the boundary condition $u_r(c, t) = 0$. Referring to the solution of Problem 5 above we have

$$R'(c) = \lambda c_1 J_0'(\lambda c) = 0$$

which defines an eigenvalue $\lambda = 0$ and positive eigenvalues λ_n . Thus

$$u(r, t) = A_0 + \sum_{n=1}^{\infty} A_n J_0(\lambda_n r) e^{-\lambda_n^2 kt}.$$

From

$$u(r, 0) = f(r) = A_0 + \sum_{n=1}^{\infty} A_n J_0(\lambda_n r)$$

we find

$$A_0 = \frac{2}{c^2} \int_0^c r f(r) dr$$

$$A_n = \frac{2}{c^2 J_0^2(\lambda_n c)} \int_0^c r J_0(\lambda_n r) f(r) dr.$$

7. Referring to Problem 5 above we have $T(t) = e^{-\lambda^2 kt}$ and $R(r) = c_1 J_0(\lambda r)$. The boundary condition $hu(1, t) + u_r(1, t) = 0$ implies $hJ_0(\lambda) + \lambda J_0'(\lambda) = 0$ which defines positive eigenvalues λ_n . Now

$$u(r, t) = \sum_{n=1}^{\infty} A_n J_0(\lambda_n r) e^{-\lambda_n^2 kt}$$

where

$$A_n = \frac{2\lambda_n^2}{(\lambda_n^2 + h^2)J_0^2(\lambda_n)} \int_0^1 r J_0(\lambda_n r) f(r) dr.$$

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8. We solve

$$\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{\partial^2 u}{\partial z^2} = 0, \quad 0 < r < 1, \quad z > 0$$

$$\left. \frac{\partial u}{\partial r} \right|_{r=1} = -hu(1, z), \quad z > 0$$

$$u(r, 0) = u_0, \quad 0 < r < 1.$$

assuming $u = RZ$ we get

$$\frac{R'' + \frac{1}{r}R'}{R} = -\frac{Z''}{Z} = -\lambda^2$$

and so

$$rR'' + R' + \lambda^2 rR = 0 \quad \text{and} \quad Z'' - \lambda^2 Z = 0.$$

Therefore

$$R(r) = c_1 J_0(\lambda r) + c_2 Y_0(\lambda r) \quad \text{and} \quad Z(z) = c_3 e^{-\lambda z} + c_4 e^{\lambda z}.$$

We use the exponential form of the solution of $Z'' - \lambda^2 Z = 0$ since the domain of the variable z is a semi-infinite interval. As usual we define $c_2 = 0$ since the temperature is surely bounded as $r \rightarrow 0$. Hence $R(r) = c_1 J_0(\lambda r)$. Now the boundary-condition $u_r(1, z) + hu(1, z) = 0$ is equivalent to

$$\lambda J'_0(\lambda) + hJ_0(\lambda) = 0. \quad (4)$$

The eigenvalues λ_n are the positive roots of (4). Finally, we must now define $c_4 = 0$ since the temperature is also expected to be bounded as $z \rightarrow \infty$. A product solution of the partial differential equation that satisfies the first boundary condition is given by

$$u_n(r, z) = A_n e^{-\lambda_n z} J_0(\lambda_n r).$$

Therefore

$$u(r, z) = \sum_{n=1}^{\infty} A_n e^{-\lambda_n z} J_0(\lambda_n r)$$

is another formal solution. At $z = 0$ we have $u_0 = A_n J_0(\lambda_n r)$. In view of (4) we use (17) and (18) of Section 12.6 in the text with the identification $b = 1$:

$$A_n = \frac{2\lambda_n^2}{(\lambda_n^2 + h^2) J_0^2(\lambda_n)} \int_0^1 r J_0(\lambda_n r) u_0 dr = \frac{2\lambda_n^2 u_0}{(\lambda_n^2 + h^2) J_0^2(\lambda_n) \lambda_n^2} t J_1(t) \Big|_0^{\lambda_n} = \frac{2\lambda_n u_0 J_1(\lambda_n)}{(\lambda_n^2 + h^2) J_0^2(\lambda_n)}. \quad (5)$$

Since $J'_0 = -J_1$ [see (5) of Section 12.6] it follows from (4) that $\lambda_n J_1(\lambda_n) = hJ_0(\lambda_n)$. Thus (5) simplifies to

$$A_n = \frac{2u_0 h}{(\lambda_n^2 + h^2) J_0^2(\lambda_n)}.$$

A solution to the boundary-value problem is then

$$u(r, z) = 2u_0 h \sum_{n=1}^{\infty} \frac{e^{-\lambda_n z}}{(\lambda_n^2 + h^2) J_0^2(\lambda_n)} J_0(\lambda_n r).$$

9. Substituting $u(r, t) = v(r, t) + \psi(r)$ into the partial differential equation gives

$$\frac{\partial^2 v}{\partial r^2} + \frac{1}{r} \frac{\partial v}{\partial r} + \psi'' + \frac{1}{r} \psi' = \frac{\partial v}{\partial t}.$$

This equation will be homogeneous provided $\psi'' + \frac{1}{r}\psi' = 0$ or $\psi(r) = c_1 \ln r + c_2$. Since $\ln r$ is unbounded as $r \rightarrow 0$ we take $c_1 = 0$. Then $\psi(r) = c_2$ and using $u(2, t) = v(2, t) + \psi(2) = 100$ we set $c_2 = \psi(r) = 100$. Referring to Problem 5 above, the solution of the boundary-value problem

$$\begin{aligned}\frac{\partial^2 v}{\partial r^2} + \frac{1}{r} \frac{\partial v}{\partial r} &= \frac{\partial v}{\partial t}, \quad 0 < r < 2, \quad t > 0, \\ v(2, t) &= 0, \quad t > 0, \\ v(r, 0) &= u(r, 0) - \psi(r)\end{aligned}$$

is

$$v(r, t) = \sum_{n=1}^{\infty} A_n J_0(\lambda_n r) e^{-\lambda_n^2 t}$$

where

$$\begin{aligned}A_n &= \frac{2}{2^2 J_1^2(2\lambda_n)} \int_0^2 r J_0(\lambda_n r) [u(r, 0) - \psi(r)] dr \\ &= \frac{1}{2 J_1^2(2\lambda_n)} \left[\int_0^1 r J_0(\lambda_n r) [200 - 100] dr + \int_1^2 r J_0(\lambda_n r) [100 - 100] dr \right] \\ &= \frac{50}{J_1^2(2\lambda_n)} \int_0^1 r J_0(\lambda_n r) dr \quad \boxed{x = \lambda_n r, \quad dx = \lambda_n dr} \\ &= \frac{50}{J_1^2(2\lambda_n)} \int_0^{\lambda_n} \frac{1}{\lambda_n^2} x J_0(x) dx \\ &= \frac{50}{\lambda_n^2 J_1^2(2\lambda_n)} \int_0^{\lambda_n} \frac{d}{dx} [x J_1(x)] dx \quad \boxed{\text{see (4) of Section 12.6 in text}} \\ &= \frac{50}{\lambda_n^2 J_1^2(2\lambda_n)} (x J_1(x)) \Big|_0^{\lambda_n} = \frac{50 J_1(\lambda_n)}{\lambda_n J_1^2(2\lambda_n)}.\end{aligned}$$

Thus

$$u(r, t) = v(r, t) + \psi(r) = 100 + 50 \sum_{n=1}^{\infty} \frac{J_1(\lambda_n) J_0(\lambda_n r)}{\lambda_n J_1^2(2\lambda_n)} e^{-\lambda_n^2 t}.$$

10. Letting $u(r, t) = u(r, t) + \psi(r)$ we obtain $r\psi'' + \psi' = -\beta r$. The general solution of this nonhomogeneous equation is found with the aid of variation of parameters: $\psi = c_1 + c_2 \ln r - \beta \frac{r^2}{4}$. In order that this solution be bounded as $r \rightarrow 0$ we define $c_2 = 0$. Using $\psi(1) = 0$ then gives $c_1 = \frac{\beta}{4}$ and so $\psi(r) = \frac{\beta}{4}(1 - r^2)$. Using $v = RT$ we find that a solution of

$$\begin{aligned}\frac{\partial^2 v}{\partial r^2} + \frac{1}{r} \frac{\partial v}{\partial r} &= \frac{\partial v}{\partial t}, \quad 0 < r < 1, \quad t > 0 \\ v(1, t) &= 0, \quad t > 0 \\ v(r, 0) &= -\psi(r), \quad 0 < r < 1\end{aligned}$$

is

$$v(r, t) = \sum_{n=1}^{\infty} A_n e^{-\lambda_n^2 t} J_0(\lambda_n r)$$

where

$$A_n = -\frac{\beta}{4} \frac{2}{J_1^2(\lambda_n)} \int_0^1 r(1 - r^2) J_0(\lambda_n r) dr$$

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and the eigenvalues are defined by $J_0(\lambda) = 0$. From the result of Problem 10, Exercises 12.6 (see also Problem 2 of this exercise set) we get

$$A_n = -\frac{\beta J_2(\lambda_n)}{\lambda_n^2 J_1^2(\lambda_n)}.$$

Thus from $u = v + \psi(r)$ it follows that

$$u(r, t) = \frac{\beta}{4}(1 - r^2) - \beta \sum_{n=1}^{\infty} \frac{J_2(\lambda_n)}{\lambda_n^2 J_1^2(\lambda_n)} e^{-\lambda_n^2 t} J_0(\lambda_n r).$$

11. (a) Writing the partial differential equation in the form

$$g \left(x \frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x} \right) = \frac{\partial^2 u}{\partial t^2}$$

and separating variables we obtain

$$\frac{xX'' + X'}{X} = \frac{T''}{gT} = -\lambda^2.$$

Then

$$xX'' + X' + \lambda^2 X = 0 \quad \text{and} \quad T'' + g\lambda^2 T = 0.$$

Letting $x = \tau^2/4$ in the first equation we obtain $dx/d\tau = \tau/2$ or $d\tau/dx = 2\tau$. Then

$$\frac{dX}{dx} = \frac{dX}{d\tau} \frac{d\tau}{dx} = \frac{2}{\tau} \frac{dX}{d\tau}$$

and

$$\begin{aligned} \frac{d^2 X}{dx^2} &= \frac{d}{dx} \left(\frac{2}{\tau} \frac{dX}{d\tau} \right) = \frac{2}{\tau} \frac{d}{dx} \left(\frac{dX}{d\tau} \right) + \frac{dX}{d\tau} \frac{d}{dx} \left(\frac{2}{\tau} \right) \\ &= \frac{2}{\tau} \frac{d}{d\tau} \left(\frac{dX}{d\tau} \right) \frac{d\tau}{dx} + \frac{dX}{d\tau} \frac{d}{d\tau} \left(\frac{2}{\tau} \right) \frac{d\tau}{dx} = \frac{4}{\tau^2} \frac{d^2 X}{d\tau^2} - \frac{4}{\tau^3} \frac{dX}{d\tau}. \end{aligned}$$

Thus

$$xX'' + X' + \lambda^2 X = \frac{\tau^2}{4} \left(\frac{4}{\tau^2} \frac{d^2 X}{d\tau^2} - \frac{4}{\tau^3} \frac{dX}{d\tau} \right) + \frac{2}{\tau} \frac{dX}{d\tau} + \lambda^2 X = \frac{d^2 X}{d\tau^2} + \frac{1}{\tau} \frac{dX}{d\tau} + \lambda^2 X = 0.$$

This is a parametric Bessel equation with solution

$$X(\tau) = c_1 J_0(\lambda\tau) + c_2 Y_0(\lambda\tau).$$

- (b) To insure a finite solution at $x = 0$ (and thus $\tau = 0$) we set $c_2 = 0$. The condition $u(L, t) = X(L)T(t) = 0$ implies $X|_{x=L} = X|_{\tau=2\sqrt{L}} = c_1 J_0(2\lambda\sqrt{L}) = 0$, which defines positive eigenvalues λ_n . The solution of $T'' + g\lambda^2 T = 0$ is

$$T(t) = c_3 \cos \lambda_n \sqrt{g} t + c_4 \sin \lambda_n \sqrt{g} t.$$

The boundary condition $u_t(x, 0) = X(x)T'(0) = 0$ implies $c_4 = 0$. Thus

$$u(\tau, t) = \sum_{n=1}^{\infty} A_n \cos \lambda_n \sqrt{g} t J_0(\lambda_n \tau).$$

From

$$u(\tau, 0) = f(\tau^2/4) = \sum_{n=1}^{\infty} A_n J_0(\lambda_n \tau)$$

we find

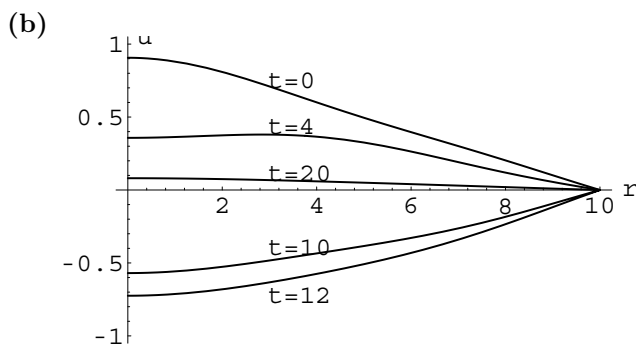
$$\begin{aligned}
 A_n &= \frac{2}{(2\sqrt{L})^2 J_1^2(2\lambda_n\sqrt{L})} \int_0^{2\sqrt{L}} \tau J_0(\lambda_n\tau) f(\tau^2/4) d\tau \quad \boxed{v = \tau/2, \, dv = d\tau/2} \\
 &= \frac{1}{2L J_1^2(2\lambda_n\sqrt{L})} \int_0^{\sqrt{L}} 2v J_0(2\lambda_n v) f(v^2) 2 dv \\
 &= \frac{2}{L J_1^2(2\lambda_n\sqrt{L})} \int_0^{\sqrt{L}} v J_0(2\lambda_n v) f(v^2) dv.
 \end{aligned}$$

12. (a) We need to solve $J_0(10\lambda) = 0$. From the Table 5.2 in Section 5.3 in the text we see that $\lambda_1 = 0.2405$, $\lambda_2 = 0.5520$, and $\lambda_3 = 0.8654$. Using a CAS to compute A_1 we find:

$$A_1 = \frac{2}{10^2 J_1^2(10\lambda_1)} \int_0^{10} r J_0(\lambda_1 r) (1 - r/10) dr = 0.7845.$$

In a similar fashion we compute $A_2 = 0.0687$ and $A_3 = 0.0531$. Then, from Example 1 in the text with $a = 1$ and $g(r) = 0$ [so that $B_n = 0$] we have

$$\begin{aligned}
 u(r, t) &\approx \sum_{n=1}^3 A_n (\cos \lambda_n t) J_0(\lambda_n r) \\
 &= 0.7845 \cos(0.2405t) J_0(0.2405r) + 0.0687 \cos(0.5520t) J_0(0.5520r) \\
 &\quad + 0.0531 \cos(0.8654t) J_0(0.8654r).
 \end{aligned}$$



13. Because of the nonhomogeneous boundary condition $u(c, t) = 200$ we use the substitution $u(r, t) = v(r, t) + \psi(r)$. This gives

$$k \left(\frac{\partial^2 v}{\partial r^2} + \frac{1}{r} \frac{\partial v}{\partial r} + \psi'' + \frac{1}{r} \psi' \right) = \frac{\partial v}{\partial t}.$$

This equation will be homogeneous provided $\psi'' + (1/r)\psi' = 0$ or $\psi(r) = c_1 \ln r + c_2$. Since $\ln r$ is unbounded as $r \rightarrow 0$ we take $c_1 = 0$. Then $\psi(r) = c_2$ and using $u(c, t) = v(c, t) + c_2 = 200$ we set $c_2 = 200$, giving $v(c, t) = 0$. Referring to Problem 5 in this section, the solution of the boundary-value problem

$$\begin{aligned}
 k \left(\frac{\partial^2 v}{\partial r^2} + \frac{1}{r} \frac{\partial v}{\partial r} \right) &= \frac{\partial v}{\partial t}, \quad 0 < r < c, \, t > 0 \\
 v(c, t) &= 0, \quad t > 0 \\
 v(r, 0) &= -200, \quad 0 < r < c
 \end{aligned}$$

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is

$$v(r, t) = \sum_{n=1}^{\infty} A_n J_0(\lambda_n r) e^{-\lambda_n^2 k t},$$

where the eigenvalues λ_n are defined by $J_0(\lambda c) = 0$, and

$$A_n = \frac{2}{c^2 J_1^2(\lambda_n c)} \int_0^c r J_0(\lambda_n r) (-200) dr = -\frac{400}{c^2 J_1^2(\lambda_n c)} \int_0^c r J_0(\lambda_n r) dr.$$

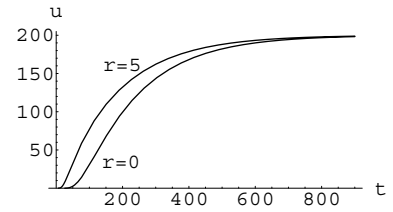
Taking $c = 10$ and $k = 0.1$ we have

$$u(r, t) = v(r, t) + 200 = 200 + \sum_{n=1}^{\infty} A_n J_0(\lambda_n r) e^{-0.1 \lambda_n^2 t},$$

where the eigenvalues λ_n are defined by $J_0(10\lambda) = 0$ and

$$A_n = -\frac{4}{J_1^2(10\lambda_n)} \int_0^{10} r J_0(\lambda_n r) dr.$$

Using a CAS we find that the first five eigenvalues are $\lambda_1 = 0.240483$, $\lambda_2 = 0.552008$, $\lambda_3 = 0.865373$, $\lambda_4 = 1.17915$, and $\lambda_5 = 1.49309$. From the graph we see that $u(5, t) \approx 100$ when $t = 132$, $u(0, t) \approx 100$ when $t = 198$, $u(5, t) \approx 200$ when $t = 808$, and $u(0, t) \approx 200$ when $t = 885$.



14. (a) The boundary-value problem is

$$a^2 \left(\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} \right) = \frac{\partial^2 u}{\partial t^2}, \quad 0 < r < 1, \quad t > 0$$

$$u(1, t) = 0, \quad t > 0$$

$$u(r, 0) = 0, \quad \frac{\partial u}{\partial t} \Big|_{t=0} = \begin{cases} -v_0, & 0 \leq r < b \\ 0, & b \leq r < 1 \end{cases}, \quad 0 < r < 1,$$

and the solution is

$$u(r, t) = \sum_{n=1}^{\infty} (A_n \cos a\lambda_n t + B_n \sin a\lambda_n t) J_0(\lambda_n r),$$

where the eigenvalues λ_n are defined by $J_0(\lambda) = 0$ and $A_n = 0$ since $f(r) = 0$. The coefficients B_n are given by

$$\begin{aligned} B_n &= \frac{2}{a\lambda_n J_1^2(\lambda_n)} \int_0^c r J_0(\lambda_n r) g(r) dr = -\frac{2v_0}{a\lambda_n J_1^2(\lambda_n)} \int_0^b r J_0(\lambda_n r) dr \\ &\quad \boxed{\text{let } x = \lambda_n r} \\ &= -\frac{2v_0}{a\lambda_n J_1^2(\lambda_n)} \int_0^{\lambda_n b} \frac{x}{\lambda_n} J_0(x) \frac{1}{\lambda_n} dx = -\frac{2v_0}{a\lambda_n^3 J_1^2(\lambda_n)} \int_0^{\lambda_n b} x J_0(x) dx \\ &= -\frac{2v_0}{a\lambda_n^3 J_1^2(\lambda_n)} (x J_1(x)) \Big|_0^{\lambda_n b} = -\frac{2v_0}{a\lambda_n^3 J_1^2(\lambda_n)} (\lambda_n b J_1(\lambda_n b)) \\ &= -\frac{2v_0 b}{a\lambda_n^2} \frac{J_1(\lambda_n b)}{J_1^2(\lambda_n)}. \end{aligned}$$

Thus,

$$u(r, t) = \frac{-2v_0 b}{a} \sum_{n=1}^{\infty} \frac{1}{\lambda_n^2} \frac{J_1(\lambda_n b)}{J_1^2(\lambda_n)} \sin(a\lambda_n t) J_0(\lambda_n r).$$

- (b) The standing wave $u_n(r, t)$ is given by $u_n(r, t) = B_n \sin(a\lambda_n t) J_0(\lambda_n r)$, which has frequency $f_n = a\lambda_n/2\pi$, where λ_n is the n th positive zero of $J_0(x)$. The fundamental frequency is $f_1 = a\lambda_1/2\pi$. The next two frequencies are

$$f_2 = \frac{a\lambda_2}{2\pi} = \frac{\lambda_2}{\lambda_1} \left(\frac{a\lambda_1}{2\pi} \right) = \frac{5.520}{2.405} f_1 = 2.295 f_1$$

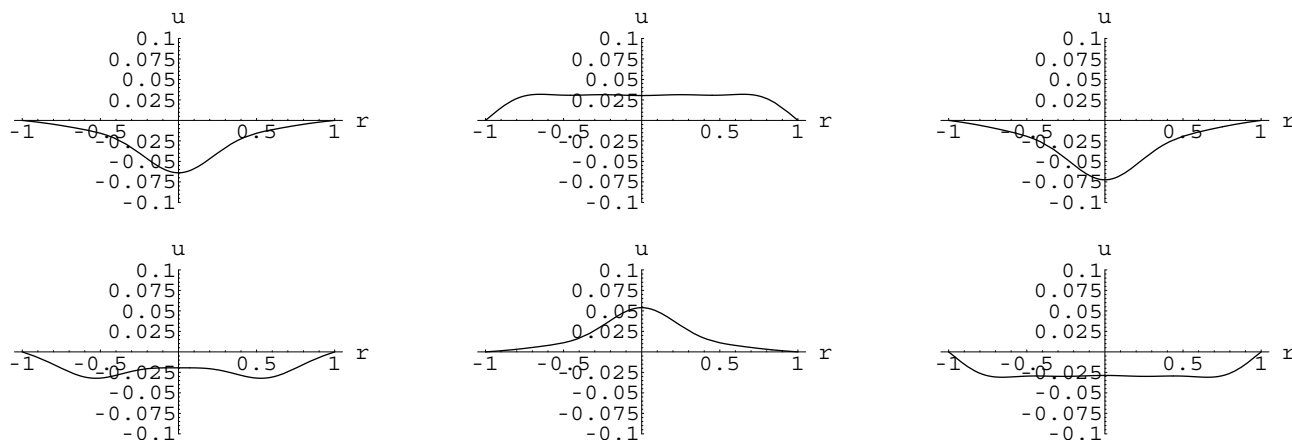
and

$$f_3 = \frac{a\lambda_3}{2\pi} = \frac{\lambda_3}{\lambda_1} \left(\frac{a\lambda_1}{2\pi} \right) = \frac{8.654}{2.405} f_1 = 3.598 f_1.$$

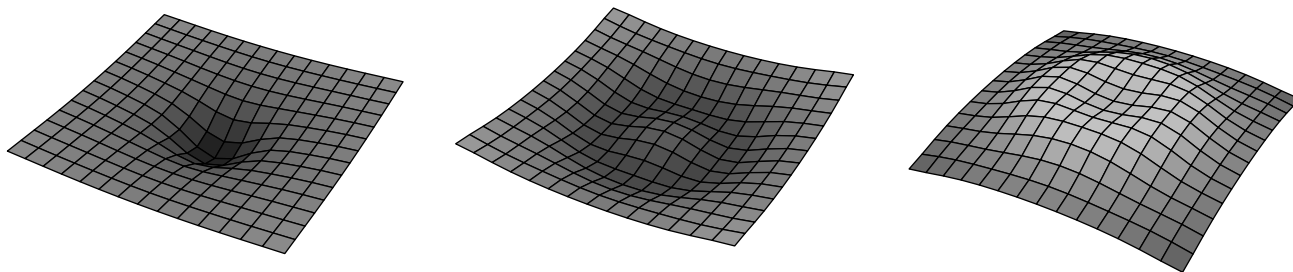
- (c) With $a = 1$, $b = \frac{1}{4}$, and $v_0 = 1$, the solution becomes

$$u(r, t) = -\frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{\lambda_n^2} \frac{J_1(\lambda_n/4)}{J_1(\lambda_n)} \sin(\lambda_n t) J_0(\lambda_n r).$$

The graphs of $S_5(r, t)$ for $t = 1, 2, 3, 4, 5, 6$ are shown below.



- (d) Three frames from the movie are shown.



15. (a) First we see that

$$\frac{R''\Theta + \frac{1}{r}R'\Theta + \frac{1}{r^2}R\Theta''}{R\Theta} = \frac{T''}{a^2T} = -\lambda^2.$$

This gives $T'' + a^2\lambda^2T = 0$. Then from

$$\frac{R'' + \frac{1}{r}R' + \lambda^2R}{-R/r^2} = \frac{\Theta''}{\Theta} = -\nu^2$$

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we get $\Theta'' + \nu^2\Theta = 0$ and $r^2R'' + rR' + (\lambda^2r^2 - \nu^2)R = 0$.

(b) The general solutions of the differential equations in part (a) are

$$T = c_1 \cos a\lambda t + c_2 \sin a\lambda t$$

$$\Theta = c_3 \cos \nu\theta + c_4 \sin \nu\theta$$

$$R = c_5 J_\nu(\lambda r) + c_6 Y_\nu(\lambda r).$$

(c) Implicitly we expect $u(r, \theta, t) = u(r, \theta + 2\pi, t)$ and so Θ must be 2π -periodic. Therefore $\nu = n$, $n = 0, 1, 2, \dots$. The corresponding eigenfunctions are $1, \cos \theta, \cos 2\theta, \dots, \sin \theta, \sin 2\theta, \dots$. Arguing that $u(r, \theta, t)$ is bounded as $r \rightarrow 0$ we then define $c_6 = 0$ and so $R = c_3 J_n(\lambda r)$. But $R(c) = 0$ gives $J_n(\lambda c) = 0$; this equation defines the eigenvalues λ_n . For each n , $\lambda_{ni} = x_{ni}/c$, $i = 1, 2, 3, \dots$.

$$\begin{aligned} \text{(d)} \quad u(r, \theta, t) &= \sum_{i=1}^n (A_{0i} \cos a\lambda_{0i}t + B_{0i} \sin a\lambda_{0i}t) J_0(\lambda_{0i}r) \\ &\quad + \sum_{n=1}^{\infty} \sum_{i=1}^{\infty} \left[(A_{ni} \cos a\lambda_{ni}t + B_{ni} \sin a\lambda_{ni}t) \cos n\theta \right. \\ &\quad \left. + (C_{ni} \cos a\lambda_{ni}t + D_{ni} \sin a\lambda_{ni}t) \sin n\theta \right] J_n(\lambda_{ni}r) \end{aligned}$$

Exercises 14.3

1. To compute

$$A_n = \frac{2n+1}{2c^n} \int_0^\pi f(\theta) P_n(\cos \theta) \sin \theta d\theta$$

we substitute $x = \cos \theta$ and $dx = -\sin \theta d\theta$. Then

$$A_n = \frac{2n+1}{2c^n} \int_1^{-1} F(x) P_n(x) (-dx) = \frac{2n+1}{2c^n} \int_{-1}^1 F(x) P_n(x) dx$$

where

$$F(x) = \begin{cases} 0, & -1 < x < 0 \\ 50, & 0 < x < 1 \end{cases} = 50 \begin{cases} 0, & -1 < x < 0 \\ 1, & 0 < x < 1 \end{cases}.$$

The coefficients A_n are computed in Example 3 of Section 12.6. Thus

$$\begin{aligned} u(r, \theta) &= \sum_{n=0}^{\infty} A_n r^n P_n(\cos \theta) \\ &= 50 \left[\frac{1}{2} P_0(\cos \theta) + \frac{3}{4} \left(\frac{r}{c} \right) P_1(\cos \theta) - \frac{7}{16} \left(\frac{r}{c} \right)^3 P_3(\cos \theta) + \frac{11}{32} \left(\frac{r}{c} \right)^5 P_5(\cos \theta) + \dots \right]. \end{aligned}$$

2. In the solution of the Cauchy-Euler equation,

$$R(r) = c_1 r^n + c_2 r^{-(n+1)},$$

we define $c_1 = 0$ since we expect the potential u to be bounded as $r \rightarrow \infty$. Hence

$$u_n(r, \theta) = A_n r^{-(n+1)} P_n(\cos \theta)$$

$$u(r, \theta) = \sum_{n=0}^{\infty} A_n r^{-(n+1)} P_n(\cos \theta).$$

When $r = c$ we have

$$f(\theta) = \sum_{n=0}^{\infty} A_n c^{-(n+1)} P_n(\cos \theta)$$

so that

$$A_n = c^{n+1} \frac{(2n+1)}{2} \int_0^\pi f(\theta) P_n(\cos \theta) \sin \theta d\theta.$$

The solution of the problem is then

$$u(r, \theta) = \sum_{n=0}^{\infty} \left(\frac{2n+1}{2} \int_0^\pi f(\theta) P_n(\cos \theta) \sin \theta d\theta \right) \left(\frac{c}{r} \right)^{n+1} P_n(\cos \theta).$$

3. The coefficients are given by

$$\begin{aligned} A_n &= \frac{2n+1}{2c^n} \int_0^\pi \cos \theta P_n(\cos \theta) \sin \theta d\theta = \frac{2n+1}{2c^n} \int_0^\pi P_1(\cos \theta) P_n(\cos \theta) \sin \theta d\theta \\ &\quad \boxed{x = \cos \theta, \quad dx = -\sin \theta d\theta} \\ &= \frac{2n+1}{2c^n} \int_{-1}^1 P_1(x) P_n(x) dx. \end{aligned}$$

Since $P_n(x)$ and $P_m(x)$ are orthogonal for $m \neq n$, $A_n = 0$ for $n \neq 1$ and

$$A_1 = \frac{2(1)+1}{2c^1} \int_{-1}^1 P_1(x) P_1(x) dx = \frac{3}{2c} \int_{-1}^1 x^2 dx = \frac{1}{c}.$$

Thus

$$u(r, \theta) = \frac{r}{c} P_1(\cos \theta) = \frac{r}{c} \cos \theta.$$

4. The coefficients are given by

$$A_n = \frac{2n+1}{2c^n} \int_0^\pi (1 - \cos 2\theta) P_n(\cos \theta) \sin \theta d\theta.$$

These were computed in Problem 18 of Section 12.6. Thus

$$u(r, \theta) = \frac{4}{3} P_0(\cos \theta) - \frac{4}{3} \left(\frac{r}{c} \right)^2 P_2(\cos \theta).$$

5. Referring to Example 1 in the text we have

$$\Theta = P_n(\cos \theta) \quad \text{and} \quad R = c_1 r^n + c_2 r^{-(n+1)}.$$

Since $u(b, \theta) = R(b)\Theta(\theta) = 0$,

$$c_1 b^n + c_2 b^{-(n+1)} = 0 \quad \text{or} \quad c_1 = -c_2 b^{-2n-1},$$

and

$$R(r) = -c_2 b^{-2n-1} r^n + c_2 r^{-(n+1)} = c_2 \left(\frac{b^{2n+1} - r^{2n+1}}{b^{2n+1} r^{n+1}} \right).$$

Then

$$u(r, \theta) = \sum_{n=0}^{\infty} A_n \frac{b^{2n+1} - r^{2n+1}}{b^{2n+1} r^{n+1}} P_n(\cos \theta)$$

where

$$\frac{b^{2n+1} - a^{2n+1}}{b^{2n+1} a^{n+1}} A_n = \frac{2n+1}{2} \int_0^\pi f(\theta) P_n(\cos \theta) \sin \theta d\theta.$$

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6. Referring to Example 1 in the text we have

$$R(r) = c_1 r^n \quad \text{and} \quad \Theta(\theta) = P_n(\cos \theta).$$

Now $\Theta(\pi/2) = 0$ implies that n is odd, so

$$u(r, \theta) = \sum_{n=0}^{\infty} A_{2n+1} r^{2n+1} P_{2n+1}(\cos \theta).$$

From

$$u(c, \theta) = f(\theta) = \sum_{n=0}^{\infty} A_{2n+1} c^{2n+1} P_{2n+1}(\cos \theta)$$

we see that

$$A_{2n+1} c^{2n+1} = (4n+3) \int_0^{\pi/2} f(\theta) \sin \theta P_{2n+1}(\cos \theta) d\theta.$$

Thus

$$u(r, \theta) = \sum_{n=0}^{\infty} A_{2n+1} r^{2n+1} P_{2n+1}(\cos \theta)$$

where

$$A_{2n+1} = \frac{4n+3}{c^{2n+1}} \int_0^{\pi/2} f(\theta) \sin \theta P_{2n+1}(\cos \theta) d\theta.$$

7. Referring to Example 1 in the text we have

$$\begin{aligned} r^2 R'' + 2r R' - \lambda^2 R &= 0 \\ \sin \theta \Theta'' + \cos \theta \Theta' + \lambda \sin \theta \Theta &= 0. \end{aligned}$$

Substituting $x = \cos \theta$, $0 \leq \theta \leq \pi/2$, the latter equation becomes

$$(1-x^2) \frac{d^2 \Theta}{dx^2} - 2x \frac{d\Theta}{dx} + \lambda^2 \Theta = 0, \quad 0 \leq x \leq 1.$$

Taking the solutions of this equation to be the Legendre polynomials $P_n(x)$ corresponding to $\lambda^2 = n(n+1)$ for $n = 1, 2, 3, \dots$, we have $\Theta = P_n(\cos \theta)$. Since

$$\left. \frac{\partial u}{\partial \theta} \right|_{\theta=\pi/2} = \Theta'(\pi/2) R(r) = 0$$

we have

$$\Theta'(\pi/2) = -(\sin \pi/2) P_n'(\cos \pi/2) = -P_n'(0) = 0.$$

As noted in the hint, $P_n'(0) = 0$ only if n is even. Thus $\Theta = P_n(\cos \theta)$, $n = 0, 2, 4, \dots$. As in Example 1, $R(r) = c_1 r^n$. Hence

$$u(r, \theta) = \sum_{n=0}^{\infty} A_{2n} r^{2n} P_{2n}(\cos \theta).$$

At $r = c$,

$$f(\theta) = \sum_{n=0}^{\infty} A_{2n} c^{2n} P_{2n}(\cos \theta).$$

Using Problem 19 in Section 12.6, we obtain

$$c^{2n} A_{2n} = (4n+1) \int_{\pi/2}^0 f(\theta) P_{2n}(\cos \theta) (-\sin \theta) d\theta$$

and

$$A_{2n} = \frac{4n+1}{c^{2n}} \int_0^{\pi/2} f(\theta) \sin \theta P_{2n}(\cos \theta) d\theta.$$

8. Referring to Example 1 in the text we have

$$R(r) = c_1 r^n + c_2 r^{-(n-1)} \quad \text{and} \quad \Theta(\theta) = P_n(\cos \theta).$$

Since we expect $u(r, \theta)$ to be bounded as $r \rightarrow \infty$, we define $c_1 = 0$. Also $\Theta(\pi/2) = 0$ implies that n is odd, so

$$u(r, \theta) = \sum_{n=0}^{\infty} A_{2n+1} r^{-2(n+1)} P_{2n+1}(\cos \theta).$$

From

$$u(c, \theta) = f(\theta) = \sum_{n=0}^{\infty} A_{2n+1} c^{-2(n+1)} P_{2n+1}(\cos \theta)$$

we see that

$$A_{2n+1} c^{-2(n+1)} = (4n+3) \int_0^{\pi/2} f(\theta) \sin \theta P_{2n+1}(\cos \theta) d\theta.$$

Thus

$$u(r, \theta) = \sum_{n=0}^{\infty} A_{2n+1} r^{-2(n+1)} P_{2n+1}(\cos \theta)$$

where

$$A_{2n+1} = (4n+3) c^{2(n+1)} \int_0^{\pi/2} f(\theta) \sin \theta P_{2n+1}(\cos \theta) d\theta.$$

9. Checking the hint, we find

$$\frac{1}{r} \frac{\partial^2}{\partial r^2}(ru) = \frac{1}{r} \frac{\partial}{\partial r} \left[r \frac{\partial u}{\partial r} + u \right] = \frac{1}{r} \left[r \frac{\partial^2 u}{\partial r^2} + \frac{\partial u}{\partial r} + \frac{\partial u}{\partial r} \right] = \frac{\partial^2 u}{\partial r^2} + \frac{2}{r} \frac{\partial u}{\partial r}.$$

The partial differential equation then becomes

$$\frac{\partial^2}{\partial r^2}(ru) = r \frac{\partial u}{\partial t}.$$

Now, letting $ru(r, t) = v(r, t) + \psi(r)$, since the boundary condition is nonhomogeneous, we obtain

$$\frac{\partial^2}{\partial r^2} [v(r, t) + \psi(r)] = r \frac{\partial}{\partial t} \left[\frac{1}{r} v(r, t) + \psi(r) \right]$$

or

$$\frac{\partial^2 v}{\partial r^2} + \psi''(r) = \frac{\partial v}{\partial t}.$$

This differential equation will be homogeneous if $\psi''(r) = 0$ or $\psi(r) = c_1 r + c_2$. Now

$$u(r, t) = \frac{1}{r} v(r, t) + \frac{1}{r} \psi(r) \quad \text{and} \quad \frac{1}{r} \psi(r) = c_1 + \frac{c_2}{r}.$$

Since we want $u(r, t)$ to be bounded as r approaches 0, we require $c_2 = 0$. Then $\psi(r) = c_1 r$. When $r = 1$

$$u(1, t) = v(1, t) + \psi(1) = v(1, t) + c_1 = 100,$$

and we will have the homogeneous boundary condition $v(1, t) = 0$ when $c_1 = 100$. Consequently, $\psi(r) = 100r$.

The initial condition

$$u(r, 0) = \frac{1}{r} v(r, 0) + \frac{1}{r} \psi(r) = \frac{1}{r} v(r, 0) + 100 = 0$$

Exercises 14.3

implies $v(r, 0) = -100r$. We are thus led to solve the new boundary-value problem

$$\frac{\partial^2 v}{\partial r^2} = \frac{\partial v}{\partial t}, \quad 0 < r < 1, \quad t > 0,$$

$$v(1, t) = 0, \quad \lim_{r \rightarrow 0} \frac{1}{r} v(r, t) < \infty,$$

$$v(r, 0) = -100r.$$

Letting $v(r, t) = R(r)T(t)$ and separating variables leads to

$$R'' + \lambda^2 R = 0 \quad \text{and} \quad T' + \lambda^2 T = 0$$

with solutions

$$R(r) = c_3 \cos \lambda r + c_4 \sin \lambda r \quad \text{and} \quad T(t) = c_5 e^{-\lambda^2 t}.$$

The boundary conditions are equivalent to $R(1) = 0$ and $\lim_{r \rightarrow 0} \frac{1}{r} R(r) < \infty$. Since

$$\lim_{r \rightarrow 0} \frac{1}{r} R(r) = \lim_{r \rightarrow 0} \frac{c_3 \cos \lambda r}{r} + \lim_{r \rightarrow 0} \frac{c_4 \sin \lambda r}{r} = \lim_{r \rightarrow 0} \frac{c_3 \cos \lambda r}{r} + c_4 \lambda < \infty$$

we must have $c_3 = 0$. Then $R(r) = c_4 \sin \lambda r$, and $R(1) = 0$ implies $\lambda = n\pi$ for $n = 1, 2, 3, \dots$. Thus

$$v_n(r, t) = A_n e^{-n^2 \pi^2 t} \sin n\pi r$$

for $n = 1, 2, 3, \dots$. Using the condition $\lim_{r \rightarrow 0} \frac{1}{r} R(r) < \infty$ it is easily shown that there are no eigenvalues for $\lambda = 0$, nor does setting the common constant to $+\lambda^2$ when separating variables lead to any solutions. Now, by the superposition principle,

$$v(r, t) = \sum_{n=1}^{\infty} A_n e^{-n^2 \pi^2 t} \sin n\pi r.$$

The initial condition $v(r, 0) = -100r$ implies

$$-100r = \sum_{n=1}^{\infty} A_n \sin n\pi r.$$

This is a Fourier sine series and so

$$\begin{aligned} A_n &= 2 \int_0^1 (-100r \sin n\pi r) dr = -200 \left[-\frac{r}{n\pi} \cos n\pi r \Big|_0^1 + \int_0^1 \frac{1}{n\pi} \cos n\pi r dr \right] \\ &= -200 \left[-\frac{\cos n\pi}{n\pi} + \frac{1}{n^2 \pi^2} \sin n\pi r \Big|_0^1 \right] = -200 \left[-\frac{(-1)^n}{n\pi} \right] = \frac{(-1)^n 200}{n\pi}. \end{aligned}$$

A solution of the problem is thus

$$\begin{aligned} u(r, t) &= \frac{1}{r} v(r, t) + \frac{1}{r} \psi(r) = \frac{1}{r} \sum_{n=1}^{\infty} (-1)^n \frac{20}{n\pi} e^{-n^2 \pi^2 t} \sin n\pi r + \frac{1}{r} (100r) \\ &= \frac{200}{\pi r} \sum_{n=1}^{\infty} \frac{(-1)^n}{n} e^{-n^2 \pi^2 t} \sin n\pi r + 100. \end{aligned}$$

10. Referring to Problem 9 we have

$$\frac{\partial^2 v}{\partial r^2} + \psi''(r) = \frac{\partial v}{\partial t}$$

where $\psi(r) = c_1 r$. Since

$$u(r, t) = \frac{1}{r}v(r, t) + \frac{1}{r}\psi(r) = \frac{1}{r}v(r, t) + c_1$$

we have

$$\frac{\partial u}{\partial r} = \frac{1}{r}v_r(r, t) - \frac{1}{r^2}v(r, t).$$

When $r = 1$,

$$\left. \frac{\partial u}{\partial r} \right|_{r=1} = v_r(1, t) - v(1, t)$$

and

$$\left. \frac{\partial u}{\partial r} \right|_{r=1} + hu(1, t) = v_r(1, t) - v(1, t) + h[v(1, t) + \psi(1)] = v_r(1, t) + (h-1)v(1, t) + hc_1.$$

Thus the boundary condition

$$\left. \frac{\partial u}{\partial r} \right|_{r=1} + hu(1, t) = hu_1$$

will be homogeneous when $hc_1 = hu_1$ or $c_1 = u_1$. Consequently $\psi(r) = u_1 r$. The initial condition

$$u(r, 0) = \frac{1}{r}v(r, 0) + \frac{1}{r}\psi(r) = \frac{1}{r}v(r, 0) + u_1 = u_0$$

implies $v(r, 0) = (u_0 - u_1)r$. We are thus led to solve the new boundary-value problem

$$\begin{aligned} \frac{\partial^2 v}{\partial r^2} &= \frac{\partial v}{\partial t}, \quad 0 < r < 1, \quad t > 0, \\ v_r(1, t) + (h-1)v(1, t) &= 0, \quad t > 0, \\ \lim_{r \rightarrow 0} \frac{1}{r}v(r, t) &< \infty, \\ v(r, 0) &= (u_0 - u_1)r. \end{aligned}$$

Separating variables as in Problem 9 leads to

$$R(r) = c_3 \cos \lambda r + c_4 \sin \lambda r \quad \text{and} \quad T(t) = c_5 e^{-\lambda^2 t}.$$

The boundary conditions are equivalent to

$$R'(1) + (h-1)R(1) = 0 \quad \text{and} \quad \lim_{r \rightarrow 0} \frac{1}{r}R(r) < \infty.$$

As in Problem 6 we use the second condition to determine that $c_3 = 0$ and $R(r) = c_4 \sin \lambda r$. Then

$$R'(1) + (h-1)R(1) = c_4 \lambda \cos \lambda + c_4 (h-1) \sin \lambda = 0$$

and the eigenvalues λ_n are the consecutive nonnegative roots of $\tan \lambda = \lambda/(1-h)$. Now

$$v(r, t) = \sum_{n=1}^{\infty} A_n e^{-\lambda_n^2 t} \sin \lambda_n r.$$

From

$$v(r, 0) = (u_0 - u_1)r = \sum_{n=1}^{\infty} A_n \sin \lambda_n r$$

we obtain

$$A_n = \frac{\int_0^1 (u_0 - u_1)r \sin \lambda_n r \, dr}{\int_0^1 \sin^2 \lambda_n r \, dr}.$$

Exercises 14.3

We compute the integrals

$$\int_0^1 r \sin \lambda_n r \, dr = \left(\frac{1}{\lambda_n^2} \sin \lambda_n r - \frac{1}{\lambda_n} \cos \lambda_n r \right) \Big|_0^1 = \frac{1}{\lambda_n^2} \sin \lambda_n - \frac{1}{\lambda_n} \cos \lambda_n$$

and

$$\int_0^1 \sin^2 \lambda_n r \, dr = \left(\frac{1}{2} r - \frac{1}{4\lambda_n} \sin 2\lambda_n r \right) \Big|_0^1 = \frac{1}{2} - \frac{1}{4\lambda_n} \sin 2\lambda_n.$$

Using $\lambda_n \cos \lambda_n = -(h-1) \sin \lambda_n$ we then have

$$\begin{aligned} A_n &= (u_0 - u_1) \frac{\frac{1}{\lambda_n^2} \sin \lambda_n - \frac{1}{\lambda_n} \cos \lambda_n}{\frac{1}{2} - \frac{1}{4\lambda_n} \sin 2\lambda_n} = (u_0 - u_1) \frac{4 \sin \lambda_n - 4\lambda_n \cos \lambda_n}{2\lambda_n^2 - \lambda_n \sin 2\lambda_n} \\ &= 2(u_0 - u_1) \frac{\sin \lambda_n + (h-1) \sin \lambda_n}{\lambda_n^2 + (h-1) \sin \lambda_n \sin \lambda_n} = 2(u_0 - u_1) h \frac{\sin \lambda_n}{\lambda_n^2 + (h-1) \sin^2 \lambda_n}. \end{aligned}$$

Therefore

$$u(r, t) = \frac{1}{r} v(r, t) + \frac{1}{r} \psi(r) = u_1 + 2(u_0 - u_1) h \sum_{n=1}^{\infty} \frac{\sin \lambda_n \sin \lambda_n r}{r[\lambda_n^2 + (h-1) \sin^2 \lambda_n]} e^{-\lambda_n^2 t}.$$

11. We write the differential equation in the form

$$a^2 \frac{1}{r} \frac{\partial^2}{\partial r^2} (ru) = \frac{\partial^2 u}{\partial t^2} \quad \text{or} \quad a^2 \frac{\partial^2}{\partial r^2} (ru) = r \frac{\partial^2 u}{\partial t^2},$$

and then let $v(r, t) = ru(r, t)$. The new boundary-value problem is

$$\begin{aligned} a^2 \frac{\partial^2 v}{\partial r^2} &= \frac{\partial^2 v}{\partial t^2}, \quad 0 < r < c, \quad t > 0 \\ v(c, t) &= 0, \quad t > 0 \\ v(r, 0) &= rf(r), \quad \frac{\partial v}{\partial t} \Big|_{t=0} = rg(r). \end{aligned}$$

Letting $v(r, t) = R(r)T(t)$ and separating variables we obtain

$$\begin{aligned} R'' + \lambda^2 R &= 0 \\ T'' + a^2 \lambda^2 T &= 0 \end{aligned}$$

and

$$\begin{aligned} R(r) &= c_1 \cos \lambda r + c_2 \sin \lambda r \\ T(t) &= c_3 \cos a\lambda t + c_4 \sin a\lambda t. \end{aligned}$$

Since $u(r, t) = v(r, t)/r$, in order to insure boundedness at $r = 0$ we define $c_1 = 0$. Then $R(r) = c_2 \sin \lambda r$ and the condition $R(c) = 0$ implies $\lambda = n\pi/c$. Thus

$$v(r, t) = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi a}{c} t + B_n \sin \frac{n\pi a}{c} t \right) \sin \frac{n\pi}{c} r.$$

From

$$v(r, 0) = rf(r) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{c} r$$

we see that

$$A_n = \frac{2}{c} \int_0^c rf(r) \sin \frac{n\pi}{c} r \, dr.$$

From

$$\left. \frac{\partial v}{\partial t} \right|_{t=0} = rg(r) = \sum_{n=1}^{\infty} \left(B_n \frac{n\pi a}{c} \right) \sin \frac{n\pi}{c} r$$

we see that

$$B_n = \frac{c}{n\pi a} \cdot \frac{2}{c} \int_0^c rg(r) \sin \frac{n\pi}{c} r dr = \frac{2}{n\pi a} \int_0^c rg(r) \sin \frac{n\pi}{c} r dr.$$

12. Proceeding as in Example 1 we obtain

$$\Theta(\theta) = P_n(\cos \theta) \quad \text{and} \quad R(r) = c_1 r^n + c_2 r^{-(n+1)}$$

so that

$$u(r, \theta) = \sum_{n=0}^{\infty} (A_n r^n + B_n r^{-(n+1)}) P_n(\cos \theta).$$

To satisfy $\lim_{r \rightarrow \infty} u(r, \theta) = -Er \cos \theta$ we must have $A_n = 0$ for $n = 2, 3, 4, \dots$. Then

$$\lim_{r \rightarrow \infty} u(r, \theta) = -Er \cos \theta = A_0 \cdot 1 + A_1 r \cos \theta,$$

so $A_0 = 0$ and $A_1 = -E$. Thus

$$u(r, \theta) = -Er \cos \theta + \sum_{n=0}^{\infty} B_n r^{-(n+1)} P_n(\cos \theta).$$

Now

$$u(c, \theta) = 0 = -Ec \cos \theta + \sum_{n=0}^{\infty} B_n c^{-(n+1)} P_n(\cos \theta)$$

so

$$\sum_{n=0}^{\infty} B_n c^{-(n+1)} P_n(\cos \theta) = Ec \cos \theta$$

and

$$B_n c^{-(n+1)} = \frac{2n+1}{2} \int_0^\pi Ec \cos \theta P_n(\cos \theta) \sin \theta d\theta.$$

Now $\cos \theta = P_1(\cos \theta)$ so, for $n \neq 1$,

$$\int_0^\pi \cos \theta P_n(\cos \theta) \sin \theta d\theta = 0$$

by orthogonality. Thus $B_n = 0$ for $n \neq 1$ and

$$B_1 = \frac{3}{2} Ec^3 \int_0^\pi \cos^2 \theta \sin \theta d\theta = Ec^3.$$

Therefore,

$$u(r, \theta) = -Er \cos \theta + Ec^3 r^{-2} \cos \theta.$$

Chapter 14 Review Exercises

1. We have

$$\begin{aligned} A_0 &= \frac{1}{2\pi} \int_0^\pi u_0 d\theta + \frac{1}{2\pi} \int_\pi^{2\pi} (-u_0) d\theta = 0 \\ A_n &= \frac{1}{c^n \pi} \int_0^\pi u_0 \cos n\theta d\theta + \frac{1}{c^n \pi} \int_\pi^{2\pi} (-u_0) \cos n\theta d\theta = 0 \\ B_n &= \frac{1}{c^n \pi} \int_0^\pi u_0 \sin n\theta d\theta + \frac{1}{c^n \pi} \int_\pi^{2\pi} (-u_0) \sin n\theta d\theta = \frac{2u_0}{c^n n \pi} [1 - (-1)^n] \end{aligned}$$

and so

$$u(r, \theta) = \frac{2u_0}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n} \left(\frac{r}{c}\right)^n \sin n\theta.$$

2. We have

$$\begin{aligned} A_0 &= \frac{1}{2\pi} \int_0^{\pi/2} d\theta + \frac{1}{2\pi} \int_{3\pi/2}^{2\pi} d\theta = \frac{1}{2} \\ A_n &= \frac{1}{c^n \pi} \int_0^{\pi/2} \cos n\theta d\theta + \frac{1}{c^n \pi} \int_{3\pi/2}^{2\pi} \cos n\theta d\theta = \frac{1}{c^n n \pi} \left[\sin \frac{n\pi}{2} - \sin \frac{3n\pi}{2} \right] \\ B_n &= \frac{1}{c^n \pi} \int_0^{\pi/2} \sin n\theta d\theta + \frac{1}{c^n \pi} \int_{3\pi/2}^{2\pi} \sin n\theta d\theta = \frac{1}{c^n n \pi} \left[\cos \frac{3n\pi}{2} - \cos \frac{n\pi}{2} \right] \end{aligned}$$

and so

$$u(r, \theta) = \frac{1}{2} + \frac{1}{\pi} \sum_{n=1}^{\infty} \left(\frac{r}{c}\right)^n \left[\frac{\sin \frac{n\pi}{2} - \sin \frac{3n\pi}{2}}{n} \cos n\theta + \frac{\cos \frac{3n\pi}{2} - \cos \frac{n\pi}{2}}{n} \sin n\theta \right].$$

3. The conditions $\Theta(0) = 0$ and $\Theta(\pi) = 0$ applied to $\Theta = c_1 \cos \lambda\theta + c_2 \sin \lambda\theta$ give $c_1 = 0$ and $\lambda = n$, $n = 1, 2, 3, \dots$, respectively. Thus we have the Fourier sine-series coefficients

$$A_n = \frac{2}{\pi} \int_0^\pi u_0(\pi\theta - \theta^2) \sin n\theta d\theta = \frac{4u_0}{n^3 \pi} [1 - (-1)^n].$$

Thus

$$u(r, \theta) = \frac{4u_0}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n^3} r^n \sin n\theta.$$

4. In this case

$$A_n = \frac{2}{\pi} \int_0^\pi \sin \theta \sin n\theta d\theta = \frac{1}{\pi} \int_0^\pi [\cos(1-n)\theta - \cos(1+n)\theta] d\theta = 0, \quad n \neq 1.$$

For $n = 1$,

$$A_1 = \frac{2}{\pi} \int_0^\pi \sin^2 \theta d\theta = \frac{1}{\pi} \int_0^\pi (1 - \cos 2\theta) d\theta = 1.$$

Thus

$$u(r, \theta) = \sum_{n=1}^{\infty} A_n r^n \sin n\theta$$

reduces to

$$u(r, \theta) = r \sin \theta.$$

5. We solve

$$\begin{aligned}\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} &= 0, \quad 0 < \theta < \frac{\pi}{4}, \quad \frac{1}{2} < r < 1, \\ u(r, 0) &= 0, \quad u(r, \pi/4) = 0, \quad \frac{1}{2} < r < 1, \\ u(1/2, \theta) &= u_0, \quad u_r(1, \theta) = 0, \quad 0 < \theta < \frac{\pi}{4}.\end{aligned}$$

Proceeding as in Example 1 in Section 14.1 in the text we obtain the separated equations

$$\begin{aligned}r^2 R'' + rR' - \lambda^2 R &= 0 \\ \Theta'' + \lambda^2 \Theta &= 0\end{aligned}$$

with solutions

$$\begin{aligned}\Theta(\theta) &= c_1 \cos \lambda \theta + c_2 \sin \lambda \theta \\ R(r) &= c_3 r^\lambda + c_4 r^{-\lambda}.\end{aligned}$$

Applying the boundary conditions $\Theta(0) = 0$ and $\Theta(\pi/4) = 0$ gives $c_1 = 0$ and $\lambda = 4n$ for $n = 1, 2, 3, \dots$. From $R_r(1) = 0$ we obtain $c_3 = c_4$. Therefore

$$u(r, \theta) = \sum_{n=1}^{\infty} A_n (r^{4n} + r^{-4n}) \sin 4n\theta.$$

From

$$u(1/2, \theta) = u_0 = \sum_{n=1}^{\infty} A_n \left(\frac{1}{2^{4n}} + \frac{1}{2^{-4n}} \right) \sin 4n\theta$$

we find

$$A_n \left(\frac{1}{2^{4n}} + \frac{1}{2^{-4n}} \right) = \frac{2}{\pi/4} \int_0^{\pi/4} u_0 \sin 4n\theta \, d\theta = \frac{2u_0}{n\pi} [1 - (-1)^n]$$

or

$$A_n = \frac{2u_0}{n\pi(2^{4n} + 2^{-4n})} [1 - (-1)^n].$$

Thus the steady-state temperature in the plate is

$$u(r, \theta) = \frac{2u_0}{\pi} \sum_{n=1}^{\infty} \frac{[r^{4n} + r^{-4n}][1 - (-1)^n]}{n[2^{4n} + 2^{-4n}]} \sin 4n\theta.$$

6. We solve

$$\begin{aligned}\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} &= 0, \quad r > 1, \quad 0 < \theta < \pi, \\ u(r, 0) &= 0, \quad u(r, \pi) = 0, \quad r > 1, \\ u(1, \theta) &= f(\theta), \quad 0 < \theta < \pi.\end{aligned}$$

Separating variables we obtain

$$\begin{aligned}\Theta(\theta) &= c_1 \cos \lambda \theta + c_2 \sin \lambda \theta \\ R(r) &= c_3 r^\lambda + c_4 r^{-\lambda}.\end{aligned}$$

Applying the boundary conditions $\Theta(0) = 0$, and $\Theta(\pi) = 0$ gives $c_1 = 0$ and $\lambda = n$ for $n = 1, 2, 3, \dots$. Assuming $f(\theta)$ to be bounded, we expect the solution $u(r, \theta)$ to also be bounded as $r \rightarrow \infty$. This requires that $c_3 = 0$. Therefore

$$u(r, \theta) = \sum_{n=1}^{\infty} A_n r^{-n} \sin n\theta.$$

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From

$$u(1, \theta) = f(\theta) = \sum_{n=1}^{\infty} A_n \sin n\theta$$

we obtain

$$A_n = \frac{2}{\pi} \int_0^{\pi} f(\theta) \sin n\theta \, d\theta.$$

7. Letting $u(r, t) = R(r)T(t)$ and separating variables we obtain

$$\frac{R'' + \frac{1}{r}R' - hR}{R} = \frac{T'}{T} = \mu$$

so

$$R'' + \frac{1}{r}R' - (\mu + h)R = 0 \quad \text{and} \quad T' - \mu T = 0.$$

From the second equation we find $T(t) = c_1 e^{\mu t}$. If $\mu > 0$, $T(t)$ increases without bound as $t \rightarrow \infty$. Thus we assume $\mu \leq 0$. Since $h > 0$ we can take $\mu = -\lambda^2 - h$. Then

$$R'' + \frac{1}{r}R' + \lambda^2 R = 0$$

is a parametric Bessel equation with solution

$$R(r) = c_1 J_0(\lambda r) + c_2 Y_0(\lambda r).$$

Since Y_0 is unbounded as $r \rightarrow 0$ we take $c_2 = 0$. Then $R(r) = c_1 J_0(\lambda r)$ and the boundary condition $u(1, t) = R(1)T(t) = 0$ implies $J_0(\lambda) = 0$. This latter equation defines the positive eigenvalues λ_n . Thus

$$u(r, t) = \sum_{n=1}^{\infty} A_n J_0(\lambda_n r) e^{(-\lambda_n^2 - h)t}.$$

From

$$u(r, 0) = 1 = \sum_{n=1}^{\infty} A_n J_0(\lambda_n r)$$

we find

$$\begin{aligned} A_n &= \frac{2}{J_1^2(\lambda_n)} \int_0^1 r J_0(\lambda_n r) \, dr && \boxed{x = \lambda_n r, \, dx = \lambda_n \, dr} \\ &= \frac{2}{J_1^2(\lambda_n)} \int_0^{\lambda_n} \frac{1}{\lambda_n^2} x J_0(x) \, dx. \end{aligned}$$

From recurrence relation (4) in Section 12.6 of the text we have

$$x J_0(x) = \frac{d}{dx} [x J_1(x)].$$

Then

$$A_n = \frac{2}{\lambda_n^2 J_1^2(\lambda_n)} \int_0^{\lambda_n} \frac{d}{dx} [x J_1(x)] \, dx = \frac{2}{\lambda_n^2 J_1^2(\lambda_n)} \left(x J_1(x) \right) \Big|_0^{\lambda_n} = \frac{2\lambda_n J_1(\lambda_n)}{\lambda_n^2 J_1^2(\lambda_n)} = \frac{2}{\lambda_n J_1(\lambda_n)}$$

and

$$u(r, t) = 2e^{-ht} \sum_{n=1}^{\infty} \frac{J_0(\lambda_n r)}{\lambda_n J_1(\lambda_n)} e^{-\lambda_n^2 t}$$

8. Proceeding in the usual manner we find

$$u(r, t) = \sum_{n=1}^{\infty} A_n \cos a\lambda_n t J_0(\lambda_n r)$$

where the eigenvalues are defined by $J_0(\lambda) = 0$. Thus the initial condition gives

$$u_0 J_0(x_k r) = \sum_{n=1}^{\infty} A_n J_0(\lambda_n r)$$

and so

$$A_n = \frac{2}{J_1^2(\lambda_n)} \int_0^1 r (u_0 J_0(x_k r)) J_0(\lambda_n r) dr.$$

But $J_0(\lambda) = 0$ implies that the eigenvalues are the positive zeros of J_0 , that is, $\lambda_n = x_n$, $n = 1, 2, 3, \dots$. Therefore

$$A_n = \frac{2u_0}{J_1^2(\lambda_n)} \int_0^1 r J_0(\lambda_k r) J_0(\lambda_n r) dr = 0, \quad n \neq k$$

by orthogonality. For $n = k$,

$$A_k = \frac{2u_0}{J_1^2(\lambda_k)} \int_0^1 r J_0^2(\lambda_k r) dr = u_0$$

by (6) of Section 12.6. Thus the solution $u(r, t)$ reduces to one term when $n = k$, and

$$u(r, t) = u_0 \cos a \lambda_k t J_0(\lambda_k r) = u_0 \cos a x_k t J_0(x_k r).$$

9. Referring to Example 2 in Section 14.2 we have

$$R(r) = c_1 J_0(\lambda r) + c_2 Y_0(\lambda r)$$

$$Z(z) = c_3 \cosh \lambda z + c_4 \sinh \lambda z$$

where $c_2 = 0$ and $J_0(2\lambda) = 0$ defines the positive eigenvalues λ_n . From $Z'(0) = 0$ we obtain $c_4 = 0$. Then

$$u(r, z) = \sum_{n=1}^{\infty} A_n \cosh \lambda_n z J_0(\lambda_n r).$$

From

$$u(r, 4) = 50 = \sum_{n=1}^{\infty} A_n \cosh 4\lambda_n J_0(\lambda_n r)$$

we obtain (as in Example 1 of Section 14.1)

$$A_n \cosh 4\lambda_n = \frac{2(50)}{4J_1^2(2\lambda_n)} \int_0^2 r J_0(\lambda_n r) dr = \frac{50}{\lambda_n J_1(2\lambda_n)}.$$

Thus the temperature in the cylinder is

$$u(r, z) = 50 \sum_{n=1}^{\infty} \frac{\cosh \lambda_n z J_0(\lambda_n r)}{\lambda_n \cosh 4\lambda_n J_1(2\lambda_n)}.$$

10. Using $u = RZ$ and $-\lambda^2$ as a separation constant leads to

$$r^2 R'' + rR' + \lambda^2 r^2 R = 0, \quad R'(1) = 0, \quad \text{and} \quad Z'' - \lambda^2 Z = 0.$$

Thus

$$R(r) = c_1 J_0(\lambda r) + c_2 Y_0(\lambda r)$$

$$Z(z) = c_3 \cosh \lambda z + c_4 \sinh \lambda z$$

for $\lambda > 0$. Arguing that $u(r, z)$ is bounded as $r \rightarrow 0$ we defined $c_2 = 0$. Since the eigenvalues are defined by $J'_0(\lambda) = 0$ we know that $\lambda = 0$ is an eigenvalue. The solutions are then

$$R(r) = c_1 + c_2 \ln r \quad \text{and} \quad Z(z) = c_3 z + c_4$$

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where $c_2 = 0$. Thus a formal solution is

$$u(r, z) = A_0 z + B_0 + \sum_{n=1}^{\infty} (A_n \sinh \lambda_n z + B_n \cosh \lambda_n z) J_0(\lambda_n r).$$

Finally, the specified conditions $z = 0$ and $z = 1$ give, in turn,

$$B_0 = 2 \int_0^1 r f(r) dr$$

$$B_n = \frac{2}{J_0^2(\lambda_n)} \int_0^1 r f(r) J_0(\lambda_n r) dr$$

$$A_0 = -B_0 + 2 \int_0^1 r g(r) dr$$

$$A_n = \frac{1}{\sinh \lambda_n} \left[-B_n \cosh \lambda_n + \frac{2}{J_0^2(\lambda_n)} \int_0^1 r g(r) J_0(\lambda_n r) dr \right].$$

11. Referring to Example 1 in Section 14.3 of the text we have

$$u(r, \theta) = \sum_{n=0}^{\infty} A_n r^n P_n(\cos \theta).$$

For $x = \cos \theta$

$$u(1, \theta) = \begin{cases} 100 & 0 < \theta < \pi/2 \\ -100 & \pi/2 < \theta < \pi \end{cases} = 100 \begin{cases} -1, & -1 < x < 0 \\ 1, & 0 < x < 1 \end{cases} = g(x).$$

From Problem 22 in Exercise 12.6 we have

$$u(r, \theta) = 100 \left[\frac{3}{2} r P_1(\cos \theta) - \frac{7}{8} r^3 P_3(\cos \theta) + \frac{11}{16} r^5 P_5(\cos \theta) + \cdots \right].$$

12. Since

$$\frac{1}{r} \frac{\partial^2}{\partial r^2} (ru) = \frac{1}{r} \frac{\partial}{\partial r} \left[r \frac{\partial u}{\partial r} + u \right] = \frac{1}{r} \left[r \frac{\partial^2 u}{\partial r^2} + \frac{\partial u}{\partial r} + \frac{\partial u}{\partial r} \right] = \frac{\partial^2 u}{\partial r^2} + \frac{2}{r} \frac{\partial u}{\partial r}$$

the differential equation becomes

$$\frac{1}{r} \frac{\partial^2}{\partial r^2} (ru) = \frac{\partial^2 u}{\partial t^2} \quad \text{or} \quad \frac{\partial^2}{\partial r^2} (ru) = r \frac{\partial^2 u}{\partial t^2}.$$

Letting $v(r, t) = ru(r, t)$ we obtain the boundary-value problem

$$\frac{\partial^2 v}{\partial r^2} = \frac{\partial^2 v}{\partial t^2}, \quad 0 < r < 1, \quad t > 0$$

$$\left. \frac{\partial v}{\partial r} \right|_{r=1} - v(1, t) = 0, \quad t > 0$$

$$v(r, 0) = r f(r), \quad \left. \frac{\partial v}{\partial t} \right|_{t=0} = r g(r), \quad 0 < r < 1.$$

If we separate variables using $v(r, t) = R(r)T(t)$ then we obtain

$$R(r) = c_1 \cos \lambda r + c_2 \sin \lambda r$$

$$T(t) = c_3 \cos \lambda t + c_4 \sin \lambda t.$$

Since $u(r, t) = v(r, t)/r$, in order to insure boundedness at $r = 0$ we define $c_1 = 0$. Then $R(r) = c_2 \sin \lambda r$. Now the boundary condition $R'(1) - R(1) = 0$ implies $\lambda \cos \lambda - \sin \lambda = 0$. Thus, the eigenvalues λ_n are the positive solutions of $\tan \lambda = \lambda$. We now have

$$v_n(r, t) = (A_n \cos \lambda_n t + B_n \sin \lambda_n t) \sin \lambda_n r.$$

For the eigenvalue $\lambda = 0$,

$$R(r) = c_1 r + c_2 \quad \text{and} \quad T(t) = c_3 t + c_4,$$

and boundedness at $r = 0$ implies $c_2 = 0$. We then take

$$v_0(r, t) = A_0 t r + B_0 r$$

so that

$$v(r, t) = A_0 t r + B_0 r + \sum_{n=1}^{\infty} (a_n \cos \lambda_n t + B_n \sin \lambda_n t) \sin \lambda_n r.$$

Now

$$v(r, 0) = r f(r) = B_0 r + \sum_{n=1}^{\infty} A_n \sin \lambda_n r.$$

Since $\{r, \sin \lambda_n r\}$ is an orthogonal set on $[0, 1]$,

$$\int_0^1 r \sin \lambda_n r \, dr = 0 \quad \text{and} \quad \int_0^1 \sin \lambda_n r \sin \lambda_m r \, dr = 0$$

for $m \neq n$. Therefore

$$\int_0^1 r^2 f(r) \, dr = B_0 \int_0^1 r^2 \, dr = \frac{1}{3} B_0$$

and

$$B_0 = 3 \int_0^1 r^2 f(r) \, dr.$$

Also

$$\int_0^1 r f(r) \sin \lambda_n r \, dr = A_n \int_0^1 \sin^2 \lambda_n r \, dr$$

and

$$A_n = \frac{\int_0^1 r f(r) \sin \lambda_n r \, dr}{\int_0^1 \sin^2 \lambda_n r \, dr}.$$

Now

$$\int_0^1 \sin^2 \lambda_n r \, dr = \frac{1}{2} \int_0^1 (1 - \cos 2\lambda_n r) \, dr = \frac{1}{2} \left[1 - \frac{\sin 2\lambda_n}{2\lambda_n} \right] = \frac{1}{2} [1 - \cos^2 \lambda_n].$$

Since $\tan \lambda_n = \lambda_n$,

$$1 + \lambda_n^2 = 1 + \tan^2 \lambda_n = \sec^2 \lambda_n = \frac{1}{\cos^2 \lambda_n}$$

and

$$\cos^2 \lambda_n = \frac{1}{1 + \lambda_n^2}.$$

Then

$$\int_0^1 \sin^2 \lambda_n r \, dr = \frac{1}{2} \left[1 - \frac{1}{1 + \lambda_n^2} \right] = \frac{\lambda_n^2}{2(1 + \lambda_n^2)}$$

and

$$A_n = \frac{2(1 + \lambda_n^2)}{\lambda_n^2} \int_0^1 r f(r) \sin \lambda_n r \, dr.$$

Similarly, setting

$$\left. \frac{\partial v}{\partial t} \right|_{t=0} = r g(r) = A_0 r + \sum_{n=1}^{\infty} B_n \lambda_n \sin \lambda_n r$$

Chapter 14 Review Exercises

we obtain

$$A_0 = 3 \int_0^1 r^2 g(r) dr$$

and

$$B_n = \frac{2(1 + \lambda_n^2)}{\lambda_n^3} \int_0^1 r g(r) \sin \lambda_n r dr.$$

Therefore, since $v(r, t) = ru(r, t)$ we have

$$u(r, t) = A_0 t + B_0 + \sum_{n=1}^{\infty} (A_n \cos \lambda_n t + B_n \sin \lambda_n t) \frac{\sin \lambda_n r}{r},$$

where the λ_n are solutions of $\tan \lambda = \lambda$ and

$$A_0 = 3 \int_0^1 r^2 g(r) dr$$

$$B_0 = 3 \int_0^1 r^2 f(r) dr$$

$$A_n = \frac{2(1 + \lambda_n^2)}{\lambda_n^2} \int_0^1 r f(r) \sin \lambda_n r dr$$

$$B_n = \frac{2(1 + \lambda_n^2)}{\lambda_n^3} \int_0^1 r g(r) \sin \lambda_n r dr$$

for $n = 1, 2, 3, \dots$.

- 13.** We note that the differential equation can be expressed in the form

$$\frac{d}{dx} [xu'] = -\lambda^2 xu.$$

Thus

$$u_n \frac{d}{dx} [xu'_m] = -\lambda_m^2 xu_m u_n$$

and

$$u_m \frac{d}{dx} [xu'_n] = -\lambda_n^2 xu_n u_m.$$

Subtracting we obtain

$$u_n \frac{d}{dx} [xu'_m] - u_m \frac{d}{dx} [xu'_n] = (\lambda_n^2 - \lambda_m^2) xu_m u_n$$

and

$$\int_a^b u_n \frac{d}{dx} [xu'_m] dx - \int_a^b u_m \frac{d}{dx} [xu'_n] dx = (\lambda_n^2 - \lambda_m^2) \int_a^b xu_m u_n dx.$$

Using integration by parts this becomes

$$\begin{aligned} & u_n xu'_m \Big|_a^b - \int_a^b xu'_m u'_n dx - u_m xu'_n \Big|_a^b + \int_a^b xu'_n u'_m dx \\ &= b[u_n(b)u'_m(b) - u_m(b)u'_n(b)] - a[u_n(a)u'_m(a) - u_m(a)u'_n(a)] \\ &= (\lambda_n^2 - \lambda_m^2) \int_a^b xu_m u_n dx. \end{aligned}$$

Since

$$u(x) = Y_0(\lambda a)J_0(\lambda x) - J_0(\lambda a)Y_0(\lambda x)$$

we have

$$u_n(b) = Y_0(\lambda_n a)J_0(\lambda_n b) - J_0(\lambda_n a)Y_0(\lambda_n b) = 0$$

by the definition of the λ_n . Similarly $u_m(b) = 0$. Also

$$u_n(a) = Y_0(\lambda a)J_0(\lambda_n a) - J_0(\lambda_n a)Y_0(\lambda a) = 0$$

and $u_m(a) = 0$. Therefore

$$\int_a^b x u_m u_n dx = \frac{1}{\lambda_n^2 - \lambda_m^2} (b[u_n(b)u'_m(b) - u_m(b)u'_n(b)] - a[u_n(a)u'_m(a) - u_m(a)u'_n(a)]) = 0$$

and the $u_n(x)$ are orthogonal with respect to the weight function x .

14. Letting $u(r, t) = R(r)T(t)$ and separating variables we obtain

$$rR'' + R' + \lambda^2 rR = 0$$

$$T' + \lambda^2 T = 0,$$

with solutions

$$R(r) = c_1 J_0(\lambda r) + c_2 Y_0(\lambda r)$$

$$T(t) = c_3 e^{-\lambda^2 t}.$$

Now the boundary conditions imply

$$R(a) = 0 = c_1 J_0(\lambda a) + c_2 Y_0(\lambda a)$$

$$R(b) = 0 = c_1 J_0(\lambda b) + c_2 Y_0(\lambda b)$$

so that

$$c_2 = -\frac{c_1 J_0(\lambda a)}{Y_0(\lambda a)}$$

and

$$c_1 J_0(\lambda b) - \frac{c_1 J_0(\lambda a)}{Y_0(\lambda a)} Y_0(\lambda b) = 0$$

or

$$Y_0(\lambda a)J_0(\lambda b) - J_0(\lambda a)Y_0(\lambda b) = 0.$$

This equation defines λ_n for $n = 1, 2, 3, \dots$. Now

$$R(r) = c_1 J_0(\lambda r) - c_1 \frac{J_0(\lambda a)}{Y_0(\lambda a)} Y_0(\lambda r) = \frac{c_1}{Y_0(\lambda a)} [Y_0(\lambda a)J_0(\lambda r) - J_0(\lambda a)Y_0(\lambda r)]$$

and

$$u_n(r, t) = A_n [Y_0(\lambda_n a)J_0(\lambda_n r) - J_0(\lambda_n a)Y_0(\lambda_n r)] e^{-\lambda_n^2 t} = A_n u_n(r) e^{-\lambda_n^2 t}.$$

Thus

$$u(r, t) = \sum_{n=1}^{\infty} A_n u_n(r) e^{-\lambda_n^2 t}.$$

From the initial condition

$$u(r, 0) = f(r) = \sum_{n=1}^{\infty} A_n u_n(r)$$

we obtain

$$A_n = \frac{\int_a^b r f(r) u_n(r) dr}{\int_a^b r u_n^2(r) dr}.$$

15 Integral Transform Method

Exercises 15.1

1. (a) The result follows by letting $\tau = u^2$ or $u = \sqrt{\tau}$ in $\operatorname{erf}(\sqrt{t}) = \frac{2}{\sqrt{\pi}} \int_0^{\sqrt{t}} e^{-u^2} du$.

(b) Using $\mathcal{L}\{t^{-1/2}\} = \frac{\sqrt{\pi}}{s^{1/2}}$ and the first translation theorem, it follows from the convolution theorem that

$$\begin{aligned}\mathcal{L}\{\operatorname{erf}(\sqrt{t})\} &= \frac{1}{\sqrt{\pi}} \mathcal{L}\left\{\int_0^t \frac{e^{-\tau}}{\sqrt{\tau}} d\tau\right\} = \frac{1}{\sqrt{\pi}} \mathcal{L}\{1\} \mathcal{L}\{t^{-1/2}e^{-t}\} = \frac{1}{\sqrt{\pi}} \frac{1}{s} \mathcal{L}\{t^{-1/2}\} \Big|_{s \rightarrow s+1} \\ &= \frac{1}{\sqrt{\pi}} \frac{1}{s} \frac{\sqrt{\pi}}{\sqrt{s+1}} = \frac{1}{s\sqrt{s+1}}.\end{aligned}$$

2. Since $\operatorname{erfc}(\sqrt{t}) = 1 - \operatorname{erf}(\sqrt{t})$ we have

$$\mathcal{L}\{\operatorname{erfc}(\sqrt{t})\} = \mathcal{L}\{1\} - \mathcal{L}\{\operatorname{erf}(\sqrt{t})\} = \frac{1}{s} - \frac{1}{s\sqrt{s+1}} = \frac{1}{s} \left[1 - \frac{1}{\sqrt{s+1}}\right].$$

3. By the first translation theorem,

$$\mathcal{L}\{e^t \operatorname{erf}(\sqrt{t})\} = \mathcal{L}\{\operatorname{erf}(\sqrt{t})\} \Big|_{s \rightarrow s-1} = \frac{1}{s\sqrt{s+1}} \Big|_{s \rightarrow s-1} = \frac{1}{\sqrt{s}(s-1)}.$$

4. By the first translation theorem and the result of Problem 2,

$$\begin{aligned}\mathcal{L}\{e^t \operatorname{erfc}(\sqrt{t})\} &= \mathcal{L}\{\operatorname{erfc}(\sqrt{t})\} \Big|_{s \rightarrow s-1} = \left(\frac{1}{s} - \frac{1}{s\sqrt{s+1}}\right) \Big|_{s \rightarrow s-1} = \frac{1}{s-1} - \frac{1}{\sqrt{s}(s-1)} \\ &= \frac{\sqrt{s}-1}{\sqrt{s}(s-1)} = \frac{\sqrt{s}-1}{\sqrt{s}(\sqrt{s}+1)(\sqrt{s}-1)} = \frac{1}{\sqrt{s}(\sqrt{s}+1)}.\end{aligned}$$

5. From table entry 3 and the first translation theorem we have

$$\begin{aligned}\mathcal{L}\left\{e^{-Gt/C} \operatorname{erf}\left(\frac{x}{2}\sqrt{\frac{RC}{t}}\right)\right\} &= \mathcal{L}\left\{e^{-Gt/C} \left[1 - \operatorname{erfc}\left(\frac{x}{2}\sqrt{\frac{RC}{t}}\right)\right]\right\} \\ &= \mathcal{L}\{e^{-Gt/C}\} - \mathcal{L}\left\{e^{-Gt/C} \operatorname{erfc}\left(\frac{x}{2}\sqrt{\frac{RC}{t}}\right)\right\} \\ &= \frac{1}{s+G/C} - \frac{e^{-x\sqrt{RC}}\sqrt{s}}{s} \Big|_{s \rightarrow s+G/C} \\ &= \frac{1}{s+G/C} - \frac{e^{-x\sqrt{RC}}\sqrt{s+G/C}}{s+G/C} = \frac{C}{Cs+G} \left(1 - e^{x\sqrt{RCs+RG}}\right).\end{aligned}$$

6. We first compute

$$\begin{aligned}
 \frac{\sinh a\sqrt{s}}{s \sinh \sqrt{s}} &= \frac{e^{a\sqrt{s}} - e^{-a\sqrt{s}}}{s(e^{\sqrt{s}} - e^{-\sqrt{s}})} = \frac{e^{(a-1)\sqrt{s}} - e^{-(a+1)\sqrt{s}}}{s(1 - e^{-2\sqrt{s}})} \\
 &= \frac{e^{(a-1)\sqrt{s}}}{s} \left[1 + e^{-2\sqrt{s}} + e^{-4\sqrt{s}} + \dots \right] - \frac{e^{-(a+1)\sqrt{s}}}{s} \left[1 + e^{-2\sqrt{s}} + e^{-4\sqrt{s}} + \dots \right] \\
 &= \left[\frac{e^{-(1-a)\sqrt{s}}}{s} + \frac{e^{-(3-a)\sqrt{s}}}{s} + \frac{e^{-(5-a)\sqrt{s}}}{s} + \dots \right] \\
 &\quad - \left[\frac{e^{-(1+a)\sqrt{s}}}{s} + \frac{e^{-(3+a)\sqrt{s}}}{s} + \frac{e^{-(5+a)\sqrt{s}}}{s} + \dots \right] \\
 &= \sum_{n=0}^{\infty} \left[\frac{e^{-(2n+1-a)\sqrt{s}}}{s} - \frac{e^{-(2n+1+a)\sqrt{s}}}{s} \right].
 \end{aligned}$$

Then

$$\begin{aligned}
 \mathcal{L} \left\{ \frac{\sinh a\sqrt{s}}{s \sinh \sqrt{s}} \right\} &= \sum_{n=0}^{\infty} \left[\mathcal{L} \left\{ \frac{e^{-(2n+1-a)\sqrt{s}}}{s} \right\} - \mathcal{L} \left\{ -\frac{e^{-(2n+1+a)\sqrt{s}}}{s} \right\} \right] \\
 &= \sum_{n=0}^{\infty} \left[\operatorname{erfc} \left(\frac{2n+1-a}{2\sqrt{t}} \right) - \operatorname{erfc} \left(\frac{2n+1+a}{2\sqrt{t}} \right) \right] \\
 &= \sum_{n=0}^{\infty} \left(\left[1 - \operatorname{erf} \left(\frac{2n+1-a}{2\sqrt{t}} \right) \right] - \left[1 - \operatorname{erf} \left(\frac{2n+1+a}{2\sqrt{t}} \right) \right] \right) \\
 &= \sum_{n=0}^{\infty} \left[\operatorname{erf} \left(\frac{2n+1+a}{2\sqrt{t}} \right) - \operatorname{erf} \left(\frac{2n+1-a}{2\sqrt{t}} \right) \right].
 \end{aligned}$$

7. Taking the Laplace transform of both sides of the equation we obtain

$$\begin{aligned}
 \mathcal{L}\{y(t)\} &= \mathcal{L}\{1\} - \mathcal{L} \left\{ \int_0^t \frac{y(\tau)}{\sqrt{t-\tau}} d\tau \right\} \\
 Y(s) &= \frac{1}{s} - Y(s) \frac{\sqrt{\pi}}{\sqrt{s}} \\
 \frac{\sqrt{s} + \sqrt{\pi}}{\sqrt{s}} Y(s) &= \frac{1}{s} \\
 Y(s) &= \frac{1}{\sqrt{s}(\sqrt{s} + \sqrt{\pi})}.
 \end{aligned}$$

Thus

$$y(t) = \mathcal{L}^{-1} \left\{ \frac{1}{\sqrt{s}(\sqrt{s} + \sqrt{\pi})} \right\} = e^{\pi t} \operatorname{erfc}(\sqrt{\pi t}). \quad \boxed{\text{By entry 5 in the table}}$$

8. Using entries 3 and 5 in the table, we have

$$\begin{aligned}
 &\mathcal{L} \left\{ -e^{ab} e^{b^2 t} \operatorname{erfc} \left(b\sqrt{t} + \frac{a}{2\sqrt{t}} \right) + \operatorname{erfc} \left(\frac{a}{2\sqrt{t}} \right) \right\} \\
 &= -\mathcal{L} \left\{ e^{ab} e^{b^2 t} \operatorname{erfc} \left(b\sqrt{t} + \frac{a}{2\sqrt{t}} \right) \right\} + \mathcal{L} \left\{ \frac{a}{2\sqrt{t}} \right\} \\
 &= -\frac{e^{-a\sqrt{s}}}{\sqrt{s}(\sqrt{s} + b)} + \frac{e^{-a\sqrt{s}}}{s}
 \end{aligned}$$

Exercises 15.1

$$\begin{aligned}
 &= e^{-a\sqrt{s}} \left[\frac{1}{s} - \frac{1}{\sqrt{s}(\sqrt{s}+b)} \right] = e^{-a\sqrt{s}} \left[\frac{1}{s} - \frac{\sqrt{s}}{s(\sqrt{s}+b)} \right] \\
 &= e^{-a\sqrt{s}} \left[\frac{\sqrt{s}+b-\sqrt{s}}{s(\sqrt{s}+b)} \right] = \frac{be^{-a\sqrt{s}}}{s(\sqrt{s}+b)}.
 \end{aligned}$$

$$\begin{aligned}
 9. \quad \int_a^b e^{-u^2} du &= \int_a^0 e^{-u^2} du + \int_0^b e^{-u^2} du = \int_0^b e^{-u^2} du - \int_0^a e^{-u^2} du \\
 &= \frac{\sqrt{\pi}}{2} \operatorname{erf}(b) - \frac{\sqrt{\pi}}{2} \operatorname{erf}(a) = \frac{\sqrt{\pi}}{2} [\operatorname{erf}(b) - \operatorname{erf}(a)]
 \end{aligned}$$

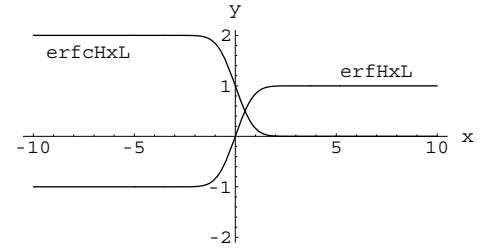
10. Since $f(x) = e^{-x^2}$ is an even function,

$$\int_{-a}^a e^{-u^2} du = 2 \int_0^a e^{-u^2} du.$$

Therefore,

$$\int_{-a}^a e^{-u^2} du = \sqrt{\pi} \operatorname{erf}(a).$$

11. The function $\operatorname{erf}(x)$ is symmetric with respect to the origin, while $\operatorname{erfc}(x)$ appears to be symmetric with respect to the point $(0, 1)$. From the graph it appears that $\lim_{x \rightarrow -\infty} \operatorname{erf}(x) = -1$ and $\lim_{x \rightarrow -\infty} \operatorname{erfc}(x) = 2$.



Exercises 15.2

1. The boundary-value problem is

$$\begin{aligned}
 a^2 \frac{\partial^2 u}{\partial x^2} &= \frac{\partial^2 u}{\partial t^2}, \quad 0 < x < L, \quad t > 0, \\
 u(0, t) &= 0, \quad u(L, t) = 0, \quad t > 0, \\
 u(x, 0) &= A \sin \frac{\pi}{L} x, \quad \left. \frac{\partial u}{\partial t} \right|_{t=0} = 0.
 \end{aligned}$$

Transforming the partial differential equation gives

$$\frac{d^2 U}{dx^2} - \left(\frac{s}{a} \right)^2 U = -\frac{s}{a^2} A \sin \frac{\pi}{L} x.$$

Using undetermined coefficients we obtain

$$U(x, s) = c_1 \cosh \frac{s}{a} x + c_2 \sinh \frac{s}{a} x + \frac{As}{s^2 + a^2 \pi^2 / L^2} \sin \frac{\pi}{L} x.$$

The transformed boundary conditions, $U(0, s) = 0$, $U(L, s) = 0$ give in turn $c_1 = 0$ and $c_2 = 0$. Therefore

$$U(x, s) = \frac{As}{s^2 + a^2 \pi^2 / L^2} \sin \frac{\pi}{L} x$$

and

$$u(x, t) = A \mathcal{L}^{-1} \left\{ \frac{s}{s^2 + a^2 \pi^2 / L^2} \right\} \sin \frac{\pi}{L} x = A \cos \frac{a\pi}{L} t \sin \frac{\pi}{L} x.$$

2. The transformed equation is

$$\frac{d^2 U}{dx^2} - s^2 U = -2 \sin \pi x - 4 \sin 3\pi x$$

and so

$$U(x, s) = c_1 \cosh sx + c_2 \sinh sx + \frac{2}{s^2 + \pi^2} \sin \pi x + \frac{4}{s^2 + 9\pi^2} \sin 3\pi x.$$

The transformed boundary conditions, $U(0, s) = 0$ and $U(1, s) = 0$ give $c_1 = 0$ and $c_2 = 0$. Thus

$$U(x, s) = \frac{2}{s^2 + \pi^2} \sin \pi x + \frac{4}{s^2 + 9\pi^2} \sin 3\pi x$$

and

$$\begin{aligned} u(x, t) &= 2 \mathcal{L}^{-1} \left\{ \frac{1}{s^2 + \pi^2} \right\} \sin \pi x + 4 \mathcal{L}^{-1} \left\{ \frac{1}{s^2 + 9\pi^2} \right\} \sin 3\pi x \\ &= \frac{2}{\pi} \sin \pi t \sin \pi x + \frac{4}{3\pi} \sin 3\pi t \sin 3\pi x. \end{aligned}$$

3. The solution of

$$a^2 \frac{d^2 U}{dx^2} - s^2 U = 0$$

is in this case

$$U(x, s) = c_1 e^{-(x/a)s} + c_2 e^{(x/a)s}.$$

Since $\lim_{x \rightarrow \infty} u(x, t) = 0$ we have $\lim_{x \rightarrow \infty} U(x, s) = 0$. Thus $c_2 = 0$ and

$$U(x, s) = c_1 e^{-(x/a)s}.$$

If $\mathcal{L}\{u(0, t)\} = \mathcal{L}\{f(t)\} = F(s)$ then $U(0, s) = F(s)$. From this we have $c_1 = F(s)$ and

$$U(x, s) = F(s) e^{-(x/a)s}.$$

Hence, by the second translation theorem,

$$u(x, t) = f\left(t - \frac{x}{a}\right) \mathcal{U}\left(t - \frac{x}{a}\right).$$

4. Expressing $f(t)$ in the form $(\sin \pi t)[1 - \mathcal{U}(t - 1)]$ and using the result of Problem 3 we find

$$\begin{aligned} u(x, t) &= f\left(t - \frac{x}{a}\right) \mathcal{U}\left(t - \frac{x}{a}\right) \\ &= \sin \pi \left(t - \frac{x}{a}\right) \left[1 - \mathcal{U}\left(t - \frac{x}{a} - 1\right)\right] \mathcal{U}\left(t - \frac{x}{a}\right) \\ &= \sin \pi \left(t - \frac{x}{a}\right) \left[\mathcal{U}\left(t - \frac{x}{a}\right) - \mathcal{U}\left(t - \frac{x}{a}\right) \mathcal{U}\left(t - \frac{x}{a} - 1\right)\right] \\ &= \sin \pi \left(t - \frac{x}{a}\right) \left[\mathcal{U}\left(t - \frac{x}{a}\right) - \mathcal{U}\left(t - \frac{x}{a} - 1\right)\right] \end{aligned}$$

Now

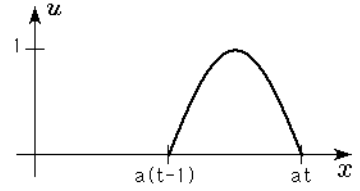
$$\begin{aligned} \mathcal{U}\left(t - \frac{x}{a}\right) - \mathcal{U}\left(t - \frac{x}{a} - 1\right) &= \begin{cases} 0, & 0 \leq t < x/a \\ 1, & x/a \leq t \leq x/a + 1 \\ 0, & t > x/a + 1 \end{cases} \\ &= \begin{cases} 0, & x < a(t - 1) \text{ or } x > at \\ 1, & a(t - 1) \leq x \leq at \end{cases} \end{aligned}$$

Exercises 15.2

so

$$u(x, t) = \begin{cases} 0, & x < a(t-1) \text{ or } x > at \\ \sin \pi(t - x/a), & a(t-1) \leq x \leq at. \end{cases}$$

The graph is shown for $t > 1$.



5. We use

$$U(x, s) = c_1 e^{-(x/a)s} - \frac{g}{s^3}.$$

Now

$$\mathcal{L}\{u(0, t)\} = U(0, s) = \frac{A\omega}{s^2 + \omega^2}$$

and so

$$U(0, s) = c_1 - \frac{g}{s^3} = \frac{A\omega}{s^2 + \omega^2} \quad \text{or} \quad c_1 = \frac{g}{s^3} + \frac{A\omega}{s^2 + \omega^2}.$$

Therefore

$$U(x, s) = \frac{A\omega}{s^2 + \omega^2} e^{-(x/a)s} + \frac{g}{s^3} e^{-(x/a)s} - \frac{g}{s^3}$$

and

$$\begin{aligned} u(x, t) &= A \mathcal{L}^{-1} \left\{ \frac{\omega e^{-(x/a)s}}{s^2 + \omega^2} \right\} + g \mathcal{L}^{-1} \left\{ \frac{e^{-(x/a)s}}{s^3} \right\} - g \mathcal{L}^{-1} \left\{ \frac{1}{s^3} \right\} \\ &= A \sin \omega \left(t - \frac{x}{a} \right) \mathcal{U} \left(t - \frac{x}{a} \right) + \frac{1}{2} g \left(t - \frac{x}{a} \right)^2 \mathcal{U} \left(t - \frac{x}{a} \right) - \frac{1}{2} g t^2. \end{aligned}$$

6. Transforming the partial differential equation gives

$$\frac{d^2 U}{dx^2} - s^2 U = -\frac{\omega}{s^2 + \omega^2} \sin \pi x.$$

Using undetermined coefficients we obtain

$$U(x, s) = c_1 \cosh sx + c_2 \sinh sx + \frac{\omega}{(s^2 + \pi^2)(s^2 + \omega^2)} \sin \pi x.$$

The transformed boundary conditions $U(0, s) = 0$ and $U(1, s) = 0$ give, in turn, $c_1 = 0$ and $c_2 = 0$. Therefore

$$U(x, s) = \frac{\omega}{(s^2 + \pi^2)(s^2 + \omega^2)} \sin \pi x$$

and

$$\begin{aligned} u(x, t) &= \omega \sin \pi x \mathcal{L}^{-1} \left\{ \frac{1}{(s^2 + \pi^2)(s^2 + \omega^2)} \right\} \\ &= \frac{\omega}{\omega^2 - \pi^2} \sin \pi x \mathcal{L}^{-1} \left\{ \frac{1}{\pi} \frac{\pi}{s^2 + \pi^2} - \frac{1}{\omega} \frac{\omega}{s^2 + \omega^2} \right\} \\ &= \frac{\omega}{\pi(\omega^2 - \pi^2)} \sin \pi t \sin \pi x - \frac{1}{\omega^2 - \pi^2} \sin \omega t \sin \pi x. \end{aligned}$$

7. We use

$$U(x, s) = c_1 \cosh \frac{s}{a} x + c_2 \sinh \frac{s}{a} x.$$

Now $U(0, s) = 0$ implies $c_1 = 0$, so $U(x, s) = c_2 \sinh(s/a)x$. The condition $E \frac{dU}{dx} \Big|_{x=L} = F_0$ then yields $c_2 = F_0 a / E s \cosh(s/a)L$ and so

$$\begin{aligned} U(x, s) &= \frac{aF_0}{Es} \frac{\sinh(s/a)x}{\cosh(s/a)L} = \frac{aF_0}{Es} \frac{e^{(s/a)x} - e^{-(s/a)x}}{e^{(s/a)L} + e^{-(s/a)L}} \\ &= \frac{aF_0}{Es} \frac{e^{(s/a)(x-L)} - e^{-(s/a)(x+L)}}{1 + e^{-2sL/a}} \\ &= \frac{aF_0}{E} \left[\frac{e^{-(s/a)(L-x)}}{s} - \frac{e^{-(s/a)(3L-x)}}{s} + \frac{e^{-(s/a)(5L-x)}}{s} - \dots \right] \\ &\quad - \frac{aF_0}{E} \left[\frac{e^{-(s/a)(L+x)}}{s} - \frac{e^{-(s/a)(3L+x)}}{s} + \frac{e^{-(s/a)(5L+x)}}{s} - \dots \right] \\ &= \frac{aF_0}{E} \sum_{n=0}^{\infty} (-1)^n \left[\frac{e^{-(s/a)(2nL+L-x)}}{s} - \frac{e^{-(s/a)(2nL+L+x)}}{s} \right] \end{aligned}$$

and

$$\begin{aligned} u(x, t) &= \frac{aF_0}{E} \sum_{n=0}^{\infty} (-1)^n \left[\mathcal{L}^{-1} \left\{ \frac{e^{-(s/a)(2nL+L-x)}}{s} \right\} - \mathcal{L}^{-1} \left\{ \frac{e^{-(s/a)(2nL+L+x)}}{s} \right\} \right] \\ &= \frac{aF_0}{E} \sum_{n=0}^{\infty} (-1)^n \left[\left(t - \frac{2nL+L-x}{a} \right) \mathcal{U} \left(t - \frac{2nL+L-x}{a} \right) \right. \\ &\quad \left. - \left(t - \frac{2nL+L+x}{a} \right) \mathcal{U} \left(t - \frac{2nL+L+x}{a} \right) \right]. \end{aligned}$$

8. We use

$$U(x, s) = c_1 e^{-(x/a)s} + c_2 e^{(x/a)s} - \frac{v_0}{s^2}.$$

Now $\lim_{x \rightarrow \infty} \frac{dU}{dx} = 0$ implies $c_2 = 0$, and $U(0, s) = 0$ then gives $c_1 = v_0/s^2$. Hence

$$U(x, s) = \frac{v_0}{s^2} e^{-(x/a)s} - \frac{v_0}{s^2}$$

and

$$u(x, t) = v_0 \left(t - \frac{x}{a} \right) \mathcal{U} \left(t - \frac{x}{a} \right) - v_0 t.$$

9. Transforming the partial differential equation gives

$$\frac{d^2 U}{dx^2} - s^2 U = -s x e^{-x}.$$

Using undetermined coefficients we obtain

$$U(x, s) = c_1 e^{-sx} + c_2 e^{sx} - \frac{2s}{(s^2 - 1)^2} e^{-x} + \frac{s}{s^2 - 1} x e^{-x}.$$

The transformed boundary conditions $\lim_{x \rightarrow \infty} U(x, s) = 0$ and $U(0, s) = 0$ give, in turn, $c_2 = 0$ and $c_1 = 2s/(s^2 - 1)^2$. Therefore

$$U(x, s) = \frac{2s}{(s^2 - 1)^2} e^{-sx} - \frac{2s}{(s^2 - 1)^2} e^{-x} + \frac{s}{s^2 - 1} x e^{-x}.$$

From entries (13) and (26) in the table we obtain

$$\begin{aligned} u(x, t) &= \mathcal{L}^{-1} \left\{ \frac{2s}{(s^2 - 1)^2} e^{-sx} - \frac{2s}{(s^2 - 1)^2} e^{-x} + \frac{s}{s^2 - 1} x e^{-x} \right\} \\ &= 2(t - x) \sinh(t - x) \mathcal{U}(t - x) - t e^{-x} \sinh t + x e^{-x} \cosh t. \end{aligned}$$

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10. We use

$$U(x, s) = c_1 e^{-xs} + c_2 e^{xs} + \frac{s}{s^2 - 1} e^{-x}.$$

Now $\lim_{x \rightarrow \infty} u(x, t) = 0$ implies $\lim_{x \rightarrow \infty} U(x, s) = 0$, so we define $c_2 = 0$. Then

$$U(x, s) = c_1 e^{-xs} + \frac{s}{s^2 - 1} e^{-x}.$$

Finally, $U(0, s) = 1/s$ gives $c_1 = 1/s - s/(s^2 - 1)$. Thus

$$U(x, s) = \frac{1}{s} - \frac{s}{s^2 - 1} e^{-xs} + \frac{s}{s^2 - 1} e^{-x}$$

and

$$\begin{aligned} u(x, t) &= -\mathcal{L}^{-1} \left\{ \frac{s}{s^2 - 1} e^{-(x/a)s} \right\} + \mathcal{L}^{-1} \left\{ \frac{s}{s^2 - 1} \right\} e^{-x} \\ &= -\cosh \left(t - \frac{x}{a} \right) \mathcal{U} \left(t - \frac{x}{a} \right) + e^{-x} \cosh t. \end{aligned}$$

11. (a) We use

$$U(x, s) = c_1 e^{-\sqrt{s/k}x} + c_2 e^{\sqrt{s/k}x}.$$

Now $\lim_{x \rightarrow \infty} u(x, t) = 0$ implies $\lim_{x \rightarrow \infty} U(x, s) = 0$, so we define $c_2 = 0$. Then

$$U(x, s) = c_1 e^{-\sqrt{s/k}x}.$$

Finally, from $U(0, s) = u_0/s$ we obtain $c_1 = u_0/s$. Thus

$$U(x, s) = u_0 \frac{e^{-\sqrt{s/k}x}}{s}$$

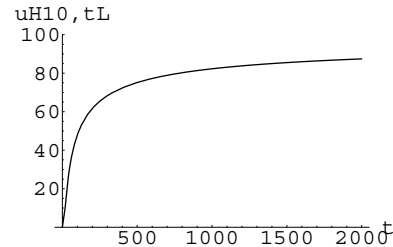
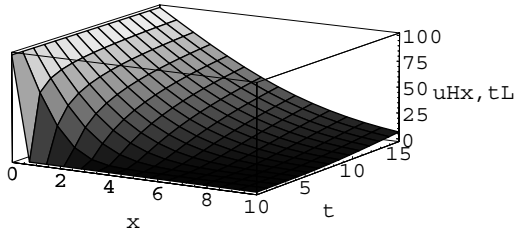
and

$$u(x, t) = u_0 \mathcal{L}^{-1} \left\{ \frac{e^{-\sqrt{s/k}x}}{s} \right\} = u_0 \mathcal{L}^{-1} \left\{ \frac{e^{-(x/\sqrt{k})\sqrt{s}}}{s} \right\} = u_0 \operatorname{erfc} \left(\frac{x}{2\sqrt{kt}} \right).$$

Since $\operatorname{erfc}(0) = 1$,

$$\lim_{t \rightarrow \infty} u(x, t) = \lim_{t \rightarrow \infty} u_0 \operatorname{erfc}(x/2\sqrt{kt}) = u_0.$$

(b)



12. (a) Transforming the partial differential equation and using the initial condition gives

$$k \frac{d^2 U}{dx^2} - sU = 0.$$

Since the domain of the variable x is an infinite interval we write the general solution of this differential equation as

$$U(x, s) = c_1 e^{-\sqrt{s/k}x} + c_2 e^{\sqrt{s/k}x}.$$

Transforming the boundary conditions gives $U'(0, s) = -A/s$ and $\lim_{x \rightarrow \infty} U(x, s) = 0$. Hence we find $c_2 = 0$ and $c_1 = A\sqrt{k}/s\sqrt{s}$. From

$$U(x, s) = A\sqrt{k} \frac{e^{-\sqrt{s/k}x}}{s\sqrt{s}}$$

we see that

$$u(x, t) = A\sqrt{k} \mathcal{L}^{-1} \left\{ \frac{e^{-\sqrt{s/k}x}}{s\sqrt{s}} \right\}.$$

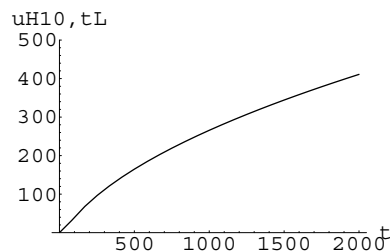
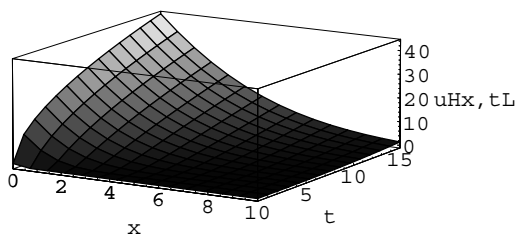
With the identification $a = x/\sqrt{k}$ it follows from entry 49 of the table in Appendix III that

$$\begin{aligned} u(x, t) &= A\sqrt{k} \left\{ 2\sqrt{\frac{t}{\pi}} e^{-x^2/4kt} - \frac{x}{\sqrt{k}} \operatorname{erfc}\left(\frac{x}{2\sqrt{kt}}\right) \right\} \\ &= 2A\sqrt{\frac{kt}{\pi}} e^{-x^2/4kt} - Ax \operatorname{erfc}\left(\frac{x}{2\sqrt{kt}}\right). \end{aligned}$$

Since $\operatorname{erfc}(0) = 1$,

$$\lim_{t \rightarrow \infty} u(x, t) = \lim_{t \rightarrow \infty} \left(2A\sqrt{\frac{kt}{\pi}} e^{-x^2/4kt} - Ax \operatorname{erfc}\left(\frac{x}{2\sqrt{kt}}\right) \right) = \infty.$$

(b)



13. We use

$$U(x, s) = c_1 e^{-\sqrt{s}x} + c_2 e^{\sqrt{s}x} + \frac{u_1}{s}.$$

The condition $\lim_{x \rightarrow \infty} u(x, t) = u_1$ implies $\lim_{x \rightarrow \infty} U(x, s) = u_1/s$, so we define $c_2 = 0$. Then

$$U(x, s) = c_1 e^{-\sqrt{s}x} + \frac{u_1}{s}.$$

From $U(0, s) = u_0/s$ we obtain $c_1 = (u_0 - u_1)/s$. Thus

$$U(x, s) = (u_0 - u_1) \frac{e^{-\sqrt{s}x}}{s} + \frac{u_1}{s}$$

and

$$u(x, t) = (u_0 - u_1) \mathcal{L}^{-1} \left\{ \frac{e^{-x\sqrt{s}}}{s} \right\} + u_1 \mathcal{L}^{-1} \left\{ \frac{1}{s} \right\} = (u_0 - u_1) \operatorname{erfc}\left(\frac{x}{2\sqrt{t}}\right) + u_1.$$

14. We use

$$U(x, s) = c_1 e^{-\sqrt{s}x} + c_2 e^{\sqrt{s}x} + \frac{u_1 x}{s}.$$

The condition $\lim_{x \rightarrow \infty} u(x, t)/x = u_1$ implies $\lim_{x \rightarrow \infty} U(x, s)/x = u_1/s$, so we define $c_2 = 0$. Then

$$U(x, s) = c_1 e^{-\sqrt{s}x} + \frac{u_1 x}{s}.$$

From $U(0, s) = u_0/s$ we obtain $c_1 = u_0/s$. Hence

$$U(x, s) = u_0 \frac{e^{-\sqrt{s}x}}{s} + \frac{u_1 x}{s}$$

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and

$$u(x, t) = u_0 \mathcal{L}^{-1} \left\{ \frac{e^{-x\sqrt{s}}}{s} \right\} + u_1 x \mathcal{L}^{-1} \left\{ \frac{1}{s} \right\} = u_0 \operatorname{erfc} \left(\frac{x}{2\sqrt{t}} \right) + u_1 x.$$

15. We use

$$U(x, s) = c_1 e^{-\sqrt{s}x} + c_2 e^{\sqrt{s}x} + \frac{u_0}{s}.$$

The condition $\lim_{x \rightarrow \infty} u(x, t) = u_0$ implies $\lim_{x \rightarrow \infty} U(x, s) = u_0/s$, so we define $c_2 = 0$. Then

$$U(x, s) = c_1 e^{-\sqrt{s}x} + \frac{u_0}{s}.$$

The transform of the remaining boundary conditions gives

$$\left. \frac{dU}{dx} \right|_{x=0} = U(0, s).$$

This condition yields $c_1 = -u_0/s(\sqrt{s} + 1)$. Thus

$$U(x, s) = -u_0 \frac{e^{-\sqrt{s}x}}{s(\sqrt{s} + 1)} + \frac{u_0}{s}$$

and

$$\begin{aligned} u(x, t) &= -u_0 \mathcal{L}^{-1} \left\{ \frac{e^{-x\sqrt{s}}}{s(\sqrt{s} + 1)} \right\} + u_0 \mathcal{L}^{-1} \left\{ \frac{1}{s} \right\} \\ &= u_0 e^{x+t} \operatorname{erfc} \left(\sqrt{t} + \frac{x}{2\sqrt{t}} \right) - u_0 \operatorname{erfc} \left(\frac{x}{2\sqrt{t}} \right) + u_0 \end{aligned} \quad \boxed{\text{By (5) in the table in 15.1.}}$$

16. We use

$$U(x, s) = c_1 e^{-\sqrt{s}x} + c_2 e^{\sqrt{s}x}.$$

The condition $\lim_{x \rightarrow \infty} u(x, t) = 0$ implies $\lim_{x \rightarrow \infty} U(x, s) = 0$, so we define $c_2 = 0$. Hence

$$U(x, s) = c_1 e^{-\sqrt{s}x}.$$

The remaining boundary condition transforms into

$$\left. \frac{dU}{dx} \right|_{x=0} = U(0, s) - \frac{50}{s}.$$

This condition gives $c_1 = 50/s(\sqrt{s} + 1)$. Therefore

$$U(x, s) = 50 \frac{e^{-\sqrt{s}x}}{s(\sqrt{s} + 1)}$$

and

$$u(x, t) = 50 \mathcal{L}^{-1} \left\{ \frac{e^{-x\sqrt{s}}}{s(\sqrt{s} + 1)} \right\} = -50 e^{x+t} \operatorname{erfc} \left(\sqrt{t} + \frac{x}{2\sqrt{t}} \right) + 50 \operatorname{erfc} \left(\frac{x}{2\sqrt{t}} \right).$$

17. We use

$$U(x, s) = c_1 e^{-\sqrt{s}x} + c_2 e^{\sqrt{s}x}.$$

The condition $\lim_{x \rightarrow \infty} u(x, t) = 0$ implies $\lim_{x \rightarrow \infty} U(x, s) = 0$, so we define $c_2 = 0$. Hence

$$U(x, s) = c_1 e^{-\sqrt{s}x}.$$

The transform of $u(0, t) = f(t)$ is $U(0, s) = F(s)$. Therefore

$$U(x, s) = F(s) e^{-\sqrt{s}x}$$

and

$$u(x, t) = \mathcal{L}^{-1} \left\{ F(s) e^{-x\sqrt{s}} \right\} = \frac{x}{2\sqrt{\pi}} \int_0^t \frac{f(t-\tau) e^{-x^2/4\tau}}{\tau^{3/2}} d\tau.$$

18. We use

$$U(x, s) = c_1 e^{-\sqrt{s}x} + c_2 e^{\sqrt{s}x}.$$

The condition $\lim_{x \rightarrow \infty} u(x, t) = 0$ implies $\lim_{x \rightarrow \infty} U(x, s) = 0$, so we define $c_2 = 0$. Then $U(x, s) = c_1 e^{-\sqrt{s}x}$. The transform of the remaining boundary condition gives

$$\left. \frac{dU}{dx} \right|_{x=0} = -F(s)$$

where $F(s) = \mathcal{L}\{f(t)\}$. This condition yields $c_1 = F(s)/\sqrt{s}$. Thus

$$U(x, s) = F(s) \frac{e^{-\sqrt{s}x}}{\sqrt{s}}.$$

Using entry (44) of the table and the convolution theorem we obtain

$$u(x, t) = \mathcal{L}^{-1} \left\{ F(s) \cdot \frac{e^{-\sqrt{s}x}}{\sqrt{s}} \right\} = \frac{1}{\sqrt{\pi}} \int_0^t f(\tau) \frac{e^{-x^2/4(t-\tau)}}{\sqrt{t-\tau}} d\tau.$$

19. Transforming the partial differential equation gives

$$\frac{d^2 U}{dx^2} - sU = -60.$$

Using undetermined coefficients we obtain

$$U(x, s) = c_1 e^{-\sqrt{s}x} + c_2 e^{\sqrt{s}x} + \frac{60}{s}.$$

The condition $\lim_{x \rightarrow \infty} u(x, t) = 60$ implies $\lim_{x \rightarrow \infty} U(x, s) = 60/s$, so we define $c_2 = 0$. The transform of the remaining boundary condition gives

$$U(0, s) = \frac{60}{s} + \frac{40}{s} e^{-2s}.$$

This condition yields $c_1 = \frac{40}{s} e^{-2s}$. Thus

$$U(x, s) = \frac{60}{s} + 40e^{-2s} \frac{e^{-\sqrt{s}x}}{s}.$$

Using entry (46) of the table in Appendix III and the second translation theorem we obtain

$$u(x, t) = \mathcal{L}^{-1} \left\{ \frac{60}{s} + 40e^{-2s} \frac{e^{-\sqrt{s}x}}{s} \right\} = 60 + 40 \operatorname{erfc} \left(\frac{x}{2\sqrt{t-2}} \right) \mathcal{U}(t-2).$$

20. The solution of the transformed equation

$$\frac{d^2 U}{dx^2} - sU = -100$$

by undetermined coefficients is

$$U(x, s) = c_1 e^{\sqrt{s}x} + c_2 e^{-\sqrt{s}x} + \frac{100}{s}.$$

From the fact that $\lim_{x \rightarrow \infty} U(x, s) = 100/s$ we see that $c_1 = 0$. Thus

$$U(x, s) = c_2 e^{-\sqrt{s}x} + \frac{100}{s}. \quad (1)$$

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Now the transform of the boundary condition at $x = 0$ is

$$U(0, s) = 20 \left[\frac{1}{s} - \frac{1}{s} e^{-s} \right].$$

It follows from (1) that

$$\frac{20}{s} - \frac{20}{s} e^{-s} = c_2 + \frac{100}{s} \quad \text{or} \quad c_2 = -\frac{80}{s} - \frac{20}{s} e^{-s}$$

and so

$$\begin{aligned} U(x, s) &= \left(-\frac{80}{s} - \frac{20}{s} e^{-s} \right) e^{-\sqrt{s}x} + \frac{100}{s} \\ &= \frac{100}{s} - \frac{80}{s} e^{-\sqrt{s}x} - \frac{20}{s} e^{-\sqrt{s}x} e^{-s}. \end{aligned}$$

Thus

$$\begin{aligned} u(x, t) &= 100 \mathcal{L}^{-1} \left\{ \frac{1}{s} \right\} - 80 \mathcal{L}^{-1} \left\{ \frac{e^{-\sqrt{s}x}}{s} \right\} - 20 \mathcal{L}^{-1} \left\{ \frac{e^{-\sqrt{s}x}}{s} e^{-s} \right\} \\ &= 100 - 80 \operatorname{erfc}(x/2\sqrt{t}) - 20 \operatorname{erfc}(x/2\sqrt{t-1}) \mathcal{U}(t-1). \end{aligned}$$

21. Transforming the partial differential equation gives

$$\frac{d^2 U}{dx^2} - sU = 0$$

and so

$$U(x, s) = c_1 e^{-\sqrt{s}x} + c_2 e^{\sqrt{s}x}.$$

The condition $\lim_{x \rightarrow -\infty} u(x, t) = 0$ implies $\lim_{x \rightarrow -\infty} U(x, s) = 0$, so we define $c_1 = 0$. The transform of the remaining boundary condition gives

$$\left. \frac{dU}{dx} \right|_{x=1} = \frac{100}{s} - U(1, s).$$

This condition yields

$$c_2 \sqrt{s} e^{\sqrt{s}} = \frac{100}{s} - c_2 e^{\sqrt{s}}$$

from which it follows that

$$c_2 = \frac{100}{s(\sqrt{s} + 1)} e^{-\sqrt{s}}.$$

Thus

$$U(x, s) = 100 \frac{e^{-(1-x)\sqrt{s}}}{s(\sqrt{s} + 1)}.$$

Using entry (49) of the table in Appendix III we obtain

$$u(x, t) = 100 \mathcal{L}^{-1} \left\{ \frac{e^{-(1-x)\sqrt{s}}}{s(\sqrt{s} + 1)} \right\} = 100 \left[-e^{1-x+t} \operatorname{erfc} \left(\sqrt{t} + \frac{1-x}{\sqrt{t}} \right) + \operatorname{erfc} \left(\frac{1-x}{2\sqrt{t}} \right) \right].$$

22. Transforming the partial differential equation gives

$$k \frac{d^2 U}{dx^2} - sU = -\frac{r}{s}.$$

Using undetermined coefficients we obtain

$$U(x, s) = c_1 e^{-\sqrt{s/k}x} + c_2 e^{\sqrt{s/k}x} + \frac{r}{s^2}.$$

The condition $\lim_{x \rightarrow \infty} \frac{\partial u}{\partial x} = 0$ implies $\lim_{x \rightarrow \infty} \frac{dU}{dx} = 0$, so we define $c_2 = 0$. The transform of the remaining boundary condition gives $U(0, s) = 0$. This condition yields $c_1 = -r/s^2$. Thus

$$U(x, s) = r \left[\frac{1}{s^2} - \frac{e^{-\sqrt{s/k}x}}{s^2} \right].$$

Using entries (3) and (46) of the table in Appendix III and the convolution theorem we obtain

$$u(x, t) = r \mathcal{L}^{-1} \left\{ \frac{1}{s^2} - \frac{1}{s} \cdot \frac{e^{-\sqrt{s/k}x}}{s} \right\} = rt - r \int_0^t \operatorname{erfc} \left(\frac{x}{2\sqrt{k\tau}} \right) d\tau.$$

23. The solution of

$$\frac{d^2 U}{dx^2} - sU = -u_0 - u_0 \sin \frac{\pi}{L} x$$

is

$$U(x, s) = c_1 \cosh(\sqrt{s}x) + c_2 \sinh(\sqrt{s}x) + \frac{u_0}{s} + \frac{u_0}{s + \pi^2/L^2} \sin \frac{\pi}{L} x.$$

The transformed boundary conditions $U(0, s) = u_0/s$ and $U(L, s) = u_0/s$ give, in turn, $c_1 = 0$ and $c_2 = 0$. Therefore

$$U(x, s) = \frac{u_0}{s} + \frac{u_0}{s + \pi^2/L^2} \sin \frac{\pi}{L} x$$

and

$$u(x, t) = u_0 \mathcal{L}^{-1} \left\{ \frac{1}{s} \right\} + u_0 \mathcal{L}^{-1} \left\{ \frac{1}{s + \pi^2/L^2} \right\} \sin \frac{\pi}{L} x = u_0 + u_0 e^{-\pi^2 t/L^2} \sin \frac{\pi}{L} x.$$

24. The transform of the partial differential equation is

$$k \frac{d^2 U}{dx^2} - hU + h \frac{u_m}{s} = sU - u_0$$

or

$$k \frac{d^2 U}{dx^2} - (h + s)U = -h \frac{u_m}{s} - u_0.$$

By undetermined coefficients we find

$$U(x, s) = c_1 e^{\sqrt{(h+s)/k}x} + c_2 e^{-\sqrt{(h+s)/k}x} + \frac{hu_m + u_0 s}{s(s + h)}.$$

The transformed boundary conditions are $U'(0, s) = 0$ and $U'(L, s) = 0$. These conditions imply $c_1 = 0$ and $c_2 = 0$. By partial fractions we then get

$$U(x, s) = \frac{hu_m + u_0 s}{s(s + h)} = \frac{u_m}{s} - \frac{u_m}{s + h} + \frac{u_0}{s + h}.$$

Therefore,

$$u(x, t) = u_m \mathcal{L}^{-1} \left\{ \frac{1}{s} \right\} - u_m \mathcal{L}^{-1} \left\{ \frac{1}{s + h} \right\} + u_0 \mathcal{L}^{-1} \left\{ \frac{1}{s + h} \right\} = u_m - u_m e^{-ht} + u_0 e^{-ht}.$$

25. We use

$$U(x, s) = c_1 \cosh \sqrt{\frac{s}{k}} x + c_2 \sinh \sqrt{\frac{s}{k}} x + \frac{u_0}{s}.$$

The transformed boundary conditions $\left. \frac{dU}{dx} \right|_{x=0} = 0$ and $U(1, s) = 0$ give, in turn, $c_2 = 0$ and

Exercises 15.2

$c_1 = -u_0/s \cosh \sqrt{s/k}$. Therefore

$$\begin{aligned}
 U(x, s) &= \frac{u_0}{s} - \frac{u_0 \cosh \sqrt{s/k} x}{s \cosh \sqrt{s/k}} = \frac{u_0}{s} - u_0 \frac{e^{\sqrt{s/k} x} + e^{-\sqrt{s/k} x}}{s(e^{\sqrt{s/k}} + e^{-\sqrt{s/k}})} \\
 &= \frac{u_0}{s} - u_0 \frac{e^{\sqrt{s/k}(x-1)} + e^{-\sqrt{s/k}(x+1)}}{s(1 + e^{-2\sqrt{s/k}})} \\
 &= \frac{u_0}{s} - u_0 \left[\frac{e^{-\sqrt{s/k}(1-x)}}{s} - \frac{e^{-\sqrt{s/k}(3-x)}}{s} + \frac{e^{-\sqrt{s/k}(5-x)}}{s} - \dots \right] \\
 &\quad - u_0 \left[\frac{e^{-\sqrt{s/k}(1+x)}}{s} - \frac{e^{-\sqrt{s/k}(3+x)}}{s} + \frac{e^{-\sqrt{s/k}(5+x)}}{s} - \dots \right] \\
 &= \frac{u_0}{s} - u_0 \sum_{n=0}^{\infty} (-1)^n \left[\frac{e^{-(2n+1-x)\sqrt{s/k}}}{s} + \frac{e^{-(2n+1+x)\sqrt{s/k}}}{s} \right]
 \end{aligned}$$

and

$$\begin{aligned}
 u(x, t) &= u_0 \mathcal{L}^{-1} \left\{ \frac{1}{s} \right\} - u_0 \sum_{n=0}^{\infty} (-1)^n \left[\mathcal{L}^{-1} \left\{ \frac{e^{-(2n+1-x)\sqrt{s/k}}}{s} \right\} - \mathcal{L}^{-1} \left\{ \frac{e^{-(2n+1+x)\sqrt{s/k}}}{s} \right\} \right] \\
 &= u_0 - u_0 \sum_{n=0}^{\infty} (-1)^n \left[\operatorname{erfc} \left(\frac{2n+1-x}{2\sqrt{kt}} \right) - \operatorname{erfc} \left(\frac{2n+1+x}{2\sqrt{kt}} \right) \right].
 \end{aligned}$$

26. We use

$$c(x, s) = c_1 \cosh \sqrt{\frac{s}{D}} x + c_2 \sinh \sqrt{\frac{s}{D}} x.$$

The transform of the two boundary conditions are $c(0, s) = c_0/s$ and $c(1, s) = c_0/s$. From these conditions we obtain $c_1 = c_0/s$ and

$$c_2 = c_0(1 - \cosh \sqrt{s/D})/s \sinh \sqrt{s/D}.$$

Therefore

$$\begin{aligned}
 c(x, s) &= c_0 \left[\frac{\cosh \sqrt{s/D} x}{s} + \frac{(1 - \cosh \sqrt{s/D})}{s \sinh \sqrt{s/D}} \sinh \sqrt{s/D} x \right] \\
 &= c_0 \left[\frac{\sinh \sqrt{s/D} (1-x)}{s \sinh \sqrt{s/D}} + \frac{\sinh \sqrt{s/D} x}{s \sinh \sqrt{s/D}} \right] \\
 &= c_0 \left[\frac{e^{\sqrt{s/D}(1-x)} - e^{-\sqrt{s/D}(1-x)}}{s(e^{\sqrt{s/D}} - e^{-\sqrt{s/D}})} + \frac{e^{\sqrt{s/D} x} - e^{-\sqrt{s/D} x}}{s(e^{\sqrt{s/D}} - e^{-\sqrt{s/D}})} \right] \\
 &= c_0 \left[\frac{e^{-\sqrt{s/D} x} - e^{-\sqrt{s/D}(2-x)}}{s(1 - e^{-2\sqrt{s/D}})} + \frac{e^{\sqrt{s/D}(x-1)} - e^{-\sqrt{s/D}(x+1)}}{s(1 - e^{-2\sqrt{s/D}})} \right] \\
 &= c_0 \frac{(e^{-\sqrt{s/D} x} - e^{-\sqrt{s/D}(2-x)})}{s} \left(1 + e^{-2\sqrt{s/D}} + e^{-4\sqrt{s/D}} + \dots \right) \\
 &\quad + c_0 \frac{(e^{\sqrt{s/D}(x-1)} - e^{-\sqrt{s/D}(x+1)})}{s} \left(1 + e^{-2\sqrt{s/D}} + e^{-4\sqrt{s/D}} + \dots \right)
 \end{aligned}$$

$$\begin{aligned}
&= c_0 \sum_{n=0}^{\infty} \left[\frac{e^{-(2n+x)\sqrt{s/D}}}{s} - \frac{e^{-(2n+2-x)\sqrt{s/D}}}{s} \right] \\
&\quad + c_0 \sum_{n=0}^{\infty} \left[\frac{e^{-(2n+1-x)\sqrt{s/D}}}{s} - \frac{e^{-(2n+1+x)\sqrt{s/D}}}{s} \right]
\end{aligned}$$

and

$$\begin{aligned}
c(x, t) &= c_0 \sum_{n=0}^{\infty} \left[\mathcal{L}^{-1} \left\{ \frac{e^{-\frac{(2n+x)\sqrt{s}}{\sqrt{D}}}}{s} \right\} - \mathcal{L}^{-1} \left\{ \frac{e^{-\frac{(2n+2-x)\sqrt{s}}{\sqrt{D}}}}{s} \right\} \right] \\
&\quad + c_0 \sum_{n=0}^{\infty} \left[\mathcal{L}^{-1} \left\{ \frac{e^{-\frac{(2n+1-x)\sqrt{s}}{\sqrt{D}}}}{s} \right\} - \mathcal{L}^{-1} \left\{ \frac{e^{-\frac{(2n+1+x)\sqrt{s}}{\sqrt{D}}}}{s} \right\} \right] \\
&= c_0 \sum_{n=0}^{\infty} \left[\operatorname{erfc} \left(\frac{2n+x}{2\sqrt{Dt}} \right) - \operatorname{erfc} \left(\frac{2n+2-x}{2\sqrt{Dt}} \right) \right] \\
&\quad + c_0 \sum_{n=0}^{\infty} \left[\operatorname{erfc} \left(\frac{2n+1-x}{2\sqrt{Dt}} \right) - \operatorname{erfc} \left(\frac{2n+1+x}{2\sqrt{Dt}} \right) \right].
\end{aligned}$$

Now using $\operatorname{erfc}(x) = 1 - \operatorname{erf}(x)$ we get

$$\begin{aligned}
c(x, t) &= c_0 \sum_{n=0}^{\infty} \left[\operatorname{erf} \left(\frac{2n+2-x}{2\sqrt{Dt}} \right) - \operatorname{erf} \left(\frac{2n+x}{2\sqrt{Dt}} \right) \right] \\
&\quad + c_0 \sum_{n=0}^{\infty} \left[\operatorname{erf} \left(\frac{2n+1+x}{2\sqrt{Dt}} \right) - \operatorname{erf} \left(\frac{2n+1-x}{2\sqrt{Dt}} \right) \right].
\end{aligned}$$

27. We use

$$U(x, s) = c_1 e^{-\sqrt{RCs+RG}x} + c_2 e^{\sqrt{RCs+RG}x} + \frac{Cu_0}{Cs+G}.$$

The condition $\lim_{x \rightarrow \infty} \partial u / \partial x = 0$ implies $\lim_{x \rightarrow \infty} dU/dx = 0$, so we define $c_2 = 0$. Applying $U(0, s) = 0$ to

$$U(x, s) = c_1 e^{-\sqrt{RCs+RG}x} + \frac{Cu_0}{Cs+G}$$

gives $c_1 = -Cu_0/(Cs+G)$. Therefore

$$U(x, s) = -Cu_0 \frac{e^{-\sqrt{RCs+RG}x}}{Cs+G} + \frac{Cu_0}{Cs+G}$$

and

$$\begin{aligned}
u(x, t) &= u_0 \mathcal{L}^{-1} \left\{ \frac{1}{s+G/C} \right\} - u_0 \mathcal{L}^{-1} \left\{ \frac{e^{-x\sqrt{RC}\sqrt{s+G/C}}}{s+G/C} \right\} \\
&= u_0 e^{-Gt/C} - u_0 e^{-Gt/C} \operatorname{erfc} \left(\frac{x\sqrt{RC}}{2\sqrt{t}} \right) \\
&= u_0 e^{-Gt/C} \left[1 - \operatorname{erfc} \left(\frac{x}{2} \sqrt{\frac{RC}{t}} \right) \right] \\
&= u_0 e^{-Gt/C} \operatorname{erf} \left(\frac{x}{2} \sqrt{\frac{RC}{t}} \right).
\end{aligned}$$

Exercises 15.2

28. We use

$$U(x, s) = c_1 e^{-\sqrt{s+h}x} + c_2 e^{\sqrt{s+h}x}.$$

The condition $\lim_{x \rightarrow \infty} u(x, t) = 0$ implies $\lim_{x \rightarrow \infty} U(x, s) = 0$, so we take $c_2 = 0$. Therefore

$$U(x, s) = c_1 e^{-\sqrt{s+h}x}.$$

The Laplace transform of $u(0, t) = u_0$ is $U(0, s) = u_0/s$ and so

$$U(x, s) = u_0 \frac{e^{-\sqrt{s+h}x}}{s}$$

and

$$u(x, t) = u_0 \mathcal{L}^{-1} \left\{ \frac{e^{-\sqrt{s+h}x}}{s} \right\} = u_0 \mathcal{L}^{-1} \left\{ \frac{1}{s} e^{-\sqrt{s+h}x} \right\}.$$

From the first translation theorem,

$$\mathcal{L}^{-1} \left\{ e^{-\sqrt{s+h}x} \right\} = e^{-ht} \mathcal{L}^{-1} \left\{ e^{-x\sqrt{s}} \right\} = e^{-ht} \frac{x}{2\sqrt{\pi t^3}} e^{-x^2/4t}.$$

Thus, from the convolution theorem we obtain

$$u(x, s) = \frac{u_0 x}{2\sqrt{\pi}} \int_0^t \frac{e^{-h\tau - x^2/4\tau}}{\tau^{3/2}} d\tau.$$

29. (a) Letting $C(x, s) = \mathcal{L} \{c(x, t)\}$ we obtain

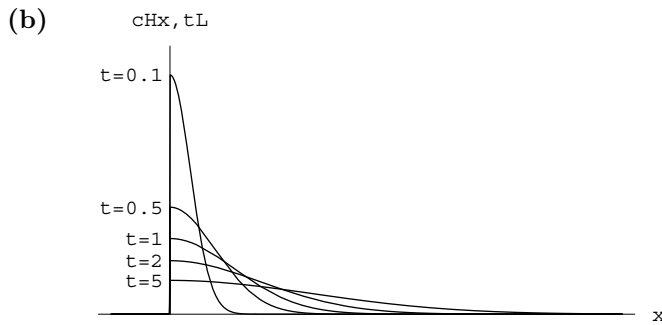
$$\frac{d^2 C}{dx^2} - \frac{s}{k} C = 0 \quad \text{subject to} \quad \frac{dC}{dx} \Big|_{x=0} = -A.$$

The solution of this initial-value problem is

$$C(x, s) = A\sqrt{k} \frac{e^{-(x/\sqrt{k})\sqrt{s}}}{\sqrt{s}},$$

so that

$$c(x, t) = A\sqrt{\frac{k}{\pi t}} e^{-x^2/4kt}.$$



(c) $\int_0^\infty c(x, t) dx = Ak \operatorname{erf} \frac{x}{2\sqrt{kt}} \Big|_0^\infty = Ak(1 - 0) = Ak$

30. (a) We use

$$U(x, s) = c_1 e^{-(s/a)x} + c_2 e^{(s/a)x} + \frac{v_0^2 F_0}{(a^2 - v_0^2)s^2} e^{-(s/v_0)x}.$$

The condition $\lim_{x \rightarrow \infty} u(x, t) = 0$ implies $\lim_{x \rightarrow \infty} U(x, s) = 0$, so we must define $c_2 = 0$. Consequently

$$U(x, s) = c_1 e^{-(s/a)x} + \frac{v_0^2 F_0}{(a^2 - v_0^2)s^2} e^{-(s/v_0)x}.$$

The remaining boundary condition transforms into $U(0, s) = 0$. From this we find

$$c_1 = -v_0^2 F_0 / (a^2 - v_0^2) s^2.$$

Therefore, by the second translation theorem

$$U(x, s) = -\frac{v_0^2 F_0}{(a^2 - v_0^2) s^2} e^{-(s/a)x} + \frac{v_0^2 F_0}{(a^2 - v_0^2) s^2} e^{-(s/v_0)x}$$

and

$$\begin{aligned} u(x, t) &= \frac{v_0^2 F_0}{a^2 - v_0^2} \left[\mathcal{L}^{-1} \left\{ \frac{e^{-(x/v_0)s}}{s^2} \right\} - \mathcal{L}^{-1} \left\{ \frac{e^{-(x/a)s}}{s^2} \right\} \right] \\ &= \frac{v_0^2 F_0}{a^2 - v_0^2} \left[\left(t - \frac{x}{v_0} \right) \mathcal{U} \left(t - \frac{x}{v_0} \right) - \left(t - \frac{x}{a} \right) \mathcal{U} \left(t - \frac{x}{a} \right) \right]. \end{aligned}$$

(b) In the case when $v_0 = a$ the solution of the transformed equation is

$$U(x, s) = c_1 e^{-(s/a)x} + c_2 e^{(s/a)x} - \frac{F_0}{2as} x e^{-(s/a)x}.$$

The usual analysis then leads to $c_1 = 0$ and $c_2 = 0$. Therefore

$$U(x, s) = -\frac{F_0}{2as} x e^{-(s/a)x}$$

and

$$u(x, t) = -\frac{x F_0}{2a} \mathcal{L}^{-1} \left\{ \frac{e^{-(x/a)s}}{s} \right\} = -\frac{x F_0}{2a} \mathcal{U} \left(t - \frac{x}{a} \right).$$

Exercises 15.3

1. From formulas (5) and (6) in the text,

$$A(\alpha) = \int_{-1}^0 (-1) \cos \alpha x \, dx + \int_0^1 (2) \cos \alpha x \, dx = -\frac{\sin \alpha}{\alpha} + 2 \frac{\sin \alpha}{\alpha} = \frac{\sin \alpha}{\alpha}$$

and

$$\begin{aligned} B(\alpha) &= \int_{-1}^0 (-1) \sin \alpha x \, dx + \int_0^1 (2) \sin \alpha x \, dx \\ &= \frac{1 - \cos \alpha}{\alpha} - 2 \frac{\cos \alpha - 1}{\alpha} = \frac{3(1 - \cos \alpha)}{\alpha}. \end{aligned}$$

Hence

$$f(x) = \frac{1}{\pi} \int_0^\infty \frac{\sin \alpha \cos \alpha x + 3(1 - \cos \alpha) \sin \alpha x}{\alpha} d\alpha.$$

2. From formulas (5) and (6) in the text,

$$A(\alpha) = \int_\pi^{2\pi} 4 \cos \alpha x \, dx = 4 \frac{\sin 2\pi\alpha - \sin \pi\alpha}{\alpha}$$

and

$$B(\alpha) = \int_\pi^{2\pi} 4 \sin \alpha x \, dx = 4 \frac{\cos \pi\alpha - \cos 2\pi\alpha}{\alpha}.$$

Exercises 15.3

Hence

$$\begin{aligned}
 f(x) &= \frac{4}{\pi} \int_0^\infty \frac{(\sin 2\pi\alpha - \sin \pi\alpha) \cos \alpha x + (\cos \pi\alpha - \cos 2\pi\alpha) \sin \alpha x}{\alpha} d\alpha \\
 &= \frac{4}{\pi} \int_0^\infty \frac{\sin 2\pi\alpha \cos \alpha x - \cos 2\pi\alpha \sin \alpha x - \sin \pi\alpha \cos \alpha x + \cos \pi\alpha \sin \alpha x}{\alpha} d\alpha \\
 &= \frac{4}{\pi} \int_0^\infty \frac{\sin \alpha(2\pi - x) - \sin \alpha(\pi - x)}{\alpha} d\alpha.
 \end{aligned}$$

3. From formulas (5) and (6) in the text,

$$\begin{aligned}
 A(\alpha) &= \int_0^3 x \cos \alpha x dx = \left. \frac{x \sin \alpha x}{\alpha} \right|_0^3 - \frac{1}{\alpha} \int_0^3 \sin \alpha x dx \\
 &= \left. \frac{3 \sin 3\alpha}{\alpha} + \frac{\cos \alpha x}{\alpha^2} \right|_0^3 = \frac{3\alpha \sin 3\alpha + \cos 3\alpha - 1}{\alpha^2}
 \end{aligned}$$

and

$$\begin{aligned}
 B(\alpha) &= \int_0^3 x \sin \alpha x dx = -\left. \frac{x \cos \alpha x}{\alpha} \right|_0^3 + \frac{1}{\alpha} \int_0^3 \cos \alpha x dx \\
 &= -\left. \frac{3 \cos 3\alpha}{\alpha} + \frac{\sin \alpha x}{\alpha^2} \right|_0^3 = \frac{\sin 3\alpha - 3\alpha \cos 3\alpha}{\alpha^2}.
 \end{aligned}$$

Hence

$$\begin{aligned}
 f(x) &= \frac{1}{\pi} \int_0^\infty \frac{(3\alpha \sin 3\alpha + \cos 3\alpha - 1) \cos \alpha x + (\sin 3\alpha - 3\alpha \cos 3\alpha) \sin \alpha x}{\alpha^2} d\alpha \\
 &= \frac{1}{\pi} \int_0^\infty \frac{3\alpha(\sin 3\alpha \cos \alpha x - \cos 3\alpha \sin \alpha x) + \cos 3\alpha \cos \alpha x + \sin 3\alpha \sin \alpha x - \cos \alpha x}{\alpha^2} d\alpha \\
 &= \frac{1}{\pi} \int_0^\infty \frac{3\alpha \sin \alpha(3 - x) + \cos \alpha(3 - x) - \cos \alpha x}{\alpha^2} d\alpha.
 \end{aligned}$$

4. From formulas (5) and (6) in the text,

$$\begin{aligned}
 A(\alpha) &= \int_{-\infty}^\infty f(x) \cos \alpha x dx \\
 &= \int_{-\infty}^0 0 \cdot \cos \alpha x dx + \int_0^\pi \sin x \cos \alpha x dx + \int_\pi^\infty 0 \cdot \cos \alpha x dx \\
 &= \frac{1}{2} \int_0^\pi [\sin(1 + \alpha)x + \sin(1 - \alpha)x] dx \\
 &= \frac{1}{2} \left[-\frac{\cos(1 + \alpha)x}{1 + \alpha} - \frac{\cos(1 - \alpha)x}{1 - \alpha} \right]_0^\pi \\
 &= -\frac{1}{2} \left[\frac{\cos(1 + \alpha)\pi - 1}{1 + \alpha} + \frac{\cos(1 - \alpha)\pi - 1}{1 - \alpha} \right] \\
 &= -\frac{1}{2} \left[\frac{\cos(1 + \alpha)\pi - \alpha \cos(1 + \alpha)\pi + \cos(1 - \alpha)\pi + \alpha \cos(1 - \alpha)\pi - 2}{1 - \alpha^2} \right] \\
 &= \frac{1 + \cos \alpha\pi}{1 - \alpha^2},
 \end{aligned}$$

and

$$\begin{aligned} B(\alpha) &= \int_0^\pi \sin x \sin \alpha x \, dx = \frac{1}{2} \int_0^\pi [\cos(1-\alpha)x - \cos(1+\alpha)x] \, dx \\ &= \frac{1}{2} \left[\frac{\sin(1-\alpha)\pi}{1-\alpha} - \frac{\sin(1+\alpha)\pi}{1+\alpha} \right] = \frac{\sin \alpha \pi}{1-\alpha^2}. \end{aligned}$$

Hence

$$\begin{aligned} f(x) &= \frac{1}{\pi} \int_0^\infty \frac{\cos \alpha x + \cos \alpha x \cos \alpha \pi + \sin \alpha x \sin \alpha \pi}{1-\alpha^2} \, d\alpha \\ &= \frac{1}{\pi} \int_0^\infty \frac{\cos \alpha x + \cos \alpha(x-\pi)}{1-\alpha^2} \, d\alpha. \end{aligned}$$

5. From formula (5) in the text,

$$A(\alpha) = \int_0^\infty e^{-x} \cos \alpha x \, dx.$$

Recall $\mathcal{L}\{\cos kt\} = s/(s^2 + k^2)$. If we set $s = 1$ and $k = \alpha$ we obtain

$$A(\alpha) = \frac{1}{1+\alpha^2}.$$

Now

$$B(\alpha) = \int_0^\infty e^{-x} \sin \alpha x \, dx.$$

Recall $\mathcal{L}\{\sin kt\} = k/(s^2 + k^2)$. If we set $s = 1$ and $k = \alpha$ we obtain

$$B(\alpha) = \frac{\alpha}{1+\alpha^2}.$$

Hence

$$f(x) = \frac{1}{\pi} \int_0^\infty \frac{\cos \alpha x + \alpha \sin \alpha x}{1+\alpha^2} \, d\alpha.$$

6. From formulas (5) and (6) in the text,

$$\begin{aligned} A(\alpha) &= \int_{-1}^1 e^x \cos \alpha x \, dx \\ &= \frac{e(\cos \alpha + \alpha \sin \alpha) - e^{-1}(\cos \alpha - \alpha \sin \alpha)}{1+\alpha^2} \\ &= \frac{2(\sinh 1) \cos \alpha - 2\alpha(\cosh 1) \sin \alpha}{1+\alpha^2} \end{aligned}$$

and

$$\begin{aligned} B(\alpha) &= \int_{-1}^1 e^x \sin \alpha x \, dx \\ &= \frac{e(\sin \alpha - \alpha \cos \alpha) - e^{-1}(-\sin \alpha - \alpha \cos \alpha)}{1+\alpha^2} \\ &= \frac{2(\cosh 1) \sin \alpha - 2\alpha(\sinh 1) \cos \alpha}{1+\alpha^2}. \end{aligned}$$

Hence

$$f(x) = \frac{1}{\pi} \int_0^\infty [A(\alpha) \cos \alpha x + B(\alpha) \sin \alpha x] \, d\alpha.$$

7. The function is odd. Thus from formula (11) in the text

$$B(\alpha) = 5 \int_0^1 \sin \alpha x \, dx = \frac{5(1 - \cos \alpha)}{\alpha}.$$

Exercises 15.3

Hence from formula (10) in the text,

$$f(x) = \frac{10}{\pi} \int_0^\infty \frac{(1 - \cos \alpha) \sin \alpha x}{\alpha} d\alpha.$$

8. The function is even. Thus from formula (9) in the text

$$A(\alpha) = \pi \int_1^2 \cos \alpha x dx = \pi \left(\frac{\sin 2\alpha - \sin \alpha}{\alpha} \right).$$

Hence from formula (8) in the text,

$$f(x) = 2 \int_0^\infty \frac{(\sin 2\alpha - \sin \alpha) \cos \alpha x}{\alpha} d\alpha.$$

9. The function is even. Thus from formula (9) in the text

$$\begin{aligned} A(\alpha) &= \int_0^\pi x \cos \alpha x dx = \frac{x \sin \alpha x}{\alpha} \Big|_0^\pi - \frac{1}{\alpha} \int_0^\pi \sin \alpha x dx \\ &= \frac{\pi \alpha \sin \pi \alpha}{\alpha} + \frac{1}{\alpha^2} \cos \alpha x \Big|_0^\pi = \frac{\pi \alpha \sin \pi \alpha + \cos \pi \alpha - 1}{\alpha^2}. \end{aligned}$$

Hence from formula (8) in the text

$$f(x) = \frac{2}{\pi} \int_0^\infty \frac{(\pi \alpha \sin \pi \alpha + \cos \pi \alpha - 1) \cos \alpha x}{\alpha^2} d\alpha.$$

10. The function is odd. Thus from formula (11) in the text

$$\begin{aligned} B(\alpha) &= \int_0^\pi x \sin \alpha x dx = -\frac{x \cos \alpha x}{\alpha} \Big|_0^\pi + \frac{1}{\alpha} \int_0^\pi \cos \alpha x dx \\ &= -\frac{\pi \cos \pi \alpha}{\alpha} + \frac{1}{\alpha^2} \sin \alpha x \Big|_0^\pi = \frac{-\pi \alpha \cos \pi \alpha + \sin \pi \alpha}{\alpha^2}. \end{aligned}$$

Hence from formula (10) in the text,

$$f(x) = \frac{2}{\pi} \int_0^\infty \frac{(-\pi \alpha \cos \pi \alpha + \sin \pi \alpha) \sin \alpha x}{\alpha^2} d\alpha.$$

11. The function is odd. Thus from formula (11) in the text

$$\begin{aligned} B(\alpha) &= \int_0^\infty (e^{-x} \sin x) \sin \alpha x dx \\ &= \frac{1}{2} \int_0^\infty e^{-x} [\cos(1 - \alpha)x - \cos(1 + \alpha)x] dx \\ &= \frac{1}{2} \int_0^\infty e^{-x} \cos(1 - \alpha)x dx - \frac{1}{2} \int_0^\infty e^{-x} \cos(1 + \alpha)x dx. \end{aligned}$$

Now recall

$$\mathcal{L}\{\cos kt\} = \int_0^\infty e^{-st} \cos kt dt = s/(s^2 + k^2).$$

If we set $s = 1$, and in turn, $k = 1 - \alpha$ and then $k = 1 + \alpha$, we obtain

$$B(\alpha) = \frac{1}{2} \frac{1}{1 + (1 - \alpha)^2} - \frac{1}{2} \frac{1}{1 + (1 + \alpha)^2} = \frac{1}{2} \frac{(1 + \alpha)^2 - (1 - \alpha)^2}{[1 + (1 - \alpha)^2][1 + (1 + \alpha)^2]}.$$

Simplifying the last expression gives

$$B(\alpha) = \frac{2\alpha}{4 + \alpha^4}.$$

Hence from formula (10) in the text

$$f(x) = \frac{4}{\pi} \int_0^\infty \frac{\alpha \sin \alpha x}{4 + \alpha^4} d\alpha.$$

12. The function is odd. Thus from formula (11) in the text

$$B(\alpha) = \int_0^\infty x e^{-x} \sin \alpha x dx.$$

Now recall

$$\mathcal{L}\{t \sin kt\} = -\frac{d}{ds} \mathcal{L}\{\sin kt\} = 2ks/(s^2 + k^2)^2.$$

If we set $s = 1$ and $k = \alpha$ we obtain

$$B(\alpha) = \frac{2\alpha}{(1 + \alpha^2)^2}.$$

Hence from formula (10) in the text

$$f(x) = \frac{4}{\pi} \int_0^\infty \frac{\alpha \sin \alpha x}{(1 + \alpha^2)^2} d\alpha.$$

13. For the cosine integral,

$$A(\alpha) = \int_0^\infty e^{-kx} \cos \alpha x dx = \frac{k}{k^2 + \alpha^2}.$$

Hence

$$f(x) = \frac{2}{\pi} \int_0^\infty \frac{k \cos \alpha x}{k^2 + \alpha^2} d\alpha = \frac{2k}{\pi} \int_0^\infty \frac{\cos \alpha x}{k^2 + \alpha^2} d\alpha.$$

For the sine integral,

$$B(\alpha) = \int_0^\infty e^{-kx} \sin \alpha x dx = \frac{\alpha}{k^2 + \alpha^2}.$$

Hence

$$f(x) = \frac{2}{\pi} \int_0^\infty \frac{\alpha \sin \alpha x}{k^2 + \alpha^2} d\alpha.$$

14. From Problem 13 the cosine and sine integral representations of e^{-kx} , $k > 0$, are respectively,

$$e^{-kx} = \frac{2k}{\pi} \int_0^\infty \frac{\cos \alpha x}{k^2 + \alpha^2} d\alpha \quad \text{and} \quad e^{-kx} = \frac{2}{\pi} \int_0^\infty \frac{\alpha \sin \alpha x}{k^2 + \alpha^2} d\alpha.$$

Hence, the cosine integral representation of $f(x) = e^{-x} - e^{-3x}$ is

$$e^{-x} - e^{-3x} = \frac{2}{\pi} \int_0^\infty \frac{\cos \alpha x}{1 + \alpha^2} d\alpha - \frac{2(3)}{\pi} \int_0^\infty \frac{\cos \alpha x}{9 + \alpha^2} d\alpha = \frac{4}{\pi} \int_0^\infty \frac{3 - \alpha^2}{(1 + \alpha^2)(9 + \alpha^2)} \cos \alpha x d\alpha.$$

The sine integral representation of f is

$$e^{-x} - e^{-3x} = \frac{2}{\pi} \int_0^\infty \frac{\alpha \sin \alpha x}{1 + \alpha^2} d\alpha - \frac{2}{\pi} \int_0^\infty \frac{\alpha \sin \alpha x}{9 + \alpha^2} d\alpha = \frac{16}{\pi} \int_0^\infty \frac{\alpha \sin \alpha x}{(1 + \alpha^2)(9 + \alpha^2)} d\alpha.$$

15. For the cosine integral,

$$A(\alpha) = \int_0^\infty x e^{-2x} \cos \alpha x dx.$$

But we know

$$\mathcal{L}\{t \cos kt\} = -\frac{d}{ds} \frac{s}{(s^2 + k^2)} = \frac{(s^2 - k^2)}{(s^2 + k^2)^2}.$$

If we set $s = 2$ and $k = \alpha$ we obtain

$$A(\alpha) = \frac{4 - \alpha^2}{(4 + \alpha^2)^2}.$$

Exercises 15.3

Hence

$$f(x) = \frac{2}{\pi} \int_0^\infty \frac{(4 - \alpha^2) \cos \alpha x}{(4 + \alpha^2)^2} d\alpha.$$

For the sine integral,

$$B(\alpha) = \int_0^\infty x e^{-2x} \sin \alpha x dx.$$

From Problem 12, we know

$$\mathcal{L}\{t \sin kt\} = \frac{2ks}{(s^2 + k^2)^2}.$$

If we set $s = 2$ and $k = \alpha$ we obtain

$$B(\alpha) = \frac{4\alpha}{(4 + \alpha^2)^2}.$$

Hence

$$f(x) = \frac{8}{\pi} \int_0^\infty \frac{\alpha \sin \alpha x}{(4 + \alpha^2)^2} d\alpha.$$

16. For the cosine integral,

$$\begin{aligned} A(\alpha) &= \int_0^\infty e^{-x} \cos x \cos \alpha x dx \\ &= \frac{1}{2} \int_0^\infty e^{-x} [\cos(1 + \alpha)x + \cos(1 - \alpha)x] dx \\ &= \frac{1}{2} \frac{1}{1 + (1 + \alpha)^2} + \frac{1}{2} \frac{1}{1 + (1 - \alpha)^2} \\ &= \frac{1}{2} \frac{1 + (1 - \alpha)^2 + 1 + (1 + \alpha)^2}{[1 + (1 + \alpha)^2][1 + (1 - \alpha)^2]} \\ &= \frac{2 + \alpha^2}{4 + \alpha^4}. \end{aligned}$$

Hence

$$f(x) = \frac{2}{\pi} \int_0^\infty \frac{(2 + \alpha^2) \cos \alpha x}{4 + \alpha^4} d\alpha.$$

For the sine integral,

$$\begin{aligned} B(\alpha) &= \int_0^\infty e^{-x} \cos x \sin \alpha x dx \\ &= \frac{1}{2} \int_0^\infty e^{-x} [\sin(1 + \alpha)x - \sin(1 - \alpha)x] dx \\ &= \frac{1}{2} \frac{1 + \alpha}{1 + (1 + \alpha)^2} - \frac{1}{2} \frac{1 - \alpha}{1 + (1 - \alpha)^2} \\ &= \frac{1}{2} \left[\frac{(1 + \alpha)[1 + (1 - \alpha)^2] - (1 - \alpha)[1 + (1 + \alpha)^2]}{[1 + (1 + \alpha)^2][1 + (1 - \alpha)^2]} \right] \\ &= \frac{\alpha^3}{4 + \alpha^4}. \end{aligned}$$

Hence

$$f(x) = \frac{2}{\pi} \int_0^\infty \frac{\alpha^3 \sin \alpha x}{4 + \alpha^4} d\alpha.$$

17. By formula (8) in the text

$$f(x) = 2\pi \int_0^\infty e^{-\alpha} \cos \alpha x \, d\alpha = \frac{2}{\pi} \frac{1}{1+x^2}, \quad x > 0.$$

18. From the formula for sine integral of $f(x)$ we have

$$\begin{aligned} f(x) &= \frac{2}{\pi} \int_0^\infty \left(\int_0^\infty f(x) \sin \alpha x \, dx \right) \sin \alpha x \, d\alpha \\ &= \frac{2}{\pi} \left[\int_0^1 1 \cdot \sin \alpha x \, d\alpha + \int_1^\infty 0 \cdot \sin \alpha x \, d\alpha \right] \\ &= \frac{2}{\pi} \frac{(-\cos \alpha x)}{x} \Big|_0^1 = \frac{2}{\pi} \frac{1 - \cos x}{x}. \end{aligned}$$

19. (a) From formula (7) in the text with $x = 2$, we have

$$\frac{1}{2} = \frac{2}{\pi} \int_0^\infty \frac{\sin \alpha \cos \alpha}{\alpha} \, d\alpha = \frac{1}{\pi} \int_0^\infty \frac{\sin 2\alpha}{\alpha} \, d\alpha.$$

If we let $\alpha = x$ we obtain

$$\int_0^\infty \frac{\sin 2x}{x} \, dx = \frac{\pi}{2}.$$

(b) If we now let $2x = kt$ where $k > 0$, then $dx = (k/2)dt$ and the integral in part (a) becomes

$$\int_0^\infty \frac{\sin kt}{kt/2} (k/2) \, dt = \int_0^\infty \frac{\sin kt}{t} \, dt = \frac{\pi}{2}.$$

20. With $f(x) = e^{-|x|}$, formula (16) in the text is

$$C(\alpha) = \int_{-\infty}^\infty e^{-|x|} e^{i\alpha x} \, dx = \int_{-\infty}^\infty e^{-|x|} \cos \alpha x \, dx + i \int_{-\infty}^\infty e^{-|x|} \sin \alpha x \, dx.$$

The imaginary part in the last line is zero since the integrand is an odd function of x . Therefore,

$$C(\alpha) = \int_{-\infty}^\infty e^{-|x|} \cos \alpha x \, dx = 2 \int_0^\infty e^{-x} \cos \alpha x \, dx = \frac{2}{1+\alpha^2}$$

and so from formula (15) in the text,

$$f(x) = \frac{1}{\pi} \int_{-\infty}^\infty \frac{\cos \alpha x}{1+\alpha^2} \, d\alpha = \frac{2}{\pi} \int_0^\infty \frac{\cos \alpha x}{1+\alpha^2} \, d\alpha.$$

This is the same result obtained from formulas (8) and (9) in the text.

Exercises 15.4

For the boundary-value problems in this section it is sometimes useful to note that the identities

$$e^{i\alpha} = \cos \alpha + i \sin \alpha \quad \text{and} \quad e^{-i\alpha} = \cos \alpha - i \sin \alpha$$

imply

$$e^{i\alpha} + e^{-i\alpha} = 2 \cos \alpha \quad \text{and} \quad e^{i\alpha} - e^{-i\alpha} = 2i \sin \alpha.$$

Exercises 15.4

1. Using the Fourier transform, the partial differential equation becomes

$$\frac{dU}{dt} + k\alpha^2 U = 0 \quad \text{and so} \quad U(\alpha, t) = ce^{-k\alpha^2 t}.$$

Now

$$\mathcal{F}\{u(x, 0)\} = U(\alpha, 0) = \mathcal{F}\{e^{-|x|}\}.$$

We have

$$\mathcal{F}\{e^{-|x|}\} = \int_{-\infty}^{\infty} e^{-|x|} e^{i\alpha x} dx = \int_{-\infty}^{\infty} e^{-|x|} (\cos \alpha x + i \sin \alpha x) dx = \int_{-\infty}^{\infty} e^{-|x|} \cos \alpha x dx.$$

The integral

$$\int_{-\infty}^{\infty} e^{-|x|} \sin \alpha x dx = 0$$

since the integrand is an odd function of x . Continuing we obtain

$$\mathcal{F}\{e^{-|x|}\} = 2 \int_0^{\infty} e^{-x} \cos \alpha x dx = \frac{2}{1 + \alpha^2}.$$

But $U(\alpha, 0) = c = \frac{2}{1 + \alpha^2}$ gives

$$U(\alpha, t) = \frac{2e^{-k\alpha^2 t}}{1 + \alpha^2}$$

and so

$$\begin{aligned} u(x, t) &= \frac{2}{2\pi} \int_{-\infty}^{\infty} \frac{e^{-k\alpha^2 t} e^{-i\alpha x}}{1 + \alpha^2} d\alpha = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{e^{-k\alpha^2 t}}{1 + \alpha^2} (\cos \alpha x - i \sin \alpha x) d\alpha \\ &= \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{e^{-k\alpha^2 t} \cos \alpha x}{1 + \alpha^2} d\alpha = \frac{2}{\pi} \int_0^{\infty} \frac{e^{-k\alpha^2 t} \cos \alpha x}{1 + \alpha^2} d\alpha. \end{aligned}$$

2. Since the domain of x is $(-\infty, \infty)$ we transform the differential equation using the Fourier transform:

$$-k\alpha^2 U(\alpha, t) = \frac{du}{dt}$$

$$\frac{du}{dt} + k\alpha^2 U(\alpha, t) = 0$$

$$U(\alpha, t) = ce^{-k\alpha^2 t}. \tag{1}$$

The transform of the initial condition is

$$\begin{aligned} \mathcal{F}\{u(x, 0)\} &= \int_{-\infty}^{\infty} u(x, 0) e^{i\alpha x} dx = \int_{-1}^0 (-100e^{i\alpha x}) dx + \int_0^1 100e^{i\alpha x} dx \\ &= -100 \frac{1 - e^{-i\alpha}}{i\alpha} + 100 \frac{e^{i\alpha} - 1}{i\alpha} = 100 \frac{e^{i\alpha} + e^{-i\alpha} - 2}{i\alpha} \\ &= 100 \frac{2 \cos \alpha - 2}{i\alpha} = 200 \frac{\cos \alpha - 1}{i\alpha}. \end{aligned}$$

Thus

$$U(\alpha, 0) = 200 \frac{\cos \alpha - 1}{i\alpha},$$

and since $c = U(\alpha, 0)$ in (1) we have

$$U(\alpha, t) = 200 \frac{\cos \alpha - 1}{i\alpha} e^{-k\alpha^2 t}.$$

Applying the inverse Fourier transform we obtain

$$\begin{aligned}
 u(x, t) &= \mathcal{F}^{-1}\{U(\alpha, t)\} \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} 200 \frac{\cos \alpha - 1}{i\alpha} e^{-k\alpha^2 t} e^{-i\alpha x} d\alpha \\
 &= \frac{100}{\pi} \int_{-\infty}^{\infty} 200 \frac{\cos \alpha - 1}{i\alpha} e^{-k\alpha^2 t} (\cos \alpha x - i \sin \alpha x) d\alpha \\
 &= \frac{100}{\pi} \int_{-\infty}^{\infty} \underbrace{\frac{\cos \alpha x (\cos \alpha - 1)}{i\alpha} e^{-k\alpha^2 t}}_{\text{odd function}} d\alpha - \frac{100}{\pi} \int_{-\infty}^{\infty} \underbrace{\frac{\sin \alpha x (\cos \alpha - 1)}{\alpha} e^{-k\alpha^2 t}}_{\text{even function}} d\alpha \\
 &= \frac{200}{\pi} \int_0^{\infty} \frac{\sin \alpha x (1 - \cos \alpha)}{\alpha} e^{-k\alpha^2 t} d\alpha.
 \end{aligned}$$

3. Using the Fourier transform, the partial differential equation becomes

$$\frac{dU}{dt} + k\alpha^2 U = 0 \quad \text{and so} \quad U(\alpha, t) = ce^{-k\alpha^2 t}.$$

Now

$$\mathcal{F}\{u(x, 0)\} = U(\alpha, 0) = \sqrt{\pi} e^{-\alpha^2/4}$$

by the given result. This gives $c = \sqrt{\pi} e^{-\alpha^2/4}$ and so

$$U(\alpha, t) = \sqrt{\pi} e^{-(\frac{1}{4} + kt)\alpha^2}.$$

Using the given Fourier transform again we obtain

$$u(x, t) = \sqrt{\pi} \mathcal{F}^{-1}\{e^{-(1+4kt)\alpha^2/4}\} = \frac{1}{\sqrt{1+4kt}} e^{-x^2/(1+4kt)}.$$

4. (a) We use $U(\alpha, t) = ce^{-k\alpha^2 t}$. The Fourier transform of the boundary condition is $U(\alpha, 0) = F(\alpha)$. This gives $c = F(\alpha)$ and so $U(\alpha, t) = F(\alpha)e^{-k\alpha^2 t}$. By the convolution theorem and the given result, we obtain

$$u(x, t) = \mathcal{F}^{-1}\{F(\alpha) \cdot e^{-k\alpha^2 t}\} = \frac{1}{2\sqrt{k\pi t}} \int_{-\infty}^{\infty} f(\tau) e^{-(x-\tau)^2/4kt} d\tau.$$

- (b) Using the definition of f and the solution is part (a) we obtain

$$u(x, t) = \frac{u_0}{2\sqrt{k\pi t}} \int_{-1}^1 e^{-(x-\tau)^2/4kt} d\tau.$$

If $u = \frac{x-\tau}{2\sqrt{kt}}$, then $d\tau = -2\sqrt{kt} du$ and the integral becomes

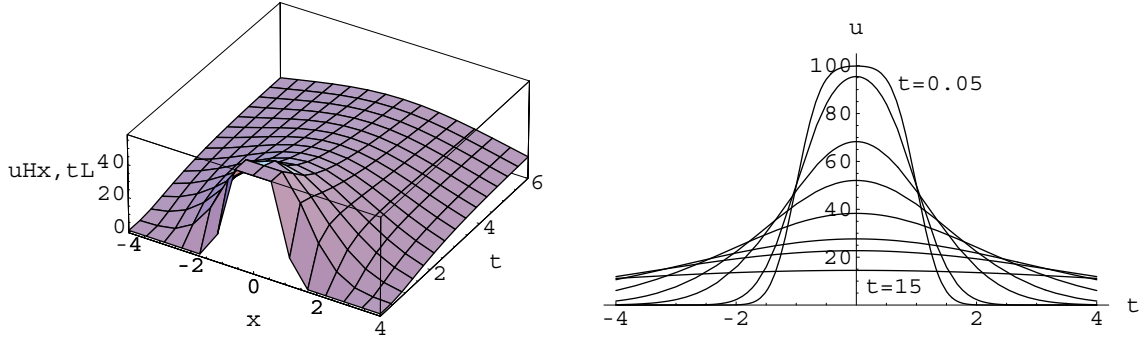
$$u(x, t) = \frac{u_0}{\sqrt{\pi}} \int_{(x-1)/2\sqrt{kt}}^{(x+1)/2\sqrt{kt}} e^{-u^2} du.$$

Using the result in Problem 9, Exercises 15.1, we have

$$u(x, t) = \frac{u_0}{2} \left[\operatorname{erf}\left(\frac{x+1}{2\sqrt{kt}}\right) - \operatorname{erf}\left(\frac{x-1}{2\sqrt{kt}}\right) \right].$$

Exercises 15.4

5.



Since $\operatorname{erf}(0) = 0$ and $\lim_{x \rightarrow \infty} \operatorname{erf}(x) = 1$, we have

$$\lim_{t \rightarrow \infty} u(x, t) = 50[\operatorname{erf}(0) - \operatorname{erf}(0)] = 0$$

and

$$\lim_{x \rightarrow \infty} u(x, t) = 50[\operatorname{erf}(\infty) - \operatorname{erf}(\infty)] = 50[1 - 1] = 0.$$

6. (a) Using the Fourier sine transform, the partial differential equation becomes

$$\frac{dU}{dt} + k\alpha^2 U = k\alpha u_0.$$

The general solution of this linear equation is

$$U(\alpha, t) = ce^{-k\alpha^2 t} + \frac{u_0}{\alpha}.$$

But $U(\alpha, 0) = 0$ implies $c = -u_0/\alpha$ and so

$$U(\alpha, t) = u_0 \frac{1 - e^{-k\alpha^2 t}}{\alpha}$$

and

$$u(x, t) = \frac{2u_0}{\pi} \int_0^\infty \frac{1 - e^{-k\alpha^2 t}}{\alpha} \sin \alpha x \, d\alpha.$$

(b) The solution of part (a) can be written

$$u(x, t) = \frac{2u_0}{\pi} \int_0^\infty \frac{\sin \alpha x}{\alpha} \, d\alpha - \frac{2u_0}{\pi} \int_0^\infty \frac{\sin \alpha x}{\alpha} e^{-k\alpha^2 t} \, d\alpha.$$

Using $\int_0^\infty \frac{\sin \alpha x}{\alpha} \, d\alpha = \pi/2$ the last line becomes

$$u(x, t) = u_0 - \frac{2u_0}{\pi} \int_0^\infty \frac{\sin \alpha x}{\alpha} e^{-k\alpha^2 t} \, d\alpha.$$

7. Using the Fourier sine transform we find

$$U(\alpha, t) = ce^{-k\alpha^2 t}.$$

Now

$$\mathcal{F}_S\{u(x, 0)\} = U(\alpha, 0) = \int_0^1 \sin \alpha x \, dx = \frac{1 - \cos \alpha}{\alpha}.$$

From this we find $c = (1 - \cos \alpha)/\alpha$ and so

$$U(\alpha, t) = \frac{1 - \cos \alpha}{\alpha} e^{-k\alpha^2 t}$$

and

$$u(x, t) = \frac{2}{\pi} \int_0^\infty \frac{1 - \cos \alpha}{\alpha} e^{-k\alpha^2 t} \sin \alpha x \, d\alpha.$$

8. Since the domain of x is $(0, \infty)$ and the condition at $x = 0$ involves $\partial u / \partial x$ we use the Fourier cosine transform:

$$-k\alpha^2 U(\alpha, t) - k u_x(0, t) = \frac{dU}{dt}$$

$$\frac{dU}{dt} + k\alpha^2 U(\alpha, t) = kA$$

$$U(\alpha, t) = ce^{-k\alpha^2 t} + \frac{A}{\alpha^2}.$$

Since

$$\mathcal{F}\{u(x, 0)\} = U(\alpha, 0) = 0$$

we find $c = -A/\alpha^2$, so that

$$U(\alpha, t) = A \frac{1 - e^{-k\alpha^2 t}}{\alpha^2}.$$

Applying the inverse Fourier cosine transform we obtain

$$u(x, t) = \mathcal{F}_C^{-1}\{U(\alpha, t)\} = \frac{2A}{\pi} \int_0^\infty \frac{1 - e^{-k\alpha^2 t}}{\alpha^2} \cos \alpha x \, d\alpha.$$

9. Using the Fourier cosine transform we find

$$U(\alpha, t) = ce^{-k\alpha^2 t}.$$

Now

$$\mathcal{F}_C\{u(x, 0)\} = \int_0^1 \cos \alpha x \, dx = \frac{\sin \alpha}{\alpha} = U(\alpha, 0).$$

From this we obtain $c = (\sin \alpha)/\alpha$ and so

$$U(\alpha, t) = \frac{\sin \alpha}{\alpha} e^{-k\alpha^2 t}$$

and

$$u(x, t) = \frac{2}{\pi} \int_0^\infty \frac{\sin \alpha}{\alpha} e^{-k\alpha^2 t} \cos \alpha x \, d\alpha.$$

10. Using the Fourier sine transform we find

$$U(\alpha, t) = ce^{-k\alpha^2 t} + \frac{1}{\alpha}.$$

Now

$$\mathcal{F}_S\{u(x, 0)\} = \mathcal{F}_S\{e^{-x}\} = \int_0^\infty e^{-x} \sin \alpha x \, dx = \frac{\alpha}{1 + \alpha^2} = U(\alpha, 0).$$

From this we obtain $c = \alpha/(1 + \alpha^2) - 1/\alpha$. Therefore

$$U(\alpha, t) = \left(\frac{\alpha}{1 + \alpha^2} - \frac{1}{\alpha} \right) e^{-k\alpha^2 t} + \frac{1}{\alpha} = \frac{1}{\alpha} - \frac{e^{-k\alpha^2 t}}{\alpha(1 + \alpha^2)}$$

and

$$u(x, t) = \frac{2}{\pi} \int_0^\infty \left(\frac{1}{\alpha} - \frac{e^{-k\alpha^2 t}}{\alpha(1 + \alpha^2)} \right) \sin \alpha x \, d\alpha.$$

Exercises 15.4

11. (a) Using the Fourier transform we obtain

$$U(\alpha, t) = c_1 \cos \alpha at + c_2 \sin \alpha at.$$

If we write

$$\mathcal{F}\{u(x, 0)\} = \mathcal{F}\{f(x)\} = F(\alpha)$$

and

$$\mathcal{F}\{u_t(x, 0)\} = \mathcal{F}\{g(x)\} = G(\alpha)$$

we first obtain

$$c_1 = F(\alpha) \text{ from } U(\alpha, 0) = F(\alpha) \text{ and then } c_2 = G(\alpha)/\alpha a \text{ from } \left. \frac{dU}{dt} \right|_{t=0} = G(\alpha). \text{ Thus}$$

$$U(\alpha, t) = F(\alpha) \cos \alpha at + \frac{G(\alpha)}{\alpha a} \sin \alpha at$$

and

$$u(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left(F(\alpha) \cos \alpha at + \frac{G(\alpha)}{\alpha a} \sin \alpha at \right) e^{-i\alpha x} d\alpha.$$

(b) If $g(x) = 0$ then $c_2 = 0$ and

$$\begin{aligned} u(x, t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\alpha) \cos \alpha at e^{-i\alpha x} d\alpha \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\alpha) \left(\frac{e^{\alpha at i} + e^{-\alpha at i}}{2} \right) e^{-i\alpha x} d\alpha \\ &= \frac{1}{2} \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} F(\alpha) e^{-i(x-at)\alpha} d\alpha + \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\alpha) e^{-i(x+at)\alpha} d\alpha \right] \\ &= \frac{1}{2} [f(x-at) + f(x+at)]. \end{aligned}$$

12. Using the Fourier sine transform we obtain

$$U(\alpha, t) = c_1 \cos \alpha at + c_2 \sin \alpha at.$$

Now

$$\mathcal{F}_S\{u(x, 0)\} = \mathcal{F}_S\{xe^{-x}\} = \int_0^{\infty} xe^{-x} \sin \alpha x dx = \frac{2\alpha}{(1+\alpha^2)^2} = U(\alpha, 0).$$

Also,

$$\mathcal{F}_S\{u_t(x, 0)\} = \left. \frac{dU}{dt} \right|_{t=0} = 0.$$

This last condition gives $c_2 = 0$. Then $U(\alpha, 0) = 2\alpha/(1+\alpha^2)^2$ yields $c_1 = 2\alpha/(1+\alpha^2)^2$. Therefore

$$U(\alpha, t) = \frac{2\alpha}{(1+\alpha^2)^2} \cos \alpha at$$

and

$$u(x, t) = \frac{4}{\pi} \int_0^{\infty} \frac{\alpha \cos \alpha at}{(1+\alpha^2)^2} \sin \alpha x d\alpha.$$

13. Using the Fourier cosine transform we obtain

$$U(x, \alpha) = c_1 \cosh \alpha x + c_2 \sinh \alpha x.$$

Now the Fourier cosine transforms of $u(0, y) = e^{-y}$ and $u(\pi, y) = 0$ are, respectively, $U(0, \alpha) = 1/(1 + \alpha^2)$ and $U(\pi, \alpha) = 0$. The first of these conditions gives $c_1 = 1/(1 + \alpha^2)$. The second condition gives

$$c_2 = -\frac{\cosh \alpha \pi}{(1 + \alpha^2) \sinh \alpha \pi}.$$

Hence

$$U(x, \alpha) = \frac{\cosh \alpha x}{1 + \alpha^2} - \frac{\cosh \alpha \pi \sinh \alpha x}{(1 + \alpha^2) \sinh \alpha \pi} = \frac{\sinh \alpha \pi \cosh \alpha x - \cosh \alpha \pi \sinh \alpha x}{(1 + \alpha^2) \sinh \alpha \pi} = \frac{\sinh \alpha(\pi - x)}{(1 + \alpha^2) \sinh \alpha \pi}$$

and

$$u(x, t) = \frac{2}{\pi} \int_0^\infty \frac{\sinh \alpha(\pi - x)}{(1 + \alpha^2) \sinh \alpha \pi} \cos \alpha y \, d\alpha.$$

14. Since the boundary condition at $y = 0$ now involves $u(x, 0)$ rather than $u'(x, 0)$, we use the Fourier sine transform. The transform of the partial differential equation is then

$$\frac{d^2 U}{dx^2} - \alpha^2 U + \alpha u(x, 0) = 0 \quad \text{or} \quad \frac{d^2 U}{dx^2} - \alpha^2 U = -\alpha.$$

The solution of this differential equation is

$$U(x, \alpha) = c_1 \cosh \alpha x + c_2 \sinh \alpha x + \frac{1}{\alpha}.$$

The transforms of the boundary conditions at $x = 0$ and $x = \pi$ in turn imply that $c_1 = 1/\alpha$ and

$$c_2 = \frac{\cosh \alpha \pi}{\alpha \sinh \alpha \pi} - \frac{1}{\alpha \sinh \alpha \pi} + \frac{\alpha}{(1 + \alpha^2) \sinh \alpha \pi}.$$

Hence

$$\begin{aligned} U(\alpha, x) &= \frac{1}{\alpha} - \frac{\cosh \alpha x}{\alpha} + \frac{\cosh \alpha \pi}{\alpha \sinh \alpha \pi} \sinh \alpha x - \frac{\sinh \alpha x}{\alpha \sinh \alpha \pi} + \frac{\alpha \sinh \alpha x}{(1 + \alpha^2) \sinh \alpha \pi} \\ &= \frac{1}{\alpha} - \frac{\sinh \alpha(\pi - x)}{\alpha \sinh \alpha \pi} - \frac{\sinh \alpha x}{\alpha(1 + \alpha^2) \sinh \alpha \pi}. \end{aligned}$$

Taking the inverse transform it follows that

$$u(x, y) = \frac{2}{\pi} \int_0^\infty \left(\frac{1}{\alpha} - \frac{\sinh \alpha(\pi - x)}{\alpha \sinh \alpha \pi} - \frac{\sinh \alpha x}{\alpha(1 + \alpha^2) \sinh \alpha \pi} \right) \sin \alpha y \, d\alpha.$$

15. Using the Fourier cosine transform with respect to x gives

$$U(\alpha, y) = c_1 e^{-\alpha y} + c_2 e^{\alpha y}.$$

Since we expect $u(x, y)$ to be bounded as $y \rightarrow \infty$ we define $c_2 = 0$. Thus

$$U(\alpha, y) = c_1 e^{-\alpha y}.$$

Now

$$\mathcal{F}_C\{u(x, 0)\} = \int_0^1 50 \cos \alpha x \, dx = 50 \frac{\sin \alpha}{\alpha}$$

and so

$$U(\alpha, y) = 50 \frac{\sin \alpha}{\alpha} e^{-\alpha y}$$

and

$$u(x, y) = \frac{100}{\pi} \int_0^\infty \frac{\sin \alpha}{\alpha} e^{-\alpha y} \cos \alpha x \, d\alpha.$$

Exercises 15.4

16. The boundary condition $u(0, y) = 0$ indicates that we now use the Fourier sine transform. We still have $U(\alpha, y) = c_1 e^{-\alpha y}$, but

$$\mathcal{F}_S\{u(x, 0)\} = \int_0^1 50 \sin \alpha x \, dx = 50(1 - \cos \alpha)/\alpha = U(\alpha, 0).$$

This gives $c_1 = 50(1 - \cos \alpha)/\alpha$ and so

$$U(\alpha, y) = 50 \frac{1 - \cos \alpha}{\alpha} e^{-\alpha y}$$

and

$$u(x, y) = \frac{100}{\pi} \int_0^\infty \frac{1 - \cos \alpha}{\alpha} e^{-\alpha y} \sin \alpha x \, d\alpha.$$

17. We use the Fourier sine transform with respect to x to obtain

$$U(\alpha, y) = c_1 \cosh \alpha y + c_2 \sinh \alpha y.$$

The transforms of $u(x, 0) = f(x)$ and $u(x, 2) = 0$ give, in turn, $U(\alpha, 0) = F(\alpha)$ and $U(\alpha, 2) = 0$. The first condition gives $c_1 = F(\alpha)$ and the second condition then yields

$$c_2 = -\frac{F(\alpha) \cosh 2\alpha}{\sinh 2\alpha}.$$

Hence

$$\begin{aligned} U(\alpha, y) &= F(\alpha) \cosh \alpha y - \frac{F(\alpha) \cosh 2\alpha \sinh \alpha y}{\sinh 2\alpha} \\ &= F(\alpha) \frac{\sinh 2\alpha \cosh \alpha y - \cosh 2\alpha \sinh \alpha y}{\sinh 2\alpha} \\ &= F(\alpha) \frac{\sinh \alpha(2 - y)}{\sinh 2\alpha} \end{aligned}$$

and

$$u(x, y) = \frac{2}{\pi} \int_0^\infty F(\alpha) \frac{\sinh \alpha(2 - y)}{\sinh 2\alpha} \sin \alpha x \, d\alpha.$$

18. The domain of y and the boundary condition at $y = 0$ suggest that we use a Fourier cosine transform. The transformed equation is

$$\frac{d^2 U}{dx^2} - \alpha^2 U - u_y(x, 0) = 0 \quad \text{or} \quad \frac{d^2 U}{dx^2} - \alpha^2 U = 0.$$

Because the domain of the variable x is a finite interval we choose to write the general solution of the latter equation as

$$U(x, \alpha) = c_1 \cosh \alpha x + c_2 \sinh \alpha x.$$

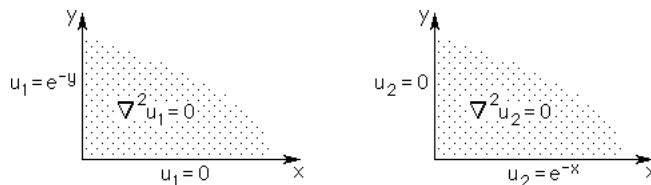
Now $U(0, \alpha) = F(\alpha)$, where $F(\alpha)$ is the Fourier cosine transform of $f(y)$, and $U'(\pi, \alpha) = 0$ imply $c_1 = F(\alpha)$ and $c_2 = -F(\alpha) \sinh \alpha \pi / \cosh \alpha \pi$. Thus

$$U(x, \alpha) = F(\alpha) \cosh \alpha x - F(\alpha) \frac{\sinh \alpha \pi}{\cosh \alpha \pi} \sinh \alpha x = F(\alpha) \frac{\cosh \alpha(\pi - x)}{\cosh \alpha \pi}.$$

Using the inverse transform we find that a solution to the problem is

$$u(x, y) = \frac{2}{\pi} \int_0^\infty F(\alpha) \frac{\cosh \alpha(\pi - x)}{\cosh \alpha \pi} \cos \alpha y \, d\alpha.$$

19. We solve two boundary-value problems:



Using the Fourier sine transform with respect to y gives

$$u_1(x, y) = \frac{2}{\pi} \int_0^\infty \frac{\alpha e^{-\alpha x}}{1 + \alpha^2} \sin \alpha y \, d\alpha.$$

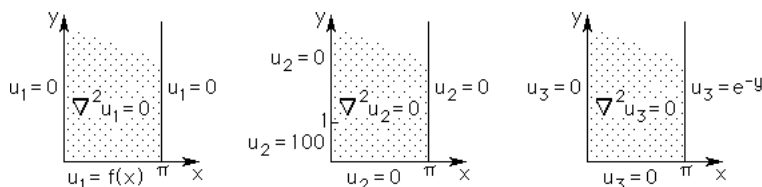
The Fourier sine transform with respect to x yields the solution to the second problem:

$$u_2(x, y) = \frac{2}{\pi} \int_0^\infty \frac{\alpha e^{-\alpha y}}{1 + \alpha^2} \sin \alpha x \, d\alpha.$$

We define the solution of the original problem to be

$$u(x, y) = u_1(x, y) + u_2(x, y) = \frac{2}{\pi} \int_0^\infty \frac{\alpha}{1 + \alpha^2} [e^{-\alpha x} \sin \alpha y + e^{-\alpha y} \sin \alpha x] \, d\alpha.$$

20. We solve the three boundary-value problems:



Using separation of variables we find the solution of the first problem is

$$u_1(x, y) = \sum_{n=1}^{\infty} A_n e^{-ny} \sin nx \quad \text{where} \quad A_n = \frac{2}{\pi} \int_0^\pi f(x) \sin nx \, dx.$$

Using the Fourier sine transform with respect to y gives the solution of the second problem:

$$u_2(x, y) = \frac{200}{\pi} \int_0^\infty \frac{(1 - \cos \alpha) \sinh \alpha(\pi - x)}{\alpha \sinh \alpha \pi} \sin \alpha y \, d\alpha.$$

Also, the Fourier sine transform with respect to y gives the solution of the third problem:

$$u_3(x, y) = \frac{2}{\pi} \int_0^\infty \frac{\alpha \sinh \alpha x}{(1 + \alpha^2) \sinh \alpha \pi} \sin \alpha y \, d\alpha.$$

The solution of the original problem is

$$u(x, y) = u_1(x, y) + u_2(x, y) + u_3(x, y).$$

21. Using the Fourier transform with respect to x gives

$$U(\alpha, y) = c_1 \cosh \alpha y + c_2 \sinh \alpha y.$$

The transform of the boundary condition $\frac{\partial u}{\partial y} \Big|_{y=0} = 0$ is $\frac{dU}{dy} \Big|_{y=0} = 0$. This condition gives $c_2 = 0$. Hence

$$U(\alpha, y) = c_1 \cosh \alpha y.$$

Now by the given information the transform of the boundary condition $u(x, 1) = e^{-x^2}$ is

Exercises 15.4

$U(\alpha, 1) = \sqrt{\pi} e^{-\alpha^2/4}$. This condition then gives $c_1 = \sqrt{\pi} e^{-\alpha^2/4} \cosh \alpha$. Therefore

$$U(\alpha, y) = \sqrt{\pi} \frac{e^{-\alpha^2/4} \cosh \alpha y}{\cosh \alpha}$$

and

$$\begin{aligned} U(x, y) &= \frac{1}{2\sqrt{\pi}} \int_{-\infty}^{\infty} \frac{e^{-\alpha^2/4} \cosh \alpha y}{\cosh \alpha} e^{-i\alpha x} d\alpha \\ &= \frac{1}{2\sqrt{\pi}} \int_{-\infty}^{\infty} \frac{e^{-\alpha^2/4} \cosh \alpha y}{\cosh \alpha} \cos \alpha x d\alpha \\ &= \frac{1}{\sqrt{\pi}} \int_0^{\infty} \frac{e^{-\alpha^2/4} \cosh \alpha y}{\cosh \alpha} \cos \alpha x d\alpha. \end{aligned}$$

22. Entries 42 and 43 of the table in Appendix III imply

$$\int_0^{\infty} e^{-st} \frac{\sin at}{t} dt = \arctan \frac{a}{s}$$

and

$$\int_0^{\infty} e^{-st} \frac{\sin at \cos bt}{t} dt = \frac{1}{2} \arctan \frac{a+b}{s} + \frac{1}{2} \arctan \frac{a-b}{s}.$$

Identifying $\alpha = t$, $x = a$, and $y = s$, the solution of Problem 16 is

$$\begin{aligned} u(x, y) &= \frac{100}{\pi} \int_0^{\infty} \frac{1 - \cos \alpha}{\alpha} e^{-\alpha y} \sin \alpha x d\alpha \\ &= \frac{100}{\pi} \left[\int_0^{\infty} \frac{\sin \alpha x}{\alpha} e^{-\alpha y} d\alpha - \int_0^{\infty} \frac{\sin \alpha x \cos \alpha}{\alpha} e^{-\alpha y} d\alpha \right] \\ &= \frac{100}{\pi} \left[\arctan \frac{x}{y} - \frac{1}{2} \arctan \frac{x+1}{y} - \frac{1}{2} \arctan \frac{x-1}{y} \right]. \end{aligned}$$

Exercises 15.5

1. We show that $\frac{1}{4} \overline{F}_4 F_4 = I$:

$$\frac{1}{4} \overline{F}_4 F_4 = \frac{1}{4} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 4 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 4 \end{pmatrix} = I.$$

Thus $F_4^{-1} = \frac{1}{4} \overline{F}_4$.

2. We have

$$\int_{-\infty}^{\infty} f(x) \delta_{\epsilon}(x-a) dx = \frac{1}{2\epsilon} \int_{a-\epsilon}^{a+\epsilon} f(x) dx = \frac{1}{2\epsilon} f(c)(2\epsilon) = f(c)$$

by the mean value theorem for integrals.

3. By the sifting property,

$$\mathcal{F}\{\delta(x)\} = \int_{-\infty}^{\infty} \delta(x) e^{i\alpha x} dx = e^{i\alpha 0} = 1.$$

4. We already know that $f * \delta = \delta * f$. Then, by the sifting property,

$$(f * \delta)(x) = \int_{-\infty}^{\infty} f(\tau) \delta(x - \tau) d\tau = \int_{-\infty}^{\infty} f(\tau) \delta(\tau - x) d\tau = f(x).$$

5. Using integration by parts with $u = f(x)$ and $dv = \delta'(x - a)$ we find

$$\int_{-\infty}^{\infty} f(x) \delta'(x - a) dx = - \int_{-\infty}^{\infty} f'(x) \delta(x - a) dx = -f'(a)$$

by the sifting property.

6. Using a CAS we find

$$\mathcal{F}\{g(x)\} = \frac{1}{2}[\text{sign}(A - \alpha) + \text{sign}(A + \alpha)]$$

where $\text{sign}(t) = 1$ if $t > 0$ and $\text{sign } t = -1$ if $t < 0$. Thus

$$\mathcal{F}\{g(x)\} = \begin{cases} 1, & -A < \alpha < A \\ 0, & \text{elsewhere.} \end{cases}$$

7. Using

$$\begin{aligned} \omega_8 &= \frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2} & \omega_8^5 &= -\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2} \\ \omega_8^2 &= i & \omega_8^6 &= -i \\ \omega_8^3 &= -\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2} & \omega_8^7 &= \frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2} \\ \omega_8^4 &= -1 & \omega_8^8 &= 1 \end{aligned}$$

we have

$$F8 = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & \frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2} & i & -\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2} & -1 & -\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2} & -i & \frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2} \\ 1 & i & -1 & -i & 1 & i & -1 & -i \\ 1 & -\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2} & -i & \frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2} & -1 & \frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2} & i & -\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2} \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & -\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2} & i & \frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2} & -1 & \frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2} & -i & -\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2} \\ 1 & -i & -1 & i & 1 & -i & -1 & i \\ 1 & \frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2} & -i & -\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2} & -1 & -\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2} & i & \frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2} \end{pmatrix}.$$

In factored form

$$F8 = \begin{pmatrix} I_4 & D_4 \\ I_4 & -D_4 \end{pmatrix} \begin{pmatrix} F_4 & 0 \\ 0 & F_4 \end{pmatrix} P,$$

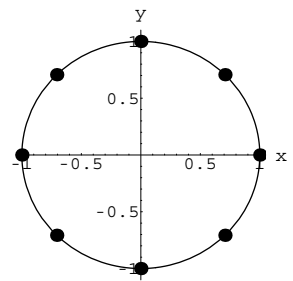
where I_4 is the 4×4 identity matrix.

$$D_4 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \sqrt{2}/2 + i\sqrt{2}/2 & 0 & 0 \\ 0 & 0 & i & 0 \\ 0 & 0 & 0 & -\sqrt{2}/2 + i\sqrt{2}/2 \end{pmatrix},$$

and P is the 8×8 matrix with 1 in positions $(1, 1)$, $(2, 3)$, $(3, 5)$, $(4, 7)$, $(5, 2)$, $(6, 4)$, $(7, 6)$, and $(8, 8)$.

Exercises 15.5

8. The 8th roots of unity, $\omega_8^1, \omega_8^2, \dots, \omega_8^8$ are shown in the solution of Problem 7 above. The points in the complex plane are equally spaced on the perimeter of the unit circle.

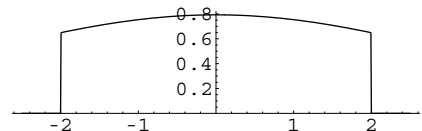


9. The Fourier transform of $g(x) = (\sin 2x)/\pi x$ is

$$G(\alpha) = \begin{cases} 1, & -2 < \alpha < 2 \\ 0, & \text{elsewhere.} \end{cases}$$

This implies that $(f * g)(x) = \mathcal{F}^{-1}\{F(\alpha)G(\alpha)\}$ is band-limited.

The graph of $F(\alpha)G(\alpha)$, which is identical to the graph of $\mathcal{F}(f * g)$, is shown.



10. For $N = 6$,

$$F_6 = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1/2 + \sqrt{3}i/2 & -1/2 + \sqrt{3}i/2 & -1 & -1/2 - \sqrt{3}i/2 & 1/2 - \sqrt{3}i/2 \\ 1 & -1/2 + \sqrt{3}i/2 & -1/2 - \sqrt{3}i/2 & 1 & -1/2 + \sqrt{3}i/2 & -1/2 - \sqrt{3}i/2 \\ 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & -1/2 - \sqrt{3}i/2 & -1/2 + \sqrt{3}i/2 & 1 & -1/2 - \sqrt{3}i/2 & -1/2 + \sqrt{3}i/2 \\ 1 & 1/2 - \sqrt{3}i/2 & -1/2 - \sqrt{3}i/2 & -1 & -1/2 + \sqrt{3}i/2 & 1/2 + \sqrt{3}i/2 \end{pmatrix}.$$

If, for example, $f = (2, 0, 1, 6, 2, 3)$, then

$$c = \frac{1}{6} \overline{F}_6 f = \begin{pmatrix} 7/3 \\ -2/3 + \sqrt{3}i/3 \\ 5/6 - \sqrt{3}i/6 \\ -2/3 \\ 5/6 - \sqrt{3}i/6 \\ -2/3 - i/\sqrt{3} \end{pmatrix}.$$

Chapter 5 Review Exercises

1. The partial differential equation and the boundary conditions indicate that the Fourier cosine transform is appropriate for the problem. We find in this case

$$u(x, y) = \frac{2}{\pi} \int_0^\infty \frac{\sinh \alpha y}{\alpha(1 + \alpha^2) \cosh \alpha \pi} \cos \alpha x \, d\alpha.$$

2. We use the Laplace transform and undetermined coefficients to obtain

$$U(x, s) = c_1 \cosh \sqrt{s} x + c_2 \sinh \sqrt{s} x + \frac{50}{s + 4\pi^2} \sin 2\pi x.$$

The transformed boundary conditions $U(0, s) = 0$ and $U(1, s) = 0$ give, in turn, $c_1 = 0$ and $c_2 = 0$. Hence

$$U(x, s) = \frac{50}{s + 4\pi^2} \sin 2\pi x$$

and

$$u(x, t) = 50 \sin 2\pi x \mathcal{L}^{-1} \left\{ \frac{1}{s + 4\pi^2} \right\} = 50e^{-4\pi^2 t} \sin 2\pi x.$$

3. The Laplace transform gives

$$U(x, s) = c_1 e^{-\sqrt{s+h}x} + c_2 e^{\sqrt{s+h}x} + \frac{u_0}{s+h}.$$

The condition $\lim_{x \rightarrow \infty} \partial u / \partial x = 0$ implies $\lim_{x \rightarrow \infty} dU/dx = 0$ and so we define $c_2 = 0$. Thus

$$U(x, s) = c_1 e^{-\sqrt{s+h}x} + \frac{u_0}{s+h}.$$

The condition $U(0, s) = 0$ then gives $c_1 = -u_0/(s+h)$ and so

$$U(x, s) = \frac{u_0}{s+h} - u_0 \frac{e^{-\sqrt{s+h}x}}{s+h}.$$

With the help of the first translation theorem we then obtain

$$\begin{aligned} u(x, t) &= u_0 \mathcal{L}^{-1} \left\{ \frac{1}{s+h} \right\} - u_0 \mathcal{L}^{-1} \left\{ \frac{e^{-\sqrt{s+h}x}}{s+h} \right\} = u_0 e^{-ht} - u_0 e^{-ht} \operatorname{erfc} \left(\frac{x}{2\sqrt{t}} \right) \\ &= u_0 e^{-ht} \left[1 - \operatorname{erfc} \left(\frac{x}{2\sqrt{t}} \right) \right] = u_0 e^{-ht} \operatorname{erf} \left(\frac{x}{2\sqrt{t}} \right). \end{aligned}$$

4. Using the Fourier transform and the result $\mathcal{F} \{ e^{-|\alpha|} \} = 1/(1 + \alpha^2)$ we find

$$\begin{aligned} u(x, t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{1 - e^{-\alpha^2 t}}{\alpha^2(1 + \alpha^2)} e^{-i\alpha x} d\alpha \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{1 - e^{-\alpha^2 t}}{\alpha^2(1 + \alpha^2)} \cos \alpha x d\alpha \\ &= \frac{1}{\pi} \int_0^{\infty} \frac{1 - e^{-\alpha^2 t}}{\alpha^2(1 + \alpha^2)} \cos \alpha x d\alpha. \end{aligned}$$

5. The Laplace transform gives

$$U(x, s) = c_1 e^{-\sqrt{s}x} + c_2 e^{\sqrt{s}x}.$$

The condition $\lim_{x \rightarrow \infty} u(x, t) = 0$ implies $\lim_{x \rightarrow \infty} U(x, s) = 0$ and so we define $c_2 = 0$. Thus

$$U(x, s) = c_1 e^{-\sqrt{s}x}.$$

The transform of the remaining boundary condition is $U(0, s) = 1/s^2$. This gives $c_1 = 1/s^2$. Hence

$$U(x, s) = \frac{e^{-\sqrt{s}x}}{s^2} \quad \text{and} \quad u(x, t) = \mathcal{L}^{-1} \left\{ \frac{1}{s} \frac{e^{-\sqrt{s}x}}{s} \right\}.$$

Using

$$\mathcal{L}^{-1} \left\{ \frac{1}{s} \right\} = 1 \quad \text{and} \quad \mathcal{L}^{-1} \left\{ \frac{e^{-\sqrt{s}x}}{s} \right\} = \operatorname{erfc} \left(\frac{x}{2\sqrt{t}} \right),$$

it follows from the convolution theorem that

$$u(x, t) = \int_0^t \operatorname{erfc} \left(\frac{x}{2\sqrt{\tau}} \right) d\tau.$$

6. The Laplace transform and undetermined coefficients gives

$$U(x, s) = c_1 \cosh sx + c_2 \sinh sx + \frac{s-1}{s^2 + \pi^2} \sin \pi x.$$

Chapter 5 Review Exercises

The conditions $U(0, s) = 0$ and $U(1, s) = 0$ give, in turn, $c_1 = 0$ and $c_2 = 0$. Thus

$$U(x, s) = \frac{s-1}{s^2 + \pi^2} \sin \pi x$$

and

$$\begin{aligned} u(x, t) &= \sin \pi x \mathcal{L}^{-1} \left\{ \frac{s}{s^2 + \pi^2} \right\} - \frac{1}{\pi} \sin \pi x \mathcal{L}^{-1} \left\{ \frac{\pi}{s^2 + \pi^2} \right\} \\ &= (\sin \pi x) \cos \pi t - \frac{1}{\pi} (\sin \pi x) \sin \pi t. \end{aligned}$$

7. The Fourier transform gives the solution

$$\begin{aligned} u(x, t) &= \frac{u_0}{2\pi} \int_{-\infty}^{\infty} \left(\frac{e^{i\alpha\pi} - 1}{i\alpha} \right) e^{-i\alpha x} e^{-k\alpha^2 t} d\alpha \\ &= \frac{u_0}{2\pi} \int_{-\infty}^{\infty} \frac{e^{i\alpha(\pi-x)} - e^{-i\alpha x}}{i\alpha} e^{-k\alpha^2 t} d\alpha \\ &= \frac{u_0}{2\pi} \int_{-\infty}^{\infty} \frac{\cos \alpha(\pi-x) + i \sin \alpha(\pi-x) - \cos \alpha x + i \sin \alpha x}{i\alpha} e^{-k\alpha^2 t} d\alpha. \end{aligned}$$

Since the imaginary part of the integrand of the last integral is an odd function of α , we obtain

$$u(x, t) = \frac{u_0}{2\pi} \int_{-\infty}^{\infty} \frac{\sin \alpha(\pi-x) + \sin \alpha x}{\alpha} e^{-k\alpha^2 t} d\alpha.$$

8. Using the Fourier cosine transform we obtain

$$U(x, \alpha) = c_1 \cosh \alpha x + c_2 \sinh \alpha x.$$

The condition $U(0, \alpha) = 0$ gives $c_1 = 0$. Thus

$$U(x, \alpha) = c_2 \sinh \alpha x.$$

Now

$$\mathcal{F}_C\{u(\pi, y)\} = \int_1^2 \cos \alpha y dy = \frac{\sin 2\alpha - \sin \alpha}{\alpha} = U(\pi, \alpha).$$

This last condition gives $c_2 = (\sin 2\alpha - \sin \alpha)/\alpha \sinh \alpha\pi$. Hence

$$U(x, \alpha) = \frac{\sin 2\alpha - \sin \alpha}{\alpha \sinh \alpha\pi} \sinh \alpha x$$

and

$$u(x, y) = \frac{2}{\pi} \int_0^\infty \frac{\sin 2\alpha - \sin \alpha}{\alpha \sinh \alpha\pi} \sinh \alpha x \cos \alpha y d\alpha.$$

9. We solve the two problems

$$\frac{\partial^2 u_1}{\partial x^2} + \frac{\partial^2 u_1}{\partial y^2} = 0, \quad x > 0, \quad y > 0,$$

$$u_1(0, y) = 0, \quad y > 0,$$

$$u_1(x, 0) = \begin{cases} 100, & 0 < x < 1 \\ 0, & x > 1 \end{cases}$$

and

$$\frac{\partial^2 u_2}{\partial x^2} + \frac{\partial^2 u_2}{\partial y^2} = 0, \quad x > 0, \quad y > 0,$$

$$u_2(0, y) = \begin{cases} 50, & 0 < y < 1 \\ 0, & y > 1 \end{cases}$$

$$u_2(x, 0) = 0.$$

Using the Fourier sine transform with respect to x we find

$$u_1(x, y) = \frac{200}{\pi} \int_0^\infty \left(\frac{1 - \cos \alpha}{\alpha} \right) e^{-\alpha y} \sin \alpha x \, d\alpha.$$

Using the Fourier sine transform with respect to y we find

$$u_2(x, y) = \frac{100}{\pi} \int_0^\infty \left(\frac{1 - \cos \alpha}{\alpha} \right) e^{-\alpha x} \sin \alpha y \, d\alpha.$$

The solution of the problem is then

$$u(x, y) = u_1(x, y) + u_2(x, y).$$

10. The Laplace transform gives

$$U(x, s) = c_1 \cosh \sqrt{s} x + c_2 \sinh \sqrt{s} x + \frac{r}{s^2}.$$

The condition $\frac{\partial u}{\partial x} \Big|_{x=0} = 0$ transforms into $\frac{dU}{dx} \Big|_{x=0} = 0$. This gives $c_2 = 0$. The remaining condition $u(1, t) = 0$ transforms into $U(1, s) = 0$. This condition then implies $c_1 = -r/s^2 \cosh \sqrt{s}$. Hence

$$U(x, s) = \frac{r}{s^2} - r \frac{\cosh \sqrt{s} x}{s^2 \cosh \sqrt{s}}.$$

Using geometric series and the convolution theorem we obtain

$$\begin{aligned} u(x, t) &= r \mathcal{L}^{-1} \left\{ \frac{1}{s^2} \right\} - r \mathcal{L}^{-1} \left\{ \frac{\cosh \sqrt{s} x}{s^2 \cosh \sqrt{s}} \right\} \\ &= rt - r \sum_{n=0}^{\infty} (-1)^n \left[\int_0^t \operatorname{erfc} \left(\frac{2n+1-x}{2\sqrt{\tau}} \right) d\tau + \int_0^t \operatorname{erfc} \left(\frac{2n+1+x}{2\sqrt{\tau}} \right) d\tau \right]. \end{aligned}$$

11. The Fourier sine transform with respect to x and undetermined coefficients give

$$U(\alpha, y) = c_1 \cosh \alpha y + c_2 \sinh \alpha y + \frac{A}{\alpha}.$$

The transforms of the boundary conditions are

$$\frac{dU}{dy} \Big|_{y=0} = 0 \quad \text{and} \quad \frac{dU}{dy} \Big|_{y=\pi} = \frac{B\alpha}{1+\alpha^2}.$$

The first of these conditions gives $c_2 = 0$ and so

$$U(\alpha, y) = c_1 \cosh \alpha y + \frac{A}{\alpha}.$$

The second transformed boundary condition yields $c_1 = B/(1+\alpha^2) \sinh \alpha \pi$. Therefore

$$U(\alpha, y) = \frac{B \cosh \alpha y}{(1+\alpha^2) \sinh \alpha \pi} + \frac{A}{\alpha}$$

and

$$u(x, y) = \frac{2}{\pi} \int_0^\infty \left(\frac{B \cosh \alpha y}{(1+\alpha^2) \sinh \alpha \pi} + \frac{A}{\alpha} \right) \sin \alpha x \, d\alpha.$$

12. Using the Laplace transform gives

$$U(x, s) = c_1 \cosh \sqrt{s} x + c_2 \sinh \sqrt{s} x.$$

Chapter 5 Review Exercises

The condition $u(0, t) = u_0$ transforms into $U(0, s) = u_0/s$. This gives $c_1 = u_0/s$. The condition $u(1, t) = u_0$ transforms into $U(1, s) = u_0/s$. This implies that $c_2 = u_0(1 - \cosh \sqrt{s})/s \sinh \sqrt{s}$. Hence

$$\begin{aligned} U(x, s) &= \frac{u_0}{s} \cosh \sqrt{s} x + u_0 \left[\frac{1 - \cosh \sqrt{s}}{s \sinh \sqrt{s}} \right] \sinh \sqrt{s} x \\ &= u_0 \left[\frac{\sinh \sqrt{s} \cosh \sqrt{s} x - \cosh \sinh \sqrt{s} \sinh \sqrt{s} x + \sinh \sqrt{s} x}{s \sinh \sqrt{s}} \right] \\ &= u_0 \left[\frac{\sinh \sqrt{s} (1 - x) + \sinh \sqrt{s} x}{s \sinh \sqrt{s}} \right] \\ &= u_0 \left[\frac{\sinh \sqrt{s} (1 - x)}{s \sinh \sqrt{s}} + \frac{\sinh \sqrt{s} x}{s \sinh \sqrt{s}} \right] \end{aligned}$$

and

$$\begin{aligned} u(x, t) &= u_0 \left[\mathcal{L}^{-1} \left\{ \frac{\sinh \sqrt{s} (1 - x)}{s \sinh \sqrt{s}} \right\} + \mathcal{L}^{-1} \left\{ \frac{\sinh \sqrt{s} x}{s \sinh \sqrt{s}} \right\} \right] \\ &= u_0 \sum_{n=0}^{\infty} \left[\operatorname{erf} \left(\frac{2n+2-x}{2\sqrt{t}} \right) - \operatorname{erf} \left(\frac{2n+x}{2\sqrt{t}} \right) \right] \\ &\quad + u_0 \sum_{n=0}^{\infty} \left[\operatorname{erf} \left(\frac{2n+1+x}{2\sqrt{t}} \right) - \operatorname{erf} \left(\frac{2n+1-x}{2\sqrt{t}} \right) \right]. \end{aligned}$$

13. Using the Fourier transform gives

$$U(\alpha, t) = c_1 e^{-k\alpha^2 t}.$$

Now

$$u(\alpha, 0) = \int_0^{\infty} e^{-x} e^{i\alpha x} dx = \frac{e^{(i\alpha-1)x}}{i\alpha-1} \Big|_0^{\infty} = 0 - \frac{1}{i\alpha-1} = \frac{1}{1-i\alpha} = c_1$$

so

$$U(\alpha, t) = \frac{1+i\alpha}{1+\alpha^2} e^{-k\alpha^2 t}$$

and

$$u(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{1+i\alpha}{1+\alpha^2} e^{-k\alpha^2 t} e^{-i\alpha x} d\alpha.$$

Since

$$\frac{1+i\alpha}{1+\alpha^2} (\cos \alpha x - i \sin \alpha x) = \frac{\cos \alpha x + \alpha \sin \alpha x}{1+\alpha^2} + \frac{i(\alpha \cos \alpha x - \sin \alpha x)}{1+\alpha^2}$$

and the integral of the product of the second term with $e^{-k\alpha^2 t}$ is 0 (it is an odd function), we have

$$u(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{\cos \alpha x + \alpha \sin \alpha x}{1+\alpha^2} e^{-k\alpha^2 t} d\alpha.$$

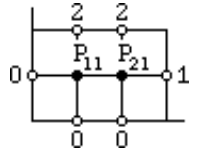
16 Numerical Solutions of Partial Differential Equations

Exercises 16.1

1. The figure shows the values of $u(x, y)$ along the boundary. We need to determine u_{11} and u_{21} . The system is

$$\begin{aligned} u_{21} + 2 + 0 + 0 - 4u_{11} &= 0 & \text{or} & & -4u_{11} + u_{21} &= -2 \\ 1 + 2 + u_{11} + 0 - 4u_{21} &= 0 & & & u_{11} - 4u_{21} &= -3. \end{aligned}$$

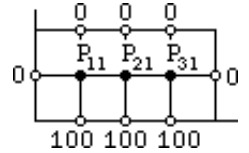
Solving we obtain $u_{11} = 11/15$ and $u_{21} = 14/15$.



2. The figure shows the values of $u(x, y)$ along the boundary. We need to determine u_{11} , u_{21} , and u_{31} . By symmetry $u_{11} = u_{31}$ and the system is

$$\begin{aligned} u_{21} + 0 + 0 + 100 - 4u_{11} &= 0 & \text{or} & & -4u_{11} + u_{21} &= -100 \\ u_{31} + 0 + u_{11} + 100 - 4u_{21} &= 0 & & & 2u_{11} - 4u_{21} &= -100. \\ 0 + 0 + u_{21} + 100 - 4u_{31} &= 0 & & & & \end{aligned}$$

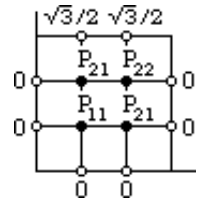
Solving we obtain $u_{11} = u_{31} = 250/7$ and $u_{21} = 300/7$.



3. The figure shows the values of $u(x, y)$ along the boundary. We need to determine u_{11} , u_{21} , u_{12} , and u_{22} . By symmetry $u_{11} = u_{21}$ and $u_{12} = u_{22}$. The system is

$$\begin{aligned} u_{21} + u_{12} + 0 + 0 - 4u_{11} &= 0 & \text{or} & & 3u_{11} + u_{12} &= 0 \\ 0 + u_{22} + u_{11} + 0 - 4u_{21} &= 0 & & & u_{11} - 3u_{12} &= -\frac{\sqrt{3}}{2} \\ u_{22} + \sqrt{3}/2 + 0 + u_{11} - 4u_{12} &= 0 & & & & \\ 0 + \sqrt{3}/2 + u_{12} + u_{21} - 4u_{22} &= 0 & & & & \end{aligned}$$

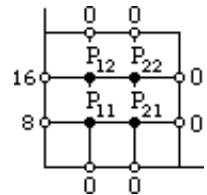
Solving we obtain $u_{11} = u_{21} = \sqrt{3}/16$ and $u_{12} = u_{22} = 3\sqrt{3}/16$.



4. The figure shows the values of $u(x, y)$ along the boundary. We need to determine u_{11} , u_{21} , u_{12} , and u_{22} . The system is

$$\begin{aligned} u_{21} + u_{12} + 8 + 0 - 4u_{11} &= 0 & \text{or} & & -4u_{11} + u_{21} + u_{12} &= -8 \\ 0 + u_{22} + u_{11} + 0 - 4u_{21} &= 0 & & & u_{11} - 4u_{21} + u_{22} &= 0 \\ u_{22} + 0 + 16 + u_{11} - 4u_{12} &= 0 & & & u_{11} - 4u_{12} + u_{22} &= -16 \\ 0 + 0 + u_{12} + u_{21} - 4u_{22} &= 0 & & & u_{21} + u_{12} - 4u_{22} &= 0. \end{aligned}$$

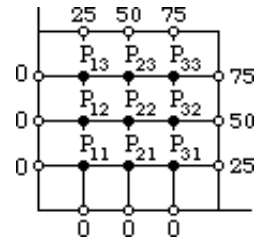
Solving we obtain $u_{11} = 11/3$, $u_{21} = 4/3$, $u_{12} = 16/3$, and $u_{22} = 5/3$.



Exercises 16.1

5. The figure shows the values of $u(x, y)$ along the boundary. For Gauss-Seidel the coefficients of the unknowns u_{11} , u_{21} , u_{31} , u_{12} , u_{22} , u_{32} , u_{13} , u_{23} , u_{33} are shown in the matrix

$$\begin{bmatrix} 0 & .25 & 0 & .25 & 0 & 0 & 0 & 0 & 0 \\ .25 & 0 & .25 & 0 & .25 & 0 & 0 & 0 & 0 \\ 0 & .25 & 0 & 0 & 0 & .25 & 0 & 0 & 0 \\ .25 & 0 & 0 & 0 & .25 & 0 & .25 & 0 & 0 \\ 0 & .25 & 0 & .25 & 0 & .25 & 0 & .25 & 0 \\ 0 & 0 & .25 & 0 & .25 & 0 & 0 & 0 & .25 \\ 0 & 0 & 0 & .25 & 0 & 0 & 0 & .25 & 0 \\ 0 & 0 & 0 & 0 & .25 & 0 & .25 & 0 & .25 \\ 0 & 0 & 0 & 0 & 0 & .25 & 0 & .25 & 0 \end{bmatrix}$$



- The constant terms in the equations are 0, 0, 6.25, 0, 0, 12.5, 6.25, 12.5, 37.5. We use 25 as the initial guess for each variable. Then $u_{11} = 6.25$, $u_{21} = u_{12} = 12.5$, $u_{31} = u_{13} = 18.75$, $u_{22} = 25$, $u_{32} = u_{23} = 37.5$, and $u_{33} = 56.25$.
6. The coefficients of the unknowns are the same as shown above in Problem 5. The constant terms are 7.5, 5, 20, 10, 0, 15, 17.5, 5, 27.5. We use 32.5 as the initial guess for each variable. Then $u_{11} = 21.92$, $u_{21} = 28.30$, $u_{31} = 38.17$, $u_{12} = 29.38$, $u_{22} = 33.13$, $u_{32} = 44.38$, $u_{13} = 22.46$, $u_{23} = 30.45$, and $u_{33} = 46.21$.
7. (a) Using the difference approximations for u_{xx} and u_{yy} we obtain

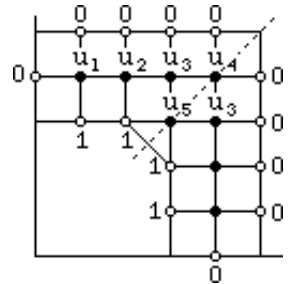
$$u_{xx} + u_{yy} = \frac{1}{h^2}(u_{i+1,j} + u_{i,j+1} + u_{i-1,j} + u_{i,j-1} - 4u_{ij}) = f(x, y)$$

so that

$$u_{i+1,j} + u_{i,j+1} + u_{i-1,j} + u_{i,j-1} - 4u_{ij} = h^2 f(x, y).$$

- (b) By symmetry, as shown in the figure, we need only solve for u_1 , u_2 , u_3 , u_4 , and u_5 . The difference equations are

$$\begin{aligned} u_2 + 0 + 0 + 1 - 4u_1 &= \frac{1}{4}(-2) \\ u_3 + 0 + u_1 + 1 - 4u_2 &= \frac{1}{4}(-2) \\ u_4 + 0 + u_2 + u_5 - 4u_3 &= \frac{1}{4}(-2) \\ 0 + 0 + u_3 + u_3 - 4u_4 &= \frac{1}{4}(-2) \\ u_3 + u_3 + 1 + 1 - 4u_5 &= \frac{1}{4}(-2) \end{aligned}$$



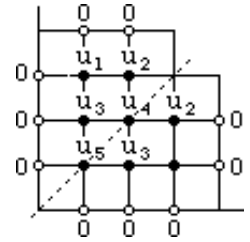
or

$$\begin{aligned} u_1 &= 0.25u_2 + 0.375 \\ u_2 &= 0.25u_1 + 0.25u_3 + 0.375 \\ u_3 &= 0.25u_2 + 0.25u_4 + 0.25u_5 + 0.125 \\ u_4 &= 0.5u_3 + 0.125 \\ u_5 &= 0.5u_3 + 0.625. \end{aligned}$$

Using Gauss-Seidel iteration we find $u_1 = 0.5427$, $u_2 = 0.6707$, $u_3 = 0.6402$, $u_4 = 0.4451$, and $u_5 = 0.9451$.

8. By symmetry, as shown in the figure, we need only solve for u_1, u_2, u_3, u_4 , and u_5 . The difference equations are

$$\begin{aligned}
 u_2 + 0 + 0 + u_3 - 4u_1 &= -1 & u_1 &= 0.25u_2 + 0.25u_3 + 0.25 \\
 0 + 0 + u_1 + u_4 - 4u_2 &= -1 & u_2 &= 0.25u_1 + 0.25u_4 + 0.25 \\
 u_4 + u_1 + 0 + u_5 - 4u_3 &= -1 & \text{or } u_3 &= 0.25u_1 + 0.25u_4 + 0.25u_5 + 0.25 \\
 u_2 + u_2 + u_3 + u_3 - 4u_4 &= -1 & u_4 &= 0.5u_2 + 0.5u_3 + 0.25 \\
 u_3 + u_3 + 0 + 0 - 4u_5 &= -1 & u_5 &= 0.5u_3 + 0.25.
 \end{aligned}$$



Using Gauss-Seidel iteration we find $u_1 = 0.6157$, $u_2 = 0.6493$, $u_3 = 0.8134$, $u_4 = 0.9813$, and $u_5 = 0.6567$.

Exercises 16.2

1. We identify $c = 1$, $a = 2$, $T = 1$, $n = 8$, and $m = 40$. Then $h = 2/8 = 0.25$, $k = 1/40 = 0.025$, and $\lambda = 2/5 = 0.4$.

TIME	X=0.25	X=0.50	X=0.75	X=1.00	X=1.25	X=1.50	X=1.75
0.000	1.0000	1.0000	1.0000	1.0000	0.0000	0.0000	0.0000
0.025	0.6000	1.0000	1.0000	0.6000	0.4000	0.0000	0.0000
0.050	0.5200	0.8400	0.8400	0.6800	0.3200	0.1600	0.0000
0.075	0.4400	0.7120	0.7760	0.6000	0.4000	0.1600	0.0640
0.100	0.3728	0.6288	0.6800	0.5904	0.3840	0.2176	0.0768
0.125	0.3261	0.5469	0.6237	0.5437	0.4000	0.2278	0.1024
0.150	0.2840	0.4893	0.5610	0.5182	0.3886	0.2465	0.1116
0.175	0.2525	0.4358	0.5152	0.4835	0.3836	0.2494	0.1209
0.200	0.2248	0.3942	0.4708	0.4562	0.3699	0.2517	0.1239
0.225	0.2027	0.3571	0.4343	0.4275	0.3571	0.2479	0.1255
0.250	0.1834	0.3262	0.4007	0.4021	0.3416	0.2426	0.1242
0.275	0.1672	0.2989	0.3715	0.3773	0.3262	0.2348	0.1219
0.300	0.1530	0.2752	0.3448	0.3545	0.3101	0.2262	0.1183
0.325	0.1407	0.2541	0.3209	0.3329	0.2943	0.2166	0.1141
0.350	0.1298	0.2354	0.2990	0.3126	0.2787	0.2067	0.1095
0.375	0.1201	0.2186	0.2790	0.2936	0.2635	0.1966	0.1046
0.400	0.1115	0.2034	0.2607	0.2757	0.2488	0.1865	0.0996
0.425	0.1036	0.1895	0.2438	0.2589	0.2347	0.1766	0.0945
0.450	0.0965	0.1769	0.2281	0.2432	0.2211	0.1670	0.0896
0.475	0.0901	0.1652	0.2136	0.2283	0.2083	0.1577	0.0847
0.500	0.0841	0.1545	0.2002	0.2144	0.1961	0.1487	0.0800

(continued)

Exercises 16.2

(continued)

TIME	X=0.25	X=0.50	X=0.75	X=1.00	X=1.25	X=1.50	X=1.75
0.525	0.0786	0.1446	0.1876	0.2014	0.1845	0.1402	0.0755
0.550	0.0736	0.1354	0.1759	0.1891	0.1735	0.1320	0.0712
0.575	0.0689	0.1269	0.1650	0.1776	0.1632	0.1243	0.0670
0.600	0.0645	0.1189	0.1548	0.1668	0.1534	0.1169	0.0631
0.625	0.0605	0.1115	0.1452	0.1566	0.1442	0.1100	0.0594
0.650	0.0567	0.1046	0.1363	0.1471	0.1355	0.1034	0.0559
0.675	0.0532	0.0981	0.1279	0.1381	0.1273	0.0972	0.0525
0.700	0.0499	0.0921	0.1201	0.1297	0.1196	0.0914	0.0494
0.725	0.0468	0.0864	0.1127	0.1218	0.1124	0.0859	0.0464
0.750	0.0439	0.0811	0.1058	0.1144	0.1056	0.0807	0.0436
0.775	0.0412	0.0761	0.0994	0.1074	0.0992	0.0758	0.0410
0.800	0.0387	0.0715	0.0933	0.1009	0.0931	0.0712	0.0385
0.825	0.0363	0.0671	0.0876	0.0948	0.0875	0.0669	0.0362
0.850	0.0341	0.0630	0.0823	0.0890	0.0822	0.0628	0.0340
0.875	0.0320	0.0591	0.0772	0.0836	0.0772	0.0590	0.0319
0.900	0.0301	0.0555	0.0725	0.0785	0.0725	0.0554	0.0300
0.925	0.0282	0.0521	0.0681	0.0737	0.0681	0.0521	0.0282
0.950	0.0265	0.0490	0.0640	0.0692	0.0639	0.0489	0.0265
0.975	0.0249	0.0460	0.0601	0.0650	0.0600	0.0459	0.0249
1.000	0.0234	0.0432	0.0564	0.0610	0.0564	0.0431	0.0233

2.

(x,y)	exact	approx	abs error
(0.25,0.1)	0.3794	0.3728	0.0066
(1,0.5)	0.1854	0.2144	0.0290
(1.5,0.8)	0.0623	0.0712	0.0089

3. We identify $c = 1$, $a = 2$, $T = 1$, $n = 8$, and $m = 40$. Then $h = 2/8 = 0.25$, $k = 1/40 = 0.025$, and $\lambda = 2/5 = 0.4$.

TIME	X=0.25	X=0.50	X=0.75	X=1.00	X=1.25	X=1.50	X=1.75
0.000	1.0000	1.0000	1.0000	1.0000	0.0000	0.0000	0.0000
0.025	0.7074	0.9520	0.9566	0.7444	0.2545	0.0371	0.0053
0.050	0.5606	0.8499	0.8685	0.6633	0.3303	0.1034	0.0223
0.075	0.4684	0.7473	0.7836	0.6191	0.3614	0.1529	0.0462
0.100	0.4015	0.6577	0.7084	0.5837	0.3753	0.1871	0.0684
0.125	0.3492	0.5821	0.6428	0.5510	0.3797	0.2101	0.0861
0.150	0.3069	0.5187	0.5857	0.5199	0.3778	0.2247	0.0990
0.175	0.2721	0.4652	0.5359	0.4901	0.3716	0.2329	0.1078
0.200	0.2430	0.4198	0.4921	0.4617	0.3622	0.2362	0.1132
0.225	0.2186	0.3809	0.4533	0.4348	0.3507	0.2358	0.1160
0.250	0.1977	0.3473	0.4189	0.4093	0.3378	0.2327	0.1166
0.275	0.1798	0.3181	0.3881	0.3853	0.3240	0.2275	0.1157
0.300	0.1643	0.2924	0.3604	0.3626	0.3097	0.2208	0.1136
0.325	0.1507	0.2697	0.3353	0.3412	0.2953	0.2131	0.1107
0.350	0.1387	0.2495	0.3125	0.3211	0.2808	0.2047	0.1071
0.375	0.1281	0.2313	0.2916	0.3021	0.2666	0.1960	0.1032
0.400	0.1187	0.2150	0.2725	0.2843	0.2528	0.1871	0.0989
0.425	0.1102	0.2002	0.2549	0.2675	0.2393	0.1781	0.0946
0.450	0.1025	0.1867	0.2387	0.2517	0.2263	0.1692	0.0902
0.475	0.0955	0.1743	0.2236	0.2368	0.2139	0.1606	0.0858
0.500	0.0891	0.1630	0.2097	0.2228	0.2020	0.1521	0.0814
0.525	0.0833	0.1525	0.1967	0.2096	0.1906	0.1439	0.0772
0.550	0.0779	0.1429	0.1846	0.1973	0.1798	0.1361	0.0731
0.575	0.0729	0.1339	0.1734	0.1856	0.1696	0.1285	0.0691
0.600	0.0683	0.1256	0.1628	0.1746	0.1598	0.1214	0.0653
0.625	0.0641	0.1179	0.1530	0.1643	0.1506	0.1145	0.0617
0.650	0.0601	0.1106	0.1438	0.1546	0.1419	0.1080	0.0582
0.675	0.0564	0.1039	0.1351	0.1455	0.1336	0.1018	0.0549
0.700	0.0530	0.0976	0.1270	0.1369	0.1259	0.0959	0.0518
0.725	0.0497	0.0917	0.1194	0.1288	0.1185	0.0904	0.0488
0.750	0.0467	0.0862	0.1123	0.1212	0.1116	0.0852	0.0460
0.775	0.0439	0.0810	0.1056	0.1140	0.1050	0.0802	0.0433
0.800	0.0413	0.0762	0.0993	0.1073	0.0989	0.0755	0.0408
0.825	0.0388	0.0716	0.0934	0.1009	0.0931	0.0711	0.0384
0.850	0.0365	0.0674	0.0879	0.0950	0.0876	0.0669	0.0362
0.875	0.0343	0.0633	0.0827	0.0894	0.0824	0.0630	0.0341
0.900	0.0323	0.0596	0.0778	0.0841	0.0776	0.0593	0.0321
0.925	0.0303	0.0560	0.0732	0.0791	0.0730	0.0558	0.0302
0.950	0.0285	0.0527	0.0688	0.0744	0.0687	0.0526	0.0284
0.975	0.0268	0.0496	0.0647	0.0700	0.0647	0.0495	0.0268
1.000	0.0253	0.0466	0.0609	0.0659	0.0608	0.0465	0.0252

(x,y)	exact	approx	abs error
(0.25,0.1)	0.3794	0.4015	0.0221
(1,0.5)	0.1854	0.2228	0.0374
(1.5,0.8)	0.0623	0.0755	0.0132

Exercises 16.2

4. We identify $c = 1$, $a = 2$, $T = 1$, $n = 8$, and $m = 20$. Then $h = 2/8 = 0.25$, $h = 1/20 = 0.05$, and $\lambda = 4/5 = 0.8$.

TIME	X=0.25	X=0.50	X=0.75	X=1.00	X=1.25	X=1.50	X=1.75
0.00	1.00	1.00	1.00	1.00	0.00	0.00	0.00
0.05	0.20	1.00	1.00	0.20	0.80	0.00	0.00
0.10	0.68	0.36	0.36	1.32	-0.32	0.64	0.00
0.15	-0.12	0.62	1.13	-0.76	1.76	-0.64	0.51
0.20	0.56	0.44	-0.79	2.77	-2.18	2.20	-0.82
0.25	0.01	-0.44	3.04	-4.03	5.28	-3.72	2.25
0.30	-0.36	2.70	-5.41	9.07	-9.37	8.26	-4.33
0.35	2.38	-6.24	12.67	-17.26	19.49	-15.91	9.20
0.40	-6.42	15.78	-26.40	36.08	-38.23	32.50	-18.25
0.45	16.47	-35.72	57.33	-73.35	77.80	-64.68	36.94
0.50	-38.46	80.48	-121.66	152.12	-157.11	130.60	-73.91
0.55	87.46	-176.38	259.07	-314.28	320.44	-263.18	148.83
0.60	-193.58	383.05	-547.97	652.17	-654.23	533.32	-299.84
0.65	422.59	-823.07	1156.96	-1353.07	1340.93	-1083.25	606.56
0.70	-912.01	1757.48	-2435.09	2810.16	-2753.61	2207.94	-1230.53
0.75	1953.19	-3732.17	5115.16	-5837.05	5666.65	-4512.08	2504.67
0.80	-4157.65	7893.99	-10724.47	12127.68	-11679.29	9244.30	-5112.47
0.85	8809.78	-16642.09	22452.02	-25199.62	24105.16	-18979.99	10462.92
0.90	-18599.54	34994.69	-46944.58	52365.51	-49806.79	39042.46	-21461.75
0.95	39155.48	-73432.11	98054.91	-108820.40	103010.45	-80440.31	44111.02
1.00	-82238.97	153827.58	-204634.95	226144.53	-213214.84	165961.36	-90818.86

(x,y)	exact	approx	abs error
(0.25,0.1)	0.3794	0.6800	0.3006
(1,0.5)	0.1854	152.1152	151.9298
(1.5,0.8)	0.0623	9244.3042	9244.2419

In this case $\lambda = 0.8$ is greater than 0.5 and the procedure is unstable.

5. We identify $c = 1$, $a = 2$, $T = 1$, $n = 8$, and $m = 20$. Then $h = 2/8 = 0.25$, $k = 1/20 = 0.05$, and $\lambda = 4/5 = 0.8$.

TIME	X=0.25	X=0.50	X=0.75	X=1.00	X=1.25	X=1.50	X=1.75
0.00	1.0000	1.0000	1.0000	1.0000	0.0000	0.0000	0.0000
0.05	0.5265	0.8693	0.8852	0.6141	0.3783	0.0884	0.0197
0.10	0.3972	0.6551	0.7043	0.5883	0.3723	0.1955	0.0653
0.15	0.3042	0.5150	0.5844	0.5192	0.3812	0.2261	0.1010
0.20	0.2409	0.4171	0.4901	0.4620	0.3636	0.2385	0.1145
0.25	0.1962	0.3452	0.4174	0.4092	0.3391	0.2343	0.1178
0.30	0.1631	0.2908	0.3592	0.3624	0.3105	0.2220	0.1145
0.35	0.1379	0.2482	0.3115	0.3208	0.2813	0.2056	0.1077
0.40	0.1181	0.2141	0.2718	0.2840	0.2530	0.1876	0.0993
0.45	0.1020	0.1860	0.2381	0.2514	0.2265	0.1696	0.0904
0.50	0.0888	0.1625	0.2092	0.2226	0.2020	0.1523	0.0816
0.55	0.0776	0.1425	0.1842	0.1970	0.1798	0.1361	0.0732
0.60	0.0681	0.1253	0.1625	0.1744	0.1597	0.1214	0.0654
0.65	0.0599	0.1104	0.1435	0.1544	0.1418	0.1079	0.0582
0.70	0.0528	0.0974	0.1268	0.1366	0.1257	0.0959	0.0518
0.75	0.0466	0.0860	0.1121	0.1210	0.1114	0.0851	0.0460
0.80	0.0412	0.0760	0.0991	0.1071	0.0987	0.0754	0.0408
0.85	0.0364	0.0672	0.0877	0.0948	0.0874	0.0668	0.0361
0.90	0.0322	0.0594	0.0776	0.0839	0.0774	0.0592	0.0320
0.95	0.0285	0.0526	0.0687	0.0743	0.0686	0.0524	0.0284
1.00	0.0252	0.0465	0.0608	0.0657	0.0607	0.0464	0.0251

(x,y)	exact	approx	abs error
(0.25,0.1)	0.3794	0.3972	0.0178
(1,0.5)	0.1854	0.2226	0.0372
(1.5,0.8)	0.0623	0.0754	0.0131

6. We identify $c = 15/88 \approx 0.1705$, $a = 20$, $T = 10$, $n = 10$, and $m = 10$. Then $h = 2$, $k = 1$, and $\lambda = 15/352 \approx 0.0426$.

TIME	X=2	X=4	X=6	X=8	X=10	X=12	X=14	X=16	X=18
0	30.0000	30.0000	30.0000	30.0000	30.0000	30.0000	30.0000	30.0000	30.0000
1	28.7216	30.0000	30.0000	30.0000	30.0000	30.0000	30.0000	30.0000	28.7216
2	27.5521	29.9455	30.0000	30.0000	30.0000	30.0000	30.0000	29.9455	27.5521
3	26.4800	29.8459	29.9977	30.0000	30.0000	30.0000	29.9977	29.8459	26.4800
4	25.4951	29.7089	29.9913	29.9999	30.0000	29.9999	29.9913	29.7089	25.4951
5	24.5882	29.5414	29.9796	29.9995	30.0000	29.9995	29.9796	29.5414	24.5882
6	23.7515	29.3490	29.9618	29.9987	30.0000	29.9987	29.9618	29.3490	23.7515
7	22.9779	29.1365	29.9373	29.9972	29.9998	29.9972	29.9373	29.1365	22.9779
8	22.2611	28.9082	29.9057	29.9948	29.9996	29.9948	29.9057	28.9082	22.2611
9	21.5958	28.6675	29.8670	29.9912	29.9992	29.9912	29.8670	28.6675	21.5958
10	20.9768	28.4172	29.8212	29.9862	29.9985	29.9862	29.8212	28.4172	20.9768

Exercises 16.2

7. We identify $c = 15/88 \approx 0.1705$, $a = 50$, $T = 10$, $n = 10$, and $m = 10$. Then $h = 5$, $k = 1$, and $\lambda = 3/440 \approx 0.0068$.

TIME	X=5	X=10	X=15	X=20	X=25	X=30	X=35	X=40	X=45
0	30.0000	30.0000	30.0000	30.0000	30.0000	30.0000	30.0000	30.0000	30.0000
1	29.7955	30.0000	30.0000	30.0000	30.0000	30.0000	30.0000	30.0000	29.7955
2	29.5937	29.9986	30.0000	30.0000	30.0000	30.0000	30.0000	29.9986	29.5937
3	29.3947	29.9959	30.0000	30.0000	30.0000	30.0000	30.0000	29.9959	29.3947
4	29.1984	29.9918	30.0000	30.0000	30.0000	30.0000	30.0000	29.9918	29.1984
5	29.0047	29.9864	29.9999	30.0000	30.0000	30.0000	29.9999	29.9864	29.0047
6	28.8136	29.9798	29.9998	30.0000	30.0000	30.0000	29.9998	29.9798	28.8136
7	28.6251	29.9720	29.9997	30.0000	30.0000	30.0000	29.9997	29.9720	28.6251
8	28.4391	29.9630	29.9995	30.0000	30.0000	30.0000	29.9995	29.9630	28.4391
9	28.2556	29.9529	29.9992	30.0000	30.0000	30.0000	29.9992	29.9529	28.2556
10	28.0745	29.9416	29.9989	30.0000	30.0000	30.0000	29.9989	29.9416	28.0745

8. We identify $c = 50/27 \approx 1.8519$, $a = 20$, $T = 10$, $n = 10$, and $m = 10$. Then $h = 2$, $k = 1$, and $\lambda = 25/54 \approx 0.4630$.

TIME	X=2	X=4	X=6	X=8	X=10	X=12	X=14	X=16	X=18
0	18.0000	32.0000	42.0000	48.0000	50.0000	48.0000	42.0000	32.0000	18.0000
1	16.1481	30.1481	40.1481	46.1481	48.1481	46.1481	40.1481	30.1481	16.1481
2	15.1536	28.2963	38.2963	44.2963	46.2963	44.2963	38.2963	28.2963	15.1536
3	14.2226	26.8414	36.4444	42.4444	44.4444	42.4444	36.4444	26.8414	14.2226
4	13.4801	25.4452	34.7764	40.5926	42.5926	40.5926	34.7764	25.4452	13.4801
5	12.7787	24.2258	33.1491	38.8258	40.7407	38.8258	33.1491	24.2258	12.7787
6	12.1622	23.0574	31.6460	37.0842	38.9677	37.0842	31.6460	23.0574	12.1622
7	11.5756	21.9895	30.1875	35.4385	37.2238	35.4385	30.1875	21.9895	11.5756
8	11.0378	20.9636	28.8232	33.8340	35.5707	33.8340	28.8232	20.9636	11.0378
9	10.5230	20.0070	27.5043	32.3182	33.9626	32.3182	27.5043	20.0070	10.5230
10	10.0420	19.0872	26.2620	30.8509	32.4400	30.8509	26.2620	19.0872	10.0420

9. We identify $c = 260/159 \approx 1.6352$, $a = 100$, $T = 10$, $n = 10$, and $m = 10$. Then $h = 10$, $k = 1$, and $\lambda = 13/795 \approx 0.0164$.

TIME	X=10	X=20	X=30	X=40	X=50	X=60	X=70	X=80	X=90
0	8.0000	16.0000	24.0000	32.0000	40.0000	32.0000	24.0000	16.0000	8.0000
1	8.0000	16.0000	23.6075	31.3459	39.2151	31.6075	23.7384	15.8692	8.0000
2	8.0000	15.9936	23.2279	30.7068	38.4452	31.2151	23.4789	15.7384	7.9979
3	7.9999	15.9812	22.8606	30.0824	37.6900	30.8229	23.2214	15.6076	7.9937
4	7.9996	15.9631	22.5050	29.4724	36.9492	30.4312	22.9660	15.4769	7.9874
5	7.9990	15.9399	22.1606	28.8765	36.2228	30.0401	22.7125	15.3463	7.9793
6	7.9981	15.9118	21.8270	28.2945	35.5103	29.6500	22.4610	15.2158	7.9693
7	7.9967	15.8791	21.5037	27.7261	34.8117	29.2610	22.2112	15.0854	7.9575
8	7.9948	15.8422	21.1902	27.1709	34.1266	28.8733	21.9633	14.9553	7.9439
9	7.9924	15.8013	20.8861	26.6288	33.4548	28.4870	21.7172	14.8253	7.9287
10	7.9894	15.7568	20.5911	26.0995	32.7961	28.1024	21.4727	14.6956	7.9118

10. (a) With $n = 4$ we have $h = 1/2$ so that $\lambda = 1/100 = 0.01$.

(b) We observe that $\alpha = 2(1 + 1/\lambda) = 202$ and $\beta = 2(1 - 1/\lambda) = -198$. The system of equations is

$$\begin{aligned} -u_{01} + \alpha u_{11} - u_{21} &= u_{20} - \beta u_{10} + u_{00} \\ -u_{11} + \alpha u_{21} - u_{31} &= u_{30} - \beta u_{20} + u_{10} \\ -u_{21} + \alpha u_{31} - u_{41} &= u_{40} - \beta u_{30} + u_{20}. \end{aligned}$$

Now $u_{00} = u_{01} = u_{40} = u_{41} = 0$, so the system is

$$\begin{aligned} \alpha u_{11} - u_{21} &= u_{20} - \beta u_{10} \\ -u_{11} + \alpha u_{21} - u_{31} &= u_{30} - \beta u_{20} + u_{10} \\ -u_{21} + \alpha u_{31} &= -\beta u_{30} + u_{20} \end{aligned}$$

or

$$\begin{aligned} 202u_{11} - u_{21} &= \sin \pi + 198 \sin \frac{\pi}{2} = 198 \\ -u_{11} + 202u_{21} - u_{31} &= \sin \frac{3\pi}{2} + 198 \sin \pi + \sin \frac{\pi}{2} = 0 \\ -u_{21} + 202u_{31} &= 198 \sin \frac{3\pi}{2} + \sin \pi = -198. \end{aligned}$$

(c) The solution of this system is $u_{11} \approx 0.9802$, $u_{21} = 0$, $u_{31} \approx -0.9802$. The corresponding entries in Table 16.3 in the text are 0.9768, 0, and -0.9768 .

11. (a) The differential equation is $k \frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}$ where $k = K/\gamma\rho$. If we let $u(x, t) = v(x, t) + \psi(x)$, then

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 v}{\partial x^2} + \psi'' \quad \text{and} \quad \frac{\partial u}{\partial t} = \frac{\partial v}{\partial t}.$$

Substituting into the differential equation gives

$$k \frac{\partial^2 v}{\partial x^2} + k\psi'' = \frac{\partial v}{\partial t}.$$

Requiring $k\psi'' = 0$ we have $\psi(x) = c_1x + c_2$. The boundary conditions become

$$u(0, t) = v(0, t) + \psi(0) = 20 \quad \text{and} \quad u(20, t) = v(20, t) + \psi(20) = 30.$$

Letting $\psi(0) = 20$ and $\psi(20) = 30$ we obtain the homogeneous boundary conditions in v : $v(0, t) = v(20, t) = 0$. Now $\psi(0) = 20$ and $\psi(20) = 30$ imply that $c_1 = 1/2$ and $c_2 = 20$. The steady-state solution is $\psi(x) = \frac{1}{2}x + 20$.

(b) To use the Crank-Nicholson method we identify $c = 375/212 \approx 1.7689$, $a = 20$, $T = 400$, $n = 5$, and $m = 40$. Then $h = 4$, $k = 10$, and $\lambda = 1875/1696 \approx 1.1055$.

Exercises 16.2

TIME	X=4.00	X=8.00	X=12.00	X=16.00
0.00	50.0000	50.0000	50.0000	50.0000
10.00	32.7433	44.2679	45.4228	38.2971
20.00	29.9946	36.2354	38.3148	35.8160
30.00	26.9487	32.1409	34.0874	32.9644
40.00	25.2691	29.2562	31.2704	31.2580
50.00	24.1178	27.4348	29.4296	30.1207
60.00	23.3821	26.2339	28.2356	29.3810
70.00	22.8995	25.4560	27.4554	28.8998
80.00	22.5861	24.9481	26.9482	28.5859
90.00	22.3817	24.6176	26.6175	28.3817
100.00	22.2486	24.4022	26.4023	28.2486
110.00	22.1619	24.2620	26.2620	28.1619
120.00	22.1055	24.1707	26.1707	28.1055
130.00	22.0687	24.1112	26.1112	28.0687
140.00	22.0447	24.0724	26.0724	28.0447
150.00	22.0291	24.0472	26.0472	28.0291
160.00	22.0190	24.0307	26.0307	28.0190
170.00	22.0124	24.0200	26.0200	28.0124
180.00	22.0081	24.0130	26.0130	28.0081
190.00	22.0052	24.0085	26.0085	28.0052
200.00	22.0034	24.0055	26.0055	28.0034
210.00	22.0022	24.0036	26.0036	28.0022
220.00	22.0015	24.0023	26.0023	28.0015
230.00	22.0009	24.0015	26.0015	28.0009
240.00	22.0006	24.0010	26.0010	28.0006
250.00	22.0004	24.0007	26.0007	28.0004
260.00	22.0003	24.0004	26.0004	28.0003
270.00	22.0002	24.0003	26.0003	28.0002
280.00	22.0001	24.0002	26.0002	28.0001
290.00	22.0001	24.0001	26.0001	28.0001
300.00	22.0000	24.0001	26.0001	28.0000
310.00	22.0000	24.0001	26.0001	28.0000
320.00	22.0000	24.0000	26.0000	28.0000
330.00	22.0000	24.0000	26.0000	28.0000
340.00	22.0000	24.0000	26.0000	28.0000
350.00	22.0000	24.0000	26.0000	28.0000

We observe that the approximate steady-state temperatures agree exactly with the corresponding values of $\psi(x)$.

12. We identify $c = 1$, $a = 1$, $T = 1$, $n = 5$, and $m = 20$. Then $h = 0.2$, $k = 0.04$, and $\lambda = 1$. The values below were obtained using *Excel*, which carries more than 12 significant digits. In order to see evidence of instability use $0 \leq t \leq 2$.

TIME	X=0.2	X=0.4	X=0.6	X=0.8	TIME	X=0.2	X=0.4	X=0.6	X=0.8
0.00	0.5878	0.9511	0.9511	0.5878	1.04	0.0000	0.0000	0.0000	0.0000
0.04	0.3633	0.5878	0.5878	0.3633	1.08	0.0000	0.0000	0.0000	0.0000
0.08	0.2245	0.3633	0.3633	0.2245	1.12	0.0000	0.0000	0.0000	0.0000
0.12	0.1388	0.2245	0.2245	0.1388	1.16	0.0000	0.0000	0.0000	0.0000
0.16	0.0858	0.1388	0.1388	0.0858	1.20	-0.0001	0.0001	-0.0001	0.0001
0.20	0.0530	0.0858	0.0858	0.0530	1.24	0.0001	-0.0002	0.0002	-0.0001
0.24	0.0328	0.0530	0.0530	0.0328	1.28	-0.0004	0.0006	-0.0006	0.0004
0.28	0.0202	0.0328	0.0328	0.0202	1.32	0.0010	-0.0015	0.0015	-0.0010
0.32	0.0125	0.0202	0.0202	0.0125	1.36	-0.0025	0.0040	-0.0040	0.0025
0.36	0.0077	0.0125	0.0125	0.0077	1.40	0.0065	-0.0106	0.0106	-0.0065
0.40	0.0048	0.0077	0.0077	0.0048	1.44	-0.0171	0.0277	-0.0277	0.0171
0.44	0.0030	0.0048	0.0048	0.0030	1.48	0.0448	-0.0724	0.0724	-0.0448
0.48	0.0018	0.0030	0.0030	0.0018	1.52	-0.1172	0.1897	-0.1897	0.1172
0.52	0.0011	0.0018	0.0018	0.0011	1.56	0.3069	-0.4965	0.4965	-0.3069
0.56	0.0007	0.0011	0.0011	0.0007	1.60	-0.8034	1.2999	-1.2999	0.8034
0.60	0.0004	0.0007	0.0007	0.0004	1.64	2.1033	-3.4032	3.4032	-2.1033
0.64	0.0003	0.0004	0.0004	0.0003	1.68	-5.5064	8.9096	-8.9096	5.5064
0.68	0.0002	0.0003	0.0003	0.0002	1.72	14.416	-23.326	23.326	-14.416
0.72	0.0001	0.0002	0.0002	0.0001	1.76	-37.742	61.067	-61.067	37.742
0.76	0.0001	0.0001	0.0001	0.0001	1.80	98.809	-159.88	159.88	-98.809
0.80	0.0000	0.0001	0.0001	0.0000	1.84	-258.68	418.56	-418.56	258.685
0.84	0.0000	0.0000	0.0000	0.0000	1.88	677.24	-1095.8	1095.8	-677.245
0.88	0.0000	0.0000	0.0000	0.0000	1.92	-1773.1	2868.9	-2868.9	1773.1
0.92	0.0000	0.0000	0.0000	0.0000	1.96	4641.9	-7510.8	7510.8	-4641.9
0.96	0.0000	0.0000	0.0000	0.0000	2.00	-12153	19663	-19663	12153
1.00	0.0000	0.0000	0.0000	0.0000					

Exercises 16.3

1. (a) Identifying $h = 1/4$ and $k = 1/10$ we see that $\lambda = 2/5$.

TIME	X=0.25	X=0.5	X=0.75
0.00	0.1875	0.2500	0.1875
0.10	0.1775	0.2400	0.1775
0.20	0.1491	0.2100	0.1491
0.30	0.1066	0.1605	0.1066
0.40	0.0556	0.0938	0.0556
0.50	0.0019	0.0148	0.0019
0.60	-0.0501	-0.0682	-0.0501
0.70	-0.0970	-0.1455	-0.0970
0.80	-0.1361	-0.2072	-0.1361
0.90	-0.1648	-0.2462	-0.1648
1.00	-0.1802	-0.2591	-0.1802

Exercises 16.3

(b) Identifying $h = 2/5$ and $k = 1/10$ we see that $\lambda = 1/4$.

TIME	X=0.4	X=0.8	X=1.2	X=1.6
0.00	0.0032	0.5273	0.5273	0.0032
0.10	0.0194	0.5109	0.5109	0.0194
0.20	0.0652	0.4638	0.4638	0.0652
0.30	0.1318	0.3918	0.3918	0.1318
0.40	0.2065	0.3035	0.3035	0.2065
0.50	0.2743	0.2092	0.2092	0.2743
0.60	0.3208	0.1190	0.1190	0.3208
0.70	0.3348	0.0413	0.0413	0.3348
0.80	0.3094	-0.0180	-0.0180	0.3094
0.90	0.2443	-0.0568	-0.0568	0.2443
1.00	0.1450	-0.0768	-0.0768	0.1450

(c) Identifying $h = 1/10$ and $k = 1/25$ we see that $\lambda = 2\sqrt{2}/5$.

TIME	X=0.1	X=0.2	X=0.3	X=0.4	X=0.5	X=0.6	X=0.7	X=0.8	X=0.9
0.00	0.0000	0.0000	0.0000	0.0000	0.0000	0.5000	0.5000	0.5000	0.5000
0.04	0.0000	0.0000	0.0000	0.0000	0.0800	0.4200	0.5000	0.5000	0.4200
0.08	0.0000	0.0000	0.0000	0.0256	0.2432	0.2568	0.4744	0.4744	0.2312
0.12	0.0000	0.0000	0.0082	0.1126	0.3411	0.1589	0.3792	0.3710	0.0462
0.16	0.0000	0.0026	0.0472	0.2394	0.3076	0.1898	0.2108	0.1663	-0.0496
0.20	0.0008	0.0187	0.1334	0.3264	0.2146	0.2651	0.0215	-0.0933	-0.0605
0.24	0.0071	0.0657	0.2447	0.3159	0.1735	0.2463	-0.1266	-0.3056	-0.0625
0.28	0.0299	0.1513	0.3215	0.2371	0.2013	0.0849	-0.2127	-0.3829	-0.1223
0.32	0.0819	0.2525	0.3168	0.1737	0.2033	-0.1345	-0.2580	-0.3223	-0.2264
0.36	0.1623	0.3197	0.2458	0.1657	0.0877	-0.2853	-0.2843	-0.2104	-0.2887
0.40	0.2412	0.3129	0.1727	0.1583	-0.1223	-0.3164	-0.2874	-0.1473	-0.2336
0.44	0.2657	0.2383	0.1399	0.0658	-0.3046	-0.2761	-0.2549	-0.1565	-0.0761
0.48	0.1965	0.1410	0.1149	-0.1216	-0.3593	-0.2381	-0.1977	-0.1715	0.0800
0.52	0.0466	0.0531	0.0225	-0.3093	-0.2992	-0.2260	-0.1451	-0.1144	0.1300
0.56	-0.1161	-0.0466	-0.1662	-0.3876	-0.2188	-0.2114	-0.1085	0.0111	0.0602
0.60	-0.2194	-0.2069	-0.3875	-0.3411	-0.1901	-0.1662	-0.0666	0.1140	-0.0446
0.64	-0.2485	-0.4290	-0.5362	-0.2611	-0.2021	-0.0969	0.0012	0.1084	-0.0843
0.68	-0.2559	-0.6276	-0.5625	-0.2503	-0.1993	-0.0298	0.0720	0.0068	-0.0354
0.72	-0.3003	-0.6865	-0.5097	-0.3230	-0.1585	0.0156	0.0893	-0.0874	0.0384
0.76	-0.3722	-0.5652	-0.4538	-0.4029	-0.1147	0.0289	0.0265	-0.0849	0.0596
0.80	-0.3867	-0.3464	-0.4172	-0.4068	-0.1172	-0.0046	-0.0712	-0.0005	0.0155
0.84	-0.2647	-0.1633	-0.3546	-0.3214	-0.1763	-0.0954	-0.1249	0.0665	-0.0386
0.88	-0.0254	-0.0738	-0.2202	-0.2002	-0.2559	-0.2215	-0.1079	0.0385	-0.0468
0.92	0.2064	-0.0157	-0.0325	-0.1032	-0.3067	-0.3223	-0.0804	-0.0636	-0.0127
0.96	0.3012	0.1081	0.1380	-0.0487	-0.2974	-0.3407	-0.1250	-0.1548	0.0092
1.00	0.2378	0.3032	0.2392	-0.0141	-0.2223	-0.2762	-0.2481	-0.1840	-0.0244

2. (a) In Section 13.4 the solution of the wave equation is shown to be

$$u(x, t) = \sum_{n=1}^{\infty} (A_n \cos n\pi t + B_n \sin n\pi t) \sin n\pi x$$

where

$$A_n = 2 \int_0^1 \sin \pi x \sin n\pi x \, dx = \begin{cases} 1, & n = 1 \\ 0, & n = 2, 3, 4, \dots \end{cases}$$

and

$$B_n = \frac{2}{n\pi} \int_0^1 0 \, dx = 0.$$

Thus $u(x, t) = \cos \pi t \sin \pi x$.

- (b) We have $h = 1/4$, $k = 0.5/5 = 0.1$ and $\lambda = 0.4$. Now $u_{0,j} = u_{4,j} = 0$ or $j = 0, 1, \dots, 5$, and the initial values of u are $u_{1,0} = u(1/4, 0) = \sin \pi/4 \approx 0.7071$, $u_{2,0} = u(1/2, 0) = \sin \pi/2 = 1$, $u_{3,0} = u(3/4, 0) = \sin 3\pi/4 \approx 0.7071$. From equation (6) in the text we have

$$u_{i,1} = 0.8(u_{i+1,0} + u_{i-1,0}) + 0.84u_{i,0} + 0.1(0).$$

Then $u_{1,1} \approx 0.6740$, $u_{2,1} = 0.9531$, $u_{3,1} = 0.6740$. From equation (3) in the text we have for $j = 1, 2, 3, \dots$

$$u_{i,j+1} = 0.16u_{i+1,j} + 2(0.84)u_{i,j} + 0.16u_{i-1,j} - u_{i,j-1}.$$

The results of the calculations are given in the table.

TIME	x=0.25	x=0.50	x=0.75
0.0	0.7071	1.0000	0.7071
0.1	0.6740	0.9531	0.6740
0.2	0.5777	0.8169	0.5777
0.3	0.4272	0.6042	0.4272
0.4	0.2367	0.3348	0.2367
0.5	0.0241	0.0340	0.0241

(c)

i, j	approx	exact	error
1, 1	0.6740	0.6725	0.0015
1, 2	0.5777	0.5721	0.0056
1, 3	0.4272	0.4156	0.0116
1, 4	0.2367	0.2185	0.0182
1, 5	0.0241	0.0000	0.0241
2, 1	0.9531	0.9511	0.0021
2, 2	0.8169	0.8090	0.0079
2, 3	0.6042	0.5878	0.0164
2, 4	0.3348	0.3090	0.0258
2, 5	0.0340	0.0000	0.0340
3, 1	0.6740	0.6725	0.0015
3, 2	0.5777	0.5721	0.0056
3, 3	0.4272	0.4156	0.0116
3, 4	0.2367	0.2185	0.0182
3, 5	0.0241	0.0000	0.0241

Exercises 16.3

3. (a) Identifying $h = 1/5$ and $k = 0.5/10 = 0.05$ we see that $\lambda = 0.25$.

TIME	X=0.2	X=0.4	X=0.6	X=0.8
0.00	0.5878	0.9511	0.9511	0.5878
0.05	0.5808	0.9397	0.9397	0.5808
0.10	0.5599	0.9059	0.9059	0.5599
0.15	0.5256	0.8505	0.8505	0.5256
0.20	0.4788	0.7748	0.7748	0.4788
0.25	0.4206	0.6806	0.6806	0.4206
0.30	0.3524	0.5701	0.5701	0.3524
0.35	0.2757	0.4460	0.4460	0.2757
0.40	0.1924	0.3113	0.3113	0.1924
0.45	0.1046	0.1692	0.1692	0.1046
0.50	0.0142	0.0230	0.0230	0.0142

- (b) Identifying $h = 1/5$ and $k = 0.5/20 = 0.025$ we see that $\lambda = 0.125$.

TIME	X=0.2	X=0.4	X=0.6	X=0.8
0.00	0.5878	0.9511	0.9511	0.5878
0.03	0.5860	0.9482	0.9482	0.5860
0.05	0.5808	0.9397	0.9397	0.5808
0.08	0.5721	0.9256	0.9256	0.5721
0.10	0.5599	0.9060	0.9060	0.5599
0.13	0.5445	0.8809	0.8809	0.5445
0.15	0.5257	0.8507	0.8507	0.5257
0.18	0.5039	0.8153	0.8153	0.5039
0.20	0.4790	0.7750	0.7750	0.4790
0.23	0.4513	0.7302	0.7302	0.4513
0.25	0.4209	0.6810	0.6810	0.4209
0.28	0.3879	0.6277	0.6277	0.3879
0.30	0.3527	0.5706	0.5706	0.3527
0.33	0.3153	0.5102	0.5102	0.3153
0.35	0.2761	0.4467	0.4467	0.2761
0.38	0.2352	0.3806	0.3806	0.2352
0.40	0.1929	0.3122	0.3122	0.1929
0.43	0.1495	0.2419	0.2419	0.1495
0.45	0.1052	0.1701	0.1701	0.1052
0.48	0.0602	0.0974	0.0974	0.0602
0.50	0.0149	0.0241	0.0241	0.0149

4. We have $\lambda = 1$. The initial values of n are $u_{1,0} = u(0.2, 0) = 0.16$, $u_{2,0} = u(0.4) = 0.24$, $u_{3,0} = 0.24$, and $u_{4,0} = 0.16$. From equation (6) in the text we have

$$u_{i,1} = \frac{1}{2}(u_{i+1,0} + u_{i-1,0}) + 0u_{i,0} + k \cdot 0 = \frac{1}{2}(u_{i+1,0} + u_{i-1,0}).$$

Then, using $u_{0,0} = u_{5,0} = 0$, we find $u_{1,1} = 0.12$, $u_{2,1} = 0.2$, $u_{3,1} = 0.2$, and $u_{4,1} = 0.12$.

5. We identify $c = 24944.4$, $k = 0.00020045$ seconds = 0.20045 milliseconds, and $\lambda = 0.5$. Time in the table is expressed in milliseconds.

TIME	X=10	X=20	X=30	X=40	X=50
0.00000	0.1000	0.2000	0.3000	0.2000	0.1000
0.20045	0.1000	0.2000	0.2750	0.2000	0.1000
0.40089	0.1000	0.1938	0.2125	0.1938	0.1000
0.60134	0.0984	0.1688	0.1406	0.1688	0.0984
0.80178	0.0898	0.1191	0.0828	0.1191	0.0898
1.00223	0.0661	0.0531	0.0432	0.0531	0.0661
1.20268	0.0226	-0.0121	0.0085	-0.0121	0.0226
1.40312	-0.0352	-0.0635	-0.0365	-0.0635	-0.0352
1.60357	-0.0913	-0.1011	-0.0950	-0.1011	-0.0913
1.80401	-0.1271	-0.1347	-0.1566	-0.1347	-0.1271
2.00446	-0.1329	-0.1719	-0.2072	-0.1719	-0.1329
2.20491	-0.1153	-0.2081	-0.2402	-0.2081	-0.1153
2.40535	-0.0920	-0.2292	-0.2571	-0.2292	-0.0920
2.60580	-0.0801	-0.2230	-0.2601	-0.2230	-0.0801
2.80624	-0.0838	-0.1903	-0.2445	-0.1903	-0.0838
3.00669	-0.0932	-0.1445	-0.2018	-0.1445	-0.0932
3.20713	-0.0921	-0.1003	-0.1305	-0.1003	-0.0921
3.40758	-0.0701	-0.0615	-0.0440	-0.0615	-0.0701
3.60803	-0.0284	-0.0205	0.0336	-0.0205	-0.0284
3.80847	0.0224	0.0321	0.0842	0.0321	0.0224
4.00892	0.0700	0.0953	0.1087	0.0953	0.0700
4.20936	0.1064	0.1555	0.1265	0.1555	0.1064
4.40981	0.1285	0.1962	0.1588	0.1962	0.1285
4.61026	0.1354	0.2106	0.2098	0.2106	0.1354
4.81070	0.1273	0.2060	0.2612	0.2060	0.1273
5.01115	0.1070	0.1955	0.2851	0.1955	0.1070
5.21159	0.0821	0.1853	0.2641	0.1853	0.0821
5.41204	0.0625	0.1689	0.2038	0.1689	0.0625
5.61249	0.0539	0.1347	0.1260	0.1347	0.0539
5.81293	0.0520	0.0781	0.0526	0.0781	0.0520
6.01338	0.0436	0.0086	-0.0080	0.0086	0.0436
6.21382	0.0156	-0.0564	-0.0604	-0.0564	0.0156
6.41427	-0.0343	-0.1043	-0.1107	-0.1043	-0.0343
6.61472	-0.0931	-0.1364	-0.1578	-0.1364	-0.0931
6.81516	-0.1395	-0.1630	-0.1942	-0.1630	-0.1395
7.01561	-0.1568	-0.1915	-0.2150	-0.1915	-0.1568
7.21605	-0.1436	-0.2173	-0.2240	-0.2173	-0.1436
7.41650	-0.1129	-0.2263	-0.2297	-0.2263	-0.1129
7.61695	-0.0824	-0.2078	-0.2336	-0.2078	-0.0824
7.81739	-0.0625	-0.1644	-0.2247	-0.1644	-0.0625
8.01784	-0.0526	-0.1106	-0.1856	-0.1106	-0.0526
8.21828	-0.0440	-0.0611	-0.1091	-0.0611	-0.0440
8.41873	-0.0287	-0.0192	-0.0085	-0.0192	-0.0287
8.61918	-0.0038	0.0229	0.0867	0.0229	-0.0038
8.81962	0.0287	0.0743	0.1500	0.0743	0.0287
9.02007	0.0654	0.1332	0.1755	0.1332	0.0654
9.22051	0.1027	0.1858	0.1799	0.1858	0.1027
9.42096	0.1352	0.2160	0.1872	0.2160	0.1352
9.62140	0.1540	0.2189	0.2089	0.2189	0.1540
9.82185	0.1506	0.2030	0.2356	0.2030	0.1506
10.02230	0.1226	0.1822	0.2461	0.1822	0.1226

Exercises 16.3

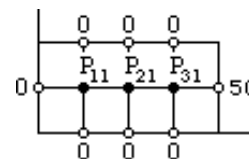
6. We identify $c = 24944.4$, $k = 0.00010022$ seconds = 0.10022 milliseconds, and $\lambda = 0.25$. Time in the table is expressed in milliseconds.

TIME	X=10	X=20	X=30	X=40	X=50
0.00000	0.2000	0.2667	0.2000	0.1333	0.0667
0.10022	0.1958	0.2625	0.2000	0.1333	0.0667
0.20045	0.1836	0.2503	0.1997	0.1333	0.0667
0.30067	0.1640	0.2307	0.1985	0.1333	0.0667
0.40089	0.1384	0.2050	0.1952	0.1332	0.0667
0.50111	0.1083	0.1744	0.1886	0.1328	0.0667
0.60134	0.0755	0.1407	0.1777	0.1318	0.0666
0.70156	0.0421	0.1052	0.1615	0.1295	0.0665
0.80178	0.0100	0.0692	0.1399	0.1253	0.0661
0.90201	-0.0190	0.0340	0.1129	0.1184	0.0654
1.00223	-0.0435	0.0004	0.0813	0.1077	0.0638
1.10245	-0.0626	-0.0309	0.0464	0.0927	0.0610
1.20268	-0.0758	-0.0593	0.0095	0.0728	0.0564
1.30290	-0.0832	-0.0845	-0.0278	0.0479	0.0493
1.40312	-0.0855	-0.1060	-0.0639	0.0184	0.0390
1.50334	-0.0837	-0.1237	-0.0974	-0.0150	0.0250
1.60357	-0.0792	-0.1371	-0.1275	-0.0511	0.0069
1.70379	-0.0734	-0.1464	-0.1533	-0.0882	-0.0152
1.80401	-0.0675	-0.1515	-0.1747	-0.1249	-0.0410
1.90424	-0.0627	-0.1528	-0.1915	-0.1595	-0.0694
2.00446	-0.0596	-0.1509	-0.2039	-0.1904	-0.0991
2.10468	-0.0585	-0.1467	-0.2122	-0.2165	-0.1283
2.20491	-0.0592	-0.1410	-0.2166	-0.2368	-0.1551
2.30513	-0.0614	-0.1349	-0.2175	-0.2507	-0.1772
2.40535	-0.0643	-0.1294	-0.2154	-0.2579	-0.1929
2.50557	-0.0672	-0.1251	-0.2105	-0.2585	-0.2005
2.60580	-0.0696	-0.1227	-0.2033	-0.2524	-0.1993
2.70602	-0.0709	-0.1219	-0.1942	-0.2399	-0.1889
2.80624	-0.0710	-0.1225	-0.1833	-0.2214	-0.1699
2.90647	-0.0699	-0.1236	-0.1711	-0.1972	-0.1435
3.00669	-0.0678	-0.1244	-0.1575	-0.1681	-0.1115
3.10691	-0.0649	-0.1237	-0.1425	-0.1348	-0.0761
3.20713	-0.0617	-0.1205	-0.1258	-0.0983	-0.0395
3.30736	-0.0583	-0.1139	-0.1071	-0.0598	-0.0042
3.40758	-0.0547	-0.1035	-0.0859	-0.0209	0.0279
3.50780	-0.0508	-0.0889	-0.0617	0.0171	0.0552
3.60803	-0.0460	-0.0702	-0.0343	0.0525	0.0767
3.70825	-0.0399	-0.0478	-0.0037	0.0840	0.0919
3.80847	-0.0318	-0.0221	0.0297	0.1106	0.1008
3.90870	-0.0211	0.0062	0.0648	0.1314	0.1041
4.00892	-0.0074	0.0365	0.1005	0.1464	0.1025
4.10914	0.0095	0.0680	0.1350	0.1558	0.0973
4.20936	0.0295	0.1000	0.1666	0.1602	0.0897
4.30959	0.0521	0.1318	0.1937	0.1606	0.0808
4.40981	0.0764	0.1625	0.2148	0.1581	0.0719
4.51003	0.1013	0.1911	0.2291	0.1538	0.0639
4.61026	0.1254	0.2164	0.2364	0.1485	0.0575
4.71048	0.1475	0.2373	0.2369	0.1431	0.0532
4.81070	0.1659	0.2526	0.2315	0.1379	0.0512
4.91093	0.1794	0.2611	0.2217	0.1331	0.0514
5.01115	0.1867	0.2620	0.2087	0.1288	0.0535

Chapter 16 Review Exercises

1. Using the figure we obtain the system

$$\begin{aligned} u_{21} + 0 + 0 + 0 - 4u_{11} &= 0 \\ u_{31} + 0 + u_{11} + 0 - 4u_{21} &= 0 \\ 50 + 0 + u_{21} + 0 - 4u_{31} &= 0. \end{aligned}$$



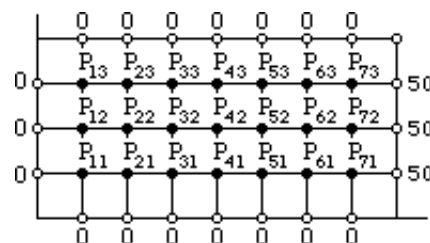
By Gauss-Elimination then,

$$\left[\begin{array}{ccc|c} -4 & 1 & 0 & 0 \\ 1 & -4 & 1 & 0 \\ 0 & 1 & -4 & -50 \end{array} \right] \xrightarrow[\text{operations}]{\text{row}} \left[\begin{array}{ccc|c} 1 & -4 & 1 & 0 \\ 0 & 1 & -4 & -50 \\ 0 & 0 & 1 & 13.3928 \end{array} \right].$$

The solution is $u_{11} = 0.8929$, $u_{21} = 3.5714$, $u_{31} = 13.3928$.

2. By symmetry we observe that $u_{i,1} = u_{i,3}$ for $i = 1, 2, \dots, 7$. We then use Gauss-Seidel iteration with an initial guess of 7.5 for all variables to solve the system

$$\begin{aligned} u_{11} &= 0.25u_{21} + 0.25u_{12} \\ u_{21} &= 0.25u_{31} + 0.25u_{22} + 0.25u_{11} \\ u_{31} &= 0.25u_{41} + 0.25u_{32} + 0.25u_{21} \\ u_{41} &= 0.25u_{51} + 0.25u_{42} + 0.25u_{31} \\ u_{51} &= 0.25u_{61} + 0.25u_{52} + 0.25u_{41} \\ u_{61} &= 0.25u_{71} + 0.25u_{62} + 0.25u_{51} \\ u_{71} &= 12.5 + 0.25u_{72} + 0.25u_{61} \\ u_{12} &= 0.25u_{22} + 0.5u_{11} \\ u_{22} &= 0.25u_{32} + 0.5u_{21} + 0.25u_{12} \\ u_{32} &= 0.25u_{42} + 0.5u_{31} + 0.25u_{22} \\ u_{42} &= 0.25u_{52} + 0.5u_{41} + 0.25u_{32} \\ u_{52} &= 0.25u_{62} + 0.5u_{51} + 0.25u_{42} \\ u_{62} &= 0.25u_{72} + 0.5u_{61} + 0.25u_{52} \\ u_{72} &= 12.5 + 0.5u_{71} + 0.25u_{62}. \end{aligned}$$



After 30 iterations we obtain $u_{11} = u_{13} = 0.1765$, $u_{21} = u_{23} = 0.4566$, $u_{31} = u_{33} = 1.0051$, $u_{41} = u_{43} = 2.1479$, $u_{51} = u_{53} = 4.5766$, $u_{61} = u_{63} = 9.8316$, $u_{71} = u_{73} = 21.6051$, $u_{12} = 0.2494$, $u_{22} = 0.6447$, $u_{32} = 1.4162$, $u_{42} = 3.0097$, $u_{52} = 6.3269$, $u_{62} = 13.1447$, $u_{72} = 26.5887$.

3. (a)

TIME	X=0.0	X=0.2	X=0.4	X=0.6	X=0.8	X=1.0
0.00	0.0000	0.2000	0.4000	0.6000	0.8000	0.0000
0.01	0.0000	0.2000	0.4000	0.6000	0.5500	0.0000
0.02	0.0000	0.2000	0.4000	0.5375	0.4250	0.0000
0.03	0.0000	0.2000	0.3844	0.4750	0.3469	0.0000
0.04	0.0000	0.1961	0.3609	0.4203	0.2922	0.0000
0.05	0.0000	0.1883	0.3346	0.3734	0.2512	0.0000

Chapter 16 Review Exercises

(b)

TIME	X=0.0	X=0.2	X=0.4	X=0.6	X=0.8	X=1.0
0.00	0.0000	0.2000	0.4000	0.6000	0.8000	0.0000
0.01	0.0000	0.2000	0.4000	0.6000	0.8000	0.0000
0.02	0.0000	0.2000	0.4000	0.6000	0.5500	0.0000
0.03	0.0000	0.2000	0.4000	0.5375	0.4250	0.0000
0.04	0.0000	0.2000	0.3844	0.4750	0.3469	0.0000
0.05	0.0000	0.1961	0.3609	0.4203	0.2922	0.0000

(c) The table in part (b) is the same as the table in part (a) shifted downward one row.

17 Functions of a Complex Variable

Exercises 17.1

1. $3 + 3i$
2. $-4i$
3. $i^8 = (i^2)^4 = (-1)^4 = 1$
4. $i^{11} = i(i^2)^5 = i(-1)^5 = -i$
5. $7 - 13i$
6. $-3 - 9i$
7. $-7 + 5i$
8. $-7 + 8i$
9. $11 - 10i$
10. $\frac{3}{4} + \frac{2}{3}i$
11. $-5 + 12i$
12. $-2 - 2i$
13. $-2i$
14. $\frac{i}{1+i} \cdot \frac{1-i}{1-i} = \frac{i+1}{2} = \frac{1}{2} + \frac{1}{2}i$
15. $\frac{2-4i}{3+5i} \cdot \frac{3-5i}{3-5i} = \frac{-14-22i}{34} = -\frac{7}{17} - \frac{11}{17}i$
16. $\frac{10-5i}{6+2i} \cdot \frac{6-2i}{6-2i} = \frac{50-50i}{40} = \frac{5}{4} - \frac{5}{4}i$
17. $\frac{9+7i}{1+i} \cdot \frac{1-i}{1-i} = \frac{16-2i}{2} = 8 - i$
18. $\frac{3-i}{11-2i} \cdot \frac{11+2i}{11+2i} = \frac{35-5i}{125} = \frac{7}{25} - \frac{1}{25}i$
19. $\frac{2-11i}{6-i} \cdot \frac{6+i}{6+i} = \frac{23-64i}{37} = \frac{23}{37} - \frac{64}{37}i$
20. $\frac{4+3i}{3+4i} \cdot \frac{3-4i}{3-4i} = \frac{24-7i}{25} = \frac{24}{25} - \frac{7}{25}i$
21. $(1+i)(10+10i) = 10(1+i)^2 = 20i$
22. $[(1+i)(1-i)]^2(1-i) = 4 - 4i$
23. $20 + 23i + \frac{1}{2-i} \cdot \frac{2+i}{2+i} = 20 + 23i + \frac{2}{5} + \frac{1}{5}i = \frac{102}{5} + \frac{116}{5}i$
24. $(2+3i)(-i)^2 = -2 - 3i$
25. $\frac{i}{9+7i} \cdot \frac{9-7i}{9-7i} = \frac{7+9i}{130} = \frac{7}{130} + \frac{9}{130}i$
26. $\frac{1}{6+8i} \cdot \frac{6-8i}{6-8i} = \frac{6-8i}{84} = \frac{1}{14} - \frac{2}{21}i$
27. $\frac{x}{x^2+y^2}$
28. $x^2 - y^2$
29. $-2y - 4$
30. 0
31. $\sqrt{(x-1)^2 + (y-3)^2}$
32. $\sqrt{36x^2 + 16y^2}$
33. $2x + 2yi = -9 + 2i$ implies $2x = -9$ and $2y = 2$. Hence $z = -\frac{9}{2} + i$.
34. $-x + 3yi = -7 + 6i$ implies $-x = -7$ and $3y = 6$. Hence $z = 7 + 2i$.
35. $x^2 - y^2 + 2xyi = 0 + i$ implies $x^2 - y^2 = 0$ and $2xy = 1$. Now $y = x$ implies $2x^2 = 1$ and so $x = \pm 1/\sqrt{2}$. The choice $y = -x$ gives $-2x^2 = 1$ which has no real solution. Hence $z = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i$ and $z = -\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i$.

Exercises 17.1

36. $x^2 - y^2 - 4x + (-2xy - 4y)i = 0 + 0i$ implies $x^2 - y^2 - 4x = 0$ and $y(-2x - 4) = 0$. If $y = 0$ then $x(x - 4) = 0$ and so $z = 0$ and $z = 4$. If $-2x - 4 = 0$ or $x = -2$ then $12 - y^2 = 0$ or $y = \pm 2\sqrt{3}$. This gives $z = -2 + 2\sqrt{3}i$ and $z = -2 - 2\sqrt{3}i$.
37. $|10 + 8i| = \sqrt{164}$ and $|11 - 6i| = \sqrt{157}$. Hence $11 - 6i$ is closer to the origin.
38. $|\frac{1}{2} - \frac{1}{4}i| = \frac{\sqrt{5}}{4}$ and $|\frac{2}{3} + \frac{1}{6}i| = \frac{\sqrt{17}}{6}$. Since $\frac{\sqrt{5}}{4} < \frac{\sqrt{17}}{6}$, $\frac{1}{2} - \frac{1}{4}i$ is closer to the origin.
39. $|z_1 - z_2| = |(x_1 - x_2) + i(y_1 - y_2)| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$ which is the distance formula in the plane.
40. By the triangle inequality, $|z + 6 + 8i| \leq |z| + |6 + 8i|$. On the circle, $|z| = 2$ and so $|z + 6 + 8i| \leq 2 + \sqrt{100} = 12$.

Exercises 17.2

1. $2(\cos 2\pi + i \sin 2\pi)$
2. $10(\cos \pi + i \sin \pi)$
3. $3\left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}\right)$
4. $6\left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)$
5. $\sqrt{2}\left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}\right)$
6. $5\sqrt{2}\left(\cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4}\right)$
7. $2\left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right)$
8. $4\left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3}\right)$
9. $\frac{3\sqrt{2}}{2}\left(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4}\right)$
10. $6\left[\cos\left(-\frac{\pi}{6}\right) + i \sin\left(-\frac{\pi}{6}\right)\right]$
11. $z = -\frac{5\sqrt{3}}{2} - \frac{5}{2}i$
12. $z = -8 + 8i$
13. $z = 5.5433 + 2.2961i$
14. $z = 8.0902 + 5.8779i$
15. $z_1 z_2 = 8\left[\cos\left(\frac{\pi}{8} + \frac{3\pi}{8}\right) + i \sin\left(\frac{\pi}{8} + \frac{3\pi}{8}\right)\right] = 8i$; $\frac{z_1}{z_2} = \frac{1}{2}\left[\cos\left(\frac{\pi}{8} - \frac{3\pi}{8}\right) + i \sin\left(\frac{\pi}{8} - \frac{3\pi}{8}\right)\right] = \frac{\sqrt{2}}{4} - \frac{\sqrt{2}}{4}i$
16. $z_1 z_2 = \sqrt{6}\left[\cos\left(\frac{\pi}{4} + \frac{\pi}{12}\right) + i \sin\left(\frac{\pi}{4} + \frac{\pi}{12}\right)\right] = \frac{\sqrt{6}}{2} + \frac{3\sqrt{2}}{2}i$
 $\frac{z_1}{z_2} = \frac{\sqrt{6}}{3}\left[\cos\left(\frac{\pi}{4} - \frac{\pi}{12}\right) + i \sin\left(\frac{\pi}{4} - \frac{\pi}{12}\right)\right] = \frac{\sqrt{2}}{2} + \frac{\sqrt{6}}{6}i$
17. $\left[3\sqrt{2}\left(\cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4}\right)\right]\left[10\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)\right] = 30\sqrt{2}\left[\cos\left(\frac{7\pi}{4} + \frac{\pi}{3}\right) + i \sin\left(\frac{7\pi}{4} + \frac{\pi}{3}\right)\right]$
 $= 40.9808 + 10.9808i$
18. $\left[4\sqrt{2}\left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}\right)\right]\left[\sqrt{2}\left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4}\right)\right] = 8\left[\cos\left(\frac{\pi}{4} + \frac{3\pi}{4}\right) + i \sin\left(\frac{\pi}{4} + \frac{3\pi}{4}\right)\right] = -8$
19. $\frac{\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}}{2\sqrt{2}\left[\cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4}\right]} = \frac{\sqrt{2}}{4}\left[\cos\left(\frac{3\pi}{2} - \frac{7\pi}{4}\right) + i \sin\left(\frac{3\pi}{2} - \frac{7\pi}{4}\right)\right] = \frac{1}{4} - \frac{1}{4}i$

$$20. \frac{2\sqrt{2} \left[\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right]}{2 \left[\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right]} = \sqrt{2} \left[\cos \left(\frac{\pi}{3} - \frac{2\pi}{3} \right) + i \sin \left(\frac{\pi}{3} - \frac{2\pi}{3} \right) \right] = \frac{\sqrt{2}}{2} - \frac{\sqrt{6}}{2} i$$

$$21. 2^9 \left[\cos \frac{9\pi}{3} + i \sin \frac{9\pi}{3} \right] = -512$$

$$22. (2\sqrt{2})^5 \left[\cos \left(-\frac{5\pi}{4} \right) + i \sin \left(-\frac{5\pi}{4} \right) \right] = -128 + 128i$$

$$23. \left(\frac{\sqrt{2}}{2} \right)^{10} \left[\cos \frac{10\pi}{4} + i \sin \frac{10\pi}{4} \right] = \frac{1}{32} i$$

$$24. (2\sqrt{2})^4 \left[\cos \frac{8\pi}{3} + i \sin \frac{8\pi}{3} \right] = -32 + 32\sqrt{3} i$$

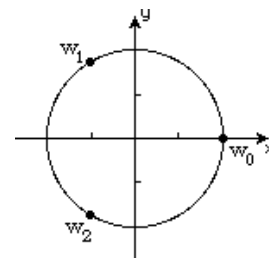
$$25. \cos \frac{12\pi}{8} + i \sin \frac{12\pi}{8} = -i$$

$$26. (\sqrt{3})^6 \left[\cos \frac{12\pi}{9} + i \sin \frac{12\pi}{9} \right] = -\frac{27}{2} - \frac{27\sqrt{3}}{2} i$$

$$27. 8^{1/3} = 2 \left[\cos \frac{2k\pi}{3} + i \sin \frac{2k\pi}{3} \right], \quad k = 0, 1, 2$$

$$w_0 = 2[\cos 0 + i \sin 0] = 2; \quad w_1 = 2 \left[\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right] = -1 + \sqrt{3} i$$

$$w_2 = 2 \left[\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3} \right] = -1 - \sqrt{3} i$$



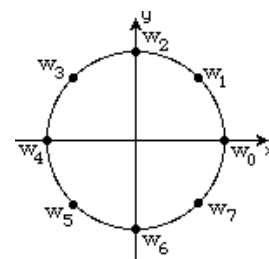
$$28. (1)^{1/8} = \cos \frac{k\pi}{4} + i \sin \frac{k\pi}{4}, \quad k = 0, 1, 2, \dots, 7$$

$$w_0 = \cos 0 + i \sin 0 = 1; \quad w_1 = \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} i$$

$$w_2 = \cos \frac{\pi}{2} + i \sin \frac{\pi}{2} = i; \quad w_3 = \cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} = -\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} i$$

$$w_4 = \cos \pi + i \sin \pi = -1; \quad w_5 = \cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} = -\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} i$$

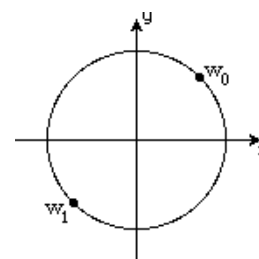
$$w_6 = \cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2} = -i; \quad w_7 = \cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4} = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} i$$



$$29. (i)^{1/2} = \cos \left(\frac{\pi}{4} + k\pi \right) + i \sin \left(\frac{\pi}{4} + k\pi \right), \quad k = 0, 1$$

$$w_0 = \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} i$$

$$w_1 = \cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} = -\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} i$$



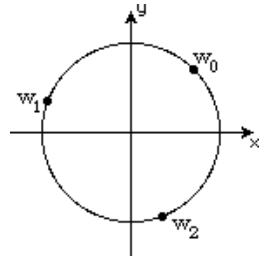
Exercises 17.2

30. $(-1 + i)^{1/3} = 2^{1/6} \left[\cos \left(\frac{\pi}{4} + \frac{2k\pi}{3} \right) + i \sin \left(\frac{\pi}{4} + \frac{2k\pi}{3} \right) \right], \quad k = 0, 1, 2$

$$w_0 = 2^{1/6} \left[\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right] = \frac{1}{\sqrt[3]{2}} + \frac{1}{\sqrt[3]{2}} i = 0.7937 + 0.7937i$$

$$w_1 = 2^{1/6} \left[\cos \frac{11\pi}{12} + i \sin \frac{11\pi}{12} \right] = -1.0842 + 0.2905i$$

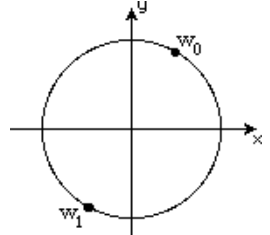
$$w_2 = 2^{1/6} \left[\cos \frac{19\pi}{12} + i \sin \frac{19\pi}{12} \right] = 0.2905 - 1.0842i$$



31. $(-1 + \sqrt{3}i)^{1/2} = 2^{1/2} \left[\cos \left(\frac{\pi}{3} + k\pi \right) + i \sin \left(\frac{\pi}{3} + k\pi \right) \right], \quad k = 0, 1$

$$w_0 = 2^{1/2} \left[\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right] = \frac{\sqrt{2}}{2} + \frac{\sqrt{6}}{2} i$$

$$w_1 = 2^{1/2} \left[\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3} \right] = -\frac{\sqrt{2}}{2} - \frac{\sqrt{6}}{2} i$$

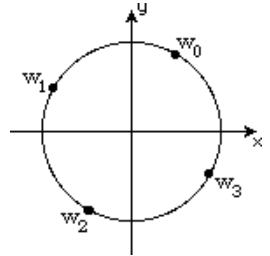


32. $(-1 - \sqrt{3}i)^{1/4} = 2^{1/4} \left[\cos \left(\frac{\pi}{3} + \frac{k\pi}{2} \right) + i \sin \left(\frac{\pi}{3} + \frac{k\pi}{2} \right) \right], \quad k = 0, 1, 2, 3$

$$w_0 = 2^{1/4} \left[\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right] = 2^{1/4} \left[\frac{1}{2} + \frac{\sqrt{3}}{2} i \right]$$

$$w_1 = 2^{1/4} \left[\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right] = 2^{1/4} \left[-\frac{\sqrt{3}}{2} + \frac{1}{2} i \right]$$

$$w_2 = 2^{1/4} \left[\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3} \right] = 2^{1/4} \left[-\frac{1}{2} - \frac{\sqrt{3}}{2} i \right]; \quad w_3 = 2^{1/4} \left[\cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6} \right] = 2^{1/4} \left[\frac{\sqrt{3}}{2} - \frac{1}{2} i \right]$$



33. The solutions are the four fourth roots of -1 ;

$$w_k = \cos \frac{\pi + 2k\pi}{4} + i \sin \frac{\pi + 2k\pi}{4}, \quad k = 0, 1, 2, 3.$$

We have

$$w_1 = \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} i$$

$$w_3 = \cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} = -\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} i$$

$$w_2 = \cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} = -\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} i$$

$$w_4 = \cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4} = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} i.$$

34. $(z^4 - 1)^2 = 0$ is the same as $(z - i)^2(z + i)^2(z - 1)^2(z + 1)^2 = 0$. Thus $z_1 = 1$, $z_2 = -1$, $z_3 = i$, and $z_4 = -i$ are roots of multiplicity two.

$$\begin{aligned} 35. \quad & \left(\cos \frac{\pi}{9} + i \sin \frac{\pi}{9} \right)^{12} \left[2 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right) \right]^5 = 2^5 \left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3} \right) \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right) \\ & = 32 \left[\cos \left(\frac{4\pi}{3} + \frac{5\pi}{6} \right) + i \sin \left(\frac{4\pi}{3} + \frac{5\pi}{6} \right) \right] \\ & = 32 \left(\cos \frac{13\pi}{6} + i \sin \frac{13\pi}{6} \right) = 32 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right) = 16\sqrt{3} + 16i \end{aligned}$$

$$36. \quad \frac{\left[8 \left(\cos \frac{3\pi}{8} + i \sin \frac{3\pi}{8} \right) \right]^3}{\left[2 \left(\cos \frac{\pi}{16} + i \sin \frac{\pi}{16} \right) \right]^{10}} = \frac{2^9}{2^{10}} \left[\cos \left(\frac{9\pi}{8} - \frac{10\pi}{16} \right) + i \left(\frac{9\pi}{8} - \frac{10\pi}{16} \right) \right] = \frac{1}{2} \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right) = \frac{1}{2} i$$

37. We have

$$(\cos 2\theta + i \sin \theta)^2 = \cos 2\theta + i \sin 2\theta$$

Also

$$(\cos \theta + i \sin \theta)^2 = \cos^2 \theta - \sin^2 \theta + (2 \sin \theta \cos \theta)i.$$

Equating real and imaginary parts gives

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta, \quad \sin 2\theta = 2 \sin \theta \cos \theta.$$

38. We have

$$(\cos \theta + i \sin \theta)^3 = \cos 3\theta + i \sin 3\theta.$$

Also

$$\begin{aligned} (\cos \theta + i \sin \theta)^3 &= \cos^3 \theta + 3 \cos^2 \theta (i \sin \theta) + 3 \cos \theta (i \sin \theta)^2 + (i \sin \theta)^3 \\ &= \cos^3 \theta - 3 \cos \theta \sin^2 \theta + (3 \cos^2 \theta \sin \theta - \sin^3 \theta)i. \end{aligned}$$

Equating real and imaginary parts gives

$$\cos 3\theta = \cos^3 \theta - 3 \cos \theta \sin^2 \theta, \quad \sin 3\theta = 3 \cos^2 \theta \sin \theta - \sin^3 \theta.$$

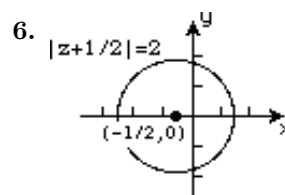
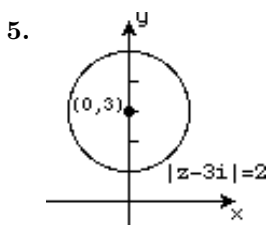
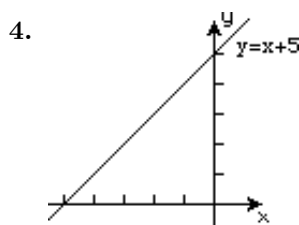
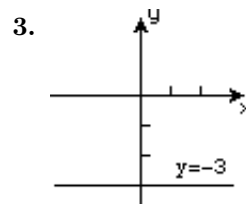
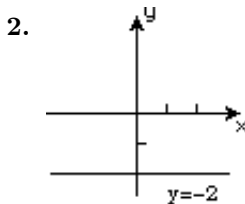
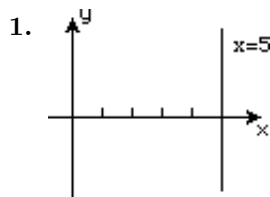
39. (a) $\text{Arg}(z_1) = \pi$, $\text{Arg}(z_2) = \frac{\pi}{2}$, $\text{Arg}(z_1 z_2) = -\frac{\pi}{2}$, $\text{Arg}(z_1) + \text{Arg}(z_2) = \frac{3\pi}{2} \neq \text{Arg}(z_1 z_2)$

(b) $\text{Arg}(z_1/z_2) = -\frac{\pi}{2}$, $\text{Arg}(z_1) - \text{Arg}(z_2) = \pi - \frac{\pi}{2} = \frac{\pi}{2} \neq \text{Arg}(z_1/z_2)$

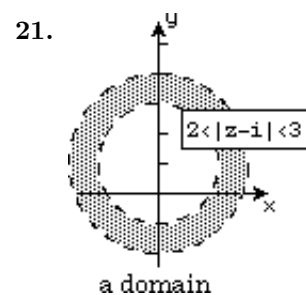
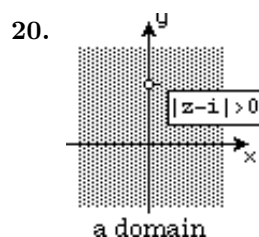
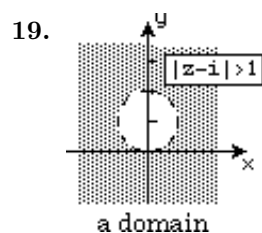
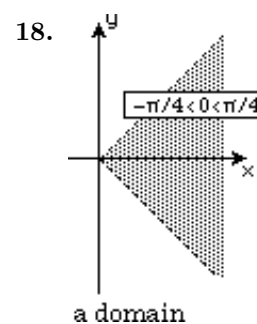
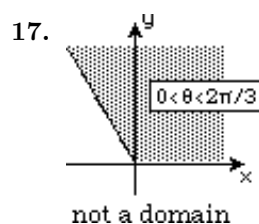
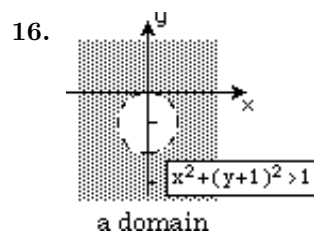
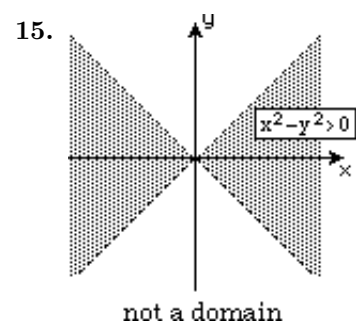
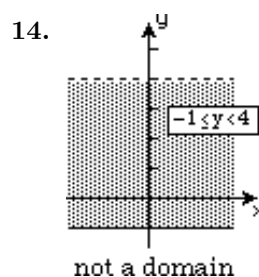
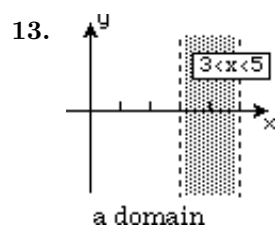
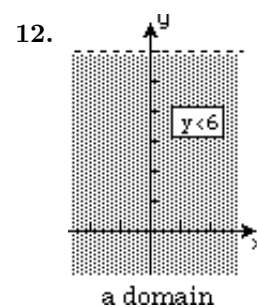
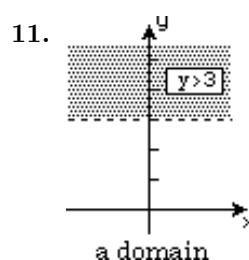
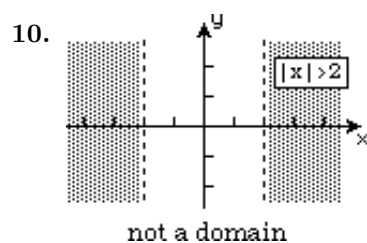
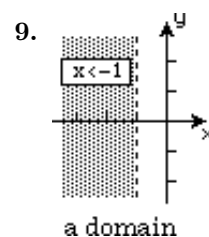
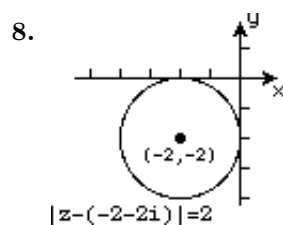
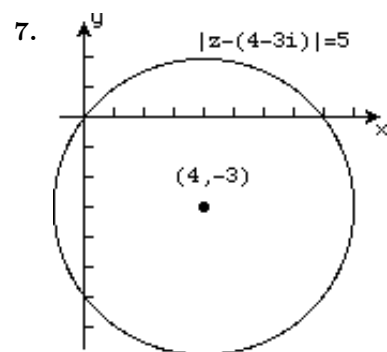
40. (a) If we take $\arg(z_1) = \pi$ and $\arg(z_2) = \pi/2$ then $\arg(z_1) + \arg(z_2) = 3\pi/2$ is an argument of the product $z_1 z_2 = -5i$. With these same arguments we see that $\arg(z_1) - \arg(z_2) = \pi/2$ is an argument of the quotient $z_1/z_2 = \frac{1}{5}i$.

(b) If we take $\arg(z_1) = \pi$ and $\arg(z_2) = -\pi/2$ then $\arg(z_1) + \arg(z_2) = \pi/2$ is an argument of the product $z_1 z_2 = 5i$. With these same arguments we see that $\arg(z_1) - \arg(z_2) = 3\pi/2$ is an argument of the quotient $z_1/z_2 = -\frac{1}{5}i$.

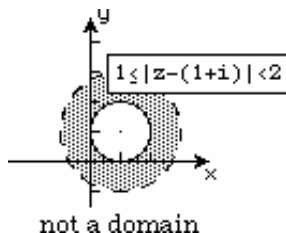
Exercises 17.3



Exercises 17.3



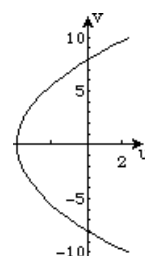
22.



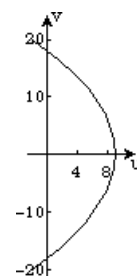
23. The given equation is equivalent to $(x+1)^2 + y^2 = x^2 + (y-1)^2$. This simplifies to $y = -x$ which describes a straight line through the origin.
24. $|\operatorname{Re}(z)| = |x|$ is the same as $\sqrt{x^2}$ and $|z| = \sqrt{x^2 + y^2}$. Since $y^2 \geq 0$ the inequality $\sqrt{x^2} \leq \sqrt{x^2 + y^2}$ is true for all complex numbers.
25. The given equation simplifies to the equation $x^2 - y^2 = 1$ which is a hyperbola with center at the origin.
26. Since $|z - i|$ and $|z - (-i)|$ represent distances from the point (x, y) to i and $-i$, respectively, the equation is the distance formula definition of an ellipse with foci at $(0, 1)$ and $(0, -1)$.

Exercises 17.4

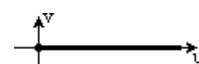
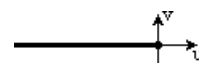
1. Substituting $y = 2$ into $u = x^2 - y^2$, $v = 2xy$ gives the parametric equations $u = x^2 - 4$, $v = 4x$. Using $x = v/4$ the first equation gives $u = v^2/16 - 4$. The graph is the parabola shown.



2. Substituting $x = -3$ into $u = x^2 - y^2$, $v = 2xy$ gives the parametric equations $u = 9 - y^2$, $v = -6y$. Using $y = -v/6$ the first equation gives $u = 9 - v^2/36$. The graph is the parabola shown.

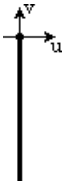


3. $x = 0$ gives $u = -y^2$, $v = 0$. Since $-y^2 \leq 0$ for all real values of y , the image is the origin and the negative u -axis.
4. $y = 0$ gives $u = x^2$, $v = 0$. Since $x^2 \geq 0$ for all real values of x , the image is the origin and the positive u -axis.
5. $y = x$ gives $u = 0$, $v = 2x^2$. Since $x^2 \geq 0$ for all real values of x , the image is the origin and the positive v -axis.



Exercises 17.4

6. $y = -x$ gives $u = 0$, $v = -2x^2$. Since $-x^2 \leq 0$ for all real values of x , the image is the origin and the negative v -axis.



7. $f(z) = (6x - 5) + i(6y + 9)$ 8. $f(z) = (7x - 9y - 3) + i(7y - 9x + 2)$
9. $f(z) = (x^2 - y^2 - 3x) + i(2xy - 3y + 4)$ 10. $f(z) = (3x^2 - 3y^2 + 2x) + i(-6xy + 2y)$
11. $f(z) = (x^3 - 3xy^2 - 4x) + i(3x^2y - y^3 - 4y)$ 12. $f(z) = (x^4 - 6x^2y^2 + y^4) + i(4x^3y - 4xy^3)$
13. $f(z) = \left(x + \frac{x}{x^2 + y^2}\right) + i\left(y - \frac{y}{x^2 + y^2}\right)$ 14. $f(z) = \frac{x^2 + y^2 + x}{(x + 1)^2 + y^2} + i\frac{y}{(x + 1)^2 + y^2}$
15. (a) $f(0 + 2i) = -4 + i$ (b) $f(2 - i) = 3 - 9i$ (c) $f(5 + 3i) = 1 + 86i$
16. (a) $f(1 + i) = 3 - 2i$ (b) $f(2 - i) = \frac{7}{2} + 10i$ (c) $f(1 + 4i) = 3 - 32i$
17. (a) $f(4 - 6i) = 14 - 20i$ (b) $f(-5 + 12i) = -13 + 43i$ (c) $f(2 - 7i) = 3 - 26i$
18. (a) $f(0 + \frac{\pi}{4}i) = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$ (b) $f(-1 - \pi i) = -e^{-1}$ (c) $f(3 + \frac{\pi}{3}i) = \frac{1}{2}e^3 + \frac{\sqrt{3}}{2}e^3i$
19. $\lim_{z \rightarrow i} (4z^3 - 5z^2 + 4z + 1 - 5i) = 6 - 5i$
20. $\lim_{z \rightarrow 1-i} \frac{5z^2 - 2z + 2}{z + 1} = \frac{5(1-i)^2 - 2(1-i) + 2}{2-i} = \frac{8}{5} - \frac{16}{5}i$
21. $\lim_{z \rightarrow i} \frac{z^4 - 1}{z - i} = \lim_{z \rightarrow i} \frac{(z^2 - 1)(z - i)(z + i)}{z - i} = -4i$
22. $\lim_{z \rightarrow 1+i} \frac{z^2 - 2z + 2}{z^2 - 2i} = \lim_{z \rightarrow 1+i} \frac{[z - (1+i)][z - (1-i)]}{[z - (1+i)][z - (-1-i)]} = \frac{1}{2} + \frac{1}{2}i$
23. Along the y -axis, $\lim_{z \rightarrow 0} \frac{x + iy}{x - iy} = \lim_{y \rightarrow 0} \frac{iy}{-iy} = -1$, whereas along the x -axis, $\lim_{z \rightarrow 0} \frac{x + iy}{x - iy} = \lim_{x \rightarrow 0} \frac{x}{x} = 1$.
24. Along the line $x = 1$, $\lim_{z \rightarrow 1} \frac{x + y - 1}{z - 1} = \lim_{y \rightarrow 0} \frac{y}{iy} = \frac{1}{i} = -i$, whereas along the x -axis, $\lim_{z \rightarrow 1} \frac{x + y - 1}{z - 1} = \lim_{x \rightarrow 1} \frac{x - 1}{x - 1} = 1$.
25. $f'(z) = \lim_{\Delta z \rightarrow 0} \frac{(z + \Delta z)^2 - z^2}{\Delta z} = \lim_{\Delta z \rightarrow 0} \frac{2z\Delta z + (\Delta z)^2}{\Delta z} = \lim_{\Delta z \rightarrow 0} (2z + \Delta z) = 2z$
26. $f'(z) = \lim_{\Delta z \rightarrow 0} \frac{\frac{1}{z + \Delta z} - \frac{1}{z}}{\Delta z} = \lim_{\Delta z \rightarrow 0} \frac{-\Delta z}{(\Delta z)z(z + \Delta z)} = \lim_{\Delta z \rightarrow 0} \frac{-1}{z(z + \Delta z)} = -\frac{1}{z^2}$
27. $f'(z) = 12z^2 - (6 + 2i)z - 5$
28. $f'(z) = 20z^3 - 3iz^2 + (16 - 2i)z$
29. $f'(z) = (2z + 1)(2z - 4) + 2(z^2 - 4z + 8i) = 6z^2 - 14z - 4 + 8i$

30. $f'(z) = (z^5 + 3iz^3)(4z^2 + 3iz^2 + 4z - 6i) + (z^4 + iz^3 + 2z^2 - 6iz)(5z^4 + 9iz^2)$

31. $f'(z) = 6z(z^2 - 4i)^2$

32. $f'(z) = 6(2z - 1/z)^5(2 + 1/z^2)$

33. $f'(z) = \frac{(2z+i)3 - (3z-4+8i)2}{(2z+1)^2} = \frac{8-13i}{(2z+i)^2}$

34. $f'(z) = \frac{(z^3+1)(10z-1) - (5z^2-z)3z^2}{(z^3+1)^2} = \frac{-5z^4+2z^3+10z-1}{(z^3+1)^2}$

35. $3i$

36. $0, 2-5i$

37. $-2i, 2i$

38. $3-4i, 3+4i$

39. We have

$$\lim_{\Delta z \rightarrow 0} \frac{\overline{z + \Delta z} - \bar{z}}{\Delta z} = \lim_{\Delta z \rightarrow 0} \frac{\overline{\Delta z}}{\Delta z}.$$

If we let $\Delta z \rightarrow 0$ along a horizontal line then $\Delta z = \Delta x$, $\overline{\Delta z} = \Delta x$, and

$$\lim_{\Delta z \rightarrow 0} \frac{\overline{\Delta z}}{\Delta z} = \lim_{\Delta x \rightarrow 0} \frac{\Delta x}{\Delta x} = 1.$$

If we let $\Delta z \rightarrow 0$ along a vertical line then $\Delta z = i\Delta y$, $\overline{\Delta z} = -i\Delta y$, and

$$\lim_{\Delta z \rightarrow 0} \frac{\overline{\Delta z}}{\Delta z} = \lim_{\Delta y \rightarrow 0} \frac{-i\Delta y}{i\Delta y} = -1.$$

Since these two limits are not equal, $f(z) = \bar{z}$ cannot be differentiable at any z .

40. We have
$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{(z + \Delta z)(\bar{z} + \overline{\Delta z}) - z\bar{z}}{\Delta z} = \lim_{\Delta z \rightarrow 0} \left(\bar{z} + z \frac{\overline{\Delta z}}{\Delta z} + \overline{\Delta z} \right).$$

If $z = 0$, then the above limit becomes

$$f'(0) = \lim_{\Delta z \rightarrow 0} \overline{\Delta z} = 0.$$

If $z \neq 0$ then we first let $\Delta z \rightarrow 0$ along a horizontal line so that $\Delta z = \Delta x$ and $\overline{\Delta z} = \Delta x$. Thus,

$$f'(z) = \lim_{\Delta z \rightarrow 0} \left(\bar{z} + z \frac{\Delta x}{\Delta x} + \Delta x \right) = \bar{z} + z.$$

Next we let $\Delta z \rightarrow 0$ along a vertical line so that $\Delta z = i\Delta y$, $\overline{\Delta z} = -i\Delta y$. Thus

$$f'(z) = \lim_{\Delta y \rightarrow 0} \left(\bar{z} + z \frac{-i\Delta y}{i\Delta y} + i\Delta y \right) = \bar{z} - z.$$

We must have $\bar{z} + z = \bar{z} - z$ which implies $z = 0$. This is a contradiction to the assumption that $z \neq 0$. Hence $f(z) = |z|^2$ is differentiable only at $z = 0$.

41. Each linear equation in the system $\frac{dx}{dt} = 2x$, $\frac{dy}{dt} = 2y$ can be solved directly. We obtain $x(t) = c_1 e^{2t}$ and $y(t) = c_2 e^{2t}$.

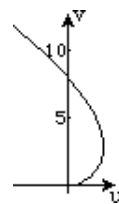
42. The system $\frac{dx}{dt} = -y$, $\frac{dy}{dt} = x$ can be solved as in Section 3.11. We obtain $x(t) = c_1 \cos t + c_2 \sin t$, $y(t) = c_1 \sin t - c_2 \cos t$.

43. The equations in the system $\frac{dx}{dt} = \frac{x}{x^2 + y^2}$, $\frac{dy}{dt} = \frac{y}{x^2 + y^2}$ can be divided to give $\frac{dy}{dx} = \frac{y}{x}$. By separation of variables we obtain $y = cx$.

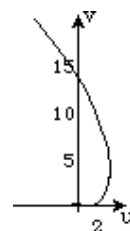
44. Each equation in the system $\frac{dx}{dt} = x^2$, $\frac{dy}{dt} = -y^2$ can be solved directly by separation of variables. We obtain $x(t) = \frac{-1}{t + c_1}$, $y = \frac{1}{t + c_2}$.

Exercises 17.4

45. If $y = \frac{1}{2}x^2$ the equations $u = x^2 - y^2$, $v = 2xy$ give $u = x^2 - \frac{1}{4}x^4$, $v = x^3$. With the aid of a computer, the graph of these parametric equations is shown.



46. If $y = (x-1)^2$ the equations $u = x^2 - y^2$, $v = 2xy$ give $u = x^2 - (x-1)^4$, $v = 2x(x-1)^2$. With the aid of a computer the graph of these parametric equations is shown.



Exercises 17.5

- $u = x^3 - 3xy^2$, $v = 3x^2y - y^3$; $\frac{\partial u}{\partial x} = 3x^2 - 3y^2 = \frac{\partial v}{\partial y}$, $\frac{\partial u}{\partial y} = -6xy = -\frac{\partial v}{\partial x}$
- $u = 3x^2 - 3y^2 + 5x$, $v = 6xy + 5y - 6$; $\frac{\partial u}{\partial x} = 6x + 5 = \frac{\partial v}{\partial y}$, $\frac{\partial u}{\partial y} = -6y = -\frac{\partial v}{\partial x}$
- $u = x$, $v = 0$; $\frac{\partial u}{\partial x} = 1$, $\frac{\partial v}{\partial y} = 0$. Since $1 \neq 0$, f is not analytic at any point.
- $u = y$, $v = x$; $\frac{\partial u}{\partial x} = 0 = \frac{\partial v}{\partial y}$, $\frac{\partial u}{\partial y} = 1$, $-\frac{\partial v}{\partial x} = -1$. Since $1 \neq -1$, f is not analytic at any point.
- $u = -2x + 3$, $v = 10y$; $\frac{\partial u}{\partial x} = -2$, $\frac{\partial v}{\partial y} = 10$. Since $-2 \neq 10$, f is not analytic at any point.
- $u = x^2 - y^2$, $v = -2xy$; $\frac{\partial u}{\partial x} = 2x$, $\frac{\partial v}{\partial y} = -2x$; $\frac{\partial u}{\partial y} = -2y$, $-\frac{\partial v}{\partial x} = 2y$

The Cauchy-Riemann equations hold only at $(0,0)$. Since there is no neighborhood about $z = 0$ within which f is differentiable we conclude f is nowhere analytic.

- $u = x^2 + y^2$, $v = 0$; $\frac{\partial u}{\partial x} = 2x$, $\frac{\partial v}{\partial y} = 0$; $\frac{\partial u}{\partial y} = 2y$, $-\frac{\partial v}{\partial x} = 0$

The Cauchy-Riemann equations hold only at $(0,0)$. Since there is no neighborhood about $z = 0$ within which f is differentiable we conclude f is nowhere analytic.

- $u = \frac{x}{x^2 + y^2}$, $v = \frac{y}{x^2 + y^2}$; $\frac{\partial u}{\partial x} = \frac{y^2 - x^2}{(x^2 + y^2)^2}$, $\frac{\partial v}{\partial y} = \frac{x^2 - y^2}{(x^2 + y^2)^2}$; $\frac{\partial u}{\partial y} = -\frac{2xy}{(x^2 + y^2)^2} = \frac{\partial v}{\partial x}$

The Cauchy-Riemann equations hold only at $(0,0)$. Since there is no neighborhood about $z = 0$ within which f is differentiable, we conclude f is nowhere analytic.

- $u = e^x \cos y$, $v = e^x \sin y$; $\frac{\partial u}{\partial x} = e^x \cos y = \frac{\partial v}{\partial y}$; $\frac{\partial u}{\partial y} = -e^x \sin y = -\frac{\partial v}{\partial x}$. f is analytic for all z .
- $u = x + \sin x \cosh y$, $v = y + \cos x \sinh y$; $\frac{\partial u}{\partial x} = 1 + \cos x \cosh y = \frac{\partial v}{\partial y}$; $\frac{\partial u}{\partial y} = \sin x \sinh y = -\frac{\partial v}{\partial x}$. f is analytic for all z .
- $u = e^{x^2 - y^2} \cos 2xy$, $v = e^{x^2 - y^2} \sin 2xy$; $\frac{\partial u}{\partial x} = -2ye^{x^2 - y^2} \sin 2xy + 2xe^{x^2 - y^2} \cos 2xy = \frac{\partial v}{\partial y}$

$$\frac{\partial u}{\partial y} = -2xe^{x^2-y^2} \sin 2xy - 2ye^{x^2-y^2} \cos 2xy = -\frac{\partial v}{\partial x}. \quad f \text{ is analytic for all } z.$$

$$12. \quad u = 4x^2 + 5x - 4y^2 + 9, \quad v = 8xy + 5y - 1; \quad \frac{\partial u}{\partial x} = 8x + 5 = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -8y = -\frac{\partial v}{\partial x}. \quad f \text{ is analytic for all } z.$$

$$13. \quad u = \frac{x-1}{(x-1)^2 + y^2}, \quad v = -\frac{y}{(x-1)^2 + y^2}; \quad \frac{\partial u}{\partial x} = \frac{y^2 - (x-1)^2}{[(x-1)^2 + y^2]^2} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{2y(x-1)}{[(x-1)^2 + y^2]^2} = -\frac{\partial v}{\partial x}$$

f is analytic in any domain not containing $z = 1$.

$$14. \quad u = \frac{x^3 + xy^2 + x}{x^2 + y^2}, \quad v = \frac{x^2y + y^3 - y}{x^2 + y^2}; \quad \frac{\partial u}{\partial x} = \frac{x^4 + 2x^2y^2 - x^2 + y^2 + y^4}{(x^2 + y^2)^2} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = \frac{-2xy}{(x^2 + y^2)^2} = -\frac{\partial v}{\partial x}$$

f is analytic in any domain not containing $z = 0$.

$$15. \quad \frac{\partial u}{\partial x} = 3 = b = \frac{\partial v}{\partial y}; \quad \frac{\partial u}{\partial y} = -1 = -a = -\frac{\partial v}{\partial x}. \quad f \text{ is analytic for all } z \text{ when } b = 3, a = 1.$$

16. The Cauchy-Riemann equations yield the system

$$\begin{array}{ll} 2x + ay = dx + 2y & (2-d)x + (2-a)y = 0 \\ ax + 2by = -2cx - dy & (a+2c)x + (2b+d)y = 0. \end{array} \quad \text{or}$$

The system holds for $z = x + iy$ whenever $2-d=0$, $2-a=0$, $a+2c=0$, and $2b+d=0$. That is, f is analytic for all z when $a=2$, $b=-1$, $c=-1$, and $d=2$.

$$17. \quad u = x^2 + y^2, \quad v = 2xy; \quad \frac{\partial u}{\partial x} = 2x, \quad \frac{\partial v}{\partial y} = 2x; \quad \frac{\partial u}{\partial y} = 2y, \quad -\frac{\partial v}{\partial x} = -2y$$

u and v are continuous and have continuous first partial derivatives. The Cauchy-Riemann equations are satisfied for any x and for $y = 0$, that is, for points on the real axis. The function f is differentiable but not analytic along this axis; there is no neighborhood about any point $z = x$ within which f is differentiable.

$$18. \quad u = 3x^2y^2, \quad v = -6x^2y^2; \quad \frac{\partial u}{\partial x} = 6xy^2, \quad \frac{\partial v}{\partial y} = -12x^2y; \quad \frac{\partial u}{\partial y} = 6x^2y, \quad -\frac{\partial v}{\partial x} = 12xy^2$$

u and v are continuous and have continuous first partial derivatives. The Cauchy-Riemann equations are satisfied whenever $6xy(y+2x) = 0$ and $6xy(x-2y) = 0$. The point satisfying $y+2x=0$ and $x-2y=0$ is $z=0$. The points that satisfy $6xy=0$ are the points along the y -axis ($x=0$) or along the x -axis ($y=0$). The function f is differentiable but not analytic on either axis; there is no neighborhood about any point $z=x$ or $z=iy$ within which f is differentiable.

$$19. \quad u = x^3 + 3xy^2 - x, \quad v = y^3 + 3x^2y - y; \quad \frac{\partial u}{\partial x} = 3x^2 + 3y^2 - 1, \quad \frac{\partial v}{\partial y} = 3y^2 + 3x^2 - 1; \quad \frac{\partial u}{\partial y} = 6xy, \quad -\frac{\partial v}{\partial x} = -6xy.$$

u and v are continuous and have continuous first partial derivatives. The Cauchy-Riemann equations are satisfied whenever $6xy = -6xy$ or $12xy = 0$. The points satisfying $12xy = 0$ are the points along the y -axis ($x=0$) or along the x -axis ($y=0$). The function f is differentiable but not analytic on either axis; there is no neighborhood about any point $z=x$ or $z=iy$ within which f is differentiable.

$$20. \quad u = x^2 - x + y, \quad v = y^2 - 5y - x; \quad \frac{\partial u}{\partial x} = 2x - 1, \quad \frac{\partial v}{\partial y} = 2y - 5; \quad \frac{\partial u}{\partial y} = 1, \quad -\frac{\partial v}{\partial x} = 1$$

u and v are continuous and have continuous first partial derivatives. The Cauchy-Riemann equations are satisfied whenever $2x-1=2y-5$ or for points on the line $y=x+2$. The function f is differentiable but not analytic on this line; there is no neighborhood about any point $z=x+(x+2)i$ within which f is differentiable.

21. Since f is entire,

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = e^x \cos y + ie^x \sin y = f(z).$$

Exercises 17.5

22. Since f is entire,

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = -2ye^{x^2-y^2} \sin 2xy + 2xe^{x^2-y^2} \cos 2xy + i(2ye^{x^2-y^2} \cos 2xy + 2xe^{x^2-y^2} \sin 2xy).$$

23. $\frac{\partial^2 u}{\partial x^2} = 0$, $\frac{\partial^2 u}{\partial y^2} = 0$ gives $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$. Thus u is harmonic. Now $\frac{\partial u}{\partial x} = 1 = \frac{\partial v}{\partial y}$ implies $v = y + h(x)$, $\frac{\partial u}{\partial y} = 0 = -\frac{\partial v}{\partial x}$ implies $0 = -h'(x)$, and so $h(x) = C$ (a constant.) Therefore $f(z) = x + i(y + C)$.

24. $\frac{\partial^2 u}{\partial x^2} = 0$, $\frac{\partial^2 u}{\partial y^2} = 0$ gives $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$. Thus u is harmonic. Now $\frac{\partial u}{\partial x} = 2 - 2y = \frac{\partial v}{\partial y}$ implies $v = 2y - y^2 + h(x)$, $\frac{\partial u}{\partial y} = -2x = -\frac{\partial v}{\partial x} = -h'(x)$ implies $h'(x) = 2x$ or $h(x) = x^2 + C$. Therefore $f(z) = 2x - 2xy + i(2y - y^2 + x^2 + C)$.

25. $\frac{\partial^2 u}{\partial x^2} = 2$, $\frac{\partial^2 u}{\partial y^2} = -2$ gives $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$. Thus u is harmonic. Now $\frac{\partial u}{\partial x} = 2x = \frac{\partial v}{\partial y}$ implies $v = 2xy + h(x)$, $\frac{\partial u}{\partial y} = -2y = -\frac{\partial v}{\partial x} = -2y - h'(x)$ implies $h'(x) = 0$ or $h(x) = C$. Therefore $f(z) = x^2 - y^2 + i(2xy + C)$.

26. $\frac{\partial^2 u}{\partial x^2} = -24xy$, $\frac{\partial^2 u}{\partial y^2} = 24xy$ gives $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$. Thus u is harmonic. Now $\frac{\partial u}{\partial x} = 4y^3 - 12x^2y + 1 = \frac{\partial v}{\partial y}$ implies $v = y^4 - 6x^2y^2 + y + h(x)$, $\frac{\partial u}{\partial y} = 12xy^2 - 4x^3 = -\frac{\partial v}{\partial x} = 12xy^2 - h'(x)$ implies $h'(x) = 4x^3$ or $h(x) = x^4 + C$. Therefore $f(z) = 4xy^3 - 4x^3y + x + i(y^4 - 6x^2y^2 + y + x^4 + C)$.

27. $\frac{\partial^2 u}{\partial x^2} = \frac{2y^2 - 2x^2}{(x^2 + y^2)^2}$, $\frac{\partial^2 u}{\partial y^2} = \frac{2x^2 - 2y^2}{(x^2 + y^2)^2}$ gives $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$. Thus u is harmonic. Now $\frac{\partial u}{\partial x} = \frac{2x}{x^2 + y^2} = \frac{\partial v}{\partial y}$ implies $v = 2 \tan^{-1} \frac{y}{x} + h(x)$, $\frac{\partial u}{\partial y} = \frac{2y}{x^2 + y^2} = -\frac{\partial v}{\partial x} = \frac{2y}{x^2 + y^2} - h'(x)$ implies $h'(x) = 0$ or $h(x) = C$. Therefore $f(z) = \log_e(x^2 + y^2) + i \left(\tan^{-1} \frac{y}{x} + C \right)$, $z \neq 0$.

28. $\frac{\partial^2 u}{\partial x^2} = 2e^x \cos y + e^x(x \cos y - y \sin y)$, $\frac{\partial^2 u}{\partial y^2} = e^x(-x \cos y + y \sin y - 2 \cos y)$ gives $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$. Thus u is harmonic. Now $\frac{\partial u}{\partial x} = e^x \cos y + e^x(x \cos y - y \sin y) = \frac{\partial v}{\partial y}$. Integrating by parts with respect to y implies

$$v = e^x \sin y + e^x(x \sin y + y \cos y - \sin y) + h(x) = xe^x \sin y + ye^x \cos y + h(x),$$

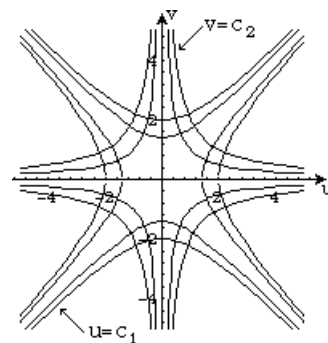
and

$$\frac{\partial u}{\partial y} = -xe^x \sin y - ye^x \cos y - e^x \sin y = -\frac{\partial v}{\partial x} = -xe^x \sin y - e^x \sin y - ye^x \cos y + h'(x)$$

implies $h'(x) = 0$ or $h(x) = C$. Therefore

$$f(z) = e^x(x \cos y - y \sin y) + ie^x(x \sin y + y \cos y + C).$$

29. The level curves $u(x, y) = c_1$ and $v(x, y) = c_2$ are the families of hyperbolas $x^2 - y^2 = c_1$ and $2xy = c_2$, respectively. The graphs of these families are displayed on the same axes in the figure.



30. $f(z) = \frac{x}{x^2 + y^2} - i \frac{y}{x^2 + y^2}$. The level curves $u(x, y) = c_1$ and $v(x, y) = c_2$ are the family of circles $x = c_1(x^2 + y^2)$ and $-y = c_2(x^2 + y^2)$, with the exception that $(0, 0)$ is not on the circumference of any circle.
31. $f(z) = x + \frac{x}{x^2 + y^2} + i \left(y - \frac{y}{x^2 + y^2} \right)$. The level curve $v(x, y) = 0$ is described by $y - \frac{y}{x^2 + y^2} = 0$ or $y(x^2 + y^2 - 1) = 0$. We see that either $y = 0$ or $x^2 + y^2 = 1$. Thus $v(x, y) = 0$ gives either the x -axis (without the origin $(0, 0)$) or the unit circle $x^2 + y^2 = 1$.
32. If $\nabla u = \frac{\partial u}{\partial x} \mathbf{i} + \frac{\partial u}{\partial y} \mathbf{j}$ and $\nabla v = \frac{\partial v}{\partial x} \mathbf{i} + \frac{\partial v}{\partial y} \mathbf{j}$, then $\nabla u \cdot \nabla v = \frac{\partial u}{\partial x} \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \frac{\partial v}{\partial y}$. By the Cauchy-Riemann equations this becomes

$$\nabla u \cdot \nabla v = \frac{\partial v}{\partial y} \frac{\partial v}{\partial x} + \left(-\frac{\partial v}{\partial x} \right) \frac{\partial v}{\partial y} = 0.$$

Since the gradients of u and v are orthogonal vectors, the level curves $u(x, y) = c_1$ and $u(x, y) = c_2$ are orthogonal families.

Exercises 17.6

1. $e^{\frac{\pi}{6}i} = \cos \frac{\pi}{6} + i \sin \frac{\pi}{6} = \frac{\sqrt{3}}{2} + \frac{1}{2}i$
2. $e^{-\frac{\pi}{3}i} = \cos \frac{\pi}{3} - i \sin \frac{\pi}{3} = \frac{1}{2} - \frac{\sqrt{3}}{2}i$
3. $e^{-1+\frac{\pi}{4}i} = e^{-1} \cos \frac{\pi}{4} + ie^{-1} \sin \frac{\pi}{4} = e^{-1} \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i \right)$
4. $e^{2-\frac{\pi}{2}i} = e^2 \cos \left(-\frac{\pi}{2} \right) + ie^2 \sin \left(-\frac{\pi}{2} \right) = -e^2i$
5. $e^{\pi+\pi i} = e^{\pi} \cos \pi + ie^{\pi} \sin \pi = -e^{\pi}$
6. $e^{-\pi+\frac{3\pi}{2}i} = e^{-\pi} \cos \frac{3\pi}{2} + ie^{-\pi} \sin \frac{3\pi}{2} = -e^{-\pi}i$
7. $e^{1.5+2i} = e^{1.5} \cos 2 + ie^{1.5} \sin 2 = -1.8650 + 4.0752i$
8. $e^{-0.3+0.5i} = e^{-0.3} \cos 0.5 + ie^{-0.3} \sin 0.5 = 0.6501 + 0.3552i$
9. $e^{5i} = \cos 5 + i \sin 5 = 0.2837 - 0.9589i$
10. $e^{-0.23-i} = e^{-0.23} \cos(-1) + ie^{-0.23} \sin(-1) = 0.4293 - 0.6686i$
11. $e^{\frac{11\pi}{12}i} = \cos \frac{11\pi}{12} + i \sin \frac{11\pi}{12} = -0.9659 + 0.2588i$

Exercises 17.6

12. $e^{5+\frac{5\pi}{2}i} = e^5 \cos \frac{5\pi}{2} + ie^5 \sin \frac{5\pi}{2} = e^5 i$
13. $e^{-iz} = e^{y-xi} = e^y \cos x - ie^y \sin x$
14. $e^{2\bar{z}} = e^{2x-2yi} = e^{2x} \cos 2y - ie^{2x} \sin 2y$
15. $e^{z^2} = e^{x^2-y^2+2xyi} = e^{x^2-y^2} \cos 2xy + ie^{x^2-y^2} \sin 2xy$
16. $e^{1/z} = e^{x/(x^2+y^2)-iy/(x^2+y^2)} = e^{x/(x^2+y^2)} \cos \frac{y}{x^2+y^2} - ie^{x/(x^2+y^2)} \sin \frac{y}{x^2+y^2}$
17. $|e^z|^2 = e^{2x} \cos^2 y + e^{2x} \sin^2 y = e^{2x}(\cos^2 y + \sin^2 y) = e^{2x}$ implies $|e^z| = e^x$.
18. $\frac{e^{z_1}}{e^{z_2}} = \frac{e^{x_1} \cos y_1 + ie^{x_1} \sin y_1}{e^{x_2} \cos y_2 + ie^{x_2} \sin y_2} = \frac{(e^{x_1} \cos y_1 + ie^{x_1} \sin y_1)(e^{x_2} \cos y_2 - ie^{x_2} \sin y_2)}{e^{2x_2}}$
 $= e^{x_1-x_2}[(\cos y_1 \cos y_2 + \sin y_1 \sin y_2) + i(\sin y_1 \cos y_2 - \cos y_1 \sin y_2)]$
 $= e^{x_1-x_2}[\cos(y_1 - y_2) + i \sin(y_1 - y_2)] = e^{x_1-x_2+i(y_1-y_2)} = e^{(x_1+iy_1)-(x_2+iy_2)} = e^{z_1-z_2}$
19. $e^{z+\pi i} = e^{x+(y+\pi)i} = e^x[\cos(y+\pi) + i \sin(y+\pi)] = e^x[\cos(y-\pi) + i \sin(y-\pi)] = e^{x+(y-\pi)i} = e^{z-\pi i}$
20. $(e^z)^n = (e^x[\cos y + i \sin y])^n = e^{nx}[\cos y + i \sin y]^n = e^{nx}[\cos ny + i \sin ny] = e^{nz}$, n an integer
21. $u = e^x \cos y$, $v = -e^x \sin y$; $\frac{\partial u}{\partial x} = e^x \cos y$, $\frac{\partial v}{\partial y} = -e^x \cos y$; $\frac{\partial u}{\partial y} = -e^x \sin y$, $-\frac{\partial v}{\partial x} = e^x \sin y$

Since the Cauchy-Riemann equations are not satisfied at any point, f is nowhere analytic.

22. (a) $u = e^{x^2-y^2} \cos 2xy$, $v = e^{x^2-y^2} \sin 2xy$; $\frac{\partial u}{\partial x} = -2ye^{x^2-y^2} \sin 2xy + 2xe^{x^2-y^2} \cos 2xy = \frac{\partial v}{\partial y}$;

$$\frac{\partial u}{\partial y} = -2xe^{x^2-y^2} \sin 2xy - 2ye^{x^2-y^2} \cos 2xy = -\frac{\partial v}{\partial x}$$

Since u , v , and their first partial derivatives are continuous, and u and v satisfy the Cauchy-Riemann equations everywhere, the function f is differentiable everywhere. Hence f is entire.

- (b) $\frac{\partial^2 u}{\partial x^2} = -4y^2 e^{x^2-y^2} \cos 2xy - 4xy e^{x^2-y^2} \sin 2xy - 4xy e^{x^2-y^2} \sin 2xy + \cos 2xy[4x^2 e^{x^2-y^2} + 2e^{x^2-y^2}]$;
 $\frac{\partial^2 u}{\partial y^2} = -4x^2 e^{x^2-y^2} \cos 2xy + 4xy e^{x^2-y^2} \sin 2xy + 4xy e^{x^2-y^2} \sin 2xy + \cos 2xy[4y^2 e^{x^2-y^2} - 2e^{x^2-y^2}]$

Since $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ the function u is harmonic throughout the z -plane.

23. $\ln(-5) = \log_e 5 + i(\pi + 2n\pi) = 1.16094 + (\pi + 2n\pi)i$
24. $\ln(-ei) = \log_e e + i\left(-\frac{\pi}{2} + 2n\pi\right) = 1 + \left(-\frac{\pi}{2} + 2n\pi\right)i$
25. $\ln(-2 + 2i) = \log_e 2\sqrt{2} + i\left(\frac{3\pi}{4} + 2n\pi\right) = 1.0397 + \left(\frac{3\pi}{4} + 2n\pi\right)i$
26. $\ln(1 + i) = \log_e \sqrt{2} + i\left(\frac{\pi}{4} + 2n\pi\right) = 0.3466 + \left(\frac{\pi}{4} + 2n\pi\right)i$
27. $\ln(\sqrt{2} + \sqrt{6}i) = \log_e 2\sqrt{2} + i\left(\frac{\pi}{3} + 2n\pi\right) = 1.0397 + \left(\frac{\pi}{3} + 2n\pi\right)i$
28. $\ln(-\sqrt{3} + i) = \log_e 2 + i\left(\frac{5\pi}{6} + 2n\pi\right) = 0.6932 + \left(\frac{5\pi}{6} + 2n\pi\right)i$
29. $\text{Ln}(6 - 6i) = \log_e 6\sqrt{2} + i\left(-\frac{\pi}{4}\right) = 2.1383 - \frac{\pi}{4}i$
30. $\text{Ln}(-e^3) = \log_e e^3 + \pi i = 3 + \pi i$
31. $\text{Ln}(-12 + 5i) = \log_e 13 + i\left(\tan^{-1}\left(-\frac{5}{12}\right) + \pi\right) = 2.5649 + 2.7468i$

$$32. \operatorname{Ln}(3-4i) = \log_e 5 + i \tan^{-1} \left(-\frac{4}{3} \right) = 1.6094 - 0.9273i$$

$$33. \operatorname{Ln}(1 + \sqrt{3}i)^5 = \operatorname{Ln}(16 - 16\sqrt{3}i) = \log_e 32 - \frac{\pi}{3}i = 3.4657 - \frac{\pi}{3}i$$

$$34. \operatorname{Ln}(1+i)^4 = \operatorname{Ln}(-4) = \log_e 4 + \pi i = 1.3863 + \pi i$$

$$35. z = \ln(4i) = \log_e 4 + i \left(\frac{\pi}{2} + 2n\pi \right) = 1.3863 + \left(\frac{\pi}{2} + 2n\pi \right) i$$

$$36. \frac{1}{z} = \ln(-1) = \log_e 1 + i(\pi + 2n\pi) = (2n+1)\pi i \quad \text{and so} \quad z = -\frac{i}{(2n+1)\pi}.$$

$$37. z - 1 = \ln(-ie^2) = \log_e e^2 + i \left(\frac{3\pi}{2} + 2n\pi \right) = 2 + \left(\frac{3\pi}{2} + 2n\pi \right) i \quad \text{and so} \quad z = 3 \left(\frac{3\pi}{2} + 2n\pi \right) i.$$

$$38. \text{By the quadratic formula, } e^z = -\frac{1}{2} + \frac{\sqrt{3}}{2}i \text{ or } e^z = -\frac{1}{2} - \frac{\sqrt{3}}{2}i. \text{ Hence}$$

$$z = \ln \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i \right) = \left(\frac{2\pi}{3} + 2n\pi \right) i \quad \text{or} \quad z = \ln \left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i \right) = \left(\frac{4\pi}{3} + 2n\pi \right) i.$$

$$39. (-i)^{4i} = e^{4i \ln(-i)} = e^{4i[\log_e 1 + i(-\frac{\pi}{2} + 2n\pi)]} = e^{(2-8n)\pi}$$

$$40. 3^{1/\pi} = e^{\frac{1}{\pi} \ln 3} = e^{\frac{1}{\pi}[\log_e 3 + 2n\pi i]} = e^{-2n} \left[\cos \left(\frac{1}{\pi} \log_e 3 \right) + i \sin \left(\frac{1}{\pi} \log_e 3 \right) \right] = e^{-2n}[0.9395 + 0.3426i]$$

$$41. (1+i)^{(1+i)} = e^{(1+i) \ln(1+i)} = e^{(1+i)[\log_e \sqrt{2} + i(\frac{\pi}{4} + 2n\pi)]} \\ = e^{\log_e \sqrt{2} - (\frac{\pi}{4} + 2n\pi)} \left[\cos \left(\frac{\pi}{4} + \log_e \sqrt{2} \right) + i \sin \left(\frac{\pi}{4} + \log_e \sqrt{2} \right) \right] = e^{-2n\pi}[0.2740 + 0.5837i]$$

$$42. (1-i)^{2i} = e^{2i \ln(1-i)} = e^{2i[\log_e \sqrt{2} + i(-\frac{\pi}{4} + 2n\pi)]} = e^{\frac{\pi}{2} - 4n\pi} [\cos(\log_e 2) + i \sin(\log_e 2)] = e^{-4n\pi}[3.7004 + 3.0737i]$$

$$43. (-1)^{-\frac{2i}{\pi}} = e^{-\frac{2i}{\pi} \ln(-1)} = e^{-\frac{2i}{\pi}(\pi i)} = e^2 = 7.3891$$

$$44. (1-i)^{2i} = e^{2i \ln(1-i)} = e^{2i[\log_e \sqrt{2} - \frac{\pi}{4}i]} = e^{\frac{\pi}{2}} [\cos(\log_e 2) + i \sin(\log_e 2)] = 3.7004 + 3.0737i$$

$$45. \text{If } z_1 = i \text{ and } z_2 = -1 + i \text{ then}$$

$$\operatorname{Ln}(z_1 z_2) = \operatorname{Ln}(-1-i) = \log_e \sqrt{2} - \frac{3\pi}{4}i,$$

whereas

$$\operatorname{Ln} z_1 + \operatorname{Ln} z_2 = \frac{\pi}{2}i + \left(\log_e \sqrt{2} + \frac{3\pi}{4}i \right) = \log_e \sqrt{2} + \frac{5\pi}{4}i.$$

$$46. \text{If } z_1 = -i \text{ and } z_2 = i \text{ then}$$

$$\operatorname{Ln}(z_1/z_2) = \operatorname{Ln}(-1) = \pi i, \quad \text{whereas} \quad \operatorname{Ln} z_1 - \operatorname{Ln} z_2 = -\frac{\pi}{2}i - \frac{\pi}{2}i = -\pi i.$$

$$47. \text{(a) The statement is false.}$$

$$\operatorname{Ln}(-1+i)^2 = \operatorname{Ln}(-2i) = \log_e 2 - \frac{\pi}{2}i, \quad \text{whereas} \quad 2\operatorname{Ln}(-1+i) = 2 \left(\log_e \sqrt{2} + \frac{3\pi}{4}i \right) = \log_e 2 + \frac{3\pi}{2}i.$$

$$\text{(b) The statement is false.}$$

$$\operatorname{Ln} i^3 = \operatorname{Ln}(-i) = -\frac{\pi}{2}i, \quad \text{whereas} \quad 3\operatorname{Ln} i = \frac{3\pi}{2}i.$$

$$\text{(c) The statement is true. If we take } \arg(-i) = \frac{3\pi}{2} \text{ then } \ln i^3 = \ln(-i) = \frac{3\pi}{2}i \text{ for } n=0. \text{ Also, } 3\ln i = 3 \left(\frac{\pi}{2}i \right).$$

$$48. \text{(a) } (i^i)^2 = (e^{i \ln i})^2 = [e^{-(\frac{\pi}{2} + 2n\pi)}]^2 = e^{-(\pi + 4n\pi)} \quad \text{and} \quad i^{2i} = e^{2i \ln i} = e^{-(\pi + 4n\pi)}$$

$$\text{(b) } (i^2)^i = (-1)^i = e^{i \ln(-1)} = e^{-(\pi + 2n\pi)}, \quad \text{whereas} \quad i^{2i} = e^{-(\pi + 4n\pi)}$$

Exercises 17.6

49. Since $|z| = \sqrt{x^2 + y^2}$ and $\text{Arg} z = \tan^{-1} \frac{y}{x}$ for $x > 0$ we have

$$\text{Ln} z = \log_e |z| + i \text{Arg} z = \log_e (x^2 + y^2)^{1/2} + i \tan^{-1} \frac{y}{x} = \frac{1}{2} \log_e (x^2 + y^2) + i \tan^{-1} \frac{y}{x}.$$

50. (a) $u = \log_e (x^2 + y^2); \quad \frac{\partial^2 u}{\partial x^2} = \frac{2(y^2 - x^2)}{(x^2 + y^2)^2}, \quad \frac{\partial^2 u}{\partial y^2} = \frac{2(x^2 - y^2)}{(x^2 + y^2)^2}$

Since $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ the function u is harmonic in any domain not containing the point $(0, 0)$.

(b) $v = \tan^{-1} \frac{y}{x}; \quad \frac{\partial^2 v}{\partial x^2} = \frac{2xy}{(x^2 + y^2)^2}, \quad \frac{\partial^2 v}{\partial y^2} = \frac{-2xy}{(x^2 + y^2)^2}$

Since $\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$ the function v is harmonic in any domain not containing the point $(0, 0)$.

Exercises 17.7

- $\cos(3i) = \cosh 3 = 10.0677$
- $\sin(-2i) = i \sinh(-2) = -3.6269i$
- $\sin\left(\frac{\pi}{4} + i\right) = \sin \frac{\pi}{4} \cosh(1) + i \cos \frac{\pi}{4} \sinh(1) = 1.0911 + 0.8310i$
- $\cos(2 - 4i) = \cos(2) \cosh(-4) - \sin(2) \sinh(-4) = -11.3642 - 24.8147i$
- $\tan(i) = \frac{\sin(i)}{\cos(i)} = \frac{i \sinh(1)}{\cosh(1)} = 0.7616i$
- $\cot\left(\frac{\pi}{2} + 3i\right) = \frac{\cos(\frac{\pi}{2} + 3i)}{\sin(\frac{\pi}{2} + 3i)} = \frac{-i \sinh(3)}{\cosh(3)} = -0.9951i$
- $\sec(\pi + i) = \frac{1}{\cos(\pi + i)} = \frac{1}{-\cosh(1)} = -0.6481$
- $\csc(1 + i) = \frac{1}{\sin(1 + i)} = \frac{1}{\sin(1) \cosh(1) + i \cos(1) \sinh(1)} = 0.6215 - 0.3039i$
- $\cosh(\pi i) = \cos(i(\pi i)) = \cos(-\pi) = \cos \pi = -1$
- $\sinh\left(\frac{3\pi}{2} i\right) = -i \sin\left(i\left(\frac{3\pi}{2} i\right)\right) = -i \sin\left(-\frac{3\pi}{2}\right) = i \sin \frac{3\pi}{2} = -i$
- $\sinh\left(1 + \frac{\pi}{3} i\right) = \sinh(1) \cos \frac{\pi}{3} + i \cosh(1) \sin \frac{\pi}{3} = 0.5876 + 1.3363i$
- $\cosh(2 + 3i) = \cosh(2) \cos(3) + i \sinh(2) \sin(3) = -3.7245 + 0.5118i$
- $\sin\left(\frac{\pi}{2} + i \ln 2\right) = \sin \frac{\pi}{2} \cosh(\ln 2) + i \cos \frac{\pi}{2} \sinh(\ln 2) = \frac{e^{\ln 2} + e^{\ln 2^{-1}}}{2} = \frac{2 + \frac{1}{2}}{2} = \frac{5}{4}$
- $\cos\left(\frac{\pi}{2} + i \ln 2\right) = \cos \frac{\pi}{2} \cosh(\ln 2) - i \sin \frac{\pi}{2} \sinh(\ln 2) = -i \cdot \frac{e^{\ln 2} - e^{\ln 2^{-1}}}{2} = -i \cdot \frac{2 - \frac{1}{2}}{2} = -\frac{3}{4}i$
- $\frac{e^{iz} - e^{-iz}}{2i} = 2$ gives $e^{2(iz)} - 4ie^{iz} - 1 = 0$. By the quadratic formula, $e^{iz} = 2i \pm \sqrt{3}i$ and so

$$iz = \ln[(2 \pm \sqrt{3})i]$$

$$z = -i \left[\log_e(2 \pm \sqrt{3}) + \left(\frac{\pi}{2} + 2n\pi\right)i \right] = \frac{\pi}{2} + 2n\pi - i \log_e(2 \pm \sqrt{3}), \quad n = 0, \pm 1, \pm 2, \dots$$

16. $\frac{e^{iz} + e^{-iz}}{2} = -3i$ gives $e^{2(iz)} + 6ie^{iz} + 1 = 0$. By the quadratic formula, $e^{iz} = -3i \pm \sqrt{10}i$ and so $iz = \ln[-3 \pm \sqrt{10}i]$. Hence

$$\begin{aligned} z &= -i \left[\log_e(\sqrt{10} - 3) + \left(\frac{\pi}{2} + 2n\pi \right) i \right] & \text{or} & & z &= -i \left[\log_e(\sqrt{10} + 3) + \left(\frac{3\pi}{2} + 2n\pi \right) i \right] \\ z &= \frac{\pi}{2} + 2n\pi - i \log_e(\sqrt{10} - 3) & \text{or} & & z &= \frac{3\pi}{2} + 2n\pi - i \log_e(\sqrt{10} + 3) \end{aligned}$$

$$n = 0, \pm 1, \pm 2, \dots$$

17. $\frac{e^z - e^{-z}}{2} = i$ gives $e^{2z} - 2ie^z - 1 = 0$. By the quadratic formula, $e^z = -i$ and so

$$z = \ln(-i) = \log_e 1 + \left(-\frac{\pi}{2} + 2n\pi \right) i = \left(-\frac{\pi}{2} + 2n\pi \right) i, \quad n = 0, \pm 1, \pm 2, \dots$$

18. $\frac{e^z - e^{-z}}{2} = -1$ gives $e^{2z} + 2e^z - 1 = 0$. By the quadratic formula, $e^z = -1 \pm \sqrt{2}$, and so

$$\begin{aligned} z &= \ln(-1 \pm \sqrt{2}) \\ z &= \log_e(\sqrt{2} - 1) + 2n\pi i \quad \text{or} \quad z = \log_e(\sqrt{2} + 1) + (\pi + 2n\pi)i, \end{aligned}$$

$$n = 0, \pm 1, \pm 2, \dots$$

19. $\cos z = \sin z$ gives $\tan z = 1$. One solution is $z = \frac{\pi}{4}$. Since $\tan z$ is π -periodic, $z = \frac{\pi}{4} + n\pi$, $n = 0, \pm 1, \pm 2, \dots$ are also solutions. That these are the only solutions can be proved by solving

$$\frac{e^{iz} + e^{-iz}}{2} = \frac{e^{iz} - e^{-iz}}{2i}$$

by the method illustrated in Problems 15-18.

20. $\cos z = i \sin z$ gives $e^{iz} + e^{-iz} = e^{iz} - e^{-iz}$ or $e^{-iz} = 0$. Since this last equation has no solutions, the original equation has no solutions.
21. $\cos z = \cosh 2$ implies $\cos x \cosh y - i \sin x \sinh y = \cosh 2 + 0i$ and so we must have $\cos x \cosh y = \cosh 2$ and $\sin x \sinh y = 0$. The last equation has solutions $x = n\pi$, $n = 0, \pm 1, \pm 2, \dots$, or $y = 0$. For $y = 0$ the first equation becomes $\cos x = \cosh 2$. Since $\cosh 2 > 1$ this equation has no solutions. For $x = n\pi$ the first equation becomes $(-1)^n \cosh y = \cosh 2$. Since $\cosh y > 0$ we see n must be even, say, $n = 2k$, $k = 0, \pm 1, \pm 2, \dots$. Now $\cosh y = \cosh 2$ implies $y = \pm 2$. Solutions of the original equation are then

$$z = 2k\pi \pm 2i, \quad k = 0, \pm 1, \pm 2, \dots$$

22. $\sin z = i \sinh 2$ implies $\sin x \cosh y + i \cos x \sinh y = 0 + i \sinh 2$ and so we must have $\sin x \cosh y = 0$ and $\cos x \sinh y = \sinh 2$. Since $\cosh y > 0$ for all real numbers, the first equation has only the solutions $x = n\pi$, $n = 0, \pm 1, \pm 2, \dots$. For $x = n\pi$ the second equation becomes $(-1)^n \sinh y = \sinh 2$. If n is even, $\sinh y = \sinh 2$ implies $y = 2$ ($\sinh y$ is one-to-one.) If n is odd, $-\sinh y = \sinh 2$ implies $\sinh y = -\sinh(-2)$ and so $y = -2$. Solutions of the original equation are then

$$z = 2k\pi + 2i, \quad z = (2k + 1)\pi - 2i, \quad k = 0, \pm 1, \pm 2, \dots$$

23. $\cos z = \frac{e^{i(x+iy)} + e^{-i(x+iy)}}{2} = \frac{1}{2}(e^{-y}e^{ix} + e^ye^{-ix}) = \frac{1}{2}[e^{-y}(\cos x + i \sin x) + e^y(\cos x - i \sin x)]$
- $$= \cos x \left(\frac{e^y + e^{-y}}{2} \right) - i \sin x \left(\frac{e^y - e^{-y}}{2} \right) = \cos x \cosh y - i \sin x \sinh y$$

Exercises 17.7

24. $\sinh z = \frac{e^{x+iy} - e^{-x-iy}}{2} = \frac{1}{2}(e^x e^{iy} - e^{-x} e^{-iy}) = \frac{1}{2}[e^x(\cos y + i \sin y) - e^{-x}(\cos y - i \sin y)]$
 $= \left(\frac{e^x - e^{-x}}{2}\right) \cos y + i \left(\frac{e^x + e^{-x}}{2}\right) \sin y = \sinh x \cos y + i \cosh x \sin y$
25. $\cosh z = \frac{e^{x+iy} + e^{-x-iy}}{2} = \frac{1}{2}(e^x e^{iy} + e^{-x} e^{-iy}) = \frac{1}{2}[e^x(\cos y + i \sin y) + e^{-x}(\cos y - i \sin y)]$
 $= \left(\frac{e^x + e^{-x}}{2}\right) \cos y + i \left(\frac{e^x - e^{-x}}{2}\right) \sin y = \cosh x \cos y + i \sinh x \sin y$
26. $|\sinh z|^2 = \sinh^2 x \cos^2 y + \cosh^2 x \sin^2 y = \sinh^2 x \cos^2 y + (1 + \sinh^2 x) \sin^2 y$
 $= \sinh^2 x (\cos^2 y + \sin^2 y) + \sin^2 y = \sinh^2 x + \sin^2 y$
27. $|\cosh z|^2 = \cosh^2 x \cos^2 y + \sinh^2 x \sin^2 y = (1 + \sinh^2 x) \cos^2 y + \sinh^2 x \sin^2 y$
 $= \cos^2 y + \sinh^2 x (\cos^2 y + \sin^2 y) = \cos^2 y + \sinh^2 x$
28. $\cos^2 z + \sin^2 z = \left(\frac{e^{1z} + e^{-iz}}{2}\right)^2 + \left(\frac{e^{iz} - e^{-iz}}{2i}\right)^2 = \frac{1}{4}[e^{2iz} + 2 + e^{-2iz} - (e^{2iz} - 2 + e^{-2iz})] = \frac{4}{4} = 1$
29. $\cosh^2 z - \sinh^2 z = \left(\frac{e^z + e^{-z}}{2}\right)^2 - \left(\frac{e^z - e^{-z}}{2}\right)^2 = \frac{1}{4}[e^{2z} + 2 + e^{-2z} - (e^{2z} - 2 + e^{-2z})] = \frac{4}{4} = 1$
30. $\tan z = \frac{\sin z}{\cos z} = \frac{\sin z \cos z}{|\cos z|^2} = \frac{[\sin x \cosh y + i \cos x \sinh y][\cos x \cosh y + i \sin x \sinh y]}{\cos^2 x + \sinh^2 y}$
 $= \frac{(\sin x \cos x \cosh^2 y - \sin x \cos x \sinh^2 y)}{\cos^2 x + \sinh^2 y} + i \frac{\cos^2 x \sinh y \cosh y + \sin^2 x \sinh y \cosh y}{\cos^2 x + \sinh^2 y}$
 $= \frac{\sin x \cos x (\cosh^2 y - \sinh^2 y)}{\cos^2 x + \sinh^2 y} + i \frac{\sin y \cosh y (\cos^2 x + \sin^2 x)}{\cos^2 x + \sinh^2 y}$
 $= \frac{\sin x \cos x}{\cos^2 x + \sinh^2 y} + i \frac{\sinh y \cosh y}{\cos^2 x + \sinh^2 y} = \frac{\sin 2x}{2(\cos^2 x + \sinh^2 y)} + i \frac{\sinh 2y}{2(\cos^2 x + \sinh^2 y)}$

But

$$2 \cos^2 x + 2 \sinh^2 y = (2 \cos^2 x - 1) + (2 \sinh^2 y + 1) = \cos 2x + \cosh 2y.$$

Therefore $\tan z = u + iz$ where

$$u = \frac{\sin 2x}{\cos 2x + \cosh 2y}, \quad v = \frac{\sinh 2y}{\cos 2x + \cosh 2y}.$$

31. $\tanh(z + \pi i) = \frac{\sinh(x + (y + \pi)i)}{\cosh(x + (y + \pi)i)} = \frac{\sinh x \cos(y + \pi) + i \cosh x \sin(y + \pi)}{\cosh x \cos(y + \pi) + i \sinh x \sin(y + \pi)}$
 $= \frac{-[\sinh x \cos y + i \cosh x \sinh y]}{-[\cosh x \cos y + i \sinh x \sin y]} = \frac{-\sinh z}{-\cosh z} = \tanh z$
32. (a) $\overline{\sin z} = \sin x \cosh y - i \cos x \sinh y = \sin x \cosh(-y) + i \cos x \sinh(-y) = \sin(x - iy) = \sin \bar{z}$
(b) $\overline{\cos z} = \cos x \cosh y + i \sin x \sinh y = \cos x \cosh(-y) - i \sin x \sinh(-y) = \cos(x - iy) = \cos \bar{z}$

Exercises 17.8

$$1. \sin^{-1}(-i) = -i \ln(1 \pm \sqrt{2}) = \begin{cases} 2n\pi - i \log_e(1 + \sqrt{2}) \\ (2n+1)\pi - i \log_e(\sqrt{2} - 1) \end{cases}$$

$$\text{Since } \sqrt{2} - 1 = 1/(\sqrt{2} + 1) \text{ we can have } \sin^{-1}(-i) = \begin{cases} 2n\pi - i \log_e(1 + \sqrt{2}) \\ (2n+1)\pi + i \log_e(1 + \sqrt{2}) \end{cases}.$$

This can be written compactly as

$$\sin^{-1}(-i) = n\pi + (-1)^{n+1}i \log_e(1 + \sqrt{2}), \quad k = 0, \pm 1, \pm 2, \dots$$

$$2. \sin^{-1} \sqrt{2} = -i \ln[\sqrt{2} \pm 1]i = 2n\pi + \frac{\pi}{2} - i \log_e(\sqrt{2} \pm 1) = 2n\pi + \frac{\pi}{2} \pm i \log_e(1 + \sqrt{2}), \quad n = 0, \pm 1, \pm 2, \dots$$

$$3. \sin^{-1} 0 = -i \ln(\pm 1) = \begin{cases} 2n\pi + i \log_e 1 \\ (2n+1)\pi + i \log_e 1 \end{cases} = \begin{cases} 2n\pi \\ (2n+1)\pi \end{cases} = n\pi, \quad n = 0, \pm 1, \pm 2, \dots$$

$$4. \sin^{-1} \frac{13}{5} = -i \ln \left[\left(\frac{13}{5} \pm \frac{12}{5} \right) i \right] = \begin{cases} 2n\pi + \frac{\pi}{2} - i \log_e 5 \\ 2n\pi + \frac{\pi}{2} - i \log_e \frac{1}{5} \end{cases} = 2n\pi + \frac{\pi}{2} \pm i \log_e 5, \quad n = 0, \pm 1, \pm 2, \dots$$

$$5. \cos^{-1} 2 = -i \ln(2 \pm \sqrt{3}) = \begin{cases} 2n\pi - i \log_e(2 + \sqrt{3}) \\ 2n\pi - i \log_e(2 - \sqrt{3}) \end{cases}$$

Since $2 - \sqrt{3} = 1/(2 + \sqrt{3})$ this can be written compactly as

$$\cos^{-1} 2 = 2n\pi \pm i \log_e(2 + \sqrt{3}), \quad k = 0, \pm 1, \pm 2, \dots$$

$$6. \cos^{-1} 2i = -i \ln[(2 \pm \sqrt{5})i] = \begin{cases} 2n\pi - \frac{\pi}{2} + i \log_e(2 + \sqrt{5}) \\ 2n\pi + \frac{\pi}{2} - i \log_e(2 + \sqrt{5}) \end{cases}, \quad n = 0, \pm 1, \pm 2, \dots$$

$$7. \cos^{-1} \frac{1}{2} = -i \ln \left(\frac{1}{2} \pm \frac{\sqrt{3}}{2} i \right) = \begin{cases} 2n\pi + \frac{\pi}{3} - i \log_e 1 \\ 2n\pi + \frac{\pi}{3} - i \log_e 1 \end{cases} = 2n\pi \pm \frac{\pi}{2}, \quad n = 0, \pm 1, \pm 2, \dots$$

$$8. \cos^{-1} \frac{5}{3} = -i \ln \left(\frac{5}{3} \pm \frac{4}{3} \right) = \begin{cases} 2n\pi - i \log_e 3 \\ 2n\pi - i \log_e \frac{1}{3} \end{cases} = 2n\pi \pm i \log_e 3, \quad n = 0, \pm 1, \pm 2, \dots$$

$$9. \tan^{-1} 1 = \frac{i}{2} \ln \frac{i+1}{i-1} = \frac{i}{2} \ln(-i) = -n\pi + \frac{\pi}{4} + \frac{i}{2} \log_e 1 = \frac{\pi}{4} - n\pi, \quad n = 0, \pm 1, \pm 2, \dots$$

Note that this can also be written as $\tan^{-1} 1 = \frac{\pi}{4} + n\pi, \quad n = 0, \pm 1, \pm 2, \dots$

$$10. \tan^{-1} 3i = \frac{i}{2} \ln \left(\frac{4i}{-2i} \right) = \frac{i}{2} \ln(-2) = -\frac{\pi}{2} - n\pi + i \log_e \sqrt{2}, \quad n = 0, \pm 1, \pm 2, \dots$$

$$11. \sinh^{-1} \frac{4}{3} = \ln \left(\frac{4}{3} \pm \frac{5}{3} \right) = \begin{cases} \log_e 3 + 2n\pi i \\ \log_e \frac{1}{3} + (2n+1)\pi i \end{cases} = (-1)^n \log_e 3 + n\pi i, \quad n = 0, \pm 1, \pm 2, \dots$$

$$12. \cosh^{-1} i = \ln[(1 \pm \sqrt{2})i] = \begin{cases} \log_e(1 + \sqrt{2}) + (\frac{\pi}{2} + 2n\pi)i \\ \log_e(\sqrt{2} - 1) + (-\frac{\pi}{2} + 2n\pi)i \end{cases}, \quad n = 0, \pm 1, \pm 2, \dots$$

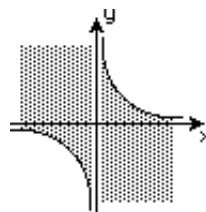
$$13. \tanh^{-1}(1 + 2i) = \frac{1}{2} \ln \frac{2+2i}{-2i} = \frac{1}{2} \ln(-1 + i) = \frac{1}{2} \left[\log_e \sqrt{2} + \left(\frac{3\pi}{4} + 2n\pi \right) i \right] = \frac{1}{4} \log_e 2 + \left(\frac{3\pi}{8} + n\pi \right) i$$

$$14. \tanh^{-1}(-\sqrt{3}i) = \frac{1}{2} \ln \frac{1-\sqrt{3}i}{1+\sqrt{3}i} = \frac{1}{2} \ln \left(-\frac{1}{2} - \frac{\sqrt{3}}{2} i \right) = \frac{1}{2} \left[\log_e 1 + \left(\frac{4\pi}{3} + 2n\pi \right) i \right] = \left(\frac{2\pi}{3} + n\pi \right) i,$$

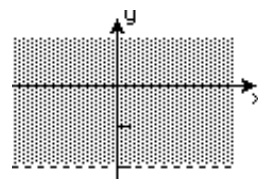
$n = 0, \pm 1, \pm 2, \dots$

Chapter 17 Review Exercises

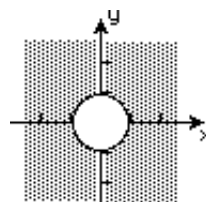
1. 0; 32
2. third
3. $-7/25$
4. $-8i$
5. $4/5$
6. The closed annular region between the circles $|z + 2| = 1$ and $|z + 2| = 3$. These circles have center at $z = -2$.
7. False. $\text{Arg}[(-1 + i) + (-1 - i)] = \text{Arg}(-2) = \pi$
8. $-5\pi/6$
9. $z = \ln(2i) = \log_e 2 + i \left(\frac{\pi}{2} + 2n\pi \right), n = 0, \pm 1, \pm 2, \dots$
10. True
11. $(1 + i)^{2+i} = e^{(2+i)[\log_e 2 + \frac{\pi}{4}i]} = e^{(\log_e 2 - \frac{\pi}{4}) + i(\log_e \sqrt{2} + \frac{\pi}{2})} = e^{\log_e 2 - \frac{\pi}{4}} \left[\cos \left(\log_e \sqrt{2} + \frac{\pi}{2} \right) + i \sin \left(\log_e \sqrt{2} + \frac{\pi}{2} \right) \right]$
 $= -0.3097 + 0.8577i$
12. $f(-1 + i) = -33 + 26i$
13. False
14. $2\pi i$
15. $\text{Ln}(-ie^3) = \log_e e^3 + \left(-\frac{\pi}{2} \right) i = 3 - \frac{\pi}{2} i$
16. True
17. $58 - 4i$
18. $-\frac{1}{13} - \frac{17}{13}i$
19. $-8 + 8i$
20. $4e^{\pi i/12} = 4 \left(\cos \frac{\pi}{12} + i \sin \frac{\pi}{12} \right) = 3.8637 + 1.0353i$
21. The region satisfying $xy \leq 1$ is shown in the figure.



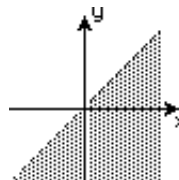
22. The region satisfying $y + 5 > 3$ or $y > -2$ is shown in the figure.



23. The region satisfying $|z| \geq 1$ is shown in the figure.



24. The region satisfying $y < x$ is shown in the figure.



25. Ellipse with foci $(0, -2)$ and $(0, 2)$

$$26. \left| \frac{z-w}{1-z\bar{w}} \right|^2 = \frac{z-w}{1-z\bar{w}} \cdot \frac{\bar{z}-\bar{w}}{1-\bar{z}w} = \frac{z\bar{z}-z\bar{w}-w\bar{z}+w\bar{w}}{1-\bar{z}w-z\bar{w}+z\bar{z}w\bar{w}} = \frac{1-z\bar{w}-w\bar{z}+|w|^2}{1-\bar{z}w-z\bar{w}+|w|^2} = 1,$$

since $|z|^2 = z\bar{z} = 1$ and $|w| \neq 1$.

27. The four fourth roots of $1-i$ are given by

$$w_R = 2^{1/8} \left[\cos \left(-\frac{\pi}{16} + \frac{k\pi}{2} \right) + i \sin \left(-\frac{\pi}{16} + \frac{k\pi}{2} \right) \right], \quad n = 0, 1, 2, 3$$

$$w_0 = 2^{1/8} \left[\cos \left(-\frac{\pi}{16} \right) + i \sin \left(-\frac{\pi}{16} \right) \right] = 1.0696 - 0.2127i$$

$$w_1 = 2^{1/8} \left[\cos \frac{7\pi}{16} + i \sin \frac{7\pi}{16} \right] = 0.2127 + 1.0696i$$

$$w_2 = 2^{1/8} \left[\cos \frac{15\pi}{16} + i \sin \frac{15\pi}{16} \right] = -1.0696 + 0.2127i$$

$$w_3 = 2^{1/8} \left[\cos \frac{23\pi}{16} + i \sin \frac{23\pi}{16} \right] = -0.2127 - 1.0696i$$

28. $z^{3/2} = \frac{2}{5} + \frac{1}{5}i$ implies $z^3 = \frac{3}{25} + \frac{4}{25}i$. The three cube roots of $\frac{3}{25} + \frac{4}{25}i$ are

$$w_k = \left(\frac{1}{5} \right)^{1/3} \left[\cos \left(\frac{1}{3} \tan^{-1} \left(\frac{4}{3} \right) + \frac{2k\pi}{3} \right) + i \sin \left(\frac{1}{3} \tan^{-1} \left(\frac{4}{3} \right) + \frac{2k\pi}{3} \right) \right], \quad k = 0, 1, 2$$

$$w_0 = \left(\frac{1}{5} \right)^{1/3} [\cos(0.3091) + i \sin(0.3091)] = 0.5571 + 0.1779i$$

$$w_1 = \left(\frac{1}{5} \right)^{1/3} [\cos(2.4035) + i \sin(2.4035)] = -0.4326 + 0.3935i$$

$$w_2 = \left(\frac{1}{5} \right)^{1/3} [\cos(4.4979) + i \sin(4.4979)] = -0.1245 - 0.5714i.$$

29. Write $\frac{1+i}{\sqrt{2}} = e^{\pi i/4}$ so that

$$z^{24} = e^{6\pi i} = 1, \quad z^{20} = e^{5\pi i} = -1, \quad z^{12} = e^{3\pi i} = -1, \quad z^6 = e^{3\pi i/2} = -i.$$

Therefore

$$f\left(\frac{1+i}{\sqrt{2}}\right) = 1 - 3(-1) + 4(-1) - 5(-i) = 5i.$$

30. $\text{Im}(z - 3\bar{z}) = 4y$, $z\text{Re}(z^2) = (x^3 - xy^2) + i(x^2y - y^3)$. Thus,

$$f(z) = (4y + x^3 - xy^2 - 5x) + i(x^2y - y^3 - 5y).$$

31. $u = x^2 - y$, $v = y^2 - x$. When $x = 1$ we get the parametric equations $u = 1 - y$, $v = y^2 - 1$. Eliminating y then gives $v = (1 - u)^2 - 1 = u^2 - 2u$. This is an equation of a parabola.

32. $u = x/(x^2 + y^2)$, $v = -y/(x^2 + y^2)$. When $x = 1$ we get the parametric equations $u = 1/(1 + y^2)$, $v = -y/(1 + y^2)$. From this we find $u^2 + v^2 - u = 0$. This describes a circle with the exception that $(0, 0)$ is not on its circumference.

33. $z = z^{-1}$ gives $z^2 = 1$ or $(z - 1)(z + 1) = 0$. Thus $z = \pm 1$.

34. $\bar{z} = 1/z$ gives $z\bar{z} = 1$ or $|z|^2 = 1$. All points on the circle $|z| = 1$ satisfy the equation.

35. $\bar{z} = -z$ gives $x = -x$ or $x = 0$. All complex numbers of the form $z = 0 + iy$ (pure imaginaries) satisfy the equation.

Chapter 17 Review Exercises

36. $z^2 = \bar{z}^2$ gives $xy = -xy$ or $xy = 0$. This implies $x = 0$ or $y = 0$. All real numbers ($y = 0$) and all pure imaginary numbers ($x = 0$) satisfy the equation.

37. $u = -2xy - 5x$, $v = x^2 - 5y - y^2$; $\frac{\partial u}{\partial x} = -2y - 5 = \frac{\partial v}{\partial y}$, $\frac{\partial u}{\partial y} = -2x = -\frac{\partial v}{\partial x}$; $f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = -2y - 5 + 2xi$

38. $u = x^3 + xy^2 - 4x$, $v = 4y - y^3 - x^2y$; $\frac{\partial u}{\partial x} = 3x^2 + y^2 - 4$, $\frac{\partial v}{\partial y} = 4 - 3y^2 - x^2$, $\frac{\partial u}{\partial y} = 2xy = -\frac{\partial v}{\partial x}$

The Cauchy-Riemann equations are satisfied at all points on the circle $x^2 + y^2 = 2$. Continuity of u , v , and the first partial derivatives guarantee f is differentiable on the circle. However, f is nowhere analytic.

39. $\text{Ln}(1+i)(1-i) = \text{Ln}(2) = \log_e 2$; $\text{Ln}(1+i) = \log_e \sqrt{2} + \frac{\pi}{4}i$; $\text{Ln}(1-i) = \log_e \sqrt{2} - \frac{\pi}{4}i$. Therefore

$$\text{Ln}(1+i) + \text{Ln}(1-i) = 2 \log_e \sqrt{2} = \log_e 2 = \text{Ln}(1+i)(1-i).$$

40. $\text{Ln} \frac{1+i}{1-i} = \text{Ln } i = \log_e 1 + \frac{\pi}{2}i = \frac{\pi}{2}i$; $\text{Ln}(1+i) = \log_e \sqrt{2} + \frac{\pi}{4}i$; $\text{Ln}(1-i) = \log_e \sqrt{2} - \frac{\pi}{4}i$. Therefore

$$\text{Ln}(1+i) - \text{Ln}(1-i) = \frac{\pi}{4}i - \left(-\frac{\pi}{4}i\right) = \frac{\pi}{2}i = \text{Ln} \frac{1+i}{1-i}.$$

18 Integration in the Complex Plane

Exercises 18.1

1. $\int_C (z+3) dz = (2+4i) \left[\int_1^3 (2t+3) dt + i \int_1^3 (4t-1) dt \right] = (2+4i)[14+14i] = -28+84i$
2. $\int_C (2\bar{z}-z) dz = \int_0^2 [-t-3(t^2+2)i](-1+2ti) dt = \int_0^2 (6t^3+13t) dt + i \int_0^2 (t^2+2) dt = 50 + \frac{20}{3}i$
3. $\int_C z^2 dz = (3+2i)^3 \int_{-2}^2 t^2 dt = \frac{16}{3}(3+2i)^3 = -48 + \frac{736}{3}i$
4. $\int_C (3z^2-2z) dz = \int_0^1 (-15t^4+4t^3+3t^2-2t) dt + i \int_0^1 (-6t^5+12t^3-6t^2) dt = -2+0i = -2$
5. Using $z = e^{it}$, $-\pi/2 \leq t \leq \pi/2$, and $dz = ie^{it} dt$, $\int_C \frac{1+z}{z} dz = - \int_{-\pi/2}^{\pi/2} (1+e^{it}) dt = (2+\pi)i$.
6. $\int_C |z|^2 dz = \int_1^2 \left(2t^5 + \frac{2}{t} \right) dt - i \int_1^2 \left(t^2 + \frac{1}{t^4} \right) dt = 21 + \ln 4 - \frac{21}{8}i$
7. Using $z = e^{it} = \cos t + i \sin t$, $dz = (-\sin t + i \cos t) dt$ and $x = \cos t$,

$$\begin{aligned} \oint_C \operatorname{Re}(z) dz &= \int_0^{2\pi} \cos t (-\sin t + i \cos t) dt = - \int_0^{2\pi} \sin t \cos t dt + i \int_0^{2\pi} \cos^2 t dt \\ &= -\frac{1}{2} \int_0^{2\pi} \sin 2t dt + \frac{1}{2} i \int_0^{2\pi} (1 + \cos 2t) dt = \pi i. \end{aligned}$$

8. Using $z+i = e^{it}$, $0 \leq t \leq 2\pi$, and $dz = ie^{it} dt$,

$$\oint_C \left[\frac{1}{(z+i)^3} - \frac{5}{z+i} + 8 \right] dz = i \int_0^{2\pi} [e^{-2it} - 5 + 8e^{it}] dt = -10\pi i.$$

9. Using $y = -x+1$, $0 \leq x \leq 1$, $z = x + (-x+1)i$, $dz = (1-i) dx$,

$$\int_C (x^2 + iy^3) dz = (1-i) \int_1^0 [x^2 + (1-x)^3 i] dx = -\frac{7}{12} + \frac{1}{12}i.$$

10. Using $z = e^{it}$, $\pi \leq t \leq 2\pi$, $dz = ie^{it} dt$, $x = \cos t = (e^{it} + e^{-it})/2$, $y = \sin t = (e^{it} - e^{-it})/2i$,

$$\begin{aligned} \int_C (x^3 - iy^3) dz &= \frac{1}{8} i \int_{\pi}^{2\pi} (e^{3it} + 3e^{it} + 3e^{-it} + e^{-3it}) e^{it} dt + \frac{1}{8} i \int_{\pi}^{2\pi} (e^{3it} - 3e^{it} + 3e^{-it} - e^{-3it}) e^{it} dt \\ &= \frac{1}{8} i \int_{\pi}^{2\pi} (2e^{4it} + 6) dt = \frac{3\pi}{4} i. \end{aligned}$$

11. $\int_C e^z dz = \int_{C_1} e^z dz + \int_{C_2} e^z dz$ where C_1 and C_2 are the line segments $y=0$, $0 \leq x \leq 2$ and $y=-\pi x+2\pi$, $1 \leq x \leq 2$, respectively. Now

$$\int_{C_1} e^z dz = \int_0^2 e^x dx = e^2 - 1$$

$$\int_{C_2} e^z dz = (1-\pi i) \int_2^1 e^{x+(-\pi x+2\pi)i} dx = (1-\pi i) \int_2^1 e^{(1-\pi i)x} dx = e^{1-\pi i} - e^{2(1-\pi i)} = -e - e^2.$$

Exercises 18.1

In the second integral we have used the fact that e^z has period $2\pi i$. Thus

$$\int_C e^z dz = (e^2 - 1) + (-e - e^2) = -1 - e.$$

12. $\int_C \sin z dz = \int_{C_1} \sin z dz + \int_{C_2} \sin z dz$ where C_1 and C_2 are the line segments $y = 0, 0 \leq x \leq 1$, and $x = 1, 0 \leq y \leq 1$, respectively. Now

$$\begin{aligned}\int_{C_1} \sin z dz &= \int_0^1 \sin x dx = 1 - \cos 1 \\ \int_{C_2} \sin z dz &= i \int_0^1 \sin(1 + iy) dy = \cos 1 - \cos(1 + i).\end{aligned}$$

Thus

$$\int_C \sin z dz = (1 - \cos 1) + (\cos 1 - \cos(1 + i)) = 1 - \cos(1 + i) = (1 - \cos 1 \cosh 1) + i \sin 1 \sinh 1 = 0.1663 + 0.9889i.$$

13. We have $\int_C \operatorname{Im}(z - i) dz = \int_{C_1} (y - 1) dz + \int_{C_2} (y - 1) dz$

On $C_1, z = e^{it}, 0 \leq t \leq \pi/2, dz = ie^{it} dt, y = \sin t = (e^{it} - e^{-it})/2i,$

$$\int_{C_1} (y - 1) dz = \frac{1}{2} \int_0^{\pi/2} [e^{it} - e^{-it} - 2i]e^{it} dt = \frac{1}{2} \int_0^{\pi/2} [e^{2it} - 1 + 2ie^{it}] dt = 1 - \frac{\pi}{4} - \frac{1}{2}i.$$

On $C_2, y = x + 1, -1 \leq x \leq 0, z = x + (x + 1)i, dz = (1 + i) dx,$

$$\int_{C_2} (y - 1) dz = (1 + i) \int_0^{-1} x dx = \frac{1}{2} + \frac{1}{2}i.$$

Thus

$$\int_C \operatorname{Im}(z - i) dz = \left(1 - \frac{\pi}{4} - \frac{1}{2}i\right) + \left(\frac{1}{2} + \frac{1}{2}i\right) = \frac{3}{2} - \frac{\pi}{4}.$$

14. Using $x = 6 \cos t, y = 2 \sin t, \pi/2 \leq t \leq 3\pi/2, z = 6 \cos t + 2i \sin t, dz = (-6 \sin t + 2i \cos t) dt,$

$$\int_C dz = -6 \int_{\pi/2}^{3\pi/2} \sin t dt + 2i \int_{\pi/2}^{3\pi/2} \cos t dt = 2i(-2) = -4i.$$

15. We have $\oint_C ze^z dz = \int_{C_1} ze^z dz + \int_{C_2} ze^z dz + \int_{C_3} ze^z dz + \int_{C_4} ze^z dz$

On $C_1, y = 0, 0 \leq x \leq 1, z = x, dz = dx,$

$$\int_{C_1} ze^z dz = \int_0^1 xe^x dx = xe^x - e^x \Big|_0^1 = 1.$$

On $C_2, x = 1, 0 \leq y \leq 1, z = 1 + iy, dz = i dy,$

$$\int_{C_2} ze^z dz = i \int_0^1 (1 + iy)e^{1+iy} dy = ie^{i+1}.$$

On $C_3, y = 1, 0 \leq x \leq 1, z = x + i, dz = dx,$

$$\int_{C_3} ze^z dz = \int_1^0 (x + i)e^{x+i} dx = (i - 1)e^i - ie^{1+i}.$$

On C_4 , $x = 0$, $0 \leq y \leq 1$, $z = iy$, $dz = i dy$,

$$\int_{C_4} z e^z dz = - \int_1^0 y e^{iy} dy = (1 - i)e^i - 1.$$

Thus

$$\oint_C z e^z dz = 1 + i e^{i+1} + (i - 1)e^i - i e^{1+i} + (1 - i)e^i - 1 = 0.$$

16. We have

$$\int_C f(z) dz = \int_{C_1} f(z) dz + \int_{C_2} f(z) dz$$

On C_1 , $y = x^2$, $-1 \leq x \leq 0$, $z = x + ix^2$, $dz = (1 + 2xi) dx$,

$$\int_{C_1} f(z) dz = \int_{-1}^0 2(1 + 2xi) dx = 2 - 2i.$$

On C_2 , $y = x^2$, $0 \leq x \leq 1$, $z = x + ix^2$, $dz = (1 + 2xi) dx$,

$$\int_{C_2} f(z) dz = \int_0^1 6x(1 + 2xi) dx = 3 + 4i.$$

Thus

$$\int_C f(z) dz = 2 - 2i + 3 + 4i = 5 + 2i.$$

17. We have

$$\oint_C x dz = \int_{C_1} x dz + \int_{C_2} x dz + \int_{C_3} x dz$$

On C_1 , $y = 0$, $0 \leq x \leq 1$, $z = x$, $dz = dx$,

$$\int_{C_1} x dz = \int_0^1 x dx = \frac{1}{2}.$$

On C_2 , $x = 1$, $0 \leq y \leq 1$, $z = 1 + iy$, $dz = i dy$,

$$\int_{C_2} x dz = i \int_0^1 dy = i.$$

On C_3 , $y = x$, $0 \leq x \leq 1$, $z = x + ix$, $dz = (1 + i) dx$,

$$\int_{C_3} x dz = (1 + i) \int_1^0 x dx = -\frac{1}{2} - \frac{1}{2}i.$$

Thus

$$\oint_C x dz = \frac{1}{2} + i - \frac{1}{2} - \frac{1}{2}i = \frac{1}{2}i.$$

18. We have

$$\oint_C (2z - 1) dz = \int_{C_1} (2z - 1) dz + \int_{C_2} (2z - 1) dz + \int_{C_3} (2z - 1) dz$$

On C_1 , $y = 0$, $0 \leq x \leq 1$, $z = x$, $dz = dx$,

$$\int_{C_1} (2z - 1) dz = \int_0^1 (2x - 1) dx = 0.$$

On C_2 , $x = 1$, $0 \leq y \leq 1$, $z = 1 + iy$, $dz = i dy$,

$$\int_{C_2} (2z - 1) dz = -2 \int_0^1 y dy + i \int_0^1 dy = -1 + i.$$

On C_3 , $y = x$, $z = x + ix$, $dz = (1 + i) dx$,

$$\int_{C_3} (2z - 1) dz = (1 + i) \int_1^0 (2x - 1 + 2ix) dx = 1 - i.$$

Exercises 18.1

Thus
$$\oint_C (2z - 1) dz = 0 - 1 + i + 1 - i = 0.$$

19. We have
$$\oint_C z^2 dz = \int_{C_1} z^2 dz + \int_{C_2} z^2 dz + \int_{C_3} z^2 dz$$

On C_1 , $y = 0$, $0 \leq x \leq 1$, $z = x$, $dz = dx$,

$$\int_{C_1} z^2 dz = \int_0^1 x^2 dx = \frac{1}{3}.$$

On C_2 , $x = 1$, $0 \leq y \leq 1$, $z = 1 + iy$, $dz = i dy$,

$$\int_{C_2} z^2 dz = \int_0^1 (1 + iy)^2 i dy = -1 + \frac{2}{3} i.$$

On C_3 , $y = x$, $0 \leq x \leq 1$, $z = x + ix$, $dz = (1 + i) dx$,

$$\int_{C_3} z^2 dz = (1 + i)^3 \int_1^0 x^2 dx = \frac{2}{3} - \frac{2}{3} i.$$

Thus
$$\oint_C z^2 dz = \frac{1}{3} - 1 + \frac{2}{3} i + \frac{2}{3} - \frac{2}{3} i = 0.$$

20. We have
$$\oint_C \bar{z}^2 dz = \int_{C_1} \bar{z}^2 dz + \int_{C_2} \bar{z}^2 dz + \int_{C_3} \bar{z}^2 dz$$

On C_1 , $y = 0$, $0 \leq x \leq 1$, $z = x$, $dz = dx$,

$$\int_{C_1} \bar{z}^2 dz = \int_0^1 x^2 dx = \frac{1}{3}.$$

On C_2 , $x = 1$, $0 \leq y \leq 1$, $z = 1 + iy$, $dz = i dy$,

$$\int_{C_2} \bar{z}^2 dz = - \int_0^1 (1 - iy)^2 (-i dy) = 1 + \frac{2}{3} i.$$

On C_3 , $y = x$, $0 \leq x \leq 1$, $z = x + ix$, $dz = (1 + i) dx$,

$$\int_{C_3} \bar{z}^2 dz = (1 - i)^2 (1 + i) \int_1^0 x^2 dx = -\frac{2}{3} + \frac{2}{3} i.$$

Thus
$$\oint_C \bar{z}^2 dz = \frac{1}{3} + 1 + \frac{2}{3} i - \frac{2}{3} + \frac{2}{3} i = \frac{2}{3} + \frac{4}{3} i.$$

21. On C , $y = -x + 1$, $0 \leq x \leq 1$, $z = x + (-x + 1)i$, $dz = (1 - i) dx$,

$$\int_C (z^2 - z + 2) dz = (1 - i) \int_0^1 [x^2 - (1 - x)^2 - x + 2 + (3x - 2x^2 - 1)i] dx = \frac{4}{3} - \frac{5}{3} i.$$

22. We have
$$\int_C (z^2 - z + 2) dz = \int_{C_1} (z^2 - z + 2) dz + \int_{C_2} (z^2 - z + 2) dz$$

On C_1 , $y = 1$, $0 \leq x \leq 1$, $z = x + i$, $dz = dx$,

$$\int_{C_1} (z^2 - z + 2) dz = \int_0^1 [(x + i)^2 - x + 2 - i] dx = \frac{5}{6}.$$

On C_2 , $x = 1$, $0 \leq y \leq 1$, $z = 1 + iy$, $dz = i dy$,

$$\int_{C_2} (z^2 - z + 2) dz = i \int_1^0 [(1 + iy)^2 + 1 - iy] dy = \frac{1}{2} - \frac{5}{3} i.$$

Thus
$$\int_C (z^2 - z + 2) dz = \frac{1}{2} - \frac{5}{3}i + \frac{5}{6} = \frac{4}{3} - \frac{5}{3}i.$$

23. On C , $y = 1 - x^2$, $0 \leq x \leq 1$, $z = x + i(1 - x^2)$, $dz = (1 - 2xi) dx$,

$$\int_C (z^2 - z + 2) dz = \int_0^1 (-5x^4 + 2x^3 + 7x^2 - 3x + 1) dx + i \int_0^1 (2x^5 - 8x^3 + 3x^2 - 1) dx = \frac{4}{3} - \frac{5}{3}i.$$

24. On C , $x = \sin t$, $y = \cos t$, $0 \leq t \leq \pi/2$ or $z = ie^{-it}$, $dz = e^{-it} dt$,

$$\begin{aligned} \int_C (z^2 - z + 2) dz &= \int_0^{\pi/2} (-e^{-2it} - ie^{-it} + 2)e^{-it} dt = \int_0^{\pi/2} (-e^{-3it} - ie^{-2it} + 2e^{-it}) dt \\ &= -\frac{1}{3}ie^{-3\pi i/2} + \frac{1}{2}e^{-\pi i} + 2ie^{-\pi i/2} + \frac{1}{3}i - \frac{1}{2} - 2i = \frac{4}{3} - \frac{5}{3}i. \end{aligned}$$

25. On C , $\left| \frac{e^z}{z^2 + 1} \right| \leq \frac{|e^z|}{|z|^2 - 1} = \frac{e^5}{24}$. Thus $\left| \oint_C \frac{e^z}{z^2 + 1} dz \right| \leq \frac{e^5}{24} \cdot 10\pi = \frac{5\pi}{12} e^5$.

26. On C , $\left| \frac{1}{z^2 - 2i} \right| \leq \frac{1}{|z|^2 - |2i|} = \frac{1}{34}$. Thus $\left| \int_C \frac{1}{z^2 - 2i} dz \right| \leq \frac{1}{34} \cdot \frac{1}{2}(12\pi) = \frac{3\pi}{17}$.

27. The length of the line segment from $z = 0$ to $z = 1 + i$ is $\sqrt{2}$. In addition, on this line segment

$$|z^2 + 4| \leq |z|^2 + 4 \leq |1 + i|^2 + 4 = 6.$$

Thus $\left| \int_C (z^2 + 4) dz \right| \leq 6\sqrt{2}$.

28. On C , $\left| \frac{1}{z^3} \right| = \frac{1}{|z|^3} = \frac{1}{64}$. Thus $\left| \int_C \frac{1}{z^3} dz \right| \leq \frac{1}{64} \cdot \frac{1}{4}(8\pi) = \frac{\pi}{32}$.

29. (a)
$$\begin{aligned} \int_C dz &= \lim_{\|P\| \rightarrow 0} \sum_{k=1}^n \Delta z_k = \lim_{\|P\| \rightarrow 0} \sum_{k=1}^n (z_k - z_{k-1}) \\ &= \lim_{\|P\| \rightarrow 0} [(z_1 - z_0) + (z_2 - z_1) + (z_3 - z_2) + \cdots + (z_{n-1} - z_{n-2}) + (z_n - z_{n-1})] \\ &= \lim_{\|P\| \rightarrow 0} (z_n - z_0) = z_n - z_0 \end{aligned}$$

(b) With $z_n = -2i$ and $z_0 = 2i$, $\int_C dz = -2i - (2i) = -4i$.

30. With $z_k^* = z_k$,

$$\begin{aligned} \int_C z dz &= \lim_{\|P\| \rightarrow 0} \sum_{k=1}^n z_k(z_k - z_{k-1}) \\ &= \lim_{\|P\| \rightarrow 0} [(z_1^2 - z_1 z_0) + (z_2^2 - z_2 z_1) + \cdots + (z_n^2 - z_n z_{n-1})]. \end{aligned} \quad (1)$$

With $z_k^* = z_{k-1}$,

$$\begin{aligned} \int_C z dz &= \lim_{\|P\| \rightarrow 0} \sum_{k=1}^n z_{k-1}(z_k - z_{k-1}) \\ &= \lim_{\|P\| \rightarrow 0} [(z_0 z_1 - z_0^2) + (z_1 z_2 - z_1^2) + \cdots + (z_{n-1} z_n - z_{n-1}^2)]. \end{aligned} \quad (2)$$

Adding (1) and (2) gives

$$2 \int_C z dz = \lim_{\|P\| \rightarrow 0} (z_n^2 - z_0^2) \quad \text{or} \quad \int_C z dz = \frac{1}{2}(z_n^2 - z_0^2).$$

Exercises 18.1

31. (a) $\int_C (6z + 4) dz = 6 \int_C z dz + 4 \int_C dz = \frac{6}{2}[(2 + 3i)^2 - (1 + i)^2] + 4[(2 + 3i) - (1 + i)] = -11 + 38i$

(b) Since the contour is closed, $z_0 = z_n$ and so

$$6 \int_C z dz + 4 \int_C dz = 6[z_0^2 - z_0^2] + 4[z_0 - z_0] = 0.$$

32. For $f(z) = 1/z$, $\overline{f(z)} = 1/\bar{z}$, so on $z = 2e^{it}$, $\bar{z} = 2e^{-it}$, $dz = 2ie^{it} dt$, and

$$\oint_C \overline{f(z)} dz = \int_0^{2\pi} \frac{1}{2e^{-it}} \cdot 2ie^{it} dt = \frac{1}{2} e^{2it} \Big|_0^{2\pi} = \frac{1}{2} [e^{4\pi i} - 1] = 0.$$

Thus circulation = $\operatorname{Re} \left(\oint_C \overline{f(z)} dz \right) = 0$, net flux = $\operatorname{Im} \left(\oint_C \overline{f(z)} dz \right) = 0$.

33. For $f(z) = 2z$, $\overline{f(z)} = 2\bar{z}$, so on $z = e^{it}$, $\bar{z} = e^{-it}$, $dz = ie^{it} dt$, and

$$\oint_C \overline{f(z)} dz = \int_0^{2\pi} (e^{-it})(ie^{it} dt) = 2i \int_0^{2\pi} dt = 4\pi i.$$

Thus circulation = $\operatorname{Re} \left(\oint_C \overline{f(z)} dz \right) = 0$, net flux = $\operatorname{Im} \left(\oint_C \overline{f(z)} dz \right) = 4\pi$.

34. For $f(z) = 1/(\bar{z}-1)$, $\overline{f(z)} = 1/(z-1)$, so on $z-1 = 2e^{it}$, $dz = 2ie^{it} dt$, and

$$\oint_C \overline{f(z)} dz = \int_0^{2\pi} \frac{1}{2e^{it}} \cdot 2ie^{it} dt = i \int_0^{2\pi} dt = 2\pi i.$$

Thus circulation = $\operatorname{Re} \left(\oint_C \overline{f(z)} dz \right) = 0$, net flux = $\operatorname{Im} \left(\oint_C \overline{f(z)} dz \right) = 2\pi$.

35. For $f(z) = \bar{z}$, $\overline{f(z)} = z$ so on the square we have

$$\oint_C \overline{f(z)} dz = \int_{C_1} z dz + \int_{C_2} z dz + \int_{C_3} z dz + \int_{C_4} z dz$$

where C_1 is $y = 0$, $0 \leq x \leq 1$, C_2 is $x = 1$, $0 \leq y \leq 1$, C_3 is $y = 1$, $0 \leq x \leq 1$, and C_4 is $x = 0$, $0 \leq y \leq 1$. Thus

$$\int_{C_1} z dz = \int_0^1 x dx = \frac{1}{2}$$

$$\int_{C_2} z dz = i \int_0^1 (1 + iy) dy = -\frac{1}{2} + i$$

$$\int_{C_3} z dz = \int_1^0 (x + i) dx = -\frac{1}{2} - i$$

$$\int_{C_4} z dz = -\int_1^0 y dy = \frac{1}{2}$$

and so

$$\oint_C \overline{f(z)} dz = \frac{1}{2} + \left(-\frac{1}{2} + i\right) + \left(-\frac{1}{2} - i\right) + \frac{1}{2} = 0$$

$$\text{circulation} = \operatorname{Re} \left(\oint_C \overline{f(z)} dz \right) = \operatorname{Re}(0) = 0$$

$$\text{net flux} = \operatorname{Im} \left(\oint_C \overline{f(z)} dz \right) = \operatorname{Im}(0) = 0.$$

Exercises 18.2

1. $f(z) = z^3 - 1 + 3i$ is a polynomial and so is an entire function.
2. z^2 is entire and $\frac{1}{z-4}$ is analytic within and on the circle $|z| = 1$.
3. $f(z) = \frac{z}{2z+3}$ is discontinuous at $z = -3/2$ but is analytic within and on the circle $|z| = 1$.
4. $f(z) = \frac{z-3}{z^2+2z+2}$ is discontinuous at $z = -1+i$ and at $z = -1-i$ but is analytic within and on the circle $|z| = 1$.
5. $f(z) = \frac{\sin z}{(z^2-25)(z^2+9)}$ is discontinuous at $z = \pm 5$ and at $z = \pm 3i$ but is analytic within and on the circle $|z| = 1$.
6. $f(z) = \frac{e^z}{2z^2+11z+15}$ is discontinuous at $z = -5/2$ and at $z = -3$ but is analytic within and on the circle $|z| = 1$.
7. $f(z) = \tan z$ is discontinuous at $z = \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \dots$ but is analytic within and on the circle $|z| = 1$.
8. $f(z) = \frac{z^2-9}{\cosh z}$ is discontinuous at $\frac{\pi}{2}i, \pm \frac{3\pi}{2}i, \dots$ but is analytic within and on the circle $|z| = 1$.
9. By the principle of deformation of contours we can choose the more convenient circular contour C_1 defined by $|z| = 1$. Thus

$$\oint_C \frac{1}{z} dz = \oint_{C_1} \frac{1}{z} dz = 2\pi i$$

by (4) of Section 18.2.

10. By the principle of deformation of contours we can choose the more convenient circular contour C_1 defined by $|z - (-1-i)| = \frac{1}{16}$. Thus

$$\oint_C \frac{5}{z+1+i} dz = 5 \oint_{C_1} \frac{1}{z - (-1-i)} dz = 5(2\pi i) = 10\pi i$$

by (4) of Section 18.2.

11. By Theorem 18.4 and (4) of Section 18.2,

$$\oint_C \left(z + \frac{1}{z} \right) dz = \oint_C z dz + \oint_C \frac{1}{z} dz = 0 + 2\pi i = 2\pi i.$$

12. By Theorem 18.4 and (4) of Section 18.2,

$$\oint_C \left(z + \frac{1}{z^2} \right) dz = \oint_C \frac{1}{z} dz + \oint_C \frac{1}{z^2} dz = 0 + 0 = 0.$$

13. Since $f(z) = \frac{z}{z^2 - \pi^2}$ is analytic within and on C it follows from Theorem 18.4 that $\oint_C \frac{z}{z^2 - \pi^2} dz = 0$.

14. By (4) of Section 18.2, $\oint_C \frac{10}{(z+i)^4} dz = 0$.

15. By partial fractions, $\oint_C \frac{2z+1}{z(z+1)} dz = \oint_C \frac{1}{z} dz + \oint_C \frac{1}{z+1} dz$.

Exercises 18.2

(a) By Theorem 18.4 and (4) of Section 18.2,

$$\oint_C \frac{1}{z} dz + \oint_C \frac{1}{z+1} dz = 2\pi i + 0 = 2\pi i.$$

(b) By writing $\oint_C = \oint_{C_1} + \oint_{C_2}$ where C_1 and C_2 are the circles $|z| = 1/2$ and $|z+1| = 1/2$, respectively, we have by Theorem 18.4 and (4) of Section 18.2,

$$\begin{aligned} \oint_C \frac{1}{z} dz + \oint_C \frac{1}{z+1} dz &= \oint_{C_1} \frac{1}{z} dz + \oint_{C_1} \frac{1}{z+1} dz + \oint_{C_2} \frac{1}{z} dz + \oint_{C_2} \frac{1}{z+1} dz \\ &= 2\pi i + 0 + 0 + 2\pi i = 4\pi i. \end{aligned}$$

(c) Since $f(z) = \frac{2z+1}{z(z+1)}$ is analytic within and on C it follows from Theorem 18.4 that

$$\oint_C \frac{2z+1}{z^2+z} dz = 0.$$

16. By partial fractions, $\oint_C \frac{2z}{z^2+3} dz = \oint_C \frac{1}{z+\sqrt{3}i} dz + \oint_C \frac{1}{z-\sqrt{3}i} dz$.

(a) By Theorem 18.4,

$$\oint_C \frac{1}{z+\sqrt{3}i} dz + \oint_C \frac{1}{z-\sqrt{3}i} dz = 0 + 0 = 0.$$

(b) By Theorem 18.4 and (4) of Section 18.2,

$$\oint_C \frac{1}{z+\sqrt{3}i} dz + \oint_C \frac{1}{z-\sqrt{3}i} dz = 0 + 2\pi i = 2\pi i.$$

(c) By writing $\oint_C = \oint_{C_1} + \oint_{C_2}$ where C_1 and C_2 are the circles $|z+\sqrt{3}i| = 1/2$ and $|z-\sqrt{3}i| = 1/2$, respectively, we have by Theorem 18.4 and (4) of Section 18.2,

$$\begin{aligned} \oint_C \frac{1}{z+\sqrt{3}i} dz + \oint_C \frac{1}{z-\sqrt{3}i} dz &= \oint_{C_1} \frac{1}{z+\sqrt{3}i} dz + \oint_{C_1} \frac{1}{z-\sqrt{3}i} dz + \oint_{C_2} \frac{1}{z+\sqrt{3}i} dz + \oint_{C_2} \frac{1}{z-\sqrt{3}i} dz \\ &= 2\pi i + 0 + 0 + 2\pi i = 4\pi i. \end{aligned}$$

17. By partial fractions, $\oint_C \frac{-3z+2}{z^2-8z+12} dz = \oint_C \frac{1}{z-2} dz - 4 \oint_C \frac{1}{z-6} dz$.

(a) By Theorem 18.4 and (4) of Section 18.2,

$$\oint_C \frac{1}{z-2} dz - 4 \oint_C \frac{1}{z-6} dz = 0 - 4(2\pi i) = -8\pi i.$$

(b) By writing $\oint_C = \oint_{C_1} + \oint_{C_2}$ where C_1 and C_2 are the circles $|z-2| = 1$ and $|z-6| = 1$, respectively, we have by Theorem 18.4 and (4) of Section 18.2,

$$\begin{aligned} \oint_C \frac{1}{z-2} dz - 4 \oint_C \frac{1}{z-6} dz &= \oint_{C_1} \frac{1}{z-2} dz - 4 \oint_{C_1} \frac{1}{z-6} dz + \oint_{C_2} \frac{1}{z-2} dz - 4 \oint_{C_2} \frac{1}{z-6} dz \\ &= 2\pi i - 4(0) + 0 - 4(2\pi i) = -6\pi i. \end{aligned}$$

18. (a) By writing $\oint_C = \oint_{C_1} + \oint_{C_2}$ where C_1 and C_2 are the circles $|z+2|=1$ and $|z-2i|=1$, respectively, we have by Theorem 18.4 and (4) of Section 18.2,

$$\begin{aligned}\oint_C \left(\frac{3}{z+2} - \frac{1}{z-2i} \right) dz &= \oint_{C_1} \frac{3}{z+2} dz - \oint_{C_1} \frac{1}{z-2i} dz + \oint_{C_2} \frac{3}{z+2} dz - \oint_{C_2} \frac{1}{z-2i} dz \\ &= 3(2\pi i) - 0 + 0 - 2\pi i = 4\pi i.\end{aligned}$$

19. By partial fractions,

$$\oint_C \frac{z-1}{z(z-i)(z-3i)} dz = \frac{1}{3} \oint_C \frac{1}{z} dz + \left(-\frac{1}{2} + \frac{1}{2}i \right) \oint_C \frac{1}{z-i} dz + \left(\frac{1}{6} - \frac{1}{2}i \right) \oint_C \frac{1}{z-3i} dz.$$

By Theorem 18.4 and (4) of Section 18.2,

$$\oint_C \frac{z-1}{z(z-i)(z-3i)} dz = 0 + \left(-\frac{1}{2} + \frac{1}{2}i \right) 2\pi i + 0 = \pi(-1-i).$$

20. By partial fractions,

$$\oint_C \frac{1}{z^3+2iz^2} dz = \frac{1}{4} \oint_C \frac{1}{z} dz - \frac{1}{2}i \oint_C \frac{1}{z^2} dz - \frac{1}{4} \oint_C \frac{1}{z+2i} dz.$$

By Theorem 18.4 and (4) of Section 18.2,

$$\oint_C \frac{1}{z^3+2iz^2} dz = \frac{1}{4} 2\pi i - \frac{1}{2}i(0) - \frac{1}{4}(0) = \frac{\pi}{2}i.$$

21. We have

$$\oint_C \frac{8z-3}{z^2-z} dz = \oint_{C_1} \frac{8z-3}{z^2-z} dz - \oint_{C_2} \frac{8z-3}{z^2-z} dz$$

where C_1 and C_2 are the closed portions of the curve C enclosing $z=0$ and $z=1$, respectively. By partial fractions, Theorem 18.4, and (4) of Section 18.2,

$$\begin{aligned}\oint_{C_1} \frac{8z-3}{z^2-z} dz &= 5 \oint_{C_1} \frac{1}{z-1} dz + 3 \oint_{C_1} \frac{1}{z} dz = 5(0) + 3(2\pi i) = 6\pi i \\ \oint_{C_2} \frac{8z-3}{z^2-z} dz &= 5 \oint_{C_2} \frac{1}{z-1} dz + 3 \oint_{C_2} \frac{1}{z} dz = 5(2\pi i) + 3(0) = 10\pi i.\end{aligned}$$

Thus

$$\oint_C \frac{8z-3}{z^2-z} dz = 6\pi i - 10\pi i = -4\pi i.$$

22. By choosing the more convenient contour C_1 defined by $|z-z_0|=r$ where r is small enough so that the circle C_1 lies entirely within C we can write

$$\oint_C \frac{1}{(z-z_0)^n} dz = \oint_{C_1} \frac{1}{(z-z_0)^n} dz.$$

Let $z-z_0=re^{it}$, $0 \leq t \leq 2\pi$ and $dz=ire^{it}dt$. Then for $n=1$:

$$\oint_{C_1} \frac{1}{z-z_0} dz = \int_0^{2\pi} \frac{1}{re^{it}} ire^{it} dt = i \int_0^{2\pi} dt = 2\pi i.$$

For $n \neq 1$:

$$\oint_{C_1} \frac{1}{(z-z_0)^n} dz = \frac{i}{r^{n-1}} \int_0^{2\pi} e^{(1-n)it} dt = \frac{i}{r^{n-1}} \frac{e^{(1-n)it}}{i(1-n)} \Big|_0^{2\pi} = \frac{1}{r^{n-1}(1-n)} [e^{2\pi(1-n)i} - 1] = 0$$

since $e^{2\pi(1-n)i} = 1$.

Exercises 18.2

23. Write
$$\oint_C \left(\frac{e^z}{z+3} - 3\bar{z} \right) dz = \oint_C \frac{e^z}{z+3} dz - 3 \oint_C \bar{z} dz.$$

By Theorem 18.4, $\oint_C \frac{e^z}{z+3} dz = 0$. However, since $\bar{z}$ is not analytic,

$$\oint_C \bar{z} dz = \int_0^{2\pi} e^{-it}(ie^{it} dt) = 2\pi i.$$

Thus

$$\oint_C \left(\frac{e^z}{z+3} - 3\bar{z} \right) dz = 0 - 3(2\pi i) = -6\pi i.$$

24. Write
$$\oint_C (z^2 + z + \operatorname{Re}(z)) dz = \oint_C (z^2 + z) dz + \oint_C \operatorname{Re}(z) dz.$$

By Theorem 18.4, $\oint_C (z^2 + z) dz = 0$. However, since $\operatorname{Re}(z) = x$ is not analytic,

$$\oint_C x dz = \oint_{C_1} x dz + \oint_{C_2} x dz + \oint_{C_3} x dz$$

where C_1 is $y = 0$, $0 \leq x \leq 1$, C_2 is $x = 1$, $0 \leq y \leq 2$, and C_3 is $y = 2x$, $0 \leq x \leq 1$. Thus,

$$\oint_C x dz = \int_0^1 x dx + i \int_0^2 dy + (1+2i) \int_1^0 x dx = \frac{1}{2} + 2i - \frac{1}{2}(1+2i) = i.$$

Exercises 18.3

1. (a) Choosing $x = 0$, $-1 \leq y \leq 1$ we have $z = iy$, $dz = i dy$. Thus

$$\int_C (4z - 1) dz = i \int_{-1}^1 (4iy - 1) dy = -2i.$$

(b)
$$\int_C (4z - 1) dz = \int_{-i}^i (4z - 1) dz = 2z^2 - z \Big|_{-i}^i = -2i$$

2. (a) Choosing the line $y = \frac{1}{3}x$, $0 \leq x \leq 3$ we have $z = x + \frac{1}{3}xi$, $dz = (1 + \frac{1}{3}i) dx$. Thus

$$\int_C e^z dz = \int_0^3 e^{(1+\frac{1}{3}i)x} \left(1 + \frac{1}{3}i \right) dx = e^{(1+\frac{1}{3}i)x} \Big|_0^3 = e^{3+i} - e^0 = (e^3 \cos 1 - 1) + ie^3 \sin 1.$$

(b)
$$\int_C e^z dz = \int_0^{3+i} e^z dz = e^z \Big|_0^{3+i} = e^{3+i} - e^0 = (e^3 \cos 1 - 1) + ie^3 \sin 1$$

3. The given integral is independent of the path. Thus

$$\int_C 2z dz = \int_{-2+7i}^{2-i} 2z dz = z^2 \Big|_{-2+7i}^{2-i} = 48 + 24i.$$

4. The given integral is independent of the path. Thus

$$\int_C 6z^2 dz = \int_2^{2-i} 6z^2 dz = z^3 \Big|_2^{2-i} = -15 - 24i.$$

5.
$$\int_0^{3+i} z^2 dz = \frac{1}{3} z^3 \Big|_0^{3+i} = 6 + \frac{26}{3} i$$

6. $\int_{-2i}^1 (3z^2 - 4z + 5i) dz = z^3 - 2z^2 + 5iz \Big|_{-2i}^1 = -19 - 3i$
7. $\int_{1-i}^{1+i} z^3 dz = \frac{1}{4} z^4 \Big|_{1-i}^{1+i} = 0$
8. $\int_{-3i}^{2i} (z^3 - z) dz = \frac{1}{4} z^4 - \frac{1}{2} z^2 \Big|_{-3i}^{2i} = \frac{123}{4}$
9. $\int_{-i/2}^{1-i} (2z+1)^2 dz = \frac{1}{6} (2z+1)^3 \Big|_{-i/2}^{1-i} = -\frac{7}{6} - \frac{22}{3}i$
10. $\int_1^i (iz+1)^3 dz = \frac{1}{4i} (iz+1)^4 \Big|_1^i = -i$
11. $\int_{i/2}^i e^{\pi z} dz = \frac{1}{\pi} e^{\pi z} \Big|_{i/2}^i = -\frac{1}{\pi} - \frac{1}{\pi}i$
12. $\int_{1-i}^{1+2i} ze^{z^2} dz = \frac{1}{2} e^{z^2} \Big|_{1-i}^{1+2i} = \frac{1}{2} [e^{-3+4i} - e^{-2i}] = \frac{1}{2} (e^{-3} \cos 4 - \cos 2) + \frac{1}{2} (e^{-3} \sin 4 + \sin 2)i = 0.1918 + 0.4358i$
13. $\int_{\pi}^{\pi+2i} \sin \frac{z}{2} dz = -2 \cos \frac{z}{2} \Big|_{\pi}^{\pi+2i} = -2 \left[\cos \left(\frac{\pi}{2} + i \right) - \cos \frac{\pi}{2} \right] = 2i \sin \frac{\pi}{2} \sinh 1 = 2.3504i$
14. $\int_{1-2i}^{\pi i} \cos z dz = \sin z \Big|_{1-2i}^{\pi i} = \sin \pi i - \sin(1-2i) = i \sinh \pi - [\sinh 1 \cosh 2 - i \cos 1 \sinh 2]$
 $= -\sin 1 \cosh 2 + i(\sinh \pi + \cos 1 \sinh 2) = -3.1658 + 13.5083i$
15. $\int_{\pi i}^{2\pi i} \cosh z dz = \sinh z \Big|_{\pi i}^{2\pi i} = \sinh 2\pi i - \sinh \pi i = i \sin 2\pi - i \sin \pi = 0$
16. $\int_i^{1+\frac{\pi}{2}i} \sinh 3z dz = \frac{1}{3} \cosh 3z \Big|_i^{1+\frac{\pi}{2}i} = \frac{1}{3} \left[\cosh \left(3 + \frac{3\pi}{2}i \right) - \cosh 3i \right]$
 $= \frac{1}{3} \left[\cosh 3 \cos \frac{3\pi}{2} + i \sinh 3 \sin \frac{3\pi}{2} - \cosh 3 \right] = -\frac{1}{3} \cosh 3 - \frac{1}{3} i \sinh 3 = 0.3300 - 3.3393i$
17. $\int_{-4i}^{4i} \frac{1}{z} dz = \operatorname{Ln} z \Big|_{-4i}^{4i} = \operatorname{Ln} 4i - \operatorname{Ln}(-4i) = \log_e 4 + \frac{\pi}{2}i - \left(\log_e 4 - \frac{\pi}{2}i \right) = \pi i$
18. $\int_{1+i}^{4+4i} \frac{1}{z} dz = \operatorname{Ln} z \Big|_{1+i}^{4+4i} = \operatorname{Ln}(4+4i) - \operatorname{Ln}(1+i) = \log_e 4\sqrt{2} + \frac{\pi}{4}i - \left(\log_e \sqrt{2} + \frac{\pi}{4}i \right) = \log_e 4 = 1.3863$
19. $\int_{-4i}^{4i} \frac{1}{z^2} dz = -\frac{1}{z} \Big|_{-4i}^{4i} = -\left[\frac{1}{4i} - \left(\frac{1}{-4i} \right) \right] = \frac{1}{2}i$
20. $\int_{1-i}^{1+\sqrt{3}i} \left(\frac{1}{z} + \frac{1}{z^2} \right) dz = \operatorname{Ln} z - \frac{1}{z} \Big|_{1-i}^{1+\sqrt{3}i} = \log_e 2 + \frac{\pi}{3}i - \frac{1}{1+\sqrt{3}i} - \left(\log_e \sqrt{2} - \frac{\pi}{4}i - \frac{1}{1-i} \right)$
 $= \log_e \sqrt{2} + \frac{1}{4} + i \left(\frac{7\pi}{12} + \frac{\sqrt{3}}{4} + \frac{1}{2} \right) = 0.5966 + 2.7656i$
21. Integration by parts gives

$$\int e^z \cos z dz = \frac{1}{2} e^z (\cos z + \sin z) + C$$

Exercises 18.3

and so

$$\begin{aligned}\int_{\pi}^i e^z \cos z \, dz &= \frac{1}{2} e^z (\cos z + \sin z) \Big|_{\pi}^i = \frac{1}{2} [e^i (\cos i + \sin i) - e^{\pi} (\cos \pi + \sin \pi)] \\ &= \frac{1}{2} [(\cos 1 \cosh 1 - \sin 1 \sinh 1 + e^{\pi}) + i(\cos 1 \sinh 1 + \sin 1 \cosh 1)] = 11.4928 + 0.9667i.\end{aligned}$$

22. Integration by parts gives

$$\int z \sin z \, dz = -z \cos z + \sin z + C$$

and so

$$\int_0^i z \sin z \, dz = -z \cos z + \sin z \Big|_0^i = -i \cos i + \sin i = -i \cosh 1 + i \sinh 1 = -0.3679i.$$

23. Integration by parts gives

$$\int z e^z \, dz = z e^z - e^z + C$$

and so

$$\int_i^{1+i} z e^z \, dz = e^z (z-1) \Big|_i^{1+i} = i e^{1+i} + e^i (1-i) = (\cos 1 + \sin 1 - e \sin 1) + i(\sin 1 - \cos 1 + e \cos 1) = -0.9056 + 1.7699i.$$

24. Integration by parts gives

$$\int z^2 e^z \, dz = z^2 e^z - 2z e^z + 2e^z + C$$

and so

$$\int_0^{\pi i} z^2 e^z \, dz = e^z (z^2 - 2z + 2) \Big|_0^{\pi i} = e^{\pi i} (-\pi^2 - 2\pi i + 2) - 2 = \pi^2 - 4 + 2\pi i.$$

Exercises 18.4

1. By Theorem 18.9, with $f(z) = 4$,

$$\oint_C \frac{4}{z-3i} \, dz = 2\pi i \cdot 4 = 8\pi i.$$

2. By Theorem 18.10 with $f(z) = z^2$ and $f'(z) = 2z$,

$$\oint_C \frac{z^2}{(z-3i)^2} \, dz = \frac{2\pi i}{1!} 2(3i) = -12\pi.$$

3. By Theorem 18.9 with $f(z) = e^z$,

$$\oint_C \frac{e^z}{z-\pi i} \, dz = 2\pi i e^{\pi i} = -2\pi i.$$

4. By Theorem 18.9 with $f(z) = 1 + 2e^z$,

$$\oint_C \frac{1+2e^z}{z} \, dz = 2\pi i (1 + 2e^0) = 6\pi i.$$

5. By Theorem 18.9 with $f(z) = z^2 - 3z + 4i$,

$$\oint_C \frac{z^2 - 3z + 4i}{z - (-2i)} \, dz = 2\pi i (-4 + 6i + 4i) = -\pi(20 + 8i).$$

6. By Theorem 18.9 with $f(z) = \frac{1}{3} \cos z$,

$$\oint_C \frac{\frac{1}{3} \cos z}{z - \frac{\pi}{3}} dz = 2\pi i \left(\frac{1}{3} \cos \frac{\pi}{3} \right) = \frac{\pi}{3} i.$$

7. (a) By Theorem 18.9 with $f(z) = \frac{z^2}{z + 2i}$,

$$\oint_C \frac{\frac{z^2}{z + 2i}}{z - 2i} dz = 2\pi i \left(-\frac{4}{4i} \right) = -2\pi.$$

- (b) By Theorem 18.9 with $f(z) = \frac{z^2}{z - 2i}$,

$$\oint_C \frac{\frac{z^2}{z - 2i}}{z - (-2i)} dz = 2\pi i \left(\frac{-4}{-4i} \right) = 2\pi.$$

8. (a) By Theorem 18.9 with $f(z) = \frac{z^2 + 3z + 2i}{z + 4}$,

$$\oint_C \frac{\frac{z^2 + 3z + 2i}{z + 4}}{z - 1} dz = 2\pi i \left(\frac{4 + 2i}{5} \right) = \pi \left(-\frac{4}{5} + \frac{8}{5} i \right).$$

- (b) By Theorem 18.9 with $f(z) = \frac{z^2 + 3z + 2i}{z - 1}$,

$$\oint_C \frac{\frac{z^2 + 3z + 2i}{z - 1}}{z - (-4)} dz = 2\pi i \left(\frac{4 + 2i}{-5} \right) = \pi \left(\frac{4}{5} - \frac{8}{5} i \right).$$

9. By Theorem 18.9 with $f(z) = \frac{z^2 + 4}{z - i}$,

$$\oint_C \frac{\frac{z^2 + 4}{z - i}}{z - 4i} dz = 2\pi i \left(-\frac{12}{3i} \right) = -8\pi.$$

10. By Theorem 18.9 with $f(z) = \frac{\sin z}{z + \pi i}$,

$$\oint_C \frac{\frac{\sin z}{z + \pi i}}{z - \pi i} dz = 2\pi i \left(\frac{\sin \pi i}{2\pi i} \right) = i \sinh \pi.$$

11. By Theorem 18.10 with $f(z) = e^{z^2}$, $f'(z) = 2ze^{z^2}$, and $f''(z) = 4z^2e^{z^2} + 2e^{z^2}$,

$$\oint_C \frac{e^{z^2}}{(z - i)^3} dz = \frac{2\pi i}{2!} [-4e^{-1} + 2e^{-1}] = -2\pi e^{-1} i.$$

12. By Theorem 18.10 with $f(z) = z$, $f'(z) = 1$, $f''(z) = 0$, and $f'''(z) = 0$,

$$\oint_C \frac{z}{(z - (-i))^4} dz = \frac{2\pi i}{3!} (0) = 0.$$

Exercises 18.4

13. By Theorem 18.10 with $f(z) = \cos 2z$, $f'(z) = -2 \sin 2z$, $f''(z) = -4 \cos 2z$, $f'''(z) = 8 \sin 2z$, $f^{(4)}(z) = 16 \cos 2z$,

$$\oint_C \frac{\cos 2z}{z^5} dz = \frac{2\pi i}{4!} (16 \cos 0) = \frac{4\pi}{3} i.$$

14. By Theorem 18.10 with $f(z) = e^{-z} \sin z$, $f'(z) = e^{-z} \cos z - e^{-z} \sin z$, and $f''(z) = -2e^{-z} \cos z$,

$$\oint_C \frac{e^{-z} \sin z}{z^3} dz = \frac{2\pi i}{2!} (-2e^0 \cos 0) = -2\pi i.$$

15. (a) By Theorem 18.9 with $f(z) = \frac{2z+5}{z-2}$,

$$\oint_C \frac{\frac{2z+5}{z-2}}{z} dz = 2\pi i \left(-\frac{5}{2} \right) = -5\pi i.$$

- (b) Since the circle $|z - (-1)| = 2$ encloses only $z = 0$, the value of the integral is the same as in part (a).

- (c) From Theorem 18.9 with $f(z) = \frac{2z+5}{z}$,

$$\oint_C \frac{\frac{2z+5}{z}}{z-2} dz = 2\pi i \left(\frac{9}{2} \right) = 9\pi i.$$

- (d) Since the circle $|z - (-2i)| = 1$ encloses neither $z = 0$ nor $z = 2$ it follows from the Cauchy-Goursat Theorem, Theorem 18.4, that

$$\oint_C \frac{2z+5}{z(z-2)} dz = 0.$$

16. By partial fractions,

$$\oint_C \frac{z}{(z-1)(z-2)} dz = 2 \oint_C \frac{dz}{z-2} - \oint_C \frac{dz}{z-1}.$$

- (a) By the Cauchy-Goursat Theorem, Theorem 18.4,

$$\oint_C \frac{z}{(z-1)(z-2)} dz = 0.$$

- (b) As in part (a), the integral is 0.

- (c) By Theorem 18.4, $\oint_C \frac{dz}{z-2} = 0$ whereas by Theorem 18.9, $\oint_C \frac{dz}{z-1} = 2\pi i$. Thus

$$\oint_C \frac{z}{(z-1)(z-2)} dz = -2\pi i.$$

- (d) By Theorem 18.9, $\oint_C \frac{dz}{z-1} = 2\pi i$ and $\oint_C \frac{dz}{z-2} = 2\pi i$. Thus

$$\oint_C \frac{z}{(z-1)(z-2)} dz = 2(2\pi i) - 2\pi i = 3\pi i.$$

17. (a) By Theorem 18.10 with $f(z) = \frac{z+2}{z-1-i}$ and $f'(z) = \frac{-3-i}{(z-1-i)^2}$,

$$\oint_C \frac{\frac{z+2}{z-1-i}}{z^2} dz = \frac{2\pi i}{1!} \left(\frac{-3-i}{(-1-i)^2} \right) = -\pi(3+i).$$

(b) By Theorem 18.9 with $f(z) = \frac{z+2}{z^2}$,

$$\oint_C \frac{\frac{z+2}{z^2}}{z-(1+i)} dz = 2\pi i \left(\frac{3+i}{(1+i)^2} \right) = \pi(3+i).$$

18. (a) By Theorem 18.10 with $f(z) = \frac{1}{z-4}$, $f'(z) = -\frac{1}{(z-4)^2}$, and $f''(z) = \frac{2}{(z-4)^3}$,

$$\oint_C \frac{\frac{1}{z-4}}{z^3} dz = \frac{2\pi i}{2!} \left(\frac{2}{-64} \right) = -\frac{\pi}{32} i.$$

(b) By the Cauchy-Goursat Theorem, Theorem 18.4,

$$\oint_C \frac{1}{z^3(z-4)} dz = 0.$$

19. By writing $\oint_C \left(\frac{e^{2iz}}{z^4} - \frac{z^4}{(z-i)^3} \right) dz = \oint_C \frac{e^{2iz}}{z^4} dz - \oint_C \frac{z^4}{(z-i)^3} dz$

we can apply Theorem 18.10 to each integral:

$$\oint_C \frac{e^{2iz}}{z^4} dz = \frac{2\pi i}{3!}(-8i) = \frac{8\pi}{3}, \quad \oint_C \frac{z^4}{(z-i)^3} dz = \frac{2\pi i}{2!}(-12) = -12\pi i.$$

Thus

$$\oint_C \left(\frac{e^{2iz}}{z^4} - \frac{z^4}{(z-i)^3} \right) dz = \pi \left(\frac{8}{3} + 12i \right).$$

20. By writing $\oint_C \left(\frac{\cosh z}{(z-\pi)^3} - \frac{\sin^2 z}{(2z-\pi)^3} \right) dz = \oint_C \frac{\cosh z}{(z-\pi)^3} dz - \oint_C \frac{\frac{1}{8}\sin^2 z}{(z-\frac{\pi}{2})^3} dz$

we apply Theorem 18.4 to the first integral and Theorem 18.10 to the second:

$$\oint_C \frac{\cosh z}{(z-\pi)^3} dz = 0, \quad \oint_C \frac{\frac{1}{8}\sin^2 z}{(z-\frac{\pi}{2})^3} dz = \frac{2\pi i}{2!} \left(-\frac{1}{4} \sin^2 \frac{\pi}{2} \right) = -\frac{\pi}{4} i.$$

Thus

$$\oint_C \left(\frac{\cosh z}{(z-\pi)^3} - \frac{\sin^2 z}{(2z-\pi)^3} \right) dz = \frac{\pi}{4} i.$$

21. We have $\oint_C \frac{1}{z^3(z-1)^2} dz = \oint_{C_1} \frac{\frac{1}{(z-1)^2}}{z^3} dz + \oint_{C_2} \frac{\frac{1}{z^3}}{(z-1)^2} dz$

where C_1 and C_2 are the circles $|z| = 1/3$ and $|z-1| = 1/3$, respectively. By Theorem 18.10,

$$\oint_{C_1} \frac{\frac{1}{(z-1)^2}}{z^3} dz = \frac{2\pi i}{2!}(6) = 6\pi i, \quad \oint_{C_2} \frac{\frac{1}{z^3}}{(z-1)^2} dz = \frac{2\pi i}{1!}(-3) = -6\pi i.$$

Thus

$$\oint_C \frac{1}{z^3(z-1)^2} dz = 6\pi i - 6\pi i = 0.$$

22. We have $\oint_C \frac{1}{z^2(z^2+1)} dz = \oint_{C_1} \frac{\frac{1}{z^2(z+i)}}{z-i} dz + \oint_{C_2} \frac{\frac{1}{z^2+1}}{z^2} dz$

Exercises 18.4

where C_1 and C_2 are the circles $|z - i| = 1/3$ and $|z| = 1/8$, respectively. By Theorems 18.9 and 18.10,

$$\oint_{C_1} \frac{1}{\frac{z^2(z+i)}{z-i}} dz = 2\pi i \left(\frac{1}{-2i} \right) = -\pi, \quad \oint_{C_2} \frac{1}{\frac{z^2+1}{z^2}} dz = \frac{2\pi i}{1!}(0) = 0.$$

Thus

$$\oint_C \frac{1}{z^2(z^2+1)} dz = -\pi.$$

23. We have

$$\oint_C \frac{3z+1}{z(z-2)^2} dz = \oint_{C_1} \frac{\frac{3z+1}{z}}{(z-2)^2} dz - \oint_{C_2} \frac{\frac{3z+1}{(z-2)^2}}{z} dz$$

where C_1 and C_2 are the closed portions of the curve C enclosing $z = 2$ and $z = 0$, respectively. By Theorems 18.10 and 18.9,

$$\oint_{C_1} \frac{\frac{3z+1}{z}}{(z-2)^2} dz = \frac{2\pi i}{1!} \left(-\frac{1}{4} \right) = -\frac{\pi}{2} i, \quad \oint_{C_2} \frac{\frac{3z+1}{(z-2)^2}}{z} dz = 2\pi i \left(\frac{1}{4} \right) = \frac{\pi}{2} i.$$

Thus

$$\oint_C \frac{3z+1}{z(z-2)^2} dz = -\frac{\pi}{2} i - \frac{\pi}{2} i = -\pi i.$$

24. We have

$$\oint_C \frac{e^{iz}}{(z^2+1)^2} dz = \oint_{C_1} \frac{\frac{e^{iz}}{(z+i)^2}}{(z-i)^2} dz - \oint_{C_2} \frac{\frac{e^{iz}}{(z-i)^2}}{(z-(-i))^2} dz$$

where C_1 and C_2 are the closed portions of the curve C enclosing $z = i$ and $z = -i$, respectively. By Theorem 18.10,

$$\oint_{C_1} \frac{\frac{e^{iz}}{(z+i)^2}}{(z-i)^2} dz = \frac{2\pi i}{1!} \left(\frac{-4e^{-1}}{-8i} \right) = \pi e^{-1}, \quad \oint_{C_2} \frac{\frac{e^{iz}}{(z-i)^2}}{(z-(-i))^2} dz = \frac{2\pi i}{1!} \left(\frac{0}{8i} \right) = 0.$$

Thus

$$\oint_C \frac{e^{iz}}{(z^2+1)^2} dz = \pi e^{-1}.$$

Chapter 18 Review Exercises

1. True
2. False
3. True
4. True
5. 0
6. $\pi(-16 + 8i)$
7. $\pi(6\pi - i)$
8. a constant function
9. True (Use partial fractions and write the given integral as two integrals.)
10. True
11. integer not equal to -1 ; -1
12. 12π

13. Since $f(z) = z$ is entire, $\int_C (x + iy) dz$ is independent of the path C . Thus

$$\oint_C (x + iy) dz = \int_{-4}^3 z dz = \frac{z^2}{2} \Big|_{-4}^3 = -\frac{7}{2}.$$

14. We have $\int_C (x - iy) dz = \int_{C_1} (x - iy) dz + \int_{C_2} (x - iy) dz + \int_{C_3} (x - iy) dz$

On C_1 , $x = 4$, $0 \leq y \leq 2$, $z = 4 + iy$, $dz = i dy$,

$$\int_{C_1} (4 - iy)i dy = i \int_0^2 (4 - iy) dy = i \left(4y - \frac{i}{2}y^2 \right) \Big|_0^2 = 2 + 8i.$$

On C_2 , $y = 2$, $-4 \leq x \leq 3$, $z = x + 2i$, $dz = dx$,

$$\int_{C_2} (x - 2i) dx = \int_{-4}^3 (x - 2i) dx = \frac{1}{2}x^2 - 2ix \Big|_{-4}^3 = -\frac{7}{2} - 14i.$$

On C_3 , $x = 3$, $0 \leq y \leq 2$, $z = 3 + iy$, $dz = i dy$,

$$\int_{C_3} (3 - iy)i dy = i \int_2^0 (3 - iy) dy = i \left(3y - \frac{i}{2}y^2 \right) \Big|_2^0 = -2 - 6i.$$

Thus

$$\int_C (x - iy) dz = 2 + 8i - \frac{7}{2} - 14i - 2 - 6i = -\frac{7}{2} - 12i.$$

15. $\int_C |z^2| dz = \int_0^2 (t^4 + t^2) dt + 2i \int_0^2 (t^5 + t^3) dt = \frac{136}{15} + \frac{88}{3}i$

16. $\int_C e^{\pi z} dz = \frac{1}{\pi} \int_i^{1+i} e^{\pi z} (\pi dz) = \frac{1}{\pi} e^{\pi z} \Big|_i^{1+i} = \frac{1}{\pi} (1 - e^{\pi})$

17. By the Cauchy-Goursat Theorem, Theorem 18.4, $\oint_C e^{\pi z} dz = 0$.

18. $\int_{3i}^{1-i} (4z - 6) dz = 2z^2 - 6z \Big|_{3i}^{1-i} = 12 + 20i$

19. $\int_C \sin z dz = \int_1^{1+4i} \sin z dz = -\cos z \Big|_1^{1+4i} = \cos 1 - \cos(1 + 4i) = -14.2144 + 22.9637i$

20. $\int_C (4z^3 + 3z^2 + 2z + 1) dz = \int_0^{2i} (4z^3 + 3z^2 + 2z + 1) dz = z^4 + z^3 + z^2 + z \Big|_0^{2i} = 12 - 6i$

21. On $|z| = 1$, let $z = e^{it}$, $dz = ie^{it} dt$, so that

$$\oint_C (z^{-2} + z^{-1} + z + z^2) dz = i \int_0^{2\pi} (e^{-2it} + e^{-it} + e^{it} + e^{2it}) e^{it} dt = -e^{-it} + it + \frac{1}{2}e^{2it} + \frac{1}{3}e^{3it} \Big|_0^{2\pi} = 2\pi i.$$

22. By partial-fractions and Theorem 18.9,

$$\oint_C \frac{3z + 4}{z^2 - 1} dz = \frac{7}{2} \oint_C \frac{1}{z - 1} dz - \frac{1}{2} \oint_C \frac{1}{z - (-1)} dz = \frac{7}{2}(2\pi i) - \frac{1}{2}(2\pi i) = 6\pi i.$$

23. By Theorem 18.10 with $f(z) = e^{-2z}$, $f'(z) = -2e^{-2z}$, $f''(z) = 4e^{-2z}$, and $f'''(z) = -8e^{-2z}$,

$$\oint_C \frac{e^{-2z}}{z^4} dz = \frac{2\pi i}{3!} (-8) = -\frac{8\pi}{3} i.$$

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24. By Theorem 18.10 with $f(z) = \frac{\cos z}{z-1}$ and $f'(z) = \frac{\sin z - \cos z - z \sin z}{(z-1)^2}$,

$$\oint_C \frac{\frac{\cos z}{z-1}}{z^2} dz = \frac{2\pi i}{1!} \left(\frac{-1}{1} \right) = -2\pi i.$$

25. By Theorem 18.9 with $f(z) = \frac{1}{2(z+3)}$,

$$\oint_C \frac{\frac{1}{2(z+3)}}{(z - (-1/2))} dz = 2\pi i \left(\frac{1}{5} \right) = \frac{2\pi}{5} i.$$

26. Since the function $f(z) = z/\sin z$ is analytic within and on the given simple closed contour C , it follows from the Cauchy-Goursat Theorem, Theorem 18.4, that

$$\oint_C z \csc z dz = 0.$$

27. Using the principle of deformation of contours we choose C to be the more convenient circular contour $|z+i| = \frac{1}{4}$. On this circle $z = -i + \frac{1}{4}e^{it}$ and $dz = \frac{1}{4}ie^{it} dt$. Thus

$$\oint_C \frac{z}{z+i} dz = i \int_0^{2\pi} \left(\frac{1}{4}e^{it} - i \right) dt = 2\pi.$$

28. (a) By Theorem 18.9 with $f(z) = \frac{e^{i\pi z}}{2(z-2)}$,

$$\oint_C \frac{\frac{e^{i\pi z}}{2(z-2)}}{z-1/2} dz = 2\pi i \left(\frac{e^{i\pi/2}}{-3} \right) = \frac{2\pi}{3}.$$

- (b) By Theorem 18.9 with $f(z) = \frac{e^{i\pi z}}{2z-1}$,

$$\oint_C \frac{\frac{e^{i\pi z}}{2z-1}}{z-2} dz = 2\pi i \left(\frac{e^{2\pi i}}{3} \right) = \frac{2\pi}{3} i.$$

- (c) By the Cauchy-Goursat Theorem, Theorem 18.4,

$$\oint_C \frac{e^{i\pi z}}{2z^2 - 5z + 2} dz = 0.$$

29. For $f(z) = z^n g(z)$ we have $f'(z) = z^n g'(z) + n z^{n-1} g(z)$ and so

$$\frac{f'(z)}{f(z)} = \frac{z^n g'(z) + n z^{n-1} g(z)}{z^n g(z)} = \frac{g'(z)}{g(z)} + \frac{n}{z}.$$

Thus by Theorem 18.4 and (4) of Section 18.2,

$$\oint_C \frac{f'(z)}{f(z)} dz = \oint_C \frac{g'(z)}{g(z)} dz + n \oint_C \frac{1}{z} dz = 0 + n(2\pi i) = 2n\pi i.$$

30. We have

$$\left| \int_C \operatorname{Ln}(z+1) dz \right| \leq |\max \text{ of } \operatorname{Ln}(z+1) \text{ on } C| \cdot 2,$$

where 2 is the length of the line segment. Now

$$|\operatorname{Ln}(z+1)| \leq |\log_e(z+1)| + |\operatorname{Arg}(z+1)|.$$

But $\max \operatorname{Arg}(z+1) = \pi/4$ when $z = i$ and $\max|z+1| = \sqrt{10}$ when $z = 2+i$. Thus,

$$\left| \int_C \operatorname{Ln}(z+1) dz \right| \leq \left(\frac{1}{2} \log_e 10 + \frac{\pi}{4} \right) 2 = \log_e 10 + \frac{\pi}{2}.$$

19 Series and Residues

Exercises 19.1

1. $5i, -5, -5i, 5, 5i$
2. $2 - i, 1, 2 + i, 3, 2 - i$
3. $0, 2, 0, 2, 0$
4. $1 + i, 2i, -2 + 2i, -4, -4 - 4i$
5. Converges. To see this write the general term as $\frac{3i + 2/n}{1 + i}$.
6. Converges. To see this write the general term as $\left(\frac{2}{5}\right)^n \frac{1 + n2^{-n}i}{1 + 3n5^{-n}i}$.
7. Converges. To see this write the general term as $\frac{(i + 2/n)^2}{i}$.
8. Diverges. To see this consider the term $\frac{n}{n+1}i^n$ and take n to be an odd positive integer.
9. Diverges. To see this write the general term as $\sqrt{n} \left(1 + \frac{1}{\sqrt{n}}i^n\right)$.
10. Converges. The real part of the general term converges to 0 and the imaginary part of the general term converges to π .
11. $\operatorname{Re}(z_n) = \frac{8n^2 + n}{4n^2 + 1} \rightarrow 2$ as $n \rightarrow \infty$, and $\operatorname{Im}(z_n) = \frac{6n^2 - 4n}{4n^2 + 1} \rightarrow \frac{3}{2}$ as $n \rightarrow \infty$.
12. Write $z_n = \left(\frac{1}{4} + \frac{1}{4}i\right)^n$ in polar form as $z_n = \left(\frac{\sqrt{2}}{4}\right)^n \cos n\theta + i \left(\frac{\sqrt{2}}{4}\right)^n \sin n\theta$. Now

$$\operatorname{Re}(z_n) = \left(\frac{\sqrt{2}}{4}\right)^n \cos n\theta \rightarrow 0 \text{ as } n \rightarrow \infty \quad \text{and} \quad \operatorname{Im}(z_n) = \left(\frac{\sqrt{2}}{4}\right)^n \sin n\theta \rightarrow 0 \text{ as } n \rightarrow \infty$$
 since $\sqrt{2}/4 < 1$.
13. $S_n = \frac{1}{1+2i} - \frac{1}{2+2i} + \frac{1}{2+2i} - \frac{1}{3+2i} + \frac{1}{3+2i} - \frac{1}{4+2i} + \cdots + \frac{1}{n+2i} - \frac{1}{n+1+2i} = \frac{1}{1+2i} - \frac{1}{n+1+2i}$
 Thus, $\lim_{n \rightarrow \infty} S_n = \frac{1}{1+2i} = \frac{1}{5} - \frac{2}{5}i$.
14. By partial fractions, $\frac{i}{k(k+1)} = \frac{i}{k} - \frac{i}{k+1}$ and so

$$S_n = i - \frac{i}{2} + \frac{i}{2} - \frac{i}{3} + \frac{i}{3} - \frac{i}{4} + \cdots + \frac{i}{n} - \frac{i}{n+1} = i - \frac{i}{n+1}.$$
 Thus $\lim_{n \rightarrow \infty} S_n = i$.
15. We identify $a = 1$ and $z = 1 - i$. Since $|z| = \sqrt{2} > 1$ the series is divergent.
16. We identify $a = 4i$ and $z = 1/3$. Since $|z| = 1/3 < 1$ the series converges to

$$\frac{4i}{1 - 1/3} = 6i.$$
17. We identify $a = i/2$ and $z = i/2$. Since $|z| = 1/2 < 1$ the series converges to

$$\frac{i/2}{1 - i/2} = -\frac{1}{5} + \frac{2}{5}i.$$

18. We identify $a = 1/2$ and $z = i$. Since $|z| = 1$ the series is divergent.

19. We identify $a = 3$ and $z = 2/(1 + 2i)$. Since $|z| = 2/\sqrt{5} < 1$ the series converges to

$$\frac{3}{1 - \frac{2}{1 + 2i}} = \frac{9}{5} - \frac{12}{5}i.$$

20. We identify $a = -1/(1 + i)$ and $z = i/(1 + i)$. Since $|z| = 1/\sqrt{2} < 1$ the series converges to

$$\frac{-\frac{1}{1+i}}{1 - \frac{i}{1+i}} = -1.$$

21. From

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{1}{(1-2i)^{n+2}}}{\frac{1}{(1-2i)^{n+1}}} \right| = \frac{1}{|1-2i|} = \frac{1}{\sqrt{5}}$$

we see that the radius of convergence is $R = \sqrt{5}$. The circle of convergence is $|z - 2i| = \sqrt{5}$.

22. From

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{1}{n+1} \left(\frac{i}{1+i} \right)^{n+1}}{\frac{1}{n} \left(\frac{i}{1+i} \right)^n} \right| = \lim_{n \rightarrow \infty} \frac{n}{n+1} \left| \frac{i}{1+i} \right| = \frac{1}{\sqrt{2}}$$

we see that the radius of convergence is $R = \sqrt{2}$. The circle of convergence is $|z| = \sqrt{2}$.

23. From

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{(-1)^{n+1}}{(n+1)2^{n+1}}}{\frac{(-1)^n}{n2^n}} \right| = \lim_{n \rightarrow \infty} \frac{n}{2(n+1)} = \frac{1}{2}$$

we see that the radius of convergence is $R = 2$. The circle of convergence is $|z - 1 - i| = 2$.

24. From

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{1}{(n+1)^2(3+4i)^{n+1}}}{\frac{1}{n^2(3+4i)^n}} \right| = \lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^2 \frac{1}{|3+4i|} = \frac{1}{5}$$

we see that the radius of convergence is $R = 5$. The circle of convergence is $|z + 3i| = 5$.

25. From

$$\lim_{n \rightarrow \infty} \sqrt[n]{|1 + 3i|^n} = |1 + 3i| = \sqrt{10}$$

we see that the radius of convergence is $R = 1/\sqrt{10}$. The circle of convergence is $|z - i| = 1/\sqrt{10}$.

26. From

$$\lim_{n \rightarrow \infty} \sqrt[n]{\left| \frac{1}{n^n} \right|} = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

we see that the radius of convergence is ∞ . The power series with center 0 converges absolutely for all z .

27. From

$$\lim_{n \rightarrow \infty} \sqrt[n]{\left| \frac{1}{5^{2n}} \right|} = \lim_{n \rightarrow \infty} \frac{1}{25} = \frac{1}{25}$$

we see that the radius of convergence is $R = 25$. The circle of convergence is $|z - 4 - 3i| = 25$.

Exercises 19.1

28. From
$$\lim_{n \rightarrow \infty} \sqrt[n]{\left| (-1)^n \left(\frac{1+2i}{2} \right)^n \right|} = \lim_{n \rightarrow \infty} \left| \frac{1+2i}{2} \right| = \frac{\sqrt{5}}{2}$$

we see that the radius of convergence is $R = 2/\sqrt{5}$. The circle of convergence is $|z + 2i| = 2/\sqrt{5}$.

29. The circle of convergence is $|z - i| = 2$. Since the series of absolute values

$$\sum_{k=1}^{\infty} \left| \frac{(z-1)^k}{k2^k} \right| = \sum_{k=1}^{\infty} \frac{|z-1|^k}{k2^k} = \sum_{k=1}^{\infty} \frac{2^k}{k2^k} = \sum_{k=1}^{\infty} \frac{1}{k}$$

is the divergent harmonic series. But $z = -2 + i$ is on the circle of convergence and $(z - i)^k = (-2)^k$. The series

$$\sum_{k=1}^{\infty} \frac{(-2)^k}{k2^k} = \sum_{k=1}^{\infty} \frac{(-1)^k}{k}$$

is convergent.

30. (a) The circle of convergence is $|z| = 1$. Since the series of absolute values

$$\sum_{k=1}^{\infty} \left| \frac{z^k}{k^2} \right| = \sum_{k=1}^{\infty} \frac{|z|^k}{k^2} = \sum_{k=1}^{\infty} \frac{1}{k^2}$$

converges, the given series is absolutely convergent for every z on $|z| = 1$. Since absolute convergence implies convergence, the given series converges for all z on $|z| = 1$.

(b) The circle of convergence is $|z| = 1$. On the circle, $n|z|^n \rightarrow \infty$ as $n \rightarrow \infty$. This implies $nz^n \not\rightarrow 0$ as $n \rightarrow \infty$. Thus by Theorem 19.3 the series is divergent for every z on the circle $|z| = 1$.

Exercises 19.2

1. $\frac{z}{1+z} = z[1 - z + z^2 - z^3 + \cdots] = z - z^2 + z^3 - z^4 + \cdots = \sum_{k=1}^{\infty} (-1)^{k+1} z^k; \quad R = 1$

2. $\frac{1}{4-2z} = \frac{1}{4} \cdot \left[1 + \frac{z}{2} + \frac{z^2}{2^2} + \frac{z^3}{2^3} + \cdots \right] = \frac{1}{4} \sum_{k=0}^{\infty} \frac{z^k}{2^k}; \quad R = 2$

3. Differentiating $\frac{1}{1+2z} = 1 - 2z + 2^2z^2 - 2^3z^3 + \cdots$ gives $\frac{-2}{(1+2z)^2} = -2 + 2 \cdot 2^2z - 3 \cdot 2^3z^2 + \cdots$. Thus

$$\frac{1}{(1+2z)} = 1 - 2 \cdot (2z) + 3 \cdot (2z)^2 - \cdots = \sum_{k=1}^{\infty} (-1)^{k-1} k (2z)^{k-1}. \quad R = 1/2$$

4. Using the binomial series gives

$$\frac{z}{(1-z)} = z \left[1 + 3z + \frac{3 \cdot 4}{2!} z^2 + \frac{3 \cdot 4 \cdot 5}{3!} z^3 + \cdots \right] = z + 3z^2 + \frac{3 \cdot 4}{2!} z^3 + \frac{3 \cdot 4 \cdot 5}{3!} z^4 + \cdots. \quad R = 1$$

5. Replacing z in $e^z = \sum_{k=0}^{\infty} \frac{z^k}{k!}$ by $-2z$ gives $e^{-2z} = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} (2z)^k. \quad R = \infty$

6. Replacing z in $e^z = \sum_{k=0}^{\infty} \frac{z^k}{k!}$ by $-z^2$ and multiplying the result by z gives $ze^{-z^2} = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} z^{2k+1}. \quad R = \infty$

7. Subtracting the series for e^z and e^{-z} gives $\sinh z = \frac{1}{2}(e^z - e^{-z}) = \sum_{k=0}^{\infty} \frac{z^{2k+1}}{(2k+1)!}. \quad R = \infty$

8. Adding the series for e^z and e^{-z} gives $\cosh z = \frac{1}{2}(e^z + e^{-z}) = \sum_{k=0}^{\infty} \frac{z^{2k}}{(2k)!}$. $R = \infty$

9. Replacing z in $\cos z = \sum_{k=0}^{\infty} (-1)^k \frac{z^{2k}}{(2k)!}$ by $z/2$ gives $\cos \frac{z}{2} = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} \left(\frac{z}{2}\right)^{2k}$. $R = \infty$

10. Replacing z in $\sin z = \sum_{k=0}^{\infty} (-1)^k \frac{z^{2k+1}}{(2k+1)!}$ by $3z$ gives $\sin 3z = \sum_{k=0}^{\infty} (-1)^k \frac{(3z)^{2k+1}}{(2k+1)!}$. $R = \infty$

11. Replacing z in $\sin z = \sum_{k=0}^{\infty} (-1)^k \frac{z^{2k+1}}{(2k+1)!}$ by z^2 gives $\sin z^2 = \sum_{k=0}^{\infty} (-1)^k \frac{z^{4k+2}}{(2k+1)!}$. $R = \infty$

12. Using the identity $\cos z = \frac{1}{2}(1 + \cos 2z)$ and the series $\cos z = \sum_{k=0}^{\infty} (-1)^k \frac{z^{2k}}{(2k)!}$ gives

$$\cos^2 z = \frac{1}{2} + \frac{1}{2} \sum_{k=0}^{\infty} (-1)^k \frac{(2z)^{2k}}{(2k)!} = 1 + \sum_{k=1}^{\infty} (-1)^k \frac{2^{2k-1}}{(2k)!} z^{2k}. R = \infty$$

13. Using (6) of Section 19.1,

$$\frac{1}{z} = \frac{1}{1 + (z-1)} = 1 - (z-1) + (z-1)^2 - (z-1)^3 + \cdots = \sum_{k=0}^{\infty} (-1)^k (z-1)^k. R = 1$$

14. Using (6) of Section 19.1,

$$\begin{aligned} \frac{1}{z} &= \frac{1}{1+i+(z-1-i)} = \frac{1}{1+i} \cdot \frac{1}{1+\frac{z-1-i}{1+i}} = \frac{1}{1+i} \left[1 - \frac{(z-1-i)}{1+i} + \frac{(z-1-i)^2}{(1+i)^2} - \frac{(z-1-i)^3}{(1+i)^3} + \cdots \right] \\ &= \frac{1}{1+i} - \frac{(z-1-i)}{(1+i)^2} + \frac{(z-1-i)^2}{(1+i)^3} - \frac{(z-1-i)^3}{(1+i)^4} + \cdots = \sum_{k=0}^{\infty} (-1)^k \frac{(z-1-i)^k}{(1+i)^{k+1}}. R = \sqrt{2} \end{aligned}$$

15. Using (5) of Section 19.1,

$$\begin{aligned} \frac{1}{3-z} &= \frac{1}{3-2i-(z-2i)} = \frac{1}{3-2i} \cdot \frac{1}{1-\frac{z-2i}{3-2i}} \\ &= \frac{1}{3-2i} \left[1 + \frac{z-2i}{3-2i} + \frac{(z-2i)^2}{(3-2i)^2} + \frac{(z-2i)^3}{(3-2i)^3} + \cdots \right] \\ &= \frac{1}{3-2i} + \frac{z-2i}{(3-2i)^2} + \frac{(z-2i)^2}{(3-2i)^3} + \frac{(z-2i)^3}{(3-2i)^4} + \cdots = \sum_{k=0}^{\infty} \frac{(z-2i)^k}{(3-2i)^{k+1}}. R = \sqrt{13} \end{aligned}$$

16. Using (6) of Section 19.1,

$$\begin{aligned} \frac{1}{1+z} &= \frac{1}{1-i+z+i} = \frac{1}{1-i} \cdot \frac{1}{1+\frac{z+i}{1-i}} = \frac{1}{1-i} \left[1 - \frac{z+i}{1-i} + \frac{(z+i)^2}{(1-i)^2} - \frac{(z+i)^3}{(1-i)^3} + \cdots \right] \\ &= \frac{1}{1-i} - \frac{z+i}{(1-i)^2} + \frac{(z+i)^2}{(1-i)^3} - \frac{(z+i)^3}{(1-i)^4} + \cdots = \sum_{k=0}^{\infty} (-1)^k \frac{(z+i)^k}{(1-i)^{k+1}}. R = \sqrt{2} \end{aligned}$$

Exercises 19.2

17. Using (5) of Section 19.1,

$$\begin{aligned}\frac{z-1}{3-z} &= (z-1) \cdot \frac{1}{2-(z-1)} = \frac{(z-1)}{2} \cdot \frac{1}{1-\frac{z-1}{2}} = \frac{z-1}{2} \left[1 + \frac{z-1}{2} + \frac{(z-1)^2}{2^2} + \frac{(z-1)^3}{2^3} + \cdots \right] \\ &= \frac{z-1}{2} + \frac{(z-1)^2}{2^2} + \frac{(z-1)^3}{2^3} + \frac{(z-1)^4}{2^4} + \cdots = \sum_{k=1}^{\infty} \frac{(z-1)^k}{2^k}. \quad R=2\end{aligned}$$

18. Using (5) of Section 19.1,

$$\begin{aligned}\frac{1+z}{1-z} &= -1 + \frac{2}{1-z} = -1 + \frac{2}{1-i-(z-i)} = -1 + \frac{2}{1-i} \cdot \frac{1}{1-\frac{z-i}{1-i}} \\ &= -1 + \frac{2}{1-i} \left[1 + \frac{z-i}{1-i} + \frac{(z-i)^2}{(1-i)^2} + \frac{(z-i)^3}{(1-i)^3} + \cdots \right] \\ &= -1 + \frac{2}{1-i} + \frac{2(z-i)}{(1-i)^2} + \frac{2(z-i)^2}{(1-i)^3} + \frac{2(z-i)^3}{(1-i)^4} + \cdots = -1 + \sum_{k=0}^{\infty} \frac{2(z-i)^k}{(1-i)^{k+1}}. \quad R=\sqrt{2}\end{aligned}$$

19. Using (8) of Section 19.2,

$$\cos z = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2 \cdot 1!} \left(z - \frac{\pi}{4}\right) - \frac{\sqrt{2}}{2 \cdot 2!} \left(z - \frac{\pi}{4}\right)^2 + \frac{\sqrt{2}}{2 \cdot 3!} \left(z - \frac{\pi}{4}\right)^3 + \cdots. \quad R=\infty$$

20. Using the identity $\sin z = \cos(z - \pi/2)$ and (14) of Section 19.2, $\sin z = \sum_{k=0}^{\infty} (-1)^k \frac{(z - \frac{\pi}{2})^{2k}}{(2k)!}$. $R=\infty$

21. Using $e^z = e^{3i} \cdot e^{z-3i}$ and (12) of Section 19.2, $e^z = e^{3i} \sum_{k=0}^{\infty} \frac{(z-3i)^k}{k!}$. $R=\infty$

22. Using $(z-1)e^{-2z} = e^2(z-1)e^{-2(z-1)}$ and (12) of Section 19.2,

$$(z-1)e^{-2z} = e^2 \sum_{k=0}^{\infty} \frac{(-1)^k 2^k}{k!} (z-1)^{k+1}. \quad R=\infty$$

23. Using (8) of Section 19.2, $\tan z = z + \frac{1}{3}z^3 + \frac{2}{15}z^5 + \cdots$.

24. Using (8) of Section 19.2, $e^{1/(1+z)} = e - ez + \frac{3e}{2}z^2 - \cdots$.

25. Using (5) of Section 19.1,

$$\begin{aligned}f(z) &= \frac{1}{z-2i} - \frac{1}{z-i} = -\frac{1}{2i} \cdot \frac{1}{1-z/2i} + \frac{1}{i} \cdot \frac{1}{1-z/i} \\ &= -\frac{1}{2i} \left(1 + \frac{z}{2i} + \frac{z^2}{(2i)^2} + \frac{z^3}{(2i)^3} + \cdots \right) + \frac{1}{i} \left(1 + \frac{z}{i} + \frac{z^2}{i^2} + \frac{z^3}{i^3} + \cdots \right) = -\frac{i}{2} - \frac{3}{4}z + \frac{7i}{8}z^2 + \frac{15}{16}z^3 - \cdots.\end{aligned}$$

The radius of convergence is $R=1$.

26. Using (5) and (6) of Section 19.1,

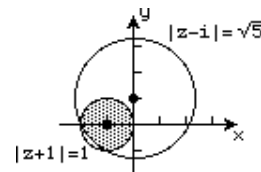
$$\begin{aligned}f(z) &= \frac{2}{z+1} - \frac{1}{z-3} = 2 \cdot \frac{1}{1+z} + \frac{1}{3} \cdot \frac{1}{1-z/3} = 2(1-z+z^2-z^3+\cdots) + \frac{1}{3} \left(1 + \frac{z}{3} + \frac{z^2}{3^2} + \frac{z^3}{3^3} + \cdots \right) \\ &= \frac{7}{3} - \frac{17}{9}z + \frac{55}{27}z^2 - \frac{161}{81}z^3 + \cdots.\end{aligned}$$

27. The distance from $2+5i$ to i is $|2+5i-i| = |2+4i| = 2\sqrt{5}$.

28. The distance from πi to 0 is $|\pi i| = \pi$.

29. The Taylor series are

$$f(z) = \sum_{k=0}^{\infty} (-1)^k (z+1)^k, \quad R = 1; \quad \text{and} \quad f(z) = \sum_{k=0}^{\infty} (-1)^k \frac{(z-i)^k}{(2+i)^{k+1}}, \quad R = \sqrt{5}.$$

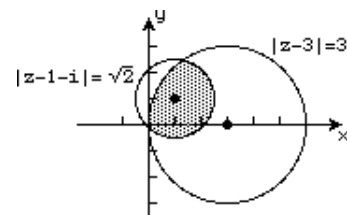


30. The series are

$$f(z) = \sum_{k=0}^{\infty} (-1)^k \frac{(z-3)^k}{3^{k+1}}, \quad R = 3$$

and

$$f(z) = \sum_{k=0}^{\infty} (-1)^k \frac{(z-1-i)^k}{(1+i)^{k+1}}, \quad R = \sqrt{2}.$$



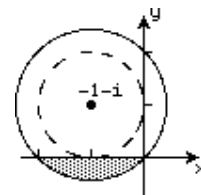
31. (a) The distance from z_0 to the branch cut is one unit.

(b) The first term of the series defined by (8) of this section is

$$f(-1+i) = \text{Ln}(-1+i) = \log_e \sqrt{2} + \frac{3\pi}{4}i = \frac{1}{2} \log_e 2 + \frac{3\pi}{4}i.$$

The subsequent terms of the series come from $f'(z) = \frac{1}{z}$, $f''(z) = -\frac{1}{z^2}$, and so on, evaluated at $-1+i$.

(c) The series converges within the circle $|z+1-i| = \sqrt{2}$. Although the series converges in the shaded region, it does not converge to (or represent) $\text{Ln } z$ in this region.



32. (a) $R = 1$, which is the distance from the origin to $z = -1$.

(b) Using (8) of this section (or integrating the series for $1/(1+z)$ we obtain for $R = 1$,

$$\text{Ln}(1+z) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} z^k.$$

(c) By replacing z in part (b) by $-z$ we obtain for $R = 1$,

$$\text{Ln}(1-z) = -\sum_{k=0}^{\infty} \frac{z^k}{k}.$$

(d) One way of obtaining the Maclaurin series for $\text{Ln} \left(\frac{1+z}{1-z} \right)$ is to use (8) of this section. Alternatively, let us write

$$\text{Ln} \left(\frac{1+z}{1-z} \right) = \text{Ln}(1+z) - \text{Ln}(1-z)$$

and subtract the series in parts (b) and (c). This gives for the common circle of convergence $|z| = 1$,

$$\text{Ln} \left(\frac{1+z}{1-z} \right) = 2z + \frac{2}{3}z^3 + \frac{2}{5}z^5 + \frac{2}{7}z^7 + \cdots = 2 \sum_{k=0}^{\infty} \frac{1}{(2k+1)} z^{2k+1}.$$

Exercises 19.2

But recall that in general $\text{Ln}(z_1/z_2) \neq \text{Ln} z_1 - \text{Ln} z_2$ since $\text{Ln} z_1$ and $\text{Ln} z_2$ could differ by a constant multiple of i . That is $\text{Ln} z_1 - \text{Ln} z_2 = Ci$, for some C . So

$$\text{Ln} \left(\frac{1+z}{1-z} \right) = \text{Ln}(1+z) - \text{Ln}(1-z) - Ci.$$

When $z = 0$ we obtain $\text{Ln} 1 = \text{Ln} 1 - \text{Ln} 1 - Ci$. Since $\text{Ln} 1 = 0$ we get $C = 0$.

33. From $e^z \approx 1 + z + \frac{z^2}{2}$ we obtain

$$e^{(1+i)/10} \approx 1 + \frac{1+i}{10} + \frac{(1+i)^2}{100} = 1.1 + 0.12i.$$

34. From $\sin z \approx z - \frac{z^3}{6}$ we obtain

$$\sin \left(\frac{1+i}{10} \right) \approx \frac{1+i}{10} - \frac{1}{6} \left(\frac{1+i}{10} \right)^3 = \frac{1}{10} + \frac{1}{10}i - \frac{1}{6} \left(\frac{-2+2i}{1000} \right) = \frac{301}{3000} + \frac{299}{3000}i.$$

35. Using the series $e^z = \sum_{k=0}^{\infty} \frac{z^k}{k!}$ we obtain $e^{-t^2} = \sum_{k=0}^{\infty} \frac{(-1)^k t^{2k}}{k!}$. Thus

$$\frac{2}{\sqrt{\pi}} \int_0^z e^{-t^2} dt = \frac{2}{\sqrt{\pi}} \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \int_0^z t^{2k} dt = \frac{2}{\sqrt{\pi}} \sum_{k=0}^{\infty} \frac{(-1)^k}{k!(2k+1)} z^{2k+1}.$$

36.
$$e^{iz} = \sum_{k=0}^{\infty} \frac{(iz)^k}{k!} = 1 + i \frac{z}{1!} - \frac{z^2}{2!} - i \frac{z^3}{3!} + \frac{z^4}{4!} + i \frac{z^5}{5!} - \frac{z^6}{6!} - i \frac{z^7}{7!} + \dots$$

$$= \left(1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \frac{z^6}{6!} + \dots \right) + i \left(\frac{z}{1!} - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} + \dots \right) = \cos z + i \sin z$$

Exercises 19.3

1. $f(z) = \frac{1}{z} \left(1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \frac{z^6}{6!} + \dots \right) = \frac{1}{z} - \frac{z}{2!} + \frac{z^3}{4!} - \frac{z^5}{6!} + \dots$
2. $f(z) = \frac{1}{z^5} \left[z - \left(z - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} + \dots \right) \right] = \frac{1}{3!z^2} - \frac{1}{5!} + \frac{z^2}{7!} - \frac{z^4}{9!} + \dots$
3. $f(z) = 1 - \frac{1}{1!z^2} + \frac{1}{2!z^4} - \frac{1}{3!z^6} + \dots$
4. $f(z) = \frac{1}{z^2} \left[1 - \left(1 + \frac{z}{1!} + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots \right) \right] = -\frac{1}{1!z} - \frac{1}{2!} - \frac{z}{3!} - \frac{z^2}{4!} - \dots$
5. $f(z) = \frac{e \cdot e^{z-1}}{z-1} = \frac{e}{z-1} \left(1 + \frac{(z-1)}{1!} + \frac{(z-1)^2}{2!} + \frac{(z-1)^3}{3!} + \dots \right) = \frac{e}{z-1} + \frac{e}{1!} + \frac{e(z-1)}{2!} + \frac{e(z-1)^2}{3!} + \dots$
6. $f(z) = z \left(1 - \frac{1}{2!z^2} + \frac{1}{4!z^4} - \frac{1}{6!z^6} + \dots \right) = z - \frac{1}{2!z} + \frac{1}{4!z^3} - \frac{1}{6!z^5} + \dots$
7. $f(z) = -\frac{1}{3z} \cdot \frac{1}{1-\frac{z}{3}} = -\frac{1}{3z} \left[1 + \frac{z}{3} + \frac{z^2}{3^2} + \frac{z^3}{3^3} + \dots \right] = -\frac{1}{3z} - \frac{1}{3^2} - \frac{z}{3^3} - \frac{z^2}{3^4} - \dots$

$$8. f(z) = \frac{1}{z^2} \cdot \frac{1}{1 - \frac{3}{z}} = \frac{1}{z^2} \left[1 + \frac{3}{z} + \frac{3^2}{z^2} + \frac{3^3}{z^3} + \cdots \right] = \frac{1}{z^2} + \frac{3}{z^3} + \frac{3^2}{z^4} + \frac{3^3}{z^5} + \cdots$$

$$9. f(z) = \frac{1}{z-3} \cdot \frac{1}{3+z-3} = \frac{1}{3(z-3)} \cdot \frac{1}{1 + \frac{z-3}{3}} = \frac{1}{3(z-3)} \left[1 - \frac{z-3}{3} + \frac{(z-3)^2}{3^2} - \frac{(z-3)^3}{3^3} + \cdots \right]$$

$$= \frac{1}{3(z-3)} - \frac{1}{3^2} + \frac{z-3}{3^3} - \frac{(z-3)^2}{3^4} + \cdots$$

$$10. f(z) = \frac{1}{z-3} \cdot \frac{1}{z-3+3} = \frac{1}{(z-3)^2} \cdot \frac{1}{1 + \frac{3}{z-3}} = \frac{1}{(z-3)^2} \left[1 - \frac{3}{z-3} + \frac{3^2}{(z-3)^2} - \frac{3^3}{(z-3)^3} + \cdots \right]$$

$$= \frac{1}{(z-3)^2} - \frac{3}{(z-3)^3} + \frac{3^2}{(z-3)^4} - \frac{3^3}{(z-3)^5} + \cdots$$

$$11. f(z) = \frac{1}{3} \left[\frac{1}{z-3} - \frac{1}{z} \right] = \frac{1}{3} \left[\frac{1}{z-4+1} - \frac{1}{4+z-4} \right] = \frac{1}{3} \left[\frac{1}{z-4} \cdot \frac{1}{1 + \frac{1}{z-4}} - \frac{1}{4} \cdot \frac{1}{1 + \frac{z-4}{4}} \right]$$

$$= \frac{1}{3} \left[\frac{1}{z-4} \left(1 - \frac{1}{z-4} + \frac{1}{(z-4)^2} - \frac{1}{(z-4)^3} + \cdots \right) - \frac{1}{4} \left(1 - \frac{z-4}{4} + \frac{(z-4)^2}{4^2} - \frac{(z-4)^3}{4^3} + \cdots \right) \right]$$

$$= \cdots - \frac{1}{3(z-4)^2} + \frac{1}{3(z-1)} - \frac{1}{12} + \frac{z-4}{3 \cdot 4^2} - \frac{(z-4)^2}{3 \cdot 4^3} + \cdots$$

$$12. f(z) = \frac{1}{3} \left[\frac{1}{z-3} - \frac{1}{z} \right] = \frac{1}{3} \left[\frac{1}{-4+z+1} - \frac{1}{z+1-1} \right] = \frac{1}{3} \left[-\frac{1}{4} \cdot \frac{1}{1 - \frac{z+1}{4}} - \frac{1}{z+1} \cdot \frac{1}{1 - \frac{1}{z+1}} \right]$$

$$= \frac{1}{3} \left[-\frac{1}{4} \left(1 + \frac{z+1}{4} + \frac{(z+1)^2}{4^2} + \frac{(z+1)^3}{4^3} + \cdots \right) - \frac{1}{z+1} \left(1 + \frac{1}{z+1} + \frac{1}{(z+1)^2} + \frac{1}{(z+1)^3} + \cdots \right) \right]$$

$$= \cdots - \frac{1}{(z+1)^2} - \frac{1}{z+1} - \frac{1}{12} - \frac{z+1}{3 \cdot 4^2} - \frac{(z+1)^2}{3 \cdot 4^3} - \cdots$$

$$13. f(z) = \frac{1}{z-2} - \frac{1}{z-1} = -\frac{1}{2} \cdot \frac{1}{1 - \frac{z}{2}} - \frac{1}{z} \cdot \frac{1}{1 - \frac{1}{z}}$$

$$= -\frac{1}{2} \left(1 + \frac{z}{2} + \frac{z^2}{2^2} + \frac{z^3}{2^3} + \cdots \right) - \frac{1}{z} \left(1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \cdots \right) = \cdots - \frac{1}{z^2} - \frac{1}{z} - \frac{1}{2} - \frac{z}{2^2} - \frac{z^2}{2^3} - \cdots$$

$$14. f(z) = \frac{1}{z-2} - \frac{1}{z-1} = \frac{1}{z} \cdot \frac{1}{1 - \frac{2}{z}} - \frac{1}{z} \cdot \frac{1}{1 - \frac{1}{z}} = \frac{1}{z} \left(1 + \frac{2}{z} + \frac{2^2}{z^2} + \frac{2^3}{z^3} + \cdots \right) - \frac{1}{z} \left(1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \cdots \right)$$

$$= \frac{1}{z^2} + \frac{2^2-1}{z^3} + \frac{2^3-1}{z^4} + \frac{2^4-1}{z^5} + \cdots$$

$$15. f(z) = \frac{1}{z-1} \cdot \frac{-1}{1-(z-1)} = \frac{-1}{z-1} [1 + (z-1) + (z-1)^2 + (z-1)^3 + \cdots] = -\frac{1}{z-1} - 1 - (z-1) - (z-1)^2 - \cdots$$

$$16. f(z) = \frac{1}{z-2} \cdot \frac{1}{1+(z-2)} = \frac{1}{z-2} [1 - (z-2) + (z-2)^2 - (z-2)^3 + \cdots] = \frac{1}{z-2} - 1 + (z-2) - (z-2)^2 + \cdots$$

Exercises 19.3

$$\begin{aligned}
 17. \quad f(z) &= \frac{1/3}{z+1} + \frac{2/3}{z-2} = \frac{1}{3(z+1)} + \frac{2}{3} \cdot \frac{1}{-3+(z+1)} = \frac{1}{3(z+1)} - \frac{2}{9} \cdot \frac{1}{1-\frac{z+1}{3}} \\
 &= \frac{1}{3(z+1)} - \frac{2}{9} \left[1 + \frac{z+1}{3} + \frac{(z+1)^2}{3^2} + \frac{(z+1)^3}{3^3} + \dots \right] \\
 &= \frac{1}{3(z+1)} - \frac{2}{9} - \frac{2(z+1)}{3^3} - \frac{2(z+1)^2}{3^4} - \dots
 \end{aligned}$$

$$\begin{aligned}
 18. \quad f(z) &= \frac{1}{3(z+1)} + \frac{2}{3} \cdot \frac{1}{(z+1)-3} = \frac{1}{3(z+1)} + \frac{2}{3(z+1)} \cdot \frac{1}{1-\frac{3}{z+1}} \\
 &= \frac{1}{3(z+1)} + \frac{2}{3(z+1)} \left(1 + \frac{3}{z+1} + \frac{3^2}{(z+1)^2} + \frac{3^3}{(z+1)^3} + \dots \right) \\
 &= \frac{1}{z+1} + \frac{2}{(z+1)^2} + \frac{2 \cdot 3}{(z+1)^3} + \frac{2 \cdot 3^2}{(z+1)^4} + \dots
 \end{aligned}$$

$$\begin{aligned}
 19. \quad f(z) &= \frac{1/3}{z+1} + \frac{2/3}{z-2} = \frac{1}{3z} \cdot \frac{1}{1+\frac{1}{z}} - \frac{1}{3} \cdot \frac{1}{1-\frac{z}{2}} = \frac{1}{3z} \left(1 - \frac{1}{z} + \frac{1}{z^2} - \frac{1}{z^3} + \dots \right) - \frac{1}{3} \left(1 + \frac{z}{2} + \frac{z^2}{2^2} + \frac{z^3}{2^3} + \dots \right) \\
 &= \dots - \frac{1}{3z^2} + \frac{1}{3z} - \frac{1}{3} - \frac{z}{3 \cdot 2} - \frac{z^2}{3 \cdot 2^2} - \dots
 \end{aligned}$$

$$\begin{aligned}
 20. \quad f(z) &= \frac{2/3}{z-2} + \frac{1}{3} \frac{1}{3+(z-2)} = \frac{2/3}{z-2} + \frac{1}{9} \cdot \frac{1}{1+\frac{z-2}{3}} = \frac{2/3}{z-2} + \frac{1}{9} \left(1 + \frac{z-2}{3} + \frac{(z-2)^2}{3^2} + \frac{(z-2)^3}{3^3} + \dots \right) \\
 &= \frac{2}{3(z-2)} + \frac{1}{9} + \frac{z-2}{3^3} + \frac{(z-2)^2}{3^4} + \dots
 \end{aligned}$$

$$21. \quad f(z) = \frac{1}{z}(1-z)^{-2} = \frac{1}{z} \left(1 + (-2)(-z) + \frac{(-2)(-3)}{2!}(-z)^2 + \frac{(-2)(-3)(-4)}{3!}(-z)^3 + \dots \right) = \frac{1}{z} + 2 + 3z + 4z^2 + \dots$$

$$\begin{aligned}
 22. \quad f(z) &= \frac{1}{z^3(1-\frac{1}{z})^2} = \frac{1}{z^3} \left(1 - \frac{1}{z} \right)^{-2} \\
 &= \frac{1}{z^3} \left(1 + (-2) \left(-\frac{1}{z} \right) + \frac{(-2)(-3)}{2!} \left(-\frac{1}{z} \right)^2 + \frac{(-2)(-3)(-4)}{3!} \left(-\frac{1}{z} \right)^3 + \dots \right) \\
 &= \frac{1}{z^3} + \frac{2}{z^4} + \frac{3}{z^5} + \frac{4}{z^6} + \dots
 \end{aligned}$$

$$\begin{aligned}
 23. \quad f(z) &= \frac{1}{(z-2)[1+(z-2)]^3} = \frac{1}{z-2} [1+(z-2)]^{-3} \\
 &= \frac{1}{z-2} \left(1 + (-3)(z-2) + \frac{(-3)(-4)}{2!}(z-2)^2 + \frac{(-3)(-4)(-5)}{3!}(z-2)^3 + \dots \right) \\
 &= \frac{1}{z-2} - 3 + 6(z-2) - 10(z-2)^2 + \dots
 \end{aligned}$$

$$\begin{aligned}
 24. \quad f(z) &= \frac{1}{(z-3)^3} \cdot \frac{-1}{1-(z-1)} = \frac{-1}{(z-1)^3} [1 + (z-1) + (z-1)^2 + (z-1)^3 + \dots] \\
 &= -\frac{1}{(z-1)^3} - \frac{1}{(z-1)^2} - \frac{1}{z-1} - 1 - (z-1) - \dots
 \end{aligned}$$

$$25. \quad f(z) = \frac{3}{z} + \frac{4}{z-1} = \frac{3}{z} - 4 \cdot \frac{1}{1-z} = \frac{3}{z} - 4(1 + z + z^2 + z^3 + \dots) = \frac{3}{z} - 4 - 4z - 4z^2 - \dots$$

26. $f(z) = \frac{4}{z-1} + 3 \cdot \frac{1}{1+(z-1)} = \frac{4}{z-1} + 3(1-(z-1)+(z-1)^2-(z-1)^3+\dots) = \frac{4}{z-1} + 3-3(z-1)+3(z-1)^2-\dots$
27. $f(z) = z + \frac{2}{z-2} = 1 + (z-1) + \frac{2}{-1+z-1} = 1 + (z-1) + \frac{2}{z-1} \cdot \frac{1}{1-\frac{1}{z-1}}$
 $= 1 + (z-1) + \frac{2}{z-1} \left(1 + \frac{1}{z-1} + \frac{1}{(z-1)^2} + \frac{1}{(z-1)^3} + \dots \right) = \dots + \frac{2}{(z-1)^2} + \frac{2}{z-1} + 1 + (z-1)$
28. $f(z) = z + \frac{2}{z-2} = \frac{2}{z-2} + 2 + (z-2)$

Exercises 19.4

1. Using $e^{2z} = \sum_{k=0}^{\infty} \frac{2^k z^k}{k!}$ we obtain

$$\frac{e^{2z} - 1}{z} = \frac{\left(1 + \frac{2}{1!}z + \frac{2^2}{2!}z^2 + \frac{2^3}{3!}z^3 + \dots \right) - 1}{z} = \frac{1}{z} \left(\frac{2}{1!}z + \frac{2^2}{2!}z^2 + \frac{2^3}{3!}z^3 + \dots \right) = \frac{2}{1!} + \frac{2^2}{2!}z + \frac{2^3}{3!}z^2 + \dots$$

From the form of the last series we see that $z = 0$ is a removable singularity. Define $f(0) = 2$.

2. Using $\sin 4z = \sum_{k=0}^{\infty} (-1)^k \frac{(4z)^{2k+1}}{(2k+1)!}$ we obtain

$$\begin{aligned} \frac{\sin 4z - 4z}{z^2} &= \frac{\left(\frac{4}{1!}z - \frac{4^3}{3!}z^3 + \frac{4^5}{5!}z^5 - \frac{4^7}{7!}z^7 + \dots \right) - 4z}{z^2} = \frac{1}{z^2} \left(-\frac{4^3}{3!}z^3 + \frac{4^5}{5!}z^5 - \frac{4^7}{7!}z^7 + \dots \right) \\ &= -\frac{4^3}{3!}z + \frac{4^5}{5!}z^3 - \frac{4^7}{7!}z^5 + \dots \end{aligned}$$

From the form of the last series we see that $z = 0$ is a removable singularity. Define $f(0) = 0$.

3. Since $f(-2+i) = f'(-2+i) = 0$ and $f''(z) = 2$ for all z , $z = -2+i$ is a zero of order two.
4. Write $f(z) = z^4 - 16 = (z^2 - 4)(z^2 + 4) = (z-2)(z+2)(z-2i)(z+2i)$ to see that 2 , -2 , $2i$, and $-2i$ are zeros of f . Now $f'(z) = 4z^3$ and $f'(2) \neq 0$, $f'(-2) \neq 0$, $f'(2i) \neq 0$, and $f'(-2i) \neq 0$. This indicates that each zero is of order one.
5. Write $f(z) = z^2(z^2 + 1) = z^2(z-i)(z+i)$ to see that 0 , i , and $-i$ are zeros of f . Now $f'(z) = 4z^3 + 2z$ and $f'(i) \neq 0$ and $f'(-i) \neq 0$. This indicates that $z = i$ and $z = -i$ are zeros of order one. However $f'(0) = 0$, but $f''(0) = 2 \neq 0$. Hence $z = 0$ is a zero of order two.
6. Write $f(z) = (z^2 + 9)/z = (z-3i)(z+3i)/z$ to see that $3i$ and $-3i$ are zeros of f . Now $f'(z) = 1 - 9/z^2$ and $f'(3i) = f'(-3i) = 2 \neq 0$. This indicates that each zero is of order one.
7. Write $f(z) = e^z(e^z - 1)$ to see that $2n\pi i$, $n = 0, \pm 1, \pm 2, \dots$ are zeros of f . Now $f'(z) = 2e^{2z} - e^z$ and $f'(2n\pi i) = 2e^{4n\pi i} - e^{2n\pi i} = 1 \neq 0$. This indicates that each zero is of order one.
8. The zeros of f are the zeros of $\sin z$, that is, $n\pi$, $n = 0, \pm 1, \pm 2, \dots$. From $f'(z) = 2 \sin z \cos z$ we see $f'(n\pi) = 0$. From $f''(z) = 2(-\sin^2 z + \cos^2 z)$ we see $f''(n\pi) \neq 0$. This indicates that each zero is of order two.
9. From $f(z) = z(1 - \cos z^2) = z \left(-\frac{z^4}{2!} + \frac{z^8}{4!} - \dots \right) = z^5 \left(-\frac{1}{2!} + \frac{z^4}{4!} - \dots \right)$
 we see that $z = 0$ is a zero of order five.

Exercises 19.4

10. From
$$f(z) = z - \sin z = \frac{z^3}{3!} - \frac{z^5}{5!} + \cdots = z^3 \left(\frac{1}{3!} - \frac{z^2}{5!} + \cdots \right)$$

we see that $z = 0$ is a zero of order three.

11. From
$$f(z) = 1 - e^{z-1} = -\frac{z-1}{1!} - \frac{(z-1)^2}{2!} - \cdots = (z-1) \left(1 - \frac{z-1}{2!} - \cdots \right)$$

we see that $z = 1$ is a zero of order one.

12. From the series $e^z = -\sum_{k=0}^{\infty} \frac{(z-\pi i)^k}{k!}$ centered at πi and

$$\begin{aligned} f(z) &= 1 - \pi i + z + e^z = 1 - \pi i + z + \left(-1 - \frac{z-\pi i}{1!} - \frac{(z-\pi i)^2}{2!} - \frac{(z-\pi i)^3}{3!} - \cdots \right) \\ &= -\frac{(z-\pi i)^2}{2!} - \frac{(z-\pi i)^3}{3!} - \cdots = (z-\pi i)^2 \left(-\frac{1}{2!} - \frac{z-\pi i}{3!} - \cdots \right) \end{aligned}$$

we see that $z = \pi i$ is a zero of order two.

13. From
$$f(z) = \frac{3z-1}{[(z-(-1+2i)][z-(-1-2i)]}$$

and Theorem 19.11 we see that $-1+2i$ and $-1-2i$ are simple poles.

14. From $f(z) = \frac{5z^2-6}{z^2}$ and Theorem 19.11 we see that 0 is a pole of order two.

15. From $f(z) = \frac{1+4i}{(z+2)(z+i)^4}$ and Theorem 19.11 we see that -2 is a simple pole and $-i$ is a pole of order four.

16. From
$$f(z) = \frac{z-1}{(z+1)^2 \left[z - \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i \right) \right] \left[z - \left(\frac{1}{2} - \frac{\sqrt{3}}{2}i \right) \right]}$$

and Theorem 19.11 we see that -1 is a pole of order two and $\frac{1}{2} + \frac{\sqrt{3}}{2}i$ and $\frac{1}{2} - \frac{\sqrt{3}}{2}i$ are simple poles.

17. Since $\sin z$ and $\cos z$ are analytic at $n\pi$, $n = 0, \pm 1, \pm 2, \dots$, $\sin z$ has zeros of order one at $n\pi$, and $\cos n\pi \neq 0$, it follows from Theorem 19.11 that the numbers $n\pi$, $n = 0, \pm 1, \pm 2, \dots$ are simple poles of $f(z) = \tan z$.

18. From $z^2 \sin \pi z = z^3 \left(\pi - \frac{\pi^3 z^2}{3!} + \cdots \right)$ we see $z = 0$ is a zero of order three. From $f(z) = \frac{\cos \pi z}{z^2 \sin \pi z}$ and Theorem 19.11 we see 0 is a pole of order three. The numbers n , $n \pm 1, \pm 2, \dots$ are simple poles.

19. From the Laurent series

$$f(z) = \frac{1 - \cosh z}{z^4} = \frac{1 - \left(1 + \frac{z^2}{2!} + \frac{z^4}{4!} + \frac{z^6}{6!} + \cdots \right)}{z^4} = -\frac{1}{2!z^2} - \frac{1}{4!} - \frac{z^2}{6!} - \cdots$$

we see that 0 is a pole of order two.

20. From the Laurent series

$$f(z) = \frac{e^z}{z^2} = \frac{\left(1 + \frac{z}{1!} + \frac{z^2}{2!} + \cdots \right)}{z^2} = \frac{1}{z^2} + \frac{1}{z} + \frac{1}{2!} + \cdots$$

we see that 0 is a pole of order two.

21. From $1 - e^z = 1 - \left(1 + \frac{z}{1!} + \frac{z^2}{2!} + \cdots \right) = z \left(-1 - \frac{z}{2!} - \cdots \right)$ we see that $z = 0$ is a zero of order one. By periodicity of e^z it follows that $z = 2n\pi i$, $n = 0, \pm 1, \pm 2, \dots$ are zeros of order one. From $f(z) = \frac{1}{1 - e^z}$ and Theorem 19.11 we see that the numbers $2n\pi i$, $n = 0, \pm 1, \pm 2, \dots$ are simple poles.

22. $z = 0$ is a removable singularity of the function $(\sin z)/z$. From $f(z) = \frac{\sin z}{z(z-1)}$ we see that only 1 is a (simple) pole.
23. The function $f(z) = \frac{\sin(1/z)}{\cos(1/z)}$ fails to be defined at $z = 0$ and at the solutions of $\cos \frac{1}{z} = 0$, that is, at $\frac{1}{z} = (2n+1)\frac{\pi}{2}$, $n = 0, \pm 1, \pm 2, \dots$. Since $z = \frac{2}{(2n+1)\pi}$, $n = 0, \pm 1, \pm 2, \dots$ we see that in any neighborhood of $z = 0$ there are points at which f is not defined and thus not analytic. Hence $z = 0$ is a non-isolated singularity.
24. From the Laurent series

$$f(z) = z^3 \left[\frac{1}{z} - \frac{1}{3!} \left(\frac{1}{z} \right)^3 + \frac{1}{5!} \left(\frac{1}{z} \right)^5 - \frac{1}{7!} \left(\frac{1}{z} \right)^7 + \dots \right] = z^2 - \frac{1}{3!} + \frac{1}{5!z^2} - \frac{1}{7!z^4} + \dots, \quad 0 < |z|,$$

we see that the principal part contains an infinite number of nonzero terms. Hence $z = 0$ is an essential singularity.

Exercises 19.5

- $$f(z) = \frac{2}{5(z-1)} \cdot \frac{1}{1 + \frac{z-1}{5}} = \frac{2}{5(z-1)} \left(1 - \frac{z-1}{5} + \frac{(z-1)^2}{5^2} - \frac{(z-1)^3}{5^3} + \dots \right)$$

$$= \frac{2/5}{z-1} - \frac{2}{25} + \frac{2(z-1)}{5^3} - \frac{2(z-1)^2}{5^4} + \dots$$

$$\text{Res}(f(z), 1) = 2/5$$
- $$f(z) = \frac{1}{z^3}(1-z)^{-3} = \frac{1}{z^3} \left(1 + (-3)(-z) + \frac{(-3)(-4)}{2!}(-z)^2 + \frac{(-3)(-4)(-5)}{3!}(-z)^3 + \dots \right)$$

$$= \frac{1}{z^3} + \frac{3}{z^2} + \frac{6}{z} + 10 + \dots$$

$$\text{Res}(f(z), 0) = 6$$
- $$f(z) = -\frac{3}{z} - \frac{1}{z-2} = -\frac{3}{z} + \frac{1}{2} \cdot \frac{1}{1 - \frac{z}{2}} = -\frac{3}{z} + \frac{1}{2} \left(1 + \frac{z}{2} + \frac{z^2}{2^2} + \frac{z^3}{2^3} + \dots \right) = -\frac{3}{z} + \frac{1}{2} + \frac{z}{2^2} + \frac{z^2}{2^3} + \dots$$

$$\text{Res}(f(z), 0) = -3$$
- $$f(z) = (z+3)^2 \left(\frac{2}{z+3} - \frac{2^3}{3!(z+3)^3} + \frac{2^5}{5!(z+3)^5} + \dots \right) = \dots + \frac{2^5}{5!(z+3)^3} - \frac{2^3}{3!(z+3)} + 2(z+3)$$

$$\text{Res}(f(z), -3) = -\frac{4}{3}$$
- $$f(z) = e^{-2/z^2} = \sum_{k=0}^{\infty} \frac{(-2/z^2)^k}{k!} = \dots - \frac{2^3}{3!z^6} + \frac{2^2}{2!z^4} - \frac{2}{1!z^2} + 1; \quad \text{Res}(f(z), 0) = 0$$
- $$f(z) = \frac{e^{-2}}{(z-2)^2} e^{-(z-2)} = \frac{e^{-2}}{(z-2)^2} \left(1 - \frac{z-2}{1!} + \frac{(z-2)^2}{2!} - \frac{(z-2)^3}{3!} + \dots \right)$$

$$= \frac{e^{-2}}{(z-2)^2} - \frac{e^{-2}}{z-2} + \frac{e^{-2}}{2} - \frac{e^{-2}(z-2)}{3!} + \dots$$

$$\text{Res}(f(z), 2) = -e^{-2}$$

Exercises 19.5

7. $\text{Res}(f(z), 4i) = \lim_{z \rightarrow 4i} (z - 4i) \cdot \frac{z}{(z - 4i)(z + 4i)} = \lim_{z \rightarrow 4i} \frac{z}{z + 4i} = \frac{1}{2}$
 $\text{Res}(f(z), -4i) = \lim_{z \rightarrow -4i} (z + 4i) \cdot \frac{z}{(z - 4i)(z + 4i)} = \lim_{z \rightarrow -4i} \frac{z}{z - 4i} = \frac{1}{2}$
8. $\text{Res}(f(z), 1/2) = \lim_{z \rightarrow 1/2} (z - 1/2) \frac{4z + 8}{2(z - 1/2)} = \lim_{z \rightarrow 1/2} (2z + 4) = 5$
9. $\text{Res}(f(z), 1) = \lim_{z \rightarrow 1} (z - 1) \frac{1}{z^2(z + 2)(z - 1)} = \lim_{z \rightarrow 1} \frac{1}{z^2(z + 2)} = \frac{1}{3}$
 $\text{Res}(f(z), -2) = \lim_{z \rightarrow -2} (z + 2) \frac{1}{z^2(z + 2)(z - 1)} = \lim_{z \rightarrow -2} \frac{1}{z^2(z - 1)} = -\frac{1}{12}$
 $\text{Res}(F(z), 0) = \frac{1}{1!} \lim_{z \rightarrow 0} \frac{d}{dz} \left[z^2 \cdot \frac{1}{z^2(z + 2)(z - 1)} \right] = \lim_{z \rightarrow 0} \frac{-2z - 1}{(z + 2)^2(z - 1)^2} = -\frac{1}{4}$
10. $\text{Res}(f(z), 1 + i) = \frac{1}{1!} \lim_{z \rightarrow 1+i} \frac{d}{dz} \left[(z - 1 - i)^2 \cdot \frac{1}{(z - 1 - i)^2(z - 1 + i)^2} \right] = \lim_{z \rightarrow 1+i} \frac{-2}{(z - 1 + i)^3} = -\frac{1}{4}i$
 $\text{Res}(f(z), 1 - i) = \frac{1}{1!} \lim_{z \rightarrow 1-i} \frac{d}{dz} \left[(z - 1 + i)^2 \cdot \frac{1}{(z - 1 - i)^2(z - 1 + i)^2} \right] = \lim_{z \rightarrow 1-i} \frac{-2}{(z - 1 - i)^3} = \frac{1}{4}i$
11. $\text{Res}(f(z), -1) = \lim_{z \rightarrow -1} (z + 1) \cdot \frac{5z^2 - 4z + 3}{(z + 1)(z + 2)(z + 3)} = \lim_{z \rightarrow -1} \frac{5z^2 - 4z + 3}{(z + 2)(z + 3)} = 6$
 $\text{Res}(f(z), -2) = \lim_{z \rightarrow -2} (z + 2) \cdot \frac{5z^2 - 4z + 3}{(z + 1)(z + 2)(z + 3)} = \lim_{z \rightarrow -2} \frac{5z^2 - 4z + 3}{(z + 1)(z + 3)} = -31$
 $\text{Res}(f(z), -3) = \lim_{z \rightarrow -3} (z + 3) \cdot \frac{5z^2 - 4z + 3}{(z + 1)(z + 2)(z + 3)} = \lim_{z \rightarrow -3} \frac{5z^2 - 4z + 3}{(z + 1)(z + 2)} = 30$
12. $\text{Res}(f(z), -3) = \lim_{z \rightarrow -3} (z + 3) \cdot \frac{2z - 1}{(z - 1)^4(z + 3)} = \lim_{z \rightarrow -3} \frac{2z - 1}{(z - 1)^4} = -\frac{7}{256}$
 $\text{Res}(f(z), 1) = \frac{1}{3!} \lim_{z \rightarrow 1} \frac{d^3}{dz^3} \left[(z - 1)^4 \cdot \frac{2z - 1}{(z - 1)^4(z + 3)} \right] = \frac{1}{6} \lim_{z \rightarrow 1} \frac{-42}{(z + 3)^4} = -\frac{7}{256}$
13. $\text{Res}(f(z), 0) = \frac{1}{1!} \lim_{z \rightarrow 0} \frac{d}{dz} \left[z^2 \cdot \frac{\cos z}{z^2(z - \pi)^3} \right] = \lim_{z \rightarrow 0} \frac{-(z - \pi) \sin z - 3 \cos z}{(z - \pi)^4} = -\frac{3}{\pi^4}$
 $\text{Res}(f(z), \pi) = \frac{1}{2!} \lim_{z \rightarrow \pi} \frac{d^2}{dz^2} \left[(z - \pi)^3 \cdot \frac{\cos z}{z^2(z - \pi)^3} \right] = \frac{1}{2} \lim_{z \rightarrow \pi} \frac{-z^2 \cos z + 4z \sin z + 6 \cos z}{z^4} = \frac{\pi^2 - 6}{2\pi^4}$
14. Using $\frac{d}{dz}(e^z - 1) = e^z$ and the result in (4),

$$\text{Res}(f(z), 2n\pi i) = \frac{e^z}{e^z} \Big|_{z=2n\pi i} = 1.$$

15. Using $\frac{d}{dz} \cos z = -\sin z$ and the result in (4),

$$\text{Res} \left(f(z), (2n + 1) \frac{\pi}{2} \right) = \frac{1}{-\sin z} \Big|_{z=(2n+1)\frac{\pi}{2}} = \frac{1}{-\sin(2n + 1)\frac{\pi}{2}} = (-1)^{n+1}.$$

16. $z = 0$ is a pole of order two. Thus by (2) and L'Hôpital's rule,

$$\text{Res}(f(z), 0) = \frac{1}{1!} \lim_{z \rightarrow 0} \frac{d}{dz} \left[z^2 \cdot \frac{1}{z \sin z} \right] = \lim_{z \rightarrow 0} \frac{\sin z - z \cos z}{\sin^2 z} = \lim_{z \rightarrow 0} \frac{\cos z + z \sin z - \cos z}{2 \sin z \cos z} = \lim_{z \rightarrow 0} \frac{z}{2 \cos z} = 0.$$

For the simple poles at $z = n\pi$, $n = \pm 1, \pm 2, \dots$ we have from (4),

$$\operatorname{Res}(f(z), n\pi) = \frac{1}{z \cos z + \sin z} \Big|_{z=n\pi} = \frac{(-1)^n}{n\pi}.$$

17. (a) $\oint_C \frac{1}{(z-1)(z+2)^2} dz = 0$ by Theorem 18.4.

(b) $\oint_C \frac{1}{(z-1)(z+2)^2} dz = 2\pi i \operatorname{Res}(f(z), 1) = \frac{2\pi}{9} i$

(c) $\oint_C \frac{1}{(z-1)(z+2)^2} dz = 2\pi i [\operatorname{Res}(f(z), 1) + \operatorname{Res}(f(z), -2)] = 2\pi i \left[\frac{1}{9} + \left(-\frac{1}{9} \right) \right] = 0$

18. (a) $\oint_C \frac{z+1}{z^2(z-2i)} dz = 2\pi i \operatorname{Res}(f(z), 0) = \pi \left(-1 + \frac{1}{2} i \right)$

(b) $\oint_C \frac{z+1}{z^2(z-2i)} dz = 2\pi i \operatorname{Res}(f(z), 2i) = \pi \left(1 - \frac{1}{2} i \right)$

(c) $\oint_C \frac{z+1}{z^2(z-2i)} dz = 2\pi i [\operatorname{Res}(f(z), 0) + \operatorname{Res}(f(z), 2i)] = 2\pi i \left[\frac{1}{4} + \frac{1}{2} i + \left(-\frac{1}{4} - \frac{1}{2} i \right) \right] = 0$

19. (a) From the Laurent series $z^3 e^{-1/z^2} = \dots - \frac{1}{3!z^3} + \frac{1}{2!z} - z + z^3$ we see $\operatorname{Res}(f(z), 0) = 1/2$. Hence

$$\oint_C z^3 e^{-1/z^2} dz = 2\pi i \operatorname{Res}(f(z), 0) = \pi i.$$

(b) $\oint_C z^3 e^{-1/z^2} dz = 2\pi i \operatorname{Res}(f(z), 0) = \pi i$

(c) $\oint_C z^3 e^{1/z^2} dz = 0$ by Theorem 18.4.

20. (a) $\oint_C \frac{1}{z \sin z} dz = 0$ by Theorem 18.4.

(b) $z = 0$ is a pole of order two (see Problem 16). Thus

$$\oint_C \frac{1}{z \sin z} dz = 2\pi i \operatorname{Res}(f(z), 0) = 2\pi i(0) = 0.$$

(c) $\oint_C \frac{1}{z \sin z} dz = 2\pi i [\operatorname{Res}(f(z), -\pi) + \operatorname{Res}(f(z), 0) + \operatorname{Res}(f(z), \pi)] = 2\pi i \left[\frac{1}{\pi} + 0 + \left(-\frac{1}{\pi} \right) \right] = 0$

21. $\oint_C \frac{1}{z^2 + 4z + 13} dz = 2\pi i \operatorname{Res}(f(z), -2 + 3i) = \frac{\pi}{3}$

22. $\oint_C 1z^3(z-1)^4 dz = 2\pi i \operatorname{Res}(f(z), 1) = -20\pi i$

23. $\oint_C \frac{z}{z^4 - 1} dz = 2\pi i [\operatorname{Res}(f(z), -1) + \operatorname{Res}(f(z), 1) + \operatorname{Res}(f(z), -i) + \operatorname{Res}(f(z), i)] = 2\pi i \left[\frac{1}{4} + \frac{1}{4} - \frac{1}{4} - \frac{1}{4} \right] = 0$

24. $\oint_C \frac{z}{(z+1)(z^2+1)} dz = 2\pi i [\operatorname{Res}(f(z), i) + \operatorname{Res}(f(z), -i)] = 2\pi i \left[\frac{1}{4} - \frac{1}{4} i + \frac{1}{4} + \frac{1}{4} i \right] = \pi i$

25. $\oint_C \frac{ze^z}{z^2 - 1} dz = 2\pi i [\operatorname{Res}(f(z), 1) + \operatorname{Res}(f(z), -1)] = 2\pi i \left[\frac{e}{2} + \frac{e^{-1}}{2} \right] = 2\pi i \cosh 1$

Exercises 19.5

26. $\oint_C \frac{e^z}{z^3 + 2z^2} dz = 2\pi i [\text{Res}(f(z), 0) + \text{Res}(f(z), -2)] = 2\pi i \left[\frac{1}{2} + \frac{e^{-2}}{4} \right] = \pi i \left(1 + \frac{1}{2} e^{-2} \right)$
27. $\oint_C \frac{\tan z}{z} dz = 2\pi i \text{Res} \left(f(z), \frac{\pi}{2} \right) = -4i$. Note: $z = 0$ is not a pole. See Example 1, Section 19.4.
28. $\oint_C \frac{\cot \pi z}{z^2} dz = 2\pi i \text{Res}(f(z), 0) = 2\pi i \left(-\frac{\pi}{3} \right) = -\frac{2\pi^2}{3} i$
 Note: $z = 0$ is a pole of order three. Use L'Hôpital's rule (or *Mathematica*) to show that
- $$\text{Res}(f(z), 0) = \frac{1}{2} \lim_{z \rightarrow 0} \frac{d^2}{dz^2} z \cot \pi z = \frac{1}{2} \lim_{z \rightarrow 0} [-2\pi \csc^2 \pi z + 2\pi^2 z \cot \pi z \csc^2 \pi z] = \frac{1}{2} \left(-\frac{2\pi}{3} \right) = -\frac{\pi}{3}.$$
29. $\oint_C \cot \pi z dz = 2\pi i [\text{Res}(f(z), 1) + \text{Res}(f(z), 2) + \text{Res}(f(z), 3)] = 2\pi i \left[\frac{1}{\pi} + \frac{1}{\pi} + \frac{1}{\pi} \right] = 6i$
30. $\oint_C \frac{2z-1}{z^2(z^3+1)} dz = 2\pi i \left[\text{Res}(f(z), 0) + \text{Res}(f(z), -1) + \text{Res} \left(f(z), \frac{1}{2} + \frac{\sqrt{3}}{2} i \right) \right]$
 $= 2\pi i \left[2 + (-1) + \left(-\frac{1}{2} - \frac{1}{6}\sqrt{3}i \right) \right] = \pi \left(\frac{\sqrt{3}}{3} + i \right)$
31. $\oint_C \frac{e^{iz} + \sin z}{(z-\pi)^4} dz = 2\pi i \text{Res}(f(z), \pi) = \pi \left(-\frac{1}{3} + \frac{1}{3} i \right)$
32. $\oint_C \frac{\cos z}{(z-1)^2(z^2+9)} dz = 2\pi i \text{Res}(f(z), 1) = 2\pi i (-0.02 \cos 1 - 0.1 \sin 1) = -0.5966i$

Exercises 19.6

1. $\int_0^{2\pi} \frac{d\theta}{1 + \frac{1}{2} \sin \theta} = \oint_C \frac{4}{z^2 + 4iz - 1} dz = (4)2\pi i \text{Res}(f(z), (\sqrt{3} - 2)i) = \frac{4\pi}{\sqrt{3}}$
2. $\int_0^{2\pi} \frac{d\theta}{10 - 6 \cos \theta} = \frac{1}{2} \cdot \left(\frac{-2}{i} \right) \oint_C \frac{dz}{(3z-1)(z-3)} = (i)2\pi i \text{Res} \left(f(z), \frac{1}{3} \right) = \frac{\pi}{4}$
3. $\int_0^{2\pi} \frac{\cos \theta}{3 + \sin \theta} d\theta = \oint_C \frac{z^2 + 1}{z(z^2 + 6iz - 1)} dz = 2\pi i [\text{Res}(f(z), 0) + \text{Res}(f(z), -3 + 2\sqrt{2}i)] = 0$
4. $\int_0^{2\pi} \frac{1}{1 + 3 \cos^2 \theta} d\theta = \frac{4}{i} \oint_C \frac{z}{3z^4 + 10z^2 + 3} dz = \left(\frac{4}{i} \right) 2\pi i \left[\text{Res} \left(f(z), \left(\frac{\sqrt{3}}{3} i \right) \right) + \text{Res} \left(f(z), -\frac{\sqrt{3}}{3} i \right) \right] = \pi$
5. $\int_0^\pi \frac{d\theta}{2 - \cos \theta} = \frac{1}{2} \int_0^{2\pi} \frac{d\theta}{2 - \cos \theta} = -\frac{1}{i} \oint_C \frac{dz}{z^2 - 4z + 1} = \left(-\frac{1}{i} \right) 2\pi i \text{Res}(f(z), 2 - \sqrt{3}) = \frac{\pi}{\sqrt{3}}$
6. $\int_0^\pi \frac{d\theta}{1 + \sin^2 \theta} = \frac{1}{2} \int_0^{2\pi} \frac{d\theta}{1 + \sin^2 \theta} = -\frac{2}{i} \oint_C \frac{z}{z^4 - 6z^2 + 1} dz$
 $= \left(-\frac{2}{i} \right) 2\pi i [\text{Res}(f(z), \sqrt{3 - 2\sqrt{2}}) + \text{Res}(f(z), -\sqrt{3 - 2\sqrt{2}})] = \frac{\pi}{\sqrt{2}}$
7. $\int_0^{2\pi} \frac{\sin^2 \theta}{5 + 4 \cos \theta} d\theta = -\frac{1}{4i} \oint_C \frac{(z^2 - 1)^2}{z^2(2z^2 + 5z + 2)} dz = \left(-\frac{1}{4i} \right) 2\pi i \left[\text{Res}(f(z), 0) + \text{Res} \left(f(z), -\frac{1}{2} \right) \right] = \frac{\pi}{4}$
8. $\int_0^{2\pi} \frac{\cos^2 \theta}{3 - \sin \theta} d\theta = \frac{1}{2i} \oint_C \frac{z^4 + 2z^2 + 1}{z^2(iz^2 + 6z - i)} dz = \left(\frac{1}{2i} \right) 2\pi i [\text{Res}(f(z), 0) + \text{Res}(f(z), 3 - 2\sqrt{2}i)] = \pi[6 - 4\sqrt{2}]$

9. We use $\cos 2\theta = (z^2 + z^{-2})/2$.

$$\int_0^{2\pi} \frac{\cos 2\theta}{5 - 4 \cos \theta} d\theta = \frac{i}{2} \oint_C \frac{z^4 + 1}{z^2(2z^2 - 5z + 2)} dz = \left(\frac{i}{2}\right) 2\pi i \left[\text{Res}(f(z), 0) + \text{Res}\left(f(z), \frac{1}{2}\right) \right] = \frac{\pi}{6}$$

$$10. \int_0^{2\pi} \frac{1}{\cos \theta + 2 \sin \theta + 3} d\theta = \frac{2}{i} \oint_C \frac{1}{(1 - 2i)z^2 + 6z + 1 + 2i} = \left(\frac{2}{i}\right) 2\pi i \text{Res}\left(f(z), -\frac{1}{5} - \frac{2}{5}i\right) = \pi$$

$$11. \int_{-\infty}^{\infty} \frac{1}{x^2 - 2x + 2} dx = 2\pi i \text{Res}(f(z), 1 + i) = \pi$$

$$12. \int_{-\infty}^{\infty} \frac{1}{x^2 - 2x + 25} dx = 2\pi i \text{Res}(f(z), 1 + 2\sqrt{6}i) = \frac{\pi}{2\sqrt{6}}$$

$$13. \int_{-\infty}^{\infty} \frac{1}{(x^2 + 4)^2} dx = 2\pi i \text{Res}(f(z), 2i) = \frac{\pi}{16}$$

$$14. \int_{-\infty}^{\infty} \frac{x^2}{(x^2 + 1)^2} dx = 2\pi i \text{Res}(f(z), i) = \frac{\pi}{2}$$

$$15. \int_{-\infty}^{\infty} \frac{1}{(x^2 + 1)^3} dx = 2\pi i \text{Res}(f(z), i) = \frac{3\pi}{8}$$

$$16. \int_{-\infty}^{\infty} \frac{x}{(x^2 + 4)^3} dx = 0 \quad (\text{The integrand is an odd function})$$

$$17. \int_{-\infty}^{\infty} \frac{2x^2 - 1}{x^4 + 5x^2 + 4} dx = 2\pi i [\text{Res}(f(z), i) + \text{Res}(f(z), 2i)] = \frac{\pi}{2}$$

$$18. \int_{-\infty}^{\infty} \frac{dx}{(x^2 + 1)^2(x^2 + 9)} = 2\pi i [\text{Res}(f(z), i) + \text{Res}(f(z), 3i)] = \frac{5\pi}{96}$$

$$19. \int_0^{\infty} \frac{x^2 + 1}{x^4 + 1} dx = \frac{1}{2} \int_{-\infty}^{\infty} \frac{x^2 + 1}{x^4 + 1} dx = \pi i \text{Res}(f(z), i) = \frac{\pi}{\sqrt{2}}$$

$$20. \int_0^{\infty} \frac{1}{x^6 + 1} dx = \frac{1}{2} \int_{-\infty}^{\infty} \frac{1}{x^6 + 1} dx = \pi i \left[\text{Res}\left(f(z), \frac{\sqrt{3}}{2} + \frac{1}{2}i\right) + \text{Res}(f(z), i) + \text{Res}\left(f(z), \frac{-\sqrt{3}}{2} + \frac{1}{2}i\right) \right] = \frac{\pi}{3}$$

$$21. \int_{-\infty}^{\infty} \frac{e^{ix}}{x^2 + 1} dx = 2\pi i \text{Res}(f(z), i) = \pi e^{-1}. \text{ Therefore, } \int_{-\infty}^{\infty} \frac{\cos x}{x^2 + 1} dx = \text{Re} \left(\int_{-\infty}^{\infty} \frac{e^{ix}}{x^2 + 1} dx \right) = \pi e^{-1}.$$

$$22. \int_{-\infty}^{\infty} \frac{e^{2ix}}{x^2 + 1} dx = 2\pi i \text{Res}(f(z), i) = \pi e^{-2}. \text{ Therefore, } \int_{-\infty}^{\infty} \frac{\cos 2x}{x^2 + 1} dx = \text{Re} \left(\int_{-\infty}^{\infty} \frac{e^{2ix}}{x^2 + 1} dx \right) = \pi e^{-2}.$$

$$23. \int_{-\infty}^{\infty} \frac{x e^{ix}}{x^2 + 1} dx = 2\pi i \text{Res}(f(z), i) = \pi e^{-1} i. \text{ Therefore, } \int_{-\infty}^{\infty} \frac{x \sin x}{x^2 + 1} dx = \text{Im} \left(\int_{-\infty}^{\infty} \frac{x e^{ix}}{x^2 + 1} dx \right) = \pi e^{-1}.$$

$$24. \int_{-\infty}^{\infty} \frac{e^{ix}}{(x^2 + 4)^2} dx = 2\pi i \text{Res}(f(z), 2i) = \frac{3e^{-2}}{16} \pi; \quad \int_{-\infty}^{\infty} \frac{\cos x}{(x^2 + 4)^2} dx = \text{Re} \left(\int_{-\infty}^{\infty} \frac{e^{ix}}{(x^2 + 4)^2} dx \right) = \frac{3e^{-2}}{16} \pi.$$

Therefore

$$\int_0^{\infty} \frac{\cos x}{(x^2 + 4)^2} dx = \frac{1}{2} \left(\frac{3e^{-2}}{16} \pi \right) = \frac{3e^{-2}}{32} \pi.$$

$$25. \int_{-\infty}^{\infty} \frac{e^{3ix}}{(x^2 + 1)^2} dx = 2\pi i \text{Res}(f(z), i) = 2\pi e^{-3}; \quad \int_{-\infty}^{\infty} \frac{\cos 3x}{(x^2 + 1)^2} dx = \text{Re} \left(\int_{-\infty}^{\infty} \frac{e^{3ix}}{(x^2 + 1)^2} dx \right) = 2\pi e^{-3}.$$

Therefore

$$\int_0^{\infty} \frac{\cos 3x}{(x^2 + 1)^2} dx = \frac{1}{2} (2\pi e^{-3}) = \pi e^{-3}.$$

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26. $\int_{-\infty}^{\infty} \frac{e^{ix}}{x^2 + 4x + 5} dx = 2\pi i \operatorname{Res}(f(z), -2 + i) = \pi e^{-1-2i}$. Therefore

$$\int_{-\infty}^{\infty} \frac{\sin x}{x^2 + 4x + 5} dx = \operatorname{Im} \left(\int_{-\infty}^{\infty} \frac{e^{ix}}{x^2 + 4x + 5} dx \right) = -\pi e^{-1} \sin 2$$

27. $\int_{-\infty}^{\infty} \frac{e^{2ix}}{x^4 + 1} dx = 2\pi i \left[\operatorname{Res} \left(f(z), \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} i \right) + \operatorname{Res} \left(f(z), -\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} i \right) \right]$

$$= 2\pi i \left[\left(-\frac{\sqrt{2}}{8} - \frac{\sqrt{2}}{8} i \right) e^{(-\sqrt{2} + \sqrt{2} i)} + \left(\frac{\sqrt{2}}{8} - \frac{\sqrt{2}}{8} i \right) e^{(-\sqrt{2} - \sqrt{2} i)} \right]$$

$$= \pi e^{-\sqrt{2}} \left[\frac{\sqrt{2}}{2} \cos \sqrt{2} + \frac{\sqrt{2}}{2} \sin \sqrt{2} \right]$$

$$\int_{-\infty}^{\infty} \frac{\cos 2x}{x^4 + 1} dx = \operatorname{Re} \left(\int_{-\infty}^{\infty} \frac{e^{2ix}}{x^4 + 1} dx \right) = \pi e^{-\sqrt{2}} \left[\frac{\sqrt{2}}{2} \cos \sqrt{2} + \frac{\sqrt{2}}{2} \sin \sqrt{2} \right]$$

Therefore

$$\int_0^{\infty} \frac{\cos 2x}{x^4 + 1} dx = \pi e^{-\sqrt{2}} \frac{\sqrt{2}}{4} (\cos \sqrt{2} + \sin \sqrt{2}).$$

28. $\int_{-\infty}^{\infty} \frac{x e^{ix}}{x^4 + 1} dx = 2\pi i \left[\operatorname{Res} \left(f(z), \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} i \right) + \operatorname{Res} \left(f(z), -\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} i \right) \right]$

$$= 2\pi i \left[-\frac{i}{4} e^{(-1/\sqrt{2} + i/\sqrt{2})} + \frac{i}{4} e^{(-1/\sqrt{2} - i/\sqrt{2})} \right] = \left(\pi e^{-1/\sqrt{2}} \sin \frac{1}{\sqrt{2}} \right) i$$

$$\int_{-\infty}^{\infty} \frac{x \sin x}{x^4 + 1} dx = \operatorname{Im} \left(\int_{-\infty}^{\infty} \frac{x e^{ix}}{x^4 + 1} dx \right) = \pi e^{-1/\sqrt{2}} \sin \frac{1}{\sqrt{2}}$$

Therefore

$$\int_0^{\infty} \frac{x \sin x}{x^4 + 1} dx = \frac{\pi}{2} e^{-1/\sqrt{2}} \sin \frac{1}{\sqrt{2}}.$$

29. $\int_{-\infty}^{\infty} \frac{e^{ix}}{(x^2 + 1)(x^2 + 9)} dx = 2\pi i [\operatorname{Res}(f(z), i) + \operatorname{Res}(f(z), 3i)] = 2\pi i \left[-\frac{i}{16} e^{-1} + \frac{i}{48} e^{-3} \right] = \frac{1}{8} e^{-1} - \frac{1}{24} e^{-3}$

30. $\int_{-\infty}^{\infty} \frac{x e^{ix}}{(x^2 + 1)(x^2 + 4)} dx = 2\pi i [\operatorname{Res}(f(z), i) + \operatorname{Res}(f(z), 2i)] = 2\pi i \left[\frac{1}{6} e^{-1} - \frac{1}{6} e^{-2} \right] = \frac{\pi}{3} (e^{-1} - e^{-2}) i;$

$$\int_{-\infty}^{\infty} \frac{x \sin x}{(x^2 + 1)(x^2 + 4)} dx = \operatorname{Im} \left(\int_{-\infty}^{\infty} \frac{x e^{ix}}{(x^2 + 1)(x^2 + 4)} dx \right) = \frac{\pi}{3} (e^{-1} - e^{-2}).$$

Therefore

$$\int_0^{\infty} \frac{x \sin x}{(x^2 + 1)(x^2 + 4)} dx = \frac{1}{2} \left[\frac{\pi}{3} (e^{-1} - e^{-2}) \right] = \frac{\pi}{6} (e^{-1} - e^{-2}).$$

31. Consider the contour integral $\oint_C \frac{e^{iz}}{z} dz$. The function $f(z) = \frac{1}{z}$ has a simple pole at $z = 0$. If we use the contour C shown in Figure 19.14, it follows from the Cauchy-Goursat Theorem that

$$\oint_C = \int_{C_R} + \int_{-R}^{-r} + \int_{-C_r} + \int_r^R = 0.$$

Taking limits as $R \rightarrow \infty$ and as $r \rightarrow 0$ and using Theorem 19.17 we then find

$$\text{P.V.} \int_{-\infty}^{\infty} \frac{e^{ix}}{x} dx - \pi i \operatorname{Res}(f(z) e^{iz}, 0) = 0 \quad \text{or} \quad \text{P.V.} \int_{-\infty}^{\infty} \frac{e^{ix}}{x} dx = \pi i.$$

Equating the imaginary parts of $\int_{-\infty}^{\infty} \frac{\cos x + i \sin x}{x} dx = 0 + \pi i$ gives

$$\int_{-\infty}^{\infty} \frac{\sin x}{x} dx = \pi.$$

- 32.** Consider the contour integral $\oint_C \frac{e^{iz}}{z(z^2 + 1)} dz$. The function $f(z) = \frac{1}{z(z^2 + 1)}$ has simple poles at $z = 0$ and at $z = i$. If we use the contour C shown in Figure 19.14, it follows from Theorem 19.14 that

$$\oint_C = \int_{C_R} + \int_{-R}^{-r} + \int_{-C_r} + \int_r^R = 2\pi i \operatorname{Res}(f(z), i).$$

Taking limits as $R \rightarrow \infty$ and as $r \rightarrow 0$ and using Theorem 19.17 we then find

$$\text{P.V.} \int_{-\infty}^{\infty} \frac{e^{ix}}{x(x^2 + 1)} dx - \pi i \operatorname{Res}(f(z)e^{iz}, 0) = 2\pi i \operatorname{Res}(f(z)e^{iz}, i)$$

or

$$\text{P.V.} \int_{-\infty}^{\infty} \frac{e^{ix}}{x(x^2 + 1)} dx = \pi i + 2\pi i \left(-\frac{e^{-1}}{2} \right).$$

Equating the imaginary parts of $\int_{-\infty}^{\infty} \frac{\cos x + i \sin x}{x(x^2 + 1)} dx = 0 + \pi(1 - e^{-1})i$ gives

$$\int_{-\infty}^{\infty} \frac{\sin x}{x(x^2 + 1)} dx = \pi(1 - e^{-1}).$$

- 33.** $\int_0^\pi \frac{d\theta}{(a + \cos \theta)^2} = \frac{1}{2} \int_0^{2\pi} \frac{d\theta}{(a + \cos \theta)^2} = \frac{2}{i} \oint_C \frac{z}{(z^2 + 2az + 1)^2} dz$ (C is $|z| = 1$) $= \frac{2}{i} \oint_C \frac{z}{(z - r_1)^2(z - r_2)^2} dz$

where $r_1 = -a + \sqrt{a^2 - 1}$, $r_2 = -a - \sqrt{a^2 - 1}$. Now

$$\oint_C \frac{z}{(z - r_1)^2(z - r_2)^2} dz = 2\pi i \operatorname{Res}(f(z), r_1) = 2\pi i \frac{a}{4(\sqrt{a^2 - 1})^3} = \frac{a\pi}{2(\sqrt{a^2 - 1})^3} i.$$

Thus

$$\int_0^\pi \frac{d\theta}{(a + \cos \theta)^2} = \frac{2}{i} \cdot \frac{a\pi}{2(\sqrt{a^2 - 1})^3} i = \frac{a\pi}{(\sqrt{a^2 - 1})^3}.$$

When $a = 2$ we obtain

$$\int_0^\pi \frac{d\theta}{(2 + \cos \theta)^2} = \frac{2\pi}{(\sqrt{3})^3} \quad \text{and so} \quad \int_0^{2\pi} \frac{d\theta}{(2 + \cos \theta)^2} = \frac{4\pi}{3\sqrt{3}}.$$

- 34.** $\int_0^{2\pi} \frac{\sin^2 \theta}{a + b \cos \theta} d\theta = \frac{i}{2b} \oint_C \frac{z^2 - 1}{z^2(z - r_1)(z - r_2)} dz$ (C is $|z| = 1$) where $r_1 = (-a + \sqrt{a^2 - b^2})/b$, $r_2 = (-a - \sqrt{a^2 - b^2})/b$. Now

$$\oint_C \frac{z^2 - 1}{z^2(z - r_1)(z - r_2)} dz = 2\pi i [\operatorname{Res}(f(z), 0) + \operatorname{Res}(f(z), r_1)] = 2\pi i \left[-\frac{2a}{b} + \frac{2\sqrt{a^2 - b^2}}{b} \right].$$

Thus

$$\int_0^{2\pi} \frac{\sin^2 \theta}{a + b \cos \theta} d\theta = \frac{2\pi}{b^2} (a - \sqrt{a^2 - b^2}), \quad a > b > 0.$$

Exercises 19.6

When $a = 5$, $b = 4$ we obtain

$$\int_0^{2\pi} \frac{\sin^2 \theta}{5 + 4 \cos \theta} d\theta = \frac{2\pi}{16}(5 - \sqrt{9}) = \frac{\pi}{4}.$$

- 35.** Consider the contour integral $\oint_C \frac{e^{az}}{1+e^z} dz$. The function $f(z) = \frac{e^{az}}{1+e^z}$ has simple poles at $z = \pi i, 3\pi i, 5\pi i, \dots$ in the upper plane. Using the contour in Figure 19.15 we have from Theorem 19.14

$$\oint_C = \int_{-r}^r + \int_{C_2} + \int_{C_3} + \int_{C_4} = 2\pi i \operatorname{Res}(f(z), \pi i) = -2\pi i e^{a\pi i}.$$

On C_2 , $z = r + iy$, $0 \leq y \leq \pi$, $dz = i dy$,

$$\left| \int_{C_2} \frac{e^{az}}{1+e^z} dz \right| \leq \frac{e^{ar}}{e^r - 1} (2\pi).$$

Because $0 < a < 1$, this last expression goes to 0 as $r \rightarrow \infty$. On C_3 , $z = x + 2\pi i$, $-r \leq x \leq r$, $dz = dx$,

$$\int_{C_3} \frac{e^{az}}{1+e^z} dz = \int_{-r}^r \frac{e^{a(x+2\pi i)}}{1+e^{x+2\pi i}} dx = -e^{2a\pi i} \int_{-r}^r \frac{e^{ax}}{1+e^x} dx.$$

On C_4 , $z = -r + iy$, $0 \leq y \leq 2\pi$, $dz = i dy$,

$$\left| \int_{C_4} \frac{e^{az}}{1+e^z} dz \right| \leq \frac{e^{-ar}}{1 - e^{-r}} (2\pi).$$

Because $0 < a$, this last expression goes to 0 as $r \rightarrow \infty$. Hence

$$\int_{-r}^r \frac{e^{ax}}{1+e^x} dx - e^{2a\pi i} \int_{-r}^r \frac{e^{ax}}{1+e^x} dx = -2\pi i e^{a\pi i}$$

gives, as $r \rightarrow \infty$,

$$(1 - e^{2a\pi i}) \int_{-\infty}^{\infty} \frac{e^{ax}}{1+e^x} dx = -2\pi i e^{a\pi i}.$$

That is

$$\int_{-\infty}^{\infty} \frac{e^{ax}}{1+e^x} dx = -\frac{2\pi i e^{a\pi i}}{1 - e^{2a\pi i}} = \frac{\pi}{\frac{e^{a\pi i} - e^{-a\pi i}}{2i}} = \frac{\pi}{\sin a\pi}.$$

- 36.** Using the Fourier sine transform with respect to y the partial differential equation becomes $\frac{d^2 U}{dx^2} - \alpha^2 U = 0$ and so

$$U(x, \alpha) = c_1 \cosh \alpha x + c_2 \sinh \alpha x.$$

The boundary condition $u(0, y)$ becomes $U(0, \alpha) = 0$ and so $c_1 = 0$. Thus $U(x, \alpha) = c_2 \sinh \alpha x$. Now to evaluate

$$U(\pi, \alpha) = \int_0^\infty \frac{2y}{y^4 + 4} \sin \alpha y dy = \int_{-\infty}^\infty \frac{y}{y^4 + 4} \sin \alpha y dy$$

we use the contour integral $\int_C \frac{ze^{i\alpha z}}{z^4 + 4} dz$ and

$$\int_{-\infty}^\infty \frac{xe^{i\alpha x}}{x^4 + 4} dx = 2\pi i [\operatorname{Res}(f(z), 1+i) + \operatorname{Res}(f(z), -1+i)] = 2\pi i \left[-\frac{1}{8}ie^{(-1+i)\alpha} + \frac{1}{8}ie^{(-1-i)\alpha} \right] = \frac{\pi}{2}(e^{-\alpha} \sin \alpha)i$$

$$\int_{-\infty}^\infty \frac{x \sin \alpha x}{x^4 + 4} dx = \operatorname{Im} \left(\int_{-\infty}^\infty \frac{xe^{i\alpha x}}{x^4 + 4} dx \right) = \frac{\pi}{2} e^{-\alpha} \sin \alpha.$$

Finally, $U(\pi, \alpha) = \frac{\pi}{2} e^{-\alpha} \sin \alpha = c_2 \sinh \alpha \pi$ gives $c_2 = \frac{\pi}{2} \frac{e^{-\alpha} \sin \alpha}{\sinh \alpha \pi}$. Hence $U(x, \alpha) = \frac{\pi}{2} \frac{e^{-\alpha} \sin \alpha}{\sinh \alpha \pi} \sinh \alpha x$ and

$$u(x, y) = \int_0^\infty \frac{e^{-\alpha} \sin \alpha}{\sinh \alpha \pi} \sinh \alpha x \sin \alpha y d\alpha.$$

Chapter 19 Review Exercises

1. True 2. False 3. False 4. True
 5. True 6. True 7. True 8. five
 9. $1/\pi$ 10. three; $-1/6$ 11. $|z - i| = \sqrt{5}$ 12. False

$$\begin{aligned} 13. \frac{e^{z(1+i)} + e^{z(1-i)}}{2} &= \frac{1}{2} \left(1 + z(1+i) + \frac{z^2}{2!}(1+i)^2 + \cdots \right) + \frac{1}{2} \left(1 + z(1-i) + \frac{z^2}{2!}(1-i)^2 + \cdots \right) \\ &= 1 + z \left[\frac{(1+i) + (1-i)}{2} \right] + \frac{z^2}{2!} \left[\frac{(1+i)^2 + (1-i)^2}{2} \right] + \cdots = 1 + \sum_{k=1}^{\infty} \frac{(\sqrt{2})^k \cos \frac{k\pi}{4}}{k!} z^k \end{aligned}$$

Here we have used $(1+i)^n = (\sqrt{2})^n e^{n\pi i/4}$ and $(1-i)^n = (\sqrt{2})^n e^{-n\pi i/4}$ so that

$$\frac{(1+i)^n + (1-i)^n}{2} = (\sqrt{2})^n \left[\frac{e^{n\pi i/4} + e^{-n\pi i/4}}{2} \right] = (\sqrt{2})^n \cos \frac{n\pi}{4}.$$

14. $\sin \frac{\pi}{z} = 0$ implies $z = \frac{1}{n}$, $n = \pm 1, \pm 2, \dots$. All singularities are isolated except the singularity $z = 0$.
 15. $f(z) = \frac{1}{z^4} \left[1 - \left(1 + \frac{iz}{1!} + \frac{i^2 z^2}{2!} + \frac{i^3 z^3}{3!} + \frac{i^4 z^4}{4!} + \cdots \right) \right] = -\frac{i}{z^3} + \frac{1}{2!z^2} + \frac{i}{3!z} - \frac{1}{4!} - \frac{iz}{5!} + \cdots$
 16. $e^{z/(z-2)} = e \cdot e^{2/(z-2)} = e \left(1 + \frac{2}{z-2} + \frac{2^2}{2!(z-2)^2} + \frac{2^3}{3!(z-2)^3} + \cdots \right) = e \sum_{k=0}^{\infty} \frac{2^k}{k!} (z-2)^{-k}$
 17. $(z-i)^2 \sin \frac{1}{z-i} = (z-i)^2 \left[\frac{1}{z-i} - \frac{1}{3!(z-i)^3} + \frac{1}{5!(z-i)^5} - \cdots \right] = \cdots + \frac{1}{5!(z-i)^3} - \frac{1}{3!(z-i)} + (z-i)$
 18. $\frac{1 - \cos z^2}{z^5} = \frac{1}{z^5} \left[1 - \left(1 - \frac{z^4}{2!} + \frac{z^8}{4!} - \frac{z^{12}}{6!} + \frac{z^{16}}{8!} - \cdots \right) \right] = \frac{1}{2!z} - \frac{z^3}{4!} + \frac{z^7}{6!} - \frac{z^{11}}{8!} + \cdots$
 19. (a) $f(z) = \frac{1}{z-3} - \frac{1}{z-1} = \frac{1}{1-z} - \frac{1}{3} \cdot \frac{1}{1-\frac{z}{3}} = (1+z+z^2+z^4+\cdots) - \frac{1}{3} \left(1 + \frac{z}{3} + \frac{z^2}{3^2} + \frac{z^3}{3^3} + \cdots \right)$

$$= \frac{2}{3} + \frac{8}{9}z + \frac{26}{27}z^2$$

 (b) $f(z) = -\frac{1}{z} \cdot \frac{1}{1-\frac{1}{z}} - \frac{1}{3} \cdot \frac{1}{1-\frac{z}{3}} = -\frac{1}{z} \left(1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \cdots \right) - \frac{1}{3} \left(1 + \frac{z}{3} + \frac{z^2}{3^2} + \frac{z^3}{3^3} + \cdots \right)$

$$= \cdots - \frac{1}{z^3} - \frac{1}{z^2} - \frac{1}{z} - \frac{1}{3} - \frac{z}{3^2} - \frac{z^2}{3^3} - \cdots$$

 (c) $f(z) = -\frac{1}{z} \cdot \frac{1}{1-\frac{1}{z}} + \frac{1}{z} \cdot \frac{1}{1-\frac{z}{3}} = -\frac{1}{z} \left(1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \cdots \right) + \frac{1}{z} \left(1 + \frac{z}{3} + \frac{z^2}{3^2} + \frac{z^3}{3^3} + \cdots \right)$

$$= \frac{2}{z^2} + \frac{8}{z^3} + \frac{26}{z^4} + \cdots$$

Chapter 19 Review Exercises

$$\begin{aligned} \text{(d)} \quad f(z) &= -\frac{1}{z-1} - \frac{1}{2} \cdot \frac{1}{1 - \frac{z-1}{2}} = -\frac{1}{z-1} - \frac{1}{2} \left(1 + \frac{z-1}{2} + \frac{(z-1)^2}{2^2} + \frac{(z-1)^3}{3!} + \cdots \right) \\ &= -\frac{1}{z-1} - \frac{1}{2} - \frac{z-1}{2^2} - \frac{(z-1)^2}{2^3} - \cdots \end{aligned}$$

$$\begin{aligned} 20. \text{ (a)} \quad f(z) &= \frac{1}{25} \left(1 - \frac{z}{5} \right)^{-2} = \frac{1}{25} \left[1 + (-2) \left(-\frac{z}{5} \right) + \frac{(-2)(-3)}{2!} \left(-\frac{z}{5} \right)^2 + \frac{(-2)(-3)(-4)}{3!} \left(-\frac{z}{5} \right)^3 + \cdots \right] \\ &= \frac{1}{25} + 2 \frac{z}{5^3} + 3 \frac{z^2}{5^4} + 4 \frac{z^3}{5^5} + \cdots \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad (z-5)^{-2} &= \frac{1}{z^2} \left(1 - \frac{5}{z} \right)^{-2} = \frac{1}{z^2} \left[1 + (-2) \left(-\frac{5}{z} \right) + \frac{(-2)(-3)}{2!} \left(-\frac{5}{z} \right)^2 + \frac{(-2)(-3)(-4)}{3!} \left(-\frac{5}{z} \right)^3 + \cdots \right] \\ &= \frac{1}{z^2} + 2 \frac{5}{z^3} + 3 \frac{5^2}{z^4} + 4 \frac{5^3}{z^5} + \cdots \end{aligned}$$

$$\text{(c)} \quad \frac{1}{(z-5)^2} \text{ is the Laurent series.}$$

$$21. \oint_C \frac{2z+5}{z(z+2)(z-1)^4} dz = 2\pi i [\text{Res}(f(z), 0) + \text{Res}(f(z), -2)] = \frac{404}{81} \pi i$$

$$22. \oint_C \frac{z^2}{(z-1)^3(z^2+4)} dz = 2\pi i \text{Res}(f(z), 1) = \frac{8\pi}{125} i$$

$$23. \oint_C \frac{1}{2 \sin z - 1} dz = 2\pi i \text{Res} \left(f(z), \frac{\pi}{6} \right) = \frac{2\pi}{\sqrt{3}} i$$

$$24. \oint_C \frac{z+1}{\sinh z} dz = 2\pi i [\text{Res}(f(z), 0) + \text{Res}(f(z), \pi i)] = 2\pi i [1 + (-\pi i - 1)] = 2\pi^2$$

$$\begin{aligned} 25. \oint_C \frac{e^{2z}}{z^4 + 2z^3 + 2z^2} dz &= 2\pi i [\text{Res}(f(z), 0) + \text{Res}(f(z), -1+i) + \text{Res}(f(z), -1-i)] \\ &= 2\pi i \left[\frac{1}{2} + \frac{e^{-2}}{4} (\cos 2 + i \sin 2) + \frac{e^{-2}}{4} (\cos 2 - i \sin 2) \right] = \pi(1 + e^{-2} \cos 2) i \end{aligned}$$

$$26. \oint_C \frac{1}{z^4 - 2z^2 + 4} dz = 2\pi i \left[\text{Res} \left(f(z), \frac{\sqrt{6}}{2} + \frac{\sqrt{2}}{2} i \right) + \text{Res} \left(f(z), -\frac{\sqrt{6}}{2} + \frac{\sqrt{2}}{2} i \right) \right] = \frac{\pi}{2\sqrt{2}}$$

$$27. \oint_C \frac{1}{z(e^z - 1)} dz = 2\pi i \text{Res}(f(z), 0) = -\pi i. \text{ Note: } z = 0 \text{ is a pole of order two, and so}$$

$$\text{Res}(f(z), 0) = \lim_{z \rightarrow 0} \frac{d}{dz} z^2 \cdot \frac{1}{z^2 \left(1 + \frac{z}{2!} + \frac{z^2}{3!} + \cdots \right)} = \lim_{z \rightarrow 0} -\frac{\left(\frac{1}{2!} + \frac{2z}{3!} + \cdots \right)}{\left(1 + \frac{z}{2!} + \frac{z^2}{3!} + \cdots \right)^2} = -\frac{1}{2}$$

$$28. \oint_C \frac{z}{(z-1)(z+1)^{10}} dz = 2\pi i [\text{Res}(f(z), 1) + \text{Res}(f(z), -1)] = 2\pi i \left[\frac{1}{2^{10}} + \left(-\frac{1}{2^{10}} \right) \right] = 0$$

29. Using two integrals,

$$\begin{aligned} \oint_C z e^{3/z} dz + \oint_C \frac{\sin z}{z^2(z-\pi)^3} dz &= 2\pi i \text{Res}(f(z), 0) + 2\pi i [\text{Res}(f(z), 0) + \text{Res}(f(z), \pi)] \\ &= 2\pi i \cdot \frac{9}{2} + 2\pi i \left[-\frac{1}{\pi^3} + \frac{2}{\pi^3} \right] = \left(9\pi + \frac{2}{\pi^2} \right) i. \end{aligned}$$

Note: In the first integral $z = 0$ is an essential singularity and the residue is obtained from the Laurent series

$$ze^{3/z} = \cdots + \frac{3^3}{3!z^2} + \frac{3^2}{2!z} + 3 + z.$$

$$30. \oint_C \csc \pi z \, dz = 2\pi i [\operatorname{Res}(f(z), 0) + \operatorname{Res}(f(z), 1) + \operatorname{Res}(f(z), 2)] = 2\pi i \left[\frac{1}{\pi} + \left(-\frac{1}{\pi} \right) + \frac{1}{\pi} \right] = 2i$$

$$31. \int_{-\infty}^{\infty} \frac{x^2}{(x^2 + 2x + 2)(x^2 + 1)^2} \, dx = 2\pi i [\operatorname{Res}(f(z), -1 + i) + \operatorname{Res}(f(z), i)] = 2\pi i \left[\frac{3}{25} - \frac{4}{25}i - \frac{3}{25} + \frac{9}{100}i \right] = \frac{7\pi}{50}$$

$$32. \int_{-\infty}^{\infty} \frac{x + ai}{x^2 + a^2} e^{ix} \, dx = \int_{-\infty}^{\infty} \frac{x \cos x - a \sin x}{x^2 + a^2} \, dx + i \int_{-\infty}^{\infty} \frac{a \cos x + x \sin x}{x^2 + a^2} \, dx = 0 + 2\pi i \operatorname{Res}(f(z), ai) = 2\pi i e^{-a}$$

Thus

$$\int_{-\infty}^{\infty} \frac{a \cos x + x \sin x}{x^2 + a^2} \, dx = 2\pi e^{-a}.$$

$$33. \int_0^{2\pi} \frac{\cos^2 \theta}{2 + \sin \theta} \, d\theta = \frac{1}{2} \oint_C \frac{z^4 + 2z^2 + 1}{z^2(z^2 + 4iz - 1)} \, dz \quad (C \text{ is } |z| = 1) = \pi i [\operatorname{Res}(f(z), 0) + \operatorname{Res}(f(z), (-2 + \sqrt{3})i)]$$

$$= \pi i [-4i + 2\sqrt{3}i] = (4 - 2\sqrt{3})\pi$$

[Note: The answer in the text is correct but not simplified.]

$$34. \int_0^{2\pi} \frac{\cos 3\theta}{5 - 4 \cos \theta} \, d\theta = -\frac{1}{2i} \oint_C \frac{z^6 + 1}{z^3(z - 2)(2z - 1)} \, dz \quad (C \text{ is } |z| = 1) = -\pi \left[\operatorname{Res}(f(z), 0) + \operatorname{Res}\left(f(z), \frac{1}{2}\right) \right]$$

$$= -\pi \left[\frac{21}{8} + \left(-\frac{65}{24} \right) \right] = \frac{\pi}{12}$$

35. The integrand of $\oint_C \frac{1 - e^{iz}}{z^2} \, dz$ has a simple pole at $z = 0$. Using a contour as in Figure 19.14 of Section 19.6 we have

$$\oint_C = \int_{C_R} + \int_{-R}^{-r} + \int_{-C_R} + \int_r^R = 0.$$

By taking limits as $R \rightarrow 0$ and as $r \rightarrow 0$, and using Theorem 19.17 we find

$$\text{P.V.} \int_{-\infty}^{\infty} \frac{1 - e^{ix}}{x^2} \, dx - \pi i \operatorname{Res}(f(z), 0) = 0.$$

Thus

$$\text{P.V.} \int_{-\infty}^{\infty} \frac{1 - \cos x - i \sin x}{x^2} \, dx = \pi.$$

Equating real parts gives

$$\text{P.V.} \int_{-\infty}^{\infty} \frac{1 - \cos x}{x^2} \, dx = \pi.$$

Finally

$$\int_0^{\infty} \frac{1 - \cos x}{x^2} \, dx = \frac{1}{2} \int_{-\infty}^{\infty} \frac{1 - \cos x}{x^2} \, dx = \frac{\pi}{2}.$$

36. We have

$$Ce^{-a^2 z^2} e^{ibz} \, dz = \int_{-r}^r + \int_{C_1} + \int_{C_2} + \int_{C_3} = 0$$

by the Cauchy-Goursat Theorem. Therefore

$$\int_{-r}^r = - \int_{C_1} - \int_{C_2} - \int_{C_3}.$$

Chapter 19 Review Exercises

Let C_1 and C_3 denote the vertical sides of the rectangle. By the ML-inequality, $\int_{C_1} \rightarrow 0$ and $\int_{C_3} \rightarrow 0$ as $r \rightarrow \infty$. On C_2 , $z = x + \frac{b}{2a^2}i$, $-r \leq x \leq r$, $dz = dx$,

$$\begin{aligned} \int_{-\infty}^{\infty} e^{-ax^2} e^{ibx} dx &= - \int_{\infty}^{-\infty} e^{-a^2(x+\frac{b}{2a^2}i)^2} e^{ib(x+\frac{b}{2a^2}i)} dx = \int_{-\infty}^{\infty} e^{-a^2x^2} e^{-b^2/4a^2} dx \\ &= \int_{-\infty}^{\infty} e^{-a^2x^2} (\cos bx + i \sin bx) dx = e^{-b^2/4a^2} \int_{-\infty}^{\infty} e^{-a^2x^2} dx. \end{aligned}$$

Using the given value of $\int_{-\infty}^{\infty} e^{-a^2x^2} dx$ and equating real and imaginary parts gives

$$\int_{-\infty}^{\infty} e^{-ax^2} \cos bx dx = \frac{\sqrt{\pi}}{a} e^{-b^2/4a^2} \quad \text{and so} \quad \int_0^{\infty} e^{-ax^2} \cos bx dx = \frac{\sqrt{\pi}}{2a} e^{-b^2/4a^2}.$$

$$\begin{aligned} \mathbf{37.} \quad a_k &= \frac{1}{2\pi i} \oint_C \frac{e^{(u/2)(z-z^{-1})}}{z^{k+1}} dz = \frac{1}{2\pi i} \int_0^{2\pi} \frac{e^{(u/2)(e^{it}-e^{-it})}}{(e^{it})^{k+1}} i e^{it} dt = \frac{1}{2\pi} \int_0^{2\pi} e^{(u/2)(2i \sin t)} e^{-kit} dt \\ &= \frac{1}{2\pi} \int_0^{2\pi} e^{-i(kt-u \sin t)} dt = \frac{1}{2\pi} \int_0^{2\pi} [\cos(kt-u \sin t) - i \sin(kt-u \sin t)] dt = \frac{1}{2\pi} \int_0^{2\pi} \cos(kt-u \sin t) dt \end{aligned}$$

since $\int_0^{2\pi} \sin(kt-u \sin t) dt = 0$. (To obtain this last result, expand the integrand and let $t = 2\pi - x$.)

20 Conformal Mappings and Applications

Exercises 20.1

1. For $w = \frac{1}{z}$, $u = \frac{x}{x^2 + y^2}$ and $v = \frac{-y}{x^2 + y^2}$. If $y = x$, $u = \frac{1}{2} \frac{1}{x}$, $v = -\frac{1}{2} \frac{1}{x}$, and so $v = -u$. The image is the line $v = -u$ (with the origin $(0, 0)$ excluded.)
2. If $y = 1$, $u = \frac{x}{x^2 + 1}$ and $v = \frac{-1}{x^2 + 1}$. It follows that $u^2 + v^2 = \frac{1}{x^2 + 1} = -v$ and so $u^2 + (v + \frac{1}{2})^2 = (\frac{1}{2})^2$. This is a circle with radius $r = \frac{1}{2}$ and center at $(0, -\frac{1}{2}) = -\frac{1}{2}i$. The circle can also be described by $|w + \frac{1}{2}i| = \frac{1}{2}$.
3. For $w = z^2$, $u = x^2 - y^2$ and $v = 2xy$. If $xy = 1$, $v = 2$ and so the hyperbola $xy = 1$ is mapped onto the line $v = 2$.
4. If $x^2 - y^2 = 4$, $u = 4$ and so the hyperbola $x^2 - y^2 = 4$ is mapped onto the vertical line $u = 4$.
5. For $w = \text{Ln } z$, $u = \log_e |z|$ and $v = \text{Arg } z$. The semi-circle $|z| = 1$, $y > 0$ may also be described by $r = 1$, $0 < \theta < \pi$. Therefore $u = 0$ and $0 < v < \pi$. The image is therefore the open line segment from $z = 0$ to $z = \pi i$.
6. If $\theta = \pi/4$, then $v = \theta = \pi/4$. In addition $u = \log_e r$ will vary from $-\infty$ to ∞ . The image is therefore the horizontal line $v = \pi/4$.
7. For $w = z^{1/2} = (re^{i\theta})^{1/2} = r^{1/2}e^{i\theta/2}$ and $\theta = \theta_0$, $w = \sqrt{r}e^{i\theta_0/2}$. Therefore $\text{Arg } w = \theta_0/2$ and so the image is the ray $\theta = \theta_0/2$.
8. If $r = 2$ and $0 \leq \theta \leq \frac{\pi}{2}$, $w = \sqrt{2}e^{i\theta/2}$. Therefore $|w| = \sqrt{2}$ and $0 \leq \text{Arg } w \leq \pi/4$. This image is a circular arc.
9. For $w = e^z$, $u = e^x \cos y$ and $v = e^x \sin y$. Therefore if $e^x \cos y = 1$, $u = 1$. The curve $e^x \cos y = 1$ is mapped into the line $u = 1$. Since $v = \frac{\sin y}{\cos y} = \tan y$, v varies from $-\infty$ to ∞ and the image is the line $u = 1$.
10. If $w = z + \frac{1}{z}$ and $z = e^{it}$, $w = e^{it} - e^{-it} = 2i \cos t$. Therefore $u = 2 \cos t$ and $v = 0$ and so the image is the closed interval $[-2, 2]$ on the u -axis.
11. The first quadrant may be described by $r > 0$, $0 < \theta < \pi/2$. If $w = 1/z$ and $z = re^{i\theta}$, $w = \frac{1}{r}e^{-i\theta}$. Therefore $\text{Arg } w = -\theta$ and so $-\pi/2 < \text{Arg } w < 0$. The image is therefore the fourth quadrant.
12. For $w = \frac{1}{z}$, $u = \frac{x}{x^2 + y^2}$ and $v = \frac{-y}{x^2 + y^2}$. The line $y = 0$ is mapped to the line $v = 0$, and, from Problem 2, the line $y = 1$ is mapped onto the circle $|w + \frac{1}{2}i| = \frac{1}{2}$. Since $f(\frac{1}{2}i) = -2i$, the region $0 \leq y \leq 1$ is mapped onto the points in the half-plane $v \leq 0$ which are on or outside the circle $|w + \frac{1}{2}i| = \frac{1}{2}$. (The image does not include the point $w = 0$.)
13. Since $w = e^{x+iy} = e^x e^{iy}$ and $\pi/4 \leq y \leq \pi/2$, $\pi/4 \leq \text{Arg } w \leq \pi/2$ and $|w| = e^x$. The image is therefore the angular wedge defined by $\pi/4 \leq \text{Arg } w \leq \pi/2$.
14. Since $w = e^{x+iy} = e^x e^{iy}$ and $0 \leq x \leq 1$, $0 \leq y \leq \pi$, we have $|w| = e^x$ and $\text{Arg } w = y$. Therefore $1 \leq |w| \leq e$ and $0 \leq \text{Arg } w \leq \pi$. These inequalities define a semi-angular region in the w -plane.
15. The mapping $w = z + 4i$ is a translation which maps the circle $|z| = 1$ to a circle of radius $r = 1$ and with center $w = 4i$. This circle may be described by $|w - 4i| = 1$.

Exercises 20.1

16. If $w = 2z - 1$ and $|z| = 1$, then, since $z = \frac{w+1}{2}$, $\left|\frac{w+1}{2}\right| = 1$ or $|w+1| = 2$. The image is a circle with center at $w = -1$ and with radius $r = 2$.
17. The mapping $w = iz$ is a rotation through 90° since $i = e^{i\pi/2}$. Therefore the strip $0 \leq y \leq 1$ is rotated through 90° and so the strip $-1 \leq u \leq 0$ is the image in the w -plane.
18. Since $w = (1+i)z = \sqrt{2}e^{i\pi/4}z$, the mapping is the composite of a rotation through 45° and a magnification by $\alpha = \sqrt{2}$. The image of the first quadrant is therefore the angular wedge $\pi/4 \leq \text{Arg } w \leq 3\pi/4$.
19. The power function $w = z^3$ changes the opening of the wedge $0 \leq \text{Arg } z \leq \pi/4$ by a factor of 3. Therefore the image region is $0 \leq \text{Arg } w \leq 3\pi/4$.
20. The power function $w = z^{1/2}$ changes the opening of the wedge $0 \leq \text{Arg } z \leq \pi/4$ by a factor of $1/2$. Therefore the image region is $0 \leq \text{Arg } w \leq \pi/8$.
21. We first let $z_1 = z - i$ to map the region $1 \leq y \leq 4$ to the region $0 \leq y_1 \leq 3$. We then let $w = e^{-i\pi/2}z_1$ to rotate this strip through -90° . Therefore $w = -i(z - i) = -iz - 1$ maps $1 \leq y \leq 4$ to the strip $0 \leq u \leq 3$.
22. The mapping $w = z - i$ lowers the strip $1 \leq y \leq 4$ one unit so that the image is $0 \leq v \leq 3$ in the w -plane.
23. We first let $z_1 = z - 1$ to map the disk $|z - 1| \leq 1$ to the disk $|z_1| \leq 1$. We then use the magnification $w = 2z_1$ to obtain $|w| \leq 2$ as the image. The composite of these two mappings is $w = 2(z - 1)$.
24. The mapping $w = iz$ will rotate the strip $-1 \leq x \leq 1$ through 90° so that the strip $-1 \leq v \leq 1$ results.
25. We first use $z_1 = e^{-i\pi/4}z$ to rotate the wedge $\pi/4 \leq \text{Arg } z \leq \pi/2$ to the wedge $0 \leq \text{Arg } z_1 \leq \pi/4$. The power function $w = z_1^4$ then changes the opening of this wedge by a factor of 4 so that the strip $0 \leq \text{Arg } w \leq \pi$ results. The composite of these two mappings is $w = (e^{-\pi/4i}z)^4 = e^{-\pi i}z^4 = -z^4$.
26. The magnification $z_1 = \frac{\pi}{4}z$ maps the strip $0 \leq y \leq 4$ to the strip $0 \leq y_1 \leq \pi$. By Example 1, Section 20.1, $w = e^{z_1}$ maps this strip onto the upper half-plane $v \geq 0$. The composite of these two mappings is $w = e^{\frac{\pi}{4}z}$.
27. By Example 1, Section 20.1, $z_1 = e^z$ maps the strip $0 \leq y \leq \pi$ onto the upper half-plane $y_1 \geq 0$, or $0 \leq \text{Arg } z_1 \leq \pi$. The power function $w = z_1^{3/2}$ changes the opening of this wedge by a factor of $3/2$ so the wedge $0 \leq \text{Arg } w \leq 3\pi/2$ results. The composite of these two mappings is $w = (e^z)^{3/2} = e^{3z/2}$.
28. The power function $z_1 = z^{2/3}$ maps the wedge $0 \leq \text{Arg } z \leq 3\pi/2$ to the upper half-plane $0 \leq \text{Arg } z_1 \leq \pi$. We then let $w = e^{-i\pi/2}z_1 + 2 = -iz_1 + 2$ to rotate the upper half-plane through -90° and then translate 2 units to the right. Therefore the composite function is $w = -iz^{2/3} + 2$ and the image region is the half-plane $u \geq 2$.
29. We may obtain the image region by (1) rotating R through 180° and (2) raising the resulting region one unit in the vertical direction. Therefore $w = e^{i\pi}z + i = -z + i$.
30. The mapping $z_1 = -(z - \pi i)$ lowers R by π units in the vertical direction and then rotates the resulting region through 180° . The image region R_1 is upper half-plane $y_1 \geq 0$. By Example 1, Section 20.1, $w = \text{Ln } z_1$ maps R_1 onto the strip $0 \leq v \leq \pi$. The composite of these two mappings is $w = \text{Ln}(\pi i - z)$.

Exercises 20.2

1. Since $f'(z) = 3(z^2 - 1)$, f is conformal at all points except $z = \pm 1$.
2. Since $f'(z) = -\sin z$, f is conformal at all points except $z = \pm n\pi$.
3. $f'(z) = 1 + e^z$ and $1 + e^z = 0$ for $z = \pm i + 2n\pi i$. Therefore f is conformal except for $z = \pi i + 2n\pi i$.
4. $f(z) = z + \operatorname{Ln} z + 1$ is analytic for all z except $z = y \leq 0$. But $f'(z) = 1 + 1/z$ and $1 + 1/z = 0$ for $z = -1$. Therefore $f(z)$ is conformal at all points except those on the branch cut $y \leq 0$.
5. $f(z) = (z^2 - 1)^{1/2} = e^{\frac{1}{2}\operatorname{Ln}(z^2 - 1)}$ is analytic for all z outside the interval $[-1, 1]$ on the real axis. This follows from the fact that $z^2 - 1 = (x^2 - y^2 - 1) + 2xyi$ and so we must exclude values of z for which $v = 2xy = 0$ and $u = x^2 - y^2 - 1 \leq 0$. Therefore $y = 0$ and $x^2 \leq 1$. $f'(z) = z/(z^2 - 1)^{1/2}$ is non-zero outside this interval. Therefore f is conformal except for $z = x$, $-1 \leq x \leq 1$.
6. The function $f(z) = \pi i - \frac{1}{2}[\operatorname{Ln}(z + 1) + \operatorname{Ln}(z - 1)]$ is analytic except on the branch cut $x - 1 \leq 0$ or $x \leq 1$, and

$$f'(z) = -\frac{1}{2} \left(\frac{1}{z+1} + \frac{1}{z-1} \right) = -\frac{z}{z^2 - 1}$$

is non-zero for $z \neq 0, \pm 1$. Therefore f is conformal except for $z = x$, $x \leq 1$.

7. $f(z) = \cos z = \sin(\pi/2 - z)$ is the composite of $z_1 = \pi/2 - z$ and $w = \sin z$. The strip $0 \leq x \leq \pi$ is mapped onto the strip $-\pi/2 \leq x \leq \pi/2$ by $z_1 = \pi/2 - z$ and $w = \sin z$ maps this strip onto the region shown in Figure 20.11(b). The horizontal segment $z(t) = t + ib$, $0 < t < \pi$ is first mapped to the horizontal segment $z_i(t) = (\pi/2 - t) - ib$, $0 < t < \pi$. This latter segment is mapped onto the lower or upper portion of the ellipse

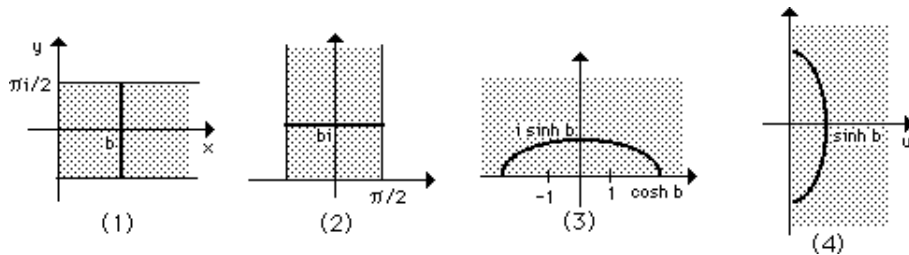
$$\frac{u^2}{\cosh^2 b} + \frac{v^2}{\sinh^2 b} = 1$$

according to whether $b > 0$ or $b < 0$. See Figure 20.11.

8. $f(z) = \sinh z = -i \sin(iz)$ is the composite of $z_1 = iz$, $z_2 = \sin z_1$, and $w = -iz_2$. The strip $-\pi/2 \leq y \leq \pi/2$, $x \geq 0$ is rotated through 90° so that the image is the strip $-\pi/2 \leq x \leq \pi/2$, $y \geq 0$. This latter strip is mapped to the upper half-plane $y_2 \geq 0$ by $z_2 = \sin z_1$, and $w = -iz_2$ rotates this upper half-plane through -90° . The final image region is the half-plane $u \geq 0$. A vertical line segment in the original strip is mapped to the right hand side of the ellipse

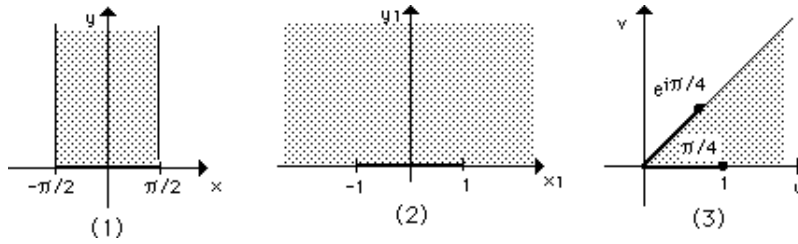
$$\frac{u^2}{\sinh^2 b} + \frac{v^2}{\cosh^2 b} = 1.$$

See the figures below.



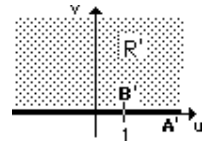
9. $f(z) = (\sin z)^{1/4}$ is the composite of $z_1 = \sin z$ and $w = z_1^{1/4}$. The region $-\pi/2 \leq x \leq \pi/2$, $y \geq 0$ is mapped to the upper half-plane $y_2 \geq 0$ by $z_1 = \sin z$ (See Example 2) and the power function $w = z_1^{1/4}$ maps this upper half-plane to the angular wedge $0 \leq \operatorname{Arg} w \leq \pi/4$. The real interval $[-\pi/2, \pi/2]$ is first mapped to $[-1, 1]$ and then to the union of the line segments from $e^{i\pi/4}$ to 0 and 0 to 1. See the figures on the following page.

Exercises 20.2

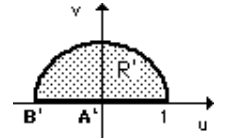


10. From Example 3, $u = (r + 1/r) \cos \theta$ and $v = (r - 1/r) \sin \theta$. If $r = 1$, $u = 2 \cos \theta$ and $v = 0$. Therefore the image of the circle $|z| = 1$ is the real interval $[-2, 2]$. If $0 < r < 1$, $u^2/a^2 + v^2/b^2 = 1$ where $a = r + 1/r > 2$ and $b = 1/r - r$. The resulting ellipse together with $[-2, 2]$ fill up the entire w -plane. Therefore the image of the region $|z| \leq 1$ is the entire w -plane. If $z = x$ and $-1 \leq x \leq 1$, $w = x + 1/x$. Therefore w is real and, from an analysis of the graph of $w = x + 1/x$ for $-1 \leq x \leq 1$, $|w| \geq 2$. Therefore the segment $[-1, 1]$ (with 0 excluded) is mapped on those points $w = u$ on the u -axis for which $|u| \geq 2$.

11. Using H-4 with $a = 2$, $w = \cos(\pi z/2)$ maps R onto the target region R' . The image of AB is shown in the figure.

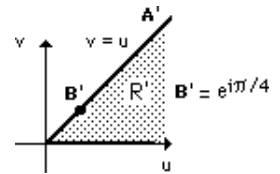


12. Using C-3 $w = e^z$ maps R onto the target region R' . The image of AB is shown in the figure.



13. Using H-5, $z_1 = \left(\frac{1+z}{1-z}\right)^2$ maps R onto the upper half-plane $y_1 \geq 0$, and $w = z_1^{1/4}$ maps this half-plane onto the target region R' . The image of AB is shown in the figure, and

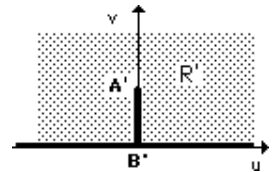
$$w = \left(\frac{1+z}{1-z}\right)^{1/2}.$$



14. Using H-1, $z_1 = i \frac{1-z}{1+z}$ maps R onto the upper half-plane $y_1 \geq 0$, and using M-4 with $a = 1$, $w = (z_1^2 - 1)^{1/2}$ maps this upper half-plane onto the target region R' . Therefore

$$w = \left[-\left(\frac{1-z}{1+z}\right)^2 - 1 \right]^{1/2}$$

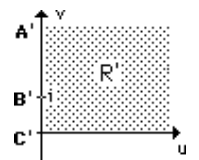
and the image of AB is shown in the figure.



15. Using H-6, $z_1 = \frac{e^{\pi/z} + e^{-\pi/z}}{e^{\pi/z} - e^{-\pi/z}}$ maps R onto the upper half-plane $y_1 \geq 0$, and $w = z_1^{1/2}$ maps this half-plane onto the target region R' . Therefore

$$w = \left(\frac{e^{\pi/z} + e^{-\pi/z}}{e^{\pi/z} - e^{-\pi/z}} \right)^{1/2}$$

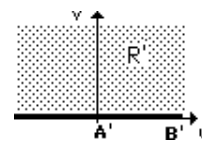
and the image of AB is shown in the figure.



16. We can translate R to the origin, magnify by 2, and then use H-1 to reach the target region

R' . Therefore $z_1 = 2(z - \frac{1}{2})$, $w = i \frac{1-z}{1+z}$ and so

$$w = i \frac{1 - (2z - 1)}{1 + (2z - 1)} = i \frac{1 - z}{z}.$$



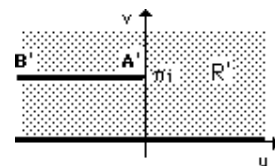
The image of AB is shown in the figure.

17. $z_1 = \text{Ln} z$ maps R onto the horizontal strip $0 \leq y_1 \leq \pi$, and, to prepare this strip for mapping by the sine function, we let $z_2 = -iz_1 - \pi/2$ to obtain the vertical strip $-\pi/2 \leq x \leq \pi/2$. Finally $w = \sin z_2$ maps this vertical strip onto the target region R' . Therefore

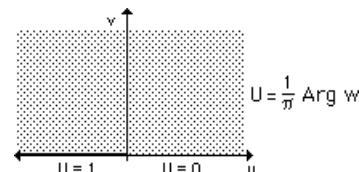
$$w = \sin \left(-i \text{Ln} z - \frac{\pi}{2} \right).$$

The image of AB is the real interval $(-\infty, -1]$.

18. Using E-9, $z_1 = \cosh z$ maps R onto the upper half-plane $y_1 \geq 0$. Using M-7, $w = z_1 + \text{Ln} z_1 + 1$ maps this half-plane onto the target region R' . Therefore $w = \cosh z + \text{Ln}(\cosh z) + 1$ and the image of AB is shown in the figure.

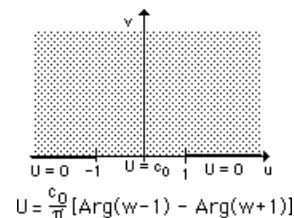


In Exercises 19-22, we find a conformal mapping $w = f(z)$ that maps the given region R onto the upper half-plane $v \geq 0$ and transfers the boundary conditions so that the resulting Dirichlet problem is as shown in the figure.



19. $f(z) = z^4$ and so $u = U(f(z)) = \frac{1}{\pi} \text{Arg} z^4$. The solution may also be written as $u(r, \theta) = 4\theta/\pi$.
20. $f(z) = \left(\frac{1+z}{1-z} \right)^2$, using H-5, and so $u = U(f(z)) = \frac{1}{\pi} \text{Arg} \left(\frac{1+z}{1-z} \right)^2$.
21. $f(z) = i \frac{1-z}{1+z}$, using H-1, and so $u = U(f(z)) = \frac{1}{\pi} \text{Arg} \left(\frac{1-z}{1+z} \right)$. The solution may also be written as $u(x, y) = \frac{1}{\pi} \tan^{-1} \left(\frac{1-x^2-y^2}{2y} \right)$.
22. $f(z) = \frac{1}{2} \left(z + \frac{1}{z} \right)$, using H-3 with $a = 1$, and so $u = U(f(z)) = \frac{1}{\pi} \text{Arg} \frac{1}{2} \left(z + \frac{1}{z} \right)$. The solution may also be written as $u(x, y) = \frac{1}{\pi} \tan^{-1} \left[\frac{y}{x} \frac{x^2 + y^2 - 1}{x^2 + y^2 + 1} \right]$.

In Exercises 23-26, we find a conformal mapping $w = f(z)$ that maps the given region R onto the upper half-plane $v \geq 0$ and transfers the boundary conditions so that the resulting Dirichlet problem is as shown in the figure.



23. $f(z) = z^2$ and $c_0 = 1$. Therefore $u = \frac{1}{\pi} [\text{Arg}(z^2 - 1) - \text{Arg}(z^2 + 1)]$.

Exercises 20.2

24. The mapping $z_1 = z^2$ maps R onto the region R_1 defined by $y_1 \geq 0$, $|z_1| \geq 1$ and shown in H-3, and $w = \frac{1}{2} \left(z_1 + \frac{1}{z_1} \right)$ maps R_1 onto the upper half-plane $v \geq 0$. Letting $c_0 = 5$,

$$u = \frac{5}{\pi} \left[\operatorname{Arg} \left(\frac{1}{2} \left[z^2 + \frac{1}{z^2} \right] - 1 \right) - \operatorname{Arg} \left(\frac{1}{2} \left[z^2 + \frac{1}{z^2} \right] + 1 \right) \right].$$

25. $f(z) = e^{\pi z}$, using H-2 with $a = 1$. Letting $c_0 = 10$,

$$u = \frac{10}{\pi} [\operatorname{Arg}(e^{\pi z} - 1) - \operatorname{Arg}(e^{\pi z} + 1)].$$

26. $f(z) = \cos(\pi z/2)$ using H-4 with $a = 2$. Letting $c_0 = 4$,

$$u = \frac{4}{\pi} [\operatorname{Arg}(\cos(\pi z/2) - 1) - \operatorname{Arg}(\cos(\pi z/2) + 1)].$$

27. (a) If $u = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2}$,

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{\partial^4 \phi}{\partial x^4} + 2 \frac{\partial^4 \phi}{\partial x^2 \partial y^2} + \frac{\partial^4 \phi}{\partial y^4} = 0$$

since ϕ is assumed to be biharmonic.

- (b) If $g = u + iv$, then $\phi = \operatorname{Re}(\bar{z}g(z)) = xu + yv$.

$$\begin{aligned} \frac{\partial^2 \phi}{\partial x^2} &= 3 \frac{\partial u}{\partial x} + x \frac{\partial^2 u}{\partial x^2} + y \frac{\partial^2 v}{\partial x^2} \\ \frac{\partial^2 \phi}{\partial y^2} &= 2 \frac{\partial v}{\partial y} + x \frac{\partial^2 u}{\partial y^2} + y \frac{\partial^2 v}{\partial y^2}. \end{aligned}$$

Since u and v are harmonic and $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$,

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 2 \frac{\partial u}{\partial x} + 2 \frac{\partial v}{\partial y} = 4 \frac{\partial u}{\partial x}.$$

Now $u_1 = \frac{\partial u}{\partial x}$ is also harmonic and so $\frac{\partial^2 u_1}{\partial x^2} + \frac{\partial^2 u_1}{\partial y^2} = 0$. But

$$\frac{\partial^2 u_1}{\partial x^2} + \frac{\partial^2 u_1}{\partial y^2} = \frac{1}{4} \left[\frac{\partial^4 \phi}{\partial x^4} + 2 \frac{\partial^4 \phi}{\partial x^2 \partial y^2} + \frac{\partial^4 \phi}{\partial y^4} \right]$$

and so ϕ is biharmonic.

Exercises 20.3

1. (a) For $T(z) = i/z$, $T(0) = \infty$, $T(1) = i$, and $T(\infty) = 0$.
 (b) If $|z| = 1$, $|w| = |i/z| = 1/|z| = 1$. Therefore the image of the circle $|z| = 1$ is the circle $|w| = 1$ in the w -plane. The circle $|z - 1| = 1$ passes through the pole at $z = 0$ and so the image is a line. Since $T(2) = \frac{1}{2}i$ and $T(1+i) = \frac{1}{2} + \frac{1}{2}i$, the image is the line $v = \frac{1}{2}$.
 (c) The disk $|z| \leq 1$ is mapped onto the disk $|w| \geq 1$.
2. (a) For $T(z) = \frac{1}{z-1}$, $T(0) = -1$, $T(1) = \infty$, and $T(\infty) = 0$.

- (b) The circle $|z| = 1$ passes through the pole at $z = 1$ and so the image is a line. Since $T(-1) = -\frac{1}{2}$ and $T(i) = -\frac{1}{2} - \frac{1}{2}i$, the image is the line $u = -\frac{1}{2}$. If $|z - 1| = 1$, $|w| = 1/|z - 1| = 1$ and the image is the circle $|w| = 1$ in the w -plane.
- (c) Since $T(0) = -1$, the image of the disk $|z| \leq 1$ is the half-plane $u = -\frac{1}{2}$.
3. (a) For $T(z) = \frac{z+1}{z-1}$, $T(0) = -1$, $T(1) = \infty$, and $T(\infty) = 1$.
- (b) The circle $|z| = 1$ passes through the pole at $z = 1$ and so the image is a line. Since $T(-1) = 0$ and $T(i) = -i$, the image is the line $u = 0$. If $|z - 1| = 1$,
- $$|w - 1| = \left| \frac{z+1}{z-1} - 1 \right| = \frac{2}{|z-1|} = 2$$
- and so the image is the circle $|w - 1| = 2$ in the w -plane.
- (c) Since $T(0) = -1$, the image of the disk $|z| \leq 1$ is the half-plane $u \leq 0$.
4. (a) For $T(z) = \frac{z-i}{z}$, $T(0) = \infty$, $T(1) = 1 - i$, and $T(\infty) = 1$.
- (b) $T(i) = 0$, $T(1) = 1 - i$ and $T(-i) = 2$. The image of $|z| = 1$ is therefore a circle which passes through 0, $1 - i$, and 2. This is the circle $|w - 1| = 1$. The circle $|z - 1| = 1$ passes through the pole at $z = 0$ and so the image is a line. Since $T(2) = 1 - \frac{1}{2}i$ and $T(1+i) = \frac{1}{2} - \frac{1}{2}i$, the image is the line $v = -\frac{1}{2}$.
- (c) $T(\frac{1}{2}) = 1 - 2i$ which is exterior to the circle $|w - 1| = 1$. Therefore the image of $|z| \leq 1$ is $|w - 1| \geq 1$, the exterior of the circle $|w - 1| = 1$ in the w -plane.

5. $S^{-1}(T(z)) = \frac{az+b}{cz+d}$ where

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \text{adj} \left(\begin{bmatrix} i & 1 \\ 1 & -1 \end{bmatrix} \right) \begin{bmatrix} 1 & 0 \\ i & -1 \end{bmatrix} = \begin{bmatrix} -1-i & 1 \\ -2 & -i \end{bmatrix}.$$

Therefore

$$S^{-1}(T(z)) = \frac{(-1-i)z+1}{-2z-i} = \frac{(1+i)z-1}{2z+i} \quad \text{and} \quad S^{-1}(w) = \frac{-w-1}{-w+i} = \frac{w+1}{w-i}.$$

6. $S^{-1}(T(z)) = \frac{az+b}{cz+d}$ where

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \text{adj} \left(\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \right) \begin{bmatrix} i & 0 \\ 1 & -2i \end{bmatrix} = \begin{bmatrix} -1+i & 2i \\ 2-i & -4i \end{bmatrix}.$$

Therefore

$$S^{-1}(T(z)) = \frac{(-1+i)z+2i}{(2-i)z-4i} \quad \text{and} \quad S^{-1}(w) = \frac{w-1}{-w+2}.$$

7. $S^{-1}(T(z)) = \frac{az+b}{cz+d}$ where

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \text{adj} \left(\begin{bmatrix} 1 & -2 \\ 1 & -1 \end{bmatrix} \right) \begin{bmatrix} 2 & -3 \\ 1 & -3 \end{bmatrix} = \begin{bmatrix} 0 & -3 \\ -1 & 0 \end{bmatrix}.$$

Therefore

$$S^{-1}(T(z)) = \frac{-3}{-z} = \frac{3}{z} \quad \text{and} \quad S^{-1}(w) = \frac{-w+2}{-w+1} = \frac{w-2}{w-1}.$$

8. $S^{-1}(T(z)) = \frac{az+b}{cz+d}$ where

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \text{adj} \left(\begin{bmatrix} 2-i & 0 \\ 1 & -1-i \end{bmatrix} \right) \begin{bmatrix} 1 & -1+i \\ i & -2 \end{bmatrix} = \begin{bmatrix} -1-i & 2 \\ 2i & -3+i \end{bmatrix}.$$

Exercises 20.3

Therefore

$$S^{-1}(T(z)) = \frac{(-1-i)z+2}{2iz-3+i} \quad \text{and} \quad S^{-1}(w) = \frac{(-1-i)w}{-w+2-i} = \frac{(1+i)w}{w-2+1}.$$

9. $T(z) = \frac{(z-z_1)(z_2-z_3)}{(z-z_3)(z_2-z_1)}$ maps z_1, z_2, z_3 to $0, 1, \infty$. Therefore $T(z) = \frac{(z+1)(-2)}{(z-2)(1)} = -2 \frac{z+1}{z-2}$ maps $-1, 0, 2$ to $0, 1, \infty$.

10. $T(z) = \frac{(z-i)i}{(z+i)(-i)} = -\frac{z-i}{z+i}$ maps $i, 0, -i$ to $0, 1, \infty$.

11. $S(w) = \frac{(w-w_1)(w_2-w_3)}{(w-w_3)(w_2-w_1)}$ maps w_1, w_2, w_3 to $0, 1, \infty$ and so S' maps $0, 1, \infty$ to w_1, w_2, w_3 . Therefore $z = \frac{(w-0)(i-2)}{(w-2)(i-0)}$ and so $w = \frac{2z}{z-1-2i}$ maps $0, 1, \infty$ to $0, i, 2$.

12. As in Exercise 11, $z = \frac{(w-1-i)(-1+i)}{(w-1+i)(-1-i)}$ and, solving for w , $w = \frac{2z-2}{(1+i)z-1+i}$ maps $0, 1, \infty$ to $1+i, 0, 1-i$.

13. Using the cross-ratio formula (7),

$$\frac{(w-i)(1)}{w(1)} = \frac{(z+1)(-1)}{(z-1)(1)}$$

and so $w = \frac{i}{2} \frac{z-1}{z}$ maps $-1, 0, 1$ to $i, \infty, 0$.

14. Using the cross-ratio formula (7),

$$\frac{(1)(-i-1)}{(w-1)(1)} = \frac{(z+1)(-1)}{(z-1)(1)}$$

and so $w = \frac{(2+i)z-i}{z+1}$ maps $-1, 0, 1$ to $\infty, -i, 1$.

15. Using the cross-ratio formula (7),

$$S(w) = \frac{(w+1)(-3)}{(w-3)(1)} = \frac{(z-1)(2i)}{(z+i)(i-1)} = T(z).$$

We can solve for w to obtain

$$w = 3 \frac{(1+i)z + (1-i)}{(-3+5i)z - 3-5i}.$$

Alternatively we can apply the matrix method to compute $w = S^{-1}(T(z))$.

16. Using the cross-ratio formula (7),

$$S(w) = \frac{(w-i)(-i-1)}{(w-1)(-2i)} = \frac{(z-1)(2i)}{(z+i)(i-1)} = T(z).$$

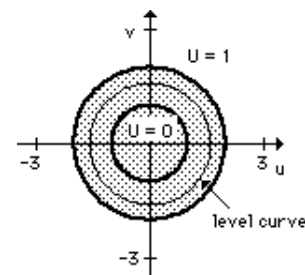
We can solve for w to obtain

$$w = \frac{(1+2i)z-i}{iz+1-2i}.$$

Alternatively we can apply the matrix method to compute $w = S^{-1}(T(z))$.

17. From Example 2, $z = \frac{w+2}{w-1}$ maps the annular region $1 < |w| < 2$ onto the region R and the circle $|w| = 1$ corresponds to the line $x = -1/2$. Solving for w , $w = \frac{z+2}{z-1}$ maps R onto the annulus and the transferred boundary conditions are shown in the figure. The solution to this new Dirichlet problem is $U = \log_e r / \log_e 2$ and so

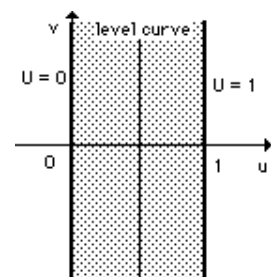
$$u = U(f(z)) = \frac{1}{\log_e 2} \log_e \left| \frac{z+2}{z-1} \right|$$



is the solution to the Dirichlet problem in Figure 20.37. The level curves are the images of the level curves of U , $|w| = r$ for $1 < r < 2$ under the mapping $z = \frac{w+2}{w-1}$. Since these circles do not pass through the pole at $w = 1$, the images are circles.

18. The mapping $T(z) = \frac{1}{2} \frac{z+1}{z}$ maps $-1, 1, 0$ to $0, 1, \infty$ and maps each of the two circles in R to lines since both circles pass through the pole at $z = 0$. Since $T(\frac{1}{2} + \frac{1}{2}i) = 1 - i$ and $T(1) = 1$, the circle $|z - \frac{1}{2}| = \frac{1}{2}$ is mapped onto the line $u = 1$. Likewise, the circle $|z + \frac{1}{2}| = \frac{1}{2}$ is mapped onto the line $u = 0$. The transferred boundary conditions are shown in the figure and $U(u, v) = u$ is the solution. The solution to the Dirichlet problem in Figure 20.38 is

$$u = U(T(z)) = \operatorname{Re} \left(\frac{1}{2} \frac{z+1}{z} \right) = \frac{1}{2} + \frac{1}{2} \frac{x}{x^2 + y^2}.$$



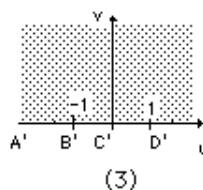
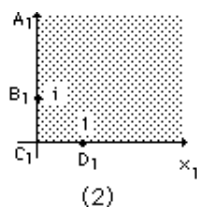
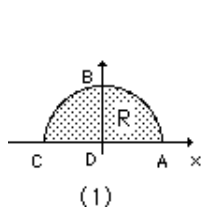
The level curves $u = c$ are the circles with centers on the x -axis which pass through the origin. The level curve $u = \frac{1}{2}$, however, is the vertical line $x = 0$.

19. The linear fractional transformation that sends $1, i, -i$ to $0, 1, -1$ satisfies the cross-ratio equation

$$\frac{(w-0)(2)}{(w+1)(1)} = \frac{(z-1)(2i)}{(z+i)(i-1)}.$$

Solving for w , $w = i \frac{1-z}{1+z} = T(z)$. Since $T(0) = i$, the image of the disk $|z| \leq 1$ is the upper half-plane $v \geq 0$.

20. The linear fractional transformation that sends $1, i, -1$ to $\infty, i, 0$ is $T(z) = \frac{1+z}{1-z}$ and so $f(z) = \left(\frac{1+z}{1-z} \right)^2$ maps $1, i, -1$ to $\infty, -1, 0$. The upper semi-circle is mapped by T to the positive imaginary axis, and the real interval $[-1, 1]$ is mapped to the positive real axis. Since $T\left(\frac{i}{2}\right) = \frac{3}{5} + \frac{4}{5}i$, the image of R under T is the first quadrant. $w = z_1^2$ doubles the size of the opening so that the image under f is the upper half-plane $v \geq 0$. See the figures below.



Exercises 20.3

$$21. T_2(T_1(z)) = \frac{a_2 T_1(z) + b_2}{c_2 T_1(z) + d_2} = \frac{a_2 \frac{a_1 z + b_1}{c_1 z + d_1} + b_2}{c_2 \frac{a_1 z + b_1}{c_1 z + d_1} + d_2} = \frac{a_1 a_2 z + a_2 b_1 + b_2 c_1 z + b_2 d_1}{a_1 c_2 z + b_1 c_2 + c_1 d_2 z + d_1 d_2} = \frac{(a_1 a_2 + c_1 b_2)z + (b_1 a_2 + d_1 b_2)}{(a_1 c_2 + c_1 d_2)z + (b_1 c_2 + d_1 d_2)}$$

$$22. |w - w_1| = \lambda |w - w_2| \implies (u - u_1)^2 + (v - v_1)^2 = \lambda^2 [(u - u_2)^2 + (v - v_2)^2]$$

The latter equation may be put in the form

$$Au^2 + Bv^2 + Cu + Dv + F = 0$$

where $A = B = 1 - \lambda^2$. If $\lambda = 1$, the line $Cu + Dv + F = 0$ results. If $\lambda > 0$ and $\lambda \neq 1$, then the equation defines a circle.

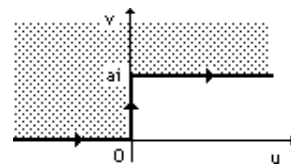
Exercises 20.4

$$1. \arg f'(t) = -\frac{1}{2} \text{Arg}(t-1) = \begin{cases} -\pi/2, & t < 1 \\ 0, & t > 1 \end{cases}. \text{ Since } f(1) = 0, \text{ the image is the first quadrant.}$$

$$2. \arg f'(t) = -\frac{1}{3} \text{Arg}(t+1) = \begin{cases} -\pi/3, & t < -1 \\ 0, & t > -1 \end{cases}. \text{ In (2), } \alpha_1 = 2\pi/3 \text{ and since } f(-1) = 0, \text{ the image is the wedge } 0 \leq \text{Arg } w \leq 2\pi/3.$$

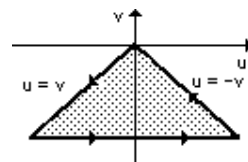
$$3. \arg f'(t) = -\frac{1}{2} \text{Arg}(t+1) + \frac{1}{2} \text{Arg}(t-1) = \begin{cases} 0, & t < -1 \\ \pi/2, & -1 < t < 1 \\ 0, & t > 1 \end{cases}$$

and $\alpha_1 = \pi/2$ and $\alpha_2 = 3\pi/2$. Since $f(-1) = 0$, the image of the upper half-plane is the region shown in the figure.



$$4. \arg f'(t) = -\frac{1}{2} \text{Arg}(t+1) - \frac{3}{4} \text{Arg}(t-1) = \begin{cases} -5\pi/4, & t < -1 \\ -3\pi/4, & -1 < t < 1 \\ 0, & t > 1 \end{cases}$$

and $\alpha_1 = \pi/2$ and $\alpha_2 = \pi/4$. Since $f(-1) = 0$, the image of the upper half-plane is the region shown in the figure.



5. Since $\alpha_1 = \alpha_2 = \alpha_3 = \pi/2$, $\alpha_i/\pi - 1 = -1/2$ and so $f'(z) = A(z+1)^{-1/2}z^{-1/2}(z-1)^{-1/2}$ for some constant A .
6. Since $\alpha_1 = \pi/3$ and $\alpha_2 = \pi/2$, $\alpha_1/\pi - 1 = -2/3$ and $\alpha_2/\pi - 1 = -1/2$ and so $f'(z) = A(z+1)^{-2/3}z^{-1/2}$ for some constant A .
7. Since $\alpha_1 = \alpha_2 = 2\pi/3$, $\alpha_i/\pi - 1 = -1/3$ and so $f'(z) = A(z+1)^{-1/3}z^{-1/3}$ for some constant A .
8. Since $\alpha_1 = 3\pi/2$ and $\alpha_2 = \pi/4$, $\alpha_1/\pi - 1 = 1/2$ and $\alpha_2/\pi - 1 = -3/4$. Therefore $f'(z) = A(z+1)^{1/2}z^{-3/4}$ for some constant A .
9. Since $\alpha_1 = \alpha_2 = \pi/2$, $f'(z) = A(z+1)^{-1/2}(z-1)^{-1/2} = A/(z^2-1)^{1/2}$. Therefore $f(z) = A \cosh^{-1} z + B$. But $f(-1) = \pi i$ and $f(1) = 0$. Since $\cosh^{-1} 1 = 0$, $B = 0$. Since $\cosh^{-1}(-1) = \pi i$, $\pi i = A(\pi i)$ and so $A = 1$. Hence $f(z) = \cosh^{-1} z$.
10. Since $\alpha_1 = 3\pi/2 = \alpha_2$, $f'(z) = A(z+1)^{1/2}(z-1)^{1/2} = A(z^2-1)^{1/2}$. Therefore

$$f(z) = A \left[\frac{z(z^2-1)^{1/2}}{2} - \frac{1}{2} \text{Ln}(z + (z^2-1)^{1/2}) \right] + B.$$

but $f(-1) = -ai$ and $f(1) = ai$. It follows that

$$ai = f(1) = B, \quad -ai = f(-1) = A \left(-\frac{\pi i}{2} \right) + B$$

and so $B = ai$ and $A = 4a/\pi$. Therefore

$$f(z) = \frac{4a}{\pi} \left[\frac{z(z^2 - 1)^{1/2}}{2} - \frac{1}{2} \operatorname{Ln}(z + (z^2 - 1)^{1/2}) \right] + ai = \frac{4a}{\pi} \left[\frac{z(z^2 - 1)^{1/2}}{2} - \frac{1}{2} \cosh^{-1} z \right] + ai.$$

11. $f'(z) = A(z+1)^{\alpha_1/\pi-1} z^{\alpha_2/\pi-1} (z-1)^{\alpha_3/\pi-1}$ from (3). Since $f(-1) = \pi i$, $\alpha_1 \rightarrow \pi$ as $w_1 \rightarrow \infty$ in the horizontal direction. Likewise $\alpha_2 \rightarrow 0$ and $\alpha_3 \rightarrow \pi$. This suggests we examine $f'(z) = Az^{-1} = A/z$. Therefore $f(z) = A \operatorname{Ln} z + B$. But $f(-1) = \pi i$ and $f(1) = 0$. It follows that $A = 1$ and $B = 0$ so that $f(z) = \operatorname{Ln} z$. We verified in Example 1, Section 20.1 that $f(z) = \operatorname{Ln} z$ maps the upper half-plane $y \geq 0$ to the horizontal strip $0 \leq v \leq \pi$.
12. From (3), $f'(z) = Az^{-3/4}(z-1)^{\alpha_2/\pi-1}$. But $\alpha_2 \rightarrow \pi$ as $\theta \rightarrow 0$. This suggests that we examine $f'(z) = Az^{-3/4}$. Therefore $f(z) = A_1 z^{1/4} + B_1$. But $f(0) = 0$ and $f(1) = 1$ so that $B_1 = 0$ and $A_1 = 1$. Hence $f(z) = z^{1/4}$ and we recognize that this power function maps the upper half-plane onto the wedge $0 \leq \operatorname{Arg} w \leq \pi/4$.
13. From (3), $f'(z) = A(z+1)^{\alpha_1/\pi-1} z^{\alpha_2/\pi-1} (z-1)^{\alpha_3/\pi-1}$. But as $u_1 \rightarrow 0$, $\alpha_1 \rightarrow \pi/2$, $\alpha_2 \rightarrow 2\pi$, and $\alpha_3 \rightarrow \pi/2$. This suggests that we examine

$$f'(z) = A(z+1)^{-1/2} z(z-1)^{-1/2} = A \frac{z}{(z^2 - 1)^{1/2}}.$$

Therefore $f(z) = A(z^2 - 1)^{1/2} + B$. But $f(-1) = f(1) = 0$ and $f(0) = ai$. This implies that $B = 0$ and $ai = Ai$ or $A = a$. Therefore $f(z) = a(z^2 - 1)^{1/2}$. By expressing $f(z)$ as the composite of $z_1 = z^2$, $z_2 = z_1 - 1$, $z_3 = z_2^{1/2}$ and $w = az_3$ we can show that the image of the upper half-plane is R' .

14. If $w(t) = u(t) + iv(t)$, then $w'(t) = u'(t) + iv'(t)$ and so

$$\tan(\arg w'(t)) = \frac{v'(t)}{u'(t)} = \frac{dv}{du}.$$

If $\arg w'(t)$ is constant, then $dv/du = m$ or $v = mu + b$ for some constants m and b .

Exercises 20.5

1. Using (3) with $x_0 = -1$, $x_1 = 0$, $x_2 = 1$ and $u_1 = -1$ and $u_2 = 1$,

$$u = -\frac{1}{\pi} \operatorname{Arg} \left(\frac{z}{z+1} \right) + \frac{1}{\pi} \operatorname{Arg} \left(\frac{z-1}{z} \right).$$

2. Using (3) with $x_0 = -2$, $x_1 = 0$, $x_2 = 1$ and $u_1 = 5$ and $u_2 = 1$,

$$u = \frac{5}{\pi} \operatorname{Arg} \left(\frac{z}{z+2} \right) + \frac{1}{\pi} \operatorname{Arg} \left(\frac{z-1}{z} \right).$$

3. The harmonic function

$$u_1 = \frac{5}{\pi} [\pi - \operatorname{Arg}(z-1)] = \begin{cases} 5, & x > 1 \\ 0, & x < 1 \end{cases}, \quad \text{and} \quad u_2 = -\frac{1}{\pi} \operatorname{Arg} \left(\frac{z+1}{z+2} \right) + \frac{1}{\pi} \operatorname{Arg} \left(\frac{z}{z+1} \right)$$

from (3) satisfies all boundary conditions except that $u_2 = 0$ for $x > 1$. Therefore $u = u_1 + u_2$ is the solution to the given Dirichlet problem.

4. The harmonic function

$$u_1 = \frac{1}{\pi} \operatorname{Arg}(z+2) = \begin{cases} 1, & x < -2 \\ 0, & x > -2 \end{cases}, \quad \text{and} \quad u_2 = -\frac{1}{\pi} \operatorname{Arg} \left(\frac{z+1}{z+2} \right) + \frac{1}{\pi} \operatorname{Arg} \left(\frac{z+1}{z-1} \right)$$

Exercises 20.5

satisfies all boundary conditions except that $u_2 = 0$ for $x < -2$. Therefore $u = u_1 + u_2$ is the solution to the given Dirichlet problem.

5. By Theorem 20.5,

$$u(x, y) = \frac{y}{\pi} \int_0^1 \frac{t^2}{(x-t)^2 + y^2} dt.$$

If we let $s = x - t$, this integral can be expressed in terms of the natural log and inverse tangent. Using *Maple* we obtain

$$u = \frac{y}{\pi} \left(\frac{y^2 - x^2}{y} \left[\tan^{-1} \left(\frac{x-1}{y} \right) - \tan^{-1} \left(\frac{x}{y} \right) \right] + x \ln \frac{(x-1)^2 + y^2}{x^2 + y^2} + 1 \right).$$

6. From Theorem 20.5,

$$u(x, y) = \frac{y}{\pi} \int_{-\infty}^{\infty} \frac{\cos t}{(x-t)^2 + y^2} dt = \frac{y}{\pi} \int_{-\infty}^{\infty} \frac{\cos(x-s)}{s^2 + y^2} ds$$

letting $s = x - t$. But $\cos(x-s) = \cos x \cos s + \sin x \sin s$. It follows that

$$u(x, y) = \frac{y \cos x}{\pi} \int_{-\infty}^{\infty} \frac{\cos s}{s^2 + y^2} ds + \frac{y \sin x}{\pi} \int_{-\infty}^{\infty} \frac{\sin s}{s^2 + y^2} ds = \frac{y \cos x}{\pi} \left(\frac{\pi e^{-y}}{y} \right) = e^{-y} \cos x, \quad y > 0.$$

7. The mapping $f(z) = z^2$ maps R onto the upper half-plane R' . The corresponding boundary value problem in R' is shown in the figure. From (3),

$$U = \frac{5}{\pi} \text{Arg}(w+1) + \frac{1}{\pi} \text{Arg} \left(\frac{w-1}{w} \right)$$

is the solution in R' . Therefore

$$u = U(f(z)) = \frac{5}{\pi} \text{Arg}(z^2+1) + \frac{1}{\pi} \text{Arg} \left(\frac{z^2-1}{z^2} \right)$$

is the solution to the original Dirichlet problem in R .

8. Using H-4 with $a = 3$, $f(z) = \cos(\pi z/3)$ maps R onto the upper half-plane R' . The corresponding Dirichlet problem in R' is shown in the figure. From (3),

$$U = \frac{1}{\pi} [\pi - \text{Arg}(w-1)] + \frac{1}{\pi} \text{Arg}(w+1)$$

is the solution in R' . Therefore

$$u = U(f(z)) = 1 + \frac{1}{\pi} [\text{Arg}(\cos(\pi z/3) + 1) - \text{Arg}(\cos(\pi z/3) - 1)]$$

is the solution to the original Dirichlet problem in R .

9. Using H-1, $f(z) = i \frac{1-z}{1+z}$ maps R onto the upper half-plane R' . The corresponding

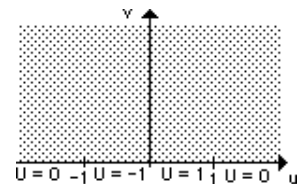
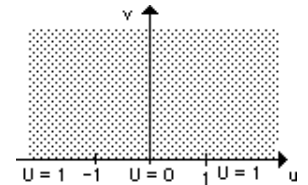
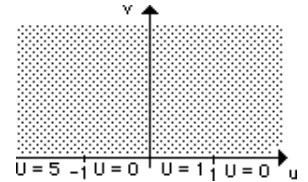
Dirichlet problem in R' is shown in the figure. From (3),

$$U = -\frac{1}{\pi} \text{Arg} \left(\frac{w}{w+1} \right) + \frac{1}{\pi} \text{Arg} \left(\frac{w-1}{w} \right) = \frac{1}{\pi} \left[\text{Arg} \left(\frac{w-1}{w} \right) - \text{Arg} \left(\frac{w}{w+1} \right) \right].$$

The harmonic function $u = U(f(z))$ may be simplified to

$$u = \frac{1}{\pi} \left[\text{Arg} \left(\frac{(1-i)z - (1+i)}{1-z} \right) - \text{Arg} \left(\frac{1-z}{-(1+i)z + 1-i} \right) \right]$$

and is the solution to the original Dirichlet problem in R .



10. Using H-5, $f(z) = \left(\frac{1+z}{1-z}\right)^2$ maps R onto the upper half-plane R' . The corresponding Dirichlet problem in R' is shown in the figure. From (3),

$$U = \frac{1}{\pi} \operatorname{Arg} \left(\frac{w}{w+1} \right) + \frac{1}{\pi} [\pi - \operatorname{Arg}(w-1)] = 1 - \frac{1}{\pi} \operatorname{Arg}(w-1) + \frac{1}{\pi} \operatorname{Arg} \left(\frac{w}{w+1} \right).$$

The harmonic function $u = U(f(z))$ may be simplified to

$$u = 1 - \frac{1}{\pi} \operatorname{Arg} \left(\frac{4z}{(1-z)^2} \right) + \frac{1}{\pi} \operatorname{Arg} \left(\frac{(1+z)^2}{2(1+z^2)} \right).$$

11. From Theorem 20.6, $u(x, y) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{t^2}{\pi^2} \frac{1-|z|^2}{|e^{it}-z|^2} dt$. Therefore,

$$u(0, 0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{t^2}{\pi^2} dt = \frac{1}{3}.$$

To estimate $u(0.5, 0)$ and $u(-0.5, 0)$ we must use a numerical integration method. With the aid of Simpson's Rule, $u(0.5, 0) = 0.1516$ and $u(-0.5, 0) = 0.5693$.

12. From Theorem 20.6, $u(x, y) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-|t|} \frac{1-|z|^2}{|e^{it}-z|^2} dt$. Therefore,

$$u(0, 0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-|t|} dt = \frac{1}{\pi} \int_0^{\pi} e^{-t} dt = \frac{1}{\pi} (1 - e^{-\pi}).$$

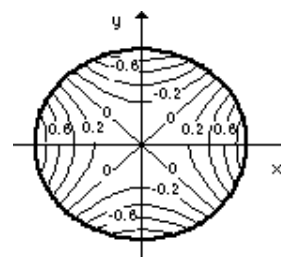
With the aid of Simpson's Rule, $u(0.5, 0) = 0.5128$ and $u(-0.5, 0) = 0.1623$.

13. $u(0, 0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} u(e^{it}) dt$. The latter integral is just the average value of $u(e^{it})$ for $-\pi \leq t \leq \pi$.

14. For $u(e^{i\theta}) = \cos 2\theta$, the Fourier series solution (6) reduces to

$$u(r, \theta) = r^2 \cos 2\theta = \operatorname{Re}(z^2) \quad \text{or} \quad u(x, y) = x^2 - y^2.$$

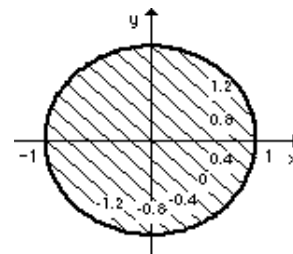
The corresponding system of level curves is shown in the figure.



15. For $u(e^{i\theta}) = \sin \theta + \cos \theta$, the Fourier series solution (6) reduces to

$$u(r, \theta) = r \sin \theta + r \cos \theta \quad \text{or} \quad u(x, y) = y + x.$$

The corresponding system of level curves is shown in the figure.

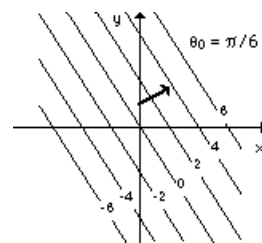


Exercises 20.6

1. $g(z) = \cos \theta_0 - i \sin \theta_0 = e^{-i\theta_0}$ is analytic everywhere and so $\operatorname{div} \mathbf{F} = 0$ and $\operatorname{curl} \mathbf{F} = \mathbf{0}$ by Theorem 20.7. A complex potential is $G(z) = e^{-i\theta_0} z$ and

$$\phi(x, y) = \operatorname{Re}(G(z)) = x \cos \theta_0 + y \sin \theta_0.$$

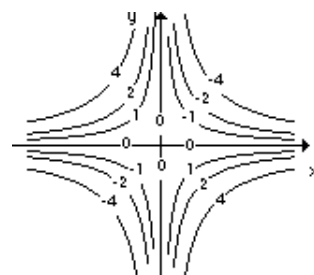
The equipotential lines (corresponding to $\theta_0 = \pi/6$) are shown in the figure.



2. $g(z) = -y + xi = i(x + iy) = iz$ is analytic everywhere and so $\operatorname{div} \mathbf{F} = 0$ and $\operatorname{curl} \mathbf{F} = \mathbf{0}$ by Theorem 20.7. A complex potential is $G(z) = \frac{i}{2} z^2$ and

$$\phi(x, y) = \operatorname{Re}\left(\frac{i}{2} z^2\right) = -xy.$$

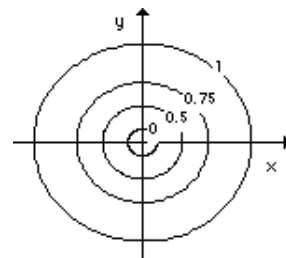
The equipotential lines $-xy = c$ are shown in the figure and are hyperbolas.



3. $g(z) = \frac{x}{x^2 + y^2} - \frac{y}{x^2 + y^2} i = \frac{1}{z}$ is analytic for $z \neq 0$ and so $\operatorname{div} \mathbf{F} = 0$ and $\operatorname{curl} \mathbf{F} = \mathbf{0}$ by Theorem 20.7. A complex potential is $G(z) = \operatorname{Ln} z$ and

$$\phi(x, y) = \operatorname{Re}(G(z)) = \frac{1}{2} \log_e(x^2 + y^2).$$

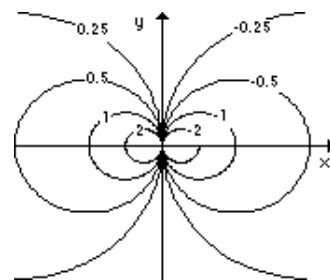
The equipotential lines $\phi(x, y) = c$ are circles $x^2 + y^2 = e^{2c}$ and are shown in the figure.



4. $g(z) = \frac{x^2 - y^2 - 2xyi}{(x^2 + y^2)^2} = \frac{1}{z^2}$ is analytic for $z \neq 0$ and so $\operatorname{div} \mathbf{F} = 0$ and $\operatorname{curl} \mathbf{F} = \mathbf{0}$ by Theorem 20.7. A complex potential is $G(z) = -\frac{1}{z}$ and

$$\phi(x, y) = \operatorname{Re}\left(-\frac{1}{z}\right) = -\frac{x}{x^2 + y^2}.$$

The equipotential lines $-\frac{x}{x^2 + y^2} = c$ can be written as $\left(x + \frac{1}{2c}\right)^2 + y^2 = \left(\frac{1}{2c}\right)^2$ for $c \neq 0$. See the figure.



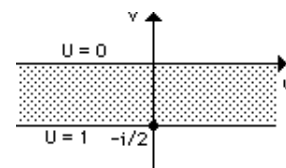
5. The mapping $f(z) = z^4$ maps the wedge $0 \leq \operatorname{Arg} z \leq \pi/4$ to the upper half-plane R' and $U = \frac{1}{\pi} \operatorname{Arg} w$ is the solution to the corresponding Dirichlet problem in R' . Therefore

$$\phi(x, y) = U(z^4) = \frac{1}{\pi} \operatorname{Arg} z^4 = \frac{4}{\pi} \operatorname{Arg} z$$

is the potential in the wedge. A complex potential is $G(z) = \frac{4}{\pi} \operatorname{Ln} z$ and, since $\phi(r, \theta) = \frac{4}{\pi} \theta$, the equipotential lines are the rays $\theta = \frac{\pi}{4} c$. Finally

$$\mathbf{F} = \overline{G'(z)} = \frac{4}{\pi} \frac{1}{\bar{z}} = \frac{4}{\pi} \left(\frac{x}{x^2 + y^2}, \frac{y}{x^2 + y^2} \right).$$

6. The function $f(z) = \frac{1}{z}$ maps the original region R to the strip $-\frac{1}{2} \leq v \leq 0$ (see Example 2, Section 20.1). The boundary conditions transfer as shown in the figure.



$U = -2v$ is the solution in the horizontal strip and so

$$\phi(x, y) = -2\operatorname{Im}\left(\frac{1}{z}\right) = \frac{2y}{x^2 + y^2}$$

is the potential in the original region R . The equipotential lines $\frac{2y}{x^2 + y^2} = c$ may be written as

$x^2 + \left(y + \frac{1}{c}\right)^2 = \left(\frac{1}{c}\right)^2$ for $c \neq 0$ and are circles. If $c = 0$, we obtain the line $y = 0$. Note that

$\phi(x, y) = \operatorname{Re}\left(\frac{2i}{z}\right)$ and so $G(z) = \frac{2i}{z}$ is a complex potential. The corresponding vector field is

$$\mathbf{F} = \overline{G'(z)} = \frac{2i}{\bar{z}^2} = \left(\frac{-4xy}{(x^2 + y^2)^2}, \frac{2(x^2 - y^2)}{(x^2 + y^2)^2} \right).$$

7. Using H-5, $w = \left(\frac{1-z}{1+z}\right)^2 = \left(\frac{z-1}{z+1}\right)^2$ maps R onto the upper half-plane R' and $U = \frac{1}{\pi}\operatorname{Arg} w$ is the solution to the corresponding Dirichlet problem in R' . Therefore

$$\mu = \frac{1}{\pi}\operatorname{Arg}\left(\frac{z-1}{z+1}\right)^2.$$

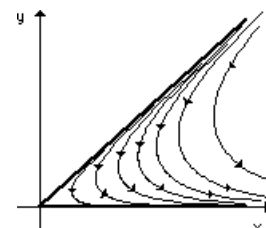
In R' the equipotential lines are rays $\theta = \theta_0$. The inverse transformation is $z = S(T(w))$ where $T(w) = w^{1/2}$ and $S(w) = \frac{1+w}{1-w}$. T maps the ray $\theta = \theta_0$ to the ray $\theta = \theta_0/2$ and S maps $\theta = \theta_0/2$ to an arc of a circle since $S(0) = 1$ and $S(\infty) = -1$ and S is a linear fractional transformation.

8. Using C-1 with $b = 2$ and $c = 4$, we have $a = \frac{3+\sqrt{5}}{2}$ and $r_0 = \frac{7-3\sqrt{5}}{2}$, so $T(z) = \frac{z-a}{az-1}$ maps R onto the annular region R' defined by $r_0 \leq |w| \leq 1$. The solution to the corresponding Dirichlet problem in R' is $U = \frac{\log_e |w|}{\log_e r_0}$ and so

$$\phi = \frac{1}{\log_e r_0} \log_e \left| \frac{z-a}{az-1} \right|.$$

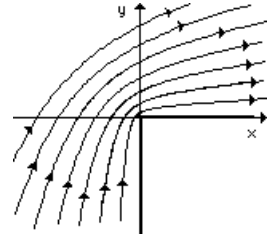
The level curves of U are circles $|w| = r$ where $r_0 < r < 1$, and the equipotential lines $\phi(x, y) = c$ are the images of these circles under the inverse transformation $T^{-1}(w) = \frac{-w+a}{-aw+1}$. T^{-1} has a pole at $w = \frac{1}{a} = \frac{2}{3+\sqrt{5}}$ (≈ 0.38) and $r_0 < \frac{1}{a} < 1$. All circles $|w| = r$ are mapped onto circles in the z -plane with the exception of $|w| = \frac{1}{a}$ which is mapped onto a line.

9. (a) $\psi(x, y) = \operatorname{Im}(z^4) = 4xy(x^2 - y^2)$ and so $\psi(x, y) = 0$ when $y = x$ and $y = 0$.
 (b) $\mathbf{V} = \overline{G'(z)} = \overline{4z^3} = 4(x^3 - 3xy^2, y^3 - 3x^2y)$
 (c) In polar coordinates $r^4 \sin 4\theta = c$ or $r = (c \csc 4\theta)^{1/4}$, for $0 < \theta < \pi/4$, are the streamlines. See the figure.

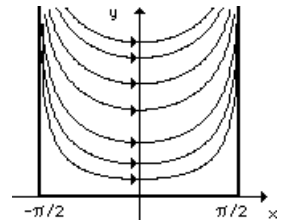


Exercises 20.6

10. (a) Since $G(re^{i\theta}) = r^{2/3}e^{i2\theta/3}$, $\psi(r, \theta) = \text{Im}(G(re^{i\theta})) = r^{2/3} \sin \frac{2\theta}{3}$. Note that $\psi = 0$ on the boundary where $\theta = 0$ and $\theta = 3\pi/2$.
- (b) $\mathbf{V} = \overline{G'(z)} = \frac{2}{3}z^{-1/3}$. Therefore, letting $z = re^{i\theta}$, $\mathbf{V} = \frac{2}{3}r^{-1/3}(\cos(\theta/3), \sin(\theta/3))$ for $0 < \theta < 3\pi/2$.
- (c) $r^{2/3} \sin(2\theta/3) = c$ implies that $r = [c \csc(2\theta/3)]^{2/3}$ for $0 < \theta < 3\pi/2$. The streamlines are shown in the figure.



11. (a) $\psi(x, y) = \text{Im}(\sin z) = \cos x \sinh y$ and $\psi(x, y) = 0$ when $x = \pm\pi/2$ or when $y = 0$.
- (b) $\mathbf{V} = \overline{G'(z)} = \overline{\cos z} = (\cos x \cosh y, \sin x \sinh y)$.
- (c) $\cos x \sinh y = c \implies y = \sinh^{-1}(c \sec x)$ and the streamlines are shown in the figure.



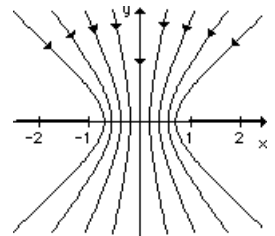
12. (a) The image of R under $w = i \sin^{-1} z$ is the horizontal strip (see E-6) $-\pi/2 \leq v \leq \pi/2$ and

$$\psi(x, y) = \text{Im}(i \sin^{-1} z) = \begin{cases} \pi/2, & x \geq 1 \\ -\pi/2, & x \leq -1 \end{cases}.$$

Each piece of boundary is therefore a streamline.

(b) $\mathbf{V} = \overline{G'(z)} = \frac{i}{(1 - z^2)^{1/2}} = \frac{-i}{(1 - \bar{z}^2)^{1/2}}$

- (c) The streamlines are the images of the lines $v = b$, $-\pi/2 < b < \pi/2$ under $z = -i \sin w$ and are therefore hyperbolas. See Example 2, Section 20.2, and the figure. Note that at $z = 0$, $v = -i$ and the flow is downward.



13. (a) If $z = re^{i\theta}$, $G(re^{i\theta}) = r^2 e^{2i\theta} + \frac{1}{r^2} e^{-2i\theta}$ and so

$$\psi(r, \theta) = \text{Im}(G(re^{i\theta})) = \left(r^2 - \frac{1}{r^2}\right) \sin 2\theta.$$

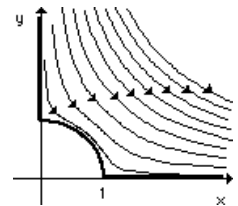
Note that $\psi = 0$ when $r = 1$ or when either $\theta = 0$ or $\theta = \pi/2$. Therefore $\psi = 0$ on the boundary of R .

- (b) $\mathbf{V} = \overline{G'(z)} = \overline{(2z - 2z^{-3})}$ and so in polar coordinates

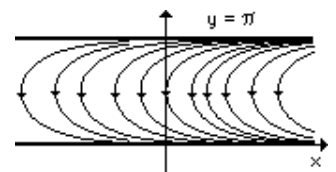
$$\mathbf{V} = 2re^{-i\theta} - \frac{2}{r^3} e^{3i\theta} = 2(r \cos \theta - r^{-3} \cos 3\theta, -r \sin \theta - r^{-3} \sin 3\theta).$$

- (c) In rectangular coordinates, the streamlines are

$$\psi(x, y) = 2xy \left[1 - \frac{1}{(x^2 + y^2)^2}\right] = c.$$



14. (a) $\psi(x, y) = \text{Im}(e^z) = e^x \sin y$ and so $\psi = 0$ when $y = 0$ or π . Therefore $\psi = 0$ on the boundary of R .
- (b) $\mathbf{V} = \overline{G'(z)} = \overline{e^z} = (e^x \cos y, -e^x \sin y)$
- (c) The streamlines are $e^x \sin y = c$ and so $x = \log_e(c \csc y)$ for $0 < y < \pi$. See the figure.

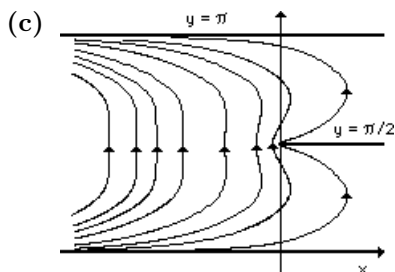


15. (a) For $f(z) = \pi i - \frac{1}{2}[\text{Ln}(z+1) + \text{Ln}(z-1)]$

$$f(t) = \pi i - \frac{1}{2}[\log_e |t+1| + \log_e |t-1| + i\text{Arg}(t+1) + i\text{Arg}(t-1)]$$

and so $\text{Im}(f(t)) = \begin{cases} 0, & t < -1 \\ \pi/2, & -1 < t < 1 \\ \pi, & t > 1 \end{cases}$. Hence $\text{Im}(G(z)) = \psi(x, y) = 0$ on the boundary of R .

(b) $x = -\frac{1}{2}[\log_e |t+1+ic| + \log_e |t-1+ic|]$, $y = \pi - \frac{1}{2}[\text{Arg}(t+1+ic) + \text{Arg}(t-1+ic)]$ for $c > 0$

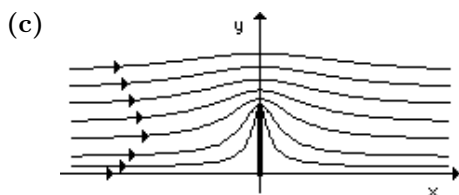


16. (a) For $f(z) = (z^2 - 1)^{1/2}$,

$$f(t) = |t^2 - 1|^{1/2} \cos\left(\frac{1}{2}\text{Arg}(t^2 - 1)\right) + i|t^2 - 1|^{1/2} \sin\left(\frac{1}{2}\text{Arg}(t^2 - 1)\right)$$

and so $f(t) = \begin{cases} |t^2 - 1|^{1/2}, & |t| > 1 \\ i|t^2 - 1|^{1/2}, & |t| < 1 \end{cases}$. Hence $\text{Im}(G(z)) = 0$ on the boundary of R .

(b) $x = |(t+ic)^2 - 1|^{1/2} \cos\left(\frac{1}{2}\text{Arg}((t+ic)^2 - 1)\right)$, $y = |(t+ic)^2 - 1|^{1/2} \sin\left(\frac{1}{2}\text{Arg}((t+ic)^2 - 1)\right)$ for $c > 0$



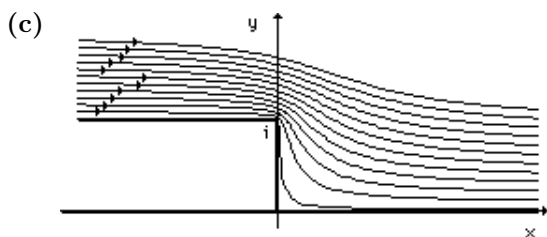
17. (a) For $f(z) = \frac{1}{\pi}[(z^2 - 1)^{1/2} + \cosh^{-1} z]$,

$$f(t) = \frac{1}{\pi} \left[(t^2 - 1)^{1/2} + \cosh^{-1} t \right] = \frac{1}{\pi} \left[(t^2 - 1)^{1/2} + \text{Ln}(t + (t^2 - 1)^{1/2}) \right]$$

and so $\text{Im}(f(t)) = \begin{cases} 1, & t < -1 \\ 0, & t > 1 \end{cases}$ and $\text{Re}(f(t)) = 0$, for $-1 < t < 1$. Hence $\text{Im}(G(z)) = \psi(x, y) = 0$ on the boundary of R .

(b) $x = \text{Re}\left(\frac{1}{\pi} \left[((t+ic)^2 - 1)^{1/2} + \cosh^{-1}(t+ic) \right]\right)$,

$y = \text{Im}\left(\frac{1}{\pi} \left[((t+ic)^2 - 1)^{1/2} + \cosh^{-1}(t+ic) \right]\right)$ for $c > 0$



Exercises 20.6

18. (a) For $f(z) = 2(z+1)^{1/2} + \text{Ln} \left(\frac{(z+1)^{1/2} - 1}{(z+1)^{1/2} + 1} \right)$,

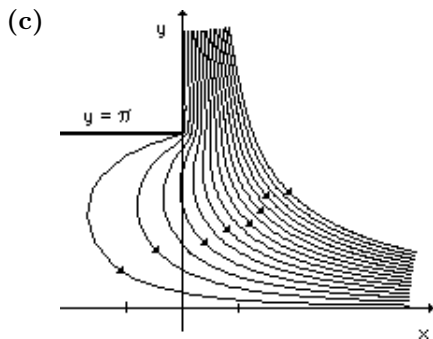
$$f(t) = 2(t+1)^{1/2} + \text{Ln} \frac{(t+1)^{1/2} - 1}{(t+1)^{1/2} + 1}.$$

If we write $(t+1)^{1/2} = |t+1|^{1/2} e^{(i/2)\text{Arg}(t+1)}$, we may conclude that

$$\text{Im}(f(t)) = \begin{cases} 0, & t > 0 \\ \pi, & -1 < t < 0 \end{cases} \quad \text{and} \quad \text{Re}(f(t)) = 0 \text{ for } t < -1.$$

Therefore $\text{Im}(G(z)) = \psi(x, y) = 0$ on the boundary of R .

$$\begin{aligned} \text{(b)} \quad x &= \text{Re} \left[2(t+ic+1)^{1/2} + \text{Ln} \frac{(t+ic+1)^{1/2} - 1}{(t+ic+1)^{1/2} + 1} \right] \\ y &= \text{Im} \left[2(t+ic+1)^{1/2} + \text{Ln} \frac{(t+ic+1)^{1/2} - 1}{(t+ic+1)^{1/2} + 1} \right] \quad \text{for } c > 0 \end{aligned}$$



19. In Example 5, $\mathbf{V} = (2x, -2y)$ and so the only stagnation point occurs at $z = 0$. In Example 6, $\mathbf{V} = 1 - 1/\bar{z}^2$ and so if $\mathbf{V} = \mathbf{0}$, $\bar{z}^2 = 1$. Therefore $z = 1, -1$ are the only stagnation points.
20. (a) $\psi(x, y) = \text{Im}(G(z)) = k\text{Arg}(z - x_1)$ and so if $\psi(x, y) = c$, $\text{Arg}(z - x_1)$ is constant. This implies that the streamlines are rays with vertex at $z = x_1$.

$$\text{(b)} \quad \mathbf{V} = \overline{G'(z)} = \frac{\overline{k}}{z - x_1} = \frac{k}{\overline{z - x_1}} = \frac{k}{|z - x_1|^2} (z - x_1).$$

The direction of the flow is determined by the sign of k , and if $k < 0$ the flow is directed towards $z = x_1$.

21. $f(z) = z^2$ maps the first quadrant onto the upper half-plane and $f(\xi_0) = f(1) = 1$. Therefore $G(z) = \text{Ln}(z^2 - 1)$ is the complex potential, and so

$$\psi(x, y) = \text{Arg}(z^2 - 1) = \tan^{-1} \left(\frac{2xy}{x^2 - y^2 - 1} \right)$$

is the streamline function where $\tan^{-1}$ is chosen to be between 0 and π . If $\psi(x, y) = c$, then $x^2 + Bxy - y^2 - 1 = 0$ where $B = -2 \cot c$. Each hyperbola in the family passes through $(1, 0)$ and the boundary of the first quadrant satisfies $\psi(x, y) = 0$.

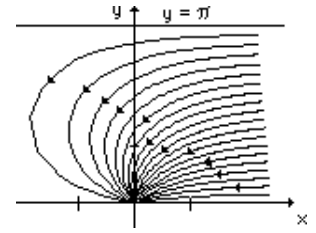
22. (a) From E-5, $f(z) = e^z$ maps the horizontal strip $0 < y < \pi$ onto the upper half-plane and $f(\xi_0) = f(0) = 1$. Therefore $G(z) = k\text{Ln}(e^z - 1)$ is a complex potential. To construct a sink at $\xi_0 = 0$, we must have $k < 0$.

$$(b) \psi(x, y) = \operatorname{Im}(k \operatorname{Ln}(e^z - 1)) = k \operatorname{Arg}(e^z - 1) = k \tan^{-1} \left(\frac{e^x \sin y}{e^x \cos y - 1} \right)$$

where $\tan^{-1}$ is chosen to be between 0 and π . If we set $k = -1$, then the streamlines $\psi(x, y) = c$, $-\pi < c < 0$, satisfy $e^x[B \cos y - \sin y] = B$ where $B = -\tan c$, and so

$$x = \log_e \left[\frac{B}{B \cos y - \sin y} \right].$$

Note that B will vary over all real values. The streamlines are also the images of rays through $w = 1$ under the inverse transformation $z = \operatorname{Ln} w$. See the figure.



$$23. \psi = \operatorname{Im}(G(z)) = k \operatorname{Arg}(z - 1) - k \operatorname{Arg}(z + 1) = k \operatorname{Arg} \left(\frac{z - 1}{z + 1} \right).$$

(See Figure 20.55 in the text). In rectangular coordinates

$$\psi(x, y) = k \tan^{-1} \left(\frac{2y}{x^2 + y^2 - 1} \right)$$

where $\tan^{-1}$ is chosen to be between 0 and π . Level curves $\psi(x, y) = c$ can be put in the form

$$x^2 + y^2 - 2By = 1 \quad \text{or} \quad x^2 + (y - B)^2 = 1 + B^2.$$

Each member of the family passes through $(1, 0)$ and $(-1, 0)$.

$$24. (a) \mathbf{V} = \frac{a + ib}{\bar{z}} = \left(\frac{ax - by}{x^2 + y^2}, \frac{bx + ay}{x^2 + y^2} \right) \quad \text{and since } (x'(t), y'(t)) = \mathbf{V}, \text{ the path of the particle satisfies}$$

$$\frac{dx}{dt} = \frac{ax - by}{x^2 + y^2}, \quad \frac{dy}{dt} = \frac{bx + ay}{x^2 + y^2}.$$

(b) Switching to polar coordinates,

$$\begin{aligned} \frac{dr}{dt} &= \frac{1}{r} \left(x \frac{dx}{dt} + y \frac{dy}{dt} \right) = \frac{1}{r} \left(\frac{ax^2 - bxy}{r^2} + \frac{bxy + ay^2}{r^2} \right) = \frac{a}{r} \\ \frac{d\theta}{dt} &= \frac{1}{r^2} \left(-y \frac{dx}{dt} + x \frac{dy}{dt} \right) = \frac{1}{r^2} \left(\frac{-axy + by^2}{r^2} + \frac{bx^2 + axy}{r^2} \right) = \frac{b}{r^2}. \end{aligned}$$

Therefore $\frac{dr}{d\theta} = \frac{a}{b} r$ and so $r = ce^{a\theta/b}$.

(c) $\frac{dr}{dt} = \frac{a}{r} < 0$ if and only if $a < 0$, and in this case r is decreasing and the curve spirals inward. $\frac{d\theta}{dt} = \frac{b}{r^2} < 0$ if and only if $b < 0$, and in this case θ is decreasing and the curve is traversed clockwise.

Chapter 20 Review Exercises

1. $f(z) = x^2 - y^2 + 2xyi$ and so the hyperbola $xy = 2$ is mapped onto the line $v = 4$.
2. $-i = e^{-i\pi/2}$ and so $f(z) = -iz$ is a rotation through -90° .
3. The wedge $0 \leq \operatorname{Arg} w \leq 2\pi/3$. See figure 20.6 in the text.
4. $f'(z) = \sinh z = 0$ for $z = \pm\pi i$, and so $f(z) = \cosh z$ is conformal except at $z = \pm n\pi i$.

Chapter 20 Review Exercises

5. True, by Theorem 20.2. $\operatorname{Arg} w$ is harmonic and is the upper half-plane $v \geq 0$.
6. A line, since $|z - 1| = 1$ passes through the pole at $z = 2$.
7. $0, 1, \infty$.
8. True. $\alpha_1 = \alpha_2 = \alpha_3 = \pi/2$
9. False. $\overline{g(z)} = P - iQ$ is analytic. See Theorem 20.7.
10. $iG(z) = -\psi(x, y) + i\phi(x, y)$ is analytic in R and $\operatorname{Im}(G(z)) = \phi(x, y)$. True
11. If $z = re^{i\theta}$ and $0 < \theta < \pi/2$, $w = \operatorname{Ln} z = \log_e r + i\theta$. Therefore $v = \theta$ and so $0 < v < \pi/2$. The image of the first quadrant is the strip $0 < v < \pi/2$ in the w -plane. Rays $\theta = \theta_0$ are mapped onto horizontal lines $v = \theta_0$.
12. First use $z_1 = z^2$ to map the first quadrant onto the upper half-plane $y_1 \geq 0$, and segment AB to the negative real axis. We then use $w = \frac{1}{\pi}(z_1 + \operatorname{Ln} z_1 + 1)$ to map this half-plane onto the target region R' . The composite transformation is

$$w = \frac{1}{\pi}[z^2 + \operatorname{Ln}(z^2) + 1]$$

and the image of AB is the ray extending to the left from $w = i$ along the line $v = 1$.

13. From H-4, $z_1 = \cos \pi z$ maps R onto the upper half-plane $y_1 \geq 0$, and AB onto the real segment $(-\infty, -1]$. Using C-4, $w = \frac{i - z_1}{1 + z_1}$ maps this upper half-plane onto the target region R' . The composite transformation is

$$w = \frac{i - \cos \pi z}{i + \cos \pi z}$$

and the image of AB is the circular arc lying in quadrants II and III.

14. The power function $w = z^4$ maps R onto the upper half-plane $v \geq 0$; the transferred boundary conditions are shown in the figure. From (2) of Section 20.5, $U = \frac{1}{\pi} \operatorname{Arg} \left(\frac{w-1}{w+1} \right)$ and so

$$u = \frac{1}{\pi} \operatorname{Arg} \left(\frac{z^4 - 1}{z^4 + 1} \right)$$

is the solution to the original Dirichlet problem in R .

15. The inversion $w = 1/z$ maps R onto the horizontal strip $-1 \leq v \leq -1/2$ and the transferred boundary conditions are shown in the figure. The solution in the strip is $U = 2v + 2$ and so

$$u = U \left(\frac{1}{z} \right) = 2 \operatorname{Im} \left(\frac{1}{z} \right) + 2 = \frac{-2y}{x^2 + y^2} + 2$$

is the solution to the original Dirichlet problem in R .

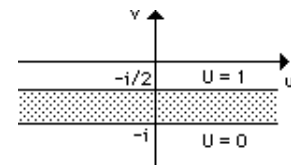
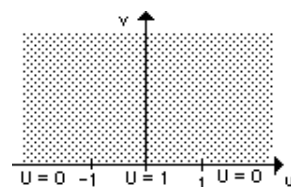
16. Using the cross ratio formula,

$$\frac{(w-i)(-i+1)}{(w+1)(-2i)} = \frac{(z-1) \cdot 1}{1 \cdot (-2)}$$

we see that $w = -\frac{z-i}{z+i}$ maps $1, -1, \infty$ to $i, -i, -1$.

17. (a) We note that as $u_1 \rightarrow +\infty$, $\alpha_1 \rightarrow 0$, $\alpha_2 \rightarrow 2\pi$, and $\alpha_3 \rightarrow 0$.

(b) Since $f'(z) = A(z+1) \mathcal{L}_{1/\pi-1} z^{\alpha_2/\pi-1} (z-1)^{\alpha_3/\pi-1}$,



this suggests that we examine

$$f'(z) = A(z+1)^{-1}z(z-1)^{-1} = A \frac{z}{z^2-1}.$$

We may write

$$f'(z) = \frac{A}{2} \left(\frac{1}{z+1} + \frac{1}{z-1} \right)$$

and so

$$f(z) = \frac{1}{2} A [\text{Ln}(z+1) + \text{Ln}(z-1)] + B.$$

(c) For t real, we can write

$$f(t) = \frac{1}{2} A [\log_e |t+1| + \log_e |t-1| + i\text{Arg}(t+1) + i\text{Arg}(t-1)] + B.$$

Since $f(0) = \frac{\pi i}{2}$, $\frac{\pi i}{2} = \frac{\pi i}{2} A + B$. For t real,

$$f(t) = \begin{cases} \frac{1}{2} A [\ln |t^2 - 1| + 2\pi i] + B, & t < -1 \\ \frac{1}{2} A \ln |t^2 - 1| + B, & t > 1 \end{cases}.$$

If A is real and we require that $\text{Im}(f(t)) = 0$ for $t < -1$, then $0 = A\pi i + \text{Im}(B)$. If $\text{Im}(f(t)) = \pi$ for $t > 1$, then $\pi = \text{Im}(B)$. All three equations are satisfied by letting $B = \pi i$ and $A = -1$. Therefore

$$f(z) = \pi i - \frac{1}{2} [\text{Ln}(z+1) + \text{Ln}(z-1)].$$

18. (a) From Theorem 20.5,

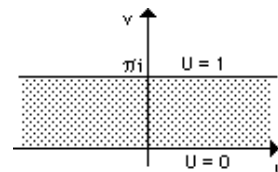
$$u(x, y) = \frac{y}{\pi} \int_{-\infty}^{\infty} \frac{\sin t}{(x-t)^2 + y^2} dt = \frac{y}{\pi} \int_{-\infty}^{\infty} \frac{\sin(x-s)}{s^2 + y^2} ds \quad (\text{letting } s = x-t).$$

But $\sin(x-s) = \sin x \cos s - \cos x \sin s$. We now proceed as in the solution to Problem 6, Section 20.5 to show that $u(x, y) = e^{-y} \sin x$.

(b) For $u(e^{i\theta}) = \sin \theta$, the Fourier Series solution (6) in Section 20.5 reduces to $u(r, \theta) = r \sin \theta$.

19. If $f(w) = w + e^w + 1$, $G(z) = f^{-1}(z)$ maps R onto the strip $0 \leq v \leq \pi$ and the transferred boundary conditions are shown in the figure. The solution for the strip is $U = v/\pi$ and so the solution in R is

$$\phi(x, y) = U(G(z)) = \frac{1}{\pi} \text{Im}(G(z)) = \frac{1}{\pi} \psi(x, y).$$



Therefore the equipotential lines $\phi(x, y) = c$ are the streamlines $\psi(x, y) = c\pi$ of the flow in Figure 20.72.

20. $G(re^{i\theta}) = -r^{1/2} \sin(\theta/2) + i[r^{1/2} \cos(\theta/2) - 1]$ and so $\psi(r, \theta) = \sqrt{r} \cos(\theta/2) = 1$. If $\psi(r, \theta) = 0$, $r \cos^2(\theta/2) = 1$ or $r \cos \theta + r = 2$ (using $\cos^2(\theta/2) = (1 + \cos \theta)/2$.) In rectangular coordinates, $x + \sqrt{x^2 + y^2} = 2$ or $y^2 = 4 - 4x$. Therefore the boundary of R is a streamline. To sketch the streamlines, note that $\psi(r, \theta) = c$ implies that $r = (c+1)^2 \sec^2(\theta/2)$. See the figure.

